

# Mastercard Network Processing

Dual Message Authorization System Guide

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# Mastercard Network Processing Dual Message Authorization System Guide

This document contains technical information and specifications for customers to understand the system functionality and to process transactions through the [Dual Message Authorization System](#). System specifications are based on Mastercard's adoption of the [International Organization for Standardization \(ISO\) 8583:1987](#) message formats.

## Audience

This document is intended for parties using the [Mastercard Network](#) to process transactions through the [Dual Message Authorization System](#). This includes acquirers, issuers, processors, and Network Enablement Partners.

This guide replaces the *Customer Interface Specification* and *Authorization Manual*.

## Product

The Mastercard Dual Message Authorization System is a global real-time message processing system and is a component of the Mastercard Network. The Mastercard Network supports real-time authorization processing followed by batch clearing processing, leading to the settlement of funds between acquirers and issuers.

## Details

| Metadata        | Value                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| Audience        | <ul style="list-style-type: none"><li>Acquirer</li><li>Issuer</li><li>Processor</li><li>Network Enablement Partner</li></ul> |
| Product         | Mastercard Network Processing and Operations                                                                                 |
| Region          | Global                                                                                                                       |
| Release version | 25.Q4                                                                                                                        |
| Type            | Technical specification guide                                                                                                |
| Publish date    | 4 November 2025                                                                                                              |

## Summary of changes, 4 November 2025

| Description of change                                                                                                                                                                                                           | Source                                                                                                                         | Where to look                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Added chapter titled Transaction Message Validation                                                                                                                                                                             | System clarification needed                                                                                                    | <a href="#">Transaction Message Validation</a>                                                                                                    |
| Revised Purpose for Administrative Advice/0620: system-generated message                                                                                                                                                        | System clarification needed                                                                                                    | <a href="#">Administrative/06xx messages</a>                                                                                                      |
| Revised flow for Unsolicited SAF after sign-on                                                                                                                                                                                  | System clarification needed                                                                                                    | <a href="#">Unsolicited SAF after sign-on</a>                                                                                                     |
| Added Values section to DE 13 (Date, Local Transaction)                                                                                                                                                                         | System clarification needed                                                                                                    | <a href="#">DE 13 (Date, Local Transaction)</a>                                                                                                   |
| Updated DE 20 (Primary Account Number [PAN] Country Code) to indicate that it is defined by and used for internal Mastercard processing and will not be present in Dual Message Authorization System messages sent to customers | <i>GLB 11347 Enhancing the Dual Message Authorization System Issuer Country Field</i>                                          | <a href="#">DE 20 (Primary Account Number [PAN] Country Code)</a>                                                                                 |
| Updated Application notes for DE 37 (Retrieval Reference Number)                                                                                                                                                                | <i>GLB 11326 Supporting Visa Authorization Advice Messages</i>                                                                 | <a href="#">DE 37 (Retrieval Reference Number)</a><br><a href="#">Subelement 04 (Acceptance Environment Response Data [Visa Only])</a>            |
| Clarified Response Code Value 61                                                                                                                                                                                                | System clarification needed                                                                                                    | <a href="#">Best practices for DE 39 values in Authorization Request Response/ 0110 response codes</a>                                            |
| Added subelement 02 (Special Acceptance Conditions Data) and subfield 01 (Mastercard One Credential Exclusion Indicator) to DE 48 (Additional Data: Private Use)                                                                | <i>AP/EUR/LAC/MEA 11242.4 Introducing Mastercard One Credential</i><br><i>US 11423.4 Introducing Mastercard One Credential</i> | <a href="#">Subelement 02 (Special Acceptance Conditions Data)</a><br><a href="#">Subfield 01 (Mastercard One Credential Exclusion Indicator)</a> |

| Description of change                                                                                                                                                                                                                                                                                                                                          | Source                                                                                                                             | Where to look                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added subfield 11 (Mastercard One Credential Account Number Indicator), subfield 12 (Mastercard One Credential Account Number), subfield 13 (Mastercard One Credential Account Number Expiration Date) to DE 48, subelement 09 (Additional PAN Data)                                                                                                           | <i>AP/EUR/LAC/MEA 11242.4 Introducing Mastercard One Credential</i><br><br><i>US 11423.4 Introducing Mastercard One Credential</i> | <a href="#">Subfield 11 (Mastercard One Credential Account Number Indicator)</a><br><br><a href="#">Subfield 12 (Mastercard One Credential Account Number)</a><br><br><a href="#">Subfield 13 (Mastercard One Credential Account Number Expiration Date)</a> |
| Updated DE 48, subelement 24 (Account Level Management [ALM] Service Data):                                                                                                                                                                                                                                                                                    | <i>AP/EUR/LAC/MEA 11242.4 Introducing Mastercard One Credential</i>                                                                | <a href="#">Subfield 05 (Mastercard One Credential Acceptance Brand)</a>                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Messages</li> <li>• ALM service names in subfield 1 (ALM Service Code)</li> <li>• Subfield 05 (Mastercard One Credential Acceptance Brand)</li> <li>• Subfield 06 (Mastercard One Credential Static/Dynamic Interchange Indicator)</li> <li>• Subfield 07 (Mastercard One Credential Issuer Account Range)</li> </ul> | <i>US 11423.4 Introducing Mastercard One Credential</i>                                                                            | <a href="#">Subfield 06 (Mastercard One Credential Static/Dynamic Interchange Indicator)</a><br><br><a href="#">Subfield 07 (Mastercard One Credential Issuer Account Range)</a>                                                                             |

| Description of change                                                                                                                                                                                                                                                                                                                                                                                                                             | Source                                                         | Where to look                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------|
| Updated the following:                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>GLB 11326 Supporting Visa Authorization Advice Messages</i> |                                                                            |
| <ul style="list-style-type: none"> <li>DE 48, subelement 04 (Acceptance Environment Response Data [Visa Only])</li> </ul>                                                                                                                                                                                                                                                                                                                         |                                                                | Subelement 04 (Acceptance Environment Response Data [Visa Only])           |
| <ul style="list-style-type: none"> <li>DE 48, subelement 05 (Additional Visa Response Data [Visa Only])</li> </ul>                                                                                                                                                                                                                                                                                                                                |                                                                | Subelement 05 (Additional Visa Response Data [Visa Only])                  |
| <ul style="list-style-type: none"> <li>DE 48, subelement 32 (Visa Merchant Identifier)</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |                                                                | Subelement 36 (Additional Visa Request Data [Visa Only])                   |
| <ul style="list-style-type: none"> <li>DE 48, subelement 36 (Additional Visa Request Data [Visa Only])</li> </ul>                                                                                                                                                                                                                                                                                                                                 |                                                                | Subelement 46 (Product ID [Visa Only])                                     |
| <ul style="list-style-type: none"> <li>DE 48, subelement 46 (Product ID)</li> </ul>                                                                                                                                                                                                                                                                                                                                                               |                                                                | Subelement 78 (Payment Service Indicators [Visa Only])                     |
| <ul style="list-style-type: none"> <li>DE 48, subelement 78 (Payment Service Indicators [Visa Only]) and these subfields:               <ul style="list-style-type: none"> <li>Subfield 1 (Spend Qualified Indicator)</li> <li>Subfield 2 (Dynamic Currency Conversion Indicator)</li> <li>Subfield 4 (Visa Checkout Indicator)</li> <li>Subfield 5 (Message Reason Code)</li> <li>Subfield 6 (Token Response Information)</li> </ul> </li> </ul> |                                                                | Subfield 1 (Spend Qualified Indicator)                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                | Subfield 2 (Dynamic Currency Conversion Indicator)                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                | Subfield 3 (U.S. Deferred Billing Indicator)                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                | Subfield 5 (Message Reason Code)                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                | Subfield 6 (Token Response Information)                                    |
| <ul style="list-style-type: none"> <li>DE 48, subelement 84 (Visa Response Codes [Visa Only])</li> </ul>                                                                                                                                                                                                                                                                                                                                          |                                                                | Subelement 84 (Visa Response Codes [Visa Only])                            |
| <ul style="list-style-type: none"> <li>DE 48, subelement 85 (Account Status [Visa Only])</li> </ul>                                                                                                                                                                                                                                                                                                                                               |                                                                | Subelement 85 (Account Status [Visa Only])                                 |
| <ul style="list-style-type: none"> <li>DE 48, subelement 86 (Relationship Participant Indicator)</li> </ul>                                                                                                                                                                                                                                                                                                                                       |                                                                | Subelement 86 (Relationship Participant Indicator [Visa Only])             |
| <ul style="list-style-type: none"> <li>DE 48, subelement 90 (Custom Payment Service Request [Visa Only])</li> </ul>                                                                                                                                                                                                                                                                                                                               |                                                                | Subelement 90 (Custom Payment Service Request [Visa Only])                 |
| <ul style="list-style-type: none"> <li>DE 48, subelement 90 (Custom Payment Service Request Response)</li> </ul>                                                                                                                                                                                                                                                                                                                                  |                                                                | Subelement 91 (Custom Payment Service Request/Transaction ID [Visa Only])  |
| <ul style="list-style-type: none"> <li>DE 48, subelement 91 (Custom Payment Service Request/Transaction ID)</li> </ul>                                                                                                                                                                                                                                                                                                                            |                                                                | Subelement 91 (Custom Payment Service Response/Transaction ID [Visa Only]) |



| Description of change                                                                                                                                                                                                                                                                                                                                                                                          | Source                                                                                                                                                                                                                                                                                | Where to look                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Subelement 91 (Custom Payment Service Response/ Transaction ID [Visa Only])</li> <li>• DE 48, subelement 94 (Commercial Card Inquiry Request)</li> <li>• DE 48, subelement 94 (Commercial Card Inquiry Response)</li> <li>• DE 48, subelement 96 (Visa Market-Specific Data Identifier)</li> <li>• DE 48, subelement 97 (Prestigious Properties Indicator)</li> </ul> |                                                                                                                                                                                                                                                                                       | <p><a href="#">Subelement 94 (Commercial Card Inquiry Request [Visa Only])</a></p> <p><a href="#">Subelement 94 (Commercial Card Inquiry Response [Visa Only])</a></p> <p><a href="#">Subelement 96 (Visa Market-Specific Data Identifier [Visa Only])</a></p> <p><a href="#">Subelement 97 (Prestigious Properties Indicator [Visa Only])</a></p> |
| Modified associated edit for DE 48, subelement 26                                                                                                                                                                                                                                                                                                                                                              | System clarification needed                                                                                                                                                                                                                                                           | <a href="#">DE 48, subelement 26</a>                                                                                                                                                                                                                                                                                                               |
| Updated messages and application notes for DE 48, subelement 32 (Mastercard Assigned ID)                                                                                                                                                                                                                                                                                                                       | <p><i>AP 11374.2 Enhancing Processing for Mastercard Installments Program with Merchant Participation in Malaysia</i></p> <p><i>MEA 11401.2 Enhancing Processing for Mastercard Installments Program with Merchant Participation in Saudi Arabia and the United Arab Emirates</i></p> | <a href="#">Subelement 32 (Mastercard Assigned ID)</a>                                                                                                                                                                                                                                                                                             |
| Added value 3 (X-Code Processing) to DE 48, subelement 42, subfield 3 (Reason for UCAF Collection Indicator Downgrade)                                                                                                                                                                                                                                                                                         | <i>GLB 11240 Expanding X-Code Processing to Support Identity Check</i>                                                                                                                                                                                                                | <a href="#">Subfield 3 (Reason for UCAF Collection Indicator Downgrade)</a>                                                                                                                                                                                                                                                                        |
| DE 48, subelement 48 (Digital Commerce Solutions Indicators)                                                                                                                                                                                                                                                                                                                                                   | <i>GLB 11248 Enhancing Digital Commerce Solutions Indicators</i>                                                                                                                                                                                                                      | <p><a href="#">Subfield 01 (Digital Commerce Domain Identifier)</a></p> <p><a href="#">Subfield 02 (Digital Commerce Program Type)</a></p> <p><a href="#">Subfield 03 (Digital Commerce Program Type Details)</a></p> <p><a href="#">Subfield 04 (Digital Commerce Check-In Details)</a></p>                                                       |

| Description of change                                                                                                                                                                                             | Source                                                                                                                                                     | Where to look                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Revised the name for DE 48, subelement 55 to Merchant Fraud Scoring Data/Approval Probability Indicator                                                                                                           | System clarification needed                                                                                                                                | <a href="#">Subelement 55 (Merchant Fraud Scoring Data/Approval Probability Indicator)</a><br><a href="#">List of DE 48 subelements</a> |
| Added Emerging Fraud Monitoring Service value combinations details to Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations) for DE 48, subelement 56 (Security Services Additional Data for Issuers) | <i>EUR/MEA 11791.1 Introducing Value Combinations to Support the Emerging Fraud Monitoring Service in Eastern Europe and the Middle/East Africa Region</i> | <a href="#">Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)</a>                                                      |
| Revised DE 48, subelement 64 (Transit Program), subfield 1 (Transit Transaction Type Indicator) section                                                                                                           | System clarification needed                                                                                                                                | <a href="#">Subfield 1 (Transit Transaction Type Indicator)</a>                                                                         |
| Revised the Values for DE 48, subelement 75 (Fraud Scoring Data), subfield 1 (Fraud Score)                                                                                                                        | System clarification needed                                                                                                                                | <a href="#">Subfield 1 (Fraud Score)</a>                                                                                                |
| Added values C67 (Merchant Presented QR) and C68 (Merchant Presented QR Refund) to DE 48, subelement 77 (Transaction Type Identifier)                                                                             | System clarification needed                                                                                                                                | <a href="#">Subelement 77 (Transaction Type Identifier)</a>                                                                             |
| Added new value 95 (Visa Money Transfer Foreign Exchange Fee [Visa Only]) and updated application notes for DE 54 (Additional Amounts), subfield 2 (Amount Type)                                                  | <i>GLB 11509 Enhancing Visa Account Funding Transaction Processing</i>                                                                                     | <a href="#">Subfield 2 (Amount Type)</a>                                                                                                |
| Added new value 06 (Product Status Inquiry Transaction Amount) to DE 54, subfield 2 (Amount Type)                                                                                                                 | <i>AP/EUR/LAC/MEA 11242.4 Introducing Mastercard One Credential</i><br><i>US 11423.4 Introducing Mastercard One Credential</i>                             | <a href="#">Subfield 2 (Amount Type)</a>                                                                                                |
| Added new value 2 (Product Status Inquiry Service [Standalone]) and associated edits to DE 61, subfield 9 (POS Transaction Status - Extended)                                                                     | <i>AP/EUR/LAC/MEA 11242.4 Introducing Mastercard One Credential</i><br><i>US 11423.4 Introducing Mastercard One Credential</i>                             | <a href="#">Subfield 9 (POS Transaction Status - Extended)</a>                                                                          |

| Description of change                                                                                                                                                      | Source                                                                                                           | Where to look                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Added value 8 (Remote terminal) to DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level) and an associated edit.                          | <i>GLB 11243 Introducing Modifications to Support Tap to More Transactions</i>                                   | <a href="#">Subfield 10 (Cardholder-Activated Terminal Level)</a><br><a href="#">Dual Message Authorization System Edits</a> |
| Added product code GPS: Mastercard Prepaid Installment Payments Card U to DE 63 (Network Data), subfield 1 (Financial Network Code)                                        | <i>GLB 11278 Introducing a Mastercard Prepaid Installment Payments Product Code</i>                              | <a href="#">Subfield 1 (Financial Network Code)</a>                                                                          |
| Added product codes MDA: Mastercard World Legend Debit, MDD: Mastercard CoBadge Debit, and MDE: Mastercard Essential Prepaid to DE 63, subfield 1 (Financial Network Code) | <i>GLB 11241 Introducing Consumer Debit and Prepaid Product Codes</i>                                            | <a href="#">Subfield 1 (Financial Network Code)</a>                                                                          |
| Updated value combinations for DE 48, subelement 71 (Valid Subfield 1, Subfield 2, and Subfield 3 value combinations)                                                      | <i>GLB 11283 Providing Relay Resistance Protocol Information to Mastercard Digital Enablement System Issuers</i> | <a href="#">Subelement 71 (Valid Subfield 1, Subfield 2, and Subfield 3 value combinations)</a>                              |

| Description of change                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Source                                                        | Where to look                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added encoding scheme details for DE 106 (Fleet Service Data) and revised the following:                                                                                                                                                                                                                                                                                                                                                                                                            | <i>GLB 11497 Clarifying Fleet Service Data Specifications</i> | <a href="#">DE 106 encoding scheme</a><br><a href="#">DE 106 subelement encoding scheme</a><br><a href="#">DE 106 (Fleet Service Data)</a><br><a href="#">Subelement 001 (Fleet Prompted Data)</a><br><a href="#">Subfield 001 (Fleet Prompt 1)</a><br><a href="#">Subfield 002 (Fleet Prompt 2)</a><br><a href="#">Subfield 003 (Fleet Prompt 3)</a><br><a href="#">Subfield 004 (Fleet Prompt 4)</a><br><a href="#">Subfield 005 (Fleet Prompt 5)</a><br><a href="#">Subelement 002 (Merchant Fleet Spend Control Override Capability Indicator)</a><br><a href="#">Subelement 003 (Fleet Spend Control Override Items)</a><br><a href="#">Subelement 004 (Fleet Fuel Information)</a><br><a href="#">Subfield 001 (Fuel Item Purchased 1)</a><br><a href="#">Subfield 002 (Fuel Item Purchased 2)</a><br><a href="#">Subfield 003 (Fuel Item Purchased 3)</a><br><a href="#">Subfield 001 (Non-Fuel Item Purchased 1)</a><br><a href="#">Subfield 002 (Non-Fuel Item Purchased 2)</a><br><a href="#">Subfield 003 (Non-Fuel Item Purchased 3)</a><br><a href="#">Subfield 004 (Non-Fuel Item Purchased 4)</a><br><a href="#">Subfield 005 (Non-Fuel Item Purchased 5)</a><br><a href="#">Subfield 006 (Non-Fuel Item Purchased 6)</a><br><a href="#">Subfield 007 (Non-Fuel Item Purchased 7)</a> |
| <ul style="list-style-type: none"> <li>Data Element (DE) 106 (Fleet Service Data)</li> <li>DE 106, subelement 001 (Fleet Prompted Data), and its subfields</li> <li>DE 106, subelement 002 (Merchant Fleet Spend Control Override Capability Indicator)</li> <li>DE 106, subelement 003 (Fleet Spend Control Override Items)</li> <li>DE 106, subelement 004 (Fleet Fuel Information), and its subfields</li> <li>DE 106, subelement 005 (Fleet Non-Fuel Information), and its subfields</li> </ul> |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Description of change                                                                                                                                                      | Source                                                        | Where to look                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                            |                                                               | <a href="#">Subfield 009 (Non-Fuel Item Purchased 9)</a><br><a href="#">Subfield 010 (Non-Fuel Item Purchased 10)</a><br><a href="#">Subfield 011 (Non-Fuel Item Purchased 11)</a><br><a href="#">Subfield 012 (Non-Fuel Item Purchased 12)</a>                                                                                                            |
| Added prompt code value 999 (No prompts issued) to the DE 106 (Fleet Service Data) subfields                                                                               | <i>GLB 11519.2 Introducing a Mastercard Fleet Prompt Code</i> | <a href="#">Subfield 001 (Fleet Prompt 1)</a><br><a href="#">Subfield 002 (Fleet Prompt 2)</a><br><a href="#">Subfield 003 (Fleet Prompt 3)</a><br><a href="#">Subfield 004 (Fleet Prompt 4)</a><br><a href="#">Subfield 005 (Fleet Prompt 5)</a>                                                                                                          |
| Updated the Application notes for DE 108 (Additional Transaction Reference Data), subelement 01 (Receiver/Recipient Data), subfield 11 (Receiver/Recipient Account Number) | System clarification needed                                   | <a href="#">Subfield 11 (Receiver/Recipient Account Number)</a>                                                                                                                                                                                                                                                                                            |
| Renamed and updated DE 116 (Mastercard Internal Use Data 2) and revised DE 119 (Additional Data: Private Use 2) with new subelements.                                      | <i>GLB 11253.2 Introducing MDES Service Indicators</i>        | <a href="#">DE 116 (Mastercard Internal Use Data 2)</a><br><a href="#">DE 119 (Additional Data: Private Use 2)</a><br><a href="#">Subelement 003 (MDES Service Indicators for Issuers)</a><br><a href="#">Subelement 004 (MDES Service Indicators for Acquirers)</a><br><a href="#">Subelement 005 (MDES Service Indicators for Issuers and Acquirers)</a> |

| Description of change                                                                                                                                                                                                                                                                                                                                                                                 | Source                                                                                                                         | Where to look                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added section on Mastercard One Credential along with support for acquirers to submit Product Status Inquiry (PSI) messages before authorization to retrieve the Mastercard One Credential Funding PAN's product code, consumer preference, and Account Level Management (ALM) service data. PSI requests can now be submitted independently or alongside other Account Status Inquiry (ASI) services | <i>AP/EUR/LAC/MEA 11242.4 Introducing Mastercard One Credential</i><br><i>US 11423.4 Introducing Mastercard One Credential</i> | <a href="#">Mastercard One Credential</a><br><a href="#">Product Status Inquiry Authorization Request/0100 message</a><br><a href="#">Account Status Inquiry with Product Status Inquiry Authorization Request/0100 message</a><br><a href="#">Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Request/0100 message</a><br><a href="#">Product Status Inquiry Authorization Response/0110 (Mastercard-Generated) message</a><br><a href="#">Account Status Inquiry with Product Status Inquiry Authorization Response/0110 message</a><br><a href="#">Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Response/0110 message</a> |
| Clarified Merchant Advice codes                                                                                                                                                                                                                                                                                                                                                                       | System clarification needed                                                                                                    | <a href="#">Merchant Advice codes</a><br><a href="#">Mastercard Authorization Optimizer</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Corrected ASCII hex values for Decimal values 205 and 206 in Extended character sets                                                                                                                                                                                                                                                                                                                  | System clarification needed                                                                                                    | <a href="#">Extended character sets</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Revised Advice Response Wait Time to a maximum of 20 seconds                                                                                                                                                                                                                                                                                                                                          | System clarification needed                                                                                                    | <a href="#">Stand-In Parameters</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Revised the values for the Indicating a recurring payment section                                                                                                                                                                                                                                                                                                                                     | System clarification needed                                                                                                    | <a href="#">Indicating a recurring payment</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Removed Mastercard Utility Payment Program                                                                                                                                                                                                                                                                                                                                                            | System clarification needed                                                                                                    | Appendix B: Services or products leveraging the system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Questions

For more information, questions or feedback about this document, click the Contact Customer Support button at the bottom of this page.

# Chapter 1 How to use this guide

This guide navigates users through the foundation of the *Dual Message Authorization System*, its functions, applicable messages, and their intended use.

## Structure of the guide

The material presents each topic with a high-level overview followed by additional details as the document progresses.

The initial chapters provide guidance for implementation of the *Dual Message Authorization System* specifications to support Mastercard account operations. The guide then describes the details of Mastercard's implementation of the *ISO 8583:1987* messaging standards. This guide also includes Mastercard data validations and supplemental reports.

The appendices provide supporting information to the chapters, including a glossary and basic information about Mastercard products and services available to customers.

This document provides the acronym phrase or word definition when the user hovers their mouse over a highlighted word. Links within the document allow the user to navigate directly to a related section, which provides supporting information for that topic.

**NOTE:** Link and hover features are available in select formats only.



## Chapter 2 High-level overview

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Introduction to Mastercard switches for processing ..... 55

Introduction to Mastercard Dual Message Authorization System processing ..... 56

Dual Message Authorization System features..... 56

Participants in Mastercard Network processing..... 57

Dual Message Authorization System basics..... 59

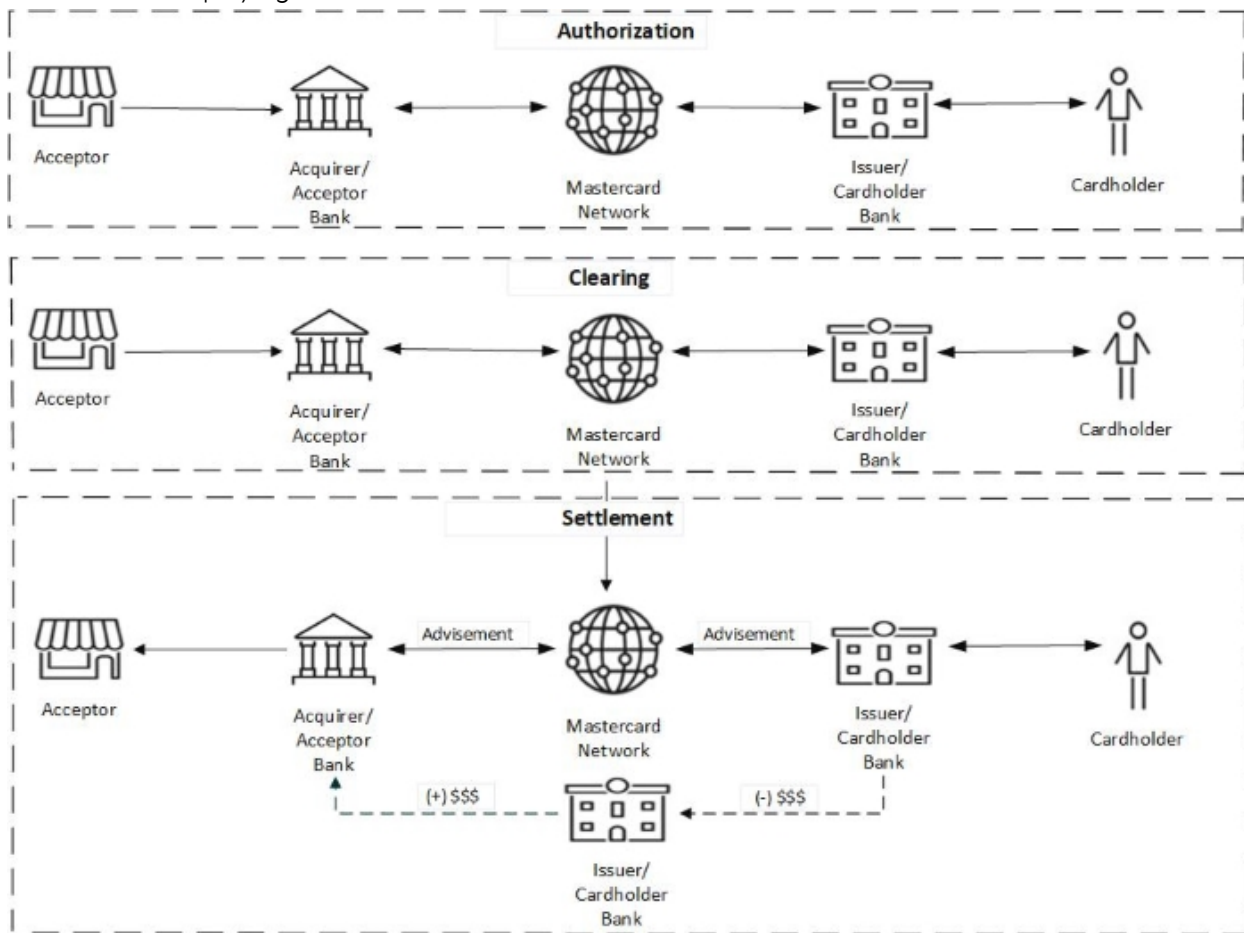
## Fundamentals of Mastercard processing

Mastercard processing supports the four-party model (cardholders, acceptors [merchant, sponsored merchant, *Automated Teller Machine (ATM)* owner, or other party], cardholder's bank and acceptor's bank). This process is explained by three main functions and the subsequent flow diagram.

- **Authorization:** Real-time communication between *acquirers* and *issuers* to approve or decline the use of a Mastercard card, access device, or account to conduct a transaction. Approval from the *issuer* creates a financial obligation from the *issuer* to the *acquirer* (for a transaction that debits funds from a cardholder's account) or from the *acquirer* to the *issuer* (for a transaction that credits funds to a cardholder's account).
- **Clearing:** Processing of multiple clearing batch processes per day (that is, clearing cycles) determine and calculate processing costs, resulting in one net amount per *acquirer* and *issuer* per clearing cycle.
- **Settlement:** Aggregation of financial positions from all clearing settlement information into one overall daily net settlement amount for each *acquirer* and *issuer*. Advisements (notifications) are created and distributed to *acquirer* and *issuer* along with Transfer Funds Orders to settlement banks for amounts to be transferred to or collected from *acquirer* and *issuer*.

### Authorization, Clearing, and Settlement flows

Process flow displaying the three main functions.



## Introduction to Mastercard switches for processing

Mastercard leverages its *Dual Message Switch* and *Single Message Switch* for the global processing of transaction data over the *Mastercard Network*.

The *Dual Message Switch* completes processing for *Authorization*, *Clearing*, and *Settlement* functions using both online and batch systems. *Authorization* processes as a real-time message, while *Clearing* and *Settlement* functions are achieved through a batch process.

The *Single Message Switch* completes processing for *Authorization*, *Clearing*, and *Settlement* functions leveraging a single online message. *Authorization* processes as a real-time message and automatically enables *Clearing* and *Settlement* functions through an internal batch process.

## Introduction to Mastercard Dual Message Authorization System processing

The Mastercard *Dual Message Authorization System* is a global real-time message processing system and is a component of the *Mastercard Network*. The *Mastercard Network* supports real-time *authorization* processing followed by batch clearing processing, leading to the *settlement* of funds between acquirers and issuers.

The *Dual Message Authorization System* processing is integrated with the *acquirer* and *issuer* processing systems. It communicates in real time to process *authorization* data to obtain an *issuer* decision.

Mastercard acquirers and issuers leverage the *Dual Message Authorization System* to exchange financial transaction data captured at the *point-of-interaction* device to obtain an *authorization* decision.

The *Dual Message Authorization System* consists of both hardware and software components. The physical computing infrastructure includes servers, routers, switches, and cabling. This hardware is configured with Mastercard's proprietary technology, which makes it a *Mastercard interface processor (MIP)*. It is physically set up and connected with an *acquirer* or *issuer* processing system in their data centers.

For more information on system access, refer to *System access information*.

## Dual Message Authorization System features

The *Dual Message Authorization System* provides many features, including but not limited to the following.

- Access for all acquirers and issuers 24 hours a day, 7 days a week.  
The *Mastercard Network* operates on a continuous basis, ensuring that cardholders can use Mastercard-branded cards and access devices anytime and that acquirers, issuers, and acceptors always have access to Mastercard authorization processing capabilities.
- Real-time authorization processing.  
The design of the *Mastercard Network* enhances processing by routing transactions along the shortest communication path to the destination. The network's design consists of many paths guaranteeing that there are always two or more paths to a given destination. If part of the network is unavailable, the multi-routing capability ensures that transactions continue toward their destinations following the shortest possible path, minimizing network congestion.
- Support for alternate routing and authorizing paths.  
At times, acquirers or issuers may experience internal communication problems, scheduled down times, or network problems. The system provides backup authorization alternatives to ensure that an authorization response occurs for every transaction.
- Security and fraud management services.

The *Dual Message Authorization System* provides services that make Mastercard transactions safe and secure.

- Currency conversion capabilities.

The *Dual Message Authorization System* provides currency conversion to acquirers and issuers to enable processing of online transaction messages in the *acquirer* or *issuer's* preferred currency.

- Supplemental reports.

Mastercard produces various authorization-related reports that display *acquirer* and *issuer* authorization activity and help acquirers and issuers to manage their back-office processes.

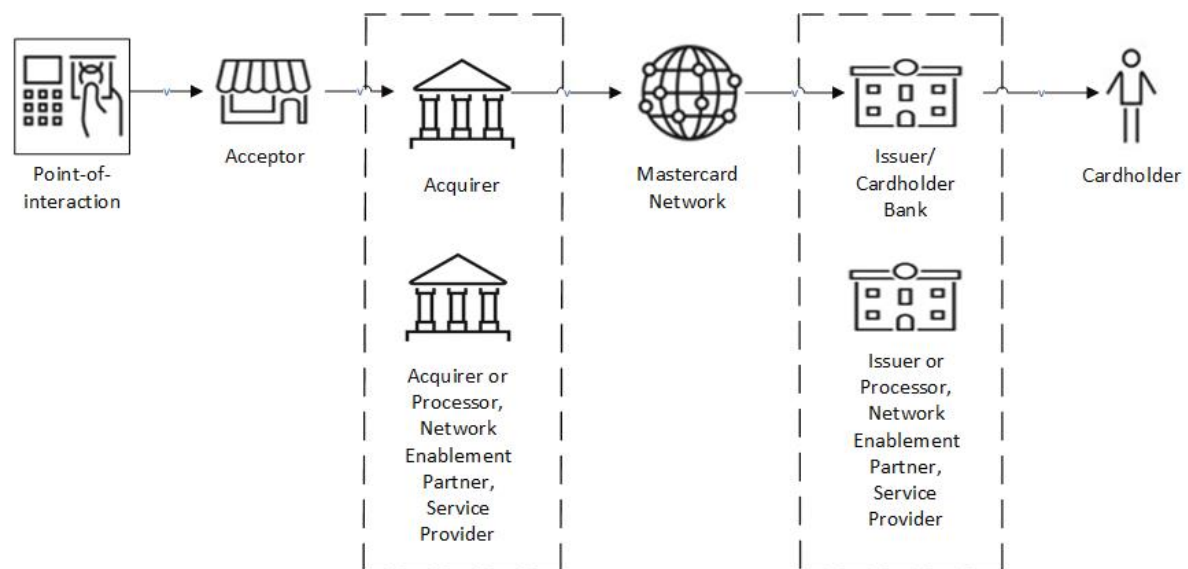
- Brand-agnostic processing capabilities

The *Dual Message Authorization System* also supports interfaces to enable non-Mastercard account processing.

## Participants in Mastercard Network processing

*Mastercard Network* processing involves various functions and types of participants. This list includes the most common participants involved in processing, along with simplified descriptions.

### Participants in Mastercard Network processing flow



## Participants and definitions

| Term                       | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acceptor                   | A person or party that agrees to accept payment cards or devices bearing a particular brand or in respect to an agreement with the issuer. Also referred to as a merchant, <a href="#">ATM</a> , bank, merchant ATM, or other party authorized to accept brand transactions.                                                                                                                                                                                                        |
| Acquirer                   | An acquirer is a licensed Mastercard customer that acquires the data relating to a transaction from the acceptor and submits that data for authorization. The acquirer also supports clearing and settlement functions to exchange funds between the issuer and the acceptor. Also referred to as an acquiring bank, acquirer bank, merchant bank, or acquiring financial institution.                                                                                              |
| Cardholder                 | A cardholder is an authorized person or party to whom an account, card, or access device has been issued for the purpose of conducting transactions that access funds through a financial account. Also referred to as a consumer or account holder.                                                                                                                                                                                                                                |
| Customer                   | A financial institution or other entity that has been approved for participation in the payment scheme.                                                                                                                                                                                                                                                                                                                                                                             |
| Issuer                     | An issuer is a licensed Mastercard customer that issues cards or accounts and is responsible for approving or declining <a href="#">authorization</a> requests. The issuer also bills cardholders and supports <a href="#">clearing</a> and <a href="#">settlement</a> functions to exchange funds between the cardholder's account and the acquirer. Also referred to as an issuing bank, issuer bank, cardholder bank, issuing financial institution, or non-bank issuing entity. |
| Network Enablement Partner | A service provider that has a direct agreement with Mastercard and performs certain functions on behalf of one or more customers.                                                                                                                                                                                                                                                                                                                                                   |
| Processor                  | A service provider that provides authorization, clearing, or settlement services on behalf of a customer. Also referred to as a third-party processor or payment processor.                                                                                                                                                                                                                                                                                                         |

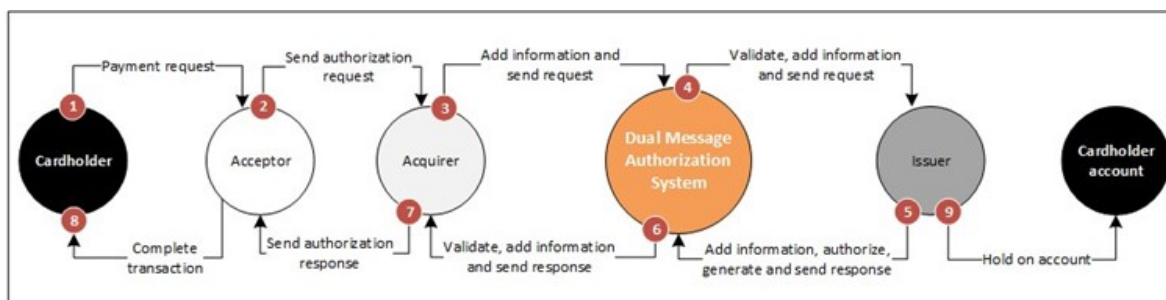
| Term             | Definition                                                                                                                                      |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Service provider | A person or party that is registered with Mastercard by a customer to perform services related to the customer's acquiring or issuing activity. |

**NOTE:** Throughout this guide, functional references to acquirer and issuer also refer to others acting as authorized agents for acquirers and issuers. Mastercard customers are responsible for the acts and omissions of their agents, including those with respect to Mastercard rules, regulations, policies, and procedures.

## Dual Message Authorization System basics

The following diagram illustrates the basic *Dual Message Authorization System* flow.

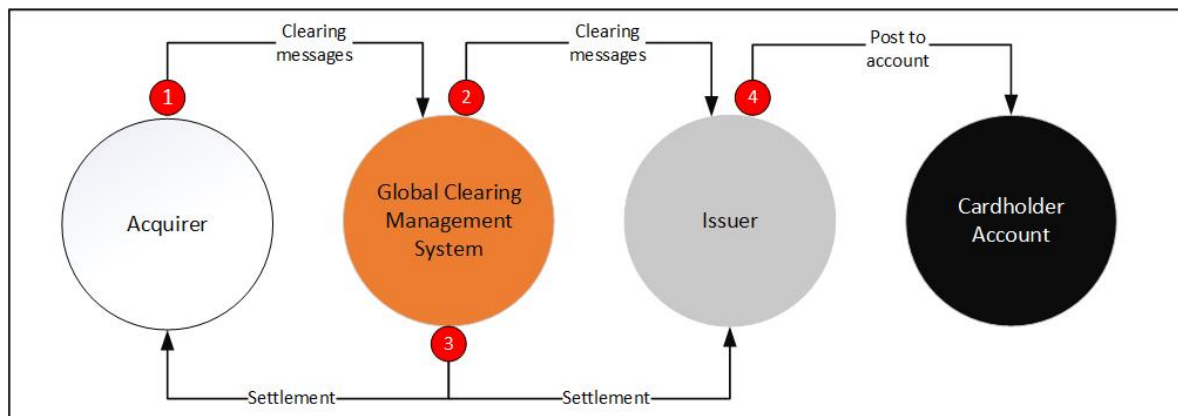
### Basic Dual Message Authorization System flow



1. The *cardholder* presents a Mastercard card to the *acceptor* to pay for goods or services.
2. The *acceptor* collects card and transaction data (for example, the *cardholder* may dip or tap the card at a terminal, enter the card data into a website payment page, place an order by phone or mail, or authorize the *acceptor* to use previously stored card data). A request for an authorization decision is sent from the *acceptor* to the *acquirer*.
3. The *acquirer* receives the request, appends additional data as needed, and sends the *authorization request* message to the Mastercard *Dual Message Authorization System*.
4. The *Dual Message Authorization System* receives the *authorization request* message and validates the format and content. Mastercard includes additional data as needed and sends the request message to the *issuer*.
5. The *issuer* receives the *authorization request* message and evaluates the transaction data and the account itself before making an authorization decision. The *issuer* generates an *authorization request response message* indicating its decision and sends the *response message* to Mastercard.
6. Mastercard receives the *authorization request response message* and validates the format and content. Mastercard includes additional data as needed and sends the *authorization request response message* to the acquirer.

7. The *acquirer* receives the *authorization request response message* and notifies the *acceptor* of the decision.
8. The *acceptor* receives the response to the decision and completes the transaction with the *cardholder*.
9. The *issuer* places a hold for the transaction amount against the *cardholder's* account or open-to-buy. The transaction will post when clearing occurs.

### Clearing and settlement process flow



*Clearing* and *settlement* will occur through a separate process. This diagram outlines this process at a high level.

1. The *acquirer* submits batched clearing messages containing the data for the completed transactions from their acceptors.
2. The Mastercard Dual Message Clearing System will forward the appropriate transaction data to each *issuer* through *batch processing*.
3. The Mastercard Dual Message Clearing System processes the messages and forwards them to the *Settlement* System. The *Settlement* System generates reporting and settles with the *acquirer* and *issuer*.
4. The *issuer* uses the clearing data received to post the transaction to the *cardholder's* account.

**NOTE: Mastercard reserves the right to record, store, and use all data transmitted by the Mastercard Dual Message Authorization System in online electronic transactions, subject to Mastercard privacy and security compliance policies and applicable laws and regulations, without further notice.**



## Chapter 3 Introduction to messages

*This section provides message definitions of all message types the [Dual Message Authorization System](#) uses followed by the detailed message flow diagrams that illustrate both the standard and exception (error condition) message processing.*

---

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## Overview

The *Dual Message Authorization System* format is based on the *ISO 8583:1987* international message standard, which is comprised of code values, *data elements*, and message types. The *ISO 8583* protocol is used by systems for the electronic exchange of transaction data for the purchase of goods or services.

The *ISO* standard defines the purpose of the message and origination through the *message type indicator (MTI)*. *MTIs* are defined below.

### ISO-defined MTI codes and related usage

| Code | Meaning                                          | Usage                                                                                                                                                                                                        |
|------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00xx | Reserved by ISO                                  | N/A                                                                                                                                                                                                          |
| 01xx | Authorization messages                           | Authorization messages are used by the <i>acquirer</i> to obtain approval or guarantee funds from the <i>issuer</i> . These messages are not intended for use in posting to the <i>cardholder's</i> account. |
| 02xx | Financial messages                               | Financial messages are used by the <i>acquirer</i> to obtain approval or guarantee funds from the <i>issuer</i> . These messages allow the approved transaction to post to the <i>cardholder's</i> account.  |
| 03xx | File action messages                             | File action messages are used to add, change, delete, or inquire into a file or record (for example, to report lost or stolen cards).                                                                        |
| 04xx | Reversal messages                                | Reversal messages are used to partially or completely negate the effects of a previous financial or authorization transaction.                                                                               |
| 05xx | Reconciliation messages (not used by Mastercard) | Reconciliation messages provide financial totals between an <i>acquirer</i> and an <i>issuer</i> .                                                                                                           |
| 06xx | Administrative messages                          | Administrative messages are used for the exchange of non-financial information between network participants.                                                                                                 |
| 07xx | Reserved by ISO                                  | N/A                                                                                                                                                                                                          |
| 08xx | Network management messages                      | Network management messages are used to control the system security and operating condition of a network and may be initiated by any network participant.                                                    |
| 09xx | Reserved by ISO                                  | N/A                                                                                                                                                                                                          |

## ISO MTI position definitions

| Position 1: Version          | Position 2: Message class | Position 3: Message function | Position 4: Message originator* |
|------------------------------|---------------------------|------------------------------|---------------------------------|
| 0: ISO 8583:1987             | 0: Reserved for ISO use   | 0: Request                   | 0: Acquirer                     |
| 1: Reserved for ISO use      | 1: Authorization          | 1: Request response          | 1: Acquirer repeat              |
| 2: Reserved for ISO use      | 2: Financial              | 2: Advice                    | 2: Card issuer                  |
| 3: Reserved for ISO use      | 3: File update            | 3: Advice response           | 3: Card issuer repeat           |
| 4: Reserved for ISO use      | 4: Reversal               | 4: Reserved for ISO use      | 4: Reserved for ISO use         |
| 5: Reserved for ISO use      | 5: Reconciliation         | 5: Reserved for ISO use      | 5: Reserved for ISO use         |
| 6: Reserved for ISO use      | 6: Administrative         | 6: Reserved for ISO use      | 6: Reserved for ISO use         |
| 7: Reserved for ISO use      | 7: Reserved for ISO use   | 7: Reserved for ISO use      | 7: Reserved for ISO use         |
| 8: Reserved for national use | 8: Network management     | 8: Reserved for ISO use      | 8: Reserved for ISO use         |
| 9: Reserved for private use  | 9: Reserved for ISO use   | 9: Reserved for ISO use      | 9: Reserved for ISO use         |

\* This position of the [MTI](#) does not indicate which party originated the specific message. This position of the [MTI](#) indicates which party originally initiated the transaction (the request and response message pair).

This guide documents the Mastercard implementation of the [ISO](#) standard for processing on Mastercard's [Dual Message Authorization System](#). The messages used by the [Dual Message Authorization System](#) are a subset of the [ISO](#)-defined messages. In some cases, Mastercard's adoption of these messages differs from [ISO](#) definitions. These are the message types supported by the [Dual Message Authorization System](#). Additional details on the use of these [MTIs](#) are included in this chapter.

## Mastercard-defined MTI codes and related usage

| Code | Meaning                                          |
|------|--------------------------------------------------|
| 01xx | <a href="#">Authorization/01xx messages</a>      |
| 03xx | <a href="#">Issuer File Update/03xx messages</a> |
| 04xx | <a href="#">Reversal/04xx messages</a>           |
| 06xx | <a href="#">Administrative/06xx messages</a>     |

| Code | Meaning                                          |
|------|--------------------------------------------------|
| 08xx | <a href="#">Network Management/08xx messages</a> |

## Mastercard message classifications

The [Mastercard Network](#) allows for the exchange of electronic data that may or may not contain account holder information or result in a financial transaction. The remaining message types are used to support administrative functions.

The message types adopted for use by the Mastercard [Dual Message Authorization System](#), as a broad classification, fall into three functional categories.

### Message type functional categories

|                                |                                                                                                                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Financial message function     | This message type function results in a financial impact to a <a href="#">cardholder</a> account. It includes instructions or data related to payments, transfers, debits, or credits.         |
| Non-financial message function | This message type function allows customers to validate account information prior to the submission of a financial message without financial impact to an account.                             |
| Supporting message function    | This message type function allows acquirers and issuers connected to the <a href="#">Mastercard Network</a> to complete administrative functions, such as file updates and network management. |

## List of Dual Message Authorization System MTIs

The following tables list the message types the [Dual Message Authorization System](#) supports and indicates the originating parties for each message type. The [Dual Message Authorization System](#) may not support all message types for each Mastercard product or service.

Acquirers and issuers are responsible for ensuring correct usage of the message types as described in the table and may be liable for the impacts of transmitting erroneous messages.

### Authorization/01xx messages

| MTI  | Description                                                                                | Acquirer | Issuer | Mastercard Network |
|------|--------------------------------------------------------------------------------------------|----------|--------|--------------------|
| 0100 | Authorization Request                                                                      | X        |        |                    |
| 0110 | Authorization Request Response                                                             |          | X      | X                  |
| 0120 | Authorization Advice: acquirer-generated                                                   | X        |        |                    |
| 0120 | Authorization Advice: system-generated                                                     |          |        | X                  |
| 0130 | Authorization Advice Response: issuer-generated (responding to an acquirer-generated 0120) |          | X      |                    |
| 0130 | Authorization Advice Response: issuer-generated (responding to a system-generated 0120)    |          | X      |                    |
| 0130 | Authorization Advice Response: system-generated                                            |          |        | X                  |
| 0180 | Authorization Response Acknowledgment                                                      | X        |        |                    |
| 0190 | Authorization Response Negative Acknowledgment                                             |          |        | X                  |

### Issuer File Update/03xx messages

| MTI  | Description                         | Acquirer | Issuer | Mastercard Network |
|------|-------------------------------------|----------|--------|--------------------|
| 0302 | Issuer File Update Request          |          | X      |                    |
| 0312 | Issuer File Update Request Response |          |        | X                  |

### Reversal/04xx messages

| MTI  | Description               | Acquirer | Issuer | Mastercard Network |
|------|---------------------------|----------|--------|--------------------|
| 0400 | Reversal Request          | X        |        |                    |
| 0410 | Reversal Request Response |          | X      | X                  |
| 0420 | Reversal Advice           |          |        | X                  |
| 0430 | Reversal Advice Response  |          | X      |                    |

### Administrative/06xx messages

| MTI  | Description                     | Acquirer | Issuer | Mastercard Network |
|------|---------------------------------|----------|--------|--------------------|
| 0600 | Administrative Request          | X        |        |                    |
| 0610 | Administrative Request Response |          | X      | X                  |
| 0620 | Administrative Advice           | X        | X      | X                  |
| 0630 | Administrative Advice Response  | X        | X      | X                  |

### Network Management/08xx messages

| MTI  | Description                                                                                                                                                                                                                                                                                                                                                                                                    | Acquirer | Issuer | Mastercard Network |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|--------------------|
| 0800 | Network Management Request: sign-on/sign-off<br>Because the <i>Mastercard Network</i> does not track session status information for an <i>acquirer</i> , <i>Dual Message Authorization System</i> sign-on/sign-off for acquirers is not required. However, acquirers may optionally send <i>Dual Message Authorization System</i> sign-on/sign-off messages (for example, if required by the vendor software). | X        | X      |                    |
| 0800 | Network Management Request: network connection status, member-generated                                                                                                                                                                                                                                                                                                                                        | X        | X      |                    |
| 0800 | Network Management Request: network connection status, system-generated                                                                                                                                                                                                                                                                                                                                        |          |        | X                  |
| 0800 | Network Management Request: host session activation/deactivation                                                                                                                                                                                                                                                                                                                                               | X        | X      | X                  |
| 0800 | Network Management Request: PEK exchange                                                                                                                                                                                                                                                                                                                                                                       |          |        | X                  |
| 0800 | Network Management Request: PEK exchange, on demand                                                                                                                                                                                                                                                                                                                                                            | X        | X      |                    |
| 0810 | Network Management Request Response: sign-on/sign-off                                                                                                                                                                                                                                                                                                                                                          |          |        | X                  |
| 0810 | Network Management Request Response: network connection status, customer-generated                                                                                                                                                                                                                                                                                                                             | X        | X      |                    |
| 0810 | Network Management Request Response: network connection status, system-generated                                                                                                                                                                                                                                                                                                                               |          |        | X                  |



| MTI  | Description                                                      | Acquirer | Issuer | Mastercard Network |
|------|------------------------------------------------------------------|----------|--------|--------------------|
| 0810 | Network Management Request: host session activation/deactivation | X        | X      | X                  |
| 0810 | Network Management Request Response: PEK exchange                | X        | X      |                    |
| 0810 | Network Management Request Response: PEK exchange on demand      |          |        | X                  |
| 0820 | Network Management Advice: PEK exchange                          |          |        | X                  |

## About message definitions

This section provides details on message definitions.

### Authorization/01xx messages

Authorization/01xx messages transmit authorization-related information. [Authorization](#) is the process of verifying if a [cardholder](#) is allowed to perform a requested action or service. All Authorization/01xx messages assume that follow-up information (for example, electronic clearing transactions) is used to complete [settlement](#) and [cardholder](#) account posting.

The [Mastercard Network](#) uses Authorization/01xx messages for:

- Authorization Request
- Authorization Request Response
- Authorization Advice
- Authorization Advice Response
- Authorization Response Acknowledgment

#### Authorization Request/0100

|          |                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | <ul style="list-style-type: none"> <li>• From an <a href="#">acquirer</a> to the <a href="#">Mastercard Network</a>.</li> <li>• From the <a href="#">Mastercard Network</a> to an <a href="#">issuer</a>.</li> </ul> |
| Purpose  | Requires issuer decisioning specific to funds availability and account status resulting in an approval or a decline.                                                                                                 |
| Response | An Authorization Request Response/0110 message is required.                                                                                                                                                          |

## Authorization Request Response/0110

|          |                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | <ul style="list-style-type: none"> <li>From an <a href="#">issuer</a> to the <a href="#">Mastercard Network</a>.</li> <li>From the <a href="#">Mastercard Network</a> to an <a href="#">acquirer</a>.</li> </ul>                                |
| Purpose  | To acknowledge an Authorization Request/0100 message informing the <a href="#">acquirer</a> of the issuer's decision.                                                                                                                           |
| Response | Acquirers may elect to acknowledge receipt of the Authorization Request/0110 message back to Mastercard through the Authorization Response Acknowledgment/0180 message. The <a href="#">acquirer</a> must be configured to perform this action. |

## About Authorization Advice messages

The [Mastercard Network](#) guarantees the delivery of all advice messages transmitted through the network. When an [advice message](#) is processed by the [Dual Message Authorization System](#), the system acknowledges receipt of the [advice message](#) to the initiator with an advice response message and forwards the advice message to the intended recipient.

If the Dual Message Authorization System cannot deliver an advice message immediately (for example, due to a system or communication failure at the receiving destination), the Dual Message Authorization System will store the message within the [store-and-forward \(SAF\)](#) queue. [SAF](#) processing automatically delivers the message to the destination when communication is restored.

The receiver of an advice message must acknowledge receipt with an [advice response message](#). When the Dual Message Authorization System receives the advice response message, Mastercard considers advice delivery complete and removes the message from SAF processing queues.

The following message types are supported by the guaranteed advice messages delivery concept:

- Authorization Advice/0120: acquirer-generated
- Authorization Advice/0120: system-generated
- Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)
- Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)

[Address Verification Service \(AVS\)](#) is a [fraud](#) deterrent service that provides greater security to merchants and cardholders. It helps to protect against fraudulent use of cards by verifying the cardholder's billing address. This functionality is supported during the process of obtaining authorization for a transaction. For more information on this service, please refer to [System functionality and services](#).

### Authorization Advice/0120: acquirer-generated

|          |                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | <ul style="list-style-type: none"><li>From the <a href="#">acquirer</a> to the <a href="#">Mastercard Network</a>.</li><li>From the Mastercard Network to the <a href="#">issuer</a>.</li></ul>                                       |
| Purpose  | Advises the <a href="#">issuer</a> of an authorization approved by the acquirer <a href="#">host</a> when unable to communicate with the Mastercard Network or advises the issuer of a change to a previously approved authorization. |
| Response | An Authorization Advice Response/0130: issuer-generated message is required.                                                                                                                                                          |

### Authorization Advice/0120: system-generated

|          |                                                                              |
|----------|------------------------------------------------------------------------------|
| Routing  | From the Mastercard Network to an issuer.                                    |
| Purpose  | Advises the issuer of an authorization decision made on their behalf.        |
| Response | An Authorization Advice Response/0130: issuer-generated message is required. |

### Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)

|          |                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | When responding to the Authorization Advice/0120: acquirer-generated message: <ul style="list-style-type: none"><li>From the issuer to the Mastercard Network.</li><li>From the Mastercard Network to the acquirer.</li></ul> |
| Purpose  | Acknowledges receipt of an Authorization Advice/0120: acquirer-generated message.                                                                                                                                             |
| Response | None                                                                                                                                                                                                                          |

### Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| Routing  | From the issuer to the Mastercard Network.                                      |
| Purpose  | Acknowledges receipt of an Authorization Advice/0120: system-generated message. |
| Response | None                                                                            |

## About Authorization Response Acknowledgment messages

For acquirers, the Dual Message Authorization System supports optional response acknowledgment messages, which indicates receipt of an Authorization Request Response/0110 message. The [acquirer](#) must be configured to perform this action.

The Dual Message Authorization System sends an Authorization Response Negative Acknowledgment/0190 message in response to a late or invalid Authorization Request Response/0110 or Reversal Request Response/0410 message from an issuer. This informs the issuer that the Dual Message Authorization System either timed out or rejected the message and initiated *Stand-In Authorization* or alternate authorization services on the issuer's behalf if applicable.

Authorization Response Acknowledgment messages include:

- Authorization Response Acknowledgment/0180
- Authorization Response Negative Acknowledgment/0190

### Authorization Response Acknowledgment/0180

|          |                                                                          |
|----------|--------------------------------------------------------------------------|
| Routing  | From the acquirer to the Mastercard Network.                             |
| Purpose  | Acknowledges receipt of the Authorization Request Response/0110 message. |
| Response | None                                                                     |

### Authorization Response Negative Acknowledgment/0190

|          |                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------|
| Routing  | From the Mastercard Network to the issuer.                                                                           |
| Purpose  | Advises issuer of a late or erroneous Authorization Request Response/0110 or Reversal Request Response/0410 message. |
| Response | None                                                                                                                 |

## Issuer File Update/03xx messages

Issuers use Issuer File Update/03xx messages to update individual data files or system parameter tables that the [Mastercard Network](#) uses to control the functionality of various Mastercard programs and services available to customers.

**NOTE:** For more information on data files and system parameter tables, refer to the [Account Management System User Manual](#).

The Issuer File Update messages include:

- Issuer File Update Request/0302
- Issuer File Update Request Response/0312

### Issuer File Update Request/0302

|          |                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------|
| Routing  | From an <i>issuer</i> to the <i>Mastercard Network</i> .                                        |
| Purpose  | Requests an Issuer File Update or inquiry action to be initiated on the <i>issuer's</i> behalf. |
| Response | An Issuer File Update Request Response/0312 message is required.                                |

### Issuer File Update Request Response/0312

|          |                                                                      |
|----------|----------------------------------------------------------------------|
| Routing  | From the <i>Mastercard Network</i> to an <i>issuer</i> .             |
| Purpose  | Indicates the result of the Issuer File Update Request/0302 message. |
| Response | None                                                                 |

## Reversal/04xx messages

Acquirers generate a Reversal Request/0400 message when the *acquirer* is unable to deliver an issuer's approved response to an acceptor. Acceptors may also request their acquirers send a Reversal Request/0400 message to cancel the full or partial amount of the original authorization amount.

The *Mastercard Network* will not attempt to match contents of the Reversal Request/0400 message with the contents of the original Authorization Request/0100 message.

The *acquirer* must send a Reversal Request/0400 message when there is an acquirer time-out. When this occurs, it is possible that an *issuer* will receive two reversals for the same *authorization request*. Issuers must decline subsequent reversal messages with the appropriate response code.

The Reversal Advice/0420 message fully reverses a previous Authorization Request/0100 message. It advises an *issuer* of a Reversal Request/0400 message responded to by the *Dual Message Authorization System*. The *Dual Message Authorization System* may generate this message on detection of an exception condition while processing an Authorization Request Response/0110 message or when an *issuer* is unavailable to respond a Reversal Request/0400 message or is delayed in providing a Reversal Request Response/0410 message.

If the *Dual Message Authorization System* cannot deliver an advice message immediately (for example, due to a system or communication failure at the receiving destination), the *Dual*

*Message Authorization System* will store the message within the *SAF* queue. *SAF* processing automatically delivers the message to the destination when communication is restored.

The receiver of an advice message must acknowledge receipt with an advice response message. When the *Dual Message Authorization System* receives the advice *response message*, Mastercard considers advice delivery complete and removes the message from *SAF* processing queues.

Reversal/04xx messages include:

- Reversal Request/0400
- Reversal Request Response/0410
- Reversal Advice/0420
- Reversal Advice Response/0430

### Reversal Request/0400

|          |                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | <ul style="list-style-type: none"> <li>• From <i>acquirer</i> to <i>Mastercard Network</i>.</li> <li>• From <i>Mastercard Network</i> to <i>issuer</i>.</li> </ul> |
| Purpose  | Reverses an earlier authorization request either fully or partially.                                                                                               |
| Response | A Reversal Request Response/0410 message is required.                                                                                                              |

### Reversal Request Response/0410

|          |                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | <ul style="list-style-type: none"> <li>• From <i>issuer</i> to <i>Mastercard Network</i>.</li> <li>• From <i>Mastercard Network</i> to <i>acquirer</i>.</li> </ul> |
| Purpose  | Advises the disposition of the Reversal Request/0400 message.                                                                                                      |
| Response | None                                                                                                                                                               |

### Reversal Advice/0420

|         |                                                          |
|---------|----------------------------------------------------------|
| Routing | From the <i>Mastercard Network</i> to an <i>issuer</i> . |
|---------|----------------------------------------------------------|

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose  | <p>Fully reverses a previous Authorization Request/0100 message. It advises an <i>issuer</i> of a Reversal Request/0400 message responded to by the <i>Mastercard Network</i>.</p> <p>The <i>Mastercard Network</i> may generate this message on detection of an exception condition while processing an Authorization Request Response/0110 message.</p> <p>The <i>Mastercard Network</i> may generate this message when an <i>issuer</i> is unavailable to respond a Reversal Request/0400 message or is delayed in providing a Reversal Request Response/0410 message.</p> |
| Response | A Reversal Advice Response/0430 message is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Reversal Advice Response/0430

|          |                                                                |
|----------|----------------------------------------------------------------|
| Routing  | From an <i>issuer</i> to the <i>Mastercard Network</i> .       |
| Purpose  | Acknowledges receipt of a Reversal Advice/0420 advice message. |
| Response | None                                                           |

## Administrative/06xx messages

Administrative/06xx messages are used to convey various types of business or technical operations information between any parties connected to the *Mastercard Network*. Depending on the context, the information may be formatted as structured or free-form text data. Information present in administrative messages can be used by the receiver to:

- Manage various Mastercard products, services, or processing arrangement
- Identify and resolve erroneously constructed messages
- Obtain proprietary or *Mastercard Network* related text messages

Administrative messages include:

- Administrative Request/0600
- Administrative Request Response/0610
- Administrative Advice/0620: system-generated
- Administrative Advice/0620: customer-generated
- Administrative Advice Response/0630

The *Dual Message Authorization System* routes messages from an originator to a receiver and, in general, does not distinguish whether the originator or receiver is an *issuer* or an *acquirer*.

The Administrative/06xx messages used by the *Dual Message Authorization System* fall under the general category of advice messages, and are subject to the Guaranteed Advice Delivery procedures.

Administrative Advice/0620: system-generated messages advise a message originator that the [Dual Message Authorization System](#) was unable to parse or successfully process an indecipherable message. The message will contain the appropriate error condition code and indicate at which point the system terminated parsing or processing. In general, messages returned in Administrative Advice/0620 messages will have improperly coded or garbled MTIs or improperly coded bit maps.

Administrative Advice/0620: customer-generated messages transmit administrative (free format) textual messages between two parties participating on the Mastercard Network.

### Administrative Request/0600

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | <ul style="list-style-type: none"> <li>• <a href="#">Acquirer</a> to <a href="#">Mastercard Network</a>.</li> <li>• <a href="#">Mastercard Network</a> to <a href="#">issuer</a>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Purpose  | Contains customer data in <a href="#">DE 113-118 (Reserved for National Use)</a> and the type of usage in <a href="#">DE 60 (Advice Reason Code)</a> . Any eligible Administrative Request/0600 message is routed to the <a href="#">issuer</a> associated to the account range within <a href="#">DE 2 (Primary Account Number [PAN])</a> and the appropriate response timer is set for the <a href="#">issuer</a> Administrative Request Response/0610 message. The response timer is configured by each <a href="#">DE 60</a> usage type and initially will be set at a default of 30 seconds for each usage type. |
| Response | An Administrative Request Response/0610 message is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Administrative Request Response/0610

|          |                                                                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type     | Interactive                                                                                                                                                                                       |
| Routing  | <ul style="list-style-type: none"> <li>• From <a href="#">issuer</a> to <a href="#">Mastercard Network</a>.</li> <li>• <a href="#">Mastercard Network</a> to <a href="#">acquirer</a>.</li> </ul> |
| Purpose  | Contains the issuer's response data to the Administrative Request/0600 request message.                                                                                                           |
| Response | None                                                                                                                                                                                              |

### Administrative Advice/0620: system-generated

|         |                                                                                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing | <ul style="list-style-type: none"> <li>• Between any two parties participating on the <a href="#">Mastercard Network</a>.</li> <li>• <a href="#">Mastercard Network</a> to message originator.</li> </ul> |
| Purpose | Used by the Mastercard Network to indicate errors or an invalid message received from the customer.                                                                                                       |



|          |                                                             |
|----------|-------------------------------------------------------------|
| Response | An Administrative Advice Response/0630 message is required. |
|----------|-------------------------------------------------------------|

### Administrative Advice/0620: customer-generated

|          |                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | <ul style="list-style-type: none"><li>Between any two parties participating on the <a href="#">Mastercard Network</a>.</li><li><a href="#">Mastercard Network</a> to message originator.</li></ul> |
| Purpose  | Used by two parties participating in a Mastercard program or service offering. Advises message originator of any issue with the original message.                                                  |
| Response | An Administrative Advice Response/0630 message is required.                                                                                                                                        |

### Administrative Advice Response/0630

|          |                                                                                        |
|----------|----------------------------------------------------------------------------------------|
| Routing  | From the receiver to the originator of the related Administrative Advice/0620 message. |
| Purpose  | Acknowledges positive receipt of an Administrative Advice/0620 advice message.         |
| Response | None                                                                                   |

## Network Management/08xx messages

Network Management/08xx messages coordinate network events or tasks and can also be used to communicate network status conditions.

These messages are used by participants on the [Dual Message Authorization System](#) and the system itself.

Network Management/08xx messages include:

- Network Management Request/0800
- Network Management Request Response/0810
- Network Management Advice/0820

Typical uses of Network Management/08xx messages include:

- Sign-on/sign-off from the [Mastercard Network](#)
- Perform a Network Connection Status to Mastercard
- Perform Host Session Activation or Deactivation
- Perform [encryption](#) key management
- [SAF](#) requests
- Advise of [SAF End of file \(EOF\)](#) conditions

The *Dual Message Authorization System* routes Network Management/08xx messages from an originator to a receiver and, in general, does not distinguish whether the originator or receiver is an *acquirer* or an *issuer*.

### Network Management Request/0800

|          |                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | Between the <i>Mastercard Network</i> and any other participant on the <i>Mastercard Network</i> .<br>May be originated by any party. |
| Purpose  | Controls the <i>Mastercard Network</i> by communicating or coordinating system condition or security.                                 |
| Response | A Network Management Request Response/0810 message is required.                                                                       |

### Network Management Request Response/0810

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| Routing  | From receiver to originator of the related Network Management Request/0800. |
| Purpose  | Acknowledges receipt of the Network Management Request/0800 message.        |
| Response | None                                                                        |

### Network Management Advice/0820

|          |                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routing  | From the <i>Mastercard Network</i> to any other participant on the <i>Mastercard Network</i> .<br><b>NOTE: May originate only from the <i>Mastercard Network</i>.</b> |
| Purpose  | Advises of a previous Network Management Request/0800 message.                                                                                                        |
| Response | None                                                                                                                                                                  |

## Message flows

This section provides details on the message flows.

### Financial transaction function flows

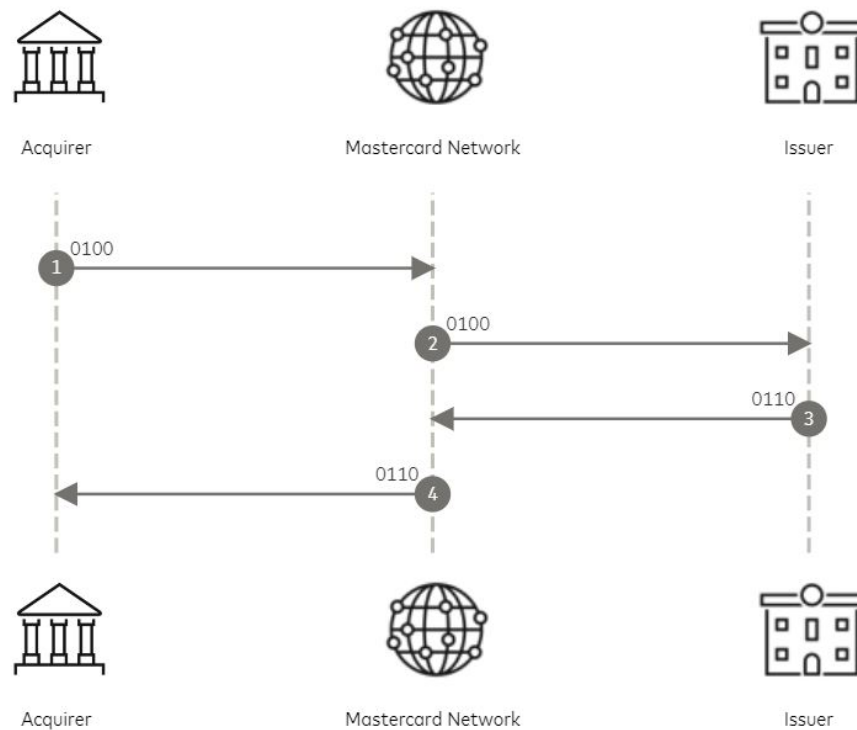
| Flow type                                                                | Description                                                                                                                                                |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standard flow                                                            | Successful completion of Authorization Request/0100 and Authorization Request Response/0110 message pair                                                   |
| Failure scenarios: communication-related flows                           | Exception occurrences which may be due to communication or problems at the <i>acquirer</i> or <i>issuer</i> system                                         |
| Failure scenarios: issuer response contains errors or late response flow | Messages contain invalid data or the response message was received after the time limit                                                                    |
| Failure scenarios: alternate issuer host configuration flows             | Europe Region: An <i>issuer</i> may designate an <i>alternate issuer host</i> for alternate processing instead of Mastercard <i>Stand-In Authorization</i> |
| Failure scenarios: no alternate processing allowed                       | Transactions for which <i>Stand-In Authorization</i> , <i>alternate issuer host</i> , or <i>X-Code</i> is not permitted                                    |
| Authorization Advice messages                                            | Provides the <i>issuer</i> information on changes to a previous authorization or action taken on their behalf for a transaction                            |
| Reversal Request and Advice messages                                     | Provides information to partially or fully reverse a previously approved transaction                                                                       |

## Standard flow

### Standard flow

This message flow describes the standard authorization transaction process.

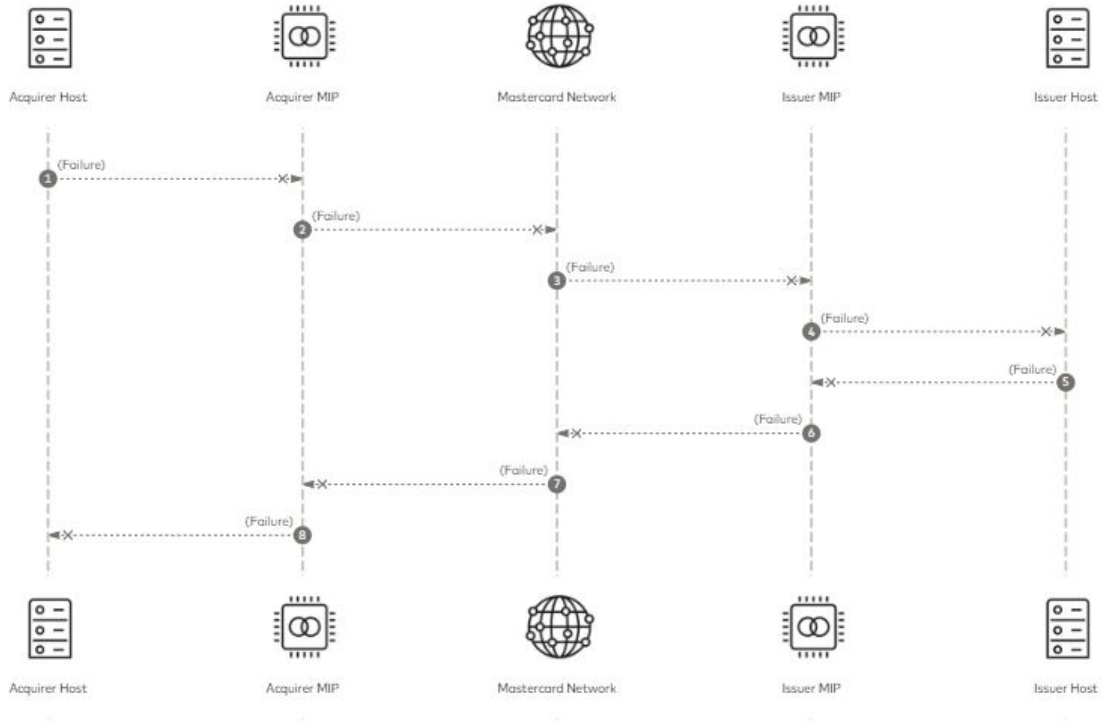
## Standard flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*.
3. The *issuer* generates an Authorization Request Response/0110 message and sends it to the *Mastercard Network*.
4. The *Mastercard Network* forwards the Authorization Request Response/0110 message to the *acquirer*.

## Failure scenarios: communication-related flows

### Failure flow



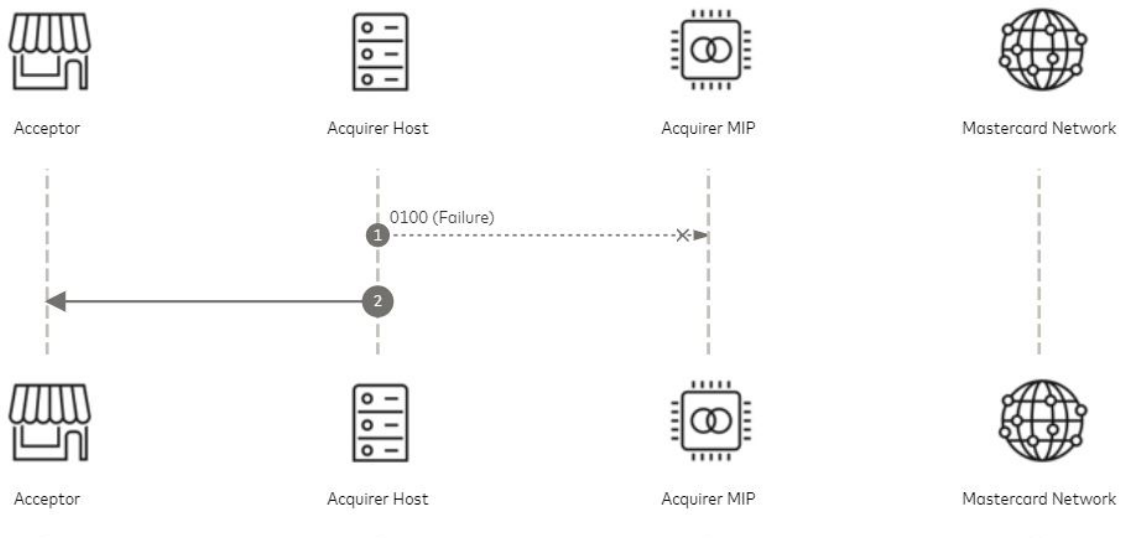
List of communication-related failure scenarios:

1. Request Failure from acquirer host to acquirer MIP
2. Request Failure from acquirer MIP to Mastercard Network
3. Request Failure from Mastercard Network to issuer MIP
4. Request Failure from issuer MIP to issuer host
5. Response Failure from issuer host to issuer MIP
6. Response Failure from issuer MIP to Mastercard Network
7. Response Failure from Mastercard Network to acquirer MIP
8. Response Failure from acquirer MIP to acquirer host

## Request Failure from acquirer host to acquirer MIP

This flow describes a failure scenario that occurs when the *acquirer host* cannot reach the *acquirer MIP* to send the transaction.

### Request failure from acquirer host to acquirer MIP flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the Mastercard Network. The *request* message is not received at the *acquirer MIP* due to a communication issue.
2. The *acquirer* completes the transaction at the point of interaction. In most cases, the *acquirer* simply denies the transaction. However, the *acquirer* may elect to approve the transaction for completion for various reasons. This approval process is known as Acquirer Host X-Code.

There are specific guidelines an *acquirer* must follow if they choose to approve the transaction on their *host*. These can be found in [System functionality and services](#) of this guide.

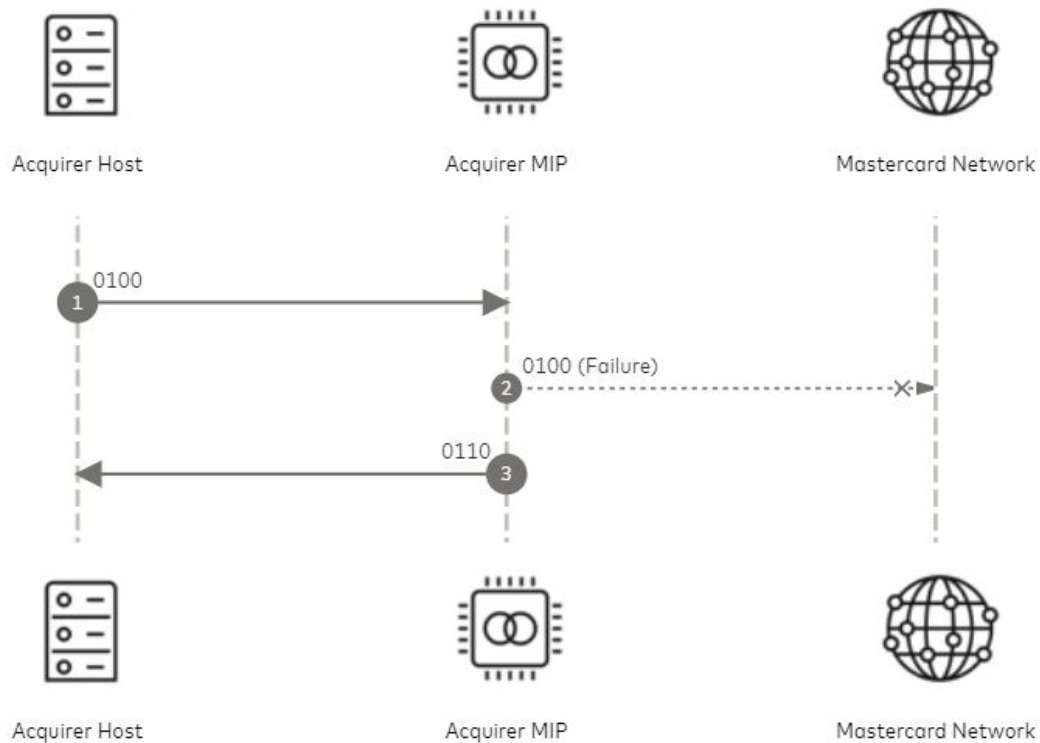
When this occurs, Mastercard recommends that the *acquirer* generate an Authorization Advice/0120: acquirer-generated message to advise the *issuer* that they approved the transaction. This allows the *issuer* to appropriately update the *cardholder's* account.

For more information on liability related to *X-Code*, refer to *Transaction Processing Rules* or *Mastercard Rules*.

## Request Failure from acquirer MIP to Mastercard Network

This flow describes a failure scenario when the [authorization request](#) has been received at the [acquirer MIP](#) but cannot be sent beyond that to the [Mastercard Network](#).

### Request failure from acquirer MIP to Mastercard Network flow



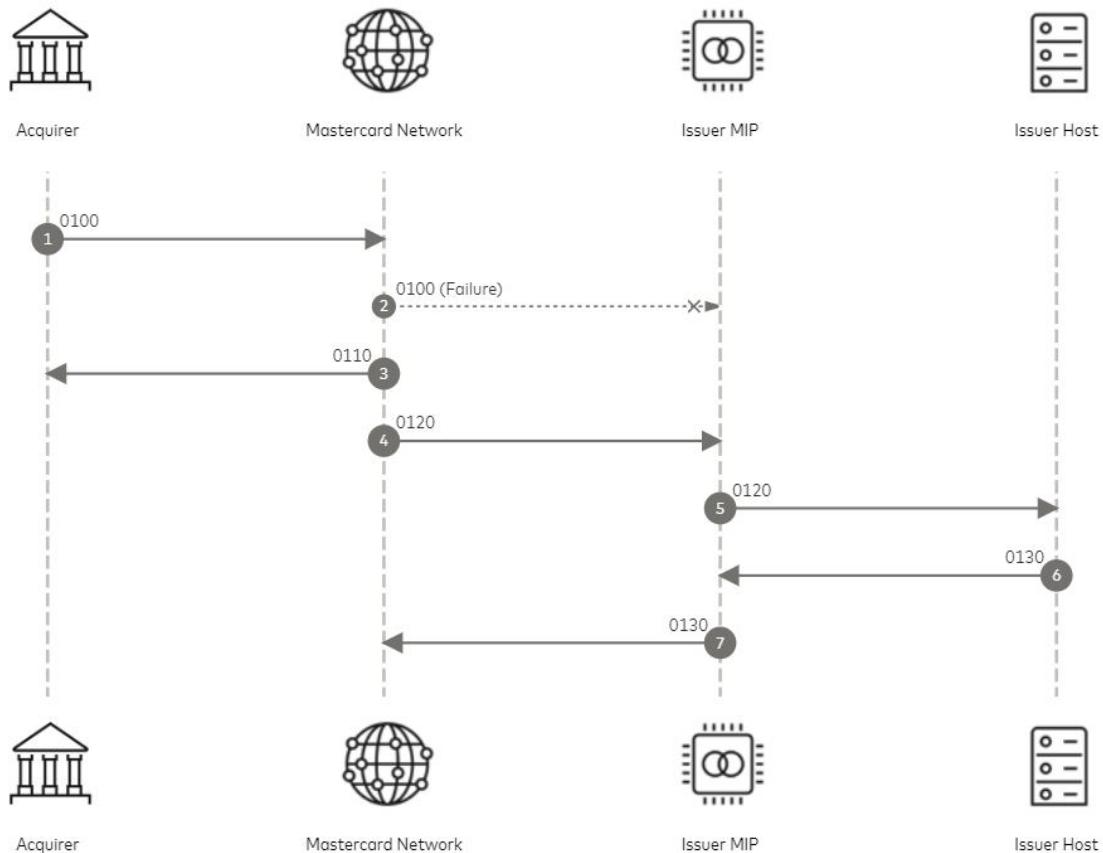
1. The [acquirer](#) initiates an Authorization Request/0100 message and sends it to the [Mastercard Network](#). The [request](#) message is received at the [acquirer MIP](#).
2. The [acquirer MIP](#) is unable to connect to the rest of the [Mastercard Network](#) due to a communication failure and cannot transmit the [request](#) message.
3. If [acquirer MIP X-Code Processing](#) is eligible for this transaction, this processing will occur and provide a decision for the transaction to the [acquirer](#) in the Authorization Request Response/0110 message.

For more information on liability related to [X-Code processing](#), refer to *Transaction Processing Rules* or *Mastercard Rules*.

## Request Failure from Mastercard Network to issuer MIP

This message flow describes when the *Mastercard Network* is unable to transmit an *authorization request* to the *issuer MIP* due to a communication failure.

### Request failure from Mastercard Network to issuer MIP flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The Mastercard Network attempts to forward the Authorization Request/0100 message to the *issuer*. The delivery of the message to the *issuer MIP* fails due to a communication failure.
3. The *Mastercard Network* detects the non-delivery of the request and if the *issuer* permits *Stand-In Authorization* or an alternate authorization process, the *Mastercard Network* sends the 0100 request message to *Stand-In Authorization*. *Stand-In Authorization* generates an Authorization Request Response/0110 message on the *issuer's* behalf based on the *issuer's* established parameters and sends it to the *acquirer*. It also generates an Authorization Advice/0120 message and forwards it to the *Mastercard Network SAF* queue for later transmission to the *issuer*.

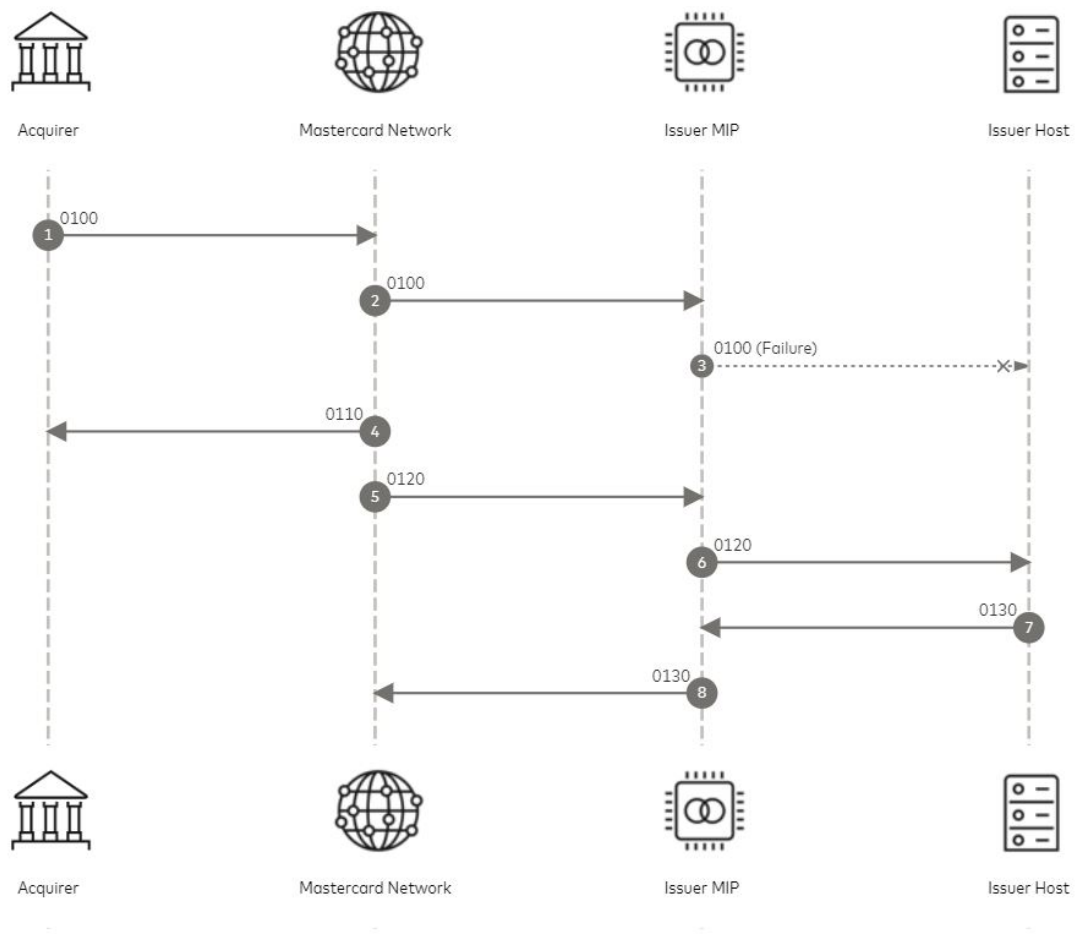


4. When communication is restored, the *Mastercard Network* will send an Authorization Advice/0120 message indicating the *Stand-In Authorization* result to the *issuer* during the *SAF* process. This message is successfully received at the *issuer MIP*.
5. The *issuer MIP* forwards the Authorization Advice/0120 *advice message* to the *issuer host*. The *issuer* uses this information to update the *cardholder* account. This action allows the *issuer* to update its *cardholder* open-to-buy and velocity data files in a timely manner.
6. The *issuer* must reply with an Authorization Advice Response/0130 Message to acknowledge receipt of the *advice message* and sends this from the *issuer host* to the *issuer MIP*.
7. The *issuer MIP* sends the *advice response message* to the *Mastercard Network* which completes this message flow.

## Request Failure from issuer MIP to issuer host

This message flow describes when the *issuer MIP* fails to transmit an *authorization request* to the *issuer host* due to a communication failure.

### Request failure from issuer MIP to issuer host flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The *Mastercard Network* successfully transmits the Authorization Request/0100 message to the *issuer MIP*.
3. The *issuer MIP* attempts to send the Authorization Request/0100 request message to the *issuer host*. The message cannot be delivered due to a communication failure.
4. The *Mastercard Network* detects the non-delivery of the request and if the *issuer* permits *Stand-In Authorization* or an alternate authorization process, the *Mastercard Network* sends the Authorization Request/0100 request message to *Stand-In Authorization*. *Stand-In Authorization* generates an Authorization Request Response/0110 message on the *issuer's*

behalf based on the *issuer's* established parameters and sends it to the *acquirer*. It also generates an Authorization Advice/0120 message and forwards it to the *Mastercard Network SAF* queue for later transmission to the *issuer*.

5. When communication is restored, the *Mastercard Network* will send the Authorization Advice/0120 message indicating the *Stand-In Authorization* result to the *issuer* during the *SAF* process. The network transmits the *advice message* to the *issuer MIP*.
6. The *issuer MIP* passes this *advice message* to the *issuer host*. This action allows the *issuer* to update its *cardholder* open-to-buy and velocity data files in a timely manner.
7. The *issuer* must reply with an Authorization Advice Response/0130 Message to acknowledge receipt of the *advice message*. This message is sent from the *issuer host* to the *issuer MIP*.

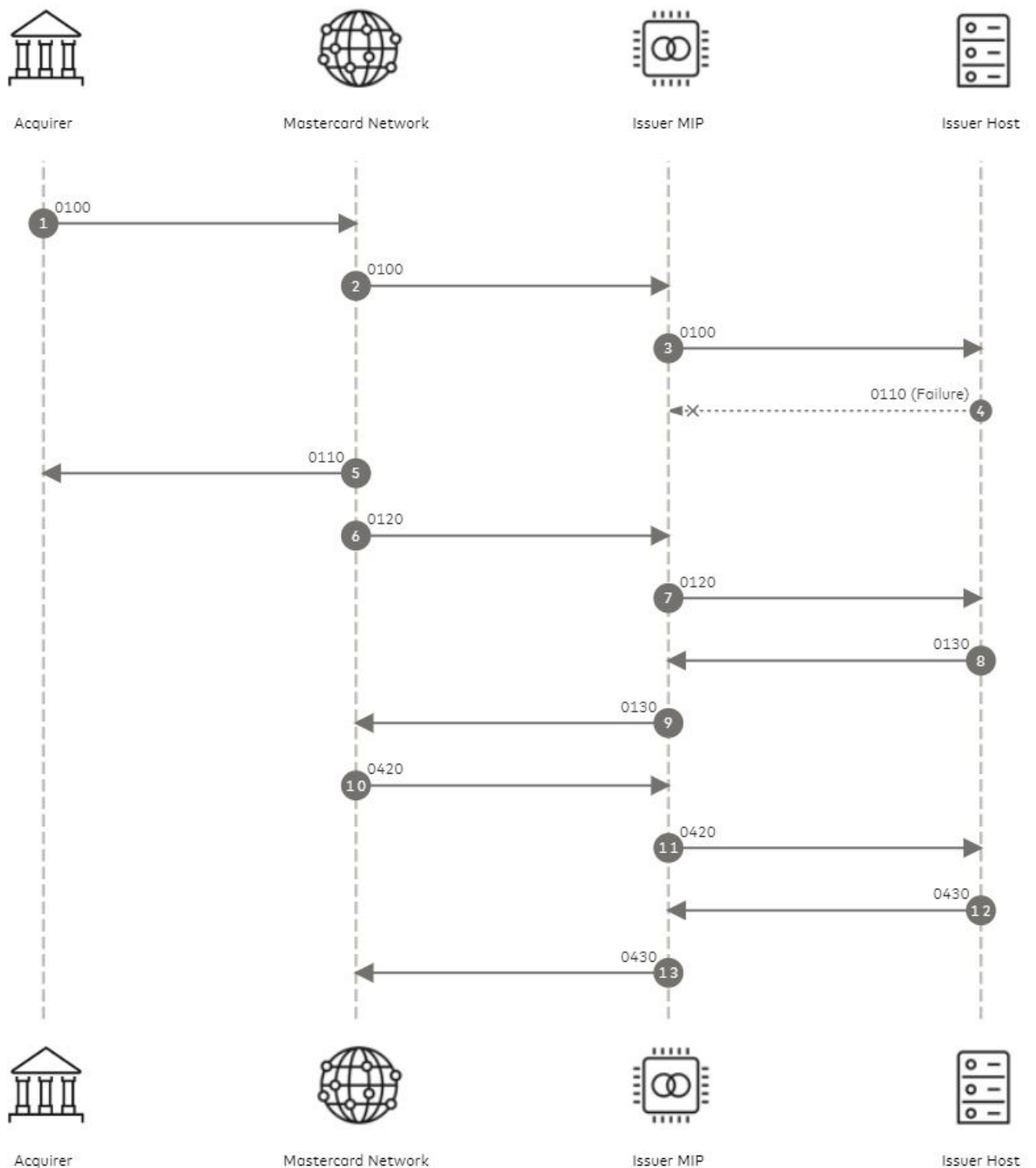
**NOTE:** Each time the *SAF* process receives an Authorization Advice Response/0130 message, it sends the next Authorization Advice/0120 message when there are multiple messages in the queue. This process repeats until there are no more advice messages. If the Mastercard Network does not receive an Authorization Advice Response/0130 message, it assumes the *issuer* did not receive the previous Authorization Advice/0120 message and ceases the *SAF* session.

8. Upon receipt of the message, the *issuer MIP* will transmit the Authorization Advice Response/0130 advice response to the *Mastercard Network* to complete the message flow.

## Response Failure from issuer host to issuer MIP

This message flow describes *exception processing* when communication fails between the *issuer host* and the *issuer MIP* during an Authorization Request Response/0110 message.

### Response failure from issuer host to issuer MIP flow



1. The *acquirer* initiates an Authorization Request/0100 message.

2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*. The message is received at the *issuer MIP*.
3. The *issuer MIP* sends the Authorization Request/0100 authorization request to the *issuer host*.
4. The *issuer* cannot transmit the Authorization Request Response/0110 message from the *issuer host* to the *issuer MIP* because of a communication failure. The *Mastercard Network* detects a time-out condition from the *issuer* on the Authorization Request Response/0110 message. If the *issuer* permits *Stand-In Authorization* or an alternate authorization process, the *Mastercard Network* sends an Authorization Request Response/0100 request message to *Stand-In Authorization*.
5. *Stand-In Authorization* generates an Authorization Request Response/0110 message on the *issuer's* behalf and forwards it to the *acquirer*. It also generates an Authorization Advice/0120 message and forwards it to the *Mastercard Network* SAF queue for later transmission to the *issuer*.

**NOTE: *Stand-In Authorization* generates and forwards Authorization Advice/0120 messages for both approved and declined Authorization Request/0100 messages.**

6. When communication re-establishes with the *issuer*, the *SAF* process transmits the Authorization Advice/0120 message to the *issuer MIP*. This *advice message* is delivered to the *issuer MIP*.
7. The *issuer MIP* transmits the Authorization Advice/0120 *advice message* to the *issuer host*. This action allows the *issuer* to update its *cardholder* open-to-buy and velocity data files in a timely manner.
8. The *issuer*, on receiving the Authorization Advice/0120 message, returns an Authorization Advice Response/0130 message from the *issuer host* to the *issuer MIP* to acknowledge receipt of the Authorization Advice/0120 *advice message*.
9. The Authorization Advice Response/0130 authorization *advice message* is transmitted from the *issuer MIP* to the *Mastercard Network*.

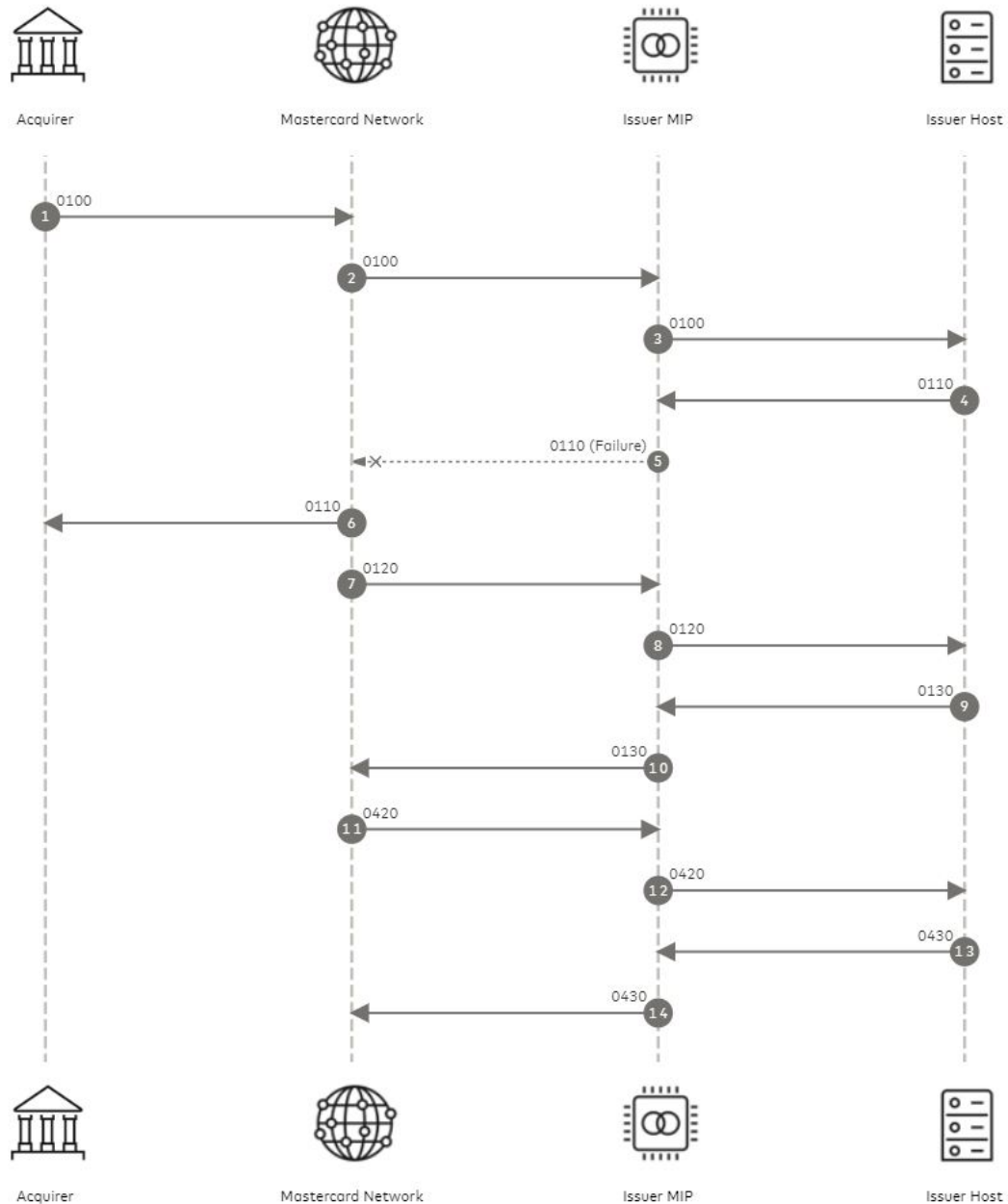
**NOTE: Each time the *SAF* process receives an Authorization Advice Response/0130 message, it sends the next Authorization Advice/0120 message when there are multiple messages in the queue. This process repeats until there are no more advice messages. If the *Mastercard Network* does not receive an Authorization Advice Response/0130 message, it assumes the *issuer* did not receive the previous Authorization Advice/0120 message and ceases the *SAF* session.**

10. The *Mastercard Network* sends a Reversal Advice/0420 message containing data indicating that the *issuer's* response timed out. This message is delivered to the *issuer MIP*.
11. The *issuer MIP* sends the Reversal Advice/0420 *advice message* to the *issuer host*.
12. When the *issuer* receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message. This message is sent from the *issuer host* to the *issuer MIP*.
13. The *issuer MIP* transmits the Reversal Advice Response/0430 message to the *Mastercard Network*.

## Response Failure from issuer MIP to Mastercard Network

This message flow describes exception processing when communication fails between the *issuer* MIP and the *Mastercard Network* during an Authorization Request Response/0110 message.

### Response failure from issuer MIP to Mastercard Network flow



1. The *acquirer* initiates an Authorization Request/0100 message.

2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*. The message is received at the *issuer MIP*.
3. The *issuer MIP* sends the Authorization Request/0100 authorization request to the *issuer host*.
4. The *issuer* evaluates the request and sends an Authorization Request Response/0110 message from the *issuer host* to the *issuer MIP*.
5. The Authorization Request Response/0110 message cannot be transmitted from the *issuer MIP* to the *Mastercard Network* because of a communications failure.
6. The *Mastercard Network* detects a time-out condition from the *issuer* on the Authorization Request Response/0110 message. If the *issuer* permits *Stand-In Authorization*, the *Mastercard Network* sends an Authorization Request/0100 message to *Stand-In Authorization*.  
*Stand-In Authorization* generates an appropriate Authorization Request Response/0110 message on the *issuer's* behalf and forwards it to the *acquirer*. It also generates an Authorization Advice/0120 message and forwards it to the *Mastercard Network SAF* queue for later transmission to the *issuer*.
7. When communication re-establishes with the *issuer*, the *SAF* process transmits the Authorization Advice/0120 message to the *issuer MIP*. This *advice message* is delivered to the *issuer MIP*.
8. The *issuer MIP* transmits the Authorization Advice/0120 *advice message* to the *issuer host*. This action allows the *issuer* to update its *cardholder* open-to-buy and velocity data files in a timely manner.
9. The *issuer*, on receiving the Authorization Advice/0120 message, returns an Authorization Advice Response/0130 message from the *issuer host* to the *issuer MIP* to acknowledge receipt of the Authorization Advice/0120 *advice message*.
10. The Authorization Advice Response/0130 authorization advice message is transmitted from the *issuer MIP* to the *Mastercard Network*.

**NOTE: Each time the *SAF* process receives an Authorization Advice Response/0130 message, it sends the next Authorization Advice/0120 message when there are multiple messages in the queue. This process repeats until there are no more *advice messages*. If the *Mastercard Network* does not receive an Authorization Advice Response/0130 message, it assumes the *issuer* did not receive the previous Authorization Advice/0120 message and ceases the *SAF* session.**

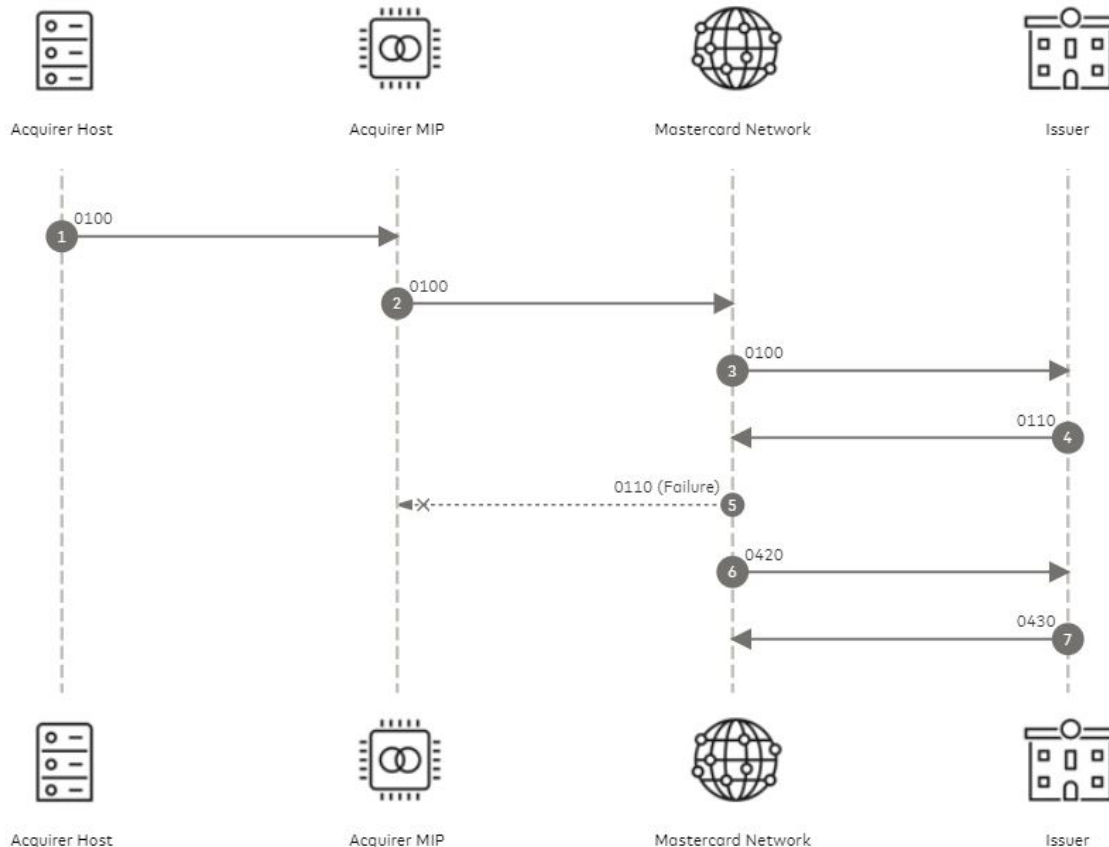
11. The *Mastercard Network* sends a Reversal Advice/0420 message containing data indicating that the *issuer's* response timed out. This message is delivered to the *issuer MIP*.
12. The *issuer* must reverse any effect to its *cardholder* or velocity files that took place during the *issuer's* approval of the initial *authorization request*.
13. When the *issuer* receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message. This message is sent from the *issuer host* to the *issuer MIP*.
14. The *issuer MIP* transmits the Reversal Advice Response/0430 message to the *Mastercard Network*.



## Response Failure from Mastercard Network to acquirer MIP

This message flow describes exception processing when communication fails between the *Mastercard Network* and the *acquirer MIP* during an Authorization Request Response/0110 message.

### Response failure from Mastercard Network to acquirer MIP flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*. The request message is received at the *acquirer MIP*.
2. The *acquirer MIP* sends the authorization request to the *Mastercard Network*.
3. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*.
4. The *issuer* generates an Authorization Request Response/0110 message providing the transaction decision and sends it to the *Mastercard Network*.
5. The *Mastercard Network* attempts to forward the Authorization Request Response/0110 message to the *acquirer MIP* but is unable to deliver the message.

6. Since the *response message* cannot be delivered, the *Mastercard Network* will create a Reversal Advice/0420 message back to the *issuer* if the Authorization Request Response/0110 message indicates an approval.  
The purpose of this advice is to let the *issuer* know that its Authorization Request Response/0110 message was not delivered, and the *issuer* should reverse or otherwise adjust its *cardholder* authorization files as necessary to reflect that the transaction could not be completed.

**NOTE: If the Authorization Request Response/0110 message indicates a decline from the *issuer*, no reversal advice is required since there is no hold against the *cardholder* account.**

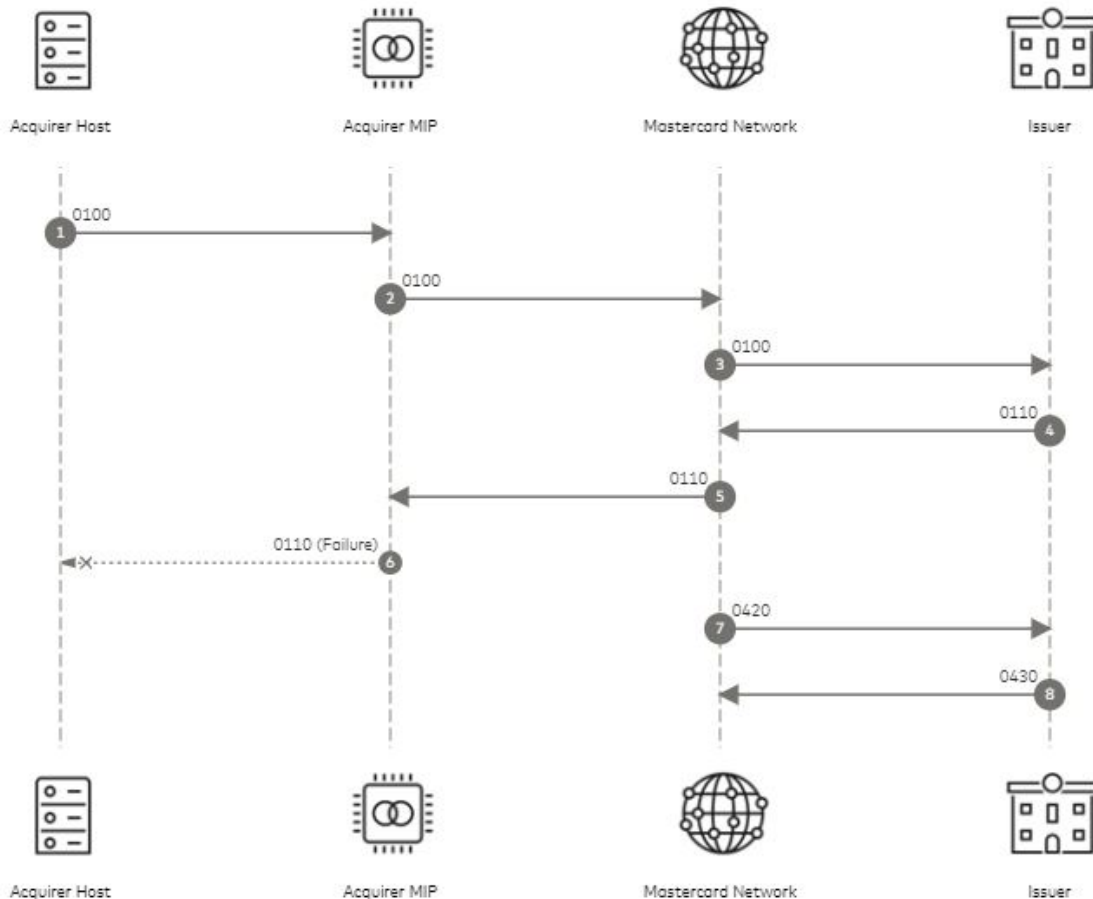
7. The *issuer* responds back to the *Mastercard Network* with a Reversal Advice Response/0430 message to acknowledge positive receipt of the Reversal Advice/0420 message.  
Meanwhile, if the *acquirer* is still operating, it detects a time-out condition on the Authorization Request Response/0110 message. When the time-out occurs, the *acquirer* must make a decision whether to authorize the transaction at the point of interaction. If the *acquirer* denies the transaction, the message flow terminates at this point.

**NOTE: Does not apply in the case of a Mastercard Hosted Mobile Phone Top-up request. If the *issuer* approves the Mastercard Hosted Mobile Phone Top-up transaction, the *Mastercard Network* will complete the Mastercard Hosted Mobile Phone Top-up request. The *acquirer* should provide the *cardholder* with notification at the ATM that the top-up request will be complete within 15 minutes.**

## Response Failure from acquirer MIP to acquirer host

This message flow describes exception processing when communication fails between the *acquirer MIP* and the *acquirer host* during an Authorization Request Response/0110 message.

### Response failure from acquirer MIP to acquirer host flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*. The request message is received at the *acquirer MIP*.
2. The *acquirer MIP* sends the authorization request to the *Mastercard Network*.
3. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*.
4. The *issuer* generates an Authorization Request Response/0110 message providing the transaction decision and sends it to the *Mastercard Network*.
5. The *Mastercard Network* delivers the Authorization Request Response/0110 response message to the *acquirer MIP*.

6. The *acquirer MIP* attempts to forward the Authorization Request Response/0110 message to the *acquirer host* and is unable to because of a communications failure.
7. The *Mastercard Network* detects the non-delivery of the *response message* and creates a Reversal Advice/0420 message back to the *issuer* if the Authorization Request Response/0110 message indicates an approval.  
The purpose of this advice is to let the issuer know that its Authorization Request Response/0110 message was not delivered, and the *issuer* should reverse or otherwise adjust its *cardholder* authorization files as necessary to reflect that the transaction could not be completed.

**NOTE: If the Authorization Request Response/0110 message indicates a decline from the *issuer*, no reversal advice is required since there is no hold against the *cardholder* account.**

8. The *issuer* responds back to the *Mastercard Network* with a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.  
Meanwhile, if the *acquirer* is still operating, it detects a time-out condition on the Authorization Request Response/0110 message. When the time-out occurs, the *acquirer* must make a decision whether to authorize the transaction at the point of interaction. If the *acquirer* denies the transaction, the message flow terminates at this point.

**NOTE: Does not apply in the case of a Mastercard Hosted Mobile Phone Top-up request. If the *issuer* approves the Mastercard Hosted Mobile Phone Top-up transaction, the *Mastercard Network* will complete the Mastercard Hosted Mobile Phone Top-up request. The *acquirer* should provide the *cardholder* with notification at the ATM that the top-up request will be complete within 15 minutes.**

## Failure scenarios: issuer response contains errors or late response flow

List of *issuer* responses that contains errors or late response failure scenarios:

- *Response from issuer contains errors*
- *Issuer response received late (after time limit)*

### Response from issuer contains errors

This message flow describes exception processing when an invalid *issuer* Authorization Request Response/0110 message is received and *Stand-In Authorization* is allowed.

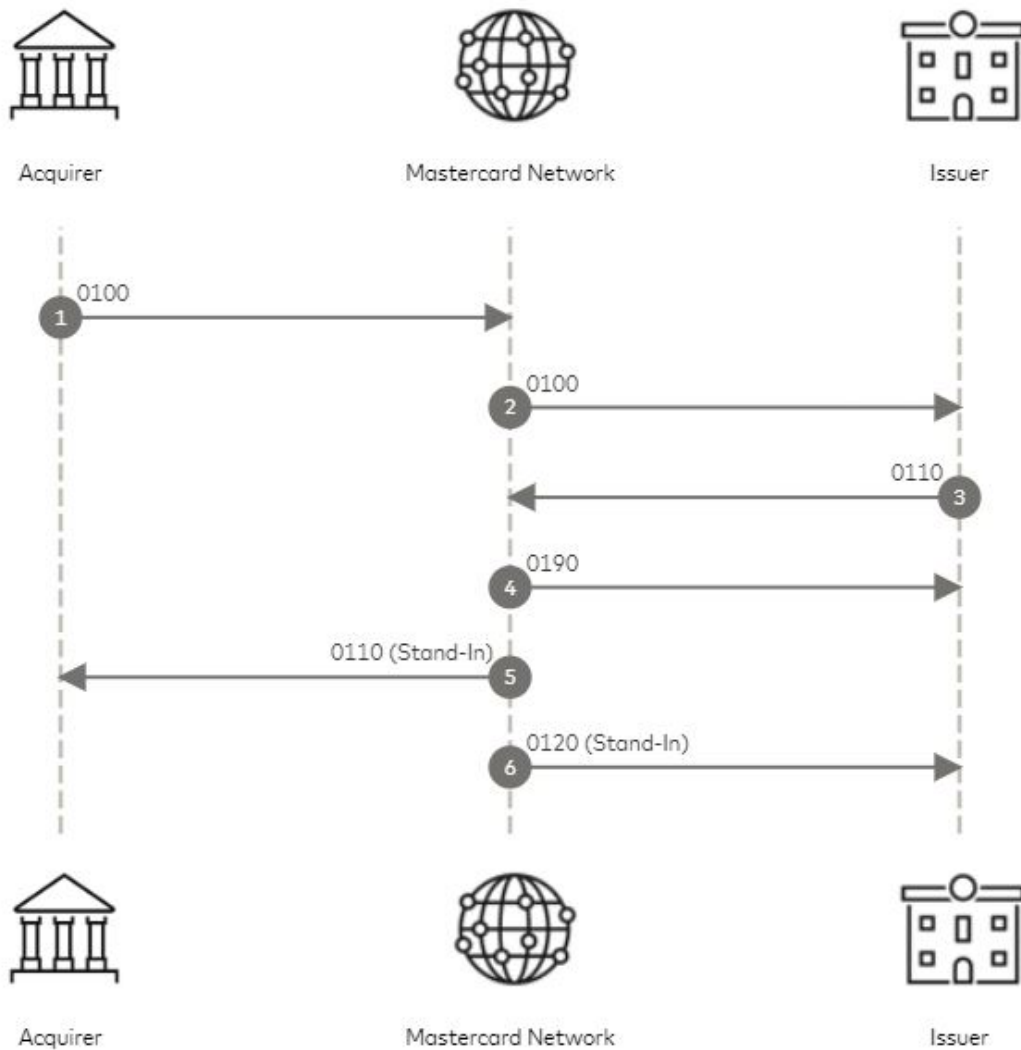
Mastercard will provide the *issuer's* response to the Authorization Request/0110 message whenever possible. If the *issuer* returns an Authorization Request Response/0110 message, Mastercard evaluates the *authorization response* to determine if the transaction can be parsed and if *DE 39 (Response Code)* can be read. If Mastercard can read *DE 39* and the *issuer* has approved the transaction, inclusive of the following *response codes*, Mastercard uses this message flow and sends the request to *Stand-In Authorization*:

- 00 Approved or completed successfully

- 10 Partial Approval
- 87 Purchase Amount Only, No Cash Back Allowed
- 91 Authorization System or issuer system inoperative
- 92 Unable to route transaction
- 96 System error (inclusive of Response Codes)

If the [response code](#) is any other than those previously listed, Mastercard replaces or removes the data responsible for the format error and forwards the remainder of the message to the [acquirer](#). Mastercard continues to log information relevant to the format error for reference purposes.

#### Response from issuer contains errors flow



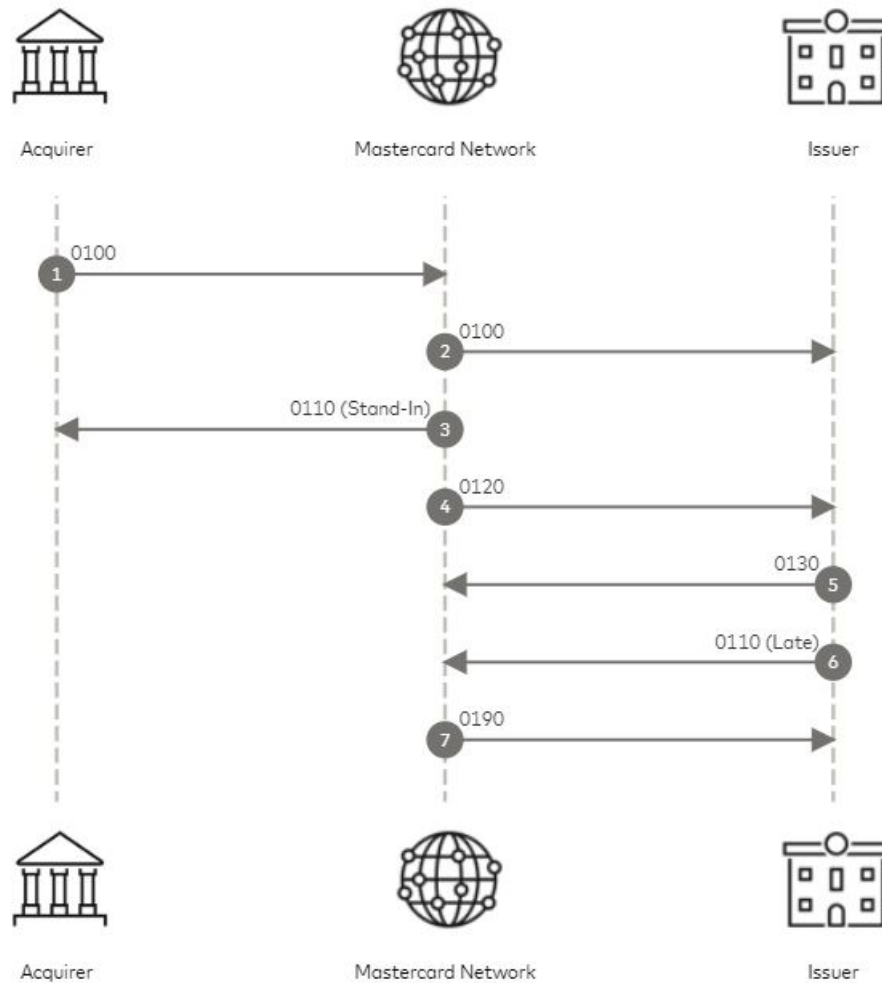
1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*.
3. The *issuer* returns an Authorization Request Response/0110 which contains invalid data. Depending on the response code provided, the transaction may route to *Stand-In Authorization*.
4. The *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message to the *issuer*.
5. The *Mastercard Network* sends the Authorization Request/0100 message to *Stand-In Authorization*. *Stand-In Authorization* generates an Authorization Request Response/0110 message and the *Mastercard Network* forwards it to the *acquirer*.
6. The *Stand-In Authorization* also generates an Authorization Advice/0120 message which is added to the *SAF* queue for the *issuer*. Once the *issuer* is available, the *advice message* is sent.

**NOTE:** There is no response necessary from the *issuer* after the *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message.

## Issuer response received late (after time limit)

This message flow describes exception processing when an Authorization Request Response/0110 message is received after the time limit and *Stand-In Authorization* is allowed.

### Issuer response received late (after time limit) flow



1. The [acquirer](#) initiates an Authorization Request/0100 message and sends it to the [Mastercard Network](#).
2. The [Mastercard Network](#) forwards the Authorization Request/0100 message to the [issuer](#).
3. The [Mastercard Network](#) detects a time-out condition from the [issuer](#) on the Authorization Request Response/0110 message. If the [issuer](#) permits *Stand-In Authorization*, the [Mastercard Network](#) sends an Authorization Request/0100 message to *Stand-In Authorization*.

*Stand-In Authorization* generates an appropriate Authorization Request Response/0110 message on the *issuer's* behalf and forwards it to the *acquirer*. It also generates an Authorization Advice/0120 message and forwards it to the *Mastercard Network SAF* queue for later transmission to the *issuer*.

4. When communication re-establishes with the *issuer*, the *SAF* process transmits the Authorization Advice/0120 message to the *issuer*.
5. The *issuer*, on receiving the Authorization Advice/0120 message, transmits an Authorization Advice Response/0130 message to the *Mastercard Network*.
6. The *issuer* sends an Authorization Request Response/0110 message after the time limit to the *Mastercard Network*.
7. The *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message to the *issuer*, indicating it has no record of a corresponding Authorization Request/0100 message.

**NOTE:** There is no response necessary from the *issuer* after the *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message.

## Failure scenarios: alternate issuer host configuration flows

List of *alternate issuer host* configuration failure scenarios:

- *Communication Failure or primary issuer not signed in*
- *Primary issuer times out*
- *Primary issuer responds after time limit and primary issuer response received first*

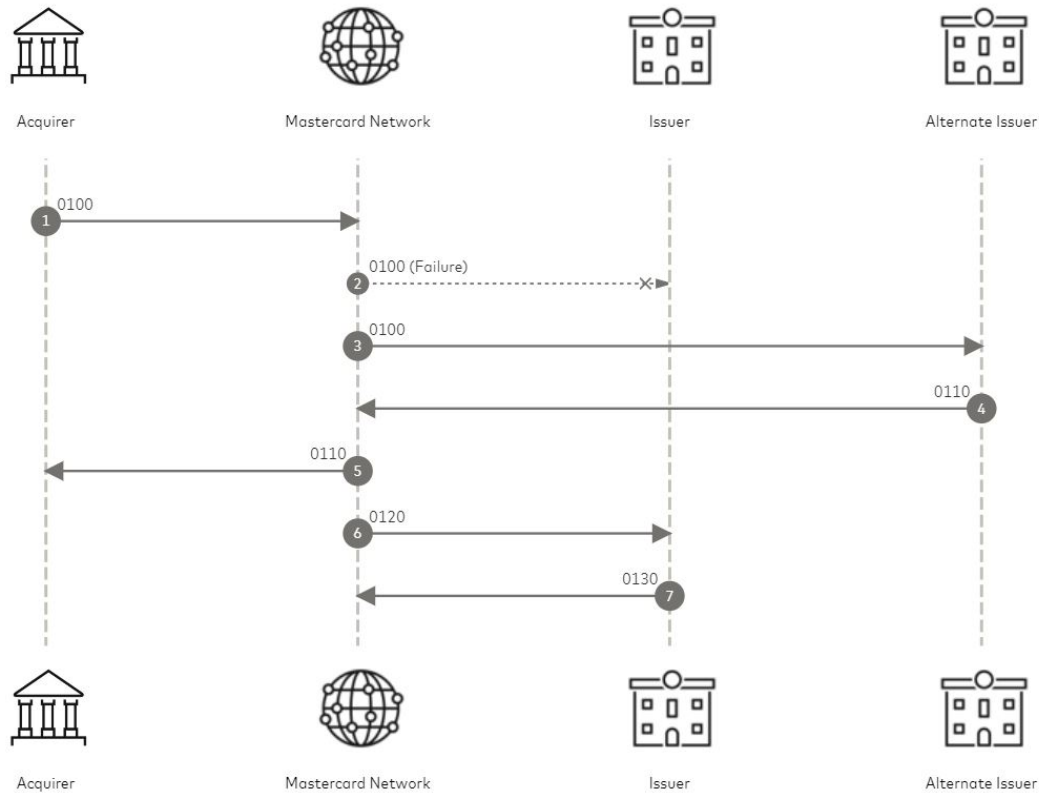
### Communication Failure or primary issuer not signed in

This message flow describes *exception processing* when the *primary issuer* is not signed in, the transaction could not be delivered to the *primary issuer*, or when the *primary issuer* does not



respond with an Authorization Request Response/0110 message within the time limit defined for the *acceptance brand*.

### Communication failure or primary issuer not signed in flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The *Mastercard Network* attempts to forward the Authorization Request/0100 message to the *primary issuer* but is unable to complete the message transmission because of a communication link failure, *issuer* failure or the *issuer* is not signed on to the *network*.
3. The *Mastercard Network* immediately sends the Authorization Request/0100 message to the *alternate issuer*.
4. The *alternate issuer* generates an Authorization Request Response/0110 message on the *primary issuer*'s behalf.
5. The *Mastercard Network* forwards the Authorization Request Response/0110 message from the *alternate issuer* to the *acquirer*.
6. The *Mastercard Network* generates an Authorization Advice/0120 message and forwards it to the *Mastercard Network SAF* process for later transmission to the *primary issuer*. This advises the *primary issuer* of the decision from the *alternate issuer*. This advises the primary

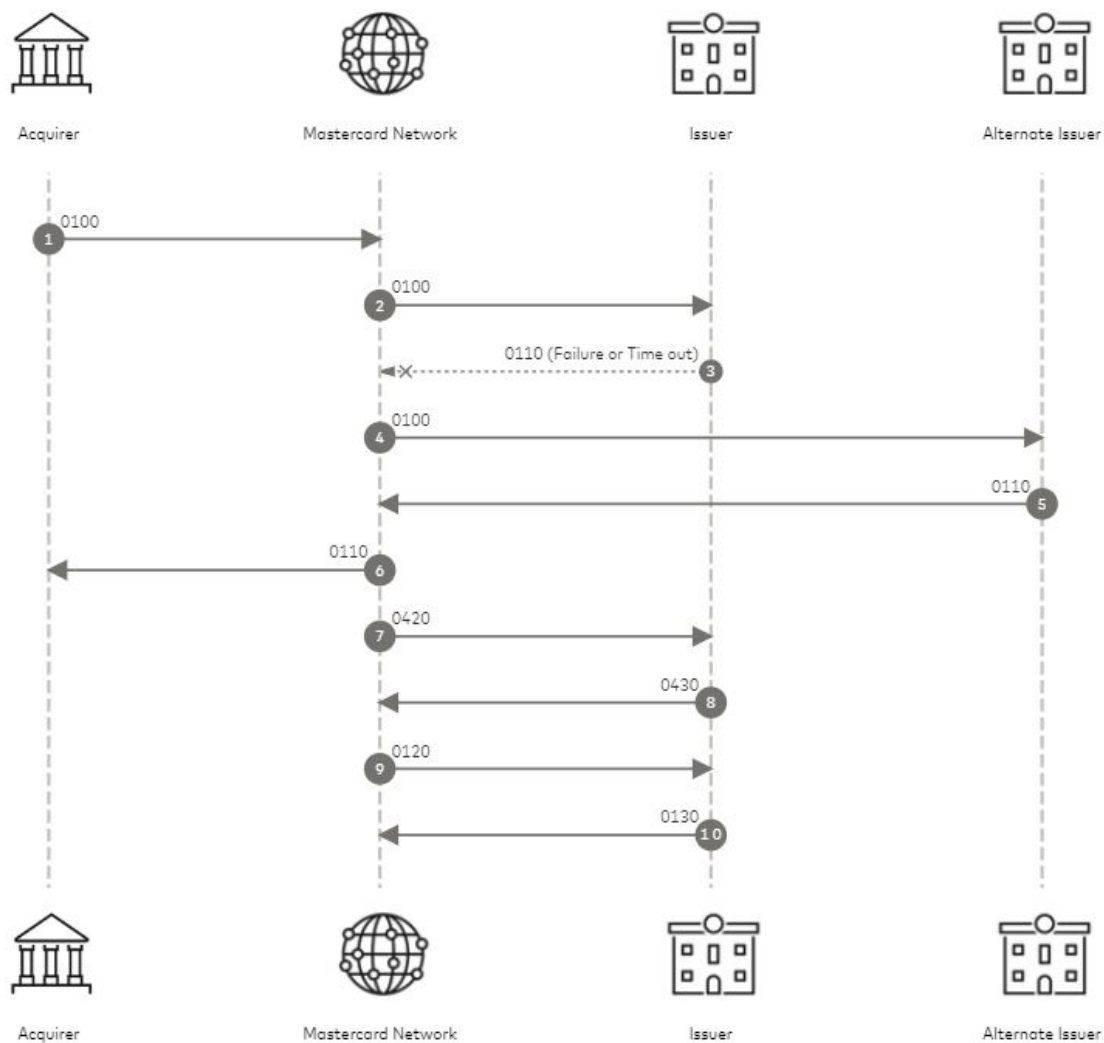
issuer of approval from the alternate issuer. No authorization advice/0120 message is sent if the transaction is declined.

- The *issuer* responds to the Authorization Advice/0120 message with an Authorization Advice Response/0130 message back to the *Mastercard Network*.

## Primary issuer times out

This message flow describes *exception processing* when no *issuer* Authorization Request Response/0110 message is received within the time limit after receipt of the Authorization Request/0100 message for transactions that are allowed to be processed by the *alternate issuer host*.

### Primary issuer times out flow

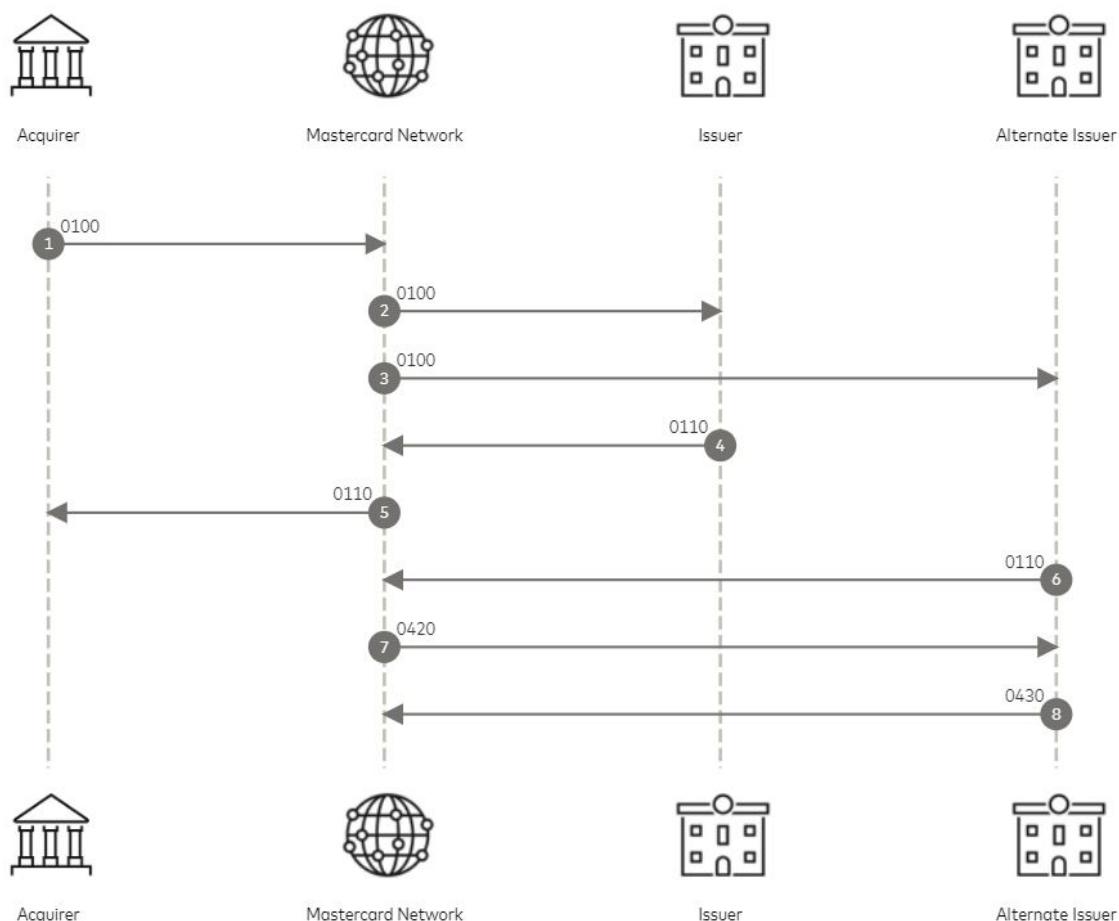


1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *primary issuer*.
3. The *Mastercard Network* does not receive the Authorization Request Response/0110 message from the *primary issuer* within the time limit.
4. The *Mastercard Network* sends the Authorization Request/0100 message to the *alternate issuer*.
5. The *alternate issuer* generates and sends the Authorization Request Response/0110 message to the *Mastercard Network*.
6. The *Mastercard Network* forwards to the *acquirer* the Authorization Request Response/0110 message received from the *alternate issuer*.
7. The *Mastercard Network* creates an Authorization Advice/0120 message and stores it in the *SAF* queue for guaranteed delivery to the *primary issuer*.
8. Upon *receipt*, the *issuer* must respond to the *SAF advice message* with an Authorization Advice Response/0130 message.
9. Since the response was not received within the time limit, the *Mastercard Network* will send an Authorization Advice/0120 message to the *primary issuer* to advise that they timed out on the Authorization Request. Even though no response was received from the *primary issuer*, this *advice message* indicates to the *primary issuer* to release any funds they may have held for the transaction.
10. When the *primary issuer* receives the Authorization Advice/0120 message, it must generate an Authorization Advice Response/0130 message to acknowledge *receipt* of the Authorization Advice/0120 message.

## Primary issuer responds after time limit and primary issuer response received first

This message flow describes *exception processing* when the Authorization Request/0100 message has already been forwarded to the *alternate issuer*, but the *primary issuer's* authorization response is received before the *alternate issuer's* response.

### Primary issuer responds after time limit and primary issuer response received first flow



1. The *acquirer* sends the Authorization Request/0100 message to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *primary issuer*.
3. If the *Mastercard Network* does not receive the Authorization Request Response/0110 message within the time limit from the *primary issuer*, the *Mastercard Network* forwards the Authorization Request/0100 message to the *alternate issuer*.
4. The *primary issuer* sends the Authorization Request Response/0110 message to the *Mastercard Network*.

5. The *Mastercard Network* receives the *primary issuer's* Authorization Request Response/0110 message before the *alternate issuer's* response and sends the *primary issuer's* Authorization Request Response/0110 message to the *acquirer*.
6. The *alternate issuer* sends the Authorization Request Response/0110 message to the *Mastercard Network*.
7. The *Mastercard Network* generates a Reversal Advice/0420 message to the *alternate issuer* to reverse the *alternate issuer's* response that was not used.
8. When the *alternate issuer* receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge *receipt* of the Reversal Advice/0420 message.

## Failure scenarios: no alternate processing allowed

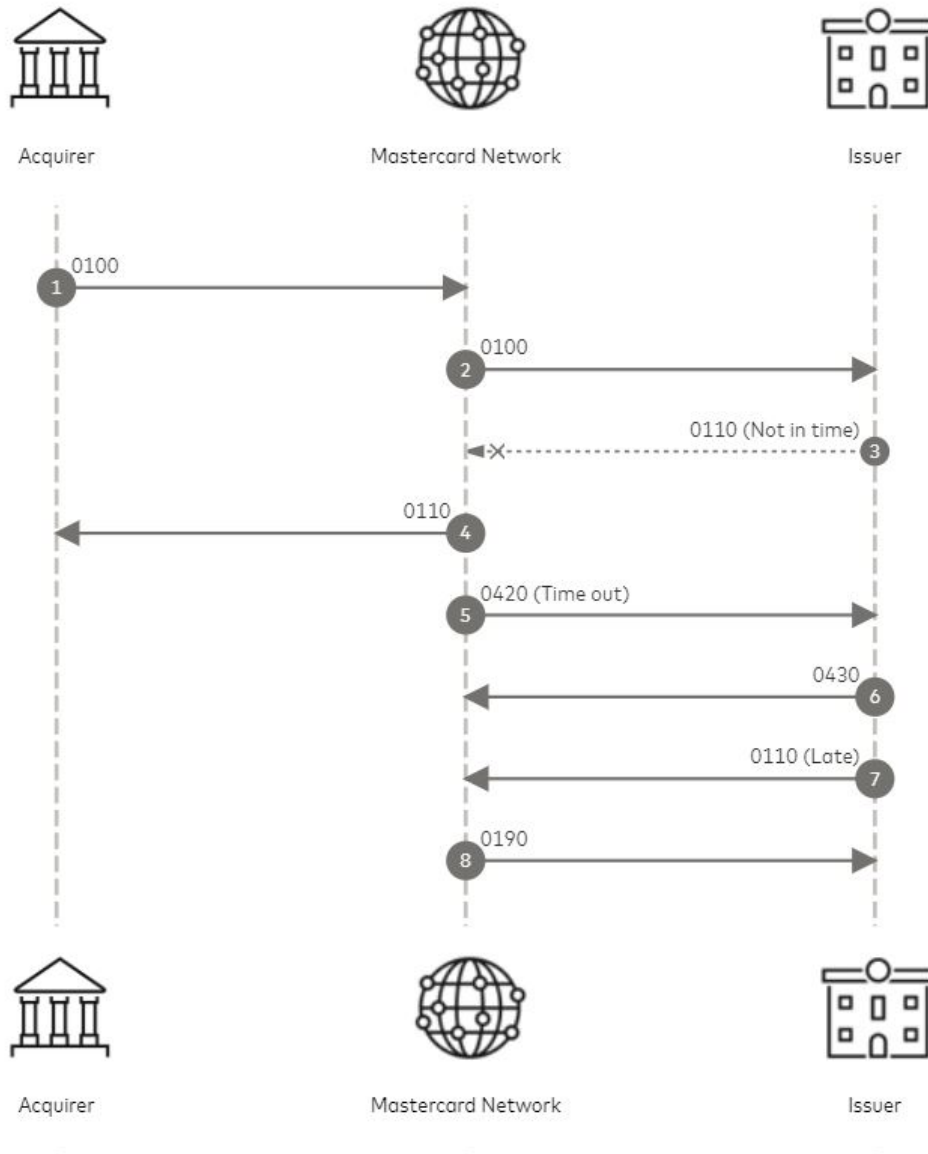
List of no alternate processing allowed failure scenarios:

- No alternate processing, issuer late response or not in time limit
- No alternate processing, issuer response contains errors

## No alternate processing, issuer late response or not in time limit

This message flow describes *exception processing* when the *issuer's* Authorization Request Response/0110 message is not received within the allowed *issuer* response time limit after *receipt* of the Authorization Request/0100 message.

### No alternate processing, issuer late response or not in time limit flow



1. The *acquirer* sends the Authorization Request/0100 message to the *Mastercard Network*.

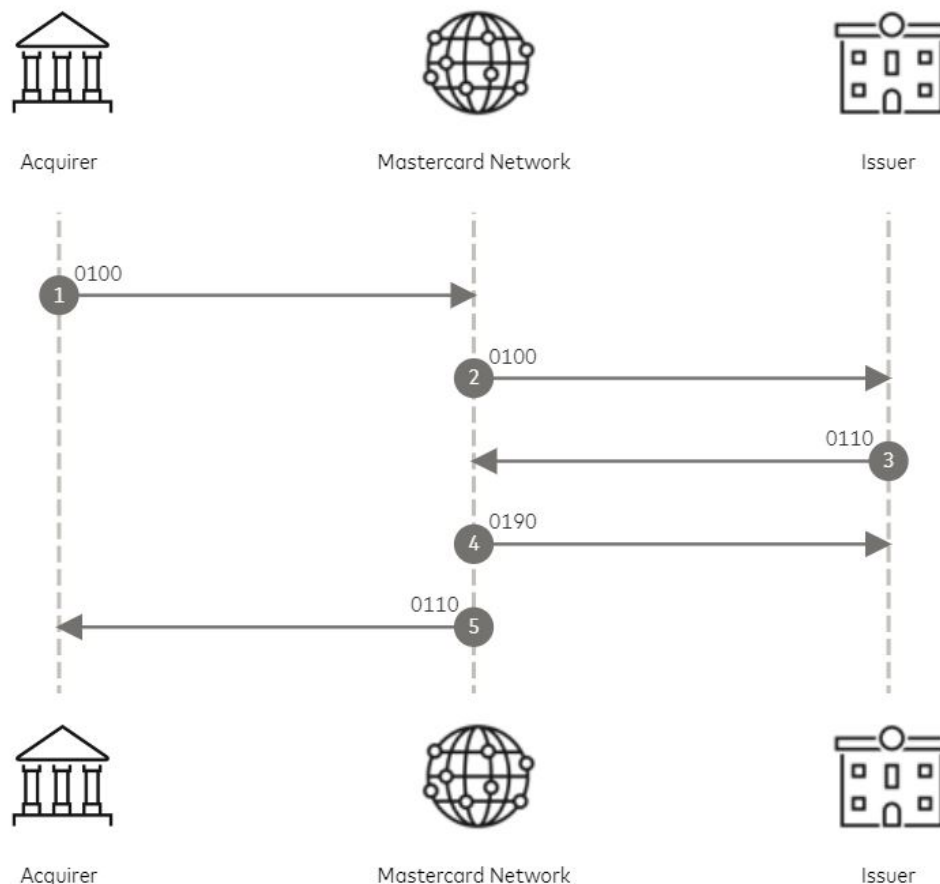
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*.
3. The *Mastercard Network* does not receive the *issuer's* Authorization Request Response/0110 within the response time allowed for the *issuer*.
4. The *acquirer* is sent an Authorization Request Response/0110 message indicating that the *issuer* could not be reached, and the transaction is declined.
5. A Reversal Advice/0420 message is sent to the *issuer* to advise the *Mastercard Network* did not receive the *issuer's* Authorization Request Response/0110 message within the allowed response time. Even though no response was received from the *issuer*, this *advice message* indicates to the *issuer* to release any funds they may have held for the transaction.
6. When the *issuer* receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge *receipt* of the Reversal Advice/0420 message.  
In case of a late response from the *issuer* the following two steps would occur.
7. The late Authorization Request Response/0110 message is received from the *issuer*.
8. The *Mastercard Network* creates an Authorization Response Negative Acknowledgment/0190 message which is sent to the *issuer*.

**NOTE:** There is no response necessary from the *issuer* after the *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message.

## No alternate processing, issuer response contains errors

This message flow describes *exception processing* when the *issuer's* Authorization Request Response/0110 message contains errors, and no alternate processing is allowed.

### No alternate processing, issuer response contains errors flow



1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*.
3. The *issuer* returns an Authorization Request Response/0110 which contains invalid data.
4. The *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message to the *issuer*.
5. The *Mastercard Network* sends the *acquirer* an Authorization Request Response/0110 message indicating that the *issuer* was not available and could not provide a response.



**NOTE:** There is no response necessary from the *issuer* after the *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message.

## Authorization Advice messages

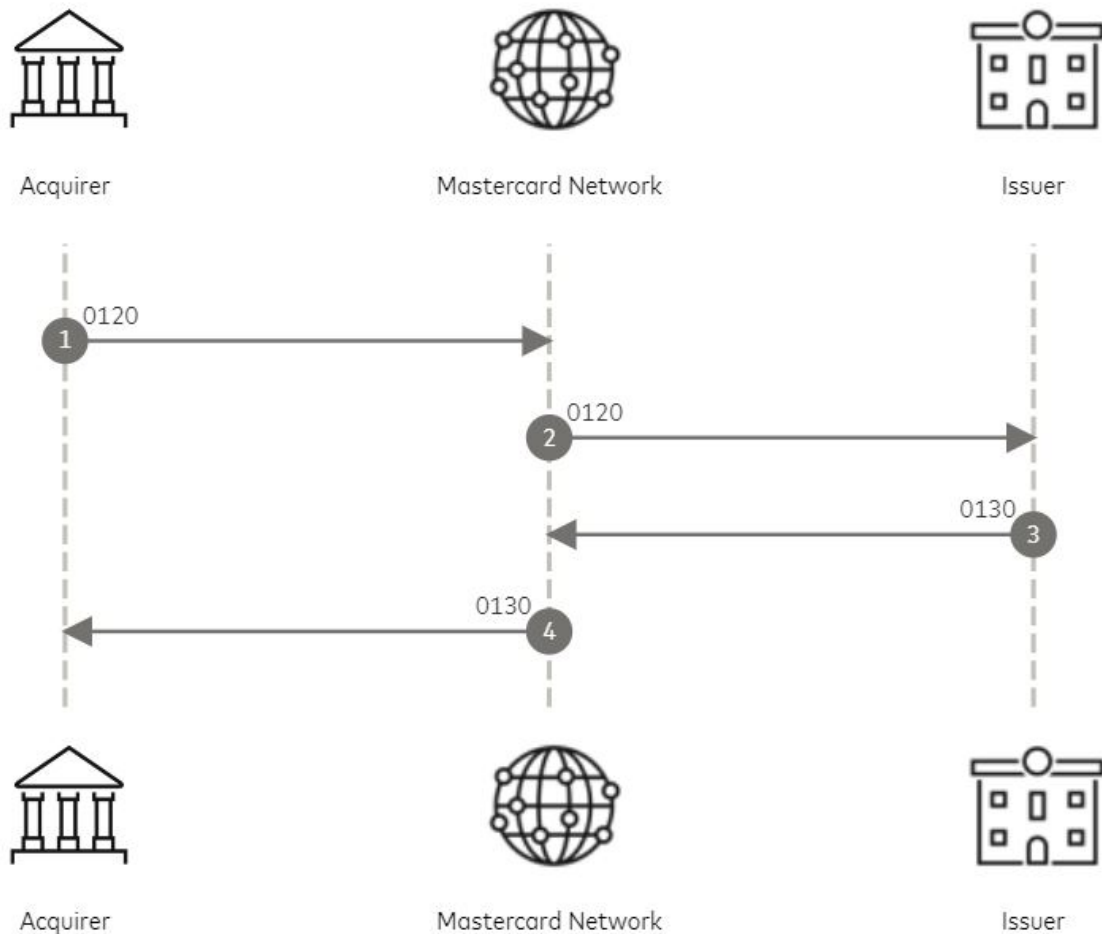
List of authorization *advice messages*:

- Acquirer-generated Authorization Advice message
- Acquirer-generated Advice message, issuer not available
- System-generated Authorization Advice message

## Acquirer-generated Authorization Advice message

This message flow describes the transaction process of the Authorization Advice/0120: acquirer-generated and Authorization Advice Response/0130: issuer-generated message when the *issuer* is available.

### Acquirer-generated authorization advice message standard flow



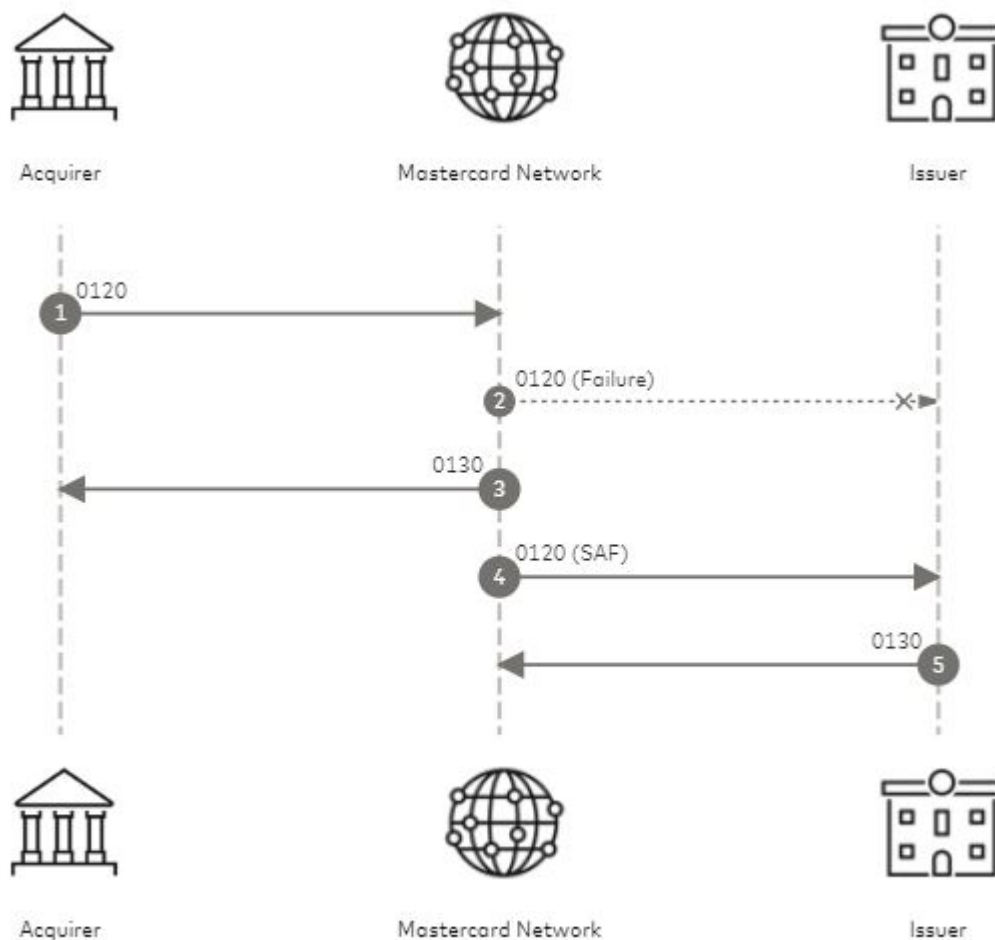
1. The *acquirer* initiates an Authorization Advice/0120: acquirer-generated message and passes the 0120 message to the *Mastercard Network*.
2. The Mastercard Network inserts DE 48, subelement 15 (Authorization System Advice Date and Time) into the message and forwards the Authorization Advice/0120: acquirer-generated message to the *issuer*.
3. The issuer returns an Authorization Advice Response/0130: issuer-generated message to the Mastercard Network to indicate positive *receipt* of the Authorization Advice/0120: acquirer-generated message. The issuer's advice *response message* will contain DE 48, subelement 15 (Authorization System Advice Date and Time).

4. The Mastercard Network forwards the issuer's Authorization Advice Response/0130 message to the acquirer.

### Acquirer-generated Advice message, issuer not available

This message flow describes the transaction process of the Authorization Advice/0120: acquirer-generated and Authorization Advice Response/0130: system-generated message when the issuer is unavailable or there is no issuer response.

#### Acquirer-generated advice message, issuer not available flow



1. The acquirer initiates an Authorization Advice/0120: acquirer-generated message and forwards the 0120 message to the Mastercard Network.
2. The Mastercard Network determines that the Authorization Advice/0120: acquirer-generated message cannot be delivered to the issuer host. The Mastercard Network routes

the Authorization Advice/0120: acquirer-generated message to the Stand-In Authorization to add to the issuer SAF queue.

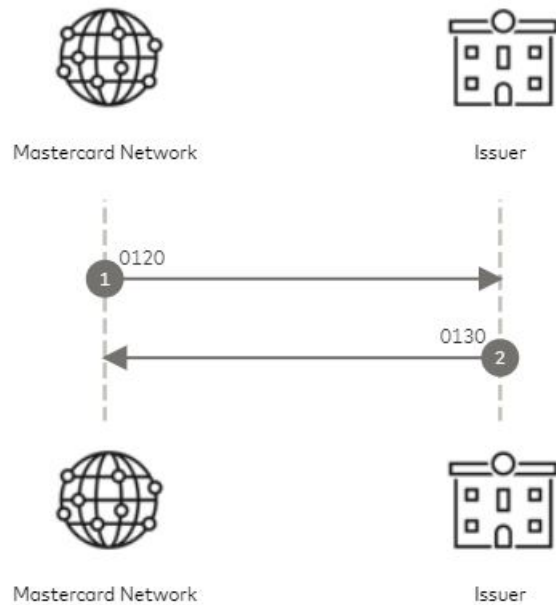
3. The Stand-In Authorization places the Authorization Advice/0120: acquirer-generated message into the SAF queue for guaranteed delivery to the issuer. The Stand-In Authorization responds to the acquirer with an Authorization Advice Response/0130: System-generated message containing DE 48, subelement 15 (Authorization System Advice Date and Time).
4. The Mastercard Network delivers the Authorization Advice/0120: acquirer-generated message to the issuer once the issuer is available.
5. The issuer will respond with an Authorization Advice Response/0130: issuer-generated message.

**NOTE: Customers in the Europe region that route to an alternate issuer host for alternate processing instead of the Stand-In Authorization will still receive an Authorization Advice/0120: acquirer-generated as described here. Alternate issuer host processing does not send Reversal Request/0400 or Authorization Advice/0120 messages to the alternate host.**

## System-generated Authorization Advice message

When Mastercard responds to an Authorization Request/0100 message on behalf of the issuer, the Mastercard Network generates an authorization response based on the issuer's parameters. The Mastercard Network also generates an Authorization Advice/0120: system-generated message and stores it in a Store-and-Forward (SAF) queue. SAF messages are forwarded to the issuer when they are available.

### System-generated authorization advice message flow



1. Stand-In Authorization generates an Authorization Advice/0120: system-generated message and sends it to the issuer.
2. The issuer returns an Authorization Advice Response/0130: issuer-generated message to the Mastercard Network to acknowledge receipt of the system-generated advice message.

### Reversal Request and Advice messages

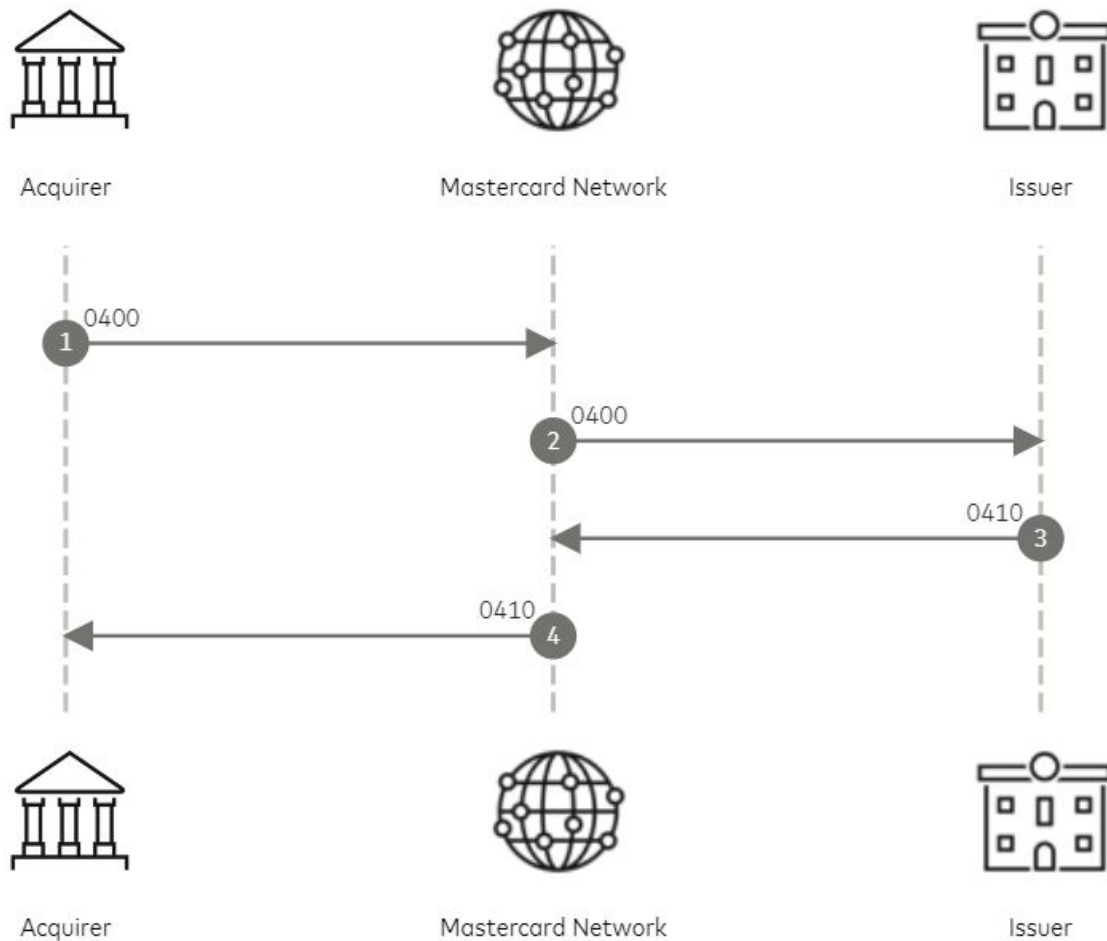
List of reversal request and advice messages flows:

- [Standard Reversal Request](#)
- [Reversal Response message failed at acquirer](#)
- [Reversal Request not delivered to issuer or issuer signed out](#)
- [Reversal Request no response from issuer or not within time limit](#)
- [Reversal Request late response from issuer](#)
- [Reversal Request response from issuer contains errors](#)
- [System-generated Reversal Advice](#)

## Standard Reversal Request

This message flow describes transaction processing of the Reversal Request/0400 and Reversal Request Response/0410 messages.

### Standard Reversal Request flow

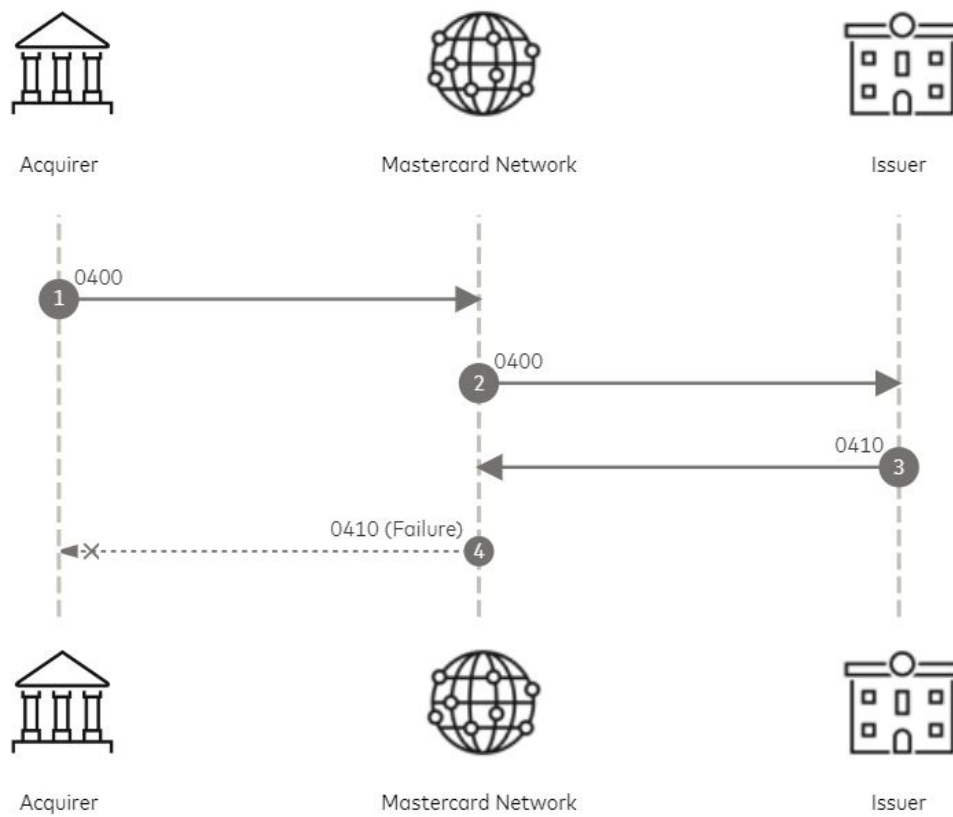


1. The acquirer initiates a Reversal Request/0400 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The issuer generates a Reversal Request Response/0410 message and sends it to the Mastercard Network.
4. The Mastercard Network forwards the Reversal Request Response/0410 message to the acquirer.

## Reversal Response message failed at acquirer

This message flow describes transaction processing when the Reversal Request Response/0410 message cannot be delivered to the acquirer.

### Reversal response message failed at acquirer flow

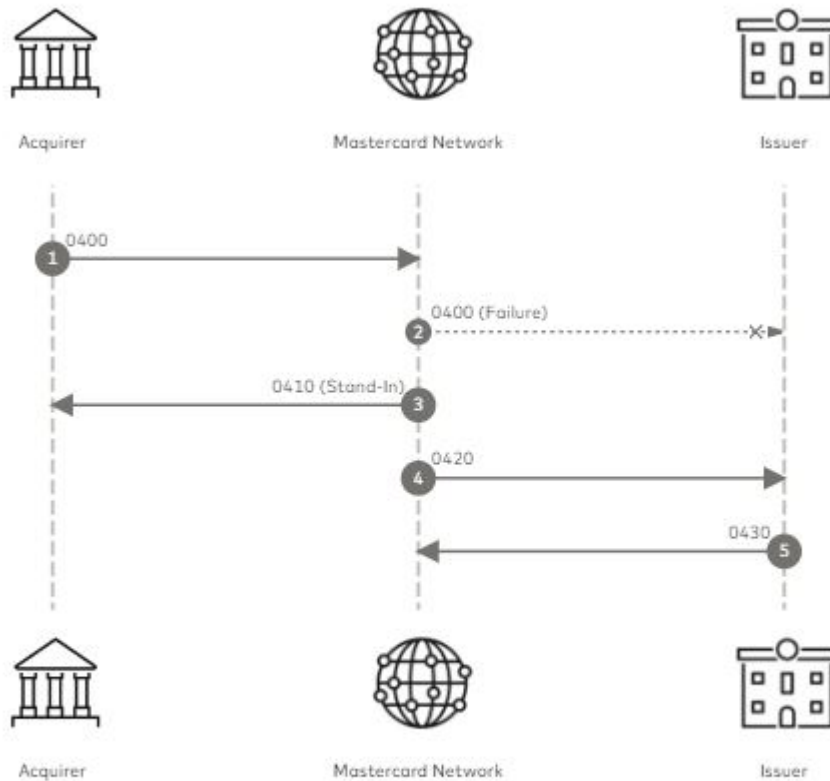


1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The issuer responds with a Reversal Request Response/0410 message to the Mastercard Network.
4. The Reversal Request Response/0410 message cannot be delivered to the acquirer's system. The Mastercard Network will store the Reversal Request/0400 and the Reversal Request Response/0410 message pair and take no further action.

## Reversal Request not delivered to issuer or issuer signed out

This message flow describes transaction processing when the issuer does not receive the Reversal Request/0400 message.

### Reversal request not delivered to issuer or issuer signed out flow



1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network attempts to forward the Reversal Request/0400 message to the issuer. If the issuer is signed out, there is no attempt to route the request to the issuer.
3. The Mastercard Network will send the Reversal Request/0400 message to the Stand-In Authorization. The Stand-In Authorization will generate the Reversal Request Response/0410 message and send it to the acquirer.
4. The Mastercard Network will create a Reversal Advice/0420 message and put it into a SAF queue for guaranteed delivery for the issuer. This advice will indicate if the issuer was signed out or if the request message could not be delivered. When the issuer is available to receive the advice message, the 0420 message will be delivered to the issuer.

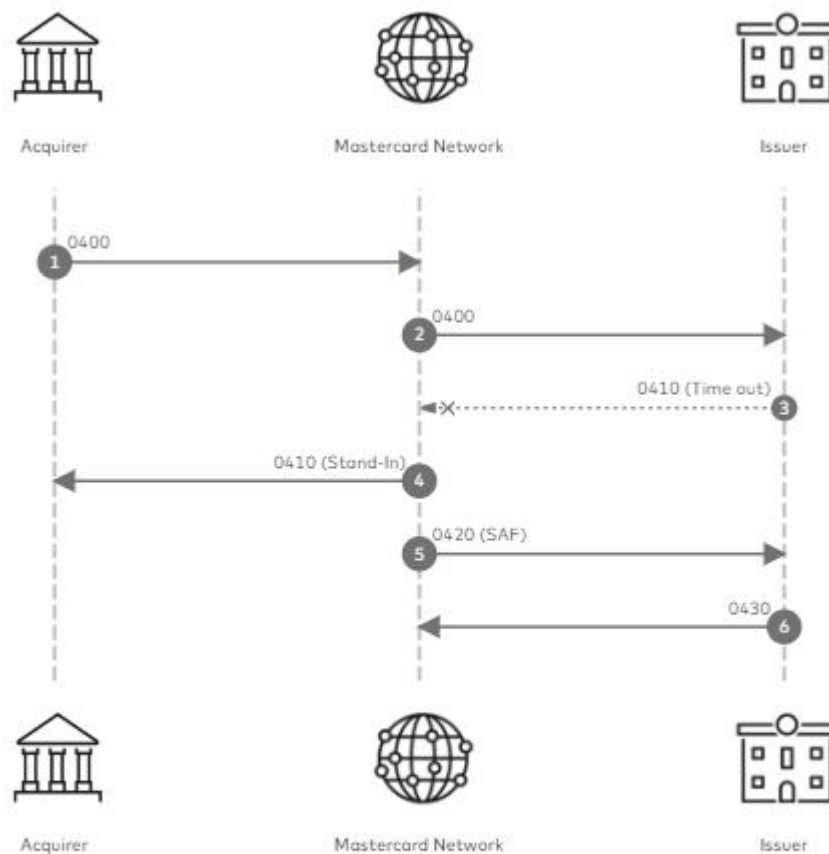


5. The issuer must reply with a Reversal Advice Response/0430 message to acknowledge receipt of the advice message.

### Reversal Request no response from issuer or not within time limit

This message flow describes transaction processing of the Reversal Request/0400 message when the issuer response is not received within the time limit.

#### Reversal request no response from issuer or not within time limit flow



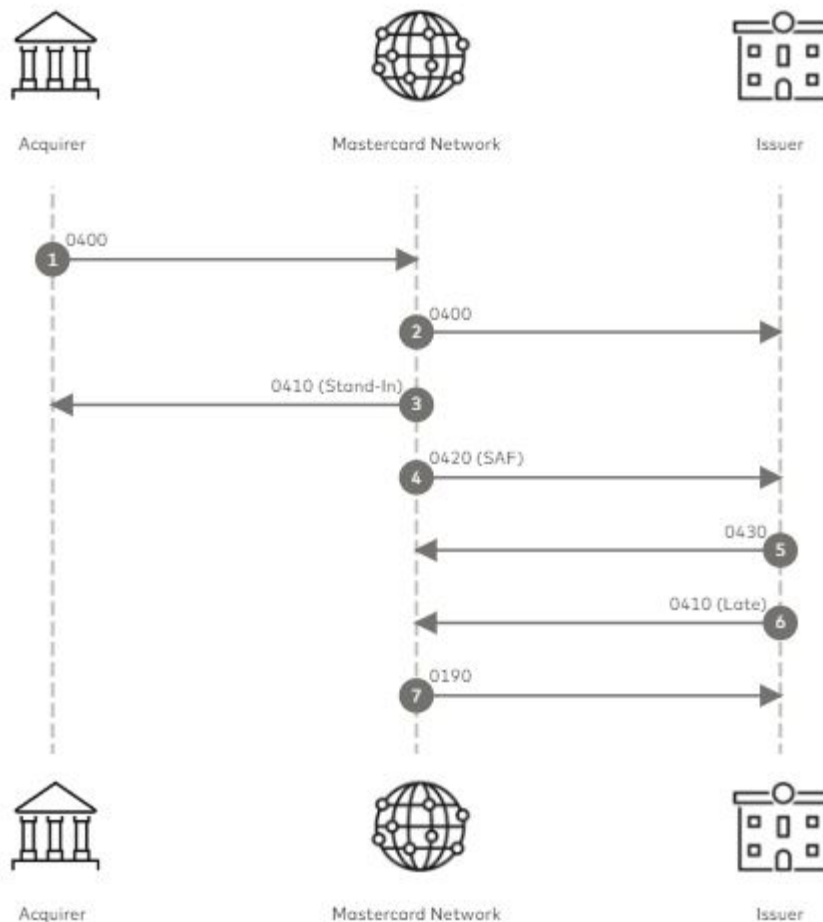
1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The Mastercard Network does not receive the issuer's Reversal Request Response/0410 message within the time limit and the Mastercard Network sends the Reversal Request/0400 message to the Stand-In Authorization.

4. The Stand-In Authorization generates the Reversal Request Response/0410 message which is then sent to the acquirer.
5. The Stand-In Authorization creates a Reversal Advice/0420 message and stores it in a SAF queue for guaranteed delivery to the issuer. When the issuer is available to receive the advice message, the Reversal Advice/0420 message will be delivered to the issuer. The message will indicate that the issuer timed out.
6. The issuer must reply with a Reversal Advice Response/0430 to acknowledge receipt of the advice message.

## Reversal Request late response from issuer

This message flow describes transaction processing when the issuer responds to the Reversal Request/0400 message after the time limit.

### Reversal request late response from issuer flow



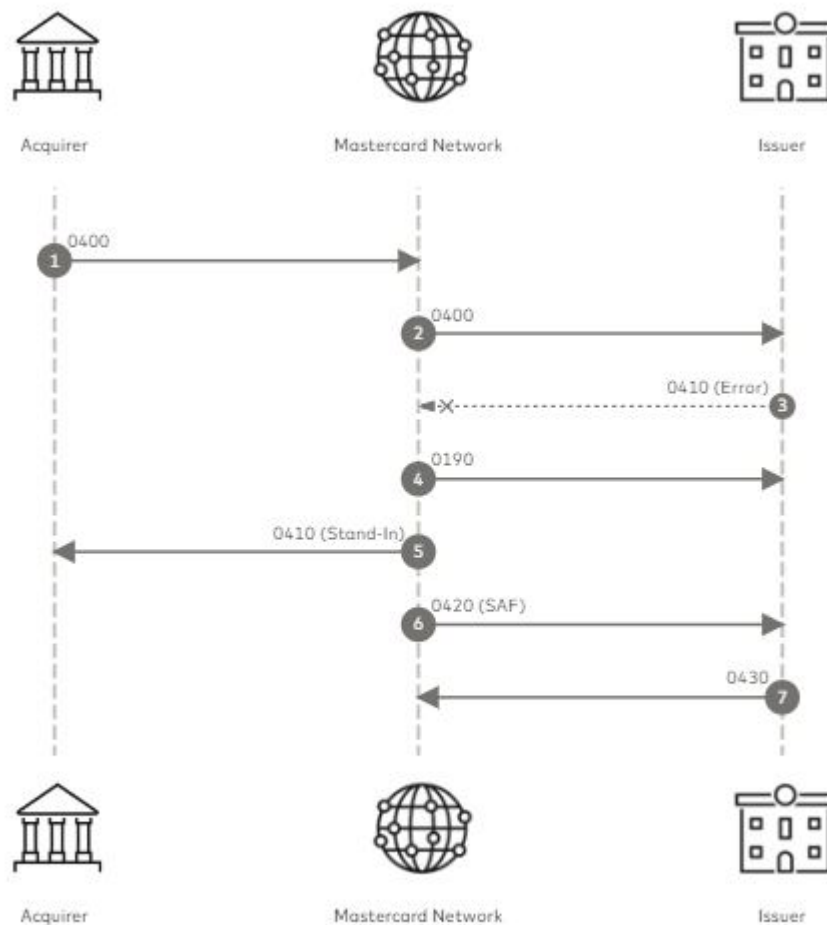
1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The Mastercard Network does not receive the Reversal Response message from the issuer within the time limit. The Reversal Request/0400 message is sent to the Stand-In Authorization. The Stand-In Authorization will generate the Reversal Request Response/0410 which is then sent to the acquirer.
4. The Stand-In Authorization generates a Reversal Advice/0420 message indicating a time-out condition and places it into the SAF queue for the issuer. When the issuer is available to receive the advice message, the 0420 message will be delivered to the issuer.
5. The issuer must reply with a Reversal Advice Response/0430 message to acknowledge receipt of the advice message.
6. The Mastercard Network receives the late Reversal Request Response/0410 message from the issuer.
7. The Mastercard Network sends the issuer an Authorization Response Negative Acknowledgment/0190 message indicating that the network cannot match the response to a pending Reversal Request message.

**NOTE: There is no response necessary from the issuer after the Mastercard Network sends an Authorization Response Negative Acknowledgment/0190 message.**

## Reversal Request response from issuer contains errors

This message flow describes transaction processing when the issuer's response message contains errors.

### Reversal request response from issuer contains errors flow



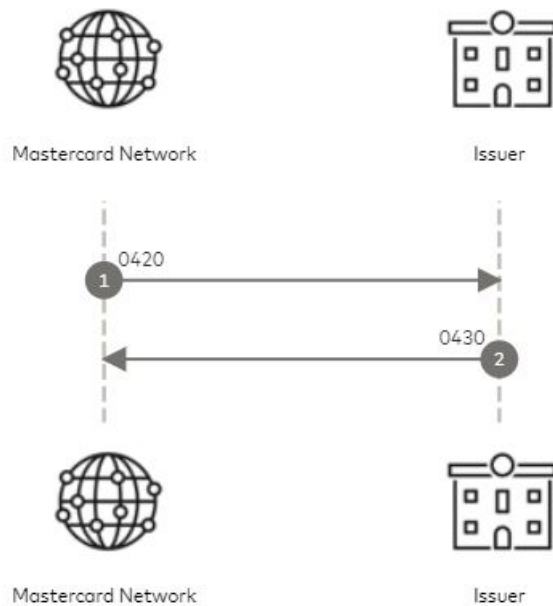
1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The issuer responds with a Reversal Request Response/0410 message to the Mastercard Network; however, the message contains errors.
4. The Mastercard Network generates an Authorization Response Negative Acknowledgment/0190 message advising the issuer of the error condition in their response.
5. The Mastercard Network will send the Reversal Request/0400 message to the Stand-In Authorization. The Stand-In Authorization will generate a Reversal Request Response/0410 message which is then sent to the acquirer.

6. The Mastercard Network will create a Reversal Advice/0420 message indicating a format error and put it into the SAF queue for the issuer. When the issuer is available to receive the advice message, the advice message will be delivered to the issuer.
7. The issuer must reply with a Reversal Advice Response/0430 message to acknowledge receipt of the advice message.

## System-generated Reversal Advice

This message flow describes transaction processing of Reversal Advice/0420 and the Reversal Advice/0430 messages.

### System-generated Reversal Advice flow



1. The Mastercard Network generates a Reversal Advice/0420: System-generated message and sends it to the issuer.
2. The issuer responds with a Reversal Advice Response/0430: Issuer-generated message to the Mastercard Network to acknowledge receipt of the system-generated advice message.

## Non-financial transaction function flows

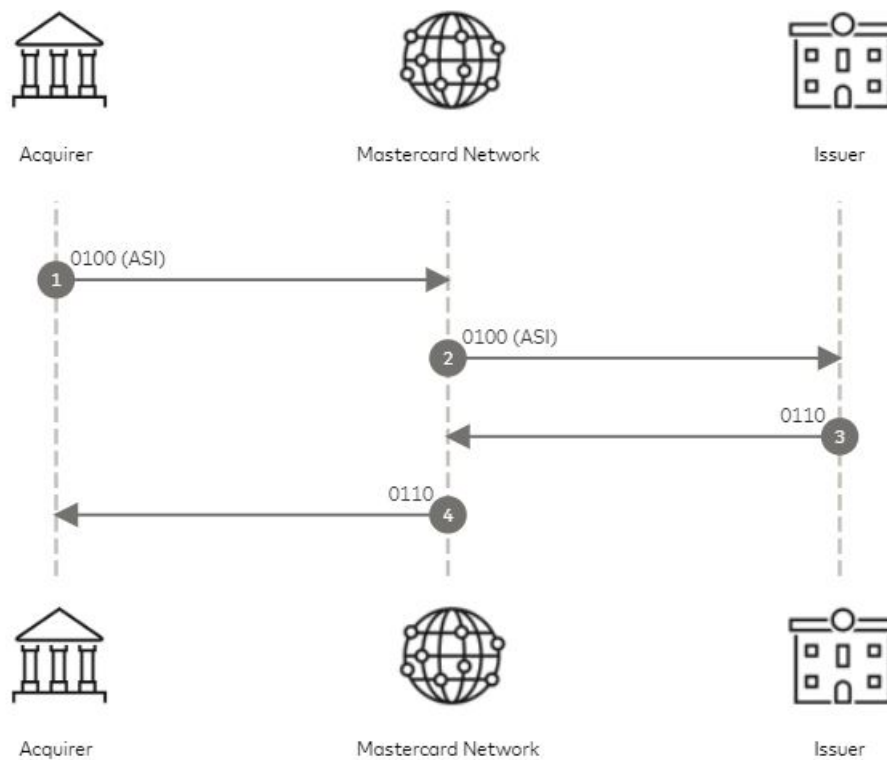
### Non-financial transaction function flows

| Flow Type                                                                                            | Description                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Authorization Request: Account Status Inquiry</a>                                        | This shows the standard flow of request/response messages when used to submit Account Status Inquiry (ASI) requests. For more information on ASI transactions, please refer to <a href="#">System functionality and services</a> of this guide. |
| <a href="#">Authorization Request: Mastercard Digital Enablement Service (MDES) pre-digitization</a> | <p>This exchange of messages between Mastercard and issuer supports the tokenization of Primary Account Numbers (PANs) through network messages.</p> <p>For more information on MDES, refer to the <i>MDES Issuer Implementation Guide</i>.</p> |
| <a href="#">Authorization Request: Chip-PIN Management</a>                                           | Messages enable cardholders to request PIN changes or PIN unblock on Chip cards. This will show successful and failure scenarios for this function.                                                                                             |
| <a href="#">Authorization Response Acknowledgment/0180</a>                                           | Optional acknowledgment message available to acquirers to respond to an Authorization Request Response/0110 message.                                                                                                                            |

## Authorization Request: Account Status Inquiry

Acquirers send Account Status Inquiry (ASI) messages to validate aspects of a cardholder account prior to sending transaction requests for authorization decisions.

### Authorization Request: Account Status Inquiry flow



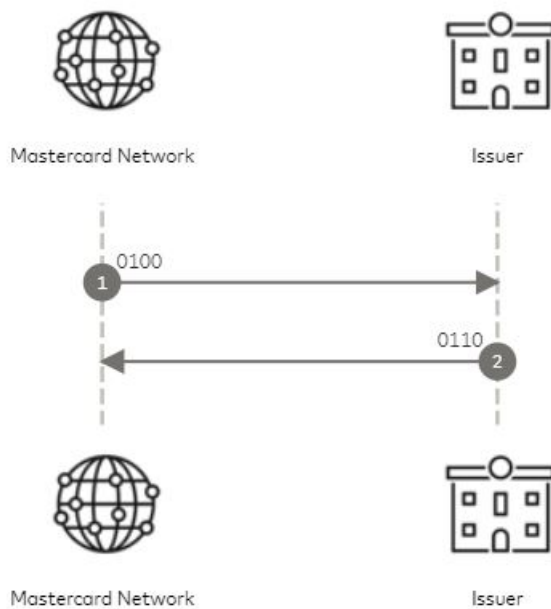
1. The acquirer initiates an Authorization Request/0100 message and sends it to the Mastercard Network. The data in the request message indicates that this is an ASI request and not requesting authorization for purchase of goods or services.
2. The Mastercard Network forwards the Authorization Request/0100 message to the issuer.
3. The issuer generates an Authorization Request Response/0110 message providing their response to the ASI and sends it to the Mastercard Network.
4. The Mastercard Network forwards the Authorization Request Response/0110 message to the acquirer.

## Authorization Request: Mastercard Digital Enablement Service (MDES) pre-digitization

The Mastercard Network sends real-time Mastercard token authorization request messages to support provisioning of a token on a cardholder's device.

For more information on MDES, refer to the [MDES Issuer Implementation Guide](#).

### Authorization Request: MDES Pre-digitization flow



1. The Mastercard Network generates an Authorization Request/0100 message as part of the pre-digitization process. The request message may also advise the MDES issuer of a status change related to the token.
2. The issuer responds with an Authorization Request Response/0110 message to the Mastercard Network to provide feedback regarding the request to tokenize the PAN.

## Authorization Request: Chip-PIN Management

List of Authorization Request: Chip PIN Management flows:

- [Standard Chip-PIN Management request](#)
- [Chip-PIN Management Request failure from acquirer](#)
- [Chip-PIN Management Response cannot be delivered to the acquirer](#)
- [Chip-PIN Management Update cannot be transmitted or applied to chip card](#)

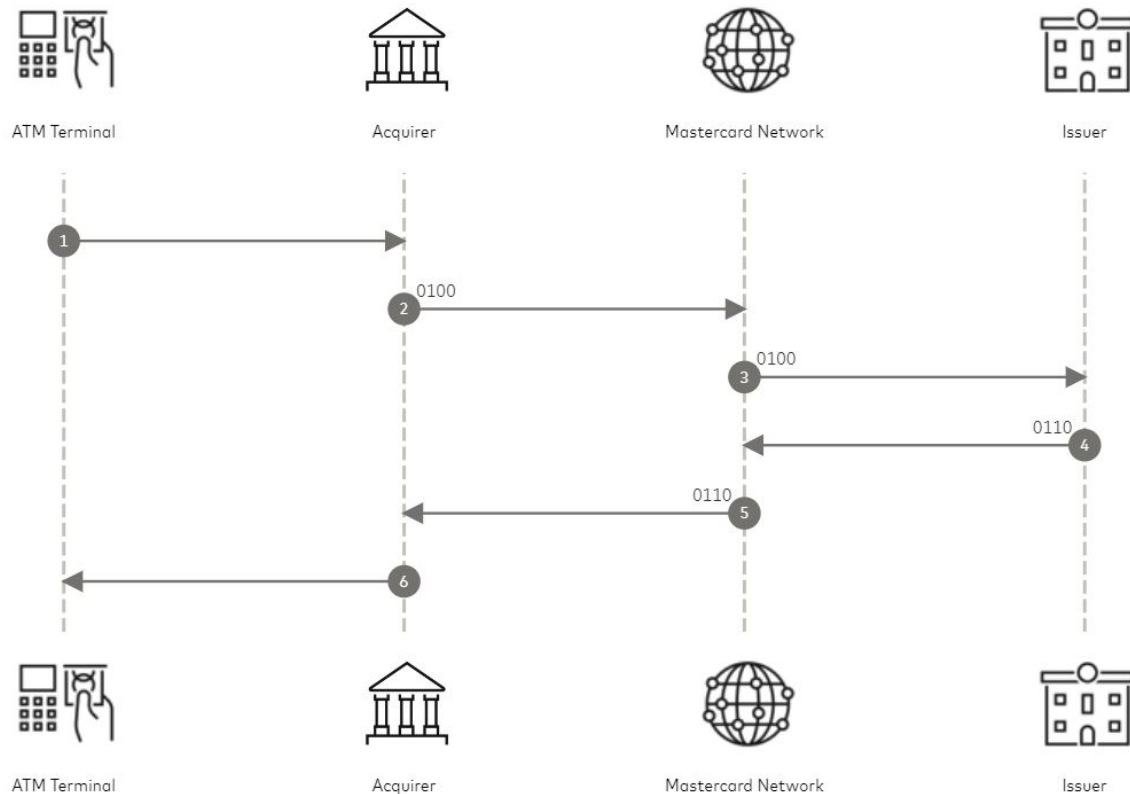


- Chip-PIN Management Request unable to connect to issuer
- PIN-management Response not received from issuer

## Standard Chip-PIN Management request

This message flow describes standard transaction processing when changing or unblocking a PIN on a chip card.

### Standard Chip-PIN Management request flow



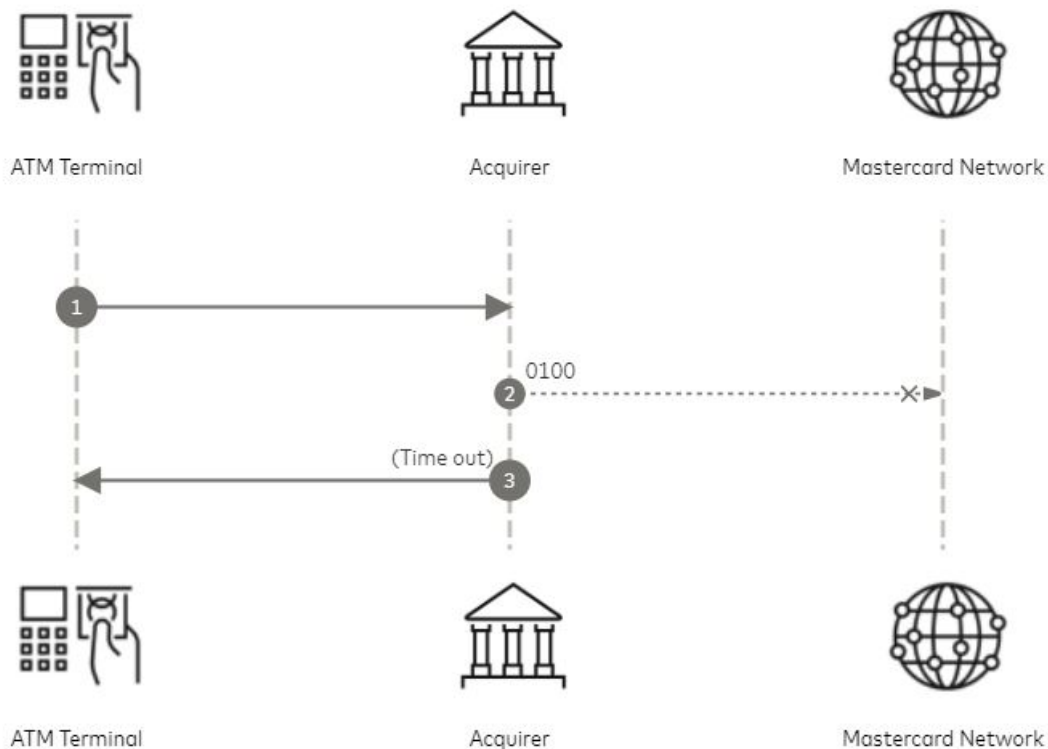
1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 message to the issuer.
4. The issuer sends an Authorization Request Response/0110 message to the Mastercard Network approving or declining the request. The issuer will make updates within the issuer host system to reflect the PIN change or PIN unblock.
5. The Mastercard Network forwards the Authorization Request Response/0110 message to the acquirer.

6. The acquirer will send the response to the ATM. A script is sent from the ATM to the card to update the chip. After the script is applied, the chip provides the result to the ATM to complete the transaction.

### Chip-PIN Management Request failure from acquirer

This message flow describes the transaction process when there is a network failure at the acquirer.

#### Chip-PIN Management Request failure from acquirer flow

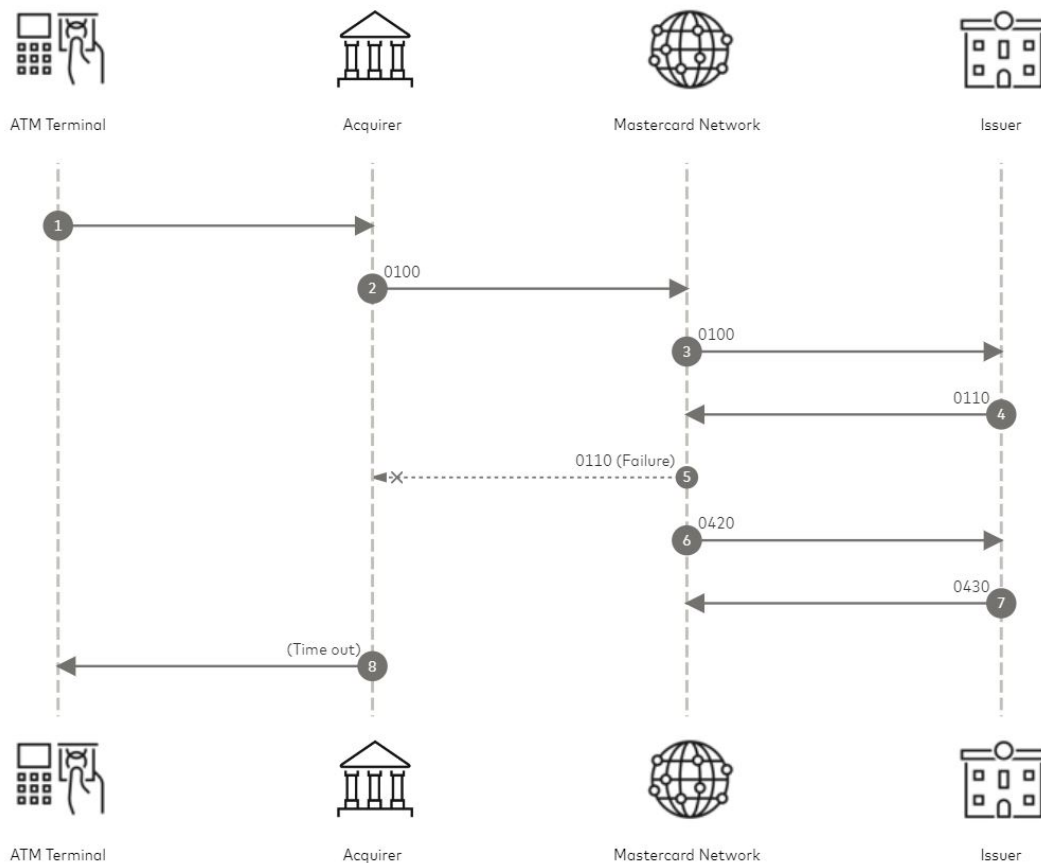


1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network. However, the request message does not reach the Mastercard Network.
3. The acquirer sends a message to the ATM advising that the service could not be completed.

## Chip-PIN Management Response cannot be delivered to the acquirer

This message flow describes the transaction process when the Mastercard Network is unable to successfully deliver the issuer's response message to the acquirer.

### Chip-PIN Management Response cannot be delivered to the acquirer flow



1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 message to the issuer.
4. The issuer sends an Authorization Request Response/0110 message to the Mastercard Network approving or declining the request. The issuer will make updates within the issuer host system to reflect the PIN change or PIN unblock.
5. The Mastercard Network fails to deliver the Authorization Request Response/0110 message to the acquirer.
6. The Mastercard Network generates a Reversal Advice/0420 message to the issuer.

7. The issuer sends a Reversal Advice Response/0430 message to the Mastercard Network acknowledging receipt of the Reversal Advice/0420 message. The issuer must also reverse the PIN change or PIN unblock applied to their system based on the Reversal Advice.

**NOTE: The Mastercard Network will only generate a Reversal Advice/0420 message if the issuer approved the original PIN management request.**

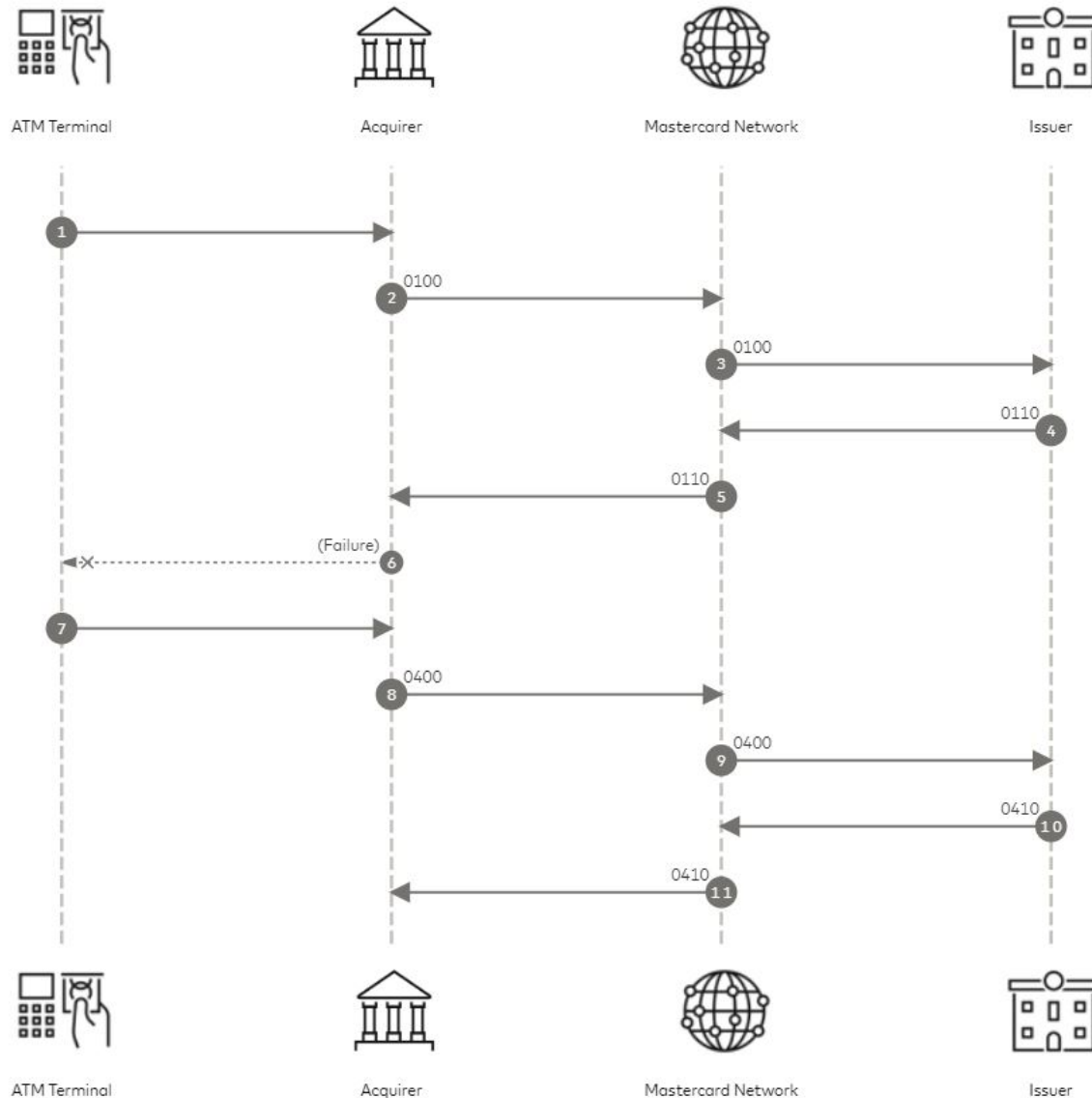
8. The request times out at the acquirer and the acquirer sends a message to the ATM advising that the service is unavailable.

**NOTE: The issuer may receive two reversals: one from the Mastercard Network to identify the undelivered response and a second from the acquirer as a result of a system timeout.**

## Chip-PIN Management Update cannot be transmitted or applied to chip card

This message flow describes the transaction process when there is a failure in transmitting the script to the chip card, or a failure in applying the script to the chip card.

### Chip-PIN Management Update cannot be transmitted or applied to chip card flow



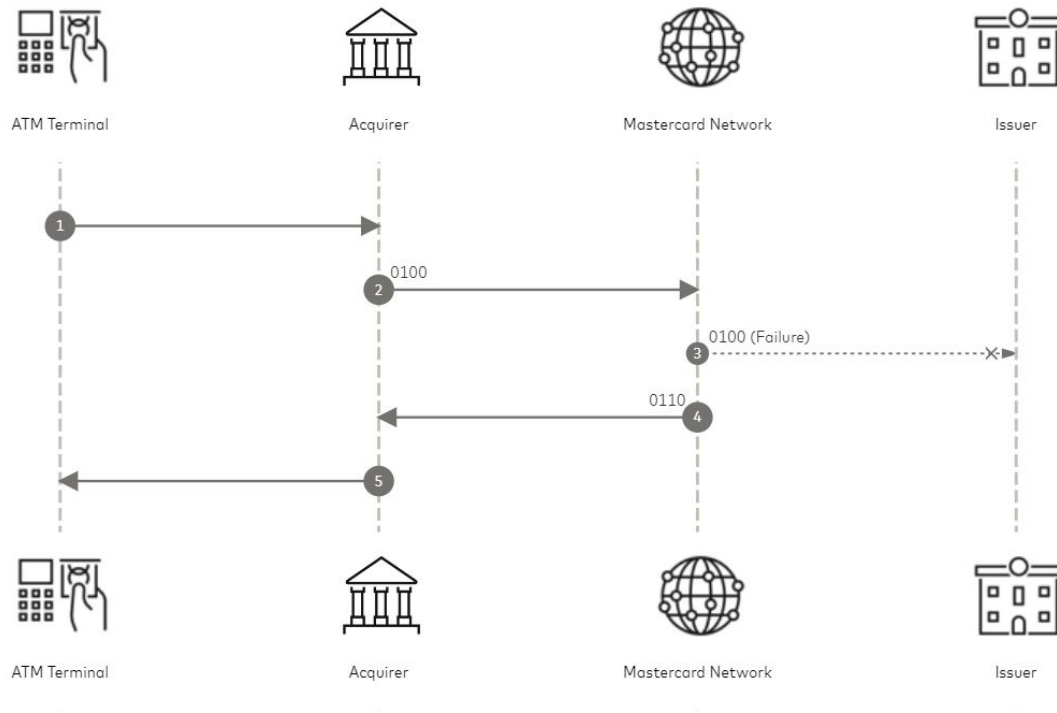
1. The cardholder initiates a Chip PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.

2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 to the issuer.
4. The issuer sends an Authorization Request Response/0110 message to the Mastercard Network approving or declining the request. The issuer will make updates within the issuer host system to reflect the PIN change or PIN unblock.
5. The Mastercard Network forwards the Authorization Request Response/0110 message to the acquirer.
6. A failure occurs for one of the following reasons:
  - The acquirer cannot send the script in DE 55 to the ATM.
  - The ATM cannot transmit the script to the chip card.
  - The chip card cannot process the script.
7. The ATM sends the processing result to the acquirer indicating the failure.
8. The acquirer generates a Reversal Request/0400 message to the Mastercard Network.
9. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
10. The issuer sends a Reversal Request Response/0410 message to the Mastercard Network. The issuer must also reverse the PIN change or PIN unblock applied to their system based on the Reversal Request.
11. The Mastercard Network forwards the Reversal Request Response/0410 message to the acquirer.

## Chip-PIN Management Request unable to connect to issuer

This message flow describes the transaction process when there is a network failure at the issuer.

### Chip-PIN Management Request unable to connect to issuer flow

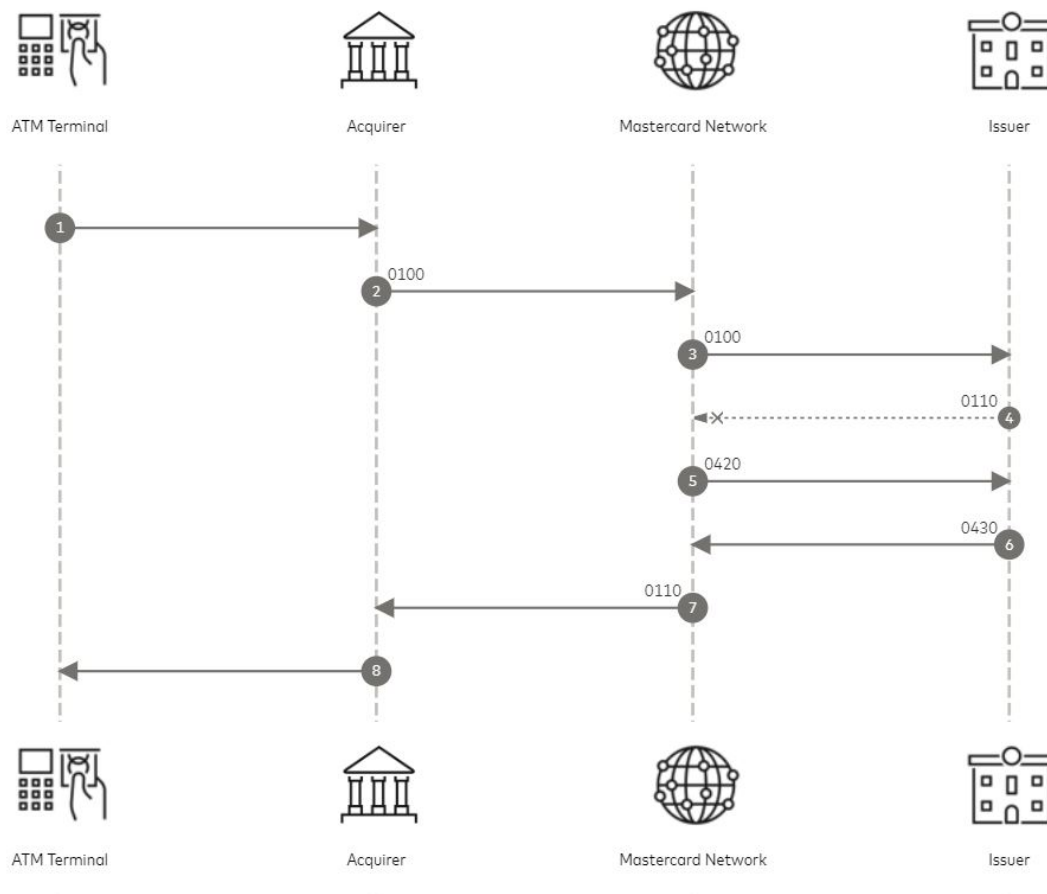


1. The cardholder initiates a Chip PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network attempts to forward the Authorization Request/0100 to the issuer. However the request message does not reach the issuer which results in a time-out for the request.
4. The Mastercard Network sends an Authorization Request Response/0110 message to the acquirer indicating the request could not be delivered.
5. The acquirer sends a message to the ATM advising that the service could not be completed.

## PIN-management Response not received from issuer

This message flow describes the transaction process when the issuer response is not delivered to the Mastercard Network or is delivered to late.

### PIN-management Response not received from issuer flow



1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 to the issuer.
4. The issuer fails to send the Authorization Request Response/0110 message to the Mastercard Network in time.
5. The Mastercard Network sends a Reversal Advice/0420 message to the issuer indicating the time-out condition.



**NOTE: If a response is received from the issuer after the allowed response time limit for ATM transactions for Mastercard and Debit Mastercard only, the Mastercard Network will send the issuer an Authorization Response Negative Acknowledgment/0190 message advising the response was late (not illustrated).**

6. The issuer sends a Reversal Advice Response/0430 message to the Mastercard Network acknowledging receipt of the Reversal Advice/0420 message. If the original transaction was a PIN change, the issuer restores the old online PIN in their host system.
7. The Mastercard Network sends the acquirer an Authorization Request Response/0110 message indicating the request was not able to be delivered.
8. The acquirer sends a message to the ATM, indicating that the service is unavailable.

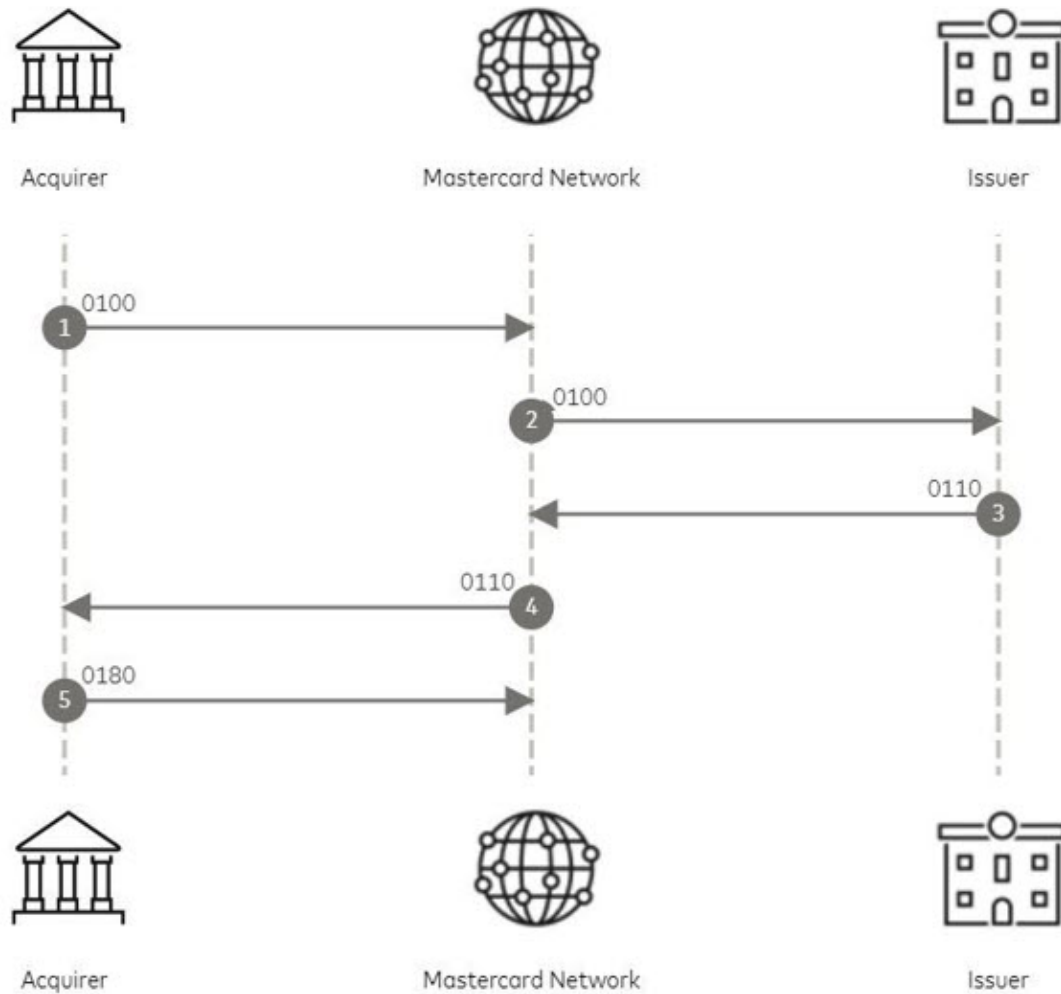
## Authorization Response Acknowledgment/0180

The Mastercard Network allows acquirers to optionally acknowledge Authorization Request Response/0110 messages. This process is accomplished using Authorization Response Acknowledgment/0180 messages.

Use of this acknowledgment can provide additional assurance that authorization responses have been delivered to the acquirer or acceptor's device. This may protect the issuer from erroneous debits against cardholder accounts due to incomplete or unsuccessful transactions.

Acquirers are not required to support Authorization Response Acknowledgment/0180, but Mastercard encourages acquirers to select this option and support these messages. Acquirers may select this option at network configuration time.

### Acquirer Response Acknowledgment flow



1. The acquirer initiates an Authorization Request/0100 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Authorization Request/0100 message to the issuer.
3. The issuer creates the Authorization Request Response/0110 message and sends it to the Mastercard Network.
4. The Mastercard Network forwards the Authorization Request Response/0110 message to the acquirer.
5. The acquirer sends an Authorization Response Acknowledgment/0180 message back to the Mastercard Network, indicating the acquirer successfully received the preceding Authorization Request Response/0110 message at the application level.

In most cases, the acquirer should send the Authorization Response Acknowledgment/0180 message to the Mastercard Network:

- Immediately after it secures, successfully receives, and logs the Authorization Request Response/0110 message
- Before the transaction is completed at the point of interaction

An Authorization Response Acknowledgment/0180 message does not necessarily indicate that the transaction was successfully completed at the point of interaction.

## Supporting transaction function flows

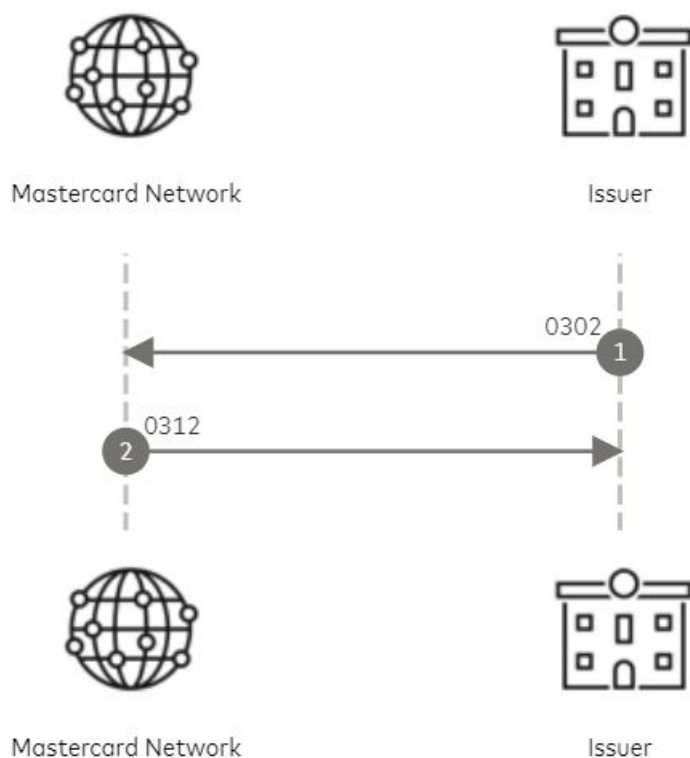
### Supporting transaction function flows

| Flow Type                       | Description                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Issuer File Update Requests     | The exchange of messages between issuers and Mastercard to update files used during authorization processing.                             |
| Administrative Request messages | Messages to transmit customer data between acquirers and issuers for specific programs. Successful and failure scenarios are represented. |
| Administrative Advice messages  | Messages exchanging free-format data between network participants or to advise originators of invalid messages.                           |
| Network management requests     | Messages coordinating system or network tasks, events or network status updates.                                                          |

## Issuer File Update Requests

This message flow describes transaction processing of the Issuer File Update Request/0302 message.

### Issuer file update requests flow



1. Issuer submits the Issuer File Update Request/0302 message to the Mastercard Network. The type of update being requested is specified by the format of the 0302 message. For more details regarding the various request layouts, refer to the *Account Management System User Manual*.
2. The Mastercard Network performs the requested issuer file update task and responds with an Issuer File Update Request Response/0312 message back to the issuer. The Response Code returned in the Issuer File Update Request Response/0312 message indicates whether the issuer file update was successfully completed.

**NOTE: The error-condition message process for Issuer File Update/03xx messages is not illustrated. If an issuer unsuccessfully forwards an Issuer File Update Request/0302 message to the Dual Message Authorization System, the issuer should retransmit the message.**

## Administrative Request messages

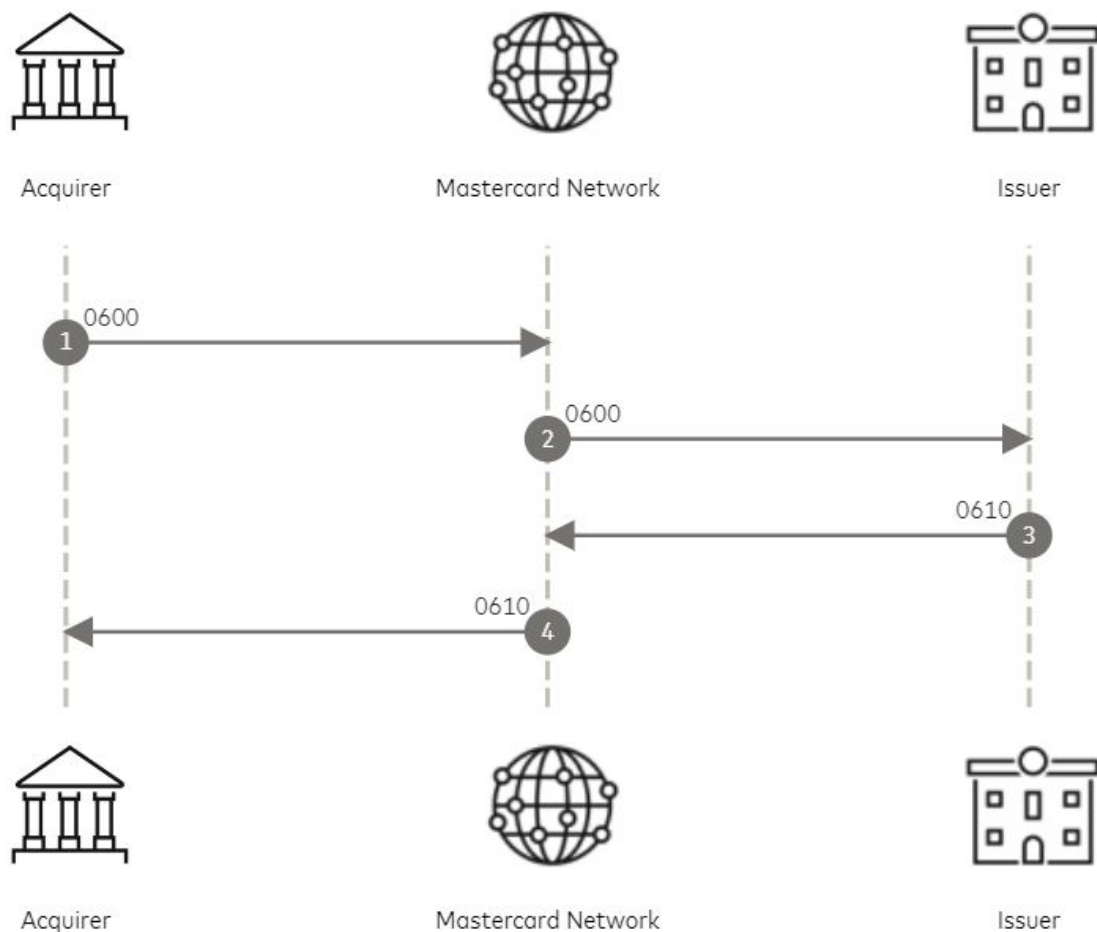
List of administrative request messages flows:

- [Standard Administrative Request](#)
- [Acquirer-edit Failure](#)
- [Communication Failure at issuer](#)
- [Communication Failure at acquirer](#)
- [Late or no issuer response](#)
- [Issuer-edit Failure](#)

### Standard Administrative Request

This message flow describes transaction processing of the Administrative Request/0600 and Administrative Request Response/0610 messages.

#### Standard Administrative request flow

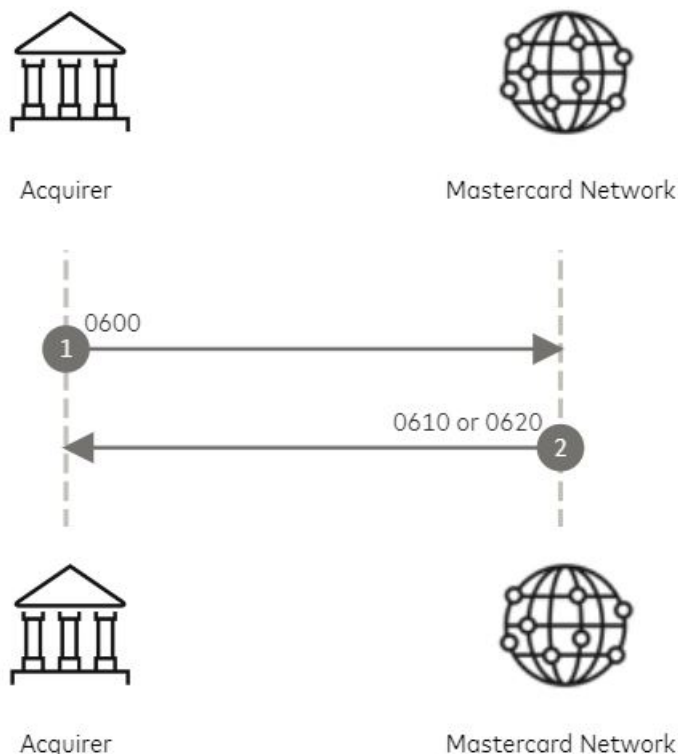


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based on the account range of the Primary Account Number (PAN).
3. The issuer responds with an Administrative Request Response/0610 message to the Mastercard Network.
4. The Mastercard Network forwards the Administrative Request Response/0610 message to the acquirer.

### Acquirer-edit Failure

This message flow describes exception processing when the Mastercard Network determines an acquirer edit failure during processing an Administrative Request/0600 message.

#### Acquirer-edit Failure flow



1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network edits the Administrative Request/0600 message and detects an acquirer edit failure. As a result, the Mastercard Network will send the acquirer either an

Administrative Request Response/0610 message or an Administrative Advice/0620 message depending on the edit errors.

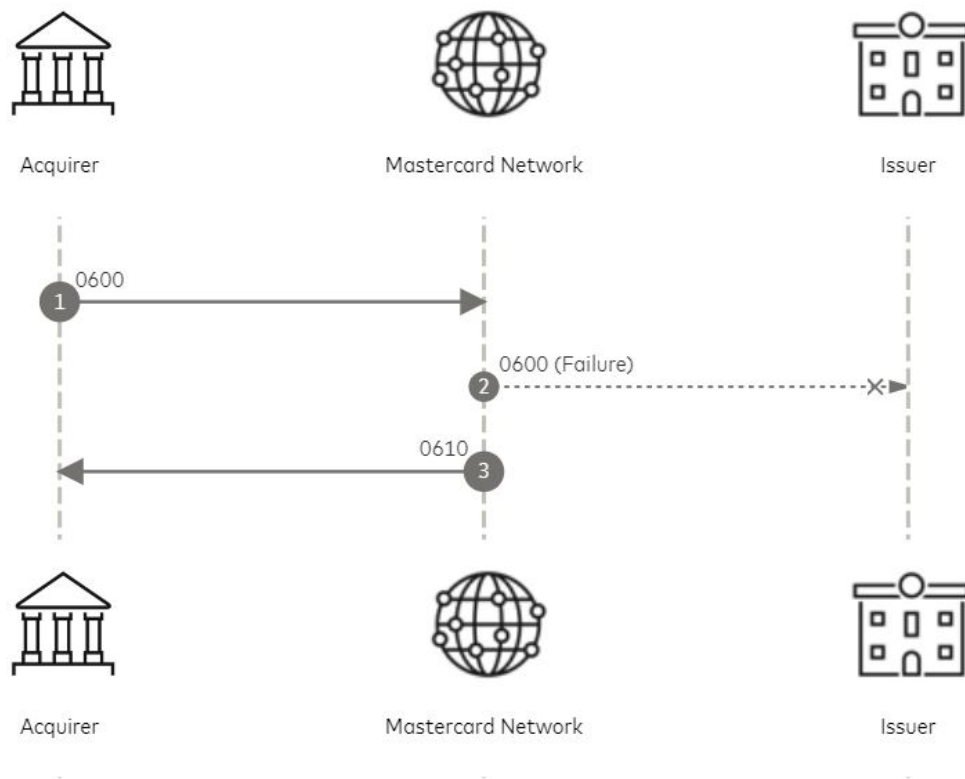
### Message types and error conditions for response messages

| Message type sent                    | Conditions for the response message                                                                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Administrative Request Response/0610 | <ul style="list-style-type: none"> <li>– Is missing mandatory data element</li> <li>– Contains data attribute error or invalid data value</li> <li>– Is from ineligible acquirer/processor</li> <li>– Is sent to ineligible issuer</li> </ul> |
| Administrative Advice/0620           | <ul style="list-style-type: none"> <li>– Cannot be parsed</li> <li>– Exceeds the maximum message length of 8k bytes</li> </ul>                                                                                                                |

## Communication Failure at issuer

This message flow describes exception processing when communication fails between the issuer and the Mastercard Network during an Administrative Request/0600 message.

### Communication Failure at issuer flow



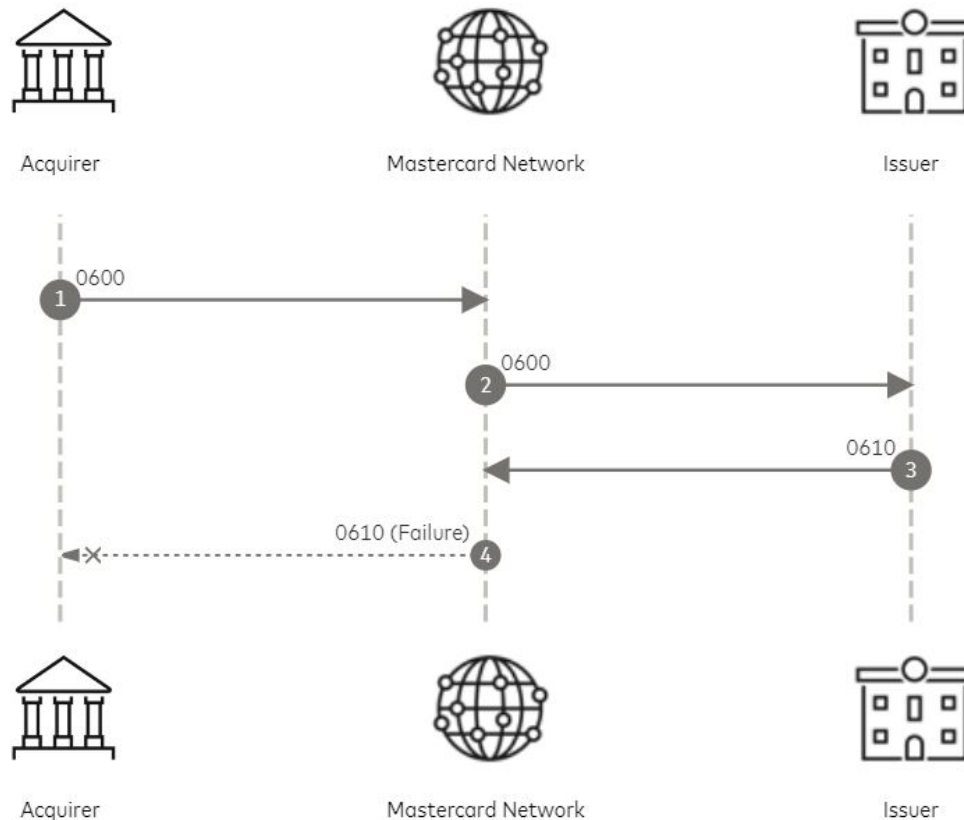
1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network attempts to forward the Administrative Request/0600 message to the issuer and is unsuccessful due to a communication failure.
3. The Mastercard Network sends the acquirer an Administrative Request Response/0610 message indicating the request could not be delivered.



## Communication Failure at acquirer

This message flow describes exception processing when communication fails between the Mastercard Network and the acquirer during an Administrative Request Response/0610 message.

### Communication Failure at acquirer flow

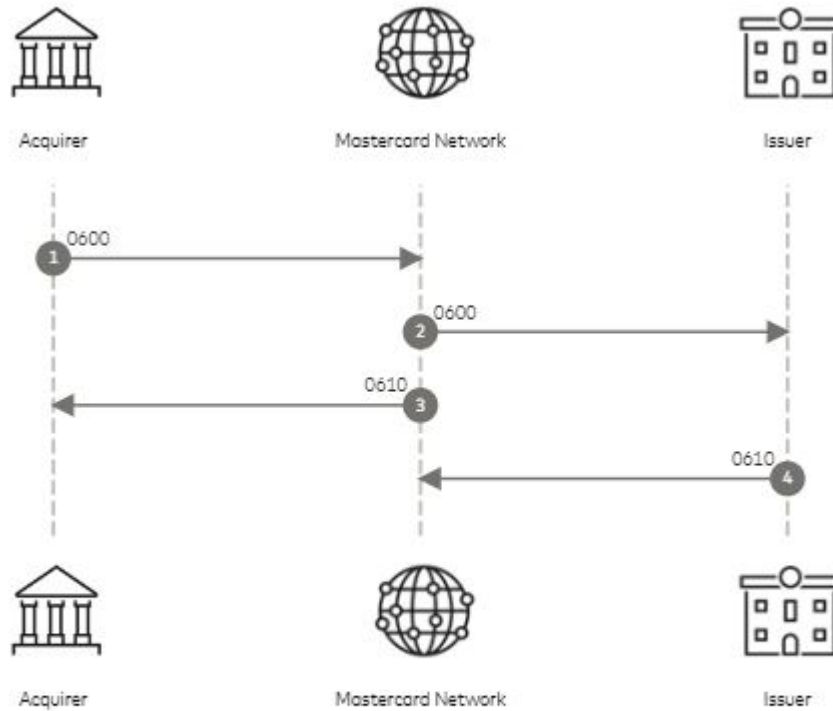


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based upon the account range in the Primary Account Number (PAN).
3. The issuer sends an Administrative Request Response/0610 message to the Mastercard Network.
4. The Mastercard Network attempts to forward the Administrative Request Response/0610 message to the acquirer and is unsuccessful because of a communication link failure or acquirer failure. The Mastercard Network logs the undelivered 0610 message and takes no further action. The acquirer host will time out the 0600 message and resend, if appropriate.

## Late or no issuer response

This message flow describes exception processing when the Mastercard Network either does not receive an Administrative Request Response/0610 message from the issuer or receives a late response.

### Late or no issuer response flow

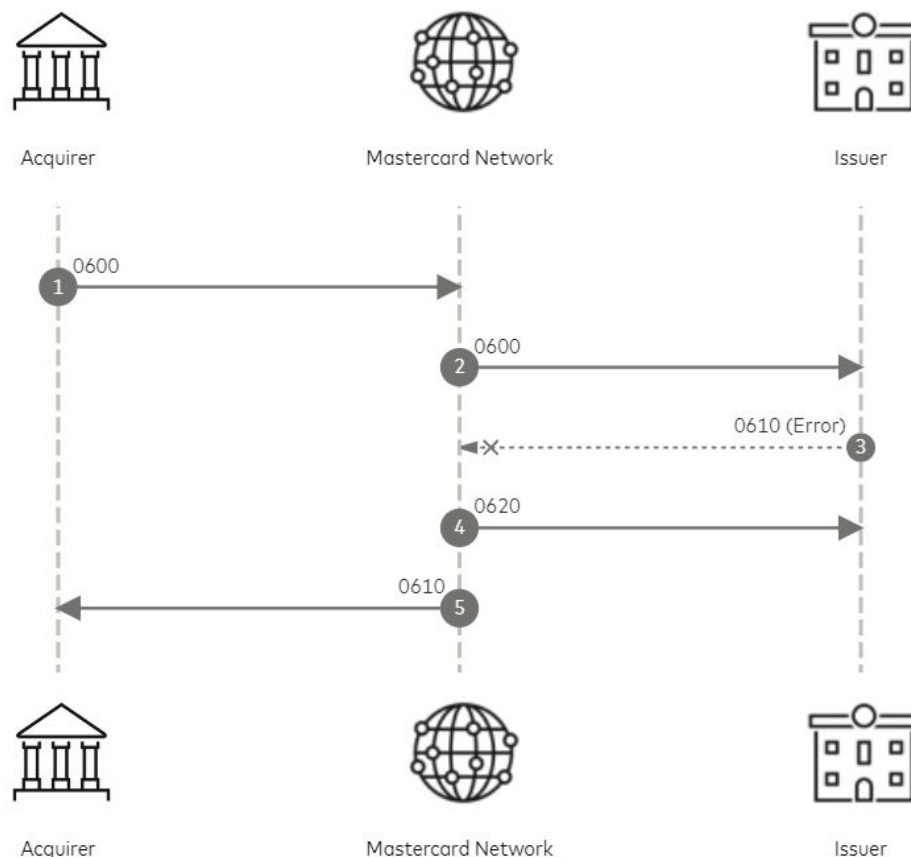


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based upon the account range contained in the PAN.
3. If the Mastercard Network receives no response from the issuer within the time limits, the Mastercard Network sends the acquirer an Administrative Request Response/0610 message indicating the administrative request message could not be delivered.
4. If the Mastercard Network receives the issuer's response after the time limit expired, the Mastercard Network logs the late issuer 0610 message and takes no further action.

## Issuer-edit Failure

This message flow describes exception processing when the Mastercard Network receives an Administrative Request Response/0610 message that contains an edit failure from the issuer.

### Issuer-edit Failure flow



1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based upon the account range contained in the PAN.
3. The issuer sends an Administrative Request Response/0610 message to the Mastercard Network. During validation, the Mastercard Network detects edit errors in the response message.
4. The Mastercard Network will generate an Administrative Advice/0620 message to the issuer indicating the following has occurred:
  - Cannot be parsed
  - Exceeds the maximum message length of 8k bytes

- Is missing mandatory data element
  - Contains a data attribute error or invalid data value
5. The Mastercard Network sends the acquirer an Administrative Request Response/0610 message indicating the administrative request was unsuccessful.

## Administrative Advice messages

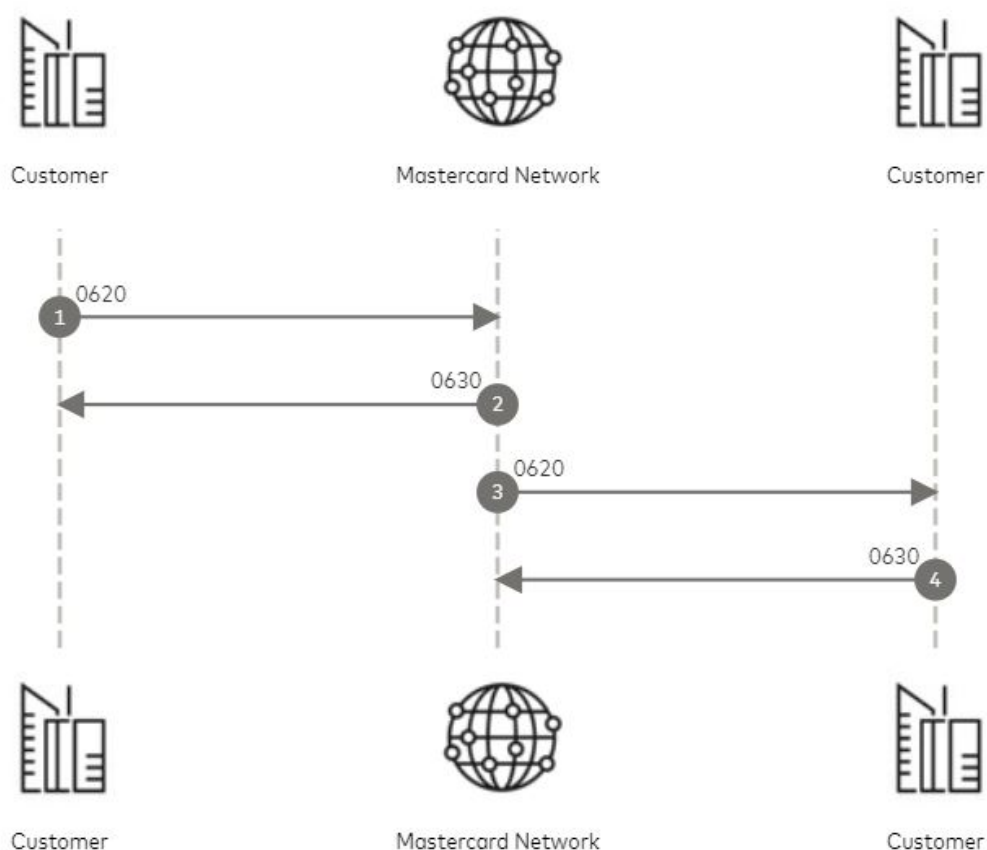
List of administrative advice message flows:

- [Standard Administrative Advice](#)
- [Failure scenario: Administrative Advice for invalid message](#)
- [MDES-notification advice](#)

## Standard Administrative Advice

This message flow describes the standard Administrative Advice/06xx message process between customers. Acquirers or issuers can originate or receive these messages.

### Standard Administrative Advice flow

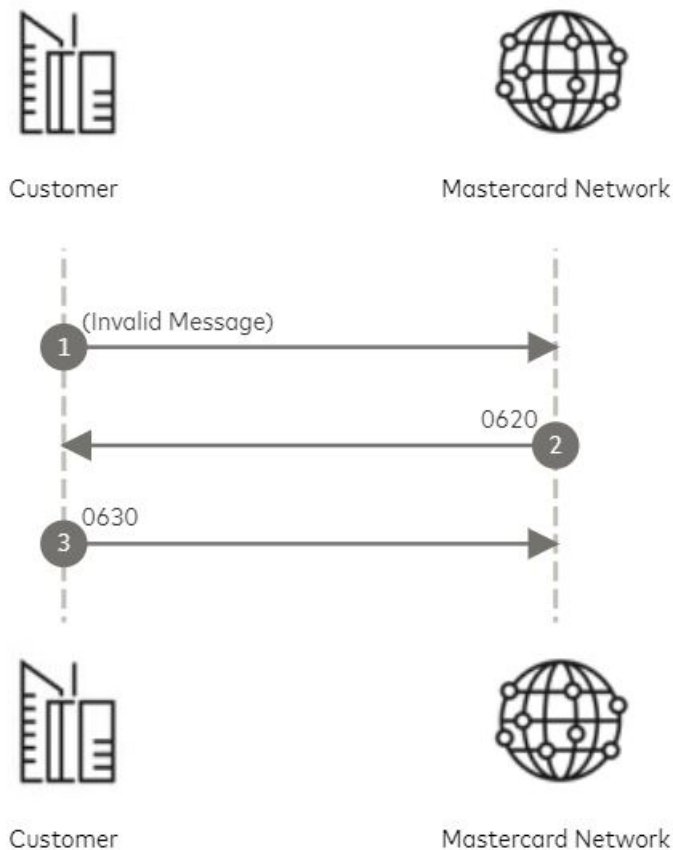


1. The originating customer generates an Administrative Advice/0620 message to the Mastercard Network destined for another customer.
2. The Mastercard Network acknowledges receipt of the Administrative Advice/0620 message by returning an Administrative Advice Response/0630 message to the originating customer.
3. The Mastercard Network forwards the Administrative Advice/0620 message to the intended receiving customer.
4. The receiving customer responds with an Administrative Advice Response/0630 to the Mastercard Network to acknowledge receipt of the advice message.

## Failure scenario: Administrative Advice for invalid message

This message flow describes when an invalid message is received and an Administrative Advice/0620: system-generated is sent.

### Failure scenario: Administrative Advice for invalid message flow



1. The originating customer sends an unidentifiable or invalid message to the Mastercard Network.
2. The Mastercard Network generates an Administrative Advice/0620 message to the originator advising where the failure occurred while parsing or processing the message.
3. The customer acknowledges receipt of the Administrative Advice/0620 message and returns an Administrative Advice Response/0630 message to the Mastercard Network.

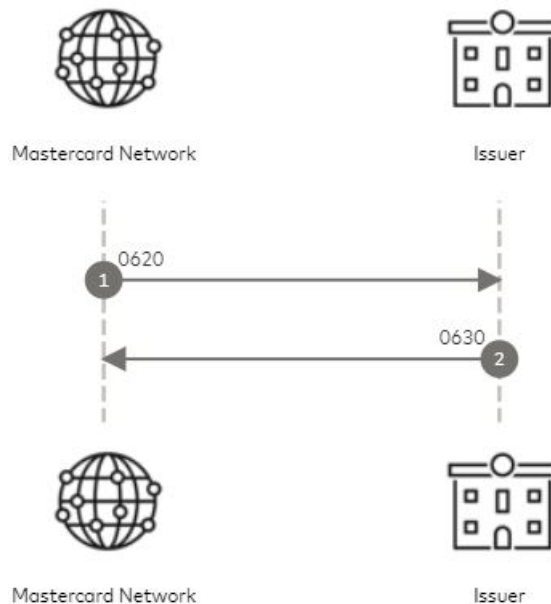
## MDES-notification advice

Mastercard supports layouts for Administrative Advice/0620 messages to communicate activation codes or issues with activation codes to MDES issuers in support of token issuance,

and to notify issuers that Mastercard has assigned a token on their behalf and of updates to a token's status.

For more information on MDES, refer to the [MDES Issuer Implementation Guide](#).

### MDES-notification advice flow



1. The Mastercard Network generates an Administrative Advice/0620 message as part of the tokenization process. Data within the message indicates that it is a MDES advice to the issuer.
2. The issuer responds with an Administrative Advice Response/0630 message to the Mastercard Network to acknowledge the receipt of the MDES related 0620 advice message.

## Network management requests

List of network management request flows:

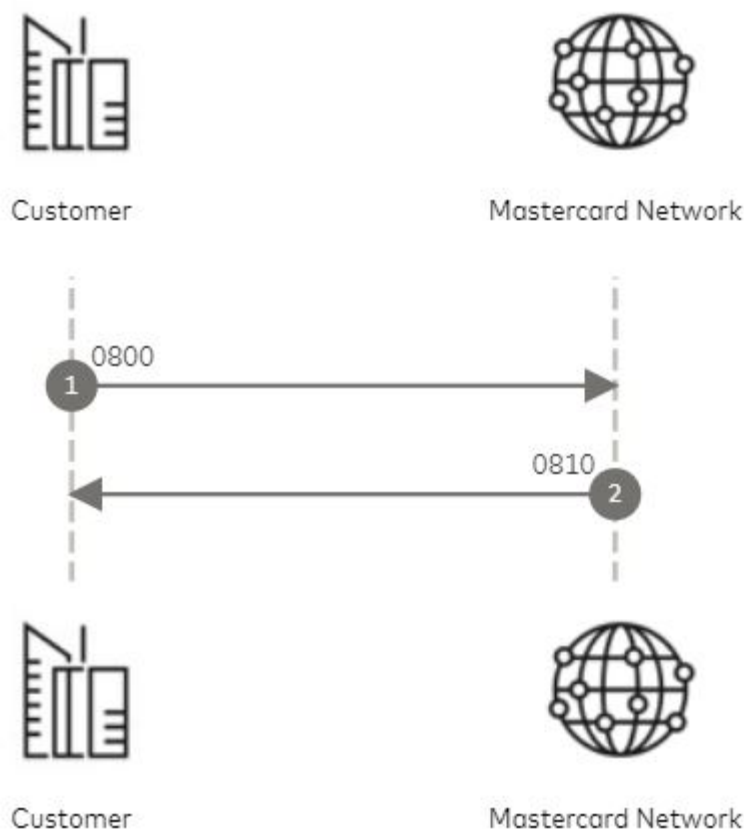
- [Network sign-on or sign-off](#)
- [Request for issuer-generated or solicited SAF](#)
- [Unsolicited SAF after sign-on](#)
- [Network-connection status request: customer-generated](#)
- [Network-connection status request: system-generated](#)
- [Host activation or deactivation](#)

- PIN Encryption Key exchange request: customer generated
- PEK-exchange request: system generated

## Network sign-on or sign-off

This message flow describes the standard Network Management Request/0800 message: sign-on/sign-off message to sign on to or sign off from the Mastercard Network. The originator of these request messages can be either an acquirer or an issuer.

### Network sign-on or sign-off flow



1. The customer creates a Network Management Request/0800: sign-on/sign-off message and sends it to the Mastercard Network.
2. After receiving the Network Management Request/0800: sign-on/sign-off message, the Mastercard Network creates a Network Management Request Response/0810: sign-on/sign-off message and sends it to the customer.



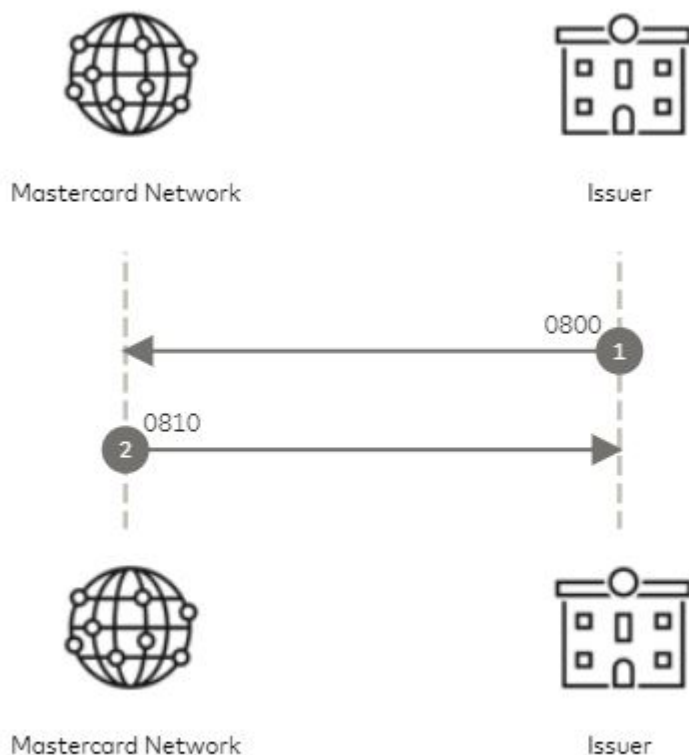
**NOTE:** Customers are reminded that they should use any one or a combination of the network connectivity management options described in this guide. Mastercard recommends using the varying levels of probes or echo tests to verify host connectivity to the Mastercard Network instead of using the sign-on request message.

## Request for issuer-generated or solicited SAF

This message flow describes the issuer-generated (solicited) SAF session using a Network Management Request/0800: SAF Request message.

When an issuer has a large number of SAF records (for example, because of an extended outage), Mastercard can deliver SAF records to the issuer through alternate methods.

### Request for issuer-generated or solicited SAF flow



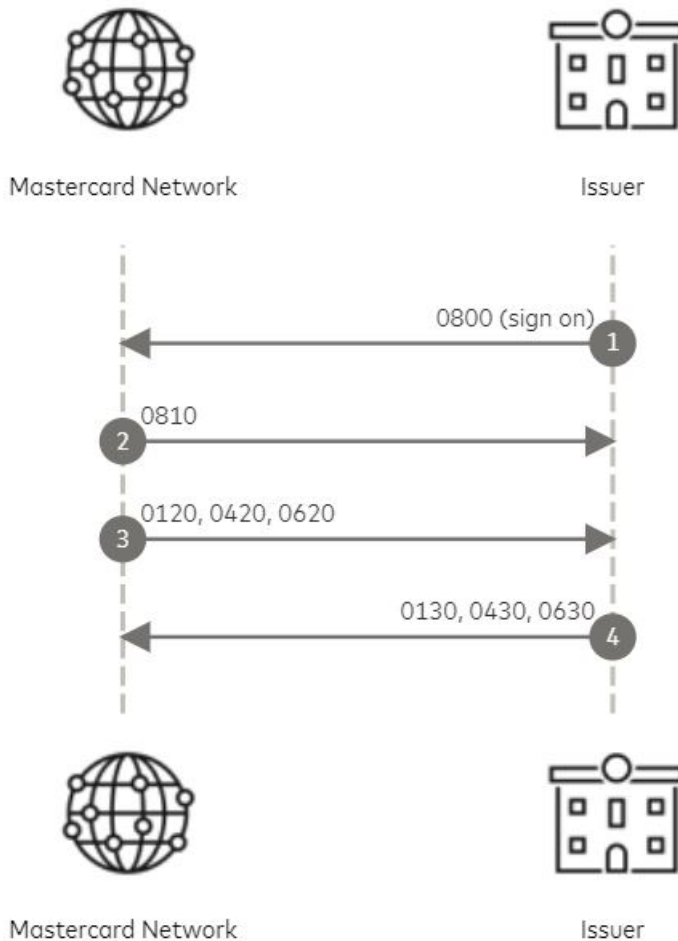
1. The issuer sends a Network Management Request/0800: SAF Request message requesting a SAF session. The request is acknowledged; however, it will not affect SAF processing.
2. The Mastercard Network generates the Network Management Request Response/0810 acknowledgment message. After the Mastercard Network responds to the issuer SAF

request message, no further processing of the issuer's request is performed. All SAF messages are processed unsolicited.

## Unsolicited SAF after sign-on

This message flow describes the unsolicited SAF session with the issuer after signing on to the Mastercard Network using a Network Management Request/0800: sign-on/sign-off message.

### Unsolicited SAF after sign-on flow



1. The issuer sends a Network Management Request/0800: sign-on/sign-off message to sign on to the Mastercard Network.
2. The Mastercard Network sends a Network Management Request Response/0810 message acknowledging the issuer's sign-on request.

3. The Stand-In Authorization checks for SAF messages in the queue and initiates a SAF session. The SAF process forwards an Authorization Advice/0120, Reversal Advice/0420, or Administrative Advice/0620 message to the issuer. The Stand-In Authorization continues to check for SAF messages periodically as long as the issuer is signed in, and forwards messages if there are any available.
4. The issuer generates an Authorization Advice Response/0130, Reversal Advice Response/0430, or Administrative Advice Response/0630 message. These response messages advise the Mastercard Network that the issuer received the previous Authorization Advice/0120, Reversal Advice/0420, or Administrative Advice/0620 message and prompt transmission of the next Authorization Advice/0120, Reversal Advice/0420, or Administrative Advice/0620 message.

**NOTE: Issuers receive SAF messages intermixed with authorization and reversal request messages.**

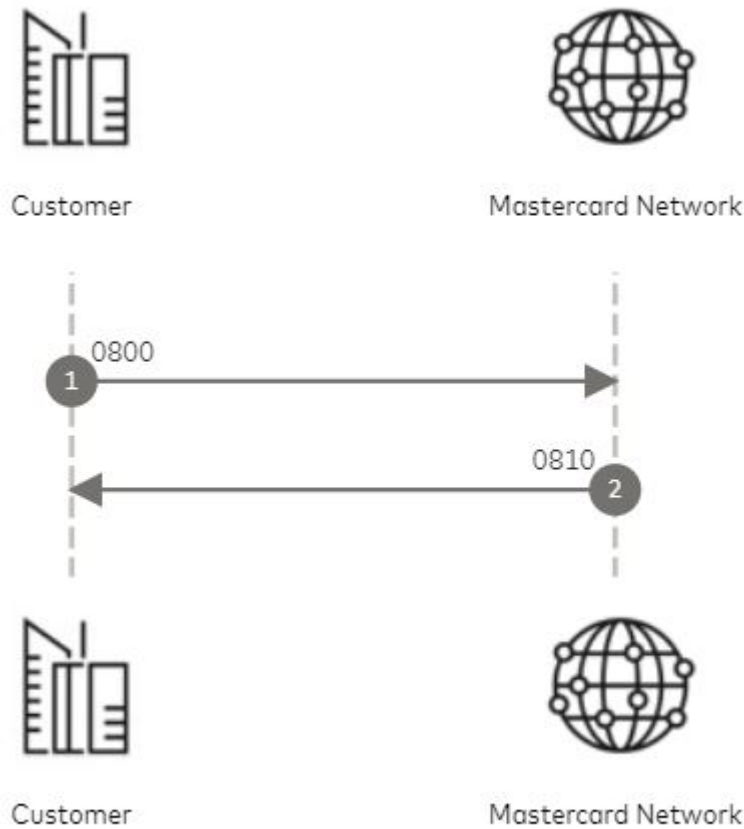
### Network-connection status request: customer-generated

This message flow describes transaction processing of the Network Management Request/0800: network connection status: customer-generated message.

Customers should send the Network Management Request Network Connection Status: echo test when they want to verify the status of their connection to the Mastercard Network instead of using a sign-on message.

Customers may not receive a Network Management Request Response/0810 message in the event Mastercard is experiencing an internal network or system issue which impacts the ability to process network management messages. When this occurs, the Mastercard customer must continue to process all financial messages. The customer must report the lack of a Network Management Request Response/0810 message to Mission Control and Operations with details about this scenario and Mastercard will investigate.

**Network-connection status request: customer-generated flow**



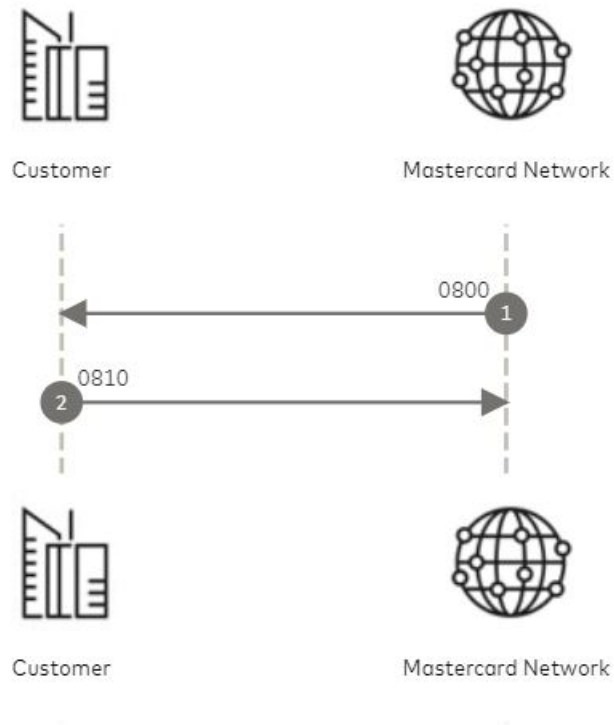
1. The customer sends a Network Management Request/0800: network connection status, customer-generated message indicating an echo test to the Mastercard Network to verify the customer's connection.
2. The Mastercard Network responds with a Network Management Request Response/0810: network connection status, system-generated message to confirm the connection is active.

**Network-connection status request: system-generated**

This message flow describes transaction processing of the Network Management Request/0800: network connection status: system-generated message.

Customers that request Mastercard to periodically check their connection status to the Mastercard Network will receive the Network Management Request/0800 message from Mastercard.

**Network-connection status: system-generated flow**

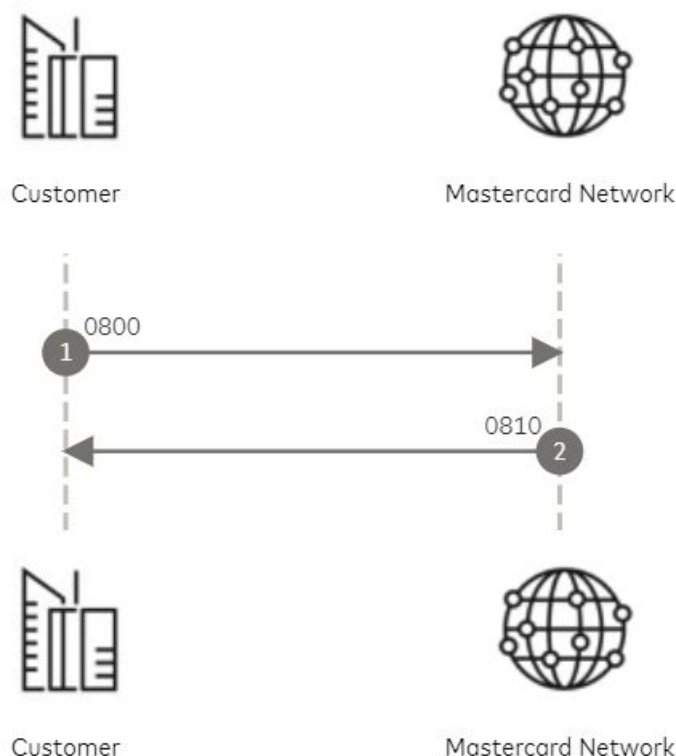


1. The Mastercard Network sends a Network Management Request/0800: network connection status: system-generated message identifying the message as an echo test to the customer to verify the customer's connection.
2. The customer responds with a Network Management Request Response/0810: network connection status: customer-generated message to confirm that the connection is active.

## Host activation or deactivation

This message flow describes transaction processing of the Network Management Request/0800: host session activation/deactivation message.

### Host activation or deactivation flow



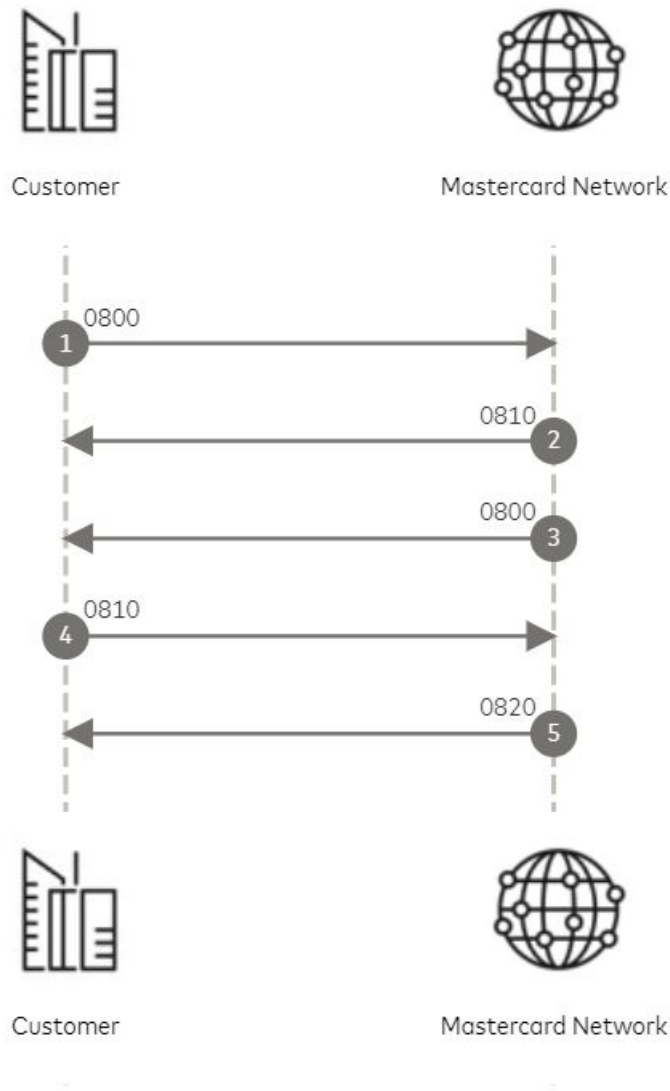
1. The customer sends a Network Management Request/0800: Host Session Activation/Deactivation message to identify session activation or session deactivation, through the value supplied in DE 70, to the Mastercard Network.
2. The Mastercard Network responds with a Network Management Request Response/0810: Host Session Activation/Deactivation message indicating successful completion.

### PIN Encryption Key exchange request: customer generated

Customers must be capable of sending a Network Management Request/0800: PEK exchange customer-generated message to the Mastercard Network requesting a new PEK. The Mastercard Network immediately initiates a PEK exchange with the customer identified in DE 33 (Forwarding Institution Identification Code) of the Network Management Request/0800: PEK exchange On Demand message. Customers should use this feature when systems problems

occur and a re-synchronization of the PEK is necessary. Mastercard exchanges PEKs with customers every 24 hours.

**PEK exchange request: customer-generated flow**



1. The customer determines there is a problem with its PEK and sends a Network Management Request/0800: PEK exchange On Demand message to the Mastercard Network requesting a new PEK.
2. The Mastercard Network responds with a Network Management Request Response/0810: PEK exchange On Demand message. The Mastercard Network uses the Key Encryption Key (KEK) to encrypt the new PEK.

3. The Mastercard Network sends a Network Management Request/0800: PEK exchange message containing the new PEK. The message contains the new PEK and the associated ID of the customer, identified in DE 2 (Primary Account Number [PAN]).  
The customer stores the new PEK and will use it for PIN encryption in subsequent Authorization Request/0100 messages.
4. The customer sends a Network Management Request Response/0810: PEK exchange message to the Mastercard Network acknowledging receipt of the Network Management Request/0800: PEK exchange message exchanging the PEK.
5. The Mastercard Network sends a Network Management Advice/0820: PEK exchange message to the customer notifying it that the new PEK is active and operational or advising of the failure. If the Network Management Advice/0820: PEK exchange message indicated that the PEK is active and operational, then subsequent Authorization Request/0100 messages must use the new PEK to translate the PIN data.

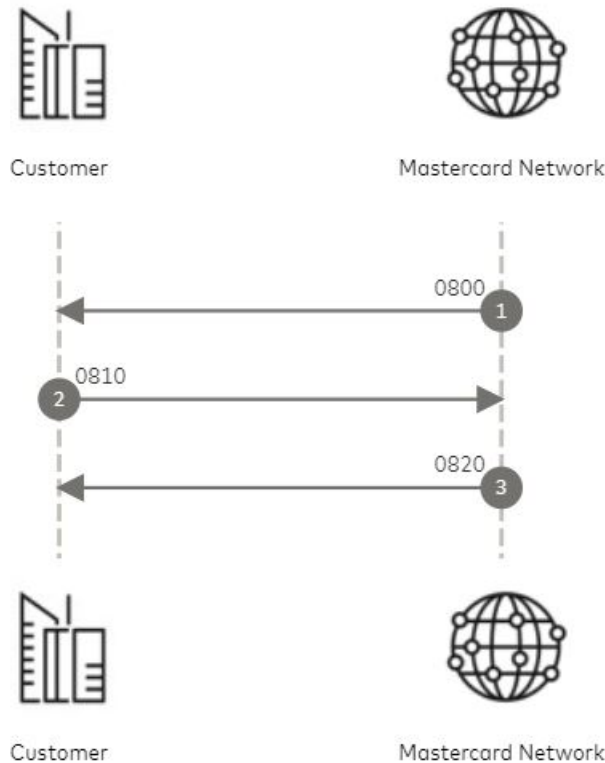
**NOTE: Error-condition message flows for Network Management/08xx messages are not illustrated. If an acquirer or issuer unsuccessfully forwards a Network Management/08xx message to the Mastercard Network, the acquirer or issuer should retransmit the message.**



## PEK-exchange request: system generated

This message flow describes transaction processing as it exchanges new PEKs with the customer every 24 hours.

### PEK exchange request: system-generated flow



1. The predetermined time has passed for exchanging a new PEK. The Mastercard Network uses the Key Encryption Key (KEK) to encrypt the new PEK. The Mastercard Network sends a Network Management Request/0800: PEK exchange message containing the PEK. The message contains the new PEK (encrypted using the KEK) in DE 48 (Additional Data: Private Use) or DE 110 (Additional Data 2) and the customer ID in DE 2 (Primary Account Number [PAN]).
2. The customer stores the new PEK. The customer sends a Network Management Request Response/0810: PEK exchange message to the Mastercard Network acknowledging receipt of the Network Management Request message exchanging the PEK.
3. The Mastercard Network sends a Network Management Advice/0820: PEK exchange message to notify the customer that the new PEK is active and operational or to advise of the failure. If the Network Management Advice/0820: PEK exchange message indicated

that the PEK is active and operational, then subsequent Authorization Request/0100 messages must use the new PEK to translate the PIN data.

## Chapter 4 General message layout information

*This section describes how the data attributes and notations are documented for use in the Dual Message Authorization System. This information provides details necessary to build or receive the messages supported by the system as described in [Data element definitions and details](#).*

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Overview

This describes how the data attributes and notations are documented for use in the Mastercard Network's Dual Message System. This information provides details necessary to build or receive the messages supported by the system as described in Data element definitions and details. The notations used to describe the data elements and message formats are:

- Data representation
- Data length
- Data field
- Justification
- Date and time attribute notations
- Entity descriptions
- Presence notations
- Program and service category notations
- Presence requirement notations

Data representation

Data representation notations indicate how data is transmitted, processed, and stored.

Data representation notations

| Notation | Description                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a        | alphabetic characters A-Z and a-z (Note that Country and Country Subdivision must be capitalized)                                                                                                                                                                                                           |
| an       | alphabetic (A-Z and a-z) and numeric (digits 0-9) characters                                                                                                                                                                                                                                                |
| ans      | alphabetic (A-Z and a-z), numeric (digits 0-9) and special characters (including spaces)                                                                                                                                                                                                                    |
| as       | alphabetic (A-Z and a-z) and special characters (including spaces)                                                                                                                                                                                                                                          |
| b        | binary<br><br>All binary data elements are constructed of bit-strings that have lengths that are an integral number of eight-bit bytes. No binary data element has a length of less than eight bits (one byte).<br><br>b-8 indicates a fixed-length binary field of eight characters (eight bytes, 64 bits) |
| n        | numeric digits 0-9                                                                                                                                                                                                                                                                                          |
| ns       | numeric digits 0-9 and special characters (including spaces)                                                                                                                                                                                                                                                |

| Notation | Description                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| s        | special characters<br><br>Special characters are any non-numeric or non-alphabetic characters (for example, @, #, &, and \$), including spaces. |

The data encoding conventions have been adapted for all Mastercard Network messages as follows:

- The system aligns all message data element fields on byte boundaries; for example, a data field cannot begin with the low order nibble or any bit other than the high-order bit of any byte.
- All data types listed in the data representation table are encoded for transmission between the Mastercard Network and processor systems using EBCDIC display character representation.
- The ASCII to EBCDIC character set table indicates alpha, numeric, and special characters sent to the Mastercard Network. When received, the Mastercard Network passes these characters through as they are received. The Extended ASCII Character Sets table presents the mapping of decimal to hexadecimal character sets when received by the Mastercard Network. Any characters not shown in the tables are mapped as a space.
- All numeric (attribute n) data elements are right-justified with leading zeros unless otherwise specified in the individual data element definitions. All other data elements are left-justified with trailing blanks unless otherwise specified.
- All Track-2 or Track-3 (attribute ans) data elements are encoded as EBCDIC representations of the hexadecimal data specified in ISO specification 7811 and 7812. Thus, a hex D (binary 1101) is encoded as an EBCDIC character D, and so on. The LLVAR or LLLVAR length specification of each data element specifies the field length in number of bytes.
- The system encodes all length subfields as numeric EBCDIC, right-justified with leading zeros unless otherwise specified.

## Data length

Data length notations define the characteristics of the data length field.

### Data length notations

| Notation  | Description                                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -digit(s) | Fixed length in number of positions.<br><br>Example: n-3 indicates a 3-position numeric data element.<br><br>Example: an-10 indicates a 10-position alphanumeric data element. |

| Notation              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ...digit(s)           | <p>Variable length field, with maximum number of positions specified.</p> <p>Example: n...11 indicates a variable-length numeric data element of 1 to 11 digits.</p> <p>Example: an...25 indicates a variable-length alphanumeric data element of 1 to 25 positions.</p>                                                                                                                                                                                                                                                           |
| LLVAR                 | <p>Present with a variable-length data element attribute, indicates that the data element contains these two fields:</p> <ul style="list-style-type: none"> <li>• LL is the length field and represents the number of positions in the variable-length data field that follows. The length field contains a value in the range of 01 to 99.</li> <li>• VAR is the variable-length data field.</li> </ul> <p>Example: an...25; LLVAR represents a variable-length alphanumeric data element with a length of 1 to 25 positions.</p> |
| LLLVAR                | <p>Present with a variable-length data element attribute, indicates that the data element contains two fields:</p> <ul style="list-style-type: none"> <li>• LLL is the length field and represents the number of positions in the variable-length data field that follows. The length field contains a value in the range 001–999.</li> <li>• VAR is the variable-length data field.</li> </ul> <p>Example: an...500; LLLVAR indicates a variable-length alphanumeric data element having a length of 1–500 positions.</p>         |
| TAGLL or TAGLLL       | <p>Integrated Product Messages (IPM) private data subelement (PDS) use only. Unique tag value assigned to the PDS followed by the fixed length value.</p>                                                                                                                                                                                                                                                                                                                                                                          |
| TAGLLVAR or TAGLLLVAR | <p>IPM PDS use only. Unique tag value assigned to the PDS followed by the variable length value.</p>                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Data field

### Data field notations

| Notation              | Description                                                                         |
|-----------------------|-------------------------------------------------------------------------------------|
| Contents of subfields | <p>Subfield number or number range</p> <p>Example: Contents of subfields 1 to 8</p> |

| Notation                | Description                                                              |
|-------------------------|--------------------------------------------------------------------------|
| Contents of position(s) | Position number or number range<br>Example: Contents of positions 1 to 8 |
| N/A                     | Not applicable                                                           |

## Justification

Data justification indicates the position of the data in the data element. A justification of not applicable (N/A) indicates that the values contained within those fields or subfields take up all positions allowed.

### Justification notations

| Notation      | Description                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------|
| Left          | Data is left-justified                                                                                                     |
| Right         | Data is right-justified                                                                                                    |
| See subfields | Data justification is defined in the subfield description indicating the subfield justification may vary between subfields |
| N/A           | Not applicable                                                                                                             |

Right-justified data fields and subfields that do not contain a value that fills the entire field must contain leading zeros. Most numeric (attribute n) data elements are right-justified; for example, amounts always have this right justification with leading zeros.

Left-justified fields and subfields that do not use the entire field with a value must have trailing spaces.

## Date and time attribute notations

Date and time notations indicate the format of the data that represents date and time.

### Date and time notations

| Notation | Description               |
|----------|---------------------------|
| MM       | Month (two digits, 01–12) |



| Notation | Description                                    |
|----------|------------------------------------------------|
| DD       | Day (two digits, 01–31)                        |
| YY       | Year (last two digits of calendar year, 00–99) |
| hh       | Hour (two digits, 00–23)                       |
| mm       | Minute (two digits, 00–59)                     |
| ss       | Second (two digits, 00–59)                     |

## Entity descriptions

Several entities may insert or modify data elements in a message as it flows from the message origin to the Mastercard Network and from the Mastercard Network to the message destination.

These entities typically include the acquirer or issuer at the origin, the Mastercard Network, and the acquirer or issuer at the destination.

### Entity description

| Notation | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Org      | <p>Originator requirements. The message originator must satisfy this data element's requirements.</p> <p>Examples of originators include:</p> <ul style="list-style-type: none"> <li>• Authorization Request/0100 message from an acquirer</li> <li>• Financial Transaction Request/0200 message from an acquirer</li> <li>• First Presentment/1240 message from an acquirer</li> </ul>                                                                                                    |
| Sys      | <p>Mastercard Network requirements. The system may insert, correct, modify, delete, or echo this data element while, for example, routing a message from the originator to the destination. The system may overwrite the data element and, as a result, overwrite any previous content.</p>                                                                                                                                                                                                |
| Dst      | <p>Destination requirements. The message destination must expect this data element (read it) and accept this data element (process it) if the originator requirements are satisfied.</p> <p>Examples of destinations include:</p> <ul style="list-style-type: none"> <li>• Authorization Request/0100 message received by an issuer</li> <li>• Financial Transaction Request/0200 message received by an issuer</li> <li>• First Presentment/1240 message received by an issuer</li> </ul> |

## Presence notations

Presence notations indicate if and how data is present. These notations appear in the originator (Org), Mastercard Network (Sys), and destination (Dst).

Mastercard uses these notation codes to indicate how message senders and receivers must understand the use of data in a message.

### Codes used by the originator or destination

| Notation | Description                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M        | Mandatory. The data element is required in the message.                                                                                                                                                         |
| C        | Conditional. The data element is required in the message if conditions described in the accompanying text apply.                                                                                                |
| O        | Optional. The data element is not required, but may be included in the message at the message initiator's option.                                                                                               |
| ME       | Mandatory Echo. The data element is required in a response message and must contain the same value (echoed) from the original request or advice message.                                                        |
| CE       | Conditional Echo. The data element is required in a response message if it was present in the original request or advice message, and it must contain the same value (echoed) from the original message.        |
| OE       | Optional Echo. The data element is not required in a response message if it was present in the original request or advice message, but the same value (echoed) may be included from the original message.       |
| •        | Not Required or Not Applicable. The data element is not required or is not applicable. The transaction originator should not include this data element if this code/symbol is present in the Org or Dst column. |

**NOTE: The echo notations do not apply to the Dual Message Clearing System.**

Mastercard uses these notation codes to indicate how message receivers must interpret the presence or absence of data managed by the Mastercard Network.

### Codes used by the Mastercard Network

| Notation | Description                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X        | System Interaction. The Mastercard Network may (or will) insert or overwrite the data element, depending on specific support services provided for individual programs and services. |

| Notation | Description                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------|
| XE       | System Echo. The Mastercard Network will contain the value from the original request or advice (echoed), if present. |
| • or P   | Pass-through. The data element is forwarded by the Mastercard Network to the destination (unmodified).               |

## Program and service category notations

Program and service category notations identify the various programs and services.

### Program and service category notations

| Notation | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MC       | Mastercard, including Debit Mastercard, activity processed on the Mastercard Network.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| MS       | Maestro activity processed on the Mastercard Network.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| CI       | Cirrus activity processed on the Mastercard Network.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| NP       | <p>Network processing services activities (often referred to as private label activity) are the Mastercard Network processes that do not fall into one of the following categories: Mastercard (MC), Visa (VI), or Travel and Entertainment (TE).</p> <p>This specification indicates the most general application data element requirements for private label activity.</p> <p>Processors that support all these general data element specifications are able to access or participate in any Dual Message Authorization System credit card or debit card gateways. For detailed information on specific gateways, refer to the appropriate program or service user guide.</p> |
| VI       | <p>Visa activity transmitted through the Mastercard Network.</p> <p>This specification includes application requirements for accepting Visa activity.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TE       | <p>Travel and Entertainment (T&amp;E) card activity transmitted through the Mastercard Network that does not fall into one of the following categories: Mastercard (MC), network processing services activity (NP), or Visa (VI).</p> <p>This specification includes application interface requirements for Travel and Entertainment (T&amp;E) card activity, including American Express, and Diners Club.</p>                                                                                                                                                                                                                                                                  |

## Presence requirement notations

Presence requirement notations describe the possible presence or usage requirements. An entity presence requirement is the three combined values in the originator (Org), Mastercard Network (Sys), and destination (Dst).

### Presence requirement notations

| Org | Sys | Dst | Description                                                                                                               |
|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------|
| M   | .   | M   | Mandatory; originator must provide the data element, and the destination must receive the data element.                   |
| M   | X   | M   | Mandatory; originator must provide the data element; Mastercard Network adjusts/appends data.                             |
| C   | P   | C   | Conditionally required; conditions are described in Comments column for originators and destinations to send and receive. |
| C   | X   | C   | Conditionally required; Mastercard Network adjusts/appends data.                                                          |
| .   | X   | C   | Mastercard Network provides the data element conditionally.                                                               |
| .   | X   | M   | Mastercard Network always provides the data element.                                                                      |
| O   | X   | C   | Optional; originator may provide the data element; Mastercard Network adjusts/appends data.                               |
| M   | X   | .   | Mandatory; originator must provide the data element; Mastercard Network does not forward data.                            |
| C   | X   | .   | Conditionally required; Mastercard Network does not forward data.                                                         |
| O   | X   | .   | Optional; Mastercard Network does not forward data.                                                                       |
| .   | X   | .   | Mastercard use only.                                                                                                      |

## Message types

## Bit map information

All Mastercard Network messages are variable-length, with a bit map scheme used as the first data elements of the message following the Message Type Identifier (MTI) to indicate the presence or absence of additional data elements in the message.

Each bit map is a 64-bit string contained within an eight-byte data element.

The Mastercard switching systems use a maximum of two-bit maps: a Primary and a Secondary Bit Map. Bits set to 1 or 0 in the Primary Bit Map indicate the presence or absence of DE 2 (Primary Account Number [PAN]) through DE 64 (Message Authentication Code [MAC]). Bits set to 1 or 0 in the Secondary Bit Map indicate the presence or absence of DE 66 (Settlement Code) through DE 128 (Message Authentication Code [MAC]).

All bit positions are interpreted from left to right within each bit map. For example, within the Primary Bit Map the left-most bit is bit number 1 and the right-most bit is bit number 64. The first bit in each bit map is 1 or 0 to indicate the presence or absence of another (immediately following) bit map data element.

Bit number 1 in the Primary Bit Map and bit number 65 in the Secondary Bit Map (that is, the first bit in each bit map) have no corresponding data elements; these bits indicate the presence or absence of additional bit map data elements in the message. If bit number 1 is 1, it indicates that the Secondary Bit Map is present and that selected data elements in the range DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) also are present in the message as indicated by bit positions in the Secondary Bit Map. Mastercard currently uses a maximum of two-bit maps (Primary and Secondary) in Mastercard Network messages, with a maximum number of message data elements ranging from DE 1 (Bit Map, Secondary) through DE 128 (Message Authentication Code [MAC]). Consequently, DE 65 (Bit Map, Extended) (the first bit in the Secondary Bit Map) must always be 0.

Each message must contain the Primary Bit Map. The Secondary Bit Map must be included in a message only when data elements in the range DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.

## Authorization Request/0100

This is the message layout for the Authorization Request/0100 message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                               |
|-------|-------------------------------|-----|-----|-----|----------------------------------------|
| -     | Message Type Identifier (MTI) | M   | P   | M   | Constant: 0100 (Authorization Request) |
| -     | Bit Map, Primary              | M   | X   | M   | Mandatory for all messages             |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                                                                                               |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Bit Map, Secondary                  | C   | X   | C   | Required data element if any DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message                                                                                   |
| 2     | Primary Account Number (PAN)        | M   | X   | M   | Contains cardholder's Primary Account Number (PAN)                                                                                                                                                                     |
| 3     | Processing Code                     | M   | P   | M   | Indicates type of transaction and the affected cardholder To or From Account Type                                                                                                                                      |
| 4     | Amount, Transaction                 | M   | P   | M   | Transaction amount in the currency of the acquirer's acceptor. DE 49 (Currency Code, Transaction) also must be present with this data element to identify the transaction currency.                                    |
| 5     | Amount, Settlement                  | •   | X   | C   | Transaction amount in the settlement currency in DE 50 (Currency Code, Settlement). The Dual Message Authorization System provides DE 05 to issuers if they choose to receive settlement amount related data elements. |
| 6     | Amount, Cardholder Billing          | •   | X   | M   | Transaction amount in the issuer currency. The Dual Message Authorization System provides this data element. DE 51 (Currency Code, Cardholder Billing) will also be present to identify the currency for this amount.  |
| 7     | Transmission Date and Time          | M   | P   | M   | Transaction time stamp; <i>Universal Time (UTC)</i> date and time that the originator initiates the message.                                                                                                           |
| 9     | Conversion Rate, Settlement         | •   | X   | C   | The Dual Message Authorization System provides DE 9 if the issuer chooses to receive settlement amount-related data elements.                                                                                          |
| 10    | Conversion Rate, Cardholder Billing | •   | X   | M   | The Dual Message Authorization System provides DE 10 with the rate used to convert the transaction currency to the cardholder billing currency.                                                                        |
| 11    | Systems Trace Audit Number (STAN)   | M   | P   | M   | Number assigned by the message initiator to uniquely identify a transaction.                                                                                                                                           |

| DE ID | Data Element Name                                            | Org | Sys | Dst | Comments                                                                                                                                                      |
|-------|--------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12    | Time, Local Transaction                                      | C   | P   | C   | Used in all customer-related reports and statements, if available.                                                                                            |
| 13    | Date, Local Transaction                                      | C   | P   | C   | Used in all customer-related reports and statements, if available.                                                                                            |
| 14    | Date, Expiration                                             | C   | X   | C   | Specifies the year and month a PAN is considered expired by the issuer.                                                                                       |
| 15    | Date, Settlement                                             | •   | X   | M   | The acquirer omits this data element, and the Mastercard Network inserts it and forwards it to the issuer.                                                    |
| 16    | Date, Conversion                                             | •   | X   | M   | Currency conversion rate file effective date. The Mastercard Network provides this data element.                                                              |
| 18    | Merchant Type                                                | M   | P   | M   | Refer to the <i>Quick Reference Booklet</i> for a listing of MCCs.                                                                                            |
| 19    | Acquiring Institution Country Code                           | •   | X   | C   | Based on Mastercard franchise data information.                                                                                                               |
| 20    | Primary Account Number (PAN) Country Code                    | C   | X   | C   | Required for messages when the PAN begins with 59.                                                                                                            |
| 22    | Point-of-Service (POS) Entry Mode                            | M   | X   | M   | Used to indicate the method used for PAN data entry to initiate a transaction.                                                                                |
| 23    | Card Sequence Number                                         | C   | X   | C   | Must be present for transactions that include DE 55 (ICC system-related data) and where the ICC provides the application PAN sequence number to the terminal. |
| 26    | Point-of-Service (POS) Personal ID Number (PIN) Capture Code | C   | P   | C   | Required when PIN data is present and the terminal PIN capture capability is other than 12 characters.                                                        |
| 28    | Amount, Transaction Fee                                      | C   | P   | C   | Must contain fee amount, if applied.                                                                                                                          |
| 32    | Acquiring Institution ID Code                                | M   | P   | M   | Contains acquirer's identification information.                                                                                                               |
| 33    | Forwarding Institution ID Code                               | C   | P   | C   | Contains the processor ID number forwarding this message to the Mastercard Network.                                                                           |

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                      |
|-------|-----------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 35    | Track 2 Data                      | C   | X   | C   | Required if the transaction entry point captured Track 2 data. Required for ATM transactions. For Maestro transactions, such as e-commerce, the Dual Message Authorization System creates Track 2 data if it is not provided by the acquirer. |
| 37    | Retrieval Reference Number        | C   | P   | C   | Required for ATM, Chip and POS card-read transactions.                                                                                                                                                                                        |
| 41    | Acceptor Terminal ID              | C   | P   | C   | Required for all ATM and card-read POS transactions.                                                                                                                                                                                          |
| 42    | Acceptor ID                       | C   | P   | C   | Mandatory for POS transaction types containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), values 00 (Purchase), 09 (Purchase with Cash Back), and 28 (Payment Transaction).                                       |
| 43    | Acceptor Name and Location        | C   | P   | C   | Required for Chip and POS card-read transactions.<br><br>Required for Mastercard Hosted Mobile Phone Top-up transactions.                                                                                                                     |
| 45    | Track 1 Data                      | C   | X   | C   | Required if the transaction entry point captured Track 1 data.                                                                                                                                                                                |
| 48    | Additional Data: Private Use      | M   | X   | M   | Contains a variety of subelements depending on the card and purpose of the request.                                                                                                                                                           |
| 49    | Currency Code, Transaction        | M   | P   | M   | Identifies the currency code of DE 4 (Amount, Transaction).                                                                                                                                                                                   |
| 50    | Currency Code, Settlement         | •   | X   | C   | Identifies the currency code of DE 5 (Amount, Settlement).<br><br>The Dual Message Authorization System provides DE 50 if the issuer chooses to receive settlement amount-related data elements.                                              |
| 51    | Currency Code, Cardholder Billing | •   | X   | M   | Defines the currency of DE 6 (Amount, Cardholder Billing), DE 8 (Amount, Cardholder Billing Fee), and DE 95 (Replacement Amounts), subfield 3 (Actual Amount, Cardholder Billing).                                                            |



| DE ID | Data Element Name                                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|---------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 52    | Personal ID Number (PIN) Data                     | C   | X   | C   | Required for ATM transactions.<br><br><b>NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 53    | Security-Related Control information              | C   | X   | C   | Required for acquirers using the Dual Message Authorization System to perform PIN translation services. Acquirers provide DE 53 to identify the PIN Block Format and key used for PIN encryption.<br><br>The Dual Message Authorization System provides issuers using the Dual Message Authorization System to perform PIN translation the PIN Block Format and key used by Mastercard to encrypt the PIN before sending the Authorization Request/0100 to the issuer.<br><br><b>NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.</b> |
| 54    | Additional Amounts                                | C   | X   | C   | The Dual Message Authorization System will forward the occurrence of each DE 54 amount type provided by the acquirer in the acquirer's currency and will provide an additional occurrence of each DE 54 amount type in the issuer's currency.                                                                                                                                                                                                                                                                                                                                                       |
| 55    | Integrated Circuit Card (ICC) System-Related Data | C   | X   | C   | Conditional data for chip-based transactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 56    | Payment Account Data                              | •   | X   | C   | When Mastercard is the BIN Controller (as defined by EMVCo) DE 56 will be present and contain the PAR value when one is associated with the PAN.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 61    | Point-of-Service (POS) Data                       | M   | X   | M   | Describes the conditions present at the point-of-interaction at the time when the originator initiates the transaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 62    | Intermediate Network Facility (INF) Data          | O   | P   | O   | May be used to provide unique acquiring network ID codes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 63    | Network Data                                      | •   | X   | M   | Contains the financial network code and reference data for the transaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                         |
|-------|-------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 104   | Digital Payment Data                      | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                             |
| 105   | Multi-Use Transaction Identification Data | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                             |
| 106   | Fleet Service Data                        | O   | P   | C   | Use of DE 106 for authorization is required only for EMV chip cards personalized using the IFSF/Conexxus specification for retrieving supplemental data from acceptors and cardholders.                                                                                                                                                                          |
| 108   | Additional Transaction Reference Data     | C   | P   | C   | Mandatory for originating institution to submit DE 108 on all Mastercard Send transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions, and Mastercard™ Merchant Presented QR Payment and Funding Transactions).<br><br>Optional on Mastercard Merchant Presented QR Refund Payment Transactions. |
| 110   | Encryption Data                           | C   | X   | C   | Includes PIN encryption data.<br><br><b>NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.</b>                                                                                                                                                                                                       |
| 112   | Additional Data (National Use)            | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                             |
| 120   | Record Data                               | C   | X   | C   | Used to transmit file record data or textual character string data.                                                                                                                                                                                                                                                                                              |
| 122   | Additional Record Data                    | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                             |
| 124   | Member-Defined Data                       | C   | X   | C   | May contain customer-defined data.                                                                                                                                                                                                                                                                                                                               |
| 125   | New PIN Data                              | C   | X   | C   | Must be present for all PIN change transactions; otherwise not present.<br><br><b>NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.</b>                                                                                                                                                             |
| 127   | Private Data                              | O   | X   | •   | Private data for message initiator's use.                                                                                                                                                                                                                                                                                                                        |

## Authorization Request Response/0110

This is the message layout for the Authorization Request Response/0110 message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------|-------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI) | M   | P   | M   | Constant: 0110 (Authorization Request Response).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| -     | Bit Map, Primary              | M   | X   | M   | Mandatory for all messages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1     | Bit Map, Secondary            | C   | X   | C   | Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2     | Primary Account Number (PAN)  | ME  | XE  | ME  | Must be the same value as in the original Authorization Request/0100.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3     | Processing Code               | ME  | XE  | ME  | Must be the same value as in the original Authorization Request/0100.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4     | Amount, Transaction           | CE  | X   | M   | Must be the same value as in the original Authorization Request/0100 except when DE 39 (Response Code) contains value 10 (Partial approval) or value 87 (Purchase Amount Only: No Cash Back Allowed).                                                                                                                                                                                                                                                                                                                                           |
| 5     | Amount, Settlement            | CE  | X   | C   | Must be the same value as in the original Authorization Request/0100, if present, except when DE 39 (Response Code) contains value 10 (Partial approval) or value 87 (Purchase Amount Only: No Cash Back Allowed). The Mastercard Network provides this data element if the acquirer chooses to receive settlement amount-related data elements.<br><br>This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing. |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6     | Amount, Cardholder Billing          | C   | X   | C   | <p>Must be the same value as in the original Authorization Request/0100, except when DE 39 (Response Code) contains value 10 (Partial approval) or value 87 (Purchase Amount Only: No Cash Back Allowed).</p> <p>This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.</p>                         |
| 7     | Transmission Date and Time          | ME  | XE  | ME  | Must be the same value as in the original Authorization Request/0100.                                                                                                                                                                                                                                                                                                                                                                |
| 9     | Conversion Rate, Settlement         | CE  | X   | C   | <p>Must be the same value as in the original Authorization Request/0100, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements.</p> <p>This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.</p> |
| 10    | Conversion Rate, Cardholder Billing | CE  | X   | C   | <p>Must be the same value as in the original Authorization Request/0100 message.</p> <p>This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.</p>                                                                                                                                                  |
| 11    | Systems Trace Audit Number (STAN)   | ME  | XE  | ME  | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                        |
| 15    | Date, Settlement                    | ME  | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                        |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                     |
|-------|-------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16    | Date, Conversion                          | CE  | X   | C   | Must be the same value as in the original Authorization Request/0100 message.<br><br>This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing. |
| 20    | Primary Account Number (PAN) Country Code | CE  | XE  | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                    |
| 28    | Amount, Transaction Fee                   | •   | XE  | CE  | Must be the same value as in the Authorization Request/0100 message, if present.                                                                                                                                                                                             |
| 32    | Acquiring Institution ID Code             | ME  | XE  | ME  | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                |
| 33    | Forwarding Institution ID Code            | CE  | XE  | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                    |
| 37    | Retrieval Reference Number                | CE  | XE  | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                    |
| 38    | Authorization ID Response                 | C   | P   | C   | Response ID that the authorizing institution or agent assigned for an approved request.                                                                                                                                                                                      |

| DE ID | Data Element Name            | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39    | Response Code                | M   | X   | M   | <p>Contains message-specific values. Refer to <a href="#">Data element definitions and details</a> for list of values by message type.</p> <p>Issuers responding with DE 39 (Response Code), value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed) will not be required to echo DE 4 (Amount, Transaction) in the Authorization Request Response/0110. Likewise, if DE 5 (Amount, Settlement) was present in the Authorization Request/0100 message to the issuer, the issuer will not be required to echo DE 5 in the Authorization Request Response/0110 message when responding with DE 39, value 10 or value 87. The issuer will provide the partial approval amount in DE 6 (Amount, Cardholder Billing).</p> |
| 41    | Acceptor Terminal ID         | CE  | XE  | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 44    | Additional Response Data     | C   | X   | C   | May be used for referral phone numbers in denied transactions, or cardholder ID information in approved transactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 48    | Additional Data: Private Use | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 49    | Currency Code, Transaction   | CE  | XE  | ME  | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 50    | Currency Code, Settlement    | CE  | X   | C   | <p>Must be the same value as in the original Authorization Request/0100 message, if present. The Mastercard Network provides this data element if the acquirer chooses to receive settlement amount-related data elements.</p> <p>This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.</p>                                                                                                                                                                                                                                                                                                                     |

| DE ID | Data Element Name                                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|---------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 51    | Currency Code, Cardholder Billing                 | CE  | X   | C   | Must be the same value as in the original Authorization Request/0100 message.<br><br>This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.                                                                                                                                                                                                 |
| 54    | Additional Amounts                                | C   | X   | C   | The Dual Message Authorization System will forward the occurrence of each DE 54 amount type as provided by the issuer in the issuer's currency and will provide an additional occurrence of each DE 54 amount type in the acquirer's currency.                                                                                                                                                                                                                               |
| 55    | Integrated Circuit Card (ICC) System-Related Data | C   | X   | C   | Conditional data for chip-based transactions.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 56    | Payment Account Data                              | C   | X   | C   | When Mastercard is the BIN Controller (as defined by EMVCo), the issuer is not required to provide DE 56 in the response message. If a PAR value is associated with the PAN, Mastercard will insert DE 56 containing the PAR value in the response message before forwarding the message to the acquirer. When the issuer is the BIN Controller and has associated a PAR value with the PAN, the issuer must include DE 56 containing the PAR value in the response message. |
| 62    | Intermediate Network Facility (INF) Data          | CE  | XE  | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                                                                                                                                                    |
| 63    | Network Data                                      | ME  | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                                                                |
| 102   | Account ID-1                                      | C   | P   | C   | May contain cardholder "from" account number.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 103   | Account ID-2                                      | C   | P   | C   | May contain cardholder "to" account number.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 105   | Multi-Use Transaction Identification Data         | C   | X   | C   | Applicable subelement data.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| DE ID | Data Element Name                     | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                |
|-------|---------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 106   | Fleet Service Data                    | C   | X   | C   | Applicable subelement data.                                                                                                                                                                                                                                                                                                                                             |
| 108   | Additional Transaction Reference Data | C   | X   | C   | <p>Mandatory for originating institution to submit DE 108 on all Mastercard Send transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions, and Mastercard™ Merchant Presented QR Payment and Funding Transactions).</p> <p>Optional on Mastercard Merchant Presented QR Refund Payment Transactions.</p> |
| 112   | Additional Data (National Use)        | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                                    |
| 120   | Record Data                           | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                                               |
| 121   | Authorizing Agent ID Code             | C   | X   | C   | Contains the Mastercard customer ID of an alternate authorizer, if authorized by other than the issuer or issuer's primary authorizer.                                                                                                                                                                                                                                  |
| 123   | Receipt Free Text                     | O   | X   | C   | <p>Optional for issuers. Present for acquirers if provided by the issuer. If not supported by the acquirer, the Dual Message Authorization System removes DE 123 from the message.</p> <p><b>NOTE: Applicable only to Swedish Domestic Authorization Switching Service (SASS) or Peru domestic POS transactions.</b></p>                                                |
| 124   | Member-Defined Data                   | C   | X   | C   | May contain issuer-defined data.                                                                                                                                                                                                                                                                                                                                        |
| 127   | Private Data                          | O   | XE  | CE  | Private data for message initiator's use. The Dual Message Authorization System will echo DE 127 from the original Authorization Request/0100 message, if present.                                                                                                                                                                                                      |



## Authorization Advice/0120: acquirer-generated

This is the message layout for the Authorization Advice/0120: acquirer-generated message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                              |
|-------|-------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | M   | P   | M   | Constant: 0120 (Authorization Advice).                                                                                                                |
| -     | Bit Map, Primary                    | M   | X   | M   | Mandatory for all messages                                                                                                                            |
| 1     | Bit Map, Secondary                  | C   | X   | C   | Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.                                 |
| 2     | Primary Account Number (PAN)        | M   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                         |
| 3     | Processing Code                     | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                         |
| 4     | Amount, Transaction                 | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message unless the message is submitted to complete a preauthorized transaction. |
| 5     | Amount, Settlement                  | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.            |
| 6     | Amount, Cardholder Billing          | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                     |
| 7     | Transmission Date and Time          | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                         |
| 9     | Conversion Rate, Settlement         | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.            |
| 10    | Conversion Rate, Cardholder Billing | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                     |
| 11    | Systems Trace Audit Number (STAN)   | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                         |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                       |
|-------|-------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------|
| 12    | Time, Local Transaction                   | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.      |
| 13    | Date, Local Transaction                   | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.      |
| 14    | Date, Expiration                          | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.      |
| 15    | Date, Settlement                          | •   | X   | M   | The Dual Message Authorization System provides this data element.                              |
| 16    | Date, Conversion                          | •   | X   | M   | The Dual Message Authorization System provides this data element.                              |
| 18    | Merchant Type                             | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                  |
| 19    | Acquiring Institution Country Code        | •   | X   | C   | Based on Mastercard franchise data information.                                                |
| 20    | Primary Account Number (PAN) Country Code | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.      |
| 22    | Point-of-Service (POS) Entry Mode         | M   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                  |
| 23    | Card Sequence Number                      | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.      |
| 28    | Amount, Transaction Fee                   | C   | P   | C   | Contains an online transaction fee as permitted by the operating rules of a bank card product. |
| 32    | Acquiring Institution ID Code             | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                  |
| 33    | Forwarding Institution ID Code            | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.      |
| 35    | Track 2 Data                              | O   | X   | O   | Must be the same value as in the original Authorization Request/0100 message, if present.      |

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                      |
|-------|-----------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37    | Retrieval Reference Number        | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                     |
| 38    | Authorization ID Response         | C   | P   | C   | Must be either the same value as in the original Authorization Request Response/0110 message, if present, or a value provided by the acquirer if the acquirer or merchant processed and approved the transaction. Must be the same value as in the original Authorization Request Response/0110 message provided by the acquirer, if present. |
| 39    | Response Code                     | M   | P   | M   | Must be either the same value as in the original Authorization Request Response/0110 message or a value provided by the acquirer if the acquirer or merchant processed and approved the transaction.                                                                                                                                          |
| 41    | Acceptor Terminal ID              | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                     |
| 42    | Acceptor ID                       | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                     |
| 43    | Acceptor Name and Location        | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                     |
| 45    | Track 1 Data                      | O   | X   | O   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                     |
| 48    | Additional Data: Private Use      | M   | X   | M   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                          |
| 49    | Currency Code, Transaction        | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                 |
| 50    | Currency Code, Settlement         | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.                                                                                                                                                                                                    |
| 51    | Currency Code, Cardholder Billing | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                                                                                                                                                                                                             |

| DE ID | Data Element Name                                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 54    | Additional Amounts                                | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.<br><br>The Dual Message Authorization System will forward the occurrence of each DE 54 amount type provided by the acquirer in the acquirer's currency and will provide an additional occurrence of each DE 54 amount type in the issuer's currency. |
| 55    | Integrated Circuit Card (ICC) System-Related Data | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                      |
| 56    | Payment Account Data                              | •   | X   | C   | When Mastercard is the BIN Controller, the PAR value is inserted when available.                                                                                                                                                                                                                                                               |
| 60    | Advice Reason Code                                | M   | P   | M   | DE 60, subfield 1 (Advice Reason Code) must contain the value 190 (acquirer approved), value 191 (Acquirer Processing System Completed Authorization Transaction), or value 192 (M/Chip Offline Advice to Issuer).                                                                                                                             |
| 61    | Point-of-Service (POS) Data                       | M   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                  |
| 62    | Intermediate Network Facility (INF) Data          | O   | P   | O   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                      |
| 63    | Network Data                                      | •   | X   | M   | The Mastercard Network provides this data element.                                                                                                                                                                                                                                                                                             |
| 102   | Account ID-1                                      | C   | P   | C   | Must be the same value as in the original Authorization Request Response/0110 message, if present.                                                                                                                                                                                                                                             |
| 103   | Account ID-2                                      | C   | P   | C   | Must be the same value as in the original Authorization Request Response/0110 message, if present.                                                                                                                                                                                                                                             |
| 105   | Multi-Use Transaction Identification Data         | C   | X   | C   | Contains the applicable subelement data.                                                                                                                                                                                                                                                                                                       |

| DE ID | Data Element Name                     | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 106   | Fleet Service Data                    | O   | P   | C   | Use of DE 106 (Fleet Service Data) for authorization is required only for EMV chip cards personalized using the IFSF/Conexus specification for retrieving supplemental data from acceptors and cardholders.                                                                                                                                                      |
| 108   | Additional Transaction Reference Data | O   | P   | C   | Mandatory for originating institution to submit DE 108 on all Mastercard Send transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions, and Mastercard™ Merchant Presented QR Payment and Funding Transactions).<br><br>Optional on Mastercard Merchant Presented QR Refund Payment Transactions. |
| 112   | Additional Data (National Use)        | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                             |
| 121   | Authorizing Agent ID Code             | C   | X   | C   | Must be the Mastercard assigned customer ID of the authorizing entity, if present.<br><br>Must be the same value as in the original Authorization Request Response/0110 message, if present.                                                                                                                                                                     |
| 122   | Additional Record Data                | C   | X   | C   | Applicable subelement data.                                                                                                                                                                                                                                                                                                                                      |
| 124   | Member-Defined Data                   | O   | X   | C   | May contain acquirer-defined data.                                                                                                                                                                                                                                                                                                                               |
| 127   | Private Data                          | O   | X   | •   | Private data for message initiator's use.                                                                                                                                                                                                                                                                                                                        |

## Authorization Advice/0120: system-generated

This is the message layout for the Authorization Advice/0120: system-generated message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                               |
|-------|-------------------------------|-----|-----|-----|----------------------------------------|
| -     | Message Type Identifier (MTI) | •   | X   | M   | Constant: 0120 (Authorization Advice). |
| -     | Bit Map, Primary              | •   | X   | M   |                                        |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                   |
|-------|-------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Bit Map, Secondary                  | •   | X   | C   | Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.                      |
| 2     | Primary Account Number (PAN)        | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                              |
| 3     | Processing Code                     | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                              |
| 4     | Amount, Transaction                 | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                              |
| 5     | Amount, Settlement                  | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. |
| 6     | Amount, Cardholder Billing          | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                          |
| 7     | Transmission Date and Time          | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                              |
| 9     | Conversion Rate, Settlement         | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. |
| 10    | Conversion Rate, Cardholder Billing | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                          |
| 11    | Systems Trace Audit Number (STAN)   | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                              |
| 12    | Time, Local Transaction             | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 13    | Date, Local Transaction             | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 14    | Date, Expiration                    | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |

| DE ID | Data Element Name                                            | Org | Sys | Dst | Comments                                                                                           |
|-------|--------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------|
| 15    | Date, Settlement                                             | •   | X   | M   | The Dual Message Authorization System provides this data element.                                  |
| 16    | Date, Conversion                                             | •   | X   | M   | The Dual Message Authorization System provides this data element.                                  |
| 18    | Merchant Type                                                | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                      |
| 19    | Acquiring Institution Country Code                           | •   | X   | C   | Based on Mastercard franchise data information.                                                    |
| 20    | Primary Account Number (PAN) Country Code                    | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.          |
| 22    | Point-of-Service (POS) Entry Mode                            | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                      |
| 23    | Card Sequence Number                                         | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.          |
| 26    | Point-of-Service (POS) Personal ID Number (PIN) Capture Code | •   | C   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.          |
| 28    | Amount, Transaction Fee                                      | •   | X   | C   | Must contain the same value as the original Authorization Request/0100 message, if present.        |
| 32    | Acquiring Institution ID Code                                | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                      |
| 33    | Forwarding Institution ID Code                               | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.          |
| 35    | Track 2 Data                                                 | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.          |
| 37    | Retrieval Reference Number                                   | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.          |
| 38    | Authorization ID Response                                    | •   | X   | C   | Must be the same value as in the original Authorization Request Response/0110 message, if present. |

| DE ID | Data Element Name                                 | Org | Sys | Dst | Comments                                                                                                                                   |
|-------|---------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| 39    | Response Code                                     | •   | X   | M   | Must be the same value as in the original Authorization Request Response/0110 message.                                                     |
| 41    | Acceptor Terminal ID                              | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 42    | Acceptor ID                                       | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 43    | Acceptor Name and Location                        | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 44    | Additional Response Data                          | •   | X   | C   | Must be the same value as in the original Authorization Request Response/0110 message, if present.                                         |
| 45    | Track 1 Data                                      | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 48    | Additional Data: Private Use                      | •   | X   | M   | Contains applicable subelement data.                                                                                                       |
| 49    | Currency Code, Transaction                        | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                              |
| 50    | Currency Code, Settlement                         | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. |
| 51    | Currency Code, Cardholder Billing                 | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                          |
| 54    | Additional Amounts                                | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 55    | Integrated Circuit Card (ICC) System-Related Data | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                  |
| 56    | Payment Account Data                              | •   | X   | C   | When Mastercard is the BIN Controller, the PAR value is inserted when available.                                                           |
| 60    | Advice Reason Code                                | •   | X   | M   | Specifies reason for this Authorization Advice/0120 message.                                                                               |



| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                     |
|-------|-------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 61    | Point-of-Service (POS) Data               | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                |
| 62    | Intermediate Network Facility (INF) Data  | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                    |
| 63    | Network Data                              | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 message. The Mastercard Network provides this data element, if required.                                                                |
| 102   | Account ID-1                              | •   | X   | C   | Must be the same value as in the original Authorization Request Response/0110 message, if present.                                                                                                           |
| 103   | Account ID-2                              | •   | X   | C   | Must be the same value as in the original Authorization Request Response/0110 message, if present.                                                                                                           |
| 105   | Multi-Use Transaction Identification Data | •   | X   | C   | Contains applicable subelement data.                                                                                                                                                                         |
| 106   | Fleet Service Data                        | •   | X   | C   | Use of DE 106 (Fleet Service Data) for authorization is required only for EMV chip cards personalized using the IFSF/Conexxus specification for retrieving supplemental data from acceptors and cardholders. |
| 108   | Additional Transaction Reference Data     | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.<br><br>Optional for Mastercard Merchant Presented QR Transactions.                                                 |
| 112   | Additional Data (National Use)            | •   | X   | C   | Contains applicable subelement data.                                                                                                                                                                         |
| 121   | Authorizing Agent ID Code                 | •   | X   | C   | Must be the same value as in the original Authorization Request Response/0110 message, if present.                                                                                                           |
| 122   | Additional Record Data                    | •   | X   | C   | Applicable subelement data.                                                                                                                                                                                  |
| 124   | Member-defined Data                       | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                    |

## Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)

This is the message layout for the Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120) message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                        |
|-------|-------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | M   | X   | •   | Constant: 0130 (Authorization Advice Response).                                 |
| -     | Bit Map, Primary                    | M   | X   | •   |                                                                                 |
| 1     | Bit Map, Secondary                  | C   | X   | •   | Must be the same value as in the Authorization Advice/0120 message, if present. |
| 2     | Primary Account Number (PAN)        | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 3     | Processing Code                     | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 4     | Amount, Transaction                 | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 5     | Amount, Settlement                  | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present. |
| 6     | Amount, Cardholder Billing          | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 7     | Transmission Date and Time          | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 9     | Conversion Rate, Settlement         | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present. |
| 10    | Conversion Rate, Cardholder Billing | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 11    | Systems Trace Audit Number (STAN)   | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 15    | Date, Settlement                    | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |
| 16    | Date, Conversion                    | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.             |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                |
|-------|-------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------|
| 20    | Primary Account Number (PAN) Country Code | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 23    | Card Sequence Number                      | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 28    | Amount, Transaction Fee                   | OE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 32    | Acquiring Institution ID Code             | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.                                                     |
| 33    | Forwarding Institution ID Code            | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 37    | Retrieval Reference Number                | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 39    | Response Code                             | M   | X   | •   | For more detail, refer to the data element definition for DE 39.                                                        |
| 41    | Acceptor Terminal ID                      | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 44    | Additional Response Data                  | C   | X   | •   | May contain additional response information for certain error conditions in original Authorization Advice/0120 message. |
| 49    | Currency Code, Transaction                | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.                                                     |
| 50    | Currency Code, Settlement                 | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 51    | Currency Code, Cardholder Billing         | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.                                                     |

| DE ID | Data Element Name                        | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                             |
|-------|------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 56    | Payment Account Data                     | C   | X   | •   | When Mastercard is the BIN Controller, issuers are not required to send PAR. Mastercard will insert the PAR in the message before sending to the acquirer, when available.<br><br>When Mastercard is not the BIN Controller, the issuer may provide the PAR in the response message when available. Mastercard will pass this value to the acquirer. |
| 62    | Intermediate Network Facility (INF) Data | CE  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message, if present.                                                                                                                                                                                                                                                                      |
| 63    | Network Data                             | ME  | XE  | •   | Must be the same value as in the Authorization Advice/0120 message.                                                                                                                                                                                                                                                                                  |

## Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)

This is the message layout for the Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                                                    |
|-------|-------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI) | M   | P   | M   | Constant: 0130 (Authorization Advice Response).                                             |
| -     | Bit Map, Primary              | M   | X   | M   |                                                                                             |
| 1     | Bit Map, Secondary            | C   | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present. |
| 2     | Primary Account Number (PAN)  | ME  | XE  | ME  | Must be the same value as in the Authorization Advice/0120: acquirer-generated.             |
| 3     | Processing Code               | ME  | XE  | ME  | Must be the same value as in the Authorization Advice/0120: acquirer-generated.             |
| 4     | Amount, Transaction           | ME  | XE  | ME  | Must be the same value as in the Authorization Advice/0120: acquirer-generated.             |

General message layout information

Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                        |
|-------|-------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Amount, Settlement                        | CE  | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.<br><br>The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements. |
| 6     | Amount, Cardholder Billing                | CE  | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                 |
| 7     | Transmission Date and Time                | ME  | XE  | ME  | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                 |
| 9     | Conversion Rate, Settlement               | CE  | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                     |
| 10    | Conversion Rate, Cardholder Billing       | CE  | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                 |
| 11    | Systems Trace Audit Number (STAN)         | ME  | XE  | ME  | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                 |
| 15    | Date, Settlement                          | ME  | X   | M   | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                 |
| 16    | Date, Conversion                          | CE  | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                     |
| 20    | Primary Account Number (PAN) Country Code | CE  | XE  | CE  | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                     |
| 23    | Card Sequence Number                      | CE  | XE  | CE  | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                     |
| 28    | Amount, Transaction Fee                   | OE  | XE  | •XE | Must be the same value as in the Authorization Advice/0120 message, if present.                                                                                                                                                                 |
| 32    | Acquiring Institution ID Code             | ME  | XE  | ME  | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                 |

General message layout information

Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)

| DE ID | Data Element Name                        | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                    |
|-------|------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 33    | Forwarding Institution ID Code           | CE  | XE  | CE  | Must be the same value as in the Authorization Advice/0120: Acquirer-generated, if present.                                                                                                                                                                                                                                                                 |
| 37    | Retrieval Reference Number               | CE  | XE  | CE  | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                                                                                                                                 |
| 39    | Response Code                            | M   | P   | M   | Response Code for the Authorization Advice Response/0130 message. For values, refer to DE 39.                                                                                                                                                                                                                                                               |
| 41    | Acceptor Terminal ID                     | CE  | XE  | CE  | Must be the same value as in the Authorization Advice/0120 acquirer-generated, if present.                                                                                                                                                                                                                                                                  |
| 44    | Additional Response Data                 | C   | P   | C   | May contain additional response information for certain error conditions in the original Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                                                                         |
| 48    | Additional Data: Private Use             | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                        |
| 49    | Currency Code, Transaction               | ME  | XE  | ME  | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                                                                                                                             |
| 50    | Currency Code, Settlement                | CE  | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                                                                                                                                 |
| 51    | Currency Code, Cardholder Billing        | CE  | X   | C   | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                                                                                                                                                                                             |
| 56    | Payment Account Data                     | C   | X   | C   | <p>When Mastercard is the BIN Controller, issuers are not required to send PAR. Mastercard will insert the PAR in the message before sending to the acquirer, when available.</p> <p>When Mastercard is not the BIN Controller, the issuer may provide the PAR in the response message when available. Mastercard will pass this value to the acquirer.</p> |
| 62    | Intermediate Network Facility (INF) Data | CE  | XE  | CE  | Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.                                                                                                                                                                                                                                                                 |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                          |
|-------|-------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 63    | Network Data                              | ME  | X   | M   | Must be the same value as in the Authorization Advice/0120: acquirer-generated.                                                                                                   |
| 105   | Multi-Use Transaction Identification Data | C   | X   | C   | Contains applicable subelement data.                                                                                                                                              |
| 106   | Fleet Service Data                        | C   | X   | C   | Contains applicable subelement data.                                                                                                                                              |
| 127   | Private Data                              | O   | XE  | CE  | Private data for message originator's use. The Dual Message Authorization System will echo in DE 127 from the original Authorization Advice/0120: acquirer-generated, if present. |

## Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                                                |
|-------|-------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI) | •   | X   | M   | Constant: 0130 (Authorization Advice Response).                                         |
| -     | Bit Map, Primary              | •   | X   | M   | Mandatory                                                                               |
| 1     | Bit Map, Secondary            | •   | X   | C   | Must be the same value as in the Authorization Advice/0120 message, if present.         |
| 2     | Primary Account Number (PAN)  | •   | XE  | ME  | Must be the same value as in the Authorization Advice/0120 message.                     |
| 3     | Processing Code               | •   | XE  | ME  | Must be the same value as in the Authorization Advice/0120 message.                     |
| 4     | Amount, Transaction           | •   | XE  | ME  | Must be the same value as in the Authorization Advice/0120 message.                     |
| 5     | Amount, Settlement            | •   | X   | C   | If enriched by Mastercard, then the acquirer would receive it and opt-in to receive it. |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                |
|-------|-------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6     | Amount, Cardholder Billing                | •   | X   | C   | Must be the same value as in the Authorization Advice/0120 message.<br><br>The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements. |
| 7     | Transmission Date and Time                | •   | XE  | ME  | Must be the same value as in the Authorization Advice/0120 message.                                                                                                                                                     |
| 9     | Conversion Rate, Settlement               | •   | X   | C   | If enriched by Mastercard, then the acquirer would receive it.                                                                                                                                                          |
| 10    | Conversion Rate, Cardholder Billing       | •   | X   | C   | If enriched by Mastercard, then the acquirer would receive it.                                                                                                                                                          |
| 11    | Systems Trace Audit Number (STAN)         | •   | XE  | ME  | Must be the same value as in the Authorization Advice/0120 message.                                                                                                                                                     |
| 15    | Date, Settlement                          | •   | X   | M   | If enriched by Mastercard, then the acquirer would receive it.                                                                                                                                                          |
| 16    | Date, Conversion                          | •   | X   | C   | If enriched by Mastercard, then the acquirer would receive it.                                                                                                                                                          |
| 20    | Primary Account Number (PAN) Country Code | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message, if present.                                                                                                                                         |
| 23    | Card Sequence Number                      | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message, if present.                                                                                                                                         |
| 28    | Amount, Transaction Fee                   | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message, if present.                                                                                                                                         |
| 32    | Acquiring Institution ID Code             | •   | XE  | ME  | Must be the same value as in the Authorization Advice/0120 message.                                                                                                                                                     |
| 33    | Forwarding Institution ID Code            | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message, if present.                                                                                                                                         |
| 37    | Retrieval Reference Number                | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message, if present.                                                                                                                                         |
| 39    | Response Code                             | •   | X   | M   | For more detail, refer to the data element definition for DE 39.                                                                                                                                                        |



| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                |
|-------|-------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------|
| 41    | Acceptor Terminal ID                      | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 44    | Additional Response Data                  | •   | X   | C   | May contain additional response information for certain error conditions in original Authorization Advice/0120 message. |
| 48    | Additional Data: Private Use              | •   | X   | C   | Contains applicable subelement data.                                                                                    |
| 49    | Currency Code, Transaction                |     | XE  | ME  | Must be the same value as in the Authorization Advice/0120 message.                                                     |
| 50    | Currency Code, Settlement                 | •   | X   | C   | If enriched by Mastercard, then the acquirer would receive it.                                                          |
| 51    | Currency Code, Cardholder Billing         | •   | X   | C   | If enriched by Mastercard, then the acquirer would receive it.                                                          |
| 56    | Payment Account Data                      | •   | X   | C   | When Mastercard is the BIN Controller, the PAR value is inserted when available.                                        |
| 62    | Intermediate Network Facility (INF) Data  | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message, if present.                                         |
| 63    | Network Data                              | •   | X   | M   | If enriched by Mastercard, then the acquirer would receive it.                                                          |
| 105   | Multi-Use Transaction Identification Data | •   | X   | C   |                                                                                                                         |
| 127   | Private Data                              | •   | XE  | CE  | Must be the same value as in the Authorization Advice/0120 message.                                                     |

## Authorization Response Acknowledgment/0180

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                |
|-------|-------------------------------|-----|-----|-----|---------------------------------------------------------|
| -     | Message Type Identifier (MTI) | M   | X   | •   | Constant: 0180 (Authorization Response Acknowledgment). |
| -     | Bit Map, Primary              | M   | X   | •   |                                                         |

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                              |
|-------|-----------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------|
| 1     | Bit Map, Secondary                | C   | X   | •   | Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message. |
| 7     | Transmission Date and Time        | ME  | XE  | •   | Must be the same value as in the original Authorization Request Response/0110 message.                                |
| 11    | Systems Trace Audit Number (STAN) | ME  | XE  | •   | Must be the same value as in the original Authorization Request Response/0110 message.                                |
| 39    | Response Code                     | M   | X   | •   | Must be 00 to indicate positive Response Acknowledgment.                                                              |
| 63    | Network Data                      | ME  | XE  | •   | Must be the same value as in the original Authorization Request Response/0110 message.                                |
| 127   | Private Data                      | C   | X   | •   | Private data for message initiator's use.                                                                             |

## Authorization Response Negative Acknowledgment/0190

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                         |
|-------|-----------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | •   | X   | M   | Constant: 0190 (Authorization Response Negative Acknowledgment).                                                                 |
| -     | Bit Map, Primary                  | •   | X   | M   |                                                                                                                                  |
| 1     | Bit Map, Secondary                | •   | X   | C   | Required data element if DE 65 (Bit Map, Extended) through DE 128 are present in the message.                                    |
| 7     | Transmission Date and Time        | •   | XE  | ME  | Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message. |
| 11    | Systems Trace Audit Number (STAN) | •   | XE  | ME  | Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message. |

| DE ID | Data Element Name        | Org | Sys | Dst | Comments                                                                                                                         |
|-------|--------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------|
| 39    | Response Code            | •   | X   | M   | Indicates specific reason for this Negative Response Acknowledgment.                                                             |
| 44    | Additional Response Data | •   | X   | C   | For a listing of values, refer to <a href="#">Data element definitions and details</a> .                                         |
| 63    | Network Data             | •   | XE  | ME  | Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message. |
| 127   | Private Data             | •   | XE  | CE  | Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message. |

## Issuer File Update Request/0302

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                   |
|-------|-----------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | M   | X   | •   | Constant: 0302 (Issuer File Update Request).                                                               |
| -     | Bit Map, Primary                  | M   | X   | •   | Mandatory.                                                                                                 |
| 1     | Bit Map, Secondary                | M   | X   | •   | Mandatory.                                                                                                 |
| 2     | Primary Account Number (PAN)      | C   | X   | •   | May contain PAN information if required for this Issuer File Update Request/0302 message.                  |
| 7     | Transmission Date and Time        | M   | X   | •   | Transaction time stamp; UTC date and time that this transaction was entered into interchange.              |
| 11    | Systems Trace Audit Number (STAN) | M   | X   | •   | Transaction trace number; must be unique value for transaction initiator within each UTC day.              |
| 33    | Forwarding Institution ID Code    | M   | X   | •   | Mastercard-assigned customer ID code. Identifies the institution submitting this file maintenance request. |
| 56    | Payment Account Data              | O   | X   | •   | DE 56 is optional when DE 101 (File Name) is MCC111 (PAN-PAR Mapping File).                                |

| DE ID | Data Element Name       | Org | Sys | Dst | Comments                                                                                                                                                       |
|-------|-------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 63    | Network Data            | •   | X   | •   | Dual Message Authorization System transaction ID code; Mastercard provides this data as a unique identifier for this Issuer File Update Request/ 0302 message. |
| 91    | Issuer File Update Code | M   | X   | •   | Issuer File Update function code.                                                                                                                              |
| 96    | Message Security Code   | M   | X   | •   | Issuer File Update password or security code used to authenticate Issuer File Update permissions.<br><br>Data must be provided in EBCDIC hexadecimal format.   |
| 101   | File Name               | M   | X   | •   | Name of file the issuer is maintaining or accessing.                                                                                                           |
| 120   | Record Data             | C   | X   | •   | Contains the specific Issuer File Update detail record data.                                                                                                   |
| 127   | Private Data            | O   | X   | •   | Private use data element, available for message initiator's optional use.                                                                                      |

## Issuer File Update Request Response/0312

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                        |
|-------|-----------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | •   | M   | M   | Constant: 0312 (Issuer File Update Request Response).                                           |
| -     | Bit Map, Primary                  | •   | X   | M   | Mandatory.                                                                                      |
| 1     | Bit Map, Secondary                | •   | X   | M   | Mandatory.                                                                                      |
| 2     | Primary Account Number (PAN)      | •   | XE  | CE  | Must be the same value as in the original Issuer File Update Request/ 0302 message, if present. |
| 7     | Transmission Date and Time        | •   | XE  | ME  | Must be the same value as in the original Issuer File Update Request/ 0302 message.             |
| 11    | Systems Trace Audit Number (STAN) | •   | XE  | ME  | Must be the same value as in the original Issuer File Update Request/ 0302 message.             |

| DE ID | Data Element Name              | Org | Sys | Dst | Comments                                                                                                                                                |
|-------|--------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 33    | Forwarding Institution ID Code | •   | XE  | ME  | Must be the same value as in the original Issuer File Update Request/0302 message, if present.                                                          |
| 39    | Response Code                  | •   | X   | M   | Indicates disposition of Issuer File Update Request/0302 message.                                                                                       |
| 44    | Additional Response Data       | •   | X   | C   | May contain additional response information, based on DE 39 (Response Code).                                                                            |
| 63    | Network Data                   | •   | X   | M   | Must be the same value as in the original Issuer File Update Request/0302.                                                                              |
| 91    | Issuer File Update Code        | •   | XE  | CE  | Must be the same value as in the original Issuer File Update Request/0302 message.                                                                      |
| 96    | Message Security Code          | •   | XE  | XE  | Must be the same value as in the original Issuer File Update Request/0302 message.                                                                      |
| 101   | File Name                      | •   | ME  | M   | Must be the same value as in the original Issuer File Update Request/0302 message.                                                                      |
| 120   | Record Data                    | •   | X   | C   | This data element is used to return file inquiry and rejects records associated with reject responses. Refer to the data element definition for DE 120. |
| 122   | Additional Record Data         | •   | X   | C   | This data element is used to return additional data resulting from a file inquiry.                                                                      |
| 127   | Private Data                   | •   | XE  | CE  | Must be the same value as in the original Issuer File Update Request/0302 message, if present.                                                          |

## Reversal Request/0400

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                           |
|-------|-------------------------------|-----|-----|-----|------------------------------------|
| -     | Message Type Identifier (MTI) | M   | P   | M   | Constant: 0400 (Reversal Request). |

| DE ID | Data Element Name            | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Bit Map, Primary             | M   | X   | M   | Mandatory.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1     | Bit Map, Secondary           | M   | X   | M   | Mandatory.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2     | Primary Account Number (PAN) | M   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                            |
| 3     | Processing Code              | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                            |
| 4     | Amount, Transaction          | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message. On a partial approval or purchase only approval, must be the same as the Authorization Request Response/0110 message. Must be a value other than all zeros.                                                                                                                                                                                |
| 5     | Amount, Settlement           | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed.                             |
| 6     | Amount, Cardholder Billing   | •   | X   | M   | The Dual Message Authorization System provides this data element. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed and the time the Reversal Request/0400 message is processed by the Dual Message Authorization System. |
| 7     | Transmission Date and Time   | M   | P   | M   | Transaction time stamp; UTC date and time that this transaction was entered into the system.                                                                                                                                                                                                                                                                                                                             |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9     | Conversion Rate, Settlement         | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed and the time the Reversal Request/0400 message is processed. |
| 10    | Conversion Rate, Cardholder Billing | •   | X   | M   | The Dual Message Authorization System provides this data element. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed and the time the Reversal Request/0400 message is processed.                                                                          |
| 11    | Systems Trace Audit Number (STAN)   | M   | P   | M   | Transaction trace number; must be unique value for transaction initiator within each UTC day.                                                                                                                                                                                                                                                                                                                                                            |
| 12    | Time, Local Transaction             | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                                                                                                                                |
| 13    | Date, Local Transaction             | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                                                                                                                                |
| 14    | Date, Expiration                    | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                                                                                                                                |
| 15    | Date, Settlement                    | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                                                                                                                                                                                                                                                                                                                        |
| 16    | Date, Conversion                    | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                                                                                                                                                                                                                                                                                                                        |
| 18    | Merchant Type                       | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                                            |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                        |
|-------|-------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19    | Acquiring Institution Country Code        | •   | X   | C   | The Dual Message Authorization System provides this data element.                                                                                                                                                                               |
| 20    | Primary Account Number (PAN) Country Code | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                       |
| 22    | Point-of-Service (POS) Entry Mode         | M   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                   |
| 23    | Card Sequence Number                      | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                       |
| 28    | Amount, Transaction Fee                   | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                       |
| 32    | Acquiring Institution ID Code             | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                   |
| 33    | Forwarding Institution ID Code            | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                       |
| 37    | Retrieval Reference Number                | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                       |
| 38    | Authorization ID Response                 | C   | P   | C   | Must be the same value as in the original Authorization Request Response/0110 message if present.                                                                                                                                               |
| 39    | Response Code                             | M   | P   | M   | Must contain value 32 for a partial reversal. For a full reversal, must contain value 06, 17, 34, or 68 as applicable, or if no other value applies, may contain the same value as in the original Authorization Request Response/0110 message. |
| 41    | Acceptor Terminal ID                      | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                       |
| 42    | Acceptor ID                               | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                       |



| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 43    | Acceptor Name and Location        | C   | P   | C   | Must be the same value as in the original Authorization Request/0100 message, if present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 48    | Additional Data: Private Use      | M   | X   | M   | <p>Must contain the transaction category code (TCC) from the original Authorization Request/0100.</p> <p>Subelement 63 (Trace ID) must be present.</p> <p>Additional DE 48 subelements may be present in Reversal Request/0400 messages. Refer to the applicable message tables present in the DE 48 subelements found in Data element definitions and details.</p> <p><b>NOTE: The Data element definitions and details chapter specify which message types apply for each DE 48 subelement. The Dual Message Authorization System only forwards DE 48 subelements to the issuer in Reversal Request/0400 messages if the DE 48 subelements have been defined for the Reversal Request/0400 message type.</b></p> |
| 49    | Currency Code, Transaction        | M   | P   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 50    | Currency Code, Settlement         | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 51    | Currency Code, Cardholder Billing | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 54    | Additional Amounts                        | C   | X   | C   | Must be the same value as in the original Authorization Request/0100 message. This data element will only be passed to the issuer when DE 54, subfield 2 (Amount Type) contains the value 40 (Amount Cash Back). The Dual Message Authorization System will forward occurrences of this amount type in both the acquirer's currency and the issuer's currency. If the original authorization was a Purchase with Cash Back request (subfield 2, value 40) and the cash back portion of the request was not approved by the issuer (Response Code 87 [Purchase Only Approval]), DE 54 should not be included in the reversal request. |
| 56    | Payment Account Data                      | •   | X   | C   | When Mastercard is the BIN Controller, the PAR value is inserted when available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 61    | Point-of-Service (POS) Data               | M   | X   | M   | Must be the same value as in the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 62    | Intermediate Network Facility (INF) Data  | O   | P   | O   | If present in an original Authorization Request/0100 message, may be present in subsequent Reversal Request/0400.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 63    | Network Data                              | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 90    | Original Data Elements                    | M   | P   | M   | Contains certain data elements from the original Authorization Request/0100 message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 95    | Replacement Amounts                       | C   | X   | C   | If this data element is provided by an acquirer in a full reversal message, it must contain all zeros to indicate a full reversal.<br><br>If this data element is provided by an acquirer in a partial reversal message, it must contain a value other than all zeros to indicate a partial reversal.                                                                                                                                                                                                                                                                                                                                |
| 105   | Multi-Use Transaction Identification Data | C   | X   | C   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| DE ID | Data Element Name                     | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                   |
|-------|---------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 108   | Additional Transaction Reference Data | O   | P   | C   | Mandatory for originating institution to submit DE 108 on all Mastercard Send Payment Transactions. Mandatory for originating institution to submit DE 108 on all Funding Transactions.<br><br>Optional for Mastercard Merchant Presented QR Transactions. |
| 112   | Additional Data (National Use)        | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                       |
| 122   | Additional Record Data                | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                       |
| 124   | Member-Defined Data                   | O   | X   | C   | Member-defined data element, available for message initiator and recipient optional use.                                                                                                                                                                   |
| 127   | Private Data                          | O   | X   | .   | Private use data element, available for message initiator's optional use.                                                                                                                                                                                  |

## Reversal Request Response/0410

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                                 |
|-------|-------------------------------|-----|-----|-----|--------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI) | M   | P   | M   | Constant: 0410 (Reversal Request Response)                               |
| -     | Bit Map, Primary              | M   | X   | M   | Mandatory                                                                |
| 1     | Bit Map, Secondary            | M   | X   | M   | Mandatory                                                                |
| 2     | Primary Account Number (PAN)  | ME  | XE  | ME  | Must be the same value as in the original Reversal Request/0400 message. |
| 3     | Processing Code               | ME  | XE  | ME  | Must be the same value as in the original Reversal Request/0400 message. |
| 4     | Amount, Transaction           | CE  | XE  | ME  | Must be the same value as in the original Reversal Request/0400 message. |

| DE ID | Data Element Name           | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-----------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Amount, Settlement          | CE  | X   | C   | <p>Must be the same value as in the original Reversal Request/0400 message, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements.</p> <p>This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.</p> |
| 6     | Amount, Cardholder Billing  | CE  | X   | C   | <p>Must be the same value as in the original Reversal Request/0400 message.</p> <p>This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.</p>                                                                                                                                                          |
| 7     | Transmission Date and Time  | ME  | XE  | ME  | <p>Must be the same value as in the original Reversal Request/0400 message.</p>                                                                                                                                                                                                                                                                                                                                                                  |
| 9     | Conversion Rate, Settlement | CE  | X   | C   | <p>Must be the same value as in the original Reversal Request/0400 message, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements.</p> <p>This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.</p> |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                         |
|-------|-------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10    | Conversion Rate, Cardholder Billing       | CE  | X   | C   | Must be the same value as in the original Reversal Request/0400 message.<br><br>This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing. |
| 11    | Systems Trace Audit Number                | ME  | XE  | ME  | Must be the same value as in the original Reversal Request/0400 message.                                                                                                                                                                                                         |
| 15    | Date, Settlement                          | ME  | X   | M   | Must be the same value as in the original Reversal Request/0400 message.                                                                                                                                                                                                         |
| 16    | Date, Conversion                          | CE  | X   | C   | Must be the same value as in the original Reversal Request/0400 message.<br><br>This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing. |
| 20    | Primary Account Number (PAN) Country Code | CE  | XE  | CE  | Must be the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                             |
| 28    | Amount, Transaction Fee                   | OE  | XE  | CE  | Must contain the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                        |
| 32    | Acquiring Institution ID Code             | ME  | XE  | ME  | Must be the same value as in the original Reversal Request/0400 message.                                                                                                                                                                                                         |
| 33    | Forwarding Institution ID Code            | CE  | XE  | CE  | Must be the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                             |
| 37    | Retrieval Reference Number                | CE  | XE  | CE  | Must be the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                             |

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|-----------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39    | Response Code                     | M   | P   | M   | Must be present in the Reversal Request Response/0410 message. Defines the disposition of a previous message or an action taken as a result of receipt of a previous message.                                                                                                                                                                                                                                                             |
| 41    | Acceptor Terminal ID              | CE  | XE  | CE  | Must be the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                                                                                                                                                                                      |
| 44    | Additional Response Data          | C   | X   | C   | Contains additional response data.                                                                                                                                                                                                                                                                                                                                                                                                        |
| 48    | Additional Data: Private Use      | M   | X   | C   | Must contain the transaction category code (TCC) and subelement 63 (Trace ID) from the Reversal Request/0400. Additional DE 48 subelements may be present in Reversal Request Response/0410 messages. Refer to the applicable message tables present in the DE 48 subelements found in Data element definitions and details.                                                                                                              |
| 49    | Currency Code, Transaction        | CE  | XE  | ME  | Must be the same value as in the original Reversal Request/0400 message.                                                                                                                                                                                                                                                                                                                                                                  |
| 50    | Currency Code, Settlement         | CE  | X   | C   | Must be the same value as in the original Reversal Request/0400 message, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements.<br><br>This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing. |
| 51    | Currency Code, Cardholder Billing | CE  | X   | C   | Must be the same value as in the original Reversal Request/0400 message.<br><br>This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.                                                                                                                                                          |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 56    | Payment Account Data                      | C   | X   | C   | When Mastercard is the BIN Controller, issuers are not required to send PAR. Mastercard will insert the PAR in the message before sending to the acquirer, when available.<br><br>When Mastercard is not the BIN Controller, the issuer may provide the PAR in the response message when available. Mastercard will pass this value to the acquirer. |
| 62    | Intermediate Network Facility (INF) Data  | CE  | XE  | CE  | Must be the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                                                                                                 |
| 63    | Network Data                              | ME  | X   | M   | Must be the same value as in the original Reversal Request/0400 message.                                                                                                                                                                                                                                                                             |
| 90    | Original Data Elements                    | ME  | XE  | ME  | Must be the same value as in the original Reversal Request/0400 message.                                                                                                                                                                                                                                                                             |
| 95    | Replacement Amounts                       | CE  | X   | C   | Must be the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                                                                                                 |
| 105   | Multi-Use Transaction Identification Data | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                 |
| 108   | Additional Transaction Reference Data     | C   | X   | C   | Mandatory for originating institution to submit DE 108 on all Mastercard Send Payment Transactions, mandatory for originating institution to submit DE 108 on all Funding Transactions.<br><br>If DE 108 is submitted but the acquirer is not ready to receive, the transaction may fail on the acquirer side.                                       |
| 112   | Additional Data (National Use)            | C   | X   | C   | Contains applicable subelement data.                                                                                                                                                                                                                                                                                                                 |
| 121   | Authorizing Agent ID Code                 | C   | X   | C   | Contains the Mastercard member ID of an alternate Stand-In Authorization, that performed Dual Message System processing on-behalf of an issuer or issuer's primary authorizer.                                                                                                                                                                       |
| 123   | Receipt Free Text                         | O   | X   | C   |                                                                                                                                                                                                                                                                                                                                                      |
| 124   | Member-Defined Data                       | O   | X   | C   | May contain customer-defined data.                                                                                                                                                                                                                                                                                                                   |

| DE ID | Data Element Name | Org | Sys | Dst | Comments                                  |
|-------|-------------------|-----|-----|-----|-------------------------------------------|
| 127   | Private Data      | O   | XE  | CE  | Private data for message initiator's use. |

## Reversal Advice/0420

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                      |
|-------|-------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI) | •   | X   | M   | Constant: 0420 (Reversal Advice).                                                                                                                                                                                                                                                             |
| -     | Bit Map, Primary              | •   | X   | M   | Mandatory.                                                                                                                                                                                                                                                                                    |
| 1     | Bit Map, Secondary            | •   | X   | M   | Mandatory.                                                                                                                                                                                                                                                                                    |
| 2     | Primary Account Number (PAN)  | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.                                                                                                                                                                                        |
| 3     | Processing Code               | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.                                                                                                                                                                                        |
| 4     | Amount, Transaction           | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message (except for a partial approval or purchase only approval). For a partial approval or purchase only approval, must be the same value as the Authorization Request Response/0110 message. |
| 5     | Amount, Settlement            | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.                                                                                                                                                    |
| 6     | Amount, Cardholder Billing    | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                                                                                                                                                             |
| 7     | Transmission Date and Time    | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.                                                                                                                                                                                        |



| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                   |
|-------|-------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| 9     | Conversion Rate, Settlement               | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. |
| 10    | Conversion Rate, Cardholder Billing       | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                          |
| 11    | Systems Trace Audit Number (STAN)         | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.                                     |
| 13    | Date, Local Transaction                   | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.                         |
| 15    | Date, Settlement                          | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                          |
| 16    | Date, Conversion                          | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                          |
| 19    | Acquiring Institution Country Code        | •   | X   | C   | The Dual Message Authorization System provides this data element.                                                                          |
| 20    | Primary Account Number (PAN) Country Code | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.                         |
| 22    | Point-of-Service (POS) Entry Mode         | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.                                     |
| 28    | Amount, Transaction Fee                   | •   | X   | C   | Must contain the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.                    |
| 32    | Acquiring Institution ID Code             | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.                                     |
| 33    | Forwarding Institution ID Code            | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.                         |
| 37    | Retrieval Reference Number                | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.                         |

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-----------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 38    | Authorization ID Response         | •   | X   | C   | Must be the same value as in the original Authorization Request Response/0110 or Reversal Request Response/0410 message, if present.                                                                                                                                                                                                                                                                                                     |
| 39    | Response Code                     | •   | X   | M   | Value 82 (Timeout at issuer), if no Authorization Request Response/0110 or Reversal Request Response/0410 is received. When provided in response to Reversal Request/0400 messages processed by the Dual Message Authorization System on behalf of the issuer, will contain the value from the original Reversal Request/0400 message. Otherwise, will be the same value as in the original Authorization Request Response/0110 message. |
| 41    | Acceptor Terminal ID              | •   | X   | C   | Must be the same value as in the original Reversal Request/0400 message, if present.                                                                                                                                                                                                                                                                                                                                                     |
| 43    | Acceptor Name and Location        | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.                                                                                                                                                                                                                                                                                                                       |
| 48    | Additional Data: Private Use      | •   | X   | C   | Must contain the transaction category code (TCC) from the original Authorization Request/0100 or Reversal Request/0400 message.<br><br><b>NOTE: The Data element definitions and details chapter specify which message types apply for each DE 48 subelement.</b>                                                                                                                                                                        |
| 49    | Currency Code, Transaction        | •   | X   | M   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.                                                                                                                                                                                                                                                                                                                                   |
| 50    | Currency Code, Settlement         | •   | X   | C   | The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.                                                                                                                                                                                                                                                                                               |
| 51    | Currency Code, Cardholder Billing | •   | X   | M   | The Dual Message Authorization System provides this data element.                                                                                                                                                                                                                                                                                                                                                                        |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                             |
|-------|-------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 54    | Additional Amounts                        | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message or the original Authorization Request Response/0110 or Reversal Request Response/0410 message. |
| 56    | Payment Account Data                      | •   | X   | C   | When Mastercard is the BIN Controller, the PAR value is inserted when available.                                                                                                                     |
| 60    | Advice Reason Code                        | •   | X   | M   | Indicates exact purpose of the Reversal Advice/0420 message.                                                                                                                                         |
| 62    | Intermediate Network Facility (INF) Data  | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present. Add system-generated advice for Reversal Advice/0420 message.                     |
| 63    | Network Data                              | •   | X   | M   | The Mastercard Network provides this data element.                                                                                                                                                   |
| 90    | Original Data Elements                    | •   | X   | M   | Must be present when the Dual Message Authorization System reverses a Reversal Request/0400 message.                                                                                                 |
| 95    | Replacement Amounts                       | •   | X   | C   | If this data element is provided by the acquirer in a partial reversal message, it must be included in the subsequent Reversal Advice/0420 message.                                                  |
| 105   | Multi-Use Transaction Identification Data | •   | X   | C   |                                                                                                                                                                                                      |
| 121   | Authorizing Agent ID Code                 | •   | X   | C   | Must be the same value as in the original Authorization Request Response/0110 or Reversal Request Response/0410 message, if present.                                                                 |
| 122   | Additional Record Data                    | •   | X   | C   |                                                                                                                                                                                                      |
| 124   | Member-Defined Data: General Use          | •   | X   | C   |                                                                                                                                                                                                      |
| 127   | Private Data                              | •   | X   | C   | Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.                                                                                   |

## Reversal Advice Response/0430

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                            |
|-------|-------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)             | M   | X   | •   | Constant: 0430 (Reversal Advice Response)                                           |
| -     | Bit Map, Primary                          | M   | X   | •   | Mandatory.                                                                          |
| 1     | Bit Map, Secondary                        | M   | X   | •   | Mandatory.                                                                          |
| 2     | Primary Account Number (PAN)              | ME  | X   | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 3     | Processing Code                           | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 4     | Amount, Transaction                       | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 5     | Amount, Settlement                        | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 6     | Amount, Cardholder Billing                | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 7     | Transmission Date and Time                | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 9     | Conversion Rate, Settlement               | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 10    | Conversion Rate, Cardholder Billing       | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 11    | Systems Trace Audit Number                | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 15    | Date, Settlement                          | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 16    | Date, Conversion                          | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 20    | Primary Account Number (PAN) Country Code | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                            |
|-------|-------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------|
| 28    | Amount, Transaction Fee                   | OE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 32    | Acquiring Institution ID Code             | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 33    | Forwarding Institution ID Code            | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 37    | Retrieval Reference Number                | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 39    | Response Code                             | M   | X   | •   | Must be present in the Reversal Advice Response/0430 message.                       |
| 44    | Additional Response Data                  | C   | X   | •   | Contains additional response data.                                                  |
| 49    | Currency Code, Transaction                | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 50    | Currency Code, Settlement                 | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 51    | Currency Code, Cardholder Billing         | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 56    | Payment Account Data                      | O   | X   | •   | Issuers may optionally provide DE 56.                                               |
| 62    | Intermediate Network Facility (INF) Data  | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 63    | Network Data                              | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 90    | Original Data Elements                    | ME  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message.             |
| 95    | Replacement Amounts                       | CE  | XE  | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |
| 105   | Multi-Use Transaction Identification Data | C   | X   | •   |                                                                                     |
| 127   | Private Data                              | C   | X   | •   | Must be the same value as in the original Reversal Advice/0420 message, if present. |

## Administrative Request/0600

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                        | Org | Sys | Dst | Comments                                                                                               |
|-------|------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)            | M   | P   | M   | Constant: 0600 (Administrative Request).                                                               |
| -     | Bit Map, Primary                         | M   | X   | M   | Mandatory.                                                                                             |
| 1     | Bit Map, Secondary                       | M   | X   | M   | Mandatory.                                                                                             |
| 2     | Primary Account Number (PAN)             | M   | P   | M   | Identifies the receiver of the Administrative Request/0600 message.                                    |
| 7     | Transmission Date and Time               | M   | P   | M   | Transaction time stamp; UTC date and time that this transaction was entered into the interchange.      |
| 11    | Systems Trace Audit Number (STAN)        | M   | P   | M   | Transaction trace number; must be a unique value for the message initiator within each UTC day         |
| 32    | Acquiring Institution ID Code            | M   | P   | M   | Identifies the acquirer of the Administrative Request/0600 message.                                    |
| 33    | Forwarding Institution ID Code           | C   | P   | C   | Identifies the acquirer processor of the Administrative Request/0600 message, if present.              |
| 41    | Acceptor Terminal ID                     | O   | P   | O   | Identifies the merchant terminal ID, if present.                                                       |
| 42    | Acceptor ID                              | O   | P   | O   | Identifies the merchant ID, if present.                                                                |
| 43    | Acceptor Name and Location               | O   | P   | O   | Contains the merchant name and address, if present.                                                    |
| 60    | Advice Reason Code                       | M   | P   | M   | Indicates specific type of message.                                                                    |
| 62    | Intermediate Network Facility (INF) Data | O   | P   | O   | May contain message initiator network trace information.                                               |
| 63    | Network Data                             | .   | X   | M   | Contains the Banknet Reference Number that the Mastercard Network provides as a unique transaction ID. |
| 127   | Private Data                             | O   | X   | .   | May contain message initiator information.                                                             |

## Administrative Request Response/0610

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                   |
|-------|-----------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | M   | P   | M   | Constant: 0610 (Administrative Request Response).                                          |
| -     | Bit Map, Primary                  | M   | X   | M   | Mandatory.                                                                                 |
| 1     | Bit Map, Secondary                | M   | X   | M   | Mandatory.                                                                                 |
| 2     | Primary Account Number (PAN)      | ME  | P   | ME  | Must be the same value as in the original Administrative Request/0600 message.             |
| 7     | Transmission Date and Time        | ME  | XE  | ME  | Must be the same value as in the original Administrative Request/0600 message.             |
| 11    | Systems Trace Audit Number (STAN) | ME  | XE  | ME  | Must be the same value as in the original Administrative Request/0600 message.             |
| 32    | Acquiring Institution ID Code     | ME  | XE  | ME  | Must be the same value as in the original Administrative Request/0600 message.             |
| 33    | Forwarding Institution ID Code    | CE  | XE  | CE  | Must be the same value as in the original Administrative Request/0600 message, if present. |
| 39    | Response Code                     | M   | P   | M   | Indicates the disposition of the original Administrative Request/0600 message.             |
| 41    | Acceptor Terminal ID              | CE  | XE  | CE  | Must be the same value as in the original Administrative Request/0600 message, if present. |
| 42    | Acceptor ID                       | CE  | XE  | CE  | Must be the same value from the original Administrative Request/0600 message, if present.  |
| 43    | Acceptor Name and Location        | CE  | XE  | CE  | Must be the same value from the original Administrative Request/0600 message, if present.  |
| 44    | Additional Response Data          | C   | P   | C   | May contain additional error code information.                                             |

| DE ID | Data Element Name                        | Org | Sys | Dst | Comments                                                                                   |
|-------|------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------|
| 60    | Advice Reason Code                       | ME  | XE  | ME  | Must be the same value from the original Administrative Request/0600 message.              |
| 62    | Intermediate Network Facility (INF) Data | CE  | XE  | CE  | Must be the same value as in the original Administrative Request/0600 message, if present. |
| 63    | Network Data                             | ME  | X   | M   | Must be the same value as in the original Administrative Request/0600 message.             |
| 127   | Private Data                             | .   | XE  | CE  | May contain message initiator information.                                                 |

## Administrative Advice/0620: member-generated

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                  |
|-------|-----------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | M   | P   | M   | Constant: 0620 (Administrative Advice).                                                                                   |
| -     | Bit Map, Primary                  | M   | X   | M   | Mandatory.                                                                                                                |
| 1     | Bit Map, Secondary                | M   | X   | M   | Mandatory.                                                                                                                |
| 7     | Transmission Date and Time        | M   | P   | M   | Transaction time stamp; UTC date and time that this transaction was entered into interchange.                             |
| 11    | Systems Trace Audit Number (STAN) | M   | P   | M   | Transaction trace number; must be unique value for transaction initiator within each UTC day.                             |
| 33    | Forwarding Institution ID Code    | M   | P   | M   | Identifies the customer, institution, or Mastercard Network facility originating this Administrative Advice/0620 message. |
| 48    | Additional Data: Private Use      | C   | P   | C   | This data element is not applicable to this message when used for Administrative textual message transmittal.             |
| 60    | Advice Reason Code                | M   | P   | M   | Must contain value 650 (Administrative textual message transmittal).                                                      |



| DE ID | Data Element Name                        | Org | Sys | Dst | Comments                                                                                                                                                                                                      |
|-------|------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 62    | Intermediate Network Facility (INF) Data | C   | P   | C   | Contains acquiring network trace information that intermediate network facilities (INFs) may require to quickly and accurately route Administrative Advice/0620 messages back to the originating institution. |
| 63    | Network Data                             | •   | X   | M   | Contains the Banknet Reference Number that the Mastercard Network provides as a unique transaction ID.                                                                                                        |
| 100   | Receiving Institution ID Code            | M   | P   | M   | Identifies the customer, institution, or Mastercard Network facility that will receive this Administrative Advice/0620 message.                                                                               |
| 120   | Record Data                              | C   | P   | C   | Contains textual message. May contain name, address, or interoffice routing information; free text format; used as an aid to insure that administration message is forwarded to a specific individual.        |
| 127   | Private Data                             | O   | X   | •   | Private use data element, available for optional use by message initiator.                                                                                                                                    |

## Administrative Advice/0620: system-generated

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                      |
|-------|-----------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | •   | M   | M   | Constant: 0620 (Administrative Advice).                                                       |
| -     | Bit Map, Primary                  | •   | X   | M   | Mandatory.                                                                                    |
| 1     | Bit Map, Secondary                | •   | X   | M   | Mandatory.                                                                                    |
| 7     | Transmission Date and Time        | •   | X   | M   | Transaction time stamp; UTC date and time that this transaction was entered into interchange. |
| 11    | Systems Trace Audit Number (STAN) | •   | X   | M   | Transaction trace number; must be unique value for transaction initiator within each UTC day. |

| DE ID | Data Element Name                        | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 33    | Forwarding Institution ID Code           | •   | X   | M   | Identifies the customer, institution, or Mastercard Network facility originating this Administrative Advice/0620 message.                                                                                                                                                                                                                                                                                                           |
| 42    | Acceptor ID                              | •   | X   | C   | Identifies the acceptor that defines the point of the transaction in both local and interchange environments. DE 42 is used as a merchant ID to uniquely identify the merchant in a POS transaction.                                                                                                                                                                                                                                |
| 48    | Additional Data: Private Use             | •   | X   | C   | If DE 60 (Advice Reason Code) = 600, DE 48 will not be present in the message.                                                                                                                                                                                                                                                                                                                                                      |
| 56    | Payment Account Data                     | •   | X   | C   | When Mastercard is the BIN Controller (as defined by EMVCo) DE 56 will be present and contain the PAR value when one is associated with the PAN. DE 56 is only present when DE 60 (Advice Reason Code) = 0251.                                                                                                                                                                                                                      |
| 60    | Advice Reason Code                       | •   | X   | M   | Indicates the specific reason for the message.                                                                                                                                                                                                                                                                                                                                                                                      |
| 62    | Intermediate Network Facility (INF) Data | •   | X   | C   | <p>If DE 60 (Advice Reason Code) = 600, INF data in the Administrative Advice/0620 message contains MCBN620060000xxx, where xxx is the MIP ID. This data must be returned in an Administrative Advice Response/0630 message.</p> <p>If DE 60 (Advice Reason Code) = 650, INF data in the Administrative Advice/0620 message contains the value RISK, and it must be returned in an Administrative Advice Response/0630 message.</p> |
| 63    | Network Data                             | •   | X   | M   | Contains the Banknet Reference Number that the Mastercard Network provides as a unique transaction ID.                                                                                                                                                                                                                                                                                                                              |
| 100   | Receiving Institution ID Code            | •   | X   | M   | Identifies the customer, institution, or Mastercard Network facility that will receive this Administrative Advice/0620.                                                                                                                                                                                                                                                                                                             |

| DE ID | Data Element Name | Org | Sys | Dst | Comments                                                                                                                                                |
|-------|-------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120   | Record Data       | •   | X   | C   | If DE 60 (Advice Reason Code) = 600, contains rejected message.<br><br>DE 120 will also be present if DE 60 (Advice Reason Code) = 0250, 0251, or 0252. |
| 127   | Private Data      | •   | O   | O   | Private use data element, available for optional use by message initiator.                                                                              |

## Administrative Advice Response/0630

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                        | Org | Sys | Dst | Comments                                                                                  |
|-------|------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)            | M   | M   | •   | Constant: 0630 (Administrative Advice Response).                                          |
| -     | Bit Map, Primary                         | M   | X   | •   | Mandatory.                                                                                |
| 1     | Bit Map, Secondary                       | M   | X   | •   | Mandatory.                                                                                |
| 7     | Transmission Date and Time               | ME  | XE  | •   | Must be the same value from the original Administrative Advice/0620 message.              |
| 11    | Systems Trace Audit Number (STAN)        | ME  | XE  | •   | Must be the same value from the original Administrative Advice/0620 message.              |
| 33    | Forwarding Institution ID                | ME  | XE  | •   | Must be the same value from the original Administrative Advice/0620 message.              |
| 39    | Response Code                            | M   | X   | •   | Indicates the disposition of the original Administrative Advice/0620 message.             |
| 44    | Additional Response Data                 | C   | X   | •   | May contain the additional error code information depending on the value in DE 39.        |
| 62    | Intermediate Network Facility (INF) Data | CE  | XE  | •   | Must be the same value as in the original Administrative Advice/0620 message, if present. |

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                                      |
|-------|-------------------------------|-----|-----|-----|-------------------------------------------------------------------------------|
| 63    | Network Data                  | ME  | XE  | •   | Must be the same value as in the original Administrative Advice/0620 message. |
| 100   | Receiving Institution ID Code | ME  | •   | •   | Must be the same value as in the original Administrative Advice/0620 message. |
| 127   | Private Data                  | O   | X   | •   | Private use data element, available for optional use by message initiator.    |

## Network Management Request/0800: sign-on/sign-off

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                         | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                     |
|-------|-------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)             | M   | M   | •   | Constant: 0800 (Network Management Request).                                                                                                                                                                                                                 |
| -     | Bit Map, Primary                          | M   | X   | •   | Mandatory.                                                                                                                                                                                                                                                   |
| 1     | Bit Map, Secondary                        | M   | X   | •   | Mandatory.                                                                                                                                                                                                                                                   |
| 2     | Primary Account Number (PAN)              | M   | X   | •   | Must contain PAN prefix for sign-on/off by prefix functions; must contain Mastercard Group Sign-on for group sign-on/off functions or both the Mastercard Group Sign-on followed by the card prefix for a prefix sign-on/sign-off within a particular group. |
| 7     | Transmission Date and Time                | M   | X   | •   | Transaction time stamp; contains UTC date and time that this transaction was entered into the system.                                                                                                                                                        |
| 11    | Systems Trace Audit Number (STAN)         | M   | X   | •   | Transaction trace number; must be unique value for transaction indicator within each UTC day.                                                                                                                                                                |
| 20    | Primary Account Number (PAN) Country Code | C   | X   | •   | Country Code is required if DE 2 (Primary Account Number [PAN]) contains a BIN beginning with 59.                                                                                                                                                            |
| 33    | Forwarding Institution ID Code            | M   | X   | •   | Identifies the customer, institution, or Mastercard Network facility originating this Network Management Request/0800 message.                                                                                                                               |

| DE ID | Data Element Name                    | Org | Sys | Dst | Comments                                                                                                                                                                                                                                     |
|-------|--------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 53    | Security-Related Control Information | C   | X   | •   | Issuers performing their own PIN validation and issuers participating in the PIN validation in Stand-In Authorization service must provide the PIN Key Index Number to be used for PIN translation. DE 53 and DE 110 are mutually exclusive. |
| 63    | Network Data                         | •   | X   | •   | Dual Message Authorization System transaction ID code that the Mastercard Network provides as a unique identifier for this transaction.                                                                                                      |
| 70    | Network Management Information Code  | M   | X   | •   | Indicates the specific purpose of this message.                                                                                                                                                                                              |
| 94    | Service Indicator                    | M   | X   | •   | For values applicable to the Network Management/0800 message, refer to <a href="#">Data element definitions and details</a> .                                                                                                                |
| 96    | Message Security Code                | M   | X   | •   | Contains the Mastercard customer password security code, allowing access to the Mastercard Network by a customer or Intermediate Network Facilities (INF) processor. Data must be provided in EBCDIC hexadecimal format.                     |
| 110   | Encryption Data                      | C   | X   | •   | Includes key management encryption data. DE 110 and DE 53 are mutually exclusive.                                                                                                                                                            |
| 127   | Private Data                         | O   | X   | •   | Private use data element, available for optional use by message initiator.                                                                                                                                                                   |

## Network Management Request/0800: network-connection status, member-generated

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                     |
|-------|-------------------------------|-----|-----|-----|----------------------------------------------|
| -     | Message Type Identifier (MTI) | M   | X   | •   | Constant: 0800 (Network Management Request). |
| -     | Bit Map, Primary              | M   | X   | •   | Mandatory.                                   |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                           |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Bit Map, Secondary                  | M   | X   | •   | Mandatory.                                                                                                                         |
| 2     | Primary Account Number (PAN)        | M   | X   | •   | Group Sign-on ID for which the network connection status is being requested.                                                       |
| 7     | Transmission Date and Time          | M   | X   | •   | Transaction time stamp; contains UTC date and time that this transaction was entered into interchange.                             |
| 11    | Systems Trace Audit Number (STAN)   | M   | X   | •   | Transaction trace number; must be unique value for transaction indicator within each UTC day.                                      |
| 33    | Forwarding Institution ID Code      | M   | X   | •   | Identifies the customer, institution, or Mastercard Network facility originating this Network Management Request/0800 message.     |
| 63    | Network Data                        | •   | X   | •   | Mastercard Network transaction ID code that the network provides as a unique identifier for this transaction.                      |
| 70    | Network Management Information Code | M   | X   | •   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message. |

## Network Management Request/0800: network-connection status, system-generated

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name             | Org | Sys | Dst | Comments                                                                                               |
|-------|-------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI) | •   | X   | M   | Constant: 0800 (Network Management Request).                                                           |
| -     | Bit Map, Primary              | •   | X   | M   | Mandatory.                                                                                             |
| 1     | Bit Map, Secondary            | •   | X   | M   | Mandatory.                                                                                             |
| 2     | Primary Account Number (PAN)  | •   | X   | M   | Group Sign-on ID for which the Mastercard Network is requesting the network connection status.         |
| 7     | Transmission Date and Time    | •   | X   | M   | Transaction time stamp; contains UTC date and time that this transaction was entered into interchange. |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                           |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 11    | Systems Trace Audit Number (STAN)   | •   | X   | M   | Transaction trace number; must be unique value for transaction indicator within each UTC day.                                      |
| 33    | Forwarding Institution ID Code      | •   | X   | M   | Identifies the Mastercard Network facility originating this Network Management Request/0800 message.                               |
| 63    | Network Data                        | •   | X   | M   | Mastercard Network transaction ID code that the network provides as a unique identifier for this transaction.                      |
| 70    | Network Management Information Code | •   | X   | M   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message. |

## Network Management Request/0800: host-session activation/deactivation

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                       |
|-------|-----------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | M   | X   | •   | Constant: 0800 (Network Management Request).                                                                                   |
| -     | Bit Map, Primary                  | M   | X   | •   | Mandatory.                                                                                                                     |
| 1     | Bit Map, Secondary                | M   | X   | •   | Mandatory.                                                                                                                     |
| 2     | Primary Account Number (PAN)      | M   | X   | •   | Group Sign-on ID for which the network connection status is being requested.                                                   |
| 7     | Transmission Date and Time        | M   | X   | •   | Transaction time stamp; contains UTC date and time that this transaction was entered into interchange.                         |
| 11    | Systems Trace Audit Number (STAN) | M   | X   | •   | Transaction trace number; must be unique value for transaction indicator within each UTC day.                                  |
| 33    | Forwarding Institution ID Code    | M   | X   | •   | Identifies the customer, institution, or Mastercard Network facility originating this Network Management Request/0800 message. |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                           |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 70    | Network Management Information Code | M   | X   | •   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message. |

## Network Management Request/0800: PEK exchange

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                                       |
|-------|-------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | •   | X   | M   | Constant: 0800 (Network Management Request).                                                                                                                   |
| -     | Bit Map, Primary                    | •   | X   | M   | Mandatory.                                                                                                                                                     |
| 1     | Bit Map, Secondary                  | •   | X   | M   | Mandatory.                                                                                                                                                     |
| 2     | Primary Account Number (PAN)        | •   | X   | M   | Member Group ID of the member receiving the working key exchange; the Mastercard Network uses this to route the message to the appropriate issuer or acquirer. |
| 7     | Transmission Date and Time          | •   | X   | M   | Date and time that this message is transmitted; specified in UTC date and time format.                                                                         |
| 11    | Systems Trace Audit Number (STAN)   | •   | X   | M   | Transaction trace number; must be unique value for all messages associated with a given transaction within each UTC day.                                       |
| 33    | Forwarding Institution ID Code      | •   | X   | M   | Constant: 002202; identifies the Mastercard Network.                                                                                                           |
| 48    | Additional Data—Private Use         | •   | O   | C   | Key Exchange Data Block for variant key change function; the KEK is used to encrypt the PEK.                                                                   |
| 63    | Network Data                        | •   | X   | M   | Mastercard Network provides this data element; which includes a Banknet Reference Number for this transaction.                                                 |
| 70    | Network Management Information Code | •   | X   | M   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.                             |



| DE ID | Data Element Name | Org | Sys | Dst | Comments                                 |
|-------|-------------------|-----|-----|-----|------------------------------------------|
| 110   | Encryption Data   | •   | X   | C   | Includes key management encryption data. |

## Network Management Request/0800: PEK exchange on demand

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                           |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | M   | M   | •   | Constant: 0800 (Network Management Request).                                                                                       |
| -     | Bit Map, Primary                    | M   | X   | •   | Mandatory.                                                                                                                         |
| 1     | Bit Map, Secondary                  | M   | X   | •   | Mandatory.                                                                                                                         |
| 2     | Primary Account Number (PAN)        | M   | X   | •   | Member Group ID of the member requesting the working key exchange.                                                                 |
| 7     | Transmission Date and Time          | M   | X   | •   | Date and time that this message is transmitted; specified in UTC date and time format.                                             |
| 11    | Systems Trace Audit Number (STAN)   | M   | X   | •   | Transaction trace number; must be unique value for all messages associated with a given transaction within each UTC day.           |
| 33    | Forwarding Institution ID Code      | M   | X   | •   | ID of the member originating this message; must be the same value as DE 2 (Primary Account Number [PAN]).                          |
| 63    | Network Data                        | •   | X   | •   | Mastercard Network provides this data element; which includes a Banknet Reference Number for this transaction.                     |
| 70    | Network Management Information Code | M   | X   | •   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message. |
| 127   | Private Data                        | O   | X   | •   | Private use data element, available for optional use by message initiator; this data is not passed to the Mastercard Network.      |

## Network Management Request Response/0810: sign-on/sign-off

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                           |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | •   | X   | M   | Constant: 0810 (Network Management Request sign-on/sign-off).                                                                      |
| -     | Bit Map, Primary                    | •   | X   | M   | Mandatory.                                                                                                                         |
| 1     | Bit Map, Secondary                  | •   | X   | M   | Mandatory.                                                                                                                         |
| 2     | Primary Account Number              | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                 |
| 7     | Transmission Date and Time          | •   | XE  | ME  |                                                                                                                                    |
| 11    | Systems Trace Audit Number (STAN)   | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                 |
| 33    | Forwarding Institution ID Code      | •   | XE  | ME  |                                                                                                                                    |
| 39    | Response Code                       | •   | X   | M   | Indicates disposition of original Network Management Request/0800 message.                                                         |
| 44    | Additional Response Data            | •   | C   | C   | May be present to provide additional information about message error conditions when the value in DE 39 is 30.                     |
| 63    | Network Data                        | •   | X   | M   | Must be the same value as in the original Network Management Request/0800: sign-on/sign-off                                        |
| 70    | Network Management Information Code | •   | XE  | ME  | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message. |
| 127   | Private Data                        | •   | XE  | CE  | Private use data element, available for optional use by message initiator; this data is not passed to the Mastercard Network.      |

## Network Management Request Response/0810: network-connection status, member-generated

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                           |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | M   | M   | •   | Constant: 0810 (Network Management Request Response).                                                                              |
| -     | Bit Map, Primary                    | M   | X   | •   | Mandatory.                                                                                                                         |
| 1     | Bit Map, Secondary                  | M   | X   | •   | Mandatory.                                                                                                                         |
| 2     | Primary Account Number (PAN)        | ME  | XE  | •   | The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.                            |
| 7     | Transmission Date and Time          | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                 |
| 11    | Systems Trace Audit Number (STAN)   | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                 |
| 33    | Forwarding Institution ID Code      | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                 |
| 39    | Response Code                       | M   | X   | •   | Indicates disposition of original Network Management Request/0800 message.                                                         |
| 63    | Network Data                        | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                 |
| 70    | Network Management Information Code | ME  | XE  | •   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message. |

## Network Management Request Response/0810: network-connection status, system-generated

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                      |
|-------|-------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | •   | X   | M   | Constant: 0810 (Network Management Request Response).                                                                                         |
| -     | Bit Map, Primary                    | •   | X   | M   | Mandatory.                                                                                                                                    |
| 1     | Bit Map, Secondary                  | •   | X   | M   | Mandatory.                                                                                                                                    |
| 2     | Primary Account Number (PAN)        | •   | XE  | ME  | The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.                                       |
| 7     | Transmission Date and Time          | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                            |
| 11    | Systems Trace Audit Number (STAN)   | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                            |
| 33    | Forwarding Institution ID Code      | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                            |
| 39    | Response Code                       | •   | X   | M   | Indicates disposition of original Network Management Request/0800 message.                                                                    |
| 44    | Additional Response Data            | •   | X   | C   | May be present to provide additional information about message error conditions when the value in DE 39 (Response Code) is 30 (Format error). |
| 63    | Network Data                        | •   | X   | M   | Must be the same value as in the original Network Management Request/0800 message.                                                            |
| 70    | Network Management Information Code | •   | XE  | ME  | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.            |

## Network Management Request Response/0810: host-session activation/deactivation

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                      |
|-------|-------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | •   | X   | M   | Constant: 0810 (Network Management Request Response).                                                                                         |
| -     | Bit Map, Primary                    | •   | X   | M   | Mandatory.                                                                                                                                    |
| 1     | Bit Map, Secondary                  | •   | X   | M   | Mandatory.                                                                                                                                    |
| 2     | Primary Account Number (PAN)        | •   | XE  | ME  | The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.                                       |
| 7     | Transmission Date and Time          | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                            |
| 11    | Systems Trace Audit Number (STAN)   | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                            |
| 33    | Forwarding Institution ID Code      | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                            |
| 39    | Response Code                       | •   | X   | M   | Indicates disposition of original Network Management Request/0800 message.                                                                    |
| 44    | Additional Response Data            | •   | X   | C   | May be present to provide additional information about message error conditions when the value in DE 39 (Response Code) is 30 (Format error). |
| 70    | Network Management Information Code | •   | XE  | ME  | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.            |

## Network Management Request Response/0810: PEK exchange

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                   |
|-------|-------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | M   | X   | •   | Constant: 0810 (Network Management Request Response).                                                                                      |
| -     | Bit Map, Primary                    | M   | X   | •   | Mandatory.                                                                                                                                 |
| 1     | Bit Map, Secondary                  | M   | X   | •   | Mandatory.                                                                                                                                 |
| 2     | Primary Account Number (PAN)        | ME  | XE  | •   | The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.                                    |
| 7     | Transmission Date and Time          | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 11    | Systems Trace Audit Number (STAN)   | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 33    | Forwarding Institution ID Code      | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 39    | Response Code                       | M   | X   | •   | Indicates disposition of original Network Management Request/0800 message.                                                                 |
| 44    | Additional Response Data            | C   | X   | •   | May be present to provide additional information on message error conditions when the value in DE 39 (Response Code) is 30 (Format error). |
| 48    | Additional Data: Private Use        | C   | X   | •   | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 63    | Network Data                        | ME  | XE  | •   | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 70    | Network Management Information Code | ME  | XE  | •   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.         |

| DE ID | Data Element Name | Org | Sys | Dst | Comments                                                                           |
|-------|-------------------|-----|-----|-----|------------------------------------------------------------------------------------|
| 110   | Additional Data-2 | O   | X   | •   | Must be the same value as in the original Network Management Request/0800 message. |

## Network Management Request Response/0810: PEK exchange on demand

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                 | Org | Sys | Dst | Comments                                                                                                                                   |
|-------|-----------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)     | •   | X   | M   | Constant: 0810 (Network Management Request Response).                                                                                      |
| -     | Bit Map, Primary                  | •   | X   | M   | Mandatory.                                                                                                                                 |
| 1     | Bit Map, Secondary                | •   | X   | M   | Mandatory.                                                                                                                                 |
| 2     | Primary Account Number (PAN)      | •   | XE  | ME  | The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.                                    |
| 7     | Transmission Date and Time        | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 11    | Systems Trace Audit Number (STAN) | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 33    | Forwarding Institution ID Code    | •   | XE  | ME  | Must be the same value as in the original Network Management Request/0800 message.                                                         |
| 39    | Response Code                     | •   | X   | M   | Indicates disposition of original Network Management Request/0800 message.                                                                 |
| 44    | Additional Response Data          | •   | X   | C   | May be present to provide additional information on message error conditions when the value in DE 39 (Response Code) is 30 (Format error). |
| 63    | Network Data                      | •   | X   | M   | Must be the same value as in the original Network Management Request/0800 message.                                                         |

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                           |
|-------|-------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 70    | Network Management Information Code | •   | XE  | ME  | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message. |

## Network Management Advice/0820: PEK exchange

Following is the list of the data elements applicable to this message.

### Message data element layout

| DE ID | Data Element Name                   | Org | Sys | Dst | Comments                                                                                                                                                                 |
|-------|-------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -     | Message Type Identifier (MTI)       | •   | X   | M   | Constant: 0820 (Network Management Advice).                                                                                                                              |
| -     | Bit Map, Primary                    | •   | X   | M   | Mandatory.                                                                                                                                                               |
| 1     | Bit Map, Secondary                  | •   | X   | M   | Mandatory.                                                                                                                                                               |
| 2     | Primary Account Number (PAN)        | •   | X   | M   | Member Group ID of the member receiving this working key exchange; the Mastercard Network uses this data to route the transaction to the appropriate issuer or acquirer. |
| 7     | Transmission Date and Time          | •   | X   | M   | Date and time that this message is transmitted; specified in UTC date and time format.                                                                                   |
| 11    | Systems Trace Audit Number (STAN)   | •   | X   | M   | Transaction trace number; must be unique value for all messages associated with a given transaction within each UTC day.                                                 |
| 33    | Forwarding Institution ID Code      | •   | X   | M   | Constant: 002202; identifies the Dual Message Authorization System.                                                                                                      |
| 48    | Additional Data: Private Use        | •   | X   | C   | Key Exchange Data Block for key exchange function; the communication key is used to encrypt the working key.                                                             |
| 63    | Network Data                        | •   | X   | M   | Mastercard Network provides this data element; which includes a Banknet Reference Number for this transaction.                                                           |
| 70    | Network Management Information Code | •   | X   | M   | Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.                                       |



| DE ID | Data Element Name | Org | Sys | Dst | Comments                                 |
|-------|-------------------|-----|-----|-----|------------------------------------------|
| 110   | Encryption Data   | •   | X   | C   | Includes key management encryption data. |

## Chapter 5 Data element definitions and details

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| Subfield 08 (Currency Code).....                                        | 1105 |
| Subfield 09 (Installment Interest Rate).....                            | 1105 |
| Subfield 10 (Installment Payment Entity).....                           | 1106 |

|                                                                       |      |
|-----------------------------------------------------------------------|------|
| Subfield 11 (Installment Interest Charged).....                       | 1106 |
| Subfield 12 (Member-Defined Data).....                                | 1107 |
| Japan: Payment Transactions.....                                      | 1108 |
| Subelement 030 (Japan Domestic POS Data).....                         | 1108 |
| Subelement 031 (Japan Domestic Response Code).....                    | 1111 |
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| Subfield 1 (Payment Option Code).....                                 | 1113 |
| Subfield 2 (Bonuses Per Year).....                                    | 1114 |
| Subfield 3 (First Bonus Month).....                                   | 1115 |
| Subfield 4 (First Bonus Amount).....                                  | 1115 |
| Subfield 5 (Second Bonus Month).....                                  | 1116 |
| Subfield 6 (Second Bonus Amount).....                                 | 1117 |
| Subfield 7 (Total Number of Installments).....                        | 1117 |
| Subfield 8 (First Installment Month).....                             | 1118 |
| Mexcta: Payment Transactions.....                                     | 1119 |
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| Subelement 005 (Issuing Bank Name (AKA Doing Business As [DBA]))..... | 1119 |
| Subelement 006 (Financial Institution ID (FIID)).....                 | 1120 |
| Subelement 007 (Installment Payment Data).....                        | 1121 |
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| Subfield 01 (Primary Cardholder Identifier).....                      | 1124 |
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| Paraguay: Payment Transactions.....                                   | 1126 |
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| Subelement 002 (Installment Payment Response Data).....               | 1128 |
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| Percta: Payment Transactions.....                                     | 1130 |
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| Subelement 008 (Installment Payment Response Data).....               | 1132 |
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| Subelement 041 (Poland VAT Data).....                                 | 1132 |
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| Subelement 017 (ATM Domestic Fee).....                                | 1134 |
| Subfield 1 (ATM Service Fee).....                                     | 1135 |
| Subfield 2 (ATM Disloyalty Fee).....                                  | 1135 |

|                                                                         |      |
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| Subfield 3 (Credit Card Fee).....                                       | 1136 |
| Taiwan: National Data.....                                              | 1136 |
| Subelement 038 (Taiwan National Data).....                              | 1136 |
| United Kingdom: Debt Repayment Transactions.....                        | 1137 |
| Subelement 033 (UK Recipient Details).....                              | 1137 |
| Subfield 1 (Recipient Last Name).....                                   | 1138 |
| Subfield 2 (Recipient Postal Code).....                                 | 1139 |
| Subfield 3 (Recipient Date of Birth).....                               | 1139 |
| Subfield 4 (Recipient Account Number).....                              | 1140 |
| DE 113 (Reserved for National Use).....                                 | 1140 |
| DE 114 (Reserved for National Use).....                                 | 1140 |
| DE 115 (Reserved for National Use).....                                 | 1140 |
| DE 116 (Mastercard Internal Use Data 2).....                            | 1141 |
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| MCC 103: Error Codes.....                                               | 1174 |
| MCC 104: Error Codes.....                                               | 1175 |



|                                                                 |      |
|-----------------------------------------------------------------|------|
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| MCC 111: Error Codes.....                                       | 1182 |
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| Subfield 002 (Acceptor Country Subdivision Code).....           | 1191 |
| Subfield 003 (Acceptor URL address).....                        | 1191 |
| Subfield 004 (Acceptor Customer Service Phone Number).....      | 1192 |
| Subfield 005 (Acceptor Phone Number).....                       | 1193 |
| Subfield 006 (Acceptor Additional Contact Information).....     | 1194 |
| Subfield 007 (Acceptor Tax ID).....                             | 1194 |
| Subfield 008 (Partner ID Code).....                             | 1195 |
| Subfield 009 (Service Location City Name).....                  | 1196 |
| Subfield 010 (Service Location Country Subdivision Code).....   | 1197 |
| Subfield 011 (Service Location Country Code).....               | 1197 |
| Subfield 012 (Service Location Postal Code).....                | 1198 |
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| DE 124 (Member-Defined Data: General Use).....                  | 1206 |
| DE 124 (Member-Defined Data: Prepaid Voucher).....              | 1208 |
| Subfield 1 (Merchant Direct Settlement Indicator).....          | 1208 |
| DE 124 (Member-Defined Data: Mastercard Send).....              | 1209 |
| DE 124 (Member-Defined Data: Brazil Maestro Only).....          | 1210 |
| Subfield 1 (Unique Reference Number).....                       | 1210 |
| Subfield 2 (Sender/Payer/User ID).....                          | 1210 |
| Subfield 3 (Sender/Payer Address).....                          | 1211 |
| Subfield 4 (Additional Sender Information).....                 | 1212 |
| Subfield 6 (Discretionary Message on Sales Slip Supported)..... | 1212 |
| Subfield 7 (Discretionary Message on Sales Slip Code).....      | 1213 |
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| Subfield 9 (Phoneshop [Phone Company ID]).....                  | 1214 |
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|                                                                  |      |
|------------------------------------------------------------------|------|
| Subfield 11 (Phoneshop [Message Security Code]).....             | 1215 |
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| DE 124 (Member-Defined Data [Colombia Domestic Use Only]).....   | 1217 |
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| Subfield 2 (Tax [IVA]).....                                      | 1218 |
| Subfield 3 (Tax Amount Base).....                                | 1218 |
| Subfield 4 (Retailer Data).....                                  | 1219 |
| Subfield 5 (Terminal Acquirer Data).....                         | 1219 |
| Subfield 6 (Acquirer Original Processing Code).....              | 1220 |
| Subfield 7 (Bill Payment/Top up Data).....                       | 1220 |
| Subfield 8 (Local POS Data).....                                 | 1221 |
| Subfield 9 (Local Response Codes).....                           | 1222 |
| Subfield 10 (Original Transaction Data).....                     | 1222 |
| Subfield 11 (IAC Tax Amount).....                                | 1223 |
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| DE 126 (Private Data).....                                       | 1226 |
| DE 127 (Private Data).....                                       | 1227 |
| DE 128 (Message Authentication Code).....                        | 1229 |

## Data element layout

This section contains the data element structure for describing data elements. This section omits values and application note information from the data element attributes of those data elements not currently used in the Mastercard Network messages.

### Attributes

| Attribute           | Description                                              |
|---------------------|----------------------------------------------------------|
| Data Representation | Annotation and data length (fixed or variable in length) |
| Data Length         | 2 or 3 positions, if variable in length                  |
| Data Field          | Contents of subelement, subfields, or N/A                |
| Subfields           | Indicates the number of subelements, subfields, or N/A   |
| Justification       | Left, Right, or N/A                                      |

Following is the usage of DE xx (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                          | Org | Sys | Dst |
|----------------------------------------------------------------------------------|-----|-----|-----|
| Applicable message types                                                         | .   | .   | .   |
| Refer to <a href="#">General message layout information</a> for the usage codes. |     |     |     |

## Values

Valid values and name of each value listed.

## Application notes

Specific application notes, conditions, and cross-edits where applicable.

## Subelement layout

This is the subelement structure for describing data element subelements.

### Attributes

| Attribute              | Description                                              |
|------------------------|----------------------------------------------------------|
| Subelement ID          | Subelement identifier                                    |
| Subelement Data Length | 2 or 3 positions, if variable in length                  |
| Data Representation    | Annotation and data length (fixed or variable in length) |
| Data Field             | Contents of subfields                                    |
| Subfields              | Indicates the number of subfields                        |
| Justification          | See subfields (may be subfield-specific)                 |

Following is the usage of subelement XX (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                          | Org | Sys | Dst |
|------------------------------------------------------------------|-----|-----|-----|
| Applicable message types                                         | .   | .   | .   |
| Refer to General message layout information for the usage codes. |     |     |     |

### Values

The list of subelement values or see subfields.

## Subfield layout

This is the subfield structure for describing data element and subelement subfields. This section omits Subfield ID and Subfield Length from the subfield attributes when not including subfield ID and length in the data.

### Attributes

| Attribute            | Description                                                                         |
|----------------------|-------------------------------------------------------------------------------------|
| Subfield ID          | Subfield identifier (omitted if not part of the data)                               |
| Subfield Data Length | 2 or 3 positions, if variable in length                                             |
| Data Representation  | Annotation and data length (fixed or variable in length)                            |
| Data Field           | Contents of positions                                                               |
| Justification        | Left, Right, or N/A, when not using justification to describe subelement subfields) |

### Values

Valid values and name of each value listed.

## Position layout

This is the position structure for describing subelement and subfield positions.

### Position layout

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| Data Representation | Annotation and data length (fixed or variable in length) |
| Data Field          | Description of data                                      |
| Values              | List of values                                           |

## Primary Bit Map

The Primary Bit Map must always be present in a message. The most frequently used data elements are indexed from [DE 1 \(Bit Map, Secondary\)](#) through [DE 64 \(Message Authentication Code \[MAC\]\)](#). Infrequently used data elements are indexed from the [DE 66 \(Settlement Code\)](#)

through [DE 128 \(Message Authentication Code \[MAC\]\)](#) message only when data elements in the range [DE 66 \(Settlement Code\)](#) through [DE 128 \(Message Authentication Code \[MAC\]\)](#) are present in the message.

### Attributes

| Attribute           | Description                             |
|---------------------|-----------------------------------------|
| Data Representation | b-64                                    |
| Data Length         | N/A                                     |
| Data Field          | Fixed length 8 bytes (for each bit map) |
| Subfields           | N/A                                     |
| Justification       | N/A                                     |

**NOTE: If both bit maps are present, the total length of the bit map data element is 16 bytes.**

### Usage

Following is the usage of the Primary Bit Map (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | M   | X   | M   |
| Authorization Request Response/0110                                                             | M   | X   | M   |
| Authorization Advice/0120: acquirer-generated                                                   | M   | X   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | M   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | M   | X   | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | X   | M   |
| Authorization Response Acknowledgment/0180                                                      | M   | X   | •   |
| Authorization Response Negative Acknowledgment/0190                                             | •   | X   | M   |
| Issuer File Update Request/0302                                                                 | M   | X   | •   |

| Message                                                                               | Org | Sys | Dst |
|---------------------------------------------------------------------------------------|-----|-----|-----|
| Issuer File Update Request Response/0312                                              | .   | X   | M   |
| Reversal Request/0400                                                                 | M   | X   | M   |
| Reversal Request Response/0410                                                        | M   | X   | M   |
| Reversal Advice/0420                                                                  | .   | X   | M   |
| Reversal Advice Response/0430                                                         | M   | X   | .   |
| Administrative Request/0600                                                           | M   | X   | M   |
| Administrative Request Response/0610                                                  | M   | X   | M   |
| Administrative Advice/0620: member-generated                                          | M   | X   | M   |
| Administrative Advice/0620: system-generated                                          | .   | X   | M   |
| Administrative Advice Response/0630                                                   | M   | X   | .   |
| Network Management Request/0800: sign-on/sign-off                                     | M   | X   | .   |
| Network Management Request/0800: network connection status, member-generated          | M   | X   | .   |
| Network Management Request/0800: network connection status, system-generated          | .   | X   | M   |
| Network Management Request/0800: host session activation/deactivation                 | M   | X   | .   |
| Network Management Request/0800: PEK exchange                                         | .   | X   | M   |
| Network Management Request/0800: PEK exchange, on demand                              | M   | X   | .   |
| Network Management Request Response/0810: host session activation/deactivation        | .   | X   | M   |
| Network Management Request Response/0810: network connection status, member-generated | M   | X   | .   |
| Network Management Request Response/0810: network connection status, system-generated | .   | X   | M   |
| Network Management Request Response/0810: PEK exchange                                | M   | X   | .   |
| Network Management Request Response/0810: PEK exchange-on demand                      | .   | X   | M   |
| Network Management Request Response/0810: sign-on/sign-off                            | .   | X   | M   |
| Network Management Advice/0820: PEK exchange                                          | .   | X   | M   |

## Values

Bits corresponding to mandatory data elements for a specific MTI must have a value of 1 to indicate the presence of the data element in the message. Otherwise, the Dual Message

Authorization System rejects the message and returns it to the message originator using an Administrative Advice/0620, with the reject reason indicated in [DE 60 \(Advice Reason Code\)](#).

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

## DE 1 (Bit Map, Secondary)

[DE 1 \(Bit Map, Secondary\)](#) is a series of 64 bits that identifies the presence (1) or absence (0) of each data element in the second segment of a message. This would include [DE 65 \(Bit Map, Extended\)](#) through [DE 128 \(Message Authentication Code \[MAC\]\)](#).

### Attributes

| Attribute           | Description                                               |
|---------------------|-----------------------------------------------------------|
| Data Representation | b-8                                                       |
| Data Length         | N/A                                                       |
| Data Field          | Indicates the presence or absence of data elements 65–128 |
| Subfields           | N/A                                                       |
| Justification       | N/A                                                       |

### Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | X   | C   |
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | C   | X   | C   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | C   | X   | •   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | X   | C   |



| Message                                                                                  | Org | Sys | Dst |
|------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Response Acknowledgment/0180                                               | C   | X   | •   |
| Authorization Response Negative Acknowledgment/0190                                      | •   | X   | C   |
| Issuer File Update Request/0302                                                          | M   | X   | •   |
| Issuer File Update Request Response/0312                                                 | •   | X   | M   |
| Reversal Request/0400                                                                    | M   | X   | M   |
| Reversal Request Response/0410                                                           | M   | X   | M   |
| Reversal Advice/0420                                                                     | •   | X   | M   |
| Reversal Advice Response/0430                                                            | M   | X   | •   |
| Administrative Request/0600                                                              | M   | X   | M   |
| Administrative Request Response/0610                                                     | M   | X   | M   |
| Administrative Advice/0620: system-generated                                             | •   | X   | M   |
| Administrative Advice/0620: member-generated                                             | M   | X   | M   |
| Administrative Advice Response/0630                                                      | M   | X   | •   |
| Network Management Request/0800: sign-on/sign-off                                        | M   | X   | •   |
| Network Management Request/0800: host session activation/<br>deactivation                | M   | X   | •   |
| Network Management Request/0800: PEK exchange                                            | •   | X   | M   |
| Network Management Request/0800: PEK exchange, on demand                                 | M   | X   | •   |
| Network Management Request/0800: network connection status,<br>member-generated          | M   | X   | •   |
| Network Management Request/0800: network connection status,<br>system-generated          | •   | X   | M   |
| Network Management Request Response/0810: host session activation/<br>deactivation       | •   | X   | M   |
| Network Management Request Response/0810: network connection<br>status, member-generated | M   | X   | •   |
| Network Management Request Response/0810: network connection<br>status, system-generated | •   | X   | M   |
| Network Management Request Response/0810: PEK exchange                                   | M   | X   | •   |
| Network Management Request Response/0810: PEK exchange, on<br>demand                     | •   | X   | M   |
| Network Management Request Response/0810: sign-on/sign-off                               | •   | X   | M   |
| Network Management Advice/0820: PEK exchange                                             | •   | X   | M   |

## Values

Bits corresponding to mandatory data elements for a specific MTI must have a value of 1 to indicate the presence of the data element in the message. Otherwise, the Dual Message Authorization System rejects the message and returns it to the message initiator using an Administrative Advice/0620, with the reject reason indicated in [DE 60 \(Advice Reason Code\)](#).

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

# DE 2 (Primary Account Number [PAN])

[DE 2 \(Primary Account Number \[PAN\]\)](#) is a series of digits used to identify a customer account or relationship.

## Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n...19; LLVAR              |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–19 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

## Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

## Applicable messages

| Message                                                                                      | Org | Sys | Dst |
|----------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                   | M   | X   | M   |
| Authorization Request Response/0110                                                          | ME  | XE  | ME  |
| Authorization Advice/0120: acquirer-generated                                                | M   | X   | M   |
| Authorization Advice/0120: system-generated                                                  | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120) | ME  | XE  | •   |

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | XE  | ME  |
| Issuer File Update Request/0302                                                                 | C   | X   | •   |
| Issuer File Update Request Response/0312                                                        | •   | XE  | CE  |
| Reversal Request/0400                                                                           | M   | X   | M   |
| Reversal Request Response/0410                                                                  | ME  | XE  | ME  |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | X   | •   |
| Administrative Request/0600                                                                     | M   | P   | M   |
| Administrative Request Response/0610                                                            | ME  | P   | ME  |
| Network Management Request/0800: sign-on/sign-off                                               | M   | X   | •   |
| Network Management Request/0800: network connection status, member-generated                    | M   | X   | •   |
| Network Management Request/0800: network connection status, system-generated                    | •   | X   | M   |
| Network Management Request/0800: PEK exchange                                                   | •   | X   | M   |
| Network Management Request/0800: PEK exchange, on demand                                        | M   | X   | •   |
| Network Management Request/0800: host session activation/deactivation                           | M   | X   | •   |
| Network Management Request Response/0810: host session activation/deactivation                  | •   | XE  | ME  |
| Network Management Request Response/0810: network connection status, member-generated           | ME  | XE  | •   |
| Network Management Request Response/0810: network connection status, system-generated           | •   | XE  | ME  |
| Network Management Request Response/0810: PEK exchange                                          | ME  | XE  | •   |
| Network Management Request Response/0810: PEK exchange on demand                                | •   | XE  | ME  |
| Network Management Request Response/0810: sign-on/sign-off                                      | •   | XE  | ME  |
| Network Management Advice/0820: PEK exchange                                                    | •   | X   | M   |

## Values

Mastercard customer ID numbers, when used in this data element, must be valid values that Mastercard assigns.

DE 2 must be at least 11 digits and contain the variable-length prefix assigned to the specific Issuer ID Number (IIN) for Administrative Request/0600 messages.

Network Management Request/0800 messages may contain the Mastercard Group Sign-on ID only, the card prefix only, or both the Mastercard Group Sign-on ID followed by the card prefix in DE 2.

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

## About Primary Account Number

[DE 2 \(Primary Account Number \[PAN\]\)](#) is a series of digits used to identify a customer account or relationship.

### Primary Account Number

This data element is used for all numeric PANs up to 19 digits long when PAN is in Authorization/01xx messages, Reversal Advice/04xx messages, and Administrative Advice/06xx messages (when required). [DE 2](#) may contain a subset of this data within Network Management/08xx messages.

This data element data consists of three primary components:

- Issuing Institution Information (IIN)
- Individual Account ID Number
- PAN check digit

When [DE 2](#) contains a 59 BIN, [DE 20 \(Primary Account Number Country Code\)](#) also must be included in the message to uniquely identify the PAN. Note that 59 numbers are not guaranteed to be unique without an accompanying Country Code.

Specific requirements for PAN composition are described in the ISO 7812 and 7813 specifications. All PANs used in Dual Message Authorization System messages must conform to the ISO PAN encoding requirements, as specified in those specifications.

The Individual Account ID Number encoded or embossed on a card may be a cardholder ID number or a "master account number" related to one or more of the cardholder's accounts.

A card issuer may return the actual number of the cardholder account(s) affected by a transaction in an appropriate response message by using [DE 102 \(Account ID-1\)](#), [DE 103 \(Account ID-2\)](#), or both. However, the PAN used in the original Request message must then remain in [DE 2](#) for all subsequent messages related to the original request (such as Responses, Advices, Reversals, Chargebacks, Retrieval Requests, and Retrieval Fulfillments).

### Issuing Institution Information

PAN may contain Issuing Institution Information (IIN) when required in Authorization/01xx, Issuer File Update/03xx, and Network Management/08xx messages. IIN may consist of either a Mastercard customer ID number, a variable-length issuer card prefix sequence, or a Mastercard

assigned Group Sign-on ID as required by a specific message. Where card prefix information is required, the Dual Message Authorization System accommodates variable-length prefix sequences from one to 11 digits.

Where a Group Sign-on ID is required, the Dual Message Authorization System accommodates the five-digit Member Group ID, which Mastercard uses to identify a grouping of issuer account ranges that use the same transaction criteria for routing to the same destination route. More than one Group Sign-on ID value may be defined for an account range if an issuer chooses to route authorization traffic to different destinations according to transaction criteria.

## DE 3 (Processing Code)

*DE 3 (Processing Code)* describes the effect of a transaction on the customer account and the type of accounts affected.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-6                       |
| Data Length         | N/A                       |
| Data Field          | Contents of subfields 1–3 |
| Subfields           | 3                         |
| Justification       | See subfields             |

### Usage

Following is the usage of *DE 3* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                      | Org | Sys | Dst |
|----------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                   | M   | P   | M   |
| Authorization Request Response/0110                                                          | ME  | XE  | ME  |
| Authorization Advice/0120: acquirer-generated                                                | M   | P   | M   |
| Authorization Advice/0120: system-generated                                                  | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120) | ME  | XE  | •   |

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | XE  | ME  |
| Reversal Request/0400                                                                           | M   | P   | M   |
| Reversal Request Response/0410                                                                  | ME  | XE  | ME  |
| Reversal Advice/0420                                                                            | .   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | .   |

## Values

See subfields.

## Application notes

Acquirers and prepaid card issuers in the U.S. region must support POS balance inquiry transaction type and functionality for all prepaid Mastercard credit and Debit Mastercard card account ranges.

The following cross edits occur.

### Authorization Request/0100 Messages edits

| IF...                                 | THEN...                                                         |
|---------------------------------------|-----------------------------------------------------------------|
| DE 48 (Transaction Category Code) = C | the Cardholder Transaction Type Code must be 01, 17, or 30.     |
| DE 48 (Transaction Category Code) = P | the Cardholder Transaction Type Code must be 28.                |
| DE 48 (Transaction Category Code) = Z | the Cardholder Transaction Type Code must be 01, 30, 91, or 92. |

### Reversal Request/0400 Messages edits

| IF...                                        | THEN...                                                         |
|----------------------------------------------|-----------------------------------------------------------------|
| <i>DE 48 (Transaction Category Code) = C</i> | the Cardholder Transaction Type Code must be 01, 17, or 30.     |
| <i>DE 48 (Transaction Category Code) = P</i> | the Cardholder Transaction Type Code must be 28.                |
| <i>DE 48 (Transaction Category Code) = Z</i> | the Cardholder Transaction Type Code must be 01, 30, 91, or 92. |

### Authorization Advice/0120: acquirer-generated messages edits

| IF...                                                                                                                                                                                                                                                                                                                 | THEN...                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">DE 3</a>, subfield 1 contains one of the following values:</p> <p>01 = Withdrawal</p> <p>28 = Payment Transaction</p> <p>30 = Balance Inquiry</p> <p>40 = Account Transfer</p> <p>91 = PIN Unblock</p> <p>92 = PIN Change</p> <p>or</p> <p>00 = Purchase and DE 48 (Transaction Category Code) = Z</p> | <p>the Dual Message Authorization System will return to the acquirer an Authorization Advice Response/0130 messages where:</p> <p>DE 39 = 30</p> <p>DE 44 = 003</p> |

### For Account Status Inquiry Service transactions

| WHEN...                                                                                                                                                                                                                                                                                                                              | THEN the Dual Message Authorization System...                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the Authorization Request/0100 message contains <a href="#">DE 61 (Point-of-Service [POS] Data)</a>, subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI])</p> <p>and</p> <p><a href="#">DE 3 (Processing Code)</a> contains a value other than 00 (Purchase), 01 (Withdrawal), or 28 (Payment)</p> | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li><a href="#">DE 39 (Response Code)</a> = 30</li> <li><a href="#">DE 44 (Additional Response Data)</a> = 003</li> </ul> |

## Subfield 1 (Cardholder Transaction Type Code)

[DE 3](#), subfield 1 (Cardholder Transaction Type Code) describes the specific transaction type.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | n-2         |

| Attribute     | Description               |
|---------------|---------------------------|
| Data Field    | Contents of positions 1–2 |
| Justification | N/A                       |

## Values

### Cardholder Account Debits

| Value | Description                | MC | NP | VI | TE | MS | CI |
|-------|----------------------------|----|----|----|----|----|----|
| 00    | Purchase                   | √  | √  | √  | √  | √  | √  |
| 01    | Withdrawal                 | √  | √  | √  | √  | √  | √  |
| 02    | Debit Adjustment           |    | √  |    |    |    |    |
| 09    | Purchase with Cash Back    | √  | √  |    |    | √  |    |
| 10    | Visa Only. Account Funding |    |    | √  |    |    |    |
| 17    | Cash Disbursement          | √  | √  | √  | √  |    |    |
| 18    | Scrip Issue                |    | √  |    |    |    |    |

### Cardholder Account Credits

| Value | Description            | MC | NP | VI | TE | MS | CI |
|-------|------------------------|----|----|----|----|----|----|
| 20    | Purchase Return/Refund | √  | √  | √  |    | √  |    |
| 21    | Deposit                |    | √  |    |    |    |    |
| 22    | Credit Adjustment      |    | √  |    |    |    |    |
| 28    | Payment Transaction    | √  | √  | √  |    | √  | √  |

### Cardholder Account Inquiries

| Value | Description     | MC | NP | VI | TE | MS | CI |
|-------|-----------------|----|----|----|----|----|----|
| 30    | Balance Inquiry | √  | √  | √  |    | √  | √  |



### Cardholder Account Transfers

| Value | Description      | MC | NP | VI | TE | MS | CI |
|-------|------------------|----|----|----|----|----|----|
| 40    | Account Transfer | √  | √  |    |    |    |    |

### Reserved Values

| Value | Description             | MC | NP | VI | TE | MS | CI |
|-------|-------------------------|----|----|----|----|----|----|
| 90    | Reserved for Future Use | √  |    |    |    |    |    |

### PIN Management Transactions

| Value | Description | MC | NP | VI | TE | MS | CI |
|-------|-------------|----|----|----|----|----|----|
| 91    | PIN Unblock | √  | √  |    |    | √  | √  |
| 92    | PIN Change  | √  | √  |    |    | √  | √  |

### Application notes

Value 20 is allowed for MDES related Authorization Request/0100 messages even if the account status is suspended or delete pending.

Value 20 is allowed for Visa gateway transactions to support the authorization of returns or refunds.

Value 20 is allowed for Mastercard and Debit Mastercard transactions and must be supported by Mastercard and Debit Mastercard issuers, to support the authorization of returns or refunds.

Effective with Release 20.Q2, value 20 must be supported by acquirers of Mastercard and Debit Mastercard transactions (support is optional for merchants).

Mastercard requires that issuers receive and respond to refund transactions involving a Mastercard, Debit Mastercard, or Maestro card.

Values 21–24 are not currently supported for use on the Dual Message Authorization System.

Value 30 is only applicable to eligible issuers.

Values 91 and 92 are only applicable to eligible acquirers and issuers.

## Subfield 2 (Cardholder From Account Type Code)

[DE 3](#), subfield 2 (Cardholder From Account Type Code) describes the cardholder account type affected for cardholder account debits and inquiries and the from account type for cardholder account transfer transactions.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3–4 |
| Justification       | N/A                       |

### Values

Subfield 2 must be one of the following values.

### Cardholder From Account type code

| Value | Description                                       | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------|----|----|----|----|----|----|
| 00    | Default Account (not specified or not applicable) | √  | √  | √  | √  | √  | √  |
| 10    | Savings Account                                   | √  | √  | √  |    | √  | √  |
| 20    | Checking Account                                  | √  | √  | √  |    | √  | √  |
| 30    | Credit Card Account                               | √  | √  | √  |    | √  | √  |
| 38    | Credit Line Account                               |    | √  |    |    |    |    |
| 39    | Corporate                                         | √  |    |    |    |    |    |
| 40    | Universal Account (Customer ID number)            |    | √  |    |    |    |    |
| 50    | Money Market Investment Account                   |    | √  |    |    |    |    |
| 60    | Stored Value Account                              | √  |    |    |    |    |    |
| 90    | Revolving Loan Account                            |    | √  |    |    |    |    |

## Subfield 3 (Cardholder To Account Type Code)

[DE 3](#), subfield 3 (Cardholder To Account Type Code) describes the cardholder account type affected for cardholder account credits and the To Account Type for cardholder account transfer transactions.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 5–6 |
| Justification       | N/A                       |

### Values

Subfield 3 must be one of the following values.

### Cardholder To Account type code

| Value | Description                                       | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------|----|----|----|----|----|----|
| 00    | Default Account (not specified or not applicable) | √  | √  | √  | √  | √  | √  |
| 10    | Savings Account                                   |    | √  |    |    | √  | √  |
| 20    | Checking Account                                  |    | √  |    |    | √  | √  |
| 30    | Credit Card Account                               | √  | √  | √  |    |    |    |
| 38    | Credit Line Account                               |    | √  |    |    |    |    |
| 40    | Universal Account                                 |    | √  |    |    |    |    |
| 50    | Money Market Investment Account                   |    | √  |    |    |    |    |
| 58    | IRA Investment Account                            |    | √  |    |    |    |    |
| 90    | Revolving Loan Account                            |    | √  |    |    |    |    |
| 91    | Installment Loan Account                          |    | √  |    |    |    |    |
| 92    | Real Estate Loan Account                          |    | √  |    |    |    |    |

## DE 4 (Amount, Transaction)

*DE 4 (Amount, Transaction)* is the amount of funds the cardholder requested in the local currency of the acquirer or source location of the transaction.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-12                       |
| Data Length         | N/A                        |
| Data Field          | Contents of positions 1–12 |
| Subfields           | N/A                        |
| Justification       | Right with leading zeros   |

### Usage

Following is the usage of *DE 4* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | M   | P   | M   |
| Authorization Request Response/0110                                                             | CE  | X   | M   |
| Authorization Advice/0120: acquirer-generated                                                   | M   | P   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | ME  | XE  | •   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | XE  | ME  |
| Reversal Request/0400                                                                           | M   | P   | M   |
| Reversal Request Response/0410                                                                  | CE  | XE  | ME  |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | •   |

## Values

Valid numeric data must be present in this data element.

The value must be zero for:

- Authorization Request/0100 and Reversal Request/0400: Balance Inquiry, except where ATM transaction fees are allowed
- Authorization Request/0100: Chargeback Period Extension Request
- Authorization Request/0100 and Reversal Request/0400: PIN Management (PIN Unblock and PIN Change), except where ATM transaction fees are allowed
- Authorization Request/0100: Card Activation Request at POS
- Authorization Request/0100: Account Status Inquiry with Authentication Indicator

[DE 28 \(Amount, Transaction Fee\)](#) must be included in DE 4 when DE 28 is present in the message.

## Application notes

[DE 4](#) cannot exceed the region or country Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payment Transaction limit.

## System edits

| WHEN...                                                                                                                                                                                                        | THEN...                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer responds with <a href="#">DE 39 (Response Code)</a> , value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed), the issuer is not required to echo <a href="#">DE 4</a> | the Dual Message Authorization System will forward to the acquirer, the Authorization Request Response/0110 message containing the partial approval amount in <a href="#">DE 4</a> . |

## For Account Status Inquiry Service Transactions

## System edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request/0100 message contains <a href="#">DE 61 (Point-of-Service [POS] Data)</a> , subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI])<br>and<br><a href="#">DE 4 (Amount, Transaction)</a> contains a value greater than zero and <a href="#">DE 3 (Processing Code)</a> contains a value of 00 (Purchase) | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 004</li> </ul> |

## For Automated Fuel Dispenser (AFD) Transactions

The contents of [DE 4](#) may equal zero when the Authorization Advice/0120 message is sent by a Europe region acquirer for an AFD transaction in Europe and contains [DE 18 \(Merchant Type\)](#) = 5542 (Fuel Dispenser, Automated) and [DE 60 \(Advice Reason Code\)](#) = 191 (Acquirer Completed Authorization Transaction).

Issuers should be prepared to receive zero amount AFD completion advices (canceling a previously approved authorization) from any acquirer as there is no Dual Message Authorization System edit restricting such advices from Europe region acquirers.

## For Authorization Chargeback Protection Period Extension Requests

### System edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | THEN the Dual Message Authorization System...                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the acquirer sends an Authorization Chargeback Protection Period Extension Authorization Request/0100 message that contains the following:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 3 (Processing Code)</a>, subfield 1 (Transaction Type), value 00 (Purchase)</li> <li>• <a href="#">DE 4 (Transaction Amount)</a>, value 0</li> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 63 (Trace ID) is present</li> <li>• <a href="#">DE 61</a>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring)</li> <li>• <a href="#">DE 61 (POS Data)</a>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)</li> </ul> <p>and the issuer is not available to respond to the request</p> | <p>sends an Authorization Request Response/0110 message where:</p> <p><a href="#">DE 39 (Response Code)</a> = 91 (Authorization System or issuer system inoperative)</p>                                                                              |
| <p>the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 3 (Processing Code)</a>, subfield 1 (Transaction Type), value 00 (Purchase)</li> <li>• <a href="#">DE 4 (Transaction Amount)</a>, value 0</li> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 63 (Trace ID) is NOT present</li> <li>• <a href="#">DE 61</a>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring), and</li> <li>• <a href="#">DE 61 (POS Data)</a>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)</li> </ul>                                                                      | <p>sends an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format Error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 048063</li> </ul> |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 3 (Processing Code)</a>, subfield 1 (Transaction Type), value 00 (Purchase)</li> <li>• <a href="#">DE 4 (Transaction Amount)</a>, value 0</li> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 63 (Trace ID) is present</li> <li>• <a href="#">DE 61</a>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring)</li> <li>• <a href="#">DE 61 (POS Data)</a>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)</li> </ul> <p>and if the values for Contactless Transit Aggregated, Transit Debt Recovery Transactions are present:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 64 (Transit Program), subfield 1 (Transit Transaction Type Individual) = 03 (Post-Authorized Aggregated), 04 (Authorized Aggregated Split Clearing), or 06 (Post-Authorized Aggregated Maestro)</li> </ul> | <p>sends an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format Error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 048064</li> </ul>                                                              |
| <p>the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 3 (Processing Code)</a>, subfield 1 (Transaction Type), value 00 (Purchase)</li> <li>• <a href="#">DE 4 (Transaction Amount)</a>, value 0</li> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 63 (Trace ID) is present</li> <li>• <a href="#">DE 61</a>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring)</li> <li>• <a href="#">DE 61 (POS Data)</a>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)</li> </ul> <p>and if the values for Transit Debt Recovery Transactions are present:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 64 (Transit Program), subfield 1 (Transit Transaction Type Individual) = 07 (Debt Recovery)</li> </ul>                                                                                                                                     | <p>sends an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format Error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 048064 (Transaction Debit Recovery) or 048095 (Installment payment)</li> </ul> |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 3 (Processing Code)</a>, subfield 1 (Transaction Type), value 00 (Purchase)</li> <li>• <a href="#">DE 4 (Transaction Amount)</a>, value 0</li> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 63 (Trace ID) is present</li> <li>• <a href="#">DE 61</a>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring)</li> <li>• <a href="#">DE 61 (POS Data)</a>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)</li> </ul> <p>or if the values for an installment payment transaction are present:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 95 (Promotion Code)</li> </ul> | <p>sends an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format Error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 048064 (Transaction Debit Recovery) or 048095 (Installment payment)</li> </ul> |
| <p>the acquirer sends an Acquirer-generated Reversal Request/0400 message for an Authorization Chargeback Protection Period Extension request where the message contains the following:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 3 (Processing Code)</a>, subfield 1 (Transaction Type), value 00 (Purchase)</li> <li>• <a href="#">DE 4 (Transaction Amount)</a>, value 0</li> <li>• <a href="#">DE 48 (Additional Data)</a>, subelement 63 (Trace ID) is present</li> <li>• <a href="#">DE 61</a>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring)</li> <li>• <a href="#">DE 61 (POS Data)</a>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)</li> </ul>                                                                                                                                                  | <p>sends an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format Error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 004 (Transaction Amount)</li> </ul>                                            |

## DE 5 (Amount, Settlement)

[DE 5 \(Amount, Settlement\)](#) is the amount of funds to be transferred between the acquirer and the issuer equal to [DE 4 \(Amount, Transaction\)](#) in the settlement currency. Mastercard programs and services use U.S. dollars as the currency of settlement.



## Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-12                       |
| Data Length         | N/A                        |
| Data Field          | Contents of positions 1–12 |
| Subfields           | N/A                        |
| Justification       | Right with leading zeros   |

## Usage

Following is the usage of [DE 5](#) (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | •   | X   | C   |
| Authorization Request Response/0110                                                             | CE  | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | •   | X   | C   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | •   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | X   | C   |
| Authorization Advice Response/0130: system generated (responding to an acquirer-generated 0120) | •   | X   | C   |
| Reversal Request/0400                                                                           | •   | X   | C   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420                                                                            | •   | X   | C   |
| Reversal Advice Response/0430                                                                   | CE  | XE  | •   |

## Values

This data element must contain valid numeric data.

## Application notes

All settlement amounts are specified in U.S. dollars. The settlement amount in authorization messages should be interpreted as Amount, Reporting because settlement of funds (for example, accounting posting or memo-posting) does not currently occur online for authorization messages.

The Dual Message Authorization System will provide this data element if the customer chooses to receive settlement amount-related data elements.

When the issuer responds with [DE 39 \(Response Code\)](#), value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed), the issuer is not required to echo [DE 5](#) if it was present in the Authorization Request/0100 message to the issuer. The Dual Message Authorization System will provide the partial approval amount in DE 5 of the Authorization Request Response/0110 to the acquirer if the acquirer chooses to receive settlement amount-related data elements.

The issuer may receive a Reversal Request/0400 message containing a different value in DE 5 than was present in DE 5 of the original Authorization Request/0100 message. This difference occurs when the currency conversion rate used by the Dual Message Authorization System changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

If [DE 4](#) converts to an amount that is more than 12 digits long in the settlement currency in DE 5, the Dual Message Authorization System rejects the transaction with a format error in DE 4. When a customer receives this error code, it should verify that the transaction amount is correct.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

## DE 6 (Amount, Cardholder Billing)

[DE 6 \(Amount, Cardholder Billing\)](#) indicates the transaction amount in the issuer's currency. It is the amount billed to the cardholder in the cardholder account currency, excluding cardholder billing fees.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-12                       |
| Data Length         | N/A                        |
| Data Field          | Contents of positions 1–12 |

| Attribute     | Description              |
|---------------|--------------------------|
| Subfields     | N/A                      |
| Justification | Right with leading zeros |

## Usage

Following is the usage of [DE 6](#) (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | .   | X   | M   |
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | .   | X   | M   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)      | ME  | XE  | .   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | X   | C   |
| Reversal Request/0400                                                                           | .   | X   | M   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420                                                                            | .   | X   | M   |
| Reversal Advice Response/0430                                                                   | CE  | XE  | .   |

## Values

This data element must contain valid numeric data.

## Application notes

The Dual Message Authorization System inserts this data element into all authorization and reversal messages. The Dual Message Authorization System also will insert the following data elements to indicate the:

- Conversion factor used: [DE 10 \(Conversion Rate, Cardholder Billing\)](#)
- Conversion date: [DE 16 \(Date, Conversion\)](#)
- Issuer's currency: [DE 51 \(Currency Code, Cardholder Billing\)](#)

Issuers must adhere to the following when providing a partial approval Authorization Request/0110 message:

- If [DE 39](#) is 87 (Purchase Amount Only, No Cash Back Allowed), DE 6 must be less than the requested amount.
- If DE 39 is 10 (Partial Approval) and transaction is Automated Fuel Dispenser, DE 6 can be less, equal, or greater than requested amount.

Where a minor unit of currency applies, amounts are expressed in the minor unit of currency without a decimal separator (example, value 100 represents USD 1.00).

This data element is defined as a mandatory echo in Authorization Request Response/0110 messages except when the issuer responds with [DE 39 \(Response Code\)](#), value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed).

The Dual Message Authorization System forwards the partial approval amount in DE 6 of the Authorization Request Response/0110 to the acquirer.

The issuer may receive a Reversal Request/0400 message containing a different value in DE 6 than was present in DE 6 of the original Authorization Request/0100 message. This difference occurs when the currency conversion rate used by the Dual Message Authorization System changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

The issuer may also receive DE 6 with a supplemental cardholder fee added for participants in the Issuer Currency Conversion Rate (ICCR) Service or Cross Border Fee Manager (CBFM) Service.

If [DE 4](#) converts to an amount that is more than 12 digits long in the settlement currency in [DE 5](#), the Dual Message Authorization System rejects the transaction with a format error in DE 4. When a customer receives this error code, it should verify that the transaction amount is correct.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

## DE 7 (Transmission Date and Time)

[DE 7 \(Transmission Date and Time\)](#) is the date and time that a message is entered into the Mastercard Network. Date and time must be expressed in Coordinated Universal Time (UTC).

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | n-10        |

| Attribute     | Description               |
|---------------|---------------------------|
| Data Length   | N/A                       |
| Data Field    | Contents of subfields 1–2 |
| Subfields     | 2                         |
| Justification | See subfields             |

## Usage

Following is the usage of DE 7 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | M   | P   | M   |
| Authorization Request Response/0110                                                             | ME  | XE  | ME  |
| Authorization Advice/0120: acquirer-generated                                                   | M   | P   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)      | ME  | XE  | •   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | XE  | ME  |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | XE  | ME  |
| Authorization Response Acknowledgment/0180                                                      | ME  | XE  | •   |
| Authorization Response Negative Acknowledgment/0190                                             | •   | XE  | ME  |
| Issuer File Update Request/0302                                                                 | M   | X   | •   |
| Issuer File Update Request Response/0312                                                        | •   | XE  | ME  |
| Reversal Request/0400                                                                           | M   | P   | M   |
| Reversal Request Response/0410                                                                  | ME  | XE  | ME  |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | •   |
| Administrative Request/0600                                                                     | M   | P   | M   |
| Administrative Request Response/0610                                                            | ME  | XE  | ME  |

| Message                                                                                | Org | Sys | Dst |
|----------------------------------------------------------------------------------------|-----|-----|-----|
| Administrative Advice/0620: member-generated                                           | M   | P   | M   |
| Administrative Advice/0620: system-generated                                           | .   | X   | M   |
| Administrative Advice Response/0630                                                    | ME  | XE  | .   |
| Network Management Request/0800: network connection status, member-generated           | M   | X   | .   |
| Network Management Request/0800: network connection status, system-generated           | .   | X   | M   |
| Network Management Request/0800: sign-on/sign-off                                      | M   | X   | .   |
| Network Management Request/0800: PEK exchange                                          | .   | X   | M   |
| Network Management Request/0800: PEK exchange, on demand                               | M   | X   | .   |
| Network Management Request/0800: host session activation/deactivation                  | M   | X   | .   |
| Network Management Request Response/0810 (network connection status, system-generated) | .   | XE  | ME  |
| Network Management Request Response/0810: network connection status, member-generated  | ME  | XE  | .   |
| Network Management Request Response/0810: PEK exchange                                 | ME  | XE  | .   |
| Network Management Request Response/0810: PEK exchange, on demand                      | .   | XE  | ME  |
| Network Management Request Response/0810: host session activation/deactivation         | .   | XE  | ME  |
| Network Management Request Response/0810: sign-on/sign-off                             | .   | XE  | ME  |
| Network Management Advice/0820: PEK exchange                                           | .   | X   | M   |

## Values

See subfields.

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 7 must remain unchanged for all messages associated with a given system transaction, which includes all responses and acknowledgments related to an original request message (such as authorization, file update, administrative, network management). For example, the same DE 7 is used in an Authorization Request/0100 and any related Authorization Request Response/0110, Authorization Advice/0120: acquirer-generated, Authorization Advice/0120: system-generated, or Reversal Advice/0420: system-generated message.

The Reversal Request/0400 message is the exception to this rule. Reversal Request/0400 messages are treated as an originating request and must have a unique DE 7 value assigned. The DE 7 value from the original 0100 request is included in [DE 90 \(Original Data Elements\)](#) for transaction matching purposes.

Each message initiator must assign a DE 7 to each originating request message. The transmission time provided in DE 7 must be within three minutes of the current time recorded by the Mastercard Network.

The combination of a message originator's [DE 11 \(Systems Trace Audit Number \[STAN\]\)](#) and DE 7 must uniquely identify any system transaction the message originator initiates on any given UTC day. These data elements together may be used as key data elements to identify and locate transaction records at some later time for the purpose of error resolution, settlement reconciliation, and retrieval requests.

## Subfield 1 (Date)

[DE 7](#), subfield 1 (Date) describes the valid date.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-4                       |
| Data Field          | Contents of positions 1–4 |
| Justification       | N/A                       |

### Values

This subfield must contain a valid date in MMDD format.

## Subfield 2 (Time)

[DE 7](#), subfield 2 (Time) describes the valid time.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-6                        |
| Data Field          | Contents of positions 5–10 |
| Justification       | N/A                        |

## Values

Time must contain a valid time in hhmmss format.

## DE 8 (Amount, Cardholder Billing Fee)

*DE 8 (Amount, Cardholder Billing Fee)* is the fee the issuer bills to the cardholder in the same currency as *DE 6 (Amount, Cardholder Billing)*.

**NOTE:** The Dual Message Authorization System currently does not use this data element.

## DE 9 (Conversion Rate, Settlement)

*DE 9 (Conversion Rate, Settlement)* is the factor used in the conversion from transaction to settlement amount. *DE 4 (Amount, Transaction)* is multiplied by DE 9 to determine *DE 5 (Amount, Settlement)*.

### Attributes

| Attribute           | Description                                                             |
|---------------------|-------------------------------------------------------------------------|
| Data Representation | n-8                                                                     |
| Data Length         | N/A                                                                     |
| Data Field          | Contents of subfields 1–2                                               |
| Subfields           | 2                                                                       |
| Justification       | Right, excluding the decimal indicator that must be the leftmost digit. |

## Usage

Following is the usage of DE 9 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | C   |
| Authorization Request Response/0110           | CE  | X   | C   |
| Authorization Advice/0120: acquirer-generated | •   | X   | C   |



| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | .   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | X   | C   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | X   | C   |
| Reversal Request/0400                                                                           | .   | X   | C   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420                                                                            | .   | X   | C   |
| Reversal Advice Response/0430                                                                   | CE  | XE  | .   |

## Values

The leftmost digit must be in the range 0–7 and denotes the number of positions that the decimal point shall be moved from the right. Example: For a value of "69972522," conversion rate is 9.972522.

## Application notes

The Dual Message Authorization System will insert this data element into a message if the customer chooses to receive settlement amount-related data elements. If the value in DE 5 (Amount, Settlement) is equal to the value in DE 4 (Amount, Transaction), the settlement conversion rate is 61000000.

When used in Authorization/01xx messages, this data element should be interpreted as Conversion Rate, Reporting because settlement (for example, account posting or memo posting) of funds does not occur online with Authorization/01xx messages and Reversal Advice/04xx messages.

Note that when this data element is present in a message, DE 5 (Amount, Settlement) and [DE 50 \(Currency Code, Settlement\)](#) also are present.

The issuer may receive a Reversal Request/0400 message containing a different value in DE 9 than was present in [DE 9](#) of the original Authorization Request/0100 message. This difference occurs when the currency conversion rate used by the Dual Message Authorization System changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 or Reversal Request/0400 message and was unable to complete currency conversion processing.

## Subfield 1 (Decimal Indicator)

*DE 9*, subfield 1 (Decimal Indicator) indicates the number of positions the decimal point should be moved from the right.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

Must be in the range of 0–7.

## Subfield 2 (Conversion Rate)

*DE 9*, subfield 2 (Conversion Rate) indicates the conversion rate.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-7                       |
| Data Field          | Contents of positions 2–8 |
| Justification       | N/A                       |

### Values

Must be a valid conversion rate.

## DE 10 (Conversion Rate, Cardholder Billing)

*DE 10 (Conversion Rate, Cardholder Billing)* is the factor used in the conversion from transaction to cardholder billing amount. DE 4 (Amount, Transaction) is multiplied by DE 10 to determine *DE 6 (Amount, Cardholder Billing)*.

### Attributes

| Attribute           | Description                                                             |
|---------------------|-------------------------------------------------------------------------|
| Data Representation | n-8                                                                     |
| Data Length         | N/A                                                                     |
| Data Field          | Contents of subfields 1–2                                               |
| Subfields           | 2                                                                       |
| Justification       | Right, excluding the decimal indicator that must be the leftmost digit. |

### Usage

Following is the usage of *DE 10* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | •   | X   | M   |
| Authorization Request Response/0110                                                             | CE  | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | •   | X   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)      | CE  | XE  | •   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | X   | C   |
| Reversal Request/0400                                                                           | •   | X   | M   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420                                                                            | •   | X   | M   |

| Message                       | Org | Sys | Dst |
|-------------------------------|-----|-----|-----|
| Reversal Advice Response/0430 | CE  | XE  | .   |

## Values

Fixed length 8 positions; the leftmost decimal indicator denotes the number of positions that the decimal point shall be moved from the right. Example: For data element value "69972522," conversion rate is 9.972522. The leftmost digit must be in the range 0–7.

## Application notes

The Dual Message Authorization System inserts this data element into all authorization, advice, and reversal messages. If the value in [DE 6 \(Amount, Cardholder Billing\)](#) is equal to the value in [DE 4 \(Amount, Transaction\)](#), the cardholder billing conversion rate is 61000000.

Note that [DE 6 \(Amount, Cardholder Billing\)](#) and [DE 51 \(Currency Code, Cardholder Billing\)](#) also are present.

The issuer may receive a Reversal Request/0400 message containing a different value in DE 10 than was present in DE 10 of the original Authorization Request/0100 message. This occurs when the currency conversion rate used by the Dual Message Authorization System changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 or Reversal Request/0400 message and was unable to complete currency conversion processing.

## Subfield 1 (Decimal Indicator)

[DE 10](#), subfield 1 (Decimal Indicator) indicates the number of positions the decimal point should be moved from the right.

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

## Values

Must be in the range of 0 through 7.

## Subfield 2 (Cardholder Billing Conversion Rate)

[DE 10](#), subfield 2 (Cardholder Billing Conversion Rate) indicates the cardholder billing conversion rate.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-7                       |
| Data Field          | Contents of positions 2–8 |
| Justification       | N/A                       |

### Values

Must be a valid conversion rate.

## DE 11 (System Trace Audit Number [STAN])

DE 11 (Systems Trace Audit Number [STAN]) is a number a message initiator assigns to uniquely identify a transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-6                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–6 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 11 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | M   | P   | M   |
| Authorization Request Response/0110                                                             | ME  | XE  | ME  |
| Authorization Advice/0120: acquirer-generated                                                   | M   | P   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)      | ME  | XE  | •   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | XE  | ME  |
| Authorization Advice Response/0130:system-generated (responding to an acquirer-generated 0120)  | •   | XE  | ME  |
| Authorization Response Acknowledgment/0180                                                      | ME  | XE  | •   |
| Authorization Response Negative Acknowledgment/0190                                             | •   | XE  | ME  |
| Issuer File Update Request/0302                                                                 | M   | X   | •   |
| Issuer File Update Request Response/0312                                                        | •   | XE  | ME  |
| Reversal Request/0400                                                                           | M   | P   | M   |
| Reversal Request Response/0410                                                                  | ME  | XE  | ME  |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | •   |
| Administrative Request/0600                                                                     | M   | P   | M   |
| Administrative Request Response/0610                                                            | ME  | XE  | ME  |
| Administrative Advice/0620: member-generated                                                    | M   | P   | M   |
| Administrative Advice/0620: system-generated                                                    | •   | X   | M   |
| Administrative Advice Response/0630                                                             | ME  | XE  | •   |
| Network Management Request/0800: network connection status, member-generated                    | M   | X   | •   |
| Network Management Request/0800: network connection status, system-generated                    | •   | X   | M   |
| Network Management Request/0800: sign-on/sign-off                                               | M   | X   | •   |
| Network Management Request/0800: PEK exchange                                                   | •   | X   | M   |
| Network Management Request/0800: PEK exchange, on demand                                        | M   | X   | •   |

| Message                                                                               | Org | Sys | Dst |
|---------------------------------------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: host session activation/deactivation                 | M   | X   | •   |
| Network Management Request Response/0810: network connection status, member-generated | ME  | XE  | •   |
| Network Management Request Response/0810: network connection status, system-generated | •   | XE  | ME  |
| Network Management Request Response/0810: sign-on/sign-off                            | •   | XE  | ME  |
| Network Management Request Response/0810: PEK exchange                                | ME  | XE  | •   |
| Network Management Request Response/0810: PEK exchange, on demand                     | •   | XE  | ME  |
| Network Management Request Response/0810: host session activation/deactivation        | •   | XE  | ME  |
| Network Management Advice/0820: PEK exchange                                          | •   | X   | M   |

## Values

This data element must not be all zeros or blanks.

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 11 must remain unchanged for all messages associated with a given system transaction, which includes all responses and acknowledgments related to an original request message (such as authorization, file update, administrative, and network management). For example, the same [DE 11](#) is used in an Authorization Request/0100 and any related Authorization Request Response/0110, Authorization Advice/0120: acquirer-generated, Authorization Advice/0120: system-generated, or Reversal Advice/0420: system-generated message.

The Reversal Request/0400 message is the exception to this rule. Reversal Request/0400 messages are treated as an originating request and must have a unique DE 11 value assigned. The DE 11 value from the original 0100 request is included in [DE 90 \(Original Data Elements\)](#) for transaction matching purposes.

Each message initiator must assign DE 11 to each originating request message.

The combination of a message originator's DE 11 and [DE 7 \(Transmission Date and Time\)](#) must uniquely identify any system transaction the message originator initiates on any given UTC day. These data elements together may be used as key data elements to identify and locate transaction records at some later time for the purpose of error resolution, settlement reconciliation, and retrieval requests.

Acquirers that process more than 999,999 transactions within a UTC day may repeat the same number.

## DE 12 (Time, Local Transaction)

*DE 12 (Time, Local Transaction)* is the local time at which the transaction takes place at the point of acceptor location.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-6                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–6 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 12 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |

### Values

DE 12 must contain the local time the transaction occurred at the point of interaction, represented in the local time zone and not UTC, in hhmmss format. For reversals and advices, the value must match the original transaction.

If the transaction time is not known at the time of authorization (for example, in the case of a preauthorization request), the acquirer must use the authorization time in this field.

### Application notes

This data element is defined and used identically within all Mastercard programs and services.



DE 12 is mandatory for all chip transactions. (If missing, the Dual Message Authorization System will not reject the message; however, a data integrity error will be reported.)

DE 12 is mandatory for all ATM transactions.

DE 12 is mandatory for all other card read transactions. (If missing, the Dual Message Authorization System will not reject the message.)

DE 12 is mandatory for Mastercard Hosted Mobile Phone Top-up transactions.

DE 12 is not required for MDES pre-digitization messages.

For more information, refer to the *MDES Technical Specifications for Dual and Single Message Systems*.

## DE 13 (Date, Local Transaction)

*DE 13 (Date, Local Transaction)* is the local month and day on which the transaction takes place at the point of acceptor location.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-4                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–4 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 13 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |

| Message               | Org | Sys | Dst |
|-----------------------|-----|-----|-----|
| Reversal Request/0400 | C   | P   | C   |
| Reversal Advice/0420  | .   | X   | C   |

## Values

Date must contain a valid date in MMDD format. This date must be specified in local time zone units, not in UTC units. The original value of this data element must not be changed. If there is a delay between the date that a transaction was initiated or completed at a point of interaction and the date that the transaction was subsequently entered into the Authorization Platform, DE 13 must remain set to the local date that the transaction was initiated at the acceptor location.

If the transaction date is not known at the time of authorization (for example, in the case of a preauthorization request), the acquirer must use the authorization date in this field.

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 13 is mandatory for all chip transactions (If missing, the Dual Message Authorization System will not reject the message; however, a data integrity error will be reported.)

DE 13 is mandatory for all ATM transactions.

DE 13 is mandatory for all other card read transactions. (If missing, the Dual Message Authorization System will not reject the message.)

DE 13 is mandatory for Mastercard Hosted Mobile Phone Top-up transactions.

DE 13 is not required for MDES pre-digitization messages.

For more information, refer to the *MDES Technical Specifications for Dual and Single Message Systems*.

## DE 14 (Date, Expiration)

[DE 14 \(Date, Expiration\)](#) specifies the year and month after which an issuer designates a cardholder's card to be expired.

## Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | n-4         |
| Data Length         | N/A         |

| Attribute     | Description               |
|---------------|---------------------------|
| Data Field    | Contents of positions 1–4 |
| Subfields     | N/A                       |
| Justification | N/A                       |

## Usage

Following is the usage of DE 14 (whether it is mandatory, conditional, optional, system-provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |

## Values

This data element must consist of a year and a month in YYMM format.

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

## DE 14 edits

| IF...                                                                                                                                                | THEN...               |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Track 1 or Track 2 (magnetic stripe) data is not present in the Authorization/01xx message                                                           | DE 14 is conditional. |
| Track 1 or Track 2 data (magnetic stripe) data is present in the Authorization/01xx message for Mastercard and Visa transactions                     | DE 14 is optional.    |
| Track 1 or Track 2 (magnetic stripe) data is not present in the Authorization/01xx message for transactions other than Mastercard, Maestro, and Visa | DE 14 is mandatory.   |

| IF...                              | THEN...                                                                                   |
|------------------------------------|-------------------------------------------------------------------------------------------|
| the expiration date is unavailable | DE 14 must not be present (applies to Mastercard and Visa only).                          |
| an EMV transaction                 | DE 14 must be populated with mandatory expiration date contained on the chip in TAG 5F24. |
| an MDES transaction                | DE 14 must be populated when provided by the merchant.                                    |

## DE 15 (Date, Settlement)

*DE 15 (Date, Settlement)* is the date (month and day) that funds will be transferred between an acquirer and an issuer or an appropriate intermediate network facility (INF).

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-4                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–4 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 15 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | M   |
| Authorization Request Response/0110           | ME  | X   | M   |
| Authorization Advice/0120: acquirer-generated | •   | X   | M   |
| Authorization Advice/0120: system-generated   | •   | X   | M   |

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)      | ME  | XE  | •   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | X   | M   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | X   | M   |
| Reversal Request/0400                                                                           | •   | X   | M   |
| Reversal Request Response/0410                                                                  | ME  | X   | M   |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | •   |

## Values

This data element must consist of a month and day in MMDD format.

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

For Authorization/01xx and Reversal Advice/04xx messages (in which there is no concurrent online settlement of funds), this data element is a date of network reporting. For example, the Dual Message Authorization System provides program- or service-specific financial network business date to inform acquirers and issuers of the reporting date to which a transaction applies. It should be interpreted as Date, Reporting.

In all cases the Dual Message Authorization System determines and inserts DE 15 in all originating Authorization/01xx and Reversal Advice/04xx messages.

The Dual Message Authorization System provides this date. Therefore, this data element must not be present in any originating request or advice messages. Any customer processing system or intermediate network facility (INF) must not change this date in any subsequent response message.

## DE 16 (Date, Conversion)

*DE 16 (Date, Conversion)* indicates the effective date of *DE 9 (Conversion Rate, Settlement)* and also *DE 10 (Conversion Rate, Cardholder Billing)* whenever these data elements are present within a message.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-4                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–4 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of *DE 16* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | •   | X   | M   |
| Authorization Request Response/0110                                                             | CE  | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | •   | X   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | X   | C   |
| Reversal Request/0400                                                                           | •   | X   | M   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420                                                                            | •   | X   | M   |

| Message                       | Org | Sys | Dst |
|-------------------------------|-----|-----|-----|
| Reversal Advice Response/0430 | CE  | XE  | .   |

### Values

This data element must consist of a month and day in MMDD format.

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

## DE 17 (Date, Capture)

[DE 17 \(Date, Capture\)](#) is the month and day the acquirer processed the transaction data.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 18 (Merchant Type)

[DE 18 \(Merchant Type\)](#) is the classification (acceptor business code/merchant category code [MCC]) of the merchant's type of business or service.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-4                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–4 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 18 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | M   | P   | M   |
| Authorization Advice/0120: acquirer-generated | M   | P   | M   |
| Authorization Advice/0120: system-generated   | .   | X   | M   |
| Reversal Request/0400                         | M   | P   | M   |

### Values

This data element must contain a valid MCC. Refer to the *Quick Reference Booklet* for valid codes.

### Application notes

### Edits

| IF...                                                                                                                                                                                                                                                                                                                                           | THEN the Dual Message Authorization System...                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the transaction is a Mastercard Electronic card Authorization Request/0100 message that occurs at magnetic stripe-reading or chip-reading terminals when the MCC represents a non-face-to-face environment, such as MCC 5542: Fuel Dispenser, Automated when PIN is present or chip is present with CAT 1.                                      | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 58 (Transaction not permitted to acquirer/terminal)</li> </ul>                       |
| the transaction is an Authorization Request/0100 or Reversal Request/0400 message and DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 01 (Withdrawal) and DE 18 is not 6010 (Member Financial Institution: Manual Cash Disbursements) or 6011 (Member Financial Institution: Automated Cash Disbursements) | sends an Authorization Request Response/0110 or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Service Parameters) = 018</li> </ul> |



| IF...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | THEN the Dual Message Authorization System...                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the OFI is registered for the Mastercard Send Payments Transaction Program and is submitting an Authorization Request/0100 for a Mastercard Send Payment Transaction where: <ul style="list-style-type: none"><li>• DE 3, subfield 1 contains value 28</li><li>• DE 18 does not contain an eligible MCC</li><li>• DE 48, subelement 77 contains an eligible TTI</li><li>• DE 61, subfield 10 criteria is met for Mastercard Send Payment Transactions<ul style="list-style-type: none"><li>– 0 = Not a CAT transaction</li><li>– 1 = Authorized Level 1 CAT: Automated dispensing machine with PIN</li><li>– 2 = Authorized Level 2 CAT: Self-service terminal</li><li>– 6 = Authorized Level 6 CAT: Electronic commerce</li><li>– 8 = Remote terminal</li></ul></li><li>• DE 108 is present</li></ul> | sends the OFI an Authorization Request Response/0110 or Reversal Request Response/0410 message where: <ul style="list-style-type: none"><li>• DE 39 = 30 (Format error)</li><li>• DE 44 = 018 (Merchant Type)</li></ul> |

## DE 19 (Acquiring Institution Country Code)

*DE 19 (Acquiring Institution Country Code)* is based on Mastercard franchise data information.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1-3 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | .   | X   | C   |
| Authorization Advice/0120: acquirer-generated | .   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | .   | X   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

## DE 20 (Primary Account Number [PAN] Country Code)

*DE 20 (Primary Account Number [PAN] Country Code)* is a code identifying the country where the card issuer is located.

### Usage

DE 20 is defined by and used for internal Mastercard processing and will not be present in Dual Message Authorization System messages sent to customers.

## DE 21 (Forwarding Institution Country Code)

*DE 21 (Forwarding Institution Country Code)* is the code of the country where the forwarding institution is located.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 22 (Point-of-Service [POS] Entry Mode)

*DE 22 (Point-of-Service [POS] Entry Mode)* indicates the method used for PAN entry to initiate a transaction and the PIN entry capabilities.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | n-3         |
| Data Length         | N/A         |

| Attribute     | Description               |
|---------------|---------------------------|
| Data Field    | Contents of subfields 1–2 |
| Subfields     | 2                         |
| Justification | See subfields             |

## Usage

Following is the usage of DE 22 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | M   | X   | M   |
| Authorization Advice/0120: acquirer-generated | M   | X   | M   |
| Authorization Advice/0120: system-generated   | •   | X   | M   |
| Reversal Request/0400                         | M   | X   | M   |
| Reversal Advice/0420                          | •   | X   | M   |

## Values

See subfields.

## Application notes

Following DE 22, subfield details are the [Dual Message Authorization System edits](#) that describe transaction-specific program and service edits for this data element.

### PIN Management Transactions

Chip PIN Management: Only subfield 1, value 05 (PAN auto entry via chip) and subfield 2, value 1 (Terminal has PIN entry capability) are valid.

Magnetic Stripe PIN Management: subfield 1, values 02 and 90 are valid for magnetic stripe PIN change

### All ATM Transactions

DE 22, subfield 2, must be value 1 (Terminal has PIN entry capability).

Short Message Service (SMS) Balance Inquiry Service

Issuers participating in the Short Message Service (SMS) Balance Inquiry service will receive DE 22, subfield 1 (POS Terminal PAN Entry Mode), value 81 (PAN entry via electronic commerce, including chip).

Subfield 1 (POS Terminal PAN Entry Mode)

[DE 22](#), subfield 1 (POS Terminal PAN Entry Mode) indicates the method used for PAN entry to initiate a transaction.

Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 1–2 |
| Justification       | N/A                       |

Values

| Value | Description                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 00    | Merchant terminal PAN entry mode unknown                                                                                                          |
| 01    | Merchant terminal PAN manual entry                                                                                                                |
| 02    | Merchant terminal PAN auto-entry via magnetic stripe                                                                                              |
| 03    | Merchant terminal PAN auto-entry via bar code reader                                                                                              |
| 04    | Merchant terminal PAN auto-entry via optical character reader (OCR)                                                                               |
| 05    | Merchant terminal PAN auto-entry via integrated circuit card                                                                                      |
| 07    | Merchant terminal PAN auto-entry via contactless M/Chip                                                                                           |
| 09    | PAN/Token entry via electronic commerce containing DSRP cryptogram in <a href="#">DE 55 (Integrated Circuit Card [ICC] System-Related Data)</a> . |
| 10    | Credential on File                                                                                                                                |

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 79    | <p>A hybrid terminal with an online connection to the acquirer failed in sending a chip fallback transaction (in which <a href="#">DE 22</a>, subfield 1 = 80) to the issuer.</p> <p>or</p> <p>A hybrid terminal with no online connection to the acquirer failed to read the chip card. The merchant is prompted to read the magnetic stripe from the card, the magstripe is successfully read and indicates a service code 2XX (or 6XX if card is domestic).</p> <p>To complete the transaction in both cases, a voice transaction takes place during which the merchant communicates the PAN and the expiry date originating from the magstripe track 2 data to the acquirer. The acquirer then sends an online transaction to the issuer in which the value of <a href="#">DE 22</a>, subfield 1 = 79 and in which <a href="#">DE 61</a> subfield 11 indicate that the terminal is chip capable.</p> <p>Refer to the <a href="#">M/Chip Requirements</a> for additional information.</p> |
| 80    | Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in DE 45 (Track 1 Data) or <a href="#">DE 35</a> (Track 2 Data) without alteration or truncation. To use this value, the acquirer must be qualified to use value 90.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 81    | PAN/Token entry via electronic commerce with optional Identity Check-AAV in <a href="#">DE 48</a> , subelement 43 (UCAF) or DSRP cryptogram in <a href="#">DE 104</a> , subelement 001 (Digital Payment Cryptogram).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 82    | Electronic Commerce PAN Auto Entry via Server (issuer, acquirer, or third party vendor system).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 90    | PAN auto-entry via magnetic stripe—the full track data has been read from the data encoded on the card and transmitted within the authorization request in <a href="#">DE 35 (Track 2 Data)</a> or <a href="#">DE 45 (Track 1 Data)</a> without alteration or truncation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 91    | PAN auto-entry via contactless magnetic stripe—the full track data has been transmitted by the acquirer within the authorization request in <a href="#">DE 35 (Track 2 Data)</a> or <a href="#">DE 45 (Track 1 Data)</a> and forwarded to the issuer without alteration or truncation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 95    | Visa only. Chip card with unreliable Card Verification Value (CVV) data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Subfield 2 (POS Terminal PIN Entry Mode)

[DE 22](#) (Point-of-Service [POS] Entry Mode), subfield 2 (POS Terminal PIN Entry Mode) describes the capability of the terminal device to support/accept PIN entry.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

### Values

| Value | Description                                                              |
|-------|--------------------------------------------------------------------------|
| 0     | Unspecified or unknown                                                   |
| 1     | Terminal has PIN entry capability                                        |
| 2     | Terminal does not have PIN entry capability                              |
| 3     | mPOS Software-based PIN entry capability                                 |
| 8     | Terminal has PIN entry capability but PIN pad is not currently operative |

## Dual Message Authorization System edits

The Dual Message Authorization System performs edits on specific programs and services.

Edits are performed on the following programs and services:

- Electronic Commerce Transactions
- Mastercard Consumer Presented QR Transactions
- Chip Transactions
- Magnetic Stripe or Chip-Read Transactions for Mastercard Electronic Card
- Contactless Magnetic Stripe Transactions
- Credential on File Transactions

### Electronic commerce transactions

The Dual Message Authorization System performs a cross-edit between DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) and DE48 (Additional

Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) and DE 61 (Point-of-Service [POS] Data).

### Authorization Request/0100 message

| WHEN....                                                                                                     | THEN the Dual Message Authorization System...          |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <a href="#">DE 22</a> , subfield 1 contains the value 81 (PAN entry via electronic commerce, including chip) | <a href="#">DE 48</a> , subelement 42 must be present. |

## Mastercard Consumer Presented QR Transactions

The Dual Message Authorization System performs the following system edits to help ensure that acquirers supporting merchants that choose to submit tokenized Mastercard Consumer Presented QR transactions are properly using value 03 (PAN auto-entry via barcode reader) in [DE 22 \(Point-of-Service \[POS\] Entry Mode\)](#), subfield 1 (POS Terminal PAN Entry Mode) in Authorization Request/0100 messages.

### Authorization Request/0100 message

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message contains <a href="#">DE 22 (Point-of-Service [POS] Entry Mode)</a>, subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader) and</p> <p><a href="#">DE 48 (Additional Data: Private Use)</a>, subelement 21 (Acceptance Data), subfield 2 (Additional Terminal Capability Indicator) is present and does not contain any of the valid value combinations of 00, 01, 10, or 11</p> | <p>sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:</p> <ul style="list-style-type: none"> <li><a href="#">DE 39 (Response Code)</a> = 30 (Format error)</li> <li><a href="#">DE 44 (Additional Response Data)</a> = 048021 (indicating the data element in error)</li> </ul> <p><b>NOTE: If <a href="#">DE 48</a>, subelement 21, subfield 2 is not present, the Dual Message Authorization System will continue processing the message.</b></p> |
| <p>an Authorization Request/0100 message contains <a href="#">DE 22 (Point-of-Service [POS] Entry Mode)</a>, subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader) and</p> <p><a href="#">DE 48 (Additional Data: Private Use)</a>, TCC (Transaction Category Code), contains value Z (ATM Cash Disbursement)</p>                                                                                                                                                        | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li><a href="#">DE 39 (Response Code)</a> = 58 (Transaction not permitted to acquirer/terminal)</li> </ul>                                                                                                                                                                                                                                                                                   |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                  | THEN the Dual Message Authorization System...                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Authorization Request/0100 message contains the following:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 22 (Point-of-Service [POS] Entry Mode)</a>, subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader); and</li> <li>• <a href="#">DE 52 (Personal ID Number [PIN] Data)</a> and/or</li> <li>• <a href="#">DE 110 (Encryption Data)</a>, Dataset 01 (PIN Encryption)</li> </ul> | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 58 (Transaction not permitted to acquirer/terminal)</li> </ul> |

## Chip transactions

If Mastercard determines through its Internal Chip Monitoring process that improperly formatted chip transactions are being submitted from acquirers not certified to send chip transactions, Mastercard will notify each acquirer before activating an edit.

For more information, refer to the *M/Chip Processing Services Service Description Guide*.

### Authorization Request/0100 message

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the Authorization Request/0100 message contains DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 05 (Merchant terminal PAN auto-entry via integrated circuit card), 07 (Merchant terminal PAN auto-entry via contactless M/Chip), or 09 (PAN/Token entry via electronic commerce containing DSRP cryptogram) and DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and Acquirer Chip testing level associated with the acquirer indicates the acquirer is approved for partial grade chip transaction</p> | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 055</li> </ul> |
| <p>the Authorization Request/0100 message contains <a href="#">DE 22</a>, subfield 1 contains value 09 and <a href="#">DE 55</a> is not present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved to send full grade chip transactions</p>                                                                                                                                                                                                                                                                                  | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format error)</li> <li>• <a href="#">DE 44</a> = 055</li> </ul>                            |



| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                             | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request/0100 message contains <a href="#">DE 22</a> , subfield 1 contains value 05 or 07 or 09 and <a href="#">DE 55</a> may or may not be present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is not permitted to send chip transactions                                                                                                                                                                 | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 022</li> </ul>                                                                                                                                                                                                                                                                     |
| the Authorization Request/0100 message contains <a href="#">DE 22</a> , subfield 1 contains value 09 and <a href="#">DE 61</a> (Point-of-Service [POS] Data), subfield 3 (POS Terminal Location) does not contain value 2 (Off premises of acceptor facility [cardholder terminal including home PC, mobile phone, PDA]), 4, (On premises of acceptor facility [cardholder terminal including home PC, mobile phone, PDA]) or 8 (Additional POS Terminal Locations) | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 061</li> </ul>                                                                                                                                                                                                                                                                     |
| the Authorization Request/0100 message or the Authorization Advice/0120 message contains <a href="#">DE 55</a> and <a href="#">DE 22 (Point-of-Service [POS] Entry Mode)</a> , subfield 1 does not contain value 03, 05, 07, 09, or 81                                                                                                                                                                                                                              | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 055</li> </ul>                                                                                                                                                                                                                       |
| <a href="#">DE 22</a> , subfield 1 contains value 05 or 07<br>and<br><a href="#">DE 55</a> may or may not be present<br>and<br>Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is not permitted to send chip transactions                                                                                                                                                                                                           | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: <ul style="list-style-type: none"> <li>• <a href="#">DE 39</a> = 30 (Format error)</li> <li>• <a href="#">DE 44</a> = 055</li> </ul>                                                                                                                                                                                                                                                                  |
| <a href="#">DE 22</a> , subfield 1, contains value 80<br>and<br>the service code in <a href="#">DE 35</a> —Track 2 Data is not 2xx or 6xx, indicating that it is not a chip card                                                                                                                                                                                                                                                                                    | replaces <a href="#">DE 22</a> , subfield 1, value 80 with value 90 before forwarding the Authorization Request/0100 message to the issuer and<br><br>Notifies the acquirer of this downgrade by populating <a href="#">DE 48</a> , subelement 74 (Additional Processing Information) in the Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>• Subfield 1 = 90 (Chip Fallback Transaction Downgrade Process)</li> <li>• Subfield 2 = C (Completed Successfully)</li> </ul> |

### Authorization Advice/0120: acquirer-generated edit

| WHEN...                                                                                                                                                                                                                                      | THEN the Dual Message Authorization System...                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>DE 22 (Point-of-Service [POS] Entry Mode)</i>, subfield 1 (POS Terminal PAN Entry Mode) contains value 80</p> <p>and</p> <p>the service code in <i>DE 35</i>—Track 2 Data is not 2xx or 6xx, indicating that it is not a chip card</p> | <p>Replaces <i>DE 22</i>, subfield 1, value 80 with value 90 before forwarding the Authorization Advice/0120 message to the issuer.</p> |

## Contactless Magnetic Stripe Transactions

The Dual Message Authorization System will perform the following edit on authorization transactions that contain a contactless magnetic stripe account number that is participating in the Contactless Mapping Service.

### Authorization Request/0100 message

| WHEN...                                                                                                                                                                                                                                                                                                                                                                | THEN the Dual Message Authorization System...                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>DE 22</i>, subfield 1 is value 91</p> <p>and</p> <p><i>DE 35</i> and <i>DE 45</i> are not present</p> <p><b>NOTE: This edit is not performed on an Authorization Advice/0120 message or a Reversal Request/0400 message because track data is not expected for a Reversal Request/0400 message and is optional for an Authorization Advice/0120 message.</b></p> | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <i>DE 39</i> = 30</li> <li>• <i>DE 44</i> = 022</li> </ul> |

## Credential on File Transactions

The Dual Message Authorization System performs the following edits.

### Authorization Request/0100 message

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Authorization Request/0100 message contains:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 22 (Point-of-Service [POS] Entry Mode)</a>, subfield 1 (POS Terminal PAN Entry Mode) = 10 (Credential on File), and</li> <li>• <a href="#">DE 61 (Point-of-Service [POS] Data)</a>, subfield 4 (POS Cardholder Presence) = 5 (Cardholder not present [Electronic order])</li> </ul> <p>and</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 6 (Amount, Cardholder Billing)</a>, subfield 10 is not equal to 6</li> </ul> | <p>sends an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• <a href="#">DE 39 (Response Code)</a> = 30 (Format error)</li> <li>• <a href="#">DE 44 (Additional Response Data)</a> = 061</li> </ul> |

## DE 23 (Card Sequence Number)

[DE 23 \(Card Sequence Number\)](#) distinguishes among separate cards having the same DE 2 (Primary Account Number [PAN]) or DE 34 (Primary Account Number [PAN] Extended). Issuers may encode chip cards with Card Sequence Numbers. Acquirers with chip-reading capability may pass this information encoded on the chip in DE 23 of Authorization Request/0100 messages.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–3 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

## Usage

Following is the usage of DE 23 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | C   | X   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | XE  | CE  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | .   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | XE  | CE  |
| Reversal Request/0400                                                                           | C   | X   | C   |

## Values

Acquirers:

DE 23 must be three positions in the numeric range 000–099. For contact chip or contactless chip transactions where DE 22 subfield 1 is 05 or 07 respectively, DE 23 must contain the value of the Application PAN Sequence Number (EMV tag 5F34) as personalized on the chip. Considering that the Application PAN Sequence Number (EMV tag 5F34) is defined as a one-byte numeric value in the EMV specification, the terminal or the acquirer software must convert this one-byte numeric value to a three-byte value with leading zeros in DE 23. For example, if the value of EMV tag 5F34 on the chip card is 2, then the value of DE 23 is 002. For chip or contactless chip transactions, if EMV tag 5F34 is not personalized (not present) on the chip, DE 23 cannot be present in the authorization message.

When DE 23 (Card Sequence Number) is present in an Authorization Request/0100 message and the value is not in the range of 000–099, then the Dual Message Authorization System forwards the Authorization Request Response/0110 where DE 39 = 30 (Format Error) and DE 44 (Additional Response Data) = 023.

If the account range is registered for the Authentication Indicator Type 1 Service, then DE 23 should be present.

Issuers:

The Dual Message Authorization System does not cross-edit the presence or absence of DE 23 and the value in DE 22 (Point-of-Service [POS] Entry Mode) in Authorization Request/0100

messages. However, if DE 22 has a value 05 (PAN auto-entry via chip) or 07 (PAN auto-entry via contactless M/Chip) and if DE 23 is present in the Authorization Request/0100, DE 23 contains the card sequence number from the chip. If DE 22 has a value other than 05 or 07 and DE 23 is present, then [DE 23](#) may contain erroneous information unrelated to the chip transaction.

Because of the potential for DE 23 to be present in Authorization/01xx messages, chip issuers must be prepared to receive store-and-forward (SAF) records containing DE 23.

Issuers must not return DE 23 in an Authorization Request Response/0110. If an issuer does so, Mastercard will delete DE 23 before passing the Authorization Request Response/0110 to the acquirer.

### **Brazil-issued account ranges with Flex Card Service enabled in Brazil, for acquirers**

The third position in [DE 23](#) will determine the routing of a Flex Card service enabled Point of Sale (POS) transaction acquired outside of Brazil, as follows:

- Values 0-4 indicates that the cardholder has selected the credit option. The transaction will be routed and processed as credit.
- Values 5-9 indicates that the cardholder has selected the debit option for processing; debit or credit interchange may apply to the transaction. Acquirers may be required to submit the corresponding interchange if Debit is selected and differentiated from credit for processing and interchange:
  - C = Credit Transaction if DE 3, subfield 2 is not equal to 10 or 20.
  - D = Debit Transaction if DE 3, subfield 2 is equal to 10 or 20.

### **Application notes**

**NOTE: The following feature only applies to tokens that use a secure element or cloud token type.**

DE 23 is not provided unless the issuer opts to receive it.

For more information, refer to the *Mastercard Digital Enablement Service Issuer Implementation Guide*.

## **DE 24 (Network International ID)**

[DE 24 \(Network International ID\)](#) identifies a single international network of card issuers.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 25 (Point-of-Service [POS] Condition Code)

[DE 25 \(Point-of-Service \[POS\] Condition Code\)](#) is an ID of the condition under which the transaction takes place at the point of interaction.

**NOTE:** The Dual Message Authorization System currently does not use this data element.

## DE 26 (Point-of-Service (POS) Personal ID Number [PIN] Capture Code)

DE 26 (Point-of-Service [POS] Personal ID Number [PIN] Capture Code) indicates the technique, maximum number, or both of PIN characters that can be accepted by the POS terminal used to construct the PIN data.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–2 |
| Subfields           | N/A                       |
| Justification       | Right                     |

### Usage

Following is the usage of DE 26 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | P   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |

## Values

### Applicable values

| Values | Description                                                                 |
|--------|-----------------------------------------------------------------------------|
| 00–03  | Invalid                                                                     |
| 04–12  | Indicates the maximum number of PIN characters that the terminal can accept |
| 13–99  | Reserved                                                                    |

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 26 must be used to indicate the maximum number of PIN characters that the acquiring terminal device (for example; ATM and POS terminal) is capable of accepting.

The Dual Message Authorization System requires that this data element be included in an Authorization Request/0100 message only when DE 52 (Personal ID Number [PIN] Data) or DE 110 (Encryption Data), Dataset 01 (PIN Encryption) is present and the maximum PIN character acceptance capability of the terminal is known to be other than 12 digits.

**NOTE: This data element is not used to specify the number of PIN characters actually accepted by a POS terminal device.**

## DE 27 (Authorization ID Response Length)

[DE 27 \(Authorization ID Response Length\)](#) is the maximum length of the authorization response that the acquirer can accommodate. The issuer or its agent is expected to limit response to this length.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

ISO and Mastercard defines [DE 38 \(Authorization ID Response\)](#) to be a six-character data element. All acquirers and issuers are expected to be able to accommodate and use the six-character data element for all authorization ID response codes.

## DE 28 (Amount, Transaction Fee)

*DE 28 (Amount, Transaction Fee)* is the fee charged (for example, by the acquirer) for transaction activity in the currency of DE 4 (Amount, Transaction).

### Attributes

| Attribute           | Description   |
|---------------------|---------------|
| Data Representation | an-9          |
| Data Length         | N/A           |
| Data Field          | N/A           |
| Subfields           | 2             |
| Justification       | See subfields |

### Usage

Following is the usage of DE 28 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | P   | C   |
| Authorization Request Response/0110                                                             | OE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated                                                   | C   | P   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | OE  | XE  | CE  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | OE  | XE  | .   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | XE  | CE  |
| Reversal Request/0400                                                                           | C   | P   | C   |
| Reversal Request Response/0410                                                                  | OE  | XE  | CE  |
| Reversal Advice/0420                                                                            | .   | X   | C   |



| Message                       | Org | Sys | Dst |
|-------------------------------|-----|-----|-----|
| Reversal Advice Response/0430 | OE  | XE  | .   |

## Values

This data element may be present in a message whenever an online transaction fee is permitted by the operating rules of a bank card product.

The credit or debit indicator (the first position of the data element) applies to the message recipient. Within acquirer-generated message types, a D (debit) fee amount indicates that the fee is to be applied as a debit to the message recipient, the issuer, (and therefore as a credit to the message originator, the acquirer).

## Application notes

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network business day, for all customers.

**NOTE: For acquirers that are approved to levy transaction fees, DE 28 must contain the transaction fee amount, and this amount also must be added to the requested amount contained in DE 4 (Amount, Transaction).**

## Edits

| WHEN...                                                                                                                                                                      | THEN the Dual Message Authorization System...                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 28 is not provided in an Authorization Request Response/0110 or Reversal Request Response/0410 or is provided but it contains a value different from the original request | populates the DE 28 value from the original Authorization Request/0100 or Reversal Request/0400 message in the response message before sending to the acquirer. |
| an Authorization Request/0100 message or Reversal Request/0400 message contains DE 28, and subfield 1 is not C or D or if subfield 2 is zeros or is greater than DE 4        | forwards to the acquirer an Authorization Request Response/0110 message or Reversal Request Response/0410 where DE 39 = 30 (Format Error) and DE 44 = 028.      |

## Subfield 1 (Debit/Credit Indicator)

[DE 28](#), subfield 1 (Debit/Credit Indicator) indicates the program type.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | a-1                    |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

| Value | Description |
|-------|-------------|
| C     | Credit      |
| D     | Debit       |

### Application notes

| WHEN an acquirer sends an Authorization Advice/0120 message and...                                              | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">DE 28</a> , subfield 1 (Debit/Credit Indicator) contains a value other than C (Credit) or D (Debit) | forwards to the acquirer an Authorization Advice Response/0130 message where: <ul style="list-style-type: none"><li><a href="#">DE 39 (Response Code)</a> = 30 (Format error).</li><li><a href="#">DE 44 (Additional Response Data)</a> = 028 (indicating the data element in error).</li></ul> |

## Subfield 2 (Amount)

*DE 28*, subfield 2 (Amount) indicates the fee amount in the currency of *DE 4 (Amount, Transaction)*.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-8                       |
| Data Field          | Contents of positions 2–9 |
| Justification       | Right with leading zeros  |

### Values

The fee amount must not contain all zeros or be greater than *DE 4 (Amount, Transaction)*.

## DE 29 (Amount, Settlement Fee)

*DE 29 (Amount, Settlement Fee)* is the fee to be transferred between the acquirer and the issuer equal to *DE 28 (Amount, Transaction Fee)* in the currency of *DE 5 (Amount, Settlement)*.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network's business day, for all customers.

## DE 30 (Amount, Transaction Processing Fee)

*DE 30 (Amount, Transaction Processing Fee)* is the fee charged (for example, by the acquirer, issuer, or INF) for the handling and routing of messages in the currency of *DE 4 (Amount, Transaction)*.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network's business day, for all customers.

## DE 31 (Amount, Settlement Processing Fee)

*DE 31 (Amount, Settlement Processing Fee)* is the fee charged (for example, by the acquirer, issuer, or INF) for the handling and routing of messages in the currency of *DE 5 (Amount, Settlement)*.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network's business day, for all customers.

## DE 32 (Acquiring Institution ID Code)

*DE 32 (Acquiring Institution ID Code)* identifies the acquiring institution (for example, merchant bank) or its agent.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-6                       |
| Data Length         | 2 positions               |
| Data Field          | Contents of positions 1–6 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 32 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | M   | P   | M   |
| Authorization Request Response/0110           | ME  | XE  | ME  |
| Authorization Advice/0120: acquirer-generated | M   | P   | M   |

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | XE  | ME  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | ME  | XE  | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | XE  | ME  |
| Reversal Request/0400                                                                           | M   | P   | M   |
| Reversal Request Response/0410                                                                  | ME  | XE  | ME  |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | •   |
| Administrative Request/0600                                                                     | M   | P   | M   |
| Administrative Request Response/0610                                                            | ME  | XE  | ME  |

## Values

A Mastercard customer ID number that Mastercard assigned to the entity acting as the acquiring institution for a transaction.

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 32 must contain a six-digit customer ID number assigned by Mastercard that identifies the institution acting as the "acquiring bank" or "merchant bank" for a transaction.

When an institution acts as the customer or intermediate network facility (INF) for an acquirer, the institution must:

- Provide the acquirer's Mastercard-assigned six-digit customer ID in DE 32
- Provide its Mastercard-assigned six-digit customer ID in DE 33 (Forwarding Institution ID Code)

The Mastercard customer ID number must be set up in the Dual Message Authorization System for participants to process Administrative Request/06xx messages.

## Errors

| IF...                                                                                                                                          | THEN the Dual Message Authorization System...                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in a Mastercard transaction, the acquirer submits a transaction with a value of 90 in DE 22 (Point-of-Service [POS] Entry Mode)                | DE 45 (Track 1 Data) or DE 35 (Track 2 Data) must be present, and DE 32 must contain the proper Mastercard customer ID number.                                                       |
| the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message contains DE 32 and DE 32 is not 6 digits in length | forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 where DE 39 = 30 (Format Error) and DE 44 = 032. |

For bridged transactions acquired on the Single Message System, issuers will receive the 6-digit customer ID number assigned by Mastercard that identifies the institution acting as the "acquiring bank" or "merchant bank" for a transaction in DE 32 (Acquiring Institution Identification Code) of the authorization message instead of a Mastercard-assigned internal ICA.

**NOTE: Transactions that have passed between platforms (Single Message System to Dual Message Authorization System, Dual Message Authorization System to Single Message System) are referred to as bridged transactions.**

## Visa Transactions

Acquirers must use their Mastercard-assigned customer ID in DE 32 of Visa-branded authorizations, and must also provide their Visa-assigned acquirer BIN to the Global Customer Service team.

## DE 33 (Forwarding Institution ID Code)

[DE 33 \(Forwarding Institution ID Code\)](#) identifies the institution forwarding a Request or Advice message in an interchange system if it is not the same institution as specified in DE 32 (Acquiring Institution ID Code). DE 33 is used within a message to contain the Mastercard six-digit customer ID number of the customer or INF responsible for directly routing that message to the Dual Message Authorization System.

## Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | n-6         |
| Data Length         | 2 positions |

| Attribute     | Description               |
|---------------|---------------------------|
| Data Field    | Contents of positions 1–6 |
| Subfields     | N/A                       |
| Justification | N/A                       |

## Usage

Following is the usage of DE 33 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | P   | C   |
| Authorization Request Response/0110                                                             | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated                                                   | C   | P   | C   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | XE  | CE  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | XE  | CE  |
| Issuer File Update Request/0302                                                                 | M   | X   | •   |
| Issuer File Update Request Response/0312                                                        | •   | XE  | ME  |
| Reversal Request/0400                                                                           | C   | P   | C   |
| Reversal Request Response/0410                                                                  | CE  | XE  | CE  |
| Reversal Advice/0420                                                                            | •   | X   | C   |
| Reversal Advice Response/0430                                                                   | CE  | XE  | •   |
| Administrative Request/0600                                                                     | C   | P   | C   |
| Administrative Request Response/0610                                                            | CE  | XE  | CE  |
| Administrative Advice/0620: system-generated                                                    | •   | X   | M   |
| Administrative Advice/0620: member-generated                                                    | M   | P   | M   |
| Administrative Advice Response/0630                                                             | ME  | XE  | •   |

| Message                                                                               | Org | Sys | Dst |
|---------------------------------------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: network connection status, system-generated          | •   | X   | M   |
| Network Management Request/0800: network connection status, member-generated          | M   | X   | •   |
| Network Management Request/0800: PEK exchange                                         | •   | X   | M   |
| Network Management Request/0800: PEK exchange, on demand                              | M   | X   | •   |
| Network Management Request/0800: host session activation/deactivation                 | M   | X   | •   |
| Network Management Request/0800: sign-on/sign-off                                     | M   | X   | •   |
| Network Management Request Response/0810: host session activation/deactivation        | •   | XE  | ME  |
| Network Management Request Response/0810: network connection status, member-generated | ME  | XE  | •   |
| Network Management Request Response/0810: network connection status, system-generated | •   | XE  | ME  |
| Network Management Request Response/0810: PEK exchange                                | ME  | XE  | •   |
| Network Management Request Response/0810: PEK exchange, on demand                     | •   | XE  | ME  |
| Network Management Request Response/0810: sign-on/sign-off                            | •   | XE  | ME  |
| Network Management Advice/0820: PEK exchange                                          | •   | X   | M   |

## Values

This data element, when used, must contain a valid Mastercard customer ID number.

## Application notes

DE 33 must be present in messages together with DE 32 whenever the Mastercard customer ID number of a customer or INF is different than the Mastercard customer ID number of the actual acquiring institution.

Automated Referral Service (ARS) store-and-forward transactions are included with other Stand-In processing transactions and are identified with a 003850 in DE 33 of the Authorization Advice/0120 message.

The Mastercard customer ID number must be set up in the Dual Message Authorization System for participants to process Administrative Request/06xx messages.

If an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message contains DE 33 and DE 33 is not 6 digits in length, then the Dual Message Authorization System will forward to the acquirer an Authorization Request Response/0110,



Authorization Advice Response/0130, or Reversal Request Response/0410 where DE 39 = 30 (Format Error) and DE 44 = 033.

## DE 34 (Primary Account Number [PAN], Extended)

*DE 34 (Primary Account Number [PAN], Extended)* identifies a customer account or relationship, and is used only when PAN begins with a 59 BIN.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 35 (Track 2 Data)

*DE 35 (Track 2 Data)* is the information encoded on track 2 of the card magnetic stripe as defined in the ISO 7813 specification, including data element separators but excluding beginning and ending sentinels and longitudinal redundancy check (LRC) characters as defined therein.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...37; LLVAR            |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–37 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Usage

Following is the usage of DE 35 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | O   | X   | O   |

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated | .   | X   | C   |

## Values

Whenever DE 35 is captured automatically at the point of interaction, DE 35 must contain whatever is encoded on the magnetic stripe (track 2) of the card, regardless of whether the card has been properly encoded with information in accordance with ISO specifications. The ISO 7810, 7811, 7812, and 7813 specifications document the international standards for encoding information on magnetic stripe cards.

For chip transactions, DE 35 carries data read from the chip as EMV tag 57 (Track 2 Equivalent Data). All ICCs issued by Mastercard customers must support EMV tag 57 (Track 2 Equivalent Data). Since January 2008, the value of the CVC in EMV tag 57 (Track 2 Equivalent Data) on the chip and the CVC value on the physical magnetic stripe must be different.

DE 35 must contain the hexadecimal digits 0 through 9 and D or = (the equal sign).

## Application notes

The account number in DE 2 (Primary Account Number [PAN]) or [DE 34 \(Primary Account Number \[PAN\], Extended\)](#) must match the account number in DE 35.

This data element is defined and used identically within all Mastercard programs and services.

DE 35 is mandatory for all chip transactions (if missing, the Dual Message Authorization System will not reject the message; however, a data integrity error will be reported).

DE 35 is mandatory for all ATM transactions. For Maestro transactions, such as e-commerce, the Dual Message Authorization System creates track 2 data if it is not provided by the acquirer.

DE 35 is optional for 0120 AFD completion advice. Refer to [Automated Fuel Dispenser Completion](#) for more detail.

For MDES tokenized transactions, issuers may opt in through MDES Manager to receive track data in DE 35 (Track 2 Data) on Dual Message Authorization System or Single Message System transactions to issuers.

The field separator character (binary 1101) is represented as the EBCDIC character D. However, because many ATM and POS devices perform nonstandard character translation while reading binary coded decimal (BCD) encoded magnetic stripe data, the EBCDIC character "=" may also be used to represent the field separator character in magnetic stripe data.

Following is Track 2 information content and format for Mastercard transactions.

### Track 2 information content and format for Mastercard transactions

| Field ID | Field name                                                                                                     | F (Fixed), V (Variable) | Maximum length                                                                 |
|----------|----------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------|
| 1        | Start Sentinel<br>These fields are encoded on the card but must be omitted within Track 2 data.                | F                       | n-1                                                                            |
| 2        | Primary Account Number                                                                                         | V                       | n...19                                                                         |
| 3        | Separator (binary)                                                                                             | F                       | ans-1                                                                          |
| 4        | Expiration Date                                                                                                | F                       | ans-4                                                                          |
| 5        | Extended Service Code                                                                                          | F                       | ans-3                                                                          |
| 6        | Discretionary Data (must include CVC 1)                                                                        | V                       | Balance of available digits not to exceed total track length of 40 characters. |
| 7        | End Sentinel<br>These fields are encoded on the card but must be omitted within Track 2 data.                  | F                       | n-1                                                                            |
| 8        | Longitudinal Redundancy Check<br>These fields are encoded on the card but must be omitted within Track 2 data. | F                       | n-1                                                                            |

Refer to the *Security Rules and Procedures* for additional Track 2 data information.

## DE 36 (Track 3 Data)

*DE 36 (Track 3 Data)* is the information encoded on track 3 of the card magnetic stripe as defined in the ISO 4909:1986 specification, including data element separators but excluding beginning and ending sentinels and LRC characters as defined therein.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 37 (Retrieval Reference Number)

*DE 37 (Retrieval Reference Number)* is a document reference number supplied by the system retaining the original source document of the transaction and assists in locating that source document or a copy thereof. DE 37 is made available for use by automated merchant POS systems that may be interconnected into the interchange system. Merchant POS systems may assign a unique receipt or sales document ID to be used to satisfy regulatory or legal requirements when the merchant performs source document capture and truncation. DE 37 may be used to relay source document reference numbers to the issuer at the time each transaction is processed.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans-12                     |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–12 |
| Subfields           | 2 (for chip transactions)  |
| Justification       | See subfields              |

### Usage

Following is the usage of DE 37 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | P   | C   |
| Authorization Request Response/0110                                                             | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated                                                   | C   | P   | C   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | XE  | CE  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | •   |

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | XE  | CE  |
| Reversal Request/0400                                                                           | C   | P   | C   |
| Reversal Request Response/0410                                                                  | CE  | XE  | CE  |
| Reversal Advice/0420                                                                            | .   | X   | C   |
| Reversal Advice Response/0430                                                                   | CE  | XE  | .   |

## Values

Number assigned by the acquirer.

When present, DE 37 must contain a value other than all zeros or all spaces.

DE 37 can have trailing spaces if the transaction is not a Visa or EMV chip transaction.

## Application notes

The Authorization System will pass DE 37, unaltered, to the receiving entity. The receiving entity must return this data element within any corresponding response.

DE 37 is mandatory for all chip transactions (DE 22 = 05x or 07x), chip fallback transactions (DE 22 = 80x), as well as contactless magnetic stripe transactions (DE 22 = 91x). If DE 37 is not present, the Dual Message Authorization System will not reject the message.

DE 37 is mandatory for all ATM transactions.

DE 37 is mandatory for all other card read transactions. (If missing, the Dual Message Authorization System will not reject the message.)

Visa transactions: Acquirers that use the Mastercard Network must populate CIS DE 37 with a 12-position, all numeric value containing a valid Julian date, in Visa Authorization Request/0100, Authorization Advice/0120 and Reversal Request/0400 messages.

The format for CIS DE 37 on Visa transactions is (ydddhnnnnnnn):

- yddd (positions 1–4) is the year and day of year, equivalent of the value from DE 7 (Transmission Date and Time), y=0–9, ddd=001–366
- hh (positions 5 and 6) is the hours value from the time in DE 7 (Transmission Date and Time)
- nnnnnn (positions 7–12) is the value from DE 11 (Systems Trace Audit Number)

Visa will only edit positions 1–4. To avoid problems with system edits that may detect and reject duplicate and reused CIS DE 37 values, Visa recommends that endpoints construct CIS DE 37, positions 5–12 from the data in CIS DE 7 and CIS DE 11.

Acquirers may construct DE 37 in the same manner for Mastercard, as is indicated above for Visa messages.

## Subfield 1 (Transaction Date and Initiator Discretionary Data)

DE 37, subfield 1 (Transaction Date and Initiator Discretionary Data) is used to pass chip data. The value information is specific to chip transactions.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | ans-7                     |
| Data Field          | Contents of positions 1–7 |
| Justification       | Left                      |

### Values

The date (MMDD) the transaction is captured at the point-of-service terminal.

If no discretionary data is included, the remaining three positions of this subfield should be zero-filled.

This subfield is left-justified with trailing zeros.

## Subfield 2 (Terminal Transaction Number)

DE 37, subfield 2 (Terminal Transaction Number) is a unique number that identifies the transaction with a specific POS terminal within a specific 24 hour time period. The value information is specific to chip transactions.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans-5                      |
| Data Field          | Contents of positions 8–12 |
| Justification       | Right                      |

### Values

The Terminal Transaction Number: A sequential number, per terminal. Only numeric data may be present in this subfield. This subfield must contain a unique number that identifies the transaction with a specific POS terminal within a specific 24 hour time period.

Mastercard recommends that this subfield contain the value of the Transaction Sequence Counter (EMV tag 9F41), if available.

This subfield is right-justified with leading zeros.

## DE 38 (Authorization ID Response)

*DE 38 (Authorization ID Response)* is a transaction response ID code that the authorizing institution assigns. DE 38 is used to transmit a card issuer's authorization code for Authorization transactions.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | ans-6                     |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–6 |
| Subfields           | N/A                       |
| Justification       | Left                      |

### Usage

Following is the usage of DE 38 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

### Values

When present, DE 38 must be left-justified and cannot contain all spaces, embedded spaces, all low values, all high values, or all zeros.

## Application notes

In general, the authorization code used in DE 38 may be any combination of alphanumeric characters. However, the following table identifies the standard conventions the major brands use.

### Authorization codes format

| Card type        | Length | Authorization ID response code format     |
|------------------|--------|-------------------------------------------|
| Mastercard       | 6      | alphanumeric, with no special characters. |
| Visa             | 6      | alphanumeric, with no special characters. |
| American Express | 6      | alphanumeric, trailing space(s).          |
| Diners Club      | 6      | numeric, trailing space(s).               |
| Discover         | 6      | numeric, trailing space(s).               |

DE 38 is mandatory in the Authorization Request Response/0110 message when DE 39 (Response Code) contains the value 00 (approved), 08 (Honor with ID), 10 (Partial Approval), 85 (Not Declined), or 87 (Purchase amount only, no cash back allowed). If the transaction is approved by Stand-In processing, a six-digit authorization approval code beginning with "8" appears in this field.

DE 38, position 6 (Account Category Code) may contain an Account Level Management account category code when the account range is participating in an Account Level Management service. For a complete valid value listing, refer to the *Account Level Management Manual*.

### Issuer instructions for ALM

If DE 48, subelement 38 is present in the Authorization Request/0100 message and the issuer approves the request, then the Authorization Request Response/0110 message, DE 38, position 6 should contain the same value as received in DE 48, subelement 38.

Failure to include the category code in DE 38, position 6 on the approved authorization will cause the authorization request to be routed to and processed by the Stand-In System.

### Acquirer instructions for ALM

Acquirers should use DE 38, position 6 if the Authorization Request Response/0110 message was approved (DE 39 = 00, 08, 10, 85, or 87). Acquirers should use this value in their clearing process as indicated by IPM Clearing Member Parameter Extract (MPE). If the issuer provides DE 38 in an Authorization Request Response/0110 message and DE 39 is not an approval (00, 08, 10, 85, or 87), the Dual Message Authorization System removes the DE 38 value from the response to the acquirer.



## DE 39 (Response Code)

*DE 39 (Response Code)* defines the disposition of a previous message or an action taken as a result of receipt of a previous message. Response codes also are used to indicate approval or decline of a transaction. In the event an authorization is declined, the response code indicates the reason for rejection and may indicate an action to be taken at the acceptor (for example, capture card).

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–2 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 39 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110                                                             | M   | X   | M   |
| Authorization Advice/0120: acquirer-generated                                                   | M   | P   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | M   | P   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | M   | X   | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | X   | M   |
| Authorization Response Acknowledgment/0180                                                      | M   | X   | •   |
| Authorization Response Negative Acknowledgment/0190                                             | •   | X   | M   |

| Message                                                                               | Org | Sys | Dst |
|---------------------------------------------------------------------------------------|-----|-----|-----|
| Issuer File Update Request Response/0312                                              | .   | X   | M   |
| Reversal Request/0400                                                                 | M   | P   | M   |
| Reversal Request Response/0410                                                        | M   | P   | M   |
| Reversal Advice/0420                                                                  | .   | X   | M   |
| Reversal Advice Response/0430                                                         | M   | X   | .   |
| Administrative Request Response/0610                                                  | M   | P   | M   |
| Administrative Advice Response/0630                                                   | M   | X   | .   |
| Network Management Request Response/0810: host session activation/deactivation        | .   | X   | M   |
| Network Management Request Response/0810: network connection status, member-generated | M   | X   | .   |
| Network Management Request Response/0810: network connection status, system-generated | .   | X   | M   |
| Network Management Request Response/0810: PEK exchange                                | M   | X   | .   |
| Network Management Request Response/0810: PEK exchange: on demand                     | .   | X   | M   |
| Network Management Request Response/0810: sign-on/sign-off                            | .   | X   | M   |

## Values

Valid values are listed by message type.

Must contain value 32 for a partial reversal. For a full reversal, must contain value 06, 17, 34, or 68 as applicable, or if no other value applies, may contain the same value as in the original Authorization Request Response/0110 message.

## Application notes

DE 39 must be present in all Request Response, Advice Response, and Acknowledgment messages. In addition, it is also present in Authorization Advice/0120 Stand-In messages to indicate the response code used in the Authorization Request Response/0110 to the original Authorization Request/0100.

The following information applies to purchase only, no cash back allowed responses from an issuer.

### Associated edits

| The issuer must provide...                                                                                                                                                                                                                                                                                                                                                                                                                                       | The acquirer will receive...                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the approved amount (purchase amount) in DE 6 (Amount, Cardholder Billing) in the issuer's cardholder billing currency. This amount must be the purchase amount as calculated by subtracting the DE 54 (Additional Amounts) cash back amount and, if applicable, DE 28 (Amount, Transaction Fee) from the DE 6 amount present in the Authorization Request/0100 message in the amount data element that corresponds to the issuer's cardholder billing currency. | the purchase-only approval amount in the acquirer's transaction currency in DE 4                                                                                                                                                                                                                                   |
| DE 38 (Authorization ID Response)                                                                                                                                                                                                                                                                                                                                                                                                                                | DE 38                                                                                                                                                                                                                                                                                                              |
| DE 39, value 87 (Purchase only, no cash back allowed)                                                                                                                                                                                                                                                                                                                                                                                                            | DE 39, value 87                                                                                                                                                                                                                                                                                                    |
| DE 51 (Currency Code, Cardholder Billing) with the issuer's cardholder billing currency code                                                                                                                                                                                                                                                                                                                                                                     | an occurrence of the original amount of the transaction in DE 54 (Additional Amounts) in the acquirer's transaction currency. The original amount is identified by DE 54, subfield 2 (Amount Type), value 57 (Original Amount), and subfield 4 (Debit or Credit Indicator), value C plus 12-digit original amount. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | an occurrence of the original amount of the transaction in DE 54 in the issuer's cardholder billing currency. The original amount is identified by DE 54, subfield 2, value 57, and subfield 4, value C plus 12-digit original amount.                                                                             |

### Associated edits

| IF the issuer provides a response code value of...                                                                      | THEN the Dual Message Authorization System...                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 87 (Purchase Amount Only, No Cash Back Allowed) in the Authorization Request Response/0110 (applies to Mastercard only) | requires that in the Authorization Request Response/0100 the amount in DE 6 (Amount, Cardholder Billing) is not less than the amount in the original DE 6 (Amount, Cardholder Billing) in the Authorization Request/0100 minus the amount in DE54 (Additional Amounts). |

### Associated edits

| IF the issuer provides a response code value of...                   | THEN the Issuer Processing System (IPS)...                                                                                                                                                                          |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 (Format Error) in the Authorization Request Response/0110 message | also provides DE 44 with six positions for data element and subelement format errors (For example, DE44 value 022 indicates DE 22, 0480nn indicates DE48 subelement nn, and 112nnn indicates DE122 subelement nnn). |

The Dual Message Authorization System responds as indicated for Mastercard transactions only.

#### Associated edits

| IF the issuer provides a response code value of...                                    | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 91, 92, or 96 in the Authorization Request Response/0110 (applies to Mastercard only) | automatically invokes Stand-In processing. The Stand-In System processes the transaction according to issuer-specified parameters that generate the issuer-specified response (for example, approve, decline, refer, or capture card). If Stand-In processing not available, invokes Xcode processing according to Product rules. |
| 00, 08, 10, 85, or 87 in the Authorization Request Response/0110                      | requires that DE 38 (Authorization Identification Response) is present.                                                                                                                                                                                                                                                           |

#### Associated edits

| Invalid values in DE 39 for a Mastercard ATM transaction... | Instead, the issuer should use the following response code... |
|-------------------------------------------------------------|---------------------------------------------------------------|
| 01, 03, 08, 12, 15, 63, 76, 77, or 87                       | 57                                                            |
| 78                                                          | 14                                                            |
| 84 or 94                                                    | any valid response code                                       |

#### Associated edits

| WHEN the acquirer or issuer...                                                                                                                                                                          | THEN the Dual Message Authorization System...                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| sends a Network Management/0800 group sign-off message (DE 70 = 062 or 064) within three minutes of the previous 0800 message for the same sign-on Member Group ID (DE 2: Primary Account Number [PAN]) | generates the Network Management Response/0810 message where DE 39 (Response Code) = 96 (System error). |

## Authorization Request Response/0110 response codes

The following DE 39 response codes are valid in this message.

### Response codes

| Value | Description                                                                                                                                                                          | Action      | MC | NP | VI | TE | MS | CI |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----|----|----|----|----|----|
| 00    | Approved or completed successfully                                                                                                                                                   | Approve     | √  | √  | √  | √  | √  | √  |
| 01    | Refer to card issuer                                                                                                                                                                 | Call Issuer | √  | √  | √  | √  | √  |    |
| 03    | Invalid merchant                                                                                                                                                                     | Decline     | √  | √  | √  | √  | √  |    |
| 04    | Capture card                                                                                                                                                                         | Capture     | √  | √  | √  | √  | √  | √  |
| 05    | Do not honor<br>Issuers should use this response code only when no other reason code applies.                                                                                        | Decline     | √  | √  | √  | √  | √  |    |
| 08    | Honor with ID                                                                                                                                                                        | Approve     | √  | √  | √  | √  | √  |    |
| 10    | Partial Approval                                                                                                                                                                     | Approve     | √  | √  | √  |    | √  |    |
| 12    | Invalid transaction                                                                                                                                                                  | Decline     | √  | √  | √  | √  | √  |    |
| 13    | Invalid amount                                                                                                                                                                       | Decline     | √  | √  | √  | √  | √  | √  |
| 14    | Invalid card number                                                                                                                                                                  | Decline     | √  | √  | √  | √  | √  | √  |
| 15    | Invalid issuer                                                                                                                                                                       | Decline     | √  | √  | √  | √  | √  |    |
| 30    | Format error                                                                                                                                                                         | Decline     | √  | √  | √  | √  | √  | √  |
| 41    | Lost card                                                                                                                                                                            | Capture     | √  | √  | √  | √  | √  | √  |
| 43    | Stolen card                                                                                                                                                                          | Capture     | √  | √  | √  | √  | √  | √  |
| 46    | Closed Account                                                                                                                                                                       | Decline     | √  | √  | √  | √  | √  | √  |
| 51    | Insufficient funds/over credit limit                                                                                                                                                 | Decline     | √  | √  | √  | √  | √  | √  |
| 54    | Expired card                                                                                                                                                                         | Decline     | √  | √  | √  | √  | √  | √  |
| 55    | Invalid PIN                                                                                                                                                                          | Decline     | √  | √  | √  | √  | √  | √  |
| 57    | Transaction not permitted to issuer/cardholder<br>Issuers should use this response code when their host blocks transactions based on specific transaction processing or fraud rules. | Decline     | √  | √  | √  |    | √  | √  |

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Action      | MC | NP | VI | TE | MS | CI |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----|----|----|----|----|----|
| 58    | Transaction not permitted to acquirer/terminal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Decline     | √  | √  | √  |    | √  | √  |
| 61    | Exceeds withdrawal amount limit<br>Issuers may also use this response code for transaction types other than ATM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Decline     | √  | √  | √  | √  | √  | √  |
| 62    | Restricted card                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Decline     | √  | √  | √  | √  | √  | √  |
| 63    | Security violation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Decline     | √  | √  | √  | √  | √  |    |
| 65    | Exceeds withdrawal count limit OR Identity Check Soft-Decline of EMV 3DS Authentication (merchant should resubmit authentication for ACS fully authenticated response or due to failed AAV Validation- OBS 05 service)<br>Issuers may also use this response code for transaction types other than ATM transactions. Issuers that opt to decline a contactless or card-not-present transaction and obtain a cardholder verification through a second transaction may use this response code. Refer to <a href="#">M/Chip Requirements for Contact and Contactless</a> for more details on contactless transactions and how to use response code 65. E-commerce merchants should trigger an authentication using Mastercard Identity Check following an authorization decline with response code 65. This could be due to an authentication not completed through an ACS function or due to a failed AAV Validation and a new authentication with new AAV & DS Transaction ID is necessary for the authorization to successfully complete AAV Validation (Dynamic Linking). | Decline     | √  | √  | √  | √  | √  | √  |
| 70    | Contact Card Issuer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Call Issuer | √  | √  |    |    | √  | √  |

| Value | Description                                                                                                                                                                                                                   | Action   | MC | NP | VI | TE | MS | CI |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----|----|----|----|----|----|
| 71    | PIN Not Changed                                                                                                                                                                                                               | Decline  | ✓  | ✓  |    |    | ✓  | ✓  |
| 72    | Account Not Yet Activated                                                                                                                                                                                                     | Decline  | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 75    | Allowable number of PIN tries exceeded                                                                                                                                                                                        | Decline  | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 76    | Invalid/nonexistent "To Account" specified                                                                                                                                                                                    | Decline  | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 77    | Invalid/nonexistent "From Account" specified                                                                                                                                                                                  | Decline  | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 78    | Invalid/nonexistent account specified (general)                                                                                                                                                                               | Decline  | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 79    | Life cycle (Mastercard use only)                                                                                                                                                                                              |          |    |    |    |    |    |    |
| 81    | Domestic Debit Transaction Not Allowed (Regional use only)                                                                                                                                                                    | Decline  | ✓  |    |    |    |    |    |
| 82    | Policy (Mastercard use only)                                                                                                                                                                                                  |          |    |    |    |    |    |    |
| 83    | Fraud/Security (Mastercard use only)                                                                                                                                                                                          |          |    |    |    |    |    |    |
| 84    | Invalid Authorization Life Cycle                                                                                                                                                                                              | Decline  |    | ✓  |    |    |    |    |
| 85    | Not declined<br><br>Valid for all zero amount transactions.                                                                                                                                                                   | Valid    | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 86    | PIN Validation not possible                                                                                                                                                                                                   | Decline  | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 87    | Purchase Amount Only, No Cash Back Allowed                                                                                                                                                                                    | Approved | ✓  |    |    |    | ✓  |    |
| 88    | Cryptographic failure<br>This could be due to a failed AAV Validation and a new authentication with new AAV & DS Transaction ID is necessary for the authorization to successfully complete AAV Validation (Dynamic Linking). | Decline  | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 89    | Unacceptable PIN—Transaction Declined—Retry                                                                                                                                                                                   | Decline  | ✓  | ✓  |    |    | ✓  | ✓  |
| 91    | Authorization System or issuer system inoperative                                                                                                                                                                             | See Note | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 92    | Unable to route transaction                                                                                                                                                                                                   | See Note | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 94    | Duplicate transmission detected                                                                                                                                                                                               | Decline  | ✓  | ✓  | ✓  | ✓  | ✓  |    |

| Value                                                                                                                                                                                                                                                                                                                                                                                                                           | Description  | Action   | MC | NP | VI | TE | MS | CI |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|----|----|----|----|----|----|
| 96                                                                                                                                                                                                                                                                                                                                                                                                                              | System error | See Note | √  | √  | √  | √  | √  | √  |
| <p>Note to the issuer: When a response code of 91, 92, or 96 is received from an issuer, Mastercard will attempt to perform alternate authorization processing (Alternate Issuer, Stand-In or X-Code Processing) according to the rules associated to the Mastercard brand and product.</p> <p>Note to the acquirer: When a response code of 91, 92, or 96 is returned to the acquirer, the transaction should be declined.</p> |              |          |    |    |    |    |    |    |

## Best practices for DE 39 values in Authorization Request Response/0110 response codes

### For Approvals

| Response Code Value | Description                        | Recommended Usage                                                                                                                                                                                                                                                                                                          |
|---------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00                  | APPROVED OR COMPLETED SUCCESSFULLY | To be used for approved transactions.                                                                                                                                                                                                                                                                                      |
| 08                  | HONOR WITH ID                      | <p>To be used when issuer needs to verify the identity of a cardholder for certain types of transactions, such as manual cash disbursements.</p> <p>This response should not be returned for ATM transactions or cardholder-activated (unattended) terminal transactions.</p>                                              |
| 10                  | PARTIAL APPROVAL                   | To be used for approvals of an amount that is less than requested. Must be used only when partial approval service is supported by merchant. Unless the transaction is identified with Processing Code 00 and MCC 5542 (Fuel Dispenser, Automated), the amount in DE 6 must be less than or equal to the requested amount. |
| 85                  | NOT DECLINED                       | To be used for account status inquiry (ASI) transactions and other non-financial zero amount transactions.                                                                                                                                                                                                                 |



| Response Code Value | Description                                | Recommended Usage                                                                                                                      |
|---------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 87                  | PURCHASE AMOUNT ONLY, NO CASH BACK ALLOWED | To be used for the purchase-only approval of a purchase with cash back transaction (Processing Code 09) when cash back is not allowed. |

### For Referral Responses

| Response Code Value | Description          | Recommended Usage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01                  | REFER TO CARD ISSUER | <p>To be used only for referral transactions/services, where the merchant/acquirer is required to call the issuer to complete the transaction:</p> <ul style="list-style-type: none"> <li>When DE 39 = 01, then DE 44 contains the phone number for "call issuer."</li> <li>Not allowed for ecommerce, Mail Order/Telephone Order (MO/TO), MCC 6011 (Automated Cash Disbursements—Customer Financial Institution), cardholder-activated terminal (CAT) transactions including MCC 5542 (Fuel Dispenser, Automated), fast food restaurant transactions (MCC 5814), or direct marketing transactions (MCCs 5960–5969). If necessary, use response code 57 (Transaction not permitted to issuer/cardholder).</li> <li>Referral transactions are rarely used as the probability of the merchant calling the issuer is very low.</li> </ul> |

### For Technical/Validation Declines

| Response Code Value | Description         | Recommended Usage                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12                  | INVALID TRANSACTION | <p>To be used in the following situations:</p> <ul style="list-style-type: none"> <li>Number of installments is not allowed.</li> <li>Type of transaction not allowed (To be used to indicate that the service attempted or type of transaction is not configured by or for issuer).</li> <li>For fallback transaction declines, use response code 58 (Transaction not permitted to acquirer/terminal).</li> </ul> |

| Response Code Value | Description                                       | Recommended Usage                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30                  | FORMAT ERROR                                      | To be used when the information in one or more data elements is not consistent or there is a failure in validation among data elements and applicable subelements. In this case, DE 44 also must be populated indicating in which data element or subelement is the problem. (For example, if the error is in one of the subelements of DE 48, 048 will be shown in DE 44 as well as the three digit subelement number causing the format error).                              |
| 71                  | PIN NOT CHANGED                                   | To be used when PIN change request was not successfully completed.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 86                  | PIN VALIDATION NOT POSSIBLE                       | To be used in case of PIN transactions where the issuer system is unable to perform PIN validation.                                                                                                                                                                                                                                                                                                                                                                            |
| 88                  | CRYPTOGRAPHIC FAILURE                             | To be used anytime cryptogram validation or related key processing is unsuccessful.                                                                                                                                                                                                                                                                                                                                                                                            |
| 91                  | AUTHORIZATION SYSTEM OR ISSUER SYSTEM INOPERATIVE | <p>To be used in the following situations:</p> <ul style="list-style-type: none"> <li>• System unavailability</li> <li>• System timeout</li> </ul> <p>In this case, Mastercard automatically invokes Stand-In processing or "X-Code processing (if allowed for card product)." The Stand-In System processes the transaction according to issuer-specified parameters that generate the issuer-specified response (for example, approve, decline, refer, or capture card).</p> |
| 92                  | UNABLE TO ROUTE THE TRANSACTION                   | To be used when transaction not able to be routed to authorizer. In this case, Mastercard automatically invokes Stand-In processing or "X-Code processing." The Stand-In System processes the transaction according to issuer-specified parameters that generate the issuer-specified response (for example, approve, decline, refer, or capture card).                                                                                                                        |
| 94                  | DUPLICATE TRANSMISSION DETECTED                   | To be used when the same transaction is submitted more than once, such as when multiple authorization requests are received for the same recurring payment transaction.                                                                                                                                                                                                                                                                                                        |
| 96                  | SYSTEM ERROR                                      | To be used for general system failures. In this case, Mastercard automatically invokes Stand-In processing or "X-Code processing." The Stand-In System processes the transaction according to issuer-specified parameters that generate the issuer-specified response (for example, approve, decline, refer, or capture card).                                                                                                                                                 |

### Fraud/Security-related Declines

| Response Code Value                                                                          | Description        | Recommended Usage                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04                                                                                           | CAPTURE CARD       | To be used in case of: <ul style="list-style-type: none"> <li>Blocked/cancelled cards due to confirmed previous fraudulent situation (other than lost stolen).</li> <li>Fraudulent activities at account level (such as identity theft or Account Take Over).</li> </ul> |
| 41                                                                                           | LOST CARD          | To be used in case of blocked/cancelled cards previous confirmed as a "lost card" by the cardholder.                                                                                                                                                                     |
| 43                                                                                           | STOLEN CARD        | To be used in case of blocked/cancelled cards previous confirmed as a "stolen card" by the cardholder.                                                                                                                                                                   |
| 55                                                                                           | INVALID PIN        | To be used in case of PIN required transactions with failure in PIN validation (wrong PIN) or missing PIN.                                                                                                                                                               |
| 63                                                                                           | SECURITY VIOLATION | To be used in case of suspected fraudulent transactions. It must be used as response code for fraud prevention real-time declines or for accounts/cards temporally blocked due to suspected fraudulent use. Should not be used where card is reported lost or stolen.    |
| <b>NOTE: For ATM, use response code 57 (Transaction not permitted to issuer/cardholder).</b> |                    |                                                                                                                                                                                                                                                                          |

### For Credit-related Declines

| Response Code Value | Description                          | Recommended Usage                                                                                                                                                                                                                                   |
|---------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 51                  | INSUFFICIENT FUNDS/OVER CREDIT LIMIT | To be used if the account has reached or exceeded its credit limit or the available balance is insufficient. This code must also be used in case of a temporary delinquency situation (cards not definitively blocked—in early delinquency period). |

### For Policy Declines

| Response Code Value | Description                                    | Recommended Usage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03                  | INVALID MERCHANT                               | <p>To be used in following situations:</p> <ul style="list-style-type: none"> <li>• Merchant restricted by issuer—risk or internal product policy; or</li> <li>• Card acceptor business code (MCC) restricted by issuer - risk or internal product policy.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 13                  | INVALID AMOUNT                                 | <p>To be used in following situations:</p> <ul style="list-style-type: none"> <li>• Amount is outside of the limits defined by issuer (general/policy limit). For individual credit limit, response code 51 (Insufficient funds/over credit limit) must be used; or</li> <li>• Amount is outside of limits defined by rule or regulation (for example, an installment payment less than the minimum amount allowed or a Gaming Payment Transaction exceeding the maximum amount allowed); or</li> <li>• Transaction exceeds issuer's policy threshold amount for unique transaction (such as USD 1,000,000 for a single transaction).</li> </ul>                                                                                                                                                                                                                              |
| 57                  | TRANSACTION NOT PERMITTED TO ISSUER/CARDHOLDER | <p>To be used in following situations:</p> <ul style="list-style-type: none"> <li>• Restriction at the product level</li> <li>• Restriction at the BIN level</li> </ul> <p>Restriction by internal policy (not related to merchants or MCC, in which case Response Code 03 (Invalid merchant) must be used)</p> <p>Restricted countries (such as OFAC list)</p> <p>Domestic Use Only cards used for cross border transactions</p> <p>Transaction without previous customer notification due to issuer policy restrictions</p> <p>Response Code used by Mastercard tools such as Fraud Rule Manager</p> <p><b>NOTE: Response Code 57—TRANSACTION NOT PERMITTED TO ISSUER/CARDHOLDER is used when an issuer transaction is blocked based on issuer set policies outside of fraud. For fraud-related transactions, response code 63 (Security violation) should be used.</b></p> |

| Response Code Value | Description                                               | Recommended Usage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 58                  | TRANSACTION NOT PERMITTED TO ACQUIRER/ TERMINAL           | <p>To be used when the transaction POS Entry Mode or the terminal capability does not match the transaction type according to issuer's policies. May be used in the case of fallback transactions.</p> <p>May also be used to indicate a merchant restriction (for example, the merchant does not participate in a private label program or is not permitted to offer installments).</p>                                                                                                                                                                                                        |
| 61                  | EXCEEDS WITHDRAWAL AMOUNT LIMIT                           | <p>To be used in case of withdrawal transactions above amount limit (general or daily) set by issuer according to internal policies.</p> <p>Code 61 should be used both when a transaction exceeds the withdrawal amount and when it exceeds the withdrawal count limit.</p>                                                                                                                                                                                                                                                                                                                    |
| 62                  | RESTRICTED CARD                                           | <p>To be used in following situations:</p> <ul style="list-style-type: none"> <li>• Restriction at card level (for product level, response code 57 must be used)</li> <li>• Card or Account temporarily blocked (for permanent blocking/cancellation, response code 46 must be used)</li> </ul> <p><b>NOTE: For fraud-related transactions, response code 63 (Security violation) should be used.</b></p>                                                                                                                                                                                       |
| 65                  | EXCEEDS WITHDRAWAL COUNT LIMIT OR REQUIRES AUTHENTICATION | <p>To be used in the following situations:</p> <ul style="list-style-type: none"> <li>• In the card-present space (contactless transactions and cash withdrawals): <ul style="list-style-type: none"> <li>– When strong customer authentication (SCA) is required, such as PIN entry</li> <li>– Or when the number of cash withdrawal transactions (ATM) is above the count limit (general or daily) set by the issuer according to internal policies.</li> </ul> </li> <li>• In the card-not-present space only when issuers require SCA to be performed on the electronic payment.</li> </ul> |
| 75                  | ALLOWABLE NUMBER OF PIN TRIES EXCEEDED                    | <p>To be used when the quantity of PIN tries exceeded the limit defined by the issuer.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### Life Cycle Declines

| Response Code Value | Description               | Recommended Usage                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14                  | INVALID CARD NUMBER       | <p>To be used in situations where:</p> <ul style="list-style-type: none"> <li>The account number (PAN) has not been issued by the issuer, is not valid, or is closed.</li> <li>Card expiration date doesn't match what is on the card.</li> </ul> <p><b>NOTE: For past dates—expired cards—use response code 54 (Expired Card).</b></p>                                                                                   |
| 46                  | Closed Account            | To be used when an issuer or cardholder permanently closes a credential or underlying account and no further authorization requests will be approved.                                                                                                                                                                                                                                                                     |
| 54                  | EXPIRED CARD              | <p>To be used in case of cards with expired dates (dates in the past)</p> <ul style="list-style-type: none"> <li>DE 48 subelement 84 (Merchant Advice Code) must equal 03, indicating "Do not try again".</li> <li>If the card is expired, but they have a new card with same PAN but updated expire date, the issuer can provide DE 48, subelement 84 with a value of 01 (New account information available).</li> </ul> |
| 72                  | Account Not Yet Activated | To be used when an authorization request is received but cannot be approved because the account has not been activated, such as in the case of a newly issued account.                                                                                                                                                                                                                                                    |

### Other Declines

| Response Code Value | Description  | Recommended Usage                                                                                                                                                                                                                                                                                                          |
|---------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05                  | DO NOT HONOR | <p>Only allowed when it is necessary to map a non-technical decline not specifically described in above tables.</p> <p>Issuers should use this response code only when no other response code applies.</p> <p>Mastercard uses response code 05 during Stand-In processing and via Fraud Rules Manager and other tools.</p> |

## Authorization Advice/0120 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description                                    | Action      | MC | NP | VI | TE | MS | CI |
|-------|------------------------------------------------|-------------|----|----|----|----|----|----|
| 00    | Approved or completed successfully             | Approve     | √  | √  | √  | √  | √  | √  |
| 01    | Refer to card issuer                           | Call Issuer | √  | √  | √  | √  | √  |    |
| 03    | Invalid merchant                               | Decline     | √  | √  | √  | √  | √  |    |
| 04    | Capture card                                   | Capture     | √  | √  | √  | √  | √  | √  |
| 05    | Do not honor                                   | Decline     | √  | √  | √  | √  | √  |    |
| 08    | Honor with ID                                  | Approve     | √  | √  | √  | √  | √  |    |
| 10    | Partial Approval                               | Approve     | √  | √  | √  |    | √  |    |
| 12    | Invalid transaction                            | Decline     | √  | √  | √  | √  | √  |    |
| 13    | Invalid amount                                 | Decline     | √  | √  | √  | √  | √  | √  |
| 14    | Invalid card number                            | Decline     | √  | √  | √  | √  | √  | √  |
| 15    | Invalid issuer                                 | Decline     | √  | √  | √  | √  | √  |    |
| 30    | Format error                                   | Decline     | √  | √  | √  | √  | √  | √  |
| 41    | Lost card                                      | Capture     | √  | √  | √  | √  | √  | √  |
| 43    | Stolen card                                    | Capture     | √  | √  | √  | √  | √  | √  |
| 46    | Closed Account                                 | Decline     | √  | √  | √  | √  | √  | √  |
| 51    | Insufficient funds/over credit limit           | Decline     | √  | √  | √  | √  | √  | √  |
| 54    | Expired card                                   | Decline     | √  | √  | √  | √  | √  | √  |
| 55    | Invalid PIN                                    | Decline     | √  | √  | √  | √  | √  | √  |
| 57    | Transaction not permitted to issuer/cardholder | Decline     | √  | √  | √  |    | √  | √  |
| 58    | Transaction not permitted to acquirer/terminal | Decline     | √  | √  | √  |    | √  | √  |
| 61    | Exceeds withdrawal amount limit                | Decline     | √  | √  | √  | √  | √  | √  |
| 62    | Restricted card                                | Decline     | √  | √  | √  | √  | √  | √  |
| 63    | Security violation                             | Decline     | √  | √  | √  | √  | √  |    |

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Action      | MC | NP | VI | TE | MS | CI |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----|----|----|----|----|----|
| 65    | Exceeds withdrawal count limit<br>Issuers may also use this response code for transaction types other than ATM transactions. Issuers that opt to decline a contactless or card-not-present transaction and obtain a cardholder verification through a second transaction may use this response code. Refer to <a href="#">M/Chip Requirements for Contact and Contactless</a> for more details on contactless transactions and how to use response code 65. E-commerce merchants should trigger an authentication using Mastercard Identity Check following an authorization decline with response code 65. | Decline     | √  | √  | √  | √  | √  | √  |
| 70    | Contact Card Issuer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Call Issuer | √  | √  |    |    | √  | √  |
| 71    | PIN Not Changed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Decline     | √  | √  |    |    | √  | √  |
| 72    | Account Not Yet Activated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Decline     | √  | √  | √  | √  | √  | √  |
| 75    | Allowable number of PIN tries exceeded                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Decline     | √  | √  | √  | √  | √  | √  |
| 76    | Invalid/nonexistent "To Account" specified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Decline     | √  | √  | √  |    | √  | √  |
| 77    | Invalid/nonexistent "From Account" specified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Decline     | √  | √  | √  |    | √  | √  |
| 78    | Invalid/nonexistent account specified (general)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Decline     | √  | √  | √  | √  | √  |    |
| 84    | Invalid Authorization Life Cycle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Decline     |    | √  |    |    |    |    |
| 86    | PIN Validation not possible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Decline     | √  | √  | √  | √  | √  | √  |
| 87    | Purchase Amount Only, No Cash Back Allowed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Approved    | √  |    |    |    | √  |    |
| 88    | Cryptographic failure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Decline     | √  | √  | √  | √  | √  | √  |
| 89    | Unacceptable PIN—Transaction Declined—Retry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Decline     | √  | √  |    |    | √  | √  |
| 91    | Authorization System or issuer system inoperative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Decline     | √  | √  | √  | √  | √  | √  |
| 92    | Unable to route transaction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Decline     | √  | √  | √  | √  | √  | √  |



| Value | Description                     | Action  | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------|---------|----|----|----|----|----|----|
| 94    | Duplicate transmission detected | Decline | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 96    | System error                    | Decline | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |

## Authorization Advice Response/0130 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description                                                                             | Action | MC | NP | VI | TE | MS | CI |
|-------|-----------------------------------------------------------------------------------------|--------|----|----|----|----|----|----|
| 00    | Approved or completed successfully                                                      | —      | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 12    | Invalid transaction                                                                     | —      | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 14    | Invalid card number                                                                     | —      | ✓  |    |    |    | ✓  | ✓  |
| 30    | Format error                                                                            | —      | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 57    | Transaction not permitted to issuer/cardholder. Valid only for Pay with Rewards Service | —      | ✓  |    |    |    |    |    |
| 94    | Duplicate Transmission detected                                                         | —      | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 96    | System error                                                                            | —      | ✓  | ✓  | ✓  |    | ✓  | ✓  |

## Authorization Response Acknowledgment/0180 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description            | Action | MC | NP | VI | TE | MS | CI |
|-------|------------------------|--------|----|----|----|----|----|----|
| 00    | Completed successfully | —      | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |

## Authorization Response Negative Acknowledgment/0190 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description            | Action | MC | NP | VI | TE | MS | CI |
|-------|------------------------|--------|----|----|----|----|----|----|
| 30    | Format error           | —      | √  | √  | √  | √  | √  | √  |
| 68    | Response received late | —      | √  | √  | √  | √  | √  | √  |
| 96    | System error           | —      | √  | √  | √  | √  | √  | √  |

## Issuer File Update Request Response/0312 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description                                       | Action | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------|--------|----|----|----|----|----|----|
| 00    | Issuer File Update action completed successfully  | —      | √  | √  | √  | √  | √  | √  |
| 25    | Unable to locate record on file (no action taken) | —      | √  | √  | √  | √  | √  | √  |
| 26    | Record already in active status                   | —      | √  | √  | √  | √  | √  | √  |
| 27    | Issuer File Update field edit error               | —      | √  | √  | √  | √  | √  | √  |
| 28    | Record permanently deleted                        | —      | √  | √  | √  | √  | √  | √  |
| 29    | Delete request less than 540 days                 | —      | √  | √  | √  | √  | √  | √  |
| 30    | Format error                                      | —      | √  | √  | √  | √  | √  | √  |
| 40    | Requested function not supported                  | —      | √  | √  | √  | √  | √  | √  |
| 63    | Security violation                                | —      | √  | √  | √  | √  | √  | √  |
| 80    | Duplicate add, action not performed               | —      | √  | √  | √  | √  | √  | √  |
| 96    | System error                                      | —      | √  | √  | √  | √  | √  | √  |

## Reversal Request/0400 response codes

The following DE 39 response codes are valid in this message.

### Response codes

| Value | Description                                    | Action  | MC | NP | VI | TE | MS | CI |
|-------|------------------------------------------------|---------|----|----|----|----|----|----|
| 00    | Approved or completed successfully             | —       | √  | √  | √  | √  | √  | √  |
| 01    | Refer to card issuer                           | —       | √  | √  | √  | √  | √  |    |
| 03    | Invalid merchant                               | —       | √  | √  | √  | √  | √  |    |
| 04    | Capture card                                   | —       | √  | √  | √  | √  | √  | √  |
| 05    | Do not honor                                   | —       | √  | √  | √  | √  | √  |    |
| 06    | Error (/0400 only)                             | —       | √  |    |    |    | √  | √  |
| 08    | Honor with ID                                  | —       | √  | √  | √  | √  |    |    |
| 10    | Partial Approval                               | —       | √  | √  | √  |    | √  |    |
| 12    | Invalid transaction                            | —       | √  | √  | √  | √  | √  |    |
| 13    | Invalid amount                                 | —       | √  | √  | √  | √  | √  | √  |
| 14    | Invalid card number                            | —       | √  | √  | √  | √  | √  | √  |
| 15    | Invalid issuer                                 | —       | √  | √  | √  | √  | √  |    |
| 17    | Customer cancellation (/0400 only)             | —       | √  |    |    |    | √  | √  |
| 30    | Format error                                   | —       | √  | √  | √  | √  | √  | √  |
| 32    | Partial reversal (/0400 only)                  | —       | √  |    |    |    | √  | √  |
| 34    | Suspect Fraud (/0400 only)                     | —       | √  | √  |    |    | √  |    |
| 41    | Lost card                                      | —       | √  | √  | √  | √  | √  | √  |
| 43    | Stolen card                                    | —       | √  | √  | √  | √  | √  | √  |
| 46    | Closed Accounts                                | Decline | √  | √  | √  | √  | √  | √  |
| 51    | Insufficient funds/over credit limit           | —       | √  | √  | √  | √  | √  | √  |
| 54    | Expired card                                   | —       | √  | √  | √  | √  | √  | √  |
| 55    | Invalid PIN                                    | —       | √  | √  | √  | √  | √  | √  |
| 57    | Transaction not permitted to issuer/cardholder | —       | √  | √  | √  |    | √  | √  |

| Value | Description                                       | Action  | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------|---------|----|----|----|----|----|----|
| 58    | Transaction not permitted to acquirer/terminal    | —       | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 61    | Exceeds withdrawal amount limit                   | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 62    | Restricted card                                   | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 63    | Security violation                                | —       | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 65    | Exceeds withdrawal count limit                    | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 68    | Response received late (/0400 only)               | —       | ✓  |    |    |    | ✓  | ✓  |
| 70    | Contact Card Issuer                               | —       | ✓  | ✓  |    |    | ✓  | ✓  |
| 71    | PIN Not Changed                                   | —       | ✓  | ✓  |    |    | ✓  | ✓  |
| 72    | Account Not Yet Activated                         | Decline | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 75    | Allowable number of PIN tries exceeded            | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 76    | Invalid/nonexistent "To Account" specified        | —       | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 77    | Invalid/nonexistent "From Account" specified      | —       | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 78    | Invalid/nonexistent account specified (general)   | —       | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 84    | Invalid Authorization Life Cycle                  | —       |    | ✓  |    |    |    |    |
| 85    | Not declined                                      | —       | ✓  | ✓  |    | ✓  | ✓  | ✓  |
| 86    | PIN Validation not possible                       | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 87    | Purchase Amount Only, No Cash Back Allowed        | —       | ✓  |    |    |    | ✓  |    |
| 88    | Cryptographic failure                             | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 89    | Unacceptable PIN—Transaction Declined—Retry       | —       | ✓  | ✓  |    |    | ✓  | ✓  |
| 91    | Authorization System or issuer system inoperative | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 92    | Unable to route transaction                       | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 94    | Duplicate transmission detected                   | —       | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 96    | System error                                      | —       | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |

## Reversal Request Response/0410 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description                                       | Action      | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------|-------------|----|----|----|----|----|----|
| 00    | Approved or completed successfully                | Approve     | √  | √  | √  | √  | √  | √  |
| 01    | Refer to card issuer                              | Call Issuer | √  | √  | √  | √  | √  |    |
| 12    | Invalid transaction                               | Decline     | √  | √  | √  | √  | √  |    |
| 13    | Invalid amount                                    | Decline     | √  | √  | √  | √  | √  | √  |
| 14    | Invalid card number                               | Decline     | √  | √  | √  | √  | √  | √  |
| 30    | Format error                                      | Decline     | √  | √  | √  | √  | √  | √  |
| 41    | Lost card                                         | Capture     | √  | √  | √  | √  | √  | √  |
| 43    | Stolen card                                       | Capture     | √  | √  | √  | √  | √  | √  |
| 57    | Transaction not permitted to issuer/cardholder    | Decline     | √  | √  | √  |    | √  | √  |
| 58    | Transaction not permitted to acquirer/terminal    | Decline     | √  | √  | √  |    | √  | √  |
| 62    | Restricted card                                   | Decline     | √  | √  | √  | √  | √  | √  |
| 63    | Security violation                                | Decline     | √  | √  | √  | √  | √  |    |
| 76    | Invalid/nonexistent "To Account" specified        | Decline     | √  | √  | √  |    | √  | √  |
| 77    | Invalid/nonexistent "From Account" specified      | Decline     | √  | √  | √  |    | √  | √  |
| 91    | Authorization System or issuer system inoperative | Decline     | √  | √  | √  | √  | √  | √  |
| 92    | Unable to route transaction                       | Decline     | √  | √  | √  | √  | √  | √  |
| 94    | Duplicate transmission detected                   | Decline     | √  | √  | √  | √  | √  | √  |
| 96    | System error                                      | Decline     | √  | √  | √  | √  | √  | √  |

## Reversal Advice/0420 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description                                    | Action      | MC | NP | VI | TE | MS | CI |
|-------|------------------------------------------------|-------------|----|----|----|----|----|----|
| 00    | Approved or completed successfully             | Approve     | √  | √  | √  | √  | √  | √  |
| 01    | Refer to card issuer                           | Call Issuer | √  | √  | √  | √  | √  |    |
| 03    | Invalid merchant                               | Decline     | √  | √  | √  | √  | √  |    |
| 04    | Capture card                                   | Capture     | √  | √  | √  | √  | √  | √  |
| 05    | Do not honor                                   | Decline     | √  | √  | √  | √  | √  |    |
| 06    | Error                                          | Decline     | √  |    |    |    | √  | √  |
| 08    | Honor with ID                                  | Approve     | √  | √  | √  | √  |    |    |
| 10    | Partial Approval                               | Approve     | √  | √  | √  |    | √  |    |
| 12    | Invalid transaction                            | Decline     | √  | √  | √  | √  | √  |    |
| 13    | Invalid amount                                 | Decline     | √  | √  | √  | √  | √  | √  |
| 14    | Invalid card number                            | Decline     | √  | √  | √  | √  | √  | √  |
| 15    | Invalid issuer                                 | Decline     | √  | √  | √  | √  | √  |    |
| 17    | Customer cancellation                          | Decline     | √  |    |    |    | √  | √  |
| 30    | Format error                                   | Decline     | √  | √  | √  | √  | √  | √  |
| 32    | Partial reversal                               | Decline     | √  |    |    |    | √  | √  |
| 34    | Suspect Fraud                                  | Decline     | √  | √  |    |    | √  |    |
| 41    | Lost card                                      | Capture     | √  | √  | √  | √  | √  | √  |
| 43    | Stolen card                                    | Capture     | √  | √  | √  | √  | √  | √  |
| 46    | Closed Account                                 | Decline     | √  | √  | √  | √  | √  | √  |
| 51    | Insufficient funds/over credit limit           | Decline     | √  | √  | √  | √  | √  | √  |
| 54    | Expired card                                   | Decline     | √  | √  | √  | √  | √  | √  |
| 55    | Invalid PIN                                    | Decline     | √  | √  | √  | √  | √  | √  |
| 57    | Transaction not permitted to issuer/cardholder | Decline     | √  | √  | √  |    | √  | √  |
| 58    | Transaction not permitted to acquirer/terminal | Decline     | √  | √  | √  |    | √  | √  |

| Value | Description                                             | Action      | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------------|-------------|----|----|----|----|----|----|
| 61    | Exceeds withdrawal amount limit                         | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 62    | Restricted card                                         | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 63    | Security violation                                      | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 65    | Exceeds withdrawal count limit                          | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 68    | Response received late                                  | Decline     | ✓  |    |    |    | ✓  | ✓  |
| 70    | Contact Card Issuer                                     | Call Issuer | ✓  | ✓  |    |    | ✓  | ✓  |
| 71    | PIN Not Changed                                         | Decline     | ✓  | ✓  |    |    | ✓  | ✓  |
| 72    | Account Not Yet Activated                               | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 75    | Allowable number of PIN tries exceeded                  | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 76    | Invalid/nonexistent "To Account" specified              | Decline     | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 77    | Invalid/nonexistent "From Account" specified            | Decline     | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 78    | Invalid/nonexistent account specified (general)         | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 82    | Timeout at issuer                                       | —           | ✓  | ✓  | ✓  |    | ✓  | ✓  |
| 84    | Invalid Authorization Life Cycle                        | Decline     |    | ✓  |    |    |    |    |
| 85    | Not declined<br><br>Valid for zero amount transactions. | Valid       | ✓  | ✓  |    | ✓  | ✓  | ✓  |
| 86    | PIN Validation not possible                             | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 87    | Purchase Amount Only, No Cash Back Allowed              | Approved    | ✓  |    |    |    | ✓  |    |
| 88    | Cryptographic failure                                   | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 89    | Unacceptable PIN—Transaction Declined—Retry             | Decline     | ✓  | ✓  |    |    | ✓  | ✓  |
| 91    | Authorization System or issuer system inoperative       | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 92    | Unable to route transaction                             | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 94    | Duplicate transmission detected                         | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| 96    | System error                                            | Decline     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |

## Reversal Advice Response/0430 Message and Administrative Advice Response/0630 response codes

The following [DE 39](#) response codes are valid in these messages.

### Response codes

| Value | Description                        | Action | MC | NP | VI | TE | MS | CI |
|-------|------------------------------------|--------|----|----|----|----|----|----|
| 00    | Approved or completed successfully | —      | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 30    | Format error                       | —      | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 94    | Duplicate Transmission detected    | —      | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| 96    | System error                       | —      | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |

## Administrative Request Response/0610 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description                                       | Action                               | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------|--------------------------------------|----|----|----|----|----|----|
| 00    | Received and processed successfully               | None                                 | ✓  | ✓  |    |    |    |    |
| 30    | Format error                                      | Determine error and resubmit request | ✓  | ✓  |    |    |    |    |
| 57    | Transaction not permitted to issuer/cardholder    | Decline                              | ✓  | ✓  |    |    |    |    |
| 58    | Transaction not permitted to acquirer/terminal    | Decline                              | ✓  | ✓  |    |    |    |    |
| 91    | Authorization System or issuer system inoperative | Resubmit request                     | ✓  | ✓  |    |    |    |    |
| 92    | Unable to route transaction                       | Determine error and resubmit request | ✓  | ✓  |    |    |    |    |



| Value | Description  | Action           | MC | NP | VI | TE | MS | CI |
|-------|--------------|------------------|----|----|----|----|----|----|
| 96    | System error | Resubmit request | √  | √  |    |    |    |    |

## Network Management Request Response/0810 response codes

The following [DE 39](#) response codes are valid in this message.

### Response codes

| Value | Description                                       | Action | MC | NP | VI | TE | MS | CI |
|-------|---------------------------------------------------|--------|----|----|----|----|----|----|
| 00    | Approved or completed successfully                | —      | √  | √  | √  | √  | √  | √  |
| 30    | Format error                                      | —      | √  | √  | √  | √  | √  | √  |
| 63    | Security violation                                | —      | √  | √  | √  | √  | √  | √  |
| 79    | Key Exchange Validation failed                    | —      | √  | √  | √  | √  | √  | √  |
| 91    | Authorization System or issuer system inoperative | —      | √  | √  | √  | √  | √  | √  |
| 94    | Duplicate SAF request                             | —      | √  | √  | √  | √  | √  | √  |
| 96    | System error                                      | —      | √  | √  | √  | √  | √  | √  |

## DE 40 (Service Restriction Code)

[DE 40 \(Service Restriction Code\)](#) identifies geographic or service availability.

**NOTE:** The Dual Message Authorization System currently does not use this data element.

## DE 41 (Acceptor Terminal ID)

*DE 41 (Acceptor Terminal ID)* is the identifier assigned by the acceptor or the acquirer to the terminal, payment gateway, or other acceptance device used to capture account data to affect a transaction at the acceptor location.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | ans-8                     |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–8 |
| Subfields           | N/A                       |
| Justification       | Left with trailing spaces |

### Usage

Following is the usage of DE 41 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | P   | C   |
| Authorization Request Response/0110                                                             | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated                                                   | C   | P   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | XE  | CE  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | .   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | XE  | CE  |
| Reversal Request/0400                                                                           | C   | P   | C   |
| Reversal Request Response/0410                                                                  | CE  | XE  | CE  |
| Reversal Advice/0420                                                                            | .   | X   | C   |

| Message                              | Org | Sys | Dst |
|--------------------------------------|-----|-----|-----|
| Administrative Request/0600          | O   | P   | O   |
| Administrative Request Response/0610 | CE  | XE  | CE  |

### Processing rules

The Acceptor Terminal ID must be present in all transactions.

Acceptor Terminal ID must uniquely identify each point-of-sale terminal and ATM terminal at the acceptor location. For electronic commerce, phone/mail order, or other card-not-present environments, the value must uniquely identify each payment gateway or other device used to capture account data to affect a transaction at the acceptor location.

**NOTE: Mastercard does not apply systematic edits to this DE.**

## DE 42 (Acceptor ID)

[DE 42 \(Acceptor ID\)](#) contains the identifier assigned by the acquirer to the acceptor and its location, or to the payment facilitator.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans-15                     |
| Data Length         | N/A                        |
| Data Field          | Contents of positions 1–15 |
| Subfields           | N/A                        |
| Justification       | Left                       |

### Usage

Following is the usage of DE 42 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | P   | C   |

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |
| Administrative Request/0600                   | O   | P   | O   |
| Administrative Request Response/0610          | CE  | XE  | CE  |
| Administrative Advice/0620: system-generated  | •   | X   | C   |

### Processing rules

The Acceptor ID must be present in all transactions and must uniquely identify each acceptor and its location.

Payment facilitator-enabled transactions must uniquely identify the payment facilitator. The Acceptor ID of the sponsored merchant must be provided in the sponsored merchant ID data element.

The following Dual Message Authorization System edit will apply:

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                      | THEN the Authorization System...                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 42 (Acceptor ID) is not present and DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type code) contains the value 00 (Purchase), 09 (Purchase with Cash Back), or 28 (Payment Transaction) in an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or a Reversal Request/0400 message | sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or a Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format Error)</li> <li>DE 44 (Additional Response Data) = 042 (which identifies DE 42 as the source of the error)</li> </ul> |

### American Express Authorizations

An American Express merchant ID consists of the 10-digit numeric value (referred to as "SE number" by American Express) or two-character alphanumeric IATA airline code (which may be followed by the IATA travel agent ID, T + 5-8 digits, separated by a space). Since DE 42 is defined as ans-15, an American Express 10-digit SE number in character format should be left-justified and followed by trailing spaces. This 10-digit number also should pass the Modulus 9 check digit routine.

## DE 43 (Acceptor Name and Location)

*DE 43 (Acceptor Name and Location)* contains the name and location of the acceptor point of interaction.

### Attributes

| Attribute           | Description                               |
|---------------------|-------------------------------------------|
| Data Representation | ans-40 (supports extended character sets) |
| Data Length         | N/A                                       |
| Data Field          | Contents of subfields 1–5                 |
| Subfields           | 5                                         |
| Justification       | See subfields                             |

### Usage

Following is the usage of DE 43 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Advice/0420                          | •   | X   | C   |
| Administrative Request/0600                   | O   | P   | O   |
| Administrative Request Response/0610          | CE  | XE  | CE  |

### Values

The data element subfields are dependent on the type of message type sent.

## Application notes

This data element is required for all Authorization Request/0100 messages for Mastercard and Visa programs and services.

For Authorization Request/0100, use of DE 43 is conditional (C) based on the program or service being processed. If the program is Mastercard or Visa, use of DE 43 is mandatory (M). Its usage is optional if the transaction involves Private Label, Travel and Entertainment, or other programs.

This data element satisfies "Regulation-E" requirements. It also is required when the DE 22 (Point-of-Service [POS] Entry Mode) is 05 (PAN auto-entry via integrated circuit card).

As ISO specifies, this data element is an alphanumeric text string of 40 characters. Mastercard requires standardized formatting of this data element as specified in the following table to meet uniform standards for printing this information on customer account statements and billing statements.

This data element is required for all Mastercard programs and services.

DE 43 is required for Mastercard Hosted Mobile Phone Top-up transactions.

## Subfield 1 (Acceptor Name)

*DE 43 (Acceptor Name and Location)*, subfield 1 (Acceptor Name) contains the name of the acceptor that should be recognizable by the cardholder and may include other descriptors, such as a store number.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans-22                     |
| Data Field          | Contents of positions 1–22 |
| Justification       | Left                       |

## Processing rules

The Acceptor Name must be present for all transactions and contain a valid merchant/ATM owner/entity name that is most recognizable to the cardholder.

Unique identifiers, invoice numbers, or additional descriptors must be placed after the merchant/ATM owner/entity name.

When a merchant is a part of a chain or franchise, the name most recognizable to the cardholder must be provided at the beginning of the field, followed by a store number or other unique identifier for each location. For a chain or franchise that has been issued a Mastercard Assigned ID, all submitted transactions must contain the Mastercard Assigned ID data element.

For any MCC in the 3000-3350 range (Airlines), the Abbreviated Airline listed in Chapter 3 of the *Quick Reference Booklet* must be provided in positions 1-9 of the field followed by the Airline Ticket Number.

For any MCC in the 3351-3500 range (Automobile/Vehicle Rentals), the first nine characters of the Description listed in Chapter 3 of the *Quick Reference Booklet* must be provided in positions 1-9 of the field and then followed by a unique identifier for each location.

For any MCC in the 3501-3999 range (Hotels and Motels), the first nine characters of the Description listed in Chapter 3 of the *Quick Reference Booklet* must be provided in positions 1-9 of the field and then followed by a unique identifier for each location.

The (ISO)/IEC 8859-1:1998 (Latin Alphabet No. 1) character set must be used.

### Program rules

Payment facilitator-enabled transactions: The sponsored merchant name must be provided. The Payment Facilitator Name or abbreviated name must be provided, followed by an asterisk (\*) and the sponsored merchant name. All submitted transactions must contain the Payment Facilitator ID submitted in the Payment Facilitator ID data element.

Staged digital wallet operator: The staged digital wallet operator name must be provided. If issuer authorization for the transaction is occurring at the time of the consumer's purchase of products or services, then the staged digital wallet operator name must be provided, followed by an asterisk (\*) and the retailer name. If the retailer is identified with an MCC in the 3000-3999 range, then the appropriate industry MCC value, rather than individual retailer MCC, if assigned, must be provided.

Mastercard® rePower™ transactions must have the value MC repower following the merchant/ATM owner/entity name providing the service.

Payment Transactions must contain the name of the payment provider. Payee information must be provided in the Additional Transaction Reference Data and Mastercard Send Receiver Data data elements.

### Country rules

Mexico-acquired transactions must contain only

- uppercase alpha characters (A-Z) (cannot contain accent characters)
- numeric characters (0-9)
- ampersand (&)
- asterisk (\*)
- space (cannot be the first character or contain more than one space between words)

## Subfield 2 (Space)

*DE 43 (Acceptor Name and Location)*, subfield 2 (Space) indicates a space character.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | ans-1                   |
| Data Field          | Contents of position 23 |
| Justification       | N/A                     |

### Value

Delimiter (space).

## Subfield 3 (Acceptor City)

*DE 43 (Acceptor Name and Location)*, subfield 3 (Acceptor City) indicates the city of the merchant.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-13                      |
| Data Field          | Contents of positions 24–36 |
| Justification       | Left                        |

### Values

Acceptor City Name must contain the name of the city, town, or village as defined by country postal standards.

For U.S. region transactions, Acceptor City Name must match the U.S. Postal Service list of acceptable cities and spellings by postal code.

### Processing rules

The Acceptor City Name must be present for all transactions and contain a valid location for the acceptor. The location of the acceptor will generally be the physical location where the transaction occurred or the business/operations location for transactions that occur virtually.



### Program rules

Payment facilitator-enabled transactions: The sponsored merchant city name must be provided.

## Subfield 4 (Space)

[DE 43 \(Acceptor Name and Location\)](#), subfield 4 (Space) indicates a space character.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | ans-1                   |
| Data Field          | Contents of position 37 |
| Justification       | N/A                     |

### Value

Delimiter (space).

## Subfield 5 (Acceptor Country Code)

[DE 43 \(Acceptor Name and Location\)](#), subfield 5 (Acceptor Country Code) contains the country code of the acceptor.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | a-3                         |
| Data Field          | Contents of positions 38-40 |
| Justification       | Left                        |

### Values

Acceptor Country Code must contain a valid three-character alphabetic code as defined in Chapter 2 of the *Quick Reference Booklet*. When territories have a defined country code in the *Quick Reference Booklet*, that country code must be used.

### Processing rules

The Acceptor Country Code must be present for all transactions.

## Program rules

Payment facilitator-enabled transactions: The sponsored merchant country code must be provided.

## DE 43, subfield 5 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

### Authorization Request/0100 message edits for DE 43, subfield 5

#### Associated edits

| If...                                                                    | Then the Dual Message Authorization System...                                                                                                                                                        |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">DE 43</a> , subfield 5 does not contain a valid country code | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• <a href="#">DE 39</a> = 30</li><li>• <a href="#">DE 44</a> = 043</li></ul> |

### Authorization Advice/0120: acquirer-generated message edits for DE 43, subfield 5

#### Associated edits

| If...                                                                    | Then the Dual Message Authorization System...                                                                                                                                                                         |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">DE 43</a> , subfield 5 does not contain a valid country code | will return to the acquirer an Authorization Advice Response/0130: system-generated message where: <ul style="list-style-type: none"><li>• <a href="#">DE 39</a> = 30</li><li>• <a href="#">DE 44</a> = 043</li></ul> |

## Reversal Request/0400 message edits for DE 43, subfield 5

### Associated edits

| If...                                                                    | Then the Dual Message Authorization System...                                                                                                                                                     |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">DE 43</a> , subfield 5 does not contain a valid country code | will return to the acquirer a Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>• <a href="#">DE 39</a> = 30</li> <li>• <a href="#">DE 44</a> = 043</li> </ul> |

## DE 44 (Additional Response Data)

[DE 44 \(Additional Response Data\)](#) provides other supplemental data that may be required in response to an authorization or other type of transaction request. This data element may also be present in any response message when DE 39 (Response Code) contains the value 30, indicating that a Format Error condition was detected in the preceding message. In this case, DE 44 will contain either a three- or six-digit value that indicates the data element number (three digits) only or the data element plus subelement numbers (six digits) in which the format error occurred. If the format error occurred in a data element that does not have subelements, DE 44 will contain a three-digit numeric value.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...25; LLVAR            |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–25 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Usage

Following is the usage of DE 44 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | P   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | C   | X   | .   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | X   | C   |
| Authorization Response Negative Acknowledgment/0190                                             | .   | X   | C   |
| Issuer File Update Request Response/0312                                                        | .   | X   | C   |
| Reversal Request Response/0410                                                                  | C   | X   | C   |
| Reversal Advice Response/0430                                                                   | C   | X   | .   |
| Administrative Request Response/0610                                                            | C   | P   | C   |
| Administrative Advice Response/0630                                                             | C   | X   | .   |
| Network Management Request Response/0810: Host Session Activation/Deactivation                  | .   | X   | C   |
| Network Management Request Response/0810: PEK exchange                                          | C   | X   | .   |
| Network Management Request Response/0810: PEK exchange–On Demand                                | .   | X   | C   |
| Network Management Request Response/0810: Network Connection Status, System-generated           | .   | X   | C   |
| Network Management Request Response/0810: sign-on/sign-off                                      | .   | X   | C   |

### Values

Contains supplemental data.

## DE 44 values by program or service

The following table lists the usage of [DE 44](#) by program or service. Usage is dependent on values in [DE 39 \(Response Code\)](#). The following table defines the additional response data that is in [DE 44](#) when the listed values are present in [DE 39](#). No data element edits are performed on the variable-length field.

### DE 44 usage by program or service

| IF DE 39 is... | THEN DE 44...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MC | NP | VI | TE | MS | CI |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| 00             | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |
| 01             | contains the phone number for "call issuer" response codes. Applies to Authorization Request Response/0110 messages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | √  | √  | √  | √  | √  |    |
| 08             | contains the ID information for "approve with ID" response codes for example: <ul style="list-style-type: none"> <li>• B = Blank Name</li> <li>• C = Cardholder's ID number or hologram message</li> <li>• X= Cardholder's last name</li> </ul> Applies to Authorization Request Response/0110 messages.                                                                                                                                                                                                                                                                                                                                                                                                                              | √  | √  | √  | √  | √  |    |
| 12             | should be accessed when the value 12 is present in <a href="#">DE 39</a> for Mastercard Corporate Fleet Card® transactions only as follows: <ul style="list-style-type: none"> <li>• 01 = Issuer determines that the ID number provided in the Authorization Request/0100 message is invalid causing the transaction to be declined</li> <li>• 02 = Issuer determines that the Driver Number provided in the Authorization Request/0100 message is invalid causing the transaction to be declined</li> <li>• 03 = Issuer determines that the Vehicle Number provided in the Authorization Request/0100 message is invalid causing the transaction to be declined.</li> </ul> Applies to Authorization Request Response/0110 messages. | √  |    |    |    |    |    |
| 25             | contains 120 indicating an error in <a href="#">DE 120 (Record Data)</a> and a three-digit code (such as 120xxx, where xxx indicates the field in the file update request where the error occurred). A listing of these codes can be found in the Error Codes section for <a href="#">DE 120</a> . Applies to Issuer File Update Request Response/0312 messages.                                                                                                                                                                                                                                                                                                                                                                      | √  | √  | √  | √  | √  | √  |

| IF DE 39 is... | THEN DE 44...                                                                                                                                                                                                                                                                                                                                        | MC | NP | VI | TE | MS | CI |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| 27             | contains a 120 indicating an error in <a href="#">DE 120</a> and a three-digit code (such as 120xxx, where xxx indicates the field in the file update request where the error occurred). A listing of these codes can be found in the Error Codes section for <a href="#">DE 120</a> . Applies to Issuer File Update Request Response/0312 messages. | √  | √  | √  | √  | √  | √  |
| 30             | contains either the data element/subelement when a format error is detected (for example, DE 44 value 022 indicates DE 22, 0480nn indicates DE 48 subelement nn, 112nnn indicates DE 122 subelement nnn). Applies to all response messages.                                                                                                          | √  | √  | √  | √  | √  | √  |
| 57             | contains 003 indicating an error in <a href="#">DE 3 (Processing Code)</a> when an issuer is unable to process a balance inquiry (30), purchase with cash back (09), purchase return (20), or payment (28) transaction type.                                                                                                                         | √  | √  | √  |    | √  | √  |
| 85             | contains the date and time used in electronic commerce transactions, after which a cardholder may re-apply for a certificate. Format is YYYYMMDDHH. HH is the 24-hour military clock.                                                                                                                                                                | √  |    | √  |    | √  |    |

## Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edit.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or Reversal Request/0400 message contains one or both of the following:</p> <ul style="list-style-type: none"><li>• <i>DE 48 (Additional Data: Private Use), DE 104 (Digital Payment Data), DE 105 (Multi-Use Transaction Identification Data), DE 106 (Fleet Service Data), DE 108 (Additional Transaction Reference Data), DE 112 (Additional Data [National Use]), DE 119 (Additional Data: Private Use 2), or DE 122 (Additional Record Data)</i> with subelements that have incorrect length and/or incorrect format (Data Representations)</li><li>• A subelement ID appears more than once within the same data element</li></ul> | <p>sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where <i>DE 44</i> is 6 positions for subelement format errors:</p> <ul style="list-style-type: none"><li>• <i>DE 39 (Response Code)</i> = 30</li><li>• <i>DE 44 (Additional Response Data)</i> = 0480nn or 1080nn (where nn is the subelement number) or 104nnn, 105nnn, 106nnn, 112nnn, 119nnn, or 122nnn (where nnn is the subelement number)</li></ul> <p>(<i>DE 44</i> is three positions for Dual Message System (Authorization) if no subelements are present, for example, for DE 22 format error: DE 44 = 022)</p> |

### Examples

- When an edit error occurs on *DE 48*, the *DE 44* data will be populated as these examples below:
  - Error on *DE 48* subelement 42, subfield 1: 048042 (no subfield information is provided)
  - Error on *DE 48* subelement 61: 048061
- *DE 48* TCC format error will be responded with *DE 44* = 048000.

For all data elements which do not contain subelements, DE 44 identifies the data element alone. For example, a format error in DE 22 is represented with a DE 44 value of 022.

## DE 45 (Track 1 Data)

*DE 45 (Track 1 Data)* is the information encoded on track 1 of the card's magnetic stripe as defined in the ISO 7813 specification, including data element separators but excluding beginning and ending sentinels and LRC characters as defined in this data element definition.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...76; LLVAR            |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–76 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Usage

Following is the usage of DE 45 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | O   | X   | O   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

### Values

Whenever DE 45 is captured automatically at the point of interaction, this data element must contain whatever is encoded on the magnetic stripe (track 1) of the card, regardless of whether the card has been properly encoded with information in accordance with ISO specifications.

The ISO 7810, 7811, 7812, and 7813 specifications document the international standards for encoding information on magnetic stripe cards.

Length subelement must not be greater than 76 bytes.

The account number in DE 2 (Primary Account Number) or DE 34 (Primary Account Number [PAN], Extended) must match the account number in DE 45.



## Application notes

This data element is defined and used identically within all Mastercard programs and services.

For MDES tokenized transactions, issuers may opt in through MDES Manager to receive track data in DE 45 (Track 1 Data) on Dual Message Authorization System or Single Message System transactions to issuers.

[DE 45](#) is optional for 0120 AFD completion advice.

## Fields in DE 45

| Field ID | Field Name                                                                                                     | F = Fixed    |                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------|
|          |                                                                                                                | V = Variable | Maximum Length                                                                 |
| 1        | Start Sentinel<br>These fields are encoded on the card but must be omitted within Track 2 data.                | F            | n-1                                                                            |
| 2        | Format Code-B (encode character B)                                                                             | F            | an-1                                                                           |
| 3        | Primary Account Number                                                                                         | V            | n...19                                                                         |
| 4        | Separator (binary)                                                                                             | F            | ans-1                                                                          |
| 5        | Cardholder name                                                                                                | V            | ans...2-26                                                                     |
| 6        | Separator (binary)                                                                                             | F            | ans-1                                                                          |
| 7        | Expiration Date                                                                                                | F            | ans-4                                                                          |
| 8        | Extended Service Code                                                                                          | F            | ans-3                                                                          |
| 9        | Discretionary Data (must include CVC 1)                                                                        | V            | Balance of available digits not to exceed total track length of 79 characters. |
| 10       | End Sentinel<br>These fields are encoded on the card but must be omitted within Track 2 data.                  | F            | n-1                                                                            |
| 11       | Longitudinal Redundancy Check<br>These fields are encoded on the card but must be omitted within Track 2 data. | F            | n-1                                                                            |

Refer to the *Security Rules and Procedures* for additional Track 1 Data information.

## DE 46 (Expanded Additional Amounts)

*DE 46 (Additional Data: ISO Use)* provides data supplemental to that already conveyed in the specific data elements in the message.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 47 (Additional Data: National Use)

*DE 47 (Additional Data—National Use)* is reserved for national organizations to define data unique to country applications.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

This data element should not be present in any Dual Message Authorization System messages. However, if encountered in Dual Message Authorization System messages routed between customers of the same country, it is passed unaltered through the network. The Dual Message Authorization System performs no editing or processing functions on this data element.

## DE 48 (Additional Data: Private Use)

DE 48 (Additional Data: Private Use) contains data associated with various Mastercard programs, products, and services. DE 48 is used to convey data that is not related to any other ISO-defined data elements. It is a variable-length data element used for multiple purposes.

### Attributes

| Attribute           | Description                                                                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Representation | ans...999; LLLVAR<br><br>Although this data element is ans, some subelements deviate from ans and contain binary data. Refer to subelement descriptions of each subelement.<br><br>The length field must be in the range 001–999. Subelements are identified by valid subelement ID and length. |
| Data Length         | 3 positions                                                                                                                                                                                                                                                                                     |
| Data Field          | Contents of subelements                                                                                                                                                                                                                                                                         |
| Subelements         | 99                                                                                                                                                                                                                                                                                              |

| Attribute     | Description     |
|---------------|-----------------|
| Justification | See subelements |

## Usage

The following applies to DE 48 usage:

- Subelements do not need to be in any particular order or sequence within DE 48.
- Subelements may contain multiple occurrences of program or service data, up to the maximum limit specified for that subelement. However, the same subelement ID cannot be present within DE 48 more than once. Messages submitted to Mastercard containing multiple occurrences of a subelement within DE 48 will be rejected.
- Message originators and receivers must be able to send and receive any subelement applicable to their business or as mandated by Mastercard.
- Message originators must not use a subelement ID that is not defined by Mastercard. A message containing a subelement not defined by Mastercard will not be rejected by Mastercard. However, the undefined subelement will be removed before being set to the message receiver. The content of such subelements will not be recognized by Mastercard or used in any Mastercard processing.
- For subelements defined with a CE (conditional echo) presence notation for an Org (originator) entity Mastercard will not reject a message from the message originator if the original subelement data is not returned as defined in a response message. Based on the presence notation for the Dst (destination) entity Mastercard will forward the subelement data as received from the message originator if it is present in a response message.

Following is the overall usage of DE 48 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | M   | X   | M   |
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | M   | X   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | X   | C   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | X   | C   |
| Reversal Request/0400                                                                           | M   | X   | M   |
| Reversal Request Response/0410                                                                  | M   | X   | C   |

| Message                                                | Org | Sys | Dst |
|--------------------------------------------------------|-----|-----|-----|
| Reversal Advice/0420                                   | .   | X   | C   |
| Administrative Advice/0620: member-generated           | C   | P   | C   |
| Administrative Advice/0620: system-generated           | .   | X   | C   |
| Network Management Request/0800: PEK exchange          | .   | O   | C   |
| Network Management Request Response/0810: PEK exchange | C   | X   | .   |
| Network Management Advice/0820: PEK exchange           | .   | X   | C   |

### Values

The length field must be in the range 001–999. Subelements are identified by valid subelement ID and length.

### Application notes

Mastercard may occasionally introduce new DE 48 subelements and subfields between releases to facilitate special processing within a country, region, or among pilot participants. Mastercard requires customers to be able to successfully process various online messages that may contain new unannounced DE 48 subelements and subfields.

DE 48 and DE 119 contain different subelement definitions. Subelements defined for DE 48 must be sent in DE 48, not DE 119.

This data element's content may vary by program and service. Additional or supplemental information that may be required in a transaction message is specified in Dual Message Authorization System edits for this data element.

For information on other uses of this data element, refer to General message layout.

## DE 48 Transaction Category Code

In Authorization/01XX and Reversal/04XX messages, the format is "LLLt," where "t" is the transaction category code (TCC). The message must contain an appropriate TCC as the first byte of data after the length within [DE 48](#). The TCC classifies major categories of transactions, such as "Retail Sale," "Cash Disbursement," and "Mail Order."

### Valid values

| TCC   | Description                                                                                                 | MC | NP | VI | TE | MS | CI |
|-------|-------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| Space | A value of space in place of a valid TCC requests that the Authorization System perform the TCC assignment. | √  | √  | √  | √  | √  | √  |
| A     | Auto/Vehicle Rental                                                                                         | √  | √  | √  | √  | √  |    |

| TCC | Description                               | MC | NP | VI | TE | MS | CI |
|-----|-------------------------------------------|----|----|----|----|----|----|
| C   | Cash Disbursement                         | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| F   | Restaurant                                | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| H   | Hotel/Motel                               | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| O   | Hospitalization, College                  | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| P   | Payment Transaction                       | ✓  | ✓  |    |    | ✓  | ✓  |
| R   | Retail Sale                               | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| T   | Phone, Mail, or Electronic Commerce Order | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| U   | Unique                                    | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |
| X   | Airline and Other Transportation Services | ✓  | ✓  | ✓  | ✓  | ✓  |    |
| Z   | ATM Cash Disbursement                     | ✓  | ✓  | ✓  | ✓  | ✓  | ✓  |

**NOTE:** Mastercard will optionally assign the Transaction Category Code (TCC) in [DE 48 \(Additional Data: Private Use\)](#) of authorization messages on behalf of an acquirer. All acquirers processing through the Mastercard Network may choose to have the Authorization System assign the TCC in authorization messages on their behalf, if TCC is submitted with a <space> value. Refer to the *Quick Reference Booklet* for a complete definition of all TCC values.

## DE 48 Subelement Encoding Scheme

The following table identifies the structure of the DE 48 subelement encoding scheme.

The DE 48 subelement encoding scheme covers these Authorization, Advice, Reversal, and Reversal Advice message types:

Authorization Request/0100

Authorization Request Response/0110

Authorization Advice/0120: acquirer-generated

Authorization Advice/0120: system-generated

Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)

Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)

Reversal Request/0400

Reversal Request Response/0410

Reversal Advice/0420

## Values

| LLL      "VAR": 999 maximum bytes (TCC+ Subelement Data) |        |                                                 |           |                         |                                                 |           |                         |
|----------------------------------------------------------|--------|-------------------------------------------------|-----------|-------------------------|-------------------------------------------------|-----------|-------------------------|
| 3 bytes                                                  | 1 byte | 2 bytes                                         | 2 bytes   | 1-99 bytes              | 2 bytes                                         | 2 bytes   | 1-99 bytes              |
|                                                          |        | SE ID + Length + Data will not exceed 103 bytes |           |                         | SE ID + Length + Data will not exceed 103 bytes |           |                         |
| Total Data Element Length                                | TCC    | First Subelement (SE) Data                      |           |                         | Second Subelement (SE) Data                     |           |                         |
|                                                          |        | SE ID                                           | SE Length | SE Variable Length Data | SE ID                                           | SE Length | SE Variable Length data |
| Mandatory                                                |        |                                                 |           |                         |                                                 |           |                         |
| 1002 maximum bytes (LLL + Subelement Data)               |        |                                                 |           |                         |                                                 |           |                         |

## Values

| Number of Bytes | Attribute                       | Description                                     |
|-----------------|---------------------------------|-------------------------------------------------|
| 3               | Total Data Element Length       | The "LLL" portion of the data element up to 999 |
| 1               | Transaction Category Code (TCC) | Must be a valid TCC or a space                  |
| 2               | Subelement ID                   | In the range 00–99                              |
| 2               | Subelement Length               | In the range of 01–99                           |
| 1...99          | Subelement Variable Length Data | Contains valid values                           |

## DE 48 Subelement Encoding Scheme in Network Management Messages

The following table identifies the structure of the [DE 48](#) subelement encoding scheme in Network Management messages.

## Values

| LLL "VAR": 999 maximum bytes (Subelement Data) |         |         |            |         |         |            |
|------------------------------------------------|---------|---------|------------|---------|---------|------------|
| 3 bytes                                        | 2 bytes | 2 bytes | 1-96 bytes | 2 bytes | 2 bytes | 1-96 bytes |

| LLL                                        | "VAR": 999 maximum bytes (Subelement Data) |           |                         |                             |           |                         |
|--------------------------------------------|--------------------------------------------|-----------|-------------------------|-----------------------------|-----------|-------------------------|
| Total Data Element Length                  | First Subelement (SE) Data                 |           |                         | Second Subelement (SE) Data |           |                         |
|                                            | SE ID                                      | SE Length | SE Variable Length Data | SE ID                       | SE Length | SE Variable Length data |
| 1002 maximum bytes (LLL + Subelement Data) |                                            |           |                         |                             |           |                         |

### Values

| Number of Bytes | Attribute                       | Description                 |
|-----------------|---------------------------------|-----------------------------|
| 3               | Total Data Element Length       | The "LLL" portion of LLLVAR |
| 2               | Subelement ID                   | In the range 00–99          |
| 2               | Subelement Length               | In the range of 01–96       |
| 1...96          | Subelement Variable Length Data | Contains valid values.      |

## List of DE 48 subelements

*DE 48* subelements are listed in numeric order. Subelements that are specific to a brand service or program are clearly indicated in the subelement title or description or both.

### Values

| Subelement ID | Subelement name                                  | Data representation |
|---------------|--------------------------------------------------|---------------------|
| 02            | Special Acceptance Conditions Data               | ans...99; LLVAR     |
| 03            | Additional Visa Request Data (Visa Only)         | an-14               |
| 04            | Acceptance Environment Response Data (Visa Only) | ans...80; LLVAR     |
| 05            | Additional Visa Response Data (Visa Only)        | ans...41; LLVAR     |
| 06            | Advanced Digital Transaction Data 1              | ans...84; LLVAR     |
| 07            | Advanced Digital Transaction Data 2              | ans...73; LLVAR     |
| 08            | Advanced Digital Transaction Data 3              | ans...98; LLVAR     |
| 09            | Virtual Card Number                              | an...99; LLVAR      |
| 11            | Key Exchange Block Data (Double-length Keys)     | an-54               |
| 11            | Key Exchange Block Data (Triple-length Keys)     | an-70               |

| <b>Subelement ID</b> | <b>Subelement name</b>                                                        | <b>Data representation</b> |
|----------------------|-------------------------------------------------------------------------------|----------------------------|
| 12                   | Routing Indicator                                                             | a-1                        |
| 13                   | Mastercard Hosted Mobile Phone Top-up Request Data                            | ans-47                     |
| 14                   | Account Type Indicator                                                        | ans-1                      |
| 15                   | Authorization System Advice Date and Time                                     | n-10                       |
| 16                   | Processor Pseudo ICA                                                          | n-7                        |
| 17                   | Authentication Indicator                                                      | n-1                        |
| 18                   | Service Parameters                                                            | ans...99; LLVAR            |
| 20                   | Cardholder Verification Method                                                | a-1                        |
| 21                   | Acceptance Data                                                               | n-11; LLVAR                |
| 22                   | Multi-Purpose Merchant Indicator                                              | ans...29; LLVAR            |
| 23                   | Payment Initiation Channel                                                    | an-2                       |
| 24                   | Account Level Management [ALM] Service Data                                   | an...65; LLVAR             |
| 25                   | Mastercard Cash Program Data                                                  | ans...14; LLVAR            |
| 26                   | Wallet Program Data                                                           | an-3                       |
| 27                   | Transaction Analysis                                                          | an...97; LLVAR             |
| 28                   | Cardless ATM Order ID                                                         | an-10                      |
| 29                   | Additional POS Terminal Locations                                             | an-1                       |
| 30                   | Token Transaction Identifier                                                  | ans...64; LLVAR            |
| 32                   | Mastercard Assigned ID                                                        | an-6                       |
| 32                   | Visa Merchant Identifier (Visa Only)                                          | an-8                       |
| 33                   | PAN Mapping File Information                                                  | an...93; LLVAR             |
| 34                   | ATC Information                                                               | an-11                      |
| 35                   | Contactless Non-Card Form Factor Request/Response                             | an-1                       |
| 36                   | Additional Visa Request Data (Visa Only)                                      | ans...94; LLVAR            |
| 37                   | Additional Merchant Data                                                      | ans...71; LLVAR            |
| 38                   | Account Category                                                              | an-1                       |
| 39                   | Account Data Compromise Information                                           | ans-30                     |
| 40                   | Electronic Commerce Merchant/Cardholder Certificate Serial Number (Visa Only) | n...40; LLVAR              |
| 41                   | Electronic Commerce Certificate Qualifying Information                        | ans...54; LLVAR            |



| <b>Subelement ID</b> | <b>Subelement name</b>                                                                             | <b>Data representation</b> |
|----------------------|----------------------------------------------------------------------------------------------------|----------------------------|
| 42                   | Electronic Commerce Indicators                                                                     | n-7 or n-19                |
| 43                   | Universal Cardholder Authentication Field (UCAF)                                                   | ans...32; LLVAR            |
| 43                   | 3-D Secure for Mastercard Identity Check                                                           | ans-28 (base 64 encoded)   |
| 43                   | Static AAV                                                                                         | ans-28                     |
| 43                   | Digital Secure Remote Payment Universal Cardholder Authentication Field [UCAF]                     | ans-28 (base 64 encoded)   |
| 43                   | 3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express] | nb-21                      |
| 44                   | Visa 3-D Secure Electronic Commerce Transaction Identifier (XID) (Visa and American Express)       | b-20                       |
| 45                   | Visa 3-D Secure Electronic Commerce Transaction Response Code (Visa and American Express)          | an-1                       |
| 46                   | Product ID (Visa Only)                                                                             | an-2                       |
| 47                   | Mastercard Payment Gateway Transaction Indicator                                                   | ans...8                    |
| 48                   | Digital Commerce Solutions Indicators                                                              | ans...73; LLVAR            |
| 49                   | Time Validation Information                                                                        | n-15                       |
| 50                   | Embedded Interchange Data                                                                          | ans...99; LLVAR            |
| 51                   | Merchant On-behalf Services                                                                        | ans...99; LLVAR            |
| 52                   | Interchange Structure Indicator                                                                    | an-2                       |
| 53                   | E-ID Request Code                                                                                  | ans...99; LLVAR            |
| 55                   | Merchant Fraud Scoring Data/Approval Probability Indicator                                         | an...32; LLVAR             |
| 56                   | Security Services Additional Data for Issuers                                                      | an...96; LLVAR             |
| 57                   | Security Services Additional Data for Acquirers                                                    | an...96; LLVAR             |
| 58                   | ATM Additional Data                                                                                | ans-33                     |
| 60                   | Additional Service Data for Issuers                                                                | ans...43; LLVAR            |
| 61                   | POS Data Extended Condition Codes                                                                  | n-5                        |
| 62                   | Real-Time Payment Information                                                                      | ans...76; LLVAR            |
| 63                   | Trace ID                                                                                           | ans-15                     |
| 64                   | Transit Program                                                                                    | n-4                        |
| 65                   | Terminal Complaint Indicator                                                                       | n-2                        |
| 66                   | Authentication Data                                                                                | ans...45; LLVAR            |

| <b>Subelement ID</b> | <b>Subelement name</b>                                     | <b>Data representation</b> |
|----------------------|------------------------------------------------------------|----------------------------|
| 67                   | Mastercard Send Information                                | ans...99; LLVAR            |
| 68                   | Financial Account Information                              | ans...68; LLVAR            |
| 71                   | On-behalf Services                                         | ans...40; LLVAR            |
| 72                   | Issuer Chip Authentication                                 | b...16; LLVAR              |
| 74                   | Additional Processing Information                          | an...30; LLVAR             |
| 75                   | Fraud Scoring Data                                         | an...32; LLVAR             |
| 76                   | Mastercard Electronic Acceptance Indicator                 | a-1                        |
| 77                   | Transaction Type Identifier                                | an-3                       |
| 78                   | Payment Service Indicators (Visa Only)                     | ans-6                      |
| 79                   | Chip CVR/TVR Bit Error Result                              | an...50; LLVAR             |
| 80                   | PIN Service Code                                           | a-2                        |
| 82                   | Address Verification Service Request                       | n-2                        |
| 83                   | Address Verification Service Response                      | a-1                        |
| 84                   | Merchant Advice Code                                       | an-2                       |
| 84                   | Visa Response Codes (Visa Only)                            | an-2                       |
| 84                   | American Express Response Codes (American Express Only)    | an-3                       |
| 85                   | Account Status (Visa Only)                                 | a-1                        |
| 86                   | Relationship Participant Indicator (Visa Only)             | a-1                        |
| 87                   | Card Validation Code Result                                | a-1                        |
| 87                   | CVV2 Response (Visa Only)                                  | a-1                        |
| 88                   | Magnetic Stripe Compliance Status Indicator                | a-1                        |
| 89                   | Magnetic Stripe Compliance Error Indicator                 | a-1                        |
| 90                   | Lodging and Auto Rental Indicator                          | a-1                        |
| 90                   | Custom Payment Service Request (Visa Only)                 | a-1                        |
| 90                   | Custom Payment Service Request Response (Visa Only)        | a-1                        |
| 91                   | Acquirer Reference Data (American Express Only)            | ans...15; LLVAR            |
| 91                   | Custom Payment Service Request Transaction ID (Visa Only)  | an...19; LLVAR             |
| 91                   | Custom Payment Service Response/Transaction ID (Visa Only) | ans...19; LLVAR            |
| 92                   | CVC 2                                                      | n-3                        |
| 92                   | CVV2 Data (Visa Only)                                      | ns-6                       |

| Subelement ID | Subelement name                                              | Data representation |
|---------------|--------------------------------------------------------------|---------------------|
| 93            | Fleet Card ID Request Data (Visa Only)                       | ans...19; LLVAR     |
| 94            | Commercial Card Inquiry Request (Visa Only)                  | ans-4               |
| 94            | Commercial Card Inquiry Response (Visa Only)                 | ans-4               |
| 95            | Mastercard Promotion Code                                    | an-6                |
| 95            | American Express CID Card Identifier (American Express Only) | n-4                 |
| 96            | Visa Market-specific Data Identifier (Visa Only)             | a-1                 |
| 97            | Prestigious Properties Indicator (Visa Only)                 | a-1                 |
| 98            | Mastercard Corporate Fleet Card ID/Driver Number             | n...6; LLVAR        |
| 99            | Mastercard Corporate Fleet Card Vehicle Number               | n...6; LLVAR        |

## Subelement 02 (Special Acceptance Conditions Data)

DE 48, subelement 02 (Special Acceptance Conditions Data) allows an acquirer to provide additional information about the transaction, the acceptor, etc. not elsewhere provided, that is relevant to the way Mastercard processes the transaction.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Subelement ID       | 02                      |
| Subelement length   | 2 positions             |
| Data representation | ans...99; LLVAR         |
| Data field          | Contents of subfield 01 |
| Subfields           | 1                       |
| Justification       | See subfields           |

### Usage

Following is the usage of DE 48, subelement 02 (whether it is mandatory, conditional, optional, system-provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120:system-generated    | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |
| Reversal Advice/0420: system-generated        | •   | X   | C   |

### Application notes

Mastercard will introduce additional DE 48, subelement 02 subfields in the future. Acquirers must provide only subfields that are relevant to the transaction. For example, in the future if six subfields exist and only subfield 05 is relevant for a given transaction, then an acquirer must provide only subfield 05. Subfields 01 to 04 and 06 must not be provided.

For example, to provide subfield 01 with a value of C, the acquirer would submit 0101C where:

- 01 = subfield 1
- 01 = length of subfield 1
- C = value of subfield 1

## Subfield 01 (Mastercard One Credential Exclusion Indicator)

DE 48, subelement 02 (Special Acceptance Conditions Data), subfield 01 (Mastercard One Credential Exclusion Indicator) allows an acquirer to instruct Mastercard to limit the conditions in which the Mastercard One Credential service will apply to qualified transactions.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield data length | 2 positions            |
| Data representation  | an-1                   |
| Data field           | Contents of position 1 |
| Justification        | N/A                    |

## Values

| Value | Description                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------|
| C     | Do not assign a Mastercard One Credential funding PAN containing the Mastercard Credit Acceptance Brand for this transaction |
| D     | Do not assign a Mastercard One Credential funding PAN containing the Debit Mastercard Acceptance Brand for this transaction  |
| M     | Do not assign a Mastercard One Credential funding PAN containing the Maestro Acceptance Brand for this transaction           |

## Edits

Mastercard will perform the following edits.

## Edits

| When the Authorization Request/0100 message contains...                                                 | Then the Dual Message Authorization System...                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 02, subfield 01 (Mastercard One Credential Exclusion Indicator) with an invalid value | will reject the transaction and send an Authorization Response/0110 message to the acquirer with: <ul style="list-style-type: none"><li>• DE 39 = 30 (Format Error)</li><li>• DE 44 = 048002</li></ul> |

## Application notes

The Dual Message Authorization System will remove DE 48, subelement 02, subfield 01 (Mastercard One Credential Exclusion Indicator) from the message if the issuer does not participate in the Mastercard One Credential service.

## Subelement 03 (Additional Visa Request Data [Visa Only])

DE 48, subelement 03 (Additional Visa Request Data [Visa Only]) will contain additional request data when provided by the acquirer.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 03                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | an-14                 |
| Data Field             | Contents of subfields |
| Subfields              | 1                     |
| Justification          | See subfields         |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request /0100         | C   | P   | C   |
| Authorization Request Response/0110 | OE  | XE  | CE  |
| Reversal Request/0400               | C   | P   | C   |
| Reversal Request Response/0410      | CE  | XE  | CE  |

### Values

See subfields.

## Subfield 01 (Plan Registration System Identifier)

DE 48, subelement 03, subfield 01 (Plan Registration System Identifier) provides a plan identifier for Visa Installment Solutions transactions.

### Attributes

| Attribute            | Description                |
|----------------------|----------------------------|
| Subfield ID          | 01                         |
| Subfield Data Length | 2 positions                |
| Data Representation  | an-10                      |
| Data Field           | Contents of positions 1–10 |
| Justification        | N/A                        |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Subelement 04 (Acceptance Environment Response Data [Visa Only])

DE 48, subelement 04 (Acceptance Environment Response Data[Visa Only]) contains the reasons that PSD2 exemptions were not accepted for Visa-branded transactions.

### Attributes

| Attributes          | Description                |
|---------------------|----------------------------|
| Subelement ID       | 04                         |
| Subelement Length   | 2                          |
| Data Representation | ans...80; LLVAR            |
| Data Field          | Contents of positions 1–80 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Usage

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |
| Authorization Advice Response/0130  | C   | X   | C   |

### Values

Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Subelement 05 (Additional Visa Response Data [Visa Only])

[DE 48](#), subelement 05 (Additional Visa Response Data [Visa Only]) will contain additional response data when provided by Visa.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 05                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...41; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 7                     |
| Justification          | See subfields         |

### Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |
| Authorization Advice Response/0130  | C   | X   | C   |



| Message                        | Org | Sys | Dst |
|--------------------------------|-----|-----|-----|
| Reversal Request Response/0410 | C   | X   | C   |

### Values

See subfields.

## Subfield 01 (Commercial Choice Program Data, Agreement ID)

[DE 48](#), subelement 05, subfield 01 (Commercial Choice Program Data, Agreement ID) contains the Commercial Choice Program data agreement ID as received from Visa.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...4; LLVAR          |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Subfield 02 (Digital Authentication Program)

[DE 48](#), subelement 05, subfield 02 (Digital Authentication Program) contains the value to indicate when a transaction qualifies for the Digital Authentication Program as received from Visa.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...2; LLVAR          |
| Data Field           | Contents of subfield 02 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Contains Authentication Program 01 (DAF indicator). Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Subfield 03 (Authorization Response Source Reason Code)

[DE 48](#), subelement 05, subfield 03 (Authorization Response Source Reason Code) contains the authorization response reason code as received from Visa.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-1                   |
| Data Field           | Contents of subfield 03 |
| Justification        | N/A                     |

### Values

Contains the response source reason code that identifies the source of the Visa Field 39 response decision. Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Subfield 04 (Account Funding Source)

DE 48, subelement 05, subfield 04 (Account Funding Source) contains the Account Funding Source product code value received from Visa.

### Attributes

| Attribute       | Description |
|-----------------|-------------|
| Subfield ID     | 04          |
| Subfield Length | 2 positions |

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | an-1                    |
| Data Field          | Contents of subfield 04 |
| Justification       | N/A                     |

## Subfield 05 (Applied Product Platform Code)

DE 48, subelement 05, subfield 05 (Applied Product Platform Code) contains the Applied Product Platform Code value received from Visa.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Subfield ID         | 05                      |
| Subfield Length     | 2 positions             |
| Data Representation | an-2                    |
| Data Field          | Contents of subfield 05 |
| Justification       | N/A                     |

## Subfield 06 (Applied Cardholder ID Method)

DE 48, subelement 05, subfield 06 (Applied Cardholder ID Method) contains the Cardholder ID Method Indicator value received from Visa.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Subfield ID         | 06                      |
| Subfield Length     | 2 positions             |
| Data Representation | an-1                    |
| Data Field          | Contents of subfield 06 |
| Justification       | N/A                     |

## Subfield 07 (Applied E-Commerce Payment Indicator/ECI Only)

DE 48, subelement 05, subfield 07 (Applied E-Commerce Payment Indicator/ECI Only) contains the Applied ECommerce Payment Indicator/ECI Only code value received from Visa.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Subfield ID         | 07                      |
| Subfield Length     | 2 positions             |
| Data Representation | n-2                     |
| Data Field          | Contents of subfield 07 |
| Justification       | N/A                     |

## Subelement 06 (Advanced Digital Transaction Data 1)

[DE 48](#), subelement 06 (Advanced Digital Transaction Data 1) provides transaction-related data captured during checkout to the issuer.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 06                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...84; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 3                     |
| Justification          | See subfields         |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | .   | X   | C   |

### Application notes

The Dual Message System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

### Subfield 01 (Binding ID)

[DE 48](#), subelement 06, subfield 01 (Binding ID) helps to uniquely identify the component of data that is bound to the credential used in the transaction.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...48; LLVAR         |
| Data Field           | Contents of subfield 01 |
| Justification        | Left                    |

### Values

The unique and fixed identifier to a component of data bound to the credential.

## Application notes

| When DE 104 (Digital Payment Data), subelement 005 (Digital Authentication Data), subfield 001 (Cardholder Verification) indicates a value of...                                                                                                                                 | Then...                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (Transaction authenticated with a Mastercard-qualified multi-factor authentication [MFA] method) in position 1<br><br><b>NOTE: In Europe region: Transaction authenticated with a Mastercard-qualified MFA method used as Strong Customer Authentication [SCA] delegation.</b> | the component of data is an identifier demonstrating the possession of a consumer device that is bound to the token. Multiple identifiers may be created for the same consumer device bound to the token. It will help establish a pattern of device possession that further confirms the authenticity of the consumer who initiated the transaction. |
| 2 (Transaction authenticated with a Mastercard-operated authentication method) in position 1                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                       |
| 4 (Device wallet transaction authenticated with a Consumer Device Cardholder Verification Method [CDCVM]) in position 1<br><br><b>NOTE: This value may not be present in current device wallet implementations.</b>                                                              | the component of data is a unique identifier to the consumer device that is bound to the token. It will help establish a pattern of device usage that further confirms the authenticity of the consumer who initiated the transaction.                                                                                                                |

## Subfield 02 (Phone Number)

[DE 48](#), subelement 06, subfield 02 (Phone Number) contains the phone number that was used to authenticate the consumer during checkout, or the consumer's preferred phone number. The presence of the phone number helps establish the consumer's authenticity when matching the phone number provided during checkout to a list of known phone numbers for the consumer.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n...15; LLVAR           |
| Data Field           | Contents of subfield 02 |
| Justification        | Left                    |

Values

The full phone number or partial phone number of the consumer used during checkout.

Subfield 03 (Device Location)

DE 48, subelement 06, subfield 03 (Device Location) contains the latitude and longitude of the device used when the cardholder was authenticated during checkout. This field helps to determine if the transaction was initiated from an unexpected location.

Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-9                   |
| Data Field           | Contents of subfield 03 |
| Justification        | N/A                     |

Values

Latitude and longitude where the device of the consumer is attempting to check out is located.

Device location latitude an-4, hexadecimal encoded degrees with two decimal places.

Device location longitude an-4, hexadecimal encoded degrees with two decimal places.

Device location lat/long Sector n-1, one of the following values:

Device location lat/long Sector n-1 values

| Value | Latitude | Longitude |
|-------|----------|-----------|
| 1     | N        | W         |
| 2     | N        | E         |
| 3     | S        | W         |
| 4     | S        | E         |

## Subelement 07 (Advanced Digital Transaction Data 2)

[DE 48](#), subelement 07 (Advanced Digital Transaction Data 2) provides transaction-related data captured during checkout to the issuer.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | n-2                   |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...73; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 2                     |
| Justification          | See subfields         |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | .   | X   | C   |

### Application notes

The Dual Message System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.



## Subfield 01 (Email Address Format)

[DE 48](#), subelement 07, subfield 01 (Email Address Format) indicates if an email address was passed in hashed or clear format during checkout.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-1                     |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Values

| Value | Description                                  | Corresponding subfield 02...     |
|-------|----------------------------------------------|----------------------------------|
| 0     | Hashed email format received during checkout | Contains a hashed email address. |
| 1     | Clear email format received during checkout  | Contains a clear email address.  |

## Subfield 02 (Email Address)

[DE 48](#), subelement 07, subfield 02 (Email Address) contains the cardholder email address. The presence of the email address helps establish the consumer's authenticity when matching the email address provided during checkout to a list of known email addresses for the consumer.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...64; LLVAR         |
| Data Field           | Contents of subfield 02 |

| Attribute     | Description |
|---------------|-------------|
| Justification | Left        |

## Values

The clear email address or hashed email address collected during checkout.

## Application notes

Hashing helps the issuer verify the cardholder's email without the merchant or wallet provider providing the full email for privacy and security reasons.

When the merchant or wallet provider provides a hashed email address, Mastercard receives the hash email address and includes it in the data sent to the issuer. The issuer may apply the hash to an email address on file for the cardholder and attempt to match it with the hashed email address provided by the merchant or wallet provider.

When the hash computed by the issuer matches the hash of the one received from the merchant or wallet provider, the comparison with the hash value may be used by the issuer to assist in determining the transaction authorization decision.

The field contains the hash resulting from the following 'emailAddressHash' algorithm, right-padded with spaces. 'emailAddress' is the lower case UTF-8 bytes of the email address.

The following two functions define the hash algorithm:

```
public String emailAddressHash (String emailAddress) {  
    String random8Bytes = 123CCB2F30BA420B  
    return random8Bytes + leastSignificant24bytes(strongerHash(emailAddress +  
    random8Bytes))  
}  
public String strongerHash (String dataToHash) {  
    String currentHash = dataToHash;  
    for (int i = 0; i < 5000; i++) {  
        currentHash = sha256(currentHash);  
    }  
    return sha256(sha256(dataToHash) + currentHash);  
}
```

**NOTE:** String random8Bytes = 123CCB2F30BA420B is a fixed value for all merchants and wallet providers.

## Hash calculation example

| Component            | Example                                                                |
|----------------------|------------------------------------------------------------------------|
| Sample email address | cardholdername@walletprovider.com                                      |
| emailAddress         | 63617264686F6C6465726E616D654077616C6C65<br>7470726F76696465722E636F6D |

| Component                 | Example                                                              |
|---------------------------|----------------------------------------------------------------------|
| random8Bytes              | 123CCB2F30BA420B                                                     |
| Email address hash output | 123CCB2F30BA420B2476A8250E0E146164385A34<br>D34CC9E3A5680D7C5252C481 |

## Subelement 08 (Advanced Digital Transaction Data 3)

[DE 48](#), subelement 08 (Advanced Digital Transaction Data 3) provides transaction-related data captured during checkout to the issuer.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 08                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...98; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 4                     |
| Justification          | See subfields         |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | .   | X   | C   |

### Application notes

The Dual Message System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

## Subfield 01 (IP Address)

DE 48, subelement 08, subfield 01 (IP Address) contains a unique string of characters that identifies a device. The presence of the IP address helps determine if the transaction was initiated from an unusual network, helping establish a pattern of safe device usage that further confirms the authenticity of the consumer who initiated the transaction.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...39; LLVAR         |
| Data Field           | Contents of subfield 01 |
| Justification        | Left                    |

### Values

The IP address collected during checkout. Mastercard supports both IPv4 and IPv6 IP address formats.

### Example values

| IP address format | Example value                           |
|-------------------|-----------------------------------------|
| IPv4              | 100.10.0.1                              |
| IPv6              | 1234:5678:9AB1:1C23:1088:5103:9012:9120 |

## Subfield 02 (Shipping Address Line 1)

[DE 48](#), subelement 08, subfield 02 (Shipping Address Line 1) helps identify the cardholder that initiated the transaction. Shipping address is one component of data that helps establish the

consumer's authenticity when matching the consumer's own billing address or set of commonly used shipping addresses.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...30; LLVAR         |
| Data Field           | Contents of subfield 02 |
| Justification        | Left                    |

### Values

The shipping address line 1 collected during checkout.

## Subfield 03 (Shipping Address Country)

[DE 48](#), subelement 08, subfield 03 (Shipping Address Country) contains the three-character alphabetic country code where the shipping address is located. Refer to the *Quick Reference Booklet* for specification format.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | a-3                     |
| Data Field           | Contents of subfield 03 |
| Justification        | N/A                     |

### Values

The shipping address country collected during checkout.

## Subfield 04 (Shipping Address Postal Code)

DE 48, subelement 08, subfield 04 (Shipping Address Postal Code) contains the postal code of the shipping address.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 04                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...10; LLVAR         |
| Data Field           | Contents of subfield 04 |
| Justification        | Left                    |

### Values

The shipping address postal code collected during checkout.

## Subelement 09 (Additional PAN Data)

DE 48, subelement 09 (Additional PAN Data) is used to provide issuers with accounts related to the PAN in DE 2 (Primary Account Number [PAN]).

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 09                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | an...99; LLVAR        |
| Data Field             | Contents of subfields |
| Subfields              | 6                     |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                                                                             | Org | Sys | Dst |
|-------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                          | .   | X   | C   |
| Authorization Request Response/0110:<br>tokenization authorization request response | C   | X   | C   |
| Authorization Advice/0120: system-generated                                         | .   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                       | .   | X   | C   |
| Reversal Request/0400                                                               | .   | X   | C   |
| Reversal Advice/0420: system-generated                                              | .   | X   | C   |

### Application notes

Issuers may only populate this subelement in Authorization Request Response/0110: Tokenization Authorization Request Response messages when DE 2 (Primary Account Number [PAN]) contains an issuer-generated consumer VCN in the related Authorization Request/0100: Tokenization Authorization Request message. Mastercard removes this subelement before forwarding the message to the acquirer.

If applicable, Mastercard will populate subfields 11-13 with Mastercard One Credential funding credential data and send it to the issuer in request messages. Mastercard removes subfields 11-13 before forwarding the message to the acquirer.

### Subfield 01 (Additional Account Number Indicator)

DE 48, subelement 09, subfield 01 (Additional Account Number Indicator) indicates the type of account number in DE 48, subelement 09, subfield 02 (Additional Account Number).

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-1                    |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Values

| Value | Description            |
|-------|------------------------|
| A     | Primary Account Number |
| V     | Virtual card number    |

## Subfield 02 (Additional Account Number)

DE 48, subelement 09, subfield 02 (Additional Account Number) contains an account number related to the PAN in DE 2.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n...19; LLVAR           |
| Data Field           | Contents of subfield 02 |
| Justification        | N/A                     |

## Subfield 03 (Additional Account Number Expiration Date)

DE 48, subelement 09, subfield 03 (Additional Account Number Expiration Date) contains the expiration date of the account number in DE 48, subelement 09, subfield 02 (Additional Account Number).

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-4                     |
| Data Field           | Contents of subfield 03 |



| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

The Additional Account Number Expiration Date is given in YYMM format.

## Subfield 11 (Mastercard One Credential Account Number Indicator)

DE 48, subelement 09, subfield 11 (Mastercard One Credential Account Number Indicator) provides the Mastercard One Credential account number indicator to the issuer.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 11                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of position 1 |
| Justification        | N/A                    |

### Values

| Value | Description                           |
|-------|---------------------------------------|
| 1     | Mastercard One Credential Funding PAN |

## Subfield 12 (Mastercard One Credential Account Number)

DE 48, subelement 09, subfield 12 (Mastercard One Credential Account Number) provides the Mastercard One Credential account number (funding PAN) to the issuer.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 12          |
| Subfield Data Length | 2 positions |

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n...19; LLVAR              |
| Data Field          | Contents of positions 1-19 |
| Justification       | N/A                        |

## Subfield 13 (Mastercard One Credential Account Number Expiration Date)

DE 48, subelement 09, subfield 13 (Mastercard One Credential Account Number Expiration Date) provides the Mastercard One Credential account number expiration date to the issuer.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 13                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | n-4, format YYMM          |
| Data Field           | Contents of positions 1-4 |
| Justification        | N/A                       |

## Subelement 10 (Encrypted PIN Block Key)

[DE 48](#), subelement 10 (Encrypted PIN Block Key) contains 16 hexadecimal characters in the range 0–9 and A–F to represent the 16 bits of a PIN block key encrypted under another key. Individual network, program, or service rules define the other encryption key.

**NOTE:** The Dual Message Authorization System currently does not use this data element.

## Subelement 11 (Key Exchange Block Data [Double-Length Keys])

[DE 48](#), subelement 11 (Key Exchange Block Data [Double-Length Keys]) contains a data block specifically formatted to contain all the control data, encrypted key data, and key check values

to complete an encryption key change operation between any two processors (for example, between a customer or *Intermediate Network Facilities (INF)* and the network).

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 11                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-54                     |
| Data Field             | Contents of subfields 1–5 |
| Subfields              | 5                         |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 11 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                | Org | Sys | Dst |
|--------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: PEK exchange          | •   | X   | C   |
| Network Management Request Response/0810: PEK exchange | C   | X   | •   |

### Values

Subelement 11 must be encoded for subfields 1–5 defined as follows.

### Subfield 1 (Key Class ID)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Field          | Contents of positions 1–2 |
| Values              | PK (PIN key)              |

#### Subfield 2 (Key Index Number)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3-4 |
| Values              | 00                        |

#### Subfield 3 (Key Cycle Number)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 5-6 |
| Values              | 00-99 (sequential)        |

#### Subfield 4 (PIN Encryption Key [PEK])

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | an-32                      |
| Data Field          | Contents of positions 7-38 |
| Values              | hex, 0-9, A-F              |

#### Subfield 5 (Key Check Value)

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | an-16                       |
| Data Field          | Contents of positions 39-54 |
| Values              | hex, 0-9, A-F               |

### Subelement 11 (Key Exchange Block Data [Triple-Length Keys])

[DE 48](#), subelement 11 (Key Exchange Block Data [Triple-Length Keys]) contains a data block specifically formatted to contain all the control data, encrypted key data, and key check values

to complete an encryption key change operation between any two processors (for example, between a customer or *INF* and the network).

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 11                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-70                     |
| Data Field             | Contents of subfields 1–5 |
| Subfields              | 5                         |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 11 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                | Org | Sys | Dst |
|--------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: PEK exchange          | •   | X   | C   |
| Network Management Request Response/0810: PEK exchange | C   | X   | •   |

### Values

Subelement 11 must be encoded for subfields 1–5 as defined below.

### Subfield 1 (Key Class ID)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Field          | Contents of positions 1–2 |
| Values              | PK (PIN key)              |

### Subfield 2 (Key Index Number)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3–4 |
| Values              | 00                        |

### Subfield 3 (Key Cycle Number)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 5–6 |
| Values              | 00–99 (sequential)        |

### Subfield 4 (PIN Encryption Key [PEK])

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | an-48                      |
| Data Field          | Contents of positions 7–54 |
| Values              | hex, 0–9, A–F              |

### Subfield 5 (Key Check Value)

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | an-16                       |
| Data Field          | Contents of positions 55–70 |
| Values              | hex, 0–9, A–F               |

## Subelement 12 (Routing Indicator)

[DE 48](#), subelement 12 (Routing Indicator) is defined and derived by the Dual Message Authorization System and passed to the issuer to indicate that alternate issuer host routing has

been invoked. Subelement 12 is applicable only to issuers that use an alternate issuer host route instead of the Stand-In System.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 12                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 12 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated | •   | X   | C   |
| Reversal Request/0400                         | •   | X   | C   |

### Values

| Value | Description                   |
|-------|-------------------------------|
| A     | Alternate issuer host routing |
| P     | Primary issuer host routing   |

## Subelement 13 (Mastercard Hosted Mobile Phone Top-Up Request Data)

[DE 48](#), subelement 13 (Mastercard Hosted Mobile Phone Top-up Request Data) contains the mobile phone number and mobile phone service provider name.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 13                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans-47                |
| Data Field             | Contents of subfields |
| Subfields              | 2                     |
| Justification          | See subfields         |

### Usage

Following is the usage of subelement 13 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | .   | C   |
| Authorization Request Response/0110 | CE  | .   | CE  |

## Subfield 1 (Mobile Phone Number)

[DE 48](#), subelement 13, subfield 1 (Mobile Phone Number) contains the phone number of the wireless phone for which the customer is purchasing extra service.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | ans-17      |



| Attribute     | Description                |
|---------------|----------------------------|
| Data Field    | Contents of positions 1–17 |
| Justification | Left                       |

### Values

Cannot contain all spaces or all zeros.

## Subfield 2 (Mobile Phone Service Provider Name)

[DE 48](#), subelement 13, subfield 2 (Mobile Phone Service Provider Name) contains the name or other identifier of the mobile phone service provider.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-30                      |
| Data Field          | Contents of positions 18–47 |
| Justification       | Left                        |

### Values

Cannot contain all spaces or all zeros.

## Subelement 14 (Account Type Indicator)

[DE 48](#), subelement 14 (Account Type Indicator) contains the value that identifies the cardholder's account type choice. If this subelement is not provided by the acquirer, Mastercard will derive the value. Subelement 14 must only be included on Brazil Flex Card Service-enabled transactions.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 14                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | ans-1                  |
| Data Field             | Contents of position 1 |

| Attribute     | Description |
|---------------|-------------|
| Subfields     | N/A         |
| Justification | N/A         |

## Usage

Following is the usage of subelement 14 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | X   | C   |
| Authorization Request Response/0110         | CE  | XE  | CE  |
| Authorization Advice/0120                   | C   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |
| Authorization Advice Response/0130          | CE  | XE  | CE  |
| Reversal Request/0400                       | C   | X   | C   |
| Reversal Request Response/0410              | CE  | XE  | CE  |
| Reversal Advice/0420                        | •   | X   | C   |

Acquirers in Brazil must be able to receive DE 48 (Additional Data: National Use), subelement 14 (Account Type Indicator) in the following response messages only if they have sent this data element in an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message:

- Authorization Request Response/0110 message
- Authorization Advice Response/0130 message
- Reversal Request Response/0410 message

## Values

C = Credit Transaction

D = Debit Transaction

## Application notes

Subelement 14 (Account Type Indicator) is used in Brazil to identify the transaction as:

- Credit: (Credit/Mastercard). The transaction will be processed by the Dual Message Authorization System with a Mastercard credit brand/product.
- Debit: (Debit/Mastercard). The transaction will be processed by the Dual Message Authorization System as a Debit Mastercard transaction with a Mastercard debit brand/product.

For Brazil issued Flex Card Service enabled POS transactions acquired outside of Brazil, Mastercard will populate [DE 48](#), subelement 14 based on the value present in DE 23, position 3.

- C = Credit Transaction (values 0, 1, 2, 3, or 4)
- D = Debit Transaction (values 5, 6, 7, 8, or 9)

For Brazil issued Flex Card Service enabled ATM transactions acquired outside of Brazil, Mastercard will populate DE 48, subelement 14 based on the value present in DE 3, subfield 2:

- C = Credit Transaction if DE 3, subfield 2 is not equal to 10 or 20.
- D = Debit Transaction if DE 3, subfield 2 is equal to 10 or 20.

For Brazil issued Flex Card Service enabled transactions acquired in Brazil, Mastercard will populate DE 48, subelement 14 based on the value present in DE 3, subfield 2:

- C = Credit Transaction if DE 3, subfield 2 is not equal to 10 or 20.
- D = Debit Transaction if DE 3, subfield 2 is equal to 10 or 20.

For Brazil issued Flex Card Service enabled card present transactions acquired outside of Brazil, if the values in DE 23 is not present then Mastercard will populate DE 48, subelement 14 with C = Credit transaction.

For Brazil issued Flex Card Service enabled card not present transactions acquired outside of Brazil, Mastercard will populate DE 48, subelement 14 with 'C' = Credit transaction.

## Subelement 15 (Authorization System Advice Date and Time)

[DE 48](#), subelement 15 (Authorization System Advice Date and Time), (in UTC units) is a transaction time stamp that the Dual Message Authorization System supplies for each Authorization Advice/0120: acquirer-generated message. It indicates the date and time that the advice is entered into the network.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 15                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n-10                      |
| Data Field             | Contents of subfields 1–2 |
| Subfields              | 2                         |

| Attribute     | Description   |
|---------------|---------------|
| Justification | See subfields |

## Usage

Following is the usage of subelement 15 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: acquirer-generated                                                   | •   | X   | M   |
| Authorization Advice/0120: system generated                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | OE  | X   | C   |

## Values

See subfields.

## Application notes

Acquirers may use the contents of subelement 15 to aid in the creation of the clearing message.

## Subfield 1 (Date)

[DE 48](#), subelement 15, subfield 1 (Date) contains the valid date of the Authorization Advice/0120: acquirer-generated message inserted by the Dual Message Authorization System.

## Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-4                       |
| Data Field          | Contents of positions 1-4 |
| Justification       | N/A                       |

## Values

This subfield contains a valid date in MMDD format.

## Subfield 2 (Time)

[DE 48](#), subelement 15, subfield 2 (Time) contains the valid time of the Authorization Advice/0120: acquirer-generated message inserted by the Dual Message Authorization System.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-6                        |
| Data Field          | Contents of positions 5–10 |
| Justification       | N/A                        |

### Values

This subfield contains a valid time in hhmmss format.

## Subelement 16 (Processor Pseudo ICA)

[DE 48](#), subelement 16 (Processor Pseudo ICA) identifies the institution submitting a request or advice that is not the same as the institution provided in [DE 32 \(Acquiring Institution ID Code\)](#) or [DE 33 \(Forwarding Institution ID Code\)](#).

### Attributes

| Attribute              | Description |
|------------------------|-------------|
| Subelement ID          | 16          |
| Subelement Data Length | 2 positions |
| Data Representation    | n-7         |
| Data Field             | N/A         |
| Subfields              | N/A         |
| Justification          | N/A         |

### Usage

Following is the usage of [DE 48](#), subelement 16 (whether it is mandatory, conditional, optional, system-provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated | •   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |

### Application notes

The Dual Message Authorization System populates this subelement based on the institution submitting the transaction and the issuer's preference for receipt of this data.

## Subelement 17 (Authentication Indicator)

[DE 48](#), subelement 17 (Authentication Indicator) is defined by the Dual Message Authorization System and is passed to the issuer to indicate that the transaction qualified for Authentication service.

### Attributes

| Attribute              | Description |
|------------------------|-------------|
| Subelement ID          | 17          |
| Subelement Data Length | 2 positions |
| Data Representation    | n-1         |
| Subfields              | N/A         |

### Usage

Following is the usage of subelement 17 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | C   |
| Authorization Request Response/0110           | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated | •   | X   | C   |

| Message                                                                                      | Org | Sys | Dst |
|----------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated                                                  | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to acquirer-generated 0120) | •   | X   | C   |

### Valid values

| Value | Description                                             |
|-------|---------------------------------------------------------|
| 1     | Transaction qualified for Authentication Service Type 1 |
| 2     | Transaction qualified for Authentication Service Type 2 |

### Application notes

#### Cardholder Authentication Service

Acquirers must support receiving subelement 17 in authorization response messages.

Issuers participating in the Cardholder Authentication Service must be prepared to receive the valid cardholder authentication result in subelement 17.

## Subelement 18 (Service Parameters)

[DE 48](#), subelement 18 (Service Parameters) defines the service parameters of the transaction.

### Attributes

| Attribute              | Value                   |
|------------------------|-------------------------|
| Subelement ID          | 18                      |
| Subelement Data Length | 2 positions             |
| Data Representation    | ans...99; LLVAR         |
| Data Field             | Contents of subfield 01 |
| Subfields              | 1                       |
| Justification          | N/A                     |

### Usage

Following is the usage of subelement 18 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | .   |

### Values

Contains the Canada Domestic Indicator value in subfield 01.

## Subfield 01 (Canada Domestic Indicator)

[DE 48](#), subelement 18, subfield 01 (Canada Domestic Indicator) indicates the merchant accepts Canada domestic Debit Mastercard cards.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 01                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-1                      |
| Data Field           | Contents of positions 1-2 |
| Justification        | N/A                       |

### Values

Y = Yes, the merchant accepts Canada domestic Debit Mastercard cards.

Example: 18050101Y—subelement 18 (Service Parameters), length 05, subfield 01 (Canada Domestic Indicator), length 01, value Y.

**NOTE:** The remaining 94 characters are reserved for future use.

## Subelement 20 (Cardholder Verification Method)

Acquirers use [DE 48](#), subelement 20 (Cardholder Verification Method) to notify Mastercard that the original transaction is signature/offline PIN-based, no CVM used, or online PIN-based for



transaction routing in Authorization Advice/0120: acquirer-generated and Reversal Request/0400 messages.

### Attributes

| Attribute              | Value                  |
|------------------------|------------------------|
| Subelement ID          | 20                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 20 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: acquirer-generated | M   | X   | •   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | M   | X   | •   |
| Reversal Advice/0420                          | •   | X   | C   |

### Values

| Value | Description                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| P     | Online PIN verification                                                                                                                           |
| S     | Can signify signature, "Offline PIN verification" (for chip transactions), "M-PIN" (for Mobile Device with PIN entry capability) or "No CVM used" |

If DE 48, subelement 20 is present and the value is not P or S, the transaction will be rejected with format error 30.

## Subelement 21 (Acceptance Data)

*DE 48 (Additional Data-Private Use)*, subelement 21 (Acceptance Data) indicates the merchant terminal's capability to support specific programs and services.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 21                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | n-11; LLVAR           |
| Data Field             | Contents of subfields |
| Subfields              | 2                     |
| Justification          | See subfields         |

### Usage

Following is the usage of subelement 21 (whether it is mandatory, conditional, optional, system-provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |
| Reversal Advice/0420                          | •   | X   | C   |

### Values

See subfields.

### Application notes

The subelement should be provided to indicate the merchant terminal's capabilities in supporting specific programs and services.

## Subfield 01 (mPOS Acceptance Device Type)

*DE 48 (Additional Data: Private Use)*, subelement 21 (Acceptance Data), subfield 01 (mPOS Acceptance Device Type) identifies the type of device used by the merchant as the terminal for accepting mPOS transactions. This subfield is required for all mPOS terminals.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n-1                    |
| Data Field           | Contents of position 1 |
| Justification        | N/A                    |

### Values

| Value | Description                                                                 |
|-------|-----------------------------------------------------------------------------|
| 0     | Dedicated mPOS Terminal with PCI compliant dongle (with or without key pad) |
| 1     | Off the Shelf Mobile Device                                                 |
| 2-9   | Reserved for Future Use                                                     |

## Subfield 02 (Additional Terminal Capability Indicator)

*DE 48*, subelement 21, subfield 02 (Additional Terminal Capability Indicator) identifies Mastercard Consumer Presented QR and barcode terminal capabilities. This subfield is optional. Customers that have opted to submit or process tokenized Mastercard Consumer Presented QR transactions are expected to support this subfield.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 02          |
| Subfield Data Length | 2 positions |

| Attribute           | Description                                                        |
|---------------------|--------------------------------------------------------------------|
| Data Representation | n-2                                                                |
| Data Field          | Contents of positions 1-2                                          |
| Justification       | N/A                                                                |
| Position 1          | Mastercard Consumer Presented QR Capability Support Indicator      |
| Position 2          | Mastercard Consumer Presented Barcode Capability Support Indicator |

### Values

| Value | Description   |
|-------|---------------|
| 0     | Not Supported |
| 1     | Supported     |

### Subfield 2: Valid Position 1 and Position 2 Value Combinations

### Valid combinations

| Valid Combinations | QR Capability | Barcode Capability |
|--------------------|---------------|--------------------|
| 00                 | Not supported | Not supported      |
| 01                 | Not supported | Supported          |
| 10                 | Supported     | Not supported      |
| 11                 | Supported     | Supported          |

## Subelement 22 (Multi-Purpose Merchant Indicator)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator) may be used for multiple purposes needed to identify the transaction and may contain data that is unique to programs, services, or other specifics about the transaction.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 22                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...29; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 6                     |
| Justification          | See subfields         |

### Usage

Following is the usage of subelement 22 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | O   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Authorization Advice Response/0130            | C   | X   | C   |
| Reversal Request/0400                         | O   | X   | C   |
| Reversal Request Response/0410                | C   | X   | C   |
| Reversal Advice/0420                          | C   | X   | C   |

### Application notes

Dual Message Authorization System will remove [DE 48](#), subelement 22, subfields 01, 02, 03, 05, and 06 if received in an Authorization Request Response/0110 message from an issuer.

Dual Message Authorization System will remove [DE 48](#), subelement 22, subfields 01, 02, 03, 04, 05, and 06 if received in an Reversal Request Response/0410 or Advice Response/0130 message from an issuer.

Refer to the subfields contained within this subelement to obtain the identifier supporting programs, services, or other specifics about the transaction.

## Subfield 01 (Low-Risk Merchant Indicator)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator), subfield 01 (Low-Risk Merchant Indicator) is used by acquirers to identify why a low-risk transaction did not use Strong Customer Authentication (SCA) to prevent an unnecessary decline by an issuer in the European Economic Area (EEA), United Kingdom (UK), and Gibraltar.

For Visa-branded transactions, [DE 48](#), subelement 22 (Multi-Purpose Merchant Indicator), subfield 01 (Low-Risk Merchant Indicator) is used to identify the transaction is from a Trusted Merchant, not needing Strong Customer Authentication (SCA) and preventing unnecessary declines by issuers in the European Economic Area (EEA), United Kingdom (UK), and Gibraltar.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-2                    |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Usage

Following is the usage of subelement 22, subfield 01 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | O   | X   | C   |

## Values

| Value |   | Description                                      |
|-------|---|--------------------------------------------------|
| 01    | = | Merchant Initiated Transaction                   |
| 02    | = | Acquirer Low-Fraud and Transaction Risk Analysis |
| 03    | = | Recurring Payment                                |
| 04    | = | Low-Value Payment                                |
| 05    | = | Strong Customer Authentication (SCA) Delegation  |
| 06    | = | Secure Corporate Payment                         |
| 07    | = | Authentication Outage Exception                  |
| 08    | = | Trusted Merchant Exemption (Visa Only)           |

## Application notes

Subfield 01 values enable acquirers to provide issuers with an indicator to identify Authorization Request/0100 messages originating from merchants that use an acquirer exemption or other reasons for not applying SCA on transactions. This data will assist issuers in determining whether to approve or decline the transaction and to justify to Local Competent Authorities the reason why the issuers did not apply SCA on the transaction as required under PSD2.

Acquirers should only submit value 08 on Visa-branded transactions. The Dual Message Authorization System drops this value if received on transactions which are not Visa-branded.

## Subfield 02 (Single Tap Indicator)

[DE 48](#), subelement 22 (Multi-Purpose Merchant Indicator), subfield 02 (Single Tap Indicator) allows acquirers to indicate that the merchant's point of interaction environment is capable of supporting single tap processing.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-1                    |
| Data Field           | Contents of Subfield 02 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

## Usage

Following is the usage of subelement 22, subfield 02 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

## Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

## Values

| Value | Description                                   |
|-------|-----------------------------------------------|
| 1 =   | Merchant POI capable of single tap processing |

## Application notes

Value 1 must be used if the POI supports the service; otherwise, the subfield is not present.

## Subfield 03 (Response to PIN Request)

[DE 48](#), subelement 22 (Multi-Purpose Merchant Indicator), subfield 03 (Response to PIN Request) allows acquirers to indicate that the transaction contains an intentionally duplicated (replayed) Application Transaction Counter (ATC) value, within the context of a single tap transaction.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-1                    |
| Data Field           | Contents of Subfield 03 |



| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Usage

Following is the usage of subelement 22, subfield 03 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Values

| Value | Description                                                            |
|-------|------------------------------------------------------------------------|
| 1     | Transaction contains an intentionally duplicated (replayed) ATC value. |

### Application notes

Value 1 must be used by acquirers only if the issuer requests a PIN in single tap mode; otherwise, the subfield is not present.

## Subfield 04 (Issuer PIN Request in a Single Tap Mode)

[DE 48 \(Additional Data: Private Use\)](#), subelement 22 (Multi-Purpose Merchant Indicator), subfield 04 (Issuer PIN Request in a Single Tap Mode) allows issuers in Europe, Middle East/Africa, Asia/Pacific, and Latin America and the Caribbean regions to request a PIN within the context of a single tap transaction scenario. Does not apply to the United States and Canada regions.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 04          |
| Subfield Data Length | 2 positions |
| Data Representation  | an-1        |

| Attribute     | Description             |
|---------------|-------------------------|
| Data Field    | Contents of Subfield 04 |
| Justification | N/A                     |

### Usage

Following is the usage of subelement 22, subfield 04 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |

### Valid values

| Value | Description                             |
|-------|-----------------------------------------|
| 1     | Issuer requests PIN in Single Tap mode. |

### Application notes

Value 1 must be used if the issuer requests a PIN in Single Tap mode; otherwise, the subfield is not present.

When the issuer participates in the On-Behalf Contactless Counters Service, Mastercard will populate subfield 04 in Authorization Request Response/0110 messages and forward to the acquirer when:

- The issuer supports duplicate ATC
- A card present transaction is declined with response code 65 (Decline: Exceeds withdrawal count limits)
- [DE 48](#), subelement 22, subfield 02 is value 1 (Merchant capable of single tap processing)

### Subfield 05 (Cardholder/Merchant Initiated Transaction Indicator)

[DE 48](#), subelement 22 (Multi-Purpose Merchant Indicator), subfield 05 (Cardholder/Merchant Initiated Transaction Indicator) allows acquirers to identify whether the transaction was

initiated by the cardholder or the merchant as well as providing more detail as to the type of stored credential being used for the transaction.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 05                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-4                    |
| Data Field           | Contents of subfield 05 |
| Justification        | N/A                     |

### Usage

Following is the usage of subelement 22, subfield 05 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice Response/0130            | CE  | XE  | CE  |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Response/0410                        | CE  | XE  | CE  |
| Reversal Advice/0420                          | •   | X   | C   |

### Values

Mastercard will ensure the [DE 48](#), subelement 22, subfield 05 values include the ones listed in the CIT and MIT indicator values and examples table.

### CIT and MIT indicator values and examples

| Indicator category         | Indicator subcategory                                      | What's included in the definition                                                                                                                                                                                                                                                      | Examples                                                                                                                                   |
|----------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| C1<br>Cardholder-Initiated | 01<br>Credential-on-file (ad-hoc)                          | Consumer agrees to store the credential on file with the merchant for future cardholder-initiated and/or merchant-initiated transactions that may occur from time to time, or consumer chooses to use a credential previously placed on file with the merchant when making a purchase. | The consumer initiates a purchase and merchant prompts consumer to store the credential-on-file for future cardholder-initiated purchases. |
|                            | 02<br>Standing Order (variable amount and fixed frequency) | Consumer agrees to store the credential-on-file and initiates a first transaction in a series intended to be for a variable amount and a fixed frequency.                                                                                                                              | The initial transaction to store the credential-on-file for a monthly utility payment.                                                     |
|                            | 03<br>Subscription (fixed amount and frequency)            | Consumer agrees to store the credential-on-file and initiates a first transaction in a series intended to be for a fixed amount and a fixed frequency.                                                                                                                                 | The initial transaction to store the credential-on-file for a monthly newspaper subscription.                                              |
|                            | 04<br>Installment                                          | Consumer agrees to store the credential-on-file to establish an installment plan and initiates a first transaction in a series. The installment plan should reflect the single purchase of goods/services with a known amount and set frequency over a specified duration.             | The initial transaction to store the credential-on-file for a purchase to be paid through installment payments.                            |

Data element definitions and details  
Subfield 05 (Cardholder/Merchant Initiated Transaction Indicator)

| Indicator category                                              | Indicator subcategory                                            | What's included in the definition                                                                                                                                                                                                                                                                                                              | Examples                                                                                                                                                                             |
|-----------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1<br>Merchant-Initiated<br>Recurring Payment or<br>Installment | 01<br>Unscheduled Credential<br>on File                          | A transaction made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store and use the cardholder's account data to initiate one or more future transactions. The transaction can be for a fixed or variable amount but does not occur at regular intervals on a scheduled date. | The consumer agrees to allow a tollway merchant to initiate transactions when the balance in the consumer account with the merchant falls below an established amount (auto top-up). |
|                                                                 | 02<br>Standing Order (variable<br>amount and fixed<br>frequency) | A transaction made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store and use the cardholder's account data for a transaction that must occur at regularly scheduled intervals for a variable amount, as specified in the agreement.                                        | Monthly utility payments.                                                                                                                                                            |

Data element definitions and details  
Subfield 05 (Cardholder/Merchant Initiated Transaction Indicator)

| Indicator category | Indicator subcategory                           | What's included in the definition                                                                                                                                                                                                                                                                                                                                                                                                  | Examples                                                                                                                                             |
|--------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | 03<br>Subscription (fixed amount and frequency) | A transaction made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store and use the cardholder's account data for a transaction that must occur at regularly scheduled intervals for a fixed amount, as specified in the agreement. The subscription agreement may include an allowance for introductory pricing or for price changes to occur from time to time. | Monthly subscription or fixed monthly service payment.                                                                                               |
|                    | 04<br>Installment                               | A transaction in a series of transactions made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store the cardholder's account data for use by the merchant to initiate one or more future transactions for a known amount over a specified duration based on a single purchase.                                                                                    | Purchase a television for \$1,000 USD, pay in four equal installments of \$250 USD (first transaction is CIT, remaining three transactions are MIT). |

| Indicator category                            | Indicator subcategory        | What's included in the definition                                                                                                                                       | Examples                                                                                                                       |
|-----------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| M2<br>Merchant-Initiated<br>Industry Practice | 05<br>Partial Shipment       | Occurs when an agreed quantity of goods ordered through e-commerce are not all available for shipment at the time of purchase. Each shipment is a separate transaction. | Consumer ordered goods that ship at different times.                                                                           |
|                                               | 06<br>Related/Delayed Charge | An additional account charge after initial services have been rendered and payment has been processed.                                                                  | Hotel mini bar charge after the cardholder has checked out of hotel.                                                           |
|                                               | 07<br>No Show Charge         | A penalty charged according to the merchant's cancellation policy.                                                                                                      | Cardholder's cancellation of a reservation without providing proper advance notice to the merchant.                            |
|                                               | 08<br>Resubmission           | Previous attempt to obtain authorization for a transaction has been declined but the issuer's response does not prohibit the merchant from trying again later.          | <ul style="list-style-type: none"> <li>Insufficient funds/over credit limit response</li> <li>Transit debt recovery</li> </ul> |

## Subfield 06 (Message Format Version Code: Acquirer)

[DE 48](#), subelement 22 (Multi-Purpose Merchant Indicator), subfield 06 (Message Format Version Code: Acquirer) contains a code that identifies the transaction message standard applied in the formatting and submission of the acceptor-related data elements.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 06                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-1                    |
| Data Field           | Contents of Subfield 06 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

## Usage

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | M   | X   | .   |
| Authorization Advice/0120: acquirer-generated | M   | X   | .   |
| Reversal Request/0400                         | M   | X   | .   |

### Valid values

| Value | Description                  |
|-------|------------------------------|
| A     | Mastercard Pre 23Q2 Release  |
| B     | Mastercard Post 23Q2 Release |

## Application notes

The Message Format Version Code: Acquirer must be present in all transactions and contain a valid value.

*AN 6022 Introduction and Standardization of Transaction Data Elements* introduced changes to the formatting, alignment, and population criteria of acceptor (Merchant/ATM Owner) name, location and related data elements and introduced additional data elements to fully capture critical Acceptor data. As there may be varying adoption timelines for upstream acquirer, processor and Merchant/ATM Owner systems, acquirers must utilize this code to identify at the transaction level for which standard was applied. Mastercard will rely on this code to modify, realign and append data into the message format standard defined in AN 6022, allowing downstream Mastercard and customer (issuer) systems to operate consistently throughout the transition period and avoid any processing or cardholder disruption.

Value A indicates the transaction does not comply with the new standards as it contains acceptor data in the pre-23Q2 release version format.

Value B indicates the transaction does comply with the new standards as it contains acceptor data in the post-23Q2 release version format.



## Subelement 23 (Payment Initiation Channel)

[DE 48](#), subelement 23 (Payment Initiation Channel) provides information about the device type used to identify mobile-initiated (m-commerce) or other EMV Contact, Magnetic stripe Contactless, or M/Chip Contactless transactions.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 23                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | an-2                   |
| Data Field             | Contents of subfield 1 |
| Subfields              | 1                      |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 23 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | P   | C   |
| Authorization Advice/0120                   | C   | P   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |
| Reversal Advice/0420                        | •   | X   | C   |

### Values

See subfield 1.

### Application notes

Acquirers in regions where it is mandated must ensure that each newly deployed or re-deployed contactless-enabled POS terminal used by their merchants transmits the device type indicator value, if present on the card or non-card form factor, in data element [DE 48 \(Additional Data:](#)

*Private Use*, subelement 23 (Payment Initiation Channel), subfield 1 (Device Type) of authorization messages.

## Subfield 1 (Device Type)

DE 48, subelement 23, subfield 1 (Device Type) indicates the type of device used at the terminal. In the scope of chip transactions it must be populated using the device type field value of chip tag 9F6E.

**NOTE: Effective immediately, new values in DE 48, subelement 23, subfield 1 contain an indication of the form factor. Existing values from 00–19 may continue to be used, and their meanings remain unchanged; however, Mastercard will not send these values in future communication as part of DE 48, subelement 23, subfield 1. New values from 20–99 exclusively indicate the form factor only without also indicating the storage technology. Mastercard has introduced new device type values from 20–33 in anticipation of new form factors for payments, with values 34–99 reserved for future use. For information on storage technology of device types (such as Trusted Execution Environment [TEE] or Secure Element), refer to the description of DE 48, subelement 33 (PAN Mapping File Information), subfield 8 (Storage Technology).**

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Field          | Contents of positions 1–2 |
| Justification       | N/A                       |

### Values

**NOTE: Some values from 00–19 may indicate not only the physical form factor but also other attributes such as device technology and payment app specifications.**

00 = Card

01 = Mobile Network Operator (MNO) controlled removable secure element (SIM or UICC) personalized for use with a mobile phone or smartphone

02 = Key Fob

03 = Watch using a contactless chip or a fixed (non-removable) secure element not controlled by the MNO

04 = Mobile Tag

05 = Wristband

06 = Mobile Phone Case or Sleeve

07 = Mobile phone or smartphone with a fixed (non-removable) secure element controlled by the MNO, for example, code division multiple access (CDMA)

08 = Removable secure element not controlled by the MNO, for example, memory card personalized for use with a mobile phone or smartphone

09 = Mobile Phone or smartphone with a fixed (non-removable) secure element not controlled by the MNO

10 = MNO controlled removable secure element (SIM or UICC) personalized for use with a tablet or e-book

11 = Tablet or e-book with a fixed (non-removable) secure element controlled by the MNO

12 = Removable secure element not controlled by the MNO, for example, memory card personalized for use with a tablet or e-book

13 = Tablet or e-book with fixed (non-removable) secure element not controlled by the MNO

14 = Mobile phone or smartphone with a payment application running in a host processor

15 = Tablet or e-book with a payment application running in a host processor

16 = Mobile phone or smartphone with a payment application running in the Trusted Execution Environment (TEE) of a host processor

17 = Tablet or e-book with a payment application running in the TEE of a host processor

18 = Watch with a payment application running in the TEE of a host processor

19 = Watch with a payment application running in a host processor

**NOTE: Values from 20–99 exclusively indicate the form factor only without also indicating the storage technology.**

## Values

| Value | Meaning           | Example                                                                                                        |
|-------|-------------------|----------------------------------------------------------------------------------------------------------------|
| 20    | Card              |                                                                                                                |
| 21    | Phone             | Mobile phone                                                                                                   |
| 22    | Tablet/e-reader   | Tablet computer or e-reader                                                                                    |
| 23    | Watch/Wristband   | Watch or wristband, including a fitness band, smart strap, disposable band, watch add-on, and security/ID band |
| 24    | Sticker           |                                                                                                                |
| 25    | PC                | PC or laptop                                                                                                   |
| 26    | Device Peripheral | Mobile phone case or sleeve                                                                                    |

| Value | Meaning             | Example                                                                       |
|-------|---------------------|-------------------------------------------------------------------------------|
| 27    | Tag                 | Key fob or mobile tag                                                         |
| 28    | Jewelry             | Ring, bracelet, necklace, and cuff links                                      |
| 29    | Fashion Accessory   | Handbag, bag charm, and glasses                                               |
| 30    | Garment             | Dress                                                                         |
| 31    | Domestic Appliance  | Refrigerator, washing machine                                                 |
| 32    | Vehicle             | Vehicle, including vehicle attached devices                                   |
| 33    | Media/Gaming Device | Media or gaming device, including a set top box, media player, and television |
| 34    | Virtual Reality     | Headset/Headsets/Glasses                                                      |

Values 35 through 99 equals Reserved for future form factors. Any value in this range may occur within form factor and transaction data without prior notice.

## Subelement 24 (Account Level Management [ALM] Service Data)

DE 48, subelement 24 (Account Level Management [ALM] Service Data) supports Regional ALM processing and Mastercard One Credential processing.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subelement ID       | 24                        |
| Subelement Length   | 2 positions               |
| Data Representation | an...65; LLVAR            |
| Data Field          | Contents of subfields 1-7 |
| Subfields           | 7                         |
| Justification       | See subfields             |

### Usage

Following is the usage of subelement 24 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                            | Org | Sys | Dst |
|----------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                         | •   | X   | C   |
| Authorization Request Response/0110                                                                | •   | X   | C   |
| Authorization Advice/0120: system-generated                                                        | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                      | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated                                               | •   | X   | C   |
| Authorization Advice Response/0130: system-generated<br>(responding to an acquirer-generated 0120) | •   | X   | C   |
| Reversal Request/0400                                                                              | •   | X   | C   |
| Reversal Request Response/0410                                                                     | •   | X   | C   |
| Reversal Advice/0420                                                                               | •   | X   | C   |
| Reversal Advice Response/0430                                                                      | O   | X   | •   |

### Values

See subfields.

### Application notes

The following subfields will be populated when applicable to the ALM program indicated by Subfield 1 (ALM Service Code):

- Subfield 2: ALM Product Graduation Plus or Registered Product Code
- Subfield 3: ALM Product Class
- Subfield 4: ALM Rate Type

Subfields that are not applicable to the designated ALM service of the registered PAN will be populated with 'XXX'. The absence of [DE 48](#), subelement 24 for opt-in customers means the account range of the PAN does not participate in ALM.

Mastercard reserves the right add new valid codes at any time. Customer must be able to accept and process any value provided within any of the sub elements for authorization and clearing processing. In the event new valid values are added, refer to the *Account Level Management Manual*.

## Subfield 1 (ALM Service Code)

DE 48, subelement 24, subfield 1 (ALM Service Code) indicates the ALM program and service.

ALM Service Code represents the Account Level Management service or combination of ALM services for which the PAN is actively registered. The ALM Service Code is assigned at the time the PAN is registered in Account Level Management and will be populated in the supported authorization messages to ensure ALM processing is applied throughout the lifecycle of the transaction.

ALM processing was not applied to the authorization message when the ALM Service Code is present with one of the following values:

- 00000 = ALM Service Unavailable
  - When the Dual Message Authorization System is unable to make a successful connection to the Account Level Management Platform for PAN registration validation, the customer will receive an ALM Service Unavailable service code value
- 00126 = PAN not registered for ALM, but the issuer's Account Range participates, transaction will process business as usual
  - When the issuer's Account Range is activated for ALM Servicing, identified as having at least one-PAN actively registered, but the PAN as provided in the authorization message is not actively registered in ALM
- All other values provided will represent an individual ALM Service or ALM Service Combinations based on the PAN's active registration

Mastercard reserves the right to add new valid values for ALM Service Code at any time. Customers must be able to accept and process any valid values provided within the authorization messages. If new valid values are added, refer to the *Account Level Management Manual*.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-5                   |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

### ALM service names

| <b>ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)</b> | <b>DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)</b> |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALM SERVICE UNAVAILABLE IN AUTHORIZATION                                                                    | 00000                                                                                                                                                                                                                                       |
| PAN NOT REGISTERED FOR ALM OR MASTERCARD ONE CREDENTIAL, BUT ISSUERS ACCOUNT RANGE IS ACTIVE                | 00126                                                                                                                                                                                                                                       |
| PAN NOT REGISTERED FOR ALM, BUT ISSUERS ACCOUNT RANGE IS ACTIVE                                             | 00126                                                                                                                                                                                                                                       |
| PAN IS REGISTERED FOR ALM, BUT THE TRANSACTION WAS PROCESSED AS DEBIT                                       | 00127                                                                                                                                                                                                                                       |
| ENHANCED VALUE                                                                                              | 00402                                                                                                                                                                                                                                       |
| ENHANCED VALUE and PRODUCT GRADUATION PLUS                                                                  | 00513                                                                                                                                                                                                                                       |
| PRODUCT GRADUATION PLUS                                                                                     | 00616                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 1                                                                  | 00703                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 2                                                                  | 00705                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 3                                                                  | 00707                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 4                                                                  | 00710                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 5                                                                  | 00717                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 1                                      | 00804                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 2                                      | 00806                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 3                                      | 00808                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 4                                      | 00811                                                                                                                                                                                                                                       |

| <b>ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)</b> | <b>DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)</b> |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 5                                      | 00818                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 1                                                                 | 00903                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 2                                                                 | 00905                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 3                                                                 | 00907                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 4                                                                 | 00910                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 5                                                                 | 00917                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 1                                     | 01004                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 2                                     | 01006                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 3                                     | 01008                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 4                                     | 01011                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 5                                     | 01018                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 1                                                           | 01103                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 2                                                           | 01105                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 3                                                           | 01107                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 4                                                           | 01110                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 5                                                           | 01117                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 1                               | 01204                                                                                                                                                                                                                                       |



| <b>ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)</b> | <b>DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)</b> |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 2                               | 01206                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 3                               | 01208                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 4                               | 01211                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 5                               | 01218                                                                                                                                                                                                                                       |
| AFFLUENT SHORTFALL WORLD ELITE MASTERCARD                                                                   | 01323                                                                                                                                                                                                                                       |
| AFFLUENT SHORTFALL WORLD ELITE MASTERCARD and PRODUCT GRADUATION PLUS                                       | 01425                                                                                                                                                                                                                                       |
| WORLD HIGH VALUE                                                                                            | 01519                                                                                                                                                                                                                                       |
| WORLD HIGH VALUE and PRODUCT GRADUATION PLUS                                                                | 01620                                                                                                                                                                                                                                       |
| AFFLUENT SHORTFALL WORLD MASTERCARD                                                                         | 01723                                                                                                                                                                                                                                       |
| AFFLUENT SHORTFALL WORLD MASTERCARD and PRODUCT GRADUATION PLUS                                             | 01825                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: WORLD QUALIFIED                                                        | 02205                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: NAME FOR GCMS PRODUCT ID MPE QUALIFIED                                 | 02207                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: WORLD ELITE QUALIFIED                                                  | 02210                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: MUSE MASTERCARD QUALIFIED                                              | 02217                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: REQUALIFICATION 1 (MUSE MASTERCARD REQUALIFIED)                        | 02301                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: REQUALIFICATION 2 (WORLD ELITE REQUALIFIED)                            | 02424                                                                                                                                                                                                                                       |
| MASTERCARD ONE CREDENTIAL                                                                                   | 50000                                                                                                                                                                                                                                       |

| <b>ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)</b>                  | <b>DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)</b> |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PAN REGISTERED FOR MASTERCARD ONE CREDENTIAL, BUT NO CONSUMER PREFERENCE MET OR TRANSACTION WAS NOT ELIGIBLE FOR THE SERVICE | 50127                                                                                                                                                                                                                                       |
| ENHANCED VALUE AND MASTERCARD ONE CREDENTIAL                                                                                 | 50402                                                                                                                                                                                                                                       |
| ENHANCED VALUE AND PRODUCT GRADUATION PLUS AND MASTERCARD ONE CREDENTIAL                                                     | 50513                                                                                                                                                                                                                                       |
| PRODUCT GRADUATION PLUS AND MASTERCARD ONE CREDENTIAL                                                                        | 50616                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 1 AND MASTERCARD ONE CREDENTIAL                                                     | 50703                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 2 AND MASTERCARD ONE CREDENTIAL                                                     | 50705                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 3 AND MASTERCARD ONE CREDENTIAL                                                     | 50707                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 4 AND MASTERCARD ONE CREDENTIAL                                                     | 50710                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 5 AND MASTERCARD ONE CREDENTIAL                                                     | 50717                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 1 AND PRODUCT GRADUATION PLUS AND MASTERCARD ONE CREDENTIAL                         | 50804                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 2 AND PRODUCT GRADUATION PLUS AND MASTERCARD ONE CREDENTIAL                         | 50806                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 3 AND PRODUCT GRADUATION PLUS AND MASTERCARD ONE CREDENTIAL                         | 50808                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 4 AND PRODUCT GRADUATION PLUS AND MASTERCARD ONE CREDENTIAL                         | 50811                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT CORE LEVEL 5 AND PRODUCT GRADUATION PLUS AND MASTERCARD ONE CREDENTIAL                         | 50818                                                                                                                                                                                                                                       |

| <b>ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)</b> | <b>DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)</b> |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 1 AND MASTERCARD ONE CREDENTIAL                                   | 50903                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 2 AND MASTERCARD ONE CREDENTIAL                                   | 50905                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 3 AND MASTERCARD ONE CREDENTIAL                                   | 50907                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 4 AND MASTERCARD ONE CREDENTIAL                                   | 50910                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 5 AND MASTERCARD ONE CREDENTIAL                                   | 50917                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 1 AND MASTERCARD ONE CREDENTIAL       | 51004                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 2 AND MASTERCARD ONE CREDENTIAL       | 51006                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 3 AND MASTERCARD ONE CREDENTIAL       | 51008                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 4 AND MASTERCARD ONE CREDENTIAL       | 51011                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 5 AND MASTERCARD ONE CREDENTIAL       | 51018                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 1 AND MASTERCARD ONE CREDENTIAL                             | 51103                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 2 AND MASTERCARD ONE CREDENTIAL                             | 51105                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 3 AND MASTERCARD ONE CREDENTIAL                             | 51107                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 4 AND MASTERCARD ONE CREDENTIAL                             | 51110                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 5 AND MASTERCARD ONE CREDENTIAL                             | 51117                                                                                                                                                                                                                                       |

| <b>ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)</b> | <b>DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)</b> |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 1 AND MASTERCARD ONE CREDENTIAL | 51204                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 2 AND MASTERCARD ONE CREDENTIAL | 51206                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 3 AND MASTERCARD ONE CREDENTIAL | 51208                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 4 AND MASTERCARD ONE CREDENTIAL | 51211                                                                                                                                                                                                                                       |
| SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 5 AND MASTERCARD ONE CREDENTIAL | 51218                                                                                                                                                                                                                                       |
| WORLD ELITE SHORTFALL AND MASTERCARD ONE CREDENTIAL                                                         | 51323                                                                                                                                                                                                                                       |
| WORLD ELITE SHORTFALL AND PRODUCT GRADUATION AND MASTERCARD ONE CREDENTIAL                                  | 51425                                                                                                                                                                                                                                       |
| WORLD HIGH VALUE AND MASTERCARD ONE CREDENTIAL                                                              | 51519                                                                                                                                                                                                                                       |
| WORLD HIGH VALUE AND PRODUCT GRADUATION AND MASTERCARD ONE CREDENTIAL                                       | 51620                                                                                                                                                                                                                                       |
| WORLD SHORTFALL AND MASTERCARD ONE CREDENTIAL                                                               | 51723                                                                                                                                                                                                                                       |
| WORLD SHORTFALL AND PRODUCT GRADUATION AND MASTERCARD ONE CREDENTIAL                                        | 51825                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: WORLD QUALIFIED AND MASTERCARD ONE CREDENTIAL                          | 52205                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: NAME FOR GCMS PRODUCT ID MPE QUALIFIED AND MASTERCARD ONE CREDENTIAL   | 52207                                                                                                                                                                                                                                       |

| <b>ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)</b>        | <b>DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)</b> |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONSUMER PRODUCT MONITORING SERVICE: WORLD ELITE QUALIFIED AND MASTERCARD ONE CREDENTIAL                           | 52210                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: MUSE MASTERCARD QUALIFIED AND MASTERCARD ONE CREDENTIAL                       | 52217                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: REQUALIFICATION 1 (MUSE MASTERCARD REQUALIFIED) AND MASTERCARD ONE CREDENTIAL | 52301                                                                                                                                                                                                                                       |
| CONSUMER PRODUCT MONITORING SERVICE: REQUALIFICATION 2 (WORLD ELITE REQUALIFIED) AND MASTERCARD ONE CREDENTIAL     | 52424                                                                                                                                                                                                                                       |

## Subfield 2 (ALM Product Graduation Plus or Registered Product Code)

*DE 48*, subelement 24, subfield 2 (ALM Product Graduation Plus or Registered Product Code) contains the licensed graduated product code or the registered product code for those transactions identified with a PAN registered and qualifying for Product Graduation Plus or Consumer Product Monitoring Service, or a combination of Account Level Management Services.

If the PAN is not actively registered in ALM Product Graduation Plus or Consumer Product Monitoring Service, the field will be populated as 'XXX'.

Mastercard reserves the right to add new product code valid values for eligible ALM Product Graduation Plus or the Consumer Product Monitoring Service at any time and customers must be able to accept, and process any valid values provided within the authorization messages. For new valid values, a complete product code listing, or country availability, refer to the *Account Level Management Manual*.

### Attributes

| <b>Attribute</b>     | <b>Description</b> |
|----------------------|--------------------|
| Subfield ID          | 02                 |
| Subfield Data Length | 2 positions        |

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-3                   |
| Data Field          | Contents of subfield 2 |
| Justification       | N/A                    |

### Values

For information about valid values for ALM, refer to the *Account Level Management Manual*.

### Subfield 3 (ALM Product Class)

[DE 48](#), subelement 24, subfield 3 (ALM Product Class) indicates the Product Class Override for the applicable ALM Service. Product Class provides the ability to vary interchange rates by product grouping, while maintaining common interchange structures.

If the PAN is not actively registered in ALM Product Graduation Plus, the field will be populated as 'XXX'.

Mastercard reserves the right to add new valid values for Product Class at any time and customers must be able to accept, and process any valid values provided within the authorization messages. In the event new valid values are added, refer to the *Account Level Management Manual*.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 03                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-3                   |
| Data Field           | Contents of subfield 3 |
| Justification        | N/A                    |

### Values

For information about valid values for ALM, refer to the *Account Level Management Manual*.

### Subfield 4 (ALM Rate Type)

[DE 48](#), subelement 24, subfield 4 (ALM Rate Type) indicates the Rate Type number for the applicable ALM Service. Rate Types provide the ability to vary interchange rates in the event that unique IRDs are not assigned for an applicable business service arrangement. PANs actively

registered for ALM Services that do not involve Product Graduation Plus will have an associated ALM Rate Type. Some ALM Services may combine Rate Types with Product Class Overrides (Subfield 3).

If an ALM Rate Type is not supported for the ALM service of the registered PAN, the field will be populated as 'XXX'.

Mastercard reserves the right to add new valid values for Rate Types at any time and customers must be able to accept, and process any valid values provided within the authorization messages. In the event new valid values are added, refer to the [Account Level Management Manual](#).

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 04                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-3                   |
| Data Field           | Contents of subfield 4 |
| Justification        | N/A                    |

### Values

| Rate type | Rate type description                      |
|-----------|--------------------------------------------|
| 801       | Enhanced Value Rate Adjustment             |
| 811       | Core Business to Level 1 Adjustment        |
| 812       | Core Business to Level 2 Adjustment        |
| 813       | Core Business to Level 3 Adjustment        |
| 814       | Core Business to Level 4 Adjustment        |
| 815       | Core Business to Level 5 Adjustment        |
| 821       | World Business to Level 1 Adjustment       |
| 822       | World Business to Level 2 Adjustment       |
| 823       | World Business to Level 3 Adjustment       |
| 824       | World Business to Level 4 Adjustment       |
| 825       | World Business to Level 5 Adjustment       |
| 831       | World Elite Business to Level 1 Adjustment |

| Rate type | Rate type description                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------|
| 832       | World Elite Business to Level 2 Adjustment                                                                                 |
| 833       | World Elite Business to Level 3 Adjustment                                                                                 |
| 834       | World Elite Business to Level 4 Adjustment                                                                                 |
| 835       | World Elite Business to Level 5 Adjustment                                                                                 |
| 850       | World High Value Adjustment                                                                                                |
| 851       | World Consumer Shortfall Adjustment                                                                                        |
| 852       | World Elite Consumer Shortfall Adjustment                                                                                  |
| 881       | Canada World Consumer Adjustment *                                                                                         |
| 882       | Canada MPE (Mastercard® branding to be announced) *                                                                        |
| 883       | Canada World Elite Adjustment *                                                                                            |
| 884       | Canada MUSE Adjustment *                                                                                                   |
| 885       | Canada MUSE Requalified Down Adjustment *                                                                                  |
| 886       | Canada World Elite Requalified Down Adjustment *                                                                           |
| XXX       | Rate Type does not apply, or has not been configured for the ALM Service Code provided in DE 48, subelement 24, subfield 1 |

**NOTE: \*Consumer Product Monitoring Service will process using Rate Types associated with a Registered Product Code in subfield 2 (ALM Product Code).**

## Subfield 05 (Mastercard One Credential Acceptance Brand)

DE 48, subelement 24, subfield 05 (Mastercard One Credential Acceptance Brand) contains the acceptance brand of the product code present in DE 48, subelement 24, subfield 2 (ALM Product Graduation Plus or Registered Product Code).

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 05                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-3                      |
| Data Field           | Contents of positions 1-3 |
| Justification        | N/A                       |



### Values

| Value | Description      |
|-------|------------------|
| MCC   | Mastercard       |
| DMC   | Debit Mastercard |
| MSI   | Maestro          |

### Application notes

In response messages for regular authorization, advice, and reversal messages for Mastercard One Credential transactions, acquirers will only receive DE 48, subelement 24, subfields 05-07 if the Mastercard One Credential service assigns a funding PAN and the transaction will have Dynamic interchange applied.

In the following messages, acquirers will receive DE 48, subelement 24, subfields 05-07 whenever the Mastercard One Credential service assigns a funding PAN, regardless of Static or Dynamic interchange:

- Product Status Inquiry Authorization Response/0110 (Mastercard-Generated) message
- Account Status Inquiry with Product Status Inquiry Authorization Response/0110 message
- Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Response/0110 message

### Subfield 06 (Mastercard One Credential Static/Dynamic Interchange Indicator)

DE 48, subelement 24, subfield 06 (Mastercard One Credential Static/Dynamic Interchange Indicator) identifies the transaction as being subject to dynamic or static interchange.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 06                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of position 1 |
| Justification        | N/A                    |

### Values

| Value | Description                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| D     | Dynamic Interchange: the transaction's interchange will be based on the Mastercard One Credential Funding PAN instead of the DE 2 primary PAN |
| S     | Static Interchange: the transaction's interchange will be based on the DE 2 primary PAN                                                       |

### Application notes

In response messages for regular authorization, advice, and reversal messages for Mastercard One Credential transactions, acquirers will only receive DE 48, subelement 24, subfields 05-07 if the Mastercard One Credential service assigns a funding PAN and the transaction will have Dynamic interchange applied.

In the following messages, acquirers will receive DE 48, subelement 24, subfields 05-07 whenever the Mastercard One Credential service assigns a funding PAN, regardless of Static or Dynamic interchange:

- Product Status Inquiry Authorization Response/0110 (Mastercard-Generated) message
- Account Status Inquiry with Product Status Inquiry Authorization Response/0110 message
- Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Response/0110 message

### Subfield 07 (Mastercard One Credential Issuer Account Range)

DE 48, subelement 24, subfield 07 (Mastercard One Credential Issuer Account Range) contains the Mastercard One Credential Funding PAN's issuer account range.

### Attributes

| Attribute            | Description                |
|----------------------|----------------------------|
| Subfield ID          | 07                         |
| Subfield Data Length | 2 positions                |
| Data Representation  | n...19; LLVAR              |
| Data Field           | Contents of positions 1-19 |
| Justification        | N/A                        |

## Application notes

In response messages for regular authorization, advice, and reversal messages for Mastercard One Credential transactions, acquirers will only receive DE 48, subelement 24, subfields 05-07 if the Mastercard One Credential service assigns a funding PAN and the transaction will have Dynamic interchange applied.

In the following messages, acquirers will receive DE 48, subelement 24, subfields 05-07 whenever the Mastercard One Credential service assigns a funding PAN, regardless of Static or Dynamic interchange:

- Product Status Inquiry Authorization Response/0110 (Mastercard-Generated) message
- Account Status Inquiry with Product Status Inquiry Authorization Response/0110 message
- Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Response/0110 message

## Subelement 25 (Mastercard Cash Program Data)

[DE 48](#), subelement 25 (Mastercard Cash Program Data) contains information necessary to process Mastercard Cash transactions, which includes message type identifiers.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 25                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...14; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 1                     |
| Justification          | See subfields         |

### Usage

Following is the usage of subelement 25 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

| Message               | Org | Sys | Dst |
|-----------------------|-----|-----|-----|
| Reversal Request/0400 | C   | X   | C   |
| Reversal Advice/0420  | .   | X   | C   |

## Values

See subfields.

## Subfield 01 (Message Identifier)

[DE 48](#), subelement 25, subfield 01 (Message Identifier) indicates the type of cash transaction.

## Attributes

| Attribute            | Description                                         |
|----------------------|-----------------------------------------------------|
| Subfield ID          | 01                                                  |
| Subfield Data Length | 2 positions                                         |
| Data Representation  | ans...10; LLVAR                                     |
| Data Field           | Contents of positions 1–2, positions 3–10 undefined |
| Justification        | N/A                                                 |

## Values

| Value | Description                                                    |
|-------|----------------------------------------------------------------|
| CM    | Confirmation message                                           |
| LP    | Linked load request with a purchase                            |
| LR    | Unlinked load request, or linked load request with no purchase |
| LU    | Linked status update                                           |

## Subelement 26 (Wallet Program Data)

*DE 48 (Additional Data: Private Use)*, subelement 26 (Wallet Program Data) contains the subfield that will identify transactions submitted using wallets on the Mastercard Digital Enablement Service (MDES) or Masterpass platform.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 26                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | an-3                   |
| Data Field             | Contents of subfield 1 |
| Subfields              | 1                      |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 26 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | .   | X   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

### Values

See subfield 1.

## Subfield 1 (Wallet Identifier)

DE 48, subelement 26, subfield 1 (Wallet Identifier) provides information about transactions initiated through the Masterpass Online platform or through the Mastercard Digital Enablement Service (MDES) and indicates who the Wallet Provider was for the transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-3                      |
| Data Field          | Contents of positions 1-3 |
| Justification       | N/A                       |

**NOTE:** The following values apply to both Masterpass and MDES transactions.

### Values

| Value | Description              |
|-------|--------------------------|
| 101   | Masterpass by Mastercard |

**NOTE:** The following values only apply to MDES transactions.

### Values

| Value | Description              |
|-------|--------------------------|
| 12A   | Sample Wallet ID         |
| 103   | Apple Pay                |
| 216   | Google Pay               |
| 217   | Samsung Pay              |
| 327   | Remote commerce programs |

### Application notes

The Wallet ID is required in Masterpass transactions submitted from certain regions or countries. The Wallet ID cannot contain all zeros, spaces, or special characters.

The Dual Message Authorization System inserts the Wallet Identifier for MDES transactions when available, which identifies the wallet that the transaction came from.

New Wallet Identifier values may be provided in this field without prior written notice.

The following tables indicate the presence of the Wallet Identifier in Masterpass and MDES transactions.

### Applicable messages

| Masterpass Messages                           | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

**NOTE: If the card in Masterpass is an MDES token, then MDES would add the correct Wallet ID to the transaction.**

### Applicable messages

| MDES Messages                                 | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | .   | X   | C   |
| Authorization Advice/0120: acquirer-generated | .   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | .   | X   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

**NOTE: For MDES, Mastercard inserts the Wallet Identifier for the wallet that the token was supplied to.**

## Subelement 27 (Transaction Analysis)

[DE 48](#), subelement 27 (Transaction Analysis) contains information about transaction analysis.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 27                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an...97; LLVAR            |
| Data Field             | Contents of subfields 1–2 |
| Subfields              | 2                         |

### Usage

Following is the usage of subelement 27 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | •   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

### Application notes

[DE 48](#), subelement 27 and subfields 1 and 2 are set up in Tag, Length, Value format:

27LL01LLVV02LLVVVVVVVVV...

### Example

| Subfield 1 | Subfield 2 |
|------------|------------|
| 0102CI     | 0206SNAOVP |



## Subfield 1 (Overview)

[DE 48](#), subelement 27, subfield 1 (Overview) provides issuers with a code that represents the overall disposition of the transaction based on the applicable transaction validation results and corresponding decision matrix.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-2                   |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

### Values

| Value | Description                          |
|-------|--------------------------------------|
| CI    | Continue processing with information |
| CW    | Continue processing with warning     |
| DI    | Decline issuer decision              |
| DS    | Decline suspicious                   |

## Subfield 2 (Test Results)

[DE 48](#), subelement 27, subfield 2 (Test Results) provides the list of failed validations. If several failed validations generate the same test result code, the code will only be reported one time in subfield 2.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 02          |
| Subfield Data Length | 2 positions |

| Attribute           | Description                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------|
| Data Representation | an...87; LLVAR; The LL length field of LLVAR will be an integral multiple of 3, not to exceed 87. |
| Data Field          | Contents of subfield 2                                                                            |
| Justification       | N/A                                                                                               |

## Values

| Value | Description                                              |
|-------|----------------------------------------------------------|
| CAM   | Invalid Card Authentication                              |
| CCH   | Cross channel                                            |
| CRN   | Consent requirement not fulfilled                        |
| CVF   | Cardholder verification (on terminal) was not successful |
| CVU   | CVM requirements not fulfilled                           |
| CVX   | Status CVM unknown                                       |
| DAF   | ODA failed                                               |
| DAU   | ODA was not performed                                    |
| DMM   | Data mismatch                                            |
| DNC   | Data not consistent with application or product          |
| EXP   | Token expired                                            |
| FER   | Format error                                             |
| FUZ   | Fuzzing                                                  |
| ICT   | Not a valid cryptogram type                              |
| NMK   | No matching key file/KDI combination                     |
| OVE   | CDCVM retry exceeded: token suspended                    |
| OVF   | CDCVM failed                                             |
| OVP   | CDCVM (was possible but) not performed                   |
| OVU   | CDCVM not performed                                      |
| PKC   | ODA compromised                                          |
| PPP   | PIN Pad Problem                                          |
| PTB   | PIN on terminal bypass                                   |
| PWE   | Possible wedge attack                                    |

| Value | Description                                     |
|-------|-------------------------------------------------|
| REP   | ATC replay: same UN                             |
| SKC   | Key compromised                                 |
| SNA   | Request service not allowed for this product    |
| UTP   | Unable to process                               |
| WOC   | Wallet overrule of Mastercard decision on CDCVM |

## Subelement 28 (Cardless ATM Order ID)

[DE 48](#), subelement 28 (Cardless ATM Order ID) is defined by Mastercard, sent by the acquirer participating in the Cardless ATM program and passed to the issuer participating in the Cardless ATM program.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 28                         |
| Subelement Data Length | 2 positions                |
| Data Representation    | an-10                      |
| Data Field             | Contents of positions 1–10 |
| Subfields              | N/A                        |
| Justifications         | N/A                        |

### Usage

Following is the usage of subelement 28 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Values

Cannot contain all spaces or all zeros.

## Subelement 29 (Additional POS Terminal Locations)

[DE 48](#), subelement 29 is required when [DE 61 \(Point-of-Service \[POS\] Data\)](#), subfield 3 (POS Terminal Location) is present with the value of 8 (Additional Terminal Operating Environments).

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 29                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | an-1                   |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 29 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Values

| Value | Description                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------|
| A     | Transaction has been initiated remotely and completed at a physical terminal on premises of the acceptor facility.  |
| B     | Transaction has been initiated remotely and completed at a physical terminal off premises of the acceptor facility. |

## Subelement 30 (Token Transaction Identifier)

DE 48, subelement 30 (Token Transaction Identifier) will contain, when available, the calculated Token Transaction Identifier to identify the transaction. The Token Transaction Identifier is to be retained and used to provide the transaction details associated with an original purchase and subsequent reversal messages. The Token Transaction Identifier is only sent to issuers participating in the Mastercard Digital Enablement Service.

### Attributes

| Attribute              | Description     |
|------------------------|-----------------|
| Subelement ID          | 30              |
| Subelement Data Length | 2 positions     |
| Data Representation    | ans...64; LLVAR |
| Subfields              | N/A             |

### Usage

Following is the usage of subelement 30 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated | •   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | •   | X   | C   |
| Reversal Advice/0420                          | •   | X   | C   |

### Values

A 44 byte base 64 value will be sent for Mastercard BIN ranges.

A 64 byte value will be sent for Visa BIN ranges.

### Application notes

DE 48, subelement 30 will not be present when DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains a value of 20 (Purchase Return/Refund).

## Subelement 32 (Mastercard Assigned ID)

DE 48, subelement 32 (Mastercard Assigned ID) contains the merchant ID assigned by Mastercard.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 32                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-6                      |
| Data Field             | Contents of positions 1–6 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 32 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | OE  | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Advice/0420                          | •   | X   | C   |

### Values

The Mastercard Assigned ID must be a value assigned by Mastercard, and is the same value in both the Authorization and Clearing Systems. Mastercard assigns one ID to the merchant regardless if the merchant is participating in multiple programs. This ID can also be assigned for transaction data integrity purposes. For all acceptors that have been issued a Mastercard Assigned ID, all submitted authorization requests, reversals, and advices must contain the Mastercard Assigned ID in DE 48, subelement 32.

This field is required when transactions are submitted for real-time substantiation. Real-time substantiation is indicated by DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator) containing a value of 1 (Merchant terminal verified the purchased items against the Inventory Information Approval System [IIAS]).

Participating merchants must submit the Mastercard Assigned ID for e-commerce transactions that are processed under the Maestro® Recurring Payments Program.

### Application notes for the Mastercard Installments Program with Merchant Participation for qualifying domestic transactions with product code GCS

Optional Echo (OE) indicates that the message initiator may provide the data element in an Authorization Request Response/0110 message. If so, the data element must contain the same value echoed from the original Authorization Request/0100 message.

The Dual Message Authorization System inserts or modifies this subelement in Authorization Request/0100 messages for qualifying installments transactions.

Issuers receive DE 48, subelement 32 in Authorization Request/0100 messages when the acquirer includes the subelement, or when Mastercard has inserted or modified the subelement in the corresponding Authorization Request/0100 message.

Acquirers receive DE 48, subelement 32 in Authorization Request Response/0110 messages when Mastercard has inserted or modified the subelement in the corresponding Authorization Request/0100 message.

## Subelement 32 (Visa Merchant Identifier [Visa Only])

*DE 48*, subelement 32 (Visa Merchant Identifier) contains the merchant ID assigned by Visa.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 32                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-8                      |
| Data Field             | Contents of positions 1–8 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 32 in Visa-branded transactions (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

### Values

The Visa Merchant Identifier (Visa Only) value is assigned by Visa and is only for use with Visa-branded transactions.

### Application notes

When the acquirer provides [DE 48](#), subelement 32 with a length greater than eight positions on a Visa-branded transaction, the Authorization System rejects the transaction sending the acquirer an Authorization Request Response/0110 message where [DE 39 \(Response Code\)](#) contains the value 30 (Format error) and [DE 44 \(Additional Response Data\)](#) contains the value 048032.

## Subelement 33 (PAN Mapping File Information)

[DE 48](#), subelement 33 (PAN Mapping File Information) supports the mapping between the virtual account data and actual account data.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 33                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an...93; LLVAR            |
| Data Field             | Contents of subfields 1–8 |
| Subfields              | 8                         |
| Justification          | See subfields             |



## Usage

Following is the usage of subelement 33 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                              | Org | Sys | Dst |
|------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                           | C   | X   | C   |
| Authorization Request Response/0110                  | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated        | C   | X   | C   |
| Authorization Advice/0120: system-generated          | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated | C   | X   | C   |
| Reversal Request/0400                                | C   | X   | C   |
| Reversal Request Response/0410                       | C   | X   | C   |
| Reversal Advice/0420                                 | .   | X   | C   |

## Values

See subfields.

## Application notes

The Dual Message Authorization System inserts this subelement when PAN Mapping Service was performed on the transaction. Subelement 33 may be used for research or audit purposes but need not be used for processing authorization messages.

Acquirers will receive subelement 33 in contact M/Chip, contactless magnetic stripe, or contactless M/Chip transactions when the issuer is participating in the Contactless Mapping Service. Acquirers will also receive subelement 33 in contactless M/Chip, contactless magnetic stripe, magnetic stripe, or Digital Secure Remote Payment transactions when the issuer is participating in the Mastercard Digital Enablement Service. Acquirers do not receive subelement 33 for the Mastercard In Control™ Purchase Control Service or Mastercard In Control Virtual Card Spend Control Service.

Issuers that issue contactless cards or devices that provide a primary account number (PAN) within a transaction different from what is embossed on the card, must ensure that the following is provided in Authorization Request Response/0110 messages when responding to MCC 4111, 4131, 4784, 7523 transactions.

- The value E in DE 48, subelement 33, subfield 1 (Account Number Indicator)

- The embossed number in DE 48, subelement 33, subfield 2 (Account Number)
- The card expiration date of the embossed number in DE 48, subelement 33, subfield 3 (Expiration Date)

Acquirers processing transactions for merchant categories 4111, 4131, 4784, and 7523 must pass the PAN Mapping information back to the merchants.

## Subfield 1 (Account Number Indicator)

[DE 48](#), subelement 33, subfield 1 (Account Number Indicator) indicates the type of PAN mapping account.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

### Values

| Value | Description                                                      |
|-------|------------------------------------------------------------------|
| C     | Mastercard Digital Enablement Service Secure Element Token       |
| E     | Embossed Account Number Provided by Issuer                       |
| F     | Mastercard Digital Enablement Service Static Token               |
| H     | Mastercard Digital Enablement Service Cloud-based Payments Token |
| L     | Pay with Rewards Loyalty Program Operator [LPO] Card             |
| M     | Primary Account Number                                           |
| P     | Contactless Account Number                                       |
| R     | Pay with Rewards Card                                            |
| V     | Virtual Card Number                                              |

## Subfield 2 (Account Number)

DE 48, subelement 33, subfield 2 (Account Number) indicates the PAN mapping account number.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 02                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n...19; LLVAR          |
| Data Field           | Contents of subfield 2 |
| Justification        | N/A                    |

### Values

| Value            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acquirer message | <p>Contains primary account number:</p> <ul style="list-style-type: none"> <li>When Mastercard performs token mapping or subfield 2 is provided when using a third party token service provider, subfield 2 contains only the last four digits of the cardholder's PAN.</li> <li>The Dual Message Authorization System will continue to send the full account number for tokenized In Control VCN transactions, and when DE 18 (Merchant Type) contains any of the following MCCs: <ul style="list-style-type: none"> <li>MCC 3246: RyanAir</li> <li>MCC 4111: Transportation-Suburban and Local Commuter Passenger, including Ferries</li> <li>MCC 4112: Passenger Railways</li> <li>MCC 4131: Bus Lines</li> <li>MCC 4511: Air Carries, Airlines- Not Else Where Classified</li> <li>MCC 4784: Bridge and Road Fees, Tolls</li> <li>MCC 7523: Automobile Parking Lots and Garages</li> </ul> </li> </ul> |
| Issuer message   | <p>Contains virtual number or token</p> <p>For Mastercard Digital Enablement Service transactions, subfield 2 is the token rather than the primary account number, if subfield 1 value is C, H, or F.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Subfield 3 (Expiration Date)

DE 48, subelement 33, subfield 3 (Expiration Date) indicates the expiration date of the PAN mapping account.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 03                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n-4; format YYMM       |
| Data Field           | Contents of subfield 3 |
| Justification        | N/A                    |

### Values

| Value            | Description                                                                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acquirer message | <p>Will contain the expiration date when</p> <ul style="list-style-type: none"> <li>The issuer provided one for a PAN mapping record added to the MCC106 PAN Mapping File, or</li> <li>A transit transaction response contains MCC 4111, 4131, 4784, and 7523, or</li> <li>The Mastercard Digital Enablement Service was applied</li> </ul> |
| Issuer message   | <p>May contain virtual card number or token expiration date only if acquirer provided in DE 14 of the authorization message.</p> <p>For contactless transit transactions, subfield 3 is the expiration date of the embossed number rather than the virtual number, if subfield 1 value is E.</p>                                            |

## Subfield 4 (Product Code)

[DE 48](#), subelement 33, subfield 4 (Product Code) may indicate the product code for subfield 2 account number.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 04                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-3                   |
| Data Field           | Contents of subfield 4 |
| Justification        | N/A                    |

### Values

Subfield 4 may contain product code for subfield 2 account number.

## Subfield 5 (Token Assurance Method)

[DE 48](#), subelement 33, subfield 5 (Token Assurance Method) contains a value indicating the confidence level of the token to PAN/cardholder binding.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 05                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n-2                    |
| Data Field           | Contents of subfield 5 |
| Justification        | N/A                    |

### Values

Contains a value indicating the confidence level of the token to PAN/cardholder relationship.

### Valid values

| TAM Values | TAM Name                                                             | Authentication Use Case/MA Services                                                                                                                          |
|------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10         | Card Issuer Account Verification                                     | Successful TAR, AS, ASI response with no authentication                                                                                                      |
| 11         | Card Issuer Interactive Cardholder Authentication – 1 Factor         | Token authenticated using MDES Authentication Service (OTP)                                                                                                  |
| 14         | Card Issuer Asserted Authentication                                  | Push Provisioning into without additional authentication beyond Issuer's own application                                                                     |
| 20         | Push Provisioned Token with Additional Interactive Authentication    | Push Provisioned token with MDES Authentication service following token creation                                                                             |
| 21         | Push Provisioned Token with Additional Identity Check Authentication | Push Provisioned token with Identity Check authentication following token creation                                                                           |
| 22         | Identity Check Risk-based Cardholder Authentication                  | Token authenticated via Identity Check Risk-based authentication (no cardholder challenge)                                                                   |
| 23         | Identity Check Interactive Cardholder Authentication                 | Token authenticated via Identity Check Interactive step-up (cardholder challenge)                                                                            |
| 24         | Token with device-asserted authentication                            | A Companion Cloud Token or Recurring Payment Cloud Token provisioned from an existing device token, where proof of authentication was provided and validated |

### Application notes

Subfield 5 is optional for acquirers in the Authorization Request/0100 message. Acquirers may receive the value back in an Authorization Request Response/0110 message.

### Subfield 6 (Token Requestor ID)

[DE 48](#), subelement 33, subfield 6 (Token Requestor ID) contains the ID assigned by the Token Service Provider to the Token Requestor.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 06          |
| Subfield Data Length | 2 positions |
| Data Representation  | n-11        |

| Attribute     | Description            |
|---------------|------------------------|
| Data Field    | Contents of subfield 6 |
| Justification | N/A                    |

### Values

Contains the ID assigned by the Token Service Provider to the Token Requestor. The Token Requestor ID is optional for all token types.

### Application notes

Subfield 6 is optional but should be included in messages sent to Mastercard by the acquirer if provided to the acquirer by the merchant. Acquirers may receive the value back in an Authorization Request Response/0110 message.

## Subfield 7 (Primary Account Number, Account Range)

[DE 48](#), subelement 33, subfield 7 (Primary Account Number, Account Range) carries either the first nine digits of the cardholder PAN, or the full Visa cardholder PAN in the authorization response for a transaction initiated with a Visa token.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 07                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n...19; LLVAR          |
| Data Field           | Contents of subfield 7 |
| Justification        | N/A                    |

### Values

The first nine digits of the Visa cardholder PAN, or the full Visa cardholder PAN. Acquirers must be aware that this information must not be forwarded to their merchants.

## Subfield 8 (Storage Technology)

[DE 48](#), subelement 33, subfield 8 (Storage Technology) describes the Storage Technology of a requested or created token.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 08                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-2                   |
| Data Field           | Contents of subfield 8 |
| Justification        | N/A                    |

### Usage

Following is the usage of [DE 48](#), subelement 33, subfield 8 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                              | Org | Sys | Dst |
|----------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                           | •   | X   | C   |
| Authorization Request Response/0110                                  | •   | X   | C   |
| Authorization Advice/0120: system-generated                          | •   | X   | C   |
| Reversal Request/0400                                                | •   | X   | C   |
| Reversal Advice/0420                                                 | •   | X   | C   |
| Authorization Request/0100: tokenization Eligibility Request (TER)   | •   | X   | C   |
| Authorization Request/0100: tokenization Authorization Request (TAR) | •   | X   | C   |
| Authorization Request/0100: tokenization Complete Notification (TCN) | •   | X   | C   |

### Values

| Value | Description   |
|-------|---------------|
| 01    | Device Memory |



| Value | Description                                              |
|-------|----------------------------------------------------------|
| 02    | Device Memory protected by Trusted Platform Module (TPM) |
| 03    | Server                                                   |
| 04    | Trusted Execution Environment (TEE)                      |
| 05    | Secure Element (SE)                                      |
| 06    | Virtual Execution Environment (VEE)                      |

### Application notes

Subfield 8 is optional but should be included in messages sent to Mastercard by the acquirer if provided to the acquirer by the merchant. Acquirers may receive the value back in an Authorization Request Response/0110 message.

## Subelement 34 (ATC Information)

[DE 48](#), subelement 34 (ATC Information) supports On-behalf Services.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 34                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | an-11                 |
| Data Field             | Contents of subfields |
| Subfields              | 3                     |
| Justification          | See subfields         |

### Usage

Following is the usage of subelement 34 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | .   | X   | C   |

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated | .   | X   | C   |

## Values

See subfields.

## Application notes

The Dual Message Authorization System inserts this subelement when the Dynamic CVC 3 Pre-validation Service (OBS 15), Dynamic CVC 3 Validation in Stand-In Processing Service (OBS 16), M/Chip Cryptogram Pre-validation Service (OBS 02), or M/Chip Cryptogram Validation in Stand-In Processing Service (OBS 03) was performed on the transaction and the validation result was V (Valid), A (Valid AC; ATC outside allowed range), E (Valid AC; ATC Replay), T (Valid ARQC/TC and ATC; TVR/CVR invalid), or G (Application Cryptogram is valid but not an ARQC; status of TVR/CVR unknown).

## Chip Services

| Value | Description                                                                        |
|-------|------------------------------------------------------------------------------------|
| V     | Valid ARQC/TC and ATC and TVR/CVR                                                  |
| A     | Valid Application Cryptogram; ATC out of range                                     |
| E     | Valid Application Cryptogram; ATC replay                                           |
| T     | Valid ARQC/TC and ATC; TVR/CVR invalid                                             |
| G     | Application Cryptogram valid but not an ARQC; ATC valid; status of TVR/CVR unknown |

## CVC Services

| Value | Description                    |
|-------|--------------------------------|
| V     | Valid CVC3 and ATC and TVR/CVR |
| A     | Valid CVC3; ATC out of range   |
| E     | Valid CVC; ATC replay          |

These results are found in [DE 48](#), subelement 71, subfield 2 (On-behalf [OB] Result 1).

Subelement 34 should be used for processing authorization messages and maintaining the ATC values on the issuer host. Subfields 1–3 are always present.

The following ATC information will be provided in Dual Message Authorization System and Single Message System financial transaction requests and advices to issuers for MDES transactions initiated with a token, when available. ATC information will not be provided in declined advices.

- [DE 48](#), subelement 34, subfield 1 (ATC Value)
- [DE 48](#), subelement 34, subfield 2 (ATC Discrepancy Value)
- [DE 48](#), subelement 34, subfield 3 (ATC Discrepancy Indicator)

## Subfield 1 (ATC Value)

[DE 48](#), subelement 34, subfield 1 (ATC Value) contains the derived full ATC Value used in the validation.

### Attributes

| Attribute           | Description                    |
|---------------------|--------------------------------|
| Data Representation | n-5                            |
| Data Field          | Contents of positions 1-5      |
| Justification       | Right-justified, leading zeros |

## Subfield 2 (ATC Discrepancy Value)

[DE 48](#), subelement 34, subfield 2 (ATC Discrepancy Value) is the differential between the transaction ATC and the maximum value allowed by the issuer when the transaction ATC is above the previous ATC, or the differential between the transaction ATC and the minimum value allowed by the issuer when the transaction ATC is below the previous ATC. ATC Discrepancy Value will be zero when the transaction ATC is within the issuer-defined limits.

### Attributes

| Attribute           | Description                    |
|---------------------|--------------------------------|
| Data Representation | n-5                            |
| Data Field          | Contents of positions 6-10     |
| Justification       | Right-justified, leading zeros |

## Subfield 3 (ATC Discrepancy Indicator)

DE 48, subelement 34, subfield 3 (ATC Discrepancy Indicator) indicates if the ATC Discrepancy Value is above, below, or within the maximum values allowed by the issuer.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | an-1                    |
| Data Field          | Contents of position 11 |
| Justification       | N/A                     |

### Values

#### Values

| Value | Description                                                            |
|-------|------------------------------------------------------------------------|
| G     | Indicates that the ATC value is greater than the maximum value allowed |
| L     | Indicates that the ATC value is lower than the minimum value allowed   |
| W     | Indicates that the ATC value is within the issuer-defined limits       |

## Subelement 34 subfield data examples

Following are examples of how data may appear in subfield 1, subfield 2, and subfield 3.

### Example 1

Last valid ATC value processed by Mastercard was 00011.

Issuer provided a maximum allowed range of 00035.

$00011 + 00035 = 00046$ , which represents the maximum ATC value, allowed in the next transaction.

The derived value of the ATC in the next transaction is 00088.

The ATC Discrepancy Value for this transaction is 42 which represents the difference between the maximum ATC value allowed 00046 and the derived ATC value from the transaction 00088.

Subelement 34 will contain the following:

- Subfield 1 = 00088

- Subfield 2 = 00042
- Subfield 3 = G

### Example 2

Last valid ATC value processed by Mastercard was 00068.

Issuer provided a minimum allowed range of 00035.

$00068 - 00035 = 00033$ , which represents the minimum ATC value allowed in the next transaction.

The derived value of the ATC in the next transaction is 00020.

The ATC Discrepancy Value for this transaction is 13, which represents the difference between the minimum ATC value allowed 00033 and the derived ATC value from the transaction 00020.

Subelement 34 will contain the following:

- Subfield 1 = 00020
- Subfield 2 = 00013
- Subfield 3 = L

### Example 3

Last valid ATC value processed by Mastercard was 00037.

Issuer provided a maximum allowed range of 00005.

$00037 + 00005 = 00042$ , which represents the maximum ATC value allowed in the next transaction.

The derived value of the ATC in the next transaction is 00040.

The ATC Discrepancy Value for this transaction is 0 because the derived value of the ATC in the next transaction 00040 is greater than the last valid ATC value 00037 and below the maximum ATC value allowed in the next transaction 00042.

Subelement 34 will contain the following:

- Subfield 1 = 00040
- Subfield 2 = 00000
- Subfield 3 = W

## Subelement 35 (Contactless Non-Card Form Factor Request/Response)

[DE 48](#), subelement 35 (Contactless Non-Card Form Factor Request/Response) supports Contactless Mapping Service.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 35                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | an-1                   |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 35 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | •   | X   | C   |
| Authorization Request Response/0110 | C   | X   | •   |

### Values

The contactless issuer will receive the Authorization Request/0100 message containing [DE 48](#), subelement 35, value R (Cardholder request for device).

The contactless issuer should respond with an Authorization Request Response/0110 message containing [DE 48](#), subelement 35, value A (Approve cardholder request for device) or D (Decline cardholder request for device).

### Application notes

Issuers choosing the Contactless Mapping Service "Processing and Issuance of devices" participation option (Passive Participation) will receive an Authorization Request/0100 message

when a cardholder requests a contactless device from the issuer-branded, secure website hosted by Mastercard. The issuer may then decide if the cardholder should receive the contactless device.

The Dual Message Authorization System will use the value in [DE 48](#), subelement 35 of the Authorization Request Response/0110 message to issue the contactless device or to decline the cardholder's request.

If an issuer chooses not to support subelement 35, the issuer-branded website will evaluate the responses provided for AVS and CVC 2 and approve or decline the cardholder request based on those values.

## Subelement 36 (Additional Visa Request Data [Visa Only])

[DE 48](#), subelement 36 (Additional Visa Request Data [Visa Only]) contains additional request data to assist Visa with its processing.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 36                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...94; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 5                     |
| Justification          | See subfields         |

### Usage

Following is the usage of subelement 36 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Reversal Request/0400                         | C   | P   | C   |

## Values

See subfields.

## Subfield 1 (Merchant Verification Value [MVV])

[DE 48](#), subelement 36, subfield 1 (Merchant Verification Value) contains the merchant verification value for subelement 36.

## Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 1                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-10                  |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

## Values

0–9 and A–F

## Subfield 2 (Business Application Identifier [Visa Only])

[DE 48](#), subelement 36, subfield 2 (Business Application Identifier [Visa Only]) identifies the intended use of a Visa-branded transaction.

## Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 2                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | a-2                    |
| Data Field           | Contents of subfield 2 |
| Justification        | N/A                    |

## Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.



### Application notes

Subfield 2 (Business Application Identifier [Visa Only]) is only for use with Visa-branded transactions.

### Subfield 3 (Acceptor Legal Business Name [Visa Only])

[DE 48](#), subelement 36, subfield 3 (Acceptor Legal Business Name [Visa Only]) identifies the acceptor legal business name.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 3                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...25; LLVAR        |
| Data Field           | Contents of subfield 3 |
| Justification        | N/A                    |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 3 (Acceptor Legal Business Name [Visa Only]) is only for use with Visa-branded transactions.

### Subfield 4 (Payment Facilitator Name [Visa Only])

[DE 48](#), subelement 36, subfield 4 (Payment Facilitator Name [Visa Only]) identifies the payment facilitator name.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 4                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...25; LLVAR        |
| Data Field           | Contents of subfield 4 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 4 (Payment Facilitator Name [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 5 (Purpose of Payment [Visa Only])

[DE 48](#), subelement 36, subfield 5 (Purpose of Payment [Visa Only]) identifies the purpose of payment.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 5                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...12; LLVAR        |
| Data Field           | Contents of subfield 5 |
| Justification        | N/A                    |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 5 (Purpose of Payment [Visa Only]) is only for use with Visa-branded transactions.

## Subelement 37 (Additional Merchant Data)

[DE 48](#), subelement 37 (Additional Merchant Data) contains subfields representing the ID of the participating service provider in a transaction and the sub-merchant ID when applicable. The service provider can be a payment facilitator, independent sales organization, or a merchant payment gateway. A transaction may involve more than one service provider.

## Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 37                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | ans...71; LLVAR           |
| Data Field             | Contents of subfields 1-5 |
| Subfields              | 5                         |
| Justification          | See subfields             |

## Usage

Following is the usage of subelement 37 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | C   | C   |

## Values

See subfields.

## Application notes

Specific application notes, conditions, and cross-edits where applicable.

Valid Combinations of [DE 48](#), subelement 37, subfields 1, 2, and 3:

- Subfields 1 and 3 only
- Subfield 2 only
- Subfields 1, 2, and 3 (Mastercard only)

## Subfield 1 (Payment Facilitator ID)

[DE 48](#), subelement 37, subfield 1 (Payment Facilitator ID) contains the Payment Facilitator ID assigned by Mastercard or Visa.

### Attributes

| Attribute            | Description                        |
|----------------------|------------------------------------|
| Subfield ID          | 01                                 |
| Subfield Data Length | 2 positions                        |
| Data Representation  | n-11                               |
| Data Field           | Contents of subfield 1             |
| Justification        | Right-justified with leading zeros |

### Values

The Payment Facilitator ID must always be a value assigned by Mastercard and must be the same value in both the Dual Message Authorization and Dual Message Clearing Systems. This ID is the Company ID that will be assigned during the time of registration with Mastercard for a Service Provider as "Payment Facilitator."

The value must be provided by the acquirer when a registered payment facilitator is involved in a transaction and should be right-justified with leading zeros. For example, if the Company ID is 123456, [DE 48](#), SE 37, SF 1 should be 00000123456.

## Subfield 2 (Independent Sales Organization ID)

[DE 48](#), subelement 37, subfield 2 (Independent Sales Organization ID) contains the Independent Sales Organization ID assigned by Mastercard.

### Attributes

| Attribute            | Description                        |
|----------------------|------------------------------------|
| Subfield ID          | 02                                 |
| Subfield Data Length | 2 positions                        |
| Data Representation  | n-11                               |
| Data Field           | Contents of subfield 2             |
| Justification        | Right-justified with leading zeros |

## Values

The Independent Sales Organization ID must always be a value assigned by Mastercard and is the same value in both the Dual Message Authorization and Dual Message Clearing Systems. This ID is the "Company ID" that is assigned during the time of registration via Mastercard Connect of a service provider as "Independent Sales Organization."

This value must be provided by the acquirer when a registered independent sales organization is involved in a transaction and should be right-justified with leading zeros. For example, if the Company ID is 123456, [DE 48](#), subelement 37, subfield 2 should be 00000123456.

For Visa transactions, the Marketplace ID must always be a value assigned by Visa.

This value must be provided by the acquirer when a Visa-registered Marketplace ID is involved in a transaction and should be right-justified with leading zeros. For example, if the Marketplace ID is 12345678, [DE 48](#), subelement 37, subfield 2 should be 00012345678.

## Subfield 3 (Sponsored Merchant ID)

[DE 48](#), subelement 37, subfield 3 (Sponsored Merchant ID) contains the identifier assigned by the acquirer or the payment facilitator to the sponsored merchant and its location.

## Attributes

| Attribute            | Description                     |
|----------------------|---------------------------------|
| Subfield ID          | 03                              |
| Subfield Data Length | 2 positions                     |
| Data Representation  | ans-15                          |
| Data Field           | Contents of subfield 3          |
| Justification        | Left-justified, trailing spaces |

## Processing rules

The sponsored merchant ID must be present for all payment facilitator-acquired transactions and must uniquely identify each sponsored merchant and its location.

## Subfield 4 (Merchant Country of Origin)

[DE 48](#), subelement 37, subfield 4 (Merchant Country of Origin) contains the ISO numeric country code for the home country of the government that owns or controls the merchant that initiated the transaction.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 04                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n-3                    |
| Data Field           | Contents of subfield 4 |
| Justification        | n/a                    |

### Values

If present, [DE 48](#), subelement 37, subfield 4 must contain a valid ISO numeric country code.

For example, embassies are physically located in countries that are not their "home country" (that is, the embassy of Mexico may be physically located in Singapore; however, the "home country" is Mexico).

### Requirements

[DE 48](#), subelement 37, subfield 4 must contain the location information of the government entity, not the location of the merchant.

### Usage

### Applicable messages

| Message Type Identifier    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | P   | C   |
| Authorization Advice/0120  | C   | P   | C   |

## Subfield 5 (Merchant Payment Gateway ID)

[DE 48](#), subelement 37, subfield 5 (Merchant Payment Gateway ID) identifies the registered Merchant Payment Gateway used by the merchant and is provided by the acquirer when a registered MPG service provider is involved in a transaction.

### Attributes

| Attribute           | Description                        |
|---------------------|------------------------------------|
| Subfield ID         | 05                                 |
| Subfield Length     | 2 positions                        |
| Data Representation | n-11                               |
| Data Field          | Contents of subfield 5             |
| Justification       | Right-justified with leading zeros |

### Values

The Merchant Payment Gateway ID is a value assigned by Mastercard. The Merchant Payment Gateway ID is assigned to Merchant Payment Gateway service provider at the time of their registration with Mastercard. The value must be right-justified with leading zeros. For example, if the Merchant Payment Gateway ID is 123456, then [DE 48](#), subelement 37, subfield 5 contains 00000123456.

## Subelement 38 (Account Category)

[DE 48](#), subelement 38 (Account Category) supports Account Level Management.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 38                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | an-1                   |
| Data Field             | Contents of position 1 |
| Subfields              | NA                     |
| Justification          | NA                     |

## Usage

Following is the usage of subelement 38 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | •   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |
| Reversal Advice/0420                        | •   | X   | C   |

For a complete valid values listing, refer to the *Account Level Management Manual*.

## Subelement 39 (Account Data Compromise Information)

[DE 48](#), subelement 39 (Account Data Compromise Information) contains confirmed or suspected account data compromise event information for account ranges.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 39                         |
| Subelement Data Length | 2 positions                |
| Data Representation    | ans-30                     |
| Data Field             | Contents of positions 1–30 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

## Usage

Following is the usage of subelement 39 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.



### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | .   | X   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |

### Positions 1–3 (Reserved for Future Use)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Field          | Contents of positions 1–3 |
| Values              | 000 = Not available       |

### Positions 4–6 (Reserved for Future Use)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Field          | Contents of positions 4–6 |
| Values              | N/A                       |

### Positions 7–12 (Case Key Code number 1)

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-6                       |
| Data Field          | Contents of positions 7–12  |
| Values              | Unique Key to Identify Case |

### Positions 13–18 (Case Key Code number 2)

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | ans-6       |

| Attribute  | Description                 |
|------------|-----------------------------|
| Data Field | Contents of positions 13–18 |
| Values     | Unique Key to Identify Case |

#### Positions 19–24 (Case Key Code number 3)

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-6                       |
| Data Field          | Contents of positions 19–24 |
| Values              | Unique Key to Identify Case |

#### Position 25 (Account Number)

| Attribute           | Description                                                         |
|---------------------|---------------------------------------------------------------------|
| Data Representation | n-1                                                                 |
| Data Field          | Contents of position 25                                             |
| Values              | 0–9 = Number of events in which the account number has been exposed |

#### Position 26 (Expiration Date)

| Attribute           | Description                                                      |
|---------------------|------------------------------------------------------------------|
| Data Representation | n-1                                                              |
| Data Field          | Contents of position 26                                          |
| Values              | 0–9 = Number of events in which expiration date has been exposed |

#### Position 27 (CVC 2)

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | n-1                     |
| Data Field          | Contents of position 27 |

| Attribute | Description                                                |
|-----------|------------------------------------------------------------|
| Values    | 0–9 = Number of events in which the CVC 2 has been exposed |

#### Position 28 (PIN)

| Attribute           | Description                                              |
|---------------------|----------------------------------------------------------|
| Data Representation | n-1                                                      |
| Data Field          | Contents of position 28                                  |
| Values              | 0–9 = Number of events in which the PIN has been exposed |

#### Position 29 (Magnetic Stripe)

| Attribute           | Description                                                          |
|---------------------|----------------------------------------------------------------------|
| Data Representation | n-1                                                                  |
| Data Field          | Contents of position 29                                              |
| Values              | 0–9 = Number of events in which the Magnetic Stripe has been exposed |

#### Position 30 (Personal Information)

| Attribute           | Description                                                                            |
|---------------------|----------------------------------------------------------------------------------------|
| Data Representation | n-1                                                                                    |
| Data Field          | Contents of position 30                                                                |
| Values              | 0–9 = Number of events in which the cardholder's personal information has been exposed |

## Subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only])

[DE 48](#), subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only]) contains certificate information on electronic commerce transactions when applicable.

### Attributes

| Attribute              | Description                                               |
|------------------------|-----------------------------------------------------------|
| Subelement ID          | 40                                                        |
| Subelement Data Length | 2 positions                                               |
| Data Representation    | n...40 (also contains binary data. See subfields.); LLVAR |
| Data Field             | Contents of subfields                                     |
| Subfields              | 2                                                         |
| Justification          | N/A                                                       |

### Usage

Following is the usage of subelement 40 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | P   | C   |
| Authorization Request Response/0110         | C   | P   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

### Values

Contains the contents of subfield 1 or subfield 2 or both.

## Subfield 1 (Merchant Certificate Serial Number)

[DE 48](#), subelement 40, subfield 1 (Merchant Certificate Serial Number) is the Merchant Certificate Serial Number in binary data.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | b...16; LLVAR          |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

### Values

Merchant Certificate Serial Number must be binary data. Be aware that the subfield ID and subfield length fields must still be given as numeric data.

## Subfield 2 (Cardholder Certificate Serial Number)

[DE 48](#), subelement 40, subfield 2 (Cardholder Certificate Serial Number) is the cardholder certificate serial number in binary format.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 02                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | b...16; LLVAR          |
| Data Field           | Contents of subfield 2 |
| Justification        | N/A                    |

### Values

Cardholder Certificate Serial Number must be binary data. Be aware that the subfield ID and subfield length fields must still be given as numeric data.

## Subelement 41 (Electronic Commerce Certificate Qualifying Information)

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 41                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...54; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 18                    |
| Justification          | N/A                   |

### Usage

Following is the usage of subelement 41 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Request Response/0410                | CE  | XE  | CE  |
| Reversal Advice/0420                          | .   | X   | C   |

### Values

See subfields.

## Subfield 1 (Additional Recurring Payments Information)

Subfield 1 (Additional Recurring Payments Information) is used in the Dual Message Authorization System to provide additional data for recurring payments. Currently this is used for transactions using India issued cards.

### Attributes

| Attribute            | Description     |
|----------------------|-----------------|
| Subfield ID          | 01              |
| Subfield Data Length | 2 positions     |
| Data Representation  | ans...26; LLVAR |
| Justification        | N/A             |

### Values

| Field Name/Positions                                  | Type         | Size | Comment                                                                                                                                                               |
|-------------------------------------------------------|--------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recurring Payment Mandate Type (Position 1)           | Numeric      | 1    | Values are:<br>1 = fixed amount Recurring Payment<br>2 = Recurring Payment with maximum amount                                                                        |
| Recurring Payment Transaction Type (Position 2)       | Numeric      | 1    | Values are:<br>1 = registration/first transaction<br>2 = subsequent transaction                                                                                       |
| Validation Flag (Position 3)                          | Numeric      | 1    | Values are:<br>0 = not validated<br>1 = validated                                                                                                                     |
| Maximum Recurring Payment Amount (Positions 4 – 13)   | Numeric      | 10   | Maximum amount agreed to by the cardholder.<br><br>Amount will be specified in DE 49 (Currency Code, Transaction).<br>Example: 1800.00 as 180000, with leading zeros. |
| Registered Transaction Identifier (Positions 14 – 23) | Alphanumeric | 10   | Unique identifier for Recurring Payments mandate                                                                                                                      |

Data element definitions and details  
Subfield 1 (Additional Recurring Payments Information)

| Field Name/Positions                             | Type    | Size | Comment                                                                                                                                                                                                         |
|--------------------------------------------------|---------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of Recurring Payments (Positions 24 – 25) | Numeric | 2    | Indicates the number of recurring payment debits. With leading zeros.<br><br>Value of 99 means that recurring payments are authorized until canceled or that the Number of Recurring Payments is not indicated. |
| Recurring Payment Frequency (Position 26)        | Numeric | 1    | Identifies frequency of the Recurring Payments debit.<br><br>Values are:<br>0 – annually<br>1 – daily<br>2 – weekly<br>3 – fortnightly<br>4 – monthly<br>5 – quarterly<br>6 – twice-yearly<br>8 – ad hoc        |

### Application notes

Subfield 1 facilitates the communication of additional data related to recurring payments between acquirers and issuers. Issuers in India and acquirers globally must comply with applicable local laws and regulations issued by the Reserve Bank of India (RBI) when processing card-based transactions. Mastercard is not enforcing edits to check the presence or content of any data.

[DE 48](#), subelement 41, subfield 1 will include and echo back additional data in Authorization Request Response/0110 message enrollment and first transaction of the standing order instructions, and subsequent recurring transactions within the Authorization Request Response/0110 message.

[DE 48](#), subelement 41, subfield 1 will include and echo back additional data to the acquirer, regardless of issuer interaction.



### Subfield 2 (Reserved for Future Use)

DE 48, subelement 41, subfield 2 (Reserved for Future Use) is for future use.

### Subfield 3 (Reserved for Future Use)

DE 48, subelement 41, subfield 3 (Reserved for Future Use) is for future use.

### Subfield 4 (Reserved for Future Use)

DE 48, subelement 41, subfield 4 (Reserved for Future Use) is for future use.

### Subfield 5 (Reserved for Future Use)

DE 48, subelement 41, subfield 5 (Reserved for Future Use) is for future use.

### Subfield 6 (Reserved for Future Use)

DE 48, subelement 41, subfield 6 (Reserved for Future Use) is for future use.

### Subfield 7 (Reserved for Future Use)

DE 48, subelement 41, subfield 7 (Reserved for Future Use) is for future use.

### Subfield 8 (Reserved for Future Use)

DE 48, subelement 41, subfield 8 (Reserved for Future Use) is for future use.

### Subfield 9 (Reserved for Future Use)

DE 48, subelement 41, subfield 9 (Reserved for Future Use) is for future use.

### Subfield 10 (Reserved for Future Use)

DE 48, subelement 41, subfield 10 (Reserved for Future Use) is for future use.

### Subfield 11 (Citizen ID)

DE 48, subelement 41, subfield 11 (Citizen ID) indicates the citizen ID.

#### Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 11          |

| Attribute            | Description         |
|----------------------|---------------------|
| Subfield Data Length | 2 positions         |
| Data Representation  | an...20             |
| Data Field           | National ID         |
| Justification        | N/A                 |
| Values               | Cardholder specific |

### Subfield 12 (Reserved for Future Use)

DE 48, subelement 41, subfield 12 (Reserved for Future Use) is for future use.

### Subfield 13 (Reserved for Future Use)

DE 48, subelement 41, subfield 13 (Reserved for Future Use) is for future use.

### Subfield 14 (Reserved for Future Use)

DE 48, subelement 41, subfield 14 (Reserved for Future Use) is for future use.

### Subfield 15 (Reserved for Future Use)

DE 48, subelement 41, subfield 15 (Reserved for Future Use) is for future use.

### Subfield 16 (Reserved for Future Use)

DE 48, subelement 41, subfield 16 (Reserved for Future Use) is for future use.

### Subfield 17 (Reserved for Future Use)

DE 48, subelement 41, subfield 17 (Reserved for Future Use) is for future use.

### Subfield 18 (Reserved for Future Use)

DE 48, subelement 41, subfield 18 (Reserved for Future Use) is for future use.

## Subelement 42 (Electronic Commerce Indicators)

DE 48, subelement 42 (Electronic Commerce Indicators) contains the electronic commerce indicators representing the security level and cardholder authentication associated with the transaction.

### Attributes

| Attribute              | Description                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------|
| Subelement ID          | 42                                                                                                 |
| Subelement Data Length | 2 positions                                                                                        |
| Data Representation    | n-7 unless an Identity Check downgrade or MDES SLI modification occurs in which case n-19 applies. |

| Attribute     | Description                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Field    | Contents of subfield 1 (except contents of subfields 1, 2, and 3 for acquirer Authorization Request Response/0110 messages if Identity Check downgrade or MDES SLI modification occurred) |
| Subfields     | 3                                                                                                                                                                                         |
| Justification | N/A                                                                                                                                                                                       |

## Usage

Subelement 42 must be present in all Authorization Request/0100 messages for electronic commerce transactions. Following is the usage of subelement 42 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | CE  | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |
| Reversal Advise/0420                          | •   | X   | C   |

## Values

If Identity Check downgrade or MDES SLI modification did not occur, contains the electronic commerce security level indicator and UCAF collection indicator data in subfield 1 that consists of a valid combination of positions 1, 2, and 3.

If Identity Check downgrade or MDES SLI modification occurred, contains contents of subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator), and subfield 3 (Reason for UCAF Collection Indicator Downgrade) for acquirer Authorization Request Response/0110 messages.

## Application notes

A Mastercard Identity Check transaction is downgraded to a non-Identity Check transaction if an Accountholder Authentication Value (AAV) is not present or is invalid in [DE 48](#), subelement 43 (Universal Cardholder Authentication Field [UCAF]).

For transactions initiated with Mastercard Digital Enablement Service (MDES) Tokens, Mastercard provides Dual Message Authorization System edits that validate and when necessary, modify the Security Level Indicator (SLI) in [DE 48](#), subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection indicator) in Authorization Request/0100 messages.

Mastercard sends the correct SLI to the issuer in Authorization Request/0100 messages for MDES transactions.

Mastercard notifies acquirers when the SLI has been corrected by inserting two additional [DE 48](#), subelement 42 indicators, as follows. All acquirers must be ready to receive [DE 48](#), subelement 42, subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator), and subfield 3 (Reason for UCAF Collection Indicator Downgrade) in the Authorization Request Response/0110 messages.

## Subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator)

[DE 48](#), subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator) indicates the electronic commerce security level and UCAF collection in positions 1, 2, and 3.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 01                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | n-3                       |
| Data Field           | Contents of positions 1-3 |
| Justification        | N/A                       |

### Values

Valid combination of the security level indicators in positions 1 and 2 and the UCAF collection indicator in position 3.

## Position 1 (Security Protocol)

### Attributes

| Attribute           | Description                                      |
|---------------------|--------------------------------------------------|
| Data Representation | n-1                                              |
| Data Field          | The electronic commerce security level indicator |

### Values

| Value | Description                                              |
|-------|----------------------------------------------------------|
| 0     | Reserved for existing Mastercard Europe/Visa definitions |
| 1     | Reserved for future use                                  |
| 2     | Channel                                                  |
| 3–8   | Reserved for future use                                  |

## Position 2 (Cardholder Authentication)

### Attributes

| Attribute           | Description                             |
|---------------------|-----------------------------------------|
| Data Representation | n-1                                     |
| Data Field          | The cardholder authentication indicator |

### Values

| Value | Description                |
|-------|----------------------------|
| 0     | Reserved for future use    |
| 1     | eCommerce / Identity Check |
| 3     | Reserved for future use    |
| 4     | Tokenized payment          |
| 5–9   | Reserved for future use    |

### Valid Combinations of Position 1 and Position 2

| Value | Description                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------|
| 21    | Channel encryption; cardholder certificate not used (this is the preferred value for Mastercard® Identity Check™) |
| 24    | Tokenized Payment                                                                                                 |

### Position 3 (UCAF Collection Indicator)

#### Attributes

| Attribute           | Description                   |
|---------------------|-------------------------------|
| Data Representation | n-1                           |
| Data Field          | The UCAF collection indicator |

#### Values

| Value | Description                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | Non-authenticated payment, Identity Check transaction with failed authentication, or Tokenized Payment with Dynamic Token Verification Code (DTVC).                                                                                                                                                                     |
| 1     | UCAF data collection is supported by the merchant, and UCAF data must be present ( <a href="#">DE 48</a> , subelement 43 must be present and contain an attempt AAV for Mastercard Identity Check)                                                                                                                      |
| 2     | UCAF data collection is supported by the merchant, and UCAF data must be present ( <a href="#">DE 48</a> , subelement 43 must contain a fully authenticated AAV, and DSRP cryptogram shall be optionally present for tokenized transactions). This includes cardholder-initiated transactions (CIT) for authentication. |
| 3     | UCAF data collection is supported by the merchant, and UCAF (Mastercard assigned Static Accountholder Authentication Value) data must be present.                                                                                                                                                                       |

**NOTE:** [DE 48](#), subelements 32 and 43 are required for Static AAV transactions.

Identifies participation in one of the following programs:

- Maestro Recurring Payments Program
- Mastercard Utility Payment Program
- Maestro Low Risk Merchant Program
- Maestro Static AAV for Masterpass

| Value | Description                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | Merchant has chosen to share authentication data within authorization; UCAF data collection must be present ( <a href="#">DE 48</a> , subelement 43 must contain an Insights AAV for Mastercard Identity Check). |
| 5     | Reserved for future use                                                                                                                                                                                          |
| 6     | Merchant Risk Based Decisioning                                                                                                                                                                                  |
| 7     | Merchant-initiated transactions ( <a href="#">DE 48</a> , subelement 43 only required for Identity Check).                                                                                                       |
| 8–9   | Reserved for future use                                                                                                                                                                                          |

### Valid Combinations of Position 1, Position 2 and Position 3

| Value | Description                                                                                                                                                                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 210   | Present in unauthenticated transactions, or in transactions where an Identity Check merchant has chosen not to undertake Identity Check, or where Identity Check failed authentication and the merchant desires to proceed with the transaction. Also present in transactions with a DTVC |
| 211   | Mastercard Identity Check transaction. UCAF data contains either an Attempts AAV or a Non-Low Risk AAV for Mastercard Identity Check from Smart Authentication                                                                                                                            |
| 212   | Mastercard Identity Check transaction. UCAF data contains a fully authenticated AAV for Mastercard Identity Check or a Low-Risk AAV for Mastercard Identity Check from Smart Authentication                                                                                               |
| 214   | Mastercard Identity Check where merchant has chosen to share data to generate insights within the authorization (used in Identity Check™ Insights or Ekata transactions). Insights or Ekata scores are provided to the Issuer in <a href="#">DE 48</a> , subelement 56                    |
| 216   | Mastercard Identity Check transaction. Merchant Risk Based Decisioning. This includes European PSD2 Acquirer Exemption transactions                                                                                                                                                       |



| Value | Description                                                                                                                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 217   | Mastercard Identity Check transaction. Merchant Initiated Transactions<br><br><b>NOTE: Mastercard Identity Check split shipments should be managed as 1 authentication, 1 authorization and multiple clearing records.</b> |
| 242   | Authenticated tokenized transaction with DSRP cryptogram. Transaction is not eligible for fraud-related chargeback by the issuer                                                                                           |
| 246   | Tokenized transaction with DSRP cryptogram                                                                                                                                                                                 |
| 247   | Tokenized Merchant Initiated Transaction                                                                                                                                                                                   |

### Application Notes

For Visa Gateway 3-D Secure transactions the following mapping applies:

- Position 3 "0" maps to Field 60 SLI = 07 (Not Authenticated)
- Position 3 "1" maps to Field 60 SLI = 06 (3-D Secure merchant, not authenticated)
- Position 3 "2" maps to Field 60 SLI = 05 (Authenticated with CAVV)

For American Express Gateway 3-D Secure transactions the following mapping applies:

- Position 3 "0" maps to Field 61 SLI = 07 (Not Authenticated)
- Position 3 "1" maps to Field 61 SLI = 06 (Attempted with AEVV)
- Position 3 "2" maps to Field 61 SLI = 05 (Authenticated with AEVV)

SLI is the level of security used when a cardmember provides payment information to the merchant during authentication.

JCB and Diners Club use the same Position 3 values (0, 1, and 2) as defined for Mastercard Identity Check.

## Subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator)

[DE 48](#), subelement 42, subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator) describes the original Security Level Indicators sent by the acquirer in the Authorization Request/0100 message before the Identity Check downgrade or MDES SLI modification occurred.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 02                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | n-3                       |
| Data Field           | Contents of positions 1-3 |
| Justification        | N/A                       |

### Positions

| Position | Description               |
|----------|---------------------------|
| 1        | Security Protocol         |
| 2        | Cardholder Authentication |
| 3        | UCAF Collection Indicator |

## Subfield 3 (Reason for UCAF Collection Indicator Downgrade)

[DE 48](#), subelement 42, subfield 3 (Reason for UCAF Collection Indicator Downgrade) describes the reason why the Authorization Request/0100 message was downgraded or modified.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 03                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n-1                    |
| Data Field           | Contents of subfield 3 |
| Justification        | N/A                    |

### Values

| Value | Description                                              |
|-------|----------------------------------------------------------|
| 0     | Missing Universal Cardholder Authentication Field (UCAF) |
| 1     | Invalid Universal Cardholder Authentication Field (UCAF) |
| 2     | Invalid Security Level Indicator                         |
| 3     | X-Code Processing                                        |

### Application notes

A Mastercard Identity Check transaction is downgraded to a non-Identity Check transaction if:

- An Accountholder Authentication Value (AAV) is not present
- An Accountholder Authentication Value (AAV) is invalid in [DE 48](#), subelement 43 (Universal Cardholder Authentication Field [UCAF])
- During X-Code processing, validation is not performed, following which a downgrade reason value 3 is provided

[DE 48](#), subelement 42, subfields 2 and 3 are only present in the Authorization Request Response/0110 message provided by Mastercard to the acquirer if an Identity Check downgrade or MDES SLI modification occurred.

The message format sent in the acquirer Authorization Request Response/0110 message will be in the format 0103xxx0203yyy0301z only if the Identity Check downgrade occurs. Otherwise, the message format will remain as it is today which is 0103xxx.

**NOTE:** [DE 48](#), subelement 42, subfields 2 and 3 will always be submitted together.

## Subelement 43 (Universal Cardholder Authentication Field [UCAF])

DE 48, subelement 43 (Universal Cardholder Authentication Field [UCAF]) contains the encoded Mastercard Identity Check AAV (Accountholder Authentication Value) resulting from all Identity Check fully authenticated or attempts transactions, data for Visa, JCB, Diners Club, or American Express transactions associated with the 3-D Secure Electronic Commerce Verification Service (if collected), or the Static AAV assigned by Mastercard for Maestro Recurring Payments Program, Mastercard Utility Payment Program, or Maestro Low Risk Merchant Program.

Digital Secure Remote Payments (DSRP) submitted as Electronic Commerce (POS entry mode [DE 22](#), subfield 1, value of 10 or 81) transactions should use [DE 104](#) for communicating the MDES and DSRP cryptogram and other Digital Payment information. Refer to the [Digital Secure Remote Payments: UCAF Formats](#) and *MDES Technical Specifications for Dual and Single Message Systems* guides for information on the UCAF formats supported by Mastercard.

## Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 43                         |
| Subelement Data Length | 2 positions                |
| Data Representation    | ans...32; LLVAR            |
| Data Field             | Contents of positions 1-32 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

## Usage

Subelement 43 must be present whenever UCAF data is collected for electronic commerce transactions. Following is the usage of subelement 43 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |
| Reversal Request Response/0410                | CE  | XE  | CE  |
| Reversal Advice/0420                          | •   | X   | C   |

## Values

Refer to the specific program or service applications for:

- [3-D Secure for Mastercard Identity Check](#)
- [3-D Secure Electronic Commerce Verification Service \(Visa or American Express\)](#)
- [Digital Secure Remote Payments](#)
- *SPA2 AAV and Verification for the Mastercard Identity Check Program*
- *MDES Technical Specifications for Dual and Single Message Systems*

## Subelement 43 (3-D Secure for Mastercard Identity Check)

DE 48 (Additional Data: Private Use), subelement 43 (3-D Secure for Mastercard Identity Check) contains UCAF data and is described here for Mastercard Implementation of 3-D Secure for Mastercard Identity Check.

### Attributes

| Attribute           | Value                      |
|---------------------|----------------------------|
| Subelement ID       | 43                         |
| Subelement Length   | 2 positions                |
| Data Representation | ans-28 (base 64 encoded)   |
| Data Field          | Contents of positions 1-28 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Values

Values are transaction-specific. Positions 1 and 2 of the Mastercard 3-D Secure SPA AAV are control bytes indicating the Mastercard Identity Check platform that created the field contents.

SLI 211 with h and SLI 212 with j are downgraded to SLI 210 (unauthenticated).

### Mastercard Identity Check Platform

|                                                                                                                                                     |                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>SPA2 Values for Smart Authentication non-low risk and Attempts Only authenticated SLI value 211</b>                                              |                                                                            |
| kE, kF, kL, kM                                                                                                                                      |                                                                            |
| <b>SPA2 Values for fully authenticated and CIT (cardholder-initiated transaction) and Smart Authentication low risk authenticated SLI value 212</b> |                                                                            |
| kA, kB, kC, kD, kG, kH, kJ, kK, kQ, kR, kS, kT                                                                                                      |                                                                            |
| <b>SPA2 Values for Insights authenticated SLI value 214</b>                                                                                         | <b>SPA2 Values for PSD2 Acquirer Exemption authenticated SLI value 216</b> |
| kX, kY, kZ                                                                                                                                          | kN, kU, kV, kW                                                             |
| <b>SPA2 Values for Recurring MIT (merchant initiated transactions) authenticated SLI value 217</b>                                                  |                                                                            |
| kO, kP                                                                                                                                              |                                                                            |

**NOTE:** Control byte kQ is defined for AAV Refresh with SLI = 212, and kF is defined for Attempts Only with SLI = 211. Attempts Only indicates that authentication was attempted by the merchant but was not performed either by the issuer or by Mastercard Identity Check Smart Authentication Stand-In solution.

**NOTE:** SPA2 AAV leading indicator values i, l, m, and n are reserved for future use.

### Application notes

Mastercard will continue to modify the SLI value in Mastercard Digital Enablement Service (MDES) transactions based on the AAV contained in [DE 48](#), subelement 43. Acquirers and issuers should expect to see Mastercard Identity Check with SecureCode SLI values in MDES transactions.

**NOTE:** Issuers must pass the correct security level indicator (SLI) value with the corresponding AAV control byte during cardholder authentication. Mastercard 3DS protocol, Identity Check Program Rules, and DW Integrity Edits do not allow sending Fully Authenticated AAVs for Smart Authentication non-low risk or Attempts Only Authentications nor Smart Authentication non-low risk or Attempts Only AAVs for Fully Authenticated Authentications. Reference: *SPA2 AAV and Verification for the Mastercard Identity Check Program*.

The following is an example of a properly coded DE 48, subelement 43 for Mastercard Identity Check for Fully Authenticated Authorization including AVS request.

T420701032124328kBMAAAAKRIacwR0SdLcB5eBIJjW

Refer to *Mastercard Identity Check Program Guide* and *SPA2 AAV and Verification for the Mastercard Identity Check Program* for more information.

**NOTE:** Issuers should not perform Identity Check validation on static AAVs in DE 48, subelement 43 nor on Mastercard Identity Check SPA2 AAV generated by Mastercard. Issuers who use the Mastercard AAV Validation Service should refer to the OBS (on-behalf services) field DE 48, subelement 71 (On-behalf Services) for services 05 and 06 services. 05 OBS AAV Verification Service is for full-time AAV verification. 06 OBS AAV Verification Service is for Stand-In Authorization when issuers perform self-validation. Issuers enroll in either the 05 OBS AAV Verification Service or the 06 OBS Stand-In AAV Verification Service.

Issuers and their ACS Providers that are sharing IAV (issuer authentication value) data within the SPA2 AAV can always read and verify this data as they have defined for the 4 hexadecimal bytes shared from their ACS Provider. The IAV can be found in hexadecimal bytes 3-6 after converting the UCAF value from base64 to hexadecimal.

The UCAF field ([DE 48](#), subelement 43) is a variable length field up to a maximum of 32 positions. The Mastercard Identity Check AAV is 28 characters in length. There must be no trailing spaces in the UCAF field.

## Subelement 43 (Digital Secure Remote Payment Universal Cardholder Authentication Field [UCAF])

DE 48, subelement 43 contains UCAF data and is described here for Mastercard Digital Secure Remote Payments (DSRP).

**NOTE: Mastercard requires acquirers to submit the DSRP cryptogram or the new token cryptogram type in DE 104 (Digital Payment Data), subelement 001 (Digital Payment Cryptogram) instead of submitting it in the existing UCAF DE 48, subelement 43 for all tokenized transactions.**

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 43                         |
| Subelement Data Length | 2 positions                |
| Data Representation    | ans-28; (base 64 encoded)  |
| Data Field             | Contents of positions 1-28 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Values

Transaction specific.

Refer to MDES Digital Secure Remote Payment Acquirer Implementation Guide for detailed information on the UCAF formats supported by Mastercard.

An MDES transaction might not contain DE 48, subelement 43 when the acquirer has submitted a dynamic expiration date ([DE 14](#)) and a dynamic token verification code (DE 48, subelement 92) in order to perform MDES validation services.

A Digital Secure Remote Payment transaction submitted for Mastercard Digital Enablement Service processing that involves an authorization request for a partial shipment or a recurring payment must contain the partial shipment verbiage in subelement 43 as follows: PARTIALbSHIPMENTbbbbbbbbbbbb or PARTIALSHIPMENT00000000000000 unless subelement 42 = 247 in which case subelement 43 is not required. This field contains 28 positions where b represents a space.

### Application notes

The format is version dependent.

**NOTE: Mastercard requires acquirers to submit the DSRP cryptogram in [DE 104 \(Digital Payment Data\)](#), subelement 001 (Digital Payment Cryptogram) and not in DE 48, subelement 43 (UCAF).**

## Subelement 43 (Static AAV)

[DE 48](#), subelement 43 is the Mastercard implementation of the Static AAV for Maestro Recurring Payments Program, Mastercard Utility Payment Program, Maestro Low Risk Merchant Program, or Maestro Static AAV for Masterpass.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subelement ID       | 43                         |
| Subelement length   | 2 positions                |
| Data Representation | ans-28                     |
| Data field          | Contents of positions 1-28 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Values

Position 1 is one of the following values:

- 3 is Transaction processed under the Maestro Recurring Payments Program
- 5 is Transaction processed under the Mastercard Utility Payment Program
- 5 is Maestro Basic Static AAV (Maestro e-commerce transaction initiated through Masterpass)

Position 2–7 is The Mastercard Assigned ID which uniquely identifies the merchant.

Position 8–28 is The merchant name, padded to the right with nines (9).

For the Maestro Low Risk Merchant Program (Europe region only), for subsequent transactions after the first Maestro e-commerce transaction with the same Maestro card, approved merchants will use [DE 48](#), subelement 43 in the Authorization Request/0100 message with the Mastercard-assigned static Accountholder Authentication Value (AAV) 5MARPPROGRAM9999999999999999 (28 positions).

### Application notes

When subelement 43 contains a Static AAV, subelement 32 (Mastercard Assigned ID) is mandatory. Issuers should not attempt Identity Check validation on a Static AAV.



## Subelement 43 (3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express])

[DE 48](#), subelement 43 (3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express]) is for 3-D Secure Electronic Commerce Verification Service (Visa, JCB, Diners Club, and American Express).

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 43                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | nb-21                 |
| Data Field             | Contents of subfields |
| Subfields              | 2                     |
| Justification          | N/A                   |

### Usage

Following is the usage of subelement 43 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | P   | C   |
| Reversal Request/0400      | C   | P   | C   |

### Values

See subfields.

### Application notes

Subfield 1 is not passed to external networks on Visa-branded and American Express-branded transactions. Those networks only receive subfield 2, in the respective fields for those message formats.

## Subfield 1 (3-D Secure Electronic Commerce Transaction Indicator)

DE 48, subelement 43, subfield 1 (3-D Secure Electronic Commerce Transaction Indicator) indicates that 3-D Secure authentication verification is requested on a non-Mastercard branded transaction.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data representation | n-1                    |
| Data field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

| Value | Description                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------|
| 8     | Non-Mastercard 3-D Secure Electronic Commerce transaction (Visa, JCB, Diners Club, or American Express) |

## Subfield 2 (3-D Secure Electronic Commerce Cardholder Authentication Verification Value [CAVV] Visa Field 126.9 and American Express Verification Value [AEVV] AMEX field 61)

DE 48, subelement 43, subfield 2 (3-D Secure Electronic Commerce Cardholder Authentication Verification Value [CAVV] Visa Field 126.9 and American Express Verification Value [AEVV] AMEX field 61) is the 3-D Secure authentication value to be verified.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data representation | b-20                      |
| Data field          | Contents of position 2-21 |
| Justification       | N/A                       |

## Values

The 3-D Secure Electronic Commerce Visa, JCB, Diners Club, CAVV or American Express AEVV value in binary format for usage 2 and 3.

## Subelement 44 (3-D Secure Electronic Commerce Transaction Identifier [XID][Visa and American Express])

[DE 48](#), subelement 44 (3-D Secure Electronic Commerce Transaction Identifier [XID][Visa and American Express]) contains the 3-D Secure Electronic Commerce Transaction Identifier (XID).

## Attributes

| Attribute              | Description                                             |
|------------------------|---------------------------------------------------------|
| Subelement ID          | 44                                                      |
| Subelement Data Length | 2 positions                                             |
| Data Representation    | b-20                                                    |
| Data Field             | The Visa or American Express XID value in binary format |
| Subfields              | N/A                                                     |
| Justification          | N/A                                                     |

## Usage

Following is the usage of subelement 44 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | CE  | XE  | CE  |
| Reversal Request/0400               | CE  | XE  | CE  |
| Reversal Request Response/0410      | C   | X   | C   |

## Values

The Visa XID value in binary. Visa field 126.8 or the American Express XID value in binary—AMEX field 61.

### Application notes

The Visa XID value is optional in authorization messages for Visa usage 3 TransStain.

Refer to *Visa Base I Technical Specifications* for the specific formats for Visa 3-D Secure Electronic Commerce Transaction Identifier (XID).

## Subelement 45 (3-D Secure Electronic Commerce Transaction Response Code [Visa and American Express])

[DE 48](#), subelement 45 (3-D Secure Electronic Commerce Transaction Response Code) is the 3-D Secure Electronic Commerce Transaction Response Code that contains the Visa Cardholder Authentication Verification Value (CAVV) or the American Express Verification Value (AEVV) results code.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 45                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-1                      |
| Data Field             | Contents of positions 1-2 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 45 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |

### Values

Refer to *Visa Dual Message Authorization System* for field 44.13 (CAVV Results Code) for a complete list of values available for this field.

Refer to *American Express Technical Specifications* for AMEX field 61, position 9, American Express Verification Value (AEVV Results Code) for a complete list of values available for this field.

## Subelement 46 (Product ID [Visa Only])

[DE 48](#), subelement 46 (Product ID) contains the Visa Product ID value (Visa Only).

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 46                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-2                      |
| Data Field             | Contents of positions 1–2 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 46 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |
| Authorization Advice Response/0130  | C   | X   | C   |
| Reversal Request Response/0410      | C   | X   | C   |

### Values

Refer to *Visa Dual Message Authorization System* for a complete list of values for field 62.23 (Product ID).

### Application notes

Acquirers should be prepared to receive subelement 46 in Authorization Request Response/0110, Authorization Advice Response/0130, and Reversal Request Response/0410 messages when one of the following occurs:

- Subelement 90 (Custom Payment Service Request [Visa]) is included in the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated messages, which the Dual Message Authorization System forwards to the Visa network.
- Subelement 90 is not included in the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message and the Authorization System sends the message non-peer-to-peer to the Visa issuer via the Visa network.

## Subelement 47 (Mastercard Payment Gateway Transaction Indicator)

[DE 48](#), subelement 47 (Mastercard Payment Gateway Transaction Indicator) indicates that the transaction is a Mastercard Payment Gateway transaction.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 47                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | ans-8                     |
| Data Field             | Contents of positions 1–8 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 47 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

### Values

MC-MPG/W

### Associated edits

| IF...                                                                                                                                                                     | THEN the Dual Message Authorization System...                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the account range does not participate in the MPG Authorization Blocking Service and <a href="#">DE 48</a> , subelement 47 is present but does not contain value MC-MPG/W | sends to the acquirer an Authorization Request Response/0110 message containing: <ul style="list-style-type: none"><li>• <a href="#">DE 39 (Response Code)</a> = 30</li><li>• <a href="#">DE 44 (Additional Data)</a> = 048047</li></ul> |

## Subelement 48 (Digital Commerce Solutions Indicators)

[DE 48](#), subelement 48 (Digital Commerce Solutions Indicators) identifies the domain, program type and optional check-in details of a digital commerce transaction.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 48                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | ans...73; LLVAR           |
| Data Field             | Contents of subfields 1-4 |
| Subfields              | 4                         |
| Justification          | N/A                       |

### Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | X   | C   |
| Authorization Request Response/0110         | C   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

## Values

See subfields.

## Application notes

[DE 48](#), subelement 48 is present in Authorization Request/0100, Authorization Request Response/0110, and Authorization Advice/0120 messages for specific digital commerce transactions.

## Subfield 01 (Digital Commerce Domain Identifier)

[DE 48](#), subelement 48 (Digital Commerce Solutions Indicators), subfield 01 (Digital Commerce Domain Identifier) identifies the entity responsible for populating the digital commerce solutions indicators defined in subfield 02 (Digital Commerce Program Type), subfield 03 (Digital Commerce Check-In Entity Type and Name), and subfield 04 (Digital Commerce Check-In Method).

## Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n-1                    |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

## Values

| Value | Description             |
|-------|-------------------------|
| 0     | Multi-Domain            |
| 1     | Mastercard              |
| 2     | Acquirer                |
| 3     | Issuer                  |
| 4-9   | Reserved for future use |

## Application notes

Mastercard populates the value of 1. Acquirers or issuers provide values 2 and 3.



## Subfield 02 (Digital Commerce Program Type)

DE 48, subelement 48 (Digital Commerce Solutions Indicators), subfield 02 (Digital Commerce Program Type) identifies the specific digital commerce program.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 02                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of subfield 2 |
| Justification        | N/A                    |

### Valid values

| Value | Description                |
|-------|----------------------------|
| 0     | Multi-Program              |
| 1     | SQR                        |
| 2     | Click to Pay               |
| 3     | In-store Biometrics        |
| 4     | Next Gen POI               |
| 5     | Mastercard QR: Pay by Link |
| 6     | Mastercard QR: X QR        |
| 7-9   | Reserved for future use    |

### Application notes

Subfield 2 is present when subfield 1 is present.

When subfield 01 is 1 (Mastercard), subfield 02 must one of the listed values:

- 1 (SQR)
- 2 (Click to Pay)
- 3 (In-store Biometrics)

When subfield 01 is 1 (Mastercard), or 2 (Acquirer), subfield 02 must be 4 (Next Gen POI).

When subfield 01 is 2 (Acquirer), subfield 02 can be set by the acquirer to values 5 and 6.

## Subfield 03 (Digital Commerce Program Type Details)

[DE 48](#), subelement 48 (Digital Commerce Solutions Indicators), subfield 03 (Digital Commerce Program Type Details) identifies entity type and entity name. All entities eligible for this subfield must be integrated with the Mastercard Next Gen POI API.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 03                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...15; LLVAR        |
| Data Field           | Contents of subfield 3 |
| Justification        | N/A                    |

### Values

Subfield 03 is an open text field of up to 15 characters.

### Application notes

When a transaction belongs to a single program type, subfield 03 will contain two characters:

- The first character indicates the program type
- The second character indicates the program subtype

A value of zero in the second character indicates the absence of a subtype. For example, a subfield 03 value of 10 indicates that the transaction type is Mastercard Merchant Presented QR: SRC (SQR) with no subtype.

For Mastercard QR: X QR and Mastercard QR: Pay by Link program types, two distinct subtypes will be applicable:

- A second character value of 1 will specify the subtype as Merchant Presented
- A second character value of 2 will specify the subtype as Consumer Presented

For example, a subfield 03 value of 61 indicates that the transaction type is Mastercard QR: X QR and subtype is Merchant Presented. Similarly, a subfield 03 value of 62 indicates the transaction type is Mastercard QR: X QR and subtype is Consumer Presented.

When a transaction belongs to more than one program type:

- Subfield 02 will have a value of 0 (Multi-Program)
- Subfield 03 will have four characters to indicate each type and subtype the transaction belongs to

For example, a subfield 03 value of 3040 indicates that the transaction type is both Biometric with no subtype and Next Gen Point-of-Interaction (POI) with no subtype.

For a full list of digital commerce solution indicators, refer to *MDES Technical Specifications for Dual and Single Message Systems*.

### Example field values

| Example program                                                                    | DE 48, subelement 48 representation                                                                                                                                                                                                                           | Comments                                                                                                                                                                  |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transaction with SQR                                                               | <ul style="list-style-type: none"> <li>• Subfield 01 = 1 (Mastercard)</li> <li>• Subfield 02 = 1 (SQR)</li> <li>• Subfield 03 = 10 (SQR)</li> <li>• Subfield 04 = Not present</li> </ul>                                                                      | Sample values when the transaction belongs to a single program type                                                                                                       |
| Transaction with XQR when the QR is Merchant presented                             | <ul style="list-style-type: none"> <li>• Subfield 01 = 2 (Acquirer)</li> <li>• Subfield 02 = 6 (XQR)</li> <li>• Subfield 03 = 61 (XQR, Merchant Presented)</li> <li>• Subfield 04 = Not present</li> </ul>                                                    | Sample values when the transaction belongs to a single program type and sub-type                                                                                          |
| Transaction with Next Gen POI and Stand-in Biometric                               | <ul style="list-style-type: none"> <li>• Subfield 01 = 1 (Mastercard)</li> <li>• Subfield 02 = 0 (Multi-Program)</li> <li>• Subfield 03 = 3040 (30 = Biometric, 40 = Next Gen POI)</li> <li>• Subfield 04 = Next Gen POI Check In details</li> </ul>          | Sample values when the transaction belongs to more than one program type (Multi-Program)                                                                                  |
| Transaction with Click to Pay and Pay By Link QR when the QR is consumer presented | <ul style="list-style-type: none"> <li>• Subfield 01 = 0 (Multi-Domain)</li> <li>• Subfield 02 = 0 (Multi-Program)</li> <li>• Subfield 03 = 2052 (20 = Click to Pay, 52 = Pay By Link QR, Consumer Presented)</li> <li>• Subfield 04 = Not Present</li> </ul> | Sample values when multiple program details are collected by multiple sources, like Click to Pay from Mastercard and Pay By Link QR, Consumer presented from the acquirer |

## Subfield 04 (Digital Commerce Check-In Details)

DE 48, subelement 48 (Digital Commerce Solutions Indicators), subfield 04 (Digital Commerce Check-In Details) is an open text field. This field identifies entity type, entity name and the methods a check-in entity applies to validate cardholder presence. All entities eligible for this subfield must be integrated with the Mastercard Next Gen POI API. This field may also include a reference number for tracking and reporting purpose.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 04                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...40; LLVAR        |
| Data Field           | Contents of subfield 4 |
| Justification        | N/A                    |

### Values

Subfield 04 is an open text field of up to 40 characters.

### Application notes

Subfield 04 is:

- Only present when subfield 01 , subfield 02, and subfield 03 present are present.
- Required for Next Gen Point-of-Interaction transactions.
- Optional for all other transaction types.

## Subelement 49 (Time Validation Information)

[DE 48](#), subelement 49 (Time Validation Information) contains time data used to calculate time validation and the results of validation when performed.

### Attributes

| Attribute              | Description              |
|------------------------|--------------------------|
| Subelement ID          | 49                       |
| Subelement Data Length | 2 positions              |
| Data Representation    | n-15                     |
| Subfields              | Contents of subfield 1-3 |

### Usage

Following is the usage of subelement 49 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | .   | X   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |

### Subfield 1 (Time Value)

[DE 48](#), subelement 49, subfield 1 (Time Value) contains the time data derived from the acquirer transaction to be used in the time validation.

### Attributes

| Attribute           | Description                    |
|---------------------|--------------------------------|
| Data Representation | n-8                            |
| Data Field          | Contents of positions 1-8      |
| Justification       | Right-justified, leading zeros |

### Subfield 2 (Time Discrepancy Value)

[DE 48](#), subelement 49, subfield 2 (Time Discrepancy Value) contains a positive value representing the differential in minutes between the transaction time data and service-calculated time.

### Attributes

| Attribute           | Description                    |
|---------------------|--------------------------------|
| Data Representation | n-5                            |
| Data Field          | Contents of positions 9-13     |
| Justification       | Right-justified, leading zeros |

## Subfield 3 (Time Discrepancy Indicator)

DE 48, subelement 49, subfield 3 (Time Discrepancy Indicator) contains a value that indicates if the time discrepancy value is below, above, or within the minimum and maximum values for the time validation window or that indicates time validation was not performed.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-2                         |
| Data Field          | Contents of positions 14-15 |
| Justification       | N/A                         |

### Values

| Value | Description                                   |
|-------|-----------------------------------------------|
| 01    | Positive value within time validation window  |
| 02    | Positive value outside time validation window |
| 03    | Negative value within time validation window  |
| 04    | Negative value outside time validation window |
| 05    | Unknown (time validation not performed)       |

## Subelement 50 (Embedded Interchange Data)

DE 48, subelement 50 provides data in support of commercial transactions using negotiated interchange rates between acquirers and issuers.

Issuers receive this subelement if they have opted in for the Embedded Flexible Interchange program and if a participating acquirer has included the subelement in the given transaction.

### Attributes

| Attribute              | Description     |
|------------------------|-----------------|
| Subelement ID          | 50              |
| Subelement Data Length | 2 positions     |
| Data Representation    | ans...99; LLVAR |

| Attribute     | Description               |
|---------------|---------------------------|
| Data Field    | Contents of subfields 1-6 |
| Subfields     | 6                         |
| Justification | See subfields             |

Following is the usage of DE 48, subelement 50 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Messages                                              | Org | Sys | Dst |
|-------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100 message                    | C   | P   | C   |
| Authorization Advice/0120 message: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120 message: system-generated   | •   | X   | C   |
| Reversal Request/0400 message                         | C   | P   | C   |
| Reversal Advice/0420 message                          | •   | X   | C   |

### Application notes

[DE 48](#), subelement 50, and its subfields are set up in tag, length, value format.

Registered acquirers provide this subelement to specify the negotiated interchange rate, amount, or both which applies to a commercial transaction.

Acquirers must use one of the valid combination of subfields within this subelement as listed in the Valid subfield combinations table.

### Valid subfield combinations

| Present                                                                                                                                         | Not present                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Interchange percentage only                                                                                                                     |                                                                                    |
| Subfield 01 (Interchange Indicator), subfield 02 (Interchange Percentage), subfield 03 (Unit Fee Amount), subfield 04 (Flat Interchange Amount) |                                                                                    |
| <ul style="list-style-type: none"> <li>subfield 01</li> <li>subfield 02</li> </ul>                                                              | <ul style="list-style-type: none"> <li>subfield 03</li> <li>subfield 04</li> </ul> |
| Interchange percentage with unit fee amount                                                                                                     |                                                                                    |

| Present                                                                                                 | Not present                                                                        |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>subfield 01</li> <li>subfield 02</li> <li>subfield 03</li> </ul> | subfield 04                                                                        |
| Flat interchange only                                                                                   |                                                                                    |
| <ul style="list-style-type: none"> <li>subfield 01</li> <li>subfield 04</li> </ul>                      | <ul style="list-style-type: none"> <li>subfield 02</li> <li>subfield 03</li> </ul> |

The Dual Message Authorization System rejects transactions that do not conform to one of the valid combinations. Subfield 05 (Payment Party) and subfield 06 (Supplemental Interchange Data) are optional subfields and can be included with any of the valid combinations.

Subfields with numeric attributes can take a value of all zeros. For example, an acquirer can use the second combination where the Interchange Percentage is present (and non-zero) and the Unit Fee Amount is all zeros. However, an acquirer cannot include Interchange Percentage, Unit Fee Amount, and Flat Interchange Amount in the same transaction even if one or more of those subfields contains all zeros.

Refer to the subfields contained within this subelement to obtain the supporting programs, services, or other specifics about the transaction.

## Subfield 1 (Interchange Indicator)

[DE 48](#), subelement 50, subfield 1 signals that a transaction uses a negotiated interchange rate or amount per the EFI program.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 1                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | a-1                    |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |



## Values

| Value | Description                                                |
|-------|------------------------------------------------------------|
| Y     | Transaction includes negotiated interchange rate or amount |

## Application note

This subfield is mandatory for all EFI program transactions.

## Subfield 2 (Interchange Percentage)

[DE 48](#), subelement 50, subfield 2 provides the negotiated interchange percentage rate per the EFI program.

## Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 2                        |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-6                      |
| Data Field           | Contents of subfield 2   |
| Justification        | Right with leading zeros |

## Values

The interchange percentage carries a percentage value with an implied three-position decimal.

The interchange percentage can take a value of 0.000%, indicating that the negotiated rate is zero. The maximum value for the interchange percentage is 100.000%.

For example, an Interchange Percentage of 1.234% is represented with DE 48, subelement 50, subfield 2, value 001234.

## Application notes

Interchange for an EFI program transaction is calculated based on [DE 4 \(Amount, Transaction\)](#), [DE 48](#) subelement 50 subfield 2 (Interchange Percentage), and [DE 48](#) subelement 50 subfield 3 (Unit Fee Amount), if present.

This subfield is conditional for EFI program transactions. It cannot be included in transactions that also contain subfield 4 (Flat Interchange Amount).

## Subfield 3 (Unit Fee Amount)

[DE 48](#), subelement 50, subfield 3 provides the negotiated unit fee amount per the EFI program.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 3                        |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-12                     |
| Data Field           | Contents of subfield 3   |
| Justification        | Right with leading zeros |

### Values

The Unit Fee Amount is expressed in the currency defined by [DE 49 \(Currency Code, Transaction\)](#).

Where a minor unit of currency applies, the Unit Fee Amount is expressed in the minor unit of currency without a decimal separator, such as, value 000000000100 represents USD 1.00.

### Application notes

Interchange for an EFI program transaction is calculated based on [DE 4 \(Amount, Transaction\)](#), [DE 48](#) subelement 50 subfield 2 (Interchange Percentage), and DE 48 subelement 50 subfield 3 (Unit Fee Amount).

This subfield is conditional for EFI program transactions. It can only be used in transactions that contain subfield 2. It cannot be included in transactions that also contain subfield 4 (Flat Interchange Amount).

## Subfield 4 (Flat Interchange Amount)

[DE 48](#), subelement 50, subfield 4 provides the negotiated flat interchange amount per the EFI program.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 4           |
| Subfield Data Length | 2 positions |
| Data Representation  | n-12        |

| Attribute     | Description              |
|---------------|--------------------------|
| Data Field    | Contents of subfield 4   |
| Justification | Right with leading zeros |

### Values

The Flat Interchange Amount is expressed in the currency defined by [DE 49 \(Currency Code, Transaction\)](#).

Where a minor unit of currency applies, the Flat Interchange Amount is expressed in the minor unit of currency without a decimal separator, such as, value 000000000100 represents USD 1.00.

A Flat Interchange Amount of zero indicates that the negotiated interchange is zero.

### Application notes

Interchange for an EFI program transaction is determined by subfield 4.

This subfield is conditional for EFI program transactions. It cannot be included in transactions that contain either subfield 2 or subfield 2 and subfield 3.

## Subfield 5 (Payment Party)

[DE 48](#), subelement 50, subfield 5 specifies whether the acquirer or issuer pays interchange on this transaction per the EFI program.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 5                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | α-1                    |
| Data Field           | Contents of subfield 5 |
| Justification        | N/A                    |

## Values

### Values

| Value | Description               |
|-------|---------------------------|
| A     | Acquirer pays interchange |
| I     | Issuer pays interchange   |

## Application notes

This subfield is optional for EFI program transactions and must be used only with EFI program transactions.

## Subfield 6 (Supplemental Interchange Data)

[DE 48](#), subelement 50, subfield 6 is a free-text field which acquirers can use to send proprietary data to issuers as part of EFI program transactions.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 6                      |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...59; LLVAR        |
| Data Field           | Contents of subfield 6 |
| Justification        | N/A                    |

## Values

Proprietary free-text data.

## Application notes

This subfield is optional for EFI program transactions and must be used only with EFI program transactions.

## Subelement 51 (Merchant On-behalf Services)

[DE 48](#), subelement 51 (Merchant On-behalf [OB] Services) notifies the acquirer of the On-behalf Service performed on the transaction, and the results. [DE 48](#) will support multiple occurrences of subelement 51 up to the maximum services for a transaction.

### Attributes

| Attribute              | Description                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------|
| Subelement ID          | 51                                                                                                     |
| Subelement Data Length | 2 positions                                                                                            |
| Data Representation    | ans...99; LLVAR<br>The "LL" length field of LLVAR must be an integral multiple of 4, not to exceed 96. |
| Data Field             | Contents of subfields 1–3                                                                              |
| Subfields              | 3                                                                                                      |
| Justification          | N/A                                                                                                    |

### Usage

Following is the usage of subelement 51 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | •   | X   | C   |

### Values

See subfields.

## Subfield 1 (Merchant On-behalf [OB] Service)

[DE 48](#), subelement 51, subfield 1 (Merchant On-behalf [OB] Service) indicates the service performed on the transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Field          | Contents of positions 1–2 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

## Values

See [Subelement 51 \(Valid Subfield 1 and Subfield 2 Value Combinations\)](#).

## Subfield 2 (Merchant On-behalf [OB] Result 1)

[DE 48](#), subelement 51, subfield 2 (Merchant On-behalf [OB] Result 1) indicates the results of the service processing.

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

## Subelement 51 (Valid Subfield 1 and Subfield 2 Value Combinations)

[DE 48](#), subelement 51 (Valid Subfield 1 and Subfield 2 Value Combinations) are the valid [DE 48](#), subelement 51, subfield 1 (Merchant On-behalf [OB] Service) and subfield 2 (Merchant On-behalf [OB] Result 1) value combinations. The contents of subfield 2 depend on the contents of subfield 1 as described here.

## Attributes

| Attribute           | Subfield 1 (Merchant On-behalf [OB] Service) | Subfield 2 (Merchant On-behalf [OB] Result 1) |
|---------------------|----------------------------------------------|-----------------------------------------------|
| Data Representation | an-2                                         | an-1                                          |
| Data Field          | Contents of positions 1–2                    | Contents of position 3                        |
| Justification       | N/A                                          | N/A                                           |

Following are the [DE 48](#), subelement 51, valid subfield 1 and subfield 2 value combinations.

## Values

| <b>DE 48, Subelement 51, Subfield 1 (Merchant On-behalf [OB] Service) Values (an-2)</b> |                                                   | <b>DE 48, Subelement 51, Subfield 2 (Merchant On-behalf [OB] Result 1) Values (an-1)</b> |                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90                                                                                      | EMS Real-time Fraud Scoring Service for Merchants | C                                                                                        | EMS Real-time Fraud Scoring Service for Merchants was performed successfully.                                                                                                                                                                                                    |
|                                                                                         |                                                   | I                                                                                        | Invalid; transaction does not qualify for the EMS Real-time Fraud Scoring Service for Merchants due to one of the following: <ul style="list-style-type: none"> <li>• Transaction is card present, or</li> <li>• Card issuance is outside of the valid issuing region</li> </ul> |
|                                                                                         |                                                   | U                                                                                        | EMS Real-time Fraud Scoring Service for Merchants was not performed successfully.                                                                                                                                                                                                |

## Application notes

If the value of subfield 1 and subfield 2 is 90C, then [DE 48](#), SE 55, subfield 1 and subfield 2 will be populated.

Subfield 2 will have values that identify the results from the EMS Real-time Fraud Scoring Service for Merchants processing result.

## Subfield 3 (Additional Information)

[DE 48](#), subelement 51, subfield 3 (Additional Information) indicates if the acquirer in the transaction is enrolled with the service and if the service was requested.

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 4 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Value | Description                                                         |
|-------|---------------------------------------------------------------------|
| N     | Not qualified for EMS Real-time Fraud Scoring Service for Merchants |
| blank | No value present                                                    |

### Application notes

Subfield 3 will have an indicator of N if the score was not delivered to the acquirer but the acquirer had requested the transaction be scored. This is based on service participation.

## Subelement 52 (Interchange Structure Indicator)

DE 48, subelement 52 (Interchange Structure Indicator) contains the Mastercard-provided interchange structure for point-of-sale (POS) purchase, purchase with cash back, purchase return/refund, and payment transactions initiated on the Dual Message System (Authorization).

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 52                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-2                      |
| Data Field             | Contents of positions 1–2 |
| Subfields              | N/A                       |

### Usage

Following is the usage of subelement 52 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.



### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Response/0110                                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | •   | X   | C   |

### Values

| Description                     | Value |
|---------------------------------|-------|
| Small Business Level 1          | B1    |
| Small Business Level 2          | B2    |
| Small Business Level 3          | B3    |
| Small Business Level 4          | B4    |
| Small Business Level 5          | B5    |
| Commercial Business to Business | BB    |
| Bill Pay                        | BP    |
| Core Value                      | CC    |
| Unregulated Consumer Debit      | CD    |
| Unregulated Consumer Prepaid    | CP    |
| Enhanced Value                  | EV    |
| Commercial Flex Program         | FP    |
| Fleet                           | FT    |
| Humanitarian                    | HM    |
| World High Value                | HV    |
| Installment Payments            | IP    |
| Large Market                    | LM    |
| Unregulated Commercial Debit    | MD    |
| Unregulated Commercial Prepaid  | MP    |
| Unknown Interchange Structure   | NA    |
| Commercial Payments Account     | PA    |

| Description                                        | Value |
|----------------------------------------------------|-------|
| Private Label                                      | PL    |
| Prepaid Commercial Payments Account                | PP    |
| Regulated Consumer Debit Base                      | RB    |
| Regulated Commercial Debit Base                    | RC    |
| Regulated Consumer Prepaid with Fraud Adjustment   | RD    |
| Regulated Commercial Prepaid with Fraud Adjustment | RE    |
| Regulated Consumer Debit with Fraud Adjustment     | RF    |
| Regulated Commercial Debit with Fraud Adjustment   | RG    |
| Regulated Consumer Prepaid Base                    | RP    |
| Regulated Commercial Prepaid Base                  | RR    |
| Variable Interchange Programs                      | VP    |
| World Elite                                        | WE    |
| World                                              | WO    |

### Application notes

The Dual Message Authorization System inserts the Interchange Structure Indicator in eligible Authorization Response/0110 and Authorization Advice Response/0130 messages in the U.S. region.

The Interchange Structure Indicator is applicable only to the U.S. region for Purchase, Purchase with Cash Back, Purchase Return/Refund, and Payment Transaction types.

ATM and cash disbursement transactions are not eligible for the Interchange Structure Indicator.

The Interchange Structure Indicator is not applicable to the U.S. territories of American Samoa, Guam, Northern Mariana Islands, Puerto Rico, and U.S. Virgin Islands, which are located in the Asia/Pacific and Latin America and the Caribbean regions.

## Subelement 53 (E-ID Request Code)

DE 48, subelement 53 (E-ID Request Code) enables the issuer to identify that additional cardholder information is being requested by the acquirer. This request code is sent by an acquirer to an issuer in an Account Status Inquiry (ASI) message.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Subelement ID       | 53                     |
| Subelement Length   | 2 positions            |
| Data Representation | ans...99; LLVAR        |
| Data Field          | Contents of subfield 1 |
| Subfields           | 1                      |

### Usage

Following is the usage of subelement 53 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Values

See subfields.

### Application notes

Acquirers that support E-ID will include this subelement in an ASI, queuing the issuer to perform further processing outside of the Mastercard Network. This subelement is valid for Europe region activity only.

Subelement 53 (and associated subfield 1) is effective as of 12 June 2018.

For card-not-present transactions, the acquirer should not send subelement 53.

## Subfield 1 (E-ID Request Value)

[DE 48](#), subelement 53, subfield 1 (E-ID Request Value) informs the issuer of the proposed action on the cardholder account.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 1                         |
| Subfield Data Length | 2 positions               |
| Data Representation  | ans-2                     |
| Data Field           | Contents of positions 1–2 |
| Justification        | N/A                       |

### Values

| Value | Description                                                                                   |
|-------|-----------------------------------------------------------------------------------------------|
| 01    | Informs the issuer to pull cardholder personal data for E-ID/Health ID Verification Purposes. |
| 02    | Informs the issuer to verify and send the age of the cardholder.                              |
| 03–99 | Reserved for Future Use                                                                       |

### Application notes

Issuers must:

- Validate that a valid value is present in the message; and
- Send this cardholder information through a network separate from the Mastercard Network.

## Subelement 55 (Merchant Fraud Scoring Data/Approval Probability Indicator)

[DE 48](#), subelement 55 (Merchant Fraud Scoring Data/Approval Probability Indicator) indicates the fraud score on a fraud scoring service transaction when the acquirer requested the transaction to be scored.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subelement ID       | 55                        |
| Subelement Length   | 2 positions               |
| Data Representation | an...32; LLVAR            |
| Data Field          | Contents of subfields 1–5 |
| Subfields           | 5                         |
| Justification       | See subfields             |

### Usage

Following is the usage of subelement 55 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | .   | X   | C   |

### Application notes

The Dual Message Authorization System inserts this subelement when EMS Real-time Fraud Scoring Service for Merchants is performed on the transaction.

Mastercard will require all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

**NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.**

## Subfield 1 (Merchant Fraud Score/ASI Probability Indicator)

[DE 48](#), subelement 55, subfield 1 (Merchant Fraud Score/ASI Probability Indicator) indicates the transaction risk score or the ASI probability indicator.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 01                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-3                      |
| Data Field           | Contents of positions 1–3 |
| Justification        | N/A                       |

### Values

The EMS Real-time Fraud Scoring Service for Merchants provides the risk score of 001–998, where 001 indicates the least likely fraudulent transaction and 998 indicates the most likely fraudulent transaction. For ASI transactions, the subfield provides a three-digit number from 111 to 999 in response to an ASI request submitted with a future transaction amount value.

### Application notes

Acquirers submitting Visa-branded transactions receive the merchant fraud scoring data from Visa in the response messages.

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Subfield 2 (Reason Code)

[DE 48](#), subelement 55, subfield 2 (Reason Code) contains the reason code.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 02                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-2                      |
| Data Field           | Contents of positions 1–2 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

The EMS Real-time Fraud Scoring Service for Merchants provides the score reason code, an alphanumeric code identifying the data used to derive the fraud score. For ASI transactions, the reason code is an alphanumeric code providing additional detail about the value provided for the ASI Probability Indicator. For ASI, the possible values are:

- 00: Successful
- 01: Not Enough Data (Dormant or New Card)

### Application notes

Participating acquirers may contact the Risk Solutions Team for a list of the specific score reason codes that apply to their institution.

## Subfield 3 (Reserved for Future Use)

[DE 48](#), subelement 55, subfield 3 (Reserved for Future Use) is reserved for future use.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 03                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-3                      |
| Data Field           | Contents of positions 1–3 |
| Justification        | N/A                       |

### Values

Reserved for Future Use.

## Subfield 4 (Reserved for Future Use)

DE 48, subelement 55, subfield 4 (Reserved for Future Use) is reserved for future use.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 04                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-2                      |
| Data Field           | Contents of positions 1–2 |
| Justification        | N/A                       |

### Values

Reserved for Future Use.

## Subfield 5 (Reserved for Future Use)

DE 48, subelement 55, subfield 5 (Reserved for Future Use) is reserved for future use.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 05                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-2                      |
| Data Field           | Contents of positions 1–2 |
| Justification        | N/A                       |

### Values

Reserved for Future Use.

## Subelement 56 (Security Services Additional Data for Issuers)

DE 48, subelement 56 (Security Services Additional Data for Issuers) supports Mastercard embedded security services for issuers. Issuers can receive up to 16 instances of subelement 56.



Each instance has the same format for each service. Issuers must code to receive the 16 instances that the service supports.

### Attributes

| Attribute              | Description                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Subelement ID          | 56                                                                                                    |
| Subelement Data Length | 2 positions                                                                                           |
| Data Representation    | an...96; LLVAR<br>The "LL" length field of LLVAR must be an integral multiple of 6, not to exceed 96. |
| Data Field             | Contents of subfields                                                                                 |
| Subfields              | 2                                                                                                     |
| Justification          | N/A                                                                                                   |

### Usage

Following is the usage of subelement 56 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | •   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

### Values

See subfields.

### Application notes

Mastercard will require all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement please refer to the *Global Safety and Security Standards Roadmap*.

**NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.**

## Subfield 1 (Security Services Indicator)

[DE 48](#), subelement 56, subfield 1 (Security Services Indicator) provides the security services indicator for issuers.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-3                      |
| Data Field          | Contents of positions 1–3 |
| Justification       | N/A                       |

### Values

A three-character code identifying the real-time monitoring service used.

## Subfield 2 (Security Services Data)

[DE 48](#), subelement 56, subfield 2 (Security Services Data) contains additional data supporting the issuer fraud score.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-3                      |
| Data Field          | Contents of positions 4–6 |
| Justification       | N/A                       |

### Values

A three-character value will be populated.

Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)

DE 48, subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations) following is the valid DE 48, subelement 56, subfield 1 (Security Services Indicator) and subfield 2 (Security Services Data) value combinations.

Service Data Content for Decision Intelligence Service: Authorization IQ Feature

Following is the Service Data Content for the Authorization IQ feature of the Decision Intelligence service. Customers that support Decision Intelligence will receive AQV (GDV-based Spend Ranking), AQF (Frequency-based Spend Ranking), and AQS (Segment Qualifier) on all transactions.

**NOTE: Positions 2 and 3 use the same reason code definitions.**

Service data content

| Subfield 1 (an-3)                  | Subfield 2 (an-3)                                        |                                                                         |                                                          |
|------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------|
|                                    | Position 1                                               | Position 2                                                              | Position 3                                               |
| Spending Dimension Insights        | Overall Account Spending Insights                        | Channel Spending Insights                                               | Account Transaction Insights                             |
| AQV: GDV-based Spend Ranking       | New (card activity on the network less than 60 days old) | 0                                                                       | New (card activity on the network less than 60 days old) |
| Or                                 |                                                          | Or                                                                      | 0                                                        |
| AQF: Frequency-based Spend Ranking |                                                          | None (No network activity in the last 12 months or Dormant Card Number) |                                                          |
|                                    | High = 1                                                 |                                                                         | High = 1                                                 |
|                                    | Med-1 = 4                                                |                                                                         | Med-1 = 4                                                |
|                                    | Med-2 = 5                                                |                                                                         | Med-2 = 5                                                |
|                                    | Med-3 = 6                                                |                                                                         | Med-3 = 6                                                |
|                                    | Low = 9                                                  |                                                                         | Low = 9                                                  |
| AQS: Segment Qualifier             | Refer to the Segment Definitions table that follows.     |                                                                         |                                                          |

Segment Definitions for Decision Intelligence Service: Authorization IQ Feature

The Authorization IQ feature of the Decision Intelligence service supports the following series of segments under which each transaction is qualified for dimension analysis.

**NOTE: New segments may be added to the Authorization IQ feature of the Decision Intelligence service at any time.**

## Segment Definitions

| Segment                                      | Segment Qualifier ID                                |
|----------------------------------------------|-----------------------------------------------------|
| Channel: Card Present: Domestic Segments     | 000: Card activity on network less than 60 days old |
| Travel                                       | 001                                                 |
| Retail                                       | 002                                                 |
| Gaming                                       | 003                                                 |
| Gambling                                     | 004                                                 |
| Education/Healthcare                         | 005                                                 |
| Cash (for example: ATM)                      | 006                                                 |
| Fuel                                         | 007                                                 |
| Utilities                                    | 008                                                 |
| Leisure and Entertainment                    | 009                                                 |
| Digital Goods                                | 010                                                 |
| Professional Services                        | 011                                                 |
| Reserved for Future Use                      | 012–198                                             |
| Other Card Present: Domestic                 | 199                                                 |
| Channel: Card Present: Cross-Border Segments | 200: Card activity on network less than 60 days old |
| Travel                                       | 201                                                 |
| Retail                                       | 202                                                 |
| Gaming                                       | 203                                                 |
| Gambling                                     | 204                                                 |
| Education/Healthcare                         | 205                                                 |
| Cash (for example: ATM)                      | 206                                                 |
| Fuel                                         | 207                                                 |
| Utilities                                    | 208                                                 |
| Leisure and Entertainment                    | 209                                                 |
| Digital Goods                                | 210                                                 |
| Professional Services                        | 211                                                 |
| Reserved for Future Use                      | 212–398                                             |
| Other Card Present: Cross-Border             | 399                                                 |

Data element definitions and details  
Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)

| Segment                                          | Segment Qualifier ID                                                             |
|--------------------------------------------------|----------------------------------------------------------------------------------|
| Channel: Card Not Present: Domestic Segments     | 400: Card activity on network less than 60 days old                              |
| Travel                                           | 401                                                                              |
| Retail                                           | 402                                                                              |
| Gaming                                           | 403                                                                              |
| Gambling                                         | 404                                                                              |
| Education/Healthcare                             | 405                                                                              |
| Cash (for example: ATM)                          | 406                                                                              |
| Utilities                                        | 407                                                                              |
| Leisure and Entertainment                        | 408                                                                              |
| Digital Goods                                    | 409                                                                              |
| Professional Services                            | 410                                                                              |
| Reserved for Future Use                          | 411–598                                                                          |
| Other Card Not Present: Domestic                 | 599                                                                              |
| Channel: Card Not Present: Cross-Border Segments | 600: Card activity on network less than 60 days old                              |
| Travel                                           | 601                                                                              |
| Retail                                           | 602                                                                              |
| Gaming                                           | 603                                                                              |
| Gambling                                         | 604                                                                              |
| Education/Healthcare                             | 605                                                                              |
| Cash (for example: ATM)                          | 606                                                                              |
| Utilities                                        | 607                                                                              |
| Leisure and Entertainment                        | 608                                                                              |
| Digital Goods                                    | 609                                                                              |
| Professional Services                            | 610                                                                              |
| Reserved for Future Use                          | 611–798                                                                          |
| Other Card Not Present: Cross-Border             | 799                                                                              |
| Reserved for Future Use                          | 800–899                                                                          |
| Dormant Card Number                              | 900: Previously active card with no activity on the network in the last 120 days |
| Reserved for Future Use                          | 901–999                                                                          |

## Service Data Content for Digital Transaction Insights: Payment Insights

Following is the Service Data Content for the Digital Transaction Insights feature of the Decision Intelligence service.

### Service data content

| Subfield 2 (an-3)                                                                                                                                                                                    |                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Subfield 1 (an-3)                                                                                                                                                                                    |                                                                        |
| Position 1                                                                                                                                                                                           | Position 2                                                             |
| Position 3                                                                                                                                                                                           |                                                                        |
| Security Services Indicator                                                                                                                                                                          |                                                                        |
| Risk Level                                                                                                                                                                                           | Reason Code 1                                                          |
| Reason Code 2                                                                                                                                                                                        |                                                                        |
| AIQ: Digital Transaction Insights Payment Insights                                                                                                                                                   | 0–9 where higher values indicate higher degree of risk.                |
| The following codes only apply to the Reason Code 1 field. Contact your Mastercard representative for more information about the Reason Code 2 field.                                                |                                                                        |
| <ul style="list-style-type: none"><li>Position 2 = a Mastercard-determined reason code</li><li>Position 3 = a Merchant Trust Assessment (merchant's level of trust on EMV 3DS transaction)</li></ul> |                                                                        |
| NOTE: This information is subject to change.                                                                                                                                                         |                                                                        |
|                                                                                                                                                                                                      | Reason Code                                                            |
|                                                                                                                                                                                                      | Description                                                            |
|                                                                                                                                                                                                      | A                                                                      |
|                                                                                                                                                                                                      | Risk Event - Suspicious Account Activity                               |
|                                                                                                                                                                                                      | B                                                                      |
|                                                                                                                                                                                                      | Risk Event - Unknown Device/Account Relationship                       |
|                                                                                                                                                                                                      | C                                                                      |
|                                                                                                                                                                                                      | Risk Event - Device or Profile information associated with fraud       |
|                                                                                                                                                                                                      | D                                                                      |
|                                                                                                                                                                                                      | Risk Event - Recent High Risk change to Device or Profile information  |
|                                                                                                                                                                                                      | E                                                                      |
|                                                                                                                                                                                                      | Risk Event - Recent change to Device or Profile Information            |
|                                                                                                                                                                                                      | F                                                                      |
|                                                                                                                                                                                                      | Risk Event - PAN associated with fraud event                           |
|                                                                                                                                                                                                      | G                                                                      |
|                                                                                                                                                                                                      | New Account or Insufficient Data                                       |
|                                                                                                                                                                                                      | H                                                                      |
|                                                                                                                                                                                                      | Merchant/Acquirer: Merchant (fraud) risk high (assessed by Mastercard) |
|                                                                                                                                                                                                      | I                                                                      |
|                                                                                                                                                                                                      | Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)  |
|                                                                                                                                                                                                      | J                                                                      |
|                                                                                                                                                                                                      | Environment: Good/Known IP                                             |

Data element definitions and details  
Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)

| Subfield 1 (an-3)           | Subfield 2 (an-3) |               |                                                                                                                              |
|-----------------------------|-------------------|---------------|------------------------------------------------------------------------------------------------------------------------------|
|                             | Position 1        | Position 2    | Position 3                                                                                                                   |
| Security Services Indicator | Risk Level        | Reason Code 1 | Reason Code 2                                                                                                                |
|                             |                   | K             | Cardholder: Billing address - prior history established                                                                      |
|                             |                   | L             | Cardholder: Email address - prior history established                                                                        |
|                             |                   | M             | Cardholder: Phone Number - prior history established                                                                         |
|                             |                   | N             | Cardholder: Shipping address - prior history established                                                                     |
|                             |                   | O             | Cardholder: Card number (PAN) behavior established high trust in the current transaction                                     |
|                             |                   | P             | Environment: Device known                                                                                                    |
|                             |                   | Q             | Environment: Account established on Device                                                                                   |
|                             |                   | R             | Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity) |
|                             |                   | S             | More than one Cardholder category established                                                                                |
|                             |                   | T             | More than one Merchant/Acquirer category established                                                                         |
|                             |                   | U             | More than one Environment category established                                                                               |
|                             |                   | V             | Co-occurring established link between cardholder and Merchant/Acquirer                                                       |
|                             |                   | W             | Co-occurring established link between cardholder and Environment                                                             |
|                             |                   | X             | Co-occurring established link between Merchant/Acquirer and Environment                                                      |
|                             |                   | Y             | All three categories established                                                                                             |

| Subfield 1 (an-3)           | Subfield 2 (an-3) |               |                                                                                                                                                                                                                                                                                                  |
|-----------------------------|-------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Position 1        | Position 2    | Position 3                                                                                                                                                                                                                                                                                       |
| Security Services Indicator | Risk Level        | Reason Code 1 | Reason Code 2                                                                                                                                                                                                                                                                                    |
|                             |                   | Z             | Mastercard has received a minimum of three identity elements from the merchant for this transaction and independently established the trust of the transaction. This flag is only available for DTI and may be populated if a merchant shares data for the Mastercard First Party Trust program. |

**NOTE:** The above codes depict an ordering from negative to positive. This list may be further refined. Reason codes A–H reflect riskier, or less information while reason codes I–Z reflect positive information.

**NOTE:** Identity Check Insights requests can also be submitted with Mastercard Digital Enablement Service (MDES) tokens.

### Service Data Content for Consumer Controls

Following is the Service Data Content for Consumer Controls.

### Service Data Content

| Subfield 1 (an-3)      | Subfield 2 (an-3) |                              |
|------------------------|-------------------|------------------------------|
| Service Code           | Reason Code       | Description                  |
| INC: Consumer Controls | AAL               | Alert All Transactions       |
|                        | ABD               | Alert Budget                 |
|                        | ACB               | Alert Cross-border           |
|                        | ACH               | Alert Transaction Channel    |
|                        | AFL               | Alert Filter                 |
|                        | AGE               | Alert Geo Location           |
|                        | AMC               | Alert Merchant Category Code |
|                        | ATA               | Alert Transaction Amount     |
|                        | DBD               | Decline Budget               |
|                        | DCB               | Decline Cross-border         |
|                        | DCD               | Decline Card Disabled        |



| Subfield 1 (an-3) | Subfield 2 (an-3) |                                |
|-------------------|-------------------|--------------------------------|
| Service Code      | Reason Code       | Description                    |
|                   | DCH               | Decline Transaction Channel    |
|                   | DFL               | Decline Filter                 |
|                   | DGE               | Decline Geo Location           |
|                   | DMC               | Decline Merchant Category Code |
|                   | DTA               | Decline Transaction Amount     |
|                   | NAT               | No Action Taken                |

### Digital Transaction Insights: Identity Insights

Identity insight reason codes offer a more specific view about the source of the Mastercard risk analysis, with values from AA–ZZ representing these reasons, with AA as the highest risk to ZZ as the lowest risk.

#### Reason codes

| Subfield 1 (an-3)              | Subfield 2 (an-3)                                       |                                                                                                                                                                                                                                                                                                                                                                                              |               |
|--------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                | Position 1                                              | Position 2                                                                                                                                                                                                                                                                                                                                                                                   | Position 3    |
| Security Services Indicator    | Risk Level                                              | Reason Code 1                                                                                                                                                                                                                                                                                                                                                                                | Reason Code 2 |
| DII: Digital Identity Insights | 0–9 where higher values indicate higher degree of risk. | <p>The following codes apply to both Reason Code fields 1 and 2, where:</p> <ul style="list-style-type: none"> <li>Position 2 = determines the severity of the reason code. A is high, J is medium, and R is low.</li> <li>Position 3 = determines the physical attribute the risk is associated with.</li> <li>Use positions 2 and 3 in combination for a holistic view of risk.</li> </ul> |               |

This information is subject to change. AQ–IZ, JG–QZ, and RQ–ZZ are currently not defined and are reserved for future use.

#### Reason codes

| Reason code | Description                     |
|-------------|---------------------------------|
| AA          | Consumer identity is suspicious |

Data element definitions and details  
Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)

| Reason code | Description                                                   |
|-------------|---------------------------------------------------------------|
| AB          | Increased risk associated with email                          |
| AC          | Increased risk associated with email and IP                   |
| AD          | Increased risk associated with email and address              |
| AE          | Increased risk associated with email and phone                |
| AF          | Increased risk associated with email and name                 |
| AG          | Increased risk associated with IP                             |
| AH          | Increased risk associated with IP and phone                   |
| AI          | Increased risk associated with IP and address                 |
| AJ          | Increased risk associated with IP and name                    |
| AK          | Increased risk associated with address                        |
| AL          | Increased risk associated with address and phone              |
| AM          | Increased risk associated with address and name               |
| AN          | Increased risk associated with name                           |
| AO          | Increased risk associated with name and phone                 |
| AP          | Increased risk associated with phone                          |
| AQ          | Increased risk associated with device and email               |
| AR          | Increased risk associated with device and IP                  |
| AS          | Increased risk associated with device and address             |
| AT          | Increased risk associated with device and name                |
| AU          | Increased risk associated with device and phone               |
| AV          | Increased risk associated with payment instrument and email   |
| AW          | Increased risk associated with payment instrument and email   |
| AX          | Increased risk associated with payment instrument and address |
| AY          | Increased risk associated with payment instrument and name    |
| AZ          | Increased Risk associated with payment instrument and phone   |
| JA          | New consumer or not sufficient information                    |
| JB          | Medium risk signals associated with email                     |
| JC          | Medium risk signals associated with address                   |

| Reason code | Description                                              |
|-------------|----------------------------------------------------------|
| JD          | Medium risk signals associated with phone                |
| JE          | Medium risk signals associated with name                 |
| JF          | Medium risk signals associated with IP                   |
| RA          | Consumer trust established across all PII elements       |
| RB          | Established credibility for email                        |
| RC          | Established credibility for email and IP                 |
| RD          | Established credibility for email and address            |
| RE          | Established credibility for email and phone              |
| RF          | Established credibility for email and name               |
| RG          | Established credibility for IP                           |
| RH          | Established credibility for IP and phone                 |
| RI          | Established credibility for IP and address               |
| RJ          | Established credibility for IP and name                  |
| RK          | Established credibility for address                      |
| RL          | Established credibility for address and phone            |
| RM          | Established credibility for address and name             |
| RN          | Established credibility for name                         |
| RO          | Established credibility for name and phone               |
| RQ          | Established credibility associated with device and email |
| RR          | Established credibility associated with device and IP    |

### Service Data Content for Digital Transaction Insights: Authentication Risk Analysis

Following is the Service Data Content for the Authentication Risk Analysis (ARA) feature of the Decision Intelligence service.

### Service data content

| Subfield 2<br>(an-3)                       |                                           |                                         |                                            |                                                                                                                                                                                                               |
|--------------------------------------------|-------------------------------------------|-----------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield 1<br>(an-3)                       | Position 1                                | Position 2                              | Position 3                                 | Description                                                                                                                                                                                                   |
| <b>Security<br/>Services<br/>Indicator</b> | <b>Reason code</b>                        |                                         |                                            |                                                                                                                                                                                                               |
| ARA:<br>Authentication<br>Risk Analysis    | A-Z<br>assessing<br>risk of<br>cardholder | A-Z<br>assessing<br>risk of<br>merchant | A-Z<br>assessing<br>risk of<br>environment |                                                                                                                                                                                                               |
| ARA                                        | A                                         | A                                       | A                                          | Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard)                                                                                                                               |
|                                            | A                                         | A                                       | B                                          | Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP                                                                                                                |
|                                            | A                                         | A                                       | C                                          | Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information                                                                     |
|                                            | A                                         | A                                       | D                                          | Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information                                                                               |
|                                            | A                                         | A                                       | E                                          | Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship                                                                                          |
|                                            | A                                         | A                                       | F                                          | Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship                                                                 |
|                                            | A                                         | A                                       | G                                          | Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity) |

| Subfield 2<br>(an-3)              |             |            |            |                                                                                                                                                                                                                    |
|-----------------------------------|-------------|------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield 1<br>(an-3)              | Position 1  | Position 2 | Position 3 | Description                                                                                                                                                                                                        |
| Security<br>Services<br>Indicator | Reason code |            |            |                                                                                                                                                                                                                    |
|                                   | A           | A          | H          | PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship                                                                                           |
|                                   | A           | A          | I          | New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship                                                                                          |
|                                   | A           | A          | J          | Card number (PAN) behavior established high trust in the current transaction, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship                                              |
|                                   | A           | A          | K          | New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity) |
|                                   | A           | A          | L          | New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard)                                                                                                                               |
|                                   | A           | A          | M          | New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information                                                                               |
|                                   | A           | A          | N          | Cardholder Shipping address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information                                                |
|                                   | A           | A          | O          | New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information                                                                     |

| Subfield 2<br>(an-3)              |             |            |            |                                                                                                                                                                                                       |
|-----------------------------------|-------------|------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield 1<br>(an-3)              | Position 1  | Position 2 | Position 3 | Description                                                                                                                                                                                           |
| Security<br>Services<br>Indicator | Reason code |            |            |                                                                                                                                                                                                       |
|                                   | A           | A          | P          | Cardholder Phone number prior history established, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship                                                            |
|                                   | A           | A          | Q          | Cardholder Card number (PAN) behavior established high trust in the current transaction, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information |
|                                   | A           | A          | R          | Cardholder Phone number prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information                                       |
|                                   | A           | A          | S          | New Account or Insufficient Data, Merchant (fraud) risk high (assessed by Mastercard)                                                                                                                 |
|                                   | A           | A          | T          | Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information                                    |
|                                   | A           | A          | U          | Cardholder Phone number prior history established, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP                                                                                  |
|                                   | A           | A          | V          | Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information                                              |
|                                   | A           | A          | W          | Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP                                                                               |

| Subfield 2<br>(an-3)              |             |            |            |                                                                                                                                                                                                                                                            |
|-----------------------------------|-------------|------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield 1<br>(an-3)              | Position 1  | Position 2 | Position 3 | Description                                                                                                                                                                                                                                                |
| Security<br>Services<br>Indicator | Reason code |            |            |                                                                                                                                                                                                                                                            |
|                                   | A           | A          | X          | Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard)                                                                                                                                                   |
|                                   | A           | A          | Y          | PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP                                                                                                                                                         |
|                                   | A           | A          | Z          | PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)                                          |
|                                   | A           | B          | A          | PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information                                                                                                                        |
|                                   | A           | B          | B          | Limited information and refer to DTI reason code                                                                                                                                                                                                           |
|                                   | A           | B          | C          | Card number (PAN) behavior established high trust in the current transaction, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information                                                        |
|                                   | A           | B          | D          | Cardholder Billing address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)  |
|                                   | A           | B          | E          | Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity) |

| Subfield 2<br>(an-3)              |             |            |            |                                                                                                                                                                                                                                                        |
|-----------------------------------|-------------|------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield 1<br>(an-3)              | Position 1  | Position 2 | Position 3 | Description                                                                                                                                                                                                                                            |
| Security<br>Services<br>Indicator | Reason code |            |            |                                                                                                                                                                                                                                                        |
|                                   | A           | B          | F          | Cardholder Phone number prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity) |
|                                   | A           | B          | G          | Cardholder Phone number prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)                                                                                                                               |
|                                   | A           | B          | H          | Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information                                                                           |
|                                   | A           | B          | I          | Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)                                                                                                                           |
|                                   | A           | B          | J          | Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP                                                                                                            |
|                                   | A           | B          | K          | Cardholder Phone number prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information                                                                               |

**NOTE:** The above codes depict the current list of reason codes and an appropriate ordering from negative to positive. This list may be further refined in future.



## Emerging Fraud Monitoring Service

### Emerging Fraud Monitoring Service value combinations

| Subfield 1 | Subfield 2                                                                   |                             |                                                                    |
|------------|------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------|
|            | Position 1<br>Risk tier, where higher values indicate higher degrees of risk | Position 2<br>Risk Category | Position 3<br>Suggested Action to either Decline (D) or Review (R) |
| EFM        | 0 to 9                                                                       | A to Z                      | D or R                                                             |

### Reason codes

| Reason code | Category             | Description                                             |
|-------------|----------------------|---------------------------------------------------------|
| A           | Emerging High risk   | Terminal Parameter Manipulation                         |
| B           | Emerging High risk   | Benign MCC                                              |
| C           | Emerging High risk   | Authentication Not Performed/<br>Unverified User Action |
| D           | Emerging High risk   | Recurring Merchant Misrepresentation                    |
| E           | Emerging High risk   | Address Verification Risk/AVS Bypass                    |
| F           | Emerging High risk   | Credential takeover                                     |
| G           | Emerging High risk   | Untrusted MCC codes                                     |
| H           | Emerging High risk   | Past Suspicion Fraud                                    |
| I           | Emerging High risk   | State-Specific Fraudulent Activity                      |
| J           | Emerging High risk   | Cross border fraud                                      |
| K           | Emerging High risk   | Tokenization Exploitation Fraud                         |
| L           | Undetected High risk | Terminal Parameter Manipulation                         |
| M           | Undetected High risk | Benign MCC                                              |
| N           | Undetected High risk | Authentication Not Performed/<br>Unverified User Action |
| O           | Undetected High risk | Recurring Merchant Misrepresentation                    |

Data element definitions and details  
Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)

| Reason code | Category             | Description                                             |
|-------------|----------------------|---------------------------------------------------------|
| P           | Undetected High risk | Address Verification Risk/AVS Bypass                    |
| Q           | Undetected High risk | Credential takeover                                     |
| R           | Undetected High risk | Untrusted MCC codes                                     |
| S           | Undetected High risk | Past Suspicion Fraud                                    |
| T           | Undetected High risk | State-Specific Fraudulent Activity                      |
| U           | Undetected High risk | Cross border fraud                                      |
| V           | Undetected High risk | Tokenization Exploitation Fraud                         |
| W           | Evolved High risk    | Terminal Parameter Manipulation                         |
| X           | Evolved High risk    | Benign MCC                                              |
| Y           | Evolved High risk    | Authentication Not Performed/<br>Unverified User Action |
| Z           | Evolved High risk    | Recurring Merchant Misrepresentation                    |
| "0"         | Evolved High risk    | Address Verification Risk/AVS Bypass                    |
| 1           | Evolved High risk    | Credential takeover                                     |
| 2           | Evolved High risk    | Untrusted MCC codes                                     |
| 3           | Evolved High risk    | Past Suspicion Fraud                                    |
| 4           | Evolved High risk    | State-Specific Fraudulent Activity                      |
| 5           | Evolved High risk    | Cross border fraud                                      |
| "6"         | Evolved High risk    | Tokenization Exploitation Fraud                         |
| 7           | Miscellaneous        | --                                                      |
| 8           | Future use           | --                                                      |
| 9           | Future use           | --                                                      |

## Emerging Fraud Monitoring Service Example Output

### Sample output: EFM9KD

| Output | Identifier                                     | Description                                                  |
|--------|------------------------------------------------|--------------------------------------------------------------|
| EFM    | Identifier for Emerging Fraud Monitoring alert | Static Field                                                 |
| 9      | Model output for Risk Tier                     | High likelihood that this transaction presents a fraud risk  |
| K      | Model output for Fraud Category                | Emerging High Risk Category, Tokenization Exploitation Fraud |
| D      | Model output for Suggested Action              | EFM recommends that this transaction be Declined             |

## Subelement 57 (Security Services Additional Data for Acquirers)

[DE 48](#), subelement 57 (Security Services Additional Data for Acquirers) supports Mastercard embedded security services for acquirers. Acquirers can receive up to 16 instances of subfields one and two within subelement 57. Each instance has the same format for each service. Acquirers must code to receive the 16 instances that the service supports.

### Attributes

| Attribute              | Description                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Subelement ID          | 57                                                                                                    |
| Subelement Data Length | 2 positions                                                                                           |
| Data Representation    | an...96; LLVAR<br>The "LL" length field of LLVAR must be an integral multiple of 6, not to exceed 96. |
| Data Field             | Contents of subfields                                                                                 |
| Subfields              | 2                                                                                                     |
| Justification          | N/A                                                                                                   |

### Usage

Following is the usage of subelement 57 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | .   | X   | C   |

### Values

See subfields.

## Merchant Risk Predict

Merchant Risk Predict provides predictive, real-time, AI-generated merchant fraud detection scores to acquirers on dual message authorization transactions. The seven scores sit within the authorization message to acquirers and provide feedback on merchant health, compliance, and chargeback risks.

### Subfield 1 (Security Services Indicator)

[DE 48](#), subelement 57, subfield 1 (Security Services Indicator) provides the security services indicator for acquirers.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-3                      |
| Data Field          | Contents of positions 1–3 |
| Justification       | N/A                       |

### Values

A three-character code identifying the Merchant Risk Predict score.

## Subfield 2 (Security Services Data)

[DE 48](#), subelement 57, subfield 2 (Security Services Data) contains a likelihood value for each Merchant Risk Predict score.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-3                      |
| Data Field          | Contents of positions 4–6 |
| Justification       | N/A                       |

### Values

A three-character value will be populated for each score indicating the likelihood of occurrence for each fraud type.

## Subelement 58 (ATM Additional Data)

[DE 48](#), subelement 58 (ATM Additional Data) only applies to Swedish Domestic Authorization Switching Service (SASS). Subelement 58 contains watermark data captured at an ATM. Watermark data is a card authentication technology supported by Swedish ATMs.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 58                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans-33                |
| Data Field             | Contents of subfields |
| Subfields              | 8                     |
| Justification          | N/A                   |

### Usage

Following is the usage of subelement 58 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | O   | X   | C   |
| Reversal Request/0400      | O   | X   | C   |
| Reversal Advice/0420       | .   | X   | C   |

### Values

See subfields.

### Subfield 1 (ATM Time)

[DE 48](#), subelement 58, subfield 1 (ATM Time) indicates the ATM time.

### Attributes

| Attribute           | Description                      |
|---------------------|----------------------------------|
| Data Representation | n-4                              |
| Data Field          | Contents of positions 1–4 (hhmm) |
| Justification       | N/A                              |

### Subfield 2 (ATM Date)

[DE 48](#), subelement 58, subfield 2 (ATM Date) indicates the ATM date.

### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | n-6                                 |
| Data Field          | Contents of positions 5–10 (YYMMDD) |
| Justification       | N/A                                 |

## Subfield 3 (Watermark)

DE 48, subelement 58, subfield 3 (Watermark) is the watermark value.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-12                        |
| Data Field          | Contents of positions 11–22 |
| Justification       | N/A                         |

## Subfield 4 (Mark 1)

DE 48, subelement 58, subfield 4 (Mark 1) indicates the card and the watermark status.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-2                       |
| Data Field          | Contents of positions 23–24 |
| Justification       | N/A                         |

### Values

| Value | Description                          |
|-------|--------------------------------------|
| 0-    | National card                        |
| 4-    | National card, foreign currency      |
| 8-    | International card                   |
| C-    | International card, foreign currency |
| -0    | Watermark readable                   |
| -B    | Watermark unreadable                 |
| -C    | Watermark missing                    |
| -D    | Test mode                            |

| Value | Description |
|-------|-------------|
| -E    | Test mode   |

## Subfield 5 (Mark 2)

[DE 48](#), subelement 58, subfield 5 (Mark 2) indicates the reason code for reversal from ATM.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | an-2                        |
| Data Field          | Contents of positions 25–26 |
| Justification       | N/A                         |

### Values

| Value | Description                                             |
|-------|---------------------------------------------------------|
| 00    | Dispensing error for bank notes or receipts             |
| 02    | Error in response                                       |
| 08    | Failure to return card, response received from host     |
| 48    | Failure to return card, single reversal                 |
| 04    | Timeout—card not picked up, response received from host |
| 80    | Timeout—card not picked up, response received from host |
| 44    | Timeout—card not picked up, single reversal             |
| 40    | Single reversal, unknown reason                         |
| 06    | Error in response and timeout picking up card           |
| 0A    | Error in response and failure to return card            |



## Subfield 6 (Mark 3)

DE 48, subelement 58, subfield 6 (Mark 3) indicates stock of bank notes and receipt status.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-2                       |
| Data Field          | Contents of positions 27–28 |
| Justification       | N/A                         |

### Values

| Value | Description                              |
|-------|------------------------------------------|
| 0-    | Both SEK 100 and SEK 500 notes available |
| 4-    | SEK 500 notes not available              |
| 8-    | SEK 100 notes not available              |
| C-    | No money available                       |
| -0    | Receipt OK                               |
| -4    | Receipt low                              |
| -8    | Receipt paper empty                      |
| -C    | Receipt technical error                  |

## Subfield 7 (Card Swallowed Status)

DE 48, subelement 58, subfield 7 (Card Swallowed Status) indicates if the ATM took the card or not.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | n-1                     |
| Data Field          | Contents of position 29 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

#### Values

| Value | Description        |
|-------|--------------------|
| 0     | Card not swallowed |
| 1     | Card swallowed     |

### Subfield 8 (Posting Date)

[DE 48](#), subelement 58, subfield 8 (Posting Date) is the posting date.

#### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-4                        |
| Data Field          | Contents of position 30–33 |
| Justification       | N/A                        |

### Subelement 60 (Additional Service Data For Issuers)

[DE 48](#) subelement 60 (Additional Service Data For Issuers) provides data in support of specific issuer services. Issuers receive this subelement if they have opted in for any of the supported services and if those services were performed on the given transaction.

#### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 60                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | ans...43; LLVAR           |
| Data Field             | Contents of subfields 1-4 |

| Attribute     | Description   |
|---------------|---------------|
| Subfields     | 4             |
| Justification | See subfields |

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated | •   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | •   | X   | C   |

### Usage

Following is the usage of [DE 48](#) subelement 60 in applicable messages, regardless if it is mandatory, conditional, optional, system provided, or not required.

### Application notes

[DE 48](#), subelement 60, and subfields 1 through 4 are set up in Tag, Length, Value (TLV) format.

The Dual Message Authorization System inserts this subelement when selected issuer services are performed. The Dual Message Authorization System will remove [DE 48](#), subelement 60 if received in Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 messages from the acquirer.

Refer to the subfields contained within this subelement to obtain the supporting programs, services, or other specifics about the transaction

### Subfield 1 (Reference Conversion Rate)

[DE 48](#), subelement 60, subfield 1 (Reference Conversion Rate) contains the European Central Bank (ECB) reference conversion rate between currencies indicated by DE 49 (Currency Code, Transaction) and DE 51 (Currency Code, Cardholder Billing).

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 01          |
| Subfield Data Length | 2 positions |

| Attribute           | Description                                                              |
|---------------------|--------------------------------------------------------------------------|
| Data Representation | n-8                                                                      |
| Data Field          | Contents of positions 1-8                                                |
| Justification       | Right, excluding the decimal indicator which must be the leftmost digit. |

### Values

The data field for [DE 48](#), subelement 60, subfield 01 is formatted similarly as the data fields for [DE 9 \(Conversion Rate, Settlement\)](#) and [DE 10 \(Conversion Rate, Cardholder Billing\)](#). Position 1 indicates the number of positions the decimal point should be moved from the right. Must be in the range of 0-7.

Positions 2-8 indicates the conversion rate.

The leftmost digit must be in the range 0–7 and denotes the number of positions that the decimal point shall be moved from the right. Example: For a value of 69972522, reference conversion rate is 9.972522 percent.

### Application notes

For European Economic Area (EEA) issuers, the Dual Message Authorization System inserts this subfield for eligible transactions if the account range participates in Cross Border Fee Manager (CBFM) or CBFM ECB Rate Service.

## Subfield 2 (Reference Conversion Rate Differential Sign Indicator)

[DE 48](#), subelement 60, subfield 2 (Reference Conversion Rate Differential Sign Indicator) establishes whether the difference of [DE 10 \(Conversion Rate, Cardholder Billing\)](#) and [DE 48](#), subelement 60, subfield 1 (Reference Conversion Rate) is a positive or negative number.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 02                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | a-1                    |
| Data Field           | Contents of position 1 |
| Justification:       | N/A                    |

## Values

[DE 48](#), subelement 60, subfield 2 provides the algebraic sign of the Reference Conversion Rate Differential.

A value of C (for Credit) indicates that [DE 10 \(Conversion Rate, Cardholder Billing\)](#) is larger than or equal to [DE 48](#), subelement 60, subfield 1 (Reference Conversion Rate).

A value of D (for Debit) indicates that [DE 10 \(Conversion Rate, Cardholder Billing\)](#) is smaller than [DE 48](#), subelement 60, subfield 1 (Reference Conversion Rate).

## Application notes

For EEA issuers, the Dual Message Authorization System inserts this subfield for eligible transactions if the account range participates in CBFM or CBFM ECB Rate Service.

## Subfield 3 (Reference Conversion Rate Percent Differential)

[DE 48](#), subelement 60, subfield 3 (Reference Conversion Rate Percent Differential) provides the difference of DE 10 (Conversion Rate, Cardholder Billing) and [DE 48](#), subelement 60, subfield 1 (Reference Conversion Rate), expressed as a percentage of [DE 48](#), subelement 60, subfield 1.

## Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 03                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | n-6                       |
| Data Field           | Contents of positions 1-6 |
| Justification:       | N/A                       |

## Values

[DE 48](#), subelement 60, subfield 3 provides a percentage of [DE 48](#), subelement 60, subfield 1 (Reference Conversion Rate) to communicate the effective currency conversion markup of the transaction.

The Reference Conversion Percent Rate Differential carries a percentage value with an implied two-position decimal.

[DE 10](#) is typically larger than [DE 48](#), subelement 60, subfield 1. Consequently, the difference of these values is typically positive. [DE 48](#), subelement 60, subfield 2 (Reference Conversion Rate Differential) can only represent zero and positive values by definition. The Reference Conversion Rate Differential and Reference Conversion Rate Differential Sign Indicator can be used together to indicate positive and negative differentials.

Example 1: A differential of +123.45 percent is represented with DE 48 SE 60 SF 2 = C and DE 48 SE 60 SF 3 = 012345

Example 2: A differential of 0.00 percent is represented with DE 48 SE 60 SF 2 = C and DE 48 SE 60 SF 3 = 000000

Example 3: A differential of -1.23 percent is represented with DE 48 SE 60 SF 2 = D and DE 48 SE 60 SF 3 = 000123

### Application notes

For EEA issuers, the Dual Message Authorization System inserts this subfield for eligible transactions if the account range participates in CBFM or CBFM ECB Rate Service.

## Subfield 4 (Currency Rate Unique Identifier)

[DE 48](#), subelement 60, subfield 4 (Currency Rate Unique Identifier) provides a unique identifier for the currency rate dataset used by the Mexico Central Bank (MCB) rate service.

### Attributes

| Attribute            | Description                |
|----------------------|----------------------------|
| Subfield ID          | 04                         |
| Subfield Data Length | 2 positions                |
| Data Representation  | an-12                      |
| Data Field           | Contents of positions 1-12 |
| Justification        | N/A                        |

The data field for DE 48, subelement 60, subfield 4 contains four pieces of information with construction YYMMDDUUUUUU.

### Values

| Where... | Is...                                                                              |
|----------|------------------------------------------------------------------------------------|
| YY       | The two-digit year at the time of conversion, right-justified with leading zeros.  |
| MM       | The two-digit month at the time of conversion, right-justified with leading zeros. |
| DD       | The two-digit day at the time of conversion, right-justified with leading zeros.   |
| UUUUUU   | A six-digit alphanumeric ID that will be unique each day.                          |

### Application notes

The Dual Message Authorization System inserts this subfield if the account range participates in the Mexico Central Bank (MCB) rate service and if the service is applied to the transaction.

## Subelement 61 (POS Data Extended Condition Codes)

[DE 48](#), subelement 61 (POS Data Extended Condition Codes) indicates whether the merchant terminal supports a specific program or service.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 61                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n-5                       |
| Data Field             | Contents of positions 1–5 |
| Subfields              | 5                         |
| Justification          | See subfields             |

### Usage

Following is the usage of subelement 61 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | C   | C   |

### Values

See subfields.

### Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting specific programs and services.

## Subfield 1 (Partial Approval Terminal Support Indicator)

[DE 48](#), subelement 61, subfield 1 (Partial Approval Terminal Support Indicator) indicates if the merchant terminal supports partial approvals.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

| Value | Description                                                     |
|-------|-----------------------------------------------------------------|
| 0     | Merchant terminal does not support receipt of partial approvals |
| 1     | Merchant terminal supports receipt of partial approvals         |

## Subfield 2 (Purchase Amount Only Terminal Support Indicator)

[DE 48](#), subelement 61, subfield 2 (Purchase Amount Only Terminal Support Indicator) indicates if the merchant terminal supports purchase only approvals.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 2 |
| Justification       | N/A                    |

### Values

| Value | Description                                                           |
|-------|-----------------------------------------------------------------------|
| 0     | Merchant terminal does not support receipt of purchase only approvals |



| Value | Description                                                   |
|-------|---------------------------------------------------------------|
| 1     | Merchant terminal supports receipt of purchase only approvals |

### Subfield 3 (Real-time Substantiation Indicator)

[DE 48](#), subelement 61, subfield 3 (Real-time Substantiation Indicator) indicates if the merchant terminal verified the purchased items against the Inventory Information Approval System (IIAS).

#### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

#### Values

| Value | Description                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | Merchant terminal did not verify the purchased items against an Inventory Information Approval System (IIAS)                                                                                                                 |
| 1     | Merchant terminal verified the purchased items against an Inventory Information Approval System (IIAS)                                                                                                                       |
| 2     | Merchant claims exemption from using an IIAS based on the IRS 90 percent rule                                                                                                                                                |
| 4     | Transaction was submitted as real-time substantiated but from a non-IIAS-certified merchant. Mastercard uses this value to notify the issuer that the merchant could not be substantiated. Acquirers may not use this value. |

## Subfield 4 (Merchant Transaction Fraud Scoring Indicator)

[DE 48](#), subelement 61, subfield 4 (Merchant Transaction Fraud Scoring Indicator) indicates if the acquirer requested the transaction to be scored by Expert Monitoring for Merchants.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 4 |
| Justification       | N/A                    |

### Values

| Value | Description              |
|-------|--------------------------|
| 0     | No action required       |
| 1     | Transaction to be scored |

### Application notes

Only a value of zero is passed to the issuer on the Authorization Request/0100 message.

Mastercard will require all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

**NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.**

## Subfield 5 (Final Authorization Indicator)

[DE 48](#), subelement 61, subfield 5 (Final Authorization Indicator) is used to identify final authorization messages.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 5 |
| Justification       | N/A                    |

### Values

| Value | Description                             |
|-------|-----------------------------------------|
| 0     | Normal Authorization/Undefined Finality |
| 1     | Final Authorization                     |

### Application notes

Final authorization is an authorization request for an amount greater than zero, which meets the following conditions:

- Authorization is requested for the final transaction amount.
- The transaction is not expected to be canceled after the authorization request is approved in full (excluding non-completion for technical reasons such as telecommunications failure or terminal failure).

Any transaction corresponding to an authorization identified as a final authorization must be presented within seven calendar days of the authorization approval date. An authorization request is properly identified as a final authorization when [DE 61 \(Point-of-Service \[POS\] Data\)](#), subfield 7 (POS Transaction Status) contains value 0 and [DE 48 \(Additional Data\)](#), subelement 61 (POS Data Extended Condition Codes), subfield 5 contains value 1.

In the Europe region, all authorization messages must be coded as either preauthorization or final authorization requests; undefined coding is not permitted.

When submitting an Acquirer-generated Authorization Advice/0120, subfield 5 (Final Authorization Indicator) should be populated with the same value as provided in the original Authorization Request/0100 message.

Normal authorization/undefined finality is an authorization request for an amount greater than zero where a final amount may or may not be known and the transaction is not expected to be canceled after the authorization request is approved in full.

An authorization request is identified as normal or undefined finality authorization when [DE 61 \(Point-of-Service \[POS\] Data\)](#), subfield 7 (POS Transaction Status) contains value 0 and [DE 48 \(Additional Data\)](#), subelement 61 (POS Data Extended Condition Codes), subfield 5 contains value 0 or is not present.

When [DE 61 \(Point-of-Service \[POS\] Data\)](#), subfield 7 (POS Transaction Status) contains value 4 (Preauthorized request), [DE 48 \(Additional Data\)](#), subelement 61 (POS Data Extended Condition Codes), subfield 5 must either contain value 0 (Normal Authorization/Undefined Finality) or not be present.

## Subelement 62 (Real-Time Payment Information)

[DE 48](#), subelement 62 (Real-Time Payment Information) contains settlement-related information associated with specific Mastercard programs that offer the participants real-time settlement options information.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subelement ID       | 62                        |
| Subelement Length   | 2 positions               |
| Data Representation | ans...76; LLVAR           |
| Data Field          | Contents of subfields 1–6 |
| Subfields           | See subfields             |
| Justification       | N/A                       |

### Usage

Following is the usage of subelement 62 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |
| Authorization Advice Response/0130  | C   | P   | C   |

| Message                        | Org | Sys | Dst |
|--------------------------------|-----|-----|-----|
| Reversal Request Response/0410 | C   | P   | C   |

## Values

See subfields.

## Application notes

This subelement and subfields are provided by the Account Holding Institution (AHI) in the Authorization Response messages and are sent unaltered to the Acquiring Institution by Mastercard.

## Subfield 1 (Merchant Direct Participation)

[DE 48](#), subelement 62, subfield 1 (Merchant Direct Participation) indicates whether or not the merchant is a direct participant in the real-time payment network and the party to which their transactions are settled. Transactions processed by a direct merchant are settled directly with the merchant, whereas those processed by an indirect merchant are settled with the Acquiring Institution.

## Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

## Values

| Value | Description                                                                          |
|-------|--------------------------------------------------------------------------------------|
| Y     | Merchant's account used for settlement via a real-time payment network.              |
| N     | Acquiring Institution's account used for settlement via a real-time payment network. |

## Subfield 2 (Type of Transaction)

[DE 48](#), subelement 62, subfield 2 (Type of Transaction) indicates whether or not a transaction will be settled via a real-time payment network and if so, what type of settlement will occur.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 02                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of subfield 2 |
| Justification        | N/A                    |

### Values

| Value | Description                                                            |
|-------|------------------------------------------------------------------------|
| D     | Intracountry/Domestic real-time payment settlement transaction         |
| C     | Intercountry real-time payment settlement transaction                  |
| R     | Intraregional real-time payment settlement transaction                 |
| X     | Interregional real-time payment settlement transaction                 |
| Z     | Other real-time payment settlement not elsewhere defined               |
| N     | Acquirer/merchant does not participate in real-time payment settlement |

## Subfield 3 (Real-Time Payment Rules)

[DE 48](#), subelement 62, subfield 3 (Real-Time Payment Rules) indicates which payment network rules and requirements govern the transaction.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 03          |
| Subfield Data Length | 2 positions |

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans...10; LLVAR        |
| Data Field          | Contents of subfield 3 |
| Justification       | N/A                    |

### Values

| Value | Description               |
|-------|---------------------------|
| PBA   | Mastercard Pay by Account |

## Subfield 4 (Real-Time Payment Settlement Network ID and Name)

[DE 48](#), subelement 62, subfield 4 (Real-Time Payment Settlement Network ID and Name) indicates the real-time payment network through which settlement fulfillment will occur.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 04                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n...3; LLVAR           |
| Data Field           | Contents of subfield 4 |
| Justification        | N/A                    |

### Values

| Value | Description     |
|-------|-----------------|
| 01    | On-us Payment   |
| 02    | Faster Payments |
| 03    | UPI             |
| 04    | IMPS            |
| 05    | TCH             |

| Value | Description             |
|-------|-------------------------|
| 06    | Real-time Payment       |
| 07–99 | Reserved for future use |

**NOTE:** Customers must be able to support any values between 01–99. New values can be added at any time.

## Subfield 5 (Real-Time Payment Funds Transfer Position)

[DE 48](#), subelement 62, subfield 5 (Real-Time Payment Funds Transfer Position) indicates the funds transfer position for a transaction settled through a real-time payment network.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 05                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | n-2                    |
| Data Field           | Contents of subfield 5 |
| Justification        | N/A                    |

### Values

| Value | Description                                            |
|-------|--------------------------------------------------------|
| 00    | Funds Transfer not initiated                           |
| 01    | Funds transferred to Acquiring Institution or merchant |
| 02    | Funds requested from Acquiring Institution or merchant |
| 03–99 | Reserved for future use                                |



## Subfield 6 (DSS Reference Number)

[DE 48](#), subelement 62, subfield 6 (DSS Reference Number) indicates the unique number for every bank transaction.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 06                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...35; LLVAR        |
| Data Field           | Contents of subfield 6 |
| Justification        | N/A                    |

## Subelement 63 (Trace ID)

[DE 48 \(Additional Data: Private Use\)](#) subelement 63 (Trace ID) contains data from [DE 63 \(Network Data\)](#) subfield 1 (Financial Network Code) and subfield 2 (Banknet Reference Number) and [DE 15 \(Date, Settlement\)](#) that is in the original Authorization Request Response/0110 message.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 63                         |
| Subelement Data Length | 2 positions                |
| Data Representation    | ans-15                     |
| Data Field             | Contents of positions 1–15 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Usage

Following is the usage of subelement 63 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | M   | P   | M   |
| Reversal Request Response/0410                | ME  | XE  | ME  |
| Reversal Advice/0420                          | •   | X   | M   |

### Values

Contains the contents of positions 1–15 as defined below.

### Values of Positions 1–9 (Network Data)

| Attribute           | Value                     |
|---------------------|---------------------------|
| Data Representation | ans-9                     |
| Data Field          | Contents of positions 1–9 |

### Values of Positions 1–9

Contents of [DE 63](#), subfield 1 (Financial Network Code) and subfield 2 (Banknet Reference Number) in the original Authorization Request Response/0110 message. The Banknet Reference number is six characters.

| Value         | Description       |
|---------------|-------------------|
| Positions 1–3 | DE 63, subfield 1 |
| Positions 4–9 | DE 63, subfield 2 |

### Values of Positions 10–15 (Date Settlement)

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | ans-6       |

| Attribute  | Description                 |
|------------|-----------------------------|
| Data Field | Contents of positions 10–15 |

### Values of Positions 10–15

Contents of [DE 15 \(Date, Settlement\)](#) in the original Authorization Request Response/0110 message. The four-digit Settlement Date is in MMDD format followed by two spaces.

### Application notes

[DE 48 \(Additional Data: Private Use\)](#), subelement 63 must be present in Reversal Request/0400 messages and must contain data in positions 1–15, otherwise the message will be rejected with a format error response where [DE 39 \(Response Code\)](#) is 30 and DE 44 is 048063. [DE 48 \(Additional Data: Private Use\)](#), subelement 63 may contain a value of zeros in Reversal Request/0400 messages or Reversal Advice/0420 messages only when an Authorization Request Response/0110 was not successfully received or processed by the acquirer, or if the customer does not have the original reference number or the settlement date from the authorization.

[DE 48 \(Additional Data: Private Use\)](#), subelement 63 must be provided in Authorization Request/0100 messages to identify an incremental preauthorization and a zero amount Authorization Chargeback Protection Period Extension request.

[DE 48 \(Additional Data: Private Use\)](#), subelement 63 has been added to the acquirer-generated authorization advice/0120 layout to further assist issuers in matching an AFD completion advice to the original preauthorization. For additional details, refer to [Clearing AFD Transactions](#) in the Automated Fuel Dispenser Completion section of the Program and Service Format Requirements chapter.

### Application note: European Economic Area (EEA) only

**NOTE: The following information applies in the EEA only, in support of Payment Service Directive 2 (PSD2) Regulatory Technical Standards (RTS) regarding Strong Customer Authentication (SCA).**

Mastercard uses DE 48, subelement 63 (Trace ID) to provide a reference to the first authorization for recurring payment and merchant-initiated transactions in Dual Message Authorization Request/0100, acquirer-generated advice/0120 and Reversal Request/0400 messages.

## Subelement 64 (Transit Program)

[DE 48](#), subelement 64 (Transit Program) is used to identify transit transactions in authorization messages.

**NOTE: Transit Program field is not limited to MCC 4111, MCC 4131, and MCC 4784.**

## Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 64                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n-4                       |
| Data Field             | Contents of subfields 1–2 |
| Subfields              | 2                         |
| Justification          | N/A                       |

## Usage

Following is the usage of subelement 64 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

## Values

Contains the Transit Program values in subfields 1 and 2.

## Subfield 1 (Transit Transaction Type Indicator)

[DE 48](#), subelement 64, subfield 1 (Transit Transaction Type Indicator) indicates the transit transaction type identifier. This is a mandatory data element for transit transactions. For more information about transit transactions, refer to the *Global Transit Implementation Guide* or any regional or country specific variant.

**NOTE: Values 3, 4, and 6 for Transit Transaction Type Indicators subelement 64, subfield 1 are limited to MCC 4111, MCC 4131, and MCC 4784.**

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 1-2 |
| Justification       | N/A                       |

### Values

| Value | Description                          |
|-------|--------------------------------------|
| 01    | Prefunded                            |
| 02    | Real-time Authorized                 |
| 03    | Post-Authorized Aggregated           |
| 04    | Authorized Aggregated Split Clearing |
| 05    | Other                                |
| 06    | Post-authorized Aggregated Maestro   |
| 07    | Debt Recovery                        |
| 08-99 | Reserved for Future Use              |

## Subfield 2 (Transportation Mode Indicator)

[DE 48](#), subelement 64, subfield 2 (Transportation Mode Indicator) indicates the transportation mode for a transit transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3-4 |
| Justification       | N/A                       |

### Values

| Value | Description                                       |
|-------|---------------------------------------------------|
| 00    | Unknown                                           |
| 01    | Urban Bus                                         |
| 02    | Interurban Bus                                    |
| 03    | Light Train Mass Transit (Underground Metro, LTR) |
| 04    | Train                                             |
| 05    | Commuter Train                                    |
| 06    | Water Borne Vehicle                               |
| 07    | Toll                                              |
| 08    | Parking                                           |
| 09    | Taxi                                              |
| 10    | High Speed Train                                  |
| 11    | Rural Bus                                         |
| 12    | Express Commuter Train                            |
| 13    | Para Transit                                      |
| 14    | Self Drive Vehicle                                |
| 15    | Coach                                             |
| 16    | Locomotive                                        |
| 17    | Powered Motor Vehicle                             |
| 18    | Trailer                                           |
| 19    | Regional Train                                    |
| 20    | Inter City                                        |
| 21    | Funicular Train                                   |
| 22    | Cable Car                                         |
| 23–99 | Reserved for Future Use                           |

## Subelement 65 (Terminal Compliant Indicator)

[DE 48](#), subelement 65 (Terminal Compliant Indicator) contains two subfields that identify whether or not a POS terminal from which the transaction originated is Terminal Line

Encryption (TLE) and Unique Key Per Terminal (UKPT)/Derived Unique Key Per Terminal (DUKPT) compliant. This field must be present in card-present, POS transactions acquired in India.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 65                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n-2                       |
| Data Field             | Contents of subfields 1–2 |
| Subfields              | 2                         |
| Justification          | See subfields             |

### Usage

Following is the usage of subelement 65 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |

### Values

Contains the Terminal Compliant Indicator values in subfields 1 and 2.

## Subfield 1 (TLE Compliant)

[DE 48](#), subelement 65, subfield 1 indicates whether or not the POS terminal from which the transaction occurred is Terminal Line Encryption (TLE)-certified.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

| Value | Description   |
|-------|---------------|
| 1     | Not Certified |
| 2     | Certified     |

## Subfield 2 (UKPT/DUKPT Compliant)

[DE 48](#), subelement 65, subfield 2 indicates whether or not the POS terminal from which the transaction occurred is Unique Key Per Terminal/Derived Unique Key Per Terminal (UKPT/ DUKPT)-certified.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 2 |
| Justification       | N/A                    |



### Values

| Value | Description   |
|-------|---------------|
| 1     | Not Certified |
| 2     | Certified     |

## Subelement 66 (Authentication Data)

*DE 48 (Additional Data)*, subelement 66 is populated by the acquirer and passed to the issuer to indicate the Program Protocol being used.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 66                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...45; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 2                     |

### Usage

Following is the usage of subelement 66 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | •   | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |

Values

See subfields.

Application notes

This is a conditional field and must be provided by the acquirers' part of the authorization messages if they are participating and have previously authenticated using the Mastercard Identity Check program. This field will not be edited to ensure that these elements were provided on the transaction. If it is provided by the acquirer or processor, it must meet the specifications defined.

Subfield 1 (Program Protocol)

DE 4, subelement 66 (Authentication Data), subfield 1 indicates the Program Protocol.

Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

Values

| Value | Description                |
|-------|----------------------------|
| 1     | EMV 3-D Secure Version 2.1 |
| 2     | EMV 3-D Secure Version 2.2 |
| 3     | EMV 3-D Secure Version 2.3 |
| 4     | EMV 3-D Secure Version 2.4 |
| 5     | EMV 3-D Secure Version 2.5 |
| 6     | EMV 3-D Secure Version 2.6 |
| 7     | EMV 3-D Secure Version 2.7 |
| 8     | EMV 3-D Secure Version 2.8 |

| Value | Description                |
|-------|----------------------------|
| 9     | EMV 3-D Secure Version 2.9 |

**NOTE: EMV 3DS specification versions are detailed by major.minor.patch.editorial (2.3.1.1). Therefore, all patch and editorial versions beyond 2.3.1.1 should be considered the same as 2.3 for the purposes of this table.**

## Subfield 2 (Directory Server Transaction ID)

[DE 48](#), subelement 66 (Authentication Data), subfield 2 allows the acquirer to pass the Directory Server Transaction ID during authorization in order to link authentication and authorization data. The Directory Server Transaction ID is generated by the EMV 3DS Mastercard Directory Server during the authentication transaction and passed back to the merchant with the authentication results.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 02                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans-36                 |
| Data Field           | Contents of subfield 2 |
| Justification        | N/A                    |

### Values

The Directory Server Transaction ID is a Universally Unique Transaction ID which can be provided by the processors/acquirers as part of the authentication transaction.

Example of a Directory Server Transaction ID: f38e6948-5388-41a6-bca4-b49723c19437

### Application notes

This is a conditional field and must be provided by the acquirer if they are participating and have previously authenticated using the Mastercard EMV 3DS (3DS2) Identity Check program. This field will not be edited to ensure these elements were provided on the transaction. If provided by the acquirer or processor, it must meet the specifications defined.

The Directory Server Transaction ID must not be provided as an all caps value.

## Subelement 67 (Mastercard Send Information)

DE 48, subelement 67 (Mastercard Send Information) contains subfield representing the Sanctions Score.

### Attributes

| Attribute              | Description              |
|------------------------|--------------------------|
| Subelement ID          | 67                       |
| Subelement Data Length | 2 positions              |
| Data Representation    | ans...99; LLVAR          |
| Data Field             | Contents of subfield 1-2 |
| Subfields              | 2                        |
| Justification          | See subfields            |

### Usage

Following is the usage of subelement 67 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | .   | X   | C   |
| Authorization Request Response/0110         | .   | X   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |

### Values

See subfields.

## Subfield 1 (Sanctions Score)

[DE 48](#), subelement 67, subfield 1 (Sanctions Score) contains the Sanctions Score populated by Mastercard.

### Attributes

| Attribute            | Description                        |
|----------------------|------------------------------------|
| Subfield ID          | 01                                 |
| Subfield Data Length | 2 positions                        |
| Data Representation  | n-3                                |
| Data Field           | Contents of subfield 1             |
| Justification        | Right-justified with leading zeros |

### Values

For issuers who opt-in to the Sanctions Scoring service, a Sanctions Score value will be populated by Mastercard on all cross-border Mastercard Send Payment Transactions globally and domestic Mastercard Send Payment Transactions in Egypt, Canada, across Europe and the United States.

The Sanctions Scoring service automatically checks the Sender name (consumer, business, government, and non-government) against select sanctions watchlists such as the OFAC Specially Designated Nationals and Blocked Persons List, UN List, and EU List.

Based on the likelihood of a match of the sender's name to a sanctioned individual or entity, a score of 000 or a score from 070 to 099 is generated and included in DE 48 (Additional Data: Private Use), subelement 67 (Mastercard Send Information), subfield 1 (Sanctions Score) of the authorization request message delivered to the issuer.

A higher score indicates a closer match to names on the applicable screening lists, while lower scores indicate a less likely match. When a score cannot be determined, the value will be 999. A score of 000 indicates that the transaction passes the Sanctions Scoring check.

**NOTE: If the Sanctions Score is 100 or if the Sanctions Score produced by the Sanctions Scoring service is greater than or equal to the Blocking Score set up by the issuer in the opt-in service, then the authorization request will be declined and an advice will be sent to the issuer with the value of 33F (Mastercard Send Issuer Blocking—Sanctions Score limit exceeded) in [DE 48](#), subelement 71 (On-behalf [OB] Services), subfield 1 (On-behalf [OB] Service) and subfield 2 (On-behalf [OB] Result 1).**

## Subfield 2 (Sanctions Score Additional Information)

DE 48, subelement 67, subfield 2 (Sanctions Score Additional Information) allows for communication of additional information regarding the Sanctions Scoring Service.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 02                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...58; LLVAR        |
| Data Field           | Contents of subfield 2 |
| Justification        | N/A                    |

### Values

Subfield 2 contains a list of watchlist(s) associated with the sanctions score generated by Mastercard, which are separated with a " | " delimiter.

## Subelement 68 (Financial Account Information)

DE 48, subelement 68 (Financial Account Information) contains the financial account information associated with the Mastercard Pay by Account token.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 68                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | ans...68; LLVAR        |
| Data Field             | Contents of subfield 1 |
| Subfields              | 1                      |

### Usage

Following is the usage of subelement 68 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                                                                                           | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                                        | •   | X   | C   |
| Authorization Request/0100: account status inquiry (merchant-initiated message and MDES pre-digitization message) | •   | X   | C   |
| Authorization Request/0100: tokenization eligibility                                                              | •   | X   | C   |
| Authorization Request/0100: tokenization authorization                                                            | •   | X   | C   |
| Authorization Request/0100: activation code notification                                                          | •   | X   | C   |
| Authorization Request/0100: tokenization complete notification                                                    | •   | X   | C   |
| Authorization Request/0100: tokenization event notification                                                       | •   | X   | C   |
| Authorization Advice/0120: system-generated                                                                       | •   | X   | C   |
| Authorization Advice Response/0130: system-generated                                                              | •   | X   | C   |
| Reversal Request/0400                                                                                             | •   | X   | C   |
| Reversal Advice/0420                                                                                              | •   | X   | C   |
| Administrative Advice/0620: issuer token notification advice for activation code notification                     | •   | X   | C   |
| Administrative Advice/0620: issuer token notification advice for tokenization complete notification               | •   | X   | C   |
| Administrative Advice/0620: issuer token notification advice for tokenization event notification                  | •   | X   | C   |

### Values

See subfields.

### Application notes

The [DE 48](#), subelement 68 information is added to the message by Mastercard and forwarded to the Account Holding Institution (AHI). Acquiring Institutions will not receive this information.

## Subfield 1 (Financial Account Identifier)

[DE 48](#), subelement 68, subfield 1 (Financial Account Identifier) contains the bank account information or other financial account information associated with the Mastercard Pay by Account token at the time the transaction occurred.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...64; LLVAR        |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

### Values

Financial account identifier must be a minimum of 9 characters and a maximum of 64 characters.

## Subelement 71 (On-behalf Services)

[DE 48](#), subelement 71 (On-behalf Services [OBS]) notifies the issuer of the On-behalf Service performed on the transaction and the results. Subelement 71 will support up to ten services for a transaction.

### Attributes

| Attribute              | Description                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------|
| Subelement ID          | 71                                                                                                     |
| Subelement Data Length | 2 positions                                                                                            |
| Data Representation    | ans...40; LLVAR<br>The "LL" length field of LLVAR must be an integral multiple of 4, not to exceed 40. |
| Data Field             | Contents of subfields 1–3                                                                              |
| Subfields              | 3                                                                                                      |
| Justification          | N/A                                                                                                    |



## Usage

Following is the usage of subelement 71 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | •   | X   | C   |
| Authorization Request Response/0110           | •   | X   | C   |
| Authorization Advice/0120: acquirer-generated | •   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | •   | X   | C   |
| Reversal Advice/0420                          | •   | X   | C   |

## Values

See subfields.

## Subfield 1 (On-behalf [OB] Service)

[DE 48](#), subelement 71, subfield 1 (On-behalf [OB] Service) indicates the service performed on the transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Field          | Contents of positions 1–2 |
| Justification       | N/A                       |

## Values

Refer to [Subelement 71 \(Valid Subfield 1 and Subfield 2 Value Combinations\)](#) and [Subelement 71 \(Valid Subfield 1 and Subfield 3 Value Combinations\)](#).

## Subfield 2 (On-behalf [OB] Result 1)

[DE 48](#), subelement 71, subfield 2 (On-behalf [OB] Result 1) indicates the results of the service processing.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

### Values

Refer to [Subelement 71 \(Valid Subfield 1 and Subfield 2 Value Combinations\)](#).

## Subfield 3 (On-behalf Result 2)

[DE 48](#), subelement 71, subfield 3 (On-behalf Result 2) identifies the results of the service processing.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 4 |
| Justification       | N/A                    |

### Values

Mastercard use only. May contain a space or a value.

Refer to [Subelement 71 \(Valid Subfield 1 and Subfield 3 Value Combinations\)](#).

## Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

[DE 48 \(Additional Data: Private Use\)](#) subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations) are the valid [DE 48](#), subelement 71 (On-behalf Services [OBS]), subfield 1 (On-behalf [OB] Service) and subfield 2 (OB Result 1) value combinations. The contents of subfield 2 depend on the contents of subfield 1 as described here.

## Attributes

| Attribute           | Subfield 1 (OB Service)   | Subfield 2 (OB Result 1) |
|---------------------|---------------------------|--------------------------|
| Data Representation | an-2                      | an-1                     |
| Data Field          | Contents of positions 1–2 | Contents of position 3   |
| Justification       | Left-justified            | N/A                      |

**NOTE: DE 48 (Additional Data: Private Use), subelement 71 will contain all applicable, on-behalf service results that were performed on a given transaction; examples include Mastercard Digital Enablement Service (MDES) service results and Mastercard Identity Check service results.**

## Values

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                                                                                                                 |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 Chip to Magnetic Stripe Conversion Service               | C Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 05 [PAN auto-entry via chip] or 79 [Chip card/PAN entry via manual entry] |
|                                                             | M Conversion of the M/Chip fallback transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 80-PAN auto-entry with magnetic stripe                           |
|                                                             | S Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 07-PAN auto-entry via contactless M/Chip                                  |
| 02 M/Chip Cryptogram Pre-validation Service                 | A Valid Application Cryptogram (AC); ATC outside allowed range                                                                                                                                               |
|                                                             | E Valid Application Cryptogram; ATC Replay                                                                                                                                                                   |
|                                                             | F Format Error                                                                                                                                                                                               |
|                                                             | G Application Cryptogram is valid but not an ARQC nor a TC, status of TVR/CVR unknown                                                                                                                        |
|                                                             | I Invalid Cryptogram                                                                                                                                                                                         |
|                                                             | K No matching key file for this PAN, PAN expiry date and KDI combination                                                                                                                                     |
|                                                             | T Valid ARQC/TC and ATC; TVR/CVR invalid                                                                                                                                                                     |

Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) |                                                                                                                                                                                                                                                                           | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1) |                                                                                                                                                                |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03                                                          | M/Chip Cryptogram Validation in Stand-In Processing                                                                                                                                                                                                                       | U                                                            | Unable to process                                                                                                                                              |
|                                                             |                                                                                                                                                                                                                                                                           | V                                                            | Valid                                                                                                                                                          |
|                                                             |                                                                                                                                                                                                                                                                           | X                                                            | Security platform time out                                                                                                                                     |
|                                                             |                                                                                                                                                                                                                                                                           | Z                                                            | Security platform processing error                                                                                                                             |
|                                                             |                                                                                                                                                                                                                                                                           | A                                                            | Valid Application Cryptogram (AC); ATC outside allowed range                                                                                                   |
|                                                             |                                                                                                                                                                                                                                                                           | E                                                            | Valid Application Cryptogram; ATC Replay                                                                                                                       |
|                                                             |                                                                                                                                                                                                                                                                           | F                                                            | Format Error                                                                                                                                                   |
|                                                             |                                                                                                                                                                                                                                                                           | G                                                            | Application Cryptogram is valid but not an ARQC nor a TC, status of TVR/CVR unknown                                                                            |
|                                                             |                                                                                                                                                                                                                                                                           | I                                                            | Invalid Cryptogram                                                                                                                                             |
|                                                             |                                                                                                                                                                                                                                                                           | K                                                            | No matching key file for this PAN, PAN expiry date and KDI combination                                                                                         |
| 04                                                          | M/Chip Cryptogram Regeneration Service                                                                                                                                                                                                                                    | T                                                            | Valid ARQC/TC and ATC; TVR/CVR invalid                                                                                                                         |
|                                                             |                                                                                                                                                                                                                                                                           | U                                                            | Unable to process                                                                                                                                              |
|                                                             |                                                                                                                                                                                                                                                                           | V                                                            | Valid                                                                                                                                                          |
|                                                             |                                                                                                                                                                                                                                                                           | X                                                            | Security platform time out                                                                                                                                     |
|                                                             |                                                                                                                                                                                                                                                                           | Z                                                            | Security platform processing error                                                                                                                             |
|                                                             |                                                                                                                                                                                                                                                                           | C                                                            | Regeneration of the M/Chip cryptogram was completed                                                                                                            |
| 05                                                          | Mastercard® Identity Check™ AAV Verification Service<br><br>Refer to <a href="#">Subelement 71 (OBS 05 Result Codes)</a> for detailed information on OBS 05 result codes as they relate to SPA2 AAV, DS Transaction ID, and Amount Check (Authorization: Authentication). | A                                                            | DS Txn ID not provided; AAV, amount and currency verified                                                                                                      |
|                                                             |                                                                                                                                                                                                                                                                           | B                                                            | DS Txn ID not provided; AAV and currency verified; amount 0-20% different                                                                                      |
|                                                             |                                                                                                                                                                                                                                                                           | C                                                            | DS Txn ID not provided; AAV and currency verified; amount greater than 20% different                                                                           |
|                                                             |                                                                                                                                                                                                                                                                           | D                                                            | DS Transaction ID does not match with PAN and SPA2 AAV within the authentication record                                                                        |
|                                                             |                                                                                                                                                                                                                                                                           | E                                                            | Format Error; AAV is not correctly formatted or failed authentication                                                                                          |
|                                                             |                                                                                                                                                                                                                                                                           | F                                                            | Failed MAC key validation; Dynamic Linking validation not provided.                                                                                            |
|                                                             |                                                                                                                                                                                                                                                                           | I                                                            | Invalid AAV; Dynamic Linking was not able to find a matching authentication due to expired time, previously used or not authenticated through Directory Server |

Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)            | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                                                                      |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                        | K No matching key file for this PAN. SPA2 AAV provided is not generated by Mastercard Identity Check (i.e. Leading Indicator & IAV only).                         |
|                                                                        | M Monetary Currency mismatch between authorization and authentication                                                                                             |
|                                                                        | P Passed MAC key Validation; Dynamic Linking validation not provided.                                                                                             |
|                                                                        | S DS Txn ID present; AAV and currency verified; amount 0-20% different.                                                                                           |
|                                                                        | T DS Txn ID present; AAV and currency verified; amount greater than 20% different.                                                                                |
|                                                                        | U Unable to process                                                                                                                                               |
|                                                                        | V Valid                                                                                                                                                           |
|                                                                        | X Security platform time out or Authentication Challenge (Results Request) not received yet; Retry authorization if declined for failed AAV Validation.           |
|                                                                        | Z Security platform processing error; Authentication Response not received yet; Retry authorization if declined for failed AAV Validation.                        |
| 06 Mastercard® Identity Check™ AAV Verification in Stand-In Processing | A DS Txn ID not provided; AAV, amount and currency verified.                                                                                                      |
|                                                                        | B DS Txn ID not provided; AAV and currency verified; amount 0-20% different.                                                                                      |
|                                                                        | C DS Txn ID not provided; AAV and currency verified; amount greater than 20% different.                                                                           |
|                                                                        | D DS Transaction ID does not match with PAN and SPA2 AAV within the authentication record                                                                         |
|                                                                        | E Format Error; AAV is not correctly formatted or failed authentication.                                                                                          |
|                                                                        | F Failed MAC key Validation; Dynamic Linking validation not provided.                                                                                             |
|                                                                        | I Invalid AAV; Dynamic Linking was not able to find a matching authentication due to expired time, previously used or not authenticated through Directory Server. |
|                                                                        | K No matching key file for this PAN. SPA2 AAV provided is not generated by Mastercard Identity Check (i.e. Leading Indicator & IAV only).                         |

Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                                                            |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | M Monetary Currency mismatch between authorization and authentication                                                                                   |
|                                                             | P Passed MAC key Validation; Dynamic Linking validation not provided                                                                                    |
|                                                             | S DS Txn ID present; AAV and currency verified; amount 0-20% different.                                                                                 |
|                                                             | T DS Txn ID present; AAV and currency verified; amount greater than 20% different.                                                                      |
|                                                             | U Unable to process                                                                                                                                     |
|                                                             | V Valid                                                                                                                                                 |
|                                                             | X Security platform time out or Authentication Challenge (Results Request) not received yet; Retry authorization if declined for failed AAV Validation. |
|                                                             | Z Security platform processing error or Authentication Response not received yet; Retry authorization if declined for failed AAV Validation.            |
| 08 Online PIN Pre-validation (Europe Only)                  | I Invalid PIN                                                                                                                                           |
|                                                             | P Mandatory PVV not on file                                                                                                                             |
|                                                             | R PIN retry exceeded (invalid PIN)                                                                                                                      |
|                                                             | U Unable to process                                                                                                                                     |
|                                                             | V Valid                                                                                                                                                 |
| 09 Online PIN Validation in Stand-In (Europe only)          | I Invalid PIN                                                                                                                                           |
|                                                             | P Mandatory PVV not on file                                                                                                                             |
|                                                             | R PIN retry exceeded (invalid PIN)                                                                                                                      |
|                                                             | U Unable to process                                                                                                                                     |
|                                                             | V Valid                                                                                                                                                 |
| 10 CVC 1 Validation Stand-In Service                        | I Invalid CVC 1                                                                                                                                         |
|                                                             | U Unable to process                                                                                                                                     |
|                                                             | V Valid                                                                                                                                                 |

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) |                                                                            | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11                                                          | CVC 1 Pre-Validation Service                                               | I                                                                                                                                                                                                                                                | Invalid CVC 1                                                                                                                                                                                                                                                                          |
|                                                             |                                                                            | K                                                                                                                                                                                                                                                | No matching key file for this PAN, PAN Expiry date combination                                                                                                                                                                                                                         |
|                                                             |                                                                            | U                                                                                                                                                                                                                                                | Unable to process                                                                                                                                                                                                                                                                      |
|                                                             |                                                                            | V                                                                                                                                                                                                                                                | Valid                                                                                                                                                                                                                                                                                  |
|                                                             |                                                                            | <b>NOTE: Issuers that enroll account ranges in the OBS 11 (CVC 1 Pre-Validation Service) that are currently enrolled in OBS 10 (CVC 1 Validation Stand-In Service) must request the removal of those account ranges from the OBS 10 service.</b> |                                                                                                                                                                                                                                                                                        |
| 14                                                          | Contactless Mapping Service                                                | C                                                                                                                                                                                                                                                | Conversion of contactless account number to PAN was completed                                                                                                                                                                                                                          |
|                                                             |                                                                            | I                                                                                                                                                                                                                                                | Invalid                                                                                                                                                                                                                                                                                |
|                                                             |                                                                            | U                                                                                                                                                                                                                                                | Unable to process                                                                                                                                                                                                                                                                      |
| 15                                                          | Dynamic CVC 3 Pre-validation (with or without Contactless Mapping Service) | A                                                                                                                                                                                                                                                | ATC outside allowed range (applicable when ATC value is dynamic [varying] value)                                                                                                                                                                                                       |
|                                                             |                                                                            | E                                                                                                                                                                                                                                                | CVC 3 ATC Replay                                                                                                                                                                                                                                                                       |
|                                                             |                                                                            | I                                                                                                                                                                                                                                                | Invalid CVC 3                                                                                                                                                                                                                                                                          |
|                                                             |                                                                            | K                                                                                                                                                                                                                                                | No matching key file for this PAN, PAN expiry date, and KDI combination                                                                                                                                                                                                                |
|                                                             |                                                                            | N                                                                                                                                                                                                                                                | Unpredictable Number Mismatch (applicable when the UN is dynamic [varying] value)<br><br>(Indicates that the number/length in the discretionary data in <a href="#">DE 45</a> or <a href="#">DE 35</a> does not match the number/length provided by the issuer during personalization) |
|                                                             |                                                                            | U                                                                                                                                                                                                                                                | Unable to process                                                                                                                                                                                                                                                                      |
|                                                             |                                                                            | V                                                                                                                                                                                                                                                | Valid                                                                                                                                                                                                                                                                                  |
|                                                             |                                                                            | X                                                                                                                                                                                                                                                | Security platform time out                                                                                                                                                                                                                                                             |
| 16                                                          | Dynamic CVC 3 Validation in Stand-In Processing                            | Z                                                                                                                                                                                                                                                | Security platform system error                                                                                                                                                                                                                                                         |
|                                                             |                                                                            | A                                                                                                                                                                                                                                                | ATC outside allowed range (applicable when ATC value is dynamic [varying] value)                                                                                                                                                                                                       |
|                                                             |                                                                            | E                                                                                                                                                                                                                                                | CVC 3 ATC Replay                                                                                                                                                                                                                                                                       |
|                                                             |                                                                            | I                                                                                                                                                                                                                                                | Invalid CVC 3                                                                                                                                                                                                                                                                          |
|                                                             |                                                                            | K                                                                                                                                                                                                                                                | No matching key file for this PAN, PAN expiry date, and KDI combination                                                                                                                                                                                                                |

Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | N Unpredictable Number Mismatch (applicable when the UN is dynamic [varying] value)<br><br>(Indicates that the number/length in the discretionary data in <a href="#">DE 45</a> or <a href="#">DE 35</a> does not match the number/length provided by the issuer during personalization)                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                             | U Unable to process                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                             | V Valid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                             | X Security platform time out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                             | Z Security platform system error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 17 In Control Virtual Card Service                          | A Virtual Card Number (expiration date does not match)<br>B Virtual Card Number (expiration date expired)<br>C Virtual Card Number Virtual CVC 2 does not match<br>D In Control Validity Period Limit<br>E In Control Transaction Amount Limit Check<br>F In Control Cumulative Amount Limit Check<br>G In Control Transaction Number Usage<br>H In Control Merchant ID Limit<br>I In Control Invalid Virtual Card Number—Real Card Number mapping relationship<br>J In Control MCC Limit<br>K In Control Database Status Bad<br>L In Control Geographic Restriction<br>M In Control Transaction Type Restriction<br>P In Control Transaction Time/Date Restriction<br>U Unable to process<br>V Valid |
| 18 Fraud Scoring Service                                    | C Fraud Scoring Service was performed successfully                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                             | U Fraud Scoring Service was not performed successfully                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 20 In Control RCN Spend Control Service                     | D In Control Validity Period Limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                             | E In Control Transaction Amount Limit Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                             | F In Control Cumulative Amount Limit Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)                 | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                                                    |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                             | G In Control Transaction Number Usage                                                                                                           |
|                                                                             | H In Control Merchant ID Limit                                                                                                                  |
|                                                                             | J In Control MCC Limit                                                                                                                          |
|                                                                             | K In Control Database Status Bad                                                                                                                |
|                                                                             | L In Control Geographic Restriction                                                                                                             |
|                                                                             | M In Control Transaction Type Restriction                                                                                                       |
|                                                                             | P In Control Transaction Time/Date Restriction                                                                                                  |
|                                                                             | U Unable to process                                                                                                                             |
|                                                                             | V Valid                                                                                                                                         |
| 25 Account Data Compromise Information                                      | Y Compromised Event Data Found                                                                                                                  |
|                                                                             | N Compromised Event Data Not Found                                                                                                              |
|                                                                             | U Unable to process                                                                                                                             |
| 26 Point-of-Interaction Service                                             | C Enrichment completed for Installments                                                                                                         |
|                                                                             | F PAN does not qualify for installments                                                                                                         |
|                                                                             | U Unable to Process                                                                                                                             |
| 31 Chip CVC to CVC 1 Conversion Service—<br>CVC 1 Key/Decision Matrix Only  | C Chip CVC Validated Successfully; Conversion of Chip CVC to CVC 1 Performed Successfully                                                       |
|                                                                             | F Track Data Formatted Incorrectly                                                                                                              |
|                                                                             | I Chip CVC Invalid; Conversion of Chip CVC to CVC 1 Not Performed                                                                               |
|                                                                             | K Issuer CVC 1 Key Record not Found for Account Range / Expiry Date combination; Service Not Performed                                          |
|                                                                             | U Unable to process                                                                                                                             |
| 32 Chip CVC to CVC 1 Conversion Service—<br>Separate Keys/Decision Matrices | C Chip CVC Validated Successfully; Conversion of Chip CVC to CVC 1 Performed Successfully                                                       |
|                                                                             | F Track Data Formatted Incorrectly                                                                                                              |
|                                                                             | I Chip CVC Invalid; Conversion of Chip CVC to CVC 1 Not Performed                                                                               |
|                                                                             | K Issuer Chip CVC Key Record not found and Issuer CVC 1 Key Record not found for Account Range / Expiry Date combination; Service Not Performed |
|                                                                             | L Issuer CVC 1 Key Record not Found for Account Range / Expiry Date combination; Service Not Performed                                          |

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M Issuer Chip CVC Key Record not Found for Account Range / Expiry Date combination; Service Not Performed         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | U Unable to process                                                                                               |
| 33 Send Blocking Service<br>OBS 33 result code values A, B, D, E, F, I, J, K, L, M, N, O, P, Q, R, S, U, V, W, X, Y apply to Send transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payment Transactions). OBS 33 result code value G apply to Mastercard Send Payment Transactions and Gaming and Gambling Payment Transactions. OBS 33 result code value T applies to Funding Transactions. OBS 33 result code value H applies to other transactions, such as coded with legacy TTI or miscoded. | A Send issuer blocking: Transaction limit not allowed for the Transaction Type Identifier (TTI)                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | B Send issuer blocking: Merchant not allowed for the Transaction Type Identifier (TTI)                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | D Send issuer blocking: Country not allowed for the Transaction Type Identifier (TTI)                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | E Send issuer blocking: Domestic activity only allowed for the Transaction Type Identifier (TTI)                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | F Send issuer blocking: Sanctions Score limit exceeded for the Transaction Type Identifier (TTI)                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | G Send Mastercard blocking: Transaction limit not allowed for the Transaction Type Identifier (TTI) and Country   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | H Mastercard blocking: General miscoding                                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | I Send Mastercard blocking: Reserved for future use                                                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | J Send Mastercard blocking: Transaction Count exceeded for the Transaction Type Identifier (TTI)                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | K Send Mastercard blocking: Aggregate transaction amount limit exceeded for the Transaction Type Identifier (TTI) |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | L Send issuer blocking: Transaction Count exceeded for the Transaction Type Identifier (TTI)                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M Send issuer blocking: Aggregate transaction amount limit exceeded for the Transaction Type Identifier (TTI)     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | N Send issuer monitoring: Transaction Count exceeded for the Transaction Type Identifier (TTI)                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | O Send issuer monitoring: Aggregate transaction amount limit exceeded for the Transaction Type Identifier (TTI)   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | P Send issuer monitoring: Transaction amount limit exceeded for the Transaction Type Identifier (TTI)             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Q Send issuer monitoring: Sanctions Score exceeded                                                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R Send issuer blocking: Invalid Card                                                                              |

Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)                 | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                       |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                                                             | S Send Mastercard blocking: Product code invalid for the Transaction Type Identifier (TTI) or the use case         |
|                                                                             | T Send Mastercard blocking: Transaction Amount limit not allowed for the Funding Transaction Type Identifier (TTI) |
|                                                                             | U Unable to process                                                                                                |
|                                                                             | V Valid                                                                                                            |
|                                                                             | W Send Mastercard blocking: Geographical restriction                                                               |
|                                                                             | X Send Mastercard blocking: Reserved for future use                                                                |
|                                                                             | Y Send Mastercard blocking: Data quality controls                                                                  |
| 37 Mastercard Merchant Presented QR Blocking Service                        | D Mastercard Merchant Presented QR Blocking-Transaction Amount Limit Exceeded                                      |
|                                                                             | E Mastercard Merchant Presented QR Blocking-Cumulative Transaction Amount Or Count Limit Exceeded                  |
|                                                                             | F Mastercard Merchant Presented QR Blocking-Domestic Activity Only                                                 |
|                                                                             | U Unable to Process                                                                                                |
|                                                                             | V Valid                                                                                                            |
| 50 MDES PAN Mapping service                                                 | C Conversion of Token to PAN completed successfully                                                                |
|                                                                             | F Format Error                                                                                                     |
|                                                                             | I Invalid Token                                                                                                    |
|                                                                             | U Unable to process                                                                                                |
| 51 MDES Pre-Validation Service for Secure Element tokens (ICC or DSRP data) | A ATC outside allowed range (applicable when ATC value is dynamic [varying] value)                                 |
|                                                                             | E ATC Replay                                                                                                       |
|                                                                             | F Format Error                                                                                                     |
|                                                                             | G Application Cryptogram is valid but not an ARQC nor a TC                                                         |
|                                                                             | I Invalid Cryptogram                                                                                               |
|                                                                             | K No matching key file for this PAN, PAN expiry date and KDI combination                                           |
|                                                                             | T Valid cryptogram and ATC; TVR/CVR invalid. Cryptogram type (ARQC/AAC/TC) not checked                             |
|                                                                             | U Unable to process                                                                                                |

Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)                                                                                                                                                                                                          | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 52 MDES Pre-Validation Service for Secure Element tokens (CVC3 data)                                                                                                                                                                                                 | V Valid ARQC/TC and ATC and TVR/CVR                                                                                          |
|                                                                                                                                                                                                                                                                      | X Security platform time out                                                                                                 |
|                                                                                                                                                                                                                                                                      | Z Security platform system error                                                                                             |
|                                                                                                                                                                                                                                                                      | A ATC outside allowed range (applicable when ATC value is dynamic [varying] value)                                           |
|                                                                                                                                                                                                                                                                      | E CVC 3 ATC Replay                                                                                                           |
|                                                                                                                                                                                                                                                                      | H Invalid Time Validation                                                                                                    |
|                                                                                                                                                                                                                                                                      | I Invalid CVC 3                                                                                                              |
|                                                                                                                                                                                                                                                                      | K No matching key file for this PAN, PAN expiry date, and KDI combination                                                    |
|                                                                                                                                                                                                                                                                      | N Unpredictable Number Length Indicator Mismatch                                                                             |
|                                                                                                                                                                                                                                                                      | U Unable to process                                                                                                          |
| 54 Mastercard Digital Enablement Service Digital Payment Data Validation Service<br>Refer to <a href="#">Subelement 71 (Valid Subfield 1 and Subfield 3 Value Combinations)</a> for more information on <a href="#">DE 48</a> , subelement 71, subfield 1, value 54. | V Valid                                                                                                                      |
|                                                                                                                                                                                                                                                                      | X Security platform time out                                                                                                 |
|                                                                                                                                                                                                                                                                      | Z Security platform system error                                                                                             |
|                                                                                                                                                                                                                                                                      | A If the transaction amount in <a href="#">DE 4</a> is less than or equal to the approximate amount in the cryptogram        |
|                                                                                                                                                                                                                                                                      | B If the transaction amount in <a href="#">DE 4</a> is greater than the approximate amount in the cryptogram by 0% to 19.99% |
| 55 Merchant Validation Service                                                                                                                                                                                                                                       | C If the transaction amount in <a href="#">DE 4</a> is greater than the approximate amount in the cryptogram by 20% or more  |
|                                                                                                                                                                                                                                                                      | D If the transaction amount in <a href="#">DE 4</a> is not zero and the approximate amount in the cryptogram is zero         |
|                                                                                                                                                                                                                                                                      | M The submitted merchant data is a match to Mastercard's merchant data                                                       |
|                                                                                                                                                                                                                                                                      | N The submitted merchant data is not a match to Mastercard's merchant data                                                   |
| 61 MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)                                                                                                                                                                                 | D ATC Invalid-Not in list of currently active Single-Use Keys                                                                |
|                                                                                                                                                                                                                                                                      | E ATC Replay                                                                                                                 |
|                                                                                                                                                                                                                                                                      | F Format Error                                                                                                               |
|                                                                                                                                                                                                                                                                      | I Invalid MD AC and UMD AC                                                                                                   |
|                                                                                                                                                                                                                                                                      | K No matching key file for this PAN, PAN expiry date and KDI combination                                                     |

Data element definitions and details  
Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) |                                                                                            | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1) |                                                                                         |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|
|                                                             |                                                                                            | L                                                            | Invalid MD AC; Valid UMD AC                                                             |
|                                                             |                                                                                            | M                                                            | Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended) |
|                                                             |                                                                                            | P                                                            | Valid MD AC; Invalid UMD AC (Invalid Mobile PIN)                                        |
|                                                             |                                                                                            | T                                                            | Invalid TVR/CVR                                                                         |
|                                                             |                                                                                            | U                                                            | Unable to process                                                                       |
|                                                             |                                                                                            | V                                                            | Valid                                                                                   |
|                                                             |                                                                                            | X                                                            | Security platform time out                                                              |
|                                                             |                                                                                            | Z                                                            | Security platform system error                                                          |
| 62                                                          | MDES Pre-Validation Service for Cloud tokens (CVC3 data)                                   | D                                                            | ATC Invalid-Not in list of currently active Single-Use Keys                             |
|                                                             |                                                                                            | E                                                            | ATC Replay                                                                              |
|                                                             |                                                                                            | F                                                            | Format Error                                                                            |
|                                                             |                                                                                            | H                                                            | Invalid Time Validation                                                                 |
|                                                             |                                                                                            | I                                                            | Invalid MD AC and UMD AC                                                                |
|                                                             |                                                                                            | K                                                            | No matching key file for this PAN, PAN expiry date and KDI combination                  |
|                                                             |                                                                                            | L                                                            | Invalid MD AC; Valid UMD AC                                                             |
|                                                             |                                                                                            | M                                                            | Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended) |
|                                                             |                                                                                            | N                                                            | Unpredictable Number Length Indicator Mismatch                                          |
|                                                             |                                                                                            | P                                                            | Valid MD AC; Invalid UMD AC (Invalid Mobile PIN)                                        |
|                                                             |                                                                                            | U                                                            | Unable to process                                                                       |
|                                                             |                                                                                            | V                                                            | Valid                                                                                   |
| 71                                                          | Token Service Provider Cloud Based Payments Chip Validation Service in Stand-In Processing | X                                                            | Security platform time out                                                              |
|                                                             |                                                                                            | Z                                                            | Security platform system error                                                          |
|                                                             |                                                                                            | E                                                            | ATC Replay                                                                              |
|                                                             |                                                                                            | F                                                            | Format Error                                                                            |
|                                                             |                                                                                            | I                                                            | Invalid MD AC and UMD AC                                                                |
|                                                             |                                                                                            | K                                                            | No matching key file for this PAN, PAN expiry date and KDI combination                  |
|                                                             |                                                                                            | L                                                            | Invalid MD AC; Valid UMD AC                                                             |
|                                                             |                                                                                            | P                                                            | Valid MD AC; Invalid UMD AC (Invalid Mobile PIN)                                        |

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) | DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1) |
|-------------------------------------------------------------|--------------------------------------------------------------|
|                                                             | T Invalid TVR/CVR                                            |
|                                                             | U Unable to process                                          |
|                                                             | V Valid                                                      |
|                                                             | X Security platform time out                                 |
|                                                             | Z Security platform system error                             |

### Subelement 71 (Valid Subfield 1 and Subfield 3 Value Combinations)

DE 48, subelement 71 (Valid Subfield 1 and Subfield 3 Value Combinations) are the valid DE 48, subelement 71 (On-behalf Services [OBS]), subfield 1 (On-behalf [OB] Service) and subfield 3 (OB Result 2) value combinations. The contents of subfield 3 depend on the contents of subfield 1 as described here.

#### Attributes

| Attribute           | Subfield 1 (OB Service)   | Subfield 3 (OB Result 2) |
|---------------------|---------------------------|--------------------------|
| Data Representation | an-2                      | ans-1                    |
| Data Field          | Contents of positions 1–2 | Contents of position 4   |
| Justification       | Left-justified            | N/A                      |

Values

Values

| DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2) |                                                                               | DE 48, Subelement 71, Subfield 3 (OB Result 2-RCA Identifier Validation) Values (ans-1) |                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 54 =                                                        | Mastercard Digital Enablement Service Digital Payment Data Validation Service | M =                                                                                     | In Europe: Transaction authenticated with Mastercard qualified multifactor authentication method as Strong Customer Authentication, and valid merchant dynamic linking<br><br>Outside of Europe: Transaction authenticated with Mastercard qualified multifactor authentication method, and valid merchant dynamic linking |
|                                                             |                                                                               | N =                                                                                     | In Europe: Merchant dynamic linking was not performed or was invalid<br><br>Outside of Europe: Merchant dynamic linking was not performed or was invalid                                                                                                                                                                   |

Subelement 71 (Valid Subfield 1, Subfield 2, and Subfield 3 value combinations)

DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1, Subfield 2, and Subfield 3 value combinations) are the valid DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Service [OBS]), subfield 1 (On-behalf [OB] Service Indicator), subfield 2 (OB Result 1), subfield 3 (On-behalf Result 2) value combinations.

### Value combinations

| Subfield 1: On-behalf (OB) Service | Subfield 2: On-behalf Result 1                                                                                          | Subfield 3: On-behalf Result 2 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 50 = MDES PAN Mapping service      | C = Conversion of Token to PAN completed successfully<br>F = Format Error<br>I = Invalid Token<br>U = Unable to process | SPACE                          |

### Value combinations

| Transaction Description                                                    | Subfield 1: On-behalf (OB) Service                                                                                                                                        | Subfield 2: On-behalf Result 1                    | Subfield 3: On-behalf Result 2    |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------|
| Tokenized transactions that carry e-commerce Dynamic Token Validation Code | 61 - MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)                                                                                    | V = Valid<br>I = Invalid<br>U = Unable to Process | T = Dynamic Token Validation Code |
| Tokenized transactions that carry e-commerce DSRP cryptogram               | 61 - MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)                                                                                    | V = Valid<br>I = Invalid<br>U = Unable to Process | SPACE = DSRP Cryptogram           |
| Tokenized contactless transactions when RRP is not performed               | 51: MDES Pre-Validation Service for Secure Element tokens (ICC or DSRP data)<br><br>61: MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data) | V = Valid                                         | SPACE = RRP not performed         |



| Transaction Description                                  | Subfield 1: On-behalf (OB) Service                                                                                                                                        | Subfield 2: On-behalf Result 1 | Subfield 3: On-behalf Result 2                                                                |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------|
| Tokenized contactless transactions when RRP is performed | 51: MDES Pre-Validation Service for Secure Element tokens (ICC or DSRP data)<br><br>61: MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data) | V = Valid                      | A = RRP performed within the established limits<br><br>B = RRP performed, time limit exceeded |

## Application notes

Issuers participating in MDES:

- Receive OBS 50 (MDES PAN Mapping service and corresponding result codes in [DE 48 \(Additional Data: Private Use\)](#), subelement 71 indicating the status of PAN mapping for both static and cloud tokenized transactions.
- Receive OBS 61 (MDES Pre-Validation Service for Cloud and Static tokens [ICC, DSRP, or DTVC data] in [DE 48 \(Additional Data: Private Use\)](#), subelement 71, subfield 1 (On-behalf [OB] Service) to indicate that MDES attempted to validate dynamic data submitted by acquirer for both static and cloud tokenized transactions and a valid value in [DE 48 \(Additional Data: Private Use\)](#), subelement 71, subfield 2 (On-behalf Result 1) to indicate the validation outcome of both dynamic data in static and cloud token transactions.
- Receive a new result code T in [DE 48 \(Additional Data: Private Use\)](#), subelement 71, subfield 3 (On-behalf Result 2) to indicate that the acquirer submitted dynamic token verification code (DTVC) and the validation outcome in On-Behalf Result 1 pertains to DTVC.
- Prepare to process static or cloud token transactions without a CVC 2, DTVC validation results, or both, in the authorization request, as in some cases this information might not be provided to the issuer. However, when CVC 2 is provided, issuers should validate it.
- Issuers should be aware that Mastercard attempts to validate the DTVC submitted by acquirers during transaction processing.
  - When DTVC can be validated, Mastercard will send OBS 61 in [DE 48 \(Additional Data: Private Use\)](#), subelement 71 and suppress the DTVC from [DE 48 \(Additional Data: Private Use\)](#), subelement 92 (CVC 2) in Authorization Request/0100 and in Financial Transaction Request/0200 messages.
  - When the validity of the CVC 2 value cannot be determined, Mastercard will send the CVC 2 in [DE 48 \(Additional Data: Private Use\)](#), subelement 92 (CVC 2) to issuers for decisioning. Issuers will not receive OBS 61 in such transactions. In some use cases, the CVC 2 may reflect the value associated with the consumer's primary account number.

## Subelement 71 (OBS 05 Result Codes)

[DE 48](#), subelement 71 (OBS 05 Result Codes) provides detailed information on the result codes that can be present in [DE 48 \(Additional Data—Private Use\)](#), subelement 71 (On-behalf Services

[OBS]], subfield 2 (OB Result 1) when [DE 48](#), subelement 71, subfield 1 (On-behalf [OB] Service) value = 05 (Mastercard Identity Check™ AAV Verification Service).

### OBS 05 Result Codes

| Code | Description                                                                                                                                                                             | SPA2 AAV                  | DS Transaction ID                     | Amount Check (Authorization : Authentication) |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------|-----------------------------------------------|
| A    | AAV and Amount Checked<br>Low risk assumption for approval.<br><br>Applies only to SPA2 AAVs.                                                                                           | Found associated with PAN | Not Present in Authorization          | < or = authentication amount                  |
| B    | Balance to Verify<br>Low risk assumption for approval.<br><br>Applies only to SPA2 AAVs.                                                                                                | Found associated with PAN | Not Present in Authorization          | 0%–19.99% of authentication amount            |
| C    | Consider the Amount<br>Applies only to SPA2 AAVs.<br><br>Amount discrepancy greater than 20%; other approval decision factors should be considered.                                     | Found associated with PAN | Not Present in Authorization          | 20% or > of authentication amount             |
| D    | DS Transaction ID Failed<br>Applies only to SPA2 AAVs.<br><br>Service unavailable or discrepancy with critical validation fields; other approval decision factors should be considered. | Found associated with PAN | Not Found associated with PAN and AAV | N/A                                           |
| F    | Failed MAC key Validation; No Dynamic Linking available.                                                                                                                                | NA                        | NA                                    | NA                                            |

| Code | Description                                                                                                                                                              | SPA2 AAV                      | DS Transaction ID                 | Amount Check<br>(Authorization :<br>Authentication) |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------|-----------------------------------------------------|
| I    | Invalid AAV<br>Service unavailable or discrepancy with critical validation fields; other approval decision factors should be considered.                                 | Not Found associated with PAN | N/A                               | N/A                                                 |
| K    | Key Not on File<br>This value usually indicates the issuer is performing Self-Validation on SPA1 AAVs and have not shared their ACS Issuer SPA1 AAV key with Mastercard. |                               |                                   |                                                     |
| M    | Monetary Currency mismatch between authorization and authentication records.                                                                                             |                               |                                   |                                                     |
| P    | Passed MAC key Validation; No Dynamic Linking available                                                                                                                  | NA                            | NA                                | NA                                                  |
| S    | DS Transaction ID Present – Balance to Verify<br>Low risk assumption for approval.<br>Applies only to SPA2 AAVs.                                                         | Found associated with PAN     | Found associated with PAN and AAV | 0%–19.99% of authentication amount                  |
| T    | Transaction ID Present – Consider the Amount<br>Applies only to SPA2 AAVs.<br>Amount discrepancy greater than 20%; other approval decision factors should be considered. | Found associated with PAN     | Found associated with PAN and AAV | 20% or > of authentication amount                   |
| U    | Service Unavailable<br>Service unavailable or discrepancy with critical validation fields; other approval decision factors should be considered.                         |                               |                                   |                                                     |

| Code | Description                                                                                                                                                                   | SPA2 AAV                  | DS Transaction ID                 | Amount Check<br>(Authorization :<br>Authentication) |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------|-----------------------------------------------------|
| V    | Valid – All Data Passes<br>Low risk assumption for approval.                                                                                                                  | Found associated with PAN | Found associated with PAN and AAV | < or = authentication amount                        |
| X    | Security Platform Timeout<br>Amount discrepancy greater than 20%; other approval decision factors should be considered.<br>Applies only to SPA1 AAVs and 06 service.          |                           |                                   |                                                     |
| Z    | Security Platform Processing Error<br>Amount discrepancy greater than 20%; other approval decision factors should be considered.<br>Applies only to SPA1 AAVs and 06 service. |                           |                                   |                                                     |

## Subelement 72 (Issuer Chip Authentication)

[DE 48](#), subelement 72 (Issuer Chip Authentication) carries data used during cryptogram processing.

### Attributes

| Attribute              | Description                                                                    |
|------------------------|--------------------------------------------------------------------------------|
| Subelement ID          | 72                                                                             |
| Subelement Data Length | 2 positions                                                                    |
| Data Representation    | b...16; LLVAR (the "LL" length field of LLVAR must be between 8–16 positions.) |
| Data Field             | Contents of subelement                                                         |
| Subfields              | N/A                                                                            |
| Justification          | N/A                                                                            |

### Usage

Following is the usage of subelement 72 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | .   | X   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |

### Values

Mastercard generated.

### Application notes

Issuers no longer need to echo [DE 48](#), subelement 72 in their response messages.

## Subelement 74 (Additional Processing Information)

[DE 48](#), subelement 74 (Additional Processing Information) provides additional information about chip transaction processing and results.

### Attributes

| Attribute              | Description                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------|
| Subelement ID          | 74                                                                                                   |
| Subelement Data Length | 2 positions                                                                                          |
| Data Representation    | an...30; LLVAR<br>The "LL" length field of LLVAR must be an integral multiple of 3 not to exceed 30. |
| Data Field             | Contents of subfields                                                                                |
| Subfields              | 2                                                                                                    |
| Justification          | See subfields                                                                                        |

### Usage

Following is the usage of subelement 74 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | O   | X   | C   |

Values

See subfields.

Application notes

[DE 48](#), subelement 74 is not applicable in the 0120/0130, 0400/0410 and 0420/0430 messages. If present, it will be removed before forwarding to its destination.

Chip Cryptogram Validation

Issuers performing chip cryptogram validation may include their validation results in [DE 48](#), subelement 74 in the Authorization Request Response/0110 message.

Chip Fallback Transaction Downgrade

The Dual Message Authorization System will include [DE 48](#), subelement 74 in the Authorization Request Response/0110 message when the transaction is eligible for the Chip Fallback Transaction Downgrade. In which case [DE 48](#), subelement 74, subfield 1 value will be "90" and subfield 2 value will be "C". Refer to the [DE 22: Point-of-Service \(POS\) Entry Mode](#) section for additional details.

Associated edits

| When the issuer sends an Authorization Request Response/0110 message containing DE 48, subelement 74 and... | THEN the Dual Message Authorization System...                                                                                   |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Subfield 1 does not contain 50                                                                              | Will remove <a href="#">DE 48</a> , subelement 74 before forwarding the message to the acquirer.                                |
| Subfield 2 contains an invalid value                                                                        | Will replace the invalid subfield 2 value with an X to indicate an unknown issue before forwarding the message to the acquirer. |

| When the issuer sends an Authorization Request Response/0110 message containing DE 48, subelement 74 and...                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <p><i>DE 48 (Additional Data—Private Use)</i>, subelement 74 (Additional Processing Information) in the Authorization Request Response/0110 is received from the issuer contains:</p> <ul style="list-style-type: none"> <li>Subfield 1 = value 90 (Chip Fallback Transaction Downgrade Process)</li> <li>Subfield 2 = value C (Completed Successfully)</li> </ul> | Removes this data from transaction before forwarding to the acquirer. |

## Subfield 1 (Processing Indicator)

*DE 48*, subelement 74, subfield 1 (Processing Indicator) indicates the transaction processing type.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Field          | Contents of positions 1-2 |
| Justification       | N/A                       |

### Values

| Value | Description                                                                       |
|-------|-----------------------------------------------------------------------------------|
| 02    | Mastercard On-behalf Service: M/Chip Cryptogram Pre-validation                    |
| 03    | Mastercard On-behalf Service: M/Chip Cryptogram Validation in Stand-In Processing |
| 50    | Issuer Chip Validation                                                            |
| 90    | Chip Fall-back Transaction Downgrade Process                                      |

## Subfield 2 (Processing Information)

[DE 48](#), subelement 74, subfield 2 (Processing Information) contains additional information about the issue incurred during chip transaction processing.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

### Values

| Value | Description                                                                                    |
|-------|------------------------------------------------------------------------------------------------|
| A     | Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range |
| C     | Completed Successfully                                                                         |
| E     | Valid Application Cryptogram; ATC Replay                                                       |
| F     | Format error in DE 55, DE 48, subelement 43, or DE 104, subelement 001                         |
| G     | Application Cryptogram is valid but is not an ARQC                                             |
| I     | Application Cryptogram invalid                                                                 |
| K     | No matching key file for this PAN, PAN expiry date and KDI combination                         |
| T     | Application Cryptogram is valid but TVR/CVR was invalid                                        |
| U     | Application Cryptogram could not be validated due to technical error                           |
| X     | Issuer provided incorrect subfield 2 value                                                     |
| X     | Security platform time out                                                                     |
| Z     | Security platform processing error                                                             |



## Subelement 74 (Valid Subfield 1 and Subfield 2 Value Combinations)

[DE 48](#), subelement 74 (Valid Subfield 1 and Subfield 2 Value Combinations) is the valid [DE 48](#), subelement 74, subfield 1 (Processing Indicator) and subfield 2 (Processing Information) value combinations. The contents of subfield 2 depend on the contents of subfield 1 as described here.

### Valid combinations

| IF subfield 1 contains... | THEN subfield 2 may contain...  |
|---------------------------|---------------------------------|
| 02                        | A, E, F, G, I, K, T, U, X, or Z |
| 03                        | A, E, F, G, I, K, T, U, X, or Z |
| 50                        | A, E, F, G, I, T, U, or X       |
| 90                        | C                               |

## Subelement 75 (Fraud Scoring Data)

[DE 48](#), subelement 75 (Fraud Scoring Data) indicates the fraud score on a fraud scoring service transaction.

Mastercard provides two scoring solutions: Expert Monitoring for Issuers or Decision Intelligence. A score will be included in Authorization Request/0100 and Authorization Advice/0120: system-generated messages to the issuer when the Expert Monitoring or Decision Intelligence scoring was performed on a transaction.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 75                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an...32; LLVAR            |
| Data Field             | Contents of subfields 1–5 |
| Subfields              | 5                         |
| Justification          | See subfields             |

### Usage

Following is the usage of subelement 75 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | .   | X   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |

### Values

Contents of subfields 1–2 and applicable subfields 3–5.

### Application notes

The Dual Message Authorization System inserts this subelement when one of the Fraud Scoring Services is performed or when both a Fraud Scoring Service and the Fraud Rule Manager Service are performed on the transaction.

**NOTE: When a rule adjusted score is provided in subfield 3, at least one or more rule reason code values will be provided in subfields 4–5. However, rule reason code values may be provided in subfields 4 or 5 with or without a rule adjusted score in subfield 3.**

**NOTE: Issuers participating in a scoring service will receive subfields 1–2. If those issuers choose to participate in the Fraud Rule Manager Service then those issuers must also prepare to receive subfields 3–5.**

**NOTE: [DE 48](#), subelement 75 will not be included in Authorization Request/0100 messages to the issuer when Mastercard is unable to perform the Fraud Scoring Service.**

Expert Monitoring for Issuers and Decision Intelligence services evaluate fraud detection elements to produce a score that indicates a level of risk in the transaction. Both services utilize the same [DE 48](#), subelement 75 field with dual message authorization transactions.

For additional details on these services, refer to the *Issuer Security Solutions Utilities Guide* and Mastercard Security Solutions Guide for Issuer Services.

### Subfield 1 (Fraud Score)

[DE 48](#), subelement 75, subfield 1 (Fraud Score) indicates the transaction risk score. The value in DE 48, subelement 75, subfield 1 is the model score as indicated in DE 48, subelement 75, subfield 3 (Model Score) unless adjusted by SafetyNet or Rules (Customer Rules or On-behalf of Rules).

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 01                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-3                      |
| Data Field           | Contents of positions 1–3 |
| Justification        | N/A                       |

### Values

Fraud Scoring System provides the three position transaction risk score from 001-998. 001 indicates the least likely fraudulent transaction and 998 indicates the most likely fraudulent transaction.

## Subfield 2 (Fraud Reason Code)

[DE 48](#), subelement 75, subfield 2 (Fraud Reason Code) indicates the key factors that influenced the fraud score. This value is the model reason code as indicated in [DE 48](#), subelement 75, subfield 4 (Model Reason Code) unless overwritten by SafetyNet or Rules (Customer Rules or On-behalf of Rules).

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 02                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-2                      |
| Data Field           | Contents of positions 1–2 |
| Justification        | N/A                       |

### Values

The Fraud Scoring Service provides the score reason code, an alphanumeric code identifying the data used to derive the fraud score.

**NOTE: Participating issuers may contact the Risk Solutions team for a list of the specific score reason codes that apply to their institution.**

## Subfield 3 (Model Score)

[DE 48](#), subelement 75, subfield 3 (Model Score) contains the model risk score. This score is used in Rule Authoring, Rule Discovery, and Decision Intelligence (DI) Dashboard.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 03                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-3                      |
| Data Field           | Contents of positions 1–3 |
| Justification        | N/A                       |

### Values

The Fraud Rule Manager Service provides a rules score of 000 through 999, where 000 indicates the least likely fraudulent transaction and 999 indicates the most likely fraudulent transaction.

## Subfield 4 (Model Reason Code)

[DE 48](#), subelement 75, subfield 4 (Model Reason Code) indicates the key factors that influenced the model fraud score.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 04                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | an-2                      |
| Data Field           | Contents of positions 1–2 |
| Justification        | N/A                       |

### Values

The Fraud Rule Manager Service provides the rule reason code, an alphanumeric code where the information provided gives the data used to derive the rules adjusted score.

Subfield 5 (Reserved for Future Use)

DE 48, subelement 75, subfield 5 (Reserved for Future Use) is reserved for future use.

Subelement 76 (Mastercard Electronic Acceptance Indicator)

DE 48, subelement 76 (Mastercard Electronic Acceptance Indicator) identifies that the transaction is a Mastercard Electronic card transaction. It indicates that the acquirer participates or does not participate in Mastercard Electronic card processing.

Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 76                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

Usage

Following is the usage of subelement 76 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | O   | X   | .   |
| Authorization Advice/0120: acquirer-generated | O   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

## Values

| Value | Description                                                                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C     | Mastercard only participant (not considered a Mastercard Electronic card transaction).                                                                                                                                              |
| E     | Acquirer and its merchant both participate in Mastercard Electronic card processing (considered as a Mastercard Electronic transaction).                                                                                            |
| M     | Acquirer participates in Mastercard Electronic card processing, but the merchant that processed this specific transaction does not participate in Mastercard Electronic (considered not to be a Mastercard Electronic transaction). |
| U     | Unidentified acquirer. It is unknown if the acquirer is a Mastercard Electronic card participant.                                                                                                                                   |

## Application notes

If acquirers do not populate subelement 76 with a valid value or a value is not present, Mastercard will populate [DE 48](#), subelement 76 on behalf of the acquirer based on the acquirer's participation in Mastercard Electronic Card Program and forward the transaction to the issuer.

## Acquirers

When the merchant participates in the Mastercard Electronic Card Program, the participating acquirer should send [DE 48](#), subelement 76 with a value of E in the Authorization Request/0100 message.

When the merchant does not participate in the Mastercard Electronic Card Program, the participating acquirer should send [DE 48](#), subelement 76 with a value of M in the Authorization Request/0100 message.

When participating acquirers do not provide a value or provide an incorrect value in subelement 76, Mastercard will default subelement 76 to the value of E in the Authorization Request/0100 message.

## Issuers

Issuers participating in Mastercard Electronic Card must be prepared to receive [DE 48](#), subelement 76 with values of C, E, or U in the Authorization Request/0100 and Authorization Advice/0120 messages.

Issuers participating in Mastercard Electronic Card must be prepared to send [DE 48](#), subelement 76 with values of C, E, or U in Authorization Request Response/0110 messages.

**NOTE: Mastercard will convert the subelement value of M to the value C in the Authorization Request/0100 sent to the issuer. Therefore, the issuer will not receive a value of M.**

## Subelement 77 (Transaction Type Identifier)

[DE 48](#), subelement 77 (Transaction Type Identifier) indicates the type of additional transaction purpose.

**NOTE: All Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions) require the presence of this identifier.**

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 77                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-3                      |
| Data Field             | Contents of positions 1-3 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 77 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

**NOTE: This subelement is mandatory when [DE 18](#)=MCC is any of the following: MCC 4829, MCC 6536, MCC 6537, MCC 6538, and MCC 6540.**

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | P   | C   |
| Authorization Request Response/0110         | OE  | X   | •   |
| Authorization Advice/0120: system-generated | •   | X   | C   |
| Reversal Request/0400                       | C   | P   | C   |
| Reversal Request Response/0410              | OE  | X   | •   |
| Reversal Advice/0420                        | •   | X   | C   |

## Values

| Value | Description                                         |
|-------|-----------------------------------------------------|
| C04   | Gaming Re-pay                                       |
| C07   | General Person-to-Person Transfer                   |
| C51   | Mastercard Send Indicator (Reserved for Future Use) |
| C52   | General Transfer to Own Account                     |
| C53   | Agent Cash Out                                      |
| C54   | Payment of Own Credit Card Bill                     |
| C55   | Business Disbursement                               |
| C56   | Government/Non-Profit Disbursement                  |
| C57   | Rapid Merchant Settlement                           |
| C58   | Cash in at ATM                                      |
| C59   | Cash in at point of Sale                            |
| C60   | Fast Refund to Original Card                        |
| C61   | Mastercard Send Indicator (Reserved for Future Use) |
| C62   | Mastercard Send Indicator (Reserved for Future Use) |
| C63   | Mastercard Send Indicator (Reserved for Future Use) |
| C64   | Mastercard Send Indicator (Reserved for Future Use) |
| C65   | General Business to Business Transfer               |
| C66   | Mastercard Send Indicator (Reserved for Future Use) |
| C67   | Merchant Presented QR                               |
| C68   | Merchant Presented QR                               |
| F07   | General Person-to-Person Transfer                   |
| F08   | Person-to-Person Transfer to Card Account           |
| F52   | General Transfer to Own Account                     |
| F53   | Agent Cash Out                                      |
| F54   | Payment of Own Credit Card Bill                     |
| F55   | Business Disbursement                               |
| F61   | Transfer to Own Staged Digital Wallet Account       |
| F64   | Transfer to Own Debit or Prepaid Account            |



| Value | Description                           |
|-------|---------------------------------------|
| F65   | General Business-to-Business Transfer |
| P10   | Purchase repayment                    |
| P11   | Business-to-Business Payment          |
| P70   | Cryptocurrency                        |
| P71   | High-risk Securities                  |

### Application notes

If [DE 3](#), subfield 1 contains value 28 (Payment Transaction), then [DE 48](#), subelement 77 must be present.

Usage of value C04 is limited to eligible acquirers and issuers in eligible countries. Refer to the *Mastercard Gaming and Gambling Payments Program Standards*.

Usage of value C07 and payment type codes C52–C59, C60–C61, C64–C65, and F07–F65 is limited to eligible countries and eligible acquirers. Refer to the *Mastercard Send and Funding Transactions Program Standards*.

**NOTE: Usage of value C67 is limited to Mastercard Merchant Presented QR transactions. Usage of value C68 is limited to Mastercard Merchant Presented QR refund transactions.**

Usage of value P11 (Business-to-Business Payment) is intended to identify transactions between a buyer and a supplier who does not accept card-based payments, via a Business Payment Service Provider who remits the payment to the supplier.

## Subelement 78 (Payment Service Indicators [Visa Only])

[DE 48](#), subelement 78 (Payment Service Indicators [Visa Only]) contains subfields which represent various Visa payment service indicators.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 78                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | ans-6                     |
| Data Field             | Contents of subfields 1–6 |
| Subfields              | 6                         |

| Attribute     | Description   |
|---------------|---------------|
| Justification | See subfields |

## Usage

Following is the usage of subelement 78 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice Response/0130            | C   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Request Response/0410                | C   | X   | C   |

## Values

See subfields.

## Application notes

Acquirers are not required to populate this subelement in the Authorization Request/0100 message if they do not support any of the Visa payment indicators in subfields 2 through 6 as noted in the following sections.

## Subfield 1 (Spend Qualified Indicator)

[DE 48](#), subelement 78, subfield 1 (Spend Qualified Indicator) provides the Visa spend qualification assessment.

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 1 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Value | Description                                                                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Space | Space will be sent back in the Authorization Request Response/0110 message to the acquirer when Visa field 62.25 is not present in the Visa 0110 message response |
| N     | Spend assessment threshold defined by Visa has not been met                                                                                                       |
| B     | Basic spend assessment threshold level met                                                                                                                        |
| Q     | Highest spend assessment threshold level met                                                                                                                      |

### Application notes

The value for this field is mapped from Visa Field 62.25 Spend Qualified Indicator. Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*. Acquirers will need to populate subfield 1 with a space if any other Payment Service Indicator is submitted in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages. Space will be sent back in subfield 1 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110, Authorization Advice Response/0130, and Reversal Request Response/0410 messages.

**NOTE: The Spend Qualified Indicator is used by acquirers for Visa's Account Level Processing (ALP) program.**

## Subfield 2 (Dynamic Currency Conversion Indicator)

[DE 48](#), subelement 78, subfield 2 (Dynamic Currency Conversion Indicator) indicates a transaction for which Dynamic Currency Conversion (DCC) is performed by the merchant at the point of sale.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 2 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Value | Description                                                                                                                        |
|-------|------------------------------------------------------------------------------------------------------------------------------------|
| Y     | Dynamic Currency Conversion was performed at the point of sale.                                                                    |
| Space | No Dynamic Currency Conversion or not echoed in the Authorization Request Response/0110 or Reversal Request Response/0410 message. |

### Application notes

If the merchant performs dynamic currency conversion at the point of sale, acquirers must send this value in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated and in Reversal Request/0400 messages in case of Reversal. This value will be mapped to Visa Field 126.19—Dynamic Currency Conversion Indicator.

Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* manual.

The acquirer will need to populate subfield 2 with a space if DCC was not performed and any other Payment Service Indicator is submitted in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated and in Reversal Request/0400 messages. Space will be sent back in subfield 2 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110, Authorization Advice Response/0130, and Reversal Request Response/0410 messages.

### Subfield 3 (U.S. Deferred Billing Indicator)

[DE 48](#), subelement 78, subfield 3 (U.S. Deferred Billing Indicator) indicates a transaction for which the billing for merchandise occurred after the merchandise was delivered to the cardholder. This applies to U.S. region acquirers only.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

## Values

| Value | Description                                                                      |
|-------|----------------------------------------------------------------------------------|
| D     | U.S. Deferred Billing Indicator                                                  |
| Space | No Deferred Billing or not echoed in Authorization Request Response/0110 message |

## Application notes

This code is provided by the merchant through United States region acquirers in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages to indicate that a Visa card is to be billed on a deferred basis, that is, the cardholder is to be billed for merchandise already received. This value will be mapped to Visa Field 126.12—Service Indicators, position 3 (Deferred Billing Indicator).

Refer to the Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*.

Acquirers will need to populate subfield 3 with a space if Deferred Billing is not applicable and any other Payment Service Indicator is submitted in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated and Reversal Request/0400 message. A space will be sent back in subfield 3 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110, Authorization Advice Response/0130, and Reversal Request Response/0410 messages.

## Subfield 4 (Visa Checkout Indicator)

[DE 48](#), subelement 78, subfield 4 (Visa Checkout Indicator) indicates that the transaction was processed through Visa Checkout.

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 4 |
| Justification       | N/A                    |

## Values

| Value | Description                                                                       |
|-------|-----------------------------------------------------------------------------------|
| Y     | Transaction is processed through Visa Checkout                                    |
| Space | No Visa Checkout or not echoed in the Authorization Request Response/0110 message |

## Application notes

If the transaction is performed through Visa Checkout, acquirers must send this value in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages. This value will be mapped to Visa Field 126.18—Agent Unique Account Result.

Refer to the Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*.

The acquirer will need to populate subfield 4 with a space if Visa Checkout was not performed and any other Payment Service Indicator is submitted in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated and Reversal Request/0400 messages. A space will be sent back in subfield 4 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110, Authorization Advice Response/0130, and Reversal Request Response/0410 messages.

## Subfield 5 (Message Reason Code)

[DE 48](#), subelement 78, subfield 5 (Message Reason Code) provides the message reason code of the transaction.

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 5 |
| Justification       | N/A                    |

## Values

| Value | Description                                                       |
|-------|-------------------------------------------------------------------|
| Space | Default (No message reason code associated with this transaction) |

| Value | Description               |
|-------|---------------------------|
| 0     | Incremental authorization |
| 1     | Resubmission              |
| 2     | Delayed charges           |
| 3     | Reauthorization           |
| 4     | No show                   |
| 5     | Account top up            |
| 6     | Deferred Authorization    |
| 7     | AFD Completion Advice     |

### Application notes

Acquirers must populate a space in Subfield 6 for Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages, if DE 48 Subelement 78 is present. Otherwise, Mastercard will decline the transaction back to the acquirer.

Acquirers must send the value based on the transaction in the Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages. This value will be mapped to Visa Field 63.3 Message Reason Code. Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*.

### Subfield 6 (Token Response Information)

[DE 48](#), subelement 78, subfield 6 (Token Response Information) provides the Token Response Information received for Visa-branded authorization transactions from VisaNet (VIP) in the authorization request response.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data representation | ans-1                  |
| Data field          | Contents of position 6 |
| Justification       | N/A                    |

### Values

| Value | Description   |
|-------|---------------|
| Space | N/A           |
| 1     | Token program |

### Application notes

Acquirers must populate a space in Subfield 6 for Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages, if DE 48 Subelement 78 is present. Otherwise, Mastercard will decline the transaction back to the acquirer.

The value for this field is provided by Visa in the Authorization Request Response/0110 and Authorization Advice Response/0130 messages. This field contains a value that can identify transactions eligible for token services. Acquirers will need to retain the value in this field from authorization request response messages and send it in clearing transactions to Visa.

Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*.

## Subelement 79 (Chip CVR/TVR Bit Error Results)

[DE 48](#), subelement 79 (Chip CVR/TVR Bit Error Results) provides the Terminal Verification Results (TVR) and Card Verification Results (CVR) bitmask, and expected values registered by the issuer. This serves as notification of bit validation errors detected in the CVR/TVR within the Issuer Application data during M/Chip Cryptogram Validation processing.

### Attributes

| Attribute              | Description                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Subelement ID          | 79                                                                                                    |
| Subelement Data Length | 2 positions                                                                                           |
| Data Representation    | an...50; LLVAR<br>The "LL" length field of LLVAR must be an integral multiple of 5, not to exceed 50. |
| Data Field             | Contents of subfields 1–4                                                                             |
| Subfields              | 4                                                                                                     |
| Justification          | N/A                                                                                                   |



## Usage

Following is the usage of subelement 79 (whether it is mandatory, conditional, optional, system provided, or not required) in all applicable messages.

## Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | •   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

## Values

See subfields.

## Application notes

The Dual Message Authorization System will stop verification of CVR/TVR bits when 10 errors are detected.

Refer to the following example. In the example:

- The binary representation of the TVR Bit Mask for Byte 3 is "10101000"
- The binary representation of the Expected Result for Byte 3 is "00000000"
- The binary representation of the Validation Result for Byte 3 is "10000000"

The Validation Result indicates that the cardholder verification was not successful.

Refer to the *M/Chip Processing Services: Service Description* document for recommended bit mask settings.

## Binary representations

|                   | Byte 1 | Byte 2 | Byte 3 | Byte 4 | Byte 5 |
|-------------------|--------|--------|--------|--------|--------|
| TVR Bit Mask: Hex | 00     | 00     | A8     | 00     | 00     |
| Expected Result   | 00     | 00     | 00     | 00     | 00     |
| Validation Result | 00     | 00     | 80     | 00     | 00     |

### Subfields

| Subfield   | Description           | Value |
|------------|-----------------------|-------|
| Subfield 1 | CVR or TVR Identifier | T     |
| Subfield 2 | Byte ID               | 03    |
| Subfield 3 | Bit Identifier        | 8     |
| Subfield 4 | Value of Bit in Error | 1     |

### Subfield 1 (CVR or TVR Identifier)

[DE 48](#), subelement 79, subfield 1 (CVR or TVR Identifier) indicates whether bit reported in error is part of CVR or TVR.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

| Value | Description |
|-------|-------------|
| C     | CVR         |
| T     | TVR         |

## Subfield 2 (Byte ID)

[DE 48](#), subelement 79, subfield 2 (Byte ID) identifies the byte number of the associated bit reported in error.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-2                      |
| Data Field          | Contents of positions 2–3 |
| Justification       | N/A                       |

### Values

01–99

## Subfield 3 (Bit Identifier)

[DE 48](#), subelement 79, subfield 3 (Bit Identifier) identifies the bit number in error within the byte identified in subfield 2.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 4 |
| Justification       | N/A                    |

### Values

1–8

## Subfield 4 (Value of Bit in Error)

[DE 48](#), subelement 79, subfield 4 (Value of Bit in Error) identifies the value of the bit in error that was submitted in transaction.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 5 |
| Justification       | N/A                    |

### Values

| Value | Description           |
|-------|-----------------------|
| 0     | The bit is turned off |
| 1     | The bit is turned on  |

## Subelement 80 (PIN Service Code)

[DE 48](#), subelement 80 (PIN Service Code) indicates the results of PIN processing by the Dual Message Authorization System.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 80                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | a-2                       |
| Data Field             | Contents of positions 1–2 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

## Usage

Subelement 80 is provided by the Dual Message Authorization System in the Authorization Request/0100 message whenever PIN data is present in the Authorization Request/0100 message.

Following is the usage of subelement 80 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | .   | X   | C   |
| Authorization Request Response/0110         | .   | X   | C   |
| Authorization Advice/0120: system-generated | C   | .   | C   |

## Values

| Value | Description                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------|
| PD    | The Dual Message Authorization System dropped the PIN (Mastercard Use Only for Credit Transactions with PIN) |
| PV    | The Dual Message Authorization System verified the PIN.                                                      |
| TV    | The Dual Message Authorization System translated the PIN for issuer verification.                            |
| PI    | The Dual Message Authorization System was unable to verify the PIN.                                          |
| TI    | The Dual Message Authorization System was unable to translate the PIN.                                       |

## Subelement 82 (Address Verification Service Request)

[DE 48](#), subelement 82 (Address Verification Service Request) indicates that verification of the cardholder billing address is requested in the authorization message.

## Attributes

| Attribute              | Description |
|------------------------|-------------|
| Subelement ID          | 82          |
| Subelement Data Length | 2 positions |

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 1–2 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

## Usage

Subelement 82 must be present in the authorization request message whenever cardholder address verification is requested by the acquirer.

Following is the usage of subelement 82 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | X   | C   |
| Authorization Request Response/0110         | CE  | XE  | CE  |
| Authorization Advice/0120: system-generated | •   | X   | C   |

## Values

| Value | Description                        |
|-------|------------------------------------|
| 52    | AVS and Authorization Request/0100 |

## Application notes

Mastercard all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

**NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.**

## Subelement 83 (Address Verification Service Response)

DE 48, subelement 83 (Address Verification Service Response) contains the AVS verification response code in the Authorization Request Response/0110 message.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 83                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Subelement 83 must be provided by the issuer in the Authorization Request Response/0110 message whenever AVS is requested by the acquirer.

Following is the usage of subelement 83 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110         | C   | X   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |

### Values

| Value | Description                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | Address matches, postal code does not.                                                                                                          |
| B     | Visa only. Street address match. Postal code not verified because of incompatible formats. (Acquirer sent both street address and postal code.) |
| C     | Visa only. Street address and postal code not verified because of incompatible formats. (Acquirer sent both street address and postal code.)    |

| Value | Description                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| D     | Visa only. Street address and postal code match.                                                                                                 |
| F     | Visa only. Street address and postal code match. Applies to U.K. only.                                                                           |
| G     | Visa only. Non-AVS participant outside the U.S.; address not verified for international transaction.                                             |
| I     | Visa only. Address information not verified for international transaction.                                                                       |
| M     | Street addresses and postal code match.                                                                                                          |
| N     | Neither address nor postal code matches.                                                                                                         |
| P     | Visa only. Postal codes match. Street address not verified because of incompatible formats. (Acquirer sent both street address and postal code.) |
| R     | Retry, system unable to process.                                                                                                                 |
| S     | AVS currently not supported. U.S. issuers will not be able to send through this response code.                                                   |
| U     | No data from issuer/Dual Message Authorization System                                                                                            |
| W     | For U.S. addresses, nine-digit postal code matches, address does not; for address outside the U.S., postal code matches, address does not.       |
| X     | For U.S. addresses, nine-digit postal code and address matches; for addresses outside the U.S., postal code and address match.                   |
| Y     | For U.S. addresses, five-digit postal code and address matches.                                                                                  |
| Z     | For U.S. addresses, five-digit postal code matches, address does not.                                                                            |

### Application notes

Mastercard all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

**NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.**



## Subelement 84 (Merchant Advice Code)

DE 48, subelement 84 (Merchant Advice Code) contains the merchant advice code.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 84                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-2                      |
| Data Field             | Contents of positions 1–2 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Subelement 84 is optionally provided by the issuer in the Authorization Request Response/0110 message.

Following is the usage of subelement 84 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                          | Org | Sys | Dst |
|----------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110                                              | O   | X   | C   |
| Authorization Request Response/0110: tokenization authorization request response | C   | C   | .   |
| Authorization Advice/0120: system-generated                                      | .   | X   | C   |
| Reversal Request Response/0410                                                   | O   | P   | C   |

### Values

| Value | Description                                  |
|-------|----------------------------------------------|
| 01    | New account information available            |
| 02    | Cannot approve at this time, try again later |

| Value | Description                                                         |
|-------|---------------------------------------------------------------------|
| 03    | Do not try again                                                    |
| 04    | Token requirements not fulfilled for this token type                |
| 05    | Negotiated value not approved                                       |
| 21    | Payment Cancellation (Mastercard use only)                          |
| 22    | Merchant does not qualify for product code                          |
| 24    | Retry after 1 hour (Mastercard use only)                            |
| 25    | Retry after 24 hours (Mastercard use only)                          |
| 26    | Retry after 2 days (Mastercard use only)                            |
| 27    | Retry after 4 days (Mastercard use only)                            |
| 28    | Retry after 6 days (Mastercard use only)                            |
| 29    | Retry after 8 days (Mastercard use only)                            |
| 30    | Retry after 10 days (Mastercard use only)                           |
| 40    | Consumer non-reloadable prepaid card                                |
| 41    | Consumer single-use virtual card number                             |
| 42    | Sanctions Scoring Service: Score Exceeds Applicable Threshold Value |
| 43    | Consumer multi-use virtual card number                              |

### Application notes

Merchant Advice Code value of 43 (Consumer Multi-Use Virtual Card Number) may be used on all approval Authorization Request Response/0110 messages and decline Authorization Request Response/0110 messages only when DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains a value of 8 (Account Status Inquiry Service [ASI]).

## Subelement 84 (Visa Response Codes [Visa Only])

DE 48, subelement 84 (Visa Response Codes [Visa Only]) contains the Visa Field 39 (Response Codes) as received from Visa.

### Attributes

| Attribute              | Description |
|------------------------|-------------|
| Subelement ID          | 84          |
| Subelement Data Length | 2 positions |

| Attribute           | Description              |
|---------------------|--------------------------|
| Data Representation | an-2                     |
| Data Field          | Contents of position 1–2 |
| Subfields           | N/A                      |
| Justification       | N/A                      |

### Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |
| Authorization Advice Response/0130  | C   | X   | C   |
| Reversal Request Response/0410      | C   | X   | C   |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for the list of valid values and definitions.

## Subelement 84 (American Express Response Codes (American Express Only))

DE 48, subelement 84 (American Express Response Codes [American Express only]) contains the American Express Data Field 39 (Action Codes) as received from American Express.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 84                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-3                      |
| Data Field             | Contents of positions 1-3 |
| Subfields              | N/A                       |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

### Applicable messages

| Attributes                          | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |

### Values

Refer to *American Express Global Credit Authorization Guide (ISO)* for further instructions.

## Subelement 85 (Account Status (Visa Only))

DE 48, subelement 85 (Account Status [Visa Only]) identifies the account range as regulated or non-regulated interchange.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Subelement ID       | 85                     |
| Subelement Length   | 2 positions            |
| Data Representation | a-1                    |
| Data Field          | Contents of position 1 |
| Subfields           | N/A                    |
| Justification       | N/A                    |

### Usage

Following is the usage of subelement 85 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |
| Authorization Advice Response/0130  | C   | P   | C   |

### Values

| Value | Description              |
|-------|--------------------------|
| R     | Account is regulated     |
| N     | Account is non-regulated |

## Subelement 86 (Relationship Participant Indicator [Visa Only])

DE 48, subelement 86 (Relationship Participant Indicator) indicates the transaction originates from a cardholder-merchant relationship.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Subelement ID       | 86                     |
| Subelement Length   | 2 positions            |
| Data Representation | a-1                    |
| Data Field          | Contents of position 1 |
| Subfields           | N/A                    |
| Justification       | N/A                    |

### Usage

Subelement 86 contains an indicator to denote a transaction on a Visa account for a cardholder with a merchant relationship.

Following is the usage of subelement 86 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice Response/0130            | CE  | XE  | CE  |

### Values

| Value | Description                          |
|-------|--------------------------------------|
| C     | Credential on File                   |
| I     | Installment Payment                  |
| R     | Merchant/Cardholder Periodic Billing |

### Application notes

Visa Credential-on-File transactions: DE 48, subelement 86 value of C (Credential on File) is conditional when DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is a value of 10 (Credential on File). Acquirers must submit a value of C when a merchant is initiating the first transaction in a series on behalf of the cardholder using credentials stored on file. For an unscheduled credential-on-file that is merchant-initiated transaction and does not occur at a scheduled interval, acquirers must submit a value of C. Stored credential transactions initiated by cardholders must not submit value C.

## Subelement 87 (Card Validation Code Result)

DE 48, subelement 87 (Card Validation Code Result) indicates the CVC 1, CVC 2, or CVC 3 result code.

### Attributes

| Attribute              | Description |
|------------------------|-------------|
| Subelement ID          | 87          |
| Subelement Data Length | 2 positions |
| Data Representation    | a-1         |

| Attribute     | Description            |
|---------------|------------------------|
| Data Field    | Contents of position 1 |
| Subfields     | N/A                    |
| Justification | N/A                    |

## Usage

Subelement 87 must be provided by the issuer in the Authorization Request Response/0110 message whenever CVC 2 verification is requested by the acquirer.

Subelement 87 is optional whenever DE 45 (Track 1 Data) or DE 35 (Track 2 Data) is present in the Authorization Request/0100 message and CVC 1 is invalid.

Subelement 87 is optionally provided in the Authorization Request Response/0110 message when an issue is encountered during CVC 3 validation.

Following is the usage of subelement 87 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | •   | X   | C   |
| Authorization Request Response/0110         | C   | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

## Values

### CVC 1

| Value | Description                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y     | Invalid CVC 1 (only if DE 35 (Track 2 Data) or DE 45 (Track 1 Data) is present in the Authorization Request/0100 message.)<br><br>Refer to Application notes section that follows for Stand-In System considerations. |

## CVC 2

| Value | Description                                          |
|-------|------------------------------------------------------|
| M     | Valid CVC 2 (match)                                  |
| N     | Invalid CVC 2 (non-match)                            |
| P     | CVC 2 not processed (issuer temporarily unavailable) |
| U     | CVC 2 Unverified: Mastercard Use Only                |

## CVC 3

| Value | Description                                           |
|-------|-------------------------------------------------------|
| E     | Length of unpredictable number was not a valid length |
| P     | Unable to process                                     |
| Y     | Invalid                                               |

## American Express CID (Card Identifier)

Refer to DE 48 (Additional Data: Private Use), Subelement 95: American Express CID Card Identifier (American Express Only).

| Value | Description             |
|-------|-------------------------|
| Y     | Valid CID (match)       |
| N     | Invalid CID (non-match) |
| U     | CID not processed       |

## Application notes

If the issuer participates in the CVC 1 validation service and the transaction meets the criteria for the CVC 1 test, the Stand-In System considers the results of the CVC 1 validation in subelement 87 when responding to the Authorization Request/0100 message. The issuer can control the response that the Stand-In System uses when the CVC 1 value is invalid or validation cannot be performed. The default value is 05 (Do Not Honor). To change these values, contact Key Management Services at [key\\_management@Mastercard.com](mailto:key_management@Mastercard.com).

If an issuer's BIN is temporarily unable to receive the CVC 2 value, the Dual Message System Authorization inserts the value: U = CVC 2 unverified (Mastercard use only).



## Subelement 87 (CVV2 Response [Visa Only])

DE 48, subelement 87 (CVV2 Response) indicates the CVV2 response on a Visa account.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 87                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 87 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |

### Values

| Value | Description                                                                            |
|-------|----------------------------------------------------------------------------------------|
| M     | CVV2 match                                                                             |
| N     | CVV2 no match                                                                          |
| P     | Not processed                                                                          |
| S     | CVV2 is on the card, but the merchant has indicated that CVV2 is not present.          |
| U     | Issuer is not Visa-certified for CVV2, has not provided Visa encryption keys, or both. |

## Subelement 88 (Magnetic Stripe Compliance Status Indicator)

DE 48, subelement 88 (Magnetic Stripe Compliance Status Indicator) indicates that the Dual Message Authorization System replaced the DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) value of 90 or 91 with value of 02.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 88                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Subelement 88 is provided by the Dual Message Authorization System whenever the Dual Message Authorization System replaces the DE 22, subfield 1, value of 90 or 91 to a value of 02 due to magnetic stripe compliance downgrade of the acquirer.

Following is the usage of subelement 88 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | •   | X   | C   |
| Authorization Request Response/0110         | OE  | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

### Values

| Value | Description                                                                                 |
|-------|---------------------------------------------------------------------------------------------|
| Y     | Dual Message Authorization System replaced DE 22, subfield 1, value 90 or 91 with value 02. |

## Subelement 89 (Magnetic Stripe Compliance Error Indicator)

DE 48, subelement 89 (Magnetic Stripe Compliance Error Indicator) indicates magnetic stripe compliance errors.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 89                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Subelement 89 is provided by the Dual Message Authorization System whenever errors are detected while editing transactions for magnetic stripe compliance. Following is the usage of subelement 89 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | •   | X   | C   |
| Authorization Request Response/0110         | OE  | X   | C   |
| Authorization Advice/0120: system-generated | •   | X   | C   |

### Values

| Value | Description                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | Track 1 or Track 2 not present in the message                                                                                                         |
| B     | Track 1 and Track 2 present in the message                                                                                                            |
| C     | DE 2 (Primary Account Number [PAN]) not equal in track data                                                                                           |
| D     | DE 14 (Expiration Date) not equal in track data                                                                                                       |
| E     | Service code invalid in track data                                                                                                                    |
| F     | Field separator(s) invalid in track data                                                                                                              |
| G     | A field within the track data has an invalid length                                                                                                   |
| H     | DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is 80, 90, or 91 when DE 48, Transaction Category Code (TCC) is T |
| I     | DE 61 (Point-of-Service Data), subfield 4 (POS Cardholder Presence) is 1, 2, 3, 4, or 5                                                               |
| J     | DE 61 Point-of-Service Data), subfield 5 (POS Card Presence) is 1                                                                                     |

## Subelement 90 (Lodging and Auto Rental Indicator)

DE 48, subelement 90 (Lodging and Auto Rental Indicator) indicates the presence of Lodging and Auto Rental Service interchange program.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 90                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Subelement 90 is provided whenever the cardholder is a preferred customer of the merchant.

Following is the usage of subelement 90 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | P   | C   |
| Authorization Request Response/0110         | CE  | XE  | CE  |
| Authorization Advice/0120: system-generated | .   | X   | C   |

### Values

| Value | Description                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------|
| P     | Cardholder is enrolled in a merchant preferred customer program and magnetic stripe data may be absent |

## Subelement 90 (Custom Payment Service Request [Visa Only])

DE 48, subelement 90 (Custom Payment Service Request [Visa Only]) contains the Authorization Characteristics Indicator (Visa Field 62.1).

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 90                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Subelement 90 is used to indicate a custom payment service transaction on a Visa account whenever applicable.

Following is the usage of subelement 90 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Reversal Request/0400                         | C   | P   | C   |

### Values

| Value | Description                                      |
|-------|--------------------------------------------------|
| I     | Incremental authorization                        |
| P     | Preferred customer                               |
| R     | Recurring payment                                |
| Y     | Request for Custom Payment Service participation |

### Application notes

Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* for a list of all customer request values in Visa field 62.1.

## Subelement 90 (Custom Payment Service Request Response [Visa Only])

DE 48, subelement 90 (Custom Payment Service Request Response) contains the Authorization Characteristics Indicator (Visa Field 62.1).

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 90                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |

| Attribute     | Description |
|---------------|-------------|
| Subfields     | N/A         |
| Justification | N/A         |

## Usage

Following is the usage of subelement 90 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |
| Authorization Advice Response/0130  | C   | P   | C   |
| Reversal Request Response/0410      | C   | P   | C   |

## Values

Acquirer receives Visa customer response code. See Application notes.

## Application notes

If the request qualifies for Visa customer and is approved, refer to the Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* for a complete list of customer-qualified codes in Visa Field 62.1. If the original request does not qualify for customer, Visa returns an "N" or "T" in the response.

## Subelement 91 (Acquirer Reference Data [American Express Only])

DE 48, subelement 91 (Acquirer Reference Data) contains the 15-digit Transaction Identifier (TID), a unique American Express tracking number. The TID is used to identify and track a card customer transaction throughout its life cycle.

## Attributes

| Attribute              | Description     |
|------------------------|-----------------|
| Subelement ID          | 91              |
| Subelement Data Length | 2 positions     |
| Data Representation    | ans...15; LLVAR |

| Attribute  | Description                |
|------------|----------------------------|
| Data Field | Contents of positions 1–15 |
| Subfields  | N/A                        |

### Usage

Following is the usage of subelement 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

### Values

A unique 15 digit TID.

## Subelement 91 (Custom Payment Service Request/Transaction ID [Visa Only])

DE 48, subelement 91 (Custom Payment Service Request/Transaction ID) indicates the presence of Custom Payment Service Request response data.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 91                         |
| Subelement Data Length | 2 positions                |
| Data Representation    | an...19; LLVAR             |
| Data Field             | Contents of positions 1–19 |
| Subfields              | N/A                        |

### Usage

Subelement 91 is used to provide transaction ID and/or Validation Code when subelement 90 indicates an incremental authorization or when submitting a reversal on a Visa account.



Following is the usage of subelement 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

#### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Reversal Request/0400                         | C   | P   | C   |

#### Values

Subelement 91 must contain the transaction ID (Visa Field 62.2, 15 bytes numeric). Optionally, the acquirer may submit 19 bytes such that the 15-byte numeric transaction ID is followed by the four-byte alphanumeric validation code for Visa Field 62.3.

## Subelement 91 (Custom Payment Service Response/Transaction ID [Visa Only])

DE 48, subelement 91 (Custom Payment Service Response/Transaction ID) provides authorization response data when custom payment service (subelement 90) is requested on a Visa account.

#### Attributes

| Attribute              | Description                                                            |
|------------------------|------------------------------------------------------------------------|
| Subelement ID          | 91                                                                     |
| Subelement Data Length | 2 positions                                                            |
| Data Representation    | ans...19; LLVAR                                                        |
| Data Field             | Contents of positions 1-4, 1-15, or 1-19 depending on the length field |
| Subfields              | N/A                                                                    |

#### Usage

Following is the usage of subelement 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |
| Authorization Advice Response/0130  | C   | X   | C   |
| Reversal Request Response/0410      | C   | X   | C   |

### Values

| Length Field Value       | Subelement 91 Value                                                                                                                                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| When length field is 04: | Subelement 91 contains the Visa customer validation code (Visa field 62.3, four bytes alphanumeric). Refer to the Visa's <i>VisaNet Authorization-Only Online Messages Technical Specifications</i> for more information on Visa customer validation codes. |
| When length field is 15: | Subelement 91 contains the transaction ID (Visa field 62.2, 15 bytes numeric).                                                                                                                                                                              |
| When length field is 19: | Subelement 91 contains the transaction ID (Visa field 62.2, 15 byte numeric) and customer validation code (Visa field 62.3, four bytes alphanumeric).                                                                                                       |

## Subelement 92 (CVC 2)

DE 48, subelement 92 (CVC 2) contains the CVC 2 value from the signature panel of the card.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 92                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n-3                       |
| Data Field             | Contents of positions 1–3 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

## Usage

Subelement 92 contains the CVC 2 value from the signature panel of the card when applicable.

Following is the usage of subelement 92 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | .   | C   |
| Authorization Request Response/0110           | CE  | .   | CE  |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |

## Values

Acquirers must not use 000 as a default when sending this subelement. Acquirers must only provide this subelement when requesting CVC 2 verification.

## Application notes

The contents of DE 48, subelement 92, when populated, are passed to the Dual Message Authorization System issuer in transactions acquired through the Single Message System.

For Mastercard Digital Enablement Service (MDES) transactions, DE 48, subelement 92 must be populated when provided by the merchant; it might contain a Dynamic Token Verification code.

## Subelement 92 (CVV2 Data (Visa Only))

DE 48, subelement 92 (CVV2 Data) consists of the Visa CVV2 presence ID, response type, and CVV2 value.

## Attributes

| Attribute              | Description                         |
|------------------------|-------------------------------------|
| Subelement ID          | 92                                  |
| Subelement Data Length | 2 positions                         |
| Data Representation    | ns-6                                |
| Data Field             | Contents of positions 1, 2, and 3–6 |

| Attribute     | Description   |
|---------------|---------------|
| Subfields     | 3             |
| Justification | See subfields |

## Usage

Subelement 92 contains the CVV2 value from the signature panel of the card when applicable.

Following is the usage of subelement 92 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | CE  | XE  | CE  |

## Values

### Subfield 1 (CVV2 Presence ID)

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

## Values

| Value | Description                                                   |
|-------|---------------------------------------------------------------|
| 0     | Merchant did not provide CVV2 or it was deliberately bypassed |
| 1     | CVV2 value present                                            |
| 2     | CVV2 is on card, but not legible                              |
| 9     | Cardholder states no CVV2 is on card                          |

## Subfield 2 (CVV2 Response Code)

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 2 |
| Justification       | N/A                    |

### Values

| Value | Description                                                                      |
|-------|----------------------------------------------------------------------------------|
| 0     | Only the normal response code in DE 39 should be returned by the issuer          |
| 1     | The normal response code and CVV2 response code should be returned by the issuer |

## Subfield 3 (CVV2 Value)

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | ns-4                      |
| Data Field          | Contents of positions 3–6 |
| Justification       | Right, with leading zeros |

### Values

CVV2 value from the signature panel of the card. Blank or space-fill as needed. No other special character is allowed.

## Subelement 93 (Fleet Card ID Request Data (Visa Only))

DE 48, subelement 93 (Fleet Card ID Request Data) contains the Fleet Card ID information on a Visa account, when applicable.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 93                    |
| Subelement Data Length | 2 positions           |
| Data Representation    | ans...19; LLVAR       |
| Data Field             | Contents of subfields |
| Subfields              | 2                     |
| Justification          | N/A                   |

### Usage

Following is the usage of subelement 93 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | P   | C   |

### Values

See subfields.

## Subfield 1 (Fleet Card ID Request Indicator)

DE 48, subelement 93, subfield 1 (Fleet Card ID Request Indicator) identifies the Fleet Card ID.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | ans-2       |

| Attribute  | Description              |
|------------|--------------------------|
| Data Field | Contents of position 1–2 |

### Values

\$\$ is Fleet Card ID

## Subfield 2 (Optional Free-form Informational Text)

DE 48, subelement 93, subfield 2 (Optional Free-form Informational Text) provides additional Point-of-Service (POS) information.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...17                   |
| Data Field          | Contents of positions 3–19 |

### Values

Additional Point-of-Service (POS) information (optional).

## Subelement 94 (Commercial Card Inquiry Request [Visa Only])

DE 48, subelement 94 (Commercial Card Inquiry Request) contains an indicator requesting a commercial card inquiry on a Visa account, when applicable.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 94                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | ans-4                     |
| Data Field             | Contents of subfields 1-2 |
| Subfields              | 2                         |
| Justification          | N/A                       |

## Usage

Following is the usage of subelement 94 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |

## Subfields

### Subfield 1 (Card Request Indicator)

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | ans-3                     |
| Data Field          | Contents of positions 1–3 |
| Justification       | N/A                       |

### Values

| Value                    | Description             |
|--------------------------|-------------------------|
| !01 (hexadecimal 5AF0F1) | Commercial Card Inquiry |

### Subfield 2 (Merchant Request for Commercial Card Type)

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 4 |
| Justification       | N/A                    |



### Values

| Value    | Description       |
|----------|-------------------|
| 0 (zero) | Request Indicator |

## Subelement 94 (Commercial Card Inquiry Response [Visa Only])

DE 48, subelement 94 (Commercial Card Inquiry Response) contains the commercial card inquiry response data as a result of a commercial card inquiry on a Visa account.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 94                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | ans-4                     |
| Data Field             | Contents of subfields 1-2 |
| Subfields              | 2                         |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 94 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |
| Authorization Advice Response/0130  | C   | P   | C   |

## Subfields

### Subfield 1 (Card Request Indicator)

| Attribute           | Description                                           |
|---------------------|-------------------------------------------------------|
| Data Representation | ans-3                                                 |
| Data Field          | Contents of positions 1–3 (Hexadecimal value: 5AF0F1) |
| Justification       | N/A                                                   |

### Values

| Value                    | Description             |
|--------------------------|-------------------------|
| !01 (hexadecimal 5AF0F1) | Commercial Card Inquiry |

### Subfield 2 (Visa Commercial Card Response)

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | ans-1                  |
| Data Field          | Contents of position 4 |
| Justification       | N/A                    |

### Values

Visa Commercial Card Response contains 0 (zero) when Visa has determined the transaction is not using a commercial card. Otherwise, Visa provides an alphabetical value indicating the type of Visa commercial purchasing card used. Refer to Field 48, Usage 27 in Visa's VisaNet Authorization-Only Online Messages Technical Specifications for the complete set of values.

## Subelement 95 (Mastercard Promotion Code)

DE 48, subelement 95 (Mastercard Promotion Code) contains the promotion code to identify unique use of a Mastercard product for a specific program or service established between issuers and merchants.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 95                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | an-6                      |
| Data Field             | Contents of positions 1–6 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Following is the usage of subelement 95 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request /0100                   | C   | P   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Request Response/0410                | CE  | XE  | CE  |
| Reversal Advice/0420                          | •   | X   | C   |

Program or service specific.

### Values

| Value  | Description                                                      |
|--------|------------------------------------------------------------------|
| ARGCTA | Installment payment transaction within Argentina                 |
| AGROF1 | Mastercard Agro Card                                             |
| BNDES1 | Brazil intracountry transactions using the Mastercard BNDES Card |
| CHLCTA | Installment payment transaction within Chile                     |
| COLCTA | Installment payment transaction within Colombia                  |
| GREECE | Installment payment transaction within Greece                    |
| HGMINS | Installment payment transaction for Georgia                      |
| MCGARS | Identifies Global Automated Service (GARS) Stand-In activity     |
| MCINST | Installment payment transaction                                  |
| MEXCTA | Installment payment transaction within Mexico                    |
| PARCEL | Installment payment transaction within Brazil                    |
| PERCTA | Installment payment transaction within Peru                      |
| PHINST | Installment payment transaction within Philippines               |
| PRYCTA | Installment payment transaction within Paraguay                  |
| URYCTA | Installment payment transaction within Uruguay                   |

## Subelement 95 (American Express CID Card Identifier (American Express Only))

DE 48, subelement 95 (American Express CID Card Identifier) contains the American Express Card Identifier (CID) number from the face of the American Express card, when applicable.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 95                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n-4                       |
| Data Field             | Contents of positions 1-4 |
| Subfields              | N/A                       |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Usage

Following is the usage of subelement 95 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | CE  | XE  | CE  |

### Values

The four-digit Card Identifier (CID) number on the front of the American Express card.

## Subelement 96 (Visa Market-Specific Data Identifier [Visa Only])

DE 48, subelement 96 (Visa Market-Specific Data Identifier) contains the market-specific data identifier.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 96                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 96 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org  | Sys  | Dst  |
|-----------------------------------------------|------|------|------|
| Authorization Request/0100                    | C    | P    | C    |
| Authorization Request Response/0110           | CE/C | XE/X | CE/C |
| Authorization Advice/0120: acquirer-generated | C    | P    | C    |
| Authorization Advice Response/0130            | CE   | XE/X | CE   |
| Reversal Request/0400                         | C    | P    | C    |
| Reversal Request Response/0410                | CE/C | XE/X | CE/C |

### Values

Refer to Field 62.4 (Market-Specific Data Identifier) in Visa's VisaNet Authorization-Only Online Messages Technical Specifications for the complete set of values.

## Subelement 97 (Prestigious Properties Indicator [Visa Only])

DE 48, subelement 97 (Prestigious Properties Indicator) contains the prestigious property indicator. For participants in the Visa Prestigious Lodging program (conditional).

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 97                     |
| Subelement Data Length | 2 positions            |
| Data Representation    | a-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Following is the usage of subelement 97 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice Response/0130            | CE  | XE  | CE  |

### Values

| Value | Description             |
|-------|-------------------------|
| D     | Visa established limits |
| B     | Visa established limits |
| S     | Visa established limits |

## Subelement 98 (Mastercard Corporate Fleet Card ID/Driver Number)

DE 48, subelement 98 (Mastercard Corporate Fleet Card ID/Driver Number) allows the corporate customer to verify the user of the card and enables more detailed reporting. It contains the ID of the user of a Fleet card, when applicable.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 98                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n...6; LLVAR              |
| Data Field             | Contents of positions 1-6 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

## Usage

Following is the usage of subelement 98 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

## Values

Cardholder unique.

## Application notes

If DE 48, subelement 98 (Mastercard Corporate Fleet Card ID/Driver Number) was present in the original 0100 AFD authorization message, subelement 98 must be the same value in acquirer-generated 0120 completion advice.

## Subelement 99 (Mastercard Corporate Fleet Card Vehicle Number)

DE 48, subelement 99 (Mastercard Corporate Fleet Card Vehicle Number) allows the corporate customer to verify the user of the card and enables more detailed reporting. It contains the ID of the vehicle used in conjunction with a Fleet card purchase, when applicable.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 99                        |
| Subelement Data Length | 2 positions               |
| Data Representation    | n...6; LLVAR              |
| Data Field             | Contents of positions 1-6 |
| Subfields              | N/A                       |
| Justification          | N/A                       |



## Usage

Following is the usage of subelement 99 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

## Values

Vehicle-specific.

## Application notes

If subelement 99 was present in the original 0100 AFD authorization message, subelement 99 must be the same value in acquirer-generated 0120 completion advice.

# DE 48, Dual Message Authorization System Edits

The Dual Message Authorization System performs edits as described in this section.

## DE 48, Proper Formatting

The Dual Message Authorization System will perform the following system edit to verify proper formatting of DE 48.

| WHEN...                                                                                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120—Acquirer-generated, or Reversal Request/0400 message contains DE 48 (Additional Data—Private Use) with subelements that have incorrect length and/or incorrect format (Data Representations), or a subelement ID appears more than once within the same data element. | <p>Sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 44 is 6 positions for subelement format errors:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 0480nn</li> </ul> |

## Examples

- When an edit error occurs on DE 48, the DE 44 data will be populated as these examples below:
  - Error on DE 48 subelement 42, subfield 1: 048042 (no subfield information is provided)
  - Error on DE 48 subelement 61: 048061
- DE 48 TCC format error will be responded with DE 44 = 048000.

For non-DE 48/DE 108/DE 112 format errors, DE 44 will only have DE information and no subelement information. For example, the format error in DE 22 will have DE 44 as 022.

## DE 48, TCC

The Dual Message Authorization System performs the following edit.

### Edit

| WHEN...                                                                                                                                            | THEN the Authorization System...                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message contains DE 48, TCC with an invalid value in position 1 | <p>Sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 = 048000</li> </ul> |

## DE 48, TCC and DE 3

The Dual Message Authorization System performs the following edits.

### Associated edits

| WHEN TCC contains...    | THEN the first two positions of DE 3 (Processing Code) must contain one of the following values... |
|-------------------------|----------------------------------------------------------------------------------------------------|
| C = Cash Disbursement   | 01 = Withdrawal<br>17 = Cash Disbursement<br>30 = Balance Inquiry                                  |
| P = Payment Transaction | 28 = Payment Transaction                                                                           |

| WHEN TCC contains...      | THEN the first two positions of DE 3 (Processing Code) must contain one of the following values...                                                  |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Z = ATM Cash Disbursement | 00 = Purchase<br>01 = Withdrawal<br>30 = Balance Inquiry<br>91 = PIN Unblock<br>92 = PIN Change<br>Only eligible acquirers may use values 91 or 92. |

#### Associated edits

| IF this edit... | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fails           | Sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30</li> <li>• DE 44 (Additional Response Data) = 003</li> </ul> |
| passes          | goes to the next edit                                                                                                                                                                                                                                                            |

#### Associated edits

| WHEN...              | THEN the Dual Message Authorization System...                                                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TCC contains a space | assigns in an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 message, a TCC value based on the acceptor business code/merchant category code (MCC) in DE 18 (Merchant Type) except under the following conditions |

### Associated edits

| WHEN...                                                                                                                                                       | THEN the Dual Message Authorization System...                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type) contains value 00 (Purchase)                                                                 | assigns TCC value T (Phone, Mail, or Electronic Commerce Order). |
| and                                                                                                                                                           |                                                                  |
| DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) contains value 00 (PAN entry mode unknown) or value 01 (PAN manual entry) |                                                                  |
| and                                                                                                                                                           |                                                                  |
| DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence) does not contain value 0 (Cardholder Present)                                       |                                                                  |

## DE 48, subelement 14

The Dual Message Authorization System will perform the following system edits.

### Combo Cards with a Single PAN

| WHEN...                                                                                                                                                                                                                                             | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request/0100 contains DE 48 (Additional Data: Private Use), subelement 14 (Account Type Indicator) with values other than C (Credit Transaction) or D (Debit Transaction)                                                         | Sends the acquirer an Authorization Request Response/0110 with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048014 (indicating the data element in error)</li> </ul>                                    |
| the Authorization Advice/0120: acquirer-generated or the Reversal Request/0400 message contains DE 48 (Additional Data: Private Use), subelement 14 (Account Type Indicator) with values other than C (Credit Transaction) or D (Debit Transaction) | forwards the acquirer an Authorization Advice Response/0130 or a Reversal Request Response/0410 with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048 (indicating the data element in error)</li> </ul> |

## DE 48, Subelement 22

The Dual Message Authorization System will perform the following system edits.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | THEN the Dual Message Authorization System...                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), subelement 22 (Multi-Purpose Merchant Indicator), subfield 01 (Low-Risk Merchant Indicator), subfield 02 (Single Tap Indicator), or subfield 03 (Response to PIN Request) in the Authorization Request/0100 message is not alphanumeric or is an invalid length                                                                                                                                                                          | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048022 (indicating the data element in error)</li> </ul> |
| the acquirer provides DE 48, subelement 22, subfield 01, subfield 02, or subfield 03 in the Authorization Request/0100 message and the issuer is not in the European Economic Area (EEA)                                                                                                                                                                                                                                                                                      | will removed subfield 01, subfield 02, or subfield 03 in the message sent to the issuer.                                                                                                                                                                       |
| the Dual Message System receives an Authorization Request/0100 message that includes a valid value in DE 48, subelement 22, subfield 02 indicating the merchant terminal supports single tap processing a response, and DE 39 (Response Code) = 65 (Decline: Exceeds withdrawal count limits) from the On-Behalf Contactless Counters Management system                                                                                                                       | sends the acquirer an Authorization Request Response/0110 containing the results. When the issuer supports duplicate ATC, the Dual Message Authorization System will add DE 48, subelement 22, subfield 04 (Issuer PIN Request in a Single Tap Mode).          |
| the Dual Message System receives a response from the issuer with DE 39 (Response Code) = 65 (Decline: Exceeds withdrawal count limits): <ul style="list-style-type: none"> <li>And the Authorization Request/0100 message includes a valid value in DE 48, subelement 22, subfield 02 indicating the merchant terminal supports single tap processing</li> <li>And DE 48, subelement 22, subfield 04 is present in the Authorization Request Response/0110 message</li> </ul> | sends the acquirer the Authorization Request Response/0110 message with DE 48, subelement 22, subfield 04 to indicate to the acquirer that an issuer supports a PIN in a single tap mode in a subsequent Authorization Request/0100 message.                   |
| the Dual Message System receives DE 48, subelement 22, subfield 04 in an Authorization Request Response/0110 message and subfield 04 does not contain alphanumeric data or has an invalid length                                                                                                                                                                                                                                                                              | will consider the response from the issuer invalid and process consistent with current invalid issuer response processing.                                                                                                                                     |
| the Dual Message System receives DE 48, subelement 22, subfield 01 in an acquirer-generated advice (Authorization Advice/0120 or Reversal Request/0400 message)                                                                                                                                                                                                                                                                                                               | removes the subfield 01 from the Authorization Advice/0120 or the Reversal Request/0400 message before sending to the issuer.                                                                                                                                  |

| WHEN...                                                                                                                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| the Dual Message System receives DE 48, subelement 22, subfield 02 or subfield 03 in a transaction message: <ul style="list-style-type: none"> <li>• Authorization Advice/0120</li> <li>• Reversal Request/0400 message</li> </ul> | removes the subfield 02 or subfield 03 from the Authorization Advice/0120 or the Reversal Request/0400 message before sending to the issuer. |
| the acquirer provides DE 48, subelement 22, subfield 01 in an Authorization Request/0100 message for an ATM transaction                                                                                                            | removes the subfield 01 from the Authorization Request/0100 message before sending to the Issuer.                                            |
| the Dual Message System receives DE 48, subelement 22, subfield 02 or subfield 03 in an ATM transaction message: <ul style="list-style-type: none"> <li>• Authorization Request/0100</li> </ul>                                    | removes the subfield 02 or subfield 03 from the Authorization Request/0100 message before sending to the Issuer.                             |
| the Dual Message System receives DE 48, subelement 22, subfield 01, subfield 02, or subfield 03 in an Authorization Request Response/0110 message                                                                                  | removes subfield 01, subfield 02, or subfield 03 from the Authorization Request Response/0110 message before sending to the acquirer.        |

## DE 48, subelement 26

The Dual Message Authorization System performs the following edits.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                             | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request/0100 or Authorization Advice/0120 message includes DE 48 (Additional Data: Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier) contains any non-alphanumeric characters or contains all spaces or all zeros | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130: system-generated where: <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 048026 (indicating the data element in error)</li> </ul> |

## DE 48, subelement 35

The Dual Message Authorization System performs the following edits.

### Associated edits

Following are Authorization Request/0100 message edits.

| WHEN...                                                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer sends an Authorization Request/0100 message containing DE 48 (Additional Data—Private Use), subelement 35 (Contactless Non-Card Form Factor Request/Response) | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>• DE 39 = 30 (Format error)</li> <li>• DE 44 = 048035</li> </ul> |

### Associated edits

The issuer-branded website will perform the following processing based on the Authorization Request Response/0110 messages.

| WHEN...                                                                                                                                                                                                                                                                                          | THEN the issuer-branded website...                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request Response/0110 message contains DE 48, subelement 35, value A                                                                                                                                                                                                           | arranges for the request contactless card or device to be sent to the cardholder                                                                                                                                       |
| the Authorization Request Response/0110 message contains DE 48, subelement 35, value D                                                                                                                                                                                                           | declines the cardholder's request for the contactless card or device and instructs the cardholder to contact the issuer for more information                                                                           |
| the Authorization Request/0100 message contains DE 48, subelement 35 and the Authorization Request Response/0110 message does not contain DE 48, subelement 35<br>or<br>the Authorization Request Response/0110 message containing DE 48, subelement 35 does not contain a valid value of A or D | arranges for the requested contactless card or device to be sent to the cardholder when:<br>Subelement 83 (Address Verification Service Response) is X or Y<br>and<br>Subelement 87 (Card Validation Code Result) is M |

## DE 48, subelement 37

The Dual Message Authorization System performs the following edits.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                  | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data: Private Use), subelement 37 (Additional Merchant Data), subfields 1 (Payment Facilitator ID) or 2 (Independent Sales Organization ID) has values other than 0–9 or contains all zeros in the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = value 30 (Format error)</li> <li>DE 44 (Additional Response Data) = value 048037 (indicating the data element in error)</li> </ul> |
| DE 48, subelement 37, subfield 3 (Sponsored Merchant) contains all spaces or all zeros in the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message                                                                                                                        | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: <ul style="list-style-type: none"> <li>DE 39 = value 30 (Format error)</li> <li>DE 44 = value 048037</li> </ul>                                                                                   |

### Merchant Country of Origin Identifier

| WHEN...                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data), subelement 37 (Additional Merchant Data), subfield 4 (Merchant Country of Origin) is non-numeric.                 | sends the acquirer an Authorization Request Response/0110 message with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048037</li> </ul> |
| DE 48 (Additional Data), subelement 37 (Additional Merchant Data) subfield 4 (Merchant Country of Origin) is not a valid ISO country code. | sends the acquirer an Authorization Request Response/0110 message with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048037</li> </ul> |
| DE 48 (Additional Data), subelement 37 (Additional Merchant Data), subfield 4 (Merchant Country of Origin) is not 3 numeric digits.        | sends the acquirer an Authorization Request Response/0110 message with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048037</li> </ul> |



### Associated edits

| When...                                                                                                                                                                                                                                  | THEN the Dual Message Authorization System...                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message contains DE 48 (Additional Data), subelement 37 (Additional Merchant Data), subfield 5 (Merchant Payment Gateway ID) with a length that is not 11 | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: <ul style="list-style-type: none"> <li>DE 39 = value 30 (Format error)</li> <li>DE 44 = value 048037</li> </ul> |
| an Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message contains DE 48, subelement 37, subfield 5 with non-numeric characters                                                                             | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: <ul style="list-style-type: none"> <li>DE 39 = value 30 (Format error)</li> <li>DE 44 = value 048037</li> </ul> |

## DE 48, subelement 38

The Dual Message Authorization System performs the following edits.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Dual Message Authorization System finds that the entire account range no longer participates in Enhanced Value, Product Graduation, or High Value and<br>at least one cardholder account within the account range previously participated in Enhanced Value, Product Graduation, or High Value | adds DE 48, subelement 38 (Account Category), value Z to the Authorization Request/0100 message for the following transaction types: <ul style="list-style-type: none"> <li>Purchases as indicated by DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) = 00</li> <li>Purchase with Cash Back as indicated by DE 3, subfield 1 = 09</li> </ul> and<br>the Dual Message Authorization System forwards the Authorization Request/0100 message to the issuer. |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the Dual Message Authorization System provides DE 48, subelement 38 in the Authorization Request/0100 message to the issuer</p> <p>and</p> <p>the issuer populates DE 39 (Response Code) in the Authorization Request Response/0110 message with one of the following values that indicate an authorization approval:</p> <ul style="list-style-type: none"> <li>• 00 (Approved or completed successfully)</li> <li>• 08 (Honor with ID)</li> <li>• 10 (Partial Approval)</li> <li>• 87 (Purchase Amount Only, No Cash Back )</li> </ul> <p>and</p> <p>DE 38 (Authorization ID Response), position 6 in the Authorization Request Response/0110 message does not equal the value of DE 48, subelement 38 of the original Authorization Request/0100 message that was provided to the issuer by the Dual Message Authorization System.</p> | <p>sends an Authorization Response Negative Acknowledgment/0190 to the issuer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30</li> <li>• DE 44 (Additional Response Data) = 038</li> </ul> <p>and</p> <p>sends the Authorization Request/0100 message to the Stand-in System for processing.</p> <p>Stand-In will ensure that DE 38, position 6 of the Authorization Request Response/0110 message matches the value provided by the Dual Message Authorization System in DE 48, subelement 38 of the original Authorization Request/0100 message.</p> |
| <p>the Dual Message Authorization System provides DE 48, subelement 38 in the Authorization Request/0100 message to the issuer</p> <p>and</p> <p>the issuer populates DE 39 (Response Code) in the Authorization Request Response/0110 message with the value 85 (Not declined)</p> <p>and</p> <p>DE 38 is present</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>sends an Authorization Response Negative Acknowledgment/0190 to the issuer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30</li> <li>• DE 44 (Additional Response Data) = 039</li> </ul> <p>and</p> <p>Sends the Authorization Request/0100 message to the Stand-in System for processing.</p>                                                                                                                                                                                                                                                       |

| WHEN...                                                                                                                                                                                                                              | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 38, position 6 in the Authorization Request Response/0110 message does not equal the value of DE 48, subelement 38 of the original Authorization Request/0100 message that was provided to the issuer by the Authorization System | <p>sends an Authorization Response Negative Acknowledgment/0190 to the issuer where:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 039</li> </ul> <p>and</p> <p>Sends the Authorization Request/0100 message to the Stand-in System for processing.</p> <p>Stand-In will ensure that DE 38, position 6 of the Authorization Request Response/0110 message matches the value provided by the Dual Message Authorization System in DE 48, subelement 38 of the original Authorization Request/0100 message.</p> |
| DE 48, subelement 38 is included in the Authorization Request Response/0110 message from the issuer                                                                                                                                  | removes DE 48, subelement 38 from the Authorization Request Response/0110 message before providing the 0110 message to the acquirer.                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## DE 48, Subelement 42 and Subelement 43

The Dual Message Authorization System performs these edits on Authorization Request/0100 and Authorization Advice/0120 messages.

### Associated edits

| WHEN DE 48, subelement 42, subfield 01, position 3 is... | THEN DE 48, subelement 43... |
|----------------------------------------------------------|------------------------------|
| 0 or is not present                                      | cannot contain UCAF data.    |
| 1, 2, 3, 4, 5, 6, or 7                                   | must contain UCAF data.      |

### Associated edits

| IF this edit... | THEN the Authorization System...                                                                                                                                                    |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fails           | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>DE 39 = 30 (Format error)</li> <li>DE 44 = 048042</li> </ul> |

| IF this edit...                                    | THEN the Authorization System...                                                                                                     |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| is missing or has an invalid AAV leading indicator | downgrades the edit which downgrades missing or invalid AAV leading indicator with associated Identity Check SLI value to SLI = 210. |
| passes                                             | proceeds to the next edit.                                                                                                           |

#### Associated edits

| WHEN DE 48, subelement 43... | THEN DE 48, subelement 42, subfield 01, position 3 must equal... |
|------------------------------|------------------------------------------------------------------|
| contains UCAF data           | 1, 2, 3, 4, 5, 6, or 7                                           |
| does not contain UCAF data   | 0 or not be present                                              |

#### Associated edits

| IF this edit...                                    | THEN the Authorization System...                                                                                                                                                                              |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fails                                              | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 (Response Code) = 30 (Format error)<br>DE 44 (Additional Response Data) = 048043 (indicating the data element in error) |
| is missing or has an invalid AAV leading indicator | downgrades the edit which downgrades missing or invalid AAV leading indicator with associated Identity Check SLI value to SLI = 210.                                                                          |
| passes                                             | the Dual Message Authorization System goes to the next edit.                                                                                                                                                  |

The Dual Message Authorization System performs the following edits on Authorization Request/0100 messages.

## DSRP transactions initiated with MDES tokens

### Associated edits

| WHEN acquirer submits...                                                            |                                        | THEN Mastercard will validate/<br>modify Security Level Indicator<br>(SLI) to... |
|-------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------|
| UCAF DE 48, SE 43                                                                   | Digital Payment Data DE 104,<br>SE 001 | SLI DE 48, SE 42, SF 1                                                           |
| DSRP Cryptogram                                                                     | N/A                                    | SLI based on DSRP cryptogram                                                     |
| N/A                                                                                 | DSRP Cryptogram                        | SLI based on DSRP cryptogram                                                     |
| Identity Check Fully<br>Authenticated or Merchant<br>Attempt AAV                    | DSRP Cryptogram                        | SLI based on AAV                                                                 |
| Identity Check Fully<br>Authenticated AAV or Identity<br>Check Merchant Attempt AAV | N/A                                    | SLI based on AAV                                                                 |

## DSRP merchant-initiated transactions, where SLI value is not 247

### Associated edits

| WHEN...                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DSRP transaction initiated with a Secure Element or Cloud token, DE 22, subfield 1 = 81 (POS entry mode) and the Authorization Request/0100 contains DE 48, subelement 42, not equal to 247 and DE 48, subelement 43 and DE 104, subelement 001 is not present | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048042</li> </ul> |

## Non-DSRP transactions initiated with an MDES token and submitted with a merchant attempt Identity Check AAV

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                               | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a transaction initiated with a Secure Element or Cloud token, DE 22, subfield 1 = 81 (POS entry mode) and the Authorization Request/0100 request contains DE 48, subelement 43, pos1 = h (Merchant Attempt Identity Check AAV)*<br><br>*Mastercard Identity Check is no longer generating h SPA1 AAVs | if dynamic expiration date and dynamic token verification code was successfully validated by the MDES service, then the Dual Message Authorization System goes to the next edit, otherwise sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 0480nn (where nn is the subelement number)</li> </ul> |

## DE 48, subelement 43 (Static AAV)

The following edits on Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages are performed on transactions submitted under the Maestro Recurring Payments Program, the Mastercard Utility Payment Program, and the Maestro Low Risk Merchant Program.

### Associated edits

| WHEN the Mastercard assigned static AAV in DE 48, subelement 43, position 1 begins with... | THEN the Dual Message Authorization System...                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 and the acceptance brand is not Maestro                                                  | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br><br>DE 39 (Response Code) = 30 (Format error)<br><br>DE 44 = 048043 (identifying the data element in error) |
| 5 and the acceptance brand is not Maestro                                                  | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br><br>DE 39 = 30<br><br>DE 44 = 048043                                                                        |

| <b>WHEN the Mastercard assigned static AAV in DE 48, subelement 43, position 1 begins with...</b> | <b>THEN the Dual Message Authorization System...</b>                                                                                                   |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| any value other than 3 or 5 and the acceptance brands are Mastercard, Maestro, or both            | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br><br>DE 39 = 30<br><br>DE 44 = 048043 |

## DE 48, subelement 42 and DE 61

The Dual Message Authorization System performs the following edits.

### Associated edits

The following edits apply on Electronic Commerce transactions.

| <b>WHEN...</b>                                                                                                          | <b>THEN...</b>                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| DE 48, subelement 42, subfield 1, position 3 contains value 1, 2, or 3, and<br><br>DE 61, subfield 10, contains value 6 | DE 61, subfield 4 contains value 4 or 5 and<br><br>DE 61, subfield 7, must not contain value 2 |

### Associated edits

| <b>IF this edit...</b> | <b>THEN the Dual Message Authorization System...</b>                                                                    |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| fails                  | generates a format error, indicated by:<br><br>DE 39 (Response Code) = 30<br><br>DE 44 (Additional Response Data) = 061 |
| passes                 | the Dual Message Authorization System goes to the next edit.                                                            |

### Associated edits

The following edits apply on Identity Check Phone Order transactions.

| WHEN...                                                                                                      | THEN...                                                                                           |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| DE 48, subelement 42, subfield 1, position 3 contains value 1 or 2 and<br>DE 61, subfield 7 contains value 2 | DE 61, subfield 4 must contain a value 3 or 4 and<br>DE 61, subfield 10 must not contain value 6. |

#### Associated edits

| IF this edit... | THEN the Dual Message Authorization System...                       |
|-----------------|---------------------------------------------------------------------|
| fails           | generates a format error indicated by:<br>DE 39 = 30<br>DE 44 = 061 |
| passes          | the Dual Message Authorization System goes to the next edit.        |

## DE 48, subelement 50

Edits for DE 48 (Additional Data, Private Use), subelement 50 (Embedded Interchange Data) are as noted.

#### Associated edits

| When...                                                                                                                               | Then the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer provides DE 48, subelement 50, but does not participate in the Embedded Flexible Interchange (EFI) program               | sends the acquirer Authorization Request Response/0110 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 58</li> </ul>                                                                                            |
| the acquirer provides DE 48, subelement 50, but DE 02 indicates a Primary Account Number that does not participate in the EFI program | sends the acquirer Authorization Request Response/0110 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 = 57</li> </ul>                                                                                                            |
| the acquirer provides DE 48, subelement 50 but DE 03, subfield 1 is not 00 (Purchase), 20 (Refund)                                    | sends the acquirer Authorization Request Response/0110 message, Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 003</li> </ul> |



| When...                                                                             | Then the Dual Message Authorization System...                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer provides DE 48, subelement 50 with an invalid combination of subfields | sends the acquirer Authorization Request Response/0110 message, Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048050</li> </ul> |

#### DE 48, subelement 50, subfield 01 (Interchange Indicator)

| When ...                                                                                                                                                                    | Then the Dual Message Authorization System...                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, without DE 48, subelement 50, subfield 01 | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 048050</li> </ul> |
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 01 with a value other than 'Y'   | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 048050</li> </ul> |

#### DE 48, subelement 50, subfield 02 (Interchange Percentage)

##### Associated edits

| When ...                                                                                                                                                                    | Then the Dual Message Authorization System...                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 02 with a length that is not six | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 048050</li> </ul> |

| When ...                                                                                                                                                                                        | Then the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 02 containing non-numeric characters                 | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data = 048050</li> </ul> |
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 02 containing a value greater than 100.000% (100000) | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data = 048050</li> </ul> |

#### DE 48, subelement 50, subfield 03 (Unit Fee Amount)

| When ...                                                                                                                                                                        | Then the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 03 with a length that is not twelve  | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data = 048050</li> </ul> |
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 03 containing non-numeric characters | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data = 048050</li> </ul> |

#### DE 48, subelement 50, subfield 04 (Flat Interchange Amount)

| When ...                                                                                                                                                                       | Then the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 04 with a length that is not twelve | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data = 048050</li> </ul> |

| When ...                                                                                                                                                                        | Then the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 04 containing non-numeric characters | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data = 048050</li> </ul> |

#### DE 48, subelement 50, subfield 05 (Payment Party)

| When ...                                                                                                                                                                         | Then the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 05 with a value other than 'A' or 'I' | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data = 048050</li> </ul> |

## DE 48, Subelement 61

The Dual Message Authorization System performs the following edits.

#### Associated edits

| WHEN...                                                     | THEN the Dual Message Authorization System...                                         |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------|
| DE 48, subelement 61 is present and DE 4 contains all zeros | drops DE 48, subelement 61 from the Authorization Request/0100 message to the issuer. |

#### Associated edits

| WHEN DE 48, subelement 61, subfield 3 is...                                                               | THEN the Dual Message Authorization System...                                                      |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1 (Merchant terminal verified the purchase items against an Inventory Information Approval System [IIAS]) | validates that DE 48, subelement 32 (if present) contains a valid Mastercard Assigned ID for IIAS. |

### Associated edits

| IF the...                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mastercard Assigned ID is valid                                                              | <p>updates the value in DE 48, subelement 61, subfield 3 as follows:</p> <ul style="list-style-type: none"> <li>• If the issuer participates in real-time substantiation, sends DE 48, subelement 61, subfield 3, value 1 (Merchant terminal verified the purchased items against an IIAS)</li> <li>• If the issuer does not participate in real-time substantiation sends DE 48, subelement 61, subfield 3, value 0 (Merchant terminal did not verify the purchased items against an IIAS).</li> </ul>                               |
| Mastercard Assigned ID is not valid or not present in the Authorization Request/0100 message | <p>updates the value in DE 48, subelement 61, subfield 3 as follows:</p> <ul style="list-style-type: none"> <li>• If the issuer participates in real-time substantiation, sends DE 48, subelement 61, subfield 3, value 4 (Transaction was submitted as real-time substantiated but from a non-IIAS certified merchant).</li> <li>• If the issuer does not participate in real-time substantiation sends DE 48, subelement 61, subfield 3, value 0 (Merchant terminal did not verify the purchased items against an IIAS).</li> </ul> |

### Associated edits

The following edits are on Authorization Request/0100 and Authorization Advice/0120—Acquirer-generated messages.

| WHEN...                                                                                                                                                                                                                                                                                                                                                               | THEN the Dual Message Authorization System...                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the acquirer sends DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator), value 2 (Merchant claims exemption from IIAS rules based on the IRS 90 percent rule) in an Authorization Request/0100 message or Authorization Advice/0120: acquirer-generated message</p> <p>and</p> <p>the issuer does not participate in real-time substantiation</p> | <p>forwards the issuer the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message containing DE 48, subelement 61, subfield 3, value 0 (Merchant terminal did not verify the purchased items against an IIAS).</p> |

| WHEN...                                                                                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer sends DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator), value 2 (Merchant claims exemption from IAS rules based on the IRS 90 percent rule) in an Authorization Request/0100 message or Authorization Advice/0120: acquirer-generated message<br><br>and<br><br>the issuer does participate in real-time substantiation | forwards the issuer the Authorization Request/0100 or Authorization Advice/0120—Acquirer-generated message containing DE 48, subelement 61, subfield 3, value 2 (Merchant claims exemption from using the IAS, based on the IRS 90 percent rule). |
| the acquirer sends DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator), value other than 0, 1, or 2 in Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message                                                                                                                                              | declines the Authorization: acquirer-generated messages where: <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30</li> <li>• DE 44 (Response Data) = 048061</li> </ul>                                                           |

## DE 48, subelement 66

The Dual Message Authorization System will perform the following system edits.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                          | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 or Authorization Advice/0120 message contains DE 48 (Additional Data: Private Use), subelement 66 (Authentication Data) subfield 1 (Program Protocol) that is the incorrect length or incorrect data representation (alphanumeric) | sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message with: <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format Error)</li> <li>• DE 44 (Additional Response Data) = 048066</li> </ul>                             |
| an Authorization Request/0100 or Authorization Advice/0120 message contains DE 48, subelement 66 (Authentication Data) subfield 2 (Directory Server Transaction ID) that is not 36 bytes in length or filled with spaces                                         | reject the message and forward to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message with: <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format Error)</li> <li>• DE 44 (Additional Response Data) = 048066</li> </ul> |

## DE 48, subelement 77

The Dual Message Authorization System will perform the following system edits.

### Bill Pay

#### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                     | THEN the Dual Message Authorization System...                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request/0100 message or the Reversal Request/0400 message contains DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) with value C91, C92, C93, or C94 and DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is not value 00 (Purchase) | will reject the transaction with <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 12 (Invalid Transaction)</li> </ul> |

### High-Risk Purchase

#### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                       | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) is present and contains value P70 (Cryptocurrency) but the MCC is not 6051 (Quasi Cash—Merchant)<br>or<br>DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) is present and contains value P71 (High-risk Securities) but the MCC is not MCC 6211 (Securities—Brokers/Dealers) | rejects the message and sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 048077 (indicating the data element in error)</li> </ul> |

## Mastercard Send and Funding Transactions

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and</p> <p>The originating institution identified by the ICA provided in DE 32 (Acquiring Institution ID Code) has not registered to process transactions via the Mastercard Send platform</p>                              | <p>rejects the transaction and sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 58 (Transaction not permitted to issuer/cardholder).</p>                                                  |
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and</p> <p>the brand product code associated with the transaction does not participate in the Mastercard Funding Program</p>                                                                                                | <p>rejects the transaction and sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 12 (Invalid Transaction).</p>                                                                             |
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and</p> <p>the Unique Transaction reference number (DE 108 [Additional Transaction Reference Data], subelement 03 [Transaction Reference Data], subfield 1 [Unique Transaction Reference]) contains all zeros or spaces</p> | <p>sends the originating institution an Authorization Request Response/0110 message to include the following:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108003 (Additional Data: Private Use)</li> </ul> |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund</p> <p>and</p> <p>the issuer account range does not participate in Mastercard Send Payment Transactions Program, Mastercard Funding Transaction Program or both.</p>                                                                                                                                                                                                 | <p>sends the originating institution an Authorization Request Response/0110 message to include the following: DE 39 (Response Code) = 57 (Transaction not permitted to issuer or cardholder)</p>                                                                                             |
| <p>the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li> <li>DE 18 (Merchant Type) does not contain MCC 6538</li> <li>DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65</li> </ul>                  | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 018 (indicating the data element in error)</li> </ul> |
| <p>the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li> <li>DE 18 (Merchant Type) does not contain MCC 4829 or MCC 6540</li> <li>DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65</li> </ul> | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 018 (indicating the data element in error)</li> </ul> |



| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"> <li>• DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li> <li>• DE 18 (Merchant Type) contains MCC 6538</li> <li>• DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65</li> <li>• DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction—Quasi Cash Disbursement).</li> </ul>                    | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)</li> </ul> |
| <p>the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"> <li>• DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li> <li>• DE 18 (Merchant Type) contains MCC 4829 or MCC 6540</li> <li>• DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65</li> <li>• DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction: Quasi Cash Disbursement).</li> </ul> | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)</li> </ul> |

DE 48, subelement 78

The Dual Message Authorization System performs the following edits.

Associated edits

| WHEN DE 48, Subelement 78 is...     | THEN DE 2...                                           |
|-------------------------------------|--------------------------------------------------------|
| D (U.S. Deferred Billing Indicator) | DE 2 (Primary Account Number [PAN]) must be a Visa PAN |

Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | THEN the Dual Message Authorization System...                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization System receives an Authorization Request/0100 and Reversal Request/0400 for a Visa-branded, Dual Message System (Authorization) transaction with DE 48 (Additional Data—Private Use), subelement 78 (Payment Service Indicators [Visa Only]) where: <ul style="list-style-type: none"><li>Subfield 1 (Spend Qualified Indicator) contains a value other than Space</li><li>Subfield 2 (Dynamic Currency Conversion Indicator) contains a value other than Y or Space</li><li>Subfield 3 (U.S. Deferred Billing Indicator) contains a value other than D or Space</li><li>Subfield 4 (Visa Checkout Indicator) contains a value other than Y or Space</li><li>Subfield 5 (Message Reason Code) contains a value other than Space, 0, 1, 2, 3, 4, 5</li><li>Subfield 6 (Reserved for Future Use) contains a value other than Space</li></ul> | sends an Authorization Request Response/0110 and Reversal Request/0400 message where: <ul style="list-style-type: none"><li>DE 39 (Response Code) = 30 (Format error)</li><li>DE 44 (Additional Response Data) = 048078 (indicating the data element in error)</li></ul> |

## DE 48, subelement 82

The Dual Message Authorization System performs the following edits.

### Associated edits

| WHEN DE 48, Subelement 82 is...                                                                                                                                                      | THEN the Dual Message Authorization System...                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 51 (AVS-Only)                                                                                                                                                                        | sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 12 (Invalid Transaction)                                                                                  |
| 52 (AVS Request) and DE 39 is 00 (Approved) or 08 (Honor with ID) or 85 (Not declined) and DE 48 SE 83 (AVS Response) is present with a value not equal to X, Y, A, W, Z, N, U, R, S | sends the issuer an Authorization Response Negative Acknowledgment/0190 message where: <ul style="list-style-type: none"> <li>DE 39 = 30 (Format Error)</li> <li>DE 44 = 048083</li> </ul> |

## DE 48, subelement 84

The Dual Message Authorization System performs the following edits.

### Associated edits

| WHEN...                                                                                                 | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 42 and/or DE 48, subelement 43 are invalid due to token requirements that are not met | returns to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 0480nn (where nn is the subelement number)</li> <li>DE 48 (Additional Data—Private Use), subelement 84 = 04</li> </ul> <p><b>NOTE: Issuers do not receive decline advice messages for transactions that are rejected back to acquirers for format errors (DE 39 = 30).</b></p> |

## DE 48, subelement 86

The Dual Message Authorization System performs the following edits.

### Associated edits

| WHEN DE 48, Subelement 86 is... | THEN DE 2...                                           |
|---------------------------------|--------------------------------------------------------|
| R (Relationship Participant)    | DE 2 (Primary Account Number [PAN]) must be a Visa PAN |

## DE 48, subelement 95

The Dual Message Authorization System performs the following system edit on Authorization Request/0100 and Reversal Request/0400 messages for Brazil installment payment transactions.

### Associated edits

| WHEN...                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), subelement 95 (Mastercard Promotion Code) contains the value PARCEL, | rejects the transaction sending the acquirer an Authorization Request Response/0110 or Reversal Request Response/0410 where:                                                                                    |
| And                                                                                                       | <ul style="list-style-type: none"> <li>DE 39 (Response Code) contains value 30 (Format error)</li> <li>DE 44 (Additional Response Data) contains value 112001 (indicating the data element in error)</li> </ul> |
| DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data) is not present         |                                                                                                                                                                                                                 |

## DE 48, in Authorization Request Response

The following edit applies to DE 48 and the Authorization Request Response/0110 message.

### Associated edits

| WHEN the issuer...                                                                                                                                                  | THEN the Authorization Request Response/0110 message must...                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| provides response data, such as AVS response, Merchant Advice Code, CVC 2 response, and CVC 1 response, in DE 48 of the Authorization Request Response/0110 message | contain the TCC as the first byte of data within DE 48<br>and<br>contain all the subelements present in DE 48 of the original Authorization Request/0100 message that also are defined in the Authorization Request Response/0110 message. |
| does not provide response data                                                                                                                                      | not contain DE 48, including the TCC.                                                                                                                                                                                                      |

## DE 49 (Currency Code, Transaction)

DE 49 (Currency Code, Transaction) is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction) and DE 95 (Replacement Amounts), subfield 1 (Actual Amount, Transaction).

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–3 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 49 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | M   | P   | M   |
| Authorization Request Response/0110                                                             | CE  | XE  | ME  |
| Authorization Advice/0120: acquirer-generated                                                   | M   | P   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | XE  | ME  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | ME  | XE  | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | XE  | ME  |
| Reversal Request/0400                                                                           | M   | P   | M   |
| Reversal Request Response/0410                                                                  | CE  | XE  | ME  |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | •   |

### Values

All currency codes must be selected from the numeric ISO standard currency codes. ISO standard currency codes identify DE 49. A list of valid values is available in the *Quick Reference Booklet*.

### Application notes

This data element is defined and used identically within all Mastercard programs and services. Acquirers will receive an Authorization Request Response/0110 message containing a format error in DE 39 when the currency code in DE 49 is a currency code other than those in the ISO Standard Currency Codes table, invalid or pseudo codes will not be accepted.

## DE 50 (Currency Code, Settlement)

DE 50 (Currency Code, Settlement) defines the currency of DE 5 (Amount, Settlement), DE 29 (Amount, Settlement Fee), and DE 95 (Replacement Amounts), subfield 2 (Actual Amount, Settlement).

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–3 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 50 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | .   | X   | C   |
| Authorization Request Response/0110                                                             | CE  | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | .   | X   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | .   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | X   | C   |
| Reversal Request/0400                                                                           | .   | X   | C   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420                                                                            | .   | X   | C   |

| Message                       | Org | Sys | Dst |
|-------------------------------|-----|-----|-----|
| Reversal Advice Response/0430 | CE  | XE  | .   |

### Values

All currency codes must be selected from the numeric ISO standard currency codes.

ISO standard currency codes identify DE 50. A list of valid values is in the *Quick Reference Booklet*.

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

As of the date of this publication, all Mastercard programs and services use U.S. dollars (840) as the currency of settlement for programs and services that the Dual Message Authorization System supports.

The Dual Message Authorization System includes this data element if the customer chooses to receive settlement amount-related data elements.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

## DE 51 (Currency Code, Cardholder Billing)

DE 51 (Currency Code, Cardholder Billing) defines the currency of DE 6 (Amount, Cardholder Billing), DE 8 (Amount, Cardholder Billing Fee), and DE 95 (Replacement Amounts), subfield 3 (Actual Amount, Cardholder Billing).

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–3 |
| Subfields           | N/A                       |
| Justification       | N/A                       |



## Usage

Following is the usage of DE 51 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | •   | X   | M   |
| Authorization Request Response/0110                                                             | CE  | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | •   | X   | M   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (Responding to an acquirer-generated 0120) | CE  | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | ME  | XE  | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | X   | C   |
| Reversal Request/0400                                                                           | •   | X   | M   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420                                                                            | •   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | •   |

## Values

All currency codes must be selected from the numeric ISO standard currency codes.

ISO standard currency codes identify DE 51. A list of valid values is in the *Quick Reference Booklet*.

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

## DE 52 (Personal ID Number [PIN] Data)

DE 52 (Personal ID Number [PIN] Data) contains a number assigned to a cardholder intended to uniquely identify that cardholder at the point of interaction. The use of the PIN is subject to bilateral agreement. The data element may contain the PIN itself or a derivative. This data element transmits PIN information from acquirers to issuers (or to the network) for PIN verification or validation.

### Attributes

| Attribute           | Description                              |
|---------------------|------------------------------------------|
| Data Representation | b-8                                      |
| Data Length         | N/A                                      |
| Data Field          | Contents of bit positions 1–64 (8 bytes) |
| Subfields           | N/A                                      |
| Justification       | N/A                                      |

### Usage

Following is the usage of DE 52 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Values

The network supports PINs from four to 12 characters long.

The encrypted PIN block always is eight bytes (64 bits) long regardless of the original PIN length.

CIS uses the Data Encryption Standard (DES) algorithm (ISO 9564-2) to translate PIN data. For purposes of PIN translation, DES requires that a 16-digit hexadecimal number and working key be provided. DE 52 houses the 16-digit number in the customer's PIN block format.

### Application notes

Because of strict security requirements implemented within the network environment, PINs are never transmitted "in the clear" as character data. In addition, PIN data is never included in

Advice or Reversal messages. The primary reason for this is that PIN data is highly sensitive information that is never stored (even in encrypted form) as a permanent component of a transaction, for security reasons. The rules, bylaws, and procedures established for individual programs and services dictate the specific requirements for PIN usage.

This data element is supported for Mastercard transactions.

The Dual Message Application System may perform PIN verification or validation services on behalf of customers that elect to use this optional service. Refer to the appropriate user guide and *Security Rules and Procedures* to determine specific PIN verification/validation options that may be selected for individual Mastercard programs and services.

DE 52 is mandatory for all ATM transactions.

### Associated edits

| IF...                                                                                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 message contains DE 52 (PIN Data) and DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is not 02, 05, 07, 80, 81, 82, 90, or 91 | rejects the message and forwards the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 = 30</li> <li>DE 44 = 052</li> </ul> |

**NOTE: DE 52, DE 53 and DE 125 are mutually exclusive with DE 110 Dataset01**

## DE 52 Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following edits for PIN translation failures.

### Associated edits

| When...                                                                                                                                                                                                                                                 | THEN the Dual Message Authorization System...                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer uses the Authorization System for PIN translation and DE 52 (Personal ID Number [PIN] Data) is present and DE 53 (Security-Related Control Information), subfield 4 (PIN Key Index Number) contains a value not known to Mastercard, or... | sends the issuer an Authorization Request Response/0110 message where DE 48 (Additional Data: Private Use), subelement 80 (PIN Service Code) has a value of TI (The Authorization System was unable to translate the PIN) |
| an Authorization System Security Translation Platform fails                                                                                                                                                                                             |                                                                                                                                                                                                                           |

## DE 53 (Security-Related Control Information)

DE 53 (Security-Related Control Information) is used with PIN data to provide specific information about PIN block encoding and PIN data encryption to assist the issuer (or its agent) in processing PINs entered at the point of interaction.

### Attributes

| Attribute           | Description           |
|---------------------|-----------------------|
| Data Representation | n-16                  |
| Data Length         | N/A                   |
| Data Field          | Contents of subfields |
| Subfields           | 6                     |
| Justification       | See subfields         |

### Usage

Following is the usage of DE 53 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                           | Org | Sys | Dst |
|---------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                        | C   | X   | C   |
| Network Management Request/0800: sign-on/sign-off | C   | X   | .   |

### Values

See subfields.

### Application notes

Use of DE 53 in online PIN transactions is specific to the PIN translation process performed on the dual message authorization network for Europe Region Acquirers and Issuers. It is not required to be provided in online PIN transactions for messages from non-Europe Region Acquirers.

**NOTE: DE 52, DE 53 and DE 125 are mutually exclusive with DE 110 Dataset01.**

## Subfield 1 (PIN Security Type Code)

DE 53, subfield 1 (PIN Security Type Code) indicates the type of security processing used for the PIN data.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 1–2 |
| Justification       | N/A                       |

### Values

| Value | Description             |
|-------|-------------------------|
| 97    | Multiple (indexed) keys |

## Subfield 2 (PIN Encryption Type Code)

DE 53, subfield 2 (PIN Encryption Type Code) indicates the type of security processing used for the PIN data.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3–4 |
| Justification       | N/A                       |

### Values

| Value | Description    |
|-------|----------------|
| 01    | DES encryption |

## Subfield 3 (PIN Block Format Code)

DE 53, subfield 3 (PIN Block Format Code) indicates the type of PIN block format used.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 5–6 |
| Justification       | N/A                       |

### Values

| Value | Description  |
|-------|--------------|
| 01    | ANSI 1       |
| 10    | ISO Format 0 |

## Subfield 4 (PIN Key Index Number)

DE 53, subfield 4 (PIN Key Index Number) indicates the specific PIN key to be used when more than one key is available in a PIN key set.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data representation | n-4                        |
| Data field          | Contents of positions 7–10 |
| Justification       | N/A                        |

### Values

0001–0099

## Subfield 5 (Reserved for Future Use)

DE 53, subfield 5 (Reserved) is reserved for future use.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-2                         |
| Data Field          | Contents of positions 11–12 |
| Justification       | N/A                         |

### Values

Not used, default to zero.

## Subfield 6 (Reserved for Future Use)

DE 53, subfield 6 (Reserved) is reserved for future use.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-4                         |
| Data Field          | Contents of positions 13–16 |
| Justification       | N/A                         |

### Values

Not used, default to zero.

## DE 54 (Additional Amounts)

DE 54 (Additional Amounts) provides information on up to three amount types and related account data.

### Attributes

| Attribute           | Description                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| Data Representation | an...240; LLLVAR<br><br>The "LLL" length field of LLLVAR must be an integral multiple of 20, not to exceed 240. |
| Data Length         | 3 positions                                                                                                     |
| Data Field          | Contents of subfields 1–5                                                                                       |
| Subfields           | 5                                                                                                               |
| Justification       | See subfields                                                                                                   |

### Usage

Following is the usage of DE 54 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

### Values

See subfields.



## Application notes

### General

An "occurrence" is defined as one set of the four DE 54 subfields. Depending on the conditions of the message, DE 54 can be sent by the acquirer, issuer, or the Dual Message Authorization System.

Mastercard Network programs and services do not use more than two "additional amounts" amount types within a single Dual Message Authorization System message, except for healthcare amount types which use three.

The message originator can send only two occurrences of the DE 54 subfields in a message. One occurrence is the first amount type in the message originator's currency. The other occurrence is the second amount type in the message originator's currency. The Dual Message Authorization System will then provide an additional occurrence of each amount type to the message recipient in the message recipient's currency. Therefore, the message recipient can receive a maximum of four occurrences of the DE 54 subfields, except for healthcare amount types where the recipient could receive up to six occurrences. Each amount type will have two occurrences, one in the message originator's currency, and one in the message recipient's currency.

When the Authorization Request Response/0110 message contains DE 39, value 10 (Partial Approval) or 87 (Purchase Amount Only, No Cash Back Allowed), the issuer can send only one occurrence of the DE 54 subfields in the issuer's currency that is not equal to DE 54, subfield 2 (Amount Type) value 57 (Original Amount). The Dual Message Authorization System will provide two occurrences of the amount type sent by the issuer to the acquirer, one in the acquirer's currency, and one in the issuer's currency. The Dual Message Authorization System will also provide two occurrences of the DE 54 subfields where DE 54, subfield 2 (Amount Type) is 57 and DE 54, subfield 4 (Debit or Credit Indicator) is C plus the 12-digit original amount. One occurrence will be in the acquirer's currency and one occurrence will be in the issuer's currency. The exception is that if an occurrence of DE 54, subfield 2 value 58 (POI Amount) is sent, then the Dual Message Authorization System will not provide an additional occurrence in the issuer's currency.

### Partial approvals

The Authorization Request Response/0110 message contains DE 54, subfield 2, value 57 only if the acquirer can process partial approvals and if the issuer is approving part of the total transaction amount. Issuers may include one additional occurrence of DE 54 subfields in the issuer's currency with an amount type value of 02 to indicate the available balance of the account. If the issuer approves the entire transaction amount, DE 54, subfield 2, value 57 will not be present in the authorization request response and the issuer may provide more than one occurrence of DE 54 subfields.

### Account balance response

The account balance response enables the issuers to include balance information as part of the response to a financial authorization request. Issuers can add account balance response information when responding to authorization requests for prepaid accounts and in ATM transactions. The account balance response is an "unsolicited" transmission of account balance

data to the point-of-sale (POS) terminal or ATM. Account balance information returned in the Authorization Request Response/0110 must be provided in the same currency as the Cardholder Billing Currency (DE 51).

### **Account status inquiry**

Amount type 05 is only applicable for Account Status Inquiry (ASI) Authorization transactions. The issuer will not receive an occurrence of DE 54 with amount type 05.

### **Purchase with cash back**

Issuers are not required to return DE 54 in Purchase with Cash Back Authorization Request Response/0110 messages. If DE 54 is returned by the issuer, it will be dropped before the response is forwarded to the acquirer. The Authorization Request Response/0110 message only contains DE 54, subfield 2, value 40 when the transaction is processed by X-Code or declined by Mastercard with a format error condition (DE 39 = 30). When reversing a Purchase with Cash Back Authorization where only the purchase amount of the transaction was approved by the issuer (DE 39 equals 87 [Purchase Only Approval]), acquirers are not required to include the cash portion of the original authorization request in DE 54 of the Reversal Request/0400 message. Also, DE 4 (Amount, Transaction) of the reversal request should only contain the actual Purchase Amount of the original authorization request.

### **Cash back without purchase in participating countries**

Cash Back without Purchase is supported for intracountry transactions in participating countries and is identified in the Authorization Request/0100 messages containing DE 3, subfield 1 (Cardholder Transaction Type), value 09 (Purchase with Cash Back) with the presence of DE 54, subfield 2, value 40. For Cash Back without Purchase transactions, DE 54 must be present and equal the same amount in DE 4.

### **Real-time substantiation**

The Authorization Request/0100 or Authorization Advice/0120 messages contain DE 54, subfield 2, values 10, 11, or 12 to indicate the healthcare, prescription, or vision Rx amounts. DE 54, subfield 2, values 10, 11, and 12 are sent only to the issuers that participate in Real-time Substantiation.

Mastercard supports DE 54 as follows for acquirers and issuers:

- Overall size allowed by Mastercard = 240 bytes or 12 occurrences
- Number of occurrences an acquiring customer can send = 4
- Number of occurrences stored in USD currency = 4
- Number of occurrences converted to issuers currency = 4
- Number of occurrences sent to issuer = 8 (4 in acquirer currency, 4 in issuer currency)

The healthcare purchase information passed in DE 54 when these accounts are used at certain U.S. merchant locations is only required in purchase transaction authorization requests. It is not required and should not be expected to be present in refund transaction (purchase return) authorization requests.

## Purchase of goods or services with cash back

Refer to [Program and Service Format Requirements, Purchase of Goods or Services with Cash Back, Dual Message Authorization System Edits](#).

## POI currency conversion

When POI currency conversion (also known as Dynamic Currency Conversion [DCC]) is performed, Mastercard suggests that acquirers provide the pre-conversion currency and amount in DE 54 with an amount type value of 58 (POI Amount) in subfield 2 for Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages.

The processing criteria for DCC transactions are as follows:

- The transaction currency code in DE 49 (Currency Code, Transaction) must not be equal to the pre-conversion currency code in DE 54, subfield 3 (Currency Code).
- DE 54 must contain subfield 2 (Amount Type), value 58 (POI Amount), along with all other DE 54 subfields.
- The transaction amount provided in DE 54, subfield 4 (Debit or Credit Indicator) cannot contain all zeros.
- DE 54, subfield 3 must contain a valid, pre-conversion currency code.
- Issuers are not required to return an occurrence of DE 54 with subfield 2, value 58 in Authorization Request Response/0110 message. If it is returned by the issuer, it will be dropped before the response is forwarded to the acquirer.
- For a Purchase with Cash Back transaction to be subjected to DCC, it must contain two occurrences of DE 54 in Authorization Request/0100 or Authorization Advice/0120 messages: one occurrence with a subfield 2 value 58 and another occurrence with a subfield 2 value of 40 (Amount Cash Back).
- Mastercard will allow acquirers to send one occurrence of DE 54 with a subfield 2 value 58 in Reversal Request/0400 messages.
- Mastercard will forward an occurrence of DE 54 with subfield 2, value of 58 in Reversal Advice/0420 messages to the issuers if the acquirer has provided an occurrence of DE 54 with subfield 2, value 58 in Authorization Request/0100 or Reversal Request/0400 message, except when a Reversal Advice/0420 message is sent to the issuer when an Authorization Request Response/0110 message from the issuer is not delivered to the acquirer.

For partial Reversal Request/0400 messages subject to DCC, acquirers may optionally supply the remainder of the transaction amount to be authorized in pre-conversion currency in DE 54, subfield 4 (Debit or Credit Indicator).

## Visa account funding

Acquirers submitting Visa Account Funding Transactions may include a value of 95 (Visa Money Transfer Foreign Exchange Fee [Visa Only]) to identify foreign exchange fees associated with the money transfer transactions when processing an Authorization Request/0100 or a Reversal Request/0400 message by populating it in DE 54, subfield 2. Mastercard will remove any occurrence of the Visa Money Transfer Foreign Exchange Fee (Visa Only) from:

- Authorization Request Response/0110 messages
- Non-Visa-branded transactions

## Subfield 1 (Account Type)

DE 54, subfield 1 (Account Type) contains the two-digit code as defined in DE 3 (Processing Code), subfield 2 (Cardholder "From Account" Type Code) or subfield 3 (Cardholder "To Account" Type Code).

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 1–2 |
| Justification       | N/A                       |

### Values

| Value | Description          |
|-------|----------------------|
| 00    | No account specified |
| 10    | Savings account      |
| 20    | Checking account     |
| 30    | Credit card account  |

The valid values for this field are the same values as defined for [DE 3 \(Processing Code\)](#), subfield 2 (Cardholder "From Account" Type Code).

## Subfield 2 (Amount Type)

DE 54, subfield 2 (Amount Type) indicates the type of amount applied.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3–4 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |
| Required as   | Conditional |

## Values

| Value | Description                                               |
|-------|-----------------------------------------------------------|
| 01    | Ledger balance                                            |
| 02    | Available balance                                         |
| 03    | Amount owing                                              |
| 04    | Amount due                                                |
| 05    | Future transaction amount                                 |
| 06    | Product Status Inquiry Transaction Amount                 |
| 10    | Healthcare eligibility amount                             |
| 11    | Prescription eligibility amount                           |
| 12    | Vision Rx eligibility amount                              |
| 17    | Prepaid online bill pay fee amount                        |
| 40    | Cash back                                                 |
| 44    | Amount Gratuity                                           |
| 57    | Original Amount                                           |
| 58    | POI Amount                                                |
| 59    | Limit/Balance amount available from Mastercard In Control |
| 80    | Copay Amount                                              |
| 90    | Available Credit                                          |
| 91    | Credit Limit                                              |
| 92    | Outstanding hold amounts                                  |
| 95    | Visa Money Transfer Foreign Exchange Fee (Visa Only)      |

## Subfield 3 (Currency Code)

DE 54, subfield 3 (Currency Code) is a three-digit code that must contain a valid numeric code.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Field          | Contents of positions 5–7 |
| Justification       | Right                     |

### Values

Refer to the Quick Reference Booklet.

## Subfield 4 (Debit or Credit Indicator)

DE 54, subfield 4 (Debit or Credit Indicator) indicates the amount is a credit or debit indicator.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | a-1                    |
| Data Field          | Contents of position 8 |
| Justification       | Right                  |

### Values

| Value | Description                       |
|-------|-----------------------------------|
| C     | Credit amount or positive balance |
| D     | Debit amount or negative balance  |

## Subfield 5 (Amount)

DE 54, subfield 5 (Amount) contains the additional amount.

### Attributes

| Attribute           | Description                        |
|---------------------|------------------------------------|
| Data Representation | n-12                               |
| Data Field          | Contents of positions 9–20         |
| Justification       | Right-justified with leading zeros |

## DE 54 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the acquirer sends an Authorization Request/0100 or an Authorization Advice/0120 message in which:</p> <ul style="list-style-type: none"> <li>An occurrence of DE 54 (Additional Amounts) contains a subfield 2 (Amount Type) value of 58 (Additional Amount, POI Amount) and</li> <li>An occurrence of DE 54 contains a subfield 3 (Currency Code) value that is equal to the DE 49 (Currency Code, Transaction) value</li> </ul> <p>or</p> <ul style="list-style-type: none"> <li>An occurrence of DE 54 contains an invalid currency code in subfield 3.</li> </ul> | <p>will reject the transaction where:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format Error)</li> <li>DE 44 (Additional Response Data) = 054 (indicating the data element in error)</li> </ul> |
| <p>the acquirer sends an Authorization Request/0100 or an Authorization Advice/0120 message in which:</p> <ul style="list-style-type: none"> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 09 (Purchase with Cash Back), and</li> <li>DE 54, subfield 2 contains value 40 (Amount Cash Back), and</li> <li>DE 54, subfield 3 contains a value that does not match the value in DE 49</li> </ul>                                                                                                                                | <p>will reject the transaction where:</p> <ul style="list-style-type: none"> <li>DE 39 = 30</li> <li>DE 44 = 054</li> </ul>                                                                                                  |

## DE 55 (Integrated Circuit Card [ICC] System-Related Data)

DE 55 (Integrated Circuit Card [ICC] System-Related Data) contains binary data that only the issuer, the issuer agent, or MDES processes; it is used locally by the payment application on the chip at a chip-capable terminal. This data element is present in chip full-grade transactions and can be present in DSRP transactions.

### Attributes

| Attribute           | Description                                         |
|---------------------|-----------------------------------------------------|
| Data Representation | b...255; LLLVAR<br>The "LLL" length field of LLLVAR |
| Data Length         | 3 positions                                         |
| Data Field          | Contents of subelements                             |
| Subelements         | Number of subelements depends on message type       |
| Justification       | N/A                                                 |

### Usage

Following is the usage of DE 55 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

### Values

See subelements.

### Application notes

For ICC information, refer to *M/Chip Requirements*.



Refer to [DE 55: Authorization System Edits](#) for a description of the edits performed on this data element.

For DSRP information, refer to Mastercard Cloud-Based Payments Product description.

For MDES tokenized transactions, issuers may opt in through MDES Manager to receive track data in DE 55 (Integrated Circuit Card [ICC] System-Related Data) on Dual Message System (Authorization) or Single Message System transactions to issuers.

## DE 55 (Subelement Encoding Scheme)

DE 55 contains chip data formatted in accordance with EMV specifications. Reference Book 3 of the EMV specification for details regarding coding of tag-length-value (TLV) data objects. The chip data in DE 55 consists of a series of subelements in a TLV format.

### Positions in DE 55

| Data Length    | Description                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1–3            | DE 55 Total Length                                                                                                                                                  |
| 4 or 4–5       | First subelement ID, in binary representation; the length is either one or two positions depending on the definition of the subelement ID in the EMV specification. |
| 5 or 5–6       | First subelement length; the value of the length subelement is always one; position depends on the subelement ID length.                                            |
| 6–xxx or 7–xxx | First subelement variable length data; the starting position depends on the subelement ID length.                                                                   |

Positions of the subelement length and variable length data depend on the subelements used. Subelement ID (Tag), Length, and Variable Length Data may be repeated as needed until all chip data has been presented.

## DE 55 Subelements

The following tables list the required and optional subelements in Authorization Request/0100 and Authorization Request Response/0110 messages that contain chip data.

**NOTE: Subelements that are not listed as either required or optional must not be included in DE 55.**

### Required subelements in Authorization Request/0100

For Authorization requests related to Chip full-grade transactions and requests related to e-commerce transactions with EMV compliant ICC data, acquirers must provide the following data in DE 55 subelements.

## Subelement descriptions

| Subelement Description                                                                                                               | Tag Value | Component | Each Component Length | Total Subelement Length |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------------------|-------------------------|
| Application Cryptogram (AC)                                                                                                          | 9F26      | ID        | 2                     | 11                      |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 8                     |                         |
| Cryptogram Information Data                                                                                                          | 9F27      | ID        | 2                     | 4                       |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 1                     |                         |
| Issuer Application Data (IAD)<br>(Mandatory if the corresponding data object [EMV tag 9F10] is provided by the card to the terminal) | 9F10      | ID        | 2                     | 4–35                    |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 1–32                  |                         |
| Unpredictable Number                                                                                                                 | 9F37      | ID        | 2                     | 7                       |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 4                     |                         |
| Application Transaction Counter                                                                                                      | 9F36      | ID        | 2                     | 5                       |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 2                     |                         |
| Terminal Verification Result (TVR)                                                                                                   | 95        | ID        | 1                     | 7                       |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 5                     |                         |
| Transaction Date                                                                                                                     | 9A        | ID        | 1                     | 5                       |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 3                     |                         |
| Transaction Type                                                                                                                     | 9C        | ID        | 1                     | 3                       |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 1                     |                         |
| Amount Authorized                                                                                                                    | 9F02      | ID        | 2                     | 9                       |
|                                                                                                                                      |           | length    | 1                     |                         |
|                                                                                                                                      |           | data      | 6                     |                         |

| Subelement Description                                                                   | Tag Value                  | Component | Each Component Length | Total Subelement Length |
|------------------------------------------------------------------------------------------|----------------------------|-----------|-----------------------|-------------------------|
| Transaction Currency Code                                                                | 5F2A                       | ID        | 2                     | 5                       |
|                                                                                          |                            | length    | 1                     |                         |
|                                                                                          |                            | data      | 2                     |                         |
| Application Interchange Profile                                                          | 82                         | ID        | 1                     | 4                       |
|                                                                                          |                            | length    | 1                     |                         |
|                                                                                          |                            | data      | 2                     |                         |
| Terminal Country Code                                                                    | 9F1A                       | ID        | 2                     | 5                       |
|                                                                                          |                            | length    | 1                     |                         |
|                                                                                          |                            | data      | 2                     |                         |
| Cardholder Verification Method (CVM) Results                                             | 9F34                       | ID        | 2                     | 6                       |
|                                                                                          |                            | length    | 1                     |                         |
|                                                                                          |                            | data      | 3                     |                         |
| NOTE: The presence of 9F34 is mandatory for all authorization messages containing DE 55. |                            |           |                       |                         |
| Terminal Capabilities                                                                    | 9F33<br>(U.S. region only) | ID        | 2                     | 6                       |
|                                                                                          |                            | length    | 1                     |                         |
|                                                                                          |                            | data      | 3                     |                         |
| Dedicated File Name                                                                      | 84                         | ID        | 1                     | 7–18                    |
|                                                                                          |                            | length    | 1                     |                         |
|                                                                                          |                            | data      | 5–16                  |                         |
| Amount Other                                                                             | 9F03                       | ID        | 2                     | 9                       |
|                                                                                          |                            | length    | 1                     |                         |
|                                                                                          |                            | data      | 6                     |                         |

For the tag values the:

- Hexadecimal representation is given here. Every two positions of hexadecimal data is one byte of binary data.
- Lengths are in binary format.
- Total subelement length is the sum of the subelement's ID, length, and data subfields.

When cash back is not permitted by product rules, 9F03 may be absent, or present with a zero value. When cash back is permitted by product rules:

- If there is a cash back amount, 9F03 carries the amount and presence is mandatory
- If there is no cash back amount, the value of 9F03 is zero. 9F03 may be absent, or present with a zero value.

### Optional subelements in Authorization Request/0100

When DE 55 is present in the Authorization Request/0100 message, the following subelements are optional in DE 55.

### Subelement descriptions

| Subelement Description                                                                                                                           | Tag Value | Component | Each Component Length | Total Subelement Length |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------------------|-------------------------|
| Application Primary Account Number (PAN) Sequence Number                                                                                         | 5F34      | ID        | 2                     | 4                       |
|                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                  |           | data      | 1                     |                         |
| Application Selection Registered Proprietary Data<br><i>For technical information, refer to M/Chip Requirements for Contact and Contactless.</i> | 9F0A      | ID        | 2                     | 3–Var                   |
|                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                  |           | data      | Var                   |                         |
| Terminal Type                                                                                                                                    | 9F35      | ID        | 2                     | 4                       |
|                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                  |           | data      | 1                     |                         |
| Interface Device (IFD) Serial Number                                                                                                             | 9F1E      | ID        | 2                     | 11                      |
|                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                  |           | data      | 8                     |                         |
| Transaction Category Code                                                                                                                        | 9F53      | ID        | 2                     | 4                       |
|                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                  |           | data      | 1                     |                         |
| Application Version Number                                                                                                                       | 9F09      | ID        | 2                     | 5                       |
|                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                  |           | data      | 2                     |                         |
| Transaction Sequence Counter                                                                                                                     | 9F41      | ID        | 2                     | 5–7                     |
|                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                  |           | data      | 2–4                   |                         |

| Subelement Description         | Tag Value                       | Component | Each Component Length | Total Subelement Length |
|--------------------------------|---------------------------------|-----------|-----------------------|-------------------------|
| Terminal Capabilities          | 9F33<br>(non-U.S. regions only) | ID        | 2                     | 6                       |
|                                |                                 | length    | 1                     |                         |
|                                |                                 | data      | 3                     |                         |
| Third Party Data               | 9F6E                            | ID        | 2                     | 8–35                    |
|                                |                                 | length    | 1                     |                         |
|                                |                                 | data      | 5–32                  |                         |
| Authenticated Application Data | 9F60                            | ID        | 2                     | 4–130                   |
|                                |                                 | length    | 1                     |                         |
|                                |                                 | data      | var-127               |                         |
| Kernel Identifier-Terminal     | 96                              | ID        | 1                     | 10                      |
|                                |                                 | length    | 1                     |                         |
|                                |                                 | data      | 8                     |                         |

### Optional subelements in Authorization Request Response/0110

Issuers may provide the following DE 55 subelements in Authorization Request Response/0110 messages related to:

- Chip full-grade transactions
- E-commerce transactions with EMV-compliant ICC data

If any of these subelements are present in the Authorization Request Response/0110 message, the acquirer must pass the subelements, unaltered, to the IC card.

### Subelement descriptions

| Subelement Description                                                                              | Tag Value | Component | Each Component Length | Total Subelement Length |
|-----------------------------------------------------------------------------------------------------|-----------|-----------|-----------------------|-------------------------|
| Issuer Authentication Data (Provides data to be transmitted to the card for issuer authentication.) | 91        | ID        | 1                     | 10–18                   |
|                                                                                                     |           | length    | 1                     |                         |
|                                                                                                     |           | data      | 8–16                  |                         |

| Subelement Description                                                                                                                                                                                                                                                                                                                           | Tag Value | Component | Each Component Length | Total Subelement Length |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------------------|-------------------------|
| Issuer Script Template 1<br><br>(Allows the issuer to provide a post-issuance command for transmission to the card. Present if issuer sends commands to ICC.)<br><br>Issuers must not send more than one template '71' script with each Authorization Request Response/0110, although each script template may contain multiple script commands. | 71        | ID        | 1                     | 3–129                   |
|                                                                                                                                                                                                                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                                                                                                                                                                                                                  |           | data      | 1–127                 |                         |
| <b>NOTE: Either tag 71 or tag 72 should be included in any single authorization request response message but not both.</b>                                                                                                                                                                                                                       |           |           |                       |                         |
| Issuer Script Template 2<br><br>(Allows the issuer to provide a post-issuance command for transmission to the card. Present if issuer sends commands to ICC.)<br><br>Issuers must not send more than one template '72' script with each Authorization Request Response/0110, although each script template may contain multiple script commands. | 72        | ID        | 1                     | 3–129                   |
|                                                                                                                                                                                                                                                                                                                                                  |           | length    | 1                     |                         |
|                                                                                                                                                                                                                                                                                                                                                  |           | data      | 1–127                 |                         |
| <b>NOTE: Either tag 71 or tag 72 should be included in any single authorization request response message but not both.</b>                                                                                                                                                                                                                       |           |           |                       |                         |

## DE 55 Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits.

If Mastercard determines through its Internal Chip Monitoring process that improperly formatted chip transactions are being submitted from acquirers not certified to send chip transactions, Mastercard will notify each acquirer before activating an edit. The Dual Message Authorization System will perform the following edits on the Authorization Request/0100 message.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22, subfield 1 contains value 05 or 07, and<br>DE 55 is present<br>and<br>Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved for partial grade chip transactions                                                       | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30</li><li>• DE 44 (Additional Response Data) = 055</li></ul> |
| DE 22, subfield 1 contains value 05<br>and<br>DE 55 is not present<br>and<br>Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved to send full grade chip transactions<br>and<br>DE 3, subfield 1 does not contain value 20 | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30</li><li>• DE 44 (Additional Response Data) = 055</li></ul> |
| DE 22, subfield 1 contains value 07<br>and<br>DE 55 is not present<br>and<br>Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved to send full grade chip transactions<br>and<br>DE 3, subfield 1 does not contain value 20 | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30</li><li>• DE 44 (Additional Response Data) = 055</li></ul>         |

### Associated edits

| WHEN...                                                                                                                                                                                                                                  | THEN The Dual Message Authorization System...                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 55 is present in the Authorization Request/0100 message or the Authorization Advice/0120 message<br>and<br>DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 is not 03, 05, 07, 09, or 81                                         | sends the Authorization Request/0100 message or Authorization Advice/0120 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 055</li> </ul> |
| DE 55 is not present in the Authorization Request/0100 message for a Chip PIN Management transaction<br>and<br>DE 3 (Processing Code) subfield 1 is not 92 and DE 22 (Point-of-Service [POS] Entry Mode) subfield 1 is not 02, 80, or 90 | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 055</li> </ul>                 |
| DE 55 is greater than 255 characters in length in the Authorization Request/0100 message                                                                                                                                                 | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 055</li> </ul>                         |

The Authorization System will perform the following edits on an Authorization Request Response/0110 message.

### Associated edits

| WHEN...                                                            | THEN the Dual Message Authorization System...                                                                                                                                                                       |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 55 is echoed in the Authorization Request Response/0110 message | sends the issuer an Authorization Response Negative Acknowledgment/0190 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 055</li> </ul> |



### Associated edits

| WHEN the issuer...                                                                                                         | THEN the Dual Message Authorization System...                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sends an Authorization Request Response/0110 message where DE 55, subelement 71 is present and exceeds 127 bytes in length | sends an Authorization Response Negative Acknowledgment/0190 message to the issuer where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30</li><li>• DE 44 (Additional Response Data) = 055</li></ul> |

### Associated edits

| IF...                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 55 is present in the Authorization Request Response/0110 and DE 55 was not present in the original Authorization Request/0100 | sends an Authorization Response Negative Acknowledgment/0190 message to the issuer where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30</li><li>• DE 44 (Additional Response Data) = 055</li></ul> |

## Dual Message Authorization System Edits: Cardholder Authentication Service

The Dual Message Authorization System will perform the following system edits related to the Cardholder Authentication Service.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Request/0100 message contains:</p> <ul style="list-style-type: none"> <li>DE 2 (Primary Account Number [PAN]) setup for Type 1 Authentication Service and</li> <li>DE 4 (Amount, Transaction) has a value of zero and</li> <li>DE 23 (Card Sequence Number) contains a value of 01X (X can be any number from 0 to 9 inclusive) and</li> <li>DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and</li> <li>DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: <ul style="list-style-type: none"> <li>CVR byte1, bit1 is present with a value other than 1 (successful) or</li> <li>CVR byte2, bit2 is present with a value other than 1 (biometric).</li> </ul> </li> </ul> | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 05 (Do not honor) and</li> </ul> <p>Sends the issuer an Authorization Advice/0120 message with DE 60 (Advice Reason Code):</p> <ul style="list-style-type: none"> <li>Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer) and</li> <li>Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not present)</li> </ul> |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | THEN the Dual Message Authorization System...                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Advice/0120: acquirer-generated message contains:</p> <ul style="list-style-type: none"> <li>• DE 2 (Primary Account Number [PAN]) setup for Type 1 Authentication Service and</li> <li>• DE 4 (Amount, Transaction) has a value of zero and</li> <li>• DE 22 (Point-of-Service [POS] Entry Mode) is 05 or 07 and</li> <li>• DE 23 (Card Sequence Number) contains a value of 01X (X can be any number from 0 to 9 inclusive) and</li> <li>• DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: <ul style="list-style-type: none"> <li>– CVR byte1, bit1 is present with a value other than 1 (successful) or</li> <li>– CVR byte2, bit2 is present with a value other than 1 (biometric).</li> </ul> </li> </ul> <p>and</p> <ul style="list-style-type: none"> <li>• DE 60 (Advice Reason Code) subfield 1 is 192 and</li> <li>• DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI])</li> </ul> | <p>sends the acquirer an Authorization Advice Response/0130 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format Error) and</li> <li>• DE 44 (Additional Response Data) = 055 (indicating the data element in error)</li> </ul> |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Request/0100 message contains:</p> <ul style="list-style-type: none"> <li>DE 2 (Primary Account Number [PAN]) setup for Type 2 Authentication Service and</li> <li>DE 4 (Amount, Transaction) has a value of zero and</li> <li>DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and</li> <li>DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: <ul style="list-style-type: none"> <li>CVR byte1, bit1 is present with a value other than 1 (successful) or</li> <li>CVR byte2, bit2 is present with a value other than 1 (biometric).</li> </ul> </li> </ul>                                                                                                                                           | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 05 (Do not honor) and</li> </ul> <p>Sends the issuer an Authorization Advice/0120 message with DE 60 (Advice Reason Code):</p> <ul style="list-style-type: none"> <li>Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer) and</li> <li>Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not present)</li> </ul> |
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Advice/0120: acquirer-generated message contains:</p> <ul style="list-style-type: none"> <li>DE 2 (Primary Account Number [PAN]) setup for Type 2 Authentication Service and</li> <li>DE 4 (Amount, Transaction) has a value of zero and</li> <li>DE 22 (Point-of-Service [POS] Entry Mode) = 05 or 07 and</li> <li>DE 60 (Advice Reason Code) subfield 1 = 192</li> <li>DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and</li> <li>DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: <ul style="list-style-type: none"> <li>CVR byte1, bit1 is present with a value other than 1 (successful) or</li> <li>CVR byte2, bit2 is present with a value other than 1 (biometric).</li> </ul> </li> </ul> | <p>sends the acquirer an Authorization Advice Response/0130 message where:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format Error) and</li> <li>DE 44 (Additional Response Data) = 055 (indicating the data element in error)</li> </ul>                                                                                                                                                                                                                                   |

## DE 56 (Payment Account Data)

DE 56 (Payment Account Data) contains unique, non-financial reference information associated with the PAN or token used to initiate the transaction.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | an...37; LLLVAR         |
| Data Length         | 3 positions             |
| Data Field          | Contents of subelements |
| Subelements         | 1                       |
| Justification       | See subelement          |

### Usage

Following is the usage of DE 56 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                                           | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                                        | •   | X   | C   |
| Authorization Request/0100: Account Status inquiry (merchant initiated message and MDES pre-digitization message) | •   | X   | C   |
| Authorization Request/0100: Tokenization Complete Notification*                                                   | •   | X   | C   |
| Authorization Request Response/0110                                                                               | C   | X   | C   |
| Authorization Request Response/0110: Tokenization Authorization Request Response*                                 | C   | X   | C   |
| Authorization Request Response/0110: Account Status Inquiry*                                                      | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                                     | •   | X   | C   |
| Authorization Advice/0120: system-generated                                                                       | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to acquirer-generated 0120)                      | C   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)                      | C   | X   | •   |

| Message                                                                                                                | Org | Sys | Dst |
|------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)                        | •   | X   | C   |
| Issuer File Update Request/0302                                                                                        | O   | X   | •   |
| Reversal Request/0400                                                                                                  | •   | X   | C   |
| Reversal Request Response/0410                                                                                         | C   | X   | C   |
| Reversal Advice/0420                                                                                                   | •   | X   | C   |
| Reversal Advice Response/0430                                                                                          | O   | X   | •   |
| Administrative Advice/0620: system-generated: issuer token notification advice for tokenization complete notification* | •   | X   | C   |

\* MDES pre-digitization message

## Subelement 01 (Payment Account Data)

DE 56, subelement 01 (Payment Account Data) contains the applicable subfields to carry the unique Payment Account Data specifically associated with the PAN or token used to initiate the transaction.

### Attributes

|                        | Attribute      | Value                     |
|------------------------|----------------|---------------------------|
| Subelement ID          | n-2            | 01                        |
| Subelement Data Length | n-2            | ...33                     |
| Data Representation    | an...33; LLVAR |                           |
| Data Field             |                | Contents of positions 1–2 |
| Number of subfields    | 1              | Payment Account Reference |

## Subfield 01 (Payment Account Reference [PAR])

DE 56, subelement 01, subfield 01 (Payment Account Reference [PAR]) contains the assigned PAR value. A PAR is a unique value associated with a single PAN and attributed to all tokens

associated with that PAN. A PAR can be used to link transactions containing PANs or tokens associated with the same underlying payment account.

Attributes

|                      | Attribute      | Value |
|----------------------|----------------|-------|
| Subfield ID          | n-2            | 01    |
| Subfield Data Length | n-2            | ...29 |
| Data Representation  | an...29; LLVAR |       |
| Justification        | N/A            |       |

Positions and descriptions

| Position | Description                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------|
| 1–4      | BIN Controller Identifier: A four-character value assigned by EMVCo                                    |
| 5–29     | A unique 25-character alphanumeric uppercase value generated by the BIN Controller and linked to a PAN |

Application notes

Mastercard supports PAR for all acquirers and issuers for MDES, MDES Lite, and non-MDES transactions in DE 56, subfield 01, subelement 01.

PAR is not added to Mastercard ATM transactions.

DE 57–DE 59 (Reserved for National Use)

DE 57–DE 59 are reserved for future use.

**NOTE: The Dual Message Authorization System currently does not use these data elements.**

## DE 60 (Advice Reason Code)

DE 60 (Advice Reason Code) indicates to the receiver of an Advice message the specific reason for the transmission of the Advice message.

### Attributes

| Attribute           | Description           |
|---------------------|-----------------------|
| Data Representation | ans...060; LLLVAR     |
| Data Length         | 3 positions           |
| Data Field          | Contents of subfields |
| Subfields           | 3                     |
| Justification       | See subfields         |

### Usage

Following is the usage of DE 60 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: acquirer-generated | M   | P   | M   |
| Authorization Advice/0120: system-generated   | •   | X   | M   |
| Reversal Advice/0420                          | •   | X   | M   |
| Administrative Request/0600                   | M   | P   | M   |
| Administrative Request Response/0610          | ME  | XE  | ME  |
| Administrative Advice/0620: member-generated  | M   | P   | M   |
| Administrative Advice/0620: system-generated  | •   | X   | M   |

### Values

See subfields.

### Application notes

Not all Advice Reason Codes may be used within all programs and services.



## Subfield 1 (Advice Reason Code)

DE 60, subfield 1 contains the advice reason code.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Field          | Contents of positions 1-3 |
| Justification       | N/A                       |

### Values

This subfield is mandatory for all advice messages and indicates the general purpose of the advice message.

The following tables displays the general values for subfields 1 and 2 based on message type for all products, though each code does not pertain to all products.

### DE 60, subfield 1 values, in Authorization Advice/0120

The following values are valid in Authorization Advice/0120 messages.

### Values

| Code | Description                                                                                                                                                                                                                            | MC | NP | VI | TE | MS | CI |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| 100  | Alternate Issuer Route: Issuer selected option<br>This secondary route is to either the Stand-In System or the optional alternate route.<br><br>This secondary route is to either the Stand-In System or the optional alternate route. | √  | √  | √  |    | √  | √  |
| 101  | Alternate Issuer Route: Issuer signed out<br>This secondary route is to either the Stand-In System or the optional alternate route.                                                                                                    | √  | √  | √  |    | √  | √  |
| 102  | Alternate Issuer Route: Issuer timed out<br>This secondary route is to either the Stand-In System or the optional alternate route.                                                                                                     | √  | √  | √  |    | √  | √  |
| 103  | Alternate Issuer Route: Issuer unavailable<br>This secondary route is to either the Stand-In System or the optional alternate route.                                                                                                   | √  | √  | √  |    | √  | √  |

| Code | Description                                                                                                                                  | MC | NP | VI | TE | MS | CI |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| 105  | Transaction processed through X-Code                                                                                                         | √  | √  | √  |    |    |    |
| 107  | PIN processing error                                                                                                                         | √  |    |    |    | √  | √  |
| 108  | Alternate Issuer Route: MIP Error<br>This secondary route is to either the Stand-In System or the optional alternate route.                  | √  | √  | √  |    | √  | √  |
| 109  | Alternate Issuer Route: Issuer Edit Response Error<br>This secondary route is to either the Stand-In System or the optional alternate route. | √  | √  | √  |    | √  | √  |
| 111  | Alternate Issuer Route: Issuer Host System Error<br>This secondary route is to either the Stand-In System or the optional alternate route.   | √  | √  | √  |    | √  | √  |
| 112  | Alternate Route: Network Not Dispatched Error<br>This secondary route is to either the Stand-In System or the optional alternate route.      | √  | √  | √  |    | √  | √  |
| 113  | Alternate Route: Issuer Undelivered<br>This secondary route is to either the Stand-In System or the optional alternate route.                | √  | √  | √  |    | √  | √  |
| 114  | Alternate Route: Direct Down Option<br>This secondary route is to either the Stand-In System or the optional alternate route.                | √  | √  | √  |    | √  | √  |
| 115  | Transaction Processed via On-behalf Service Decision                                                                                         | √  |    |    |    | √  | √  |
| 116  | Invalid Merchant                                                                                                                             | √  |    |    |    |    |    |
| 120  | Transaction Blocking                                                                                                                         | √  |    |    |    | √  | √  |
| 121  | Account Lookup Service                                                                                                                       | √  | √  |    |    | √  |    |
| 126  | Pay with Rewards Processing Advice to Issuer                                                                                                 | √  |    |    |    |    |    |
| 140  | Unable to convert contactless or virtual account number                                                                                      | √  |    |    |    | √  |    |
| 141  | Mastercard Digital Enablement Service Advice to Issuer                                                                                       | √  |    |    |    | √  |    |
| 142  | TSP Advice to Issuer                                                                                                                         | √  |    |    |    | √  |    |
| 150  | Send and other transactions Blocking Service Advice to Issuers                                                                               | √  |    |    |    | √  | √  |
| 151  | In Control Processing Advice to Issuer (Mastercard Merchant Presented QR)                                                                    | √  |    |    |    |    |    |
| 160  | Authentication Advice to Issuer                                                                                                              | √  |    |    |    | √  |    |
| 180  | CAT Risk Level 3                                                                                                                             | √  | √  |    |    |    |    |

| Code | Description                                                          | MC | NP | VI | TE | MS | CI |
|------|----------------------------------------------------------------------|----|----|----|----|----|----|
| 190  | Acquirer Processing System (APS) Approved                            | √  | √  | √  |    | √  | √  |
| 191  | Acquirer Processing System (APS) Completed Authorization Transaction | √  | √  |    | √  | √  |    |
| 192  | M/Chip Offline Advice to Issuer                                      | √  |    |    |    | √  |    |
| 200  | In Control Processing Advice to Issuer                               | √  |    |    |    |    |    |
| 300  | Transaction processed with On-Demand Decisioning                     | √  |    |    |    | √  | √  |
| 650  | Administrative textual message transmittal                           | √  | √  | √  |    | √  | √  |

## DE 60, Authorization Advice/0120 edits

The Dual Message Authorization System performs the following system edits on Authorization Advice/0120: acquirer-generated messages.

### Associated edits

| WHEN...                                                                                                                              | THEN the Dual Message Authorization System...                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 60 in the Authorization Advice/0120: acquirer-generated messages is not the value 190 (Acquirer Processing System (APS) Approved) | sends the acquirer an Authorization Advice Response/0130 message where:                                                                                           |
| or                                                                                                                                   | <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format Error)</li> <li>DE 44 (Additional Response Data) = 060 (Invalid DE 60 value)</li> </ul> |
| 191 Acquirer Completed Authorization Transaction)                                                                                    |                                                                                                                                                                   |
| or                                                                                                                                   |                                                                                                                                                                   |
| 192 (M/Chip Offline Advice to Issuer)                                                                                                |                                                                                                                                                                   |

## DE 60, subfield 1 values, in Reversal Advice/0420

The following values are valid in Reversal Advice/0420 messages.

### Values

| Code | Description                                                | MC | NP | VI | TE | MS | CI |
|------|------------------------------------------------------------|----|----|----|----|----|----|
| 400  | Banknet advice: Acquirer error; unable to deliver response | √  | √  | √  | √  | √  | √  |

| Code | Description                                                                                                                         | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| 401  | Banknet advice: Acquirer error; no Acquirer Authorization Response Acknowledgment/0180 or Financial Transaction Acknowledgment/0280 | √  | √  | √  |    | √  | √  |
| 402  | Issuer Time-out                                                                                                                     | √  | √  | √  |    | √  | √  |
| 403  | Issuer Sign-out                                                                                                                     | √  | √  | √  |    | √  | √  |
| 409  | Issuer Response Error                                                                                                               | √  | √  | √  |    | √  | √  |
| 410  | Reversal message provided by a system other than Banknet                                                                            | √  |    |    |    | √  | √  |
| 413  | Issuer Undelivered                                                                                                                  | √  | √  | √  |    | √  | √  |

## DE 60, subfield 1 values, in Administrative Request/0600

Following are the valid values in Administrative Request/0600 messages.

### Values

| Code | Description                                                             | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------------------------------------------|----|----|----|----|----|----|
| 650  | Administrative message containing customer application and account data | √  | √  |    |    |    |    |

## DE 60, subfield 1 values, in Administrative Request Response/0610

Following are valid values in Administrative Request Response/0610 messages.

### Values

| Code | Description                                                             | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------------------------------------------|----|----|----|----|----|----|
| 650  | Administrative message containing customer application and account data | √  | √  |    |    |    |    |

## DE 60, subfield 1 Values, in Administrative Advice/0620

Following are valid values in Administrative Advice/0620 messages.

### Values

| Code | Description                                                                                                                                                                           | MC | NP | VI | TE | MS | CI |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| 600  | Message unreadable or indecipherable or contains invalid data. Subfield 2 (Advice Detail Code) may contain the bit map number of the data element where message scanning was aborted) | √  | √  | √  | √  | √  | √  |
| 650  | Administrative textual message transmittal                                                                                                                                            | √  | √  | √  |    | √  | √  |

## DE 60, subfield 1 Value, in Send Blocking Service

These values are valid in Send Blocking Service transactions.

### Values

| Code | Description                                        | MC | NP | VI | TE | MS | CI |
|------|----------------------------------------------------|----|----|----|----|----|----|
| 150  | Send Blocking service processing advice to issuers | X  |    |    |    | X  | X  |

## Subfield 2 (Advice Detail Code)

DE 60, subfield 2 (Advice Detail Code) is optional, depending on the primary Advice Reason Code; if used, it provides additional (specific) information as to the exact nature of the Advice message. Advice Detail Codes are determined individually for each program and service. The Advice Detail Codes for Mastercard activity appear below. Refer to the appropriate documentation for information on codes for non-Mastercard activity.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data representation | n-4                       |
| Data field          | Contents of positions 4-7 |
| Justification       | N/A                       |

### Values

| Value | Description                                                               |
|-------|---------------------------------------------------------------------------|
| 0000  | Accept                                                                    |
| 0001  | Reject: negative file                                                     |
| 0002  | Reject: capture card                                                      |
| 0003  | Reject: issuer not participating                                          |
| 0004  | Reject: invalid PIN                                                       |
| 0005  | Reject: ATM                                                               |
| 0006  | Reject: transaction limit test                                            |
| 0009  | Reject: Invalid Time Validation                                           |
| 0041  | Reject: Payment Cancellation                                              |
| 0059  | Reject: Format error                                                      |
| 0078  | Reject: M/Chip Biometric Data not Present                                 |
| 0100  | Reject: Account Balance Listing, transaction amount exceeds current limit |
| 0101  | Reject: Dynamic Decisioning, AI model decline                             |
| 0216  | Reject: DTVC Invalid/Expired                                              |
| 0217  | Reject: Channel not Authorized by Cardholder                              |

### DE 60, subfield 2 values, in Authorization Advice/0120: system-generated

The following values are valid in Authorization Advice/0120—System-generated messages.

Reason codes 0007–1611 apply only to the Authorization Advice/0120: system-generated message.

### Values

| Value | Description                                   |
|-------|-----------------------------------------------|
| 0007  | Reject: Premium listing cumulative limit test |
| 0008  | Reject: merchant suspicious indicator test    |
| 0009  | Reject: Invalid Time Validation               |
| 0010  | Reject: Stand-In Monitoring Rule              |

| Value | Description                                                                                  |
|-------|----------------------------------------------------------------------------------------------|
| 0011  | Reject: day number 1/number of transactions                                                  |
| 0012  | Reject: day number 2/number of transactions                                                  |
| 0013  | Reject: day number 3/number of transactions                                                  |
| 0014  | Reject: day number 4/number of transactions                                                  |
| 0015  | Reject: day number 1/amount                                                                  |
| 0016  | Reject: day number 2/amount                                                                  |
| 0017  | Reject: day number 3/amount                                                                  |
| 0018  | Reject: day number 4/amount                                                                  |
| 0019  | Reject: extended Cash Advance cumulative amount                                              |
| 0020  | Reject: card number in blocked range                                                         |
| 0021  | Reject: Premium Listing transaction limit test                                               |
| 0028  | Reject: invalid CVC 1                                                                        |
| 0029  | Reject: expired card                                                                         |
| 0031  | Reject: unable to decrypt/encrypt PIN data. (Used with Advice Reason Code 107)               |
| 0036  | Reject: CVC 1 Unable to process                                                              |
| 0047  | Reject: CVC 1 No matching key file for this PAN, PAN expiry date combination; status unknown |
| 0216  | Reject: DTVC Invalid/Expired                                                                 |
| 1000  | Accept: On-Demand Decisioning Rule                                                           |
| 1001  | Reject: On-Demand Decisioning Rule                                                           |

## DE 60, subfield 2 values, in Administrative Advice/0620

Following are the valid values in Administrative Advice/0620 messages.

### Values

| Values | Description                               |
|--------|-------------------------------------------|
| 0029   | MIP-generated risk Advice transactions    |
| 0030   | Member-generated risk Advice transactions |
| 0250   | Activation Code Notification              |
| 0251   | Tokenization Complete Notification        |

| Values | Description                     |
|--------|---------------------------------|
| 0252   | Tokenization Event Notification |

## DE 60, subfield 2 values, in Customer Service Messages

Following are the valid values in Customer Service messages.

### Values

| Value | Description                          |
|-------|--------------------------------------|
| 0080  | Consumer application request         |
| 0081  | Consumer application status inquiry  |
| 0082  | Consumer user lookup inquiry         |
| 0083  | Consumer account lookup inquiry      |
| 0084  | Consumer account maintenance request |
| 0085  | Consumer counteroffer reply          |
| 0086  | Consumer preapproved offer inquiry   |
| 0090  | Business application request         |
| 0091  | Business application status inquiry  |
| 0092  | Business user lookup inquiry         |
| 0093  | Business account lookup inquiry      |
| 0094  | Business account maintenance request |
| 0095  | Business counteroffer reply          |
| 0096  | Business pre-approved offer inquiry  |

## DE 60, subfield 2 values, in Dynamic CVC 3 Validation

Following are valid values in Dynamic CVC 3 Validation transactions.

### Values

| Value | Description                                                                     |
|-------|---------------------------------------------------------------------------------|
| 0037  | Reject: No matching key file for this PAN, PAN expiry date, and KDI combination |
| 0038  | Reject: Security Platform Time Out                                              |



| Value | Description                                 |
|-------|---------------------------------------------|
| 0040  | Reject: Security Platform System Error      |
| 0042  | Reject: CVC 3 Unable to process             |
| 0043  | Reject: ATC outside allowed range           |
| 0044  | Reject: CVC 3 Invalid                       |
| 0045  | Reject: CVC 3 Unpredictable number mismatch |
| 0046  | Reject: ATC Replay                          |

## DE 60, subfield 2 values, in Mastercard In Control Service

The following values are valid in Mastercard In Control™ Service transactions.

### Values

| Value | Description                                                               |
|-------|---------------------------------------------------------------------------|
| 0060  | Reject: Virtual Card Number (expiration date does not match)              |
| 0061  | Reject: Virtual Card Number (expiration date expired)                     |
| 0062  | Reject: Virtual CVC 2 does not match                                      |
| 0063  | Reject: Validity Period Limit: In Control                                 |
| 0064  | Reject: Transaction Amount Limit Check                                    |
| 0065  | Reject: Cumulative Amount Limit Check                                     |
| 0066  | Reject: Transaction Number Usage                                          |
| 0067  | Reject: Merchant ID Limit                                                 |
| 0068  | Reject: Invalid Virtual Card Number–Real Card Number Mapping Relationship |
| 0069  | Reject: MCC Limit                                                         |
| 0070  | Reject: Database Status Bad                                               |
| 0071  | Reject: Decline Other                                                     |
| 0072  | Reject: Geographic Restriction                                            |
| 0073  | Reject: Transaction Type Restriction                                      |
| 0075  | Reject: Transaction Time/Date Restriction                                 |
| 0076  | Reject: Sanctions Score Restriction                                       |
| 0077  | Reject: Mastercard Send and Funding Transaction Count Check               |

## DE 60, subfield 2 values, in M/Chip On-Behalf Services

Following are valid values in M/Chip On-Behalf Services transactions.

### Values

| Value | Description                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 0032  | Reject: Chip Data Processing Error                                                                                                         |
| 0033  | Reject: PIN or CHIP Validation Failed                                                                                                      |
| 0034  | Reject: Chip validation failed                                                                                                             |
| 0035  | Reject: TVR/CVR validation failed                                                                                                          |
| 0037  | Reject: No matching key File for this PAN, PAN expiry date and KDI combination—Validation of ARQC and CVR/TR not performed, status unknown |
| 0038  | Reject: Security platform time out                                                                                                         |
| 0039  | Reject: Cryptogram not ARQC                                                                                                                |
| 0040  | Reject: Security platform processing error                                                                                                 |
| 0043  | Reject: ATC outside allowed range                                                                                                          |
| 0046  | Reject: ATC Replay                                                                                                                         |
| 0054  | Reject: Track Data Format Error                                                                                                            |
| 0055  | Reject: Chip CVC Invalid                                                                                                                   |
| 0056  | Reject: Chip CVC/CVC 1 Unable to process                                                                                                   |

## DE 60, subfield 2 values, in Mastercard Digital Enablement Service

Following are valid values in Mastercard Digital Enablement Service.

### Values

| Value | Description                                                         |
|-------|---------------------------------------------------------------------|
| 0201  | Reject: Invalid Token—Primary Account Number mapping relationship   |
| 0202  | Reject: Token in suspended status                                   |
| 0203  | Reject: Token deactivated                                           |
| 0204  | Reject: ATC Invalid—Not in List of Currently Active Single Use Keys |
| 0205  | Reject: ATC Replay                                                  |

| Value | Description                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------|
| 0206  | Reject: Invalid MD AC and UMD AC (Invalid Mobile PIN)                                                   |
| 0207  | Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit not Reached, Token not Suspended) |
| 0208  | Reject: Invalid MD AC; Valid UMD AC                                                                     |
| 0209  | Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended)         |
| 0210  | Reject: Unpredictable Number Length Indicator Mismatch                                                  |
| 0211  | Reject: TVR/CVR validation failed                                                                       |
| 0212  | Reject: Unable to Process                                                                               |
| 0213  | Reject: Invalid Token                                                                                   |
| 0215  | Reject: Declined by Transaction Analysis                                                                |
| 0032  | Reject: Chip Data Processing Error                                                                      |
| 0034  | Reject: Chip validation failed                                                                          |
| 0035  | Reject: TVR/CVR validation failed                                                                       |
| 0039  | Reject: Cryptogram not ARQC                                                                             |
| 0042  | Reject: CVC 3 Unable to process                                                                         |
| 0043  | Reject: CVC 3 ATC outside allowed range                                                                 |
| 0044  | Reject: CVC 3 Invalid                                                                                   |
| 0045  | Reject: CVC 3 Unpredictable number mismatch                                                             |
| 0046  | Reject: CVC 3 ATC Replay                                                                                |
| 0059  | Reject: Advice Code when DE 39 = 30 (Format Error)                                                      |

## DE 60, subfield 2 values, in Mastercard Merchant Presented QR Service

Following are valid values in Mastercard Merchant Presented QR Service transactions.

### Values

| Value | Description                     |
|-------|---------------------------------|
| 0064  | Reject: Transaction Limit Check |
| 0065  | Reject: Cumulative Limit Check  |

| Value | Description                    |
|-------|--------------------------------|
| 0072  | Reject: Geographic Restriction |

## DE 60, subfield 2 values, in Pay with Rewards

Following are valid values in Mastercard Pay with Rewards service.

### Values

| Value | Description                                                     |
|-------|-----------------------------------------------------------------|
| 0120  | Reject: Pay with Rewards—Insufficient points balance            |
| 0121  | Reject: Pay with Rewards—Redemption rule(s) failed              |
| 0122  | Reject: Pay with Rewards service was not performed successfully |
| 0123  | Reject: Pay with Rewards—Account not registered                 |
| 0124  | Reject: Pay with Rewards System error                           |

## DE 60, subfield 2 values, in PIN Validation

Following are the valid values in PIN Validation transactions.

### Values

| Value | Description                                                |
|-------|------------------------------------------------------------|
| 0050  | Reject: Unable to Process                                  |
| 0051  | Reject: Invalid PIN                                        |
| 0052  | Reject: Mandatory PIN Verification Value (PVV) not on file |
| 0052  | Reject: PIN Retry Exceeded (invalid PIN)                   |

## DE 60, subfield 2 values, in Private Label Processing

Following are valid values in Private Label Processing transactions.

### Values

| Value | Description                                                |
|-------|------------------------------------------------------------|
| 0116  | Reject: Merchant not allowed for Private Label transaction |

## DE 60, subfield 2 values, in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 0300  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0301  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 0310  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 0311  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (Global)

Following are valid values in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (global) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 0400  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0401  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 0410  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 0411  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in MCC, CAT Level, and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values for MCC, CAT Level, and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 0100  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0101  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |

| Value | Description                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| 0110  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction     |
| 0111  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction |

## DE 60, subfield 2 values, in MCC and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values in MCC and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 0500  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0501  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 0510  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 0511  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in MCC and Promotion Code in Failed Parameter Combinations (Global)

Following are valid values in MCC and Promotion Code in Failed Parameter Combinations (global) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 0600  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0601  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 0610  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 0611  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in TCC and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 0700  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0701  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |



| Value | Description                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| 0710  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction     |
| 0711  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction |

## DE 60, subfield 2 values, in TCC and Promotion Code in Failed Parameter Combinations (Global)

Following are valid values in TCC and Promotion Code in Failed Parameter Combinations (global) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 0800  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0801  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 0810  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 0811  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in MCC and CAT Level in Failed Parameter Combinations (Country-Specific)

Following are valid values in MCC and CAT Level in Failed Parameter Combinations (country-specific) transactions.

### Values

| Values | Description                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------------------|
| 0900   | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 0901   | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 0910   | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 0911   | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in MCC and CAT Level in Failed Parameter Combinations (Global)

Following are valid values in MCC and CAT Level in Failed Parameter Combinations (global) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1000  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 1001  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |

| Value | Description                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| 1010  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction     |
| 1011  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction |

## DE 60, subfield 2 values, in TCC and CAT Level in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC and CAT Level in Failed Parameter Combinations (country-specific) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1100  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 1101  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 1110  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 1111  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in TCC and CAT Level in Failed Parameter Combinations (Global)

Following are valid values in TCC and CAT Level in Failed Parameter Combinations (global) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1200  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 1201  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 1210  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 1211  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in MCC in Failed Parameter Combinations (Country-Specific)

Following are valid values in MCC in Failed Parameter Combinations (country-specific) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1300  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 1301  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |

| Value | Description                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| 1310  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction     |
| 1311  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction |

## DE 60, subfield 2 values, in MCC in Failed Parameter Combinations (Global)

Following are valid values in MCC in Failed Parameter Combinations (global) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1400  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 1401  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 1410  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 1411  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in TCC in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC in Failed Parameter Combinations (country-specific) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1500  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 1501  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |
| 1510  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card present at point of interaction       |
| 1511  | Reject: transaction limit test<br>Customer-defined limits caused transaction to fail; card not present at point of interaction   |

## DE 60, subfield 2 values, in TCC in Failed Parameter Combinations (Global)

Following are valid values in TCC in Failed Parameter Combinations (global) transactions.

### Values

| Value | Description                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1600  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card present at point of interaction     |
| 1601  | Reject: transaction limit test<br>Mastercard default limits caused transaction to fail; card not present at point of interaction |

| Value | Description                                                                                                                        |
|-------|------------------------------------------------------------------------------------------------------------------------------------|
| 1610  | Reject: transaction limit test<br><br>Customer-defined limits caused transaction to fail; card present at point of interaction     |
| 1611  | Reject: transaction limit test<br><br>Customer-defined limits caused transaction to fail; card not present at point of interaction |

## DE 60, subfield 2 values, in Miscellaneous Processing

Following are valid values in miscellaneous processing transactions.

### Values

| Value | Description                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------|
| 1700  | Accept: via X-Code Processing                                                                                          |
| 1701  | Reject: via X-Code Processing                                                                                          |
| 1702  | Reject: capture card via X-Code Processing                                                                             |
| 1903  | Reject: via MIP Transaction Blocking                                                                                   |
| 2000  | Reject: PIN data present in Authorization Request/0100 message (used in conjunction with Reversal Advice/0420 message) |
| 9999  | Reject: failed validation test because transaction type or authorization method was invalid                            |

## DE 60, subfield 2 values, in Send and other transactions blocking service

These values are valid in Send and other transactions blocking service transactions.

### Values

| Code | Description                               |
|------|-------------------------------------------|
| 5111 | Single txn limit - dom - core txn type    |
| 5112 | Single txn limit - dom - special txn type |
| 5113 | Single txn limit - xb - core txn type     |

| Code | Description                                       |
|------|---------------------------------------------------|
| 5114 | Single txn limit - xb - special txn type          |
| 5115 | Reserved for future use                           |
| 5121 | Daily accum limit - dom - core txn type           |
| 5122 | Daily accum limit - dom - special txn type        |
| 5123 | Daily accum limit - xb - core txn type            |
| 5124 | Daily accum limit - xb - special txn type         |
| 5125 | Daily accum limit - dom+xb - core txn type        |
| 5126 | Daily accum limit - dom+xb - special txn type     |
| 5127 | Daily accum limit - dom+xb - group of txn types   |
| 5128 | Reserved for future use                           |
| 5131 | Weekly accum limit - dom - core txn type          |
| 5132 | Weekly accum limit - dom - special txn type       |
| 5133 | Weekly accum limit - xb - core txn type           |
| 5134 | Weekly accum limit - xb special txn type          |
| 5135 | Weekly accum limit - dom+xb - core txn type       |
| 5136 | Weekly accum limit - dom+xb - special txn type    |
| 5137 | Weekly accum limit - dom+xb - group of txn types  |
| 5138 | Reserved for future use                           |
| 5141 | Monthly accum limit - dom - core txn type         |
| 5142 | Monthly accum limit - dom - special txn type      |
| 5143 | Monthly accum limit - xb - core txn type          |
| 5144 | Monthly accum limit - xb - special txn type       |
| 5145 | Monthly accum limit - dom+xb - core txn type      |
| 5146 | Monthly accum limit - dom+xb - special txn type   |
| 5147 | Monthly accum limit - dom+xb - group of txn types |
| 5148 | Reserved for future use                           |
| 5150 | TTI - Domestic not allowed                        |
| 5151 | TTI - ISS - XB not allowed                        |
| 5152 | TTI - Acceptor - XB not allowed                   |
| 5153 | Special txn type - Domestic not allowed           |
| 5154 | Special txn type - ISS - XB not allowed           |



| Code | Description                                  |
|------|----------------------------------------------|
| 5155 | Special txn type - Acceptor - XB not allowed |
| 5156 | MCC not allowed for country                  |
| 5157 | MCC-TTI combination not allowed for country  |
| 5158 | Country code not allowed                     |
| 5159 | Transaction currency not allowed             |
| 5160 | Other restrictions                           |
| 5161 | Reserved for future use                      |
| 5162 | Reserved for future use                      |
| 5163 | Reserved for future use                      |
| 5170 | Product code is not eligible for TTI         |
| 5171 | Account not allowed for TTI                  |
| 5172 | Account not allowed for Special txn type     |
| 5173 | Reserved for future use                      |
| 5197 | Missing TTI                                  |
| 5198 | Legacy or restricted use TTI                 |
| 5199 | TC- MCC- TTI combination not allowed         |

## Subfield 3 (Advice Detail Text)

Subfield 3 (Advice Detail Text) is optional and may be used to contain textual information supplementary to the Advice Detail Code. Advice Detail Text data is determined individually for each program and service.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...53                   |
| Data Field          | Contents of positions 8–60 |
| Justification       | Left                       |

### Values

A two character alphanumeric value that identifies which rule is responsible for ODD. Issuers should be prepared to receive any two character value in DE 60, subfield 3 (Advice Details Text).

For more information on valid values for Send Blocking Service transactions, refer to *Mastercard Send and Funding Transactions Implementation and Operations for MIP Connections*.

## DE 61 (Point-of-Service [POS] Data)

DE 61 (Point-of-Service [POS] Data) indicates the conditions that exist at the point of service at the time of the transaction. Note that DE 61 supersedes and replaces the ISO-specified DE 25 (Point-of-Service [POS] Condition Code) which is not used in the *Mastercard Network Processing Dual Message Authorization System Guide*.

### Attributes

| Attribute           | Description           |
|---------------------|-----------------------|
| Data Representation | ans...026; LLLVAR     |
| Data Length         | 3 positions           |
| Data Field          | Contents of subfields |
| Subfields           | 14                    |
| Justification       | See subfields         |

### Usage

Following is the usage of DE 61 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | M   | X   | M   |
| Authorization Advice/0120: acquirer-generated | M   | X   | M   |
| Authorization Advice/0120: system-generated   | .   | X   | M   |
| Reversal Request/0400                         | M   | X   | M   |

### Values

See subfields.

Three basic categories of data include:

- POS Condition Code: Indicating POS terminal data (mandatory) in subfields 1–11

- POS Authorization Life Cycle: Indicating the transaction is a preauthorization request and the specific amount of days for which the preauthorization will remain valid (the number of days the issuer or its agent will guarantee or hold funds) in subfield 12
- POS geographic reference: Indicating the specific merchant location of the transaction in subfields 13–14

### Application notes

This data element will be edited as described in the [Dual Message Authorization System Edits](#).

## Subfield 1 (POS Terminal Attendance)

DE 61, subfield 1 (POS Terminal Attendance) indicates if the acceptor is attending the terminal.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

| Value | Description                                                                           |
|-------|---------------------------------------------------------------------------------------|
| 0     | Attended Terminal                                                                     |
| 1     | Unattended terminal (cardholder-activated terminal [CAT], home PC, mobile phone, PDA) |
| 2     | No terminal used (voice/audio response unit [ARU] authorization); server              |

## Subfield 2 (Reserved for Future Use)

DE 61, subfield 2 (Reserved) is reserved.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | n-1         |

| Attribute     | Description            |
|---------------|------------------------|
| Data Field    | Contents of position 2 |
| Justification | N/A                    |

### Values

Zero-filled.

## Subfield 3 (POS Terminal Location)

DE 61, subfield 3 (POS Terminal Location) indicates the terminal location.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

### Values

| Value | Description                                                                                  |
|-------|----------------------------------------------------------------------------------------------|
| 0     | On premises of acceptor facility                                                             |
| 1     | Off premises of acceptor facility (merchant terminal—remote location)                        |
| 2     | Off premises of acceptor facility (cardholder terminal including home PC, mobile phone, PDA) |
| 3     | No terminal used (voice/ARU authorization); server                                           |
| 4     | On premises of acceptor facility (cardholder terminal including home PC, mobile phone, PDA)  |
| 8     | Additional POS Terminal Locations                                                            |

**NOTE: For more details, refer to DE 48, [Subelement 29 \(Additional POS Terminal Locations\)](#).**

## Subfield 4 (POS Cardholder Presence)

DE 61, subfield 4 (POS Cardholder Presence) indicates whether the cardholder is present at the point of service and explains the condition if the cardholder is not present.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 4 |
| Justification       | N/A                    |

### Values

| Value | Description                                                                      |
|-------|----------------------------------------------------------------------------------|
| 0     | Cardholder present                                                               |
| 1     | Cardholder not present, unspecified                                              |
| 2     | Cardholder not present (mail/facsimile order)                                    |
| 3     | Cardholder not present (phone or Automated Response Unit [ARU])                  |
| 4     | Recurring transactions                                                           |
| 5     | Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA]) |

## Subfield 5 (POS Card Presence)

DE 61, subfield 5 (POS Card Presence) indicates if the card is present at the point of service.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 5 |
| Justification       | N/A                    |

### Values

| Value | Description      |
|-------|------------------|
| 0     | Card present     |
| 1     | Card not present |

## Subfield 6 (POS Card Capture Capabilities)

DE 61, subfield 6 (POS Card Capture Capabilities) indicates whether the terminal has card capture capabilities.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 6 |
| Justification       | N/A                    |

### Values

| Value | Description                                             |
|-------|---------------------------------------------------------|
| 0     | Terminal/operator does not have card capture capability |
| 1     | Terminal/operator has card capture capability           |

## Subfield 7 (POS Transaction Status)

DE 61, subfield 7 (POS Transaction Status) indicates the purpose or status of the request.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 7 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Value | Description                                                           |
|-------|-----------------------------------------------------------------------|
| 0     | Normal request (original presentment)                                 |
| 1     | Deferred Authorization                                                |
| 2     | Identity Check Phone Order                                            |
| 3     | ATM Installment Inquiry                                               |
| 4     | Preauthorized request                                                 |
| 5     | Time Based Payment Authorization Request—Brazil domestic transactions |
| 6     | ATC Update                                                            |
| 8     | Account Status Inquiry Service (ASI)                                  |
| 9     | Tokenization Request/Notification                                     |

### Application notes

For DE 61, subfield 7, value 8, see separate description of Account Status Inquiry Service (both Purchase ASI and Payment ASI) and Product Inquiry Service.

## Subfield 8 (POS Transaction Security)

DE 61, subfield 8 (POS Transaction Security) indicates the acceptor's confidence in the transaction security level.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 8 |
| Justification       | N/A                    |

### Values

| Value | Description                                   |
|-------|-----------------------------------------------|
| 0     | No security concern                           |
| 1     | Suspected fraud (merchant suspicious—code 10) |
| 2     | ID verified                                   |

## Subfield 9 (POS Transaction Status - Extended)

DE 61, subfield 9 (POS Transaction Status - Extended) indicates the purpose of status of the request.

### Attributes

| Attributes          | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 9 |
| Justification       | N/A                    |

### Values

| Value | Description                                 |
|-------|---------------------------------------------|
| 0     | Normal request (not for Validation request) |
| 1     | Validation request                          |
| 2     | Product Status Inquiry Service (Standalone) |

### Application Notes

When DE 61 (Point of Service [POS] Data), subfield 9 (POS Transaction Status - Extended) contains value 1 (Validation Request), then DE 61 (Point of Service [POS] Data), subfield 07 (POS Transaction Status) must contain a value 0 (Normal request) or a value 8 (Account Status Inquiry Service [ASI]).

DE 61 (Point of Service [POS] Data), subfield 9 (POS Transaction Status - Extended) and DE 61 (Point of Service [POS] Data), subfield 7 (POS Transaction Status) in conjunction with DE 108



(Additional Transaction Reference Data), subelement 07, subfield 01 identifies a Name Validation Service request.

Validation requests are not eligible for alternate processing (Stand-In or X-Code processing). When an issuer is not available, or is offline, Validation requests will be declined using the Authorization Request Response/0110 message containing DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

Mastercard will perform the following edits.

### Authorization Request edits

| When the Authorization Request/0100 message with...                                                                                                                                                                                                                                     | Then the Dual Message System (Authorization) will...                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 61, subfield 7 (POS Transaction Status) contains value 0 and DE 61, subfield 9 (POS Transaction Status - Extended) contains value other than 0 or 1                                                                                                                                  | reject the message and send an Authorization Response/0110 message to the acquirer with:<br><br>DE 39 = 30 (Format error)<br><br>DE 44 = 061      |
| DE 61, subfield 7 (POS Transaction Status) (POS Transaction Status) contains value other than 8 or 0 and DE 61, subfield 9 (POS Transaction Status - Extended) (POS Transaction Status - Extended) contains value 1                                                                     | reject the message and send an Authorization Response/0110 message to the acquirer with:<br><br>DE 39 = 30 (Format error)<br><br>DE 44 = 061      |
| DE 61, subfield 7 (POS Transaction Status) contains value 0 and DE 61, subfield 9 (POS Transaction Status - Extended) contains 1 and DE 3 (Processing Code) contains value 28 and DE 4 (Amount, Transaction) contains value other than zeroes                                           | reject the message and send an Authorization Response/0110 message to the acquirer with:<br><br>DE 39 = 30 (Format error)<br><br>DE 44 = 004      |
| DE 61, subfield 7 (POS Transaction Status) contains value 0 and DE 61, subfield 9 (POS Transaction Status - Extended) contains value 1 and DE 3 (Processing Code) contains value other than 28                                                                                          | will reject the message and send an Authorization Response/0110 message to the acquirer with:<br><br>DE 39 = 30 (format error)<br><br>DE 44 = 003 |
| DE 61, subfield 7 (POS Transaction Status) contains value 0 and DE 61, subfield 9 (POS Transaction Status - Extended) contains 1 and DE 3 (Processing Code) contains value 28 and DE 108, subelement 07 (Data Validation Request), subfield 01 (Name Validation Request) is not present | reject the message and send an Authorization Response/0110 message to the acquirer with:<br><br>DE 39 = 30 (Format error)<br><br>DE 44 = 108007   |

| When the Authorization Request/0100 message with...                                                                                                                                                                                                                                                                                                          | Then the Dual Message System (Authorization) will...                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| DE 61, subfield 7 (POS Transaction Status) contains value 0 and DE 61, subfield 9 (POS Transaction Status - Extended) contains 1 and DE 3 (Processing Code) contains value 28 and DE 108, subelement 07 (Data Validation Request), subfield 01 (Name Validation Request) is not 10 or 11                                                                     | reject the message and send an Authorization Response/0110 message to the acquirer with:<br>DE 39 = 30 (Format error)<br>DE 44 = 108007 |
| DE 61 subfield 7 (POS Transaction Status) contains value 0 and DE 61, subfield 9 (POS Transaction Status - Extended) contains 1 and DE 3 (Processing Code) contains value 28 and DE 108, subelement 07 (Data Validation Request), subfield 01 (Name Validation Request) contains value 10 and DE 108, subelement 01 (Receiver/Recipient Data) is not present | reject the message and send an Authorization Response/0110 message to the acquirer with:<br>DE 39 = 30 (Format error)<br>DE 44 = 108001 |
| DE 61, subfield 7 (POS Transaction Status) and DE 3 (Processing Code) contain value 28 and value 0 and DE 61, subfield 9 (POS Transaction Status - Extended) contains 1 and DE 108, subelement 07 (Data Validation Request), subfield 01 (Name Validation Request) contains value 11 and DE 108, subelement 02 (Sender Data) is not present                  | reject the message and send an Authorization Response/0110 message to the acquirer with:<br>DE 39 = 30 (Format error)<br>DE 44 = 108002 |
| DE 61, subfield 7 (POS Transaction Status) contains value 8 and DE 3 (Processing Code) contains value 28 or 00 and DE 108, subelement 07 (Data Validation Request), subfield 01 (Name Validation Request) is 10 and DE 108, subelement 01 (Receiver/Recipient Data) is not present                                                                           | not forward DE 108, subelement 07 (Data Validation Request) to the issuer                                                               |
| DE 61, subfield 7 (POS Transaction Status) contains value 8 and DE 3 (Processing Code) contains value 28 or 00 and DE 108, subelement 07 (Data Validation Request), subfield 01 (Name Validation Request) is 11 and DE 108, subelement 02 (Sender Data) is not present                                                                                       | not forward DE 108, subelement 07 (Data Validation Request) to the issuer                                                               |
| DE 61, subfield 7 (POS Transaction Status) contains value other than 8 and DE 61, subfield 9 (POS Transaction Status - Extended) contains 0 and DE 108, subelement 07 (Data Validation Request), subfield 01 (Name Validation Request) is present                                                                                                            | not forward DE 108, subelement 07 (Data Validation Request) to the issuer                                                               |

| When the Authorization Request/0100 message with...                                                                                                                                           | Then the Dual Message System (Authorization) will...                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 61, subfield 7 (POS Transaction Status) contains values other than 0<br>and<br>DE 61, subfield 9 (POS Transaction Status - Extended) contains value 2                                      | will reject the transaction and send an Authorization Response/0110 message to the acquirer with: <ul style="list-style-type: none"> <li>• DE 39 = 30 (Format Error)</li> <li>• DE 44 = 061</li> </ul> |
| DE 61, subfield 7 contains value 0<br>and<br>DE 61, subfield 9 contains value 2<br>and<br>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains values other than 00 | will reject the transaction and send an Authorization Response/0110 message to the acquirer with: <ul style="list-style-type: none"> <li>• DE 39 = 30 (Format Error)</li> <li>• DE 44 = 003</li> </ul> |
| DE 61, subfield 7 contains value 0<br>and<br>DE 61, subfield 9 contains value 2<br>and<br>Issuer Account Range is not eligible for the Product Inquiry service                                | will reject the transaction and send an Authorization Response/0110 message to the acquirer with: <ul style="list-style-type: none"> <li>• DE 39 = 12 (Invalid Transaction)</li> </ul>                 |
| DE 61, subfield 7 contains value 0<br>and<br>DE 61, subfield 9 contains value 2<br>and<br>DE 3, subfield 1 contains 00<br>and<br>DE 4 (Amount, Transaction) is not zero                       | will reject the transaction and send an Authorization Response/0110 message to the acquirer with: <ul style="list-style-type: none"> <li>• DE 39 = 30 (Format Error)</li> <li>• DE 44 = 004</li> </ul> |
| DE 61, subfield 9 contains value 2<br>and<br>The Message Type Identifier (MTI) is 0120 (Authorization Advice: acquirer-generated) or 400 (Reversal Request)                                   | will reject the transaction and send a response message to the acquirer with: <ul style="list-style-type: none"> <li>• DE 39 = 12 (Invalid Transaction)</li> </ul>                                     |

### Authorization Advice and Reversal Request edits

| Authorization Advice/0120, Reversal Request/0400, message with...      | Then the Dual Message System (Authorization) will...                                                                                                                           |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 61, subfield 9 (POS Transaction Status - Extended) contains value 1 | reject the message and send an Authorization Advice Response/0130, or Reversal Request Response/0410 message to the acquirer with:<br>DE 39 = 30 (Format error)<br>DE 44 = 061 |

## Subfield 10 (Cardholder-Activated Terminal Level)

DE 61, subfield 10 (Cardholder-Activated Terminal Level) indicates the type of cardholder activated terminal used by the cardholder to initiate the transaction.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | n-1                     |
| Data Field          | Contents of position 10 |
| Justification       | N/A                     |

### Values

| Value | Description                                                   |
|-------|---------------------------------------------------------------|
| 0     | Not a CAT transaction                                         |
| 1     | Authorized Level 1 CAT: automated dispensing machine with PIN |
| 2     | Authorized Level 2 CAT: self-service terminal                 |
| 3     | Authorized Level 3 CAT: limited-amount terminal               |
| 4     | Authorized Level 4 CAT: in-flight commerce                    |
| 5     | Reserved                                                      |
| 6     | Authorized Level 6 CAT: electronic commerce                   |
| 7     | Authorized Level 7 CAT: transponder transaction               |

| Value                                                                             | Description            |
|-----------------------------------------------------------------------------------|------------------------|
| 8                                                                                 | Remote terminal        |
| 9                                                                                 | MPOS Acceptance Device |
| <b>NOTE: Mobile POS are always attended devices and not cardholder activated.</b> |                        |

### Application notes

When the recipient of the money transfer (Payment Transaction; DE 3 subfield 1 = 09) is not present, the transaction is not considered a CAT level transaction. In that case the CAT level = 0.

## Subfield 11 (POS Card Data Terminal Input Capability Indicator)

DE 61, subfield 11 (POS Card Data Terminal Input Capability Indicator) indicates the set of methods supported by the terminal for the input of account number, card, or mobile device data.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | n-1                     |
| Data Field          | Contents of position 11 |
| Justification       | N/A                     |

### Values

| Value | Description                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | Input capability unknown or unspecified.                                                                                                                                                                             |
| 1     | No terminal used (voice/ARU authorization); server.                                                                                                                                                                  |
| 2     | Terminal supports magnetic stripe input only.                                                                                                                                                                        |
| 3     | Terminal supports contactless EMV input. The terminal may also support one or more other card input types, including contactless magstripe input, EMV contact chip input, magnetic stripe input and key entry input. |

| Value | Description                                                                                                                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | Contactless Mag Stripe (Proximity Chip)<br><br>Terminal supports contactless magstripe input but not contactless EMV input. The terminal also may support one or more other card input types, including EMV contact chip input, magnetic stripe input, and key entry input. |
| 5     | Terminal supports EMV contact chip input and magnetic stripe input.                                                                                                                                                                                                         |
| 6     | Terminal supports key entry input only.                                                                                                                                                                                                                                     |
| 7     | Terminal supports magnetic stripe input and key entry input.                                                                                                                                                                                                                |
| 8     | Terminal supports EMV contact chip input, magnetic stripe input and key entry input.                                                                                                                                                                                        |
| 9     | Terminal supports EMV contact chip input only.                                                                                                                                                                                                                              |

### Application notes

DE 61, subfield 11 values 3, 4 (contactless related), 5, 8, and 9 (chip contact related) can ONLY be used if the terminal is chip certified by Mastercard.

DE 61, subfield 11 is used in conjunction with DE 22 (Point of Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode). DE 22, subfield 1 values indicate the method by which the PAN was entered.

DE 61, subfield 11 indicates the terminal data input capabilities and not the specific manner in which the terminal captured the card data for a given transaction.

For example, values 5 and 8 could be used for a magnetic stripe transaction originating from an EMV contact chip capable terminal when the contact magnetic stripe reader was used instead. Similarly, value 3 would be used for a contact chip transaction originating from a contactless EMV capable terminal when the EMV contact chip reader was used.

## Subfield 12 (POS Authorization Life Cycle)

DE 61, subfield 12 (POS Authorization Life Cycle) indicates the number of days a preauthorization will stay in effect for Visa Custom Payment Service automobile rentals and hotel reservations—otherwise it must contain zeros.

### Attributes

| Attribute           | Description                     |
|---------------------|---------------------------------|
| Data Representation | n-2                             |
| Data Field          | Contents of positions 12 and 13 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Indicates the number of days the preauthorization stays in effect; ATM and Maestro POS transactions should use 01, Visa customer transactions use applicable value. Must be zero filled when not applicable.

## Subfield 13 (POS Country Code [or Sponsored Merchant Information, if applicable])

DE 61, subfield 13 (POS Country Code) indicates the country of the POS location (not the acquirer location) using ISO-specified codes.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-3                         |
| Data Field          | Contents of positions 14–16 |
| Justification       | N/A                         |

### Values

Refer to the *Quick Reference Booklet* for valid country codes.

## Subfield 14 (Acceptor Postal Code)

This subfield contains the postal code of the acceptor.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data representation | ans...10                    |
| Data field          | Contents of positions 17–26 |
| Justification       | Left                        |

## Processing rules

The Acceptor Postal Code must be present for all transactions and contain a valid code for countries that utilize postal codes. This information must be populated and formatted as defined by country postal standards.

## Program rules

Payment facilitator-enabled transactions: The sponsored merchant postal code must be provided.

# Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on data element 61 (Point-of-Service [POS] Data).

## Authorization Request/0100 Character Edit

### Associated edits

| WHEN the acquirer...                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sends an Authorization Request/0100 message containing less than nine characters                       | sends an Authorization Request Response/0110 message to the acquirer where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 061</li> </ul> |
| sends an Authorization Request/0100 message where DE 61, subfields 1–9 contain a non-numeric character | sends an Authorization Request Response/0110 message to the acquirer where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 061</li> </ul> |

## Non-ATM CAT Level 1 Transactions

### Associated edits

| WHEN the Authorization Request/0100 message contains...                                                                                                                                                                                                                                                         | THEN the Dual Message Authorization System...                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>DE 61, subfield 10, value 1 (Authorized Level 1 CAT: Automated dispensing machine with a PIN)</li> <li>DE 52 (PIN Data) is not present</li> <li>DE 55 (Integrated Circuit Card [ICC] System-related Data) is present</li> <li>DE 110 Dataset01 is not present</li> </ul> | forwards the Authorization Request/0100 message to the issuer. |



| WHEN the Authorization Request/0100 message contains...                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>DE 61, subfield 10, value 1</li> <li>DE 52 or Dataset 01 is present</li> <li>DE 55 is present</li> </ul>                                           | forwards the Authorization Request/0100 message to the issuer.                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>DE 61, subfield 10, value 1</li> <li>DE 52 or Dataset 01 is present</li> <li>DE 55 is present</li> </ul>                                           | forwards the Authorization Request/0100 message to the issuer.                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>DE 61, subfield 10, value 1</li> <li>DE 52 or Dataset 01 is present</li> <li>DE 55 is present</li> <li>DE 110 Dataset 01 is not present</li> </ul> | sends an Authorization Request Response/0110 message to the acquirer where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 052</li> </ul> |

## Electronic Commerce Transactions

### Associated edits

| WHEN the Authorization Request/0100 message contains...                  | THEN the Dual Message Authorization System...                                                                                                                                                                |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE61 subfield 4 is 5<br>and<br>DE22 subfield 1 is not 09, 10, 81, or 82  | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 061</li> </ul> |
| DE61 subfield 10 is 6<br>and<br>DE22 subfield 1 is not 09, 10, 81, or 82 | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 061</li> </ul> |

### Associated edits

| IF...                                                                                                                                                                   | THEN...                                                                                                                                  | Otherwise, the Dual Message Authorization System...                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 42, position 3 is value 1, 2, 3, 4, 5, 6, or 7<br>and<br>DE 61, subfield 4 is value 4 (Standing order/recurring transactions) or 5 (Electronic Order) | DE 61, subfield 7 must not contain value 2 (Identity Check Phone Order)<br>and<br>Subfield 10 must contain value 6 (Electronic commerce) | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 061</li> </ul> |

### Phone Order Transactions

### Associated edits

| IF...                                            | THEN...                                                                                                                                                           | Otherwise, the Dual Message Authorization System...                                                                                                                                                          |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 42, position 3 is value 1 or 2 | DE 61, subfield 4 must contain value 3 or 4<br>and<br>Subfield 7 must contain value 2 (Identity Check Phone Order)<br>and<br>Subfield 10 must not contain value 6 | sends the acquirer an Authorization Request Response/0110 where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 061</li> </ul> |

### Account Status Inquiry Service

### Associated edits

| WHEN...                                                                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) and DE 4 (Amount, Transaction) do not equal 0 and DE 3 (Processing Code) contains a value other than zero | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 004</li> </ul> |

| WHEN...                                                                                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains value 8 (Account Status Inquiry Service [ASI])<br>and<br>DE 4 (Amount, Transaction) contains a value equal to zero<br>and<br>DE 3 (Processing Code) contains a value other than 00 (Purchase), 01 (Withdrawal), and 28 (Payment Transaction) | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 003</li> </ul> |

## Contactless Chip Transaction Edits

### Associated edits

| WHEN the Authorization Request/0100, Authorization Advice/0120, Reversal Request/0400, message contains...                                                                                                                                                                                               | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the PAN entry mode (DE 22 [Point-of-Service Entry Mode], subfield 1 [POS Terminal PAN Entry Mode]) of 07 (PAN auto-entry via contactless M/Chip) and DE 61 (Point-of-Service [POS] Data), subfield 11 (POS Card Data Terminal Input Capability Indicator) is not 3 (Contactless M/Chip [Proximity Chip]) | sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 061</li> </ul> |

## Support Tap to More edits

### Associated edits

The Dual Message Authorization System validates the POI to ensure DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level) contains value 8 as one of the valid values on MoneySend payment transactions

| WHEN the originating institution...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | THEN the Authorization Platform...                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is registered for the Mastercard MoneySend Payment Transaction Program or Mastercard Gaming Payments Program and sends an Authorization Request/0100 message or a Reversal Request/0400 message where: <ul style="list-style-type: none"><li>DE 3, subfield 1 contains value 28</li><li>DE 18 contains an MCC associated with MoneySend Payment or Gaming and Gambling payment</li><li>DE 48, subelement 77 contains the Transactions Type Identifier associated with the Mastercard Send Payment or Gaming and Gambling payment program</li><li>DE 108 is present</li></ul> AND <ul style="list-style-type: none"><li>DE 61, subfield 10 does not contain one of the following values:<ul style="list-style-type: none"><li>0 = Not a CAT transaction</li><li>1 = Authorized Level 1 CAT: Automated dispensing machine with PIN</li><li>2 = Authorized Level 2 CAT: Self-service terminal</li><li>6 = Authorized Level 6 CAT: Electronic commerce</li><li>8 = Remote terminal</li></ul></li></ul> | sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message where: <ul style="list-style-type: none"><li>DE 39 = 30 (Format Error)</li><li>DE 44 = 061010 (Cardholder-Activated Terminal Level)</li></ul> |

## DE 62 (Intermediate Network Facility [INF] Data)

DE 62 (Intermediate Network Facility [INF] Data), contains "acquiring network trace information" that INFs may require to quickly and accurately route Administrative Advice/0620 messages back to the original acquiring institution. DE 62 assists acquiring INF facilities that connect directly to the Dual Message Authorization System. It allows these INFs to maintain sufficient information within a message to permit immediate online routing of chargebacks and retrieval requests without the requirement of maintaining large online reference databases containing the original transactions.

### Attributes

| Attribute           | Description       |
|---------------------|-------------------|
| Data Representation | ans...999; LLLVAR |

| Attribute     | Description                 |
|---------------|-----------------------------|
| Data Length   | 3 positions                 |
| Data Field    | Contents of positions 1–999 |
| Subfields     | N/A                         |
| Justification | N/A                         |

## Usage

Following is the usage of DE 62 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | O   | P   | O   |
| Authorization Request Response/0110                                                             | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated                                                   | O   | P   | O   |
| Authorization Advice/0120: system-generated                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | CE  | XE  | CE  |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | CE  | XE  | •   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | •   | XE  | CE  |
| Reversal Request/0400                                                                           | O   | P   | O   |
| Reversal Request Response/0410                                                                  | CE  | XE  | CE  |
| Reversal Advice/0420                                                                            | •   | X   | C   |
| Reversal Advice Response/0430                                                                   | CE  | XE  | •   |
| Administrative Request/0600                                                                     | O   | P   | O   |
| Administrative Request Response/0610                                                            | CE  | XE  | CE  |
| Administrative Advice/0620: system generated                                                    | •   | X   | C   |
| Administrative Advice/0620: member-generated                                                    | C   | P   | C   |
| Administrative Advice Response/0630                                                             | CE  | XE  | •   |

## Values

DE 62 is a free-format, variable-length alphanumeric data element that may be used to store unique acquiring network ID codes, acquiring network chaining data, or other information useful to INFs in routing online chargebacks and retrieval requests.

## Application notes

DE 62 is an optional data element that customers may place in any originating Authorization Request/0100 or Authorization Advice/0120. Subsequently, this data element (if present in an original transaction) is required to be returned without alteration in any Administrative Advice (Retrieval)/0620 pertaining to the original transaction. It contains INF network trace information that allows acquiring INFs to directly route the chargeback or retrieval request to the original acquirer.

When the Dual Message Authorization System generates an Administrative Advice/0620, it places the value MCBN620060000xxx, where xxx is the MIP ID. When a customer generates a corresponding Administrative Advice Response/0630, DE 62 must contain the same value.

When the Dual Message Authorization System generates a System Generated Advice/0420, it sets a value of MCBN620060000. When a customer generates a corresponding System Generated Advice Response/0430, DE 62 must contain the same value.

# DE 63 (Network Data)

DE 63 (Network Data) is generated by the Dual Message Authorization System for each originating message routed through the network. The receiver must retain the data element and use it in any response or acknowledgment message associated with the originating message.

## Attributes

| Attribute           | Description           |
|---------------------|-----------------------|
| Data Representation | an...050; LLLVAR      |
| Data Length         | 3 positions           |
| Data Field          | Contents of subfields |
| Subfields           | 2                     |
| Justification       | See subfields         |

## Usage

Following is the usage of DE 63 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | .   | X   | M   |
| Authorization Request Response/0110                                                             | ME  | X   | M   |
| Authorization Advice/0120: acquirer-generated                                                   | .   | X   | M   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | ME  | X   | M   |
| Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)    | ME  | XE  | .   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | X   | M   |
| Authorization Response Acknowledgment/0180                                                      | ME  | XE  | .   |
| Authorization Response Negative Acknowledgment/0190                                             | .   | XE  | ME  |
| Issuer File Update Request/0302                                                                 | .   | X   | .   |
| Issuer File Update Request Response/0312                                                        | .   | X   | M   |
| Reversal Request/0400                                                                           | .   | X   | M   |
| Reversal Request Response/0410                                                                  | ME  | X   | M   |
| Reversal Advice/0420                                                                            | .   | X   | M   |
| Reversal Advice Response/0430                                                                   | ME  | XE  | .   |
| Administrative Request/0600                                                                     | .   | X   | M   |
| Administrative Request Response/0610                                                            | ME  | X   | M   |
| Administrative Advice/0620: system-generated                                                    | .   | X   | M   |
| Administrative Advice/0620: member-generated                                                    | M   | X   | .   |
| Administrative Advice Response/0630                                                             | ME  | XE  | .   |
| Network Management Request/0800: sign-on/sign-off                                               | .   | X   | .   |
| Network Management Request/0800: network connection Status, member-generated                    | .   | X   | .   |
| Network Management Request/0800: network connection Status, system-generated                    | .   | X   | M   |
| Network Management Request/0800: PEK exchange                                                   | .   | X   | M   |
| Network Management Request/0800: PEK exchange—On Demand                                         | .   | X   | .   |

| Message                                                                               | Org | Sys | Dst |
|---------------------------------------------------------------------------------------|-----|-----|-----|
| Network Management Request Response/0810: network connection status, member-generated | ME  | XE  | .   |
| Network Management Request Response/0810: network connection status, system-generated | .   | X   | M   |
| Network Management Request Response/0810: sign-on/sign-off                            | .   | X   | M   |
| Network Management Request Response/0810: PEK exchange                                | ME  | X   | .   |
| Network Management Request Response/0810: PEK exchange, on demand                     | .   | X   | M   |
| Network Management Advice/0820: PEK exchange                                          | .   | X   | M   |

### Values

See subfields.

### Application notes

This data element contains blanks if the transaction fails edit checking and is rejected before determining the appropriate Financial Network Code.

## Subfield 1 (Financial Network Code)

DE 63, subfield 1 (Financial Network Code) identifies the specific program or service (for example, the financial network, financial program, or card program) with which the transaction is associated. DE 63 will contain the graduated product when the issuer's cardholder account participates in the Product Graduation Account Level Management service.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-3                      |
| Data Field          | Contents of positions 1–3 |
| Justification       | Left                      |

### Valid values

| Code | Description   | MC | NP | VI | TE | MS | CI |
|------|---------------|----|----|----|----|----|----|
| ACS  | Digital Debit | √  |    |    |    |    |    |



| Code | Description                                                                                                    | MC | NP | VI | TE | MS | CI |
|------|----------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| AMX  | American Express<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program. |    |    |    | √  |    |    |
| BPC  | Bill Pay Commercial                                                                                            | √  |    |    |    |    |    |
| BPD  | Business Premium Debit                                                                                         | √  |    |    |    |    |    |
| BPE  | World Elite Business Debit                                                                                     | √  |    |    |    |    |    |
| CBL  | Carte Blanche<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.    |    |    |    | √  |    |    |
| CIR  | Cirrus                                                                                                         |    |    |    |    |    | √  |
| DAG  | Gold Debit Mastercard Salary                                                                                   | √  |    |    |    |    |    |
| DAP  | Platinum Debit Mastercard Salary                                                                               | √  |    |    |    |    |    |
| DAS  | Standard Debit Mastercard Salary                                                                               | √  |    |    |    |    |    |
| DBS  | Consumer Charge Card                                                                                           | √  |    |    |    |    |    |
| DCB  | Line of Credit or Offset Linked Card                                                                           | √  |    |    |    |    |    |
| DCC  | Diners Club<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.      |    |    |    | √  |    |    |
| DCO  | Consumer Elite Card                                                                                            | √  |    |    |    |    |    |
| DLA  | Business Executive Charge Card                                                                                 | √  |    |    |    |    |    |
| DLG  | Debit Mastercard Gold: Delayed Debit                                                                           | √  |    |    |    |    |    |
| DLH  | Debit Mastercard World Embossed: Delayed Debit                                                                 | √  |    |    |    |    |    |
| DLI  | Debit Mastercard Standard ISIC Student Card: Delayed Debit                                                     | √  |    |    |    |    |    |
| DLP  | Debit Mastercard Platinum: Delayed Debit                                                                       | √  |    |    |    |    |    |
| DLS  | Debit Mastercard Card: Delayed Debit                                                                           | √  |    |    |    |    |    |
| DLU  | Debit Mastercard Unembossed: Delayed Debit                                                                     | √  |    |    |    |    |    |
| DOS  | Standard Social                                                                                                | √  |    |    |    |    |    |
| DSV  | Discover<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.         |    |    |    | √  |    |    |

| Code | Description                                                                                                              | MC | NP | VI | TE | MS | CI |
|------|--------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| DWF  | Debit Mastercard Humanitarian Prepaid<br>This product code is not currently available for issuance in the Canada region. | √  |    |    |    |    |    |
| ETA  | Mastercard Installment Payments A                                                                                        | √  |    |    |    |    |    |
| ETB  | Mastercard Installment Payments B                                                                                        | √  |    |    |    |    |    |
| ETC  | Mastercard Installment Payments C                                                                                        | √  |    |    |    |    |    |
| ETD  | Mastercard Installment Payments D                                                                                        | √  |    |    |    |    |    |
| ETE  | Mastercard Installment Payments E                                                                                        | √  |    |    |    |    |    |
| ETF  | Mastercard Installment Payments F                                                                                        | √  |    |    |    |    |    |
| ETG  | Mastercard Installment Payments G                                                                                        | √  |    |    |    |    |    |
| ETH  | Prepaid Installment Payments A                                                                                           | √  |    |    |    |    |    |
| ETI  | Prepaid Installment Payments B                                                                                           | √  |    |    |    |    |    |
| ETJ  | Prepaid Installment Payments C                                                                                           | √  |    |    |    |    |    |
| ETK  | Prepaid Installment Payments D                                                                                           | √  |    |    |    |    |    |
| ETL  | Prepaid Installment Payments E                                                                                           | √  |    |    |    |    |    |
| ETM  | Prepaid Installment Payments F                                                                                           | √  |    |    |    |    |    |
| ETN  | Prepaid Installment Payments G                                                                                           | √  |    |    |    |    |    |
| EXC  | ADP/Exchange<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.               |    | √  |    |    |    |    |
| FIA  | Mastercard® B2B VIP 27                                                                                                   | √  |    |    |    |    |    |
| FIB  | Mastercard® B2B VIP 28                                                                                                   | √  |    |    |    |    |    |
| FIC  | Mastercard® B2B VIP 29                                                                                                   | √  |    |    |    |    |    |
| FID  | Mastercard® B2B VIP 30                                                                                                   | √  |    |    |    |    |    |
| FIE  | Mastercard® B2B VIP 31                                                                                                   | √  |    |    |    |    |    |
| FIF  | Mastercard® B2B VIP 32                                                                                                   | √  |    |    |    |    |    |
| FIG  | Mastercard® B2B VIP 33                                                                                                   | √  |    |    |    |    |    |
| FIH  | Mastercard® B2B VIP 34                                                                                                   | √  |    |    |    |    |    |
| GCP  | Mastercard Installments Card Premium                                                                                     | √  |    |    |    |    |    |
| GCS  | Mastercard Installments Payment T                                                                                        | √  |    |    |    |    |    |
| GPP  | Mastercard Prepaid Installments Card Premium                                                                             | √  |    |    |    |    |    |
| GPS  | Mastercard Prepaid Installment Payments Card U                                                                           | √  |    |    |    |    |    |

| Code | Description                                                                                                          | MC | NP | VI | TE | MS | CI |
|------|----------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| HNR  | Honor<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.                  |    | √  |    |    |    |    |
| ITT  | Instant Teller<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.         |    | √  |    |    |    |    |
| JCB  | Japanese Credit Bureau<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program. |    | √  |    |    |    |    |
| LNC  | LINC New York<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.          |    | √  |    |    |    |    |
| MAB  | World Elite Mastercard® for Business                                                                                 | √  |    |    |    |    |    |
| MAC  | Mastercard Corporate World Elite®                                                                                    | √  |    |    |    |    |    |
| MAP  | Mastercard Commercial Payments Account                                                                               | √  |    |    |    |    |    |
| MAQ  | Mastercard Prepaid Commercial Payments Account                                                                       | √  |    |    |    |    |    |
| MBA  | Mastercard B2B Product 2<br>This product code is not currently available for issuance in the Canada region.          | √  |    |    |    |    |    |
| MBB  | Mastercard Prepaid Consumer                                                                                          | √  |    |    |    |    |    |
| MBC  | Mastercard Prepaid Voucher                                                                                           | √  |    |    |    |    |    |
| MBD  | Mastercard Professional Debit Business Card                                                                          | √  |    |    |    |    |    |
| MBE  | Mastercard® Electronic Business Card                                                                                 | √  |    |    |    |    |    |
| MBF  | Mastercard Alimentação (Food) (MBF–Prepaid Mastercard Food)                                                          | √  |    |    |    |    |    |
| MBG  | Mastercard B2B Product 3                                                                                             | √  |    |    |    |    |    |
| MBH  | Mastercard B2B Product 4                                                                                             | √  |    |    |    |    |    |
| MBI  | Mastercard B2B Product 5                                                                                             | √  |    |    |    |    |    |
| MBJ  | Mastercard B2B Product 6                                                                                             | √  |    |    |    |    |    |
| MBK  | Mastercard Black                                                                                                     | √  |    |    |    |    |    |
| MBL  | Mastercard® B2B Product 28                                                                                           | √  |    |    |    |    |    |
| MBM  | Mastercard® Refeição (Meal) (MBM–Prepaid Mastercard Meal)                                                            | √  |    |    |    |    |    |
| MBN  | Mastercard® B2B Product 29                                                                                           | √  |    |    |    |    |    |

| Code | Description                                                                                                                     | MC | NP | VI | TE | MS | CI |
|------|---------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| MBO  | Mastercard® B2B Product 30                                                                                                      | √  |    |    |    |    |    |
| MBP  | Mastercard Corporate Prepaid                                                                                                    | √  |    |    |    |    |    |
| MBQ  | Mastercard® B2B Product 31                                                                                                      | √  |    |    |    |    |    |
| MBR  | Mastercard® B2B Product 32                                                                                                      | √  |    |    |    |    |    |
| MBS  | Mastercard B2B Product 1                                                                                                        | √  |    |    |    |    |    |
| MBT  | Mastercard Corporate Prepaid Travel                                                                                             | √  |    |    |    |    |    |
| MBW  | World Mastercard® Black Edition: Debit                                                                                          | √  |    |    |    |    |    |
| MCA  | Mastercard Electronic Basic Card                                                                                                | √  |    |    |    |    |    |
| MCB  | Mastercard BusinessCard® Card Mastercard Corporate Card®                                                                        | √  |    |    |    |    |    |
| MCC  | Mastercard Card                                                                                                                 | √  |    |    |    |    |    |
| MCE  | Mastercard Electronic™ Card                                                                                                     | √  |    |    |    |    |    |
| MCF  | Mastercard Corporate Fleet Card®                                                                                                | √  |    |    |    |    |    |
| MCG  | Gold Mastercard® Card                                                                                                           | √  |    |    |    |    |    |
| MCH  | Mastercard Premium Charge                                                                                                       | √  |    |    |    |    |    |
| MCI  | Mastercard Select Debit                                                                                                         | √  |    |    |    |    |    |
| MCM  | Mastercard Corporate Meeting Card<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program. | √  |    |    |    |    |    |
| MCO  | Mastercard Corporate                                                                                                            | √  |    |    |    |    |    |
| MCP  | Mastercard Corporate Purchasing Card®                                                                                           | √  |    |    |    |    |    |
| MCS  | Mastercard Standard Card                                                                                                        | √  |    |    |    |    |    |
| MCT  | Titanium Mastercard®                                                                                                            | √  |    |    |    |    |    |
| MCU  | Mastercard Unembossed                                                                                                           | √  |    |    |    |    |    |
| MCV  | Merchant-Branded Program                                                                                                        | √  |    |    |    |    |    |
| MCW  | World Mastercard® Card                                                                                                          | √  |    |    |    |    |    |
| MDA  | Mastercard World Legend Debit                                                                                                   | √  |    |    |    |    |    |
| MDB  | Debit Mastercard BusinessCard Card                                                                                              | √  |    |    |    |    |    |
| MDD  | Mastercard CoBadge Debit                                                                                                        | √  |    |    |    |    |    |
| MDE  | Mastercard Essential Prepaid                                                                                                    | √  |    |    |    |    |    |
| MDG  | Gold Debit Mastercard®                                                                                                          | √  |    |    |    |    |    |

| Code | Description                                                                                                                           | MC | NP | VI | TE | MS | CI |
|------|---------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| MDH  | World Debit Embossed                                                                                                                  | √  |    |    |    |    |    |
| MDI  | Debit Other Unembossed                                                                                                                | √  |    |    |    |    |    |
| MDJ  | (Enhanced)                                                                                                                            | √  |    |    |    |    |    |
| MDK  | Debit Mastercard Other 2 Embossed                                                                                                     | √  |    |    |    |    |    |
| MDL  | Business Debit Other Embossed                                                                                                         | √  |    |    |    |    |    |
| MDM  | Middle Market Fleet Card                                                                                                              | √  |    |    |    |    |    |
| MDN  | Middle Market Purchasing Card                                                                                                         | √  |    |    |    |    |    |
| MDO  | Debit Mastercard Other                                                                                                                | √  |    |    |    |    |    |
| MDP  | Platinum Debit Mastercard®                                                                                                            | √  |    |    |    |    |    |
| MDQ  | Middle Market Corporate Card                                                                                                          | √  |    |    |    |    |    |
| MDR  | Mastercard Debit Brokerage Card                                                                                                       | √  |    |    |    |    |    |
| MDS  |                                                                                                                                       | √  |    |    |    |    |    |
| MDT  | Mastercard Business Debit                                                                                                             | √  |    |    |    |    |    |
| MDU  | Debit Mastercard Unembossed                                                                                                           | √  |    |    |    |    |    |
| MDW  | World Elite Debit Mastercard® (Mastercard Black™<br>Debit LAC region excluding Mexico)                                                | √  |    |    |    |    |    |
| MEB  | Mastercard Executive BusinessCard Card                                                                                                | √  |    |    |    |    |    |
| MEC  | Mastercard Electronic Commercial                                                                                                      | √  |    |    |    |    |    |
| MED  | Debit Mastercard Electronic                                                                                                           | √  |    |    |    |    |    |
| MEF  | Mastercard Electronic Payment Account                                                                                                 | √  |    |    |    |    |    |
| MEO  | Mastercard Corporate Executive Card®<br>This product code is not a valid value in DE 120,<br>MCC103 or MCC104, field 3, Card Program. | √  |    |    |    |    |    |
| MEP  | Premium Debit Mastercard                                                                                                              | √  |    |    |    |    |    |
| MES  | Mastercard Enterprise Solution                                                                                                        | √  |    |    |    |    |    |
| MET  | Titanium                                                                                                                              | √  |    |    |    |    |    |
| MFB  | Flex World Elite                                                                                                                      | √  |    |    |    |    |    |
| MFD  | Flex Platinum                                                                                                                         | √  |    |    |    |    |    |
| MFE  | Flex Charge World Elite                                                                                                               | √  |    |    |    |    |    |
| MFH  | Flex World                                                                                                                            | √  |    |    |    |    |    |
| MFI  | Mastercard Debit Financial Inclusion                                                                                                  | √  |    |    |    |    |    |

| Code | Description                                                  | MC | NP | VI | TE | MS | CI |
|------|--------------------------------------------------------------|----|----|----|----|----|----|
| MFL  | Flex Charge Platinum                                         | √  |    |    |    |    |    |
| MFW  | Flex Charge World                                            | √  |    |    |    |    |    |
| MGF  | Mastercard Government Commercial Card                        | √  |    |    |    |    |    |
| MGP  | Mastercard Prepaid Gold Payroll                              | √  |    |    |    |    |    |
| MGS  | Platinum Mastercard® Prepaid General Spend                   | √  |    |    |    |    |    |
| MHA  | Mastercard Healthcare Prepaid Non-tax                        | √  |    |    |    |    |    |
| MHB  | Mastercard HSA Substantiated                                 | √  |    |    |    |    |    |
| MHC  | Mastercard Healthcare Credit Non-substantiated               | √  |    |    |    |    |    |
| MHH  | Mastercard HSA Non-substantiated                             | √  |    |    |    |    |    |
| MIA  | Mastercard Unembossed Prepaid Student Card                   | √  |    |    |    |    |    |
| MIB  | Mastercard Credit Electronic Student Card                    | √  |    |    |    |    |    |
| MIC  | Mastercard Credit Standard Student Card                      | √  |    |    |    |    |    |
| MID  | Mastercard Debit Unembossed Student Card                     | √  |    |    |    |    |    |
| MIG  | Mastercard Credit Unembossed Student Card                    | √  |    |    |    |    |    |
| MIH  | Mastercard Electronic Consumer Non U.S. Student Card         | √  |    |    |    |    |    |
| MIJ  | Mastercard Debit Unembossed Non U.S. Student Card            | √  |    |    |    |    |    |
| MIK  | Mastercard Electronic Consumer Prepaid Non U.S. Student Card | √  |    |    |    |    |    |
| MIL  | Mastercard Unembossed Prepaid Non U.S. Student Card          | √  |    |    |    |    |    |
| MIO  | Mastercard Pay & Split                                       | √  |    |    |    |    |    |
| MIP  | Mastercard Debit Prepaid Student Card                        | √  |    |    |    |    |    |
| MIS  | Mastercard Debit Standard Student Card                       | √  |    |    |    |    |    |
| MIU  | Debit Mastercard Unembossed Outside US                       | √  |    |    |    |    |    |
| MKA  | Digital Gold Debit                                           | √  |    |    |    |    |    |
| MKB  | Digital Platinum Debit                                       | √  |    |    |    |    |    |
| MKC  | Digital World Debit                                          | √  |    |    |    |    |    |
| MKD  | Digital World Elite Debit                                    | √  |    |    |    |    |    |
| MKE  | Digital Platinum                                             | √  |    |    |    |    |    |
| MKF  | Digital World                                                | √  |    |    |    |    |    |

| Code | Description                                                                                                     | MC | NP | VI | TE | MS | CI |
|------|-----------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| MKG  | Digital World Elite                                                                                             | √  |    |    |    |    |    |
| MKH  | Digital World Elite Exclusive                                                                                   | √  |    |    |    |    |    |
| MLA  | Mastercard Central Travel Solutions Air                                                                         | √  |    |    |    |    |    |
| MLB  | Mastercard Brazil Benefit for Home Improvement                                                                  | √  |    |    |    |    |    |
| MLC  | Mastercard Micro-Business Card                                                                                  | √  |    |    |    |    |    |
| MLD  | Mastercard Distribution Card                                                                                    | √  |    |    |    |    |    |
| MLE  | Mastercard Pedágio Prepaid Card (MLE–Mastercard Brazil General Benefits)                                        | √  |    |    |    |    |    |
| MLF  | Mastercard Agro                                                                                                 | √  |    |    |    |    |    |
| MLL  | Mastercard Central Travel Solutions Land                                                                        | √  |    |    |    |    |    |
| MNF  | Mastercard Public Sector Commercial Card                                                                        | √  |    |    |    |    |    |
| MNW  | World Mastercard Card (Europe)                                                                                  | √  |    |    |    |    |    |
| MOC  | Standard Maestro® Social                                                                                        |    |    |    |    | √  |    |
| MOG  | Maestro® Gold<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.     |    |    |    |    | √  |    |
| MOP  | Maestro® Platinum<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program. |    |    |    |    | √  |    |
| MOW  | World Maestro                                                                                                   |    |    |    |    | √  |    |
| MPA  | Prepaid Mastercard Payroll Card                                                                                 | √  |    |    |    |    |    |
| MPB  | Mastercard Preferred BusinessCard                                                                               | √  |    |    |    |    |    |
| MPC  | Mastercard Professional Card                                                                                    | √  |    |    |    |    |    |
| MPD  | Mastercard Flex Prepaid                                                                                         | √  |    |    |    |    |    |
| MPE  | Name for GCMS Product ID MPE                                                                                    | √  |    |    |    |    |    |
| MPF  | Prepaid Mastercard Gift Card                                                                                    | √  |    |    |    |    |    |
| MPG  | Prepaid Mastercard Consumer Reloadable Card                                                                     | √  |    |    |    |    |    |
| MPJ  | Prepaid Debit Mastercard® Card Gold                                                                             | √  |    |    |    |    |    |
| MPK  | Prepaid Mastercard Government Commercial Card                                                                   | √  |    |    |    |    |    |
| MPL  | Platinum Mastercard® Card                                                                                       | √  |    |    |    |    |    |
| MPM  | Prepaid Mastercard Consumer Promotion Card                                                                      | √  |    |    |    |    |    |
| MPN  | Prepaid Mastercard Insurance Card                                                                               | √  |    |    |    |    |    |

| Code | Description                                       | MC | NP | VI | TE | MS | CI |
|------|---------------------------------------------------|----|----|----|----|----|----|
| MPO  | Prepaid Mastercard Other Card                     | √  |    |    |    |    |    |
| MPR  | Prepaid Mastercard Travel Card                    | √  |    |    |    |    |    |
| MPT  | Prepaid Mastercard Teen Card                      | √  |    |    |    |    |    |
| MPV  | Prepaid Mastercard Government Benefit Card        | √  |    |    |    |    |    |
| MPW  | Prepaid Mastercard Corporate Card                 | √  |    |    |    |    |    |
| MPX  | Prepaid Mastercard Flex Benefit Card              | √  |    |    |    |    |    |
| MPY  | Prepaid Mastercard Employee Incentive Card        | √  |    |    |    |    |    |
| MPZ  | Prepaid Mastercard Emergency Assistance Card      | √  |    |    |    |    |    |
| MRB  | Prepaid Mastercard Electronic BusinessCard        | √  |    |    |    |    |    |
| MRC  | Prepaid Mastercard Electronic Card                | √  |    |    |    |    |    |
| MRD  | Platinum Debit Mastercard® Prepaid General Spend  | √  |    |    |    |    |    |
| MRF  | European Regulated Individual Pay                 | √  |    |    |    |    |    |
| MRG  | Prepaid Mastercard Card Outside US                | √  |    |    |    |    |    |
| MRH  | Mastercard Platinum Prepaid Travel Card           | √  |    |    |    |    |    |
| MRJ  | Prepaid Mastercard Gold Card                      | √  |    |    |    |    |    |
| MRK  | Prepaid Mastercard Public Sector Commercial Card  | √  |    |    |    |    |    |
| MRL  | Prepaid Mastercard Electronic Commercial          | √  |    |    |    |    |    |
| MRO  | Mastercard Rewards Only                           | √  |    |    |    |    |    |
| MRP  | Standard Retailer Centric Payments                | √  |    |    |    |    |    |
| MRS  | Prepaid Mastercard ISIC Student Card              | √  |    |    |    |    |    |
| MRW  | Prepaid Mastercard BusinessCard Credit Outside US | √  |    |    |    |    |    |
| MSA  | Prepaid Maestro Payroll Card                      |    |    |    |    | √  |    |
| MSB  | Maestro Small Business                            |    |    |    |    | √  |    |
| MSD  | Deferred Debit Mastercard                         | √  |    |    |    |    |    |
| MSF  | Prepaid Maestro Gift Card                         |    |    |    |    | √  |    |
| MSG  | Prepaid Maestro Consumer Reloadable Card          |    |    |    |    | √  |    |
| MSI  | Maestro                                           |    |    |    |    | √  |    |
| MSJ  | Prepaid Maestro Gold                              |    |    |    |    | √  |    |
| MSM  | Prepaid Maestro Consumer Promotion Card           |    |    |    |    | √  |    |
| MSN  | Prepaid Maestro Insurance Card                    |    |    |    |    | √  |    |
| MSO  | Prepaid Maestro Other Card                        |    |    |    |    | √  |    |



| Code | Description                               | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------------|----|----|----|----|----|----|
| MSP  | Muse Mastercard                           | √  |    |    |    |    |    |
| MSQ  | Maestro Prepaid (Reserved for Future Use) |    |    |    |    | √  |    |
| MSR  | Prepaid Maestro Travel Card               |    |    |    |    | √  |    |
| MSS  | Maestro Student Card                      |    |    |    |    | √  |    |
| MST  | Prepaid Maestro Teen Card                 |    |    |    |    | √  |    |
| MSV  | Prepaid Maestro Government Benefit Card   |    |    |    |    | √  |    |
| MSW  | Prepaid Maestro Corporate Card            |    |    |    |    | √  |    |
| MSX  | Prepaid Maestro Flex Benefit Card         |    |    |    |    | √  |    |
| MSY  | Prepaid Maestro Employee Incentive Card   |    |    |    |    | √  |    |
| MSZ  | Prepaid Maestro Emergency Assistance Card |    |    |    |    | √  |    |
| MTA  | Mastercard® B2B Product 7                 | √  |    |    |    |    |    |
| MTB  | Mastercard® B2B Product 8                 | √  |    |    |    |    |    |
| MTC  | Mastercard® B2B Product 9                 | √  |    |    |    |    |    |
| MTD  | Mastercard® B2B Product 10                | √  |    |    |    |    |    |
| MTE  | Mastercard® B2B Product 11                | √  |    |    |    |    |    |
| MTF  | Mastercard® B2B Product 12                | √  |    |    |    |    |    |
| MTG  | Mastercard® B2B Product 13                | √  |    |    |    |    |    |
| MTH  | Mastercard® B2B Product 14                | √  |    |    |    |    |    |
| MTI  | Mastercard® B2B Product 15                | √  |    |    |    |    |    |
| MTJ  | Mastercard® B2B Product 16                | √  |    |    |    |    |    |
| MTK  | Mastercard® B2B Product 17                | √  |    |    |    |    |    |
| MTL  | Mastercard® B2B Product 18                | √  |    |    |    |    |    |
| MTM  | Mastercard® B2B Product 19                | √  |    |    |    |    |    |
| MTN  | Mastercard® B2B Product 20                | √  |    |    |    |    |    |
| MTO  | Mastercard® B2B Product 21                | √  |    |    |    |    |    |
| MTP  | Mastercard Platinum Prepaid Travel Card   | √  |    |    |    |    |    |
| MTQ  | Mastercard® B2B Product 22                | √  |    |    |    |    |    |
| MTR  | Mastercard® B2B Product 23                | √  |    |    |    |    |    |
| MTS  | Mastercard® B2B Product 24                | √  |    |    |    |    |    |
| MTT  | Mastercard® B2B Product 25                | √  |    |    |    |    |    |
| MTU  | Mastercard® B2B Product 26                | √  |    |    |    |    |    |

| Code | Description                          | MC | NP | VI | TE | MS | CI |
|------|--------------------------------------|----|----|----|----|----|----|
| MTV  | Mastercard® B2B Product 27           | √  |    |    |    |    |    |
| MUP  | Platinum Debit Mastercard Unembossed | √  |    |    |    |    |    |
| MUS  | Prepaid Mastercard Unembossed US     | √  |    |    |    |    |    |
| MUW  | Mastercard World Domestic Affluent   | √  |    |    |    |    |    |
| MVA  | Mastercard B2B VIP 1                 | √  |    |    |    |    |    |
| MVB  | Mastercard B2B VIP 2                 | √  |    |    |    |    |    |
| MVC  | Mastercard® B2B VIP 3                | √  |    |    |    |    |    |
| MVD  | Mastercard B2B VIP 4                 | √  |    |    |    |    |    |
| MVE  | Mastercard B2B VIP 5                 | √  |    |    |    |    |    |
| MVF  | Mastercard B2B VIP 6                 | √  |    |    |    |    |    |
| MVG  | Mastercard B2B VIP 7                 | √  |    |    |    |    |    |
| MVH  | Mastercard B2B VIP 8                 | √  |    |    |    |    |    |
| MVI  | Mastercard B2B VIP 9                 | √  |    |    |    |    |    |
| MVJ  | Mastercard B2B VIP 10                | √  |    |    |    |    |    |
| MVK  | Mastercard B2B VIP 11                | √  |    |    |    |    |    |
| MVL  | Mastercard B2B VIP 12                | √  |    |    |    |    |    |
| MVM  | Mastercard B2B VIP 13                | √  |    |    |    |    |    |
| MVN  | Mastercard B2B VIP 14                | √  |    |    |    |    |    |
| MVO  | Mastercard® B2B VIP 15               | √  |    |    |    |    |    |
| MVP  | Mastercard® B2B VIP 16               | √  |    |    |    |    |    |
| MVQ  | Mastercard® B2B VIP 17               | √  |    |    |    |    |    |
| MVR  | Mastercard® B2B VIP 18               | √  |    |    |    |    |    |
| MVS  | Mastercard® B2B VIP 19               | √  |    |    |    |    |    |
| MVT  | Mastercard® B2B VIP 20               | √  |    |    |    |    |    |
| MVU  | Mastercard® B2B VIP 21               | √  |    |    |    |    |    |
| MVV  | Mastercard® B2B VIP 22               | √  |    |    |    |    |    |
| MVW  | Mastercard® B2B VIP 23               | √  |    |    |    |    |    |
| MXV  | Mastercard® B2B VIP 24               | √  |    |    |    |    |    |
| MVY  | Mastercard® B2B VIP 25               | √  |    |    |    |    |    |
| MVZ  | Mastercard® B2B VIP 26               | √  |    |    |    |    |    |
| MWB  | World Mastercard® for Business       | √  |    |    |    |    |    |

| Code | Description                                                                                                        | MC | NP | VI | TE | MS | CI |
|------|--------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| MWD  | World Deferred                                                                                                     | √  |    |    |    |    |    |
| MWE  | World Elite™                                                                                                       | √  |    |    |    |    |    |
| MWF  | Mastercard Humanitarian Prepaid<br>This product code is not currently available for issuance in the Canada region. | √  |    |    |    |    |    |
| MWJ  | Consumer Enhanced                                                                                                  | √  |    |    |    |    |    |
| MWK  | Consumer Plus                                                                                                      | √  |    |    |    |    |    |
| MWL  | Consumer Plus Flex                                                                                                 | √  |    |    |    |    |    |
| MWO  | Mastercard Corporate World                                                                                         | √  |    |    |    |    |    |
| MWP  | Mastercard World Prepaid                                                                                           | √  |    |    |    |    |    |
| MWR  | World Retailer Centric Payment                                                                                     | √  |    |    |    |    |    |
| MXG  | Gold Debit Mastercard® Card                                                                                        | √  |    |    |    |    |    |
| NYC  | NYCE Network<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.         |    | √  |    |    |    |    |
| OLB  | Maestro Small Business: Delayed Debit                                                                              |    |    |    |    | √  |    |
| OLG  | Maestro Gold: Delayed Debit                                                                                        |    |    |    |    | √  |    |
| OLI  | ISIC Maestro Student Card: Delayed Debit                                                                           |    |    |    |    | √  |    |
| OLP  | Maestro Platinum: Delayed Debit                                                                                    |    |    |    |    | √  |    |
| OLS  | Maestro: Delayed Debit                                                                                             |    |    |    |    | √  |    |
| OLW  | World Maestro Delayed Debit                                                                                        |    |    |    |    | √  |    |
| PLU  | PLUS<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.                 |    | √  |    |    |    |    |
| PRO  | Proprietary Card<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.     |    | √  |    |    |    |    |
| PUL  | Pulse<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.                |    | √  |    |    |    |    |
| PVA  | Private Label 1                                                                                                    | √  |    |    |    |    |    |
| PVB  | Private Label 2                                                                                                    | √  |    |    |    |    |    |
| PVC  | Private Label 3                                                                                                    | √  |    |    |    |    |    |

| Code | Description                                                                                                             | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| PVD  | Private Label 4                                                                                                         | √  |    |    |    |    |    |
| PVE  | Private Label 5                                                                                                         | √  |    |    |    |    |    |
| PVF  | Private Label 6                                                                                                         | √  |    |    |    |    |    |
| PVG  | Private Label 7                                                                                                         | √  |    |    |    |    |    |
| PVH  | Private Label 8                                                                                                         | √  |    |    |    |    |    |
| PVI  | Private Label 9                                                                                                         | √  |    |    |    |    |    |
| PVJ  | Private Label 10                                                                                                        | √  |    |    |    |    |    |
| PVL  | Private label: generic<br>This product code is not a valid value in DE 120,<br>MCC103 or MCC104, field 3, Card Program. |    | √  |    |    |    |    |
| SAG  | Gold Mastercard® Salary-Immediate Debit                                                                                 | √  |    |    |    |    |    |
| SAL  | Standard Maestro Salary                                                                                                 |    |    |    |    | √  |    |
| SAP  | Platinum Mastercard® Salary-Immediate Debit                                                                             | √  |    |    |    |    |    |
| SAS  | Standard Mastercard Salary-Immediate Debit                                                                              | √  |    |    |    |    |    |
| SBJ  | Prepaid Installment Payments H                                                                                          | √  |    |    |    |    |    |
| SBK  | Prepaid Installment Payments I                                                                                          | √  |    |    |    |    |    |
| SBP  | Small Business Prepaid                                                                                                  | √  |    |    |    |    | √  |
| SOS  | Standard Mastercard Social-Immediate Debit                                                                              | √  |    |    |    |    |    |
| SPP  | Mastercard Installment Payments P                                                                                       | √  |    |    |    |    |    |
| SPS  | Mastercard Installment Payments S                                                                                       | √  |    |    |    |    |    |
| STR  | Star Network<br>This product code is not a valid value in DE 120,<br>MCC103 or MCC104, field 3, Card Program.           |    | √  |    |    |    |    |
| SUR  | Prepaid Mastercard Unembossed Outside US                                                                                | √  |    |    |    |    |    |
| TBE  | Business: Immediate Debit                                                                                               | √  |    |    |    |    |    |
| TCB  | Mastercard Business Card: Immediate Debit                                                                               | √  |    |    |    |    |    |
| TCC  | Mastercard (Mixed BIN): Immediate Debit                                                                                 | √  |    |    |    |    |    |
| TCE  | Mastercard Electronic: Immediate Debit                                                                                  | √  |    |    |    |    |    |
| TCF  | Mastercard Fleet Card: Immediate Debit                                                                                  | √  |    |    |    |    |    |
| TCG  | Gold Mastercard Card: Immediate Debit                                                                                   | √  |    |    |    |    |    |
| TCO  | Mastercard Corporate: Immediate Debit                                                                                   | √  |    |    |    |    |    |

| Code | Description                                                                                           | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|
| TCP  | Mastercard Purchasing Card: Immediate Debit                                                           | √  |    |    |    |    |    |
| TCS  | Mastercard Standard Card: Immediate Debit                                                             | √  |    |    |    |    |    |
| TCW  | World Signia Mastercard Card: Immediate Debit                                                         | √  |    |    |    |    |    |
| TDN  | Middle Market Mastercard Purchasing Card: Immediate Debit                                             | √  |    |    |    |    |    |
| TEB  | Mastercard Executive BusinessCard Card: Immediate Debit                                               | √  |    |    |    |    |    |
| TEC  | Mastercard Electronic Commercial: Immediate Debit                                                     | √  |    |    |    |    |    |
| TEO  | Mastercard Corporate Executive Card: Immediate Debit                                                  | √  |    |    |    |    |    |
| TIB  | ISIC Mastercard Electronic Student Card: Immediate Debit                                              | √  |    |    |    |    |    |
| TIC  | ISIC Mastercard Standard Student Card: Immediate Debit                                                | √  |    |    |    |    |    |
| TIU  | Mastercard Unembossed: Immediate Debit                                                                | √  |    |    |    |    |    |
| TLA  | Mastercard Central Travel Solutions Air: Immediate Debit                                              | √  |    |    |    |    |    |
| TNF  | Mastercard Public Sector Commercial Card: Immediate Debit                                             | √  |    |    |    |    |    |
| TNW  | Mastercard New World: Immediate Debit                                                                 | √  |    |    |    |    |    |
| TPB  | Mastercard Preferred Business Card: Immediate Debit                                                   | √  |    |    |    |    |    |
| TPC  | Mastercard Professional Card: Immediate Debit                                                         | √  |    |    |    |    |    |
| TPL  | Platinum Mastercard: Immediate Debit                                                                  | √  |    |    |    |    |    |
| TWB  | World Mastercard® Black Edition: Immediate Debit                                                      | √  |    |    |    |    |    |
| VIS  | VisaNet<br>This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program. |    |    | √  |    |    |    |
| WAA  | Mastercard® Commercial B2B Product 1                                                                  | √  |    |    |    |    |    |
| WAB  | Mastercard® Commercial B2B Product 2                                                                  | √  |    |    |    |    |    |
| WAC  | Mastercard® Commercial B2B Product 3                                                                  | √  |    |    |    |    |    |
| WAD  | Mastercard® Commercial B2B Product 4                                                                  | √  |    |    |    |    |    |
| WAE  | Mastercard® Commercial B2B Product 5                                                                  | √  |    |    |    |    |    |
| WAF  | Mastercard® Commercial B2B Product 6                                                                  | √  |    |    |    |    |    |

| Code | Description                           | MC | NP | VI | TE | MS | CI |
|------|---------------------------------------|----|----|----|----|----|----|
| WAG  | Mastercard® Commercial B2B Product 7  | √  |    |    |    |    |    |
| WAH  | Mastercard® Commercial B2B Product 8  | √  |    |    |    |    |    |
| WAI  | Mastercard® Commercial B2B Product 9  | √  |    |    |    |    |    |
| WAJ  | Mastercard® Commercial B2B Product 10 | √  |    |    |    |    |    |
| WAK  | Mastercard® Commercial B2B Product 11 | √  |    |    |    |    |    |
| WAL  | Mastercard® Commercial B2B Product 12 | √  |    |    |    |    |    |
| WAM  | Mastercard® Commercial B2B Product 13 | √  |    |    |    |    |    |
| WAN  | Mastercard® Commercial B2B Product 14 | √  |    |    |    |    |    |
| WAO  | Mastercard® Commercial B2B Product 15 | √  |    |    |    |    |    |
| WAP  | Mastercard® Commercial B2B Product 16 | √  |    |    |    |    |    |
| WAQ  | Mastercard® Commercial B2B Product 17 | √  |    |    |    |    |    |
| WAT  | Mastercard® Commercial B2B Product 18 | √  |    |    |    |    |    |
| WAU  | Mastercard® Commercial B2B Product 19 | √  |    |    |    |    |    |
| WAV  | Mastercard® Commercial B2B Product 20 | √  |    |    |    |    |    |
| WAW  | Mastercard® Commercial B2B Product 21 | √  |    |    |    |    |    |
| WAX  | Mastercard® Commercial B2B Product 22 | √  |    |    |    |    |    |
| WAY  | Mastercard® Commercial B2B Product 23 | √  |    |    |    |    |    |
| WAZ  | Mastercard® Commercial B2B Product 24 | √  |    |    |    |    |    |
| WBA  | Mastercard® Commercial B2B Product 25 | √  |    |    |    |    |    |
| WBB  | Mastercard® Commercial B2B Product 26 | √  |    |    |    |    |    |
| WBC  | Mastercard® Commercial B2B Product 27 | √  |    |    |    |    |    |
| WBD  | Mastercard® Commercial B2B Product 28 | √  |    |    |    |    |    |
| WBE  | World Mastercard® Black Edition       | √  |    |    |    |    |    |
| WBF  | Mastercard® Commercial B2B Product 29 | √  |    |    |    |    |    |
| WBG  | Mastercard® Commercial B2B Product 30 | √  |    |    |    |    |    |
| WBH  | Mastercard® Commercial B2B Product 31 | √  |    |    |    |    |    |
| WBI  | Mastercard® Commercial B2B Product 32 | √  |    |    |    |    |    |
| WBJ  | Mastercard® Commercial B2B Product 33 | √  |    |    |    |    |    |
| WBK  | Mastercard® Commercial B2B Product 34 | √  |    |    |    |    |    |
| WBL  | Mastercard® Commercial B2B Product 35 | √  |    |    |    |    |    |
| WDR  | World Debit Mastercard Rewards        | √  |    |    |    |    |    |

| Code | Description              | MC | NP | VI | TE | MS | CI |
|------|--------------------------|----|----|----|----|----|----|
| WMR  | World Mastercard Rewards | √  |    |    |    |    |    |
| WPD  | World Prepaid Debit      | √  |    |    |    | √  | √  |

## Subfield 2 (Banknet Reference Number)

DE 63, subfield 2 (Banknet Reference Number) is generated by the Authorization System for each originating message it routes. The reference number is guaranteed to be a unique value for any transaction within the specified financial network on any processing day.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data representation | an...9                     |
| Data field          | Contents of positions 4–12 |
| Justification       | Left                       |

### Values

The Banknet reference number is a minimum of six characters and a maximum of nine characters long.

## Authorization Request/0100 message edits

### Authorization Request/0100 message

| When...                                                                                                                                                                                                                                               | Then the Dual Message Authorization System...                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| an Authorization Request/0100 message is for one of the Mastercard Flex Program product codes and the Issuer Account Range country has a USD transaction threshold amount and DE 5 (Amount, Settlement) is less than the transaction threshold amount | will decline the transaction with DE 39 (Response Code) = 13 (Invalid Amount). |

### Country transaction thresholds

| Country                                                                                            | Transaction threshold |
|----------------------------------------------------------------------------------------------------|-----------------------|
| Colombia, Jamaica, Japan, New Zealand, Panama, Puerto Rico, Trinidad & Tobago, U.S. Virgin Islands | USD 10,000            |
| Hong Kong SAR, Philippines, Singapore, Thailand, Vietnam                                           | USD 6,000             |
| Indonesia                                                                                          | USD 3,000             |

|                                                                                                                                                                                                                                                                                          |                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 message is for one of the Mastercard Flex Program product codes and the Issuer Account Range country is Brazil and DE 6 (Amount, Cardholder Billing) is less than the transaction threshold of BRL 55,000                                                  | will decline the transaction with DE 39 (Response Code) = 13 (Invalid Amount).                                  |
| an Authorization Request/0100 message is for one of the Mastercard Flex Program product codes and is an intracountry transaction for Bahrain, Egypt, Ghana, Kuwait, Oman, Qatar, Saudi Arabia, or United Arab Emirates, and the MCC is 4900, 9211, 9222, 9223, 9311, 9399, 9402, or 9405 | will decline the transaction with DE 39 (Response Code) = 12 (Invalid Transaction).                             |
| an Authorization Request/0100 message is for one of the Mastercard Flex Program product codes and Issuer, acquirer and merchant country are same and is either Poland or Switzerland                                                                                                     | will decline the transaction with DE 39 (Response Code) = 57 (Transaction not Permitted to Issuer/ Cardholder). |



| When...                                                                                                                                                                              | Then the Dual Message Authorization System...                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 message is for one of the Mastercard Flex Program product codes and issuer, acquirer or merchant is not one of the permitted countries for the program | will decline the transaction with DE 39 (Response Code) = 57 (Transaction not Permitted to Issuer/Cardholder). |

## DE 64 (Message Authentication Code)

DE 64 (Message Authentication Code [MAC]) validates the source and the text of the message between the sender and the receiver.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Data Representation | b-8         |
| Data Length         | N/A         |
| Data Field          | N/A         |
| Subfields           | N/A         |
| Justification       | N/A         |

### Usage

May contain message authentication code as defined by ISO standards.

### Values

The last bit position within any bit map is reserved for DE 64. If authentication is to be used on a message, the MAC information is indicated by the final bit of the final bit map of that message. The final bit of all preceding bit maps shall contain 0; for example, there shall be only one DE 64 per message and that DE 64 must be the last data element of the message.

## DE 65 (Bit Map, Extended)

DE 65 (Bit Map, Extended) is a series of eight bytes (64 bits) used to identify the presence (denoted by 1) or the absence (denoted by 0) of each data element in an extended (third) message segment.

### Attributes

| Attribute           | Description                              |
|---------------------|------------------------------------------|
| Data Representation | b-8                                      |
| Data Length         | N/A                                      |
| Data Field          | Contents of bit positions 1–64 (8 bytes) |
| Subfields           | N/A                                      |
| Justification       | N/A                                      |

### Usage

The Dual Message Authorization System defines only two message segments, the presence or absence of which is indicated by Primary and Secondary Bit Maps. DE 65 would indicate the presence of a third message segment, and must never be present in a Dual Message Authorization System message. The corresponding bit (number 65) must always be 0 in the Secondary Bit Map.

## DE 66 (Settlement Code)

DE 66 (Settlement Code) indicates the result of a reconciliation request.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 67 (Extended Payment Code)

DE 67 (Extended Payment Code) indicates the number of months that the cardholder prefers to pay for an item (the item purchased during the course of this transaction) if permitted by the card issuer.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1-2 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 67 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                                 | Org | Sys | Dst |
|---------------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                              | C   | P   | C   |
| Authorization Request Response/0110                                                                     | OE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated                                                           | C   | P   | C   |
| Authorization Advice/0120: system-generated                                                             | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120 message) | OE  | XE  | CE  |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)         | •   | XE  | CE  |
| Reversal Request/0400                                                                                   | C   | P   | C   |
| Reversal Request Response/0410                                                                          | OE  | XE  | CE  |

| Message              | Org | Sys | Dst |
|----------------------|-----|-----|-----|
| Reversal Advice/0420 | .   | X   | C   |

## DE 68 (Receiving Institution Country Code)

DE 68 (Receiving Institution Country Code) is the code of the country where the receiving institution is located.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 69 (Settlement Institution Country Code)

DE 69 (Settlement Institution Country Code) is the code of the country where the settlement institution is located.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 70 (Network Management Information Code)

DE 70 (Network Management Information Code) identifies network status.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-3                       |
| Data Length         | N/A                       |
| Data Field          | Contents of positions 1–3 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 70 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                  | Org | Sys | Dst |
|------------------------------------------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: sign-on/sign-off                                        | M   | X   | •   |
| Network Management Request/0800: host session activation/<br>deactivation                | M   | X   | •   |
| Network Management Request/0800: network connection status,<br>member-generated          | M   | X   | •   |
| Network Management Request/0800: network connection status,<br>system-generated          | •   | X   | M   |
| Network Management Request/0800: PEK exchange: on demand                                 | M   | X   | •   |
| Network Management Request/0800: PEK exchange                                            | •   | X   | M   |
| Network Management Request Response/0810: host session activation/<br>deactivation       | •   | XE  | ME  |
| Network Management Request Response/0810: network connection<br>status, member-generated | ME  | XE  | •   |
| Network Management Request Response/0810: network connection<br>status, system-generated | •   | XE  | ME  |
| Network Management Request Response/0810: PEK exchange                                   | ME  | XE  | •   |
| Network Management Request Response/0810: PEK exchange—on<br>demand                      | •   | XE  | ME  |
| Network Management Request Response/0810: sign-on/sign-off                               | •   | XE  | ME  |
| Network Management Advice/0820: PEK exchange                                             | •   | X   | M   |

### Application notes

DE 48 (Additional Data: Private Use) may be used in conjunction with DE 70 to provide complete network status or control information. The Mastercard Network uses this data element in Network Management/08xx messages to convey network control commands and network status information to and from customer information processing systems that interface directly to the Dual Message Authorization System.

This data element is defined and used identically within all Mastercard programs and services.

## Network Management Request/0800: sign-on/sign-off

The following values apply to this message.

### Values

| Code | Description                                                                    | MC | NP | VI | TE | MS | CI |
|------|--------------------------------------------------------------------------------|----|----|----|----|----|----|
| 001  | Sign-on (by prefix)                                                            | √  | √  | √  | √  | √  | √  |
| 002  | Sign-off (by prefix)                                                           | √  | √  | √  | √  | √  | √  |
| 060  | SAF session request                                                            | √  | √  | √  | √  | √  | √  |
| 061  | Group sign-on (by Mastercard group sign-on)                                    | √  | √  | √  | √  | √  | √  |
| 062  | Group sign-off (by Mastercard group sign-on)                                   | √  | √  | √  | √  | √  | √  |
| 063  | Group sign-on alternate issuer route                                           | √  | √  |    |    | √  | √  |
| 064  | Group sign-off alternate issuer route                                          | √  | √  |    |    | √  | √  |
| 065  | Prefix sign-on (by Group Sign-on ID for primary route)                         | √  | √  |    |    | √  | √  |
| 066  | Prefix sign-off (by Group Sign-on ID for primary route)                        | √  | √  |    |    | √  | √  |
| 067  | Prefix sign-on (by Group Sign-on ID for alternate issuer route) <sup>14</sup>  | √  | √  |    |    | √  | √  |
| 068  | Prefix sign-off (by Group Sign-on ID for alternate issuer route) <sup>14</sup> | √  | √  |    |    | √  | √  |

## Network Management Request/0800: Network Connection Status, Member-Generated

The following values apply to this message.

### Values

| Code | Description                         | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------|----|----|----|----|----|----|
| 270  | Network connection status—echo test | √  |    |    |    | √  | √  |

## Network Management Request/0800: Network Connection Status, System-Generated

The following values apply to this message.

### Values

| Code | Description                         | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------|----|----|----|----|----|----|
| 270  | Network connection status—echo test | √  |    |    |    | √  | √  |

## Network Management Request/0800: Host Session Activation/Deactivation

The following values apply to this message.

### Values

| Code | Description               | MC | NP | VI | TE | MS | CI |
|------|---------------------------|----|----|----|----|----|----|
| 081  | Host session activation   | √  | √  | √  | √  | √  | √  |
| 082  | Host session deactivation | √  | √  | √  | √  | √  | √  |

## Network Management Request/0800: PEK exchange

The following values apply to this message.

### Values

| Code | Description             | MC | NP | VI | TE | MS | CI |
|------|-------------------------|----|----|----|----|----|----|
| 161  | Encryption key exchange | √  | √  |    |    | √  | √  |

## Network Management Request/0800: PEK exchange-On Demand

The following values apply to this message.

### Values

| Code | Description                                               | MC | NP | VI | TE | MS | CI |
|------|-----------------------------------------------------------|----|----|----|----|----|----|
| 162  | Solicitation for Encryption Key Exchange                  | √  | √  |    |    | √  | √  |
| 163  | Solicitation for Encryption Key Exchange - TR-31 Keyblock | √  | √  | √  | √  | √  | √  |
| 166  | Load Comm Key                                             | √  | √  |    | √  | √  | √  |
| 167  | Load Previous Comm Key                                    | √  | √  |    | √  | √  | √  |

## Network Management Request Response/0810: PEK exchange

The following values apply to this message.

### Values

| Code | Description                                               | MC | NP | VI | TE | MS | CI |
|------|-----------------------------------------------------------|----|----|----|----|----|----|
| 163  | Solicitation for Encryption Key Exchange - TR-31 Keyblock | √  | √  | √  | √  | √  | √  |

## Network Management Advice/0820: PEK exchange

The following values apply to this message.

### Values

| Code | Description                                                 | MC | NP | VI | TE | MS | CI |
|------|-------------------------------------------------------------|----|----|----|----|----|----|
| 164  | Encryption TR-31 Block Key Exchange Confirmation of Success | √  | √  | √  | √  | √  | √  |
| 165  | Encryption TR-31 Block Key Exchange Advice of Failure       | √  | √  | √  | √  | √  | √  |



## DE 70 Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following system edits.

### Associated edits

| WHEN...                                                                                                                         | THEN the Dual Message Authorization System... |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| the acquirer or issuer sends 0800 request with DE 70 = 166/167 and no DE110SF10                                                 |                                               |
| the acquirer or issuer sends 0800 request with DE 70 = 166 and DE110SF10 present and doesn't match the staged key check value   | will reject with RC 96. No keys will be moved |
| the acquirer or issuer sends 0800 request with DE 70 = 167 and DE110SF10 present and doesn't match the previous key check value |                                               |

## DE 71 (Message Number)

DE 71 (Message Number) is a sequential, cyclic number the message initiator assigns to a message. Message Number is used to monitor the integrity of interchange.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 72 (Message Number Last)

DE 72 (Message Number Last) is a sequential, cyclic number the message initiator assigns to a message, used to monitor the integrity of interchange.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 73 (Date, Action)

DE 73 (Date, Action) specifies the date (year, month, and day) of a future action. In addition, a message originator may use it as a static time such as a birthdate.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 74 (Credits, Number)

DE 74 (Credits, Number) is the numeric sum of credit transactions processed.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 75 (Credits, Reversal Number)

DE 75 (Credits, Reversal Number) is the sum number of reversal credit transactions.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 76 (Debits, Number)

DE 76 (Debits, Number) is the sum number of debit transactions processed.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 77 (Debits, Reversal Number)

DE 77 (Debits, Reversal Number) is the sum number of reversal debit transactions.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 78 (Transfers, Number)

DE 78 (Transfers, Number) is the sum number of all transfer transactions processed.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 79 (Transfers, Reversal Number)

DE 79 (Transfers, Reversal Number) is the sum number of all transfer reversal transactions processed.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 80 (Inquiries, Number)

DE 80 (Inquiries, Number) is the sum number of inquiry transaction requests processed.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 81 (Authorizations, Number)

DE 81 (Authorizations, Number) is the sum number of Authorization Request/0100 and Authorization Advice/0120 messages processed.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 82 (Credits, Processing Fee Amount)

DE 82 (Credits, Processing Fee Amount) is the sum of all processing fees due to an institution or customer for services associated with handling and routing transactions. This Mastercard definition replaces the ISO standard definition.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 83 (Credits, Transaction Fee Amount)

DE 83 (Credits, Transaction Fee Amount) is the sum of all transaction fees due to an institution or customer for processing interchange transactions. This Mastercard definition replaces the ISO standard definition.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 84 (Debits, Processing Fee Amount)

DE 84 (Debits, Processing Fee Amount) is the sum of all processing fees due from an institution or customer for services associated with handling and routing transactions. This Mastercard definition replaces the ISO standard definition.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 85 (Debits, Transaction Fee Amount)

DE 85 (Debits, Transaction Fee Amount) is the sum of all transaction fees due from an institution or customer for processing interchange transactions. This Mastercard definition replaces the ISO standard definition.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 86 (Credits, Amount)

DE 86 (Credits, Amount) is the sum amount of all credit transactions processed exclusive of any fees.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 87 (Credits, Reversal Amount)

DE 87 (Credits, Reversal Amount) is the sum amount of reversal credits processed exclusive of any fees.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 88 (Debits, Amount)

DE 88 (Debits, Amount) is the sum amount of all debit transactions processed exclusive of any fees.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 89 (Debits, Reversal Amount)

DE 89 (Debits, Reversal Amount) is the sum amount of reversal debits processed exclusive of any fees.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 90 (Original Data Elements)

DE 90 (Original Data Elements) is the data elements in the original message, intended to identify a transaction for correction or reversal.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-42                      |
| Data Length         | N/A                       |
| Data Field          | Contents of subfields 1–5 |
| Subfields           | 5                         |
| Justification       | see subfields             |

### Usage

Following is the usage of DE 90 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                        | Org | Sys | Dst |
|--------------------------------|-----|-----|-----|
| Reversal Request/0400          | M   | P   | M   |
| Reversal Request Response/0410 | ME  | XE  | ME  |
| Reversal Advice/0420           | •   | X   | M   |
| Reversal Advice Response/0430  | ME  | XE  | •   |

### Values

See subfields.

### Application notes

DE 90 is mandatory in the Reversal Advice/0420 message. DE 90 is always present in the Reversal Advice/0420 message whether the Reversal Advice/0420 message is reversing an Authorization Request/0100 message or advising the issuer that Mastercard processed a Reversal Request/0400 message on behalf of the issuer.

DE 90, subfield 1 must contain the Message Type Identifier (MTI) of the original authorization message. The remaining subfields may contain valid matching values from the original authorization or may be zero-filled if they are not available.

For reversal messages, if DE 7 (Transmission Date and Time) and DE 11 (System Trace Audit Number [STAN]) from the original Authorization 0100/0110 cannot be saved, DE 90, subelements 2 and 3 may be zero-filled within the Reversal Request/0400 message.

## Subfield 1 (Original Message Type Identifier)

DE 90, subfield 1 (Original Message Type Identifier) indicates the Message Type Identifier (MTI) of the original message.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-4                       |
| Data Field          | Contents of positions 1-4 |
| Justification       | N/A                       |

### Values

MTI of original message.

## Subfield 2 (Original DE 11 [Systems Trace Audit Number])

DE 90, subfield 2 (Original DE 11 [Systems Trace Audit Number]) indicates the Systems Trace Audit Number (STAN) that was in DE 11 of the original message.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-6                        |
| Data Field          | Contents of positions 5-10 |
| Justification       | N/A                        |

### Values

STAN of original message.

## Subfield 3 (Original DE 7 [Transmission Date and Time])

DE 90, subfield 3 (Original DE 7 [Transmission Date and Time]) indicates the Transmission Date and Time that was in DE 7 of the original message.

### Attributes

| Attribute           | Description                                        |
|---------------------|----------------------------------------------------|
| Data Representation | n-10                                               |
| Data Field          | Contents of positions 11–20 (in MMDDhhmmss format) |
| Justification       | N/A                                                |

### Values

Transmission date and time of original message. This subfield must contain a valid date expressed as month (MM) and day (DD) and a valid time expressed as hours (hh), minutes (mm), and seconds (ss).

## Subfield 4 (Original DE 32 [Acquiring Institution ID Code])

DE 90, subfield 4 (Original DE 32 [Acquiring Institution ID Code]) indicates the Acquiring Institution ID Code that was in DE 32 of the original message.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-11                        |
| Data Field          | Contents of positions 21–31 |
| Justification       | Right with leading zeros    |

### Values

Acquiring Institution ID code of original message.

## Subfield 5 (Original DE 33 [Forwarding Institution ID Code])

DE 90, subfield 5 (Original DE 33 [Forwarding Institution ID Code]) indicates the Forwarding Institution ID Code that was in DE 33 of the original message.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-11                        |
| Data Field          | Contents of positions 32–42 |
| Justification       | Right with leading zeros    |

### Values

Forwarding Institution ID code of original message. If the Forwarding Institution ID code was not present in the original message, then subfield 5 must contain all zeros.

## DE 91 (Issuer File Update Code)

DE 91 (Issuer File Update Code) indicates to the system maintaining a file which procedure to follow.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Length         | N/A                    |
| Data Field          | Contents of position 1 |
| Subfields           | N/A                    |
| Justification       | N/A                    |

### Usage

Following is the usage of DE 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.



### Applicable messages

| Message                                  | Org | Sys | Dst |
|------------------------------------------|-----|-----|-----|
| Issuer File Update Request/0302          | M   | X   | .   |
| Issuer File Update Request Response/0312 | .   | XE  | ME  |

### Values

| Value | Description |
|-------|-------------|
| 1     | Add         |
| 2     | Update      |
| 3     | Delete      |
| 5     | Inquiry     |

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

## DE 92 (File Security Code)

DE 92 (File Security Code) is an Issuer File Update security code used to indicate that a message originator is authorized to update a file.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 93 (Response Indicator)

DE 93 (Response Indicator) indicates the update action a POS system takes.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 94 (Service Indicator)

DE 94 (Service Indicator) indicates the service a message recipient requires.

### Attributes

| Attribute           | Description                                                            |
|---------------------|------------------------------------------------------------------------|
| Data Representation | ans-7                                                                  |
| Data Length         | N/A                                                                    |
| Data Field          | Contents of positions 1–7. Positions 4–7 must contain spaces or zeros. |
| Subfields           | 3                                                                      |
| Justification       | See subfields                                                          |

### Usage

Following is the usage of DE 94 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                           | Org | Sys | Dst |
|---------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: sign-on/sign-off | M   | X   | .   |

### Values

This data element is used in Network Management Request/0800: sign-on/off messages to indicate that a customer is signing on or off for debit card traffic, credit card traffic, or both. It is also used to indicate if the sign-on or sign-off applies to acquirer traffic, issuer traffic, or both.

The Dual Message Authorization System allows credit card transactions and debit card transactions from a single institution to be routed to different processing systems.

### Application notes

See subfields.

## Subfield 1 (Reserved for Future Use)

DE 94, subfield 1 (Reserved for Future Use) is reserved for future use.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | an-1                   |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

### Values

| Value | Description             |
|-------|-------------------------|
| 0     | Reserved for Future Use |

## Subfield 2 (Acquirer/Issuer Indicator)

DE 94, subfield 2 (Acquirer/Issuer Indicator) indicates if the sign-on or sign-off message applies to acquirer traffic, issuer traffic, or both.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | a-1                    |
| Data Field          | Contents of position 2 |
| Justification       | N/A                    |

### Values

| Value | Description   |
|-------|---------------|
| A     | Acquirer only |
| I     | Issuer only   |

| Value | Description              |
|-------|--------------------------|
| B     | Both acquirer and issuer |

## Subfield 3 (Address Data Indicator)

DE 94, subfield 3 (Address Data Indicator) indicates how the address data is provided.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Data Representation | n-1                    |
| Data Field          | Contents of position 3 |
| Justification       | N/A                    |

### Values

| Value | Description                                                                                                                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | AVS not currently supported                                                                                                                                                                                                                                                          |
| 1     | Issuer receives complete address data                                                                                                                                                                                                                                                |
| 2     | Issuer receives condensed address data. (This supports the algorithm that uses the first five numeric digits in an address [when scanning the address from left to right].)                                                                                                          |
| 3     | Issuer receives condensed address data. (This supports the algorithm that uses up to the first five numeric digits in an address. This algorithm stops searching for a numeric after it encounters an alphabetic character or space [when scanning the address from left to right].) |
| 4     | Issuer receives condensed numeric postal code and condensed numeric address data only. (This supports the algorithm that uses the first five numeric digits in an address [when scanning the address from left to right].)                                                           |

## DE 95 (Replacement Amounts)

DE 95 (Replacement Amounts) contains the actual amount subfields necessary to perform a partial or full reversal of a financial transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-42                      |
| Data Length         | N/A                       |
| Data Field          | Contents of subfields 1–4 |
| Subfields           | 4                         |
| Justification       | See subfields             |

### Usage

Following is the usage of DE 95 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                        | Org | Sys | Dst |
|--------------------------------|-----|-----|-----|
| Reversal Request/0400          | C   | X   | C   |
| Reversal Request Response/0410 | CE  | X   | C   |
| Reversal Advice/0420           | •   | X   | C   |
| Reversal Advice Response/0430  | CE  | XE  | •   |

### Values

See subfields.

Subfield 1 (Actual Amount, Transaction) must contain valid numeric data. All other subfields must be zero-filled; if required, the Authorization System performs currency conversion.

DE 95 is necessary to perform a partial reversal of an authorization transaction. However, the Authorization System will support full reversals via usage of a Reversal Request/0400 message. Therefore, acquirers are not required to include DE 95 in the Reversal Request/0400 message. If DE 95 is included in a Reversal Request/0400 message for full reversals, DE 95 must contain a value of all zeros.

The Dual Message Authorization System requires a message initiator to generate only subfield 1 (Actual Amount, Transaction). The Dual Message Authorization System automatically calculates and inserts subfield 2 (Actual Amount, Settlement) and subfield 3 (Actual Amount, Cardholder Billing).

**NOTE: The subfield definitions the Dual Message Authorization System employs differ slightly from the ISO subfield definitions for this data element to accommodate the Mastercard automatic currency conversion service.**

Application notes

This data element is defined and used identically within all Mastercard programs and services. Mastercard does not provide currency conversion services for all programs and services.

Mastercard supports DE 95 in Reversal Request/0400 and Reversal Advice/0420 messages. DE 95 must be less than DE 4. If DE 95 contains all zeros, Mastercard will remove DE 95 before forwarding the message to the issuer.

Subfield 1 (Actual Amount, Transaction)

DE 95, subfield 1 (Actual Amount, Transaction) indicates the actual transaction amount.

Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-12                       |
| Data Field          | Contents of positions 1–12 |
| Justification       | Right with leading zeros   |

Values

Full reversal: If present, and the reversal is a full reversal, DE 95, subfield 1 must contain a value of all zeros.

Partial reversal: If the reversal is a partial reversal, DE 95, subfield 1 must contain a value less than the amount in DE 4 (Amount Transaction).

## Subfield 2 (Actual Amount, Settlement)

DE 95, subfield 2 (Actual Amount, Settlement) indicates the actual settlement amount in the settlement currency.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-12                        |
| Data Field          | Contents of positions 13–24 |
| Justification       | Right with leading zeros    |

### Values

Must contain valid numeric data. Absence of data must be indicated with zeros.

### Application notes

All settlement amounts are specified in U.S. Dollars. The Dual Message Authorization System will provide this subfield in the settlement currency (U.S. Dollars) if subfield 1 is not all zeros and if the customer chooses to receive settlement amount-related data elements; otherwise, subfield 2 is zero-filled.

## Subfield 3 (Actual Amount, Cardholder Billing)

DE 95, subfield 3 (Actual Amount, Cardholder Billing) the actual amount in the issuer currency.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-12                        |
| Data Field          | Contents of positions 25–36 |
| Justification       | Right with leading zeros    |

### Values

Must contain valid numeric data. Absence of data must be indicated with zeros. The Dual Message Authorization System will provide this subfield in the issuer's cardholder billing currency if subfield 1 is not all zeros.

## Subfield 4 (Zero Fill)

DE 95, subfield 4 (Zero Fill) indicates zeros.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-6                         |
| Data Field          | Contents of positions 37–42 |
| Justification       | N/A                         |

### Values

Must contain zeros.

## DE 96 (Message Security Code)

DE 96 (Message Security Code) is a verification between an acceptor and a card issuer that a message is authorized to update or modify a special file.

### Attributes

| Attribute           | Description                                           |
|---------------------|-------------------------------------------------------|
| Data Representation | n-8                                                   |
| Data Length         | N/A                                                   |
| Data Field          | Contents of positions 1–8 (EBCDIC hexadecimal format) |
| Subfields           | N/A                                                   |
| Justification       | N/A                                                   |

### Usage

Following is the usage of DE 96 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.



### Applicable messages

| Message                                           | Org | Sys | Dst |
|---------------------------------------------------|-----|-----|-----|
| Issuer File Update Request/0302                   | M   | X   | .   |
| Issuer File Update Request Response/0312          | .   | ME  | XE  |
| Network Management Request/0800: sign-on/sign-off | M   | X   | .   |

### Values

If an issuer has been assigned a Security Authorization Code that represents the Mastercard customer password, allowing access in the Dual Message Authorization System network by the customer or *INF* processor, this data element must contain valid security control data when used in Issuer File Update/0302 and Network Management Request/0800: sign-on/sign-off messages.

## DE 97 (Amount, Net Settlement)

DE 97 (Amount, Net Settlement) is the net value of all gross amounts.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 98 (Payee)

DE 98 (Payee), is the third-party beneficiary in a payment transaction.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 99 (Settlement Institution ID Code)

DE 99 (Settlement Institution ID Code) identifies the settlement institution or its agent.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 100 (Receiving Institution ID Code)

DE 100 (Receiving Institution ID Code) is the identity of the institution receiving a Request or Advice message in an interchange system if not the same as identified in DE 2 (Primary Account Number [PAN]) or DE 34 (Primary Account Number [PAN], Extended). The Dual Message

Authorization System uses DE 100 to determine the destination routing of Administrative/06xx messages. For these messages, DE 33 (Forwarding Institution ID Code) identifies the sender of the message; DE 100 identifies the receiver of the message.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n...11; LLVAR              |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–11 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Usage

Following is the usage of DE 100 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                      | Org | Sys | Dst |
|----------------------------------------------|-----|-----|-----|
| Administrative Advice/0620: member-generated | M   | P   | M   |
| Administrative Advice/0620: system-generated | .   | X   | M   |
| Administrative Advice Response/0630          | ME  | XE  | .   |

### Values

Must contain a valid five-digit Mastercard customer ID number. It identifies the destination customer or *INF* to receive the message.

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

Processing systems responding to originating Administrative Advice/06xx and Network Management/08xx messages must not swap the contents of DE 33 and DE 100 in the Response message to achieve proper routing of the Response to the originator.

## DE 101 (File Name)

DE 101 (File Name) is the actual or abbreviated name of the file that the issuer accesses. DE 101 is used in Issuer File Update/03xx messages to identify the specific name of a Dual Message Authorization System data file or program parameter table that is being updated by a customer's Issuer File Update Request/0302.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...17; LLVAR            |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–17 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Usage

Following is the usage of DE 101 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                  | Org | Sys | Dst |
|------------------------------------------|-----|-----|-----|
| Issuer File Update Request/0302          | M   | X   | •   |
| Issuer File Update Request Response/0312 | •   | XE  | ME  |

### Values

| Value  | Description                      |
|--------|----------------------------------|
| MCC102 | Stand-In Account File            |
| MCC103 | Electronic Warning Bulletin File |
| MCC104 | Local Stoplist File              |
| MCC105 | Payment Cancellation File        |
| MCC106 | PAN Mapping File                 |

| Value  | Description                                            |
|--------|--------------------------------------------------------|
| MCC107 | Not in use                                             |
| MCC108 | Not in use                                             |
| MCC109 | Contactless Application Transaction Counter (ATC) File |
| MCC111 | PAN-PAR (Payment Account Reference) Mapping File       |

## DE 102 (Account ID 1)

DE 102 (Account ID-1) is a series of digits that identify a customer account or relationship. Customers primarily use it for the from account in a transfer transaction. DE 102 may be used in Authorization Request Response/0110 messages to identify the specific from account that the transaction affected. DE 102 may be used for printing on cardholder transaction receipts.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...28; LLVAR            |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–28 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

### Usage

Following is the usage of DE 102 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Messages                                      | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110           | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |

## Values

The from account is the account specified in digits three and four of DE 3 (Processing Code).

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

# DE 103 (Account ID 2)

DE 103 (Account ID-2) is a series of digits that identify a customer account or relationship. Customers primarily use it for the "to" account in a transfer transaction.

## Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | ans...28; LLVAR            |
| Data Length         | 2 positions                |
| Data Field          | Contents of positions 1–28 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

## Usage

Following is the usage of DE 103 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110           | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

## Values

The to account is specified in digits five and six of DE 3 (Processing Code).

## Application notes

This data element is defined and used identically within all Mastercard programs and services.

## DE 104 (Digital Payment Data)

DE 104 (Digital Payment Data) is used in Authorization Request/0100 messages to send digital payment data for e-commerce transactions.

### Attributes

| Attribute           | Description                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Representation | ans...999; LLLVAR                                                                                                                                                         |
| Data Length         | 3 positions                                                                                                                                                               |
| Data Field          | Contents of subelements                                                                                                                                                   |
| Subelements         | 001: Digital Payment Cryptogram<br>002: Estimated Amount<br>003: Remote Commerce Acceptor Identifier<br>004: Digital Service Provider<br>005: Digital Authentication Data |
| Justification       | See "Subelements"                                                                                                                                                         |

### Usage

Following is the usage of DE 104 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Values

Transaction specific.

## DE 104: Encoding Scheme

Mastercard organizes DE 104 into a group of encoded subelements and subfields. The following table illustrates the structure of DE 104 subelements.

### Subelement structure

| LLL          | "VAR" - maximum length varies |           |                         |                             |           |                         |
|--------------|-------------------------------|-----------|-------------------------|-----------------------------|-----------|-------------------------|
| 3 bytes      | 3 bytes                       | 3 bytes   | Variable length         | 3 bytes                     | 3 bytes   | Variable length         |
| Total Length | First Subelement (SE) Data    |           |                         | Second Subelement (SE) Data |           |                         |
|              | SE ID                         | SE Length | SE Variable Length Data | SE ID                       | SE Length | SE Variable Length Data |

### Subelement structure

| Number of Bytes | Attribute                       | Value                                 |
|-----------------|---------------------------------|---------------------------------------|
| 3               | Total Length field              | The "LLL" portion of the data element |
| 3               | Subelement ID                   | In the range 001-099                  |
| 3               | Subelement Length               | Variable                              |
| 1...nnn         | Subelement Variable Length Data | Contents of Subfields                 |

The following table illustrates the structure of DE 104 subfields within a subelement.

### Subfield structure

| "VAR" - maximum length varies |           |                         |                           |           |                         |
|-------------------------------|-----------|-------------------------|---------------------------|-----------|-------------------------|
| 3 bytes                       | 3 bytes   | Variable length         | 3 bytes                   | 3 bytes   | Variable length         |
| First subfield (SF) Data      |           |                         | Second subfield (SF) Data |           |                         |
| SF ID                         | SF Length | SF variable length data | SF ID                     | SF length | SF Variable length data |

### Subfield structure

| Number of bytes | Attribute                     | Value                 |
|-----------------|-------------------------------|-----------------------|
| 3               | Subfield ID                   | In the range 001-099  |
| 3               | Subfield length               | Variable              |
| 1...nnn         | Subfield variable Length Data | Contents of subfields |

## Subelement 001 (Digital Payment Cryptogram)

DE 104, subelement 001 (Digital Payment Cryptogram) is where the DSRP Cryptogram will be sent to be processed and verified.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 001                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | ans-28; (base 64 encoded) |
| Data Field             | Contains DSRP Cryptogram  |
| Subfields              | N/A                       |
| Justification          | Left                      |

### Usage

Following is the usage of subelement 001 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |



## Values

Transaction specific. Refer to [Digital Secure Remote Payments: UCAF Formats](#) for detailed information on the formats supported by Mastercard, which continue to be applicable to the new Digital Payment Data field.

## Application notes

To be included by the acquirer for MDES electronic commerce transactions in Authorization Request/0100 messages.

To be received by the issuer when included by the acquirer in non-MDES transactions in Authorization Request/0100 messages and Reversal Request/0400 messages.

If DE 104, subelement 001 is included on any other message type it will be ignored.

**NOTE: Issuers will not receive DE 104, subelement 001 on MDES transactions.**

## Subelement 002 (Estimated Amount)

DE 104, subelement 002 (Estimated Amount) contains the estimated purchase amount of the transaction when the payment request is presented to the consumer during checkout. This amount may not always be the final amount. The estimated amount is encoded and included within the cryptogram. The estimated amount is in the transaction currency defined in DE 49 (Currency Code, Transaction).

## Attributes

| Attributes             | Description               |
|------------------------|---------------------------|
| Subelement ID          | 002                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | n-12                      |
| Data Field             | Contains estimated amount |
| Subfields              | N/A                       |
| Justification          | Right with leading zeros  |

## Usage

Following is the usage of subelement 002 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Application notes

The Authorization System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

This subelement is expected to be present in transactions initiated via wallets or remote commerce programs that support DSRP cryptogram format 3.

## Subelement 003 (Remote Commerce Acceptor Identifier)

DE 104, subelement 003 (Remote Commerce Acceptor Identifier) contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64.

### Attributes

| Attribute              | Description                                      |
|------------------------|--------------------------------------------------|
| Subelement ID          | 003                                              |
| Subelement Data Length | 3 positions                                      |
| Data Representation    | ans...150 (Base64 encoded)                       |
| Data Field             | Contains the Remote Commerce Acceptor Identifier |
| Subfields              | N/A                                              |
| Justification          | Left                                             |

### Usage

Following is the usage of subelement 003 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Application notes

Acquirers can optionally submit the Remote Commerce Acceptor Identifier in DE 104, subelement 003 within Authorization Request/0100 and Financial Transaction Request/0200 messages encoded as Base64 when provided by the merchant for MDES remote commerce transactions.

Mastercard will forward to the issuer (when opted-in to receive it). Issuers electing to use a Third Party Token Service Provider for device based wallet programs will receive the subelement in the authorization request when included by the acquirer. Applicable to Authorization Request/0100 messages.

This subelement is expected to be present in remote commerce transactions initiated via wallets or remote commerce programs that support DSRP cryptogram format 3. If present, it must be encoded as Base64.

## Subelement 004 (Digital Service Provider)

DE 104, subelement 004 (Digital Service Provider) is reserved for specific digital programs and services. It is a free-format, variable-length subelement that may be used for multiple purposes.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 004                   |
| Subelement Data Length | 3 positions           |
| Data Representation    | ans...250; LLLVAR     |
| Data Field             | Contents of subfields |
| Subfields              | 2                     |
| Justification          | See subfields         |

### Usage

Following is the usage of subelement 004 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | .   | X   | C   |

### Application notes

The Dual Message Authorization System populates this subelement in Authorization Request/0100 messages when the issuer has opted-in to receive it. Mastercard may occasionally introduce new DE 104, subelement 004 values between releases. Mastercard requires customers to code their systems with flexibility to process messages containing new values.

### Subfield 001 (Digital Service Provider ID 1)

DE 104, subelement 004 (Digital Service Provider), subfield 001 (Digital Service Provider ID 1) contains the alphanumeric value that identifies the Digital Service Provider type and entity.

This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated the cardholder in a remote commerce transaction.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 001                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | an-5                     |
| Data Field           | Contents of subfield 001 |
| Justification        | Left                     |

### Values

#### Position 1 (Digital Service Provider Type)

| Value | Description                                                              |
|-------|--------------------------------------------------------------------------|
| A     | Entity that authenticated the cardholder                                 |
| C     | Entity that provided the card on file credential used in the transaction |

| Value               | Description                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F                   | Entity that provided the credential to the merchant during checkout                                                                                                                                                                                                   |
| If position 1 is... | Then position 2-5...                                                                                                                                                                                                                                                  |
| A                   | The list of available alphanumeric authenticating entity IDs is in <i>AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values</i> and the <i>MDES Issuer Implementation Guide</i> . Both documents are on the Technical Resource Center (TRC). |
| C                   | A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).                                                                                                                                           |
| F                   | A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).                                                                                                                                           |

### Application notes

The absence of subfield 001 does not impact the authentication status of a transaction. The subfield serves as additional information and should not be used in isolation to make authorization decisions.

### Example: Formatted DE 104, subelement 004, subfield 001

104 225002 012000000000400 003 150d3d3Lm1hc3RlcmNhcmQuY29t... 004 011 001  
005111111

## Subfield 002 (Digital Service Provider ID 2)

DE 104, subelement 004 (Digital Service Provider), subfield 002 (Digital Service Provider ID 2) contains the alphanumeric value that identifies the Digital Service Provider type and entity.

This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated the cardholder in a remote commerce transaction.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 002         |
| Subfield Data Length | 3 positions |
| Data Representation  | an-5        |

| Attribute     | Description              |
|---------------|--------------------------|
| Data Field    | Contents of subfield 002 |
| Justification | Left                     |

## Values

### Position 1 (Digital Service Provider Type)

| Value | Description                                                              |
|-------|--------------------------------------------------------------------------|
| A     | Entity that authenticated the cardholder                                 |
| C     | Entity that provided the card on file credential used in the transaction |
| F     | Entity that provided the credential to the merchant during checkout      |

### Position 2-5 (Digital Service Provider Identifier)

| If position 1 is... | Then position 2-5...                                                                                                                                                                                                                                                  |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A                   | The list of available alphanumeric authenticating entity IDs is in <i>AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values</i> and the <i>MDES Issuer Implementation Guide</i> . Both documents are on the Technical Resource Center (TRC). |
| C                   | A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).                                                                                                                                           |
| F                   | A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).                                                                                                                                           |

## Application notes

The absence of subfield 002 does not impact the authentication status of a transaction. The subfield serves as additional information and should not be used in isolation to make authorization decisions.

## Subelement 005 (Digital Authentication Data)

DE 104, subelement 005 (Digital Authentication Data) is reserved for specific digital programs and services. It is a free-format, variable-length sub element that may be used for multiple purposes. The Attributes table identifies the structure of the subfield encoding scheme in the Authorization Request/0100 message.

### Attributes

| Attribute              | Description                                                                     |
|------------------------|---------------------------------------------------------------------------------|
| Subelement ID          | 005                                                                             |
| Subelement Data Length | 3 positions                                                                     |
| Data Representation    | ans...250; LLLVAR                                                               |
| Data Field             | Contents of subfields                                                           |
| Number of subfields    | Subfield 001 (Cardholder Verification)<br>Subfield 002 (Authentication Factors) |
| Justification          | See subfields                                                                   |

### Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

### Applicable messages

| Message                                               | Org | Sys | Dst |
|-------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                            | •   | X   | C   |
| Authorization Request Response/0120: system-generated | •   | X   | C   |

### Application notes

The Dual Message Authorization System populates this subelement in Authorization Request/0100 messages when the issuer has opted-in to receive it. Mastercard may occasionally introduce new DE 104, subelement 005 values between releases. Mastercard requires customers to code their systems with flexibility to process messages containing new values.

### Subfield 001 (Cardholder Verification)

DE 104, subelement 005 (Digital Authentication Data), subfield 001 (Cardholder Verification) contains an alphanumeric value to indicate the cardholder was authenticated and specifies the Cardholder Verification Method (CVM) used to authenticate the cardholder during a contactless or remote commerce transaction.

#### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 001                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | an-5                     |
| Data Field           | Contents of subfield 001 |
| Justification        | Left                     |

#### Values

#### Position 1 (Transaction authenticated with a valid Cardholder Verification Method)

| Value | Description                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Transaction authenticated with a Mastercard-qualified multi-factor authentication (MFA) method. In Europe region: Transaction authenticated with a Mastercard-qualified multi-factor authentication (MFA) method used as Strong Customer Authentication (SCA) delegation. |
| 2     | Transaction authenticated with a Mastercard-operated authentication method                                                                                                                                                                                                |
| 3     | Transaction authenticated with an Issuer authentication method                                                                                                                                                                                                            |
| 4     | Device wallet transaction authenticated with a Consumer Device Cardholder Verification Method (CDCVM). This value may not be present in current device wallet implementations.                                                                                            |



### Position 2-5 (Identifier to the Cardholder Verification Method)

| If position 1 is... | Then position 2-5 will be...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                   | A Mastercard assigned unique identifier for the MFA method. See <i>AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values</i> for the list of available alphanumeric MFA method IDs.                                                                                                                                                                                                                                                                                                                                   |
| 2                   | 0001: Mastercard authentication method with device binding and device biometrics (FIDO).<br>0002: Mastercard authentication method with device binding and passive authentication.                                                                                                                                                                                                                                                                                                                                                             |
| 3                   | 0001: Transaction authenticated by issuer through Identity Check (EMV 3-D Secure) when Mastercard is an EMV 3-D Secure requestor.<br>0002: Transaction authenticated by the issuer through MDES.<br>0003: Transaction authenticated by the issuer through a proprietary channel.                                                                                                                                                                                                                                                               |
| 4                   | Mastercard assigned unique identifier for the CDCVM method. This value may not be present in current device wallet implementations.<br><br>TRID: When the value is "4TRID" the Token Requestor ID in DE 48, subelement 33 (PAN Mapping File Information), subfield 006 (Token Requestor ID), identifies the entity that authenticated the cardholder.<br><br>DDSP: When the value is "4DDSP" the Digital Service Provider ID in DE 104, subelement 004, subfield 001 or subfield 002, identifies the entity that authenticated the cardholder. |

The list of available methods is in *AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values* and the *MDES Issuer Implementation Guide*. Both documents are on the Technical Resource Center on Mastercard Connect®.

### Application notes

The absence of subfield 001 does not impact authentication status of a transaction. The subfield serves as additional information and should not be used in isolation to make authorization decisions.

## Subfield 002 (Authentication Factors)

DE 104, subelement 005 (Digital Authentication Data), subfield 002 (Authentication Factors) contains the alphanumeric value that identifies the authentication factors used to authenticate the cardholder during a contactless or remote commerce transaction.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 002                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | an-5                     |
| Data Field           | Contents of subfield 002 |
| Justification        | Left                     |

### Values

| Position | Description                  | Values                 |
|----------|------------------------------|------------------------|
| 1-2      | First authentication factor  | value from table below |
| 3-4      | Second authentication factor | value from table below |
| 5        | Reserved for future use      | 0                      |

### Application notes

The use of authentication factors are subject to any regulatory approvals and compliance with applicable laws and regulations.

### Authentication factors by value

| Value | Authentication factor          | Description                                  | Category   |
|-------|--------------------------------|----------------------------------------------|------------|
| 00    | Not provided                   | Factor not provided                          | N/A        |
| 01    | Consumer device / device token | Token is stored securely on consumer device. | Possession |

| <b>Value</b> | <b>Authentication factor</b>                      | <b>Description</b>                                                                                                                                                                    | <b>Category</b>        |
|--------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 02           | Consumer device / device bound payment credential | Payment credential is bound to consumer device using a device binding process that ensures a unique connection between the device and an application or browser or cryptographic key. | Possession             |
| 03           | SMS OTP                                           | One Time Password or passcode sent via text message to a phone number                                                                                                                 | Possession             |
| 04           | Email OTP                                         | One Time Password or passcode sent via email to an email address only accessible from a registered device                                                                             | Possession             |
| 05           | Email OTP                                         | One Time Password or passcode sent via email to an email address accessible from any device using a knowledge factor                                                                  | Knowledge              |
| 06           | Card Validation Code 2 (CVC2)                     | Static 3-digit number printed on the back of a Mastercard-branded card                                                                                                                | Possession             |
| 07           | Application authentication with unknown factor    | Knowledge or inherence factor validated by an application (for example, issuer, merchant, or wallet application) exact factor and exact category unknown                              | Knowledge or Inherence |
| 08           | Application authentication with knowledge factor  | Knowledge factor validated by an application (for example, issuer, merchant, or wallet application) exact factor unknown                                                              | Knowledge              |

| <b>Value</b> | <b>Authentication factor</b>                     | <b>Description</b>                                                                                                         | <b>Category</b>        |
|--------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------|
| 09           | Application authentication with inherence factor | Inherence factor validated by an application (for example, issuer, merchant, or wallet application) exact factor unknown   | Inherence              |
| 0A           | Device OS authentication with unknown factor     | Knowledge or inherence factor validated by the operating system on the consumer device - exact factor and category unknown | Knowledge or Inherence |
| 0B           | Device OS authentication with knowledge factor   | Knowledge factor validated by the operating system on the consumer device - exact factor unknown                           | Knowledge              |
| 0C           | Device OS authentication with password           | Password validated by the operating system on the consumer device                                                          | Knowledge              |
| 0D           | Device OS authentication with passcode           | Passcode validated by the operating system on the consumer device                                                          | Knowledge              |
| 0E           | Device OS authentication with pattern            | Pattern validated by the operating system on the consumer device                                                           | Knowledge              |
| 0F           | Device OS authentication with inherence factor   | Inherence factor validated by the operating system on the consumer device - exact factor unknown                           | Inherence              |
| 0G           | Device OS authentication with fingerprint        | Fingerprint validated by the operating system on the consumer device                                                       | Inherence              |
| 0H           | Device OS authentication with facial recognition | Face validated by the operating system on the consumer device                                                              | Inherence              |
| 0I           | Device OS authentication with iris recognition   | Iris validated by the operating system on the consumer device                                                              | Inherence              |
| 0J           | Pattern                                          | Pattern validation outside of device OS                                                                                    | Knowledge              |

| <b>Value</b> | <b>Authentication factor</b>           | <b>Description</b>                                                                                                                               | <b>Category</b> |
|--------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| OK           | Password                               | Password validation outside of device OS                                                                                                         | Knowledge       |
| OL           | Passcode                               | Passcode validation outside of device OS                                                                                                         | Knowledge       |
| OM           | Facial recognition                     | Facial recognition outside of device OS                                                                                                          | Inherence       |
| ON           | Fingerprint                            | Fingerprint validation outside of device OS                                                                                                      | Inherence       |
| OO           | Iris recognition                       | Iris recognition outside of device OS                                                                                                            | Inherence       |
| OP           | Voice recognition                      | Voice recognition validated by an application or server                                                                                          | Inherence       |
| OQ           | Vein recognition                       | Vein recognition validated by an application or server                                                                                           | Inherence       |
| OR           | Palm geometry                          | Palm geometry validated by an application or server                                                                                              | Inherence       |
| OS           | Hand geometry                          | Hand geometry validated by an application or server                                                                                              | Inherence       |
| OT           | Behavioral biometrics                  | Biometrics that relate to behavioral processes created by the human body, for example the way cardholders tap, type, swipe or hold a device      | Inherence       |
| OU           | Behavioral-based inherence information | Information generated from multiple data points that include elements identifying, cardholder location, spending habits, and transaction history | Inherence       |

## DE 104 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

### Associated edits

| WHEN...                                                                                                                                                                                       | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 message containing DE 104 (Digital Payment Data), subelement 001 (Digital Payment Cryptogram), is of incorrect length or incorrect data representation          | will reject the message and forward to the acquirer an Authorization Request Response/0110 message with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 104001 (indicating the data element in error)</li> </ul>                |
| DE 104 (Digital Payment Data), subelement 003 (Remote Commerce Acceptor Identifier) is incorrectly formatted in an Authorization Request/0100 message and/or exceeds 150 characters in length | will reject the message and forward to the acquirer an Authorization Request Response/0110 message with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 104003 (indicating the data element and subelement in error)</li> </ul> |

## DE 105 (Multi-Use Transaction Identification Data)

DE 105 is alphanumeric, variable length using a tag-length-data encoding scheme to allow for easy addition of new subelements.

### Attributes

| Attribute           | Description                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Data Representation | ans...999; LLLVAR<br>The length field must be in the range 001-999. Subelements are identified by valid subelement ID and length. |
| Data Length         | 3 positions                                                                                                                       |
| Data Field          | Contents of subelements                                                                                                           |
| Subelements         | 1                                                                                                                                 |

| Attribute     | Description    |
|---------------|----------------|
| Justification | See subelement |

## Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | X   | C   |
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | C   | X   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | X   | C   |
| Reversal Request/0400                                                                           | C   | X   | C   |
| Reversal Request Response/0410                                                                  | C   | X   | C   |
| Reversal Advice/0420                                                                            | .   | X   | C   |
| Reversal Advice Response/0430                                                                   | C   | X   | .   |

## Encoding scheme for DE 105

Mastercard organizes DE 105 into a group of encoded subelements and subfields.

## Structure of DE 105 subelement

| LLL          | VAR: maximum length varies |                   |                            |
|--------------|----------------------------|-------------------|----------------------------|
| 3 bytes      | 3 bytes                    | 3 bytes           | Variable length            |
| Total length | First subelement data      |                   |                            |
|              | Subelement ID              | Subelement length | Subelement variable length |

### Structure of DE 105 subelement

| Number of Bytes | Attribute                       | Value                                 |
|-----------------|---------------------------------|---------------------------------------|
| 3               | Total data element length       | The "LLL" portion of the data element |
| 3               | Subelement ID                   | In the range 001–999                  |
| 3               | Subelement length               | Variable                              |
| 1...nnn         | Subelement variable length data | Contents of Subfields                 |

### Structure of DE 105 subfields within a subelement

| LLL                 | VAR: maximum length varies |                               |
|---------------------|----------------------------|-------------------------------|
| 3 bytes             | 3 bytes                    | Variable length               |
| First subfield data |                            |                               |
| Subfield ID         | Subfield length            | Subfield variable length data |

### Structure of DE 105 subfields

| Number of bytes | Attribute                     | Value                   |
|-----------------|-------------------------------|-------------------------|
| 3               | Subfield ID                   | In the range of 001-999 |
| 3               | Subfield length               | Variable                |
| 1...nnn         | Subfield variable length data | Contains valid values   |

## Subelement 001 (Transaction Link ID [TLID])

DE 105, subelement 001 (Transaction Link ID [TLID]) contains the transaction link ID that is globally unique and can be used as a single unique reference for deterministic matching and linking of all message activity.

### Attributes

| Attribute     | Description |
|---------------|-------------|
| Subelement ID | 001         |



| Attribute              | Description |
|------------------------|-------------|
| Subelement Data Length | 3 positions |
| Data Representation    | ans-22      |
| Data Field             | N/A         |
| Subfields              | N/A         |
| Justification          | N/A         |

## Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

## Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | X   | C   |
| Authorization Request Response/0110                                                             | CE  | X   | C   |
| Authorization Advice/0120: Acquirer-generated                                                   | C   | X   | C   |
| Authorization Advice/0120: System-generated                                                     | •   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (Responding to an acquirer-generated 0120) | CE  | X   | C   |
| Authorization Advice Response/0130: system-generated                                            | •   | X   | C   |
| Reversal Request/0400                                                                           | C   | X   | C   |
| Reversal Request Response/0410                                                                  | CE  | X   | C   |
| Reversal Advice/0420: System generated                                                          | •   | X   | C   |
| Reversal Advice Response/0430                                                                   | CE  | XE  | •   |

**NOTE: Authorization Request/0100 pre-digitization messages will not be assigned a TLID. These messages can be identified with DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) with a value of 9 (Tokenization Request/Notification).**

## DE 106 (Fleet Service Data)

DE 106 (Fleet Service Data) contains information pertaining to Fleet card transactions and the items being purchased.

### Attributes

| Attribute           | Description                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Data Representation | ans...999; LLLVAR<br><br>The length field must be in the range 001-999. Subelements are identified by valid subelement ID and length. |
| Data Length         | 3 positions                                                                                                                           |
| Data Field          | Contents of subelements                                                                                                               |
| Subelements         | 5                                                                                                                                     |
| Justification       | See "Subelements"                                                                                                                     |

### Usage

The following applies to DE 106 usage:

- For subelements defined with a CE (conditional echo) presence notation for an Org (originator) entity Mastercard will not reject a message from the message originator if the original subelement data is not returned as defined in a response message. Based on the presence notation for the Dst (destination) entity Mastercard will forward the subelement data as received from the message originator if it is present in a response message.
- Use of DE 106 (Fleet Service Data) for authorization is required only for EMV chip cards personalized using the IFSF/Conexus specification for retrieving supplemental data from acceptors and cardholders.

Following is the overall usage of DE 106 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | O   | P   | C   |
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | O   | P   | C   |

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | X   | C   |

### List of DE 106 subelements

DE 106 subelements are listed in numeric order. Subelements that are specific to a brand service or program are clearly indicated in the subelement title or description or both.

### Subelements

| Subelement ID | Subelement name                                             | Data representation |
|---------------|-------------------------------------------------------------|---------------------|
| 001           | Fleet Prompted Data                                         | ans...220; LLLVAR   |
| 002           | Merchant Fleet Spend, Control Override Capability Indicator | ans-1               |
| 003           | Fleet Spend Control Override Items                          | an-16               |
| 004           | Fleet Fuel Information                                      | ans...123; LLLVAR   |
| 005           | Fleet Non-Fuel Information                                  | ans...552; LLLVAR   |

## DE 106 encoding scheme

Mastercard organizes DE 106 into a group of encoded subelements and subfields. The following tables illustrate the structure of DE 106 subelements.

### DE 106 encoding scheme

Mastercard organizes DE 106 into a group of encoded subelements and subfields. The following tables illustrate the structure of DE 106 subelements.

### Structure of DE 106 subelements

| LLL          | VAR (maximum length varies) |         |                 |                        |         |                 |
|--------------|-----------------------------|---------|-----------------|------------------------|---------|-----------------|
| 3 bytes      | 3 bytes                     | 3 bytes | Variable length | 3 bytes                | 3 bytes | Variable length |
| Total length | First subelement data       |         |                 | Second subelement data |         |                 |

| LLL | VAR (maximum length varies) |                   |                                 |               |                   |                                 |
|-----|-----------------------------|-------------------|---------------------------------|---------------|-------------------|---------------------------------|
|     | Subelement ID               | Subelement length | Subelement variable length data | Subelement ID | Subelement length | Subelement variable length data |

#### Attributes of DE 106 subelements

| Number of bytes | Attribute                       | Value                                 |
|-----------------|---------------------------------|---------------------------------------|
| 3               | Total data element length       | The "LLL" portion of the data element |
| 3               | Subelement ID                   | In the range 001-999                  |
| 3               | Subelement length               | Variable                              |
| 1...nnn         | Subelement variable length data | Contents of subfields                 |

## DE 106 subelement encoding scheme

The following tables illustrate the structure of DE 106 subfields within a subelement.

#### Structure of DE 106 subfields within a subelement

| VAR (maximum length varies) |                 |                               |                      |                 |                               |
|-----------------------------|-----------------|-------------------------------|----------------------|-----------------|-------------------------------|
| 3 bytes                     | 3 bytes         | Variable length               | 3 bytes              | 3 bytes         | Variable length               |
| First subfield data         |                 |                               | Second subfield data |                 |                               |
| Subfield ID                 | Subfield length | Subfield variable length data | Subfield ID          | Subfield length | Subfield variable length data |

#### Attributes of DE 106 subfields within a subelement

| Number of bytes | Attribute                     | Value                 |
|-----------------|-------------------------------|-----------------------|
| 3               | Subfield ID                   | In the range 001-999  |
| 3               | Subfield length               | Variable              |
| 1...nnn         | Subfield variable length data | Contents of subfields |

## Subelement 001 (Fleet Prompted Data)

DE 106, subelement 001 (Fleet Prompted Data) contains up to 5 occurrences of the Prompt Code read from a card chip that indicates prompts that occur at the point of interaction and the data entered in response by the cardholder into the point-of-sale device.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 001                   |
| Subelement Data Length | 3 positions           |
| Data Representation    | ans...220; LLLVAR     |
| Data Field             | Contents of subfields |
| Subfields              | 5                     |
| Justification          | See subfields         |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | O   | P   | C   |
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | O   | P   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | X   | C   |

### Application notes

If subelement 001 was present in the original Authorization Request/0100 message for an Automated Fuel Dispenser (AFD) transaction then subelement 001 in the Authorization Advice/0120: acquirer-generated message for the AFD transaction must contain the same values.

## Subfield 001 (Fleet Prompt 1)

DE 106, subelement 001, subfield 001 (Fleet Prompt 1) contains the first occurrence of the Prompt Code and Driver Related Data.

### Attributes

| Attribute            | Description                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 001                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                |
| Data Representation  | ans...38; LLLVAR                                                                                                                                                           |
| Data Field           | Contents of positions 1-38: <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Prompt Code 1</li> <li>Positions 4-38 (ans...35): Driver Related Data 1</li> </ul> |
| Justification        | N/A                                                                                                                                                                        |

### Values

| Position                                          | Description   |
|---------------------------------------------------|---------------|
| DE 106, subelement 001, subfield 001 Position 1-3 | Prompt Code 1 |

### Prompt codes

| Prompt code | Prompt                   |
|-------------|--------------------------|
| 001         | User ID                  |
| 002         | Vehicle Number           |
| 003         | Vehicle Tag              |
| 004         | Driver ID                |
| 005         | Odometer                 |
| 009         | Work Order / P.O. Number |
| 010         | Invoice Number           |
| 011         | Trip Number              |
| 012         | Unit Number              |

| Prompt code | Prompt                  |
|-------------|-------------------------|
| 013         | Reefer Hours            |
| 015         | ZIP Code / Postal Code  |
| 016         | Replacement Car         |
| 017         | Entered Data Alpha      |
| 018         | Web Portal Data         |
| 019         | Entered Data Num        |
| 021         | Job Number              |
| 022         | Maintenance ID          |
| 023         | Department Number       |
| 024         | Trailer Number          |
| 025         | Delivery Ticket Number  |
| 026         | Hubometer               |
| 033         | Sub Fleet Number        |
| 035         | Transaction Number      |
| 036         | Control Number          |
| 038         | Reefer Temperature      |
| 040         | Driver or Vehicle Card  |
| 041         | Customer Number         |
| 042         | Additional Card Data    |
| 043         | Additional Vehicle Data |
| 044         | Engine Hours            |
| 045         | Tank Level Start        |
| 046         | Fuel Gauge Level        |
| 047         | Battery Voltage         |
| 048         | Coolant Temperature     |
| 049         | Check Engine Status     |
| 050         | Fuel Economy            |
| 051         | Engine RPM              |
| 052         | Engine Load             |
| 053         | Engine Oil Temperature  |
| 054         | Engine Time Total       |

| Prompt code | Prompt              |
|-------------|---------------------|
| 055         | Hard Breaking       |
| 056         | Hard Acceleration   |
| 058         | Idle Time           |
| 065         | Total Idle Time     |
| 067         | Engine Oil Pressure |
| 068         | Engine Oil Life     |
| 069         | Billing ID          |
| 999         | No prompts issued   |

### Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

If the Fleet card magnetic stripe or the chip Third Party Data (tag 9F6E) is encoded to indicate No Prompts Issued or chip data DF30 in tag BF0C is encoded following the International Forecourt Standards Forum (IFSF) with no prompt code values, then one of the Fleet prompt subfields 001 through 005 must contain Prompt code value 999 (No prompts issued) which will not require driver related prompted data at the point-of-sale. When this occurs, acquirers must populate the Driver Related Data field with a default value of 999.

### Subfield 002 (Fleet Prompt 2)

DE 106, subelement 01, subfield 002 (Fleet Prompt 2) contains the second occurrence of the Prompt Code and Driver Related Data.

### Attributes

| Attribute            | Description                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield ID          | 002                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                |
| Data Representation  | ans...38; LLLVAR                                                                                                                                                           |
| Data Field           | Contents of positions 1-38: <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Prompt Code 2</li> <li>Positions 4-38 (ans...35): Driver Related Data 2</li> </ul> |



| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Position                                          | Description   |
|---------------------------------------------------|---------------|
| DE 106, subelement 001, subfield 002 Position 1-3 | Prompt Code 2 |

### Prompt codes

| Prompt code | Prompt                   |
|-------------|--------------------------|
| 001         | User ID                  |
| 002         | Vehicle Number           |
| 003         | Vehicle Tag              |
| 004         | Driver ID                |
| 005         | Odometer                 |
| 009         | Work Order / P.O. Number |
| 010         | Invoice Number           |
| 011         | Trip Number              |
| 012         | Unit Number              |
| 013         | Reefer Hours             |
| 015         | ZIP Code / Postal Code   |
| 016         | Replacement Car          |
| 017         | Entered Data Alpha       |
| 018         | Web Portal Data          |
| 019         | Entered Data Num         |
| 021         | Job Number               |
| 022         | Maintenance ID           |
| 023         | Department Number        |
| 024         | Trailer Number           |
| 025         | Delivery Ticket Number   |

| Prompt code | Prompt                  |
|-------------|-------------------------|
| 026         | Hubometer               |
| 033         | Sub Fleet Number        |
| 035         | Transaction Number      |
| 036         | Control Number          |
| 038         | Reefer Temperature      |
| 040         | Driver or Vehicle Card  |
| 041         | Customer Number         |
| 042         | Additional Card Data    |
| 043         | Additional Vehicle Data |
| 044         | Engine Hours            |
| 045         | Tank Level Start        |
| 046         | Fuel Gauge Level        |
| 047         | Battery Voltage         |
| 048         | Coolant Temperature     |
| 049         | Check Engine Status     |
| 050         | Fuel Economy            |
| 051         | Engine RPM              |
| 052         | Engine Load             |
| 053         | Engine Oil Temperature  |
| 054         | Engine Time Total       |
| 055         | Hard Breaking           |
| 056         | Hard Acceleration       |
| 058         | Idle Time               |
| 065         | Total Idle Time         |
| 067         | Engine Oil Pressure     |
| 068         | Engine Oil Life         |
| 069         | Billing ID              |
| 999         | No prompts issued       |

## Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

If the Fleet card magnetic stripe or the chip Third Party Data (tag 9F6E) is encoded to indicate No Prompts Issued or chip data DF30 in tag BF0C is encoded following the International Forecourt Standards Forum (IFSF) with no prompt code values, then one of the Fleet prompt subfields 001 through 005 must contain Prompt code value 999 (No prompts issued) which will not require driver related prompted data at the point-of-sale. When this occurs, acquirers must populate the Driver Related Data field with a default value of 999.

## Subfield 003 (Fleet Prompt 3)

DE 106, subelement 001, subfield 003 (Fleet Prompt 3) contains the third occurrence of the Prompt Code and Driver Related Data.

### Attributes

| Attribute            | Description                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield ID          | 003                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                |
| Data Representation  | ans...38; LLLVAR                                                                                                                                                           |
| Data Field           | Contents of positions 1-38: <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Prompt Code 3</li> <li>Positions 4-38 (ans...35): Driver Related Data 3</li> </ul> |
| Justification        | N/A                                                                                                                                                                        |

### Values

| Position                                            | Description   |
|-----------------------------------------------------|---------------|
| DE 106, subelement 001, subfield 003, positions 1-3 | Prompt Code 3 |

### Prompt codes

| Prompt code | Prompt                   |
|-------------|--------------------------|
| 001         | User ID                  |
| 002         | Vehicle Number           |
| 003         | Vehicle Tag              |
| 004         | Driver ID                |
| 005         | Odometer                 |
| 009         | Work Order / P.O. Number |
| 010         | Invoice Number           |
| 011         | Trip Number              |
| 012         | Unit Number              |
| 013         | Reefer Hours             |
| 015         | ZIP Code / Postal Code   |
| 016         | Replacement Car          |
| 017         | Entered Data Alpha       |
| 018         | Web Portal Data          |
| 019         | Entered Data Num         |
| 021         | Job Number               |
| 022         | Maintenance ID           |
| 023         | Department Number        |
| 024         | Trailer Number           |
| 025         | Delivery Ticket Number   |
| 026         | Hubometer                |
| 033         | Sub Fleet Number         |
| 035         | Transaction Number       |
| 036         | Control Number           |
| 038         | Reefer Temperature       |
| 040         | Driver or Vehicle Card   |
| 041         | Customer Number          |
| 042         | Additional Card Data     |

| Prompt code | Prompt                  |
|-------------|-------------------------|
| 043         | Additional Vehicle Data |
| 044         | Engine Hours            |
| 045         | Tank Level Start        |
| 046         | Fuel Gauge Level        |
| 047         | Battery Voltage         |
| 048         | Coolant Temperature     |
| 049         | Check Engine Status     |
| 050         | Fuel Economy            |
| 051         | Engine RPM              |
| 052         | Engine Load             |
| 053         | Engine Oil Temperature  |
| 054         | Engine Time Total       |
| 055         | Hard Breaking           |
| 056         | Hard Acceleration       |
| 058         | Idle Time               |
| 065         | Total Idle Time         |
| 067         | Engine Oil Pressure     |
| 068         | Engine Oil Life         |
| 069         | Billing ID              |
| 999         | No prompts issued       |

### Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

If the Fleet card magnetic stripe or the chip Third Party Data (tag 9F6E) is encoded to indicate No Prompts Issued or chip data DF30 in tag BF0C is encoded following the International Forecourt Standards Forum (IFSF) with no prompt code values, then one of the Fleet prompt subfields 001 through 005 must contain Prompt code value 999 (No prompts issued) which will not require driver related prompted data at the point-of-sale. When this occurs, acquirers must populate the Driver Related Data field with a default value of 999.

## Subfield 004 (Fleet Prompt 4)

DE 106, subelement 001, subfield 004 (Fleet Prompt 4) contains the fourth occurrence of the Prompt Code and Driver Related Data.

### Attributes

| Attribute            | Description                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield ID          | 004                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                |
| Data Representation  | ans...38; LLLVAR                                                                                                                                                           |
| Data Field           | Contents of positions 1-38: <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Prompt Code 4</li> <li>Positions 4-38 (ans...35): Driver Related Data 4</li> </ul> |
| Justification        | N/A                                                                                                                                                                        |

### Values

| Position                                            | Description   |
|-----------------------------------------------------|---------------|
| DE 106, subelement 001, subfield 004, positions 1-3 | Prompt Code 4 |

### Prompt codes

| Prompt code | Prompt                   |
|-------------|--------------------------|
| 001         | User ID                  |
| 002         | Vehicle Number           |
| 003         | Vehicle Tag              |
| 004         | Driver ID                |
| 005         | Odometer                 |
| 009         | Work Order / P.O. Number |
| 010         | Invoice Number           |
| 011         | Trip Number              |
| 012         | Unit Number              |

| Prompt code | Prompt                  |
|-------------|-------------------------|
| 013         | Reefer Hours            |
| 015         | ZIP Code / Postal Code  |
| 016         | Replacement Car         |
| 017         | Entered Data Alpha      |
| 018         | Web Portal Data         |
| 019         | Entered Data Num        |
| 021         | Job Number              |
| 022         | Maintenance ID          |
| 023         | Department Number       |
| 024         | Trailer Number          |
| 025         | Delivery Ticket Number  |
| 026         | Hubometer               |
| 033         | Sub Fleet Number        |
| 035         | Transaction Number      |
| 036         | Control Number          |
| 038         | Reefer Temperature      |
| 040         | Driver or Vehicle Card  |
| 041         | Customer Number         |
| 042         | Additional Card Data    |
| 043         | Additional Vehicle Data |
| 044         | Engine Hours            |
| 045         | Tank Level Start        |
| 046         | Fuel Gauge Level        |
| 047         | Battery Voltage         |
| 048         | Coolant Temperature     |
| 049         | Check Engine Status     |
| 050         | Fuel Economy            |
| 051         | Engine RPM              |
| 052         | Engine Load             |
| 053         | Engine Oil Temperature  |
| 054         | Engine Time Total       |

| Prompt code | Prompt              |
|-------------|---------------------|
| 055         | Hard Breaking       |
| 056         | Hard Acceleration   |
| 058         | Idle Time           |
| 065         | Total Idle Time     |
| 067         | Engine Oil Pressure |
| 068         | Engine Oil Life     |
| 069         | Billing ID          |
| 999         | No prompts issued   |

### Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

If the Fleet card magnetic stripe or the chip Third Party Data (tag 9F6E) is encoded to indicate No Prompts Issued or chip data DF30 in tag BF0C is encoded following the International Forecourt Standards Forum (IFSF) with no prompt code values, then one of the Fleet prompt subfields 001 through 005 must contain Prompt code value 999 (No prompts issued) which will not require driver related prompted data at the point-of-sale. When this occurs, acquirers must populate the Driver Related Data field with a default value of 999.

### Subfield 005 (Fleet Prompt 5)

DE 106, subelement 001, subfield 005 (Fleet Prompt 5) contains the fifth occurrence of the Prompt Code and Driver Related Data.

### Attributes

| Attribute            | Description                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield ID          | 005                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                |
| Data Representation  | ans...38; LLLVAR                                                                                                                                                           |
| Data Field           | Contents of positions 1-38: <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Prompt Code 5</li> <li>Positions 4-38 (ans...35): Driver Related Data 5</li> </ul> |



| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Position                                            | Description   |
|-----------------------------------------------------|---------------|
| DE 106, subelement 001, subfield 005, positions 1-3 | Prompt Code 5 |

### Prompt codes

| Prompt code | Prompt                   |
|-------------|--------------------------|
| 001         | User ID                  |
| 002         | Vehicle Number           |
| 003         | Vehicle Tag              |
| 004         | Driver ID                |
| 005         | Odometer                 |
| 009         | Work Order / P.O. Number |
| 010         | Invoice Number           |
| 011         | Trip Number              |
| 012         | Unit Number              |
| 013         | Reefer Hours             |
| 015         | ZIP Code / Postal Code   |
| 016         | Replacement Car          |
| 017         | Entered Data Alpha       |
| 018         | Web Portal Data          |
| 019         | Entered Data Num         |
| 021         | Job Number               |
| 022         | Maintenance ID           |
| 023         | Department Number        |
| 024         | Trailer Number           |
| 025         | Delivery Ticket Number   |

| Prompt code | Prompt                  |
|-------------|-------------------------|
| 026         | Hubometer               |
| 033         | Sub Fleet Number        |
| 035         | Transaction Number      |
| 036         | Control Number          |
| 038         | Reefer Temperature      |
| 040         | Driver or Vehicle Card  |
| 041         | Customer Number         |
| 042         | Additional Card Data    |
| 043         | Additional Vehicle Data |
| 044         | Engine Hours            |
| 045         | Tank Level Start        |
| 046         | Fuel Gauge Level        |
| 047         | Battery Voltage         |
| 048         | Coolant Temperature     |
| 049         | Check Engine Status     |
| 050         | Fuel Economy            |
| 051         | Engine RPM              |
| 052         | Engine Load             |
| 053         | Engine Oil Temperature  |
| 054         | Engine Time Total       |
| 055         | Hard Breaking           |
| 056         | Hard Acceleration       |
| 058         | Idle Time               |
| 065         | Total Idle Time         |
| 067         | Engine Oil Pressure     |
| 068         | Engine Oil Life         |
| 069         | Billing ID              |
| 999         | No prompts issued       |

### Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

If the Fleet card magnetic stripe or the chip Third Party Data (tag 9F6E) is encoded to indicate No Prompts Issued or chip data DF30 in tag BF0C is encoded following the International Forecourt Standards Forum (IFSF) with no prompt code values, then one of the Fleet prompt subfields 001 through 005 must contain Prompt code value 999 (No prompts issued) which will not require driver related prompted data at the point-of-sale. When this occurs, acquirers must populate the Driver Related Data field with a default value of 999.

## Subelement 002 (Merchant Fleet Spend Control Override Capability Indicator)

DE 106, subelement 002 (Merchant Fleet Spend Control Override Capability Indicator) indicates if the merchant point-of-sale supports the Fleet Spend Control Override Capability.

### Attributes

| Attribute              | Description            |
|------------------------|------------------------|
| Subelement ID          | 002                    |
| Subelement Data Length | 3 positions            |
| Data Representation    | ans-1                  |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |
| Justification          | N/A                    |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | O   | P   | C   |

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated | •   | X   | C   |

### Values

| Value | Description                             |
|-------|-----------------------------------------|
| 0     | Does not support spend control override |
| 1     | Supports spend control override         |

### Application notes

Subelement 002 values enable acquirers to provide issuers with an indicator to identify when a merchant can support the Fleet Spend Control Override Capability. This data will assist issuers in determining when they can send revised purchase restriction values in DE 106, subelement 003 (Fleet Spend Control Override Items) in the Authorization Request Response/0110 message to override the purchase restriction values obtained from the card at the POS. This allows issuers to perform real-time adjustments to the types of products allowed to be purchased using that particular Fleet card.

## Subelement 003 (Fleet Spend Control Override Items)

DE 106, subelement 003 (Fleet Spend Control Override Items) provides merchants with information, on a transaction basis, to override purchase restrictions to allow products to be purchased that may not have otherwise been allowed.

### Attributes

| Attribute              | Description                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subelement ID          | 003                                                                                                                                                             |
| Subelement Data Length | 3 positions                                                                                                                                                     |
| Data Representation    | an-16; the 16 byte hexadecimal representation of the 8 byte binary data that conforms to the Connexus format standard defined for Purchase Restriction Tag DF32 |
| Data Field             | Contents of positions 1-16                                                                                                                                      |
| Subfields              | N/A                                                                                                                                                             |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

### Application notes

Subelement 003 enables issuers to provide acquirers with revised purchase restriction values in the Authorization Request Response/0110 message to override the purchase restriction values obtained from Connexus Purchase Restriction Tag DF32 found on the EMV card. This subelement will only be populated when DE 106, subelement 002 contains the value '1' indicating that the merchant point-of-sale supports the Fleet Spend Control Override Capability.

DE 106, subelement 003 must contain the 16 byte hexadecimal representation of the 8 byte binary data that conforms to the Connexus format standard defined for Purchase Restriction Tag DF32.

## Subelement 004 (Fleet Fuel Information)

DE 106, subelement 004 (Fleet Fuel Information) contains item level detail information of the motor fuel product(s) purchased at the POS.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 004                   |
| Subelement Data Length | 3 positions           |
| Data Representation    | ans...123; LLLVAR     |
| Data Field             | Contents of subfields |
| Subfields              | 3                     |

| Attribute     | Description   |
|---------------|---------------|
| Justification | See subfields |

## Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | O   | P   | C   |
| Authorization Advice/0120: acquirer-generated | O   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

## Application notes

DE 106, subelement 004 allows the acquirer to provide issuers with detailed information on up to three fuel products purchased.

If the fuel product being purchased and related detail is known at the time of authorization, then this field can be optionally provided in the Authorization Request/ 0100 message.

If the fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

If subelement 004 was present in the original Authorization Request/0100 message for an Automated Fuel Dispenser (AFD) transaction, then subelement 004 in the Authorization Advice/0120: acquirer-generated message for the AFD transaction must contain the same values.

## Subfield 001 (Fuel Item Purchased 1)

DE 106, subelement 004, subfield 001 (Fuel Item Purchased 1) contains detail Fleet Fuel Information of the first Fuel Item Purchased.

## Attributes

| Attribute | Description |
|-----------|-------------|
| Subfield  | 001         |

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Data Representation  | ans-35                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data Field           | <p>Contents of positions 1-35:</p> <ul style="list-style-type: none"> <li>• Positions 1-3 (n-3): Fleet Fuel Product Code 1</li> <li>• Positions 4-15 (n-12): Fleet Fuel Unit Price 1</li> <li>• Position 16 (ans-1): Fleet Fuel Unit of Measure 1</li> <li>• Positions 17-22 (n-6): Fleet Fuel Quantity 1</li> <li>• Position 23 (n-1): Fleet Fuel Quantity Exponent 1</li> <li>• Positions 24-35 (n-12): Fleet Fuel Sale Amount 1</li> </ul> |
| Justification        | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-22: Right-justified with leading zeros</p> <p>Position 23: N/A</p> <p>Positions 24-35: Right-justified with leading zeros</p>                                                                                                                                                                                     |

### Values

| Position                                            | Description               |
|-----------------------------------------------------|---------------------------|
| DE 106, subelement 004, subfield 001, positions 1-3 | Fleet Fuel Product Code 1 |

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

### Product codes

| Fleet Fuel product code | Description   |
|-------------------------|---------------|
| 001                     | Regular       |
| 002                     | Mid/Plus      |
| 003                     | Premium/Super |

| <b>Fleet Fuel product code</b> | <b>Description</b>                                             |
|--------------------------------|----------------------------------------------------------------|
| 004                            | Mid/Plus 2                                                     |
| 005                            | Premium/Super 2                                                |
| 011                            | <Deprecated> Ethanol (5.7% blend)                              |
| 012                            | <Deprecated> Mid/Plus Ethanol (5.7% blend)                     |
| 013                            | <Deprecated> Premium/Super Ethanol (5.7% blend)                |
| 014                            | <Deprecated> Ethanol (7.7% blend)                              |
| 015                            | <Deprecated> Mid/Plus Ethanol (7.7% blend)                     |
| 016                            | Green Gasoline Regular (Plant based not petroleum based)       |
| 017                            | Green Gasoline Mid/Plus (Plant based not petroleum based)      |
| 018                            | Green Gasoline Premium/Super (Plant based not petroleum based) |
| 019                            | Regular Diesel #2                                              |
| 020                            | Premium Diesel #2                                              |
| 021                            | Regular Diesel #1                                              |
| 022                            | Compressed Natural Gas                                         |
| 023                            | Liquid Propane Gas                                             |
| 024                            | Liquid Natural Gas                                             |
| 026                            | E-85                                                           |
| 027                            | Reformulated 1                                                 |
| 028                            | Reformulated 2                                                 |
| 029                            | Reformulated 3                                                 |
| 030                            | Reformulated 4                                                 |
| 031                            | Reformulated 5                                                 |
| 032                            | Diesel Off-Road (#1 and #2 Non-Taxable)                        |
| 033                            | Diesel Off-Road (Non-Taxable)                                  |
| 034                            | Biodiesel Blend Off-Road (Non-Taxable)                         |
| 035                            | Undefined Fuel - Reserved for Conexxus Future                  |
| 036                            | Racing Fuel                                                    |
| 037                            | Mid/Plus 2 (10% Blend)                                         |



| Fleet Fuel product code | Description                                                               |
|-------------------------|---------------------------------------------------------------------------|
| 038                     | Premium/Super 2 (10% Blend)                                               |
| 039                     | Mid/Plus Ethanol 2 (15% Blend)                                            |
| 040                     | Premium/Super Ethanol 2 (15% Blend)                                       |
| 041                     | <Deprecated> Premium/Super Ethanol (7.7% Blend)                           |
| 042                     | Regular Ethanol (10% Blend)                                               |
| 043                     | Mid/Plus Ethanol (10% Blend)                                              |
| 044                     | Premium/Super Ethanol (10% Blend)                                         |
| 045                     | B2 Diesel Blend 2% Biodiesel                                              |
| 046                     | B5 Diesel Blend 5% Biodiesel                                              |
| 047                     | B10 Diesel Blend 10% Biodiesel                                            |
| 048                     | B11 Diesel Blend 11% Biodiesel                                            |
| 049                     | B15 Diesel Blend 15% Biodiesel                                            |
| 050                     | B20 Diesel Blend 20% Biodiesel                                            |
| 051                     | B100 Diesel Blend 100% Biodiesel                                          |
| 052                     | B1 Diesel Blend 1% Biodiesel                                              |
| 053                     | Additized Diesel #2                                                       |
| 054                     | Additized Diesel #3                                                       |
| 060                     | Renewable Diesel ( $\geq$ R95) (meets ASTM D975 in US / EN 590 in Europe) |
| 061                     | Renewable Diesel (Biodiesel 6%-20%) (meets ASTM D7467 in US)              |
| 062                     | DEF (Diesel Exhaust Fluid)                                                |
| 063                     | Premium Diesel #1                                                         |
| 064                     | Regular Ethanol (15% Blend)                                               |
| 065                     | Mid/Plus Ethanol (15% Blend)                                              |
| 066                     | Premium/Super Ethanol (15% Blend)                                         |
| 067                     | Premium Diesel Blend <20% Biodiesel                                       |
| 068                     | Premium Diesel Blend $\geq$ 20% Biodiesel                                 |
| 069                     | B75 Diesel Blend 75% Biodiesel                                            |
| 070                     | B99 Diesel Blend 99% Biodiesel                                            |
| 099                     | Miscellaneous Fuel                                                        |

| <b>Fleet Fuel product code</b> | <b>Description</b>                           |
|--------------------------------|----------------------------------------------|
| 150                            | Jet Fuel                                     |
| 151                            | Aviation Fuel Regular                        |
| 152                            | Aviation Fuel Premium                        |
| 153                            | Aviation Fuel JP8                            |
| 154                            | Aviation Fuel 4                              |
| 155                            | Aviation Fuel 5                              |
| 156                            | Biojet (Diesel)                              |
| 157                            | Aviation Biofuel (Gasoline)                  |
| 174                            | Miscellaneous Aviation Fuel                  |
| 225                            | Marine Fuel 1                                |
| 226                            | Marine Fuel 2                                |
| 227                            | Marine Fuel 3                                |
| 228                            | Marine Fuel 4                                |
| 229                            | Marine Fuel 5                                |
| 230                            | Marine - Other                               |
| 231                            | Marine Diesel                                |
| 249                            | Miscellaneous Marine Fuel                    |
| 300                            | Kerosene – Low Sulfur                        |
| 301                            | White Gas                                    |
| 302                            | Heating Oil                                  |
| 304                            | Other Fuel (Non-Taxable)                     |
| 305                            | Kerosene – Ultra Low Sulfur                  |
| 306                            | Kerosene – Low Sulfur (Non-Taxable)          |
| 307                            | Kerosene – Ultra Low Sulfur (Non-Taxable)    |
| 308                            | EVC-1 – Level 1 charge = 110v 15 amp         |
| 309                            | EVC-2 – Level 2 charge = 240v 15-40 amp      |
| 310                            | EVC-3 – Level 3 charge = 480v 3 phase charge |
| 311                            | Biodiesel Blend 2% Off-Road (Non-Taxable)    |
| 312                            | Biodiesel Blend 5% Off-Road (Non-Taxable)    |
| 313                            | Biodiesel Blend 10% Off-Road (Non-Taxable)   |
| 314                            | Biodiesel Blend 11% Off-Road (Non-Taxable)   |

| Fleet Fuel product code | Description                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------|
| 315                     | Biodiesel Blend 15% Off-Road (Non-Taxable)                                                          |
| 316                     | Biodiesel Blend 20% Off-Road (Non-Taxable)                                                          |
| 317                     | Diesel #1 Off-Road (Non-Taxable)                                                                    |
| 318                     | Diesel #2 Off-Road (Non-Taxable)                                                                    |
| 319                     | Diesel #1 Premium Off-Road (Non-Taxable)                                                            |
| 320                     | Diesel #2 Premium Off-Road (Non-Taxable)                                                            |
| 321                     | Additive Dosage                                                                                     |
| 322                     | Ethanol Blends E16-E84                                                                              |
| 323                     | Low Octane UNL                                                                                      |
| 324                     | Blended Diesel (#1 and #2)                                                                          |
| 325                     | Off-Road Regular (Non-Taxable)                                                                      |
| 326                     | Off-Road Mid/Plus (Non-Taxable)                                                                     |
| 327                     | Off-Road Premium/Super (Non-Taxable)                                                                |
| 328                     | Off-Road Mid/Plus 2 (Non-Taxable)                                                                   |
| 329                     | Off-Road Premium/Super 2 (Non-Taxable)                                                              |
| 330                     | Recreational Fuel (90 Octane)                                                                       |
| 331                     | Hydrogen H35                                                                                        |
| 332                     | Hydrogen H70                                                                                        |
| 333                     | Renewable Diesel ( $\geq$ R95) Off-Road (Non-Taxable)<br>(meets ASTM D975 in US / EN 590 in Europe) |
| 334                     | Biodiesel Blend 1% Off-Road (Non-Taxable)                                                           |
| 335                     | Biodiesel Blend 75% Off-Road (Non-Taxable)                                                          |
| 336                     | Biodiesel Blend 99% Off-Road (Non-Taxable)                                                          |
| 337                     | Biodiesel Blend 100% Off-Road (Non-Taxable)                                                         |
| 338                     | Renewable Diesel (Biodiesel 6%-20%) Off-Road<br>(Non-taxable) (meets ASTM D7467 in US)              |
| 399                     | Miscellaneous Other Fuel                                                                            |

## Values

| Position                                         | Description                  |
|--------------------------------------------------|------------------------------|
| DE 106 subelement 004, subfield 001, position 16 | Fleet Fuel Unit of Measure 1 |

## Measurement values

| Value | Unit of measure  |
|-------|------------------|
| 1     | Gallons          |
| 2     | Liters           |
| 3     | Pounds           |
| 4     | Kilos            |
| 5     | Imperial gallons |
| 6     | Kilowatt hour    |
| 7     | Minutes          |
| 8     | Hours            |
| Space | Not applicable   |

## Application notes

Fleet Fuel Product Code contains a three-digit code that identifies the fuel item purchased.

Fleet Fuel Unit Price contains the price per unit of the item being purchased and must be right-justified with three positions to the right of the decimal.

Fleet Fuel Unit of Measure contains the code indicating the unit of measure for the fuel item purchased.

Fleet Fuel Quantity contains quantity of the fuel item purchased and must be numeric and greater than zero.

Fleet Fuel Quantity Exponent contains the exponent indicating the decimal position of the quantity of the fuel item being purchased. Value must be in the range of 0-5 and is assumed to three positions.

Fleet Fuel Sale Amount contains the sale amount of the fuel item being purchased. Value must be numeric and greater than zero and in the same currency as DE 4 (Amount, Transaction). The currency exponent for this Fleet Fuel Sale Amount is based on the exponent associated with the DE 49 (Currency Code, Transaction) field value.

## Subfield 002 (Fuel Item Purchased 2)

DE 106, subelement 004, subfield 002 (Fuel Item Purchased 2) contains detail Fleet Fuel Information of the first Fuel Item Purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 002                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Data Representation  | ans-35                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data Field           | <p>Contents of positions 1-35:</p> <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Fleet Fuel Product Code 2</li> <li>Positions 4-15 (n-12): Fleet Fuel Unit Price 2 (NUM)</li> <li>Position 16 (ans-1): Fleet Fuel Unit of Measure 2 (ANS)</li> <li>Positions 17-22 (n-6): Fleet Fuel Quantity 2</li> <li>Position 23 (n-1): Fleet Fuel Quantity Exponent 2</li> <li>Positions 24-35 (n-12): Fleet Fuel Sale Amount 2</li> </ul> |
| Justification        | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-22: Right-justified with leading zeros</p> <p>Position 23: N/A</p> <p>Positions 24-35: Right-justified with leading zeros</p>                                                                                                                                                                                     |

### Values

| Position                                            | Description               |
|-----------------------------------------------------|---------------------------|
| DE 106, subelement 004, subfield 002, positions 1-3 | Fleet Fuel Product Code 2 |

The following list of product code values is based on the Connexus Payment System Product Codes version 4.1.

## Product codes

| Fleet Fuel product code | Description                                                    |
|-------------------------|----------------------------------------------------------------|
| 001                     | Regular                                                        |
| 002                     | Mid/Plus                                                       |
| 003                     | Premium/Super                                                  |
| 004                     | Mid/Plus 2                                                     |
| 005                     | Premium/Super 2                                                |
| 011                     | <Deprecated> Ethanol (5.7% blend)                              |
| 012                     | <Deprecated> Mid/Plus Ethanol (5.7% blend)                     |
| 013                     | <Deprecated> Premium/Super Ethanol (5.7% blend)                |
| 014                     | <Deprecated> Ethanol (7.7% blend)                              |
| 015                     | <Deprecated> Mid/Plus Ethanol (7.7% blend)                     |
| 016                     | Green Gasoline Regular (Plant based not petroleum based)       |
| 017                     | Green Gasoline Mid/Plus (Plant based not petroleum based)      |
| 018                     | Green Gasoline Premium/Super (Plant based not petroleum based) |
| 019                     | Regular Diesel #2                                              |
| 020                     | Premium Diesel #2                                              |
| 021                     | Regular Diesel #1                                              |
| 022                     | Compressed Natural Gas                                         |
| 023                     | Liquid Propane Gas                                             |
| 024                     | Liquid Natural Gas                                             |
| 026                     | E-85                                                           |
| 027                     | Reformulated 1                                                 |
| 028                     | Reformulated 2                                                 |
| 029                     | Reformulated 3                                                 |
| 030                     | Reformulated 4                                                 |
| 031                     | Reformulated 5                                                 |
| 032                     | Diesel Off-Road (#1 and #2 Non-Taxable)                        |

| Fleet Fuel product code | Description                                                         |
|-------------------------|---------------------------------------------------------------------|
| 033                     | Diesel Off-Road (Non-Taxable)                                       |
| 034                     | Biodiesel Blend Off-Road (Non-Taxable)                              |
| 035                     | Undefined Fuel - Reserved for Conexxus Future                       |
| 036                     | Racing Fuel                                                         |
| 037                     | Mid/Plus 2 (10% Blend)                                              |
| 038                     | Premium/Super 2 (10% Blend)                                         |
| 039                     | Mid/Plus Ethanol 2 (15% Blend)                                      |
| 040                     | Premium/Super Ethanol 2 (15% Blend)                                 |
| 041                     | <Deprecated> Premium/Super Ethanol (7.7% Blend)                     |
| 042                     | Regular Ethanol (10% Blend)                                         |
| 043                     | Mid/Plus Ethanol (10% Blend)                                        |
| 044                     | Premium/Super Ethanol (10% Blend)                                   |
| 045                     | B2 Diesel Blend 2% Biodiesel                                        |
| 046                     | B5 Diesel Blend 5% Biodiesel                                        |
| 047                     | B10 Diesel Blend 10% Biodiesel                                      |
| 048                     | B11 Diesel Blend 11% Biodiesel                                      |
| 049                     | B15 Diesel Blend 15% Biodiesel                                      |
| 050                     | B20 Diesel Blend 20% Biodiesel                                      |
| 051                     | B100 Diesel Blend 100% Biodiesel                                    |
| 052                     | B1 Diesel Blend 1% Biodiesel                                        |
| 053                     | Additized Diesel #2                                                 |
| 054                     | Additized Diesel #3                                                 |
| 060                     | Renewable Diesel (>=R95) (meets ASTM D975 in US / EN 590 in Europe) |
| 061                     | Renewable Diesel (Biodiesel 6%-20%) (meets ASTM D7467 in US)        |
| 062                     | DEF (Diesel Exhaust Fluid)                                          |
| 063                     | Premium Diesel #1                                                   |
| 064                     | Regular Ethanol (15% Blend)                                         |
| 065                     | Mid/Plus Ethanol (15% Blend)                                        |
| 066                     | Premium/Super Ethanol (15% Blend)                                   |

| <b>Fleet Fuel product code</b> | <b>Description</b>                        |
|--------------------------------|-------------------------------------------|
| 067                            | Premium Diesel Blend <20% Biodiesel       |
| 068                            | Premium Diesel Blend >= 20% Biodiesel     |
| 069                            | B75 Diesel Blend 75% Biodiesel            |
| 070                            | B99 Diesel Blend 99% Biodiesel            |
| 099                            | Miscellaneous Fuel                        |
| 150                            | Jet Fuel                                  |
| 151                            | Aviation Fuel Regular                     |
| 152                            | Aviation Fuel Premium                     |
| 153                            | Aviation Fuel JP8                         |
| 154                            | Aviation Fuel 4                           |
| 155                            | Aviation Fuel 5                           |
| 156                            | Biojet (Diesel)                           |
| 157                            | Aviation Biofuel (Gasoline)               |
| 174                            | Miscellaneous Aviation Fuel               |
| 225                            | Marine Fuel 1                             |
| 226                            | Marine Fuel 2                             |
| 227                            | Marine Fuel 3                             |
| 228                            | Marine Fuel 4                             |
| 229                            | Marine Fuel 5                             |
| 230                            | Marine - Other                            |
| 231                            | Marine Diesel                             |
| 249                            | Miscellaneous Marine Fuel                 |
| 300                            | Kerosene – Low Sulfur                     |
| 301                            | White Gas                                 |
| 302                            | Heating Oil                               |
| 304                            | Other Fuel (Non-Taxable)                  |
| 305                            | Kerosene – Ultra Low Sulfur               |
| 306                            | Kerosene – Low Sulfur (Non-Taxable)       |
| 307                            | Kerosene – Ultra Low Sulfur (Non-Taxable) |
| 308                            | EVC-1 – Level 1 charge = 110v 15 amp      |
| 309                            | EVC-2 – Level 2 charge = 240v 15-40 amp   |



| Fleet Fuel product code | Description                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------|
| 310                     | EVC-3 – Level 3 charge = 480v 3 phase charge                                                  |
| 311                     | Biodiesel Blend 2% Off-Road (Non-Taxable)                                                     |
| 312                     | Biodiesel Blend 5% Off-Road (Non-Taxable)                                                     |
| 313                     | Biodiesel Blend 10% Off-Road (Non-Taxable)                                                    |
| 314                     | Biodiesel Blend 11% Off-Road (Non-Taxable)                                                    |
| 315                     | Biodiesel Blend 15% Off-Road (Non-Taxable)                                                    |
| 316                     | Biodiesel Blend 20% Off-Road (Non-Taxable)                                                    |
| 317                     | Diesel #1 Off-Road (Non-Taxable)                                                              |
| 318                     | Diesel #2 Off-Road (Non-Taxable)                                                              |
| 319                     | Diesel #1 Premium Off-Road (Non-Taxable)                                                      |
| 320                     | Diesel #2 Premium Off-Road (Non-Taxable)                                                      |
| 321                     | Additive Dosage                                                                               |
| 322                     | Ethanol Blends E16-E84                                                                        |
| 323                     | Low Octane UNL                                                                                |
| 324                     | Blended Diesel (#1 and #2)                                                                    |
| 325                     | Off-Road Regular (Non-Taxable)                                                                |
| 326                     | Off-Road Mid/Plus (Non-Taxable)                                                               |
| 327                     | Off-Road Premium/Super (Non-Taxable)                                                          |
| 328                     | Off-Road Mid/Plus 2 (Non-Taxable)                                                             |
| 329                     | Off-Road Premium/Super 2 (Non-Taxable)                                                        |
| 330                     | Recreational Fuel (90 Octane)                                                                 |
| 331                     | Hydrogen H35                                                                                  |
| 332                     | Hydrogen H70                                                                                  |
| 333                     | Renewable Diesel (>=R95) Off-Road (Non-Taxable)<br>(meets ASTM D975 in US / EN 590 in Europe) |
| 334                     | Biodiesel Blend 1% Off-Road (Non-Taxable)                                                     |
| 335                     | Biodiesel Blend 75% Off-Road (Non-Taxable)                                                    |
| 336                     | Biodiesel Blend 99% Off-Road (Non-Taxable)                                                    |
| 337                     | Biodiesel Blend 100% Off-Road (Non-Taxable)                                                   |
| 338                     | Renewable Diesel (Biodiesel 6%-20%) Off-Road<br>(Non-taxable) (meets ASTM D7467 in US)        |

| Fleet Fuel product code | Description              |
|-------------------------|--------------------------|
| 399                     | Miscellaneous Other Fuel |

### Values

| Position                                     | Description                  |
|----------------------------------------------|------------------------------|
| DE 106, subelement 4, subfield 2 Position 16 | Fleet Fuel Unit of Measure 2 |

### Measurement values

| Value | Unit of measure  |
|-------|------------------|
| 1     | Gallons          |
| 2     | Liters           |
| 3     | Pounds           |
| 4     | Kilos            |
| 5     | Imperial gallons |
| 6     | Kilowatt hour    |
| 7     | Minutes          |
| 8     | Hours            |
| Space | Not applicable   |

### Application notes

Fleet Fuel Product Code contains a three-digit code that identifies the fuel item purchased.

Fleet Fuel Unit Price contains the price per unit of the item being purchased and must be right-justified with three positions to the right of the decimal.

Fleet Fuel Unit of Measure contains the code indicating the unit of measure for the fuel item purchased.

Fleet Fuel Quantity contains quantity of the fuel item purchased and must be numeric and greater than zero.

Fleet Fuel Quantity Exponent contains the exponent indicating the decimal position of the quantity of the fuel item being purchased. Value must be in the range of 0-5 and is assumed to three positions.

Fleet Fuel Sale Amount contains the sale amount of the fuel item being purchased. Value must be numeric and greater than zero and in the same currency as DE 4 (Amount, Transaction). The currency exponent for this Fleet Fuel Sale Amount is based on the exponent associated with the DE 49 (Currency Code, Transaction) field value.

## Subfield 003 (Fuel Item Purchased 3)

DE 106, subelement 004, subfield 003 (Fuel Item Purchased 3) contains detail Fleet Fuel Information of the first Fuel Item Purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 003                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Data Representation  | ans-35                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Data Field           | <p>Contents of positions 1-35:</p> <ul style="list-style-type: none"> <li>• Positions 1-3 (n-3): Fleet Fuel Product Code 3</li> <li>• Positions 4-15 (n-12): Fleet Fuel Unit Price 3 (NUM)</li> <li>• Position 16 (ans-1): Fleet Fuel Unit of Measure 3</li> <li>• Positions 17-22 (n-6): Fleet Fuel Quantity 3</li> <li>• Position 23 (n-1): Fleet Fuel Quantity Exponent 3</li> <li>• Positions 24-35 (n-12): Fleet Fuel Sale Amount 3</li> </ul> |
| Justification        | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-22: Right-justified with leading zeros</p> <p>Position 23: N/A</p> <p>Positions 24-35: Right-justified with leading zeros</p>                                                                                                                                                                                           |

## Values

| Position                                            | Description               |
|-----------------------------------------------------|---------------------------|
| DE 106, subelement 004, subfield 003, positions 1-3 | Fleet Fuel Product Code 3 |

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

## Product codes

| Fleet Fuel product code | Description                                                    |
|-------------------------|----------------------------------------------------------------|
| 001                     | Regular                                                        |
| 002                     | Mid/Plus                                                       |
| 003                     | Premium/Super                                                  |
| 004                     | Mid/Plus 2                                                     |
| 005                     | Premium/Super 2                                                |
| 011                     | <Deprecated> Ethanol (5.7% blend)                              |
| 012                     | <Deprecated> Mid/Plus Ethanol (5.7% blend)                     |
| 013                     | <Deprecated> Premium/Super Ethanol (5.7% blend)                |
| 014                     | <Deprecated> Ethanol (7.7% blend)                              |
| 015                     | <Deprecated> Mid/Plus Ethanol (7.7% blend)                     |
| 016                     | Green Gasoline Regular (Plant based not petroleum based)       |
| 017                     | Green Gasoline Mid/Plus (Plant based not petroleum based)      |
| 018                     | Green Gasoline Premium/Super (Plant based not petroleum based) |
| 019                     | Regular Diesel #2                                              |
| 020                     | Premium Diesel #2                                              |
| 021                     | Regular Diesel #1                                              |
| 022                     | Compressed Natural Gas                                         |
| 023                     | Liquid Propane Gas                                             |

| <b>Fleet Fuel product code</b> | <b>Description</b>                              |
|--------------------------------|-------------------------------------------------|
| 024                            | Liquid Natural Gas                              |
| 026                            | E-85                                            |
| 027                            | Reformulated 1                                  |
| 028                            | Reformulated 2                                  |
| 029                            | Reformulated 3                                  |
| 030                            | Reformulated 4                                  |
| 031                            | Reformulated 5                                  |
| 032                            | Diesel Off-Road (#1 and #2 Non-Taxable)         |
| 033                            | Diesel Off-Road (Non-Taxable)                   |
| 034                            | Biodiesel Blend Off-Road (Non-Taxable)          |
| 035                            | Undefined Fuel - Reserved for Conexxus Future   |
| 036                            | Racing Fuel                                     |
| 037                            | Mid/Plus 2 (10% Blend)                          |
| 038                            | Premium/Super 2 (10% Blend)                     |
| 039                            | Mid/Plus Ethanol 2 (15% Blend)                  |
| 040                            | Premium/Super Ethanol 2 (15% Blend)             |
| 041                            | <Deprecated> Premium/Super Ethanol (7.7% Blend) |
| 042                            | Regular Ethanol (10% Blend)                     |
| 043                            | Mid/Plus Ethanol (10% Blend)                    |
| 044                            | Premium/Super Ethanol (10% Blend)               |
| 045                            | B2 Diesel Blend 2% Biodiesel                    |
| 046                            | B5 Diesel Blend 5% Biodiesel                    |
| 047                            | B10 Diesel Blend 10% Biodiesel                  |
| 048                            | B11 Diesel Blend 11% Biodiesel                  |
| 049                            | B15 Diesel Blend 15% Biodiesel                  |
| 050                            | B20 Diesel Blend 20% Biodiesel                  |
| 051                            | B100 Diesel Blend 100% Biodiesel                |
| 052                            | B1 Diesel Blend 1% Biodiesel                    |
| 053                            | Additized Diesel #2                             |
| 054                            | Additized Diesel #3                             |

| Fleet Fuel product code | Description                                                               |
|-------------------------|---------------------------------------------------------------------------|
| 060                     | Renewable Diesel ( $\geq$ R95) (meets ASTM D975 in US / EN 590 in Europe) |
| 061                     | Renewable Diesel (Biodiesel 6%-20%) (meets ASTM D7467 in US)              |
| 062                     | DEF (Diesel Exhaust Fluid)                                                |
| 063                     | Premium Diesel #1                                                         |
| 064                     | Regular Ethanol (15% Blend)                                               |
| 065                     | Mid/Plus Ethanol (15% Blend)                                              |
| 066                     | Premium/Super Ethanol (15% Blend)                                         |
| 067                     | Premium Diesel Blend <20% Biodiesel                                       |
| 068                     | Premium Diesel Blend $\geq$ 20% Biodiesel                                 |
| 069                     | B75 Diesel Blend 75% Biodiesel                                            |
| 070                     | B99 Diesel Blend 99% Biodiesel                                            |
| 099                     | Miscellaneous Fuel                                                        |
| 150                     | Jet Fuel                                                                  |
| 151                     | Aviation Fuel Regular                                                     |
| 152                     | Aviation Fuel Premium                                                     |
| 153                     | Aviation Fuel JP8                                                         |
| 154                     | Aviation Fuel 4                                                           |
| 155                     | Aviation Fuel 5                                                           |
| 156                     | Biojet (Diesel)                                                           |
| 157                     | Aviation Biofuel (Gasoline)                                               |
| 174                     | Miscellaneous Aviation Fuel                                               |
| 225                     | Marine Fuel 1                                                             |
| 226                     | Marine Fuel 2                                                             |
| 227                     | Marine Fuel 3                                                             |
| 228                     | Marine Fuel 4                                                             |
| 229                     | Marine Fuel 5                                                             |
| 230                     | Marine - Other                                                            |
| 231                     | Marine Diesel                                                             |
| 249                     | Miscellaneous Marine Fuel                                                 |

| Fleet Fuel product code | Description                                  |
|-------------------------|----------------------------------------------|
| 300                     | Kerosene – Low Sulfur                        |
| 301                     | White Gas                                    |
| 302                     | Heating Oil                                  |
| 304                     | Other Fuel (Non-Taxable)                     |
| 305                     | Kerosene – Ultra Low Sulfur                  |
| 306                     | Kerosene – Low Sulfur (Non-Taxable)          |
| 307                     | Kerosene – Ultra Low Sulfur (Non-Taxable)    |
| 308                     | EVC-1 – Level 1 charge = 110v 15 amp         |
| 309                     | EVC-2 – Level 2 charge = 240v 15-40 amp      |
| 310                     | EVC-3 – Level 3 charge = 480v 3 phase charge |
| 311                     | Biodiesel Blend 2% Off-Road (Non-Taxable)    |
| 312                     | Biodiesel Blend 5% Off-Road (Non-Taxable)    |
| 313                     | Biodiesel Blend 10% Off-Road (Non-Taxable)   |
| 314                     | Biodiesel Blend 11% Off-Road (Non-Taxable)   |
| 315                     | Biodiesel Blend 15% Off-Road (Non-Taxable)   |
| 316                     | Biodiesel Blend 20% Off-Road (Non-Taxable)   |
| 317                     | Diesel #1 Off-Road (Non-Taxable)             |
| 318                     | Diesel #2 Off-Road (Non-Taxable)             |
| 319                     | Diesel #1 Premium Off-Road (Non-Taxable)     |
| 320                     | Diesel #2 Premium Off-Road (Non-Taxable)     |
| 321                     | Additive Dosage                              |
| 322                     | Ethanol Blends E16-E84                       |
| 323                     | Low Octane UNL                               |
| 324                     | Blended Diesel (#1 and #2)                   |
| 325                     | Off-Road Regular (Non-Taxable)               |
| 326                     | Off-Road Mid/Plus (Non-Taxable)              |
| 327                     | Off-Road Premium/Super (Non-Taxable)         |
| 328                     | Off-Road Mid/Plus 2 (Non-Taxable)            |
| 329                     | Off-Road Premium/Super 2 (Non-Taxable)       |
| 330                     | Recreational Fuel (90 Octane)                |
| 331                     | Hydrogen H35                                 |

| Fleet Fuel product code | Description                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------|
| 332                     | Hydrogen H70                                                                                  |
| 333                     | Renewable Diesel (>=R95) Off-Road (Non-Taxable)<br>(meets ASTM D975 in US / EN 590 in Europe) |
| 334                     | Biodiesel Blend 1% Off-Road (Non-Taxable)                                                     |
| 335                     | Biodiesel Blend 75% Off-Road (Non-Taxable)                                                    |
| 336                     | Biodiesel Blend 99% Off-Road (Non-Taxable)                                                    |
| 337                     | Biodiesel Blend 100% Off-Road (Non-Taxable)                                                   |
| 338                     | Renewable Diesel (Biodiesel 6%-20%) Off-Road<br>(Non-taxable) (meets ASTM D7467 in US)        |
| 399                     | Miscellaneous Other Fuel                                                                      |

### Values

| Position                                          | Description                  |
|---------------------------------------------------|------------------------------|
| DE 106, subelement 004, subfield 003, position 16 | Fleet Fuel Unit of Measure 3 |

### Measurement values

| Value | Unit of measure  |
|-------|------------------|
| 1     | Gallons          |
| 2     | Liters           |
| 3     | Pounds           |
| 4     | Kilos            |
| 5     | Imperial gallons |
| 6     | Kilowatt hour    |
| 7     | Minutes          |
| 8     | Hours            |
| Space | Not applicable   |

### Application notes

Fleet Fuel Product Code contains a three-digit code that identifies the fuel item purchased.



Fleet Fuel Unit Price contains the price per unit of the item being purchased and must be right-justified with three positions to the right of the decimal.

Fleet Fuel Unit of Measure contains the code indicating the unit of measure for the fuel item purchased.

Fleet Fuel Quantity contains quantity of the fuel item purchased and must be numeric and greater than zero.

Fleet Fuel Quantity Exponent contains the exponent indicating the decimal position of the quantity of the fuel item being purchased. Value must be in the range of 0-5 and is assumed to three positions.

Fleet Fuel Sale Amount contains the sale amount of the fuel item being purchased. Value must be numeric and greater than zero and in the same currency as DE 4 (Amount, Transaction). The currency exponent for this Fleet Fuel Sale Amount is based on the exponent associated with the DE 49 (Currency Code, Transaction) field value.

## Subelement 005 (Fleet Non-Fuel Information)

DE 106, subelement 005 (Fleet Non-Fuel Information) contains item level detail information of the non-fuel product(s) purchased at the POS.

### Attributes

| Attribute              | Description                   |
|------------------------|-------------------------------|
| Subelement ID          | 005                           |
| Subelement Data Length | 3 positions                   |
| Data Representation    | ans...552; LLLVAR             |
| Data Field             | Contents of subfields 001-012 |
| Subfields              | 12                            |
| Justification          | See subfields                 |

### Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | O   | P   | C   |

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: acquirer-generated | O   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |

### Application notes

DE 106, subelement 005 allows the acquirer to provide issuers with detail information on up to 12 non-fuel products purchased.

If subelement 005 was present in the original Authorization Request/0100 message for an Automated Fuel Dispenser (AFD) transaction then subelement 005 in the Authorization Advice/0120: acquirer-generated message for the AFD transaction must contain the same values.

### Subfield 001 (Non-Fuel Item Purchased 1)

DE 106, subelement 005, subfield 001 (Non-Fuel Item Purchased 1) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 001                                                                                                                                                                                                                                                                                                                                                                    |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                            |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                 |
| Data Field           | Contents of positions 1-40: <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Fleet Product Code 1</li> <li>Positions 4-15 (n-12): Fleet Item Quantity 1</li> <li>Position 16 (n-1): Fleet Item Quantity Exponent 1</li> <li>Positions 17-28 (ans-12): Fleet Item Unit of Measure 1</li> <li>Positions 29-40 (n-12): Fleet Extended Item Amount 1</li> </ul> |
| Justification        | Positions 1-3: N/A<br>Positions 4-15: Right-justified with leading zeros<br>Position 16: N/A<br>Positions 17-28: Left-justified, space filled<br>Positions 29-40: Right-justified with leading zeros                                                                                                                                                                   |

## Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 001, positions 1-3   | Fleet Product Code 1         |
| DE 106 subelement 005, subfield 001, positions 17-28 | Fleet Item Unit of Measure 1 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

## Subfield 002 (Non-Fuel Item Purchased 2)

DE 106, subelement 005, subfield 002 (Non-Fuel Item Purchased 2) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 002                                                                                                                                                                                                                                                                                                                                                                                     |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                             |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                                  |
| Data Field           | <p>Contents of positions 1-40:</p> <ul style="list-style-type: none"> <li>• Positions 1-3 (n-3): Fleet Product Code 2</li> <li>• Positions 4-15 (n-12): Fleet Item Quantity 2</li> <li>• Position 16 (n-1): Fleet Item Quantity Exponent 2</li> <li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 2</li> <li>• Positions 29-40 (n-12): Fleet Extended Item Amount 2</li> </ul> |
| Justification        | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-28: Left-justified, space filled</p> <p>Positions 29-40: Right-justified with leading zero</p>                                                                                                                                                              |

### Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 002, positions 1-3   | Fleet Product Code 2         |
| DE 106 subelement 005, subfield 002, positions 17-28 | Fleet Item Unit of Measure 2 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

### Subfield 003 (Non-Fuel Item Purchased 3)

DE 106, subelement 005, subfield 003 (Non-Fuel Item Purchased 3) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield             | 003         |
| Subfield Data Length | 3 positions |
| Data Representation  | ans-40      |

| Attribute     | Description                                                                                                                                                                                                                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Field    | <p>Contents of positions 1-40:</p> <ul style="list-style-type: none"> <li>• Positions 1-3 (n-3): Fleet Product Code 3</li> <li>• Positions 4-15 (n-12): Fleet Item Quantity 3</li> <li>• Position 16 (n-1): Fleet Item Quantity Exponent 3</li> <li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 3</li> <li>• Positions 29-40 (n-12): Fleet Extended Item Amount 3</li> </ul> |
| Justification | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-28: Left-justified, space filled</p> <p>Positions 29-40: Right-justified with leading zeros</p>                                                                                                                                                             |

## Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 003, positions 1-3   | Fleet Product Code 3         |
| DE 106 subelement 005, subfield 003, positions 17-28 | Fleet Item Unit of Measure 3 |

## Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

## Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

## Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 004 (Non-Fuel Item Purchased 4)

DE 106, subelement 005, subfield 004 (Non-Fuel Item Purchased 4) contains item level detail information of the non-fuel product(s) purchased.

Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 004                                                                                                                                                                                                                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                     |
| Data Field           | Contents of positions 1-40: <ul style="list-style-type: none"><li>• Positions 1-3 (n-3): Fleet Product Code 4</li><li>• Positions 4-15 (n-12): Fleet Item Quantity 4</li><li>• Position 16 (n-1): Fleet Item Quantity Exponent 4</li><li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 4</li><li>• Positions 29-40 (n-12): Fleet Extended Item Amount 4</li></ul> |
| Justification        | Positions 1-3: N/A<br>Positions 4-15: Right-justified with leading zeros<br>Position 16: N/A<br>Positions 17-28: Left-justified, space filled<br>Positions 29-40: Right-justified with leading zeros                                                                                                                                                                       |

## Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 004, positions 1-3   | Fleet Product Code 4         |
| DE 106 subelement 005, subfield 004, positions 17-28 | Fleet Item Unit of Measure 4 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.



## Subfield 005 (Non-Fuel Item Purchased 5)

DE 106, subelement 005, subfield 005 (Non-Fuel Item Purchased 5) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 005                                                                                                                                                                                                                                                                                                                                                                                     |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                             |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                                  |
| Data Field           | <p>Contents of positions 1-40:</p> <ul style="list-style-type: none"> <li>• Positions 1-3 (n-3): Fleet Product Code 5</li> <li>• Positions 4-15 (n-12): Fleet Item Quantity 5</li> <li>• Position 16 (n-1): Fleet Item Quantity Exponent 5</li> <li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 5</li> <li>• Positions 29-40 (n-12): Fleet Extended Item Amount 5</li> </ul> |
| Justification        | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-28: Left-justified, space filled</p> <p>Positions 29-40: Right-justified with leading zeros</p>                                                                                                                                                             |

### Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 005, positions 1-3   | Fleet Product Code 5         |
| DE 106 subelement 005, subfield 005, positions 17-28 | Fleet Item Unit of Measure 5 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

## Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

## Subfield 006 (Non-Fuel Item Purchased 6)

DE 106, subelement 005, subfield 006 (Non-Fuel Item Purchased 6) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 006                                                                                                                                                                                                                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                     |
| Data Field           | Contents of positions 1-40: <ul style="list-style-type: none"><li>• Positions 1-3 (n-3): Fleet Product Code 6</li><li>• Positions 4-15 (n-12): Fleet Item Quantity 6</li><li>• Position 16 (n-1): Fleet Item Quantity Exponent 6</li><li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 6</li><li>• Positions 29-40 (n-12): Fleet Extended Item Amount 6</li></ul> |

| Attribute     | Description                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Justification | Positions 1-3: N/A<br>Positions 4-15: Right-justified with leading zeros<br>Position 16: N/A<br>Positions 17-28: Left-justified, space filled<br>Positions 29-40: Right-justified with leading zeros |

### Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 006, Positions 1-3   | Fleet Product Code 6         |
| DE 106 subelement 005, subfield 006, positions 17-28 | Fleet Item Unit of Measure 6 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

## Subfield 007 (Non-Fuel Item Purchased 7)

DE 106, subelement 005, subfield 007 (Non-Fuel Item Purchased 7) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 003                                                                                                                                                                                                                                                                                                                                                                           |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                   |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                        |
| Data Field           | <p>Contents of positions 1-40:</p> <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Fleet Product Code 7</li> <li>Positions 4-15 (n-12): Fleet Item Quantity 7</li> <li>Position 16 (n-1): Fleet Item Quantity Exponent 7</li> <li>Positions 17-28 (ans-12): Fleet Item Unit of Measure 7</li> <li>Positions 29-40 (n-12): Fleet Extended Item Amount 7</li> </ul> |
| Justification        | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-28: Left-justified, space filled</p> <p>Positions 29-40: Right-justified with leading zeros d</p>                                                                                                                                                 |

### Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 007, positions 1-3   | Fleet Product Code 7         |
| DE 106 subelement 005, subfield 007, positions 17-28 | Fleet Item Unit of Measure 7 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

### Subfield 008 (Non-Fuel Item Purchased 8)

DE 106, subelement 005, subfield 008 (Non-Fuel Item Purchased 8) contains item level detail information of the non-fuel product(s) purchased.

#### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield             | 008         |
| Subfield Data Length | 3 positions |
| Data Representation  | ans-40      |

| Attribute     | Description                                                                                                                                                                                                                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Field    | <p>Contents of positions 1-40:</p> <ul style="list-style-type: none"> <li>• Positions 1-3 (n-3): Fleet Product Code 8</li> <li>• Positions 4-15 (n-12): Fleet Item Quantity 8</li> <li>• Position 16 (n-1): Fleet Item Quantity Exponent 8</li> <li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 8</li> <li>• Positions 29-40 (n-12): Fleet Extended Item Amount 8</li> </ul> |
| Justification | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-28: Left-justified, space filled</p> <p>Positions 29-40: Right-justified with leading zeros</p>                                                                                                                                                             |

## Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 008, positions 1-3   | Fleet Product Code 8         |
| DE 106 subelement 005, subfield 008, positions 17-28 | Fleet Item Unit of Measure 8 |

## Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

## Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

## Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

## Subfield 009 (Non-Fuel Item Purchased 9)

DE 106, subelement 005, subfield 009 (Non-Fuel Item Purchased 9) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 009                                                                                                                                                                                                                                                                                                                                                                        |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                     |
| Data Field           | Contents of positions 1-40: <ul style="list-style-type: none"><li>• Positions 1-3 (n-3): Fleet Product Code 9</li><li>• Positions 4-15 (n-12): Fleet Item Quantity 9</li><li>• Position 16 (n-1): Fleet Item Quantity Exponent 9</li><li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 9</li><li>• Positions 29-40 (n-12): Fleet Extended Item Amount 9</li></ul> |
| Justification        | Positions 1-3: N/A<br>Positions 4-15: Right-justified with leading zeros<br>Position 16: N/A<br>Positions 17-28: Left-justified, space filled<br>Positions 29-40: Right-justified with leading zeros                                                                                                                                                                       |

## Values

| Position                                             | Description                  |
|------------------------------------------------------|------------------------------|
| DE 106 subelement 005, subfield 009, positions 1-3   | Fleet Product Code 9         |
| DE 106 subelement 005, subfield 009, positions 17-28 | Fleet Item Unit of Measure 9 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.



## Subfield 010 (Non-Fuel Item Purchased 10)

DE 106, subelement 005, subfield 010 (Non-Fuel Item Purchased 10) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 010                                                                                                                                                                                                                                                                                                                                                                                |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                        |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                             |
| Data Field           | <p>Contents of positions 1-40:</p> <ul style="list-style-type: none"> <li>Positions 1-3 (n-3): Fleet Product Code 10</li> <li>Positions 4-15 (n-12): Fleet Item Quantity 10</li> <li>Position 16 (n-1): Fleet Item Quantity Exponent 10</li> <li>Positions 17-28 (ans-12): Fleet Item Unit of Measure 10</li> <li>Positions 29-40 (n-12): Fleet Extended Item Amount 10</li> </ul> |
| Justification        | <p>Positions 1-3: N/A</p> <p>Positions 4-15: Right-justified with leading zeros</p> <p>Position 16: N/A</p> <p>Positions 17-28: Left-justified, space filled</p> <p>Positions 29-40: Right-justified with leading zeros</p>                                                                                                                                                        |

### Values

| Position                                             | Description                   |
|------------------------------------------------------|-------------------------------|
| DE 106 subelement 005, subfield 010, positions 1-3   | Fleet Product Code 10         |
| DE 106 subelement 005, subfield 010, positions 17-28 | Fleet Item Unit of Measure 10 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

## Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

## Subfield 011 (Non-Fuel Item Purchased 11)

DE 106, subelement 005, subfield 011 (Non-Fuel Item Purchased 11) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 011                                                                                                                                                                                                                                                                                                                                                                             |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                     |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                          |
| Data Field           | Contents of positions 1-40: <ul style="list-style-type: none"><li>• Positions 1-3 (n-3): Fleet Product Code 11</li><li>• Positions 4-15 (n-12): Fleet Item Quantity 11</li><li>• Position 16 (n-1): Fleet Item Quantity Exponent 11</li><li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 11</li><li>• Positions 29-40 (n-12): Fleet Extended Item Amount 11</li></ul> |

| Attribute     | Description                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Justification | Positions 1-3: N/A<br>Positions 4-15: Right-justified with leading zeros<br>Position 16: N/A<br>Positions 17-28: Left-justified, space filled<br>Positions 29-40: Right-justified with leading zeros |

### Values

| Position                                             | Description                   |
|------------------------------------------------------|-------------------------------|
| DE 106 subelement 005, subfield 011, Positions 1-3   | Fleet Product Code 3          |
| DE 106 subelement 005, subfield 011, positions 17-28 | Fleet Item Unit of Measure 11 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

## Subfield 012 (Non-Fuel Item Purchased 12)

DE 106, subelement 005, subfield 012 (Non-Fuel Item Purchased 12) contains item level detail information of the non-fuel product(s) purchased.

### Attributes

| Attribute            | Description                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subfield             | 012                                                                                                                                                                                                                                                                                                                                                                             |
| Subfield Data Length | 3 positions                                                                                                                                                                                                                                                                                                                                                                     |
| Data Representation  | ans-40                                                                                                                                                                                                                                                                                                                                                                          |
| Data Field           | Contents of positions 1-40: <ul style="list-style-type: none"><li>• Positions 1-3 (n-3): Fleet Product Code 12</li><li>• Positions 4-15 (n-12): Fleet Item Quantity 12</li><li>• Position 16 (n-1): Fleet Item Quantity Exponent 12</li><li>• Positions 17-28 (ans-12): Fleet Item Unit of Measure 12</li><li>• Positions 29-40 (n-12): Fleet Extended Item Amount 12</li></ul> |
| Justification        | Positions 1-3: N/A<br>Positions 4-15: Right-justified with leading zeros<br>Position 16: N/A<br>Positions 17-28: Left-justified, space filled<br>Positions 29-40: Right-justified with leading zeros                                                                                                                                                                            |

### Values

| Position                                           | Description           |
|----------------------------------------------------|-----------------------|
| DE 106 subelement 005, subfield 012, positions 1-3 | Fleet Product Code 12 |

| Position                                             | Description                   |
|------------------------------------------------------|-------------------------------|
| DE 106 subelement 005, subfield 012, positions 17-28 | Fleet Item Unit of Measure 12 |

### Fleet Item Unit of Measure codes

Refer to [Fleet Item Unit of Measure](#) for a list of the Fleet Item Unit of Measure codes and definitions.

### Conexus Payment System product codes

Refer to [Conexus Payment System product codes](#) for a list of product code values is based on the Conexus Payment System product codes.

### Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

### Fleet Item Unit of Measure

Fleet Item Unit of Measure must contain a valid code followed by spaces as defined in the Fleet Item Unit of Measure table.

### Fleet Item Unit of Measure details

| Measure Code | Name | Definition                                  |
|--------------|------|---------------------------------------------|
| ACRE         | Acre | Unit of measure equal to 4840 square yards. |
| ARES         | Are  | Unit of measure equal to 100 square meters. |

| Measure Code | Name             | Definition                                                                         |
|--------------|------------------|------------------------------------------------------------------------------------|
| CELI         | Centiliter       | Unit of volume that is equal to one hundredth of a liter.                          |
| CMET         | Centimeter       | Unit of measure that is equal to one hundredth of a meter.                         |
| EACH         | Each             | One of two or more distinct items.                                                 |
| FOOT         | Foot             | Unit of length equal to 1/3 yard.                                                  |
| GBGA         | GBGallon         | Unit of volume that is equal to 8 pints.                                           |
| GBOU         | GBOunce          | Unit of weight equal to one sixteenth of a pound.                                  |
| GBPI         | GBPint           | Unit of volume that is equal to 568 cubic centimeters.                             |
| GBQA         | GBQuart          | Unit of volume that is equal to 2 pints.                                           |
| GRAM         | Gram             | Unit of measure that is equal to one 1,000th of a kilo.                            |
| HECT         | Hectare          | Unit of measure that is equal to 10,000 square meters.                             |
| INCH         | Inch             | Measure of length equal to 2.54 cm.                                                |
| KILO         | Kilogram         | Basic unit of mass in the International System of Units (SI) equal to 1000 grams.  |
| KMET         | Kilometer        | Unit of measure that is equal to 1000 meters.                                      |
| LITR         | Liter            | Unit of volume that is equal to one thousand cubic centimeters.                    |
| METR         | Meter            | Unit of length in the metric system equal to 39.37 inches.                         |
| MILE         | Mile             | Unit of length equal to 1760 yards.                                                |
| MILI         | MilliLiter       | Unit of volume that is equal to one thousandth of a liter.                         |
| MMET         | Millimeter       | Unit of measure that is equal to one thousandth of one meter.                      |
| PIEC         | Piece            | Standard length of cloth, wallpaper, as an item for sale or amount of a substance. |
| PUND         | Pound            | Unit of weight equal to 0.454 kilograms.                                           |
| SCMT         | SquareCentimeter | Measure of a surface one centimeter by one centimeter.                             |
| SMET         | SquareMeter      | Measure of a surface one meter by one meter.                                       |

| Measure Code | Name             | Definition                                             |
|--------------|------------------|--------------------------------------------------------|
| SMIL         | SquareMillimeter | Measure of a surface one millimeter by one millimeter. |
| SQFO         | SquareFoot       | Measure of a surface one foot by one foot.             |
| SQIN         | SquareInch       | Measure of a surface one inch by one inch.             |
| SQKI         | SquareKilometer  | Measure of a surface one kilometer by one kilometer.   |
| SQMI         | SquareMile       | Measure of a surface one mile by one mile.             |
| SQYA         | SquareYard       | Measure of a surface one yard by one yard.             |
| TONS         | Ton              | Measure of weight, in Britain 2000lb (short ton).      |
| USGA         | USGallon         | Unit of volume that is equal to 8 pints.               |
| USOU         | USOunce          | Unit of weight equal to one sixteenth of a pound.      |
| USPI         | USPint           | Unit of volume that is equal to 473 cubic centimeters. |
| USQA         | USQuart          | Unit of volume that is equal to 2 pints.               |
| YARD         | Yard             | Unit of length equal to 3 feet or 0.9144 meter.        |

## Conexus Payment System product codes

The following list of product code values is based on the Conexus Payment System product codes version 4.1.

### Product codes

| Non-fuel product code | Description                    |
|-----------------------|--------------------------------|
| 100                   | General Automotive Merchandise |
| 101                   | Motor Oil                      |
| 102                   | Car Wash                       |
| 103                   | Oil Change                     |
| 104                   | Oil Filter                     |
| 105                   | Work Order                     |
| 106                   | Anti-Freeze                    |
| 107                   | Washer Fluid                   |
| 108                   | Brake Fluid                    |

| Non-fuel product code | Description                                 |
|-----------------------|---------------------------------------------|
| 109                   | Tires                                       |
| 110                   | Federal Excise Tax (Tires)                  |
| 111                   | Tire Rotation                               |
| 112                   | Batteries                                   |
| 113                   | Lube                                        |
| 114                   | Inspection                                  |
| 115                   | Labor                                       |
| 116                   | Towing                                      |
| 117                   | Road Service                                |
| 118                   | Vehicle Accessories                         |
| 119                   | Vehicle Parts                               |
| 120                   | Preventative Maintenance                    |
| 121                   | Air Conditioning Service                    |
| 122                   | Engine Service                              |
| 123                   | Transmission Service                        |
| 124                   | Brake Service                               |
| 125                   | Exhaust Service                             |
| 126                   | Body Work                                   |
| 127                   | Vehicle Glass                               |
| 128                   | Synthetic Oil                               |
| 129                   | Lamps                                       |
| 130                   | Wipers                                      |
| 131                   | Hoses                                       |
| 132                   | Tire-related (Wheel Balance, Valve Stem)    |
| 133                   | Repairs                                     |
| 134                   | Service Package                             |
| 135                   | Vehicle Parking                             |
| 136                   | Truck Tank Cleaning                         |
| 137                   | Other Lubricants                            |
| 138                   | Vehicle Fuel Additives/Treatment (Injected) |
| 139                   | Vehicle Rental                              |



| Non-fuel product code | Description                               |
|-----------------------|-------------------------------------------|
| 140                   | Air Filter                                |
| 141                   | Vehicle Prep                              |
| 142                   | Fuel System                               |
| 149                   | Miscellaneous Vehicle Products / Services |
| 175                   | Storage                                   |
| 176                   | Aircraft Ground Handling                  |
| 177                   | Aircraft Ground Power Unit                |
| 178                   | Aircraft Labor                            |
| 179                   | Aircraft Work Order                       |
| 180                   | Aircraft Maintenance                      |
| 181                   | Aircraft Service                          |
| 182                   | Transportation                            |
| 183                   | De-icing                                  |
| 184                   | Ramp Fees                                 |
| 185                   | Catering                                  |
| 186                   | Hangar Fee                                |
| 187                   | Landing Fee                               |
| 188                   | Call Out Fee                              |
| 189                   | Aircraft Rental                           |
| 190                   | Instruction Fee                           |
| 191                   | Flight Plans / Weather Brief              |
| 192                   | Charter Fee                               |
| 193                   | Communication Fee                         |
| 194                   | Aircraft Cleaning                         |
| 195                   | Cargo Handling                            |
| 196                   | Aircraft Accessories                      |
| 197                   | Pilot Supplies                            |
| 198                   | Aircraft Parking Fees                     |
| 199                   | Aircraft Tie Down Fees                    |
| 200                   | Aircraft Sanitation Fees                  |
| 201                   | Aircraft Fuel Additive                    |

| Non-fuel product code | Description                              |
|-----------------------|------------------------------------------|
| 202                   | AC Parts                                 |
| 203                   | Oxygen                                   |
| 204                   | De-fuel                                  |
| 205                   | Re-service                               |
| 206                   | Static Dissipater Additive               |
| 207                   | Corrosion Inhibitor                      |
| 208                   | Airport Fees                             |
| 209                   | Overtime Fees                            |
| 210                   | IT/Bladder                               |
| 211                   | Ground Equipment Service Fees            |
| 212                   | Secure Fees                              |
| 213                   | Flow Fee                                 |
| 224                   | Miscellaneous Aviation Products/Services |
| 250                   | Marine Service                           |
| 251                   | Marine Labor                             |
| 252                   | Marine Work Order                        |
| 253                   | Launch Fee                               |
| 254                   | Slip Rental                              |
| 299                   | Miscellaneous Marine Products/Services   |
| 400                   | General Merchandise                      |
| 401                   | General Ice                              |
| 410                   | General Tobacco                          |
| 411                   | Cigarettes                               |
| 412                   | Tobacco - Other                          |
| 420                   | General Packaged Beverage                |
| 421                   | Packaged Beverages (non-alcoholic)       |
| 422                   | Packaged Juice                           |
| 423                   | Other Packaged Beverages                 |
| 430                   | General Dispensed Beverage               |
| 431                   | Hot Dispensed Beverages                  |
| 432                   | Cold Dispensed Beverages                 |

| Non-fuel product code | Description                       |
|-----------------------|-----------------------------------|
| 433                   | Frozen Dispensed Beverages        |
| 434                   | Other Dispensed Beverages         |
| 440                   | General Snacks                    |
| 441                   | Salty Snacks                      |
| 442                   | Alternative Snacks                |
| 443                   | Sweet Snacks - Packaged           |
| 450                   | General Candy                     |
| 460                   | General Dairy                     |
| 461                   | Fluid Milk Products               |
| 462                   | Packaged Ice Cream/Novelties      |
| 463                   | Other Dairy                       |
| 470                   | General Grocery                   |
| 471                   | Groceries - Edible                |
| 472                   | Groceries - Non-Edible            |
| 473                   | Groceries - Perishable            |
| 474                   | Bread - Packaged                  |
| 475                   | Frozen Foods                      |
| 480                   | General Alcohol                   |
| 481                   | Beer - Alcoholic                  |
| 482                   | Beer - Non-Alcoholic              |
| 483                   | Wine                              |
| 484                   | Liquor                            |
| 490                   | General Deli                      |
| 491                   | Packaged Sandwiches/Deli Products |
| 492                   | Prepared Foods                    |
| 493                   | Deli Items                        |
| 500                   | General Food service              |
| 510                   | General Lottery                   |
| 511                   | Lottery - Instant                 |
| 512                   | Lottery - Online                  |
| 513                   | Lottery - Other                   |

| Non-fuel product code | Description                         |
|-----------------------|-------------------------------------|
| 520                   | General Money Order                 |
| 521                   | Money Order - Vendor Payment        |
| 522                   | Money Order - Payroll Check         |
| 523                   | Money Order - Gift Certificate      |
| 524                   | Money Order - Refund Check          |
| 525                   | Money Order - Official Check        |
| 526                   | Money Order - Rebate Check          |
| 527                   | Money Order - Dividend Check        |
| 528                   | Money Order - Utility Check         |
| 530                   | General Store Service               |
| 531                   | Home Delivery                       |
| 532                   | Prepaid Cards - Purchase            |
| 533                   | Prepaid Cards - Activation/Recharge |
| 534                   | Membership/Loyalty                  |
| 540                   | General Health & Beauty Care        |
| 550                   | General Publications                |
| 560                   | PIN Activate Prepaid Card           |
| 561                   | PIN Return Prepaid Card             |
| 562                   | Enable Device/Handset Unlock        |
| 563                   | Disable Device/Handset Lock         |
| 564                   | 3rd Party Prepaid Card Activate     |
| 565                   | 3rd Party Prepaid Card Reload       |
| 566                   | Financial Prepaid Card Activate     |
| 567                   | Financial Prepaid Card Reload       |
| 568                   | Proprietary Prepaid Card Activate   |
| 569                   | Proprietary Prepaid Card Reload     |
| 570                   | General Purpose Activate            |
| 571                   | General Purpose Reload              |
| 572                   | Real Time Recharge                  |
| 573                   | Wireless Real Time Recharge         |
| 574                   | Single Payee Bill Pay               |

| Non-fuel product code | Description                         |
|-----------------------|-------------------------------------|
| 575                   | Multiple Payee Bill Pay             |
| 600                   | Packaged DEF (Diesel Exhaust Fluid) |
| 601                   | Packaged B99                        |
| 602                   | Packaged B100                       |
| 603                   | Packaged Additive                   |
| 604                   | Packaged Kerosene                   |
| 605                   | Packaged Propane                    |
| 624                   | Miscellaneous Packaged Fuels        |
| 650                   | Scales                              |
| 651                   | Shower                              |
| 652                   | Tire Repair                         |
| 653                   | Lodging                             |
| 654                   | Wash Out                            |
| 655                   | Trailer Wash                        |
| 656                   | RV Dump Fee                         |
| 657                   | EV Charging Fee                     |
| 658                   | EV Battery Exchanges                |
| 659                   | Toll Payments                       |
| 700                   | Cannabidiol (CBD)                   |
| 900                   | Discount 1                          |
| 901                   | Discount 2                          |
| 902                   | Discount 3                          |
| 903                   | Discount 4                          |
| 904                   | Discount 5                          |
| 905                   | Coupon 1                            |
| 906                   | Coupon 2                            |
| 907                   | Coupon 3                            |
| 908                   | Coupon 4                            |
| 909                   | Coupon 5                            |
| 910                   | Lottery Pay Out - Instant           |
| 911                   | Lottery Pay Out - Online            |

| Non-fuel product code | Description                           |
|-----------------------|---------------------------------------|
| 912                   | Lottery Pay Out - Other               |
| 913                   | Split Tender                          |
| 914                   | Tax Discount/Forgiven                 |
| 915                   | Local Discount 1                      |
| 916                   | Local Discount 2                      |
| 917                   | Local Discount 3                      |
| 918                   | Local Discount 4                      |
| 919                   | Local Discount 5                      |
| 920                   | POS/Loyalty Reserved Discount 1       |
| 921                   | POS/Loyalty Reserved Discount 2       |
| 922                   | POS/Loyalty Reserved Discount 3       |
| 923                   | POS/Loyalty Reserved Discount 4       |
| 924                   | POS/Loyalty Reserved Discount 5       |
| 949                   | Miscellaneous Negative Administrative |
| 950                   | Tax 1                                 |
| 951                   | Tax 2                                 |
| 952                   | Tax 3                                 |
| 953                   | Tax 4                                 |
| 954                   | Tax 5                                 |
| 955                   | Cash Back                             |
| 956                   | Cash Back Fee                         |
| 957                   | Fee 1                                 |
| 958                   | Fee 2                                 |
| 959                   | Fee 3                                 |
| 960                   | Fee 4                                 |
| 961                   | Fee 5                                 |
| 962                   | Miscellaneous Aviation Tax            |
| 963                   | GST/HST (Canadian)/VAT 1              |
| 964                   | PST/QST (Canadian) VAT 2              |
| 965                   | SWT Rate (Canadian)                   |
| 966                   | Tax 6                                 |

| Non-fuel product code | Description                  |
|-----------------------|------------------------------|
| 967                   | Tax 7                        |
| 968                   | Tax 8                        |
| 969                   | Jet Federal Excise Tax       |
| 970                   | AvGas Federal Excise Tax     |
| 971                   | Charity                      |
| 972                   | Gratuity                     |
| 999                   | Miscellaneous Administrative |

## DE 107 (Reserved for Mastercard Use)

DE 107 (Reserved for Mastercard Use) is reserved for Mastercard use.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 108 (Additional Transaction Reference Data)

DE 108 (Additional Transaction Reference Data) provides the capability for the acquirers to send sender, receiver, and transaction level data to the issuer in Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions), Mastercard Merchant Presented QR Transactions, and Visa Account Funding Transactions.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | ans...999; LLLVAR       |
| Data Length         | 3 positions             |
| Data Field          | Contents of subelements |
| Subelements         | 11                      |
| Justification       | See "Subelements"       |

## Usage: Mastercard Send Transactions

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

**NOTE: Refer to the *Mastercard Send and Funding Transactions Program Standards* for the full list of mandatory subelements in DE 108.**

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | P   | C   |
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | O   | P   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | X   | C   |
| Reversal Request/0400                                                                           | O   | P   | C   |
| Reversal Request Response/0410                                                                  | C   | X   | C   |

### Application notes For Mastercard Send Transactions

For Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, acquirers that process transactions must send and receive DE 108 (Additional Transaction Reference Data). Issuers must receive and return DE 108 (Additional Transaction Reference Data) for all Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions.

For Funding Transactions, acquirers that process transactions must send and be prepared to receive if sending DE 108 (Additional Transaction Reference Data) for all Funding Transactions. Issuers must receive and return DE 108 (Additional Transaction Reference Data) when submitted by the acquirer on Funding Transactions.

Mastercard Send Transactions, with the exception Mastercard Merchant Presented QR transactions, will not contain the following DE 108 subelements and subfields:

- Subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number)
- Subelement 05, subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)
- Subelement 06 (QR Dynamic Code Data)



### **Application notes For Mastercard Merchant Presented QR Transactions**

For Mastercard Merchant Presented QR Funding Transactions, acquirers that process transactions must send DE 108 (Additional Transaction Reference Data). Issuers must receive DE 108 (Additional Transaction Reference Data) for all Mastercard Merchant Presented QR Funding Transactions.

Mastercard Merchant Presented QR Payment Transactions may contain the following DE 108 subelements and subfields.

- Subelement 02 (Sender Data), subfield 11 (Sender Account Number)
- Subelement 03 (Transaction Reference Data), subfield 01 (Unique Transaction Reference Number)
- Subelement 03, subfield 02 (Additional Message) (Payment Transactions only)
- Subelement 03, subfield 03 (Funding Source)
- Subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number)
- Subelement 05, subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)
- Subelement 06 (QR Dynamic Code Data)

Mastercard Merchant Presented QR Payment Transactions will not return DE 108 to the acquirer in any response message.

Customers should be aware that QR data for Mastercard Merchant Presented QR Payment Transactions, may be received in either DE 108 subelement 03, subfield 02 or DE 108, subelement 06. Subelement 06 is considered the primary location and should be considered first as the location for QR Data.

### **Application notes For Mastercard Send and Funding Transactions**

Mastercard Send Payment and Funding Transactions will not contain the following DE 108 subelements and subfields:

- Subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number)
- Subelement 05, subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)
- Subelement 06 (QR Dynamic Code Data)

### **Application notes for Account Status Inquiry Transactions related to Mastercard Send and Funding Transactions**

For Account Status Inquiry Transactions, acquirers may send and be prepared to receive DE 108 (Additional Transaction Reference Data). Issuers must be prepared to receive and return DE 108 (Additional Transaction Reference Data) with Account Status Inquiry Transactions.

Acquirers that are requesting verification of receiver/recipient or sender data will send DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 01 (Receiver/Recipient Data) or Subelement 02 (Sender Data)
- Subelement 07 (Data Verification Request)

Issuers must receive DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 01 (Receiver/Recipient Data) or Subelement 02 (Sender Data)
- Subelement 07 (Data Verification Request)

Issuers must return DE 108 (Additional Transaction Reference Data) with subelement 08 (Data Verification Response).

**Application notes for Non-Financial Authorization Transactions**

Acquirers and originators requesting validation of account owner information (Name Validation Service) will send DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 01 (Receiver/Recipient Data) or subelement 02 (Sender Data)
- Subelement 07 (Data Validation Request)

Issuers must respond with DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 08 (Data Validation Response)

Or, when an issuer participates in the on-behalf Name Match Service the response will include subelements:

- Subelement 09 (Account Information 1)
- Subelement 10 (Account Information 2)
- Subelement 11 (Account Information 3)

Issuers must be enabled for the on-behalf Name Match Service to be able to return DE 108 (Additional Transaction Reference Data) with Subelement 09 (Account Information 1), subelement 10 (Account Information 2), and subelement 11 (Account Information 3).

**DE 108 encoding scheme**

Mastercard organizes DE 108 into a group of encoded subelements and subfields. The following tables illustrate the structure of DE 108 subelements.

**Structure of DE 108 subelements**

| LLL          | VAR (maximum length varies) |                   |                                 |                        |                   |                                 |
|--------------|-----------------------------|-------------------|---------------------------------|------------------------|-------------------|---------------------------------|
| 3 bytes      | 2 bytes                     | 3 bytes           | Variable length                 | 2 bytes                | 3 bytes           | Variable length                 |
| Total length | First subelement data       |                   |                                 | Second subelement data |                   |                                 |
|              | Subelement ID               | Subelement length | Subelement variable length data | Subelement ID          | Subelement length | Subelement variable length data |

### Attributes of DE 108 subelements

| Number of bytes | Attribute                       | Value                                 |
|-----------------|---------------------------------|---------------------------------------|
| 3               | Total data element length       | The "LLL" portion of the data element |
| 2               | Subelement ID                   | In the range 01-99                    |
| 3               | Subelement length               | Variable                              |
| 1...nnn         | Subelement variable length data | Contents of subfields                 |

## DE 108 subelement encoding scheme

The following tables illustrate the structure of DE 108 subfields within a subelement.

### Structure of DE 108 subfields within a subelement

| Subelement VAR (maximum length varies) |                 |                               |                      |                 |                               |
|----------------------------------------|-----------------|-------------------------------|----------------------|-----------------|-------------------------------|
| 2 bytes                                | 2 bytes         | Variable length               | 2 bytes              | 2 bytes         | Variable length               |
| First subfield data                    |                 |                               | Second subfield data |                 |                               |
| Subfield ID                            | Subfield length | Subfield variable length data | Subfield ID          | Subfield length | Subfield variable length data |

### Attributes of DE 108 subfields within a subelement

| Number of bytes | Attribute                     | Value                 |
|-----------------|-------------------------------|-----------------------|
| 2               | Subfield ID                   | In the range 01-99    |
| 2               | Subfield length               | Variable              |
| 1...nn          | Subfield variable length data | Contents of subfields |

## DE 108 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edit to verify proper formatting of DE 108.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or Reversal Request/0400 message contains DE 108 (Additional Transaction Reference Data) with subelements that have incorrect length and/or incorrect format (Data Representations), or a subelement ID appears more than once within the same data element | forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 44 is 6 positions for subelement format errors: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Message format error)</li><li>• DE 44 (Additional Response Data) = 1080nn (where nn is the subelement number)</li></ul> |

### Examples

- When an edit error occurs on DE 108, the DE 44 data will be populated as these examples below:
  - Error on DE 108 subelement 2, subfield 1: 108002 (no subfield information is provided)

For non-DE 108 format errors, DE 44 will only have DE info and no subelement information. For example, the format error in DE 22 will have DE 44 as 022.

## Subelement 01 (Receiver/Recipient Data)

DE 108 (Additional Transaction Reference Data), subelement 01 (Receiver/Recipient Data) contains the receiver or recipient name, address, phone number, date of birth, and account details.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 01                    |
| Subelement Data Length | 3 positions           |
| Data Representation    | ans...669; LLLVAR     |
| Data Field             | Contents of subfields |

| Attribute | Description                                                        |
|-----------|--------------------------------------------------------------------|
| Subfields | Subfield 01 (Receiver/Recipient First Name)                        |
|           | Subfield 02 (Receiver/Recipient Middle Name)                       |
|           | Subfield 03 (Receiver/Recipient Last Name)                         |
|           | Subfield 04 (Receiver/Recipient Street Address)                    |
|           | Subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only]) |
|           | Subfield 05 (Receiver/Recipient City)                              |
|           | Subfield 06 (Receiver/Recipient State/Province Code)               |
|           | Subfield 07 (Receiver/Recipient Country)                           |
|           | Subfield 08 (Receiver/Recipient Postal Code)                       |
|           | Subfield 08 (Receiver/Recipient Postal Code [Visa Only])           |
|           | Subfield 09 (Receiver/Recipient Phone Number)                      |
|           | Subfield 10 (Receiver/Recipient Date of Birth)                     |
|           | Subfield 11 (Receiver/Recipient Account Number)                    |
|           | Subfield 12 (Receiver/Recipient Identification Type)               |
|           | Subfield 13 (Receiver/Recipient Identification Number)             |
|           | Subfield 13 (Receiver/Recipient Identification Number [Visa Only]) |
|           | Subfield 14 (Receiver/Recipient Identification Country Code)       |
|           | Subfield 15 (Receiver/Recipient Identification Expiration Date)    |
|           | Subfield 16 (Receiver/Recipient Nationality)                       |
|           | Subfield 17 (Receiver/Recipient Country of Birth)                  |
|           | Subfield 18 (Receiver/Recipient Account Number Type)               |
|           | Subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only]) |
|           | Subfield 20 (Receiver/Recipient Street Name [Visa Only])           |
|           | Subfield 21 (Receiver/Recipient Building Number [Visa Only])       |

| Attribute     | Description                                                                  |
|---------------|------------------------------------------------------------------------------|
|               | Subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only]) |
| Justification | See subfields                                                                |

## DE 108, subelement 01 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

### DE 108, subelement 01, subfield 01 (Receiver/Recipient First Name) and subfield 03 (Receiver/Recipient Last Name): Edit check

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Then the Dual Message Authorization System...                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>DE 3, subfield 1 contains value 28</li> <li>DE 18 contains MCC values associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>DE 48, subelement 77 contains the Transaction Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>DE 61, subfield 10 criteria are met on Moneysend Payment transaction</li> <li>DE 61, subfield 13 (POS Country Code), and the licensed country of DE 2 (Primary Account Number) do not match (indicating a cross-border transaction)</li> <li>DE 108, subelement 01, subfield 01 (Receiver First Name) and DE 108, subelement 01, subfield 03 (Receiver Last Name) are present and contain all spaces or all numeric values</li> </ul> | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Then the Dual Message Authorization System...                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC values associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transaction Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Mastercard Send Payment Transaction: <ul style="list-style-type: none"> <li>– 0 = Not a CAT transaction</li> <li>– 1 = Authorized Level 1 CAT: Automated dispensing machine with PIN</li> <li>– 2 = Authorized Level 2 CAT: Self-service terminal</li> <li>– 6 = Authorized Level 6 CAT: Electronic commerce</li> </ul> </li> <li>• DE 108 is present and all criteria are met except: <ul style="list-style-type: none"> <li>– Subelement 01 (Receiver/Recipient Data) is present and subfield 01 (Receiver First Name), OR subfield 03 (Receiver Last Name) is absent, contains all numeric, all spaces, or any combination of spaces and numbers</li> </ul> </li> </ul> | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)</li> </ul> |

**DE 108, subelement 01, subfield 06 (Receiver/Recipient State/Province Code): Edit check**

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Then the Dual Message Authorization System...                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC values associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transaction Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li> <li>• DE 108, subelement 01, subfield 07 (Receiver/Recipient Country) is U.S. or Canada</li> <li>• DE 108, subelement 01, subfield 06 (Receiver State Code) is present and contains all zeros or spaces or invalid state code for U.S. or invalid province code for Canada (Refer to the Quick Reference Booklet for valid Alpha [2 Characters] State/Province code)</li> </ul> | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)</li> </ul> |



### DE 108, Subelement 01, Subfield 07 (Receiver/Recipient Country Code): Edit check

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Then the Dual Message Authorization System...                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transaction Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li> <li>• DE 108, subelement 01, subfield 07 (Receiver/Recipient Country) is present and contains all spaces</li> </ul> | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)</li> </ul> |
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains a Transaction Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li> <li>• DE 108, subelement 01, subfield 07 (Receiver/Recipient Country) is present</li> </ul>                           | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)</li> <li>• DE 108 = 4 digits</li> </ul>   |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transaction Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li> <li>• DE 108, subelement 01, subfield 07 (Receiver/ Recipient Country) is present and is a valid country code but blocked (Part of the Mastercard Send Blocking Country List)</li> </ul> | <p>sends an Authorization Request Response/0110 message to the acquirer where DE 39 = 12 (Invalid Transaction) and sends an Authorization Advice Message/0120 to the issuer where:</p> <ul style="list-style-type: none"> <li>• DE 48, subelement 71, subfield 1, value 33 (Send Blocking Service)</li> <li>• DE 48, subelement 71, subfield 2, value W (Send Mastercard Blocking—Country not allowed for the Mastercard Send Transaction, Sender/Receiver Data)</li> <li>• DE 60, subfield 1, value 200 (In Control Processing Advice to Issuer)</li> <li>• DE 60, subfield 2, value 0072 (Reject: Geographical Restrictions)</li> </ul> |
| <p><b>NOTE: Transaction will be declined if the sender Country is subject to comprehensive geographic sanctions published by the Office of Foreign Assets Control (OFAC) <a href="http://www.treasury.gov/resource-center/sanctions/Pages/default.aspx">http://www.treasury.gov/resource-center/sanctions/Pages/default.aspx</a>. The current list of countries subject to such sanctions is Cuba, Iran, North Korea, Sudan, and Syria; however, this list is subject to change.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**DE 108, subelement 01, subfield 12 (Receiver Identification Type): Edit check**

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Then the Dual Message Authorization System...                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where: <ul style="list-style-type: none"><li>DE 3, subfield 1 contains value 28</li><li>DE 18 contains MCC values associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>DE 48, subelement 77 contains the Transaction Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li><li>DE 108 subelement 01, subfield 12 (Receiver Identification Type) is present and does not contain one of the values 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10</li></ul> | sends an Authorization Request Response/0110 message to the acquirer where: <ul style="list-style-type: none"><li>DE 39 (Response Code) = 30 (Format error)</li><li>DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)</li></ul> |

**Subfield 01 (Receiver/Recipient First Name)**

DE 108, subelement 01, subfield 01 (Receiver/Recipient First Name) contains the Receiver/Recipient first name.

**Attributes**

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...35; LLVAR         |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

## Values

If present, cannot contain all spaces or all numeric values.

## Application notes

Subfield 01 must be present for cross-border Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, and must be properly formatted.

Subfield 01 is optional for domestic Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, and Funding transactions. Mastercard network doesn't enforce system edits on this field and it is optional. However per program rules noted in the [Mastercard Send and Funding Transactions Program Standards](#) guide, it's a required field for domestic and cross-border Funding Transactions and must be provided. Not providing required data in this field may result in compliance assessments.

The first name (Consumer/Business) of the recipient/receiver is included in this subelement.

The table that follows describes the correct submission of business names.

## Application notes: Mastercard Merchant Presented QR transactions

Subfield 01 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forward to destination.

Subfield 01 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

## Name example

| Business name     | First name        | Last name         |
|-------------------|-------------------|-------------------|
| XYZ               | XYZ               | XYZ               |
| XYZ International | XYZ International | XYZ International |
| XYZ DBA MA        | XYZ DBA MA        | XYZ DBA MA        |

## Subfield 02 (Receiver/Recipient Middle Name)

DE 108, subelement 01, subfield 02 (Receiver/Recipient Middle Name) contains the middle name of the Recipient/Receiver.

## Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 02          |

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-1                   |
| Data Field           | Contents of subfield 02 |
| Justification        | N/A                     |

### Values

Valid value will consist of the middle name initial of the Receiver.

### Application notes

Subfield 02 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions) according to the published standards. Refer to the *Mastercard Send and Funding Transactions Program Standards* for the full listing of required fields and subelements.

## Subfield 03 (Receiver/Recipient Last Name)

DE 108, subelement 01, subfield 03 (Receiver/Recipient Last Name) contains the last name of the Recipient/Receiver.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...35; LLVAR         |
| Data Field           | Contents of subfield 03 |
| Justification        | N/A                     |

### Values

If present cannot contain all spaces or all numeric values.

### Application notes

**NOTE:** Refer to *Mastercard Send and Funding Transactions Program Standards* for the full listing of required fields and subelements for both funding and payment transactions.

Subfield 03 must be present for cross-border Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, and must be properly formatted.

Subfield 01 is optional for domestic Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, and Funding transactions. Mastercard network doesn't enforce system edits on this field and it is optional. However per program rules noted in the *Mastercard Send and Funding Transactions Program Standards* guide its a required field for domestic and cross-border Funding Transactions and must be provided. Not providing required data in this field may result in compliance assessments.

The last name (Consumer/Business) of the recipient/receiver is included in this subelement.

The table that follows describes the correct submission of business names.

#### Application notes: Mastercard Merchant Presented QR transactions

Subfield 03 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 03 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

#### Name example

| Business name     | First name        | Last name         |
|-------------------|-------------------|-------------------|
| XYZ               | XYZ               | XYZ               |
| XYZ International | XYZ International | XYZ International |
| XYZ DBA MA        | XYZ DBA MA        | XYZ DBA MA        |

### Subfield 04 (Receiver/Recipient Street Address)

DE 108, subelement 01, subfield 04 (Receiver/Recipient Street Address) contains the Street Address of the Recipient/Receiver.

#### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Subfield ID         | 04                      |
| Subfield length     | 2                       |
| Data representation | ans...50; LLVAR         |
| Data field          | Contents of subfield 04 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Valid value will consist of the street address of the Receiver.

### Application notes

Subfield 04 is optional for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only])

DE 108, subelement 01, subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only]) contains the street address of the receiver or recipient.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 04                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...99; LLVAR         |
| Data Field           | Contents of subfield 04 |
| Justification        | N/A                     |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 05 (Receiver/Recipient City)

DE 108, subelement 01, subfield 05 (Receiver/Recipient City) contains the city of the Receiver/Recipient.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 05                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...25; LLVAR         |
| Data Field           | Contents of subfield 05 |
| Justification        | N/A                     |

### Values

Valid location city name of the Receiver/Recipient.

### Application notes

Subfield 05 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

### Application notes: Mastercard Merchant Presented QR transactions

Subfield 05 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 05 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

## Subfield 06 (Receiver/Recipient State/Province Code)

DE 108, subelement 01, subfield 06 (Receiver/Recipient State/Province Code) contains the state/province code of the Recipient/Receiver.

### Attributes

| Attribute            | Description    |
|----------------------|----------------|
| Subfield ID          | 06             |
| Subfield Data Length | 2 positions    |
| Data Representation  | ans...3; LLVAR |



| Attribute     | Description             |
|---------------|-------------------------|
| Data Field    | Contents of subfield 06 |
| Justification | N/A                     |

### Values

Valid location state code of the Receiver/Recipient.

### Application notes

Subfield is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions). If submitted it cannot contain spaces or invalid codes when country is U.S. or Canada.

**NOTE: Data representation must be alpha (a-3) for Mastercard Send Transactions. For US and Canada, Data representation must be (a-2) for State/Province. Refer to the *Quick Reference Booklet* for valid list of State/Province codes.**

## Subfield 07 (Receiver/Recipient Country)

DE 108, subelement 01, subfield 07 (Receiver/Recipient Country) contains the country of the Recipient/Receiver.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 07                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | a-3                     |
| Data Field           | Contents of subfield 07 |
| Justification        | N/A                     |

### Values

Receiver/Recipient Country must contain a valid three-character alphabetic code and must not be part of a blocked country list. Refer to the *Quick Reference Booklet* for valid list of country codes.

### Application notes

Subfield 07 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

Transaction will be declined if the sender country is subject to comprehensive geographic sanctions published by the Office of Foreign Assets Control (OFAC), <https://ofac.treasury.gov/sanctions-programs-and-country-information>. The current list of countries subject to such sanctions is subject to change.

## Subfield 08 (Receiver/Recipient Postal Code)

DE 108, subelement 01, subfield 08 (Receiver/Recipient Postal Code) contains the postal code of the Recipient/Receiver.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Subfield ID         | 08                      |
| Subfield length     | 2                       |
| Data representation | ans...10; LLVAR         |
| Data field          | Contents of subfield 08 |
| Justification       | N/A                     |

### Values

Valid location Postal Code of the Recipient's/Receiver.

### Application notes

Subfield 08 is optional for Mastercard Send Transactions (Funding Transactions, MoneySend Payment Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 08 (Receiver/Recipient Postal Code [Visa Only])

DE 108, subelement 01, subfield 08 (Receiver/Recipient Postal Code [Visa Only]) contains the postal code of the receiver or recipient.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 08                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...16; LLVAR         |
| Data Field           | Contents of subfield 08 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 08 (Receiver/Recipient Postal Code [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 09 (Receiver/Recipient Phone Number)

DE 108, subelement 01, subfield 09 (Receiver/Recipient Phone Number) contains the phone number of the Recipient/Receiver.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 09                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...20; LLVAR         |
| Data Field           | Contents of subfield 09 |
| Justification        | N/A                     |

### Values

Valid phone number of the Recipient/Receiver.

### Application notes

Subfield 09 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

### Application notes: Mastercard Merchant Presented QR transactions

Subfield 09 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 09 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

## Subfield 10 (Receiver/Recipient Date of Birth)

DE 108, subelement 01, subfield 10 (Receiver/Recipient Date of Birth) contains the date of birth of the Receiver/Recipient.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 10                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-8                     |
| Data Field           | Contents of subfield 10 |
| Justification        | N/A                     |

### Values

Valid date of birth of the Receiver/Recipient in the format MMDDYYYY.

### Application notes

Subfield 10 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

#### Application notes: Mastercard Merchant Presented QR transactions

Subfield 10 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 10 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

## Subfield 11 (Receiver/Recipient Account Number)

DE 108, subelement 01, subfield 11 (Receiver/Recipient Account Number) contains the account number of the Recipient/Receiver.

### Attributes

| Attribute            | Description     |
|----------------------|-----------------|
| Subfield ID          | 11              |
| Subfield Data Length | 2 positions     |
| Data Representation  | ans...50; LLVAR |

| Attribute     | Description             |
|---------------|-------------------------|
| Data Field    | Contents of subfield 11 |
| Justification | N/A                     |

### Values

Valid Account Number of the Receiver/Recipient.

### Application notes

Subfield 11 (Receiver/Recipient Account Number) has a maximum length of 34 characters on Visa-branded transactions.

Subfield 11 is optional for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Mastercard Network does not enforce system edits on this field and hence is optional per specifications. However per program rules noted in the *Mastercard MoneySend and Funding Transactions Program Standards* guide, it is a required field for domestic and cross-border Funding Transactions and must be provided. Not providing required data in this field as per standards may result in compliance assessments.

### Application notes: Mastercard Merchant Presented QR transactions

Subfield 11 is mandatory for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination. If Mastercard Merchant Presented QR transactions are processed with data in DE 108, subelement 01, subfield 11 (Receiver/Recipient Account Number) exceeding the current supported length, Mastercard Merchant Presented QR edits will decline the transaction.

Subfield 11 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

## Subfield 12 (Receiver/Recipient Identification Type)

DE 108, subelement 01, subfield 12 (Receiver/Recipient Identification Type) contains the identification type of the Recipient/Receiver.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 12          |
| Subfield Data Length | 2 positions |
| Data Representation  | n-2         |

| Attribute     | Description             |
|---------------|-------------------------|
| Data Field    | Contents of subfield 12 |
| Justification | N/A                     |

## Values

If present on Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, Visa-branded funding transactions, it must contain one of the following valid values:

## Values

| Value | Description                                       |
|-------|---------------------------------------------------|
| 00    | Passport                                          |
| 01    | National Identification Card                      |
| 02    | Driver's License                                  |
| 03    | Government Issued                                 |
| 04    | Other                                             |
| 05    | Date of birth (Visa only)                         |
| 06    | Customer identification (unspecified) (Visa only) |
| 07    | Business tax ID (Visa only)                       |
| 08    | Individual tax ID (Visa only)                     |
| 09    | Company registration number (Visa only)           |
| 10    | Proxy identification (Visa only)                  |
| 11    | Social security number (Visa only)                |
| 12    | Alien registration number (Visa only)             |
| 13    | Law enforcement identification (Visa only)        |
| 14    | Military identification (Visa only)               |
| 15    | Travel identification (nonpassport) (Visa only)   |
| 16    | Email (Visa only)                                 |
| 17    | Phone number (Visa only)                          |

### Application notes

Values 03 and 04 have no corresponding values in Visa format. Acquirers should not send either value 03 or 04 with Visa-branded transactions.

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details on Visa only values.

### Subfield 13 (Receiver/Recipient Identification Number)

DE 108, subelement 01, subfield 13 (Receiver/Recipient Identification Number) contains the identification number of the Recipient/Receiver.

#### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Subfield ID         | 13                      |
| Subfield length     | 2                       |
| Data representation | ans...25; LLVAR         |
| Data field          | Contents of subfield 13 |
| Justification       | N/A                     |

#### Values

Valid identification number of the Receiver/Recipient.

### Application notes

Subfield 13 is optional for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

### Subfield 13 (Receiver/Recipient Identification Number [Visa Only])

DE 108, subelement 01, subfield 13 (Receiver/Recipient Identification Number [Visa Only]) contains the identification number of the receiver or recipient.

#### Attributes

| Attribute            | Description     |
|----------------------|-----------------|
| Subfield ID          | 13              |
| Subfield Data Length | 2 positions     |
| Data Representation  | ans...35; LLVAR |

| Attribute     | Description             |
|---------------|-------------------------|
| Data Field    | Contents of subfield 13 |
| Justification | N/A                     |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 13 (Receiver/Recipient Identification Number [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 14 (Receiver/Recipient Identification Country Code)

DE 108, subelement 01, subfield 14 (Receiver/Recipient Identification Country Code) contains the identification country code of the recipient or receiver.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 14                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-3                   |
| Data Field           | Contents of subfield 14 |
| Justification        | N/A                     |

### Values

Valid identification country code of the Receiver/Recipient.

### Application notes

Subfield 14 (Receiver/Recipient Identification Country Code) is optional for Mastercard Send Transactions, Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payment Transactions. The field has a three-character ISO country code in Visa-branded transactions.



## Subfield 15 (Receiver/Recipient Identification Expiration Date)

DE 108, subelement 01, subfield 15 (Receiver/Recipient Identification Expiration Date) contains the identification expiration date of the Receiver/Recipient.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 15                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-8                     |
| Data Field           | Contents of subfield 15 |
| Justification        | N/A                     |

### Values

Valid identification expiration date of the Receiver/Recipient in the format MMDDYYYY.

### Application notes

Subfield 15 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 16 (Receiver/Recipient Nationality)

DE 108, subelement 01, subfield 16 (Receiver/Recipient Nationality) contains the nationality of the Recipient/Receiver.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 16                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-3                   |
| Data Field           | Contents of subfield 16 |
| Justification        | N/A                     |

## Values

Nationality of the Receiver/Recipient as defined by a valid country code for the country of citizenship.

## Application notes

Subfield 16 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 17 (Receiver/Recipient Country of Birth)

DE 108, subelement 01, subfield 17 (Receiver/Recipient Country of Birth) contains the country of birth of the Recipient/Receiver.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 17                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-3                   |
| Data Field           | Contents of subfield 17 |
| Justification        | N/A                     |

## Values

Valid Country of birth of the Receiver/Recipient.

## Application notes

Subfield 17 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 18 (Receiver/Recipient Account Number Type)

DE 108, subelement 01, subfield 18 (Receiver/Recipient Account Number Type) contains the account number type of the Receiver/Recipient.

## Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 18          |

| Attribute            | Description |
|----------------------|-------------|
| Subfield Data Length | 2 positions |
| Data Representation  | n-2         |
| Justification        | n/a         |

## Values

List of valid values:

- 00: Other
- 01: RTN + Bank Account
- 02: IBAN
- 03: Card Account
- 04: Email
- 05: Phone Number
- 06: Bank account number (BAN) + bank identification code (BIC)
- 07: Wallet ID
- 08: Social Network ID

## Application notes

Acquirers are expected per participation requirements to supply Subfield 18 if the related Subfield 11 is provided. Subfield is expected to be one of the valid values for the field.

Subfield 18 is not required for Mastercard QR Transactions and if submitted will be disregarded and not be forwarded to destination.

Subfield 18 is conditional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

The Mastercard Network does not enforce system edits on this field. It is therefore conditional per specifications. However, per program rules noted in the *Mastercard Send and Funding Transactions Program Standards*, it is a required field for Funding Transactions and must be provided. Not providing required data in this field as per standards may result in compliance assessments.

## Subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only])

DE 108, subelement 01, subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only]) contains the street address line 2 of the receiver or recipient.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 19                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...99; LLVAR         |
| Data Field           | Contents of subfield 19 |
| Justification        | N/A                     |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 20 (Receiver/Recipient Street Name [Visa Only])

DE 108, subelement 01, subfield 20 (Receiver/Recipient Street Name [Visa Only]) contains the street name of the receiver or recipient.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 20                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...99; LLVAR         |
| Data Field           | Contents of subfield 20 |
| Justification        | N/A                     |

## Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Application notes

Subfield 20 (Receiver/Recipient Street Name [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 21 (Receiver/Recipient Building Number [Visa Only])

DE 108, subelement 01, subfield 21 (Receiver/Recipient Building Number [Visa Only]) contains the building number of the receiver or recipient.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 21                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...16; LLVAR         |
| Data Field           | Contents of subfield 21 |
| Justification        | N/A                     |

## Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Application notes

Subfield 21 (Receiver/Recipient Building Number [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only])

DE 108, subelement 01, subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only]) contains the minor country subdivision code of the receiver or recipient.

## Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 22          |

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...16; LLVAR         |
| Data Field           | Contents of subfield 22 |
| Justification        | N/A                     |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only]) is only for use with Visa-branded transactions.

## Subelement 02 (Sender Data)

DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data) contains the sender name, address, phone number, date of birth, and account details.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 02                    |
| Subelement Data Length | 3 positions           |
| Data Representation    | ans...368; LLVAR      |
| Data Field             | Contents of subfields |

| Attribute     | Description                                            |
|---------------|--------------------------------------------------------|
| Subfields     | Subfield 01 (Sender First Name)                        |
|               | Subfield 02 (Sender Middle Name)                       |
|               | Subfield 03 (Sender Last Name)                         |
|               | Subfield 04 (Sender Street Address)                    |
|               | Subfield 05 (Sender City)                              |
|               | Subfield 06 (Sender State/Province Code)               |
|               | Subfield 07 (Sender Country)                           |
|               | Subfield 08 (Sender Postal Code)                       |
|               | Subfield 09 (Sender Phone Number)                      |
|               | Subfield 10 (Sender Date of Birth)                     |
|               | Subfield 11 (Sender Account Number)                    |
|               | Subfield 12 (Sender Identification Type)               |
|               | Subfield 13 (Sender Identification Number)             |
|               | Subfield 13 (Sender Identification Number [Visa Only]) |
|               | Subfield 14 (Sender Identification Country Code)       |
|               | Subfield 15 (Sender Identification Expiration Date)    |
|               | Subfield 16 (Sender Nationality)                       |
|               | Subfield 17 (Sender Country of Birth)                  |
|               | Subfield 18 (Sender Account Number Type)               |
| Justification | See subfields                                          |

DE 108, subelement 02 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

DE 108, subelement 02, subfield 01 (Sender First Name), and subfield 03 (Sender Last Name): Edit check

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Then the Dual Message Authorization System... |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:                                                                                                                                                                                                                                                                                                                                                               |                                               |
| <ul style="list-style-type: none"><li>DE 3, subfield 1 contains value 28</li><li>DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li><li>DE 108, subelement 02, subfield 01 (Sender First Name), and DE 108, subelement 02, subfield 03 (Sender Last Name) are present and contain all spaces or all numeric values or any combination of spaces and numeric values</li></ul> |                                               |



| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Then the Dual Message Authorization System...                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where: sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li> <li>• DE 108, subelement 02 (Sender Data) is present and DE 108, subelement 02, subfield 01 (Sender First Name) OR subfield 03 (Sender Last Name) is absent</li> </ul> | <p>Sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 (Additional Response Data) = 108002 (Sender Data)</li> </ul> |

**DE 108, subelement 02, subfield 11 (Sender Account Number): Edit check**

| <b>When...</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Then the Dual Message Authorization System...</b>                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li> <li>• DE 108, subelement 02, subfield 11 (Sender Account Number) is present and contains all zeros, all spaces, or a combination of zeroes and spaces</li> </ul> | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108002 (Sender Data)</li> </ul> |
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li> <li>• DE108, subelement 02, subfield 11 (Sender Account Number) is not present</li> </ul>                                                                        | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108002 (Sender Data)</li> </ul> |

### DE 108, Subelement 02, Subfield 12 (Sender Identification Type): Edit check

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Then the Dual Message Authorization System...                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"><li>• DE 3, subfield 1 contains value 28</li><li>• DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>• DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>• DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li><li>• DE 108 subelement 02, subfield 12 (Sender Identification Type) is present and does not contain one of the values 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10</li></ul> | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 108002 (Sender Data)</li></ul> |

### Subfield 01 (Sender First Name)

DE 108, subelement 02, subfield 01 (Sender First Name) contains the first name of the Sender.

#### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...35; LLVAR         |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 01 must be present for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions and must be properly formatted. Subfield 01 is optional for Funding Transactions.

The first name (Consumer/Business/Government/Non-Government Organization) of the sender is included in this subelement.

The table that follows describes the correct submission of business names.

Application notes: Mastercard Merchant Presented QR transactions

Subfield 01 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forward to destination.

Subfield 01 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Name example

| Business Name     | First Name | Last Name     |
|-------------------|------------|---------------|
| XYZ               | XYZ        | XYZ           |
| XYZ International | XYZ        | International |
| XYZ DBA MA        | XYZ        | DBA MA        |

Subfield 02 (Sender Middle Name)

DE 108, subelement 02, subfield 02 (Sender Middle Name) contains the middle name initial of the Sender.

Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-1                   |
| Data Field           | Contents of subfield 02 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Valid value will consist of the middle name initial of the Sender.

### Application notes

Subfield 02 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 03 (Sender Last Name)

DE 108, subelement 02, subfield 03 (Sender Last Name) contains the last name of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...35; LLVAR         |
| Data Field           | Contents of subfield 03 |
| Justification        | N/A                     |

### Values

If present, cannot contain all spaces or numeric values.

### Application notes

Subfield 03 must be present and properly formatted for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions. Subfield 03 is optional for Funding Transactions.

The last name (Consumer/Business/Government/Non-Government Organization) of the sender is included in this subelement.

The table that follows describes the correct submission of business names.

### Application notes: Mastercard Merchant Presented QR transactions

Subfield 03 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 03 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

#### Name example

| Business name     | First name | Last name     |
|-------------------|------------|---------------|
| XYZ               | XYZ        | XYZ           |
| XYZ International | XYZ        | International |
| XYZ DBA MA        | XYZ        | DBA MA        |

### Subfield 04 (Sender Street Address)

DE 108, subelement 02, subfield 04 (Sender Street Address) contains the street address of the Sender.

#### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 04                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...50; LLVAR         |
| Data Field           | Contents of subfield 04 |
| Justification        | N/A                     |

#### Values

Valid values will consist of street address of the Sender.

#### Application notes

Subfield 04 is mandatory for Mastercard Send Payment Transactions and Gambling and Gaming Payments Transactions.

Subfield 04 is optional for Funding transactions.

## Subfield 05 (Sender City)

DE 108, subelement 02, subfield 05 (Sender City) contains the city of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 05                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...25; LLVAR         |
| Data Field           | Contents of subfield 05 |
| Justification        | N/A                     |

### Values

Valid location city name of the Sender.

### Application notes

Subfield 05 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

**NOTE: The Mastercard Network does not enforce system edits on this field. It is therefore optional per specifications. However per program rules noted in the [Mastercard Send and Funding Transactions Program Standards](#), it is a required field for Payment Transactions and must be provided. Not providing required data in this field as per standards may result in compliance assessments**

## Subfield 06 (Sender State/Province Code)

DE 108, subelement 02, subfield 06 (Sender State/Province Code) contains the state/province code of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 06                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...3; LLVAR          |
| Data Field           | Contents of subfield 06 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Valid location State code of the Sender. Cannot contain spaces or invalid code when country is U.S. or Canada.

### Application notes

Subfield 06 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions). If submitted on Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, subfield 06 cannot contain spaces or invalid code when country is U.S. or Canada.

**NOTE: Data representation must be alpha (a-3) for Mastercard Send Transactions. Refer to *Quick Reference Booklet* for list of valid country codes.**

## Subfield 07 (Sender Country)

DE 108, subelement 02, subfield 07 (Sender Country) contains the country code of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 07                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | a-3                     |
| Data Field           | Contents of subfield 07 |
| Justification        | N/A                     |

### Values

Sender Country must contain a valid three-character alphabetic code and must not be part of a blocked country list. Refer to the *Quick Reference Booklet* for a list of valid codes.

### Application notes

Subfield 07 is mandatory for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions and optional for Funding Transactions.

Subfield 07 is optional for Funding Transactions.



Transaction will be declined if the sender country is subject to comprehensive geographic sanctions published by the Office of Foreign Assets Control (OFAC), <https://ofac.treasury.gov/sanctions-programs-and-country-information>. The current list of countries subject to such sanctions is subject to change.

## Subfield 08 (Postal Code)

DE 108, subelement 02, subfield 08 (Postal Code) contains the postal code of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 08                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...10; LLVAR         |
| Data Field           | Contents of subfield 08 |
| Justification        | N/A                     |

### Values

Valid location Postal Code of the Sender.

### Application notes

Subfield 08 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 09 (Sender Phone Number)

DE 108, subelement 02, subfield 09 (Sender Phone Number) contains the phone number of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 09                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...20; LLVAR         |
| Data Field           | Contents of subfield 09 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

Valid phone number of the Sender.

### Application notes

Subfield 09 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 10 (Sender Date of Birth)

DE 108, subelement 02, subfield 10 (Sender Date of Birth) contains the date of birth of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 10                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-8                     |
| Data Field           | Contents of subfield 10 |
| Justification        | N/A                     |

### Values

Valid values will consist of the date of birth of the Sender in the format MMDDYYYY.

### Application notes

Subfield 10 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 11 (Sender Account Number)

DE 108, subelement 02, subfield 11 (Sender Account Number) contains the account number of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 11                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...50; LLVAR         |
| Data Field           | Contents of subfield 11 |
| Justification        | N/A                     |

### Values

Valid account number of the Sender.

The account number that was used to fund the money transfer (for example, the sender used a credit, debit, or other account to fund the money transfer). When cash is used to fund the money transfer, this field should be populated with a unique transaction reference number or an account number from the OI. This information will assist in identifying the sender if the receiving institution requires additional information from the originating institution for the money transfer.

### Application notes

Subfield 11 (Sender Account Number) can have a maximum length of 34 characters for Visa-branded transactions.

### Application notes: Mastercard Send Transactions

Subfield 11 must be present for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions and must be properly formatted. Subfield 11 is optional for Funding Transactions. When subfield 11 contains a combination of information as specified in subfield 18 [for example, Bank account number (BAN) + Bank Identification Code (BIC)], the data must be separated by '+' and in the order specified for Mastercard Send and Funding Transactions.

### Application notes: Mastercard Merchant Presented QR Transactions

Payment Transaction usage:

- Authorization Request/0100: Mandatory. Edit must validate. Forward to destination.
- Reversal Advice/0420: Value from Authorization Request/0100 forwarded to destination.

The enhancement introduced within DE 108, subelement 02, subfield 11 (Sender Account Number) does not apply for MCQR transactions. If MCQR transactions are processed with data in DE 108, subelement 02, subfield 11 (Sender Account Number) exceeding the current supported length, Mastercard MCQR edits will decline the transaction.

Refund payment transaction usage: Reversal Advice/0420—Value from Authorization Request/0100 forwarded to destination

Funding transaction usage: Subfield 11 is not applicable

## Subfield 12 (Sender Identification Type)

DE 108, subelement 02, subfield 12 (Sender Identification Type) contains the identification type of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 12                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-2                     |
| Data Field           | Contents of subfield 12 |
| Justification        | N/A                     |

### Values

If present on Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, Visa-branded funding transactions, it must contain one of the following valid values:

### Values

| Value | Description                  |
|-------|------------------------------|
| 00    | Passport                     |
| 01    | National Identification Card |
| 02    | Driver's License             |
| 03    | Government Issued            |
| 04    | Other                        |
| 05    | Date of birth (Visa only)    |

| Value | Description                                       |
|-------|---------------------------------------------------|
| 06    | Customer identification (unspecified) (Visa only) |
| 07    | Business tax ID (Visa only)                       |
| 08    | Individual tax ID (Visa only)                     |
| 09    | Company registration number (Visa only)           |
| 10    | Proxy identification (Visa only)                  |
| 11    | Social security number (Visa only)                |
| 12    | Alien registration number (Visa only)             |
| 13    | Law enforcement identification (Visa only)        |
| 14    | Military identification (Visa only)               |
| 15    | Travel identification (non-passport) (Visa only)  |
| 16    | Email (Visa only)                                 |
| 17    | Phone number (Visa only)                          |

### Application notes

Values 03 and 04 have no corresponding values in Visa format. Acquirers should not send either value 03 or 04 with Visa-branded transactions.

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details on Visa-only values.

## Subfield 13 (Sender Identification Number)

DE 108, subelement 02, subfield 13 (Sender Identification Number) contains the identification number of the Sender.

### Attributes

| Attribute           | Value                   |
|---------------------|-------------------------|
| Subfield ID         | 13                      |
| Subfield length     | 2                       |
| Data representation | ans...25; LLVAR         |
| Data field          | Contents of subfield 13 |
| Justification       | N/A                     |

## Values

Valid identification number of the Sender.

## Application notes

Subfield 13 is optional for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 13 (Sender Identification Number [Visa Only])

DE 108, subelement 02, subfield 13 (Sender Identification Number [Visa Only]) contains the identification number of the Sender.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 13                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...35; LLVAR         |
| Data Field           | Contents of subfield 13 |
| Justification        | N/A                     |

## Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

## Application notes

Subfield 13 (Sender Identification Number [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 14 (Sender Identification Country Code)

DE 108, subelement 02, subfield 14 (Sender Identification Country Code) contains the identification country code of the Sender.

## Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 14          |
| Subfield Data Length | 2 positions |

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | ans-3                   |
| Data Field          | Contents of subfield 14 |
| Justification       | N/A                     |

### Values

Valid identification country code of the Sender.

### Application notes

Subfield 14 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 15 (Sender Identification Expiration Date)

DE 108, subelement 02, subfield 15 (Sender Identification Expiration Date) contains the identification expiration date of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 15                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-8                     |
| Data Field           | Contents of subfield 15 |
| Justification        | N/A                     |

### Values

Valid identification expiration date of the Sender in the format MMDDYYYY.

### Application notes

Subfield 15 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 16 (Sender Nationality)

DE 108, subelement 02, subfield 16 (Sender Nationality) contains the nationality of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 16                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-3                   |
| Data Field           | Contents of subfield 16 |
| Justification        | N/A                     |

### Values

Nationality of the sender as defined by a valid country code for the country of citizenship.

### Application notes

Subfield 16 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 17 (Sender Country of Birth)

DE 108, subelement 02, subfield 17 (Sender Country of Birth) contains the country of birth of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 17                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-3                   |
| Data field           | Contents of subfield 17 |
| Justification        | N/A                     |

### Values

Valid country of birth of the Sender.



### Application notes

Subfield 17 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

### Subfield 18 (Sender Account Number Type)

DE 108, subelement 02, subfield 18 (Sender Account Number Type) contains the account number type of the Sender.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 18          |
| Subfield Data Length | 2 positions |
| Data Representation  | n-2         |
| Justification        | N/A         |

### Values

List of valid values:

- 00: Other
- 01: RTN + Bank Account
- 02: IBAN
- 03: Card Account
- 04: Email
- 05: Phone Number
- 06: Bank account number (BAN) + Bank Identification Code (BIC)
- 07: Wallet ID
- 08: Social Network ID

### Application notes

Acquirers are expected per participation requirements to supply Subfield 18 if the related Subfield 11 is provided. Subfield is expected to be one of the valid values for the field.

Subfield 18 is not required for Mastercard QR Transactions and if submitted will be disregarded and not be forwarded to destination.

Subfield 18 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payment Transactions).

The Mastercard Network does not enforce system edits on this field. It is therefore optional per specifications. However per program rules noted in the [Mastercard Send and Funding](#)

[Transactions Program Standards](#), it is a required field for Funding Transactions and must be provided. Not providing required data in this field as per standards may result in compliance assessments

### Subelement 03 (Transaction Reference Data)

DE 108, subelement 03 (Transaction Reference Data) contains the transaction reference number, funding score, participation ID, and additional data related to Mastercard Send transactions and Mastercard Merchant Presented QR transactions.

#### Attributes

| Attribute              | Description                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subelement ID          | 03                                                                                                                                                                                                                                       |
| Subelement Data Length | 3 positions                                                                                                                                                                                                                              |
| Data Representation    | ans...138; LLLVAR                                                                                                                                                                                                                        |
| Data Field             | Contents of subfields                                                                                                                                                                                                                    |
| Subfields              | Subfield 01 (Unique Transaction Reference)<br>Subfield 01 (Sender Reference Number [Visa Only])<br>Subfield 02 (Additional Message)<br>Subfield 03 (Funding Source)<br>Subfield 04 (Participant ID)<br>Subfield 05 (Transaction Purpose) |
| Justification          | See subfields                                                                                                                                                                                                                            |

## DE 108, subelement 03 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

### DE 108, Subelement 03, Subfield 01 (Unique Transaction Reference): Edit check

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Then the Dual Message Authorization System...                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment Transaction or Gaming and Gambling Payment transaction where: <ul style="list-style-type: none"><li>DE 3, subfield 1 contains value 28</li><li>DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>DE 48, subelement 77 Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li><li>DE 61, subfield 10 criteria is met for Moneysend Payment transaction</li><li>DE 108, subelement 03, subfield 01 (Unique Transaction Reference) is present and contains all spaces, all zeros, or any combination of spaces and zeros</li></ul> | sends an Authorization Request Response/0110 message to the acquirer where: <ul style="list-style-type: none"><li>DE 39 (Response Code) = 30 (Format error)</li><li>DE 44 (Additional Response Data) = 108003 (Mastercard Send Transaction Data)</li></ul> |

### Subfield 01 (Unique Transaction Reference)

DE 108, subelement 03, subfield 01 (Unique Transaction Reference) contains the Unique Transaction Reference number for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming Payments Transactions) and Mastercard Merchant Presented QR funding transactions.

#### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Subfield ID         | 01          |
| Subfield length     | 2 positions |
| Data representation | ans-19      |

| Attribute     | Description             |
|---------------|-------------------------|
| Data field    | Contents of subfield 01 |
| Justification | N/A                     |

## Values

- Values for customers that do not need to process beyond 99 transactions per second:
  - Unique Transaction Reference Number (ans-19).
  - A valid value string leads with a zero (0), followed by:
    - ICA (n-6)
    - Year (n-1)
    - Julian Date (n-3)
    - Time hhmmss (n-6)
    - Transaction Sequence Number for customers who do not need to process beyond 99 transactions: (01-99) (n-2)

Example: 0555555801215305401

- Values for customers that need to process beyond 99 transactions per second:
  - Unique Transaction Reference Number (ans-19).
  - A valid value string contains:
    - ICA (n-6)
    - Year (n-1)
    - Julian Date (n-3)
    - Time hhmmss (n-6)
    - Transaction Sequence Number for customers who need to process beyond 99 transactions: (001-999) (n-3)

Example: 5555558012153054001

## Application notes

The Mastercard Network does not enforce system edits on this field. However, per program rules noted in the *Mastercard MoneySend and Funding Transactions Program Standards* guide, it is a required field for Funding Transactions (coded with a TTI) and MoneySend Payment and Gaming Payments Transactions. Not providing required data in this field per the standards may result in a compliance assessment.

Payment Transaction usage:

- Authorization Request/0100: Optional. If present, an edit must validate. Forward to destination if edit validates
- Reversal Advice/0420: Optional. If present, forward the value from the Authorization Request/0100 to the destination

Refund Payment Transaction usage:

- Authorization Request/0100: Optional. If present, an edit must validate. Forward to destination if edit validates
- Reversal Advice/0420: Value from Authorization Request/0100 forwarded to destination

Funding Transaction usage:

- Authorization Request/0100: Mandatory. If present, edit must validate. Forward to destination if edit validates
- Authorization Request/0100: Optional. If present, an edit must validate. Forward to destination if edit validates
- Reversal Request/0400: Optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Advice/0420: Value from Authorization Request/0100 forwarded to destination

## Subfield 01 (Sender Reference Number [Visa Only])

DE 108, subelement 03, subfield 01 (Sender Reference Number [Visa Only]) contains the sender reference number.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...16; LLVAR         |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

### Application notes

Subfield 01 (Sender Reference Number [Visa Only]) is only for use with Visa-branded transactions.

## Subfield 02 (Additional Message)

DE 108, subelement 03, subfield 02 (Additional Message) contains the additional message.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...65; LLVAR         |
| Data Field           | Contents of subfield 02 |
| Justification        | N/A                     |

### Application notes

Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

### Application notes: Mastercard Merchant Presented QR Transactions

Payment Transaction usage:

- Authorization Request/0100: optional. If present, edit must validate. Forward to destination if edit is validated
- Reversal Advice/0420: optional. If present, value from Authorization Request/0100 forwarded to destination

Refund Payment Transaction usage: subfield 02 is not applicable

Funding Transaction usage:

- Authorization Advice/0120: optional. If present, edit must validate. Forward to destination if edit is validated
- Reversal Request/0400: optional. If present, edit must validate. Forward to destination if edit is validated
- Reversal Advice/0420: value from Authorization Request/0100 forwarded to destination

## Subfield 03 (Funding Source)

DE 108, subelement 03, subfield 03 (Funding Source) contains the information representing the Mastercard Send transaction funding source.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-2                     |
| Data Field           | Contents of subfield 03 |
| Justification        | N/A                     |

### Values

Must be present for MPQR payment authorization request messages and, when present, must contain one of the following valid values.

### Valid values

| Value | Description          |
|-------|----------------------|
| 01    | Credit               |
| 02    | Debit                |
| 03    | Prepaid              |
| 04    | Deposit Account      |
| 05    | Mobile Money Account |
| 06    | Cash                 |
| 07    | Other                |

### Application notes

Subfield 03 is optional for Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions on Authorization Request/0100 and Reversal Request/0400 messages.

Mastercard will not perform a valid values check on subfield 03 within the Authorization Request/0100 message. Acquirers must send DE 108, subelement 03, subfield 03 on MPQR payment Authorization Request messages.

### Application notes for Mastercard Merchant Presented QR Transactions

| Transaction                                         | Usage                                                          |
|-----------------------------------------------------|----------------------------------------------------------------|
| Payment Transaction:<br>Authorization Request/0100  | Mandatory                                                      |
| Payment Transaction:<br>Reversal Advice/0420        | Value from Authorization Request/0100 forwarded to destination |
| Refund Payment Transaction:<br>Reversal Advice/0420 | Value from Authorization Request/0100 forwarded to destination |
| Funding Transaction                                 | Subfield 03 is not applicable                                  |

### Subfield 04 (Participation ID)

DE 108, subelement 03, subfield 04 (Participation ID) contains participation ID details of the Sender.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 04                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...30; LLVAR         |
| Data Field           | Contents of subfield 04 |
| Justification        | N/A                     |

### Values

Contains the participation ID of the Sender.

### Application notes

Subfield 04 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).



Subfield 05 (Transaction Purpose)

DE 108, subelement 03, subfield 05 (Transaction Purpose) contains Mastercard Send transaction purpose details.

Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 05                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-2                     |
| Data Field           | Contents of subfield 05 |
| Justification        | N/A                     |

Values

| Value | Description                                                                           |
|-------|---------------------------------------------------------------------------------------|
| 00    | Family Support                                                                        |
| 01    | Regular Labor Transfers (expatriates)                                                 |
| 02    | Travel & Tourism                                                                      |
| 03    | Education                                                                             |
| 04    | Hospitalization & Medical Treatment                                                   |
| 05    | Emergency Need                                                                        |
| 06    | Savings                                                                               |
| 07    | Gifts                                                                                 |
| 08    | Other                                                                                 |
| 09    | Salary                                                                                |
| 10    | Crowd lending                                                                         |
| 11    | Crypto currency                                                                       |
| 12    | Refund to original card                                                               |
| 13    | Refund to new card                                                                    |
| 14    | Transfer to another person or organization in the same Staged Digital Wallet (Brazil) |

| Value | Description              |
|-------|--------------------------|
| 15    | Boleto Payment (Brazil)  |
| 16    | High Risk Securities     |
| 17    | Recycling Deposit Return |
| 18    | Value Added Tax Payment  |

### Application notes

Subfield 05 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions) and MPQR payment transactions.

Mastercard will not perform a valid values check on subfield 05 within the Authorization Request/0100 message.

## Subelement 04 (Language Description)

DE 108 (Additional Transaction Reference Data), subelement 04 (Language Description) contains additional language details supported by the customer.

### Attributes

| Attribute              | Description                                                        |
|------------------------|--------------------------------------------------------------------|
| Subelement ID          | 04                                                                 |
| Subelement Data Length | 3 positions                                                        |
| Data Representation    | ans...061; LLLVAR                                                  |
| Data Field             | Contents of subfields                                              |
| Subfields              | Subfield 1 (Language Identification)<br>Subfield 2 (Language Data) |
| Justification          | See subfields                                                      |

## Subfield 01 (Language Identification)

DE 108, subelement 04, subfield 01 (Language Identification) contains information about the language selected by the customer.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans-3                   |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Application notes

Subfield 01 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subfield 02 (Language Data)

DE 108, subelement 04, subfield 02 (Language Data) contains additional Mastercard Send information in the language selected by the customer.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | b...50; LLVAR           |
| Data Field           | Contents of subfield 02 |
| Justification        | N/A                     |

### Application notes

Subfield 02 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

## Subelement 05 (Digital Account Information)

DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information) contains data specific to Mastercard Merchant Presented QR transactions.

### Attributes

| Attribute              | Description                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Subelement ID          | 05                                                                                                                        |
| Subelement Data Length | 3 positions                                                                                                               |
| Data Representation    | ans...099; LLLVAR                                                                                                         |
| Data Field             | Contents of subfields                                                                                                     |
| Subfields              | Subfield 01 (Digital Account Reference Number)<br>Subfield 02 (Mastercard Merchant Presented QR Receiving Account Number) |
| Justification          | See subfields                                                                                                             |

### Application notes

Subelement 05 is used only for Mastercard Merchant Presented QR transactions.

## Subfield 01 (Digital Account Reference Number)

DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) is provided in a Mastercard Merchant Presented QR Payment Transaction to allow Receiving Institutions the ability to initiate a refund to the consumer via the Originating Institution.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n...19; LLVAR           |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Application notes: Mastercard Merchant Presented QR Transactions

| Transaction                                               | Usage                                                                                                                                                                                  |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Payment Transaction usage:<br>Authorization Request/0100: | Optional. If present, edit must validate. If not present, Mastercard will populate with value from DE 108, subelement 02, subfield 11 (Sender Account Number); Forward to destination. |
| Payment Transaction usage:<br>Reversal Advice/0420        | Optional. If present, value from Authorization Request/0100 forwarded to destination.                                                                                                  |
| Refund Payment Transaction usage:<br>Reversal Advice/0420 | Value from Authorization Request/0100 forwarded to destination.                                                                                                                        |
| Funding Transaction usage:                                | Subfield 01 is not applicable                                                                                                                                                          |

### Subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)

DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 02 (Mastercard Merchant Presented QR Receiving Account Number) provided in a Mastercard Merchant Presented QR Funding Transaction contains the receiving account number.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...34; LLVAR         |
| Data Field           | Contents of subfield 02 |
| Justification        | N/A                     |

### Values

Valid Account Number of the Receiver.

### Application notes: Mastercard Merchant Presented QR Transactions

| Transaction                                              | Usage                                                                                 |
|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| Payment Transaction usage                                | Subfield 02 is not applicable                                                         |
| Refund Payment Transaction usage                         | Subfield 02 is not applicable                                                         |
| Funding Transaction usage:<br>Authorization Request/0100 | Optional. If present, edit must validate. Forward to destination if edit validates.   |
| Funding Transaction usage:<br>Authorization Advice/0120  | Optional. If present, edit must validate. Forward to destination if edit validates.   |
| Funding Transaction usage:<br>Reversal Request/0400      | Optional. If present, edit must validate. Forward to destination if edit validates.   |
| Funding Transaction usage:<br>Reversal Advice/0420       | Optional. If present, value from Authorization Request/0100 forwarded to destination. |

## Subelement 06 (QR Dynamic Code Data)

DE 108 (Additional Transaction Reference Data), subelement 06 (QR Dynamic Code Data) is available for regional data with limited Mastercard editing or data validation.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 06                        |
| Subelement Data Length | 3 positions               |
| Data Representation    | ans...237; LLLVAR         |
| Data Field             | Contents of subelement 06 |
| Subfields              | N/A                       |
| Justification          | Left                      |

### Values

If present, must be alphanumeric or special characters and a valid length.

**Application notes: Mastercard Merchant Presented QR Transactions**

| Transaction                                              | Usage                                                                                 |
|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| Payment Transaction usage:<br>Authorization Request/0100 | Optional. If present, edit must validate. Forward to destination if edit validates.   |
| Payment Transaction usage:<br>Reversal Advice/0420       | Optional. If present, value from Authorization Request/0100 forwarded to destination. |
| Refund Payment Transaction usage                         | Subelement 06 is not applicable.                                                      |
| Funding Transaction usage:<br>Authorization Request/0100 | Optional. If present, edit must validate. Forward to destination if edit validates.   |
| Funding Transaction usage:<br>Authorization Advice/0120  | Optional. If present, edit must validate. Forward to destination if edit validates.   |
| Funding Transaction usage:<br>Reversal Request/0400      | Optional. If present, edit must validate. Forward to destination if edit validates.   |
| Funding Transaction usage:<br>Reversal Advice/0420       | Optional. If present, value from Authorization Request/0100 forwarded to destination. |

## Subelement 07 (Data Validation Request)

DE 108 (Additional Transaction Reference Data), subelement 07 (Data Validation Request) indicates that account status verification is requested in the Authorization message.

**Attributes**

| Attribute              | Description             |
|------------------------|-------------------------|
| Subelement ID          | 07                      |
| Subelement Data Length | 3 positions             |
| Data Representation    | an....6;LLLVAR          |
| Data Field             | Contents of subfield 01 |
| Subfields              | 1                       |
| Justification          | N/A                     |

## Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | O   | X   | C   |
| Authorization Request Response/0110 | CE  | XE  | CE  |

## Authorization Request/0100 message edits for DE 108, subelement 07

Mastercard will perform the following edits.

## Associated edits

| When...                                                                                                                                                                                                                                                                                                      | Then the Dual Message Authorization System...                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 message contains DE 108 (Additional Transaction Reference Data) with multiple instances of subelement 07 or its subfield.                                                                                                                                                      | will reject the message and send an Authorization Request Response/0110 message to the acquirer with: <ul style="list-style-type: none"><li>DE 39 (Response Code) = 30 (Format Error)</li><li>DE 44 (Additional Response Data) = 108007 (indicating DE 108 SE07 is in error)</li></ul> |
| DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 10 (Receiver Name Validation Request) and DE 108, subelement 01, subfield 01 (Receiver/Recipient First name), subfield 02 (Receiver/Recipient Middle Name), and subfield 03 (Receiver/Recipient Last Name) are not present. | will not forward DE 108, subelement 07 to the issuer                                                                                                                                                                                                                                   |
| DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 11 (Sender Name Validation Request) and DE 108, subelement 02, subfield 01 (Sender First name), subfield 02 (Sender Middle Name), and subfield 03 (Sender Last Name), are not present.                                      | will not forward DE 108, subelement 07 to the issuer                                                                                                                                                                                                                                   |



## Subfield 01 (Name Validation Request)

DE 108, subelement 07, subfield 01 (Name Validation Request) indicates the type of validation being requested from the originator.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 01                        |
| Subfield Data Length | 2 positions               |
| Data Representation  | n-2                       |
| Data Field           | Contents of positions 1-2 |
| Justification        | N/A                       |

### Values

| Value | Description                      |
|-------|----------------------------------|
| 10    | Receiver Name Validation Request |
| 11    | Sender Name Validation Request   |

### Application notes

Subfield 01 is optional for non-financial authorization transactions. If subfield 01 is present, name information must be provided in DE 108, subelement 01 (Receiver/Recipient Data) or subelement 02 (Sender Data).

## Subelement 08 (Data Validation Response)

DE 108 (Additional Transaction Reference Data), subelement 08 (Data Validation Response) contains the Account Status verification response in the Authorization request response message.

### Attributes

| Attribute              | Description |
|------------------------|-------------|
| Subelement ID          | 08          |
| Subelement Data Length | 3 positions |

| Attribute           | Description                                                                                |
|---------------------|--------------------------------------------------------------------------------------------|
| Data Representation | an...010; LLLVAR                                                                           |
| Data Field          | Contents of subfields 01-02                                                                |
| Subfields           | Subfield 01 (Receiver/Recipient Service Response)<br>Subfield 02 (Sender Service Response) |
| Justification       | See subfields                                                                              |

## Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |

If provided, Mastercard will remove subelement 08 from the Authorization Request/0100.

## Authorization Response/0110 message edits for DE 108, subelement 08

Mastercard will perform the following edits.

## Associated edits

| When...                                                                                                                  | Then the Dual Message Authorization System...                                                           |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 11 (Sender Name Validation Request) and | will not forward DE 108, subelement 08, subfield 01 (Receiver Name Validation Response) to the acquirer |
| DE 108, subelement 08, subfield 01 (Receiver Name Validation Response) includes a value from the issuer                  |                                                                                                         |

| When...                                                                                                                    | Then the Dual Message Authorization System...                                                        |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 10 (Receiver Name Validation Request) and | will not forward DE 108 subelement 08, subfield 02 (Sender Name Validation Response) to the acquirer |
| DE 108, subelement 08, subfield 02 (Sender Name Validation Response) includes a value from the issuer                      |                                                                                                      |

## Subfield 01 (Receiver Name Validation Response)

DE 108, subelement 08, subfield 01 (Receiver Name Validation Response) contains the name validation result from the issuer.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of position 1 |
| Justification        | N/A                    |

## Values

| Position | Attribute | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | an-1      | <p>Receiver Name Validation Response</p> <p>Contains the results of an attempt to verify that the following data present in the validation request message matched the data on file with the issuer:</p> <ul style="list-style-type: none"><li>Receiver First Name (DE 108, subelement 01, subfield 01)</li><li>Receiver Middle Name (DE 108, subelement 01, subfield 02)</li><li>Receiver Last Name (DE 108, subelement 01, subfield 03)</li></ul> <p>Valid values include:</p> <ul style="list-style-type: none"><li>A = Match</li><li>B = Partial Match</li><li>C = No Match</li><li>U = Unverified</li></ul> |

## Application notes

Subfield 01 is conditional for non-financial transactions.

## Application notes for Non-Financial Transactions

If DE 108, subelement 07, subfield 01 is received with value '10' in a non-financial Authorization request, and DE 108 subelement 01 (Receiver/Recipient Data), subfield 01 (Receiver/Recipient First name), subfield 02 (Receiver/Recipient Middle Name), and subfield 03 (Receiver/Recipient Last Name) are present, then the issuer should include DE 108, subelement 08, subfield 01 in the response. Issuer provided response is not edited and will be forwarded to the destination. When an issuer opts in for the Name Match service, Mastercard will issue a response in DE 108, subelement 08, subfield 01 on the issuer's behalf. Should a participating issuer provide a value in this subfield, it will be disregarded and replaced with the value provided by Mastercard.

## Subfield 02 (Sender Name Validation Response)

DE 108, subelement 08, subfield 02 (Sender Name Validation Response) contains the originator name validation results for the sender.

## Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 02          |

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield Data Length | 2 positions            |
| Data Representation  | an-1                   |
| Data Field           | Contents of position 1 |
| Justification        | N/A                    |

### Values

| Position | Attribute | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | an-1      | <p>Sender Name Validation Response</p> <p>Contains the results of an attempt to verify that the following data present in the validation request message matched the data on file with the issuer:</p> <ul style="list-style-type: none"><li>• Sender First Name (DE 108, subelement 02, subfield 01)</li><li>• Sender Middle Name (DE 108, subelement 02, subfield 02)</li><li>• Sender Last Name (DE 108, subelement 02, subfield 03)</li></ul> <p>Valid values include:</p> <ul style="list-style-type: none"><li>• A = Match</li><li>• B = Partial Match</li><li>• C = No Match</li><li>• U = Unverified</li></ul> |

### Application notes

Subfield 02 is conditional for non-financial Authorization transactions.

#### Application notes for Non-Financial Authorization Transactions

If DE 108, subelement 07, subfield 01 is received with value '11' in a non-financial Authorization request, and DE 108 subelement 02 (Sender Data), subfield 01 (Sender First name), subfield 02 (Sender Middle Name), and subfield 03 (Sender Last Name) are present, then the issuer should include DE 108, subelement 08, subfield 02 in the response. Issuer provided response is not edited and will be forwarded to the destination. When an issuer opts in for the Name Match service, Mastercard will issue a response in Data Element 108, subelement 08, subfield 02 on the issuer's behalf. Should a participating issuer provide a value in this subfield, it will be disregarded and replaced with the value provided by Mastercard.

## Subelement 09 (Account Owner Information 1)

DE 108, subelement 09 (Account Owner Information 1) contains the account owner name, sequence number and classification. The account owner information is used for the Name Match Service.

### Attributes

| Attribute           | Description                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subelement ID       | 09                                                                                                                                                                                                                      |
| Data Representation | ans...129; LLLVAR                                                                                                                                                                                                       |
| Subelement Length   | 3                                                                                                                                                                                                                       |
| Data Field          | Contents of subfields 01-05                                                                                                                                                                                             |
| Subfields           | Subfield 01 (Account Owner First Name)<br>Subfield 02 (Account Owner Middle Name)<br>Subfield 03 (Account Owner Last Name)<br>Subfield 04 (Account Owner Sequence Number)<br>Subfield 05 (Account Owner Classification) |
| Justification       | See subfields                                                                                                                                                                                                           |

Following is the usage of DE 108, subelement 09 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Usage

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | .   |

### Application notes

If provided, Mastercard will remove DE 108, subelement 09 from the Authorization Request/0100.

## Authorization Response/0110 message edits for DE 108, subelement 09

Mastercard will perform the following edits.

### Associated edits

| When...                                                                                                                                                                                                                  | Then the Dual Message Authorization System...                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request Response/0110 message contains data in DE 108, subelement 09 (Account Owner Information 1) defined with an incorrect length and/or is incorrectly formatted (Data Representations).             | will remove DE 108, subelement 09 (Account Owner Information 1) and the message continues processing.                                                                       |
| An Authorization Request Response/0110 message contains DE 108, subelement 09 (Account Owner Information 1) with subfields that have multiple instances of the same subelement or subfield within the same data element. | sends an Authorization Response Negative Acknowledgment/0190 message to the issuer with: <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 108</li></ul> |

### Subfield 01 (Account Owner First Name)

DE 108, subelement 09, subfield 01 (Account Owner First Name) contains the first name of the account owner.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 01                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |

### Values

If present, cannot contain all spaces or all numeric values.

### Application notes

Subfield 01 is optional if the issuer is opted-in for the on-behalf Name Match Service.

### Subfield 02 (Account Owner Middle Name)

DE 108, subelement 09, subfield 02 (Account Owner Middle Name) contains the middle name of the account owner.

#### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 02                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |

#### Values

If present, cannot contain all spaces or all numeric values.

### Application notes

Subfield 02 is optional if the issuer is opted-in for the on-behalf Name Match Service.

### Subfield 03 (Account Owner Last Name)

DE 108, subelement 09, subfield 03 (Account Owner Last Name) contains the last name of the account owner.

#### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 03                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |



**Values**

If present, cannot contain all spaces or all numeric values.

**Application notes**

Subfield 03 is optional if the issuer is opted-in for the on-behalf Name Match Service.

**Subfield 04 (Account Owner Sequence Number)**

DE 108, subelement 09, subfield 04 (Account Owner Sequence Number) contains the sequence number of the account owner.

**Attributes**

| Attribute           | Description               |
|---------------------|---------------------------|
| Subfield ID         | 04                        |
| Data Representation | n-3                       |
| Subfield Length     | 2                         |
| Data Field          | Contents of positions 1-3 |
| Justification       | N/A                       |

**Values**

If present, must be numeric.

**Application notes**

Subfield 04 is optional if the issuer is opted-in for the on-behalf Name Match Service.

**Subfield 05 (Account Owner Classification)**

DE 108, subelement 09, subfield 05 (Account Owner Classification) contains the classification category (Personal or Business) of the account owner.

**Attributes**

| Attribute           | Description |
|---------------------|-------------|
| Subfield ID         | 05          |
| Data Representation | a-1         |
| Subfield Length     | 2           |

| Attribute     | Description            |
|---------------|------------------------|
| Data Field    | Contents of position 1 |
| Justification | N/A                    |

If present, must contain one of these following valid values.

### Values

| Value | Description |
|-------|-------------|
| P     | Personal    |
| B     | Business    |

### Application notes

Subfield 05 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subelement 10 (Account Owner Information 2)

DE 108, subelement 10 (Account Owner Information 2) contains the details of the account owner for a given PAN. The account owner information is used for the on-behalf Name Match service.

### Attributes

| Attribute           | Description                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subelement ID       | 10                                                                                                                                                                                                                      |
| Data Representation | ans...129; LLLVAR                                                                                                                                                                                                       |
| Subelement Length   | 3                                                                                                                                                                                                                       |
| Data Field          | Contents of subfields 01-05                                                                                                                                                                                             |
| Subfields           | Subfield 01 (Account Owner First Name)<br>Subfield 02 (Account Owner Middle Name)<br>Subfield 03 (Account Owner Last Name)<br>Subfield 04 (Account Owner Sequence Number)<br>Subfield 05 (Account Owner Classification) |
| Justification       | See subfields                                                                                                                                                                                                           |

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

### Usage

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | .   |

### Application notes

If provided from originator, Mastercard will remove DE 108, subelement 10 from the Authorization Request/0100 message.

## Authorization Response/0110 message edits for DE 108, subelement 10

Mastercard will perform the following edits.

### Associated edits

| When...                                                                                                                                                                                                                  | Then the Dual Message Authorization System...                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request Response/0110 message contains data in DE 108, subelement 10 (Account Owner Information 2) defined with an incorrect length and/or is incorrectly formatted (Data Representations).             | will remove DE 108, subelement 10 (Account Owner Information 2) and the message continues processing.                                                                       |
| an Authorization Request Response/0110 message contains subfields in DE 108, subelement 10 (Account Owner Information 2) defined with an incorrect length and/or is incorrectly formatted (Data Representations)         | will remove the incorrect subfields in DE 108, subelement 10 (Account Owner Information 2) and the message continues processing.                                            |
| an Authorization Request Response/0110 message contains DE 108, subelement 10 (Account Owner Information 2) with subfields that have multiple instances of the same subelement or subfield within the same data element. | sends an Authorization Response Negative Acknowledgment/0190 message to the issuer with: <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 108</li></ul> |

## Subfield 01 (Account Owner First Name)

DE 108, subelement 10, subfield 01 (Account Owner First Name) contains the first name of the account owner.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 01                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |

### Values

If present, cannot contain all spaces or all numeric values.

### Application notes

Subfield 01 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subfield 02 (Account Owner Middle Name)

DE 108, subelement 10, subfield 02 (Account Owner Middle Name) contains the middle name of the account owner.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 02                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |

### Values

If present, cannot contain all spaces or all numeric values.

### Application notes

Subfield 02 is optional if the issuer is opted-in for the on-behalf Name Match Service.

### Subfield 03 (Account Owner Last Name)

DE 108, subelement 10, subfield 03 (Account Owner Last Name) contains the last name of the account owner.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 03                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |

### Values

If present, cannot contain all spaces or all numeric values.

### Application notes

Subfield 03 is optional if the issuer is opted-in for the on-behalf Name Match Service.

### Subfield 04 (Account Owner Sequence Number)

DE 108, subelement 10, subfield 04 (Account Owner Sequence Number) contains the sequence number of the account owner.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subfield ID         | 04                        |
| Data Representation | n-3                       |
| Subfield Length     | 2                         |
| Data Field          | Contents of positions 1-3 |
| Justification       | N/A                       |

## Values

If present, must be numeric.

## Application notes

Subfield 04 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subfield 05 (Account Owner Classification)

DE 108, subelement 10, subfield 05 (Account Owner Classification) contains the classification category (Personal or Business) of the account owner.

## Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Subfield ID         | 05                     |
| Data Representation | a-1                    |
| Subfield Length     | 2                      |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

If present, must contain one of these valid values.

## Values

| Value | Description |
|-------|-------------|
| P     | Personal    |
| B     | Business    |

## Application notes

Subfield 05 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subelement 11 (Account Owner Information 3)

DE 108, subelement 11 (Account Owner Information 3) contains the details of the account owner for a given PAN. The account owner information is used for the name validation service.

### Attributes

| Attribute           | Description                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subelement ID       | 11                                                                                                                                                                                                                      |
| Data Representation | ans...129; LLLVAR                                                                                                                                                                                                       |
| Subelement Length   | 3                                                                                                                                                                                                                       |
| Data Field          | Contents of subfields 01-05                                                                                                                                                                                             |
| Subfields           | Subfield 01 (Account Owner First Name)<br>Subfield 02 (Account Owner Middle Name)<br>Subfield 03 (Account Owner Last Name)<br>Subfield 04 (Account Owner Sequence Number)<br>Subfield 05 (Account Owner Classification) |
| Justification       | See subfields                                                                                                                                                                                                           |

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

### Usage

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | .   |

## Authorization Response/0110 message edits for DE 108, subelement 11

Mastercard will perform the following edits.

### Associated edits

| When...                                                                                                                                                                                                                  | Then the Dual Message Authorization System...                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request Response/0110 message contains subfields in DE 108, subelement 11 (Account Owner Information 3) defined with an incorrect length and/or is incorrectly formatted (Data Representations).        | will remove DE 108, subelement 11 (Account Owner Information 3) and the message continues processing.                                                                       |
| an Authorization Request Response/0110 message contains subfields in DE 108, subelement 09 (Account Owner Information 1) defined with an incorrect length and/or is incorrectly formatted (Data Representations)         | will remove the incorrect subfields in DE 108, subelement 11 (Account Owner Information 1) and the message continues processing.                                            |
| an Authorization Request Response/0110 message contains DE 108, subelement 11 (Account Owner Information 3) with subfields that have multiple instances of the same subelement or subfield within the same data element. | sends an Authorization Response Negative Acknowledgment/0190 message to the issuer with: <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 108</li></ul> |

### Subfield 01 (Account Owner First Name)

DE 108, subelement 11, subfield 01 (Account Owner First Name) contains the first name of the account owner.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 01                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |



### Values

If present, cannot contain all spaces or all numeric values.

### Application notes

Subfield 01 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subfield 02 (Account Owner Middle Name)

DE 108, subelement 11, subfield 02 (Account Owner Middle Name) contains the middle name of the account owner.

### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subfield ID         | 02                         |
| Data Representation | ans...35; LLVAR            |
| Subfield Length     | 2                          |
| Data Field          | Contents of positions 1-35 |
| Justification       | N/A                        |

### Values

Valid values will consist of the middle name of the Account Owner.

### Application notes

Subfield 02 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subfield 03 (Account Owner Last Name)

DE 108, subelement 11, subfield 03 (Account Owner Last Name) contains the last name of the account owner.

### Attributes

| Attribute           | Description     |
|---------------------|-----------------|
| Subfield ID         | 03              |
| Data Representation | ans...35; LLVAR |
| Subfield Length     | 2               |

| Attribute     | Description                |
|---------------|----------------------------|
| Data Field    | Contents of positions 1-35 |
| Justification | N/A                        |

### Values

If present, cannot contain all spaces or all numeric values.

### Application notes

Subfield 03 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subfield 04 (Account Owner Sequence Number)

DE 108, subelement 11, subfield 04 (Account Owner Sequence Number) contains the sequence number of the account owner.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subfield ID         | 04                        |
| Data Representation | n-3                       |
| Subfield Length     | 2                         |
| Data Field          | Contents of positions 1-3 |
| Justification       | N/A                       |

### Values

If present, must be numeric.

### Application notes

Subfield 04 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## Subfield 05 (Account Owner Classification)

DE 108, subelement 11, subfield 05 (Account Owner Classification) contains the classification category (Personal or Business) of the account owner.

### Attributes

| Attribute           | Description            |
|---------------------|------------------------|
| Subfield ID         | 05                     |
| Data Representation | a-1                    |
| Subfield Length     | 2                      |
| Data Field          | Contents of position 1 |
| Justification       | N/A                    |

If present, must contain one of these valid values.

### Values

| Value | Description |
|-------|-------------|
| P     | Personal    |
| B     | Business    |

### Application notes

Subfield 05 is optional if the issuer is opted-in for the on-behalf Name Match Service.

## DE 109 (Reserved for ISO Use)

DE 109 (Reserved for ISO Use) is reserved for ISO use.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 110 (Additional Data-2)

DE 110 (Additional Data-2) is reserved for use based on product type.

### Attributes

| Attribute           | Description                  |
|---------------------|------------------------------|
| Data Representation | ans...999; LLLVAR            |
| Data Length         | 3 positions, value = ...999  |
| Data Field          | Contents of subelements 9–10 |
| Subelements         | 2                            |
| Justification       | See Subelements              |

### Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

### Applicable messages

| Message                                                | Org | Sys | Dst |
|--------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: PEK exchange          | •   | X   | C   |
| Network Management Request Response/0810: PEK exchange | O   | X   | •   |
| Network Management Advice/0820: PEK exchange           | •   | X   | C   |

### Application notes

DE 110 provides supplemental data in a message when a specific ISO-designated data element is not available. It is a free-format, variable-length alphanumeric data element that may be used for multiple purposes. This data element's content may vary by program and service.

The following tables provide formats and descriptions for the subelements in DE 110. Currently, there are two subelements but as subelements are added, the subelement sequence will not have to be in the order of ID value.

## Subelement 9 (ANSI X9 TR-31 Key Block Key [128-bit Key Block Protection Key])

DE 110, subelement 9 (ANSI X9 TR-31 Key Block Key [128-bit Key Block Protection Key]) contains the attributes provided in a Network Management Request/0800 message.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 09                        |
| Subelement Data Length | 3 positions               |
| Data Representation    | an-80                     |
| Data Field             | Contents of subelement 09 |
| Subfield               | N/A                       |
| Justification          | N/A                       |

### Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

### Applicable messages

| Message                                                | Org | Sys | Dst |
|--------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: PEK exchange          | •   | X   | C   |
| Network Management Request Response/0810: PEK exchange | O   | X   | •   |
| Network Management Request Advice/0820: PEK exchange   | •   | •   | •   |

## Subelement 9 (ANSI X9 TR-31 Key Block Key [192-bit Key Block Protection Key])

DE 110, subelement 9 (ANSI X9 TR-31 Key Block Key [192-bit Key Block Protection Key]) contains the attributes provided in a Network Management Request/0800 message.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 09                        |
| Subelement Data Length | 3 positions               |
| Data Representation    | an-96                     |
| Data Field             | Contents of subelement 09 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

### Applicable messages

| Message                                                | Org | Sys | Dst |
|--------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: PEK exchange          | •   | X   | C   |
| Network Management Request Response/0810: PEK exchange | O   | X   | •   |
| Network Management Request Advice/0820: PEK exchange   | •   | •   | •   |

## Subelement 10 (Key Check Value)

DE 110, subelement 10 (Key Check Value) contains the attributes in a Network Management Request/0800 message.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 10                        |
| Subelement Data Length | 3 positions               |
| Data Representation    | an-6                      |
| Data Field             | Contents of subelement 10 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

### Applicable messages

| Message                                                | Org | Sys | Dst |
|--------------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: PEK exchange          | •   | X   | C   |
| Network Management Request Response/0810: PEK exchange | O   | C   | •   |
| Network Management Request Advice/0820: PEK exchange   | •   | X   | C   |

## DE 110 (Encryption Data)

DE 110 (Encryption Data) is used in PIN encryption data and key management encryption data.

### Attributes

| Attribute           | Description     |
|---------------------|-----------------|
| Data Representation | b...999; LLLVAR |

| Attribute      | Description         |
|----------------|---------------------|
| Data Length    | b...999 (EBCDIC)    |
| Dataset ID     | 1 byte binary value |
| Dataset Length | 2 byte binary value |
| Data Field     | Dataset Tags        |
| Tag contents   | Values              |
| Justification  | N/A                 |

### Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

### Applicable messages

| Message                                           | Org | Sys | Dst |
|---------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                        | C   | X   | C   |
| Network Management Request/0800: PEK exchange     | .   | X   | C   |
| Network Management Request/0800: sign-on/sign-off | C   | X   | .   |
| Network Management Advice/0820: PEK exchange      | .   | X   | C   |

### Application notes

The length fields in the Tag Length Value (TLV) are coded according to ISO 8825. If the TLV-coded version of a dataset is used the length of the dataset itself is always coded as a two byte binary value.

- The Dataset ID is the first component in a dataset.
- Dataset ID is a one byte binary identifier given to each dataset within a DE 110 field.
- Dataset length is a two byte binary subfield that contains the total length of the TLV element(s) within the dataset.
- The Tag field contains one byte length hexadecimal code that identifies the content of the Value field.
- The Tag length field follows the ISO 8825 spec. n-byte field that defines the length of the Value field.
- The Value field is a variable-length field that will contain the data specified by the Tag field.



**DE 110 layout**

DE 110 (Encryption Data) is a usage layout to support the PCI Standards Security Council's mandate for ISO Format 4 PIN Blocks. The layout follows DE 110 specifications in ISO/DIS 13492.

**Hierarchy of DE 110 for PIN encryption data and key exchange data**

DE 110 contains:

- Dataset ID
- Dataset length
- Tags

Each Tag contains:

- Tag ID
- Tag length
- Tag value

**Entity descriptions**

| Entity         | Description                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| Dataset ID     | A one byte hex value specific for each data set<br>Example: 0x01                                                                   |
| Dataset length | A two byte hex value. Example: If the length is 24 it will be represented as 0x00 18. This length is the accumulation of all Tags. |
| Tag ID         | A one byte hex specific for each Tag: for example, 0x80                                                                            |

| Entity     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tag length | <p>The length field is coded according to ISO 8825. The Dataset length is always coded as a two byte binary value. ISO 8825 defines two forms length octets.</p> <ul style="list-style-type: none"> <li>Definite form</li> <li>Indefinite form (not supported by Mastercard)</li> </ul> <p>The definite form has a short and long form.</p> <p>Short form: Tag length consists of a single octet or byte. This form can only be used if the number of octets in the Tag data is less than or equal to 127 bytes. It is encoded as:</p> <ul style="list-style-type: none"> <li>Bit 8 is zero</li> <li>Bits 7 to 1 encode the number of octets or bytes in the contents octets (Tag data) as an unsigned binary integer with bit 7 as the most significant bit.</li> </ul> <p>Example: If the Tag data is 38 bytes the Tag length is encoded as:</p> <pre>00100110</pre> <p>Long form: the Tag length consists of an initial octet and one or more additional octets.</p> <p>The initial octet will be encoded as:</p> <ul style="list-style-type: none"> <li>Bit 8 is one</li> <li>Bits 7 to 1 shall encode the number of additional octets in the length octets, as an unsigned binary integer with bit 7 as the most significant bit the value 11111111 must not be used.</li> </ul> <p>Coding for additional octets:</p> <ol style="list-style-type: none"> <li>First additional octet bits 8 to 1</li> <li>Second additional octet bits 8 to 1</li> <li>Each additional octet bits 8 to 1</li> </ol> <p>Octets will be encoded as unsigned binary integer equal to the number of octets in the Tag data, with bit 8 of the first additional octet as the most significant bit.</p> <p>Examples:</p> <ul style="list-style-type: none"> <li>If the Tag data is 201 bytes the Tag length would be encoded as<br/> <pre>10000001 11001001</pre> <ul style="list-style-type: none"> <li>10000001: Initial byte which tells how many additional length bytes.</li> <li>11001001: The actual length byte.</li> </ul> </li> <li>If the tag data is 402 bytes the Tag length would be encoded as<br/> <pre>10000010 00000001 10010010</pre> </li> </ul> |

| Entity    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tag value | <p>If the data representation is <i>b</i> it is binary</p> <ul style="list-style-type: none"> <li>• <i>b</i>-1 is a value of 21 specified by 0x15</li> <li>• <i>b</i>-2 is a value of 21 specified by 0x0015</li> <li>• <i>b</i>-8 is a value of 21 specified by 0x00000000000000015</li> </ul> <hr/> <p>If the data representation is <i>n</i>, only right padded Binary-Coded Decimal (BCD) is allowed and only values 0 to 9.</p> <ul style="list-style-type: none"> <li>• <i>n</i>-4 is a value of 112 stored in BCD as 0x01, 0x12</li> <li>• <i>n</i>-8 is a value of 112 stored in BCD as 0x00, 0x00, 0x01, 0x12</li> </ul> |

## Dataset 01 (PIN Encryption)

DE 110, Dataset 01 (PIN Encryption) contains the personal identification number (PIN), a unique identifier used by the cardholder at the point-of-interaction (POI).

### Attributes

| Attribute           | Description                                                                                                                                                                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset ID          | b-1                                                                                                                                                                                                                                                                                                   |
| Dataset Length      | b-2                                                                                                                                                                                                                                                                                                   |
| Data Representation | b...999; LLLVAR                                                                                                                                                                                                                                                                                       |
| Data Field          | cryptogram                                                                                                                                                                                                                                                                                            |
| Number of Tags      | 7                                                                                                                                                                                                                                                                                                     |
|                     | TAG 80 (Control), Mandatory<br>TAG 81 (Key-set Identifier), Mandatory<br>TAG 83 (Algorithm), Conditional<br>TAG 86 (Key Index), Conditional<br>TAG 87 (PIN Block Format), Mandatory<br>TAG 88 (Encrypted PIN Block), Mandatory<br>TAG 89 (Additional Encrypted PIN Block [New PIN Data]), Conditional |

### Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

**Applicable messages**

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

**TAG 80 (Control)**

DE 110, Dataset 01, TAG 80 (Control) field identifies the key management scheme and associated structure of the remainder of the data element.

**Attributes**

| Attribute           | Description |
|---------------------|-------------|
| Tag ID              | 80          |
| Tag Data Length     | b-1         |
| Data Representation | b-1         |
| Justification       | N/A         |
| Required            | Mandatory   |

**Values**

| Value | Description                               |
|-------|-------------------------------------------|
| 00    | Default value for the Dual Message System |

**TAG 81 (Key-set Identifier)**

DE 110, Dataset 01, TAG 81 (Key-set Identifier) Key-set identifier field is used with DUKPT key management exclusively. The key-set identifier uniquely identifies the base derivation key as defined in the *American National Standard for Financial Services ...Retail Financial Services Symmetric Key Management Part 1: Using Symmetric Techniques (ANSI X9.24 Part 1)*.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Tag ID              | 81          |
| Tag Data Length     | b-1         |
| Data Representation | b-8         |
| Justification       | N/A         |
| Required            | Mandatory   |

### Values

| Value            | Description                               |
|------------------|-------------------------------------------|
| 0000000000000000 | Default value for the Dual Message System |

## TAG 83 (Algorithm)

DE 110, Dataset 01, TAG 83 (Algorithm) selects the encryption algorithm used to encipher the keys contained in the associated key management data element.

### Attributes

| Attribute           | Description                       |
|---------------------|-----------------------------------|
| Tag ID              | 83                                |
| Tag Data Length     | b-1                               |
| Data Representation | n-2 (Packed binary coded decimal) |
| Justification       | N/A                               |
| Required            | Conditional                       |

### Values

| Value | Description                     |
|-------|---------------------------------|
| 01    | Data Encryption Algorithm (DEA) |

| Value | Description                             |
|-------|-----------------------------------------|
| 03    | Triple Data Encryption Algorithm (TDEA) |
| 05    | Advanced Encryption Standard (AES)      |

### Application notes

TAG 83 is for online PIN transactions performed on the Authorization System for acquirers and issuers in the Europe region. It is not a requirement for acquirers outside of the Europe region.

## TAG 86 (Key Index)

DE 110, Dataset 01, TAG 86 (Key Index) indicates the specific PIN key to be used when more than one key is available in a PIN key set.

### Attributes

| Attribute           | Description                       |
|---------------------|-----------------------------------|
| Tag ID              | 86                                |
| Tag Data Length     | b-1                               |
| Data Representation | n-2 (Packed binary coded decimal) |
| Justification       | N/A                               |
| Required            | Conditional                       |

### Values

| Value | Description                                     |
|-------|-------------------------------------------------|
| 00-99 | PIN key index value for the Dual Message System |

### Application notes

TAG 86 is for online PIN transactions performed on the Authorization System for acquirers and issuers in the Europe region. It is not a requirement for acquirers outside of the Europe region.

## TAG 87 (PIN Block Format)

DE 110, Dataset 01, TAG 87 (PIN Block Format) is the format of the PIN Block defined in ISO 9564-1.

### Attributes

#### Attributes

| Attribute           | Description                       |
|---------------------|-----------------------------------|
| Tag ID              | 87                                |
| Tag Data Length     | b-1                               |
| Data Representation | n-2 (Packed binary coded decimal) |
| Justification       | N/A                               |
| Required            | Mandatory                         |

#### Values

| Value | Description            |
|-------|------------------------|
| 10    | ISO PIN Block Format 0 |
| 11    | ISO PIN Block Format 1 |
| 14    | ISO PIN Block Format 4 |

## TAG 88 (Encrypted PIN Block)

DE 110, Dataset 01, TAG 88 (Encrypted PIN Block) contains PIN block encryption as defined in ISO 9564-1.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Tag ID              | 88          |
| Tag Data Length     | b-1         |
| Data Representation | b-8 or b-16 |
| Justification       | N/A         |

| Attribute | Description |
|-----------|-------------|
| Required  | Mandatory   |

### Values

| Value                 | Description |
|-----------------------|-------------|
| 8 byte of PIN Block   | Format 0    |
| 16 bytes of PIN Block | Format 4    |

## TAG 89 (Additional Encrypted PIN Block)

DE 110, Dataset 01, TAG 89 (Additional Encrypted PIN Block) is used for PIN change transaction.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Tag ID              | 89          |
| Tag Data Length     | b-1         |
| Data Representation | b-8 or b-16 |
| Justification       | N/A         |
| Required            | Conditional |

### Values

| Value                            | Description |
|----------------------------------|-------------|
| 8 bytes of additional PIN Block  | Format 0    |
| 16 bytes of additional PIN Block | Format 4    |



## Dataset 04 (Key Exchange)

Dataset 04 (Key Exchange) is a method in cryptography by which cryptographic keys are exchanged between two parties, allowing use of a cryptographic algorithm.

### Attributes

| Attribute           | Description                            |
|---------------------|----------------------------------------|
| Dataset ID          | b-1                                    |
| Dataset Length      | b-2                                    |
| Data Representation | b...999; LLLVAR                        |
| Data Field Contains | cryptogram                             |
| Number of Tags      | 6                                      |
|                     | TAG 80 (Control), Mandatory            |
|                     | TAG 81 (Key-set Identifier), Mandatory |
|                     | TAG 83 (Algorithm), Mandatory          |
|                     | TAG 86 (Key Index), Mandatory          |

### Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

### Applicable messages

| Message                                           | Org | Sys | Dst |
|---------------------------------------------------|-----|-----|-----|
| Network Management Request/0800: PEK exchange     | •   | X   | C   |
| Network Management Request/0800: sign-on/sign-off | C   | X   | •   |
| Network Management Advice/0820: PEK exchange      | •   | X   | C   |

## TAG 80 (Control)

DE 110, Dataset 04, TAG 80 (Control) field identifies the key management scheme and associated structure of the remainder of the data element.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Tag ID              | 80          |
| Tag Data Length     | b-1         |
| Data Representation | b-1         |
| Justification       | N/A         |
| Required            | Mandatory   |

### Values

| Value | Description                               |
|-------|-------------------------------------------|
| 00    | Default value for the Dual Message System |

## TAG 81 (Key-set Identifier)

DE 110, Dataset 04, TAG 81 (Key-set Identifier) is a number that uniquely identifies a key-set.

### Attributes

| Attribute           | Description |
|---------------------|-------------|
| Tag ID              | 81          |
| Tag Data Length     | b-1         |
| Data Representation | b-8         |
| Justification       | N/A         |
| Required            | Mandatory   |

### Values

| Value            | Description                               |
|------------------|-------------------------------------------|
| 0000000000000000 | Default value for the Dual Message System |

## TAG 83 (Algorithm)

DE 110, Dataset 04, TAG 83 (Algorithm) defines the encryption algorithm used to encipher the keys contained in the associated key management data element.

### Attributes

| Attribute           | Description                       |
|---------------------|-----------------------------------|
| Tag ID              | 83                                |
| Tag Data Length     | b-1                               |
| Data Representation | n-2 (Packed binary coded decimal) |
| Justification       | N/A                               |
| Required            | Mandatory                         |

### Values

| Value | Description                             |
|-------|-----------------------------------------|
| 01    | Data Encryption Algorithm (DEA)         |
| 03    | Triple Data Encryption Algorithm (TDEA) |
| 05    | Advanced Encryption Standard (AES)      |

## TAG 86 (Key Index)

DE 110, Dataset 04, TAG 86 (Key Index) is used when multiple keys are identified with the same key set identifier; for example, key rotation.

### Attributes

| Attribute           | Description                       |
|---------------------|-----------------------------------|
| Tag ID              | 86                                |
| Tag Data Length     | b-1                               |
| Data Representation | n-2 (Packed binary coded decimal) |
| Justification       | N/A                               |
| Required            | Mandatory                         |

### Values

| Value | Description                                                                                    |
|-------|------------------------------------------------------------------------------------------------|
| 00    | Network Management Request/0800: PEK exchange<br>Network Management Request/0820: PEK exchange |
| 00-99 | Network Management Request/0800: sign-on/sign-off                                              |

## Edits

The Dual Message Authorization System will perform the following edits.

### Authorization Request/0100 message

Dual Message Authorization System edits for DE 110 (Encryption Data) are listed.

#### Associated edits

| When...                                                                                                                                                                                            | Then the Dual Message Authorization System...                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 110, Dataset 01 (PIN Encryption), and TAG 80 (Key Management scheme) is not present in Dataset 01                                                                                               | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 110 (indicating DE 110 is in error)</li></ul> |
| DE 110, Dataset 01, and TAG 81 (Key Set Identifier) is not present in Dataset 01                                                                                                                   | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 110 (indicating DE 110 is in error)</li></ul> |
| DE 110, Dataset 01, and TAG 87 (PIN Block Format Code) is not present in Dataset 01                                                                                                                | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 110 (indicating DE 110 is in error)</li></ul> |
| DE 110, Dataset 01, and TAG 88 (Encrypted PIN Block) is not present in Dataset 01                                                                                                                  | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 110 (indicating DE 110 is in error)</li></ul> |
| an Authorization Request/0100 message contains DE 110, Dataset 01 and DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is not 02, 05, 07, 80, 81, 82, 90, or 91 | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 110 (indicating DE 110 is in error)</li></ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Then the Dual Message Authorization System...                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 message of DE 03 (Processing Code), is 92 (PIN Change) and does not contain DE 110, Dataset 01, TAG 89 (New PIN Data)                                                                                                                                                                                                                                                                                                                                                                                                                                  | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = DE 125 (New PIN Data)</li> </ul>               |
| an Authorization Request/0100 message with DE110, Dataset 01, Tag 88 is present and acquirer keys managed by Authorization, TAG 83 or TAG 86 not present                                                                                                                                                                                                                                                                                                                                                                                                                             | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 110 (indicating DE 110 is in error)</li> </ul> |
| an Authorization Request/0100 message DE 110, Dataset 01, TAG 87 is not 01, 02, 03, 10, 11, 14, or 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | will return to the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 110 (indicating DE 110 is in error)</li> </ul> |
| an Authorization Request/0100 message contains both: <ul style="list-style-type: none"> <li>DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader) and</li> <li>DE 52 (Personal ID Number [PIN] Data) or DE 110, Dataset 01, TAG 88</li> </ul>                                                                                                                                                                                                                                                            | rejects the message and sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 58 (Transaction not permitted to acquirer/terminal)</li> </ul>                                |
| an Authorization Request/0100 message for a Mastercard Electronic card contains:<br><br>DE 22, subfield 1, value 02, 03, 05, 07, 09, 80, 90, or 91, and DE 18 (Merchant Type) MCC value is 5542 (Fuel Dispensers, Automated), and DE 52 (Personal ID Number [PIN] Data), and DE 110, Dataset 01, TAG 88 are not present<br><br>or<br><br>DE 55 (Integrated Circuit Card [ICC] System-related Data) is present, and DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder Activated Terminal Level) is not 1 (Authorized Level 1 CAT: Automated dispensing machine with a PIN) | sends an Authorization Request Response/0110 message to the acquirer where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 58 (Transaction not permitted to acquirer/terminal)</li> </ul>                                                     |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Authorization Request/0100 message for a Cardless ATM transaction:</p> <ul style="list-style-type: none"> <li>• Contains DE 18 (Merchant Type), value 6011 (Member Financial Institution: Automated Cash Disbursements), and</li> <li>• DE 48 (Additional Data: Private Data), Transaction Category Code (TCC), value Z (ATM), and</li> <li>• DE 22, subfield 1 (POS Terminal PAN Entry Mode) is present and contains a value other than 09 (PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55), but DE 52, and</li> <li>• DE 110, Dataset 01, TAG 88 is not present</li> </ul> | <p>rejects the message and sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 52 (indicating the data element in error)</li> </ul> |
| <p>an Authorization Request/0100 message if</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 91 (PIN unblock) or 92 (PIN Change) and</li> <li>• DE 52 and DE 110, Dataset 01, TAG 88 is not present</li> </ul>                                                                                                                                                                                                                                                                                                                                                                      | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 52 (indicating the data element in error).</li> </ul>                        |
| <p>the Authorization Request/0100 message contains</p> <ul style="list-style-type: none"> <li>• DE 61, subfield 10, value 1 and</li> <li>• DE 52 and DE 110, Dataset 01, TAG 88 is not present and</li> <li>• DE 55 is present</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       | <p>forwards the Authorization Request/0100 message to the issuer.</p>                                                                                                                                                                                                                         |
| <p>the Authorization Request/0100 message contains</p> <ul style="list-style-type: none"> <li>• DE 61, subfield 10, value 1 and</li> <li>• DE 52 or DE 110, Dataset 01, TAG 88 is present and</li> <li>• DE 55 is present</li> </ul>                                                                                                                                                                                                                                                                                                                                                                            | <p>forwards the Authorization Request/0100 message to the issuer.</p>                                                                                                                                                                                                                         |
| <p>the Authorization Request/0100 message contains</p> <ul style="list-style-type: none"> <li>• DE 61, subfield 10, value 1 and</li> <li>• DE 52 or DE 110, Dataset 01, TAG 88 is present and</li> <li>• DE 55 is not present</li> </ul>                                                                                                                                                                                                                                                                                                                                                                        | <p>forwards the Authorization Request/0100 message to the issuer.</p>                                                                                                                                                                                                                         |

| When...                                                                                                                                                        | Then the Dual Message Authorization System...                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an authorization Request/0100 message, DE 3, subfield 1 contains value 92 (PIN Change) and DE 125 (New PIN Data) and DE 110, Dataset 01, TAG 89 is not present | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 125 (indicating the data element in error).</li> </ul> |

## DE 111 (Reserved for ISO Use)

DE 111 (Reserved for ISO Use) is reserved for ISO use.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 112 (Additional Data [National Use])

DE 112 (Additional Data [National Use]) is reserved for national organizations to define data unique to specific networks or specific programs and services. DE 112 provides other supplemental data in a message when a specific ISO-designated data element is not available. It is a variable-length, alphanumeric data element used for information on transactions between customers.

### Attributes

| Attribute           | Value                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------|
| Data Representation | a-1 (for Hungary domestic usage only for processing transactions for Employee Benefits Program). |
| Data Length         | 3 positions                                                                                      |
| Data Field          | Contents of subelements                                                                          |
| Subelements         | Determined by program                                                                            |
| Justification       | N/A                                                                                              |

### Usage

Following is the usage of DE 112 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.



### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | C   | X   | C   |
| Authorization Request Response/0110                                                             | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated                                                   | C   | X   | C   |
| Authorization Advice/0120: system-generated                                                     | .   | X   | C   |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | C   | X   | C   |
| Reversal Request/0400                                                                           | C   | X   | C   |
| Reversal Request Response/0410                                                                  | C   | X   | C   |

### Values

See subfields.

### Application notes

**NOTE: Mastercard does not perform currency conversion on any amounts in DE 112.**

## DE 112 (Encoding Scheme)

Mastercard organizes DE 112 into a group of encoded subelements. The following table illustrates the structure of DE 112.

### DE structure

| VAR: maximum length varies by program; refer to Data Representation previously for maximum length by program |                            |           |                         |                             |           |                         |
|--------------------------------------------------------------------------------------------------------------|----------------------------|-----------|-------------------------|-----------------------------|-----------|-------------------------|
| LLL                                                                                                          | 3 bytes                    | 3 bytes   | variable by program     | 3 bytes                     | 3 bytes   | variable by program     |
| Total Length                                                                                                 | First Subelement (SE) Data |           |                         | Second Subelement (SE) Data |           |                         |
|                                                                                                              | SE ID                      | SE Length | SE Variable Length Data | SE ID                       | SE Length | SE Variable Length Data |

| Number of Bytes | Attribute                       | Description                                                             |
|-----------------|---------------------------------|-------------------------------------------------------------------------|
| 3               | Total Length field              | The “LLL” portion of the data element up to maximum allowed by program. |
| 3               | Subelement ID                   | In the range 000–099                                                    |
|                 |                                 | 000–069      Defined universally for all programs and services.         |
|                 |                                 | 070–099      Defined for individual programs and services.              |
| 3               | Subelement Length               | Varies by program.                                                      |
| 1...nnn         | Subelement Variable Length Data | Contains valid values.                                                  |

## DE 112 (Dual Message Authorization System Edits)

The Dual Message Authorization System will perform the following system edit to verify proper formatting of DE 112.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or Reversal Request/0400 message contains DE 112 (Additional Data [National Use]) with subelements that have incorrect length and/or incorrect format (Data Representations), or a subelement ID appears more than once within the same data element | sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 44 is 6 positions for subelement format errors: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30</li> <li>DE 44 (Additional Response Data) = 112nnn</li> </ul> |
| DE 112, subelement 045 (Additional Commercial Issuer Data) is not properly formatted                                                                                                                                                                                                                                               | rejects the Authorization Request Response/0110 message and respond back to the Issuer with an Authorization Response Negative Acknowledgment/0190 message with DE 39 (Response Code) = 30 (Format error) and DE 44 (Additional Response Data) = 112.                                                                             |
| DE 112, subelement 045 (Additional Commercial Issuer Data) is provided for non-commercial cards                                                                                                                                                                                                                                    | removes the subelement before forwarding the Authorization Request Response/0110 message to the acquirer.                                                                                                                                                                                                                         |

| WHEN...                                                         | THEN the Dual Message Authorization System... |
|-----------------------------------------------------------------|-----------------------------------------------|
| Stand-In processing occurs on a commercial card B2B transaction | will not provide DE 112, subelement 045.      |

### Examples

- When an edit error occurs on DE 112, the DE 44 data will be populated as these examples below:
  - Error on DE 112 subelement 45, subfield 1: 11245 (no subfield information is provided)

For non-DE 112 format errors, DE 44 will only have DE info and no subelement information. For example, the format error in DE 22 will have DE 44 as 022.

## Subelement 042 (Acceptor Tax ID)

Subelement 042 (Acceptor Tax ID) will be used in the Dual Message Authorization System to support DE 112 in any received Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message.

### Attributes

| Attributes          | Value                              |
|---------------------|------------------------------------|
| Subelement ID       | 042                                |
| Data Length         | 3 positions                        |
| Data Length Value   | Variable length, 001–012 positions |
| Data Representation | an...012; LLLVAR                   |
| Data Field          | Contents of subelement 042         |
| Justification       | N/A                                |

### Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | .   |
| Authorization Advice/0120: acquirer-generated | C   | X   | .   |

| Message                | Org | Sys | Dst |
|------------------------|-----|-----|-----|
| Reversal Request /0400 | C   | X   | .   |

### Values

| Field name  | Comment                                          |
|-------------|--------------------------------------------------|
| Description | Tax ID                                           |
| Base ID     | n-11 (no dots or dashes), left filled with zeros |
| Check digit | an-1                                             |

### Application notes

This field is required when purchase, purchase with cash back, and refund transactions are submitted by acquirers for merchants in Chile and Paraguay.

The length value is defined as variable length for all markets. However, Chile and Paraguay should use 12 positions exclusively with leading zeros as shown in the examples.

### Examples

| Tax ID       | Format required |
|--------------|-----------------|
| 18.638.773-K | 00018638773K    |
| 7.932.677-1  | 000079326771    |
| 763.567-4    | 000007635674    |

## Subelement 045 (Additional Commercial Issuer Data)

DE 112, subelement 045 is used in the Dual Message Authorization System to provide data required by acquirers and issuers for intracountry processing to be compliant with local regulation for corporate issuer data for B2B transactions.

DE 112, subelement 045 must not be provided in cross-border transactions.

### Attributes

| Attribute     | Description |
|---------------|-------------|
| Subelement ID | 045         |

| Attribute              | Description                                                                                                                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Subelement Data Length | 3 positions                                                                                                                       |
| Data Representation    | ans...075; LLLVAR                                                                                                                 |
| Data Field             | Contents of subfields 01-03                                                                                                       |
| Subfields              | Subfield 01 (Taxpayer Country Code)<br>Subfield 02 (Taxpayer Identification Number)<br>Subfield 03 (Tax Registration Reason Code) |

## Usage

This is the usage of DE 112, subelement 045 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message type identifier             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

## Subfield 01 (Taxpayer Country Code)

DE 112, subelement 045, subfield 01 is for 3-position alphabetic taxpayer country code of the business buyer, legal entity responsible for the cardholder contract with the issuer of the business or corporate card.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | a...3; LLVAR            |
| Data Field           | Contents of subfield 01 |
| Justification        | Left                    |

## Subfield 02 (Taxpayer Identification Number)

DE 112, subelement 045, subfield 02 is for unique tax identification code of the business buyer, assigned by the tax authority.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...30; LLVAR         |
| Data Field           | Contents of subfield 02 |
| Justification        | N/A                     |

## Subfield 03 (Tax Registration Reason Code)

DE 112, subelement 045, subfield 03 is for tax register reason code of the business buyer, assigned by the tax authority.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 03                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | ans...30; LLVAR         |
| Data Field           | Contents of subfield 03 |
| Justification        | N/A                     |

## Subelement 055 (Brazil Benefits Payment Data)

DE 112, subelement 055 (Brazil Benefits Payment Data) contains data to support processing of intracountry transactions for the Mastercard Benefits Payment Scheme in Brazil.

### Attributes

| Attributes             | Description                   |
|------------------------|-------------------------------|
| Subelement ID          | 055                           |
| Subelement Data Length | 3 positions                   |
| Data Representation    | an...11; LLVAR                |
| Data Field             | Contents of subfields 01 - 02 |
| Subfields              | 2                             |
| Justification          | See subfields                 |

### Usage

Following is the usage of subelement 055 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | C   | P   | C   |

### Values

See subfields.

## Subfield 01 (PAT Eligibility)

DE 112, subelement 055, subfield 01 is provided by acquirers to identify merchants that are PAT (Programa de Alimentação do Trabalhador) eligible when a Brazil domestic transaction is processed as part of the Mastercard Benefits Payment Scheme.

### Attributes

| Attributes           | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | a-1                     |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Values

| Value | Description  |
|-------|--------------|
| Y     | Eligible     |
| N     | Not Eligible |

## Subfield 02 (Benefit Program Code)

DE 112, subelement 055, subfield 02 is provided by issuers to identify the specific benefit program applicable to a Brazil domestic transaction when processed within the Mastercard Benefits Payment Scheme.

### Attributes

| Attributes           | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-2                     |
| Data Field           | Contents of subfield 02 |



| Attributes    | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Value | Description                                     |
|-------|-------------------------------------------------|
| 01    | Meal Voucher: Employee Nutrition Program (PAT)  |
| 02    | Food Voucher: Employee Nutrition Program (PAT)  |
| 03    | Culture Voucher: Worker's Culture Program (PCT) |
| 04    | Meal Voucher: Consolidation of Labor Laws (CLT) |
| 05    | Food Voucher: Consolidation of Labor Laws (CLT) |

### Application notes

Acquirers should use subfield 01 (PAT Eligibility) in the Authorization Request/0100 message to indicate merchant PAT eligibility.

Issuers should use subfield 02 (Benefit Program Code) in the Authorization Request Response/0110 message to indicate the specific benefit program applicable to a transaction.

It is not required for issuers to echo back the subfield 01 in the Authorization Request Response/0110 message, however if provided, it will be passed to the acquirer.

## Subelement 60 (Mastercard Network Gateway Services Platform [MNGS] Transport Data) (Mastercard use only)

DE 112, subelement 060 (Mastercard Network Gateway Services Platform [MNGS] Transport Data) (Mastercard use only), is reserved solely for MNGS (Mastercard Network Gateway Services Platform) transport data.

### Attributes

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement ID          | 060                         |
| Subelement Data Length | 3 positions                 |
| Data Representation    | n-399                       |
| Data Field             | Contents of positions 1–399 |

| Attribute     | Description |
|---------------|-------------|
| Subfields     | N/A         |
| Justification | N/A         |

### Usage

Following is the usage of subelement 060 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Request/0100                  | C   | P   | C   |
| Authorization Request Response/0110         | C   | P   | C   |
| Authorization Advice/0120: system-generated | .   | X   | C   |
| Reversal Request/0400                       | C   | P   | C   |
| Reversal Request Response/0410              | C   | P   | C   |

### Values

This subfield contains MNGS specific application transport data.

### Application notes

MNGS acquired transactions requiring certain services fulfillment will populate subelement 60 to carry application specific data.

## All Regions: Installment Payment Transactions

Following are the technical specifications for Authorization/01xx messages for the global Mastercard Installment Payment Service. Refer to the [Mastercard Installment Payment Service User Guide](#) for additional details.

### Subelement 021 (Installment Payment Data 1)

Subelement 021 (Installment Payment Data 1) is obsolete.

**NOTE: The Dual Message Authorization System currently does not use this subelement.**

## Subelement 022 (Installment Payment Data)

Subelement 022 (Installment Payment Data) is used in the Authorization System for installment payment transactions.

### Attributes

| Attribute              | Description             |
|------------------------|-------------------------|
| Subelement ID          | 022                     |
| Subelement Data Length | 3 positions             |
| Data Representation    | ans...040; LLLVAR       |
| Data Field             | Contents of subfield 01 |
| Subfields              | 1                       |
| Justification          | See subfields           |

### Usage

Following is the usage of subelement 022 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | .   | X   | C   |
| Authorization Request Response/0110 | C   | X   | C   |

## Subfield 01 (Installment Plan ID)

DE 112, subelement 022, subfield 01 (Installment Plan ID) contains a unique value that identifies a specific Mastercard installment transaction during authorization processing, for participating issuers that opt in to receive it.

### Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 01          |

| Attribute           | Description             |
|---------------------|-------------------------|
| Subfield length     | 2 positions             |
| Data Representation | ans-36                  |
| Data Field          | Contents of subfield 01 |
| Justification       | N/A                     |

### Values

Mastercard will populate DE 112, subelement 022, subfield 01 with a unique, 36 position Mastercard Installments Payment Services installment plan ID value.

### Example value

3d56f5a7-db48-41f0-875e-30920e8465f3

### Application notes

Acquirers and issuers must never include DE 112, subelement 022, subfield 01 in messages sent to Mastercard.

Mastercard will add DE 112, subelement 022, subfield 01 to Authorization Request/0100 messages for qualified transactions associated with issuers that participate in Mastercard Installments Payment Services. If an issuer includes DE 112, subelement 022, subfield 01 in an Authorization Request Response/0110 message, Mastercard will forward data to the acquirer. However, the acquirer should ignore this data when processing response messages.

## Subelement 023 (Installment Payment Data 3)

Subelement 023 (Installment Payment Data 3) is obsolete.

**NOTE: The Dual Message Authorization System currently does not use this subelement.**

### Alternate Processing

Installment payments are not eligible for alternate processing (Stand-In System or X-Code System). When an issuer is not available, or offline, the Installment Payment transactions will be

declined using the Authorization Request Response/0110 message containing DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

## Brazil: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Brazil payment transactions within Brazil.

### Subelement 012 (Brazil Commercial and Financing Data)

DE 112, subelement 012 (Brazil Commercial and Financing Data) contains additional data to support financing data provided by the acquirer, merchant, or issuer.

**NOTE: For data requirements for Brazil intracountry transactions using the Mastercard BNDES card, refer to the section that follows. The following table applies to non-BNDES transactions only.**

#### Attributes

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement ID          | 012                         |
| Subelement Data Length | 3 positions                 |
| Data Representation    | ans-170                     |
| Data Field             | Contents of positions 1–170 |
| Subfields              | N/A                         |
| Justification          | N/A                         |

#### Usage

#### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |
| Reversal Request/0400      | C   | X   | C   |

## Values

| Field Name/Positions                             | Type         | Size      | Comment                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------|--------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Financing Type<br>(Positions 1–20)               | Alpha        | 20        | Type of loan operation, agreed between Merchant and issuer. Examples: AGRO-CUSTEIO, AGRO-INVEST, FINAME, CBN.                                                                                                                                                                                                                                           |
| Buyer ID Type<br>(Position 21)                   | Alpha        | 1         | Possible values F or J:<br><br>F = pessoa Física (person/consumer)<br><br>J = pessoa Jurídica (company/commercial)                                                                                                                                                                                                                                      |
| Buyer ID<br>(Positions 22–35)                    | Numeric      | 14        | Merchant to fill in with cardholder's CPF (Cadastro de Pessoa Física, a registration number provided by the Brazil government to all persons living in Brazil) or company's CNPJ (Cadastro Nacional da Pessoa Jurídica, registration number provided by the Brazil government to all merchants) number or issuers to supply with company's CNPJ number. |
| Buyer Phone Number<br>(Positions 36–46)          | Numeric      | 11        | Merchant to fill in with buyer phone number.                                                                                                                                                                                                                                                                                                            |
| Purchase Identification<br>(Positions 47–66)     | Alphanumeric | 20        | Invoice data or purchase request data, depending on agreement between merchant and issuer.                                                                                                                                                                                                                                                              |
| Installments Cycle<br>(Positions 67–68)          | Numeric      | 2         | Different amortization types than monthly. For future use, it does not affect transaction clearing and settlement.                                                                                                                                                                                                                                      |
| Interest Rate<br>(Positions 69–86)               | Numeric      | 18 (12.6) | Interest rate on loan. For future use.                                                                                                                                                                                                                                                                                                                  |
| Grace Period<br>(Positions 87–89)                | Numeric      | 3         | Number of amortization cycles to wait before charging the installments. For future use, it does not affect transaction clearing and settlement.                                                                                                                                                                                                         |
| Grace Period Cycle<br>(Positions 90–92)          | Numeric      | 3         | Allow different grace period types than monthly. For future use, it does not affect transaction clearing and settlement.                                                                                                                                                                                                                                |
| Grace Period Interest Rate<br>(Positions 93–110) | Numeric      | 18 (12.6) | Interest rate on grace period. For future use, it does not affect transaction clearing and settlement.                                                                                                                                                                                                                                                  |
| Reference Field 1<br>(Positions 111–130)         | Numeric      | 20        | Acquirer to populate with merchant's CNPJ number.                                                                                                                                                                                                                                                                                                       |

| Field Name/Positions                     | Type         | Size | Comment                                                                 |
|------------------------------------------|--------------|------|-------------------------------------------------------------------------|
| Reference Field 2<br>(Positions 131–150) | Alpha        | 20   | Data exchange between merchant and issuer, to identify the transaction. |
| Reference Field 3<br>(Positions 151–170) | Alpha        | 20   | Data exchange between merchant and issuer, to identify the transaction. |
| Filler Data<br>(Positions 171–227)       | Alphanumeric | 57   | Reserved for future Brazil domestic enhancements.                       |

### Mastercard Brazil BNDES Card Data Requirements

Mastercard has enhanced the Mastercard Brazilian Development Bank (BNDES) Card to increase card issuance and usage in Brazil and support alternative financing options for small and medium-sized businesses.

Mastercard supports the presence of additional data fields for Brazil intracountry transactions submitted with product code MLD (Mastercard Distribution Card). This addition meets BNDES requirements, in terms of addendum data processed and sent by the BNDES generic module, through the acquirers.

The Mastercard BNDES Card is a solution to access a credit line with subsidized rates by BNDES to finance investments for micro-, small-, and medium-sized businesses established in Brazil.

This product is developed for Brazil domestic use only and transactions are sent to the Mastercard Network using the MLD (Mastercard Distribution Card) product code with a specific promotion code (BNDES1) in all transactions. These are issuer-financed installment (up to 48 installments) transactions, made exclusively through the BNDES Operations Portal.

BNDES has enhanced its Operations Portal's Generic Module, which requires acquirers to send additional data through the Mastercard Network.

**NOTE: All Brazil domestic transactions using the Mastercard BNDES card must contain DE 63 (Network Data), subfield 1 (Financial Network Code), value MLD (Mastercard Distribution Card) in the authorization message. The specific promotion code value BNDES1 must be included in DE 48 (Additional Data—Private Use), subelement 95 (Mastercard Promotion Code).**

In issuer-financed installment billing, the acquirer submits a single clearing record for the full transaction amount. The issuer then bills the cardholder for the installments in accordance with the terms agreed by the cardholder at the point of sale.

**NOTE: A customer's CNPJ number is the registration number provided by the Brazil government to all merchants.**

DE 112, subelement 012 (Brazil Commercial and Financing Data) contains additional data to support Brazil intracountry, consumer credit transactions submitted with value MLD (Mastercard Distribution Card) in DE 63 (Network Data), subfield 1 (Financial Network Code).

For Brazil intracountry transactions using the Mastercard BNDES card, the following data must be included.

### Attributes

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement ID          | 012                         |
| Subelement Data Length | 3 positions                 |
| Data Representation    | ans-227                     |
| Data Field             | Contents of positions 1–227 |
| Subfields              | N/A                         |
| Justification          | N/A                         |

### Values

- Data Element: DE 112 (Additional Data [National Use])
- Subelement: Subelement 012 (Brazil Commercial and Financing Data)

### Positions and value

| Positions                             | Value                                   |
|---------------------------------------|-----------------------------------------|
| Positions 1–20 (Financing Type)       | CBN                                     |
| Position 21 (Buyer ID Type)           | J                                       |
| Positions 22–35 (Buyer ID)            | CNPJ number (Buyer Company Tax ID)      |
| Positions 36–110                      | Spaces                                  |
| Positions 111–130 (Reference Field 1) | CNPJ number (Merchant Brazilian Tax ID) |
| Positions 131–227                     | Spaces                                  |

## Subelement 013 (Crediário First Simulation)

DE 112 (Additional Data [National Use]), subelement 013 contains the issuer's payment conditions for the first Crediário simulation.

### Attributes

| Attribute     | Description |
|---------------|-------------|
| Subelement ID | 013         |



| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement Data Length | 3 positions                 |
| Data Representation    | n-131                       |
| Data Field             | Contents of positions 1–131 |
| Subfields              | N/A                         |
| Justification          | N/A                         |

### Usage

Following is the usage of subelement 013 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

### Values

| Field Name/Positions                         | Type    | Size | Comment                                                                                                                                                     |
|----------------------------------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment Plan Type<br>(Positions 1–2)     | Numeric | 2    | Identifies whether this is an issuer, a merchant, or a Crediario financed installment plan.<br><br>Value 25 = Crediario                                     |
| Total Number of Installments (Positions 3–4) | Numeric | 2    | Number of installments the cardholder chose to pay.<br><br>From 02–99                                                                                       |
| Installment Amount<br>(Positions 5–16)       | Numeric | 12   | Value that the cardholder is going to pay per month. Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places. |

| Field Name/Positions                                 | Type    | Size | Comment                                                                                                                                                                                               |
|------------------------------------------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Amount (Positions 17–28)                       | Numeric | 12   | Total Installment Amount (Monthly installment amount multiplied by the Number of Installments).<br><br>Transaction total amount issuer calculates, including calculated interest; two decimal places. |
| Monthly Interest Rate (Positions 29–32)              | Numeric | 4    | Interest Rates the Issuer calculates and charges per month; two decimal places.                                                                                                                       |
| Annual Interest Rate (Positions 33–37)               | Numeric | 5    | Interest Rates the issuer calculates and charges per year; two decimal places.                                                                                                                        |
| Monthly Total Effective Cost (CET) (Positions 38–49) | Numeric | 12   | Total amount charged monthly to cardholder including the payments, interest, fees, insurance, and other amounts; two decimal places.                                                                  |
| % Total Effective Cost (Positions 50–61)             | Numeric | 12   | Total Effective Cost, expressed as a percentage; two decimal places.                                                                                                                                  |
| Date of first installment (Positions 62–67)          | Numeric | 6    | Date of the first installment payment in yymmdd (year, month, day) format.                                                                                                                            |
| Taxes (Positions 68–74)                              | Numeric | 7    | All taxes, if applicable, that the cardholder is paying, including IOF; two decimal places.                                                                                                           |
| % Taxes of Total Amount (Positions 75–78)            | Numeric | 4    | Percentage of the whole purchase price that the taxes represent.                                                                                                                                      |
| Fee (Positions 79–85)                                | Numeric | 7    | All fees, if applicable, that the cardholder is paying; two decimal places.                                                                                                                           |
| % Fee of Total Amount (Positions 86–89)              | Numeric | 4    | Percentage of the whole purchase price that the fees represent.                                                                                                                                       |
| Insurance (Positions 90–96)                          | Numeric | 7    | Any insurance, if applicable, that the cardholder is paying; two decimal places.                                                                                                                      |
| % Insurance of Total Amount (Positions 97–100)       | Numeric | 4    | Percentage of the whole purchase price that the insurance represents.                                                                                                                                 |
| Other (Positions 101–107)                            | Numeric | 7    | Any other added values, if applicable, that the cardholder is paying; two decimal places.                                                                                                             |
| % Other of Total Amount (Positions 108–111)          | Numeric | 4    | Percentage of the whole purchase price that any other added values represent.                                                                                                                         |
| Total Amount to Merchant (Positions 112–123)         | Numeric | 12   | Total amount that will be paid to the merchant (purchase amount); two decimal places.                                                                                                                 |

| Field Name/Positions                                        | Type    | Size | Comment                                                           |
|-------------------------------------------------------------|---------|------|-------------------------------------------------------------------|
| % Amount to Merchant of Total Amount<br>(Positions 124–127) | Numeric | 4    | Percentage of the total amount that will be paid to the merchant. |
| Reserved for Future Use<br>(Positions 128–131)              | Numeric | 4    |                                                                   |

### Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting the first option for the Crediário installment program and services.

Information not populated or optional to a specific field should be zero-filled.

### Subelement 014 (Crediário Second Simulation)

DE 112 (Additional Data [National Use]), subelement 014 contains the issuer's payment conditions for the second Crediário simulation.

### Attributes

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement ID          | 014                         |
| Subelement Data Length | 3 positions                 |
| Data Representation    | n-131                       |
| Data Field             | Contents of positions 1–131 |
| Subfields              | N/A                         |
| Justification          | N/A                         |

### Usage

Following is the usage of subelement 014 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | .   | C   |

| Message                                     | Org | Sys | Dst |
|---------------------------------------------|-----|-----|-----|
| Authorization Advice/0120: system-generated | .   | X   | C   |

## Values

| Field Name/Positions                                 | Type    | Size | Comment                                                                                                                                                                                               |
|------------------------------------------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment Plan Type<br>(Positions 1–2)             | Numeric | 2    | Identifies whether this is an issuer, a merchant, or a Crediario financed installment plan.<br><br>Value 25 = Crediario                                                                               |
| Total Number of Installments (Positions 3–4)         | Numeric | 2    | Number of installments the cardholder chose to pay.<br><br>From 02–99                                                                                                                                 |
| Installment Amount<br>(Positions 5–16)               | Numeric | 12   | Value that the cardholder is going to pay per month. Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places.                                           |
| Total Amount (Positions 17–28)                       | Numeric | 12   | Total Installment Amount (Monthly installment amount multiplied by the Number of Installments).<br><br>Transaction total amount issuer calculates, including calculated interest; two decimal places. |
| Monthly Interest Rate<br>(Positions 29–32)           | Numeric | 4    | Interest Rates the Issuer calculates and charges per month; two decimal places.                                                                                                                       |
| Annual Interest Rate<br>(Positions 33–37)            | Numeric | 5    | Interest Rates the issuer calculates and charges per year; two decimal places.                                                                                                                        |
| Monthly Total Effective Cost (CET) (Positions 38–49) | Numeric | 12   | Total amount charged monthly to cardholder including the payments, interest, fees, insurance, and other amounts; two decimal places.                                                                  |
| % Total Effective Cost<br>(Positions 50–61)          | Numeric | 12   | Total Effective Cost, expressed as a percentage; two decimal places.                                                                                                                                  |
| Date of first installment<br>(Positions 62–67)       | Numeric | 6    | Date of the first installment payment in yymmdd (year, month, day) format.                                                                                                                            |
| Taxes (Positions 68–74)                              | Numeric | 7    | All taxes, if applicable, that the cardholder is paying, including IOF; two decimal places.                                                                                                           |

| Field Name/Positions                                     | Type    | Size | Comment                                                                                   |
|----------------------------------------------------------|---------|------|-------------------------------------------------------------------------------------------|
| % Taxes of Total Amount (Positions 75–78)                | Numeric | 4    | Percentage of the whole purchase price that the taxes represent.                          |
| Fee (Positions 79–85)                                    | Numeric | 7    | All fees, if applicable, that the cardholder is paying; two decimal places.               |
| % Fee of Total Amount (Positions 86–89)                  | Numeric | 4    | Percentage of the whole purchase price that the fees represent.                           |
| Insurance (Positions 90–96)                              | Numeric | 7    | Any insurance, if applicable, that the cardholder is paying; two decimal places.          |
| % Insurance of Total Amount (Positions 97–100)           | Numeric | 4    | Percentage of the whole purchase price that the insurance represents.                     |
| Other (Positions 101–107)                                | Numeric | 7    | Any other added values, if applicable, that the cardholder is paying; two decimal places. |
| % Other of Total Amount (Positions 108–111)              | Numeric | 4    | Percentage of the whole purchase price that any other added values represent.             |
| Total Amount to Merchant (Positions 112–123)             | Numeric | 12   | Total amount that will be paid to the merchant (purchase amount); two decimal places.     |
| % Amount to Merchant of Total Amount (Positions 124–127) | Numeric | 4    | Percentage of the total amount that will be paid to the merchant.                         |
| Reserved for Future Use (Positions 128–131)              | Numeric | 4    |                                                                                           |

### Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting the second option for the Crediário installment program and services.

Information not populated or optional to a specific field should be zero-filled.

### Subelement 015 (Crediário Third Simulation)

DE 112 (Additional Data [National Use]), subelement 015 contains the issuer's payment conditions for the third Crediário simulation.

### Attributes

| Attribute     | Description |
|---------------|-------------|
| Subelement ID | 015         |

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement Data Length | 3 positions                 |
| Data Representation    | n-131                       |
| Data Field             | Contents of positions 1–131 |
| Subfields              | N/A                         |
| Justification          | N/A                         |

## Usage

Following is the usage of subelement 015 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

## Values

| Field Name/Positions                         | Type    | Size | Comment                                                                                                                                                     |
|----------------------------------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment Plan Type<br>(Positions 1–2)     | Numeric | 2    | Identifies whether this is an issuer, a merchant, or a Crediario financed installment plan.<br><br>Value 25 = Crediario                                     |
| Total Number of Installments (Positions 3–4) | Numeric | 2    | Number of installments the cardholder chose to pay.<br><br>From 02–99                                                                                       |
| Installment Amount<br>(Positions 5–16)       | Numeric | 12   | Value that the cardholder is going to pay per month. Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places. |

| Field Name/Positions                                 | Type    | Size | Comment                                                                                                                                                                                               |
|------------------------------------------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Amount (Positions 17–28)                       | Numeric | 12   | Total Installment Amount (Monthly installment amount multiplied by the Number of Installments).<br><br>Transaction total amount issuer calculates, including calculated interest; two decimal places. |
| Monthly Interest Rate (Positions 29–32)              | Numeric | 4    | Interest Rates the Issuer calculates and charges per month; two decimal places.                                                                                                                       |
| Annual Interest Rate (Positions 33–37)               | Numeric | 5    | Interest Rates the issuer calculates and charges per year; two decimal places.                                                                                                                        |
| Monthly Total Effective Cost (CET) (Positions 38–49) | Numeric | 12   | Total amount charged monthly to cardholder including the payments, interest, fees, insurance, and other amounts; two decimal places.                                                                  |
| % Total Effective Cost (Positions 50–61)             | Numeric | 12   | Total Effective Cost, expressed as a percentage; two decimal places.                                                                                                                                  |
| Date of first installment (Positions 62–67)          | Numeric | 6    | Date of the first installment payment in yymmdd (year, month, day) format.                                                                                                                            |
| Taxes (Positions 68–74)                              | Numeric | 7    | All taxes, if applicable, that the cardholder is paying, including IOF; two decimal places.                                                                                                           |
| % Taxes of Total Amount (Positions 75–78)            | Numeric | 4    | Percentage of the whole purchase price that the taxes represent.                                                                                                                                      |
| Fee (Positions 79–85)                                | Numeric | 7    | All fees, if applicable, that the cardholder is paying; two decimal places.                                                                                                                           |
| % Fee of Total Amount (Positions 86–89)              | Numeric | 4    | Percentage of the whole purchase price that the fees represent.                                                                                                                                       |
| Insurance (Positions 90–96)                          | Numeric | 7    | Any insurance, if applicable, that the cardholder is paying; two decimal places.                                                                                                                      |
| % Insurance of Total Amount (Positions 97–100)       | Numeric | 4    | Percentage of the whole purchase price that the insurance represents.                                                                                                                                 |
| Other (Positions 101–107)                            | Numeric | 7    | Any other added values, if applicable, that the cardholder is paying; two decimal places.                                                                                                             |
| % Other of Total Amount (Positions 108–111)          | Numeric | 4    | Percentage of the whole purchase price that any other added values represent.                                                                                                                         |
| Total Amount to Merchant (Positions 112–123)         | Numeric | 12   | Total amount that will be paid to the merchant (purchase amount); two decimal places.                                                                                                                 |

| Field Name/Positions                                        | Type    | Size | Comment                                                           |
|-------------------------------------------------------------|---------|------|-------------------------------------------------------------------|
| % Amount to Merchant of Total Amount<br>(Positions 124–127) | Numeric | 4    | Percentage of the total amount that will be paid to the merchant. |
| Reserved for Future Use<br>(Positions 128–131)              | Numeric | 4    |                                                                   |

### Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting the third option for the Crediário installment program and services.

Information not populated or optional to a specific field should be zero-filled.

### Subelement 018 (Brazil Post-Dated Transaction Data)

DE 112, subelement 018 (Brazil Post-Dated Transaction Data) contains additional data to support post-dated transaction data provided by the acquirer.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 018                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans-39                     |
| Data Field             | Contents of positions 1–39 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Usage

Following is the usage of subelement 018 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | CE  | P   | C   |



| Message               | Org | Sys | Dst |
|-----------------------|-----|-----|-----|
| Reversal Request/0400 | C   | .   | C   |

### Values

| Field Name                          | Positions | Attribute | Field Description                                                                                                                                                                                                                                 |
|-------------------------------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service Code                        | 1–2       | n-2       | Type of post-dated transaction.<br>Valid Values:<br><ul style="list-style-type: none"> <li>30 = Post-Dated with Guarantee</li> <li>31 = Post-Dated without Guarantee</li> </ul> Must be present and equal to 30 or 31 when DE 61, subfield 7 = 5. |
| Number of Installments              | 3         | n-1       | Valid Values:<br>If present, value will always be 1.                                                                                                                                                                                              |
| Guarantee                           | 4         | a-1       | Valid Values:<br><ul style="list-style-type: none"> <li>Y = Yes</li> <li>N = No</li> </ul>                                                                                                                                                        |
| Guarantee Amount                    | 5–12      | n-8       | Amount of guarantee to be settled; assumed to be a credit to the issuer.<br>Valid Values:<br>Must be 00000000 if Position 1–2 = 31 (Post-Dated without Guarantee).                                                                                |
| Post Settlement Date                | 13–18     | n-6       | Proposed settlement date (expected date for completion message arrival).<br>Format: MMDDYY                                                                                                                                                        |
| Original Mastercard Settlement Date | 19–24     | n-6       | Contains zeros on authorization message.<br>Format: MMDDYY                                                                                                                                                                                        |
| Original Banknet Reference Number   | 25–33     | n-9       | Original Banknet reference number assigned by Banknet to original authorization request; contains zeros on authorization request/0100 message.                                                                                                    |
| Authorization Code                  | 34–39     | n-6       | Contains the online authorization code provided by the issuer on the original authorization response.                                                                                                                                             |

## Dual Message Authorization System Edits

### Post-Dated Transactions

The Dual Message Authorization System performs the following system edits on DE 112, subelement 018 (Brazil Post-Dated Transaction Data), on authorization messages submitted for Post-Dated transactions in Brazil.

| WHEN...                                                                                                                                                                                                                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request/0100 message contains DE 112 (Additional Data—National Use), subelement 018 (Brazil Post-Dated Transaction Data) and the issuer account range is not set up for the Post-Dated Payment service or the issuer and acquirer country codes are not the same                                                         | will reject the transaction with DE 39 (Response Code) = 57 (Transaction not permitted to issuer/ cardholder)                                                                                                              |
| the Authorization Request/0100 message contains DE 112 (Additional Data—National Use), subelement 018 (Brazil Post-Dated Transaction Data), subfield 1 (Service Code) contains value 30 or 31 and DE 61 (Point of Service Data), subfield 7 (POS Transaction Status) does not contain value 5 (Time Based Payment Authorization Request)   | will reject the transaction with <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 061 (indicating the data element in error)</li> </ul>    |
| the Authorization Request/0100 message contains DE 61 (Point of Service Data), subfield 7 (POS Transaction Status) contains a value 5 (Time Based Payment Authorization Request) and DE 112 (Additional Data—National Use), subelement 018 (Brazil Post-Dated Transaction Data), subfield 1 (Service Code) does not contain value 30 or 31 | will reject the transaction with <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 112018 (indicating the data element in error)</li> </ul> |

### Alternate Processing

Post-Dated transactions are not eligible for Stand-In or X-Code Processing. If an issuer is not available, or offline, the transactions will be declined using the Authorization Request Response/ 0110 message containing DE 39 (Response Code), value 91 (Dual Message Authorization System or issuer system inoperative).

## Subelement 019 (Original Purchase Amount)

DE 112 (Additional Data [National Use]), subelement 019 contains the amount (two decimal places) that the cardholder wants to finance. Subelement 019 is to be used only for Crediário Account Status Inquiry (ASI) transactions.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 019                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | n-12                       |
| Data Field             | Contents of positions 1–12 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Usage

Following is the usage of subelement 019 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | C   | P   | C   |

### Values

This subelement must contain valid numeric data.

### Application notes

This subelement should be provided to indicate the merchant terminal's capabilities for displaying the original purchase amount for the Crediário installment program.

Information not populated or optional to a specific field should be zero-filled.

## Brazil: Merchant Fraud Scoring Data

Following are the technical specifications for Authorization Request/0100 and Authorization Request Response/0110 messages that contain merchant fraud scoring data for card-not-present (CNP) transactions from acquirers in Brazil.

### Subelement 028 (Merchant Fraud Score Data)

DE 112, subelement 028 (Merchant Fraud Score Data) should be used by acquirers in Brazil to send merchant fraud score data.

#### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 028                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | ns-4                      |
| Data Field             | Contents of positions 1–4 |
| Subfields              | N/A                       |
| Justification          | see values                |

#### Usage

Following is the usage of subelement 028 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

#### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | O   | P   | C   |
| Authorization Request Response/0110 | •   | X   | C   |

#### Values

Acquirers may perform merchant fraud scoring either by themselves or through payment gateways. Acquirers in Brazil can forward the merchant fraud scoring information to issuers using this subelement. Issuers can use this data to know that merchant fraud scoring is done by the acquirer and use the data to make approval decision for CNP transaction.

Merchant Fraud Score data provided by acquirers in Brazil should be in the range -999 to +999. Merchant fraud score should be right-justified with leading zeros if the value is less than three digits.

### Application notes

Issuers are not expected to echo DE 112, subelement 028 (Merchant Fraud Score Data) in the Authorization Request Response/0110 message. Mastercard will echo DE 112, subelement 028 (Merchant Fraud Score Data) from Authorization Request/0100 to the acquirers in the Authorization Request Response/0110 message.

## Canada domestic: Transactions

Following are the technical specifications for the Authorization/01xx messages for Canada domestic transactions.

### Subelement 047 (Payment Consent ID)

DE 112 (Additional Data [National Use]), Subelement 047 (Payment Consent ID), is reserved solely for Interac's KONEK commerce platform payment consent ID.

#### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Subelement ID       | 047                        |
| Subelement Length   | 3                          |
| Data Representation | ans-36                     |
| Data Field          | Contents of positions 1-36 |
| Subfields           | N/A                        |
| Justification       | N/A                        |

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

#### Usage

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | P   | C   |

## Application notes

An acquirer submitting a transaction with the KONEK payment consent ID will include the KONEK payment consent ID in the Authorization Request/0100 message that the issuer will validate before approving the payment transaction in the Canada region.

## Chile: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Chile payment transactions within Chile.

### Subelement 010 (Installment Payment Data)

DE 112, Chile, subelement 010 is used in the Dual Message Authorization System for installment payment transactions within Chile.

Subelement 010 must be present if DE 48 (Additional Data: Private Use), subelement 95 (Mastercard Promotion Code) contains the value CHLCTA.

Subelement 010 contains the following subfields used in Chile Domestic Switch (ChIDS) processing. For details on data requirements for these subfields refer to the [Chile Domestic Switch Implementation Guide](#).

- Subfield 1: Total Purchase Amount
- Subfield 2: Number of Installments
- Subfield 3: Installment Type
- Subfield 4: Installment Amount
- Subfield 5: Transaction Interest Rate
- Subfield 6: Deferred Period
- Subfield 7: Deferred Period 1: Installment Amount
- Subfield 8: Deferred Period 1: Interest Rate
- Subfield 9: Deferred Period 1: Period Identifier
- Subfield 10: Deferred Period 2 Installment Amount
- Subfield 11: Deferred Period 2 Interest Rate
- Subfield 12: Deferred Period 2 Period Identifier
- Subfield 13: Deferred Period 3 Installment Amount
- Subfield 14: Deferred Period 3 Interest Rate
- Subfield 15: Deferred Period 3 Period Identifier
- Subfield 16: Simulation Flag
- Subfield 17: Grace Period Flag
- Subfield 18: Grace Period
- Subfield 19: Deferred Period Simulation Flag
- Subfield 20: Reserved for Annual Equivalent Charge Rate (CAE)

## Subelement 042 (Acceptor Tax ID)

Subelement 042 (Acceptor Tax ID) will be used in the Dual Message Authorization System to support DE 112 in any received Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message.

### Attributes

| Tax ID                 | Description      |
|------------------------|------------------|
| Subelement ID          | 042              |
| Subelement Data Length | 3 positions      |
| Data Representation    | an...012; LLLVAR |

### Usage

Following is the usage of subelement 042 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages:

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | .   |
| Authorization Advice/0120: acquirer-generated | C   | X   | .   |
| Reversal Request /0400                        | C   | X   | .   |

### Application notes

This field is required when purchase, purchase with cash back, and refund transactions are submitted by acquirers for merchants in Chile and Paraguay.

### Attributes

| Attribute           | Description                        |
|---------------------|------------------------------------|
| Data Representation | an                                 |
| Data Length         | 3 positions                        |
| Length Value        | Variable length, 001–012 positions |

| Attribute     | Description                |
|---------------|----------------------------|
| Data Field    | Contents of Subelement 042 |
| Justification | N/A                        |
| Description   | Tax ID                     |

## Colombia: Domestic Transactions

Following are the technical specifications for Authorization/01xx messages for Colombia domestic transactions.

### Subelement 035 (Issuer Fee Inquiry Indicator)

DE 112, Colombia, subelement 035 (Issuer Fee Inquiry Indicator) is populated by the acquirer and passed to the issuer to indicate that a transaction qualifies as an Issuer Fee Inquiry.

#### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 035                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | n-2                       |
| Data Field             | Contents of positions 1–2 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

#### Usage

Following is the usage of subelement 035 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

#### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | P   | C   |



### Valid values

| Value | Description                          |
|-------|--------------------------------------|
| 01    | Indicates Issuer Fee Inquiry Request |

## Subelement 036 (Issuer Fee Amount)

DE 112, Colombia, subelement 036 (Issuer Fee Amount) is populated by the issuer with the fee amount that would be associated with the intended ATM cash withdrawal.

### Attributes

| Attribute              | Description                        |
|------------------------|------------------------------------|
| Subelement ID          | 036                                |
| Subelement Data Length | 3 positions                        |
| Data Representation    | n-12                               |
| Data Field             | Contents of positions 1–12         |
| Subfields              | N/A                                |
| Justification          | Right-justified with leading zeros |

### Usage

Following is the usage of subelement 036 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

### Values

| Value              | Description      |
|--------------------|------------------|
| Valid numeric data | Ex: 000000000123 |

**NOTE:** Amounts are expressed without a decimal separator. For currencies that support exponents, users and systems are responsible for placing the decimal separator appropriately. Currently only for use in Colombia, where currency exponent = 2.

## Colombia: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Colombia payment transactions within Colombia.

### Subelement 010 (Installment Payment Data)

DE 112, Colombia, subelement 010 (Installment Payment Data) is used in the Dual Message Authorization System for Colombia payment transactions.

Subelement 010 must be present if DE 48, subelement 95 (Mastercard Promotion Code) contains the value COLCTA.

### Attributes

| Attribute              | Description                         |
|------------------------|-------------------------------------|
| Subelement ID          | 010                                 |
| Subelement Data Length | 3 positions                         |
| Data Representation    | ans-2                               |
| Data Field             | Contents of positions 1–2           |
| Subfields              | N/A                                 |
| Justification          | Left-justified with trailing spaces |

### Usage

Conditional.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request /0400                        | C   | P   | C   |

### Values

Example Values:

XX = The number of installments (01–XX)

Maximum number of installments accepted in Colombia may vary (99 as maximum value allowed).

### Subelement 011 (Customer ID)

DE 112, Colombia, subelement 011 (Customer ID) is the customer associated with a voice, card not present, or manual transaction.

### Attributes

| Attribute              | Description                         |
|------------------------|-------------------------------------|
| Subelement ID          | 011                                 |
| Subelement Data Length | 3 positions                         |
| Data Representation    | ans-11                              |
| Data Field             | Contents of positions 1–11          |
| Subfields              | N/A                                 |
| Justification          | Left-justified with trailing spaces |

### Usage

Conditional.

### Applicable messages

| Messages                                      | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | P   | C   |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request /0400                        | C   | P   | C   |

### Values

Example Values:

Contains customer ID information.

## Cuotas: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Cuotas payment transactions within Argentina and Uruguay.

### Subelement 001 (Installment Payment Data)

DE 112, Cuotas, subelement 001 (Installment Payment Data) is used in the Dual Message Authorization System for Cuotas payment transactions. Subelement 001 must be present if DE 48, subelement 95 (Mastercard Promotion Code) contains the value ARGCTA, or URYCTA.

### Attributes

| Attribute              | Description                                                                  |
|------------------------|------------------------------------------------------------------------------|
| Subelement ID          | 001                                                                          |
| Subelement Data Length | 3 positions                                                                  |
| Data Representation    | ans...4                                                                      |
| Data Field             | Contents of positions 1–4 where:<br>XX = positions 1–2<br>YY = positions 3–4 |
| Subfields              | N/A                                                                          |
| Justification          | N/A                                                                          |

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | C   | P   | C   |

Values

Cuotas Payment Transactions in the Authorization Request/0100

Used for various Cuotas transactions: From acquirer to issuer in the following format:

|    |                                                                                                                                                                                                                                                                                           |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XX | Cuotas plan type:<br>20 = Issuer-financed<br>21 = Merchant-financed<br>22 = Acquirer-financed<br>23 = Average payment financing<br>24 = Consumer financing (Purchase)<br>25 = Consumer financing (Manual Cash Advance)<br>80 = ATM Installment Inquiry<br>81 = ATM Installment Withdrawal |
| YY | The total number of Cuotas                                                                                                                                                                                                                                                                |

Cuotas Payment Transactions in the Authorization Request Response/0110

Used for various Cuotas transactions: From issuer to acquirer in the following format:

|    |                                                                                                                                                                                                                                                                                                                                  |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XX | <p>Cuotas plan type:</p> <p>20 = Issuer-financed</p> <p>21 = Merchant-financed</p> <p>22 = Acquirer-financed</p> <p>23 = Average payment financing</p> <p>24 = Consumer financing (Purchase)</p> <p>25 = Consumer financing (Manual Cash Advance)</p> <p>80 = ATM Installment Inquiry</p> <p>81 = ATM Installment Withdrawal</p> |
| YY | The total number of Cuotas.                                                                                                                                                                                                                                                                                                      |

## Subelement 003 (Installment Payment Response Data)

DE 112, Cuotas, subelement 003 (Installment Payment Response Data) is used in the Dual Message Authorization System and Cuotas payment transactions to provide issuer response data only for consumer financing plan types 24 and 25.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 003                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans...55                   |
| Data Field             | Contents of positions 1–55 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Usage

### Applicable messages

| Messages                            | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

From issuer to acquirer, upon transaction approval; all amounts in transaction currency; use the following format:

### Values

| Positions | Length | Description                                                                                                   |
|-----------|--------|---------------------------------------------------------------------------------------------------------------|
| 1–12      | 12     | Installment amount including any issuer-calculated interest, insurance, or other charges; two decimal places. |
| 13–17     | 5      | Annual nominal interest percentage rate or all zeros if nominal rate not applicable; two decimal places.      |
| 18–22     | 5      | Annual actual interest percentage rate or all zeros if actual rate not applicable; two decimal places.        |
| 23–27     | 5      | Insurance percentage rate or all zeros if insurance not applicable; two decimal places.                       |
| 28–39     | 12     | Insurance amount or all zeros if insurance not applicable; two decimal places.                                |
| 40–51     | 12     | Issuing charge amount or all zeros if issuing charge not applicable; two decimal places.                      |
| 52–53     | 2      | Total number of installments, or all zeros if total number of installments not applicable.                    |
| 54–55     | 2      | Reserved for future use; insert all zeros.                                                                    |

## Subelement 027 (ATM Credit Card Cash Advance Installments)

DE 112, Cuotas: Payment Transactions, subelement 027 (ATM Credit Card Cash Advance Installments) is used in the Dual Message Authorization System for credit card cash advance installment transactions performed at the ATM. Subelement 027 contains details of the cash advance installment transaction.

### Attributes

| Attribute              | Description                                                                       |
|------------------------|-----------------------------------------------------------------------------------|
| Subelement ID          | 027                                                                               |
| Subelement Data Length | 3 positions                                                                       |
| Data Representation    | an-4 (Authorization Request/0100)<br>an-137 (Authorization Request Response/0110) |

| Attribute     | Description                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------|
| Data Field    | Contents of subfields<br>Subfields 1–2 (Authorization Request/0100)<br>Subfields 3–9 (Authorization Request Response/0110) |
| Subfields     | 9                                                                                                                          |
| Justification | See subfields                                                                                                              |

### Usage

Following is the usage of subelement 027 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | C   | P   | C   |

### Values

See subfields.

### Subfield 1 (Transaction Type)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 1 (Transaction Type) contains the type of ATM credit card cash advance installment transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subfield ID         | 1                         |
| Data Representation | an-2                      |
| Data Field          | Contents of positions 1–2 |
| Justification       | N/A                       |



### Values

| Value | Description                |
|-------|----------------------------|
| 80    | ATM Installment Inquiry    |
| 81    | ATM Installment Withdrawal |

### Subfield 2 (Requested Number of Installments)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 2 (Requested Number of Installments) contains the requested number of payments required to payoff the credit card cash advance.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subfield ID         | 2                         |
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3–4 |
| Justification       | N/A                       |

### Values

Valid values 01–99.

### Subfield 3 (Approved Number of Installments)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 3 (Approved Number of Installments) contains the number of installment payments approved by the issuer.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subfield ID         | 3                         |
| Data Representation | n-2                       |
| Data Field          | Contents of positions 5–6 |

| Attribute     | Description |
|---------------|-------------|
| Justification | Left        |

### Values

Valid values 01–99.

### Subfield 4 (Installment Amount)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 4 (Installment Amount) contains the monthly payment amount.

#### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subfield ID         | 4                         |
| Data Representation | n-12                      |
| Data Field          | Contents of positions 7–8 |
| Justification       | Right with leading zeros  |

### Subfield 5 (Total Transaction Amount)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 5 (Total Transaction Amount) contains the total amount of the credit card cash advance and the interest charged by the issuer.

#### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Subfield ID         | 5                           |
| Data Representation | n-12                        |
| Data Field          | Contents of positions 19–30 |
| Justification       | Right with leading zeros    |

## Subfield 6 (Yearly Interest Rate)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 6 (Yearly Interest Rate) contains the yearly interest rate that the issuer charges the cardholder for the installment payment.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Subfield ID         | 6                           |
| Data Representation | n-4                         |
| Data Field          | Contents of positions 31–34 |
| Justification       | Right with leading zeros    |

## Subfield 7 (Currency Code)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 7 (Currency Code) contains the currency code the issuer will be charging the cardholder for repayment.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Subfield ID         | 7                           |
| Data Representation | n-3                         |
| Data Field          | Contents of positions 35–37 |
| Justification       | N/A                         |

## Subfield 8 (Member-Defined Data)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 8 (Member-defined Data) contains member data.

### Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 8           |

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | an-25                       |
| Data Field          | Contents of positions 38–62 |
| Justification       | N/A                         |

### Subfield 9 (Member-Defined Data)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 9 (Member-defined Data) contains member data.

#### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Subfield ID         | 9                           |
| Data Representation | an-25                       |
| Data Field          | Contents of positions 63–87 |
| Justification       | N/A                         |

## Europe Region and Philippines: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for the Europe Region and Philippines payment transactions.

### Subelement 009 (Installment Payment Data)

DE 112, subelement 009 (Installment Payment Data) is used in the Dual Message Authorization System for installment payment transactions in Georgia and Philippines.

DE 112, subelement 009 only applies when DE 48 (Additional Data: Private Use), subelement 95 (Mastercard Promotion Code) contains the value HGMINS or PHINST.

**NOTE: Customers in the Europe region should refer to global release announcements and the [Mastercard Installment Payment Service User Guide](#) for information about installment payment processing. Mastercard will no longer publish this information in announcements specific to the Europe region as this service is now available globally. Refer to [All Regions: Installment Payment Transactions](#) for more information.**

## Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 009                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | n-33                       |
| Data Field             | Contents of Positions 1–33 |
| Subfields              | N/A                        |
| Justification          | Right with leading zeros   |

## Usage

Following is the usage of subelement 009 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | CE  | P   | CE  |
| Reversal Request/0400               | C   | P   | C   |
| Reversal Request Response/0410      | CE  | P   | CE  |

## Values

| Value                                  | Description                                                                                                                                     |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment type (positions 1–2)       | Installment type contains a type of installment. Valid values include 20 (Issuer-financed), 21 (Merchant-financed), and 22 (Acquirer-financed). |
| Number of installments (positions 3–4) | Number of installments contains the number of installment payments selected by the cardholder. Valid values are 01–99.                          |

| Value                                           | Description                                                                                                                                             |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interest rate (positions 5–9)                   | Interest rate contains the interest rate (2 decimal) that the issuer charges the cardholder for the installment payments. Default rate is 0% ("00000"). |
| First installment amount (positions 10–21)      | First installment amount contains the amount of the first installment in transaction currency.                                                          |
| Subsequent installment amount (positions 22–33) | Subsequent installment amount contains the amount of the subsequent installment in transaction currency.                                                |

**NOTE: DE 112 (Additional Data [National Use]), subelement 009 (Installment Payment Data) is a conditional echo in the Authorization Request Response/0110 message for Philippines installment payment transactions for finance type 21 (merchant-financed).**

## Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on data element 112, subelement 009 (Installment Payment Data), on authorization messages submitted for installment payment transactions in Georgia.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data: Private Use), subelement 95 (Mastercard Promotion Code) contains the value HGMINS<br>and<br>DE 112 (Additional Data [National Use]), subelement 009 (Installment Payment Data) is not present in an Authorization Request/0100 message | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 112</li> </ul> |

| WHEN...                                                                                       | THEN the Dual Message Authorization System...                                                                                                                |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 95 contains the value HGMINS and                                            | sends the issuer an Authorization Response Negative Acknowledgment/0190 message where:                                                                       |
| DE 112, subelement 009 is present in the Authorization Request/0100 message and               | <ul style="list-style-type: none"> <li>DE 39 = 30</li> <li>DE 44 (Additional Response Data) = 112 (indicating the data element in error)</li> </ul>          |
| DE 112, subelement 009 is not present back in the Authorization Request Response/0110 message | Sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 91 (Authorization System or issuer system inoperative) or 05 (Do not honor) |

## Subelement 020 (Domestic Acceptor Tax ID)

Subelement 020 (Domestic Acceptor Tax ID) is used in the Croatian Authorization System for payment transactions.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 020                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans-20; LLVAR              |
| Data Field             | Contents of Positions 1–20 |
| Subfields              | N/A                        |
| Justification          | Left with trailing spaces  |

### Usage

Following is the usage of subelement 020 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | CE  | P   | CE  |
| Reversal Request/0400               | C   | P   | C   |

| Message                        | Org | Sys | Dst |
|--------------------------------|-----|-----|-----|
| Reversal Request Response/0410 | CE  | P   | CE  |

### Values

Values are assigned by the acquirer.

### Application notes

This field is required when installment transactions are submitted by participating acquirers in Croatia.

## France: Payment Transactions

The following are technical specifications for payment transactions in France. Acquirer hosts must be able to provide the specific information to issuers through the Mastercard Authorization Request/0100 and Authorization Advice/0120 messages using data element (DE) 112 (Additional Data [National Use]) for any transaction occurring at a merchant in France.

### Subelement 020 (Domestic Acceptor Tax ID)

DE 112, France, subelement 020 (Domestic Acceptor Tax ID) is used in the Dual Message Authorization System for French payment transactions. Acquirers send DE 112, subelement 020 to identify the acceptor tax ID (SIRET).

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 020                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans...14; LLLVAR           |
| Data Field             | Contents of positions 1–14 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

French payment transactions in the Authorization Request/0100 and Authorization Advice/0120, SIRET value:



### Values

| Positions | Length | Description                |
|-----------|--------|----------------------------|
| 1-3       | 3      | To identify subelement 020 |
| 4-6       | 3      | Overall subelement length  |
| 7-20      | 14     | Acceptor Tax ID            |

## Subelement 096 (Transit Offer ID)

DE 112, France, subelement 096 (Transit Offer ID) is used in the Dual Message Authorization System for French payment transactions. Acquirers send DE 112, subelement 096 to identify the cardholder transit journey.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 096                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans...20; LLLVAR           |
| Data Field             | Contents of positions 1-20 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

French payment transactions in the Authorization Request/0100 and Authorization Advice/0120, Transit Offer ID value:

### Values

| Positions | Length | Description                |
|-----------|--------|----------------------------|
| 1-3       | 3      | To identify subelement 096 |
| 4-6       | 3      | Overall subelement length  |
| 7-27      | 20     | Transit Offer ID           |

## Subelement 097 (Application Selection ID)

DE 112, France, subelement 097 (Application Selection ID) is used in the Dual Message Authorization System for French payment transactions. Acquirers send DE 112, subelement 097 to indicate if the application was selected by the merchant or by the cardholder.

### Attributes

| Attribute              | Description             |
|------------------------|-------------------------|
| Subelement ID          | 097                     |
| Subelement Data Length | 3 positions             |
| Data Representation    | n-1                     |
| Data Field             | Contents of positions 1 |
| Subfields              | N/A                     |
| Justification          | N/A                     |

### Values

0 = Selection by default

1 = Cardholder selection

2-9 = Reserved for future use

French payment transactions in the Authorization Request/0100 and Authorization Advice/0120, Application Selection ID value:

### Values

| Positions | Length | Description             |
|-----------|--------|-------------------------|
| 1-3       | 3      | Identify subelement 097 |
| 4-6       | 3      | Subelement length       |
| 7         | 1      | Application ID          |

## Subelement 098 (Brand ID)

DE 112, France, subelement 098 (Brand ID) is used in the Dual Message Authorization System for French payment transactions. Acquirers send DE 112, subelement 098 to identify the brand selected at a POI for a transaction..

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 098                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | n-2                       |
| Data Field             | Contents of positions 1-2 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Values

French payment transactions in the Authorization Request/0100 and Authorization Advice/0120, Application Selection ID value:

00: CB

01: VISA

02: Vpay

03: Electron

04: Mastercard

05: Maestro

06: JCB

07: Discover

08: UPI

09: American Express

10-79: reserved for future use

80-99: reserved for private use

**NOTE: If a subelement is null or empty, the merchant or acquirer must not populate this subelement.**

## Greece: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for installment payment transactions in Greece.

### Subelement 006 (Installment Payment Data)

DE 112, Greece, subelement 006 (Installment Payment Data) is used in the Dual Message Authorization System for Greece payment transactions. It must be present in Authorization Request/0100 and Authorization Request Response/0110 messages when DE 48, subelement 95 (Mastercard Promotion Code) contains the value GREECE.

#### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 006                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans...10                   |
| Data Field             | Contents of positions 1–10 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

#### Usage

Greece payment transactions in the Authorization Request/0100 and Authorization Request Response/0110:

#### Applicable messages

| Messages                   | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | P   | C   |
| Reversal Request/0400      | C   | P   | C   |

### Values

| Positions | Length | Description                                                                                                                                                                                                                                                                    |
|-----------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1–2       | 2      | Type of Credit:<br>21 = Merchant Financed                                                                                                                                                                                                                                      |
| 3–4       | 2      | Number of Installments<br>• 01 to 99                                                                                                                                                                                                                                           |
| 5–7       | 3      | Grace Period Before First Payment in UZZ format where:<br>• U indicates unit type<br>– N = No grace period<br>– D = Days<br>– W = Weeks<br>– M = Months<br>• ZZ indicates the number of days, weeks or months<br>– 00 to 99                                                    |
| 8–10      | 3      | Transaction Currency Code. Must be same value as provided in DE 49 (Currency Code, Transaction).<br><br>This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in Amount Transaction DE 4.<br><br>• 978 = Euro (EUR) |

## Subelement 008 (Installment Payment Response Data)

DE 112, Greece, subelement 008 (Installment Payment Response Data) is used in the Dual Message Authorization System for Greece payment transactions in the Authorization Request Response/0110.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 008                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans...23                   |
| Data Field             | Contents of positions 1–23 |

| Attribute     | Description |
|---------------|-------------|
| Subfields     | N/A         |
| Justification | N/A         |

Greece payment transactions in the Authorization Request Response/0110:

### Values

| Positions | Length | Description                                                                                                                                                                                 |
|-----------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1–12      | 12     | Installment amount (with two decimal places) including any issuer-calculated interest, insurance, or other charges.                                                                         |
| 13–18     | 6      | Due date of first installment (in DDMMYY format).                                                                                                                                           |
| 19–21     | 3      | Financing Currency Code. This is the currency in which the issuer will finance the transaction.<br><br>It specifies the currency using in Installment Amount above.<br><br>978 = Euro (EUR) |
| 22–23     | 2      | Payment Plan. Reserved for future use. Insert all zeros.                                                                                                                                    |

## Hungary: Benefits Program

Following are the technical specifications for transactions in Hungary.

### Subelement 013 (Program ID)

DE 112, subelement 013 (Program ID) contains data to support processing of intracountry transactions for Employee Benefits Program in Hungary.

### Attributes

| Attributes             | Description            |
|------------------------|------------------------|
| Subelement ID          | 013                    |
| Settlement Data Length | 3 positions            |
| Data Representation    | α-1                    |
| Data Field             | Contents of position 1 |
| Subfields              | N/A                    |

| Attributes    | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Value | Description        |
|-------|--------------------|
| K     | Program identifier |

## India: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for installment payment transactions in India.

### Subelement 044 (Installment and Recurring Payment Data: Visa Only)

DE 112, subelement 044 (Installment and Recurring Payment Data: Visa Only) is used by the Dual Message Authorization System for installment and recurring payment transactions for accounts issued in India. This subelement is used only with Visa-branded transactions.

### Attributes

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement ID          | 044                         |
| Subelement Data Length | 3 positions                 |
| Data Representation    | an...117; LLLVAR            |
| Data Field             | Contents of subfields 01-12 |
| Subfields              | 12                          |
| Justification          | See subfields               |

### Usage

Following is the usage of this subelement and whether it is mandatory, conditional, optional, system-provided, or not required in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | CE  | P   | CE  |
| Reversal Request/0400               | C   | P   | C   |
| Reversal Request Response/0410      | CE  | P   | CE  |

### Application notes

DE 112, subelement 044 and its subfields are set up in Tag, Length, Value format. The Dual Message Authorization System removes this value if received on transactions that are not Visa-branded. The Dual Message Authorization System performs no editing on this subelement's values.

### Subfield 01 (Transaction Type)

DE 112, subelement 044, subfield 01 (Transaction Type) defines characteristics based on the installment or recurring payment transaction type.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-2                     |
| Data Field           | Contents of subfield 01 |
| Justification        | N/A                     |

### Values

| Position | Value                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------|
| 1        | <ul style="list-style-type: none"> <li>1 = Recurring payment and</li> <li>2 = Installment payment.</li> </ul> |



| Position | Value                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2        | <ul style="list-style-type: none"> <li>1 = Registration/first transaction</li> <li>2 = Subsequent transaction</li> <li>3 = Modification, and</li> <li>4 = Cancellation.</li> </ul> |

## Subfield 02 (Recurring Payment Validation)

DE 112, subelement 044, subfield 02 (Recurring Payment Validation) identifies if recurring payment transaction has been validated by the originator.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-1                     |
| Data Field           | Contents of subfield 02 |
| Justification        | N/A                     |

### Values

| Value | Description   |
|-------|---------------|
| 0     | Not validated |
| 1     | Validated     |

## Subfield 03 (Maximum Recurring Payment Amount)

DE 112, subelement 044, subfield 03 (Maximum Recurring Payment Amount) defines the maximum amount authorized by the cardholder for a recurring payment transaction.

### Attributes

| Attribute   | Description |
|-------------|-------------|
| Subfield ID | 03          |

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield Data Length | 2 positions              |
| Data Representation  | n-12                     |
| Data Field           | Contents of subfield 03  |
| Justification        | Right with leading zeros |

### Values

The maximum recurring payment amount is expressed in the currency defined by the currency code (subfield 08).

Where a minor unit of currency applies, the maximum recurring payment amount is expressed in the minor unit of currency without a decimal separator (example, value 100 represents INR 1.00).

### Subfield 04 (Registered Transaction Identifier)

DE 112, subelement 044, subfield 04 (Registered Transaction Identifier) provides a unique identifier for the installment and recurring payment mandate.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 04                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-10                   |
| Data Field           | Contents of subfield 04 |
| Justification        | N/A                     |

### Values

Any combination of 10 alphanumeric characters is valid.

## Subfield 05 (Number of Payments)

DE 112, subelement 044, subfield 05 (Number of Payments) indicates the number of installment or recurring payment debits.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 05                       |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-2                      |
| Data Field           | Contents of subfield 05  |
| Justification        | Right with leading zeros |

### Values

| Value    | Description                                 |
|----------|---------------------------------------------|
| 00 to 97 | Number of payment debits with leading zeros |
| 98       | Number of payments is not indicated         |
| 99       | Payments are authorized until canceled      |

## Subfield 06 (Payment Frequency)

DE 112, subelement 044, subfield 06 (Payment Frequency) indicates the frequency of payment debits.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 06                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-1                     |
| Data Field           | Contents of subfield 06 |
| Justification        | N/A                     |

### Values

| Value | Description  |
|-------|--------------|
| 0     | Annually     |
| 1     | Daily        |
| 2     | Weekly       |
| 3     | Biweekly     |
| 4     | Monthly      |
| 5     | Quarterly    |
| 6     | Twice yearly |
| 8     | Ad hoc       |

### Subfield 07 (Total Installment Amount)

DE 112, subelement 044, subfield 07 (Total Installment Amount) indicates the total purchase amount corresponding to an installment payment.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 07                       |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-12                     |
| Data Field           | Contents of subfield 07  |
| Justification        | Right with leading zeros |

### Values

The total installment amount is expressed in the currency defined by the Currency Code (subfield 08).

Where a minor unit of currency applies, the total installment amount is expressed in the minor unit of currency without a decimal separator (example, value 100 represents INR 1.00).

## Subfield 08 (Currency Code)

DE 112, subelement 044, subfield 08 (Currency Code) indicates the ISO currency code associated with the maximum recurring payment amount (subfield 03) or total installment amount (subfield 07).

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 08                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | n-3                     |
| Data Field           | Contents of subfield 08 |
| Justification        | N/A                     |

### Values

All currency codes must be selected from the numeric ISO standard currency codes. A list of valid values is available in the [Quick Reference Booklet](#).

Example: A subfield 08 value of 356 indicates the Indian rupee.

## Subfield 09 (Installment Interest Rate)

DE 112, subelement 044, subfield 09 (Installment Interest Rate) indicates the interest rate which applies to the installment payment.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 09                       |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-4                      |
| Data Field           | Contents of subfield 09  |
| Justification        | Right with leading zeros |

### Values

Carries a percentage value with an implied two-position decimal.

Example: An interest rate of 1.23 percent is represented with DE 112, subelement 044, subfield 09 = 0123

## Subfield 10 (Installment Payment Entity)

DE 112, subelement 044, subfield 10 (Installment Payment Entity) identifies the party responsible for managing the splitting and posting of installment payments.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 10                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-1                    |
| Data Field           | Contents of subfield 10 |
| Justification        | N/A                     |

### Values

| Value | Description         |
|-------|---------------------|
| 0     | Issuer              |
| 1     | Merchant            |
| 2     | Acquirer            |
| 3     | Payment facilitator |

## Subfield 11 (Installment Interest Charged)

DE 112, subelement 044, subfield 11 (Installment Interest Charged) indicates whether interest is charged to an installment.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 11          |
| Subfield Data Length | 2 positions |

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | an-1                    |
| Data Field          | Contents of subfield 11 |
| Justification       | N/A                     |

### Values

| Value | Description                          |
|-------|--------------------------------------|
| 0     | Regular payment                      |
| 1     | Installment payment without interest |
| 2     | Installment payment with interest    |

## Subfield 12 (Member-Defined Data)

DE 112, subelement 044, subfield 12 (Member-Defined Data) is used to pass data between the acquirer and issuer as defined by the message originator.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 12                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an-20                   |
| Data Field           | Contents of subfield 12 |
| Justification        | N/A                     |

### Values

This subfield is optional for transactions that contain installment and recurring payment data.

## Japan: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Japan National Data payment transactions within Japan.

### Subelement 030 (Japan Domestic POS Data)

Subelement 030 (Japan Domestic POS Data) is used in the Dual Message Authorization System to provide data required by acquirers and issuers in Japan for intracountry processing.

#### Attributes

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement ID          | 030                         |
| Subelement Data Length | 3 positions                 |
| Data Representation    | ans-138                     |
| Data Field             | Contents of Positions 1–138 |
| Subfields              | N/A                         |
| Justification          | Right with leading zeros    |

#### Usage

Following is the usage of subelement 030 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

#### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | C   | P   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Authorization Advice Response/0130            | CE  | XE  | CE  |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Request Response/0410                | CE  | XE  | CE  |
| Reversal Advice/0420: system-generated        | •   | X   | C   |



| Message                       | Org | Sys | Dst |
|-------------------------------|-----|-----|-----|
| Reversal Advice Response/0430 | CE  | X   | .   |

## Values

| Value                           | Positions | Format                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terminal ID                     | 01–13     | ans-13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| JIS II/ISO                      | 14–82     | ans-69<br>If ISO track data is populated here it must be padded with trailing spaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Product Code                    | 83–89     | ans-7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Reserved for future use         | 90        | ans-1<br>space filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Acquirer Company Code           | 91–97     | an-7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Issuer Company Code             | 98–104    | an-7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Authorization Transmission Mode | 105       | n-1<br><ul style="list-style-type: none"> <li>1 = Memory</li> <li>2 = Online</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Entry Indicator                 | 106       | n-1<br><ul style="list-style-type: none"> <li>1 = Back Stripe ISO</li> <li>2 = Front Stripe JIS II</li> <li>3 = Manual</li> <li>4 = N/A (Domestic Private Label)</li> <li>5 = IC Chip Data ISO</li> <li>6 = IC Chip Data JIS II</li> </ul> For Reversal Request/0400 messages, if the card is read using the ISO Track (Entry Mode 1), then the track data must be placed in the field JIS II/ISO instead of DE 35 (Track 2 Data) as DE 35 is not present in reversal messages. Note that the ISO track data must be padded with trailing spaces when using the field JIS II/ISO due to its reduced length. |

| Value                                                                                                                                                                                                                                                                                                                                                               | Positions | Format                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authorization Type <ul style="list-style-type: none"> <li>• 1 = Normal</li> <li>• 2 = Authorization Reservations</li> <li>• 3 = Post-Approval Authorization</li> <li>• 4 = Card Validity Check</li> <li>• 5 = Reversal</li> <li>• 6 = Refund</li> <li>• 7 = Reversal of Authorization Reservation</li> <li>• 8 = Reversal of Post-Approval Authorization</li> </ul> | 107       | n-1                                                                                                                                                                                                                                                                                                    |
| Approval Number for Post-Approval Authorization                                                                                                                                                                                                                                                                                                                     | 108–113   | n-6                                                                                                                                                                                                                                                                                                    |
| Sales Slip Number                                                                                                                                                                                                                                                                                                                                                   | 114–118   | n-5                                                                                                                                                                                                                                                                                                    |
| Sales Slip Number for Reversal/Refund                                                                                                                                                                                                                                                                                                                               | 119–123   | n-5<br><br>Sales Slip Number for Reversal/Refund is not required when DE 03 (Processing Code), subfield 1 (Cardholder Transaction Type Code) does not equal 20 (Refund/Correction) in an authorization request message. When required, and must be all zeros when a meaningful value is not available. |
| Tax Amount                                                                                                                                                                                                                                                                                                                                                          | 124–130   | n-7<br><br>Format: nnnnnnn                                                                                                                                                                                                                                                                             |
| Local Transaction Date                                                                                                                                                                                                                                                                                                                                              | 131–138   | n-8<br><br>Format: YYYYMMDD                                                                                                                                                                                                                                                                            |

### Application notes

Because DE 112, subelement 030 contains unique data specific to acquirers and issuers in Japan, Mastercard does not enforce the use of DE 112, subelement 030.

If provided, Mastercard does not validate the contents of DE 112, subelement 030.

If provided, Mastercard will remove subelement 030 from the Authorization Request/0100, Authorization Request Response/0110, Authorization Advice/0120, Authorization Advice Response/0130, Reversal Request/ 0400, and Reversal Request Response/0410 messages when the either acquirer or the issuer country code is not 392 (Japan).

## Subelement 031 (Japan Domestic Response Code)

Subelement 031 (Japan Domestic Response Code) is used in the Dual Message Authorization System to provide data required by acquirers and issuers in Japan for intracountry processing.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 031                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | ans-3                     |
| Data Field             | Contents of Positions 1–3 |
| Subfields              | N/A                       |
| Justification          | Right with leading zeros  |

### Usage

Following is the usage of subelement 031 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | X   | C   |
| Reversal Request Response/0410      | C   | X   | C   |

### Values

| Value      | Positions | Format |
|------------|-----------|--------|
| Error Code | 01–03     | ans-3  |

### Application notes

If present, Mastercard will remove subelement 031 from the Authorization Request Response/0110 and Reversal Request Response/0410 messages when the acquirer or the issuer country code is not 392 (Japan).

## Subelement 032 (Japan Payment Options)

Subelement 032 (Japan Payment Options) is used in the Dual Message Authorization System to provide data for Japan domestic transactions.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 032                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | n-36                      |
| Data Field             | Contents of subfields 1–8 |
| Subfields              | 8                         |
| Justification          | See subfields             |

### Usage

Following is the usage of subelement 032 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | O   | .   | C   |
| Authorization Request Response/0110           | CE  | XE  | CE  |
| Authorization Advice/0120: acquirer-generated | O   | .   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Authorization Advice Response/0130            | CE  | XE  | CE  |
| Reversal Request/0400                         | C   | P   | C   |
| Reversal Request Response/0410                | CE  | XE  | CE  |
| Reversal Advice/0420: system-generated        | .   | X   | C   |
| Reversal Advice Response/0430                 | CE  | X   | .   |

### Values

See subfields.

## Application notes

If subelement 032 is present then all subfields must also be present.

Mastercard provides subelement 032 to facilitate communication of data between the acquirer and issuer. Mastercard does not enforce edits to check the presence of any data. Transactions that do not include DE 112, subelement 032 data will not be rejected.

Mastercard will remove subelement 032 from the Authorization Request/0100, Authorization Request Response/0110, Authorization Advice/0120, Authorization Advice Response/0130, Reversal Request/0400, and Reversal Request Response/0410 messages when either the acquirer or the issuer country code is not 392 (Japan).

## Subfield 1 (Payment Option Code)

DE 112, subelement 032, subfield 1 (Payment Option Code) indicates how the cardholder wants to pay for the transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 1–2 |
| Justification       | Left                      |

### Values

| Value | Description                                                                     |
|-------|---------------------------------------------------------------------------------|
| 10    | Single Payment                                                                  |
| 21    | Bonus Pay                                                                       |
| 22    | Bonus Pay (with number of bonuses)                                              |
| 23    | Bonus Pay (with bonus month)                                                    |
| 24    | Bonus Pay (with number of bonuses and bonus month)                              |
| 31    | Bonus Pay with Installments                                                     |
| 32    | Bonus Pay with Installments (with bonus amount)                                 |
| 33    | Bonus Pay with Installments (with number of bonuses, bonus month)               |
| 34    | Bonus Pay with Installments (with number of bonuses, bonus month, bonus amount) |

| Value | Description                            |
|-------|----------------------------------------|
| 61    | Installments (Twice Pay/Three or More) |
| 80    | Revolving                              |

### Application notes

This field is required and must contain one of the listed values.

### Subfield 2 (Bonuses Per Year)

DE 112, subelement 032, subfield 2 (Bonuses Per Year) indicates if the cardholder receives one or two bonuses per year.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 3–4 |
| Justification       | Right with leading zeros  |

### Values

| Value | Description          |
|-------|----------------------|
| 01    | One bonus per year   |
| 02    | Two bonuses per year |
| 00    | Not Applicable       |

### Application notes

This field must contain values 01 or 02 for Payment Option Codes 22, 24, 33, or 34; otherwise zero filled.

### Subfield 3 (First Bonus Month)

DE 112, subelement 032, subfield 3 (First Bonus Month) identifies month of the next bonus payment received by the cardholder following the purchase date.

#### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-2                       |
| Data Field          | Contents of positions 5–6 |
| Justification       | Right with leading zeros  |

#### Values

| Value | Description    |
|-------|----------------|
| 01–12 | Month          |
| 00    | Not Applicable |

#### Application notes

This field must contain a value 01 to 12 for Payment Option Codes 23, 24, 33, or 34; otherwise zero filled.

### Subfield 4 (First Bonus Amount)

DE 112, subelement 032, subfield 4 (First Bonus Amount) indicates the amount of first bonus payment. The currency is the same as the transaction currency.

#### Attributes

| Attribute           | Description                |
|---------------------|----------------------------|
| Data Representation | n-12                       |
| Data Field          | Contents of positions 7–18 |
| Justification       | Right with leading zeros   |

### Values

| Value                     | Description                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------|
| 000000000000–999999999999 | Amount of first bonus payment. The currency is the same as the transaction currency. Default is all zeros. |

### Application notes

This field must contain a Bonus Amount for Payment Option Codes 32 and 34; otherwise zero filled.

### Subfield 5 (Second Bonus Month)

DE 112, subelement 032, subfield 5 (Second Bonus Month) identifies the month of the second bonus payment received by the cardholder following the purchase date.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-2                         |
| Data Field          | Contents of positions 19–20 |
| Justification       | Right with leading zeros    |

### Values

| Value | Description    |
|-------|----------------|
| 01–12 | Month          |
| 00    | Not Applicable |

### Application notes

This field must contain a value of 01 to 12 when Payment Option Code is 24, 33, or 34 and subfield 2 (Bonuses Per Year) indicates 2; otherwise it should be zero filled.



## Subfield 6 (Second Bonus Amount)

DE 112, subelement 032, subfield 6 (Second Bonus Amount) indicates the amount of second bonus payment. The currency is the same as the transaction currency.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-12                        |
| Data Field          | Contents of positions 21–32 |
| Justification       | Right with leading zeros    |

### Values

| Value                     | Description                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| 000000000000–999999999999 | Amount of second bonus payment. The currency is the same as the transaction currency. Default is all zeros. |

### Application notes

This field should contain a Bonus Amount when Payment Option Code is 34 and subfield 2 (Bonuses Per Year) indicates 2; otherwise it should be zero filled.

## Subfield 7 (Total Number of Installments)

DE 112, subelement 032, subfield 7 (Total Number of Installments) indicates the number of installments.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-2                         |
| Data Field          | Contents of positions 33–34 |
| Justification       | Right with leading zeros    |

### Values

| Value | Description            |
|-------|------------------------|
| 01–99 | Number of installments |
| 00    | Not Applicable         |

### Application notes

This field must contain a value of 01 to 99 when Payment Option Code is 31, 32, 33, 34, or 61; otherwise zero filled.

### Subfield 8 (First Installment Month)

DE 112, subelement 032, subfield 8 (First Installment Month) indicates the month in which the first installment will be paid.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-2                         |
| Data Field          | Contents of positions 35–36 |
| Justification       | Right with leading zeros    |

### Values

| Value | Description    |
|-------|----------------|
| 01–12 | Month          |
| 00    | Not Applicable |

### Application notes

This field must contain a value of 01 to 12 when Payment Option Code is 31, 32, 33, 34, or 61; otherwise zero filled.

## Mexcta: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Mexcta payment transactions within Mexico.

### Subelement 004 (Credit Line Usage Fee (CLUF))

DE 112, Mexcta, subelement 004 (Credit Line Usage Fee [CLUF]) contains the currency code and fee amount associated with the CLUF for domestic credit card cash advance (CCCA) ATM transactions in Authorization Request Response/0110 and Authorization Reversal Request/0400 Messages.

#### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 004                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | n-11                       |
| Data Field             | Contents of positions 1-11 |
| Subfields              | N/A                        |
| Justification          | Right-justified            |

#### Values

Numeric currency code (positions 1-3).

Fee amount (positions 4-11).

### Subelement 005 (Issuing Bank Name (AKA Doing Business As [DBA]))

DE 112, Mexcta, subelement 005 (Issuing Bank Name [AKA Doing Business As (DBA)]) contains the issuing bank name (DBA) for ATM transactions where the Credit Line Usage Fee (CLUF) applies in Authorization Request Response/0110 and Authorization Reversal Request/0400 Messages.

#### Attributes

| Attribute     | Description |
|---------------|-------------|
| Subelement ID | 005         |

| Attribute              | Description                     |
|------------------------|---------------------------------|
| Subelement Data Length | 3 positions                     |
| Data Representation    | ans-20                          |
| Data Field             | Contents of positions 1–20      |
| Subfields              | N/A                             |
| Justification          | Left-justified, trailing spaces |

## Subelement 006 (Financial Institution ID (FIID))

DE 112, Mexcta, subelement 006 (Financial Institution ID [FIID]) is used in Credit Line Usage Fee (CLUF) transactions on domestic ATM credit card cash advance transactions in the Dual Message Authorization System in Authorization Request Response/0110 and Authorization Reversal Request/0400 Messages.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 006                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans-10                     |
| Data Field             | Contents of positions 1–10 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Values

Valid FIID.

## Subelement 007 (Installment Payment Data)

DE 112, Mexcta, subelement 007 (Installment Payment Data) is used in the Dual Message Authorization System for Mexcta payment transactions.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 007                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | ans-9                     |
| Data Field             | Contents of positions 1–9 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

### Values

#### Mexcta Payment Transactions in the Authorization Request/0100

| Value | Description                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XX    | Installment pay plan types (positions 1–2). Valid values include: <ul style="list-style-type: none"> <li>00 (No promotion)</li> <li>03 (Without interest for the cardholder)</li> <li>05 (With interest for the cardholder)</li> <li>07 (Buy today, pay later)</li> </ul>     |
| YY    | The total number of installments (positions 3–4). Valid value is 01–99.                                                                                                                                                                                                       |
| ZZ    | Grace period before first payment (positions 5–6). Valid values is 00–99.                                                                                                                                                                                                     |
| NNN   | Transaction Currency Code (positions 7–9). Same value as DE 49 (Currency Code, Transaction).<br><br>This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction).<br><br>484 = Mexican Peso |

### Mexcta Payment Transactions in the Authorization Request Response/0110

| Value | Description                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XX    | Installment pay plan types (positions 1–2). Valid values include: <ul style="list-style-type: none"><li>• 00 (No promotion)</li><li>• 03 (Without interest for the cardholder)</li><li>• 05 (With interest for the cardholder)</li><li>• 07 (Buy today, pay later)</li></ul>  |
| YY    | The total number of installments (positions 3–4). Valid value is 01–99.                                                                                                                                                                                                       |
| ZZ    | Grace period before first payment (positions 5–6). Valid value is 00–99.                                                                                                                                                                                                      |
| NNN   | Transaction Currency Code (positions 7–9). Same value as DE 49 (Currency Code, Transaction).<br><br>This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction).<br><br>484 = Mexican Peso |

### Subelement 008 (Installment Payment Response Data)

DE 112, Mexcta, subelement 008 (Installment Payment Response Data) is used in the Dual Message Authorization System and Mexcta payment transactions in the Authorization Request Response/0110.

#### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 008                        |
| Subelement Data Length | 23 positions               |
| Data Representation    | ans-23                     |
| Data Field             | Contents of positions 1–23 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Values

| Value                                           | Description                                                                                                                                                |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment amount (positions 1–12)             | Installment amount (with two decimal places) including any issuer-calculated interest, insurance, or other charges.                                        |
| Due Date of First Installment (positions 13–18) | Due Date of First Installment (in binary format: ddmmyyyy)                                                                                                 |
| Finance currency code (positions 19–21)         | This is the currency in which the issuer will finance the transaction.<br>This specifies the currency used in installment amount, above 484 (Mexican Peso) |
| Payment Plan (positions 22–23)                  | Reserved for future use. Insert all zeros.                                                                                                                 |

## Netherlands: IBAN (Account Inquiry)

Following are the technical specifications for Account Status Inquiry (ASI) Authorization Request Response/0110 messages that enable customers in the Netherlands to share customer data such as the International Bank Account Number (IBAN).

### Subelement 037 (Additional Cardholder Information)

DE 112 (Additional Data [National Use]), subelement 037 contains cardholder information to help track debit card activity.

### Attributes

| Attribute              | Description                                                                                |
|------------------------|--------------------------------------------------------------------------------------------|
| Subelement ID          | 037                                                                                        |
| Subelement Data Length | 003 positions                                                                              |
| Data Representation    | an...076; LLLVAR                                                                           |
| Data Field             | Contents of subfields 1–2                                                                  |
| Subfields              | Subfield 01: Primary Cardholder Identifier<br>Subfield 02: Secondary Cardholder Identifier |

### Usage

Following is the usage of subelement 37 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | C   | P   | C   |

### Values

See subfields.

### Application notes

Issuers will add DE 112, SE 037 as part of the ASI Authorization Request Response/0110 messages for the Netherlands and add the IBAN and the four-digit card number on the respective subfields. This will be done only after successfully validating that the ASI Authorization Request/0100 message is received by the issuer is a domestic transit ASI request.

### Subfield 01 (Primary Cardholder Identifier)

DE 112, subelement 037 (Additional Cardholder Information), subfield 01 (Primary Cardholder Identifier) is used by customers to share the unique cardholder identifier.

### Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 01                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an...34; LLVAR          |
| Data Field           | Contents of subfield 01 |
| Justification        | Left                    |

### Values

For Netherlands intracountry transit activity issuers will populate this subfield with the 18-digit International Bank Account Number (IBAN) within the ASI Authorization Request Response/0110 message to the acquirer.

Issuers will populate this field only upon successful validation of ASI Authorization Request/0100 message and then respond back with an ASI Authorization Request Response/0110 message with this additional DE 112, SE 037, SF 01.



## Example

The sample below is an example of an International Bank Account Number (IBAN) for the Netherlands.

## Sample

|                   |                      |
|-------------------|----------------------|
| IBAN              | NL91ABAB0XX71XX300   |
| ISO Country Code  | NL (The Netherlands) |
| IBAN Check Digits | 91                   |
| BBAN              | ABAB0XX71XX300       |
| Bank Identifier   | ABAB                 |
| Account Number    | 0XX71XX300           |

The IBAN consists of a two-letter country code, two check digits and a Basic Bank Account Number (BBAN). A BBAN includes information about the domestic bank and account number. The IBAN print format adds one space after every four characters whereas the electronic format contains no spaces.

## Subfield 02 (Secondary Cardholder Identifier)

DE 112, subelement 037 (Additional Cardholder Information), subfield 02 (Secondary Cardholder Identifier) is used by customers to share the unique cardholder identifier.

## Attributes

| Attribute            | Description             |
|----------------------|-------------------------|
| Subfield ID          | 02                      |
| Subfield Data Length | 2 positions             |
| Data Representation  | an...34; LLVAR          |
| Data Field           | Contents of subfield 02 |
| Justification        | Left                    |

## Values

For Netherlands intracountry transit-related ASI Authorization Request Response/0110 messages will be populated with the four-digit card number information of the cardholder, provided by the issuer to the acquirer, which is then provided to the merchant.

## Paraguay: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Paraguay payment transactions within Paraguay.

### Subelement 042 (Acceptor Tax ID)

Subelement 042 (Acceptor Tax ID) will be used in the Dual Message Authorization System to support DE 112 in any received Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message.

#### Attributes

| Attribute              | Description      |
|------------------------|------------------|
| Subelement ID          | 042              |
| Subelement Data Length | 3 positions      |
| Data Representation    | ans...12; LLLVAR |

#### Usage

Following is the usage of subelement 042 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages:

#### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | •   |
| Authorization Advice/0120: acquirer-generated | C   | X   | •   |
| Reversal Request /0400                        | C   | X   | •   |

#### Application notes

This field is required when purchase, purchase with cash back, and refund transactions are submitted by acquirers for merchants in Chile and Paraguay.

## Parcelas: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Parcelas payment transactions within Brazil.

### Subelement 001 (Installment Payment Data)

DE 112, Parcelas, subelement 001 (Installment Payment Data) is used in the Dual Message Authorization System for Parcelas payment transactions.

#### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 001                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | ans-4                     |
| Data Field             | Contents of positions 1–4 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

#### Values

#### Parcelas Payment Transactions in the Authorization Request/0100

Used for both issuer- or merchant-financed Parcelas transactions: From acquirer to issuer requiring approval; use the following format:

|    |                              |
|----|------------------------------|
| XX | Parcelas plan type:          |
|    | 20 = Issuer-financed         |
|    | 21 = Merchant-financed       |
|    | 25 = Crediário               |
| YY | The total number of Parcelas |

#### Parcelas Payment Transactions in the Authorization Request Response/0110

Used for both issuer- or merchant-financed Parcelas transactions: From acquirer to issuer requiring approval. Echoing this subelement in the Authorization Request Response is unique to Brazil processing. Use the following format:

|    |                              |
|----|------------------------------|
| XX | Parcelas plan type:          |
|    | 20 = Issuer-financed         |
|    | 21 = Merchant-financed       |
|    | 25 = Crediário               |
| YY | The total number of Parcelas |

## Subelement 002 (Installment Payment Response Data)

DE 112, Parcelas, subelement 002 (Installment Payment Response Data) is used in the Dual Message Authorization System and Parcelas payment transactions.

### Attributes

| Attribute              | Description                                                 |
|------------------------|-------------------------------------------------------------|
| Subelement ID          | 002                                                         |
| Subelement Data Length | 3 positions                                                 |
| Data Representation    | ans-32 if format 1 is used and<br>ans-4 if format 2 is used |
| Data Field             | Contents of positions 1–32 or 1–4                           |
| Subfields              | N/A                                                         |
| Justification          | N/A                                                         |

### Values

#### Format 1: Issuer-Financed Parcelas transactions in the Authorization Request Response/0110

Issuer acknowledgment format 1, from issuer to acquirer, upon transaction approval; all amounts are express in transaction currency; contents are as follows:

| Value           | Description                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Positions 1–4   | Parcelas information, same as subelement 001.                                                          |
| Positions 5–16  | Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places. |
| Positions 17–28 | Transaction total amount issuer calculates, including calculated interest; two decimal places.         |

| Value           | Description                                                  |
|-----------------|--------------------------------------------------------------|
| Positions 29–32 | Monthly interest rate issuer calculates; two decimal places. |

### Format 2: Merchant-Financed Parcelas transactions in the Authorization Request Response/0110

Issuer acknowledgment format 2, from issuer to acquirer, upon transaction approval; Parcelas information copied from subelement 001 as follows:

| Value | Description                  |
|-------|------------------------------|
| 21    | Merchant-financed            |
| YY    | The total number of Parcelas |

## Subelement 016 (Additional Installment Payment Response Data)

DE 112, Parcelas, subelement 016 (Additional Installment Payment Response Data) is used in the Dual Message Authorization System and Parcelas payment transactions.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 016                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans-74                     |
| Data Field             | Contents of positions 1–74 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

All Authorization Request Response/0110 messages in response to Parcelas Authorization Request/0100 messages must include additional information to identify the total effective cost of installment transactions.

### Values

| Value          | Description                      |
|----------------|----------------------------------|
| Positions 1–12 | Total Effective Cost (TEC) (N12) |

| Value           | Description                               |
|-----------------|-------------------------------------------|
| Positions 13–19 | Taxes (N7)                                |
| Positions 20–23 | % Taxes of Total Amount (N4)              |
| Positions 24–30 | Fee (N7)                                  |
| Positions 31–34 | % Fee of Total Amount (N4)                |
| Positions 35–41 | Insurance (N7)                            |
| Positions 42–45 | % Insurance of Total Amount (N4)          |
| Positions 46–52 | Other (N7)                                |
| Positions 53–56 | % Other of Total Amount (N4)              |
| Positions 57–63 | Total Amount to Merchant (N7)             |
| Positions 64–67 | % Amount to Merchant of Total Amount (N4) |
| Positions 68–74 | % Total Effect Cost (N7)                  |

## Percta: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Percta payment transactions within Peru.

### Subelement 007 (Installment Payment Data)

DE 112, Percta, Subelement 007 (Installment Payment Data) is used in the Dual Message Authorization System for Percta payment transactions.

#### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 007                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | ans-8                     |
| Data Field             | Contents of positions 1–8 |
| Subfields              | N/A                       |
| Justification          | N/A                       |

## Values

### Percta Payment Transactions in the Authorization Request/0100

| Value | Description                                                                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XX    | Type of credit (positions 1–2). Valid value is 20 (Issuer-financed) In the case of Peru, issuer finance is the only finance model.                                                                                                                                                                     |
| YY    | The total number of installments (positions 3–4). Valid value is 01–99.                                                                                                                                                                                                                                |
| Z     | Grace period before first payment (position 5). Valid values are: <ul style="list-style-type: none"> <li>0 = No grace period</li> <li>1 = One month</li> <li>2 = Two months</li> </ul>                                                                                                                 |
| NNN   | Transaction Currency Code (positions 6–8). Echo of DE 49 (Currency Code, Transaction)<br><br>This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction).<br><br>604 = Peru Nuevo Sol<br>840 = United States dollar |

### Percta Payment Transactions in the Authorization Request Response/0110

| Value | Description                                                                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XX    | Type of credit (positions 1–2). Valid value is 20 (Issuer-financed) In the case of Peru, issuer finance is the only finance model.                                                                                                                                                                     |
| YY    | The total number of installments (positions 3–4). Valid value is 01–99.                                                                                                                                                                                                                                |
| Z     | Grace period before first payment (position 5). Valid values are: <ul style="list-style-type: none"> <li>0 = No grace period</li> <li>1 = One month</li> <li>2 = Two months</li> </ul>                                                                                                                 |
| NNN   | Transaction Currency Code (positions 6–8). Echo of DE 49 (Currency Code, Transaction)<br><br>This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction).<br><br>604 = Peru Nuevo Sol<br>840 = United States dollar |

## Subelement 008 (Installment Payment Response Data)

DE 112, Percta, subelement 008 (Installment Payment Response Data) is used in the Dual Message Authorization System and Percta payment transactions.

### Attributes

| Attribute              | Description                |
|------------------------|----------------------------|
| Subelement ID          | 008                        |
| Subelement Data Length | 3 positions                |
| Data Representation    | ans-23                     |
| Data Field             | Contents of positions 1–23 |
| Subfields              | N/A                        |
| Justification          | N/A                        |

### Values

| Value                                           | Description                                                                                                                                                                                |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment amount (positions 1–12)             | Installment amount (with two decimal places) including any issuer-calculated interest, insurance, or other charges.                                                                        |
| Due Date of First Installment (positions 13–18) | Due Date of First Installment (in binary format: ddmmyyyy)                                                                                                                                 |
| Finance currency code (positions 19–21)         | This is the currency in which the issuer will finance the transaction.<br>This specifies the currency used in installment amount, above 604 (Peru Nuevo Sol) or 840 (United States dollar) |
| Payment Plan (positions 22–23)                  | Reserved for future use. Insert all zeros.                                                                                                                                                 |

## Poland: Value-Added Tax specifications

Following are technical specifications for Poland Value-Added Tax (VAT) Split payment transactions in the Authorization Request/0100 and Authorization Response/0110.

### Subelement 041 (Poland VAT Data)

DE 112, Poland, subelement 041 (Poland VAT Data) is used in the Dual Message Authorization System for Poland VAT split payment transaction. It should be present in Authorization



Request/0100 and Authorization Request Response/0110 messages when processing a VAT Split payment transaction.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Subelement ID       | 041                       |
| Data Length         | 3 positions               |
| Data Representation | ans-84                    |
| Data Field          | Contents of position 1-84 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Values

| Positions | Length | Description     | Comments                            |
|-----------|--------|-----------------|-------------------------------------|
| 1-12      | 12     | VAT Amount      | Right-justified with leading zeros  |
| 13-24     | 12     | Net Amount      | Right-justified with leading zeros  |
| 25-44     | 20     | Merchant Tax ID | Right-justified with leading spaces |
| 45-84     | 40     | Invoice Number  | Right-justified with leading spaces |

Poland VAT Split payment transactions in the Authorization Request/0100 and Authorization Response/0110.

## Spain: Domestic ATM Transactions

Following are the technical specifications for Authorization/01xx messages for domestic ATM transactions within Spain.

### Subelement 017 (ATM Domestic Fee)

DE 112, subelement 017 (ATM Domestic Fee) contains fees charged for ATM domestic transactions in Spain.

#### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 017                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | n...48; LLLVAR            |
| Data Field             | Contents of subfields 1–3 |
| Subfields              | 3                         |
| Justification          | See subfields.            |

#### Usage

Following is the usage of subelement 017 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

#### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | C   | P   | C   |
| Authorization Request Response/0110 | C   | P   | C   |
| Reversal Request/0400               | C   | P   | C   |

#### Values

See subfields.

## Subfield 1 (ATM Service Fee)

DE 112, subelement 017, subfield 1 (ATM Service Fee) contains fee amount in the currency of DE 49 (Currency Code, Transaction) charged to the issuer to compensate the ATM acquirer for the ATM cost incurred for ATM domestic transactions in Spain.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 01                       |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-12                     |
| Data Field           | Contents of subfield 1   |
| Justification        | Right with leading zeros |

### Values

ATM Service Fee amount in transaction currency.

## Subfield 2 (ATM Disloyalty Fee)

DE 112, subelement 017, subfield 2 (ATM Disloyalty Fee) contains fee amount in the currency of DE 51 (Currency Code, Cardholder Billing), charged by the issuer to the cardholder for using another bank's ATM for ATM domestic transactions in Spain.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 02                       |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-12                     |
| Data Field           | Contents of subfield 2   |
| Justification        | Right with leading zeros |

### Values

ATM Disloyalty Fee amount in Cardholder Billing currency.

### Subfield 3 (Credit Card Fee)

DE 112, subelement 017, subfield 3 (Credit Card Fee) contains fee amount in the currency of DE 51 (Currency Code, Cardholder Billing) charged by the issuer to the cardholder for the grant of credit for ATM domestic transactions in Spain.

#### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 03                       |
| Subfield Data Length | 2 positions              |
| Data Representation  | n-12                     |
| Data Field           | Contents of subfield 3   |
| Justification        | Right with leading zeros |

#### Values

Credit Card Fee amount in Cardholder Billing currency.

## Taiwan: National Data

Following are the technical specifications specific to Taiwan domestic transactions.

### Subelement 038 (Taiwan National Data)

DE 112, subelement 038 (Taiwan National Data) is used in the authorization message to pass national data between acquirers and issuers for intracountry processing.

#### Attributes

| Attribute              | Description                 |
|------------------------|-----------------------------|
| Subelement ID          | 038                         |
| Subelement Data Length | 3 positions                 |
| Data Representation    | ans..256; LLLVAR            |
| Data Field             | Contents of positions 1–256 |
| Subfields              | N/A                         |
| Justification          | N/A                         |

## Usage

Following is the usage of subelement 038 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | O   | .   | C   |
| Authorization Request Response/0110 | O   | .   | C   |
| Reversal Request/0400               | O   | .   | C   |
| Reversal Request Response/0410      | O   | .   | C   |

## Values

Market agreed values for installment data and other national specific data fields.

## United Kingdom: Debt Repayment Transactions

Following are the technical specifications for Authorization/01xx messages for United Kingdom debt repayment transactions within the U.K.

### Subelement 033 (UK Recipient Details)

DE 112, subelement 033 (UK Recipient Details) supports the transmission of recipient details in an authorization request from acquirer to issuer.

### Attributes

| Attribute              | Description               |
|------------------------|---------------------------|
| Subelement ID          | 033                       |
| Subelement Data Length | 3 positions               |
| Data Representation    | ans...89; LLLVAR          |
| Data Field             | Contents of subfields 1–4 |
| Subfields              | 4                         |
| Justification          | See subfields             |

## Usage

Following is the usage of subelement 033 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

## Values

See subfields.

### Subfield 1 (Recipient Last Name)

DE 112, subelement 033 (UK Recipient Details), subfield 1 (Recipient Last Name) contains the last name of the recipient.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 01                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ans...35; LLVAR        |
| Data Field           | Contents of subfield 1 |
| Justification        | N/A                    |

## Values

Contains valid last name of the recipient (only the first six letters of the last name are required).

## Subfield 2 (Recipient Postal Code)

DE 112, subelement 033 (UK Recipient Details), subfield 2 (Recipient Postal Code) contains the postal code of the recipient.

### Attributes

| Attribute            | Description                         |
|----------------------|-------------------------------------|
| Subfield ID          | 02                                  |
| Subfield Data Length | 10 positions                        |
| Data Representation  | ans-10                              |
| Data Field           | Contents of subfield 2              |
| Justification        | left-justified with trailing spaces |

### Values

Contains valid location postal code of the recipient. First part represents the district, then space-filled.

## Subfield 3 (Recipient Date of Birth)

DE 112, subelement 033 (UK Recipient Details), subfield 3 (Recipient Date of Birth) contains the date of birth of the recipient.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 03                     |
| Subfield Data Length | 08 positions           |
| Data Representation  | n-8                    |
| Data Field           | Contents of subfield 3 |
| Justification        | N/A                    |

### Values

Contains valid date of birth of the recipient in the format YYYYMMDD.

## Subfield 4 (Recipient Account Number)

DE 112, subelement 033 (UK Recipient Details), subfield 4 (Recipient Account Number) contains valid account number of the recipient.

### Attributes

| Attribute            | Description            |
|----------------------|------------------------|
| Subfield ID          | 04                     |
| Subfield Data Length | 2 positions            |
| Data Representation  | ns...20; LLVAR         |
| Data Field           | Contents of subfield 4 |
| Justification        | N/A                    |

### Values

Contains the recipient account number.

## DE 113 (Reserved for National Use)

DE 113 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 114 (Reserved for National Use)

DE 114 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 115 (Reserved for National Use)

DE 115 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**NOTE: The Dual Message Authorization System currently does not use this data element.**



## DE 116 (Mastercard Internal Use Data 2)

DE 116 (Mastercard Internal Use Data 2) is reserved by Mastercard for internal use only. Mastercard will never send DE 116 to external message receivers. External message originators must not include DE 116 in any messages sent to Mastercard. If DE 116 is sent to Mastercard, it will be ignored.

## DE 117 (Reserved for National Use)

DE 117 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 118 (Reserved for National Use)

DE 118 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**NOTE: The Dual Message Authorization System currently does not use this data element.**

## DE 119 (Additional Data: Private Use 2)

DE 119 (Additional Data: Private Use 2) contains data associated with various Mastercard programs, products, and services. DE 119 is used to convey data that is not related to any other ISO-defined data elements. It is a variable-length data element used for multiple purposes.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | ans...999; LLLVAR       |
| Data Element Length | 3 positions             |
| Data Field          | Contents of subelements |
| Subelements         | 3                       |
| Justification       | See Subelements         |

## Usage

The following applies to DE 119 usage:

- Subelements do not need to be in any particular order or sequence within DE 119.
- Customers must be able to send and receive all subelements available within DE 119.
- Subelements 900-999 are reserved for Mastercard internal use.

Following is the usage of DE 119 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

## Applicable messages

| Message                                | Org | Sys | Dst |
|----------------------------------------|-----|-----|-----|
| Authorization Request/<br>0100         | •   | X   | C   |
| Authorization Request<br>Response/0110 | •   | X   | C   |

## Application notes

Mastercard may introduce new DE 119 subelements and subfields between releases to facilitate special processing within a country, region, or among pilot participants. Mastercard requires customers to be able to successfully process various online messages that may contain new unannounced DE 119 subelements and subfields.

DE 119 and DE 48 contain different subelement definitions. Subelements defined for DE 119 must be sent in DE 119, not DE 48.

For information on other uses of this data element, refer to the message layouts.

## Special notes

Acquirers and issuers must be able to send and receive all DE 119 subelements. Responding to DE 119 subelements is conditional, or optional, per individual subelement.

Responding to DE 119 subelements may depend on brand, purpose of the request, and conditional to the individual program or service agreement between the Single Message System and the issuer.

## DE 119 encoding scheme

Mastercard organizes DE 119 into a group of encoded subelements and subfields. The following tables illustrate the structure of DE 119 subelements.

### Structure of DE 119 subelements

| LLL          | VAR (maximum length varies) |                   |                                 |                        |                   |                                 |
|--------------|-----------------------------|-------------------|---------------------------------|------------------------|-------------------|---------------------------------|
| 3 bytes      | 3 bytes                     | 3 bytes           | Variable length                 | 3 bytes                | 3 bytes           | Variable length                 |
| Total length | First subelement data       |                   |                                 | Second subelement data |                   |                                 |
|              | Subelement ID               | Subelement length | Subelement variable length data | Subelement ID          | Subelement length | Subelement variable length data |

### Attributes of DE 119 subelements

| Number of bytes | Attribute                       | Value                                 |
|-----------------|---------------------------------|---------------------------------------|
| 3               | Total data element length       | The "LLL" portion of the data element |
| 3               | Subelement ID                   | In the range 001-999                  |
| 3               | Subelement length               | Variable                              |
| 1...nnn         | Subelement variable length data | Contents of subfields                 |

## DE 119 subelement encoding scheme

The following tables illustrate the structure of DE 119 subfields within a subelement.

### Structure of DE 119 subfields within a subelement

| VAR (maximum length varies) |         |                 |                      |         |                 |
|-----------------------------|---------|-----------------|----------------------|---------|-----------------|
| 3 bytes                     | 3 bytes | Variable length | 3 bytes              | 3 bytes | Variable length |
| First subfield data         |         |                 | Second subfield data |         |                 |

| VAR (maximum length varies) |                 |                               |             |                 |                               |
|-----------------------------|-----------------|-------------------------------|-------------|-----------------|-------------------------------|
| Subfield ID                 | Subfield length | Subfield variable length data | Subfield ID | Subfield length | Subfield variable length data |

#### Attributes of DE 119 subfields within a subelement

| Number of bytes | Attribute                     | Value                 |
|-----------------|-------------------------------|-----------------------|
| 3               | Subfield ID                   | In the range 001-999  |
| 3               | Subfield length               | Variable              |
| 1...nnn         | Subfield variable length data | Contents of subfields |

## Subelement 003 (MDES Service Indicators for Issuers)

DE 119, subelement 003 (MDES Service Indicators for Issuers) provides indicators to issuers about issuer-related services that were applied to the transaction.

#### Attributes

| Attribute           | Description                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| Subelement ID       | 003                                                                                                           |
| Subelement Length   | 3                                                                                                             |
| Data Representation | ans...98; LLLVAR<br>The "LLL" length field of the LLLVAR must be an integral multiple of 2, not to exceed 098 |
| Justification       | N/A                                                                                                           |

#### Usage

Following is the usage of DE 119, subelement 003 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | .   | X   | C   |

### Application notes

The Dual Message Authorization System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

This string consists of a repeating 2-byte structure, where each 2-byte segment serves as an indicator of the service performed.

## Subelement 004 (MDES Service Indicators for Acquirers)

DE 119, subelement 004 (MDES Service Indicators for Acquirers) provides indicators to acquirers about acquirer-related services that were applied to the transaction.

### Attributes

| Attribute           | Description                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| Subelement ID       | 004                                                                                                           |
| Subelement Length   | 3                                                                                                             |
| Data Representation | ans...98; LLLVAR<br>The "LLL" length field of the LLLVAR must be an integral multiple of 2, not to exceed 098 |
| Justification       | N/A                                                                                                           |

### Usage

Following is the usage of DE 119, subelement 004 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | .   | X   | C   |

### Application notes

The Dual Message Authorization System inserts this subelement into Authorization Request Response/0110 messages.

## Subelement 005 (MDES Service Indicators for Issuers and Acquirers)

DE 119, subelement 005 (MDES Service Indicators for Issuers and Acquirers) provides indicators to both issuers and acquirers about common services that were applied to the transaction.

### Attributes

| Attribute           | Description                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| Subelement ID       | 005                                                                                                            |
| Subelement Length   | 3                                                                                                              |
| Data Representation | ans...098; LLLVAR<br>The "LLL" length field of the LLLVAR must be an integral multiple of 2, not to exceed 098 |
| Justification       | N/A                                                                                                            |

### Usage

Following is the usage of DE 119, subelement 005 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request/0100          | •   | X   | C   |
| Authorization Request Response/0110 | •   | X   | C   |

### Application notes

The Dual Message Authorization System inserts this subelement into:

- Authorization Request/0100 messages when the issuer has opted-in to receive it
- Authorization Request Response/0110 messages

This string consists of a repeating 2-byte structure, where each 2-byte segment serves as an indicator of the service performed.

Reserved for future Service Indicators.

Refer to *MDES Technical Specifications for Dual and Single Message Systems* for more information.

## DE 120 (Record Data)

DE 120 (Record Data) is a variable-length data element used for transmitting file record data or textual character string data in various message types.

### Attributes

| Attribute           | Description                                          |
|---------------------|------------------------------------------------------|
| Data Representation | ans...999; LLLVAR (supports extended character sets) |
| Data Length         | 3 positions                                          |
| Data Field          | Contents of subfields 1, or 2, or 3, or 4            |
| Subfields           | 4                                                    |
| Justification       | See subfields                                        |

### Usage

Following is the usage of DE 120 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                      | Org | Sys | Dst |
|----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                   | C   | X   | C   |
| Authorization Request Response/0110          | C   | X   | C   |
| Issuer File Update Request/0302              | C   | X   | .   |
| Issuer File Update Request Response/0312     | .   | X   | C   |
| Administrative Advice/0620: system-generated | .   | X   | C   |
| Administrative Advice/0620: member-generated | C   | P   | C   |

## Values

See subfields.

## Application notes

The following applies to DE 120.

### Authorization Request/0100: AVS messages

When DE 120 is present in Authorization Request/0100 messages, DE 120 contains Address Verification Service data. Only one of the four possible subfields will be present in a message from an acquirer or to an issuer. In Authorization Request Response/0110: AVS messages, DE 120 contains the AVS data originally provided in the Authorization Request/0100 message.

### Issuer File Update Request/0302 messages

When DE 120 is present in Issuer File Update Request/0302 messages, DE 120 contains the new, actual file record data used in "add" or "change" file maintenance requests. In Issuer File Update Request Response/0312 messages, DE 120 contains actual record data for file maintenance requests.

### Administrative Advice/0620 messages

When DE 120 is present in Administrative Advice/0620 messages, DE 120 contains the following:

- In Administrative Advice/0620 messages, DE 120 includes the new subfields Token Requestor ID, Wallet ID, and Device Type.
- In Administrative Advice/0620: System-generated message where DE 60 is 600, DE 120 contains the first 200 bytes of an indecipherable message.
- In Administrative Advice/0620: Member-generated message where DE 60 is 650, DE 120 contains member-provided free format textual data.

## Subfield 01 (AVS Service Indicator 1)

DE 120, subfield 01 (AVS Service Indicator 1) contains AVS data in this format for an issuer whose AVS Service Indicator status is 1.

### Attributes

| Attribute            | Description                |
|----------------------|----------------------------|
| Subfield ID          | 01                         |
| Subfield Data Length | 2 positions                |
| Data Representation  | ans...29                   |
| Data Field           | Contents of positions 1–29 |



| Attribute     | Description           |
|---------------|-----------------------|
| Justification | Left, trailing spaces |

### Values

| Value                                             | Description                |
|---------------------------------------------------|----------------------------|
| Positions 1–9: Postal Code                        |                            |
| Postal Code                                       | Cardholder postal/ZIP code |
| Positions 10–29: Address (Mastercard)             |                            |
| Address                                           | Cardholder address         |
| Positions 10–29: Address (Visa, American Express) |                            |
| Address                                           | Cardholder address         |

### Application notes

Some merchants or acquirers are currently limited to supporting only numeric data.

The Authorization System will not truncate the address data passed from the acquirer if greater than 29 positions. DE 120 is an LLLVAR field; therefore, it will indicate the length of the data being passed. The issuer should code to accept the data if it is more than 29 positions.

## Subfield 02 (AVS Service Indicator 2)

DE 120, subfield 02 (AVS Service Indicator 2) contains AVS data in this format for an issuer whose AVS Service Indicator status is 2.

### Attributes

| Attribute            | Description                |
|----------------------|----------------------------|
| Subfield ID          | 02                         |
| Subfield Data Length | 2 positions                |
| Data Representation  | an-14                      |
| Data Field           | Contents of positions 1–14 |
| Justification        | Left, blank-filled         |

### Values

| Value                                 | Description                |
|---------------------------------------|----------------------------|
| Positions 1–9: Postal Code            |                            |
| Postal Code                           | Cardholder postal/ZIP code |
| Positions 10–14: Address (Mastercard) |                            |
| Address                               | Cardholder address         |

### Application notes

Issuer receives condensed address data. (This supports the algorithm that uses the first five numeric digits in an address [when scanning the address from left to right].)

## Subfield 03 (AVS Service Indicator 3)

DE 120, subfield 03 (AVS Service Indicator 3) contains AVS data in this format for an issuer whose AVS Service Indicator status is 3.

### Attributes

| Attribute            | Description                |
|----------------------|----------------------------|
| Subfield ID          | 03                         |
| Subfield Data Length | 2 positions                |
| Data Representation  | an-14                      |
| Data Field           | Contents of positions 1–14 |
| Justification        | Left, blank-filled         |

### Values

| Value                                 | Description                |
|---------------------------------------|----------------------------|
| Positions 1–9: Postal Code            |                            |
| Postal Code                           | Cardholder postal/ZIP code |
| Positions 10–14: Address (Mastercard) |                            |

| Value   | Description        |
|---------|--------------------|
| Address | Cardholder address |

### Application notes

The issuer receives condensed address data. (This subfield supports the algorithm that uses up to the first five numeric digits that appear before the first alphabetic character or space in the address when scanning the address from left to right.)

## Subfield 04 (AVS Service Indicator 4)

DE 120, subfield 04 (AVS Service Indicator 4) contains AVS data in this format for an issuer whose AVS Service Indicator status is 4.

### Attributes

| Attribute            | Description                |
|----------------------|----------------------------|
| Subfield ID          | 04                         |
| Subfield Data Length | 2 positions                |
| Data Representation  | an-14                      |
| Data Field           | Contents of positions 1–14 |
| Justification        | Left, blank-filled         |

### Values

| Value                                 | Description                |
|---------------------------------------|----------------------------|
| Positions 1–9: Postal Code            |                            |
| Postal Code                           | Cardholder postal/ZIP code |
| Positions 10–14: Address (Mastercard) |                            |
| Address                               | Cardholder address         |

### Application notes

Issuer supports AVS, receives condensed numeric postal code and condensed numeric address only. (This supports the algorithm that uses only the numeric digits in a postal code and the first five numeric digits in an address [when scanning the address from left to right].)

## Online file maintenance

DE 120 (Record Data), contains the record detail for the file type identified in DE 101 (File Name). The record detail contained in DE 120 depends on the value provided in DE 91 (Issuer File Update Code). The following table lists the allowed DE 91 values for each DE 101 file type.

### Applicable MCCs

| DE 101 (File Name)                                        | DE 91 (Issuer File Update Code) |          |          |           |
|-----------------------------------------------------------|---------------------------------|----------|----------|-----------|
|                                                           | 1 Add                           | 2 Update | 3 Delete | 5 Inquiry |
| MCC102 (Stand-In Account File)                            | √                               | √        | √        | √         |
| MCC103 (Electronic Warning Bulletin File)                 | √                               |          | √        | √         |
| MCC104 (Local Stoplist File)                              | √                               |          | √        | √         |
| MCC105 (Payment Cancellation File)                        | √                               | √        | √        | √         |
| MCC106 (PAN Mapping File)                                 | √                               | √        | √        |           |
| MCC109 (Application Transaction Counter [ATC] File)       | √                               |          |          | √         |
| MCC111 (PAN-PAR [Payment Account Reference] Mapping File) |                                 | √        |          |           |

### MCC 102: Stand-In Account File

DE 120 (Record Data) contains Stand-In Account file data when DE 101 (File Name) contains MCC102.

### Associated edits

Following are the MCC102 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

| When DE 91 Contains... | Include These Fields in DE 120... |
|------------------------|-----------------------------------|
| 1 = Add                | use all appropriate fields        |
| 2 = Update             | Use all appropriate fields        |
| 3 = Delete             | Use field 1                       |
| 5 = Inquiry            | Use field 1                       |

## Fields

Following is the DE 120 Layout for MCC102.

| Field ID | Field Name                | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                            |
|----------|---------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Primary Account Number    | an-19      | <p>Number that is embossed or encoded or both on the card. Customers may only input account numbers for BINs assigned to the associated Mastercard-assigned customer ID.</p> <p>Format: Sixteen digit Mastercard account number, followed by three spaces.</p>                                   |
| 2        | Entry Reason              | an-1       | <p>Reason for listing this card. Valid values:</p> <p>C = Credit</p> <p>F = Fraud</p> <p>G = ATM Premium Listing</p> <p>L = Lost</p> <p>O = Other</p> <p>P = Capture Card</p> <p>S = Stolen</p> <p>U = Unauthorized Use</p> <p>V = Premium Listing</p> <p>X = Counterfeit</p>                    |
| 3        | Date Last Update Activity | n-6        | <p>Inquiry only and is ignored on add, change, and delete. (This field is returned in the 0312 response message only for inquiry requests.)</p> <p>Date of the last maintenance activity occurring on this account.</p> <p>Format: MMDDYY</p> <p>MM = month</p> <p>DD = day</p> <p>YY = year</p> |

| Field ID | Field Name                          | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                        |
|----------|-------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4        | Time Last Update Activity           | n-4        | <p>Inquiry only and is ignored on add, change, and delete. (This field returned in the 0312 response message only for inquiry requests.)</p> <p>Time of the last maintenance activity occurring on this account.</p> <p>Format: hhmm</p> <p>hh = hour</p> <p>mm = minute</p> |
| 5        | PIN Length                          | n-2        | No longer applicable. Valid values: zeros.                                                                                                                                                                                                                                   |
| 6        | Premium Listing Accumulative Limit  | n-12       | Valid only for entry reason V. For other reason codes, use 12 zeros.                                                                                                                                                                                                         |
| 7        | Premium Listing Limit Currency Code | n-3        | Valid only for Entry Reason V. For other reason codes, use three zeros.                                                                                                                                                                                                      |
| 8        | Issuer-defined Purge Date           | n-8        | The issuer-defined purge date must be at least 180 days after the account list date in YYYYMMDD format. This field is optional.                                                                                                                                              |
| 9        | Card Sequence Number                | n-3        | The card sequence number of the listed card. Only required for card-level support.                                                                                                                                                                                           |
| 10       | Card Expiration Date                | n-6        | The expiration date of the listed card in YYYYMM format. Only required for card-level support.                                                                                                                                                                               |

## MCC 103: Electronic Warning Bulletin File

DE 120 (Record Data) contains Electronic Warning Bulletin file data when DE 101 (File Name) contains MCC103.

### Fields

Following is the DE 120 Layout for MCC103.

| Field ID | Field Name             | Attributes | Comments/Valid Values                                                                                                                                                                                                                          |
|----------|------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Primary Account Number | an-19      | Number that is embossed or encoded on the card. Customers can only input account numbers for BINs assigned to the associated Mastercard-assigned customer ID.<br><br>Format: Sixteen digit Mastercard account number followed by three spaces. |
| 2        | Customer ID            | n-6        | Mastercard customer ID assigned to the BIN.<br><br>Format: right-justified, zero-filled                                                                                                                                                        |
| 3        | Card Program           | a-3        | Type of card bearing the account number. Refer to DE 63 (Network Data), for the list of card program values allowed. Card programs not allowed for MCC103 are footnoted in DE 63, subfield 1.                                                  |
| 4        | Response Code          | n-2        | Response Code this account listing should prompt for an authorization request. Valid value: 04 capture card                                                                                                                                    |
| 5        | Entry Reason           | a-1        | Reason for listing this card. Valid values:<br><br>C = Credit<br>F = Fraud<br>O = Other<br>X = Counterfeit                                                                                                                                     |
| 6        | Filler                 | an-25      | Reserved for future AMS enhancements.                                                                                                                                                                                                          |

| Field ID | Field Name                     | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|--------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7        | Regional Indicator/ Purge Date | an-7       | <p>Issuer-requested region and purge date(s) associated with this account number listing. Field is non-positional (excludes regions and purge dates that do not apply). Field may occur up to six times for that many regions and purge dates. Must be entered in ascending order (1, A,..., E).</p> <p>Format: RYYMMDD</p> <p>R = Region</p> <p>Valid region values:</p> <p>1 = United States</p> <p>A = Canada</p> <p>B = Latin America/Caribbean</p> <p>C = Asia/Pacific</p> <p>D = Europe</p> <p>E = Middle East/Africa</p> <p>YY = purge date year</p> <p>MM = purge date month</p> <p>DD = purge date day</p> <p><b>NOTE: To purge an account on the card expiration date, enter that date as the purge date. For additional information on purge dates, refer to the <a href="#">Account Management System User Manual</a>.</b></p> |

## MCC 104: Local Stoplist File

DE 120 (Record Data) contains Local Stoplist file data when DE 101 (File Name) contains MCC104.

### Associated edits

Following are the MCC104 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).



| When DE 91 Contains...                                                                          | Include These Fields in DE 120...              |
|-------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1 = Add                                                                                         | Use all applicable fields.                     |
| <b>NOTE: Add will update an existing record if the system finds a match on the master file.</b> |                                                |
| 1 = Update                                                                                      | Use all applicable fields that require update. |
| 1 = Delete (deletes account from one region)                                                    | Use fields one.                                |
| 1 = Delete (deletes account from one country)                                                   | Use fields one, seven, eight, and ten.         |
| 1 = Delete (deletes account from one subcountry)                                                | Use fields one, seven, eight, nine, and ten.   |
| 3 = Delete (deletes account from all regions, countries, or subcountries)                       | Use field one.                                 |
| 5 = Inquiry                                                                                     | Use field one.                                 |

## Fields

Following is the DE 120 Layout for MCC104.

| Field ID | Field Name             | Attributes | Comments/Valid Values                                                                                                                                                                                                   |
|----------|------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Primary Account Number | an-19      | Number that is embossed, encoded, or both on the card. Customers may input only those account numbers that contain their assigned prefixes.<br><br>Format: 16-digit Mastercard account number followed by three spaces. |
| 2        | Customer ID            | n-6        | Mastercard-assigned customer ID, a Mastercard customer may input only an account number with a BIN assigned to it.<br><br>Format: Right-justified, zero-filled                                                          |
| 3        | Card Program           | a-3        | Type of card bearing the account number. Refer to DE 63 (Network Data), for the list of card program values allowed. Card programs not allowed for MCC104 are footnoted in DE 63, subfield 1.                           |
| 4        | Response Code          | n-2        | Response Code this account listing should prompt for an authorization request. Valid value: 04 capture card                                                                                                             |

| Field ID | Field Name   | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|--------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5        | Entry Reason | a-1        | Reason for listing this card.<br><br>Valid values:<br><br>C = Credit (Prompts an Authorization response of capture card at regional level)<br>OR (Prompts an Authorization response of decline at the country/subcountry level)<br><br>F = Fraud<br><br>O = Other<br><br>X = Counterfeit                                                                                                                              |
| 6        | Filler       | an-25      | Reserved for future AMS enhancements.                                                                                                                                                                                                                                                                                                                                                                                 |
| 7        | Region       | an-1       | The region associated with this account listing. If a country or subcountry listing is entered, this field must contain the issuer's region as follows:<br><br>Valid region values:<br><br>1 = United States<br><br>A = Canada<br><br>B = Latin America/Caribbean<br><br>C = Asia/Pacific<br><br>D = Europe<br><br>E = Middle East/Africa<br><br>Region values must be entered in ascending order (1, A, B, C, D, E). |
| 8        | Country      | an-3       | If applicable, contains the country associated with this account listing. Must be a participating country within the issuer's region; values are three-digit country code or spaces.<br><br>Refer to the <a href="#">Quick Reference Booklet</a> for participating countries. Enter spaces if no country is specified.                                                                                                |

| Field ID | Field Name | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                      |
|----------|------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9        | Subcountry | an-2       | <p>If applicable, contains the subcountry associated with this account listing. Must be a participating subcountry within the issuer's country; values are two-digit subcountry code or spaces.</p> <p>Refer to the <a href="#">Account Management System User Manual</a>. Enter spaces if no subcountry is specified.</p> |
| 10       | Purge Date | n-6        | <p>Purge date (format YYMMDD) associated with this account number listing as follows:</p> <p>YY = purge date year</p> <p>MM = purge date month</p> <p>DD = purge date day</p>                                                                                                                                              |
| 11       | Region     | an-1       | <p>If applicable, second region associated with this account listing. If a second country or subcountry listing is entered, this field must contain the issuer's region. Region values must be entered in ascending order (1, A, B, C, D, E).</p>                                                                          |
| 12       | Country    | an-3       | <p>If applicable, second country associated with this account listing; must be a participating country within the issuer's region.</p>                                                                                                                                                                                     |
| 13       | Subcountry | an-2       | <p>If applicable, second subcountry associated with this account listing; must be a participating subcountry within the issuer's country.</p>                                                                                                                                                                              |
| 14       | Purge Date | n-6        | <p>If applicable, second purge date associated with this account number listing.</p>                                                                                                                                                                                                                                       |
| 15       | Region     | an-1       | <p>If applicable, third region associated with this account listing; if a third country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E).</p>                                                                            |
| 16       | Country    | an-3       | <p>If applicable, third country associated with this account listing; must be a participating country within the issuer's region.</p>                                                                                                                                                                                      |

| Field ID | Field Name | Attributes | Comments/Valid Values                                                                                                                                                                                                                      |
|----------|------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17       | Subcountry | an-2       | If applicable, third subcountry associated with this account listing; must be a participating subcountry within the issuer's country.                                                                                                      |
| 18       | Purge Date | n-6        | If applicable, third purge date associated with this account number listing.                                                                                                                                                               |
| 19       | Region     | an-1       | If applicable, fourth region associated with this account listing; if a fourth country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E). |
| 20       | Country    | an-3       | If applicable, fourth country associated with this account listing; must be a participating country within the issuer's region.                                                                                                            |
| 21       | Subcountry | an-2       | If applicable, fourth subcountry associated with this account listing; must be a participating subcountry within the issuer's country.                                                                                                     |
| 22       | Purge Date | n-6        | If applicable, fourth purge date associated with this account number listing.                                                                                                                                                              |
| 23       | Region     | an-1       | If applicable, fifth region associated with this account listing; if a fifth country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E).   |
| 24       | Country    | an-3       | If applicable, fifth country associated with this account listing; must be a participating country within the issuer's region.                                                                                                             |
| 25       | Subcountry | an-2       | If applicable, fifth subcountry associated with this account listing; must be a participating subcountry within the issuer's country.                                                                                                      |
| 26       | Purge Date | n-6        | If applicable, fifth purge date associated with this account number listing.                                                                                                                                                               |
| 27       | Region     | an-1       | If applicable, sixth region associated with this account listing; if a sixth country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E).   |

| Field ID | Field Name | Attributes | Comments/Valid Values                                                                                                                 |
|----------|------------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 28       | Country    | an-3       | If applicable, sixth country associated with this account listing; must be a participating country within the issuer's region.        |
| 29       | Subcountry | an-2       | If applicable, sixth subcountry associated with this account listing; must be a participating subcountry within the issuer's country. |
| 30       | Purge Date | n-6        | If applicable, sixth purge date associated with this account number listing.                                                          |

## MCC 105: Payment Cancellation File

DE 120 (Record Data) contains Payment Cancellation file data when DE 101 (File Name) contains MCC105.

### Associated edits

Following are the MCC105 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

| When DE 91 Contains... | Include These Fields in DE 120...                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| 1 = Add                | Use all applicable fields; fields 1–5 are mandatory, fields 6–13 are optional                             |
| 2 = Update             | Use all applicable fields; fields 1–5 are mandatory, fields 6–7 are optional. Fields 8–13 not applicable. |
| 3 = Delete             | Use fields 1, 4, and 5; fields 2 and 3 must be filled with spaces. Fields 6–13 not applicable.            |
| 5 = Inquiry            | Use fields 1, 4, and 5; fields 2 and 3 must be filled with spaces. Fields 6–13 not applicable.            |

### Fields

Following is the modified DE 120 Layout for MCC105. Amount fields 6 and 7, when provided, must include a minor unit of currency. For example, USD 100.50 entered as 000000010050. Fields 8–9 are sent in the 0312 response message.

| Field ID | Field Name                | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Primary Account Number    | an-19      | Number that is embossed, encoded, or both on the card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned<br><br>Format: Left-justified, with trailing spaces                                                                                                                  |
| 2        | Entry Reason              | an-1       | Reason for listing this card. Valid value:<br><br>A = Payment Cancellation                                                                                                                                                                                                                                                                    |
| 3        | Issuer-defined Purge Date | n-8        | The issuer-defined purge date must be no more than 15 months beyond the account list date in YYYYMMDD format.                                                                                                                                                                                                                                 |
| 4        | Acquirer ID               | n-6        | Valid acquirer ICA.                                                                                                                                                                                                                                                                                                                           |
| 5        | Acceptor ID               | ans-15     | Must be unique per acquirer ICA. Must be a value other than spaces.<br><br><b>NOTE: This field is case sensitive.</b>                                                                                                                                                                                                                         |
| 6        | Transaction Amount (low)  | n-12       | Optional field. Right-justified with leading zeros.<br><br>Indicates a single transaction amount or the start of a transaction amount range.<br><br>Transaction amount must be stated in the transaction currency and not the cardholder billing currency.                                                                                    |
| 7        | Transaction Amount (high) | n-12       | Optional field. Right-justified with leading zeros.<br><br>Indicates the end of a transaction amount range. Do not use this field if specifying a single amount.<br><br>If used, must be greater than Transaction Amount (low).<br><br>Transaction amount must be stated in the transaction currency and not the cardholder billing currency. |
| 8        | Activity Date             | n-6        | Format is MMDDYY (Month, Day, Year); Provide spaces in request message if fields beyond Field ID 9 are present. System populates data in the response message.                                                                                                                                                                                |

| Field ID | Field Name         | Attributes | Comments/Valid Values                                                                                                                                    |
|----------|--------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9        | Activity Time      | n-4        | Format is hhmm (hour, minute); Provide spaces in request message if fields beyond Field ID 9 are present. System populates data in the response message. |
| 10       | Acceptor ID Code 2 | ans-15     | Must be unique per acquirer ICA.                                                                                                                         |
| 11       | Acceptor ID Code 3 | ans-15     | Must be unique per acquirer ICA.                                                                                                                         |
| 12       | Acceptor ID Code 4 | ans-15     | Must be unique per acquirer ICA.                                                                                                                         |
| 13       | Acceptor ID Code 5 | ans-15     | Must be unique per acquirer ICA.                                                                                                                         |

## DE 120 (When Blocking Recurring Payments)

Following are various uses of DE 120 for MCC105:

- Blocking all recurring payment arrangements with all merchants

A block of all recurring payments arrangements can only be accomplished by setting up blocks for each merchant and acquirer combination.

- Blocking all recurring payment arrangements with one merchant, or blocking a cardholder's only recurring payment with one merchant
  - Fields 1–5 are required for Add
  - Fields 6–7 should not be provided
- Blocking one of many recurring payment arrangements with one merchant

The one recurring payment arrangement can only be blocked if its transaction amount is different than any of the other recurring payment arrangements with a particular merchant.

The recurring payment arrangement to be blocked has a fixed (same amount each billing) transaction amount, such as an insurance premium billing

- Fields 1–5 are required for Add
- Field 6 should be provided to specify the fixed transaction amount
- Field 7 should not be provided

The recurring payment arrangement to be blocked has a variable (different amount each billing) transaction amount, such as a long distance telephone billing. For example, if the variable billing previous amounts were between USD 15.00 and USD 50.00, field 6 should contain USD 15.00 and field 7 should contain USD 50.00. This results in transactions being blocked within this specified amount range.

- Fields 1–5 are required for Add
- Field 6 is provided to specify the lowest transaction amount billable
- Field 7 is provided to specify the highest transaction amount billable

Usage of the amount range should only be used if the billing amount of the recurring payment to be blocked is unique and does not overlap with other recurring payments from the same merchant that should not be blocked. If one recurring payment arrangement billing amount was typically between USD 15.00 and USD 50.00 but another recurring payment from the same merchant either had a variable amount that sometimes fell within USD 15.00 to USD 50.00 or had a fixed amount within USD 15.00 to USD 50.00, both recurring payments are blocked.

- The following guidelines apply for the multiple Acceptor IDs:
  - Issuers will only be able to provide multiple Acceptor IDs when PAN (field 1) and Acquirer ICA (field 4) are provided in DE 120.
  - The additional Acceptor IDs must be added in sequence. For example, if codes are entered in Acceptor ID fields 5, 10, and 12, skipping field 11, the 0302 Payment Cancellation MCC105 add request will be rejected with DE 39 = 27 (Edit Error) and DE 44 will contain field number 120011.
  - If issuers provide transaction amount or transaction amount range in the 0302 Payment Cancellation add list request, the transaction amount and transaction amount range will be applicable to all the given Acceptor IDs in the message.

## MCC 106: PAN Mapping File

DE 120 (Record Data) contains Token to PAN Mapping File data when DE 101 (File Name) contains MCC106.

### Associated edits

Following are the MCC106 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

| When DE 91 Contains... | Include These Fields in DE 120...                                                                                                                                                                                                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 = Add                | Fields 1–2 mandatory, field 3 optional                                                                                                                                                                                                                                                                                                                |
| 2 = Update             | Fields 1–2 mandatory, fields 3–5 optional<br>Valid combinations of fields 1–5 are: <ul style="list-style-type: none"> <li>• Fields 1–3</li> <li>• Fields 1–3 and field 5; field 4 must contain spaces</li> <li>• Fields 1–2 and field 4; field 3 must contain spaces</li> <li>• Fields 1–2 and field 5; fields 3 and 4 must contain spaces</li> </ul> |
| 3 = Delete             | Use field 1 only. All records associated to contactless account number will be deleted.                                                                                                                                                                                                                                                               |



## Fields

Following is the DE 120 Layout for MCC106 in the Issuer File Update Request/0302 Message.

| Field ID | Field Name                               | Attributes | Comments/Valid Values                                                                                                                                                                                                                    |
|----------|------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Contactless Account Number               | an-19      | Number that is assigned to the contactless card or device and transmitted from the card or device to the X terminal.<br><br>Format: Left-justified, with trailing spaces                                                                 |
| 2        | PAN                                      | an-19      | Number that is embossed, encoded, or both on the card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned.<br><br>Format: Left-justified, with trailing spaces            |
| 3        | Card Expiration Date (PAN)               | n-4        | Expiration date that is embossed, encoded, or both on the card that represents the cardholder primary account number.<br><br>Format: YYMM                                                                                                |
| 4        | Contactless account number (replacement) | an-19      | Replacement account number that is assigned to the contactless card or device and transmitted from card or device to the contactless terminal.<br><br>Format: Left-justified, with trailing spaces                                       |
| 5        | PAN (replacement)                        | an-19      | Number that is embossed, encoded, or both on the replacement card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned<br><br>Format: Left-justified, with trailing spaces |

## Fields

The following DE 120 layout applies to Issuer File Update Request/0302 Messages requesting the replacement of a PAN associated with an individual Mastercard Digital Enablement Service (MDES) token, or with all tokens associated with current PAN, in the MCC106 PAN Mapping File.

| Field ID | Field Name                           | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|--------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Mapping File Indicator               | an-1       | <p>M = Mastercard Digital Enablement Service Tokens excluding Static Tokens</p> <p>A = All Mastercard Digital Enablement Service Tokens including Static Tokens</p> <p>Best practice for the issuer to use value A</p>                                                                                                                                                                                                                                                                                                                               |
| 2        | Replacement PAN                      | an-19      | <p>Number replacing the number that is embossed, encoded, or both on the card (the primary account number). Customers may input only account numbers for BINs assigned to the associated customer ID assigned by Mastercard.</p> <p>The issuer has the option to provide a replacement primary account number for association to the token.</p> <p>Format: Left-justified, with trailing spaces</p>                                                                                                                                                  |
| 3        | Replacement PAN Expiration Date      | n-4        | <p>Expiration date that is associated with the number replacing the number that is embossed, encoded, or both on the card that represents the cardholder primary account number.</p> <p>If the issuer has provided a replacement primary account number, this field contains the expiration date that is associated with the replacement primary account number.</p> <p>If the issuer has not provided a replacement primary account number, this field contains the expiration date of the existing primary account number.</p> <p>Format: YYMM</p> |
| 4        | Primary Account Card Sequence Number | ans-3      | <p>If the issuer has provided a replacement primary account number, this field will contain the card sequence number associated with the replacement primary account number.</p> <p>If the issuer has not provided a replacement primary account number, this field will contain the card sequence number associated with the original primary account number.</p>                                                                                                                                                                                   |

| Field ID | Field Name                               | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5        | Notify Wallet Service Provider Indicator | n-1        | <p>If the issuer has provided a replacement primary account number and replacement primary account number expiration date, this field indicates whether the Wallet Provider should be notified of the change.</p> <p>Values:</p> <p>0 = Update token mapping information and notify the Wallet Provider with the primary account number information</p> <p>1 = Update token mapping information, but do not notify the Wallet Provider with the primary account number information</p> <p>2 = Do not update token mapping information, but do update the Wallet Provider with the primary account number information</p>                                       |
| 6        | Token                                    | an-19      | <p>Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check.</p> <p>Format: Left-justified, with trailing spaces<br/>If field not present, update shall apply to all token to PAN mappings associated with the PAN as follows:</p> <ul style="list-style-type: none"> <li>This field is optional. If present, the token to PAN mapping update will only be applied to the specified token. If not present, the token to PAN mapping update will be applied to all tokens associated with the current PAN.</li> </ul> |

## Fields

The following DE 120 layout applies to Issuer File Update Request/0302 Messages requesting the suspension, deactivation, or resumption of an MDES token in the MCC106 PAN Mapping File.

| Field ID | Field Name                               | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Mapping File Indicator                   | an-1       | M = Mastercard Digital Enablement Service Tokens excluding Static Tokens<br><br>A = All Mastercard Digital Enablement Service Tokens including Static Tokens<br><br>Best practice for the issuer to use value A                                                                                                                                                |
| 2        | Action Required                          | an-1       | S = Suspend token<br><br>D = Deactivate token<br><br>C = Resume token                                                                                                                                                                                                                                                                                          |
| 3        | Notify Wallet Service Provider Indicator | n-1        | This field indicates whether the Wallet Provider should be notified of the change.<br><br>Values:<br><br>0 = Update token mapping information and notify the Wallet Provider with the primary account number information.                                                                                                                                      |
| 4        | Token—If Updating a Specific Token       | an-19      | Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check.<br><br>Format: Left-justified, with trailing spaces.<br><br>If field not present, update shall apply to all token to PAN mappings associated with the PAN. |

## Fields

The following DE 120 layout applies to Issuer File Update Request Response/0312 Messages created by Mastercard in response to an MCC106 PAN Mapping File update request.

| Field ID | Field Name                 | Attributes | Comments/Valid Values                                                                                                                                                          |
|----------|----------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Contactless account number | an-19      | Number that is assigned to the contactless card or device and transmitted from the card or device to contactless terminal.<br><br>Format: Left-justified, with trailing spaces |

| Field ID | Field Name                               | Attributes | Comments/Valid Values                                                                                                                                                                                                                     |
|----------|------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2        | PAN                                      | an-19      | Number that is embossed, encoded, or both on the card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned.<br><br>Format: Left-justified, with trailing spaces             |
| 3        | Card Expiration Date (PAN)               | n-4        | Expiration date that is embossed, encoded, or both on the actual card.<br><br>Format: YYMM                                                                                                                                                |
| 4        | Contactless account number (replacement) | an-19      | Replacement number that is assigned to the contactless card or device and transmitted from the card or device to contactless terminal.<br><br>Format: Left-justified, with trailing spaces                                                |
| 5        | PAN (replacement)                        | an-19      | Number that is embossed, encoded, or both on the replacement card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned.<br><br>Format: Left-justified, with trailing spaces |
| 6        | Date-Last-Update Activity                | n-6        | Format: MMDDYY (Month, Day, Year)                                                                                                                                                                                                         |
| 7        | Time-Last-Update Activity                | n-4        | Format: hhmm (hour, minute)                                                                                                                                                                                                               |

## MCC106—PAN Mapping File – Token Service Provider Maintenance

### Fields

The following table provides the technical details of DE 120 layout for MCC106 in the Issuer File Update Request/0302 message.

| Field ID | Field Name  | Attributes | Comments/Valid Values                                                                        |
|----------|-------------|------------|----------------------------------------------------------------------------------------------|
| 15       | KDI Value   | an-3       | Conditional<br><br>Required for Cloud Processing Type 2D.<br>Optional for other Cloud Types. |
| 16       | AIP List ID | an-5       | Conditional<br><br>Required for Cloud Processing Type 2D.<br>Optional for other Cloud Types. |

| Field ID | Field Name            | Attributes | Comments/Valid Values                                                                    |
|----------|-----------------------|------------|------------------------------------------------------------------------------------------|
| 17       | CVM List ID Primary   | an-3       | Conditional<br>Required for Cloud Processing Type 2D.<br>Optional for other Cloud Types. |
| 18       | CVM List ID Secondary | an-3       | Conditional<br>Required for Cloud Processing Type 2D.<br>Optional for other Cloud Types. |
| 19       | CVN                   | an-2       | Conditional<br>Required for Cloud Processing Type 2D.<br>Optional for other Cloud Types. |

## MCC 109: Application Transaction Counter File

DE 120 (Record Data) contains Application Transaction Counter file data when DE 101 (File Name) contains MCC109.

### Associated edits

Following are the MCC109 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

| When DE 91 Contains...                                                           | Include These Fields in DE 120... |
|----------------------------------------------------------------------------------|-----------------------------------|
| 1 = Add                                                                          | Fields 1–4 mandatory              |
| 3 = Delete                                                                       | Fields 1–4 mandatory              |
| <b>NOTE: Applicable only when submitting an R311 bulk file deletion request.</b> |                                   |
| 5 = Inquiry                                                                      | Fields 1–3 mandatory              |

### Associated edits

Following is the DE 120 Layout for MCC109 in the Issuer File Update Request/0302 message. Fields 1 through 3 must be considered contiguous to identify if an entry exists when trying to delete a record.

| Field ID | Field Name                                              | Attributes | Comments/Valid Values                                                                                                                                                             |
|----------|---------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Contactless Account Number                              | an-19      | Number that is assigned to the contactless card or device and transmitted from the card or device to the contactless terminal<br><br>Format: Left-justified, with trailing spaces |
| 2        | Card Sequence Number                                    | n-1        | May be zero                                                                                                                                                                       |
| 3        | Contactless Account Expiration Date                     | n-4        | Expiration date<br><br>Format: YYMM                                                                                                                                               |
| 4        | Contactless Application Transaction Counter (ATC) Value | n-5        | Right-justified, leading zeros                                                                                                                                                    |

### Associated edits

Following is the DE 120 Layout for MCC109 in the Issuer File Update Request Response/0312 message.

| Field ID | Field Name                                              | Attributes | Comments/Valid Values                                                                                                                                                         |
|----------|---------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Contactless Account Number                              | an-19      | Number that is assigned to the contactless card or device and transmitted from the card or device to contactless terminal<br><br>Format: Left-justified, with trailing spaces |
| 2        | Card Sequence Number                                    | n-1        | May be zero                                                                                                                                                                   |
| 3        | Contactless Account Expiration Date                     | n-4        | Expiration date<br><br>Format: YYMM                                                                                                                                           |
| 4        | Contactless Application Transaction Counter (ATC) Value | n-5        | Right-justified, leading zeros                                                                                                                                                |
| 5        | Filler                                                  | an-1       |                                                                                                                                                                               |
| 6        | Issuer Customer ID                                      | n-6        |                                                                                                                                                                               |
| 7        | Date/Time-Last-Update Activity                          | n-12       | Format: MMDDYYHHMMSS                                                                                                                                                          |
| 8        | Entry Change Indicator                                  | an-1       | Indicates how the ATC Value was last changed<br><br>B = batch entry<br>O = online entry<br>T = transaction                                                                    |

| Field ID | Field Name               | Attributes | Comments/Valid Values                                |
|----------|--------------------------|------------|------------------------------------------------------|
| 9        | Creation Date/Time Stamp | n-14       | Format: MMDDCCYYHHMMSS                               |
| 10       | Last Transaction Time    | n-4        | Last transaction time for valid ATC.<br>Format: hhmm |

## MCC 111: PAN-PAR (Payment Account Reference) Mapping File

DE 120 (Record Data) contains PAN to PAR Mapping File data when DE 101 (File Name) contains MCC111.

Only PAN replacement and PAR record deletion actions are allowed. Both are handled as an "update".

### Associated edits

Following are the MCC111 data fields that should be included in DE 120 depending on the combination of values in DE 91 (Issuer File Update Code) and DE 120 subfield 1 (Action Required).

| When DE 91 Contains...                                          | Include These Subfields in DE 120...                                                                                                                                                                            |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 = Update                                                      | Subfield 1 is mandatory                                                                                                                                                                                         |
| This is the only valid Issuer File Update Code value for MCC111 | <ul style="list-style-type: none"> <li>When subfield 1 = U (Update), subfield 2 must contain a replacement PAN</li> <li>When subfield 1 = D (Delete), all other DE 120 subfields must contain spaces</li> </ul> |

### Subfields

The following DE 120 layout applies to Issuer File Update Request/0302 messages requesting the replacement of a PAN associated with a PAR value in the MCC111 PAN-PAR Mapping File. This action should be performed when the issuer is replacing the consumer's current PAN with a new PAN.

| Subfield ID | Subfield Name   | Attributes | Comments/Valid Values                                                                                                                                                                                                               |
|-------------|-----------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | Action Required | an-1       | U = Update the PAN-PAR mapping record                                                                                                                                                                                               |
| 2           | Replacement PAN | an-19      | Number that is embossed, encoded, or both on the replacement card. Customers may only input account numbers for BINs assigned to the associated customer ID assigned by Mastercard.<br>Format: Left-justified, with trailing spaces |



### Subfields

The following DE 120 layout applies to Issuer File Update Request/0302 messages requesting the deletion of a PAN-PAR mapping record in the MCC111 PAN-PAR Mapping File. This action should be performed only when the issuer has permanently closed/cancelled the consumer's PAN.

| Subfield ID | Subfield Name   | Attributes | Comments/Valid Values                                                                          |
|-------------|-----------------|------------|------------------------------------------------------------------------------------------------|
| 1           | Action Required | an-1       | D = Delete the PAN-PAR mapping record associated with the PAN in DE 2 (Primary Account Number) |

## DE 120 (Error Codes)

Following are the DE 120 Error Codes.

### MCC 102: Error Codes

The Dual Message Authorization System returns an Issuer File Update Request Response/0312 message where DE 39 contains the value 27 and DE 44 contains the value 120xxx (xxx indicates the DE 120 field in which the error occurred).

### Related errors and error messages

| Error Code | Error Message                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | Primary Account Number (PAN), extended not numeric<br>-Or-<br>BIN in Primary Account Number (PAN), extended not on FIM<br>-Or-<br>Origin of message invalid for BIN in Primary Account Number (PAN), extended<br>-Or-<br>Check digit in Primary Account Number (PAN), extended not correct<br>-Or-<br>Primary Account Number (PAN), extended not on Account File (for delete or inquiry). |
| 120002     | Entry Reason not one of the following: (C, F, G, L, O, P, S, U, V, X).                                                                                                                                                                                                                                                                                                                    |
| 120005     | PIN length not numeric or spaces.                                                                                                                                                                                                                                                                                                                                                         |
| 120006     | Entry-Reason = V and Premium Listing Accumulative Limit Amount not numeric.                                                                                                                                                                                                                                                                                                               |

| Error Code | Error Message                     |
|------------|-----------------------------------|
| 120007     | Invalid Currency Code             |
| 120008     | Invalid Issuer-defined Purge Date |
| 120009     | Invalid Card Sequence Number      |
| 120010     | Invalid Card Expiration Date      |

## MCC 103: Error Codes

The Dual Message Authorization System returns an Issuer File Update Request Response/0312 message where DE 39 contains the value 25 or 27 and DE 44 contains the value 120xxx (xxx indicates the DE 120 field in which the error occurred).

When DE 39 contains the value 25 (Unable to locate record on file [no action taken]), DE 44 contains the following error code:

### Related errors and error messages

| Error Code | Error Message                       |
|------------|-------------------------------------|
| 120001     | Primary account number not on file. |

When DE 39 contains the value 27, DE 44 contains one of the following error codes:

### Related errors and error messages

| Error Code | Error Message                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------|
| 120001     | Primary account number not on file                                                                  |
|            | -Or-                                                                                                |
|            | Primary Account Number (PAN), extended is not numeric                                               |
|            | -Or-                                                                                                |
|            | BIN in Primary Account Number (PAN), extended is invalid                                            |
|            | -Or-                                                                                                |
|            | Check digit in cardholder number is not correct and product code is not equal to MNS (Non-standard) |
|            | -Or-                                                                                                |
|            | Origin of message invalid for BIN in Primary Account Number (PAN), Extended.                        |

| Error Code | Error Message                                                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120002     | Issuing ICA (customer ID) is not associated with Primary Account Number (PAN), Extended BIN.                                                                                                                                    |
| 120003     | Product code is not equal to a valid card program value.                                                                                                                                                                        |
| 120004     | Response code not equal to 04 (capture card).                                                                                                                                                                                   |
| 120005     | Entry reason is not C, F, O, or X.                                                                                                                                                                                              |
| 120006     | Regional indicator is not 1, A, B, C, D, or E.<br>-Or-<br>Regional indicator is not in ascending order.                                                                                                                         |
| 120007     | Region-requested purge date invalid.<br>-Or-<br>Region-requested purge date not equal to current or future date.<br>-Or-<br>Requested purge date must be at least one day beyond current date if account is not already listed. |

## MCC 104: Error Codes

The Dual Message Authorization System returns an Issuer File Update Request Response/0312 message where DE 39 contains the value 25, 27, or 30 and DE 44 contains the value 120xxx (xxx indicates the DE 120 field in which the error occurred.)

When DE 39 contains the value 25, DE 44 contains the following error code:

### Related errors and error messages

| Error Code | Error Message        |
|------------|----------------------|
| 120001     | Account not on file. |

When DE 39 contains the value 27, DE 44 contains one of the following error codes:

## Related errors and error messages

| Error Code | Error Message                                                                                                                                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | <p>Primary Account Number (PAN), Extended is not numeric</p> <p>-Or-</p> <p>BIN in Primary Account Number (PAN), Extended is invalid</p> <p>-Or-</p> <p>Check digit in cardholder number is not correct and product code is not equal to PNS (Non-standard)</p> <p>-Or-</p> <p>Origin of message invalid for BIN in Primary Account Number (PAN), Extended.</p> |
| 120002     | Issuing ICA (customer ID) is not associated with Primary Account Number (PAN)                                                                                                                                                                                                                                                                                   |
| 120003     | <p>Card Program is not equal to a valid value</p> <p>-Or-</p> <p>BIN is not set up for proprietary card program, and card program in the request equals PRO or PNS.</p>                                                                                                                                                                                         |
| 120004     | Response code is not 04.                                                                                                                                                                                                                                                                                                                                        |
| 120005     | Entry reason is not C, F, O, or X.                                                                                                                                                                                                                                                                                                                              |
| 120006     | <p>Regional indicator is not equal to 1, A, B, C, D, or E</p> <p>-Or-</p> <p>Regional indicator is not in ascending order.</p>                                                                                                                                                                                                                                  |
| 120007     | Country invalid or outside issuer region.                                                                                                                                                                                                                                                                                                                       |
| 120008     | Subcountry invalid or outside issuer country.                                                                                                                                                                                                                                                                                                                   |
| 120009     | <p>Purge date invalid</p> <p>-Or-</p> <p>Purge date not equal to current or future date</p> <p>-Or-</p> <p>Requested purge date must be at least one day beyond current date if account is not already listed.</p>                                                                                                                                              |

## MCC 105: Error Codes

If the Dual Message Authorization System detects an error in the Issuer File Update Request/0302 message, it returns an Issuer File Update Request Response/0312 message containing error codes specific to each file name. DE 39 (Response Code) contains value 27 (Issuer File Update field edit error) and DE 44 (Additional Response Data) contains the value 120xxx (where xxx indicates the DE 120 field in which the error occurred).

### Related errors and error messages

| Error Code | Error Message                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | Primary Account Number (PAN) not within account range that is a Payment Cancellation participant<br><br>-Or-<br><br>Primary Account Number (PAN) not numeric<br><br>-Or-<br><br>BIN in Primary Account Number (PAN) not valid<br><br>-Or-<br><br>Check digit in Primary Account Number (PAN) not correct<br><br>-Or-<br><br>Primary Account Number (PAN) not on Payment Cancellation File (for update, delete, or inquiry) |
| 120002     | Entry Reason is not (A)                                                                                                                                                                                                                                                                                                                                                                                                    |
| 120003     | Invalid Issuer-defined Purge Date format<br><br>-Or-<br><br>Issuer-defined Purge Date not equal to or greater than system date<br><br>-Or-<br><br>Issuer-defined Purge Date not greater than system date by 15 months                                                                                                                                                                                                      |
| 120004     | Acquirer ICA not numeric                                                                                                                                                                                                                                                                                                                                                                                                   |
| 120005     | Acceptor ID not present                                                                                                                                                                                                                                                                                                                                                                                                    |
| 120006     | Transaction Amount (low limit) not numeric                                                                                                                                                                                                                                                                                                                                                                                 |

| Error Code    | Error Message                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120007        | Transaction Amount (high limit) not numeric<br>-Or-<br>Transaction Amount (high limit) not greater than Transaction Amount (low limit)<br>-Or-<br>Transaction Amount (high limit) is present but Transaction Amount (low limit) is not present                                                                                                                                                                     |
| 120010–120013 | Invalid Acceptor ID<br><br>If multiple Acceptor ID occurrences are not provided in sequence, then Add request will be rejected with edit error on the respective field that has spaces in-between two valid Acceptor IDs.<br><br>For example: If fields 5, 10, 12 are provided with valid values and field 11 is provided with spaces, then Add request will be rejected with edit error on field 11, DE44=120011. |

## MCC 106: Error Codes

DE 39 contains the appropriate value and DE 44 contains value 120xxx (where xxx indicates the DE 120 field in which the error occurred).

When DE 39 contains value 25 (Unable to locate record on file), DE 44 contains the following error code:

### Related errors and error messages

| Error Code | Error Message                                                                      |
|------------|------------------------------------------------------------------------------------|
| 120001     | Contactless account number not on MCC106 (PAN Mapping File) (for update or delete) |
| 120002     | PAN not on MCC106 (PAN Mapping File) (for update)                                  |

When DE 39 contains value 27 (Issuer File Update Field Edit Error), DE 44 contains one of the following error codes:

### Related errors and error messages

| Error Code | Error Message                                                                                                                                                                                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | <p>Contactless account number not numeric</p> <p>-Or-</p> <p>Prefix in contactless account number not valid</p> <p>-Or-</p> <p>Check digit in contactless account number not correct</p> <p>-Or-</p> <p>Contactless account number not within account range that is a Contactless Mapping Service participant</p>                                                 |
| 120002     | <p>PAN not numeric</p> <p>-Or-</p> <p>Prefix in PAN not valid</p> <p>-Or-</p> <p>Check digit in PAN not correct</p> <p>-Or-</p> <p>PAN and contactless account number prefixes do not have same country and product</p> <p>-Or-</p> <p>PAN prefix should not be participating in the Contactless Mapping Service</p>                                              |
| 120003     | Invalid PAN card expiration date                                                                                                                                                                                                                                                                                                                                  |
| 120004     | <p>Replacement contactless account number not numeric</p> <p>-Or-</p> <p>Prefix in Replacement contactless account number not valid</p> <p>-Or-</p> <p>Check digit in Replacement contactless account number not correct</p> <p>-Or-</p> <p>Replacement contactless account number not within account range that is a Contactless Mapping Service participant</p> |

| Error Code | Error Message                                                                                |
|------------|----------------------------------------------------------------------------------------------|
| 120005     | Replacement PAN not numeric                                                                  |
|            | -Or-                                                                                         |
|            | Prefix in Replacement PAN not valid                                                          |
|            | -Or-                                                                                         |
|            | Check digit in Replacement PAN not correct                                                   |
|            | -Or-                                                                                         |
|            | Replacement PAN and contactless account number prefixes do not have same country and product |
| 120015     | -Or-                                                                                         |
|            | Replacement PAN prefix should not be participating in the Contactless Mapping Service        |
|            |                                                                                              |
|            |                                                                                              |
| 120016     | Invalid Token Type                                                                           |
|            | -Or-                                                                                         |
|            | Invalid Cloud Processing Type                                                                |
|            | -Or-                                                                                         |
| 120017     | Invalid Key Derivation Index                                                                 |
|            |                                                                                              |
|            |                                                                                              |
|            |                                                                                              |
| 120018     | Invalid Token Type                                                                           |
|            | -Or-                                                                                         |
|            | Invalid Cloud Processing Type                                                                |
|            | -Or-                                                                                         |
| 120019     | Invalid AIP List ID                                                                          |
|            |                                                                                              |
|            |                                                                                              |
|            |                                                                                              |
| 120020     | Invalid Token Type                                                                           |
|            | -Or-                                                                                         |
|            | Invalid Cloud Processing Type                                                                |
|            | -Or-                                                                                         |
| 120021     | Invalid CVM List ID Primary                                                                  |
|            |                                                                                              |
|            |                                                                                              |
|            |                                                                                              |
| 120022     | Invalid Token Type                                                                           |
|            | -Or-                                                                                         |
|            | Invalid Cloud Processing Type                                                                |
|            | -Or-                                                                                         |
| 120023     | Invalid CVM List ID Secondary                                                                |
|            |                                                                                              |
|            |                                                                                              |
|            |                                                                                              |



| Error Code | Error Message                                                                      |
|------------|------------------------------------------------------------------------------------|
| 120019     | Invalid Token Type<br>-Or-<br>Invalid Cloud Processing Type<br>-Or-<br>Invalid CVN |

When DE 39 contains value 80 (Duplicate add, action not performed), DE 44 contains the following error code:

#### Related errors and error messages

| Error Code | Error Message                                         |
|------------|-------------------------------------------------------|
| 120001     | Contactless account number is a duplicate             |
| 120004     | Replacement contactless account number is a duplicate |

### MCC 109: Error Codes

DE 39 contains the appropriate value and DE 44 contains value 120xxx (where xxx indicates the DE 120 field in which the error occurred).

When DE 39 contains value 25 (Unable to locate record on file), DE 44 contains the following error code:

#### Related errors and error messages

| Error Code | Error Message                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | Contactless Account Number/Card Sequence Number/Contactless Account Expiration Date not on Contactless Application Transaction Counter (ATC) File (MCC109) (for add or inquiry) |

When DE 39 contains value 27 (Issuer File Update Field Edit Error), DE 44 contains one of the following error codes:

### Related errors and error messages

| Error Code | Error Message                                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | Contactless Account Number not numeric<br>-Or-<br>Prefix in contactless account not valid<br>-Or-<br>Check digit in Contactless Account Number not correct |
| 120002     | Card Sequence Number not numeric                                                                                                                           |
| 120003     | Contactless Account Expiration Date invalid                                                                                                                |
| 120004     | ATC not numeric                                                                                                                                            |

When DE 39 contains value 80 (Duplicate add, action not performed), DE 44 contains the following error code:

### Related errors and error messages

| Error Code | Error Message                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------|
| 120001     | Contactless Account Number/Card Sequence Number/Contactless Account Expiration Date already exists on file (applies to Add) |

## MCC 111: Error Codes

DE 39 contains the appropriate value and DE 44 contains value 120xxx (where xxx indicates the DE 120 subfield in which the error occurred).

When DE 39 contains value 25 (Unable to locate record on file [no action taken]), DE 44 contains the following error code:

### Related errors and error messages

| Error Code | Error Message                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | The PAN in DE 2 is not associated with a PAR value in the MCC111 PAN-PAR Mapping File and therefore the action requested in subfield 1 cannot be performed. |

When DE 39 contains value 27 (Issuer File Update field edit error), DE 44 contains the following error code:

#### Related errors and error messages

| Error Code | Error Message                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120001     | <p>The Action Required value in DE 120 subfield 1 is not D (Delete) or U (Update). Value is space or other than D or U.</p> <p>-Or-</p> <p>The Action Required value in DE 120 subfield 1 is D (Delete) but the PAN in DE 2 is associated with a token in the MCC106 Token-PAN Mapping File that is in active or suspended status. Therefore the PAN-PAR mapping record cannot be deleted.</p>                                                                                                                                                                                                                                                                                                                             |
| 120002     | <p>The content of DE 120 subfield 2 is not properly formatted</p> <ul style="list-style-type: none"> <li>• Not numeric (alpha, special, or spaces)</li> <li>• Incorrect length</li> <li>• Invalid check digit</li> </ul> <p>-Or-</p> <p>The Action Required value in DE 120 subfield 1 is U (Update) but no replacement PAN is present in subfield 2.</p> <p>-Or-</p> <p>The Action Required value in DE 120 subfield 1 is U (Update) but the account range of the replacement PAN in subfield 2 is not eligible for association with a PAR value.</p> <p>-Or-</p> <p>The Action Required value in DE 120 subfield 1 is U (Update) but the replacement PAN in subfield 2 is already associated with another PAR value.</p> |

When DE 39 contains value 80 (Duplicate Add, action not performed), DE 44 contains the following error code:

#### Related errors and error messages

| Error Code | Error Message                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120002     | <p>The PAN value in DE 2 is a duplicate of the PAN value in DE 120 subfield 2. The replacement PAN in subfield 2 must be different than the current PAN in DE 2.</p> |

## DE 121 (Authorizing Agent ID Code)

DE 121 (Authorizing Agent ID Code), when used, must contain the appropriate Mastercard-assigned customer ID number that uniquely identifies the Dual Message Authorization System Stand-In Processing facility or alternate routing customer responsible for performing Stand-In Processing on-behalf of the issuer.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n...6; LLLVAR             |
| Data Length         | 3 positions               |
| Data Field          | Contents of positions 1–6 |
| Subfields           | N/A                       |
| Justification       | N/A                       |

### Usage

Following is the usage of DE 121 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request Response/0410                | C   | X   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

### Values

DE 121 must contain a valid Mastercard customer ID number. Any values reserved for Mastercard use will be in the range 000000-000999.

The following values may be present in DE 121:

- If the 0110 message was responded to by the issuer, the value may contain the issuer's Mastercard ID.
- If the 0110 message was responded to by the MIP X-Code System, the value will be 000000.
- If the 0110 message was responded to by the Stand-In System, the value will be 000001.
- If the 0110 message was responded to by an alternate issuer host route, the value will be 000002.
- If the 0110 message was responded to by the Mastercard Rewards System, the value will be 000004 (Mastercard Rewards System).
- If the 0110 message was responded to by another Mastercard on-behalf service not mentioned above, the value will be 000003.

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

When the Stand-In Processing or alternate authorizer performs an Authorization Request/0100 on behalf of an issuer, it must insert this data element into the appropriate response message (for example, Authorization Request Response/0110) and into the appropriate Authorization Advice/0120: system-generated message so that the entire transaction audit trail clearly identifies the specific authorizing agent that actually approved the transaction.

The acquirer must insert this value in the Authorization Advice/0120 message when DE 121 is present in the Authorization Request Response/0110 message.

DE 121 is not required in Authorization Request/0100 or Authorization Advice/0120 messages the original issuer or its designated primary authorizing agent initiates.

DE 121, value 000002 is applicable only to issuers that use an alternate issuer host route instead of the Stand-In System. The issuer will provide in the Authorization Request Response/0110 message DE 121, value 000002 when an alternate issuer processed the original request.

DE 121 will have the value 000003 when a decline occurred due to an on-behalf service.

DE 121 will have the value of 000003 when a transaction was decisioned by a Mastercard on-behalf service.

DE 121, value 000004 is applicable only to issuers participating in the Mastercard Pay with Rewards service and will only be present in the Authorization Advice/0120: system-generated messages notifying issuers of a declined request. The specific reason for the decline will be specified in DE 60 (Advice Reason Code).

## DE 122 (Additional Record Data)

DE 122 is a free-format, variable-length data element used for transmitting file record data in various message types.

### Attributes

| Attribute           | Description             |
|---------------------|-------------------------|
| Data Representation | ans...999; LLLVAR       |
| Data Length         | 3 positions             |
| Data Field          | Contents of subelements |
| Subfields           | 1                       |
| Justification       | See "Subelements"       |

### Usage

Following is the usage of DE 122 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Issuer File Update Request Response/0312      | •   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |
| Reversal Advice/0420                          | •   | X   | C   |

### Application notes

Subelement 001 contains the Acceptor Street Address, the Acceptor Country Subdivision Code, Acceptor URL Address, Acceptor Customer Service Phone Number, Acceptor Phone Number, Acceptor Additional Contact Information, Acceptor Tax ID, Partner ID Code, Service Location, City Name, Service Location Country Subdivision Code, Service Location Country Code, and

Service Location Postal Code, Acceptor Geographic Coordinates, Service Location Geographic Coordinates, and Acceptor Tax ID should be used for the specified message types.

## Encoding scheme for DE 122

Mastercard organizes DE 122 into a group of encoded subelements and subfields.

### Structure of DE 122 subelement

| LLL          | VAR: maximum length varies |                   |                            |
|--------------|----------------------------|-------------------|----------------------------|
| 3 bytes      | 3 bytes                    | 3 bytes           | Variable length            |
| Total length | First subelement data      |                   |                            |
|              | Subelement ID              | Subelement length | Subelement variable length |

### Values

| Number of Bytes | Attribute                       | Value                                 |
|-----------------|---------------------------------|---------------------------------------|
| 3               | Total data element length       | The "LLL" portion of the data element |
| 3               | Subelement ID                   | In the range 001–099                  |
| 3               | Subelement length               | Variable                              |
| 1...nnn         | Subelement variable length data | Contents of Subfields                 |

### Structure of DE 122 subfields within a subelement

| LLL                 | VAR: maximum length varies |                               |
|---------------------|----------------------------|-------------------------------|
| 3 bytes             | 3 bytes                    | Variable length               |
| First subfield data |                            |                               |
| Subfield ID         | Subfield length            | Subfield variable length data |

### Values

| Number of bytes | Attribute   | Value                   |
|-----------------|-------------|-------------------------|
| 3               | Subfield ID | In the range of 001-099 |

| Number of bytes | Attribute                     | Value                 |
|-----------------|-------------------------------|-----------------------|
| 3               | Subfield length               | Variable              |
| 1...nnn         | Subfield variable length data | Contains valid values |

## Subelement 001 (Additional Acceptor Data)

DE 122 (Additional Record Data), subelement 001 (Additional Acceptor Data) contains the Acceptor Street Address, the Acceptor Country Subdivision Code, Acceptor URL Address, Acceptor Customer Service Phone Number, Acceptor Phone Number, Acceptor Additional Contact Information, Acceptor Tax ID, Partner ID Code, Service Location City Name, Service Location Country Subdivision Code, Service Location Country Code, Service Location Postal Code, Acceptor Geographic Coordinates, Service Location Geographic Coordinates, and Acceptor Tax ID Name.

### Attributes

| Attribute              | Description           |
|------------------------|-----------------------|
| Subelement ID          | 001                   |
| Subelement Data Length | 3 positions           |
| Data Representation    | ans...696; LLLVAR     |
| Data Field             | Contents of subfields |



| Attribute     | Description                                              |
|---------------|----------------------------------------------------------|
| Subfields     | Subfield 001 (Acceptor Street Address)                   |
|               | Subfield 002 (Acceptor Country Subdivision Code)         |
|               | Subfield 003 (Acceptor URL address)                      |
|               | Subfield 004 (Acceptor Customer Service Phone Number)    |
|               | Subfield 005 (Acceptor Phone Number)                     |
|               | Subfield 006 (Acceptor Additional Contact Information)   |
|               | Subfield 007 (Acceptor Tax ID)                           |
|               | Subfield 008 (Partner ID Code)                           |
|               | Subfield 009 (Service Location City Name)                |
|               | Subfield 010 (Service Location Country Subdivision Code) |
|               | Subfield 011 (Service Location Country Code)             |
|               | Subfield 012 (Service Location Postal Code)              |
|               | Subfield 013 (Acceptor Geographic Coordinates)           |
|               | Subfield 014 (Service Location Geographic Coordinates)   |
|               | Subfield 015 (Acceptor Tax ID Name)                      |
| Justification | See subfields                                            |

## Usage

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | C   | X   | C   |
| Authorization Advice/0120: system-generated   | •   | X   | C   |
| Reversal Request/0400                         | C   | X   | C   |
| Reversal Advice/0420                          | •   | X   | C   |

## Subfield 001 (Acceptor Street Address)

DE 122, subelement 001, subfield 001 (Acceptor Street Address) contains the street address of the acceptor, which should include the street number, street name, and other identifiers of the precise location, such as a building or unit number.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 001                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans...048; LLLVAR        |
| Data Field           | Contents of subfield 001 |
| Justification        | Left                     |

### Values

Acceptor Street Address must contain all Address Line 1 and Address Line 2 information, including

- the name of the street or throughfare
- the locality or community
- the building or house number; and
- any apartment, unit, suite, or floor number, if applicable.

This information must be populated and formatted as defined by country postal standards.

### Processing rules

The Acceptor Street Address must be present for all transactions and contain a valid location for the acceptor. The location of the acceptor will generally be the physical location where the transaction occurred or business location for transactions that occur virtually.

If the information for Acceptor Street Address does not exist per country postal standards, then the value of "Unavailable" must be provided.

### Program rules

Payment facilitator-enabled transactions: The sponsored merchant street address must be provided.

## Subfield 002 (Acceptor Country Subdivision Code)

DE 122, subelement 001, subfield 002 (Acceptor Country Subdivision Code) contains the ISO 3166-2 code identifying the state, province, or other country subdivision of the acceptor.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 002                       |
| Subfield Data Length | 3 positions               |
| Data Representation  | ans-003                   |
| Data Field           | Contents of subfield 002  |
| Justification        | Left with trailing spaces |

### Values

Acceptor Country Subdivision Code must contain a valid code as defined in Chapter 2 of the *Quick Reference Booklet* and must be followed by trailing spaces for codes that are less than three positions.

### Processing rules

The Acceptor Country Subdivision Code must be present for all transactions when defined in Chapter 2 of the *Quick Reference Booklet*.

### Program rules

Payment facilitator-enabled transaction: The sponsored merchant country subdivision code must be provided.

## Subfield 003 (Acceptor URL address)

DE 122, subelement 001, subfield 003 (Acceptor URL Address) contains the web address of the acceptor.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 003         |
| Subfield Data Length | 3 positions |

| Attribute           | Description              |
|---------------------|--------------------------|
| Data Representation | ans...255; LLLVAR        |
| Data Field          | Contents of subfield 003 |
| Justification       | Left                     |

### Processing rules

The Acceptor URL Address must be present for all electronic commerce transactions and contain a valid website address. For all other transactions, it must be provided unless a customer service phone is provided for contact.

### Program rules

Payment facilitator-enabled transaction: The sponsored merchant acceptor URL should be provided.

## Subfield 004 (Acceptor Customer Service Phone Number)

DE 122, subelement 001, subfield 004 (Acceptor Customer Service Phone Number) contains the phone number of the acceptor that can be used for transaction inquiries.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 004                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans...016; LLLVAR        |
| Data Field           | Contents of subfield 004 |
| Justification        | Left                     |

### Values

Acceptor Customer Service Phone Number must contain a valid value defined and formatted as per local telecom or the ITU Telecommunication Standardization Sector (ITU-T) standards.

For U.S. region and Canada region transactions, if present, Acceptor Customer Service Phone Number must be 10 positions with the beginning three positions numeric.

### Application notes

The Acceptor Customer Service Phone Number should be present for all transactions and contain a valid phone number, unless a URL is provided for contact.

For U.S. region and Canada transactions Acceptor Customer Service Phone Number must be present for mail order, phone order, and standing/recurring transactions.

### Program rules

Payment facilitator-enabled transactions: The sponsored merchant customer service phone number should be provided.

### Subfield 005 (Acceptor Phone Number)

DE 122, subelement 001, subfield 005 (Acceptor Phone Number) contains the business phone number of the acceptor, if different from the acceptor customer service phone number.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 005                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans...016; LLLVAR        |
| Data Field           | Contents of subfield 005 |
| Justification        | Left                     |

### Values

Acceptor Phone Number must contain a valid value defined and formatted as per local telecom or the ITU Telecommunication Standardization Sector (ITU-T) standards.

For U.S region and Canada region transactions, if present, Acceptor Phone Number must be 10 positions with the beginning three positions numeric.

### Processing rules

The acceptor phone number can be provided when different from the acceptor customer service phone number.

## Subfield 006 (Acceptor Additional Contact Information)

DE 122, subelement 001, subfield 006 (Acceptor Additional Contact Information) contains additional information for contacting the acceptor, such as an additional phone number or a contact name.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 006                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans...025; LLLVAR        |
| Data Field           | Contents of subfield 006 |
| Justification        | Left                     |

### Processing rules

Acceptor Additional Contact Information may contain additional information, such as secondary phone numbers, customer service operation hours, and contact name.

## Subfield 007 (Acceptor Tax ID)

DE 122, subelement 001, subfield 007 (Acceptor Tax ID) contains the identifier of the acceptor issued by a taxation authority.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 007                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans-021                  |
| Data Field           | Contents of subfield 007 |
| Justification        | Left                     |

## Values

The subfield must contain a valid value formatted per issuing taxation authority standards in positions 001 through 020 with trailing spaces and position 021 with a value from the following table.

### Values

| Value | Description                        |
|-------|------------------------------------|
| Y     | Acceptor Tax ID provided           |
| N     | Acceptor Tax ID not provided       |
| R     | Refused to provide Acceptor Tax ID |

## Processing rules

Acceptor Tax ID may be optionally provided.

## Subfield 008 (Partner ID Code)

DE 122, subelement 001, subfield 008 (Partner ID Code) identifies a specific partnership agreement, generally between specific acceptors and issuers.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 008                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans...006; LLLVAR        |
| Data Field           | Contents of subfield 008 |
| Justification        | Left                     |

## Values

Acceptors and issuers may establish Partner ID codes. If present, this subfield cannot contain all low values, all high values, all zeros, or all spaces.

### Processing rules

This subfield may be used for transactions on accounts issued under co-branding agreements when one of the co-branding partners originates the transaction. It also may be used when the agreement is between the acceptor and the clearing system, such as in certain promotional discounts.

### Subfield 009 (Service Location City Name)

DE 122, subelement 001, subfield 009 (Service Location City Name) contains the name of the city where the cardholder received the services, if it is different from the location identified by the acceptor city name.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 009                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans...013; LLLVAR        |
| Data Field           | Contents of subfield 009 |
| Justification        | Left                     |

### Values

Service Location City Name must contain the name of the city, town, or village as defined by country postal standards.

### Processing rules

Service Location City Name must be present for all transactions when different from the location identified in the Acceptor City Name. Alternatively, DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

For the U.S. region, this field must match the U.S. Postal Service list of acceptable cities and spellings.



## Subfield 010 (Service Location Country Subdivision Code)

DE 122, subelement 001, subfield 010 (Service Location Country Subdivision Code) contains the code of the country subdivision where the cardholder received the services, if it is different from the location identified by the acceptor country subdivision code.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield ID          | 010                       |
| Subfield Data Length | 3 positions               |
| Data Representation  | ans-003                   |
| Data Field           | Contents of subfield 010  |
| Justification        | Left with trailing spaces |

### Values

Service Location Country Subdivision Code must contain a valid code as defined in Chapter 2 of the [Quick Reference Booklet](#) and must be followed by trailing spaces for codes that are less than three positions.

### Processing rules

The Service Location Subdivision Code must be present for all transactions when different from the location identified in the Acceptor Subdivision Code. Alternatively, DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

## Subfield 011 (Service Location Country Code)

DE 122, subelement 001, subfield 011 (Service Location Country Code) contains the code of the country where the cardholder received the services, if it is different from the location identified by the acceptor country code.

### Attributes

| Attribute            | Description |
|----------------------|-------------|
| Subfield ID          | 011         |
| Subfield Data Length | 3 positions |

| Attribute           | Description              |
|---------------------|--------------------------|
| Data Representation | ans-003                  |
| Data Field          | Contents of subfield 011 |
| Justification       | Left                     |

### Values

Acceptor Country Code must contain a valid three-character alphabetic code as defined in Chapter 2 of the [Quick Reference Booklet](#). When territories have a defined country code in the [Quick Reference Booklet](#), that country code must be used.

### Processing rules

The Service Location Country Code must be present for all transactions when different from the location identified in the Acceptor Country Code. Alternatively, DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

## Subfield 012 (Service Location Postal Code)

DE 122, subelement 001, subfield 012 (Service Location Postal Code) contains the postal code of the country where the cardholder received the services, if it is different from the location identified by the acceptor postal code.

### Attributes

| Attribute            | Description              |
|----------------------|--------------------------|
| Subfield ID          | 012                      |
| Subfield Data Length | 3 positions              |
| Data Representation  | ans...010; LLLVAR        |
| Data Field           | Contents of subfield 012 |
| Justification        | Left                     |

### Values

Service Location Postal Code must contain a valid code as defined by country postal standards.

## Processing rules

The Service Location Postal Code must be present for all transactions when different from the Acceptor Postal Code and contain a valid code for countries that utilize postal codes. This information must be populated and formatted as defined by country postal standards. Alternatively, DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

## Subfield 013 (Acceptor Geographic Coordinates)

DE 122, subelement 001, subfield 013 (Acceptor Geographic Coordinates) contains the latitude and longitude of the acceptor's location.

## Attributes

| Attribute           | Description              |
|---------------------|--------------------------|
| Subfield ID         | 013                      |
| Subfield Length     | 3                        |
| Data Representation | ns...020; LLLVAR         |
| Data Field          | Contents of subfield 013 |
| Justification       | N/A                      |

## Values

Acceptor Geographic Coordinates, when present, must contain valid latitude and longitude values, comma-separated, represented in decimal degrees to five decimal places in positions 001 through 020 with these format standards:

- Length of geographic coordinates is a range of 15 - 20 positions
- Subfield should not contain alpha characters
- Subfield contains up to two '-'
- Subfield contains precisely two '"
- Subfield contains precisely one ','
- At least one numeric position is found before the first '"
- Five numeric positions are found between first '" and ','
- At least one numeric position is found between ',' and the second '"
- Five numeric positions are found after second '"
- Latitude is greater than or equal to -90.00000 and less than or equal to 90.00000
- Longitude is greater than or equal to -180.00000 and less than or equal to 180.00000
- No space character follows the comma separator

### Latitude and longitude examples

| Location                                    | Latitude and longitude |
|---------------------------------------------|------------------------|
| Anloga, Ghana                               | 5.78381,0.72175        |
| Lagos, Portugal                             | 37.11000,-8.67880      |
| Anchorage, AK                               | 61.21630,-149.89500    |
| Pago Pago, Eastern District, American Samoa | -14.27451,-170.70616   |

### Processing rules

Acceptor Geographic Coordinates can optionally be provided for all transactions and contain valid latitude and longitude values representing the acceptor location.

If the data present in subfield 013 is not formatted according to the "Values" specification, Mastercard will remove the subfield and continue processing the transaction.

### Subfield 014 (Service Location Geographic Coordinates)

DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) contains the latitude and longitude where the cardholder received the service, if it is different from the acceptor's location.

### Attributes

| Attribute           | Description              |
|---------------------|--------------------------|
| Subfield ID         | 014                      |
| Subfield Length     | 3                        |
| Data Representation | ns...020; LLLVAR         |
| Data Field          | Contents of subfield 014 |
| Justification       | N/A                      |

### Values

Service location geographic coordinates, when present, must contain valid latitude and longitude values, comma-separated, represented in decimal degrees to five decimal places in positions 001 through 020 with these format standards:

- Length of geographic coordinates is a range of 15 - 20 positions

- Subfield should not contain alpha characters
- Subfield contains up to two '-'
- The subfield contains precisely two '.'
- Subfield contains precisely one ','
- At least one numeric position is before the first '.'
- Five numeric positions are found between the first '.' and ','
- At least one numeric position is found between ',' and the second '.'
- Five numeric positions are found after second '.'
- Latitude is greater than or equal to -90.00000 and less than or equal to 90.00000
- Longitude is greater than or equal to -180.00000 and less than or equal to 180.00000
- No space character follows the comma separator

### Latitude and longitude examples

| Location                                    | Latitude and longitude |
|---------------------------------------------|------------------------|
| Anloga, Ghana                               | 5.78381,0.72175        |
| Lagos, Portugal                             | 37.11000,-8.67880      |
| Anchorage, AK                               | 61.21630,-149.89500    |
| Pago Pago, Eastern District, American Samoa | -14.27451,-170.70616   |

### Processing rules

Service location geographic coordinates can optionally be provided instead of, or in addition to, subfields 009 through 012 (Service Location - City Name, Country Subdivision Code, Country Code, Country Postal Code) when those subfields are required.

If the data present in subfield 014 is not formatted according to the "Values" specification, Mastercard will remove the subfield and continue processing the transaction.

### Subfield 015 (Acceptor Tax ID Name)

DE 122, subelement 001, subfield 015 (Acceptor Tax ID Name) contains the Acceptor Name that matches the Tax ID for the merchant.

### Attributes

| Attribute       | Value |
|-----------------|-------|
| Subfield ID     | 015   |
| Subfield length | 3     |

| Attribute           | Value                    |
|---------------------|--------------------------|
| Data representation | ans....80; LLLVAR        |
| Data fields         | Contents of subfield 015 |
| Justification       | Left                     |

### Values

When present, the Acceptor Tax ID Name must represent the owner name of the Acceptor Tax ID present in DE 122, subelement 001, subfield 007.

### Processing rules

This field is used for Brazil intracountry messages.

## DE 122 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

### Authorization Request/0100 message edits for DE 122, subelement 1, subfields 1 through 12

#### Associated edits

| If...                                                                    | Then the Dual Message Authorization System...                                                                                                                                     |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 122, subelement 1, subfields 1-12 extend past the valid field lengths | <p>will return to the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 122001</li> </ul> |

## Authorization Advice/0120 message edits for DE 122, subelement 1, subfields 1 through 12

### Associated edits

| If...                                                                    | Then the Dual Message Authorization System...                                                                                                                                            |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 122, subelement 1, subfields 1-12 extend past the valid field lengths | will return to the acquirer an Authorization Advice Response/0130: system-generated message where: <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 122001</li></ul> |

## Reversal Request/0400 message edits for DE 122, subelement 1, subfields 1 through 12

### Associated edits

| If...                                                                    | Then the Dual Message Authorization System...                                                                                                                     |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 122, subelement 1, subfields 1-12 extend past the valid field lengths | will return to the acquirer a Reversal Request Response/0410 message where: <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 122001</li></ul> |

## DE 122, subelement 001, subfield 013 (Acceptor Geographic Coordinates)

| If...                                                                     | Then the Dual Message Authorization System...                                                                                                                                                                                                  |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 122, subelement 001, subfield 013 length is not within expected limits | will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 122001</li></ul> |

| If...                                                            | Then the Dual Message Authorization System...                                                                                                                                                                                                     |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 122, subelement 001, subfield 013 includes an alpha character | will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 122001</li> </ul> |

#### DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates)

| If...                                                                     | Then the Dual Message Authorization System...                                                                                                                                                                                                     |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 122, subelement 001, subfield 014 length is not within expected limits | will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 122001</li> </ul> |
| DE 122, subelement 001, subfield 014 includes an alpha character          | will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 122001</li> </ul> |

## DE 123 (Receipt Free Text)

DE 123 (Receipt Free Text) only applies to the Swedish Domestic Authorization Switching Service (SASS), Peru, and the Mastercard Installment Payment Service. For SASS, DE 123 contains information to be printed on a receipt (not displayed on the terminal screen) for balance inquiry and ATM transactions (where DE 3 [Processing Code] is value 01 [Withdrawal] or value 30 [Balance Inquiry]). For Peru and Mastercard Installment Payment Service, DE 123 contains a text message to be printed on point-of-sale (POS) sales receipts.

### Attributes

| Attribute           | Description       |
|---------------------|-------------------|
| Data Representation | ans...512; LLLVAR |
| Data Length         | 3 positions       |



| Attribute     | Description |
|---------------|-------------|
| Subfields     | N/A         |
| Justification | N/A         |

## Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

## Applicable messages

| Message                             | Org | Sys | Dst |
|-------------------------------------|-----|-----|-----|
| Authorization Request Response/0110 | O   | X   | C   |
| Reversal Request Response/0410      | O   | X   | C   |

## Values

Proprietary receipt free data up to 512 characters.

## Application notes

Issuers may provide DE 123 for balance inquiry transactions, combined authorization response and account balance transactions, and ATM withdrawal transactions.

## Associated edits

| WHEN...                                                                                                                                                                                                                  | THEN the Dual Message Authorization System...                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| for SASS, DE 123 is present in an Authorization Request Response/0110 message where DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) does not contain value 01 (Withdrawal) or 30 (Balance Inquiry) | sends the issuer an Authorization Response Negative Acknowledgment/0190 where: <ul style="list-style-type: none"> <li>DE 39 = 30</li> <li>DE 44 = 123</li> </ul> |

### Associated edits

| WHEN...                                              | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 123 contains a length greater than 512 characters | <p>sends the issuer an Authorization Response Negative Acknowledgment/0190 where:</p> <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 123</li></ul> <p>The Dual Message Authorization System will not send DE 123 in an Authorization Request Response/0110 message to an acquirer that is not registered to receive this data.</p> |

## DE 124 (Member-Defined Data)

DE 124 (Member-defined Data) is described for general use and Prepaid Voucher, Brazil Maestro only, Bill Payment at the ATM only, and Mastercard Digital Enablement Service use.

**NOTE:** For detailed data layout requirements when using DE 124 with the Mastercard Digital Enablement Service, refer to the [MDES Register for Issuer Pre-digitization Network Message Values](#) section and *MDES Technical Specifications for Dual and Single Message Systems*.

## DE 124 (Member-Defined Data: General Use)

DE 124 (Member-Defined Data: General Use) may be used to submit up to 299 bytes of customer-defined data. DE 124 can contain program-specific data as defined by the DE 124 subelements.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans...299; LLLVAR           |
| Data Length         | 3 positions                 |
| Data Field          | Contents of positions 1–299 |
| Subelements         | Determined by program       |
| Justification       | N/A                         |

## Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

## Applicable messages

| Message                                       | Org | Sys | Dst |
|-----------------------------------------------|-----|-----|-----|
| Authorization Request/0100                    | C   | X   | C   |
| Authorization Request Response/0110           | C   | X   | C   |
| Authorization Advice/0120: acquirer-generated | O   | X   | C   |
| Authorization Advice/0120: system-generated   | .   | X   | C   |
| Reversal Request/0400                         | O   | X   | C   |
| Reversal Request Response/0410                | O   | X   | C   |
| Reversal Advice/0420                          | .   | X   | C   |

## Application notes

DE 124 may be required in support of programs to convey information between customers in an authorization message.

## Acquirers

## Associated edits

| IF DE 124 is...                                                          | THEN...                                                                                                                                 |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| up to 299 bytes in acquirer-generated authorization messages             | the Dual Message Authorization System will not edit the content of DE 124 and will forward the entire length of the data to the issuer. |
| more than 299 bytes of data in acquirer-generated authorization messages | the Dual Message Authorization System will only send the first 299 bytes of data to the issuer.                                         |

Absolute positioning of data in DE 124 subfields is required; padding will be necessary. Customers must select all State, Province, and Country Codes from the [Quick Reference Booklet](#). If a country code is used, it must be the ISO 3-character alphabetic (not numeric) Country Code. If used, a State or Province Code should be right-justified in this subfield with one leading blank space. Customers must not use all zeros, all low values, or all high values when formatting DE 124.

Subfields 6–13: are used for Maestro credit usage in Brazil.

## Issuers

### Associated edits

| IF DE 124 is...                                                   | THEN...                                                                                                                                   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Up to 299 bytes in issuer-generated response messages             | The Dual Message Authorization System will not edit the content of DE 124 and will forward the entire length of the data to the acquirer. |
| More than 299 bytes of data in issuer-generated response messages | The Dual Message Authorization System will truncate to 299 bytes the length of the data sent to the acquirer.                             |

The Authorization Request/0100 and Authorization Request Response/0110 messages will be able to contain DE 124 data independently from one another. Therefore, issuers may:

- Send back the same value in DE 124 in the Authorization Request Response/0110 message as was present in DE 124 of the Authorization Request/0100 message.
- Send a different value in DE 124 in the Authorization Request Response/0110 message as was present in DE 124 of the Authorization Request/0100 message.
- Not send DE 124 in the Authorization Request Response/0110 message. In this case, the acquirer will not receive DE 124 in the Authorization Request Response/0110 message.
- Include DE 124 in the Authorization Request Response/0110 message even when the acquirer did not send it in the Authorization Request/0100 message.

## DE 124 (Member-Defined Data: Prepaid Voucher)

DE 124 (Member-Defined Data: Prepaid Voucher) is for use only for Mastercard prepaid meal and food voucher programs in Europe.

### Subfield 1 (Merchant Direct Settlement Indicator)

DE 124 (Member-Defined Data: Prepaid Voucher), subfield 1 (Merchant Direct Settlement Indicator) is for use only for Mastercard prepaid meal and food voucher programs in Europe.

### Attributes

| Attribute            | Description               |
|----------------------|---------------------------|
| Subfield Data Length | 3 positions               |
| Data Representation  | ans-3                     |
| Data Field           | Contents of positions 1–3 |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

B00 equals voucher transactions will be settled bilaterally by the issuer directly with the merchant.

Value B00 is valid for use only in voucher transactions in the Europe region.

An acquirer must clear the transaction via the Dual Message Clearing System by using the bilateral settlement indicator (IPM Private Data Subelement [PDS] 0165 [Settlement Indicator], subfield 1 [Settlement Indicator] = B) in the First Presentment/1240 message.

### Application notes

Issuers must include this indicator in DE 124 Authorization Request Response/0110 messages for voucher transactions they will settle directly with the merchant.

When responding to an Authorization Request/0100 message for a transaction the issuer will settle directly with the merchant, the issuer must include the Prepaid Voucher format of DE 124, with subfield 1 containing Merchant Direct Settlement Indicator value B00.

The issuer must not include this indicator in response messages for meal voucher transactions that require settlement by Mastercard via the Mastercard settlement process.

The presence of the settlement indicator value B00 in DE 124, subfield 1, will serve as indication to the acquirer that the transaction will be settled by the issuer directly with the merchant. Receipt of this indicator value in an authorization response message by the acquirer indicates the acquirer must include PDS 0165 (Settlement Indicator), subfield 1 (Settlement Indicator) with value B (Bilateral settlement agreement applies; the Dual Message Clearing System clears the transaction but does not settle it) in the related clearing First Presentment/1240 message. This value allows the acquirer to clear the message without triggering the Mastercard settlement process.

## DE 124 (Member-Defined Data: Mastercard Send)

Acquirers and issuers are able to exchange DE 124 data in Mastercard Send (funding, refund, and payment), Gaming and Gambling, and MPQR payment transaction Authorization Request/Response messages.

Acquirers and issuers must use DE 108 (Additional Transaction Reference Data) when sending and receiving Receiver and Sender information.

## DE 124 (Member-Defined Data: Brazil Maestro Only)

DE 124 (Member-Defined Data: Brazil Maestro Only) is used only for Brazil Maestro transactions.

Subfields 1 through 8: are used for Maestro credit usage in Brazil.

### Subfield 1 (Unique Reference Number)

DE 124, subfield 1 (Unique Reference Number) contains a unique reference number.

#### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-19                              |
| Data Field          | Contents of positions 1–19          |
| Justification       | Left-justified with trailing spaces |

#### Values

Valid value string will contain a leading zero (0), followed by:

ICA (n-6)

Year (n-1)

Julian Date (n-3)

Time hhmmss (n-6)

Transaction Sequence number (01–99) (n-2)

Example: 0555555801215305401

### Subfield 2 (Sender/Payer/User ID)

DE 124, subfield 2 (Sender/Payer/User ID) contains sender consumer, business, government, and non-government names, payer name, or user ID data.

#### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-24                      |
| Data Field          | Contents of positions 20–43 |

| Attribute     | Description                         |
|---------------|-------------------------------------|
| Justification | Left-justified with trailing spaces |

### Values

Sender consumer, business, government, and non-government names, payer name, or user ID value up to 24 character spaces. If the consumer name is populated, the format must be (last name, first name).

### Subfield 3 (Sender/Payer Address)

DE 124, subfield 3 (Sender/Payer Address) contains sender/payer address data.

### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-91                              |
| Data Field          | Contents of positions 44–134        |
| Justification       | Left-justified with trailing spaces |

### Values

| Field               | Data   | Edit                                                                                                                                                                                                                          |
|---------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Street address      | ans-50 | Must be present<br>Must not be all blanks or all zeros                                                                                                                                                                        |
| City                | ans-25 | Must be present if applicable                                                                                                                                                                                                 |
| State/province code | ans-3  | If country is U.S. or Canada, state must be a valid state code for U.S. or a valid province code for Canada<br>Must not be all blanks or zeros<br>For all other countries, may contain all spaces or an applicable state code |
| Country code        | ans-3  | Must be present<br>Must not be all blanks or all zeros<br>Must be a valid country code<br>Must not be a blocked country                                                                                                       |

| Field       | Data   | Edit                          |
|-------------|--------|-------------------------------|
| Postal code | ans-10 | Must be present if applicable |

**NOTE:** Transaction will be declined if the sender Country is subject to comprehensive geographic sanctions published by the Office of Foreign Assets Control (OFAC), <https://ofac.treasury.gov/sanctions-programs-and-country-information>. The current list of countries subject to such sanctions is subject to change.

## Subfield 4 (Additional Sender Information)

DE 124, subfield 4 (Additional Sender Information) contains telephone number, date of birth, or optional message.

### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-65                              |
| Data Field          | Contents of positions 135–199       |
| Justification       | Left-justified with trailing spaces |

### Values

Additional sender information is formatted as follows:

Telephone number (n-20)

Date of Birth (n-8) (MMDDYYYY)

Optional message (ans-37)

## Subfield 6 (Discretionary Message on Sales Slip Supported)

DE 124, subfield 6 (Discretionary Message on Sales Slip Supported), used in association with Maestro credit in Brazil, is sent by the acquirer in the Authorization Request/0100 message, to indicate whether the POS terminal supports the discretionary message on the sales slip.

### Attributes

| Attribute           | Description              |
|---------------------|--------------------------|
| Data Representation | an-1                     |
| Data Field          | Contents of position 1–6 |



| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Values

| Value | Description                                                                          |
|-------|--------------------------------------------------------------------------------------|
| 0     | No<br>Merchant terminal does not support the printing of messages sent by the issuer |
| 1     | Yes<br>Merchant terminal supports the printing of messages sent by the issuer        |

## Subfield 7 (Discretionary Message on Sales Slip Code)

DE 124, subfield 7 (Discretionary Message on Sales Slip Code) is sent by the issuer in the Authorization Request Response/0110 message, to indicate the number of the message to be printed on the sales slip.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | an-3                      |
| Data Field          | Contents of positions 7–9 |
| Justification       | Left                      |

### Values

Valid value will contain the number of the message to be printed on the sales slip.

## Subfield 8 (Discretionary Message on Sales Slip Content)

DE 124, subfield 8 (Discretionary Message on Sales Slip Content) contains the variable part of the message to be printed on the sales slip.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | an-10                       |
| Data Field          | Contents of positions 10–19 |
| Justification       | Left                        |

### Values

Valid value will consist of the variable part of the message to be printed on the sales slip.

## Subfield 9 (Phoneshop [Phone Company ID])

DE 124, subfield 9 (Phoneshop [Phone Company ID]) is sent by the acquirer in the Authorization Request/0100 message, to identify the telephone company that provides the service for the Phoneshop product (for example, Telefonica = 01).

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | an-2                        |
| Data Field          | Contents of positions 20–21 |
| Justification       | Left                        |

### Values

Valid value will consist of the phone company ID.

## Subfield 10 (Phoneshop [Cell Phone Number])

DE 124, subfield 10 (Phoneshop [Cell Phone Number]) is sent by the acquirer in the Authorization Request/0100 message, to indicate the cardholder cell phone number for the Phoneshop product.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-10                        |
| Data Field          | Contents of positions 22–31 |
| Justification       | Right                       |

### Values

Valid value will consist of the cell phone number.

## Subfield 11 (Phoneshop [Message Security Code])

DE 124, subfield 11 (Phoneshop [Message Security Code]) is sent by the issuer in the Authorization Request Response/0110 message, to indicate the security code that the merchant sent for validation.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | an-4                        |
| Data Field          | Contents of positions 32–35 |
| Justification       | Left                        |

### Values

Valid value will consist of the message security code.

## Subfield 12 (Merchant CNPJ Number)

DE 124, subfield 12 (Merchant CNPJ Number) is sent by the acquirer in the Authorization Request/0100 message, to indicate the Merchant CNPJ number.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-14                        |
| Data Field          | Contents of positions 36–49 |
| Justification       | Right                       |

### Values

Valid value will consist of the Merchant CNPJ Number (a registration number provided by the government to all merchants).

## Subfield 13 (Total Annual Effective Cost)

DE 124, subfield 13 (Total Annual Effective Cost) is sent by the issuer in the Authorization Request Response/0110 message, to indicate the total annual effective cost in a financing transaction. A financing transaction includes purchase or cash withdrawal using a credit card.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-6                         |
| Data Field          | Contents of positions 50–55 |
| Justification       | Right                       |

### Values

**NOTE: Optionally, issuers may submit spaces preceding the positional start of subfield 13. Data submitted for this subfield should begin in position 50.**

Valid value will consist of the total annual effective cost in a financing transaction (including interest amount, taxes, and fees charged to the cardholder). This amount is mandated by local law number 3517 (implemented to ensure the cardholder is enforced about the total effective cost of the transaction). This applies to purchase or cash withdrawal at an ATM using a credit card.

## DE 124 (Member-Defined Data [Colombia Domestic Use Only])

DE 124 (Member-defined Data [Colombia Domestic Use Only]) is used only for Colombia domestic activity transactions.

Subfields 1 through 11: are used for Colombia domestic activity.

### Subfield 1 (Card Issuer Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 1 (Card Issuer Data) defines the FIID and Logical Network associated with the card issuer.

#### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-19                              |
| Data Field          | Contents of positions 1–19          |
| Justification       | Left-justified with trailing spaces |

#### Usage

Optional.

#### Example values

FIID (1–4)

Logical Network ID (5–8) = Examples are as follow: PRO1, MDS, BNET, ASCR, PRO2

- PRO1 = Redeban Multicolor
- MDS or BNET = Mastercard
- ASCR = AScredibanco
- PRO2 = Credibanco

Constant Value (9–19) = zeros

## Subfield 2 (Tax [IVA])

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 2 (Tax [IVA]) defines the tax for Colombia domestic activity.

### Attributes

| Attribute           | Description                        |
|---------------------|------------------------------------|
| Data Representation | n-12                               |
| Data Field          | Contents of positions 20–31        |
| Justification       | Right-justified with leading zeros |

### Usage

Required. If no value, it should contain zeroes.

### Example values

Amount is in Colombian pesos and carries an implied two position decimal.

## Subfield 3 (Tax Amount Base)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 3 (Tax Amount Base) defines the tax amount base for Colombia domestic activity.

### Attributes

| Attribute           | Description                        |
|---------------------|------------------------------------|
| Data Representation | n-12                               |
| Data Field          | Contents of positions 32–43        |
| Justification       | Right-justified with leading zeros |

### Usage

Required.

### Values

Amount is in Colombian pesos and carries an implied two position decimal.

## Subfield 4 (Retailer Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 4 (Retailer Data) defines the local merchant ID for Colombia domestic activity.

### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-27                              |
| Data Field          | Contents of positions 44–70         |
| Justification       | Left-justified with trailing spaces |

### Usage

Optional.

### Example values

Each acquirer may elect to provide the merchant ID in their original format. Example to follow:

Unique merchant ID (44–53)

Spaces (54–62)

Terminal Group (63–66)

Retailer Region (67–70)

## Subfield 5 (Terminal Acquirer Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 5 (Terminal Acquirer Data) identifies information about the terminal owner for Colombia domestic activity.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans-16                      |
| Data Field          | Contents of positions 71–86 |
| Justification       | Right-justified             |

### Usage

Optional.

### Example values

FIID (71–74)

Logical Network (75–78)

Constant Value = zeros

## Subfield 6 (Acquirer Original Processing Code)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 6 (Original Processing Code) contains the original processing code for Colombia domestic activity.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | n-6                         |
| Data Field          | Contents of positions 87–92 |
| Justification       | Right-justified             |

## Subfield 7 (Bill Payment/Top up Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 7 (Bill Payment/Top up Data) provides information related to bill payment for Colombia domestic activity.

### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-36                              |
| Data Field          | Contents of positions 93–128        |
| Justification       | Left-justified with trailing spaces |

### Values

| Value                | Description                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------|
| Service Code (93–96) | Number used to identify the provider company (for example: 0001 = Bogota Power Light Company) |



| Value                        | Description                                                                            |
|------------------------------|----------------------------------------------------------------------------------------|
| Service Description (97–126) | Service Provider Bill ID (for example: Invoice number, additional billing information) |
| Originator Device (127)      | Alphanumeric (for example, P)                                                          |
| Filler (128)                 | Space                                                                                  |

## Subfield 8 (Local POS Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 8 (Local POS Data) is present when transaction is a web payment or recurring payment. This field is optional.

### Attributes

| Attribute           | Description               |
|---------------------|---------------------------|
| Data Representation | n-1                       |
| Data Field          | Contents of positions 129 |
| Justification       | N/A                       |

### Usage

Optional.

### Values

| Values | Description                        |
|--------|------------------------------------|
| 6      | Preauthorization/recurring payment |
| 7      | Web payment, Electronic orders     |

## Subfield 9 (Local Response Codes)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 9 (Local Response Codes) identifies unique response codes for Colombia domestic activity.

### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-2                               |
| Data Field          | Contents of positions 130–131       |
| Justification       | Left-justified with trailing spaces |

### Usage

Optional.

### Example values

M2 = Invalid national customer identifier

## Subfield 10 (Original Transaction Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 10 (Original Transaction Data) is present for reversal messages to help uniquely identify the original message.

### Attributes

| Attribute           | Description                         |
|---------------------|-------------------------------------|
| Data Representation | ans-42                              |
| Data Field          | Contents of positions 132–173       |
| Justification       | Left-justified with trailing spaces |

### Usage

Optional.

### Example values

| Example value                       | Example description |
|-------------------------------------|---------------------|
| Original Message Type (132–135)     | 100 or 0200         |
| Sequence Number (136–147)           |                     |
| Original Transaction Date (148–151) | MMDD                |
| Original Transaction Time (152–157) | HHMMSS              |
| Capture Date (158–161)              | MMDD                |
| Filler (162–173)                    | All zeros           |

### Subfield 11 (IAC Tax Amount)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 11 (IAC Tax Amount) defines the IAC Tax for Colombia domestic activity.

### Attributes

| Attribute           | Description                        |
|---------------------|------------------------------------|
| Data Representation | n-12                               |
| Data Field          | Contents of positions 174–185      |
| Justification       | Right-justified with leading zeros |

## DE 124 (Member-Defined Data: Ukraine Domestic Transactions)

DE 124 (Member-Defined Data: Ukraine Domestic Transactions) enables acquirers to pass merchant identification data to issuers for Ukraine domestic transactions.

- The first three positions contain the length of the remaining positions.
- The next five positions contain the identifier UKR01.
- The next two positions contain one of the tags identifying the type of ID being transmitted.
- The next two positions contain the length of the ID value being transmitted.
- The final set of positions contain the ID value (with the length as indicated by the preceding two positions).

Values

Identifier details for Ukraine domestic transactions

| Identifier | TAG | Length (LL) | Value      | Description                                                                                                                                                                                                                                           |
|------------|-----|-------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UKR01      | 01  | 08          | ABCD0001   | Unified State Register of Enterprises and Organizations of Ukraine (USREOU). The following eight characters include a valid identity number assigned to corporate entities by the Unified State Register of Enterprises and Organizations of Ukraine. |
| UKR01      | 02  | 10          | ABCDXYZ123 | Registration number of the taxpayer's record card (RNOTRC). The following 10 characters include a valid identity number assigned to private entrepreneurs, known as the registration number of the taxpayer's record card.                            |

| Identifier | TAG | Length (LL) | Value                        | Description                                                                                                                                                                                                          |
|------------|-----|-------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UKR01      | 03  | 08-13       | ABCDXYZ1 or<br>ABCDXYZ123456 | Passport. The following eight or 13 characters (depending on the type of passport) include a valid passport letter series (if applicable) and number for private entrepreneurs for whom the RNOTRC is not available. |

Examples of Ukraine domestic transactions:

- 017UKR010108ABCD0001
- 019UKR010210ABCDXYZ123
- 017UKR010308ABCDXYZ1
- 022UKR010313ABCDXYZ123456

#### Application notes

Acquirers must provide DE 124 in Authorization Request/0100 messages for transactions that may fall into the scope of the National Bank of Ukraine requirement.

Acquirers must transmit identification data for all transaction types, which may fall into the scope of the relevant National Bank of Ukraine requirements. Identification of the relevant use cases is solely the acquirer's responsibility.

## DE 125 (New PIN Data)

DE 125 (New PIN Data) consists of a binary block containing a derived encrypted value calculated from the new PIN introduced by the cardholder at the ATM offering the PIN change service.

#### Attributes

| Attribute           | Description                              |
|---------------------|------------------------------------------|
| Data Representation | b-8                                      |
| Data Length         | N/A                                      |
| Data Field          | Contents of bit positions 1-64 (8 bytes) |

| Attribute     | Description |
|---------------|-------------|
| Subfields     | N/A         |
| Justification | N/A         |

### Usage

Following is the usage of DE 125 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                    | Org | Sys | Dst |
|----------------------------|-----|-----|-----|
| Authorization Request/0100 | C   | X   | C   |

### Values

DE 125 contains the new PIN, which is formatted into one of the supported PIN block formats and is then encrypted. The PIN block format and encryption method used must be the same as the one used for the existing PIN that is stored in DE 52. DE 125 is only required in Authorization Request/0100: PIN Change messages. Otherwise, it must not be present.

### Application notes

DE 125 is mutually exclusive with DE 110, Dataset 01, TAG 89 (Additional Encrypted PIN Block); both elements cannot be present simultaneously.

## DE 126 (Private Data)

DE 126 (Private Data) is reserved for future use.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans...100; LLLVAR           |
| Data Length         | 3 positions                 |
| Data Field          | Contents of positions 1–100 |
| Subfields           | N/A                         |

| Attribute     | Description |
|---------------|-------------|
| Justification | N/A         |

### Usage

By provision of the ISO 8583:1987 specification, Mastercard defined this data element for use as Private Data available for Mastercard's optional use for additional acquirer data.

The Dual Message Authorization System does not pass data placed in this data element through to the message receiver; rather, the Dual Message Authorization System temporarily stores the data. The Dual Message Authorization System does not return this data to the message originator in any subsequent response to an original Request, Advice, Response, or Acknowledgment message.

### Values

The length subelement must be in the range 001–100.

### Application notes

This data element is defined and used identically within all Mastercard programs and services.

## DE 127 (Private Data)

DE 127 (Private Data) may contain any private-use data that the customer may want to include in a message. Any Authorization System message originator may use DE 127.

### Attributes

| Attribute           | Description                 |
|---------------------|-----------------------------|
| Data Representation | ans...100; LLLVAR           |
| Data Length         | 3 positions                 |
| Data Field          | Contents of positions 1–100 |
| Subfields           | N/A                         |
| Justification       | N/A                         |

### Usage

Following is the usage of DE 127 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

### Applicable messages

| Message                                                                                         | Org | Sys | Dst |
|-------------------------------------------------------------------------------------------------|-----|-----|-----|
| Authorization Request/0100                                                                      | O   | X   | .   |
| Authorization Request Response/0110                                                             | O   | XE  | CE  |
| Authorization Advice/0120: acquirer-generated                                                   | O   | X   | .   |
| Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120) | .   | X   | CE  |
| Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) | O   | XE  | CE  |
| Authorization Response Acknowledgment/0180                                                      | C   | X   | .   |
| Authorization Response Negative Acknowledgment/0190                                             | .   | XE  | CE  |
| Issuer File Update Request/0302                                                                 | O   | X   | .   |
| Issuer File Update Request Response/0312                                                        | .   | XE  | CE  |
| Reversal Request/0400                                                                           | O   | X   | .   |
| Reversal Request Response/0410                                                                  | O   | XE  | CE  |
| Reversal Advice/0420                                                                            | .   | X   | C   |
| Reversal Advice Response/0430                                                                   | C   | X   | .   |
| Administrative Request/0600                                                                     | O   | X   | .   |
| Administrative Request Response/0610                                                            | .   | XE  | CE  |
| Administrative Advice/0620: system-generated                                                    | .   | O   | O   |
| Administrative Advice/0620:member-generated                                                     | O   | X   | .   |
| Administrative Advice Response/0630                                                             | .   | X   | .   |
| Network Management Request/0800: sign-on/sign-off                                               | O   | X   | .   |
| Network Management Request/0800: PEK exchange, on demand                                        | O   | X   | .   |
| Network Management Request Response/0810: sign-on/sign-off                                      | .   | XE  | CE  |

### Values

The length must be in the range 001–100.

### Application notes

Data placed in DE 127 is not passed through to the message receiver; rather, the Authorization System temporarily stores the data and returns it to the message originator in any subsequent response to an original Request, Advice, Response, or Acknowledgment message.



This data element is defined and used identically within all Mastercard programs and services.

Acquirers that provide DE 127 in the Authorization Request/0100 message will receive the first twenty bytes of the value provided in DE 072 (Data Record) of IPM Fee Collection/1740–783 for interregional non-financial ATM transactions.

## DE 128 (Message Authentication Code)

DE 128 (Message Authentication Code [MAC]) validates the source and the text of the message between the sender and the receiver.

### Attributes

| Attribute           | Description                              |
|---------------------|------------------------------------------|
| Data Representation | b-8                                      |
| Data Length         | N/A                                      |
| Data Field          | Contents of bit positions 1–64 (8 bytes) |
| Subfields           | N/A                                      |
| Justification       | N/A                                      |

### Usage

May contain message authentication code as defined by ISO standards.

### Values

Not applicable.

### Application notes

This data element should not be present in any Dual Message Authorization System message.

The last bit position within any bit map is reserved for DE 128. If authentication is to be used on a message, the MAC information is indicated by the final bit of the final bit map of that message. The final bit of all preceding bit maps shall contain zero; for example, there shall be only one DE 128 per message and that DE 128 must be the last data element of the message.

# Chapter 6 Transaction Message Validation

*System edits, or validation checks, ensure data accuracy, completeness, and consistency by flagging errors or invalid data.*

---

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Overview of system validation or editing

Mastercard performs data editing on submitted messages to enhance accuracy and data quality. An edit focuses on specific transaction components, authentication of messages, or program requirements.

Examples of data conditions edited by the system are:

- Presence of mandatory data in a message (such as PAN)
- Proper formatting of a data element or value (such as ensuring the data is alphabetic or numeric)
- Inclusion of data required for a specific product or program, which may be a validation on a single element or across multiple elements

To enforce compliance, additional validation processes occur outside of system message validations. The Data Integrity Program is one example of these processes. Refer to the *Data Integrity Monitoring Program* guide for more details.

Error notification

The Dual Message Authorization System informs customers of message processing edit failures through response and advice messages.

Data elements within these messages indicate in which data element or data element plus subelement the error occurred and provide information related to error information.

Acquirers are notified of message processing edit failures as follows:

Acquirer edit notification

| Message type                        | Data element                                 |
|-------------------------------------|----------------------------------------------|
| Authorization Request Response/0110 | DE 39 (Response Code)                        |
|                                     | DE 44 (Additional Response Data), if present |
| Authorization Advice Response/0130  | DE 39 (Response Code)                        |
|                                     | DE 44 (Additional Response Data), if present |
| Reversal Request Response/0410      | DE 39 (Response Code)                        |
|                                     | DE 44 (Additional Response Data), if present |

Issuers are notified of message processing edit failures as follows:

**Issuer edit notification**

| Message type                                        | Data element                                                          |
|-----------------------------------------------------|-----------------------------------------------------------------------|
| Authorization Response Negative Acknowledgment/0190 | DE 39 (Response Code)<br>DE 44 (Additional Response Data), if present |
| Authorization Advice/0120: system-generated         | DE 60 (Advice Reason Code)                                            |

DE 44 may appear in any response message when DE 39 contains the value 30, which signifies a format error condition in the related request message. In such cases, DE 44 will contain either:

- A three-digit number representing only the data element number
- A six-digit number indicating both the data element and the subelement numbers, if the data element includes subelements

## Edit errors best practices

These best practices help minimize edit errors and provide guidance for addressing them when they occur.

### Preventing errors

To prevent errors:

- Review and analyze all release announcements for new or updated data formatting and value requirements.
- Perform as much advance data validation and correction as possible before sending messages to Mastercard. This could include verifying the BIN or account range for the PAN, country codes, currency codes, or other data.
- Customers using parameter data obtained from Mastercard-supplied tables (such as those found in the Mastercard Parameter Extract file) as part of their own transaction data validations must use the most recent versions of the tables.
- Follow any field encoding schemes as documented in the Mastercard system specifications. This includes:
  - Populating fields in supported message types, as specified by the Message Type Indicator (MTI)
  - Complying with field length and format specifications, such as numeric or alphabetic
  - Constructing subelement ID, subelement length, subfield ID, and subfield length fields in accordance with the data element-specific encoding schemes found in the Data element definitions and details chapter
  - Using the Mastercard-defined field values
- Mastercard strongly recommends using Mastercard simulators to validate new or modified processing logic.

## Reacting to errors received

If you receive an edit error:

- Use the value in DE 44 to determine where the error occurred.
- Look at the specifications for the data element or subelement in error and compare them against the data submitted in the field.
- Review the application notes for the data element, which may provide additional information on requirements for that field.

## Error numbers and messages

Details of edits are documented in the notes sections associated with data elements, subelements, and subfields in Chapter 5, Data element definitions and details.

Additional information can be found under specific services described in Chapter 6, System functionality and services, and Appendix B, Services or products leveraging the system.

## Chapter 7 System functionality and services

This section describes [Dual Message Authorization System](#) functionality or services that are fundamental to transaction processing. This section also includes information about requesting and using these services.

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## Account balance response

The account balance response enables issuers to include account balance information as part of the response to a financial authorization request. The account balance response is an unsolicited transmission of account balance data to the point-of-sale (POS) terminal or ATM. Issuers may provide solicited transmission of account balance through other means. The account balance response provides cardholders with up-to-date balance information on their prepaid cards.

This functionality applies only to ATM transactions or transactions on prepaid cards. Issuers may not use this for credit or debit accounts for POS transactions. Use for credit or debit account increases the risk of fraudulent activity and concerns for cardholder privacy.

Issuers may provide account balance information for both approved and declined authorization responses. Prepaid Debit Mastercard card issuers must provide the cardholder's account balance information in DE 54 (Additional Amounts) of the Authorization Request Response/0110 message.

When the issuer provides the available balance in the authorization response, POS and ATM terminals that are programmed to accept account balance data can display or print the available balance.

- **Approved Responses:** The available balance on an approved response is the account balance after the transaction amount has been deducted from the account.
- **Declined Responses:** For POS transactions where a prepaid card is used, this feature gives the cardholder and merchant information needed to process a split-tender transaction. The merchant can submit another authorization request for the purchase using a lower amount that can be approved by the prepaid issuer and ask the cardholder to provide another form of payment for the remaining purchase amount. The available balance returned in a declined response identifies the current account balance, without deducting the declined transaction amount.

The Dual Message Authorization System forwards the account balance response provided by the issuer to the acquirer only when it is returned in an ATM transaction or in a POS transaction for a prepaid account. Mastercard also will remove balance information from all Maestro® and POS purchase transactions that are initiated with a debit card (not identified to be prepaid related).

## Account data compromise information

Mastercard provides issuers with account data compromise (ADC) event information in authorization messages.

### Authorization Processing

The following describes authorization processing for a transaction that contains a primary account number (PAN) with and without an ADC event match.

- When an acquirer sends an Authorization Request/0100 message to the Mastercard Network, the Dual Message Authorization System searches the ADC database for a match on the PAN.
- If an ADC event match is found, the Dual Message Authorization System inserts the following values in the Authorization Request/0100 message before sending to the issuer:
  - DE 48 (Additional Data: Private Use), subelement 39 (Account Data Compromise Information) containing the account data compromise event information.

DE 48, subelement 39 contains multiple pieces of information, including up to three case key codes (unique keys identifying the case).

Case key codes can be used to locate the Mastercard Alerts Full Case ID associated with a particular ADC event (for example, MCAAlert Full Case ID: MCA2101-US-24. The data in DE48, subelement 39 is displayed in the following format: 021014, which displays five positions for the alert number and the last digit of the year (right-justify and pad string with spaces if needed).

### Authorization reports

The Authorization Parameter Summary Report (SI737010-AA) includes an ADC Event (Account Data Compromise) participant parameter in the Global Parameters section. For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#).

### For more information

For details about data requirements, refer to [Data element definitions and details](#).

## Account Management System

Account Management System (AMS) services include account-level services that issuers use to designate special handling during authorization. Issuers can list cardholder accounts as exception accounts that receive Stand-In or chargeback protection.

Issuers can identify accounts with a preferred status in Premium Listings, providing higher dollar and transaction limits for Stand-In Authorization; or identify accounts that participate in services such as Electronic Warning Bulletin, Payment Cancellation, Contactless Mapping Services, Mastercard In Control, Enhanced Value, Product Graduation, and High Value.

## Stand-In Account File

The Stand-In Account File (MCC102) provides information for use in Stand-In Authorization when making an authorization decision on behalf of the issuer.

Issuers list the following types of accounts in the Stand-In Account File:

- Preferred or premium listings that designate higher authorization limits for preferred cardholders



- Negative accounts that require a Stand-In Authorization response of decline, refer to card issuer, or capture card for lost, stolen, fraud or accounts with credit issues
- Electronic Warning Bulletin (restricted) listings for lost, stolen, and fraudulent accounts, which are distributed daily to all acquirers

## Electronic Warning Bulletin File

The Electronic Warning Bulletin File (MCC103) allows issuers to identify and manage restricted accounts for special handling.

AMS collects and distributes Mastercard restricted account listings in the Electronic Warning Bulletin (EWB) File for use in authorization processing.

Issuers list the following types of accounts in AMS:

- Lost
- Stolen
- Counterfeit
- Severe credit problems with continued activity

Listing in AMS provides issuers with the following benefits:

- Warning bulletin chargeback rights
- Capture-card response for Stand-In Authorization and X-Code Processing

## Local Stoplist File

The Local Stoplist File (MCC104) allows participating issuers to list exception accounts at the country and subcountry levels.

This gives issuers the opportunity to select the geographical area covered by each listing to target a specific fraud problem.

- The country level listing allows issuers to list in a single country.
- The subcountry level listing allows issuers to list in a defined category such as an acceptor business code (MCC) within a country.

## Payment Cancellation File

The Payment Cancellation File (MCC105) supports Payment Cancellation for card-not-present and recurring payment transactions.

Payment cancellation allows issuers to specify criteria to block card-not-present and recurring payment transactions identified in authorization messages and clearing records.

## PAN Mapping File

The PAN Mapping File (MCC106) supports multiple mapping services.

Services which use the PAN mapping file functionality are:

- Contactless Mapping Service
- Mastercard Digital Enablement Service (MDES)
- In Control Mapping
- Token Service Provider PAN Mapping

## Contactless Application Transaction Counter File

The Contactless Application Transaction Counter (ATC) File (MCC109) contains the most recent ATC values for contactless cards or devices that were personalized using dynamic values in the ATC and unpredictable number (UN).

Issuers participating in the CVC 3 Validation in Stand-In Authorization or in the CVC 3 Pre-validation Service may provide this file to Mastercard.

## Blocking ranges of accounts

Mastercard provides customers two ways to list account ranges to stop the acceptance of a range of cards.

The following table explains how to create two types of blocked listings.

### Types of blocked listings

| Type of listing          | How to list                                         | Where it goes                                                              |
|--------------------------|-----------------------------------------------------|----------------------------------------------------------------------------|
| Group or Series          | The issuer requests the Group or Series in writing. | Mastercard adds the listing to AMS.                                        |
| Selective Range Blocking | Stand-In Authorization within MC Connect.           | Mastercard adds the listing to the Stand-In Authorization processing flow. |

The following table compares the levels of protection provided by the listing types.

Levels of protection types

| Type of listing          | Levels of protection   |        |                            |                                                             |
|--------------------------|------------------------|--------|----------------------------|-------------------------------------------------------------|
|                          | Stand-In Authorization | X-Code | Under Merchant Floor Limit | Chargeback Rights under Reason Code 4807                    |
| Group or Series          | Yes                    | Yes    | Yes                        | Only in catastrophic situations when approved by Mastercard |
| Selective Range Blocking | Yes                    | Yes    | No                         | No                                                          |

For more information

For more information about AMS and AMS services, refer to the [Account Management System User Manual](#).

For more information about blocking ranges, refer to [Stand-In Authorization Reference Guide](#).

Account Status Inquiry Service

Account Status Inquiry (ASI) allows acquirers to send Account Status Inquiry transactions to validate aspects of a cardholder account.

The Account Status Inquiry Service supports purchase account status inquiry transactions and payment account status inquiry transactions for Mastercard® credit, Debit Mastercard®, and Maestro® acceptance brands.

In addition to Mastercard products, Account Status Inquiry Service transaction processing supports the following types of acceptance brands:

- American Express
- Diners
- Discover
- JCB
- Private label cards
- Visa

## Purchase Account Status Inquiry/Recurring Payment Account Status Inquiry

Purchase ASI or Recurring Payment ASI is an optional service for merchants, which allows them to validate that a cardholder account is open and the account is not listed in the Electronic Warning Bulletin, without negatively affecting a cardholder's funds availability.

The Purchase ASI or a Recurring Payment ASI service can be used:

- To establish a recurring or bill payment relationship
- To validate a card-not-present purchase before fulfillment
- Before submitting an authorization request for the full amount of a recurring payment

A Recurring Payment ASI must comply with all existing recurring payment transaction identification requirements.

### Account Status Inquiry transaction process

1. The acquirer submits an Account Status Inquiry transaction request in the Authorization Request/0100 message containing:
  - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase)
  - DE 4 (Amount, Transaction) with a transaction amount of zero
  - DE 48 (Additional Data: Private Use), subelement 82 (Address Verification Service Request), value 52 (AVS and Authorization Request/0100) for AVS requests (optional)
  - DE 48, subelement 92 (CVC 2 Value), CVC 2 value (optional)
  - DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI])

If Recurring Payment ASI: DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 4 (Standing order/recurring transactions)

2. The issuer sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) may contain value 00 (Approved or completed successfully), 85 (Not declined), or a valid business decline response. If the issuer is unable to reply, the acquirer receives a response code of 91 (Authorization System or issuer system inoperative).
3. If applicable, the issuer provides the applicable AVS response in DE 48, subelement 83 (Address Verification Service Response), and provides a valid CVC 2 response in DE 48, subelement 87 (Card Validation Code Result) in the Authorization Request Response/0110 message.

**NOTE: Product Inquiry messages do not include address verification and/or CVC 2 validation requests. See separate description of Product Inquiry Service.**

Acquirers are prohibited from placing a value of one major unit of currency or any other nominal test amount (including equivalent units of local currency such as EUR 1 or JPY 1) that does not represent an actual purchase amount in DE 4 of the Authorization Request/0100 message.

Transactions occurring at an automated fuel dispenser in the U.S. region identified with MCC 5542 may continue to submit USD 1 (or local currency equivalent) authorization requests.

Contactless transit aggregated transactions generated by the transit authority and held for a period of time before being cleared may continue to submit an authorization request for USD 1 or an amount up to the cardholder verification method (CVM) limit amount published in the *Chargeback Guide* on the day of the transaction (or local currency equivalent) authorization requests.

### **Mastercard Digital Enablement Service (MDES) Account Status Inquiries**

Account Status Inquiry messages can also be initiated with MDES tokens, in this case the appropriate MDES services will be performed when MDES data is provided.

For more information on MDES ASI messages, refer to the [MDES Technical Specifications for Dual and Single Message Systems](#).

### **Country- and currency-related data in MDES pre-digitization and ASI messages**

Issuers that use the country-level authorization service for cards designated for local use only that have chosen to receive MDES pre-digitization and ASI messages must be aware that when a funding account range is configured for local use only, Mastercard will modify the country- and currency-related data fields in the MDES pre-digitization and ASI messages to match the issuer country code and primary currency code (as applicable) of the funding account range, according to the configurations provided by the issuer during the onboarding process. The country- and currency-related data fields include the following:

- DE 43 (Acceptor Name and Location), subfield 5 (Acceptor Country Code)
- DE 49 (Currency Code, Transaction)
- DE 61 (Additional POS Data), subfield 13 (POS Country Code)

Issuers that are not configured for local use only will continue to see the MDES pre-digitization and ASI messages with the United States country and currency codes.

## **Payment Account Status Inquiry**

A Payment Account Status Inquiry (Payment ASI) enables payers to verify that issuers will accept a Payment Transaction on behalf of a payee (recipient) for the actual transaction amount for a particular recipient card account before collecting funds from the sender and initiating the Payment Transaction authorization to credit an account.

This request allows issuers to provide a response to an inquiry sent by the payer, indicating whether the issuer will accept a subsequent Payment Transaction authorization for a particular recipient account and amount. Obtaining a successful Payment ASI helps eliminate the need for a payer to return funds to the sender.

The Payment ASI helps improve the payer's experience when confirming a recipient's card account for future use. Issuers can clearly identify:

- The difference between a Payment ASI transaction and a Payment Transaction authorization, so that they can respond and act accordingly.
- The difference between a Payment ASI transaction and a Purchase ASI transaction.

How it works

Acquirers can submit a Payment ASI Transaction for a subsequent payment transaction to determine whether an issuer will post the specified amount to a particular card account before collecting funds from the sender.

Payment ASI transactions are non-financial transactions. Issuers must not credit the receiving cardholder account based on the amount provided in the Payment ASI request.

- When an issuer receives a Payment ASI transaction of zero, the issuer should confirm that the recipient’s card account exists and can be used in the future for receiving funds.
- When an issuer receives a Payment ASI transaction for an amount greater than zero, the issuer should confirm that the amount specified does not exceed a prepaid load limit or other types of limits they may have for Payment Transactions.
- When the issuer provides an approval response to the Payment ASI for the receiving account, the issuer is indicating that it is expecting to approve a subsequent Payment Transaction request message when it is received for the account.

**NOTE: The receiving issuer may reject the subsequent Payment Transaction if circumstances for the account have changed.**

Applicable edits

| WHEN...                                                                        | THEN the Dual Message Authorization System...                                                       |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| a Payment ASI qualifies for a Mastercard® Mastercard Send™ Payment Transaction | applies regular Mastercard Send Payment edits.                                                      |
| an acquirer advice Payment ASI is initiated                                    | sends the acquirer an Authorization Advice/0130 message where DE 39 = 12 (Invalid transaction).     |
| an acquirer reversal of a Payment ASI is initiated                             | sends the acquirer a Reversal Request Response/0410 message where DE 39 = 12 (Invalid transaction). |

Payment Account Status Inquiry transaction messages

Following is a list of the data elements and values applicable to Payment ASI Authorization Request/0100 messages.

## Data elements

| Data Element                                                                               | Value                      | Comment                              |
|--------------------------------------------------------------------------------------------|----------------------------|--------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)                      | 28                         | Payment                              |
| DE 4 (Amount, Transaction)                                                                 | Equal to or greater than 0 | Equal to or greater than zero        |
| DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier)          | C02, C03, C04, C07         |                                      |
| DE 48 (Additional Data: Private Use), subelement 82 (Address Verification Service Request) | 52                         | Optional AVS                         |
| DE 48 (Additional Data: Private Use), subelement 92 (CVC 2)                                | CVC 2                      | Optional CVC 2                       |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)                   | 8                          | Account Status Inquiry Service (ASI) |

## Mastercard Send Payment ASI transaction messages

Following is a list of the data elements and values applicable to Mastercard Send Payment ASI Authorization Request/0100 messages.

## Data elements

| Data Element                                                          | Value                                                                              | Comment                                                                                                        |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) | 28 (Payment Transaction)                                                           |                                                                                                                |
| DE 4 (Amount, Transaction)                                            | Equal to or greater than 0                                                         | For transaction-value limits, refer to the <i>Mastercard Send and Funding Transactions Program Standards</i> . |
| DE 18 (Merchant Type)                                                 | MCC 6536 (Mastercard Send Intracountry)<br>MCC 6537 (Mastercard Send Intercountry) |                                                                                                                |

| Data Element                                                                               | Value                                                                                                                      | Comment                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier)          | For a complete list of Transaction Type Identifiers, refer to: <a href="#">Subelement 77 (Transaction Type Identifier)</a> | Mastercard Send Funding/Payment Transaction Type Identifiers                                                                                                                                                                                                                                       |
| DE 48 (Additional Data: Private Use), subelement 82 (Address Verification Service Request) | 52 (AVS and Authorization Request/0100)                                                                                    | Optional AVS                                                                                                                                                                                                                                                                                       |
| DE 48 (Additional Data: Private Use), subelement 92 (CVC 2)                                | CVC 2                                                                                                                      | Optional CVC 2<br>If the originating institution does not have CVC data for push payment transactions on the Mastercard Network, this field should be left blank. The receiving institution must not decline Mastercard Send payment transactions on the basis that the CVC data is not populated. |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)                   | 8 (Account Status Inquiry Service [ASI])                                                                                   |                                                                                                                                                                                                                                                                                                    |
| DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level)     | 0 (Not a CAT transaction)                                                                                                  |                                                                                                                                                                                                                                                                                                    |
| DE 108 (Additional Transaction Reference Data)                                             |                                                                                                                            | DE 108 fields are required for Mastercard Send Payment ASI Transactions. For technical requirements, refer to the <i>Mastercard Send and Funding Transactions Program Standards</i> .                                                                                                              |

## Product Inquiry Service

The Product Inquiry Service allows an acquirer to send a product inquiry authorization request message to Mastercard.

**NOTE: Applies only to the U.S. region.**

As part of the authorization response to an acquirer, Mastercard will provide the product code associated with the Mastercard account number. Additionally, because product codes for card programs potentially fall under different interchange rate structures, Mastercard will also populate the authorization response message with the applicable account category code, as defined by Account Level Management.



The information received by the acquirer through a Product Inquiry Service request, along with the published Mastercard interchange rate schedule and rate criteria, can be used by an acquirer and merchant to determine the product and associated interchange rate that may be applied to a purchase transaction.

### Acquirers

Acquirers that choose to support Product Inquiry Service transactions must send Authorization Request/0100 messages containing DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service) and DE 4 (Amount, Transaction) with a transaction amount of zero when submitting Product Inquiry Service transaction requests.

**NOTE: Product Inquiry Service messages including address verification or CVC 2 validation requests will be deemed to be an attempt to mitigate fraud. As such, they will be designated as an Account Status Inquiry Service transaction and billed accordingly.**

### Issuers

Issuers will receive Authorization Request/0100 messages containing DE 61, subfield 7, value 8, and DE 4 with a transaction amount of zero. Issuers must respond to these transactions with value 00 (Approved or completed successfully), 85 (Not Declined), 05 (Do Not Honor) or other valid business decline responses in DE 39 (Response Code).

### For more information

For more information about the Product Inquiry Service, contact the Global Customer Service team.

## Address Verification Service

Address Verification Service (AVS) is a fraud deterrent service that provides greater security to merchants and cardholders. It helps to protect against fraudulent use of cards by verifying the cardholder's billing address.

For information on the Name Validation Service, refer to [Name Validation Service](#) section in the appendix.

### How AVS Works

The acquirer requests the cardholder billing address verification as part of the Authorization Request/0100. The issuer performs the verification and returns the appropriate information to the acquirer in the Authorization Request Response/0110.

### Alternate processing

The Stand-In and X-Code systems do not perform AVS. Transactions that contain an AVS request in an Account Status Inquiry Service request will receive a response indicating the service is not available. Transactions that contain an authorization request and an AVS request

will receive the appropriate authorization response in addition to a response indicating the AVS service is not available.

## Participation requirements

Issuers must have online access to participate in AVS.

### Participation requirements

| IF the customer... | AND is located...                                                    | THEN participation is...                   |
|--------------------|----------------------------------------------------------------------|--------------------------------------------|
| issues cards       | in the U.S. and Canada regions                                       | required                                   |
| issues cards       | in a region other than the U.S. that supports AVS use for the region | optional, unless mandated by local country |

## Indicating AVS participation during sign-in

When the issuer signs into the Mastercard Network, the issuer can choose AVS participation options.

The participation is contained in DE 94 (Service Indicator), subfield 3 (Address Data Indicator) of the Network Management Request/0800: sign-on/sign-off message and indicates the following options:

- Does not participate in AVS
- Receives AVS data in non-condensed format
- Receives AVS data in condensed format

**NOTE: Issuers in North America are required to ensure all U.S. and Canada issued cards support and respond to all AVS requests from merchants. It is the issuer's responsibility to ensure that their account ranges for U.S. and Canada issued cards are enabled for AVS. DE 94, subfield 3 value 0 (AVS not currently supported) must not be used. DE 48, subelement 83 (Address Verification Service Response) value U (No data from issuer/Authorization System) may be utilized where there is a declined authorization and the issuer cannot disclose the AVS response.**

## AVS process

The AVS transaction process is listed as follows.

1. The acquirer generates an Authorization Request/0100 message with non-condensed address data included in DE 120 (Record Data). When submitting AVS requests, the Authorization Request/0100 message should always contain DE 120, subfield 01 (AVS Service Indicator 1).
2. The Mastercard Network applies an algorithm to construct the address key, which the issuer uses to verify the address.

3. The issuer host compares the address key in DE 120 to the address key in the issuer database.
4. The issuer generates an Authorization Request Response/0110 message with the AVS response included in DE 48 (Additional Data: Private Use), subelement 83 (Address Verification Service Response).

## Address key

The address key is the part of the Authorization Request/0100 message that contains the address data.

To achieve a complete match, the key that the Mastercard Network sends in DE 120 (Record Data) must be the same as the key in the issuer's database. The address key consists of the following subelements in DE 120:

- Cardholder postal/ZIP code
- Cardholder address (condensed or non-condensed)

### Cardholder postal/ZIP code

The cardholder postal/ZIP code has a fixed length of 9 bytes.

Using AVS Service Indicator 4, AVS will condense the cardholder postal/ZIP code to contain only numeric values, with a maximum of nine values.

**NOTE: Issuers of Mastercard and Debit Mastercard cards in the Canada region must support the validation of Canadian postal codes in a U.S. ZIP code format to facilitate AVS checking at U.S. automated fuel dispensers (AFDs). The required format involves input of the three numeric digits of the cardholder postal code, followed by two trailing zeros to match the required ZIP code format. For example, L7M 1W9 would be entered by the cardholder as 71900. As postal codes are communicated in DE 120 (Record Data), subfield 03 (AVS Service Indicator 3), subelement 3 (AVS Usage) in a nine-position format, L7M 1W9 would be formatted as 71900bbbb, where b is blanks or spaces. For all other non U.S. or Canada region issued cards, Mastercard recommends cardholders pay at the attended terminal within the store.**

### Cardholder address

The length of the cardholder address depends on whether it is condensed or non-condensed.

**Non-condensed** Using AVS Service Indicator 1: A non-condensed cardholder address includes the address data exactly as the acquirer provides it in DE 120. A non-condensed address key includes both the cardholder postal/ZIP code and the cardholder address data exactly as the acquirer provided them in DE 120.

The Dual Message Authorization System will not truncate the address data passed from the acquirer if greater than 29 positions. DE 120 is an LLLVAR field; therefore, it will indicate the length of the data being passed.

**NOTE: Some merchants or acquirers currently are limited to supporting only numeric data.**

Condensed Using AVS Service Indicator 2: A condensed cardholder address using AVS Service Indicator 2 includes the first five numeric values at the beginning of the cardholder address (when scanning the address from left to right) provided by the acquirer in DE 120.

The condensed address key (cardholder postal/ZIP code plus cardholder address) is a fixed length of 14 bytes. It includes the entire cardholder postal/ZIP code and the condensed cardholder address. Mastercard left-justifies and blank-fills both portions of the key, as needed.

Stages of the process to condense the address key:

1. The condensing algorithm begins condensing with the numeric value in the left-most position.
2. The algorithm extracts the first five numeric values of the cardholder address (when scanning the address from left to right).  
The condensing algorithm ignores all special characters (such as hyphens [-], slashes [/ or \], and number signs [#]).
3. AVS constructs the condensed AVS key.

Condensed Using AVS Service Indicator 3: A condensed cardholder address includes as many as five numeric values at the beginning of the cardholder address provided by the acquirer in DE 120.

The condensed address key (cardholder postal/ZIP code plus cardholder address) is a fixed length of 14 bytes. It includes the entire cardholder postal/ZIP code and the condensed cardholder address. Mastercard left-justifies and blank-fills both portions of the key, as needed.

Stages of the process to condense the address key:

1. The condensing algorithm begins condensing with the numeric value in the left-most position.
2. The algorithm extracts up to five numeric values that appear before the first alphabetic character or space. The condensing algorithm:
  - Ignores all special characters (such as hyphens [-], slashes [/ or \], and number signs [#])
  - Excludes the portions of the address that could be open to misinterpretation, such as apartment numbers
3. When AVS finds a space or an alphabetic character, it stops examining the cardholder billing address.
4. AVS constructs the condensed AVS key.

Condensed Using AVS Service Indicator 4: A condensed cardholder address using AVS Service Indicator 4 includes only the numeric values in the cardholder postal/ZIP code (with a maximum of nine values), as well as the first five numeric values in the cardholder address received (when scanning the address from left to right) before forwarding the message to the issuer.

Stages of the process to condense the post/ZIP code and address data:

1. The condensing algorithm begins condensing with the numeric value in the left-most position.
2. The algorithm extracts up to nine numeric values in the cardholder postal/ZIP code and up to five numeric values in cardholder address (when scanning the address from left to right).

The condensing algorithm ignores all special characters (such as hyphens [-], slashes [/ or \], and number signs [#])

- 3. AVS constructs the condensed AVS key.

Non-condensed versus condensed

The following table provides an example of a non-condensed address key compared to a condensed address key.

Non-condensed versus condensed details

| Non-condensed                       | Condensed Using AVS Service Indicator 2 | Condensed Using AVS Service Indicator 3 | Condensed Using AVS Service Indicator 4 |
|-------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| 4J235 1520 Main Street, Apartment 3 | 4J235 ____ 15203                        | 4J235 ____ 1520                         | 4235 ____ 15203                         |

Example 1: complete match

The cardholder billing address is 1520 Main Street, Apartment 3. The issuer is using a condensed address key using AVS Service Indicator 3.

The cardholder neglects to provide the merchant with the apartment number 3, so the merchant omits the apartment number from the AVS request. However, the matching algorithm still finds a complete match.

Example 1: complete match

| Merchant enters... |                    |                 |                      |                                                                                                                                                                                                                                                                                                                                            |
|--------------------|--------------------|-----------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Postal/ZIP Code    | Cardholder Address | AVS Address Key | Issuer's Address Key | Result/Explanation                                                                                                                                                                                                                                                                                                                         |
| 63110              | 1520 Main Street   | 63110____ 1520  | 63110____ 1520       | Complete Match<br><br>The issuer builds its address key based on the AVS algorithm. Both the issuer's address key and the AVS algorithm extract only the numeric values that precede the first space in this example (1520). Therefore, when the merchant does not enter the apartment number, this still results in a complete AVS match. |

## Example 2: no match

The cardholder's address is 54 East Street; however, the cardholder provides the merchant with the address of 59 East Street.

### Example 2: no match

| Merchant enters... |                    |                 |                      |                                                                                                         |
|--------------------|--------------------|-----------------|----------------------|---------------------------------------------------------------------------------------------------------|
| Postal/ZIP Code    | Cardholder Address | AVS Address Key | Issuer's Address Key | Result/Explanation                                                                                      |
| 63110              | 59 East Street     | 63110____59__   | 63110____54__        | No Match<br><br>The issuer's address key does not match the cardholder address given by the cardholder. |

## Issuer procedures

Stages of the process that issuers follow to receive and respond to AVS requests.

1. The issuer signs on to the Dual Message Authorization System using a Network Management Request/0800: sign-on/sign-off message.

### Issuer procedures

| IF the issuer receives...                                                             | THEN the issuer signs on with value... |
|---------------------------------------------------------------------------------------|----------------------------------------|
| non-condensed AVS data                                                                | 1 in DE 94, subfield 3.                |
| condensed AVS data using AVS Service Indicator 2                                      | 2 in DE 94, subfield 3.                |
| condensed AVS data using AVS Service Indicator 3                                      | 3 in DE 94, subfield 3.                |
| condensed AVS data using AVS Service Indicator 4                                      | 4 in DE 94, subfield 3.                |
| for customers using group sign-in, the value chosen applies to all BINs in the group. |                                        |

2. When the issuer receives an AVS request, the issuer verifies the cardholder billing address, matching it against the issuer's address file.
3. The issuer generates an Authorization Request Response/0110 that includes the AVS response and an authorization response.

## Authorization Request/0100: AVS and authorization request

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                         | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 4 (Amount, Transaction)                                           | M   | •   | M   | Must be a valid amount for authorization request with AVS requests.                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code) | M   | •   | M   | Must be a valid TCC.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DE 48, subelement 82 (Address Verification Request)                  | M   | •   | M   | 52 = AVS and Authorization Request                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DE 120 (Record Data)                                                 |     |     |     | <p>The acquirer must provide the non-condensed cardholder billing address.</p> <p><b>NOTE: Note: Some merchants/acquirers are currently limited to supporting only numeric data.</b></p> <p>The issuer will receive only one occurrence of DE 120 subfields depending on the AVS Service Indicator identified in DE 94 of the Network Management Request/0800 message. The AVS Service Indicator identifies how the issuer expects to receive address data.</p> |
| DE 120, subfield 1 (AVS Service Indicator 1)                         | M   | •   | C   | <p>Postal code (an-9)</p> <p>Address (ans...20 Mastercard)</p> <p>Address (ans...20 Visa)</p> <p>Note: At a minimum the Postal Code must be provided and cannot be less than nine bytes long (Postal code left-justified and blank filled if necessary up to nine bytes).</p>                                                                                                                                                                                   |
| DE 120, subfield 2 (AVS Service Indicator 2)                         | •   | X   | C   | <p>Postal Code (an-9)</p> <p>Address (an-5)</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Data Element                                 | Org | Sys | Dst | Values/Comments                      |
|----------------------------------------------|-----|-----|-----|--------------------------------------|
| DE 120, subfield 3 (AVS Service Indicator 3) | •   | X   | C   | Postal Code (an-9)<br>Address (an-5) |
| DE 120, subfield 4 (AVS Service Indicator 4) | •   | X   | C   | Postal Code (an-9)<br>Address (an-5) |

## Authorization Request Response/0110: AVS and Authorization Request Response

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Applicable data elements

| Data Element                                                         | Org | Sys | Dst | Values/Comments                                                                                                         |
|----------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------|
| DE 4 (Transaction Amount)                                            | ME  | P   | ME  | Must be the same value as in the original Authorization Request/0100 message.                                           |
| DE 39 (Response Code)                                                | M   | P   | M   | Contains one of the response codes listed for this data element in the Data Element Definitions chapter of this manual. |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code) | ME  | P   | ME  | Must be the same value as in the original Authorization Request/0100 message.                                           |
| DE 48, subelement 82 (Address Verification Request)                  | ME  | P   | ME  | Must be the same value as in the original Authorization Request/0100 message.                                           |
| DE 48, subelement 83 (Address Verification Response)                 | M   | P   | M   | Contains one of the response codes listed for this data element in the Data Element Definitions chapter of this manual. |
| DE 120 (Record Data)                                                 | ME  | P   | ME  | Must be the same value as in the original Authorization Request/0100 message.                                           |



**NOTE: Acquirers always receive the original information in DE 120 of the Authorization Request Response/0110 (the non-condensed subfield 01 contents) that they provided in the Authorization Request/0100.**

## Network Management Request/0800: sign-on/sign-off

Following is a list of the data elements and values applicable to this message type. All mandatory Network Management Request/0800: AVS sign-on data elements apply.

### Applicable data elements

| Data Element                                                   | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 94 (Service Indicator), subfield 3 (Address Data Indicator) | M   | .   | M   | 0 = AVS not supported<br>1 = Issuer supports AVS, receives all the address data (non-condensed)<br>2 = Issuer supports AVS, receives condensed address data<br>3 = Issuer supports AVS, receives condensed address data<br>4 = Issuer supports AVS, receives condensed numeric postal code and condensed numeric address data only. |

## Alternate Processing

When the issuer is unavailable the transaction is processed by Stand-In Authorization or X-Code as follows.

### Authorization Request/0100: Request and Address Verification Service

| WHEN...                                                               | THEN the Dual Message Authorization System....                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Authorization Request/0100 contains DE 48, subelement 82 value 52 | sends an Authorization Request Response/0110 message containing DE 39, value (based on the decision for the authorization request portion of the transaction).<br><br>if the issuer supports AVS then DE 48, subelement 83 contains value R<br><br>if the issuer does not support AVS then DE 48, subelement 83 contains value S |

## DE 48 and DE 120 structure in AVS transactions

Following is the structure of DE 48 and DE 120 in AVS transactions.

### Data element structure

The following table illustrates DE 48, subelements 82 (Address Verification Service Request) option code and subelement 83 (Address Verification Service Response) result code in an AVS transaction.

|                                           |                                         |                  |                      |                  |                   |                      |                 |
|-------------------------------------------|-----------------------------------------|------------------|----------------------|------------------|-------------------|----------------------|-----------------|
| LLL                                       | VAR: 999 maximum bytes (TCC + AVS Data) |                  |                      |                  |                   |                      |                 |
| 3 bytes                                   | 1 byte                                  | 2 bytes          | 2 bytes              | 2 bytes          | 2 bytes           | 2 bytes              | 1 byte          |
| Total Data Element Length                 | TCC                                     | AVS Request Data |                      |                  | AVS Response Data |                      |                 |
|                                           |                                         | Subelement ID 82 | Subelement Length 02 | AVS Request Code | Subelement ID 83  | Subelement Length 01 | AVS Result Code |
| Mandatory                                 |                                         |                  |                      |                  |                   |                      |                 |
| 1002 maximum bytes (LLL + TCC + AVS Data) |                                         |                  |                      |                  |                   |                      |                 |

### Data element structure

The following table illustrates DE 120 contents for subfield 01 (AVS Service Indicator 1).

|                           |                                                                  |                          |                            |                                              |  |  |  |
|---------------------------|------------------------------------------------------------------|--------------------------|----------------------------|----------------------------------------------|--|--|--|
| LLL                       | VAR: 33 maximum bytes (Mastercard) – VAR–53 maximum bytes (Visa) |                          |                            |                                              |  |  |  |
| 3 bytes                   | 2 bytes                                                          | 2 bytes                  | 9 bytes                    | 1–20 bytes (Mastercard)<br>1–40 bytes (Visa) |  |  |  |
| Total Data Element Length | Subfield ID 01                                                   | Subfield Length Variable | Cardholder postal/ZIP code | Cardholder Address                           |  |  |  |

### Data element structure

The following table illustrates DE 120 contents for subfield 02–04 (AVS Service Indicator 2–4).

|                           |                              |                 |                            |                    |  |  |  |
|---------------------------|------------------------------|-----------------|----------------------------|--------------------|--|--|--|
| LLL                       | Fixed 18 bytes               |                 |                            |                    |  |  |  |
| 3 bytes                   | 2 bytes                      | 2 bytes         | 9 bytes                    | 5 bytes            |  |  |  |
| Total Data Element Length | Subfield ID 02, or 03, or 04 | Subfield Length | Cardholder postal/ZIP code | Cardholder Address |  |  |  |

### For more information

For details about data requirements, refer to [Data element definitions and details](#).

## Alternate Processing options

Stand-In Authorization and X- Code Processing are alternate processing options.

### Stand-In Authorization service

Stand-In Authorization helps issuers protect their revenue and minimize their risk, strengthening their cardholder relationships and safeguarding their brand reputation.

Stand-In Authorization is a Mastercard service that responds on behalf of the issuer to authorization requests that have been routed to the Mastercard Network when the issuer:

Does not respond

Cannot be reached

Provides an invalid response

Declines the transaction with response code 91, 92, or 96

Stand-In Authorization supports various brands, for example:

- Mastercard®, Maestro®, Cirrus®, and non-Mastercard brands
- Segments: consumer and commercial
- Products: credit and debit
- Channels: POS, CNP, and Quasi-Cash for transactions switched through the Mastercard Network.

As transactions are made and switched through the Mastercard Network, the network attempts to pass transactions to the Issuer. If the Issuer is unavailable or unresponsive, Stand-In Authorization steps in and decides the transaction on their behalf.

Stand-In Authorization is enabled, by default, for all authorization account ranges for Mastercard, Maestro and Cirrus card programs. Information on required use of Stand-In processing is available in the Transaction Processing Rules.

Issuers may list accounts in the Stand-In Account File as Negative (lost/stolen/fraud) or Premium Listings for valued/preferred cardholders. Issuers may list individual card numbers to identify exception accounts that are processed differently than regular accounts. They can list these accounts either with higher limits to ensure approval, such as the limits for Premium Listings or as Negative for lost/stolen/fraud or unpaid/over limit accounts that must be declined. If Stand-In Authorization determines that an account is listed on an exception file, Stand-In processing is modified. For a description of how to list accounts, refer to the [Account Management System User Manual](#).

Issuers must support store-and-forward (SAF) interleaving for online Stand-In advice messages.

Stand-In Authorization can assist issuers in meeting required performance standards. Failure to meet these standards could result in noncompliance assessments. Refer to the [Mastercard Rules](#) and [Transaction Processing Rules](#) for more information about standards and assessments.

## X-Code Processing

X-Code processing applies only when the acquirer is online. It can occur either at the acquirer host or acquirer MIP.

Acquirer host X-Code Processing occurs when the acquirer host cannot communicate with the acquirer MIP.

### Rules for acquirer host X-Code Processing

For acquirer host X-Code Processing, the acquirer must approve and bear liability for the transaction, after validating that the account is not listed as restricted on the Electronic Warning Bulletin file, when the transaction amount is less than or equal to USD 300, except in the circumstances listed below.

The issuer bears liability for authorization requests that it approves. The acquirer must phone the issuer for approval in any of the following instances:

- The transaction amount is more than USD 300.
- The card is not present.
- The merchant suspects that the card may be stolen or counterfeit, or the transaction or the customer causes the merchant to be suspicious.
- The transaction type is one of the following: PIN transactions, unique, mail order/telephone order, electronic commerce order, Payment Transaction, or non-financial transactions (such as balance inquiry, PIN management).

If the transaction amount is more than USD 300 and the acquirer cannot reach the issuer, the acquirer may do any of the following:

- Approve the transaction (and bear liability for the total amount)
- Decline the transaction
- Continue attempting to reach the issuer

If the acquirer has not phoned the issuer for approval, the acquirer can create an Authorization Advice/0120: Acquirer-generated message to notify the issuer of the approved authorization based on the previously stated rules for X-Code Processing.

Acquirer MIP X-Code Processing occurs when the acquirer MIP cannot communicate with the Mastercard Network or the acquirer MIP does not receive a response from the issuer or the Stand-In Authorization within the time frame established by Mastercard.

### MIP X-Code Processing rules

A subset of transactions is eligible for MIP X-Code Processing and may receive an approval. Transactions ineligible for MIP X-Code Processing are always declined.

## Transaction eligibility

| MIP X-Code authorization response | Type of transaction                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Decline                           | <p>Ineligible transaction types include:</p> <ul style="list-style-type: none"> <li>• Account Status Inquiry Service</li> <li>• ATM Installment Inquiry</li> <li>• Balance inquiries</li> <li>• Private Label transactions</li> <li>• Product Inquiry Service</li> <li>• Cash disbursement</li> <li>• Payment Transaction</li> <li>• Unique</li> </ul> |
| Do Not Honor                      | <p>Transactions that exceed the ceiling limits defined in the MIP X-Code Ceiling Limits table</p> <p>Transactions declined as a result of MIP X-Code risk assessment</p>                                                                                                                                                                               |
| Capture Card                      | Transactions for accounts that are listed on the Electronic Warning Bulletin file                                                                                                                                                                                                                                                                      |
| Approve                           | Transactions that are less than or equal to the MIP X-Code ceiling limits and do not meet any of the criteria listed above                                                                                                                                                                                                                             |

### MIP X-Code ceiling limits

Ceiling limits are assigned to each product category, and transactions with amounts above the ceiling limits are not approved.

### Ceiling limits

| Product Category                                                     | Maximum MIP X-Code Limits |
|----------------------------------------------------------------------|---------------------------|
| Transactions using Mastercard consumer credit products               | USD 950                   |
| Transactions using Mastercard consumer debit products                | USD 400                   |
| Transactions using Mastercard commercial small business products     | USD 950                   |
| Transactions using Mastercard commercial corporate products          | USD 950                   |
| Transactions using all other Mastercard, Maestro and Cirrus products | USD 0                     |

## Liability for X-Code transactions

The issuer is liable for transactions that are approved under acquirer MIP X-Code, up to the MIP X-Code ceiling limits specified.

## MIP X-Code risk assessment

For eligible transactions with amounts falling below the ceiling limits, X-Code Processing applies risk assessment measures to determine the authorization response. This includes efforts to decline transactions for customers who express a desire to prevent on-behalf approvals by Mastercard. These measures are enhanced on an ongoing basis.

# Automated Fuel Dispenser Completion

Acquirers use the Automated Fuel Dispenser (AFD) Completion messages to advise the issuer of the total amount of the Automated Fuel Dispenser (AFD) transaction.

An AFD transaction is identified with MCC 5542 and may occur at a cardholder-activated terminal (CAT) or as an e-commerce transaction.

The acquirer sends an Authorization Request/0100 message to verify a minimum availability of funds in the cardholder's account for an Automated Fuel Dispenser (AFD) transaction. If the request is not declined, the acquirer sends an Authorization Advice/0120: acquirer-generated message to the issuer, specifying the total amount of the AFD transaction. The value in DE 60, subfield 1 (Advice Reason Code) is 191 (Acquirer Processing System [APS] Completed Authorization Transaction).

The Dual Message Authorization System forwards all Authorization Advice/0120: acquirer-generated messages directly to the issuer. The issuer must respond with an Authorization Advice Response/0130: responding to an acquirer-generated 0120 message acknowledging receipt of the Authorization Advice/0120: acquirer-generated message. If the issuer is not available or does not respond within timer limits, the Authorization Advice/0120: acquirer-generated message will be sent to the Stand-In Authorization. The Stand-In Authorization responds with an Authorization Advice Response/0130: system-generated message to the acquirer to acknowledge receipt of the Authorization Advice/0120: acquirer-generated message. The advice message will be added to the SAF for later distribution to the issuer.

**NOTE: The Data Integrity Automated Fuel Dispenser (AFD) edit for acquirers of AFD merchants that are located in the U.S. region is described in *Data Integrity Monitoring Program*.**

## AFD message scenarios

Mastercard supports the following usage scenarios for AFD authorization messages.

### AFD message scenarios

| WHEN the AFD Authorization Request Response/0110 contains...                                                                    | THEN the AFD acquirer...                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code), value 00 (Approved or completed successfully)                                                            | submits the Authorization Advice/0120: acquirer-generated message containing the final fuel sale amount.                                                           |
| DE 39, value 10 (Partial Approval)                                                                                              | submits the Authorization Advice/0120: acquirer-generated message containing the final fuel sale amount that is less than or equal to the partial approval amount. |
| DE 39, value 00 (Approved or completed successfully) or value 10 (Partial Approval) and the transaction is canceled at the pump | submits a Reversal Request/0400 message, or an Authorization Advice/0120 message with a DE 04 zero amount, to cancel the AFD transaction.                          |

**NOTE: The AFD Authorization Advice/0120: acquirer-generated message is not a replacement for the First Presentment/1240 message to the clearing system on a credit transaction.**

## Authorization Request/0100: Automated Fuel Dispenser Preauthorization

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                       | Org | Sys | Dst | Values/Comments                                                                                                    |
|--------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------|
| DE 4 (Amount, Transaction)                                         | M   | •   | M   | Must be at least an amount no less than the equivalent of USD 1.                                                   |
| DE 18 (Merchant Type)                                              | M   | •   | M   | 5542 = Fuel Dispenser, Automated                                                                                   |
| DE 61 (Point of Service Data), subfield 7 (POS Transaction Status) | M   | •   | M   | May contain one of the following values:<br>0 = Normal request (original presentment)<br>4 = Preauthorized request |

| Data Element                                             | Org | Sys | Dst | Values/Comments                                                                              |
|----------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------|
| DE 61, subfield 10 (Cardholder Activated Terminal Level) | M   | •   | M   | 2 = Authorized Level 2 CAT: Self-service terminal, or 1 = Authorized Level 1 CAT if PIN used |

## Authorization Advice/0120: acquirer-generated (Automated Fuel Dispenser Completion)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

### Applicable data elements

| Data Element                                                                                                                    | Org | Sys | Dst | Values/Comments                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 4 (Amount, Transaction)                                                                                                      | M   | •   | M   | Contains the completed transaction amount and not the requested amount.                                                                          |
| DE 18 (Merchant Type)                                                                                                           | M   | •   | M   | 5542 = Fuel Dispenser, Automated                                                                                                                 |
| DE 39 (Response Code)                                                                                                           | M   | •   | M   | Valid values:<br><br>00 = Approved or completed successfully<br><br>10 = Partial Approval                                                        |
| DE 48 (Additional Data: Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time), subfield 1 (Date) | •   | X   | M   | Contains the valid date of the Authorization Advice/0120: acquirer-generated inserted by the Dual Message Authorization System in MMDD format.   |
| DE 48, subelement 15, subfield 2 (Time)                                                                                         | •   | X   | M   | Contains the valid time of the Authorization Advice/0120: acquirer-generated inserted by the Dual Message Authorization System in hhmmss format. |
| DE 48, subelement 20 (Cardholder Verification Method)                                                                           | M   | X   | •   |                                                                                                                                                  |
| DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)                                                                     | M   | •   | M   | 191 = Acquirer Processing System (APS) Completed Authorization Transaction                                                                       |



**NOTE: Acquirers must not submit DE 48, subelement 82 (Address Verification Service Request) or subelement 92 (CVC 2) as these services are not performed on the Authorization Advice/0120: acquirer-generated message and will cause the 0120 message to be rejected by the Dual Message Authorization System.**

## Authorization Advice/0120: acquirer-generated (Automated Fuel Dispenser Completion)

Customers should be aware of the critical requirements for proper processing of acceptor business code (MCC) 5542 (Fuel Dispenser, Automated) transactions.

### Critical AFD Advice Message Data

The following information provides a summary of the critical acquirer requirements for processing AFD transactions:

- Authorization Advice/0120 (Automated Fuel Dispenser Completion) messages must contain DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code), value 191 (completed authorization) for AFD transactions with an original Authorization Request/0100 message.
- The following Authorization Advice/0120 message data elements must match the value submitted within the original Authorization Request/0100 message for issuer transaction matching purposes:
  - DE 2 (Primary Account Number)
  - DE 7 (Transmission Date and Time)
  - DE 11 (System Trace Audit Number [STAN])
  - DE 32 (Acquiring Institution ID Code)
  - DE 33 (Forwarding Institution ID Code), if present in the Authorization Request/0100 message
  - DE 38 (Authorization ID Response) and DE 39 (Response Code) with the same value as received in the original Authorization Request Response/0110 message
  - DE 48 (Additional Data), subelement 63 (Trace ID) to further assist issuers in matching an AFD completion advice to the original preauthorization
  - DE 48 (Additional Data), subelement 98 (Mastercard Corporate Fleet Card® ID/Driver Number) and/or subelement 99 (Mastercard Corporate Fleet Card® Vehicle Number), with the same values as submitted in the original Authorization Request/0100 message, if present
  - DE 121 (Authorizing Agent ID Code), with same value as was received in the original Authorization Request Response/0110 message, if present

### Track Data in AFD Advice Message: acquirers

The Authorization Advice/0120 message layout shows DE 35 (Track 2 Data) and DE 45 (Track 1 Data) as optional. Acquirers of AFD merchants are reminded that the presence of track data in the Authorization Request/0100 message for a card-present transaction does not necessitate inclusion within the AFD Advice message. This data is not needed by issuers for matching an

AFD advice to an original authorization, and storage of track data for submission of the AFD Advice may not be PCI compliant.

**Track Data in AFD Advice Message: issuers**

The Authorization Advice/0120: acquirer-generated message for an AFD transaction contains the completion amount and other reference data from the original Authorization Request/0100 message data. Issuers are reminded that DE 35 and DE 45 track data is optional in an AFD Advice message and may not be present, regardless of the Point-of-Service (POS) entry mode value in DE 22. As such, the absence or presence of track data in the AFD Advice message should not result in a format error from issuers for card-present fuel purchases.

For details about data requirements, refer to [Data element definitions and details](#) section.

**Authorization Advice Response/0130: Issuer-generated (responding to an acquirer-generated 0120)**

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) data elements apply.

**Applicable data elements**

| Data Element                                                                                                                    | Org | Sys | Dst | Values/Comments                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data: Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time), subfield 1 (Date) | ME  | .   | ME  | This should match what was received by the issuer in the Authorization Advice/0120 message, which was provided by the Dual Message Authorization System. |
| DE 48 (Additional Data: Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time), subfield 2 (Time) | ME  | .   | ME  | This should match what was received by the issuer in the Authorization Advice/0120 message, which was provided by the Dual Message Authorization System. |

**Alternate Processing**

Alternate processing is provided if an issuer does not respond or is unable or unavailable to receive the Authorization Advice/0120: acquirer-generated AFD completion messages.

The Dual Message Authorization System responds to the acquirer with an Authorization Advice Response/0130 message containing DE 39 (Response Code), value 00 (Approved or completed successfully) and sends the Authorization Advice/0120: acquirer-generated message to SAF. SAF messages will be sent to the issuer when the issuer is available. DE 60 (Advice Reason

Code), subfield 1 (Advice Reason Code) keeps value 191 (Acquirer Processing System [APS] Completed Authorization Transaction) when delivered from SAF.

## Clearing AFD Transactions

Mastercard recommends that the Trace ID (DE 15 [Settlement Date] and DE 63 [Network Data]) of the completion advice response be used within the clearing presentment.

As an alternative, if the Trace ID of the original preauthorization is used for clearing presentment, that same Trace ID must be included in the AFD completion advice message within DE 48, subelement 63 (Additional Data, Trace ID).

## ATM Bill Payment service

Bill Payment transactions at the ATM help increase the acquirer's transaction volume by providing a convenient method for cardholders to initiate bill pay request transactions.

ATM Bill Payment service supports:

- Non-Europe acquired ATM Bill Payment transactions
- Europe-acquired ATM Bill Payment transactions

### Non-Europe acquired ATM Bill Payment processing

The ATM Bill Payment service allows cardholders to use their Mastercard® credit card or card at any ATM to pay utilities and other bills when the service is supported domestically.

Stages describing authorization processing when a cardholder requests an ATM Bill Payment transaction:

1. The service is initiated by the cardholder at an ATM with a need to pay a bill (for example, electric bill, cable bill, phone bill).
2. The cardholder inputs biller account information, including the ATM bill pay code to identify the biller company, and then submits the bill pay request.
3. The ATM acquirer submits the payment request as a POS transaction to the Single Message System in a Financial Transaction Request/0200 message containing:
  - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase)
  - DE 18 (Merchant Type) with a MCC data value 6539 (Funding Transaction [Excluding Mastercard Send]). MCC 6539 is limited for use only in certain countries where the service is supported domestically.
  - DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level Indicator), with one of the following values:
    - 1 (Authorized Level 1 CAT: automated dispensing machine with PIN)
    - 2 (Authorized Level 2 CAT: self-service terminal)
  - DE 124 (Member Defined Data) contains details relating to the bill being paid (optional)

4. The Single Message System converts the Financial Transaction Request/0200 message, and then forwards an Authorization Request/0100 message via the Mastercard Network to the issuer.
5. The issuer sends the Authorization Request Response/0110 message back to the Single Message System via the Mastercard Network.
6. The Single Message System converts the Authorization Request Response/0110 message, and then sends a Financial Transaction Request Response/0210 message to the ATM acquirer.
7. The acquirer provides a receipt to the cardholder that contains the bill payment information (such as amount paid, date paid, and ATM bill pay code).
8. The acquirer initiates the bill paying process, paying funds directly to the biller (such as electric company, phone company).
9. The cardholder sees the bill payment transaction in his or her monthly bankcard statement.

Non-Europe acquired ATM Bill Payment transactions are not processed by the Stand-In Authorization or X-Code Processing. Transactions are declined using the Authorization Request Response/0110 message containing DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

### **Europe-acquired ATM Bill Payment processing**

The ATM Bill Payment service allows cardholders to use their Mastercard®, Cirrus®, or Maestro® card at any ATM to pay utilities and other bills where the service is supported domestically for the Europe region.

Stages describing authorization processing when a cardholder requests an ATM Bill Payment transactions:

1. The service is initiated by the cardholder at an ATM with a need to pay a bill (for example, electric bill, cable bill, phone bill).
2. The cardholder inputs biller account information, including the ATM bill pay code to identify the biller company, and then submits the bill pay request.
3. The ATM acquirer submits the payment request as a POS transaction to Authorization Request/0100 message containing:
  - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase)
  - DE 18 (Merchant Type) with MCC data value 6050 (Quasi-Cash: Member Financial Institution) or a more precise MCC related to the nature of the bill that is being paid. For example, MCC 4900 (Utilities: Electric, Gas, Heating Oil, Sanitary, Water) for utilities bills. MCC 6011 (Member Financial Institution: Automated Cash Disbursements) must not be used.
  - DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level Indicator), value 1 (Authorized Level 1 CAT: automated dispensing machine with PIN)
  - DE 124 (Member Defined Data) contains details relating to the bill being paid (optional)
4. The Dual Message Authorization System forwards the Authorization Request/0100 message to the issuer.
5. The issuer sends the Authorization Request Response/0110 message to the acquirer.

6. The acquirer provides a receipt to the cardholder that contains the bill payment information (such as amount paid, date paid, ATM bill pay code).
7. The acquirer initiates the bill paying process, paying funds directly to the biller (such as electric company, phone company).
8. The cardholder sees the bill payment transaction in his or her monthly bankcard statement.

### To participate

Contact your regional office for information about implementation of the ATM Bill Payment service for your local market.

## Authorization Request/0100: ATM Bill Payment, Europe Acquired

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                                                     | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)                            | M   | •   | M   | 00 = Purchase                                                                                                                                                                                                                                                                                                                                 |
| DE 18 (Merchant Type)                                                                            | M   | •   | M   | Contains value 6050 = Quasi Cash: Member Financial Institution<br><br>or a more precise MCC related to the nature of the bill that is being paid, for example, MCC 4900 (Utilities: Electric, Gas, Heating Oil, Sanitary, Water) for utilities bills. MCC 6011 (Member Financial Institution: Automated Cash Disbursements) must not be used. |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                             | M   | •   | M   | U = Unique                                                                                                                                                                                                                                                                                                                                    |
| DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level Indicator) | M   | •   | M   | 0 = Not a CAT transaction<br><br>1 = Authorized Level 1 CAT: automated dispensing machine with PIN                                                                                                                                                                                                                                            |
| DE 124 (Member-defined Data)                                                                     | O   | •   | C   | Contains details relating to the bill being paid.                                                                                                                                                                                                                                                                                             |

## Authorization Request/0100: ATM Bill Payment, Non-Europe Acquired

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                                                     | Org | Sys | Dst | Values/Comments                                                                                                                                                |
|--------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)                            | M   | •   | M   | 00 = Purchase                                                                                                                                                  |
| DE 18 (Merchant Type)                                                                            | M   | •   | M   | Must contain value 6539 = Funding Transaction, Excluding Mastercard Send                                                                                       |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                             | M   | •   | M   | U = Unique                                                                                                                                                     |
| DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level Indicator) | M   | •   | M   | One of the following values:<br><br>1 = Authorized Level 1 CAT: automated dispensing machine with PIN<br><br>2 = Authorized Level 1 CAT: self-service terminal |
| DE 124 (Member-defined Data)                                                                     | O   | •   | C   | Contains details relating to the bill being paid.                                                                                                              |

## Dual Message Authorization System Edits

The Dual Message Authorization System applies the following edits on ATM Bill Payment Service transactions.

### Applicable data elements

| WHEN the acquirer...                                                                                                                                                                           | THEN the Dual Message Authorization System...                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sends an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message containing DE 18 (Merchant Type), value 6539 (Funding Transaction, Excluding Mastercard Send) | sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message with DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal). |

## ATM Credit Card Cash Advance in Installments

This service allows cardholders to initiate an inquiry at the ATM requesting a credit card cash advance to be repaid in monthly installments (for example, 6, 12, or 18 months).

With this enhancement, a cardholder can initiate an inquiry at the ATM requesting a credit card cash advance to be repaid in monthly installments (for example, 6, 12, or 18 months). The issuer can accept the installment terms requested by the cardholder, decline the request, or decline the request and offer an alternate installment schedule. The cardholder has the option to accept or reject the issuer's terms and conditions for the installment payments. If the cardholder accepts the issuer's terms and conditions for repayment, the issuer will then approve or decline the cash advance transaction. If approved, the cardholder will receive the funds, along with the installment payment details printed on the ATM receipt. The issuer will bill the cardholder for the amount of the transaction in the agreed-upon installments.

Acquirers entering this market must be able to:

- Provide ATM screens that offer an installment payment option.
- Provide the ability to print the issuer's installment payment details on the ATM receipt.

The ATM Credit Card Cash Advance in Installments service is currently available to single message transactions (Single Message System-acquired activity).

The ATM Credit Card Cash Advance in Installments service is not available to ATM acquirers in the Europe region.

## Authorization Request/0100: ATM Installment Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                             | Org | Sys | Dst | Values/Comments                                                                                                        |
|--------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)    | M   | •   | M   | 01 (Withdrawal)                                                                                                        |
| DE 4 (Amount, Transaction)                                               | M   | •   | M   | Contains the amount of the cash advance                                                                                |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) | M   | •   | M   | Identifies that the cardholder has initiated an ATM installment inquiry transaction<br><br>3 = ATM Installment Inquiry |

| Data Element                                                                                                      | Org | Sys | Dst | Values/Comments                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data)                                | C   | •   | C   | Identifies the transaction type.<br><br>80xx = ATM Installment Inquiry, Number of Installments<br><br>81xx = ATM Installment Withdrawal, Number of Installments |
| DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type)                 | C   | •   | C   | 80 = ATM Installment Inquiry                                                                                                                                    |
| DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 2 (Requested Number of Installments) | C   |     | C   | 01–99 = the number of installments requested by the cardholder                                                                                                  |

## Authorization Request Response/0110: ATM Installment Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Applicable data elements

| Data Element                                                                                      | Org | Sys | Dst | Values/Comments                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)                             | ME  | •   | ME  | 01 (Withdrawal)                                                                                                                                                 |
| DE 4 (Amount, Transaction)                                                                        | CE  | X   | CE  | Contains the amount of the cash advance                                                                                                                         |
| DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data)                | C   | •   | C   | Identifies the transaction type.<br><br>80xx = ATM Installment Inquiry, Number of Installments<br><br>81xx = ATM Installment Withdrawal, Number of Installments |
| DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type) | C   | •   | C   | 80 = ATM Installment Inquiry                                                                                                                                    |



| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                        |
|-----------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------|
| DE 112, subelement 027, subfield 2 (Requested Number of Installments) | C   | •   | C   | 01–99 = the number of installments requested by the cardholder         |
| DE 112, subelement 027, subfield 3 (Approved Number of Installments)  | C   | •   | C   | The number of installment payments approved by the issuer              |
| DE 112, subelement 027, subfield 4 (Installment Amount)               | C   | •   | C   | The monthly payment amount                                             |
| DE 112, subelement 027, subfield 5 (Total Transaction Amount)         | C   | •   | C   | The amount of the cash advance and the interest for the transaction    |
| DE 112, subelement 027, subfield 6 (Yearly Interest Rate)             | C   | •   | C   | The interest rate that will be charged to the cardholder by the issuer |
| DE 112, subelement 027, subfield 7 (Currency Code)                    | C   | •   | C   | The currency code the issuer is charging the cardholder for repayment  |
| DE 112, subelement 027, subfields 8–11 (Member-defined Data)          | C   | •   | C   | Member-defined data                                                    |

## Authorization Request/0100: ATM Installment Withdrawal

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                                       | Org | Sys | Dst | Values/Comments                                                                                                                       |
|------------------------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)              | M   | •   | M   | 01 = Withdrawal                                                                                                                       |
| DE 4 (Amount, Transaction)                                                         | M   | •   | M   | Contains the amount of the cash advance.                                                                                              |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)           | M   | •   | M   | Identifies that the cardholder has initiated an ATM installment inquiry transaction.<br><br>0 = Normal request (original presentment) |
| DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data) | C   | •   | C   | Identifies the transaction type.<br><br>81xx = ATM Installment Withdrawal, Number of Installments                                     |

| Data Element                                                                                                      | Org | Sys | Dst | Values/Comments                                                |
|-------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------|
| DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type)                 | C   | •   | C   | 81 = ATM Installment Withdrawal                                |
| DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 2 (Requested Number of Installments) | C   | •   | C   | 01–99 = The number of installments requested by the cardholder |

## Authorization Request Response/0110: ATM Installment Withdrawal

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Applicable data elements

| Data Element                                                                                      | Org | Sys | Dst | Values/Comments                                                                               |
|---------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)                             | ME  | •   | ME  | 01 = Withdrawal                                                                               |
| DE 4 (Amount, Transaction)                                                                        | CE  | X   | CE  | Contains the amount of the cash advance.                                                      |
| DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data)                | C   | •   | C   | Identifies the transaction type.<br>81xx = ATM Installment Withdrawal, Number of Installments |
| DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type) | C   | •   | C   | 81 = ATM Installment Withdrawal                                                               |
| DE 112, subelement 027, subfield 2 (Requested Number of Installments)                             | C   | •   | C   | 01–99 = The number of installments requested by the cardholder.                               |
| DE 112, subelement 027, subfield 3 (Approved Number of Installments)                              | C   | •   | C   | 01–99 = The number of installment payments approved by the issuer.                            |
| DE 112, subelement 027, subfield 4 (Installment Amount)                                           | C   | •   | C   | The cardholder's monthly payment amount.                                                      |

| Data Element                                                     | Org | Sys | Dst | Values/Comments                                                         |
|------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------|
| DE 112, subelement 027, subfield 5<br>(Total Transaction Amount) | C   | •   | C   | The amount of the cash advance and the interest for the transaction.    |
| DE 112, subelement 027, subfield 6<br>(Yearly Interest Rate)     | C   | •   | C   | The interest rate that will be charged to the cardholder by the issuer. |
| DE 112, subelement 027, subfield 7<br>(Currency Code)            | C   | •   | C   | The currency code the issuer is charging the cardholder for repayment.  |
| DE 112, subelement 027, subfields 8–11                           | C   | •   | C   | Member-defined data.                                                    |

## Authorization and Preauthorization Processing Standards

All authorizations initiated by acceptors in the Asia/Pacific, Europe, and Middle East/Africa regions must clearly be distinguished as a final authorization or a preauthorization. Authorizations initiated by acceptors in all other regions must be distinguished as a preauthorization, final authorization, or undefined authorization.

Undefined authorizations will no longer be permitted in the U. S. region effective 17 June 2025, Latin America and the Caribbean region on 18 August 2025 and the Canada region 17 September 2025.

For additional detail on Authorization and Preauthorization Processing Standards, refer to the [Transaction Processing Rules](#).

### Authorization and preauthorization processing

Clear and reliable identification of authorizations provides better information about the nature of each authorization and allows issuers to optionally apply different processing to different types of authorizations, ultimately benefiting cardholders.

Final authorization processing is mandated for transactions initiated at acceptors in the Europe region that enables issuers globally to distinguish a final authorization from a preauthorization.

**NOTE:** Refer to the [Scenarios](#) section for detailed scenarios for preauthorization, final authorization, undefined authorization, and chargeback extension request.

## Preauthorization, final authorization, and undefined authorization transactions

An authorization for an amount greater than zero must be identified as either preauthorization, final authorization, or undefined authorization, using the requirements listed here.

### Preauthorization

An authorization for an amount greater than zero should be coded as a preauthorization if it meets one of the following three conditions:

- Authorization is requested for an estimated amount.
- The requested amount of the original authorization may be adjusted if the final Transaction Amount (amount submitted for clearing) is greater than the original amount requested.
- The transaction might not be completed for reasons other than technical failure or lack of full issuer approval. For example, the transaction might not be completed when the cardholder will be offered the choice to complete the transaction with another payment means at a later time (such as, when checking out of a hotel or returning a rental car) or when the goods ordered by the cardholder might be later found to be out of stock. The risk of technical failures such as telecommunications failure or terminal failure should not be taken into account to determine if an authorization must be coded as a preauthorization.

The authorization request is properly identified as a preauthorization when DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains value 4. Acquirers must submit the clearing presentment for a transaction approved as a preauthorization within 30 calendar days of the latest authorization date (preauthorization or incremental authorization) in order to ensure chargeback protection provided by chargeback message reason code 4808: Authorization-Related Chargeback.

Acquirers in the Asia Pacific, Canada, Latin America and Caribbean, Middle East/Africa, and United States regions that do not reverse or clear approved preauthorizations within 30 calendar days of the original preauthorization date will be considered non-compliant with preauthorization processing standards. Non-compliant transactions will be reported on the Authorization Processing Integrity Acquirer Detail report. After the monitoring and reporting period ends, non-compliant preauthorizations will be subject to a non-compliance fee.

### Preauthorization scenarios

The following scenarios are examples in which acceptors can use a preauthorization to benefit from its greater processing flexibility or from its more generous payment guarantee terms.

**NOTE:** Refer to the [Scenarios](#) section for detailed scenarios for preauthorization, final authorization, undefined authorization, and chargeback extension request.

- A hotel wants to obtain a payment guarantee at check-in for the room rate, plus a provision for possible additional expenses related to the minibar or breakfast: the hotel uses a preauthorization because it supports estimated amounts.

- Another hotel wants to obtain a payment guarantee at check-in for a hotel stay of 10 days: the hotel uses a preauthorization because the associated payment guarantee is valid for up to 30 days and the clearing presentment will take longer than seven calendar days.
- An airline wants to obtain a payment guarantee at flight-booking time for certain types of flights where the ticket may take several days to issue: the airline uses a preauthorization because the associated payment guarantee is valid for up to 30 days and the clearing presentment may take longer than seven calendar days.
- An e-commerce merchant wants to obtain a payment guarantee at order time, before it can confirm that the goods are actually available in stock: the merchant uses a preauthorization because the transaction may still be canceled if the goods are later found to be out of stock.
- Another e-commerce merchant wants to obtain a payment guarantee at order time but the items ordered may take several days to ship: the merchant uses a preauthorization because the associated payment guarantee is valid for up to 30 days and the clearing presentment may take longer than seven calendar days.

**NOTE: Maestro preauthorizations are permitted only for automated fuel dispenser (AFD), Maestro contactless transit aggregated, and card-not-present (CNP) transactions.**

### Final authorization

An authorization for an amount greater than zero should be coded as a final authorization if it meets the following conditions:

- Authorization is requested for the final transaction amount that is expected to be presented for clearing.
- The transaction is not expected to be canceled after the authorization request is approved in full (excluding non-completion for reasons of technical failure). The risk of technical failures such as telecommunications failure or terminal failure should not be taken into account to determine if an authorization must be coded as a final authorization.

Any transaction corresponding to an authorization identified as a final authorization must be presented for clearing within seven calendar days of the authorization approval date.

The authorization request is properly identified as a final authorization when DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains value 0 (Normal Request) and DE 48 (Additional Data: Private Use), subelement 61 (POS Data Extended Condition Codes), subfield 5 (Final Authorization Indicator) contains value 1 (Final Authorization).

Acquirers must submit the clearing presentment for a transaction approved as a final authorization within seven calendar days of the authorization date in order to ensure chargeback protection provided by chargeback message reason code 4808: Authorization-related Chargeback.

Acquirers in Europe or Middle East/Africa regions that reverse an approved final authorization or do not clear an approved final authorization within seven calendar days of the authorization date or clear for a different currency or transaction amount than what was authorized will be considered non-compliant with final authorization processing standards. Acquirers in Asia/Pacific, Canada, Latin America and the Caribbean, or United States regions that do not clear an approved final authorization within seven calendar days of the authorization date or clear for a

different currency or transaction amount than what was authorized will be considered non-compliant with final authorization processing standards. Non-compliant transactions will be reported on the Authorization Processing Integrity Acquirer Detail report. Non-compliant final authorizations will be subject to a non-compliance fee.

In the Europe and Middle East/Africa regions, non-compliant final authorizations will be subject to a non-compliance fee if more than one percent of Final Authorizations do not meet their requirements for a given acquirer parent ICA or MCE Group ICA over the weekly billing period. If the one percent threshold is not breached, all Final Authorizations that do not meet their requirements will not be assessed a fee.

**NOTE: Acquirers outside of the Europe and Middle East/Africa regions may optionally submit final authorizations, but must comply with requirements of clearing for the amount and currency authorized, and within seven calendar days if they choose to do so.**

### Undefined authorization

An authorization for an amount greater than zero should be coded as an undefined authorization if it meets the following conditions:

- The final transaction amount may differ from the authorized amount.
- The transaction is not expected to be canceled after the authorization request is approved in full (excluding non-completion for reasons of technical failure). The risk of technical failures such as telecommunications failure or terminal failure should not be taken into account to determine authorization message coding.

The authorization request is properly identified as an undefined authorization when DE 61, subfield 7 contains value 0, and DE 48, subelement 61, subfield 5 contains value 0, or is not present. Acquirers must submit the clearing presentment for a transaction approved as an undefined authorization within seven calendar days of the authorization date in order to ensure chargeback protection provided by chargeback message reason code 4808: Authorization-Related Chargeback.

An approved undefined authorization submitted by an acquirer in the Canada, Latin America and the Caribbean, or United States region that is not reversed or cleared within seven calendar days of the original authorization date will be considered non-compliant with undefined authorization processing standards. Non-compliant transactions will be reported on the Authorization Processing Integrity Acquirer Detail report and the non-compliant undefined authorizations will be subject to a non-compliance fee.

Acquirers in the Europe, Asia/Pacific, and Middle East/Africa regions must no longer code authorizations as undefined. An authorization must be coded either as preauthorization or final authorization. Non-compliant transactions will be reported on the Authorization Processing Integrity Acquirer Detail report and the non-compliant undefined authorizations will be subject to a non-compliance fee.

Undefined authorizations will no longer be permitted in the U. S. region effective 17 June 2025, Latin America and the Caribbean region on 18 August 2025 and the Canada region 17 August 2025.

## Scenarios

The following are common processing scenarios for preauthorizations, final authorizations, and undefined authorizations assuming reversal or clearing occurs prior to the authorization message chargeback expiration date.

**NOTE: For each authorization type, if the authorization completes after the chargeback expiration date and a chargeback extension has not been requested and approved, the acquirer must either submit a new authorization request for the completed transaction amount prior to clearing or submit the transaction for clearing without processing a new authorization and risk potential chargeback for Reason 4808 (Authorization-related Chargeback). A sample scenario has also been provided for chargeback extension request message processing.**

### Preauthorization

The cardholder checks into a hotel on 1 October, and the merchant submits a preauthorization for estimated charges up to USD 1,000. The cardholder checks out of the hotel on 5 October with no further charges applied to his/her bill during the stay. The acquirer submits a clearing presentment based on notification from the merchant of preauthorization completion for the full authorized amount.

#### Scenario P1: clearing amount matches preauthorization amount

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct  | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Clear<br>1240<br>5 Oct |                                      |                  |                                     |                                                        |                        | USD<br>1,000         | MCG123<br>456<br>1001               | USD 0        |

The cardholder checks into a hotel on 1 October, and the merchant submits a preauthorization for estimated charges up to USD 1,000. The cardholder checks out of the hotel one day early. Final bill amount upon cardholder checkout is USD 900.

P2.A: The acquirer submits a clearing presentment within 24 hours of notification from the merchant of preauthorization completion.

**Scenario P2: clearing amount below preauthorization amount, P2.A**

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/MTI/<br>Date    | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct  | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Clear<br>1240<br>4 Oct |                                      |                  |                                     |                                                        |                        | USD 900              | MCG123<br>456<br>1001               | USD 0        |

P2.B: The acquirer submits partial reversal when notified by the merchant of preauthorization completion for lower than authorized amount, and clearing will not occur within 24 hours of merchant preauthorization completion notification.

**Scenario P2: clearing amount below preauthorization amount, P2.B**

| Merchant/<br>Acquirer | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|-----------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date  | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |



| Merchant/<br>Acquirer  | Authorizati<br>on | Clearing  | Issuer           |                       |                                  |         |
|------------------------|-------------------|-----------|------------------|-----------------------|----------------------------------|---------|
| RvrsI<br>0400<br>4 Oct | 4<br>NA           | USD 1,000 | 1004<br>MCG56789 | MCG123<br>456<br>1001 | USD 900                          | USD 900 |
| Clear<br>1240<br>6 Oct |                   |           |                  |                       | USD 900<br>MCG123<br>456<br>1001 | USD 0   |

The cardholder checks into a hotel on 1 October, and the merchant submits a preauthorization for estimated charges up to USD 1,000. The cardholder accrues additional USD 500 in charges. The final bill amount upon cardholder checkout is USD 1,500. The merchant submits an incremental preauthorization for the additional charges of USD 500 and notifies the acquirer of preauthorization completion of USD 1,500. The acquirer submits a clearing presentment based on notification from the merchant of preauthorization completion.

### Scenario P3: preauthorization completes for higher amount than authorized (incremental preauthorization)

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>RvrsI<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>RvrsI<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct  | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Incr.<br>0100<br>5 Oct | 4<br>NA                              | USD 500          | 1005<br>MCG56789<br>0               | MCG123<br>456<br>1001                                  |                        |                      |                                     | USD<br>1,500 |
| Clear<br>1240<br>5 Oct |                                      |                  |                                     |                                                        |                        | USD<br>1,500         | MCG123<br>456<br>1001               | USD 0        |

The cardholder checks into a hotel on 1 October, and the merchant submits a preauthorization for estimated charges up to USD 1,000. At checkout, the cardholder pays the final bill with a different card.

The merchant notifies the acquirer of the full reversal. The acquirer submits the reversal upon the merchant preauthorization cancellation notification.

#### Scenario P4: preauthorization cancellation

| Merchant/<br>Acquirer            | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|----------------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date             | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>Card 1<br>0100<br>1 Oct  | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Rvrsl<br>Card 1<br>0400<br>5 Oct | 4<br>NA                              | USD 1,000        | 1004<br>MCG56789<br>0               | MCG123<br>456<br>1001                                  |                        |                      |                                     | USD 0        |
| Auth<br>Card 2<br>0100<br>5 Oct  | 0<br>1<br>(Final Auth)               | USD 1,000        | 1005<br>MCP65432<br>1               |                                                        |                        |                      |                                     | USD<br>1,000 |

P5.A: Installment payment terms are approved with original preauthorization (that is, at time of reservation or check-in). Installment payment terms do not change at time of transaction completion. The cardholder checks into a hotel on 1 October, and the merchant submits a preauthorization for estimated charges up to USD 1,000. A preauthorization request also includes merchant installment program ID and installment payment terms in DE 48, subelement 95 and DE 112 for the cardholder to pay in five monthly payments of USD 200, which are approved by the issuer. The cardholder checks out of the hotel on 5 October with no further charges applied to his/her bill during the stay and original installment payment terms remain unchanged from original approved authorization. The acquirer submits a clearing presentment

for first installment payment based on notification from the merchant of preauthorization completion for the full authorized amount.

### Scenario P5: preauthorization with installment payments, P5.A

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct  | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Clear<br>1240<br>5 Oct |                                      |                  |                                     |                                                        |                        | USD 200              | MCG123<br>456<br>1001               | USD 0        |

P5.B: Installment payment terms are approved with original preauthorization (that is, at time of reservation or check-in). The terms change at the time of transaction completion.

The cardholder checks into a hotel on 1 October, and the merchant submits a preauthorization for estimated charges up to USD 1,000. Preauthorization also includes merchant installment program ID and terms in DE 48, subelement 95 and DE 112 for the cardholder to pay in five monthly payments of USD 200, which are approved by the issuer. The cardholder checks out of the hotel on 5 October with no further charges applied to his/her bill during the stay, but the original installment payment terms change to four monthly payments for USD 250. Due to the installment payment term change, the original authorization with the original installment payment terms must first be reversed, and a new authorization processed with the new terms for approval. Once the new authorization is approved, clearing can occur for the first installment payment amount.

**Scenario P5: preauthorization with installment payments, P5.B**

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct  | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Rvrsl<br>0400<br>5 Oct | 4<br>NA                              | USD 1,000        | 1004<br>MCG56789<br>0               | MCG123<br>456<br>1001                                  |                        |                      |                                     | USD 0        |
| Auth<br>0100<br>5 Oct  | 0<br>0<br>(Undef<br>Auth)            | USD 1,000        | 1005<br>MCP65432<br>1               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Clear<br>1240<br>5 Oct |                                      |                  |                                     |                                                        |                        | USD 250              | MCP654<br>321<br>1005               | USD 750      |

P5.C: Installment payment terms are not included with original preauthorization (that is, at time of reservation or check-in). Installment payment terms are added at the time of transaction completion.

The cardholder checks into a hotel on 1 October, and the merchant submits a preauthorization for estimated charges up to USD 1,000. The option to pay in installments is not offered until cardholder checkout. The cardholder checks out of the hotel on 5 October with no further charges applied to his/her bill during the stay. During the checkout process, the cardholder is offered the option to pay in four monthly installment payment of USD 250 each. The original authorization on 1 October must first be reversed, and a new authorization that includes the installment payment terms must be submitted with the installment program ID and terms included in DE 48, subelement 95 and DE 112 for approval. Once the new authorization is approved, clearing can occur for the first installment payment amount.

**Scenario P5: preauthorization with installment payments, P5.C**

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                                  |                                     |              |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4             | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct  | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                                  |                                     | USD<br>1,000 |
| Rvrsl<br>0400<br>5 Oct | 4<br>NA                              | USD 1,000        | 1004<br>MCG56789<br>0               | MCG123<br>456<br>1001                                  |                        |                                  |                                     | USD 0        |
| Auth<br>0100<br>5 Oct  | 0<br>0<br>(Undef<br>Auth)            | USD 1,000        | 1005<br>MCP65432<br>1               |                                                        |                        |                                  |                                     | USD<br>1,000 |
| Clear<br>1240<br>5 Oct |                                      |                  |                                     |                                                        |                        | USD 250<br>MCP654<br>321<br>1005 | USD 750                             |              |

**Final authorization**

F1.A: The cardholder makes a purchase of goods or withdrawal of cash for USD 500. The merchant notifies the acquirer of the completed authorization. The acquirer submits a clearing presentment based on notification from the merchant of authorization completion.

### Scenario F1: final authorization, F1.A

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                                 |                        |                      |                                     |         |
|------------------------|--------------------------------------|------------------|-------------------------------------|-----------------------------------------------------------------|------------------------|----------------------|-------------------------------------|---------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Auth/<br>RvrsI<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>RvrsI<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold    |
| Auth<br>0100<br>1 Oct  | 0<br>1                               | USD 500          | 1001<br>MCG12345<br>6               |                                                                 |                        |                      |                                     | USD 500 |
| Clear<br>1240<br>2 Oct |                                      |                  |                                     |                                                                 |                        | USD 500              | MCG123<br>456<br>1001               | USD 0   |

F1.B: Final authorization transaction is subsequently reversed due to technical issue such as acquirer to terminal response delivery error, ATM dispense error, or customer returns goods prior to completion being submitted for clearing, customer cancels e-commerce order, etc.

**NOTE: In the Europe and Middle/East Africa regions, the reversal of a final authorization or completion of a final authorization for an amount different from the approved final authorization amount may attract a non-compliance fee if the number of non-compliant final authorizations exceeds 1 percent of the total final authorizations processed in a weekly billing period for an acquirer parent ICA or MCE Group ICA.**

### Scenario F1: final authorization, F1.B

| Merchant/<br>Acquirer | Authorizati<br>on                    | Clearing         | Issuer                              |                                                                 |                        |                      |                                     |      |
|-----------------------|--------------------------------------|------------------|-------------------------------------|-----------------------------------------------------------------|------------------------|----------------------|-------------------------------------|------|
| Action/<br>MTI/ Date  | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Auth/<br>RvrsI<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>RvrsI<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold |

| Merchant/<br>Acquirer  | Authorizati<br>on | Clearing  | Issuer                |                                |
|------------------------|-------------------|-----------|-----------------------|--------------------------------|
| Auth<br>0100<br>1 Oct  | 0<br>1            | USD 1,000 | 1001<br>MCG12345<br>6 | USD<br>1,000                   |
| Rvrsl<br>0400<br>2 Oct | 0<br>1            | USD 1,000 | 1004<br>MCG56789<br>0 | MCG123<br>456<br>1001<br>USD 0 |

### Undefined authorization

The cardholder makes a purchase of goods for USD 500. The merchant notifies the acquirer of the completed authorization. The acquirer submits a clearing presentment based on notification from the merchant of authorization completion.

### Scenario U1: clearing amount matches authorization amount

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |         |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|---------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold    |
| Auth<br>0100<br>1 Oct  | 0<br>0<br>(Undef<br>Auth)            | USD 500          | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD 500 |
| Clear<br>1240<br>2 Oct |                                      |                  |                                     |                                                        |                        | USD 500              | MCG123<br>456<br>1001               | USD 0   |

The cardholder makes a purchase of goods for USD 500 on 1 October, and the merchant submits an authorization for USD 500. The cardholder returns an item for USD 100 prior to the merchant submitting the completed authorization for clearing. The merchant notifies the acquirer of the completed authorization for USD 400.

U2.A: The acquirer submits a clearing presentment within 24 hours of notification from the merchant of authorization completion.

**Scenario U2: clearing amount below undefined authorization amount, U2.A**

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |         |
|------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|---------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold    |
| Auth<br>0100<br>1 Oct  | 0<br>0                               | USD 500          | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD 500 |
| Clear<br>1240<br>2 Oct |                                      |                  |                                     |                                                        |                        | USD 400              | MCG123<br>456<br>1001               | USD 0   |

U2.B: The acquirer submits a partial reversal when notified by the merchant of authorization completion for lower than authorized amount, and clearing will not occur within 24 hours of merchant preauthorization completion notification.

**Scenario U2: clearing amount below undefined authorization amount, U2.B**

| Merchant/<br>Acquirer | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |         |
|-----------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|---------|
| Action/<br>MTI/ Date  | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold    |
| Auth<br>0100<br>1 Oct | 0<br>0                               | USD 500          | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD 500 |



| Merchant/<br>Acquirer | Authorizati<br>on | Clearing | Issuer   |        |         |              |
|-----------------------|-------------------|----------|----------|--------|---------|--------------|
| RvrsI                 | 0                 | USD 500  | 1004     | MCG123 | USD 400 | USD 400      |
| 0400                  | 0                 |          | MCG56789 | 456    |         |              |
| 1 Oct                 |                   |          | 0        | 1001   |         |              |
| Clear                 |                   |          |          |        | USD 400 | MCG123 USD 0 |
| 1240                  |                   |          |          |        |         | 456          |
| 6 Oct                 |                   |          |          |        |         | 1001         |

The cardholder makes an e-commerce purchase on 1 October, and the merchant submits an authorization for charges of USD 1,000. Prior to shipping, delivery charges are adjusted by USD 50. The merchant submits an additional authorization for the additional shipping charges of USD 50. The acquirer submits two clearing presentments based on notification from the merchant of authorization completion.

### Scenario U3: authorization completes for higher amount than authorized

| Merchant/<br>Acquirer | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |        |
|-----------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------|
| Action/<br>MTI/ Date  | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>RvrsI<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>RvrsI<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold   |
| Auth                  | 0                                    | USD 1,000        | 1001                                |                                                        |                        |                      |                                     | USD    |
| 0100                  | 0                                    |                  | MCG12345                            |                                                        |                        |                      |                                     | 1,000  |
| 1 Oct                 |                                      |                  | 6                                   |                                                        |                        |                      |                                     |        |
| Auth                  | 0                                    | USD 50           | 1005                                |                                                        |                        |                      |                                     | USD    |
| 0100                  | 0                                    |                  | MCG56789                            |                                                        |                        |                      |                                     | 1,050  |
| 5 Oct                 |                                      |                  | 0                                   |                                                        |                        |                      |                                     |        |
| Clear                 |                                      |                  |                                     |                                                        |                        | USD                  | MCG123                              | USD 50 |
| 1240                  |                                      |                  |                                     |                                                        |                        | 1,000                | 456                                 |        |
| 6 Oct                 |                                      |                  |                                     |                                                        |                        |                      | 1001                                |        |

| Merchant/<br>Acquirer  | Authorizati<br>on | Clearing | Issuer             |
|------------------------|-------------------|----------|--------------------|
| Clear<br>1240<br>6 Oct |                   | USD 50   | MCG567 890<br>1005 |

The cardholder makes a purchase of goods for USD 500 on 5 October, and the merchant submits an authorization for USD 500. The cardholder returns all goods purchased prior to the merchant submitting the completed authorization for clearing. The merchant notifies the acquirer of the full reversal. The acquirer submits the reversal upon merchant authorization cancellation notification.

#### Scenario U4: authorization cancellation

| Merchant/<br>Acquirer  | Authorizati<br>on                    | Clearing         | Issuer                                                                                        |
|------------------------|--------------------------------------|------------------|-----------------------------------------------------------------------------------------------|
| Action/<br>MTI/ Date   | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63<br>Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 |
| Auth<br>0100<br>1 Oct  | 0<br>0                               | USD 1,000        | 1001<br>MCG12345<br>6                                                                         |
| Rvrsl<br>0400<br>1 Oct | 0<br>0                               | USD 1,000        | 1001<br>MCG56789<br>0                                                                         |
|                        |                                      |                  | MCG123<br>456<br>1001                                                                         |
|                        |                                      |                  | USD 0                                                                                         |

#### Chargeback extension request

On 1 October, the cardholder makes a hotel reservation for a five-night stay in November. The cardholder check-in is not expected to occur until more than 30 days after the initial preauthorization date. On 30 October, the merchant submits a chargeback extension request message to allow for later clearing and extended chargeback protection. The cardholder checks into the hotel on 23 November. The cardholder checks out of the hotel on 28 November with no further charges applied to his/her bill during the stay. The acquirer submits a clearing presentment based on notification from the merchant of preauthorization completion.

**Scenario C1: approved chargeback extension request: preauthorization anticipated to complete after initial chargeback expiration date**

| Merchant/<br>Acquirer             | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|-----------------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date              | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>RvrsI<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>RvrsI<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct             | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| CB<br>Extension<br>0100<br>31 Oct | 4<br>NA                              | USD 0            | 1031<br>MCG56789<br>0               | MCG123<br>456<br>1001                                  |                        |                      |                                     | USD<br>1,000 |
| Clear<br>1240<br>28 Nov           |                                      |                  |                                     |                                                        |                        | USD<br>1,000         | MCG123<br>456<br>1001               | USD 0        |

On 1 October, the cardholder makes a hotel reservation for a five-night stay in November. The cardholder check-in is not expected to occur until more than 30 days after the initial preauthorization date. On 30 October, the merchant submits a chargeback extension request message to allow for later clearing and extended chargeback protection. The issuer declines the chargeback extension request. The authorization expires on 31 October. The cardholder checks into the hotel on 23 November. The merchant must request a new authorization due to the previous authorization being expired and not approved for extension by the issuer. The cardholder checks out of the hotel on 28 November with no further charges applied to his/her bill during the stay. The acquirer submits a clearing presentment based on notification from the merchant of preauthorization completion.

**Scenario C2: declined chargeback extension request: preauthorization anticipated to complete after initial chargeback expiration date**

| Merchant/<br>Acquirer             | Authorizati<br>on                    | Clearing         | Issuer                              |                                                        |                        |                      |                                     |              |
|-----------------------------------|--------------------------------------|------------------|-------------------------------------|--------------------------------------------------------|------------------------|----------------------|-------------------------------------|--------------|
| Action/<br>MTI/ Date              | DE 61, SF 7<br>DE 48, SE<br>61, SF 5 | Auth Amt<br>DE 4 | Auth Trace<br>ID<br>DE 15, DE<br>63 | Incr.<br>Auth/<br>Rvrsl<br>Trace ID<br>DE 48,<br>SE 63 | Auth<br>Rvrsl<br>DE 95 | Clear<br>Amt<br>DE 4 | Clear<br>Trace ID<br>DE 63,<br>SF 2 | Hold         |
| Auth<br>0100<br>1 Oct             | 4<br>NA                              | USD 1,000        | 1001<br>MCG12345<br>6               |                                                        |                        |                      |                                     | USD<br>1,000 |
| CB<br>Extension<br>0100<br>30 Oct | 4<br>NA                              | USD 0            | 1030<br>MCG56789<br>0               | MCG123<br>456<br>1001                                  |                        |                      |                                     | USD<br>1,000 |
| Auth<br>Expires<br>31 Oct         |                                      |                  |                                     |                                                        |                        |                      |                                     | USD 0        |
| Auth<br>0100<br>23 Nov            | 4<br>NA                              | USD 1,000        | 1123<br>MCG67890<br>1               |                                                        |                        |                      |                                     | USD<br>1,000 |
| Clear<br>1240<br>28 Nov           |                                      |                  |                                     |                                                        |                        | USD<br>1,000         | MCG678<br>901<br>1123               | USD 0        |

## DE 61 to DE 48 comparison

Issuers globally must recognize values 0 (Normal Authorization/Undefined Finality) and 1 (Final Authorization) in DE 48, subelement 61, subfield 5 to distinguish a final authorization from a normal authorization/undefined finality in Authorization Request/0100 and Authorization Advice/0120 messages.

When submitting Authorization Advice/0120: acquirer-generated messages, DE 48, subelement 61, subfield 5 should be populated with the same value as provided in the original Authorization Request/0100 message.

The following table illustrates the relationship between DE 61 and DE 48 and provides recommendation on the usage of DE 48, subelement 61, subfield 5 under various transaction conditions.

### Comparison table

| DE 61, SF 7        | DE 48, SE 61, SF 5 | DE 4 | DE 3                   | Definition/Comment   | Acquirer                                                                      | Issuer                                    |
|--------------------|--------------------|------|------------------------|----------------------|-------------------------------------------------------------------------------|-------------------------------------------|
| 0 (Normal Request) | 1                  | >0   | All                    | Final Authorization  | Permitted.                                                                    | Must be able to receive.                  |
| 0                  | 1                  | 0    | 00, 01, 09, 17, 20, 28 | Not permitted        | Not permitted. Dual Message Authorization System will reject (existing edit). | Not applicable. No transaction received.  |
| 0                  | 1                  | 0    | 30, 91, 92             | Not permitted        | Not permitted (not rejected by Dual Message Authorization System).            | May ignore DE 48, SE 61, SF 5.            |
| 0                  | 0, not present     | 0    | 00, 01, 09, 17, 20, 28 | Not permitted        | Not permitted. Dual Message Authorization System will reject (existing edit). | Not applicable. No transaction received.  |
| 0                  | 0, not present     | 0    | 30, 91, 92             | Normal Authorization | Permitted.                                                                    | May ignore DE 48, SE 61, SF 5 if present. |

| DE 61, SF 7                   | DE 48, SE 61,<br>SF 5 | DE 4 | DE 3 | Definition/<br>Comment                | Acquirer                                                                                                                                                     | Issuer                                                             |
|-------------------------------|-----------------------|------|------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 0                             | 0, not present        | >0   | All  | Authorization with Undefined Finality | Permitted for acceptors outside the Europe region; not permitted for acceptors in the Europe region (not rejected by the Dual Message Authorization System). | Must be able to receive.                                           |
| 4<br>(Preauthorized request)  | 1, 0, not present     | 0    | 00   | Not permitted                         | Not permitted. Dual Message Authorization System will reject when not permitted (existing edit).                                                             | No transaction received                                            |
| 4                             | 1                     | >0   | 00   | Not permitted                         | Not permitted (not rejected by the Dual Message Authorization System).                                                                                       | May ignore DE 48, SE 61, SF 5.                                     |
| 4                             | 0, not present        | >0   | 00   | Preauthorization                      | Permitted.                                                                                                                                                   | Must be able to receive. May ignore DE 48, SE 61, SF 5 if present. |
| 2<br>(SecureCode Phone Order) | 0, 1, not present     | 0    | 00   | Not permitted                         | Not permitted. Dual Message Authorization System will reject (existing edit).                                                                                | Not applicable. No transaction received.                           |
| 2                             | 0, 1, not present     | >0   | 00   | Identity Check Phone Order            | Permitted.                                                                                                                                                   | Ignore DE 48, SE 61, SF 5 if present.                              |

| DE 61, SF 7                 | DE 48, SE 61,<br>SF 5 | DE 4 | DE 3   | Definition/<br>Comment                              | Acquirer                                                      | Issuer                                   |
|-----------------------------|-----------------------|------|--------|-----------------------------------------------------|---------------------------------------------------------------|------------------------------------------|
| 3 (ATM Installment Inquiry) | 0, 1, not present     | 0    | 01     | Not permitted                                       | Not permitted. Dual Message Authorization System will reject. | Not applicable. No transaction received. |
| 3                           | 0, 1, not present     | >0   | 01     | ATM Installment Inquiry                             | Permitted.                                                    | Ignore DE 48, SE 61, SF 5 if present.    |
| 6 (ATC Update)              | 0, 1, not present     | 0    | 00     | ATC Update                                          | Permitted.                                                    | Ignore DE 48, SE 61, SF 5 if present.    |
| 6                           | 0, 1, not present     | >0   | 00     | Not permitted                                       | Not permitted. Dual Message Authorization System will reject. | Not applicable. No transaction received. |
| 8 (Account Status Inquiry)  | 0, 1, not present     | 0    | 00, 28 | Purchase and Payment Account Status Inquiry Service | Permitted.                                                    | Ignore DE 48, SE 61, SF 5 if present.    |
| 8                           | 0, 1, not present     | >0   | 28     | Payment Account Status Inquiry Service              | Permitted.                                                    | Ignore DE 48, SE 61, SF 5 if present.    |
| 8                           | 0, 1, not present     | >0   | 00     | Not permitted                                       | Not permitted. Dual Message Authorization System will reject. | Not applicable. No transaction received. |

## Data integrity

To support the gradual transition of merchants and acquirers in the Asia/Pacific, Canada, Latin America and the Caribbean, and United States regions to the new message coding standards, Mastercard has implemented two new data integrity monitoring edits to support the transition of message coding of authorizations from undefined to preauthorization or final authorization for authorization transactions processed on the Mastercard Network.

Support of the new message coding standards provides issuers the opportunity to apply different processing or cardholder messaging to different types of authorizations. Customers wanting to receive non-compliance notifications and view reporting must be registered for Data Integrity Online through Mastercard Connect®. Refer to [Data Integrity Monitoring Program](#) for additional information.

**NOTE: Acquirers in the Europe and Middle East/Africa regions are not impacted by the new data integrity programs and non-compliance assessments.**

### **Preauthorization message data integrity**

Mastercard considers customers noncompliant when preauthorization requests account for 25 percent or more of all approved domestic purchase transactions. This program will help ensure that an acquirer does not start coding all authorizations as preauthorization in order to take advantage of the extended chargeback protection period.

The following types of transactions are excluded from the compliance validation process:

- Private Label transactions and transactions of card brands other than Mastercard including Debit Mastercard, Maestro, and Cirrus
- Transactions that are approved offline
- Installment transactions
- Incremental authorizations
- Cross-border transactions

### **Undefined authorization data integrity**

The second data integrity edit will help ensure that for acquirers that process at least 100,000 approved financial domestic transactions on the Mastercard Network, not more than 50 percent of an acquirer's total domestic transaction authorizations, per ICA, per month remain coded as undefined. This program will help ensure that acquirers make the overall transition of coding authorizations as either preauthorization or final authorization for their domestic transaction activity.

The focus on domestic transactions in this edit is due to the fact that these transactions, unlike cross-border transactions, do not require currency conversion, which may result in a difference between the authorized and cleared cardholder billing amount that an issuer may use to manage cardholder balances.

The following types of transactions are excluded from the compliance validation process:

- Private Label transactions and transactions of card brands other than Mastercard including Debit Mastercard, Maestro, and Cirrus
- Transactions that are approved offline
- Installment transactions
- Cross-border transactions



## Undefined and preauthorization processing integrity programs

Acquirers in the Asia/Pacific, Canada, Latin America and the Caribbean, and United States regions are subject to a processing integrity fee under the processing integrity programs and non-compliance assessments for Not Reversed or Cleared Undefined Authorization and Preauthorization, starting with the billing invoice. The fee will be applied for each approved authorization that is not cleared or fully reversed by an acquirer within 30 calendar days of the authorization date for preauthorizations and within seven calendar days of the authorization date for undefined financial authorizations.

Acquirers in the Europe region are not impacted by the new processing integrity programs and non-compliance assessments for Not Reversed or Cleared Undefined Authorization and Preauthorization.

## Final authorization, undefined authorization, and preauthorization processing integrity programs

Acquirers in the Middle East/Africa region are subject to processing integrity fee assessments for non-compliant final authorizations, undefined authorizations, and preauthorizations.

Acquirers in the Middle East/Africa region are impacted by a preauthorization transaction fee. Acquirers are impacted by assessments for three processing integrity programs: Final Authorizations Not Meeting Requirement, Undefined Authorizations With and Without Final Authorization Characteristics, and Not Reversed or Cleared Preauthorizations.

## Installment transactions

The following section explains how installment payment authorization messages are coded.

### Issuer-financed installment transactions

Acquirers submitting issuer-financed installment transactions should consider coding authorizations as Final if the authorization can be presented for clearing within seven calendar days of the authorization date and for the amount and currency authorized.

### Merchant-financed and acquirer-financed installment transactions

Acquirers submitting merchant-financed or acquirer-financed installment transactions should consider coding authorizations as Final if the total installment amount is known and the first installment can be presented for clearing within seven calendar days of the authorization date.

Acquirers submitting merchant-financed or acquirer-financed installment transactions should consider coding authorizations as preauthorization if the installment amount is an estimate, or the first installment requires more than seven calendar days from the authorization date to present for clearing.

### For more information

For details about data requirements, refer to [Data element definitions and details](#).

## Balance inquiries

Mastercard supports several types of balance inquiry requests cardholders can perform.

- ATM and point-of-sale (POS) terminal balance inquiries
- Short Message Service balance inquiries
- Mobile Remote Payments balance inquiries

The balance inquiry service is optional for issuers.

### ATM and point-of-sale terminal balance inquiries

The following information describes ATM balance inquiries and point-of-sale (POS) terminal balance inquiries.

#### ATM balance inquiries

The ATM balance inquiry allows Mastercard cardholders to perform a balance inquiry on the Mastercard® ATM Network.

Upon receipt of an ATM balance inquiry request, participating issuers return the available balance in the authorization response. The Dual Message Authorization System performs any applicable currency conversion and forwards the available balance to the acquirer. The acquirer provides the available balance to the ATM terminal for display at the ATM and on the customer's printed receipt.

**NOTE: When responding to intracountry ATM balance inquiry requests, issuers in the United Kingdom (U.K.) may provide a maximum of two account balances: the ledger balance in addition to the available balance.**

#### Point-of-sale terminal balance inquiries

The point-of-sale (POS) balance inquiry enables cardholders and acquirers to check the remaining amount of funds on a Mastercard prepaid card or private label prepaid card. This feature makes it easier for the cardholder to completely redeem the funds on the prepaid card and reduces the potential of a declined authorization request because the purchase amount exceeds the funds available on the card. Therefore, this feature can help to reduce checkout times and lost sales.

Upon receipt of a POS balance inquiry, participating issuers return the available balance in the authorization response. The Dual Message Authorization System performs any applicable currency conversion and forwards the available balance to the acquirer. The acquirer provides the available balance to the merchant for display on the customer's printed receipt.

There are two types of POS terminal balance inquiries:

- Cardholder-initiated balance inquiry: Cardholders can ask the cashier to check a card's remaining balance before making a purchase. The cardholder and cashier can then determine

whether the entire purchase amount can be funded by the prepaid card or whether a supplemental payment method (split tender) is needed to complete the purchase.

- Automated balance inquiry: Acquirers can expedite sales by sending an automatic POS balance inquiry on a specified range of prepaid accounts issued by the issuer. The cashier can then inform customers in advance whether they will need to use a split tender to complete the purchase.

### Alternate processing

ATM and POS balance inquiries are not eligible for processing by the Stand-In Authorization or X-Code Processing. If the issuer is unavailable, the Dual Message Authorization System returns an Authorization System or issuer system inoperative response to the acquirer.

### Reversals of balance inquiry transactions

Mastercard supports reversals of balance inquiry transactions using Reversal Request/0400 and Reversal Advice/0420 messages containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 30 (Balance Inquiry). In balance inquiry transactions, DE 4 (Amount, Transaction) contains a value of all zeros.

A balance inquiry reversal is generally initiated when the balance inquiry request is not completed at the point-of-interaction. An acquirer may submit a reversal in an effort to recoup any fee that may be assessed as a result of the original balance inquiry request.

### To participate

Issuers can complete the *Point-of-Sale (POS) and ATM Balance Inquiry Participation* (Form 771) and submit them to Global Customer Service.

### For more information

For details about data requirements refer to [Data element definitions and details](#).

## Short Message Service Balance Inquiry service

The Short Message Service (SMS) Balance Inquiry service allows cardholders with SMS-enabled mobile devices to request and receive balance information on their registered cards via their mobile phone devices.

The SMS Balance Inquiry service allows card issuers to provide their cardholders with convenient real-time access to their account balance information via their mobile devices. Issuers participating in the SMS Balance Inquiry service receive these balance inquiries with DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 81 PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram).

This service is only available to issuers of Mastercard and Debit Mastercard cards in the U.S. and Europe regions.

### To participate

Issuers that want to participate in the SMS Balance Service Program must support balance inquiries.

Issuers that want to participate in the SMS Balance Service Program must complete a contract and registration form.

### For more information

For details about data requirements refer to [Data element definitions and details](#).

## Mobile Remote Payments balance inquiries

The Mobile Remote Payments program provides participating cardholders with access to their account balance information via an application on their mobile device. Issuers participating in the Mobile Remote Payments program will receive these balance inquiries in DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 82 (PAN auto entry via server).

### For more information

For details about data requirements refer to [Data element definitions and details](#).

## Balance Inquiry: ATM

Issuers that participate in this service must test with the Customer Implementation Services team. Cirrus is the controlling gateway. It forwards balance inquiry requests only to issuers who are certified for receipt. Issuers must then reply with a balance inquiry response.

## Authorization Request/0100: ATM Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element               | Org | Sys | Dst | Values/Comments                                                                                                                                            |
|----------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code)     | M   | .   | M   | Must be 30xx00                                                                                                                                             |
| DE 4 (Amount, Transaction) | M   | .   | M   | Must be zero unless an acquirer applied an ATM transaction fee for an ATM transaction in a country where application of an ATM transaction fee is allowed. |

| Data Element                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                       |
|---------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 18 (Merchant Type)                 | M   | •   | M   | MCC must be one of the following values:<br><br>6010 (Member Financial Institution—Manual Cash Disbursements)<br><br>6011 (Member Financial Institution—Automated Cash Disbursements) |
| DE 28 (Amount, Transaction Fee)       | C   | •   | C   | May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.                                               |
| DE 35 (Track 2 Data)                  | C   | •   | C   | Must omit if DE 45 present                                                                                                                                                            |
| DE 41 (Acceptor Terminal ID)          | M   | •   | M   |                                                                                                                                                                                       |
| DE 43 (Acceptor Name and Location)    | M   | •   | M   |                                                                                                                                                                                       |
| DE 45 (Track 1 Data)                  | C   | •   | C   | Must omit if DE 35 present                                                                                                                                                            |
| DE 52 (Personal ID Number [PIN] Data) | M   | •   | M   |                                                                                                                                                                                       |

## Authorization Request/0100: ATM Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request/0100 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request/0100: ATM balance inquiry message.

### Edits

| IF...                                                                                                                    | THEN the Dual Message Authorization System...                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| the issuer does not participate in ATM balance inquiry                                                                   | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 57 (Transaction not permitted to issuer/cardholder). |
| DE 03 (Processing Code), SF01 (Cardholder Transaction Type Code) =01<br>and<br>DE 18 (Merchant Type) is not 6010 or 6011 | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br>DE 44 = 018                                    |

| IF...                                                                                                         | THEN the Dual Message Authorization System...                                                    |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| if neither DE 45 (Track 1 Data) nor DE 35 (Track 2 Data) is present in the Authorization Request/0100 message | returns the Authorization Request Response/0110 message containing:<br>DE 39 = 30<br>DE 44 = 035 |
| DE 54 (Additional Amounts) is present                                                                         | returns the Authorization Request Response/0110 message containing:<br>DE 39 = 30<br>DE 44 = 054 |

## Authorization Request Response/0110: ATM Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0100 data elements apply.

### Applicable data elements

| Data Element                                          | Org | Sys | Dst | Values/Comments                                                                               |
|-------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------|
| DE 28 (Amount, Transaction Fee)                       | CE  | X   | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.     |
| DE 39 (Response Code)                                 | M   | .   | M   | 00 = Approved or completed successfully<br>or<br>85 = Not declined                            |
| DE 54 (Additional Amounts), subfield 1 (Account Type) | M   | .   | M   | Must be the same value as DE 3, subfield 2 in the original Authorization Request/0100 message |
| DE 54, subfield 2 (Amount Type)                       | M   | .   | M   | 02 = Available Balance                                                                        |
| DE 54, subfield 3 (Currency Code)                     | M   | .   | M   | Must be a valid three-digit currency code that matches the issuer's currency code.            |
| DE 54, subfield 4 (Debit or Credit Indicator)         | M   | .   | M   | C = Credit amount plus 12 digits<br>or<br>D = Debit amount plus 12 digits                     |

## Authorization Request Response/0110: ATM Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request Response/0110 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request Response (ATM balance inquiry)/0110 message.

### Edits

| IF...                                                                                                | THEN...                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code) contains the value 00, 08, or 85 and DE 54 (Additional Amounts) is not present | the Dual Message Authorization System will send to the issuer an Authorization Response Negative Acknowledgment/0190 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054 |
| DE 54, subfield 3 (Currency Code) is not a valid currency code                                       | the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054           |
| DE 54, subfield 4 (Debit or Credit Indicator) is not C plus 12 digits or D plus 12 digits            | the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054           |

## Authorization Advice/0120: Acquirer-Generated (ATM Balance Inquiry Edits)

The Dual Message Authorization System will perform the following edit on Authorization Advice/0120: acquirer-generated messages for ATM Balance Inquiry transactions.

### Edits

| IF...                                                                                                                                       | THEN...                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer submits an Authorization Advice/0120: acquirer-generated message containing DE 3 (Processing Code), value 30 (Balance Inquiry) | the Dual Message Authorization System will reject the message and forward the acquirer an Authorization Advice Response/0130 message containing:<br><br>DE 39 = 30<br>DE 44 = 003 |

## Alternate Processing

ATM Balance inquiry transactions are not eligible for alternate (Stand-In Authorization or alternate issuer host routing) or X-Code Processing.

If the primary issuer is not available to the ATM balance inquiry request, an Authorization Request Response/0110 is returned to the acquirer with DE 39 (Response Code) value 91 (Authorization System or issuer system inoperative).

## Balance Inquiry: Point-of-Sale

When attempting to make a purchase at the point of sale, cardholders uncertain of the remaining balance on a Mastercard prepaid card or a private label card can initiate a balance inquiry at the point of sale to make a more informed decision about how to use the card's funds for the purchase. POS balance inquiry helps the cardholder to completely redeem the funds on the prepaid card, reduces the potential of a declined authorization request when the purchase amount exceeds the funds available on the card, and helps to avoid extended checkout times and lost sales.

The acquirer will receive the cardholder account balance in both the issuer's currency and the transaction currency. The acquirer has the option to provide the merchant with the cardholder account balance in the issuer's currency, transaction currency, or both. The merchant will then display the appropriate account balance on the customer's printed receipt.

Acquirer participation in POS balance inquiry is optional. To request a POS balance inquiry, the acquirer provides the appropriate data elements as defined below.



Acquirers in the U.S. region must support POS balance inquiry transaction type and functionality for all prepaid Mastercard credit and Debit Mastercard card account ranges.

Prepaid card issuers in the U.S. region must support online POS balance inquiries for all prepaid Mastercard credit and Debit Mastercard account ranges.

Issuer participation in POS balance inquiry is optional. To request participation, issuers must submit the *Point-of-Sale (POS) Balance Inquiry Participation Request Form* (Form 771) to Global Customer Service.

## Authorization Request/0100: POS Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                         |
|-----------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) | M   | •   | M   | 30 (Balance Inquiry)                                                                                                                                                                                                    |
| DE 3, subfield 2 (Cardholder "From Account" Type Code)                | M   | •   | M   | 00 (Default Account)<br>or<br>30 (Credit Card Account)                                                                                                                                                                  |
| DE 3, subfield 3 (Cardholder "To Account" Type Code)                  | M   | •   | M   | 00 (Default Account)                                                                                                                                                                                                    |
| DE 4 (Amount, Transaction)                                            | M   | •   | M   | Must be zero.                                                                                                                                                                                                           |
| DE 18 (Merchant Type)                                                 | M   | •   | M   | For POS balance inquiries, the MCC must be a value other than:<br><br>6010 = Member Financial Institution: Manual Cash Disbursements<br><br>or<br><br>6011 = Member Financial Institution: Automated Cash Disbursements |

| Data Element                                                                        | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | M   | •   | M   | Must contain one of the following values:<br><br>02 = PAN auto-entry via magnetic stripe<br><br>05 = PAN auto-entry via chip<br><br>07 = PAN auto-entry via contactless M/Chip<br><br>80 = Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN.<br><br>90 = PAN auto-entry via magnetic stripe<br><br>91 = PAN auto-entry via contactless magnetic stripe |
| DE 35 (Track 2 Data)                                                                | C   | •   | C   | Track information encoded on the magnetic stripe must be presented from either Track 2 or Track 1.                                                                                                                                                                                                                                                                                                                                                                   |
| DE 41 (Acceptor Terminal ID)                                                        | M   | •   | M   | Must contain terminal ID at the acceptor location.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DE 43 (Acceptor Name and Location)                                                  | M   | •   | M   | Must contain name and location of the acceptor as known by the cardholder.                                                                                                                                                                                                                                                                                                                                                                                           |
| DE 45 (Track 1 Data)                                                                | C   | •   | C   | Track information encoded on the magnetic stripe must be presented from either Track 1 or Track 2.                                                                                                                                                                                                                                                                                                                                                                   |

## Authorization Request Response/0110: POS Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Issuers participating in POS balance inquiry and responding to an Authorization Request/0100—POS Balance Inquiry message must send an Authorization Request Response/0110 message containing balance information in DE 54 (Additional Amounts) when DE 39 contains the value 00 (Approved or completed successfully) or 85 (Not declined).

### Applicable data elements

| Data Element                                             | Org | Sys | Dst | Values/Comments                                                                                                                                                    |
|----------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code)                                    | M   | .   | M   | 00 = Approved or completed successfully<br>or<br>85 = Not declined                                                                                                 |
| DE 54 (Additional Amounts),<br>subfield 1 (Account Type) | M   | .   | M   | Must be the same value as DE 3, subfield<br>2 in the original Authorization Request/<br>0100 message<br><br>00 = Default Account<br>or<br>30 = Credit Card Account |
| DE 54, subfield 2 (Amount Type)                          | M   | .   | M   | 02 = Available Balance                                                                                                                                             |
| DE 54, subfield 3 (Currency Code)                        | M   | .   | M   | Must be a valid three-digit currency code<br>that matches the issuer's currency code.                                                                              |
| DE 54, subfield 4 (Debit or Credit<br>Indicator)         | M   | .   | M   | C = Credit amount plus 12 digits<br>or<br>D = Debit amount plus 12 digits                                                                                          |

## Authorization Request/0100: POS Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request/0100 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request/0100: POS balance inquiry message.

### Edits

| IF...                                                  | THEN the Dual Message Authorization System...                                                                                               |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer participates in POS balance inquiry         | forwards the POS balance inquiry to the issuer.                                                                                             |
| the issuer does not participate in POS balance inquiry | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 57 (Transaction not permitted to issuer/<br>cardholder). |

| IF...                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| DE 4 (Transaction Amount) is not zero                                                                                                                                                   | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 004 |
| DE 18 (Merchant Type) is 6010 or 6011                                                                                                                                                   | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 018 |
| DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is not 02, 05, 07, 10, 80, 82, 90, or 91                                                            | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 022 |
| DE 41 (Acceptor Terminal ID) is not present                                                                                                                                             | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 041 |
| DE 43 (Acceptor Name and Location) is not present                                                                                                                                       | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 043 |
| if neither DE 45 (Track 1 Data) nor DE 35 (Track 2 Data) is present in the Authorization Request/0100: POS Balance Inquiry message<br><br>except<br><br>if DE 22, subfield 1 = 10 or 81 | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 035 |
| <b>NOTE: This edit is exempt when DE 22, subfield 1 contains values 10 or 81.</b>                                                                                                       |                                                                                                          |
| DE 54 (Additional Amounts) is present                                                                                                                                                   | returns the Authorization Request Response/0110 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054 |

## Authorization Request Response/0110: POS Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request Response/0110 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request Response/0110: POS balance inquiry message.

### Edits

| IF...                                                                                           | THEN...                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code) contains the value 00 or 85 and DE 54 (Additional Amounts) is not present | the Dual Message Authorization System will send to the issuer an Authorization Response Negative Acknowledgment/0190 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054 |
| DE 39 contains a value other than 00 or 85 and DE 54 is present                                 | the Dual Message Authorization System will remove DE 54 from the Authorization Request Response/0110 message before sending the message to the acquirer                       |
| DE 54, subfield 2 (Amount Type) is not 02 (Available Balance)                                   | the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054           |
| DE 54, subfield 3 (Currency Code) is not a valid currency code                                  | the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054           |
| DE 54, subfield 4 (Debit or Credit Indicator) is not C plus 12 digits or D plus 12 digits       | the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 054           |

## Authorization Advice/0120: acquirer-generated (POS Balance Inquiry Edits)

The Dual Message Authorization System will perform the following edits on Authorization Advice/0120: POS balance inquiry acquirer-generated messages.

### Edits

| IF...                                                                                                                                       | THEN...                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The acquirer submits an Authorization Advice/0120: acquirer-generated message containing DE 3 (Processing Code), value 30 (Balance Inquiry) | The Dual Message Authorization System will reject the message and forward the acquirer an Authorization Advice Response/0130 message containing:<br><br>DE 39 = 30<br><br>DE 44 = 003 |

## Alternate Processing

POS Balance inquiry transactions are not eligible for alternate (Stand-In Authorization or alternate issuer host routing) or X-Code Processing.

If the primary issuer is not available to respond to the POS balance inquiry request, an Authorization Request Response/0110 is returned to the acquirer with DE 39 (Response Code) value 91 (Authorization System or issuer system inoperative).

## Balance Inquiry: Short Message Service

The Balance Inquiry Short Message Service (SMS) provides cardholders with real-time access to their account balance information via their mobile devices.

Issuers that want to participate in the SMS Balance Service Program must complete a contract and Issuer SMS Balance Inquiry Service information form.

Issuers participating in the SMS Balance Inquiry service will receive these balance inquiries with DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 81 PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram).

## Authorization Request/0100: Short Message Service Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                                                                                                                                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)                                                                                                                                                 | M   | •   | M   | 30 = Balance Inquiry                                                                                                                                                                   |
| DE 18 (Merchant Type)                                                                                                                                                                                                 | M   | •   | M   | 5969 = Mail Order/Telephone Order Providers                                                                                                                                            |
| DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)                                                                                                                                   | M   | •   | M   | 81 = PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram) |
| DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) | M   | •   | M   | Zero (0)                                                                                                                                                                               |

## Chip-Specific Value Constraints

The Dual Message Authorization System supports the authorization of chip transactions generated using an integrated circuit card (ICC).

Acquirers may choose to acquire chip-related data in one of the two modes:

- Chip Partial Grade mode
- Chip Full Grade mode

The content of chip-related Authorization Request/0100 messages will vary, depending on the mode used. Issuers of ICCs must be able to accept authorization requests in either mode.

## Chip Partial Grade Value Constraints

Following is a list of specific data elements and data element values required in the Authorization Request/0100 message for a chip partial grade transaction.

### Data elements and values

| DE 22, subfield 1 | DE 22, subfield 1 value | DE 35                                            | DE 52 and DE 23  |
|-------------------|-------------------------|--------------------------------------------------|------------------|
| Offline PIN       | 05                      | ICC data as EMV tag 57 (Track 2 Equivalent Data) | Not provided     |
| Online PIN        | 05                      | ICC data as EMV tag 57 (Track 2 Equivalent Data) | Must be provided |
| Signature         | 05                      | ICC data as EMV tag 57 (Track 2 Equivalent Data) | Not provided     |
| No CVM            | 05                      | ICC data as EMV tag 57 (Track 2 Equivalent Data) | Not provided     |

### Data element requirements

| Data Element | Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22        | <p>For chip transactions, acquirers must provide value 05x.</p> <p>For a chip transaction where the magnetic stripe was used as a fallback technology, acquirers must provide the value 80x. In this case, acquirers must provide the full, unaltered track data.</p> <p>For a chip transaction where manual PAN entry was used as a fallback technology, acquirers may provide the value 79x.</p>                                                                                                                                    |
| DE 23        | The card sequence number may now be provided in chip partial grade transactions where DE 55 is not present. DE 23 must be provided if EMV tag 5F34 (Application PAN Sequence Number) is present on the ICC.                                                                                                                                                                                                                                                                                                                           |
| DE 35        | <p>The ICC data as EMV tag 57 (Track 2 Equivalent Data). This data corresponds with the data stored in Track 2 of the magnetic stripe.</p> <p>For chip transactions, acquirers must provide DE 35 EMV tag 57 if the data object was present on the ICC.</p> <p>For a chip transaction where the magnetic stripe was used as a fallback technology, DE 35 must contain the actual Track 1 or Track 2 data from the magnetic stripe, or in the case of Maestro the actual Track 2 data because only Track 2 is present on the card.</p> |
| DE 52        | DE 52 must be provided if the Cardholder verification method was Online PIN.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



## Chip Full Grade Value Constraints

This mode is for acquirers who accept chip transactions and whose infrastructure allows them to provide chip-specific data. Acquirers operating in this mode must comply with all chip partial grade requirements and the chip full grade requirements.

### Data element requirements

| Data Element | Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 23        | DE 23 must be provided if EMV tag 5F34 is present on the ICC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DE 35        | <p>The ICC data as EMV tag 57 (Track 2 Equivalent Data). This data corresponds with the data stored in Track 2 of the magnetic stripe.</p> <p>For chip transactions, acquirers must provide DE 35 EMV tag 57 if the data object was present on the ICC.</p> <p>For a chip transaction where the magnetic stripe was used as a fallback technology, DE 35 must contain the actual Track 1 or Track 2 data from the magnetic stripe, or in the case of Maestro the actual Track 2 data because only Track 2 is present on the card.</p> |
| DE 37        | DE 37, subfield 2 contains the value of Transaction Sequence Counter (EMV Tag 9F41), right-justified, left-padded with zeros.                                                                                                                                                                                                                                                                                                                                                                                                         |
| DE 52        | DE 52 must be provided if the CVM is Online PIN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DE 55        | DE 55 must be provided as specified in <a href="#">General message layout information</a> . DE 55 must be tag-length-value (TLV)-encoded and must contain the information (mandatory and optional) as specified in <a href="#">Data element definitions and details</a> .                                                                                                                                                                                                                                                             |
| DE 125       | Acquirers must provide DE 125 for PIN change transactions where DE 52 also must be present.                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Contact and Contactless Chip Specific Value Constraints

Following is the usage of DE 61, subfield 11 in conjunction with DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode). DE 22, subfield 1 values indicate the method by which the PAN was entered.

### Value constraints

| DE 61,<br>subfield 11<br>value |   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3                              | = | <p>Merchant terminal PAN auto-entry via bar code reader</p> <p>Value 3 indicates that the terminal supports contactless M/Chip and contactless magnetic stripe transactions. The terminal may also support other card input types, including contact transactions.</p> <p>DE 22, subfield 1 applicable values:</p> <ul style="list-style-type: none"> <li>• 07 (PAN auto-entry via contactless M/Chip)</li> <li>• 91 (PAN auto-entry via contactless magnetic stripe)</li> </ul>                                                                                           |
| 4                              | = | <p>Merchant terminal PAN auto-entry via optical character reader (OCR)</p> <p>Value 4 indicates that the terminal supports contactless magnetic stripe transactions. The terminal may also support other card input types, including contact transactions.</p> <p>DE 22, subfield 1 applicable value:</p> <ul style="list-style-type: none"> <li>• 91 (PAN auto-entry via contactless magnetic stripe)</li> </ul>                                                                                                                                                          |
| 5                              | = | <p>Merchant terminal PAN auto-entry via integrated circuit card</p> <p>DE 22, subfield 1 applicable values:</p> <ul style="list-style-type: none"> <li>• 05 (PAN auto-entry via chip)</li> <li>• 79 (Hybrid terminal with an online connection to acquirer failed in sending chip to magnetic stripe fallback or reading the chip card)</li> <li>• 80 (Chip card at chip-capable terminal was unable to process transaction; therefore, terminal defaulted to the magstripe-read PAN)</li> <li>• 90 (PAN auto-entry via magnetic stripe—track data is required)</li> </ul> |

| DE 61,<br>subfield 11<br>value |   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8                              | = | EMV specification (compatible chip reader), magnetic stripe reader and key entry.<br>DE 22, subfield 1 applicable values: <ul style="list-style-type: none"> <li>• 01 (PAN manual entry)</li> <li>• 05 (PAN auto-entry via chip)</li> <li>• 79 (Hybrid terminal with an online connection to acquirer failed in sending chip to magstripe fallback or reading the chip card)</li> <li>• 80 (Chip card at chip-capable terminal was unable to process transaction using data on the chip)</li> <li>• 90 (PAN auto-entry via magnetic stripe—track data is required)</li> </ul> |
| 9                              | = | PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55<br>(Integrated Circuit Card [ICC] System-Related Data).<br>DE 22, subfield 1 applicable value: 05 (PAN auto-entry via chip)                                                                                                                                                                                                                                                                                                                                                                       |

## Canada Region Debit Mastercard Merchant Acceptance

The Dual Message Authorization System allows the issuance of the Debit Mastercard® brand by Canada region issuers and allows Canada region merchants to accept domestically issued Debit Mastercard cards.

The Dual Message Authorization System prevents the accidental acceptance of Canada region-issued Debit Mastercard cards and allows issuers to issue Debit Mastercard cards either as stand-alone cards or as cards bearing other acceptance marks.

### Acquirers

Acquirers in the Canada region that have established an agreement with their merchants to process Debit Mastercard transactions must support DE 48 (Additional Data—Private Use), subelement 18 (Service Parameters), subfield 01 (Canada Domestic Indicator) and DE 39 (Response Code), value 81 (Domestic Debit Transaction Not Allowed).

For merchants that accept Canada region-issued Debit Mastercard cards, acquirers in Canada must:

- Include DE 48, subelement 18 (Service Parameters) in the Authorization Request/0100 message indicating the merchant has agreed to process Debit Mastercard.
- Support DE 39, response code value 81 (Domestic Debit Transaction Not Allowed) in the Authorization Request Response/0110 message indicating a transaction has been declined.

Issuers

Issuers in the Canada region that choose to issue Debit Mastercard cards must notify Mastercard Customer Implementation Services (CIS) if the account ranges bear other acceptance marks.

Issuers in Canada must:

- Notify Customer Implementation Services (CIS):
  - When the Debit Mastercard BIN being implemented will bear other acceptance marks.
  - The point-of-interaction (POI) environments (face-to-face or non-face-to-face) in which other acceptance marks will exist.
- Ensure Debit Mastercard cards comply with the Canada region chapter in *Mastercard Rules*.
- Immediately notify Mastercard through CIS of any changes in the account ranges affecting compliance with Mastercard Rules.

Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following edits on transactions that are acquired in Canada, on a card issued in Canada when the acceptance brand of that card is Debit Mastercard.

Associated edits

For transactions with a Canada region-issued Debit Mastercard card that bears other acceptance marks with international only Debit Mastercard routing.

| WHEN...                                                                                                                           | THEN the Dual Message Authorization System...                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer | rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed). |

Associated edits

For transactions with a Canada region-issued Debit Mastercard card bearing other acceptance marks with only domestic face-to-face functionality for the other acceptance marks on the card.

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | THEN the Dual Message Authorization System...                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer</li> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is value 00 (Purchase), 09 (Purchase with Cash Back), or 28 (Payment Transaction)</li> <li>DE 61, subfield 4 (POS Cardholder Presence) is value 0 (Cardholder present) or value 1 (Cardholder not present, unspecified)</li> </ul> | rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed). |

### Associated edits

For transactions with a Canada region-issued Debit Mastercard card bearing other acceptance marks with domestic non-face-to-face functionality for the other acceptance marks on the card.

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | THEN the Dual Message Authorization System...                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer</li> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is value 00 (Purchase), 09 (Purchase with Cash Back), or 28 (Payment Transaction)</li> <li>DE 61, subfield 4 (POS Cardholder Presence) is value 2 (Mail/facsimile order), 3 (Phone/ARU order), 4 (Standing order/recurring transactions), or 5 (Electronic order [home PC, Internet, mobile phone, PDA])</li> <li>DE 48 (Additional Data: Private Use), subelement 18 (Service Parameters), subfield 1 (Canada Domestic Indicator) is not present or is value other than Y</li> </ul> | rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed). |

### Associated edits

For transactions with a Canada region-issued Debit Mastercard card not containing other acceptance marks.

| WHEN...                                                                                                                                                                                                                                                                                                                                                 | THEN the Dual Message Authorization System...                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer</li> <li>DE 48 (Additional Data: Private Use), subelement 18 (Service Parameters), subfield 1 (Canada Domestic Indicator) is not present or is a value other than Y</li> </ul> | rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed). |

## Cardholder-activated terminals

Cardholder-activated terminals (CATs) are usually unattended terminals that accept bankcards, debit, credit, and proprietary cards. These terminals are frequently installed at rail ticketing stations, gasoline stations, toll roads, parking garages, and other merchant locations.

For procedures related to authorization and other processing requirements related to each CAT level, refer to the [Transaction Processing Rules](#).

## Automated fuel dispensers

If an issuer approves an authorization request for an automated fuel dispenser (AFD) transaction identified with MCC 5542 (an AFD transaction) and occurring at a merchant located in the United States or Canada, then within 60 minutes of the time that the authorization request message was sent, the acquirer must send an Authorization Advice/0120 message advising the issuer of the final transaction amount (or a Reversal Request/0400 message, if the transaction will not be completed or will finalize for a lesser amount than requested).

If after approving an authorization request for an AFD transaction, an issuer in the United States or Canada has placed a hold on cardholder funds in excess of USD 1 or CAD 1 respectively, then within 60 minutes of receiving the acquirer's advice message, the issuer must release any hold amount that exceeds the completed transaction amount specified.

Mastercard recommends that the Trace ID (DE 15 [Settlement Date] and DE 63 [Network Data]) of the completion advice response be used within the clearing presentment.

**NOTE: As an alternative, the Trace ID of the original preauthorization may be used for clearing presentment if that same Trace ID is included in the AFD completion advice message within DE 48, subelement 63 (Additional Data, Trace ID).**

For requirements about intra-European and inter-European Maestro transactions that occur at unattended POS terminals located at petrol stations, refer to the Automated Fuel Dispenser Transactions section in Card-Present Transactions of the [Transaction Processing Rules](#).

## Electric vehicle charging transactions

A transaction occurring at an unattended POS Terminal for the purchase of electric vehicle charging services is identified with MCC 5552 (Electric Vehicle Charging) and a CAT level indicator of CAT 1 or CAT 2 (for card-present transactions) or CAT 6 (for e-commerce transactions).

Alternatively, if the primary business of the merchant is temporary parking services, then MCC 7523 (Automobile Parking Lots and Garages) may be used.

Contactless-only acceptance is permitted. A contactless-only terminal supporting a maximum vehicle charging amount that does not exceed the applicable contactless CVM limit is properly identified as CAT 2.

The transaction may be authorized either prior to or after the vehicle charging.

### Authorization before charging

Each electric vehicle charging transaction for which authorization is requested before vehicle charging begins is properly processed as follows:

1. The merchant must inform the cardholder of any estimated amount for which authorization will be requested (for example, on a screen display or sticker at the terminal) and must obtain the cardholder's consent to the amount before initiating the authorization request. The estimated amount may be the terminal's maximum dispense amount or a specific amount requested by the cardholder.
2. The acquirer's initial authorization request (0100 or 0200) message to the issuer must be identified as a preauthorization. If the preauthorization request contains the partial approval support indicator, and the issuer provides a partial approval response, then the final transaction amount must not exceed the partial approval amount provided in DE 6 (Amount, Cardholder Billing).
3. If the transaction is finalized for an amount that:
  - a. Exceeds the authorized amount, then the acquirer must send an additional (incremental) authorization request for the unauthorized amount; or
  - b. Is less than the authorized amount, then within 24 hours of finalization, the acquirer must either send a partial reversal for the excess authorized amount, or submit the transaction clearing record.
4. In the case of a transaction canceled by the cardholder, then within 24 hours, the acquirer must send a full reversal request.

### Authorization after charging

If the merchant initiates authorization after the vehicle charging is completed, then the acquirer's authorization request must be identified as a final authorization.

## IFC Blocked Gaming File

Issuers have the option to block their cardholders from participating in In-flight Commerce (IFC) gaming (Authorized Level 4 CAT: in-flight commerce) by listing their BINs in the IFC Blocked

Gaming File. Acquirers must ensure timely delivery and installation of the IFC Blocked Gaming File to gambling service providers. IFC Blocked Gaming File access is required before every gaming transaction.

Gaming transactions are not permitted at IFC terminals acquired within Europe.

**NOTE: The IFC Blocked Gaming File applies only to in-flight commerce gaming, and not to other IFC activity.**

### How to list BINS in the IFC Blocked Gaming File

To list BINs in this file, complete the *IFC Range Blocking for Gaming Transactions* (Form 161).

### Effective dates for the IFC Blocked Gaming File

Effective dates for the IFC Blocked Gaming File are the first and 15th day of each month. Mastercard must receive issuer listings at least two weeks before the effective date to be included in the file.

### Distribution of the IFC Blocked Gaming File

Distribution of the IFC Blocked Gaming File occurs two times each month. Mastercard provides it either via a Mastercard bulk file or via Mastercard File Express. For the file layout, refer to the *IFC Blocked Gaming Transactions* (Form 161).

1. Mastercard creates the IFC Blocked Gaming File at approximately 13:00 (St. Louis time) and distributes it to acquirers' MIPs.
2. Acquirers stage the file for unloading and distribute it to their gaming service providers.
3. Gaming service providers must ensure timely delivery and installation of the IFC Blocked Gaming File to the on-board servers that monitor IFC transactions.

**NOTE: Because Mastercard requires IFC Blocked Gaming File access before every gaming transaction, acquirers and gaming service providers must distribute and install the file when they receive it.**

### Mastercard processing of IFC Gaming Transactions

In-flight commerce gaming transactions go directly to the issuer.

## Cardholder Authentication Service

Mastercard offers an indicator that allows acquirers and issuers to recognize when biometrics are used to authenticate the cardholder.

To assist issuers and governments globally to reduce cost and eliminate fraud in their benefit disbursement processes, Mastercard provides an authentication indicator to support those efforts.

The Cardholder Authentication Service is an optional service for issuers.



The authentication indicator, in association with the Biometric Card solution, indicates that the cardholder is:

The user of the card

Present at the time of the transaction

The Cardholder Authentication Service is an optional service that helps participating issuers reduce fraud and increase approvals through biometric verification of the cardholder's identity. With the Cardholder Authentication Service, participating issuers are notified when a transaction is successfully authenticated through the verification of a cardholder's biological trait, such as a thumbprint, with a biometric-enabled card.

### Data Requirements

Mastercard will populate the indicator in DE 48 (Additional Data: Private Use), subelement 17 (Authentication Indicator) based on the presence of the biometric authentication result values in DE 55 (Integrated Circuit Card [ICC] System-Related Data), subelement 9F10, CVR byte1, bit1 and byte2, bit2.

The biometric authentication result values available for DE 55, subelement 9F10, CVR byte1, bit1 are as follows:

- 0: Not successful
  - 1: Successful
- Mastercard will forward the biometric verification result indicators to the issuer in the following messages:
- Authorization Request/0100
  - Authorization Advice/0120 (acquirer-generated and system-generated)

Mastercard will only populate DE 48, subelement 17 with an authentication indicator if the biometric authentication is successful.

### Authentication Service options

Mastercard will offer the following service type options to support cardholder authentication:

- Authentication Service Type 1
- Authentication Service Type 2

### Authentication Service Type 1

For a specific program, issuers may choose an account range where DE 23 (Card Sequence Number) will be used to differentiate the chip's application (Cardholder Authentication vs. Standard).

- 01x (Cardholder Authentication Application) where x is the sequence number 0 to 9; the transaction is processed for the authentication service.
- 02x (Standard Application) where x is the sequence number 0 to 9; the transaction is processed as a standard transaction.

## Authentication Service Type 2

For a specific program, issuers may choose an account range where a PAN will be used to differentiate the chip's application (Cardholder Authentication vs. Standard).

- PAN#1: The account range associated with PAN #1 will indicate that the Cardholder Authentication application on the chip was launched.
- PAN#2: The account range associated with PAN #2 will indicate that a standard CVM (Cardholder Verification Message) application on the chip was launched.

## Authentication Service Type 1 and Authentication Service Type 2

Mastercard will populate DE 48 (Additional Data: Private Use), subelement 17 (Authentication Indicator), value 1 (Transaction qualified for Authentication Service Type 1) and value 2 (Transaction qualified for Authentication Service Type 2) to indicate that the transaction was authenticated using biometrics.

The Authentication Indicator is forwarded to the acquirer in these messages:

- Authorization Request Response/0110
- Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)

The Authentication Indicator is forwarded to the issuer in these messages:

- Authorization Request/0100
- Authorization Advice/0120 (acquirer-generated and system-generated)

If the Authentication Indicator is not populated in DE 48, subelement 17, the issuer may choose to authorize or decline the transaction.

## Account Status Inquiry with Authentication Indicator

DE 48 (Additional Data: Private Use), subelement 17 (Authentication Indicator) will be populated for Authentication Service Types 1 and 2 based on the following conditions:

- Account Range is registered for Authentication Service Type 1 or 2
- DE 3 (Processing Code), value 00 (Purchase)
- DE 4 (Amount, Transaction) is zero
- Transaction is an EMV Chip with DE 22 (Point-of-Service [POS] Entry Mode) value 05 or 07
- If present, DE 55 (Integrated Circuit Card [ICC] System-Related Data), tag 9F10:
  - CVR byte 1, bit 1 is present with a value of 1 (successful)
  - CVR byte 2, bit 2 is present with a value of 1 (biometric)
- Is an Account Status Inquiry Transaction where DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status Indicator) contains value 8 (Account Status Inquiry Service)

## Authentication Indicator during transactions other than Account Status Inquiry

DE 48, subelement 17 (Authentication Indicator) will be populated for Authentication Service Types 1 and 2 based on the following conditions:

- Account Range is registered for Authentication Service Type 1 or 2
- DE 4 (Amount, Transaction) is not zero
- Transaction is an EMV Chip with DE 22 (Point-of-Service [POS] Entry Mode) value 05 or 07
- If present, DE 55 (Integrated Circuit Card [ICC] System-Related Data), tag 9F10:
  - CVR byte 1, bit 1 is present with a value of 1 (successful)
  - CVR byte 2, bit 2 is present with a value of 1 (biometric)

### Data requirements

Mastercard will add the following in DE 60 (Advice Reason Code):

- Subfield 1 (Advice Reason Code), values in Authorization Advice/0120
  - 160 (Authentication Advice to Issuer)
  - 192 (M/Chip Offline Advice to Issuer)
- Subfield 2 (Advice Detail Code), value 0078 (M/Chip Biometric Data not Present)

The Dual Message Authorization System performs the following system edits on Authorization Request/0100 messages and Authorization Advice/0120: Acquirer-generated messages.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication service; the Authorization Request/0100 message contains:</p> <ul style="list-style-type: none"> <li>• DE 2: Primary Account Number (PAN) setup for Type 1 Cardholder Authentication Service</li> <li>• DE 4 has value of zero, and</li> <li>• DE 23 (Card Sequence Number) contains a value of 01X, and</li> <li>• DE 61, subfield 7 has value of 8</li> <li>• DE 55: Integrated Circuit Card (ICC) System-Related Data is present and tag 9F10:               <ul style="list-style-type: none"> <li>– CVR byte1, bit1 is present with a value other than 1 (successful), or</li> <li>– CVR byte 2 bit 2 is present with the value other than 1 (biometric)</li> </ul> </li> </ul> | <p>sends the acquirer an Authorization Request Response/0110 message where: DE 39 (Response Code) = 05 (Do not honor)</p> <p>and</p> <p>sends an Authorization Advice/0120 message with DE 60 (Advice Reason Code):</p> <ul style="list-style-type: none"> <li>• Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer)</li> <li>• Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not Present)</li> </ul> |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication service; the Authorization Advice/0120: acquirer-generated message contains:</p> <ul style="list-style-type: none"> <li>• DE 2: Primary Account Number (PAN) setup for Type 1 Person Present Service, and</li> <li>• DE 4 has value of zero</li> <li>• DE 61, subfield 7 has value of 8</li> <li>• DE 55: Integrated Circuit Card (ICC) System-Related Data is present and tag 9F10: <ul style="list-style-type: none"> <li>– CVR byte 1, bit1 is present with a value other than 1 (successful), or</li> <li>– CVR byte 2 bit 2 is present with the value other than 1 (biometric)</li> </ul> </li> </ul> | <p>sends the acquirer an Authorization Advice Response/0130 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format Error) and</li> <li>• DE 44 = 055</li> </ul> <p>Transactions with zero as completed amount may not be adjusted</p>                                                                                                                                                                     |
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication service; the Authorization Request/0100 message contains:</p> <ul style="list-style-type: none"> <li>• DE 2: Primary Account Number (PAN) setup for Type 2 Person Present Service, and</li> <li>• DE 4 has value of zero, and</li> <li>• DE 61, subfield 7 has value of 8, and</li> <li>• DE 55: Integrated Circuit Card (ICC) System-Related Data is present and tag 9F10: <ul style="list-style-type: none"> <li>– CVR byte 1, bit 1 is present with a value other than 1 (successful), or</li> <li>– CVR byte 2, bit 2 is present with the value other than 1 (biometric)</li> </ul> </li> </ul>        | <p>sends the acquirer an Authorization Request Response/0110 message where: DE 39 (Response Code) = 05 (Do not honor)</p> <p>and</p> <p>sends an Authorization Advice/0120 message with DE 60 (Advice Reason Code):</p> <ul style="list-style-type: none"> <li>• Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer)</li> <li>• Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not Present)</li> </ul> |
| <p>an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication service; the Authorization Advice/0120: acquirer-generated message contains:</p> <ul style="list-style-type: none"> <li>• DE 2: Primary Account Number (PAN) setup for Type 2 Cardholder Authentication Service, and</li> <li>• DE 4 has value of zero</li> <li>• DE 61, subfield 7 has value of 8</li> <li>• DE 55: Integrated Circuit Card (ICC) System-Related Data is not biometric authenticated</li> </ul>                                                                                                                                                                                           | <p>sends the acquirer an Authorization Advice Response/0130 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format Error)</li> <li>• DE 44 (Additional Response Data) = 055</li> </ul> <p>Transactions with zero as completed amount may not be adjusted</p>                                                                                                                                              |

| WHEN...                                                                                                                                                                                                                                                     | THEN the Dual Message Authorization System...                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 60 in the Authorization Advice/0120: Acquirer-generated messages is not the value 190 (Acquirer Processing System (APS) Approved) or 191 (Acquirer Processing System (APS) Completed Authorization Transaction) or 192 (M/Chip Offline Advice to Issuer) | ends the acquirer an Authorization Advice Response/0130 message where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format Error)</li> <li>DE 44 (Additional Response Data) = 060</li> </ul> |

### Offline Account Status Inquiry on Account ranges set up for the Authentication Indicator

For offline EMV Chip approved Account Status Inquiry only transactions (without purchase, payment, or cash disbursement) on account ranges set up for Authentication Service Type 1 or for Authentication Service Type 2, acquirers must be prepared to send Authorization Advice/0120: Acquirer-generated messages to the issuer with:

- DE 4 (Amount, Transaction), value zero
- DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status Indicator), value 8 (Account Status Inquiry Service)
- DE 60 (Advice Reason Code), new value 192 (M/Chip Offline Advice to Issuer)
- DE 55 (Integrated Circuit Card [ICC] System-Related Data)

Authentication Service Types can be determined by checking Integrated Product Message Mastercard Parameter Extract (MPE) Table IP0040T1: Issuer Account Range containing an Authentication Indicator Value of 1 or 2.

Issuers must support and authenticate DE 55 (Integrated Circuit Card [ICC]) System Related Data).

**NOTE: Issuers that have registered for this transaction authentication service and do not participate in M/Chip Cryptogram Pre-validation Service, will need to validate the Application Cryptogram (tag 9F26 in DE 55) and chip data as in DE 55 for Authentication transactions. In addition to their standard chip data validation processing for chip transactions, issuers may verify Subelement 9F10: Issuer Application Data (IAD), Card Verification Results, Byte 2, Bit 2 will be re-used for CVM type as follows:**

- 0 Standard: non-biometric standard CVM method was used in this transaction**
- 1 Biometric: biometric CVM method was used in this transaction**

## Card Validation Code 1 verification

Card Validation Code 1 (CVC 1) verification provides added security for the card authentication process in authorization request transactions that use magnetic stripe technology.

### Card Validation Code 1

The Card Validation Code 1 (CVC 1) is a three-digit code that must be encoded on Track 1 and Track 2 in three contiguous positions in the Discretionary Data field of the magnetic stripe on all Mastercard Cards.

Issuers may calculate the CVC 1 by one of the following two methods:

- Issuer proprietary calculation: Gives the issuer the option to derive the CVC algorithmically.
- Data Encryption Standard (DES) software: Issuer may perform the calculation through a DES software application within a host system or through use of a tamper-resistant security module (TRSM).

### Card Validation Code 1 on-behalf services

Mastercard provides several Card Validation Code 1 (CVC 1) on-behalf services to support Mastercard products. The CVC 1 on-behalf services are available for transactions that are performed using magnetic stripe technology.

To support on-behalf processing of Mastercard products, the following CVC 1 on-behalf services are available:

- CVC 1 Pre-Validation Service (OBS 11): An optional, pre-validation service performed on authorization request transactions
- CVC 1 Validation Stand-In Authorization (OBS 10): Service in which Stand-In Authorization processes and validates transactions when the issuer is not available; this service is mandatory for issuers with authorization systems that support magnetic stripe transactions.

**NOTE: All issuers must have CVC 1 Validation in Stand-In performed during Stand-In Authorization. Issuers that participate in CVC 1 Pre-Validation Service (OBS 11) do not need to participate in OBS 10 as the results from OBS 11 will be used by Stand-In Authorization.**

Issuers that participate in the CVC 1 On-Behalf Prevalidation Service receive the following information in the Authorization Request/0100 message:

- CVC 1 validation results inserted in DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf Service), value 11 (CVC 1 Pre-Validation Service), and subfield 2 (On-behalf Result 1) with one of the following values:
  - V=Valid
  - I=Invalid CVC 1

- K=No matching key file for this PAN, PAN expiry date combination
- U=Unable to process

CVC 1 validation results for OBS 11

The following table describes the results of the CVC 1 validation for OBS 11 for Dual Message System connected issuers.

Associated edits

| WHEN...                                                                                                                                                                         | THEN the Dual Message Authorization System sends...                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVC 1 Pre-Validation was successfully completed, or when the issuer has elected to receive the validation results for further decisioning, regardless of the validation results | <div>the issuer an Authorization Request/0100 message containing:</div> <ul style="list-style-type: none"><li>• DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf Service), value 11 (CVC 1 Pre-Validation Service), and subfield 2, (On-behalf Result 1) values:<ul style="list-style-type: none"><li>- V (Valid)</li><li>- I (Invalid CVC 1)</li><li>- K (No matching key file for this PAN, PAN expiry date combination)</li><li>- U (Unable to process)</li></ul></li></ul> |

| WHEN...                               | THEN the Dual Message Authorization System sends...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVC 1 Pre-Validation was unsuccessful | <p>the acquirer an Authorization Request Response/0110 message containing:</p> <ul style="list-style-type: none"> <li>• DE 48 (Additional Data: Private Use), subelement 87 (Card Validation Code Result), values: <ul style="list-style-type: none"> <li>– Y = (if pre-validation not successful)</li> </ul> </li> <li>• DE 39 (Response Code) = 05 (Do not honor)</li> </ul> <p>and</p> <p>the issuer an Authorization Advice/0120 message containing one of the following sets of values:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 05 (Do not honor)</li> <li>• DE 48 (Additional Data): Private Use, subelement 87 (Card Validation Code Result) value = Y (Invalid CVC 1)</li> <li>• Reason: An applicable CVC 1 validation key is unavailable: <ul style="list-style-type: none"> <li>– DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf Service) value 11, subfield 2 (Onbehalf Result 1) with value of K (No matching key file for this PAN, PAN expiry date combination)</li> <li>– DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) with a reason code 0047: CVC 1 No matching key file for this PAN, PAN expiry date combination, status unknown</li> </ul> </li> <li>• Reason: An applicable CVC 1 validation key is unavailable: <ul style="list-style-type: none"> <li>– DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf Service) value 11, subfield 2 (On-behalf Result 1) with value of K (No matching key file for this PAN, PAN expiry date combination)</li> <li>– DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) with a reason code 0047: CVC 1 No matching key file for this PAN, PAN expiry date combination, status unknown</li> </ul> </li> <li>• Reason: Mastercard is unable to complete the CVC 1 validation service processing due to a technical issue: <ul style="list-style-type: none"> <li>– DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1</li> </ul> </li> </ul> |



| WHEN... | THEN the Dual Message Authorization System sends...                                                                                                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | (On-behalf Service), value 11, subfield 2 (On-behalf Result 1) with a value of U (Unable to process) <ul style="list-style-type: none"><li>– DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) with a reason code 0036: Reject: CVC 1 Unable to process</li></ul> |

### CVC 1 Validation results for OBS 10

The following table describes the results of the CVC 1 validation for OBS 10 for Dual Message System connected issuers.

#### Associated edits

| WHEN...                                          | THEN Mastercard sends...                                                                                                                                                                                                 |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stand-In Authorization processes the transaction | the issuer an Authorization Advice/0120 message containing: <ul style="list-style-type: none"><li>• DE 48 (Additional Data: Private Use), subelement 87 (Card Validation Code Result), value Y (Invalid CVC 1)</li></ul> |

## Participating in CVC 1 on-behalf services

CVC 1 on-behalf services are currently available to issuers that process through the Dual Message System.

**NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such product and services in a timely manner in the event of a security issue.**

### CVC 1 Validation Stand-In Authorization (OBS 10)

This service is mandatory for issuers with authorization systems that support magnetic stripe transactions.

### CVC 1 Pre-Validation Service (OBS 11)

Issuers that choose to participate in the optional CVC 1 Pre-Validation Service (OBS 11) must register by contacting their CIS representative.

**NOTE: Issuers that enroll account ranges in the CVC 1 Pre-Validation Service that are already enrolled in the existing CVC 1 Validation Stand-In Authorization must request the removal of those account ranges from the existing CVC 1 Validation Stand-In Authorization.**

## Alternate processing of invalid CVC 1 validation results

Mastercard considers the results of the CVC 1 validation in:

- DE 48, subelement 71 (On-behalf Services) when responding to the Authorization Request/0100 message for issuers participating in pre-validation services.
- DE 48, subelement 87 (Card Validation Code Result) when responding to the Authorization Request/0100 message for issuers participating in the Stand-In validation service.

Issuers participating in one of the CVC 1 on-behalf services should reference the KMS interface section defined in [On-behalf Key Management Procedures](#) and [On-behalf Key Management Interface Specifications](#) for information about CVC 1 validation keys.

## Authorization reports for CVC 1 on-behalf services

The Authorization Parameter Summary Report (SI737010-AA) lists the name of the on-behalf service in which the issuer is participating for an account range and displays the issuer selection for participating in the Card Validation Code 1 (CVC 1) processing service.

For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#). For a sample of the Daily Account File Activity Report (AM700010-DD), refer to the [Account Management System User Manual](#).

## Card Validation Code 2 verification

Card validation code 2 (CVC 2) verification provides added security for the card authentication process in authorization requests.

### CVC 2

CVC 2 is a three-digit code algorithmically derived by the issuer and indent-printed on the signature panel to the right of the account number.

CVC 2 is one of several card authentication methods currently used by Mastercard to combat fraud. The use of CVC 2 deters fraudulent use of an account number.

## Conditions for CVC 2 verification

CVC 2 verification allows acquirers to receive online verification of CVC 2 values from issuers when certain conditions apply.

- The acquirer transmits the CVC 2 value that the cardholder provided to the merchant in DE 48 (Additional Data: Private Use), subelement 92 (CVC 2) of the Authorization Request/0100 message.
- The issuer provides a valid response in DE 48, subelement 87 (Card Validation Code Result) of the Authorization Request Response/0110 message to indicate that the CVC 2 value was verified (value M or N) or not processed (value P).

## Participation requirements

Participation requirements for acquirers and issuers using CVC 2 verification are listed in table format.

### Acquirers

When cardholders provide merchants with the CVC 2 value for transactions, acquirers must include the CVC 2 value in DE 48, subelement 92 of the Authorization Request/0100 message.

To participate in CVC 2 verification, acquirers must be able to receive the following CVC 2 values in DE 48, subelement 87 of the Authorization Request Response/0110 message.

### Values

| Subelement 87 Value | Description                                         |
|---------------------|-----------------------------------------------------|
| M                   | Valid CVC 2: match                                  |
| N                   | Invalid CVC 2: non-match                            |
| P                   | CVC 2 not processed: issuer temporarily unavailable |
| U                   | CVC 2 unverified: Mastercard use only               |

The acquirer is responsible for ensuring that the merchant receives the CVC 2 response code provided by the issuer in DE 48, subelement 87 of the Authorization Request Response/0110 message.

### Issuers

Issuers must be able to receive and to process the CVC 2 value when present in DE 48, subelement 92 of the Authorization Request/0100 message, and to provide a valid CVC 2 response in DE 48, subelement 87 of the Authorization Request Response/0110 message.

CVC 2 values M, N, and P are considered valid when used by issuers. Only Mastercard can use CVC 2 value U.

The Dual Message Authorization System places a value of U in DE 48, subelement 87 of the Authorization Request Response/0110 message to indicate that the CVC 2 value was not verified, at the issuer's request, as the issuer was temporarily unable to receive the CVC 2 value.

Issuers should note that if an acquirer transmits both CVC 1 and CVC 2 data, CVC 1 processing takes precedence over CVC 2 processing.

To comply with card validation code 2 (CVC 2) requirements, acquirers and issuers must process transactions according to the guidelines in this topic. The following message flows describe authorization message processing for CVC 2 verification transaction in these scenarios when the:

- CVC 2 is verified
- CVC 2 is unverified (because the issuer was temporarily unable to receive the CVC 2 value)
- CVC 2 is processed by Stand-In Authorization
- CVC 2 is processed by X-Code Processing

Applicable edits

| WHEN the CVC 1 value is... | THEN...                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| invalid                    | issuers should respond with value Y (Invalid CVC 1) in DE 48, subelement 87. Issuers should not verify the CVC 2 value. |
| valid                      | issuers must verify the CVC 2 value and send the appropriate response to the acquirer.                                  |

Authorization Request/0100: CVC 2 Verified

Following are the stages of an Authorization Request/0100: CVC2 transaction when the value is verified.

1. The merchant provides the CVC 2 value, and the acquirer generates an Authorization Request/0100 message with the CVC 2 value in DE 48 (Additional Data: Private Use), subelement 92 (CVC 2).
2. The Dual Message Authorization System forwards the Authorization Request/0100 message to the issuer.
3. After the issuer transmits the Authorization Request Response/0110 message, the Dual Message Authorization System verifies that one of the following values appears in DE 48, subelement 87 (Card Validation Code Result):
  - M (Valid CVC 2: match)
  - N (Invalid CVC 2: non-match)
  - P (CVC 2 not processed: issuer temporarily unavailable)

### Associated edits

| IF DE 48, subelement 87 contains... | AND IF the transaction was... | THEN the Dual Message Authorization System...                                                                                                                                                                    |
|-------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| no response or an invalid response  | not approved                  | forwards the Authorization Request Response/0110 message to the acquirer with DE 48, subelement 87 = P and<br><br>the issuer will not receive an Authorization Response Negative Acknowledgment/0190 message     |
| no response or an invalid response  | approved                      | forwards the Authorization Request/0100 message to the alternate authorization service provider for processing and<br><br>the issuer will receive an Authorization Response Negative Acknowledgment/0190 message |

## Authorization Request/0100: CVC 2 Processed by Stand-In

Following are the stages of an Authorization Request/0100: CVC 2 transaction processed by Stand-In Authorization.

1. The merchant provides the CVC 2 value, and the acquirer generates an Authorization Request/0100 message with the CVC 2 value in DE 48, subelement 92.
2. Stand-In Authorization responds in the following conditions:
  - The issuer is not signed in.
  - The transaction cannot be delivered to the issuer.
  - The issuer is not responding.
  - The issuer's Authorization Request Response/0110 message fails an edit check and is rejected.
3. Stand-In Authorization generates the Authorization Request Response/0110 message.

### Associated edits

| WHEN the issuer is...                         | THEN the Dual Message Authorization System...                                                                                                                              |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| temporarily unable to receive the CVC 2 value | places value U (CVC 2 unverified: Mastercard use only) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer.               |
| temporarily unable to process the CVC 2 value | places value P (CVC 2 not processed, issuer temporarily unavailable) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer. |

4. The acquirer transmits the CVC 2 response code, provided by the issuer or Mastercard in DE 48, subelement 87 of the Authorization Request Response/0110 message, to the merchant. The acquirer is responsible for ensuring that the merchant receives the CVC 2 response code.
5. Stand-In Authorization also generates Authorization Advice/0120 messages as appropriate and stores the advice in the Store-and-Forward (SAF) queue. Stand-In Authorization will include value U or P in DE 48, subelement 87 for issuers that are temporarily unavailable or otherwise unable to process the transaction.

## Authorization Request/0100: CVC 2 Processed by X-Code

Following are the stages of an Authorization Request/0100: CVC 2 transaction processed by X-Code Processing.

1. The merchant supplies the CVC 2 value to acquirer.
2. The acquirer creates an Authorization Request/0100 message with the CVC 2 value, but the transmission is not successful.
3. The X-Code Processing by the acquirer's MIP responds based on current X-Code Processing rules.

### Associated edits

| WHEN the issuer is...                         | THEN the Dual Message Authorization System...                                                                                                                |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temporarily unable to receive the CVC 2 value | Places value U (CVC 2 unverified: Mastercard use only) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer. |

| WHEN the issuer is...                         | THEN the Dual Message Authorization System...                                                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temporarily unable to process the CVC 2 value | Places a P (CVC 2 not processed, issuer temporarily unavailable) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer. |

- The acquirer transmits the CVC 2 response code, provided by Mastercard in DE 48, subelement 87 of the Authorization Request Response/0110 message, to the merchant. The acquirer is responsible for ensuring that the merchant receives the CVC 2 response code.
- The Dual Message Authorization System sends an Authorization Advice/0120 message to the issuer.

## Authorization Request/0100: CVC 2

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Applicable data elements

| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                    |
|-----------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 14 (Date, Expiration)                                              | C   | .   | C   | Required for issuers to validate CVC 2 value                                                                                                                                       |
| DE 48 (Additional Data: Private Use), TCC (Transaction Category Code) | M   | .   | M   | Must contain the appropriate TCC code.                                                                                                                                             |
| DE 48, subelement 87 (Card Validation Code Result)                    | .   | X   | C   | If an issuer's BIN is temporarily unable to receive the CVC 2 value, the Dual Message Authorization System inserts the following value: U = CVC 2 unverified (Mastercard use only) |
| DE 48, subelement 92 (CVC 2)                                          | C   | .   | C   | Must be the three-digit CVC 2 value sent by the merchant to the acquirer.                                                                                                          |

## Authorization Request Response/0110: CVC 2

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Applicable data elements

| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data: Private Use), TCC (Transaction Category Code) | ME  | •   | ME  | Must be present for registered CVC 2 issuers. Must also be present for unregistered CVC 2 issuers if the issuer is providing additional subelements (such as AVS response.)<br><br>Must be present for all issuers.                      |
| DE 48, subelement 87 (Card Validation Code Result), CVC 1 and CVC 2   | M   | •   | M   | Contains one of the following CVC 2 response codes:<br><br>M = Valid CVC 2 (match)<br><br>N = Invalid CVC 2 (non-match)<br><br>P = CVC 2 not processed (issuer temporarily unavailable)<br><br>U = CVC 2 Unverified: Mastercard Use Only |
| DE 48, subelement 92 (CVC 2)                                          | CE  | X   | CE  | Contains the CVC 2 code provided by the merchant to the acquirer.                                                                                                                                                                        |



## Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edit for CVC 2.

### Applicable edits

| IF DE 48, subelement 87 (Card Validation Code Result) contains... | AND if the transaction was... | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| no response or an invalid response                                | approved                      | <p>Mastercard forwards the Authorization Request Response/0110 message to the acquirer with DE 48, subelement 87, value P (CVC 2 not processed, issuer temporarily unavailable)</p> <p>the issuer will not receive an Authorization Response Negative Acknowledgment/0190 message</p> |

## Authorization Request/0100: CVC 2 Unverified

Following are the stages of an Authorization Request/0100: CVC 2 unverified transaction (issuer temporarily unable to receive the CVC 2 value).

1. The merchant provides the CVC 2 value, and the acquirer generates an Authorization Request/0100 message with the CVC 2 value in DE 48, subelement 92.
2. The Dual Message Authorization System forwards the Authorization Request/0100 message to the issuer, but the issuer is temporarily unable to receive the CVC value. The MIP forwards the Authorization Request/0100 message to the issuer without DE 48, subelement 92.
3. After the issuer transmits the Authorization Request Response/0110 message, the Dual Message Authorization System places value U (CVC 2 unverified: Mastercard use only) in DE 48, subelement 87 and forwards it to the acquirer.
4. The acquirer transmits the CVC 2 response code, provided by Mastercard in DE 48, subelement 87 of the Authorization Request Response/0110 message, to the merchant. The acquirer is responsible for ensuring that the merchant receives the CVC 2 response code.

## CVC 2 DE 48 Structure

The following diagram illustrates DE 48 subelements related to CVC 2 transactions.

### CVC 2 DE 48 Structure

| LLL                       | "VAR"  |          |              |             |          |              |                             |
|---------------------------|--------|----------|--------------|-------------|----------|--------------|-----------------------------|
| 3 bytes                   | 1 byte | 2 bytes  | 2 bytes      | 3 bytes     | 2 bytes  | 2 bytes      | 1 byte                      |
| Total Data Element Length | TCC    | SE ID 92 | SE Length 03 | CVC 2 Value | SE ID 87 | SE Length 01 | Card Validation Code Result |

## Card Validation Code 3 (CVC 3) verification

Card validation code 3 (CVC 3) verification provides added security to validate whether a Mastercard® contactless card or device is legitimate.

Mastercard supports card validation code 3 (CVC 3) result data in Authorization Request Response/0110 messages using DE 48, subelement 87 (Card Validation Code Result). Issuers performing CVC 3 validation in-house on contactless transactions may use these CVC 3 values. When the CVC 3 is valid, the Authorization Request Response/0110 message should not contain DE 48, subelement 87.

## CVC 3

CVC 3 is a code algorithmically derived by a contactless card or device.

### Authorization Request Response/0110: CVC 3 Result

Mastercard encourages issuers to include DE 48, subelement 87 when they perform CVC 3 validation and there is an issue with the CVC 3. When the CVC 3 is not valid, the issuer should send the acquirer the Authorization Request Response/0110 message containing CVC 3 result data.

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

## Applicable data elements

| Data Element                                                                      | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data: Private use), subelement 87 (Card Validation Code Result) | C   | .   | C   | <p>Contains one of the following CVC 2 response codes:</p> <ul style="list-style-type: none"> <li>E = Length of unpredictable number was not a valid length</li> <li>P = Unable to process</li> <li>Y = Invalid</li> </ul> |

## CVC 3 on-behalf services

Mastercard provides CVC 3 on-behalf services to support contactless products. The CVC 3 on-behalf services are available for transactions that derive from proximity chip functionality with magnetic stripe Track 1 or Track 2 data containing a CVC 3 value.

The following CVC 3 on-behalf services are available:

- Dynamic CVC 3 Pre-validation Service: This is an optional, stand-alone service for issuers with contactless-enabled authorization systems that do not support dynamic CVC 3 validation. The Contactless Mapping Service may be used with the Dynamic CVC 3 Pre-validation Service.  
Mastercard will perform the dynamic CVC 3 validation on behalf of the issuer before forwarding the Authorization Request/0100 message. If the issuer is unavailable, Mastercard will consider the results of the dynamic CVC 3 validation when the transaction is processed by Stand-In Authorization.
- Dynamic CVC 3 Validation in Stand-In Authorization: This service is mandatory for issuers with contactless-enabled authorization systems that support dynamic CVC 3 validation in order to provide dynamic CVC 3 validation when the issuer is not available and the transaction is processed by Stand-In Authorization.

**NOTE: All issuers globally that participate in Stand-In Authorization must have Dynamic CVC 3 Validation in Stand-In performed during Stand-In Authorization.**

The issuer receives the following information in the Authorization Request/0100 message:

- The result of the CVC 3 validation inserted in DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 2 (On-behalf Result 1) with one of the following values:
  - A = ATC outside allowed range (applicable when ATC value is dynamic [varying] value)
  - E = (CVC 3 ATC Replay)
  - I = (Invalid CVC 3)
  - K = No matching key file for this PAN, PAN expiry date, and KDI combination

- N = (Unpredictable Number Mismatch)
- U = (Unable to process)
- V = (Valid)
- X = Security platform timeout
- Z = Security platform system error
- ATC information inserted in DE 48, subelement 34 (ATC Information) when DE 48, subelement 71 is value A, E, or V.

If the results are invalid, Mastercard uses the issuer's instructions from the decision matrix as set in the key file. For more information about CVC 3 validation, issuers should reference the Key Management Services (KMS) interface section in the [On-behalf Key Management Procedures](#), [On-behalf Key Management Interface Specifications](#), and [Contactless On-behalf Services Guide](#).

The acquirer receives the following CVC 3 data in the Authorization Request Response/0110 message containing DE 48, subelement 87 (Card Validation Code Result):

- E (Length of unpredictable number was not a valid length)
- P (Unable to process)
- Y (Invalid)

If the CVC 3 data is valid, the Dual Message Authorization System will not include DE 48, subelement 87 in the Authorization Request Response/0110 message to the acquirer.

The issuer receives the following information in the Authorization Advice/0120 message when the Stand-In Authorization responded on behalf of the issuer or when the issuer requested the Optional Non-valid CVC 3 Processing:

- DE 48, subelement 71 (On-behalf Services) identifies the service performed and the result.
- DE 48 (Additional Data: Private Use), subelement 34 (Dynamic CVC 3 ATC Information) is present when the on-behalf service result in subelement 71, subfield 2 (On-behalf Result 1) was value A, E, or V.
- DE 48, subelement 87 (Card Validation Code Result) is present when the CVC 3 validation result was not valid.
- DE 60 (Advice Reason Code) identifies the specific reason for the advice message.

## Optional non-valid CVC 3 processing

Mastercard provides issuers the option to have the Dual Message Authorization System respond to the Authorization Request/0100 messages on their behalf when the CVC 3 value is not valid. Issuers receive an Authorization Advice/0120: System-generated message when the Dual Message Authorization System responds.

This option only is available to issuers participating in the Contactless Mapping Service or the Dynamic CVC 3 Pre-validation Service.

## Overview

The Contactless Application Transaction Counter (ATC) file (MCC109) contains the most recent ATC values for cards supporting EMV Contact, EMV Contactless, and Contactless Magstripe transactions. It is used in the scope of EMV Cryptogram and Dynamic CVC 3 Pre-validation Services and Validation in Stand-In Services.

Issuers may need to use the MCC109 ATC file update when the ATC value of the card gets out of sync compared to the value recorded in the Mastercard MCC109 database. It allows them to cover any ATC value discrepancies that are created when transactions are not routed through the Mastercard Network.

In the context of Dynamic CVC 3 pre-validation and validation in Stand-In Services, issuers must use the MCC109 ATC file update when the ATC length value in the discretionary data of the card is fewer than four digits.

Mastercard supports entry of the ATC data in DE 101 (File Name), value (MCC109) via the following methods:

- Online File Maintenance
- Bulk File Maintenance
- Mastercard eService Maintenance

MCC109 ATC maintenance requests submitted by the Issuer File Update Request/0302 message or Mastercard eService are applied immediately. Maintenance requests submitted by bulk file are applied one time per day at 18:00 hours (Central Time).

## Alternate processing

Mastercard considers the results of the CVC 3 validation in DE 48, subelement 71 (On-behalf Services) when responding to the Authorization Request/0100 message for issuers participating in the Dynamic CVC 3 Pre-validation or Dynamic CVC 3 Validation in Stand-In Authorization. If the results are invalid, Mastercard uses the issuer's instructions from the decision matrix as set in the key file.

Issuers participating in one of the CVC 3 on-behalf services should reference the KMS interface section defined in [On-behalf Key Management Procedures](#) and [On-behalf Key Management Interface Specifications](#) for information about dynamic CVC 3 validation keys.

## Authorization reports

These reports provide information about CVC 3 on-behalf services.

- The Authorization Parameter Summary Report (SI737010-AA) lists the name of the on-behalf service in which the issuer is participating for an account range and displays the issuer selection for participating in the optional, non-valid CVC 3 processing service.
- The Daily Account File Activity Report (AM700010-DD) contains the Contactless Application Transaction Counter (ATC) File activity.

For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#). For a sample of the Daily Account File Activity Report (AM700010-DD), refer to the [Account Management System User Manual](#).

## To participate

CVC 3 on-behalf services are currently available to issuers that process through the Dual Message System.

For more information about CVC 3 on-behalf services and how to participate, refer to the [Contactless On-behalf Services Guide](#).

### For more information

For details about data requirements, refer to [Data element definitions and details](#).

## Card Validation Code 3

Mastercard supports card validation code 3 (CVC 3) result data in Authorization Request Response/0110 messages using DE 48, subelement 87 (Card Validation Code Result). Issuers performing CVC 3 validation in-house on contactless transactions may use these CVC 3 values. When the CVC 3 is valid, the Authorization Request Response/0110 message should not contain DE 48, subelement 87.

### Authorization Request/0100: CVC 3

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

The Authorization Request/0100 message will contain CVC 3 values in DE 45 (Track 1 Data) or DE 35 (Track 2 Data) following the configuration provided by the issuer. The Mastercard contactless account number is entered via contactless magnetic stripe where DE 22 (Point of Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is 91.

Mastercard will apply the appropriate algorithm to DE 45 or DE 35 using the parameter data provided by the issuer to determine the validity of the CVC 3.

## Data elements

| Data Element                                                                                                 | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (Point of Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)                          | M   | •   | M   | 91 = PAN auto-entry via contactless magnetic stripe—the full track data has been read from the data on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.<br><br>The Mastercard contactless account number is entered via contactless magnetic stripe.                                                                                                      |
| DE 35 (Track 2 Data)                                                                                         | C   | •   | C   | Contains CVC 3 values following the configuration by the issuer.                                                                                                                                                                                                                                                                                                                                                                                     |
| DE 45 (Track 1 Data)                                                                                         | C   | •   | C   | Contains CVC 3 values following the configuration by the issuer.                                                                                                                                                                                                                                                                                                                                                                                     |
| DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) | M   | •   | M   | 15 = Dynamic CVC 3 Pre-validation (with or without Contactless Mapping Service)<br><br>16 = Dynamic CVC 3 Validation in Stand-In Authorization                                                                                                                                                                                                                                                                                                       |
| DE 48, subelement 71, subfield 2 (On-behalf [OB] Result 1) when subfield 1 contains value 15 or 16           | M   | •   | M   | A = ATC outside allowed range (applicable when ATC value is dynamic [varying value])<br><br>E = CVC 3 ATC Replay<br><br>I = Invalid CVC 3<br><br>N = Unpredictable Number Mismatch (applicable when the UN is dynamic [varying value]). (Indicates that the number/length in the discretionary data in DE 45 or DE 35 does not match the number/length provided by the issuer during personalization.)<br><br>U = Unable to process<br><br>V = Valid |
| DE 48, subelement 71, subfield 3 (On-behalf [OB] Result 2)                                                   | M   | •   | M   | This subfield is for Mastercard use only.                                                                                                                                                                                                                                                                                                                                                                                                            |

## Authorization Request Response/0110: CVC 3 Result

Mastercard encourages issuers to include DE 48, subelement 87 when they perform CVC 3 validation and there is an issue with the CVC 3. When the CVC 3 is not valid, the issuer should send the acquirer the Authorization Request Response/0110 message containing CVC 3 result data. Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Applicable data elements

| Data Element                                                                      | Org | Sys | Dst | Values/Comments                                                                                   |
|-----------------------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data: Private use), subelement 87 (Card Validation Code Result) | C   | .   | C   | E = Length of unpredictable number was not a valid length<br>P = Unable to process<br>Y = Invalid |

## Contactless CVC 3 Processing Service

Mastercard provides contactless processing services to assist issuers processing contactless transactions with the validation of the Card Validation Code 3 (CVC 3) to ensure the values provided by the acquirer match the issuer's expected values.

Mastercard offers the following on-behalf services:

- **Dynamic CVC 3 Pre-validation Service:** This is an optional, stand-alone service for issuers with contactless-enabled authorization systems that do not support dynamic CVC 3 validation. The Contactless Mapping Service may be used with the Dynamic CVC 3 Pre-validation Service.

Mastercard will perform the dynamic CVC 3 validation on behalf of the issuer before forwarding the Authorization Request/0100 message. If the issuer is unavailable, Mastercard will consider the results of the dynamic CVC 3 validation when the transaction is processed by the Stand-In Authorization.

- **Dynamic CVC 3 Validation in Stand-In Authorization Service:** This is a mandated service for issuers with contactless-enabled authorization systems that support dynamic CVC 3 validation in-house that would like Mastercard to provide dynamic CVC 3 validation when the issuer is not available and the transaction is processed by the Stand-In Authorization.

The Contactless Mapping Service is for issuers with non-contactless-enabled authorization systems that are not able to support the processing of a separate contactless account number and the validation of dynamic CVC 3 value.

Participation in any CVC 3 on-behalf service or services is defined by account range. Validation will be facilitated by information the issuer provides to Mastercard. Issuers will be requested to



provide confidential key data to Mastercard that will be critical in the validation process (unless Mastercard arranges for devices to be sent to cardholders on behalf of issuers).

## Dynamic CVC 3 Application Transaction Counter (ATC) Processing

The ATC is used in the calculation of the dynamic CVC 3 cryptogram. The ATC counter value is managed by the chip on a contactless card that increments with every authorization request that occurs at a contactless-enabled terminal.

The Dual Message Authorization System supports DE 48, subelement 71 (On-behalf Service), subfield 2 (On-behalf [OB] Result 1), value E (CVC 3 ATC Replay) in the Authorization Request/0100 message to indicate to the issuer that the ATC value has been used in a previous transaction.

When the result of Dynamic CVC 3 validation in DE 48, subelement 71, subfield 2 is V (Valid) or A (ATC outside allowed range), the Dual Message Authorization System compares the transaction's ATC value to the stored list of ATC values for each contactless account number/sequence number/expiration date combination.

If the ATC used in validating the CVC 3 value is not in the stored list of ATC values, the Dual Message Authorization System adds the transaction's five-digit ATC value to the stored list of ATC values.

If the ATC used in validating the CVC 3 value is in the stored list of ATC values, the Dual Message Authorization System replaces DE 48, subelement 71, subfield 2, value V or A with value E in the Authorization Request/0100 message to the issuer. The Dual Message Authorization System does not add the transaction's ATC again to the stored list of ATC values.

Replacing the DE 48, subelement 71, subfield 2 value V or A, with value E does not affect any data contained in the message that supports value V or A. For example, DE 48, subelement 34 supports value A, but remains unchanged if value E replaces DE 48, subelement 71, subfield 2, value A.

The Dual Message Authorization System populates DE 48, subelement 87 (Card Validation Code Result) value Y (Invalid) in the Authorization Request Response/0110 message to the acquirer.

**NOTE: The Dual Message Authorization System only adds to the list of stored ATC values for transactions that have been processed online through the Mastercard Dual Message Authorization System. ATC values for transactions processed offline are not available for the Dual Message Authorization System to use in CVC 3 ATC replay processing.**

## Dynamic CVC 3 Application Transaction Counter (ATC) Information

DE 48, subelement 34 (Dynamic CVC 3 ATC Information) provides issuers with information about the ATC value derived for use in dynamic CVC 3 validation processing.

DE 48, subelement 34 will be conditionally present in Authorization Request/0100 messages and Authorization Advice/012: system-generated messages when the issuer participates in the

Dynamic CVC 3 Pre-validation service or the Dynamic CVC 3 Validation in Stand-In Authorization service.

The Dual Message Authorization System will provide DE 48, subelement 34 when the Dynamic CVC Pre-validation service or the Dynamic CVC 3 Validation in Stand-In Authorization service is performed and validation results in DE 48, subelement 71 (On-behalf Services), subfield 2 (On-behalf [OB] Result 1) contain the value A (ATC outside allowed range), E (CVC 3 ATC Replay), or V (Valid). DE 48, subelement 34 also may be sent as part of the Contactless Mapping service if implemented with Dynamic CVC 3 Pre-validation.

When responding to the Authorization Request/0100 message, issuers should consider the results of the CVC 3 validation in DE 48, subelement 71, subfield 2 and DE 48, subelement 34 (if present).

## MCC109 (Application Transaction Counter File)

The MCC109 (Application Transaction Counter [ATC] file) contains a range of ATC values for a contactless account number/card sequence number/expiration date combination for contactless cards that were personalized using dynamic values in the ATC and unpredictable number (UN). Issuers participating in the Dynamic CVC 3 Pre-validation service or the Dynamic CVC 3 Validation in Stand-In Authorization service may provide this file to Mastercard. Issuers may only add to or inquire about this file.

Mastercard supports entry of the ATC data in DE 101 (File Name), value (MCC109) via the following methods:

- Issuer File Update/0302 messages
- Bulk file request (R311)
- Mastercard eService

MCC109 (Application Transaction Counter [ATC] File) maintenance requests submitted via the Issuer File Update Request/0302 message or Mastercard eService are applied immediately. Maintenance requests submitted by bulk file are applied one time per day at 18:00 (Central Time).

**NOTE: This functionality does not apply to contactless cards and devices that were personalized to send zeros in the ATC and UN.**

## Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on Issuer File Update Request/0302 messages when DE 101 (File Name) contains value MCC109 and DE 120 (Record Data) contains the layout for the contactless ATC File.

### Edits

| Field ID | Field Name                                              | Authorization Edit                                                                                                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Contactless Account Number                              | <ul style="list-style-type: none"> <li>Contactless account number must be present</li> <li>Contactless account number is numeric</li> <li>Contactless account number prefix is valid</li> <li>Contactless account number check digit is correct</li> <li>Contactless account number required in the Application Transaction Counter (ATC) File (MCC109) for additions and inquiries</li> </ul> |
| 2        | Card Sequence Number                                    | <ul style="list-style-type: none"> <li>Card sequence number must be present</li> <li>Card sequence number is numeric</li> <li>Value may be zero</li> <li>Card sequence number is required in the ATC File (MCC109) for additions and inquiries</li> </ul>                                                                                                                                      |
| 3        | Contactless Account Expiration Date                     | <ul style="list-style-type: none"> <li>Contactless Account Expiration Date is valid YYMM format</li> <li>Contactless Account Expiration Date is required in the ATC File (MCC109) for additions and inquiries</li> </ul>                                                                                                                                                                       |
| 4        | Contactless Application Transaction Counter (ATC) Value | <ul style="list-style-type: none"> <li>Contactless ATC is numeric (may be zero)</li> <li>Contactless ATC is required in the ATC File (MCC109) for a additions and inquiries</li> </ul>                                                                                                                                                                                                         |

**NOTE: Fields 1 through 3 are mandatory for an inquiry. Fields 1 through 4 are mandatory for an add.**

## Card Validation Code Result

The acquirer will receive DE 48, subelement 87 when the CVC 3 validation result was not valid.

Acquirers may receive the following:

- Y = Indicates the ATC was determined to be outside the allowed range specified by the issuer, the ATC was determined to be a replay, or the CVC 3 value was determined to be invalid.
- E = Indicates the length of the unpredictable number was not a valid length resulting in an Unpredictable Number (UN) that was not valid.
- P = Indicates Mastercard was unable to process the CVC 3 validation.

If the CVC 3 data is valid, the Dual Message Authorization System will not include DE 48, subelement 87 in the Authorization Request Response/0110 message to the acquirer.

The issuer will receive the Authorization Advice/0120 message, DE 48, subelement 87 when CVC 3 was invalid in the Authorization Request/0100 message.

### Optional Non-Valid CVC 3 Processing

Issuers may elect to have the Dual Message Authorization System respond to Authorization Request/0100 messages on their behalf when the CVC 3 value is not valid. The issuer will receive an Authorization Advice/0120: system-generated message when the Dual Message Authorization System responds.

For information about how to participate, refer to the [Contactless On-behalf Services Guide](#).

The following table describes the process that occurs when the issuer wants the Dual Message Authorization System to respond to Authorization Request/0100 messages on its behalf during CVC 3 validation.

#### Associated edits

| WHEN the CVC 3 validation results contain a value of... | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A, E, I, N, or U                                        | <p>sends an Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"><li>• DE 39 = 05 (Do not honor)</li><li>• DE 48, subelement 87 = E, P, or Y</li></ul> <p>generates an Authorization Advice/0120 message to the issuer containing:</p> <ul style="list-style-type: none"><li>• DE 39, value 05 (Do not honor)</li><li>• DE 48, subelement 34 (if created)</li><li>• DE 48, subelement 71 with the appropriate values</li><li>• DE 48, subelement 87, value E, P, or Y</li><li>• DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code), value 115 (Transaction processed via MIP On-behalf Service Decision)</li><li>• DE 60, subfield 2 (Advice Detail Code), one of the following values:<ul style="list-style-type: none"><li>– 0042 (CVC 3 Unable to process)</li><li>– 0043 (CVC 3 ATC outside allowed range)</li><li>– 0044 (CVC 3 Invalid)</li><li>– 0045 (CVC 3 Unpredictable number mismatch)</li><li>– 0046 (CVC 3 ATC Replay)</li></ul></li></ul> |

| WHEN the CVC 3 validation results contain a value of... | THEN the Dual Message Authorization System...                        |
|---------------------------------------------------------|----------------------------------------------------------------------|
| V                                                       | forwards the Authorization Request/0100 to the issuer for processing |

## Alternate Processing

Mastercard supports issuers subscribing to the Dynamic CVC 3 Pre-validation services that are not available to respond to the Authorization Request/0100 message and issuers subscribing to the Dynamic CVC 3 Validation in Stand-In Authorization service by responding to the authorization message on behalf of the issuer. Mastercard will consider the results of the CVC 3 validation in DE 48, subelement 71 when responding to the Authorization Request/0100.

The values in DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) are dependent upon the values contained in DE 48, subelement 71, subfield 2 (OB Result 1) as described below.

### Associated edits

| IF DE 48, subelement 71, subfield 2 contains the value... | THEN DE 60, subfield 2 will contain...                                                 |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------|
| A                                                         | 0043 (CVC 3 [ATC outside allowed range])                                               |
| E                                                         | 0046 (CVC 3 ATC Replay)                                                                |
| I                                                         | 0044 (CVC 3 [Invalid])                                                                 |
| K                                                         | 0037 (Reject: No matching key file for this PAN, PAN expiry date, and KDI combination) |
| N                                                         | 0045 (CVC 3 [Unpredictable number mismatch])                                           |
| U                                                         | 0042 (CVC 3 [Unable to process])                                                       |
| X                                                         | 0038 (Reject: Security Platform Time Out)                                              |
| Z                                                         | 0040 (Reject: Security Platform System Error)                                          |

MIP X-Code Processing does not perform CVC 3 validation. MIP X-Code Processing will respond based on the type of transaction and the MIP X-Code limits defined.

## Cirrus and Maestro transaction processing

The Dual Message Authorization System supports Cirrus® (CIR) and Maestro® (MSI) transaction processing.

**NOTE: Applies only to the Europe region.**

The Dual Message Authorization System supports Cirrus® (CIR) and Maestro® (MSI) transaction processing. Mastercard supports both acquiring and issuing activity on the Dual Message Authorization System for CIR and MSI card products.

Both acquiring and issuing CIR and MSI transactions are limited to Europe region acquirers and issuers that process through the Mastercard Network.

### Cirrus

Cirrus (CIR), as a brand, is a Mastercard card brand designed exclusively for use at ATMs. Cirrus, as a network, is a wholly owned subsidiary of Mastercard International Incorporated that operates the international ATM sharing association known as the Mastercard® ATM Network. CIR is one of the brands accepted by the Cirrus® network.

The following transaction types can be initiated at the ATM and are indicated by the contents of DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) for CIR transactions:

- 00 (Purchase)
- 01 (Withdrawal)
- 28 (Payment Transaction)
- 30 (Balance Inquiry)
- 91 (PIN Unblock)
- 92 (PIN Change)

Although the Dual Message Authorization System supports CIR transaction acquiring, only specified acquirers have this functionality. To ensure transactions are submitted by eligible acquirers, the Dual Message Authorization System verifies the acquirer parameter when an authorization transaction is branded as CIR.

### Maestro

Maestro (MSI) is a Mastercard brand for debit card products that may be used both at an ATM and at a point-of-service (POS) terminal. POS transactions include face-to-face and electronic commerce (e-commerce) transactions. Following are the valid values for DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) for MSI transactions:

- 00 (Purchase)
- 01 (Withdrawal)
- 09 (Purchase with Cash Back)
- 20 (Purchase Return/Refund)
- 28 (Payment Transaction)

- 30 (Balance Inquiry)
- 91 (PIN Unblock)
- 92 (PIN Change)

For POS processing, Mastercard has defined specific POS processing criteria for MSI transactions. DE 35 (Track 2 Data) must be present in the POS transaction. For Maestro e-commerce transactions, if the acquirer could not construct DE 35, the Dual Message Authorization System will build DE 35 using the contents of DE 2 (Primary Account Number [PAN]) and DE 14 (Date, Expiration). Although the Dual Message Authorization System supports MSI transaction acquiring, only specified acquirers have this functionality. To ensure transactions are submitted by eligible acquirers, the Dual Message Authorization System verifies the acquirer parameter when an authorization transaction is branded as MSI.

**Alternate processing**

Alternate processing is optional for CIR and MSI transactions.

For issuers that choose to use Stand-In Authorization for alternate processing, Stand-In Authorization uses either the product or account range issuer-defined limits to determine how to respond to an Authorization Request/0100 message. If an issuer does not specify account range level processing, Stand-In Authorization uses the Mastercard defined product level defaults.

For the current definition of minimum transaction limits, refer to [Mastercard Rules](#) and [Transaction Processing Rules](#).

**Associated edits**

| WHEN...                                                                                                        | THEN the Dual Message Authorization System...                                                                                |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| the issuer or the Stand-In Authorization is not available to respond to the Authorization Request/0100 message | returns an Authorization Request Response/0110 message where: DE 39 = 91 (Authorization System or issuer system inoperative) |

CIR and MSI transactions are not processed by X-Code.

**For more information**

Specific processing rules and standards have been defined for CIR ATM processing and MSI ATM processing. For more information, refer to [ATM processing for Europe region acquirers](#).

# Contactless Mapping Service for Contactless M/Chip and Contact M/Chip Transactions

The Contactless Mapping Service will be performed for participating account ranges when DE 22, subfield 1 contains value 05 (PAN auto-entry via chip) or value 07 (PAN auto-entry via contactless M/Chip).

## Processing

The following process describes how the Dual Message Authorization System processes Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages for Contactless Mapping Service transactions.

### Associated edits

| WHEN the message contains...                                                                                                                   | THEN the Dual Message Authorization System...                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| DE 22, subfield 1, value 05 (PAN auto-entry via chip)<br>and<br>The Contactless Mapping Service was performed on the transaction               | changes the value in DE 22, subfield 1 from 05 to 06, indicating that the Contactless Mapping Service occurred. |
| DE 22, subfield 1, value 07 (PAN auto-entry via contactless M/Chip)<br>and<br>The Contactless Mapping Service was performed on the transaction | changes the value in DE 22, subfield 1 from 07 to 08, indicating that the Contactless Mapping Service occurred. |

The Contactless Mapping Service occurs on Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages. The following example describes the Contactless Mapping Service being performed on an Authorization Request/0100 message.

Stages describing the process that occurs when an Authorization Request/0100 transaction is generated from a contactless M/Chip or contact M/Chip card eligible for the Contactless Mapping Service.

1. When a cardholder initiates an eligible contactless M/Chip or contact M/Chip transaction on a contactless M/Chip or contact M/Chip terminal, the contactless M/Chip PAN or contact M/Chip PAN is passed to the terminal.
2. The acquirer sends an Authorization Request/0100 message containing the contactless M/Chip PAN or contact M/Chip account number to the Dual Message Authorization System.



3. The Dual Message Authorization System maps the contactless M/Chip or contact M/Chip account number to the cardholder's PAN (or funding account), and then forwards the authorization message containing the following values to the issuer:
  - DE 2 (Primary Account Number [PAN]) containing the cardholder's PAN
  - DE 14 (Date, Expiration) containing the cardholder's expiry date, if provided
  - DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)
  - DE 23 (Card Sequence Number) containing the contactless M/Chip or contact M/Chip card sequence number, if provided
  - DE 35 (Track 2 Data), if provided, is removed
  - DE 45 (Track 1 Data), if provided, is removed
  - DE 48 (Additional Data: Private Use), subelement 33 (Contactless Mapping Information) containing contactless M/Chip or contact M/Chip PAN information
  - DE 48, subelement 71 (On-behalf Services), identifying the Contactless Mapping Service and any other on-behalf services performed
  - DE 55 (Integrated Circuit Card [ICC] System-related Data), as originally sent by the terminal
4. The issuer responds with an Authorization Request Response/0110 message containing the cardholder's PAN in DE 2.
5. The Dual Message Authorization System maps the PAN back to the contactless M/Chip or contact M/Chip account number and places the cardholder's PAN in DE 48, subelement 33, and then sends the contactless M/Chip or contact M/Chip account number in DE 2 to the acquirer.
6. The acquirer forwards the contactless M/Chip or contact M/Chip account number and other authorization response information to the contactless terminal.

The Contactless Mapping Service for contactless M/Chip and contact M/Chip transactions (DE 22, subfield 1, value 05 or 07) requires participation in the Dynamic CVC 3 Pre-validation Service unless the account range comprises M/Chip cards that have not had the magnetic stripe profile programmed.

For more information about the Mastercard Contactless Mapping Service, refer to the [Contactless On-behalf Services Guide](#).

## Dual Message Authorization System Edits

The following edit will be performed on Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages to enforce the current valid expiration date edit

that exists for both contactless M/Chip and contact M/Chip transactions that are eligible for the Contactless Mapping Service.

### Associated edits

| WHEN...                                                                                                                                                                                                                                                                        | THEN the Dual Message Authorization System...                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the expiration date in DE 14, DE 35, or DE 45 (whichever data element is first to contain the expiration date) is expired<br>and<br>DE 22, subfield 1 is value 05 or 07<br>and<br>the account is part of an account range that participates in the Contactless Mapping Service | rejects the transaction and sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 39 (Response Code) is 54 (Expired Card). |

## Credential on File transaction processing

A Credential on File transaction is a transaction in which a cardholder has explicitly authorized a merchant or payment facilitator's Sponsored Merchant to store the cardholder's Mastercard® or Maestro® account information (PAN and expiration date, or tokenized PAN and expiration date), and the cardholder subsequently authorizes that same merchant to bill the cardholder's stored Mastercard or Maestro account.

DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 10 (Credential on File) identifies Mastercard-branded Credential on File transactions submitted to the Mastercard Network.

The Credential on File indicator:

- Allows a merchant to identify a pre-existing relationship with a cardholder to the issuer, using an identifier that is common across the payments industry.
- Increases approval rates for e-commerce transactions initiated with cardholder payment credentials already stored by, or on file with, the merchant or the service provider.
- Helps to more clearly identify types of Credential on File transactions for issuers, as there are only slight differences between Credential on File recurring payment transactions and Credential on File e-commerce transactions.

### Transactions identified as Credential on File

The following are categorized as Credential on File transactions and should be identified with the Credential on File indicator:

- Recurring payments
- Installment payments

- E-Commerce and Mail Order/Telephone Order (MO/TO) transactions for which the cardholder authorizes the merchant to use the Credential on File for payment
- Card-not-present transactions originating from a merchant's brick-and-mortar location for which the cardholder authorizes the merchant to use the Credential on File
  - Credential on File transactions are always initiated without the presence of a card; however, they can be cardholder-present transactions in which the cardholder authorizes the merchant to use card credentials stored during a previous consumer interaction. An example of such a transaction is when a consumer makes a purchase at a physical location of a pharmacy that supports both physical/retail and remote business (mail, phone, or e-commerce). The consumer may have previously interacted with the pharmacy through a remote channel using their Mastercard account number and provided permission to store the number for future use. If the consumer walks into one of the pharmacy's retail locations to pick up a prescription, the merchant may offer the consumer an option to pay using the stored credential. In this case, the transaction is cardholder-present, card-not-present, and requires the use of the Credential on File identifier.
- Account Status Inquiry (ASI) messages
  - The absence of the Credential on File value in a Purchase ASI or Recurring Payment ASI message should indicate a one-time request to validate card status or a request to validate prior to saving as Credential on File for future purchases or recurring payments.

## Country Level authorization

Country level authorization is an optional service that allows issuers of local-use-only cards to have the Dual Message Authorization System automatically decline all authorizations acquired outside of their country. The Dual Message Authorization System compares the data in the point-of-service (POS) country code field with the issuer's country. The Dual Message Authorization System only forwards authorizations acquired within the issuer's country to the issuer for processing. The Dual Message Authorization System declines all other transactions.

### Country Level authorization process

Stages of the process that the Dual Message Authorization System follows if the issuer designates the BIN for local-use only:

1. When the acquirer MIP receives an Authorization Request/0100 message, the Dual Message Authorization System checks whether the issuer designated the BIN for local-use-only.
2. When the issuer has designated the BIN for local-use-only, the Dual Message Authorization System compares:
  - The issuer's country on file at Mastercard to the POS Country Code, DE 61 (Point-of-Service [POS] Data), positions 14–16 of the Authorization Request/0100 message.
  - An accurate match depends on acquirers including the correct data in this field.

Associated edits

| WHEN the issuer's country...        | THEN the Dual Message Authorization System...                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| matches the POS Country Code        | continues with normal processing.                                                                            |
| does not match the POS Country Code | returns DE 39 (Response Code) value 62 (Restricted Card) in the Authorization Request Response/0110 message. |

**NOTE: Use of the Country Level Authorization service does not alter a customer's responsibility for the use of local-use-only cards outside the country of issuance, including liability for all authorized and all below-the-floor-limit transactions. In addition, Mastercard assumes no liability for improper authorizations of such transactions that may result from issuer, acquirer, or merchant errors, or from Dual Message Authorization System or Country Level Authorization service outages.**

Issuers that use the Country Level Authorization service for cards designated for local use only and have chosen to receive MDES pre-digitization and Account Status Inquiry (ASI) messages must be aware that when a funding account range is configured for local use only, Mastercard will modify the following country- and currency-related data fields in the MDES pre-digitization and ASI messages according to the configurations provided by the issuer during the onboarding process:

- DE 43 (Acceptor Name and Location), subfield 5 (Acceptor Country Code)
- DE 49 (Currency Code, Transaction)
- DE 61, subfield 13 (POS Country Code)

Currency conversion

The Dual Message Authorization System automatically provides a currency conversion service to acquirers and issuers to allow processing of Authorization/01xx and Reversal/04xx messages in the customer's preferred currency.

Acquirers may submit online messages in any valid Mastercard supported transaction currency. Acquirers and issuers will receive amount-related data elements in the acquirer's transaction currency and issuer's cardholder billing currency even if they use the same currency. Acquirers and issuers have the option to receive amount-related data elements in the settlement currency (always U.S. dollars).

## Amount-related data elements in Authorization and Reversal messages

The following table lists how acquirers and issuers will send and receive amount-related data elements in authorization and reversal messages.

### Data elements

| DE | Acquirer Sends                     | Issuer Receives                                                                                                                                                                                                                         | Issuer Returns                                                                                                                                                                                                                               | Acquirer Receives                                                                                                                                                                                                                               |
|----|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | In acquirer's transaction currency | In acquirer's transaction currency                                                                                                                                                                                                      | In acquirer's transaction currency (echo except when responding with partial approval or purchase amount only—no cash back allowed)                                                                                                          | In acquirer's transaction currency                                                                                                                                                                                                              |
| 5  | N/A                                | In U.S. dollars if issuer receives settlement amount-related data elements<br><br>Not present if issuer does not receive settlement amount-related data elements                                                                        | In U.S. dollars if issuer receives settlement amounts (echo except when responding with partial approval or purchase amount only—no cash back allowed)<br><br>Not present if issuer does not receive settlement amount-related data elements | In U.S. dollars if acquirer receives settlement amount-related data elements.<br><br>Not present if acquirer does not receive settlement amount-related data elements                                                                           |
| 6  | N/A                                | In issuer's cardholder billing currency                                                                                                                                                                                                 | In issuer's cardholder billing currency (echo except when responding with partial approval or purchase amount only—no cash back allowed)                                                                                                     | In issuer's cardholder billing currency                                                                                                                                                                                                         |
| 9  | N/A                                | Rate used to convert DE 4 amount from acquirer's transaction currency to U.S. dollars, if issuer receives settlement amount-related data elements<br><br>Not present if issuer does not receive settlement amount-related data elements | Rate used to convert DE 4 amount from acquirer's transaction currency to U.S. dollars, if issuer receives amount-related data elements (echo)<br><br>Not present if issuer does not receive settlement amount-related data elements          | Rate used to convert DE 4 amount from the acquirer's transaction currency to U.S. dollars, if acquirer receives settlement amount-related data elements<br><br>Not present if acquirer does not receive settlement amount-related data elements |

| DE | Acquirer Sends                                                                                          | Issuer Receives                                                                                                                                                                 | Issuer Returns                                                                                                                                                                         | Acquirer Receives                                                                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | N/A                                                                                                     | Rate used to convert DE 4 amount from acquirer's transaction currency to issuer's cardholder billing currency                                                                   | Rate used to convert DE 4 amount from acquirer's transaction currency to issuer's cardholder billing currency (echo)                                                                   | Rate used to convert DE 4 amount from acquirer's transaction currency to issuer's cardholder billing currency                                                                       |
| 16 | N/A                                                                                                     | Month and day conversion rate is effective                                                                                                                                      | Month and day conversion rate is effective (echo)                                                                                                                                      | Month and day conversion rate is effective                                                                                                                                          |
| 28 | Acquirer's transaction currency                                                                         | Acquirer's transaction currency                                                                                                                                                 | Acquirer's transaction currency (echo)                                                                                                                                                 | Acquirer's transaction currency                                                                                                                                                     |
| 49 | Acquirer's transaction currency code.                                                                   | Acquirer's transaction currency code                                                                                                                                            | Acquirer's transaction currency code (echo)                                                                                                                                            | Acquirer's transaction currency code                                                                                                                                                |
| 50 | N/A                                                                                                     | Settlement currency code (840) if issuer receives settlement amount-related data elements<br><br>Not present if issuer does not receive settlement amount-related data elements | Settlement currency code (840) if issuer receives settlement amount-related data elements (echo)<br><br>Not present if issuer does not receive settlement amount-related data elements | Settlement currency code (840) if acquirer receives settlement amount-related data elements<br><br>Not present if acquirer does not receive settlement amount-related data elements |
| 51 | N/A                                                                                                     | Issuer's cardholder billing currency code                                                                                                                                       | Issuer's cardholder billing currency code (echo)                                                                                                                                       | Issuer's cardholder billing currency code                                                                                                                                           |
| 54 | If applicable to the transaction, one occurrence of each amount type in acquirer's transaction currency | One occurrence of each amount type in acquirer's transaction currency<br><br>One occurrence of each amount type in issuer's cardholder billing currency                         | If applicable to the transaction, one occurrence of each amount type in issuer's cardholder billing currency (not an echo of what the acquirer sent)                                   | One occurrence of each amount type in acquirer's transaction currency<br><br>One occurrence of each amount type in issuer's cardholder billing currency                             |

| DE | Acquirer Sends                                                                                             | Issuer Receives                                                                                                                                                                                                                                                                                                                                                        | Issuer Returns                 | Acquirer Receives                                                                                                                                                                                                                                                                                                                                                          |
|----|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 95 | If applicable, DE 95, subfield 1 in acquirer's transaction currency.<br>DE 95, subfields 2–4 contain zeros | DE 95, subfield 1 in acquirer's transaction currency<br><br>DE 95, subfield 2 in U.S. dollars if issuer receives settlement amount-related data elements<br><br>DE 95, subfield 2 zero-filled if issuer does not receive settlement amount-related data elements<br><br>DE 95, subfield 3 in issuer's cardholder billing currency<br><br>DE 95, subfield 4 zero-filled | Same values as received (echo) | DE 95, subfield 1 in acquirer's transaction currency<br><br>DE 95, subfield 2 in U.S. dollars if acquirer receives settlement amount-related data elements<br><br>DE 95, subfield 2 zero-filled if acquirer does not receive settlement amount-related data elements<br><br>DE 95, subfield 3 in issuer's cardholder billing currency<br><br>DE 95, subfield 4 zero-filled |

## Dual Message System Processing

The following Dual Message System processing notes apply to currency conversion processing.

If the acquirer provides DE 5, DE 6, DE 9, DE 10, DE 16, DE 50, or DE 51, the Dual Message Authorization System will overwrite with the appropriate values in the message sent to the issuer. DE 5, DE 6, DE 9, DE 10, DE 16, DE 50, or DE 51 will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

If the settlement currency (U.S. dollars) is the same as the acquirer's currency, DE 5 will be the same value as in DE 4 and DE 9 will contain the value 61000000.

If the acquirer and issuer are the same currency, DE 6 will be the same value as in DE 4 and DE 10 will contain the value 61000000.

DE 54 will not contain additional occurrences of amount types in the settlement currency. DE 54 will only contain occurrences of amount types in U.S. dollars if it is the currency of the acquirer or issuer.

## Data Elements Acquirers Send in Request, Advice, and Reversal Messages

Acquirers are required to send the following currency conversion related data elements in authorization request, advice, and reversal messages.

### Data elements

| DE       | Note                                                                                             | 0100 | 0120 | 0400 |
|----------|--------------------------------------------------------------------------------------------------|------|------|------|
| 4        | Acquirer's transaction currency                                                                  | P    | P    | P    |
| 5        | Always U.S. dollars                                                                              |      |      |      |
| 6        | Issuer's cardholder billing currency                                                             |      |      |      |
| 9        | Factor used in conversion from transaction amount to settlement amount                           |      |      |      |
| 10       | Factor used in conversion from transaction amount to cardholder billing amount                   |      |      |      |
| 16       | Always present                                                                                   |      |      |      |
| 28       | Acquirer's transaction currency                                                                  | C    | C    | C    |
| 49       | Acquirer's transaction currency code                                                             | P    | P    | P    |
| 50       | Always 840                                                                                       |      |      |      |
| 51       | Issuer's cardholder billing currency code                                                        |      |      |      |
| 54       | Occurrence of each amount type in acquirer's transaction currency                                |      |      |      |
| 54       | Occurrence of each amount type in issuer's cardholder billing currency                           | C    | C    | C    |
| 95, sf 1 | Acquirer's transaction currency                                                                  |      |      | P    |
| 95, sf 2 | Always U.S. dollars if acquirer chooses to receive settlement amounts, zero fill if not selected |      |      | Z    |
| 95, sf 3 | Issuer's cardholder billing currency                                                             |      |      | Z    |
| 95, sf 4 | Zero fill                                                                                        |      |      | Z    |

### Table Key

C under MTI = indicates that the data element may be present in that message .

X under MTI = indicates that the data element will be present if the acquirer chooses to receive settlement amount-related data elements.

Z under 0400 = indicates that if DE 95, subfield 1 is present, subfields 2–4 are zero-filled.



## Data Elements Acquirers Receive in Request, Advice, and Reversal Messages

Acquirers are expected to receive the following currency conversion related data elements in Request, Advice, and Reversal responses.

### Data elements

| DE       | Note                                                                                             | 0110 | 0130 | 0410 |
|----------|--------------------------------------------------------------------------------------------------|------|------|------|
| 4        | Acquirer's transaction currency                                                                  | P    | P    | P    |
| 5        | Always U.S. dollars                                                                              | X    | X    | X    |
| 6        | Issuer's cardholder billing currency                                                             | P    | P    | P    |
| 9        | Factor used in conversion from transaction amount to settlement amount                           | X    | X    | X    |
| 10       | Factor used in conversion from transaction amount to cardholder billing amount                   | C    | C    | C    |
| 16       | Always present                                                                                   | C    | C    | C    |
| 28       | Acquirer's transaction currency                                                                  | C    | C    | C    |
| 49       | Acquirer's transaction currency code                                                             | P    | P    | P    |
| 50       | Always 840                                                                                       | X    | X    | X    |
| 51       | Issuer's cardholder billing currency code                                                        | C    | C    | C    |
| 54       | Occurrence of each amount type in acquirer's transaction currency                                | C    |      |      |
| 54       | Occurrence of each amount type in issuer's cardholder billing currency                           | C    |      |      |
| 95, sf 1 | Acquirer's transaction currency                                                                  |      |      | P    |
| 95, sf 2 | Always U.S. dollars if acquirer chooses to receive settlement amounts, zero fill if not selected |      |      | X    |
| 95, sf 3 | Issuer's cardholder billing currency                                                             |      |      | P    |
| 95, sf 4 | Zero fill                                                                                        |      |      | P    |

### Table Key

P under MTI = indicates that the data element may be present in that message.

X under MTI = indicates that the data element will be present if the acquirer chooses to receive settlement amount-related data elements.

## Data Elements Issuers Receive in Request, Advice, and Reversal messages

Issuers are required to receive the following data elements in authorization and reversal messages.

### Data elements

| DE       | Note                                                                           | 0100 | 0120 | 0400 | 0420 |
|----------|--------------------------------------------------------------------------------|------|------|------|------|
| 4        | Acquirer's transaction currency                                                | P    | P    | P    | P    |
| 5        | Always U.S. dollars                                                            | X    | X    | X    | X    |
| 6        | Issuer's cardholder billing currency                                           | P    | P    | P    | P    |
| 9        | Factor used in conversion from transaction amount to settlement amount         | X    | X    | X    | X    |
| 10       | Factor used in conversion from transaction amount to cardholder billing amount | P    | P    | P    | P    |
| 16       | Always present                                                                 | P    | P    | P    | P    |
| 28       | Acquirer's transaction currency                                                | P    | P    | P    | P    |
| 49       | Acquirer's transaction currency code                                           | P    | P    | P    | P    |
| 50       | Always 840                                                                     | X    | X    | X    | X    |
| 51       | Issuer's cardholder billing currency code                                      | P    | P    | P    | P    |
| 54       | Occurrence of each amount type in acquirer's transaction currency              | P    | P    | P    | P    |
| 54       | Occurrence of each amount type in issuer's cardholder billing currency         | P    | P    | P    | P    |
| 95, sf 1 | Acquirer's transaction currency                                                |      |      | P    | P    |
| 95, sf 2 | Always U.S. dollars if selected, zero fill if not selected                     |      |      | X    | X    |
| 95, sf 3 | Issuer's cardholder billing currency                                           |      |      | P    | P    |
| 95, sf 4 | Zero fill                                                                      |      |      | P    | P    |

### Table Key

P under MTI = indicates that the data element will be present in that message

X under MTI = indicates that the data element will be present if the issuer chooses to receive settlement amount-related data elements

## Data Elements Issuers Send in Request, Advice, and Reversal Responses

Issuers are expected to send the following currency conversion related data elements in Request, Advice, and Reversal responses.

### Data elements

| DE       | Note                                                                           | 0110 | 0130 | 0410 | 0430 |
|----------|--------------------------------------------------------------------------------|------|------|------|------|
| 4        | Acquirer's transaction currency                                                | P    | P    | P    | P    |
| 5        | Always U.S. dollars                                                            | X    | X    | X    | X    |
| 6        | Issuer's cardholder billing currency                                           | P    | P    | P    | P    |
| 9        | Factor used in conversion from transaction amount to settlement amount         | X    | X    | X    | X    |
| 10       | Factor used in conversion from transaction amount to cardholder billing amount | P    | P    | P    | P    |
| 16       | Always present                                                                 | P    | P    | P    | P    |
| 28       | Acquirer's transaction currency                                                | C    | C    | C    | P    |
| 49       | Acquirer's transaction currency code                                           | P    | P    | P    | P    |
| 50       | Always 840                                                                     | X    | X    | X    | X    |
| 51       | Issuer's cardholder billing currency code                                      | P    | P    | P    | P    |
| 54       | Occurrence of each amount type in acquirer's transaction currency              |      |      |      |      |
| 54       | Occurrence of each amount type in issuer's cardholder billing currency         | P    |      |      |      |
| 95, sf 1 | Acquirer's transaction currency                                                |      |      |      | P    |
| 95, sf 2 | Always U.S. dollars if selected, zero fill if not selected                     |      |      | X    | X    |
| 95, sf 3 | Issuer's cardholder billing currency                                           |      |      | P    | P    |
| 95, sf 4 | Zero fill                                                                      |      |      | P    | P    |

### Table Key

P under MTI = indicates that the data element will be present in that message

X under MTI = indicates that the issuer will return the data element in the response message if the issuer chooses to receive settlement amount-related data elements in the original request or advice message

## Alternate Processing

Following are details regarding currency conversion in alternate processing.

Authorization Advice/0120 and Reversal Advice/0420 messages that the issuer receives through Store-and-forward (SAF) processing will always contain amount-related data elements in the acquirer's transaction currency and the issuer's cardholder billing currency.

SAF messages will contain amount-related data elements in the settlement currency (U.S. dollars) according to the issuer's preference for receiving these data elements.

## Dual Message Authorization System edits

The Dual Message Authorization System ensures that amount-related data elements, if present in the following response messages from the issuer, are the same values as provided in the request messages sent to the issuer.

Issuers should echo amount-related data elements in response messages as they were received in the request messages. The exception is when an issuer provides DE 39 (Response Code), value 10 (Partial approval) or 87 (Purchase amount only, no cash back allowed) in an Authorization Request Response/0110 message. When providing either of these responses, the issuer is required to provide DE 6 and DE 51 according to the specifications for partial approvals and purchase of goods or services with cash back.

### Authorization Request Response/0110

| IF...                                                                                                                                                                                                                | THEN the Dual Message Authorization System...                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the value in DE 39 is not 10 (Partial approval) or 87 (Purchase amount only, no cash back allowed), and the amount-related data elements returned by the issuer are different from the values received by the issuer | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = the data element in error |

### Reversal Request Response/0410

| IF...                                                                                                        | THEN the Dual Message Authorization System...                                                                                                         |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| the amount-related data elements returned by the issuer are different from the values received by the issuer | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 where:<br>DE 39 = 30<br>DE 44 = the data element in error |

### Authorization Request/0100 or Authorization Advice/0120

| IF...                                                                                                                                                                                                     | THEN the Dual Message Authorization System...                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| no occurrence of DE 54, subfield 3 (Currency Code) is equal to the currency code in DE 49<br>and<br>DE 54 subfield 2 (Amount Type) is not 58                                                              | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| DE 54 contains an occurrence where subfield 2 (Amount Type) is 58 (POI Amount) for Dynamic Currency Conversion<br>and<br>subfield 3 (Currency Code) is equal to the currency code, but not equal to DE 49 | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| DE 54 contains more than two occurrences of the DE 54 subfields<br>and<br>no occurrence has a subfield 2 (Amount Type) with indicates Real-Time Substantiation                                            | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| no occurrence has a subfield 2 (Amount Type) with indicates Real-Time Substantiation                                                                                                                      | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| DE 54 contains more than one occurrence of the DE 54 subfields for a given amount type (subfield 2)                                                                                                       | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br>DE 39 = 30<br>DE 44 = 054 |

| IF...                                                                                                                                                       | THEN the Dual Message Authorization System...                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The combined length of one occurrence of DE 54 subfields 1–4 is not equal to 20 bytes                                                                       | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br><br>DE 39 = 30<br>DE 44 = 054 |
| DE 54, subfields 1–3 are not numeric                                                                                                                        | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br><br>DE 39 = 30<br>DE 44 = 054 |
| The first position in DE 54, subfield 4 is not C (Credit amount or positive balance) or D (Debit amount or negative balance), followed by 12 numeric digits | generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where:<br><br>DE 39 = 30<br>DE 44 = 054 |

### Authorization Request Response/0110

| IF...                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| no occurrence of DE 54, subfield 3 (Currency Code) is equal to the currency code in DE 51                                                               | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br><br>DE 39 = 30<br>DE 44 = 054 |
| the value in DE 39 is not 10 or 87 and DE 54 contains more than two occurrences of the DE 54 subfields                                                  | removes occurrences of DE54 beyond 2 occurrences and forwards the message to the acquirer                                                   |
| the value in DE 39 is 10 or 87 and DE 54 contains more than one occurrence of DE 54, subfield 2 (Amount Type) that is not equal to 57 (Original Amount) | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br><br>DE 39 = 30<br>DE 44 = 054 |

| IF...                                                                                              | THEN the Dual Message Authorization System...                                                                                           |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| The system removes any incorrect DE54 occurrences<br>and<br>DE54 contains more than one occurrence | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| DE 54 contains more than one occurrence of DE 54 subfields for a given amount type (subfield 2)    | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| the combined length of one occurrence of DE 54 subfields 1–4 is not equal to 20 bytes              | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| DE 54, subfields 1–3 are not numeric                                                               | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| the first position in DE 54, subfield 4 is not C or D, followed by 12 numeric digits               | generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 054 |

## Currency conversion rates

The Dual Message Authorization System uses Mastercard issued rates to perform currency conversion between two currencies.

The T057 production and T058 test bulk files contain the Currency Conversion Rate Detail Records; Mastercard-Issued Cross Rates (Rate Class F) and Mastercard-Issued USD Rates (Rate Class M). For each Mastercard-Issued Rate, the Reference Currency is indicated in field 3 of the T057 file.

Based on the T057 Currency Conversion Rate file published for the current processing day (at T+0), Mastercard looks to implement rates on the Dual Message Authorization System at 14:05 St. Louis time the same day (T+0). However, there may be limited situations, such as latency in file transmission, which may cause a delay in file delivery. The rates will be effective upon file delivery until 14:05 St. Louis time the following day (T+1).

## Currency conversion calculation

The Dual Message Authorization System selects the Mastercard-Issued Cross Rates, class record (Rate Class F) only, when the source and target currencies match the source and Reference Currency pair identified in the rate record.

All other currency conversions occur using the Mastercard-Issued USD Rates, class records (Rate Class M), for the source currency to the target currency. In cases where the target currency is not USD, the source currency is converted to USD and then to the target currency (for example, a computed cross rate).

For example:

Assume that the Dual Message Authorization System processes a transaction in which:

- Transaction currency is GBP.
- Cardholder billing currency is EUR.
- Acquirer's reconciliation currency is USD.

Also assume that a Mastercard-Issued Cross Rate has been established for the currency pair GBP: EUR and that EUR is the Reference Currency for this currency pair.

Dual Message Authorization System will calculate:

- EUR cardholder billing amount using the GBP: EUR Mastercard-Issued Cross Rate.
- USD acquirer's reconciliation amount using the GBP: USD Mastercard-Issued USD Rate.

## Electronic Commerce Processing

Electronic commerce transactions are non-face-to-face online transactions using electronic media over any public network, such as the Internet, or private network, such as an extranet. E-commerce processing allows transactions to be initiated from a cardholder-controlled device, such as a PC or mobile phone, for purchasing goods and services on the Internet.

All e-commerce transactions must be uniquely identified by the acquirer in the authorization. This permits the issuer to assess the degree of risk associated with the transaction and to support any processing requirements associated with interchange compliance.



## Customer requirements

- All issuers must be able to receive and process all e-commerce data present in the Authorization Request/0100 messages.
- All acquirers must properly identify e-commerce messages within the Authorization Request/0100 message. They must be able to receive and to process e-commerce Authorization Request Response/0110 messages.

The Authorization Request/0100 includes the security level indicator (in DE 48, subelement 42 [Electronic Commerce Indicators] subfield 1 [Electronic Commerce Security Level Indicator and UCAF Collection Indicator]) that indicates the following security attributes of electronic commerce transactions:

- Security protocol, which indicates the security protocol that was used to facilitate the transaction (for example, channel encryption).
- Cardholder authentication, which indicates the method of cardholder authentication used to facilitate the transactions.
- UCAF collection indicator, which indicates both merchant UCAF readiness and inclusion of any specific authentication data in the UCAF field (DE 48, subelement 43).

Mastercard messaging requirements regarding the security of e-commerce transactions may vary depending upon the security protocol involved in the transaction. Methods may include:

- No security protocol
- Channel encryption, which is a security layer that resides between an application and its transport layers, for example, SSL (Secure Sockets Layer) and HTTPS (Hypertext Transfer Protocol Secure)
- Mastercard® Identity Check™ Transaction using the Universal Cardholder Authentication Field (UCAF™)

If an e-commerce item that ships late is re-authorized for message reason code 4808 (Authorization-Related Chargeback) chargeback protection, this transaction will take on the security characteristics of the original authorization within a dispute resolution.

## Channel Encryption

This section describes an Authorization Request/0100: Electronic Commerce transaction that uses channel encryption protocol between the cardholder and the merchant.

1. The cardholder browses the Internet until the cardholder is ready to make a purchase from a merchant.
2. The cardholder's browser, which supports channel encryption (that is, SSL), sends the purchase and payment information to the merchant. Channel encryption protects the information over the Internet.
3. The merchant receives the purchase information and has access to all the cardholder account data that the cardholder provided.
4. The merchant requests authorization from the acquirer.

5. The acquirer generates an Authorization Request/0100 message, including both of the following:
  - The cardholder payment data
  - The appropriate data element values that identify this as an e-commerce transaction with channel encryption protocol and no cardholder authentication.
6. The issuer receives the authorization request and generates an Authorization Request Response/0110 message.
7. The acquirer receives the Authorization Request Response/0110 and sends the response back to the merchant.
8. The merchant provides acknowledgment to the cardholder (protected by channel encryption as it travels over the Internet).

**NOTE: An approved e-commerce authorization must only have one first presentment submitted for clearing unless the authorization is for airline tickets or installment purchases. In the case of airline tickets and installment purchases multiple first presentments against one approve authorization is allowed.**

## No Security Protocol

This section describes an Authorization Request/0100—Electronic Commerce transaction with no security protocol.

1. The cardholder browses the Internet until the cardholder is ready to make a purchase from a merchant. In this example, the cardholder does not have authentication of the merchant at any time.
2. The cardholder's browser sends the purchase and payment information over the Internet to the merchant. In this example, no security protocol protects the request.
3. The merchant receives the purchase information and has access to all of the cardholder account data that the cardholder provided (including payment information).
4. The merchant requests authorization from the acquirer.
5. The acquirer generates an Authorization Request/0100 message, including both of the following:
  - Cardholder payment data
  - Appropriate data element values that identify this as an e-commerce transaction with no security protocol and no cardholder authentication
6. The issuer receives the authorization request and generates an Authorization Request Response/0110 message.
7. The acquirer receives the Authorization Request Response/0110 message and sends the response back to the merchant.
8. The merchant provides acknowledgment to the cardholder (still unprotected as it travels over the Internet).

## Authorization Request/0100: Electronic Commerce Purchase

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                                                                                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)                                                                   | M   | •   | M   | 81 = PAN entry via electronic commerce, including chip                                                                                                                                                                                                                                                                                                                                                                                       |
| DE 48 (Additional Data: Private Use), TCC (Transaction Category Code)                                                                                 | M   | •   | M   | <p>Contains one of the following values:</p> <p>P = Payment Transaction</p> <p>T = Phone, Mail, or Electronic Commerce Order</p> <p>U = Unique</p> <p>X = Airline and Other Transportation Services (irrespective of whether the transaction origin is face to face or not)</p> <p><b>NOTE: Must contain a valid TCC value for the MCC present in DE 18, as described in the MCC descriptions in Chapter 3, Quick Reference Booklet.</b></p> |
| DE 48, subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only]), subfield 1 (Merchant/Certificate Serial Number) | C   | •   | C   | <p>DE 48 subelement 40 is not required for any Mastercard e-commerce programs.</p> <p>Merchant certificate serial number in binary format</p>                                                                                                                                                                                                                                                                                                |
| DE 48, subelement 40, subfield 2 (Cardholder Certificate Serial Number)                                                                               | C   | •   | C   | Cardholder certificate serial number in binary format                                                                                                                                                                                                                                                                                                                                                                                        |
| DE 48, subelement 42 (Electronic Commerce Indicators) subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator)         | M   | •   | M   | <p>Contains security level in positions 1 and 2 and UCAF collection indicator in position 3.</p> <p>Position 1 = Security Protocol</p> <p>Position 2 = Cardholder Authentication</p> <p>Position 3 = UCAF Collection Indicator</p>                                                                                                                                                                                                           |

| Data Element                                                                                                                                   | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 43 (Universal Cardholder Authentication) for Mastercard Identity Check issuer or cardholder generated authentication data    | C   | •   | C   | Authentication data generated by Identity Check-compliant solution.                                                                                                                                                                                                                              |
| DE 48, subelement 43 (Universal Cardholder Authentication) for static AAV program.                                                             | C   | •   | C   | A static Accountholder Authentication Value (AAV) assigned by Mastercard for use with this Program.                                                                                                                                                                                              |
| DE 48, subelement 43 (Universal Cardholder Authentication) for 3-D Secure Electronic Commerce Verification Service (Visa and American Express) | C   | •   | C   | Position 1 (3-D Secure Electronic Commerce Transaction Indicator) 8 = Indicates Secure Electronic Commerce Transaction<br><br>Position 2–21 (3-D Secure Electronic Commerce Cardholder Authentication Verification Value [CAVV]) = Cardholder Authentication Verification Value (CAVV)           |
| DE 48, subelement 44 (Electronic Commerce Transaction Identifier [XID] [Visa and American Express])                                            | C   | •   | C   | Contains the 3-D Secure Electronic Commerce Transaction Identifier (XID) value in binary format.                                                                                                                                                                                                 |
| DE 61 (Point-of-Service (POS) Data), subfield 3 (POS Terminal Location)                                                                        | M   | •   | M   | One of the following, as applicable:<br><br>2 = Off premises of acceptor facility (cardholder terminal including home PC, mobile phone, PDA)<br><br>4 = On premises of acceptor facility (cardholder terminal including home PC, mobile phone, PDA)<br><br>8 = Additional POS Terminal Locations |

| Data Element                                             | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 61, subfield 4 (POS Cardholder Presence)              | M   | •   | M   | <p>When the first transaction in a recurring payment arrangement is an electronic commerce transaction:</p> <ul style="list-style-type: none"> <li>• 4 = Standing order/recurring transactions</li> </ul> <p>For each subsequent transaction in the same recurring payment arrangement use:</p> <ul style="list-style-type: none"> <li>• 4 = Standing order/recurring transactions</li> </ul> <p>For all other e-commerce transactions use:</p> <ul style="list-style-type: none"> <li>• 5 = Electronic order (home PC, Internet, mobile phone, PDA)</li> </ul> <p><b>NOTE: For Intra-EEA Transactions and to Intracountry Transactions in the EEA, recurring payment arrangements must be identified as specified by the registered switch of the Customer's choice.</b></p> |
| DE 61, subfield 7 (POS Transaction Status)               | M   | •   | M   | <p>Must not contain value 2 (Identity Check Phone Order).</p> <p><b>NOTE: An e-commerce transaction may be submitted as either a preauthorization (DE 61, subfield 7 value 4) or final authorization (DE 61, subfield 7, value 0 and DE 48, subelement 61, subfield 5, value 1) by acceptors in the Europe region.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 61, subfield 10 (Cardholder-Activated Terminal Level) | M   | •   | M   | 6 = Authorized Level 6 CAT: Electronic commerce                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Authorization Request Response/0110: Electronic Commerce Purchase

When any response data is present in DE 48 (such as AVS response, CVC 2 response), the issuer uses a normal Authorization Request Response/0110, with all subelements present in the original Authorization Request/0100 message echoed back. Following is a list of the data

elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                                                                                               | Org | Sys | Dst | Values/Comments                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                                                       | CE  | •   | CE  | Must be the same value as in the original Authorization Request/0100 message.                                          |
| DE 48, subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only])                       | CE  | •   | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.                              |
| DE 48, subelement 42 (Electronic Commerce Security Level Indicator)                                                        | ME  | •   | ME  | Must be the same value as in the original Authorization Request/0100 message.                                          |
| DE 48, subelement 43 (Universal Cardholder Authentication Field [UCAF])                                                    | CE  | •   | CE  | Must be the same value as in the original Authorization Request/0100 message, if present.                              |
| DE 48, subelement 44 Electronic Commerce Transaction Identifier [XID] [Visa and American Express])                         | CE  | •   | CE  | Must be the same value as in the original Authorization Request/0100 message, if present. (Visa and American Express.) |
| DE 48, subelement 45 (Three-Domain (3-D) Secure Electronic Commerce Transaction Response Code [Visa and American Express]) | C   | •   | C   | Contains the Visa 3-D Secure CAVV Results Code, if present. (Visa and American Express).                               |

## Static Accountholder Authentication Value

Several Mastercard programs enable enrolled merchants that have met specific qualification criteria, to accept Mastercard or Maestro cards for electronic commerce (e-commerce) transactions without using Mastercard Identity Check to authenticate every transaction. Merchants are required to perform full Mastercard Identity Check authentication on the first transaction they perform for any individual cardholder.

These Mastercard programs are as follows:

- Maestro Recurring Payments Program
- Mastercard Utility Payment Program
- Maestro Low Risk Merchant Program

As part of these programs, Mastercard:

- Assigns participating merchants a static Accountholder Authentication Value (AAV) for use with transactions that are processed without Mastercard Identity Check authentication.
- Allocates participating merchants a Mastercard Assigned ID.

Mastercard static AAV UCAF data is identified in Authorization Request/0100 messages by DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator), value 3 (UCAF data collection is supported by the merchant, and UCAF [Mastercard assigned Static Accountholder Authentication Value] data must be present).

Acquirers must provide the Mastercard assigned merchant ID in DE 48, subelement 32 (Mastercard Assigned ID) of Authorization Request/0100 and Authorization Advice/0120—Acquirer-generated messages.

The combination of static AAV data submitted in the Universal Cardholder Authentication Field (DE 48, subelement 43) and the Mastercard assigned merchant ID (DE 48, subelement 32) is unique and is verified by the Dual Message Authorization System.

**NOTE: The Mastercard Identity Check AAV Verification Service and Mastercard Identity Check AAV Verification in Stand-In Authorization will not be performed on Mastercard or Maestro e-commerce transactions that are processed under the Maestro Recurring Payments Program, the Mastercard Utility Payment Program, or the Maestro Low Risk Merchant Program. These on-behalf services will continue to be performed on Mastercard and Maestro e-commerce transactions that are authenticated using Mastercard Identity Check.**

#### For more information

For information on the Identity Check program, refer to [Mastercard Identity Check](#).

## Specific scenarios for e-commerce transactions

Depending on the merchant's inventory system and the nature of the item being purchased online, an e-commerce transaction may be submitted as either a preauthorization or a final authorization by acceptors globally.

For more information about preauthorization and final authorization usage requirements, refer to [Authorization and Preauthorization Processing Standards](#), [Incremental Preauthorization](#), and the [Transaction Processing Rules](#).

The following describes the recommended processing when dealing with estimated amounts and multi-item purchases where all items may or may not be delivered or are not delivered at the same time.

#### Associated edits

| If...                                                       | Then...                 |
|-------------------------------------------------------------|-------------------------|
| the entire e-commerce purchase is canceled (for any reason) | Submit a full reversal. |

| If...                                                                                                                                     | Then...                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the initial preauthorization is for an estimated amount based on anticipated sales tax, or estimated shipping weight                      | submit partial reversal and present clearing if the effective amount is less than the preauthorization, or submit incremental preauthorization and present clearing if the effective amount is greater (Merchant could also authorize and clear the difference separately, or just present clearing for the greater amount and risk possible issuer chargeback). |
| one item within a multi-item purchase is canceled (customer cancellation or out-of-stock)                                                 | submit a partial reversal and present clearing when the remaining items ship. Refer to the following section <a href="#">Alternate processing: canceling a single item</a> .                                                                                                                                                                                     |
| the customer requests split shipment based on inventory availability, or e-commerce aggregator processes order through multiple suppliers | refer to the following section <a href="#">E-commerce split or partial shipments</a> .                                                                                                                                                                                                                                                                           |

## Alternate processing: canceling a single item

Issuers will release any hold of funds once a clearing presentment has been matched (using the Trace ID in addition to other data elements) to the original authorization. Accordingly, a merchant is not required to submit a partial reversal if the lower amount is processed by Mastercard clearing within 24 hours of finalization of the transaction, assuming multi-clearing processing is not utilized.

## E-commerce split or partial shipments

Options for managing e-commerce split or partial shipments are listed in table format.

### Associated edits

| Available Option                                                                                                                       | Comments                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| utilize multi-clearing processing with the proper partial/final presentment message reason code.                                       | this option is the best practice for processing split shipments due to fulfillment delays or multiple suppliers.                    |
| submit partial reversal for unshipped items. Present clearing for shipped items. Separately re-authorize and clear as late items ship. | merchant risks decline on re-authorization (Refer to previous note about UCAF security/interchange for re-authorized transactions). |
| do not present clearing on a multi-item purchase until all items ship.                                                                 | merchant cash flow may be impacted due to item inventory delays.                                                                    |
| secure consumer permission to bill up front, ahead of the order shipment. Present clearing for entire order upon authorization.        | issuer inquiry or chargebacks could result if merchant billing terms and conditions are not understood.                             |



| Available Option                                             | Comments                                          |
|--------------------------------------------------------------|---------------------------------------------------|
| submit separate authorizations and clear as each item ships. | each item is processed as a separate transaction. |

## Process for an e-commerce transaction

A typical e-commerce transaction is described below.

### E-commerce transaction flow

1. The cardholder accesses a merchant's website via the Internet and requests to make a purchase.
2. The merchant submits the cardholder's information to the acquirer.
3. The acquirer sends an Authorization Request/0100 message via the Mastercard Network to the issuer.
4. The issuer makes the authorization decision and replies using an Authorization Request Response/0110 message.
5. The acquirer responds to the merchant.
6. The merchant completes the transaction according to the authorization response returned.

### Partial reversal processing: Europe region

Since issuers will release any hold of funds once a clearing presentment has been matched (using the Trace ID in addition to other data elements) to the original authorization, a merchant is not required to submit a partial reversal if the remaining items are processed by Mastercard clearing within 24 hours of finalization of the transaction.

## Automated fuel dispenser transactions and e-commerce

An automated fuel dispenser transaction identified with MCC 5542 (an AFD transaction) may occur through the use of a digital application as an in-app payment. Such AFD transactions must comply with all e-commerce transaction processing requirements. In addition, AFD e-commerce transactions use the same processing flow as those occurring at cardholder-activated terminals, with respect to the use of preauthorization and advice messages.

An AFD e-commerce transaction preauthorization request may be submitted for an estimated amount, followed by an advice message advising the issuer of the final transaction amount.

The preauthorization request may be sent for an estimated amount. If approved by the issuer, then within 60 minutes of the time that the authorization request message was sent, the acquirer must send an Authorization Advice/0120 message for the final transaction amount (or a Reversal Request/0400 message if the transaction will not be completed or will finalize for a lesser amount than requested). If the issuer has placed a hold on cardholder funds in excess of USD 1 or CAD 1, as applicable, then within 60 minutes of receiving the acquirer's advice message, the issuer must release any hold amount that exceeds the final transaction amount specified.

Mastercard recommends that the Trace ID (DE 15 [Settlement Date] and DE 63 [Network Data]) of the completion advice response be used within the clearing presentment.

**NOTE: As an alternative, the Trace ID of the original preauthorization may be used for clearing presentment if that same Trace ID is included in the AFD completion advice message within DE 48, subelement 63 (Additional Data, Trace ID).**

## E-commerce payment transactions using tokens with or without cryptographic data

Mastercard allows payment transactions to process without a cryptogram in DE 48 (Additional Data: Private Use), subelement 43 (Universal Cardholder Authentication Field [UCAF]) when they also contain DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction) in the Authorization Request/0100 message.

This functionality applies to the following transaction types:

- Electronic commerce (e-commerce): Identified by DE 22 (POS Entry Mode), subfield 1 (POS Terminal PAN Entry), value 81 (PAN/Token entry via electronic commerce with optional Identity Check AAV in the UCAF field (DE 48, subelement 43) or DSRP cryptogram in the Digital Payment Data field (DE 104).
- Credential on File: Identified by DE 22, subfield 1, value 10 (Credential on File)

Mastercard will reject payment transaction requests involving tokens without cryptograms that are not identified as payment transaction (DE 3, subfield 1, value 28).

## Digital Secure Remote Payment with cryptographic data

Mastercard provides Digital Secure Remote Payment (DSRP) transactions to enable secure mobile-originated transactions for remote payments.

DSRP authorization request messages may be transmitted either with cryptographic data in the Digital Payment Data field (DE 104) or with full EMV data in DE 55.

For details about DSRP, refer to [Digital Secure Remote Payment](#).

### Digital Payment Data field

Located in DE 104 (Digital Payment Data) this field carries digital payment data, such as the Digital Secure Remote Payments (DSRP) cryptogram associated with the token and additional information as new token-related data emerges.

Mastercard Digital Enablement Service (MDES) uses DE 104, subelement 001 (Digital Payment Cryptogram) to carry the DSRP cryptogram in lieu of the UCAF field in DE 48 (Additional Data: Private Use), subelement 43 (Universal Cardholder Authentication Field [UCAF]).

## Universal Cardholder Authentication Field

The Universal Cardholder Authentication Field (Universal Cardholder Authentication Field™) is a standard, globally interoperable method of collecting cardholder authentication data at the point of interaction.

This field provides a universal, multipurpose data transport infrastructure that is used to communicate authentication information between cardholders, merchants, acquirers, and issuers.

### Accountholder Authentication Value

The Accountholder Authentication Value (AAV) is an Identity Check-specific implementation of Universal Cardholder Authentication Field™ related to issuer authentication platforms that incorporate the Secure Payment Application (SPA) algorithm. SPA is a Mastercard security method designed to authenticate cardholders when they transact online.

#### For more information

For more information, refer to [Mastercard Identity Check](#).

## Tokenized e-commerce transactions with dynamic payment credentials

To support Digital Secure Remote Payments (DSRP) Universal Cardholder Authentication Field™ (UCAF) transactions, merchants and acquirers that participate in the Mastercard Digital Enablement Service (MDES) must make changes to their systems. Mastercard has mitigated these impacts by leveraging both a Dynamic Token Verification Code and Dynamic Expiration Date, enabling the transaction to process like a traditional e-commerce payment.

Mastercard also allows the processing of MDES e-commerce transactions that contain a Mastercard Identity Check merchant attempt AAV value in the UCAF field and have been successfully authenticated by the Dynamic Token Verification Code and Dynamic Expiration Date.

#### For more information

For more information on MDES, refer to [Mastercard Digital Enablement Service](#).

## E-commerce Fraud Alerts for Issuers

E-commerce Fraud Alerts for Issuers requires acquirers to inform issuers when a merchant or acquirer has decided not to complete an approved card-not-present (CNP) transaction because of high-fraud risk. This information will enable issuers to appropriately monitor or block those cards to prevent future fraud.

This service will be required for CNP transactions, including e-commerce, mail order/telephone order (MOTO), and recurring payment transactions.

E-commerce Fraud Alerts for issuers:

- Provides issuers with a predictive indicator about cards that need to be monitored due to suspicious behavior
- Allows issuers, acquirers, and merchants to potentially reduce fraud losses, chargebacks, and associated costs by enabling issuers to intervene early on alerted card accounts
- Enables increased collaboration and communications between merchants and issuers to take fraud out of the system and help increase cardholder confidence in the system

### How it works

When an acquirer or merchant has decided not to complete an approved CNP transaction because of high-fraud risk, the acquirer must send the Dual Message Authorization System a Reversal Request/0400 message containing DE 39 (Response Code), value 34 (Suspect Fraud).

The issuer must be prepared to receive DE 39 (Response Code), value 34 (Suspect Fraud) in Reversal Request/0400 or Reversal Advice/0420 messages and generate appropriate Reversal Request Response/0410 or Reversal Advice Response/0430 messages. Issuers must release any hold placed on the cardholder's account for the amount of the original, approved Authorization Request/0100 message.

### Fees

Refer to the **Pricing and Billing Resource Center** application on Mastercard Connect® for fees and billing information.

### Reports

The Issuer Authorization Detail report (AB201010-A2) will be used to support analysis of the reversal messages containing the new code. The Stand-In Detail Authorization report (AB111010-AA) will also be modified to support the new reason code value in DE 39.

An issuer E-commerce Fraud Alerts for Issuers report will be available for Dual Message System transactions that contains detail transaction information. This report can be requested through Portfolio Analytics.

Refer to the **Pricing and Billing Resource Center** application on Mastercard Connect® for fees and billing information.

## Comparison of security protocols

A comparison of security protocols is listed in table format.

### Comparison of security protocols

| Type of Security Protocol                      |                                                                              |                                                                             |                                                                                                                                                                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature                                        | None                                                                         | Channel Encryption                                                          | Identity Check (using UCAF; dynamic AAV)                                                                                                                                                                                                                                           |
| Protection of information over the Internet    | Unprotected                                                                  | Protected                                                                   | Protected                                                                                                                                                                                                                                                                          |
| Protection of information at the merchant site | Payment information unprotected. (Merchant sees all cardholder information.) | Payment information unprotected (Merchant sees all cardholder information). | Payment information unprotected unless used in conjunction with pseudo-account number solution (Merchant sees all cardholder information. Merchant should encrypt all sensitive data behind a firewall in compliance with the Payment Card Industry (PCI) Data Security Standard). |
| Authentication of Acquirer                     | No authentication                                                            | No authentication                                                           | No authentication                                                                                                                                                                                                                                                                  |
| Authentication of Merchant for payment         | No authentication                                                            | No authentication                                                           | No authentication                                                                                                                                                                                                                                                                  |
| Authentication of Cardholder for payment       | No authentication                                                            | No authentication                                                           | Reliable authentication provided by the Mastercard issuer or by Mastercard on-behalf of the issuer                                                                                                                                                                                 |

## Expired card override

Overriding expired card tests allows issuers to override possible expiration date keying errors, to support reissued cards, and to identify transactions they may otherwise approve.

Issuers will receive all Authorization Request/0100 and Reversal Request/0400 messages containing expired expiration dates. Issuers will have the opportunity to review each transaction, and subsequently approve or decline each transaction.

**NOTE: Mastercard automatically enables issuers for expired card override. Issuers that want Mastercard to reject Mastercard Authorization Request/0100 and Reversal Request/0400 messages containing expired expiration dates must complete the *Expiration Date Test Form (Form 379)*.**

## System definition of an expired card

The Dual Message Authorization System defines an expired card as a card with an expiration date that meets one of the following criteria.

- Earlier than the current processing year and month
- 20 years later than the current processing year

Customers can locate the expiration date in DE 14 (Date, Expiration), and from the magnetic stripe track, in DE 45 (Track 1 Data), or DE 35 (Track 2 Data).

## Expired card tests

The following table provides examples of Mastercard expired card test logic.

### Expired card tests

| Transaction Date      | Card Expiration Date (MM/YY) | Authorization Request Expiration Date (YY/MM) | Dual Message Authorization System Response | Explanation of Logic                                                                                                                          |
|-----------------------|------------------------------|-----------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 0924 (September 2024) | 1024 (October 2024)          | 2410 (October 2024)                           | Accept                                     | October 2024 is later than the current processing year and month. The system accepts the date as valid.                                       |
| 0924 (September 2024) | 0223 (February 2023)         | 2309 (February 2023)                          | Decline                                    | February 2023 is earlier than the current processing year and month. The system declines the date as expired.                                 |
| 0924 (September 2024) | 0844 (August 2044)           | 3008 (August 2030)                            | Accept                                     | August 2044 is less than 20 years from September 2024. The system accepts the date as valid, because 2030 is still within 20 years from 2044. |

| Transaction Date      | Card Expiration Date (MM/YY) | Authorization Request Expiration Date (YY/MM) | Dual Message Authorization System Response | Explanation of Logic                                                                                                                                  |
|-----------------------|------------------------------|-----------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0924 (September 2024) | 0145 (January 2045))         | 4501 (January 2045)                           | Decline                                    | January 2045 is 20 years and four months from September 2024. The system declines the dates as expired, because 2045 is more than 20 years from 2024. |

**NOTE: The Dual Message Authorization System uses DE 14 (Date, Expiration), if present, for the expired card test; otherwise, DE 35 (Track 2 Data), if available; otherwise, DE 45 (Track 1 Data), if available. If both DE 14 and track data are present, the Dual Message Authorization System uses DE 14 only.**

#### Issuer is enabled for Expired Card Override

When a transaction fails the expired card tests and the issuer is enabled for Expired Card Override, the Dual Message Authorization System forwards the Authorization Request/0100 or Reversal Request/0400 message to the issuer for processing.

#### Issuer is not enabled for Expired Card Override

If the issuer is not enabled for Expired Card Override and the transaction fails the expired card test, the Dual Message Authorization System generates DE 39 (Response Code), value 54 (Expired Card) of the Authorization Request Response/0110 or Reversal Request Response/0410 message sent to the acquirer.

#### Potential inappropriate declines

If the issuer is not enabled for Expired Card Override, the Dual Message Authorization System may decline cardholder transactions that the issuer would have approved if the Authorization Request/0100 and Reversal Request/0400 messages were sent to the issuer.

#### Expired data errors

Expired card tests assume that the expiration date in the authorization message is correct. However, improper keying, misunderstanding of the date, or other human error can cause the expiration date listed in the authorization message to be incorrect.

The following table provides an example of a reason a merchant might receive an incorrect expired card authorization response.

## Expired card tests

| Transaction Date      | Card Expiration Date (MM/YY) | Authorization Request Expiration Date (YY/MM) | Dual Message Authorization System Response | Explanation of Error                                                                                                                                                                                             |
|-----------------------|------------------------------|-----------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0924 (September 2024) | 05/31/25                     | 3105 (merchant entered 0531 instead of 0525)  | Decline                                    | May 2025 is eight months from September 2024. However, the merchant entered the month (05) and the day (31), but not the year (25). The Dual Message Authorization System reads the expiration date as May 2031. |

The system declines the date as expired, because 2031 is more than 20 years from 2010. However, the expiration date is valid, and perhaps the issuer would have accepted the transaction.

## Alternate processing

If the issuer is enabled for Expired Card Override and is unavailable, the Dual Message Authorization System performs the expired card tests, either in Stand-In Authorization or during X-Code Processing.

### Stand-In Authorization

Stand-In Authorization performs an expired card test. For transactions that fail the expired card test, Stand-In Authorization:

- Returns to the acquirer DE 39 (Response Code), value 54 (Expired card) in the Authorization Request Response/0110 message.
- Stores the transaction in the Store-and-Forward queue with DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code), value 0029 (Reject: expired card) of the Authorization Advice/0120 message, available for issuer retrieval.
- Identifies transactions with code 54 in the RESPONSE column or code 0029 in the FAIL RSN column on the following reports:
  - AB201010-AA (Authorization Detail)
  - AB111010-AA (Stand-In Detail Authorization)
  - AB517010-AA (Issuer Call Referral Tiered Detail)



Refer to the [Stand-In Authorization Reference Guide](#) and [Reports and files](#) for more information.

### X-Code Processing

X-Code Processing performs an expired card test. For transactions that fail the expired card test, X-Code Processing returns to the acquirer DE 39, value 54 of the Authorization Request Response/0110, and sends an Authorization Advice/0120 to the issuer.

## Gambling transaction processing

The Mastercard Network supports Gaming Payment Transaction processing for the Mastercard and Maestro card brands in Europe and Mastercard card brands in Middle East/ Africa region countries where the crediting of gambling winnings is permitted by law. The Dual Authorization Message System also supports Gaming Payment Transaction processing for the Mastercard and Maestro brands in the United States region to transfer lottery winnings to a card.

U.S. region Mastercard and Maestro issuers must block online gambling transactions in accordance with the Standards and applicable U.S. law. Such potentially illegal online gambling transactions are declined during Stand-In Authorization processing.

## Internet gambling transactions in the U.S. region

Internet gambling transactions are identified in Authorization Request/0100 messages by these values.

- DE 18 (Merchant Type), MCC 7995 (Gambling Transactions)
- DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level Indicator), value 6 (Authorized Level 6 CAT: Electronic commerce)

The presence of both these values in an Authorization Request/0100 message indicates that the acquirer has coded the transaction as an Internet gambling transaction, and possibly a restricted transaction (that is, a payment in connection with a gambling transaction that is illegal in the U.S. region).

U.S. region Mastercard and Maestro issuers must use a method of transaction blocking or decline all such transactions on an individual basis.

Issuers may approve, on an individual basis, any Internet gambling transaction authorization requests identified with MCC 9754 (Gambling: Horse Racing, Dog Racing, Non-Sports Intrastate Internet Gambling) that involve a U.S. region cardholder. In using MCC 9754, the acquirer asserts that the transaction involves gambling activity deemed by the acquirer to be legal in the U.S. region.

Issuers are not required to validate the transaction category code (TCC) value for this purpose.

Specific processing rules and Standards for Internet gambling transactions have been defined in Rule 3.7, Integrity of Brand and Network of the [Mastercard Rules](#).

## Stand-In Authorization processing

The Stand-In Authorization limits of zero will apply for the following parameter combinations:

- MCC 7995 (Gambling Transactions)
- Cardholder-activated terminal (CAT) level 6 (Electronic Commerce Transaction)

If this parameter combination is sent to the Stand-In Authorization for processing, Stand-In Authorization will reject the Authorization Request/0100 message by returning DE 39 (Response Code), value 57 (Transaction not permitted to issuer/cardholder).

These blocking parameters are also established for account range setup.

## Gaming Payment Transaction processing in the Europe and Middle East/Africa regions

The Mastercard Network supports Gaming Payment Transactions in Financial Transaction Request/0200 and Reversal Request/0420 messages for the Mastercard and Maestro card brands in the Europe and Mastercard card brands in the Middle East/Africa region, where online gaming and the crediting of gaming winnings on cards is permitted by law.

Refer to *Transaction Processing Rules* for the full set of Europe-specific standards for gaming Payment Transactions.

Acquirers submit their online Gaming Payment Transaction Financial Transaction Request/0200 and Reversal Request/0420 messages with the following values:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type), value 28 (Payment Transaction)
- In the Europe region, DE 4 (Amount, Transaction), maximum EUR 50,000 if an acquirer provides a transaction amount exceeding EUR 50,000 or its currency equivalent, the Mastercard Network will decline the request with DE 39 (Response Code), value 13 (Invalid amount).
- In the Middle East/Africa region, DE 4 (Amount, Transaction), maximum EUR 50,000 if an acquirer provides a transaction amount exceeding EUR 50,000 or its currency equivalent, the Mastercard Network will decline the request with DE 39 (Response Code), value 13 (Invalid amount).
- DE 18 (Merchant Type), MCC 7995 (Gambling Transactions)
- DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode):
  - Value 10 (Credential on File) and DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 5 (Cardholder not present [Electronic order]), or
  - Value 81 (PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram)
- DE 48 (Additional Data: Private Use), TCC value P (Payment Transaction)
- DE 48, subelement 77 (Transaction Type Identifier), value C04 (Gaming Re-pay)

If an acquirer has not completed an implementation project to submit Gaming Payment Transactions and sends a Gaming Payment Transaction authorization request, the Mastercard Network declines the request with DE 39 (Response Code), value 58 (Transaction not permitted to acquirer or terminal).

Issuers may receive Gaming Payment Transactions if they are in Europe and Middle East/Africa region countries where online gaming and crediting of gaming winnings on cards is permitted by law. For a list of these Europe region countries, refer to Rule 8.12.1 in Europe Region Rules of the *Mastercard Rules*. For a list of these Middle East/Africa region countries, refer to Payment Transactions and Mastercard Send Payment Transactions, MEA Region, of the *Transaction Processing Rules*. If an issuer's country is not one of the listed countries that allows gaming, the Mastercard Network declines the Gaming Payment Transaction request with DE 39 (Response Code), value 57 (Transaction not permitted to issuer/cardholder).

### Alternate processing

Gaming Payment Transactions may be routed to an alternate issuer host. Gaming Payment Transactions are not routed to Stand-In Authorization or X-Code Processing.

When there is either no alternate issuer host or no response from an alternate issuer host, the Mastercard Network will decline the transaction with DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

### Acquirer implementation

Acquirers must complete an implementation project to submit Payment Transactions for gaming winnings. For more information about registering for this service, contact your Mastercard regional representative or your Customer Implementation Services (CIS) Implementation Project Manager.

### For more information

For details about data requirements, refer to *Mastercard Rules*, and *Mastercard Gaming and Gambling Payments Program Standards*.

## Gaming Payment Transaction processing in the United States region

The [Single Message System](#) supports Gaming Payment Transactions in Financial Transaction Request/0200 and Reversal Request/0420 messages for the Mastercard and Maestro brands in the [United States \(U.S.\)](#) region to transfer lottery winnings to a card.

An acquirer in the [U.S.](#) region submits a Gaming Payment Transaction Financial Transaction Request/0200 and Reversal Request/0420 message with the following values:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type), value 28 (Payment Transaction)
- DE 4 (Amount, Transaction), maximum USD 10,000
- DE 18 (Merchant Type), MCC 7800 (Government-owned Lottery [U.S. Region Only])

- DE 48 (Additional Data: Private Use), TCC value P (Payment Transaction)
- DE 48, subelement 77, value C04 (Gaming Re-pay)

**NOTE: The Gaming Payment Transaction must not be processed as electronic commerce (e-commerce).**

An issuer in the United States region may receive Gaming Payment Transactions identified as described above.

### Alternate processing

The Gaming Payment Transaction will not be routed to Stand-In Authorization or X-Code Processing. When a Gaming Payment Transaction identified using MCC 7800 is routed to:

- Stand-In Authorization, the Financial Transaction Request/0200 message will be declined with DE 39 (Response Code), value 05 (Do Not Honor); or
- The X-Code Processing, the Financial Transaction Request/0200 message will be declined with DE 39 (Response Code), value 01 (Refer to Card Issuer).

## Global Automated Referral Service

The Global Automated Referral Service (GARS) is an optional service that streamlines the process of responding to call referrals for both acquirers and issuers.

GARS is mandatory in the U.S. region and optional for customers outside the U.S. region.

GARS reduces the processing time for call referrals from an average of 20 minutes to an average of five minutes or less.

It includes the following features:

- A toll-free phone number connecting acquirers to issuers worldwide. Routing is based on the card account number entered by the acquirer (For the toll-free phone number that applies to your area, contact the Global Customer Service team).
- Issuer response time parameters to support the timely provision of authorization responses
- Stand-In Authorization for calls not answered within the required time frame or for issuers not registered for GARS, to guarantee a timely response to the acquirer
- Tracking and reporting of issuer and acquirer performance
- Performance monitoring and pricing that motivate call referral completion and eliminate unnecessary issuance of call referrals

### GARS process

When the merchant receives a refer to card issuer authorization response, the merchant calls the acquirer.

The acquirer calls GARS, which uses one of the following paths to obtain an authorization response:

- GARS connects the acquirer to the issuer
- GARS connects the acquirer to Stand-In Authorization

## Acquirer use of GARS

GARS helps acquirers to more quickly receive a response to a call referral from issuers.

### Acquirer procedures

Acquirers receiving a refer to card issuer authorization response should use the following procedure to receive an authorization decision from the issuer.

### Associated edits

| IF...                                                                                                                                                                                                    | THEN...                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| an acquirer receives an authorization response of refer to card issuer.                                                                                                                                  | dial the GARS phone number for your country (provided when you sign up for GARS) after issuance of the refer to card issuer message. |
| GARS states "Welcome to the Mastercard Global Automated Referral Service. If you make an error entry at any time, press the star key. Please enter the Acquiring ICA number followed by the pound sign." | enter the six-digit billing ICA number, followed by the pound/hash sign (#).                                                         |
| GARS states "The ICA number entered is XXXXXX. If correct, press one. If not correct, press two."                                                                                                        | press either the one (1) or two (2) key.                                                                                             |
| GARS states "Please enter the Mastercard account number followed by the pound sign."                                                                                                                     | enter the 16-digit Mastercard cardholder account number followed by the pound/hash sign (#).                                         |
| GARS validates the Mastercard account number, dials the issuer, and the issuer comes on the line.                                                                                                        | discuss the authorization transaction.                                                                                               |
| the issuer provides you with an authorization response.                                                                                                                                                  | provide the merchant with the issuer's authorization response.                                                                       |

### Stand-In Authorization procedures

If the issuer does not connect with the call within 30 seconds, GARS connects you to Stand-In Authorization. Use the following procedure to receive an authorization decision from Stand-In Authorization.

## Associated edits

| IF...                                                                                                                                                                                                                                                                                                                                             | THEN...                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GARS states "We appreciate your patience. The issuer is not available at this time. Your call will now be serviced by Mastercard Stand-In Authorization."                                                                                                                                                                                         | follow the instructions given by GARS.                                                                                                                                                                                                                                                                                                                                                                          |
| GARS states "Please enter the MCC number, followed by the pound sign."                                                                                                                                                                                                                                                                            | enter the MCC number and the pound/hash sign (#).                                                                                                                                                                                                                                                                                                                                                               |
| GARS states "Please enter the merchant number, followed by the pound sign."                                                                                                                                                                                                                                                                       | enter the merchant number and the pound/hash sign.                                                                                                                                                                                                                                                                                                                                                              |
| GARS states "Please enter the four digit expiration date in MM/YY format, followed by the pound sign."                                                                                                                                                                                                                                            | enter the four-digit expiration date in MM/YY format and the pound/hash sign.                                                                                                                                                                                                                                                                                                                                   |
| GARS states "Please enter the transaction amount in U.S. dollars and cents, followed by the pound sign."                                                                                                                                                                                                                                          | enter the transaction amount in U.S. dollars and cents and the pound/hash sign.                                                                                                                                                                                                                                                                                                                                 |
| GARS states "You entered X dollars and X cents. If correct, press 1. If not correct, press 2."                                                                                                                                                                                                                                                    | press 2 to repeat this step.                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>GARS states "Please choose one of the following transaction types:</p> <ul style="list-style-type: none"> <li>• For retail, press 1.</li> <li>• For phone, mail order, or electronic commerce, press 2.</li> <li>• For cash advance, press 3.</li> <li>• For travel, press 4.</li> <li>• For all other transaction types, press 5."</li> </ul> | <p>enter the number for the transaction type of this transaction.</p>                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                   | <p>If 5 is entered in the previous step, GARS will offer additional options:</p> <ul style="list-style-type: none"> <li>• For automobiles, press 1.</li> <li>• For food, press 2.</li> <li>• For hotel, press 3.</li> <li>• For tuition or hospital, press 4.</li> <li>• For unique, press 5.</li> <li>• For payment, press 6.</li> </ul> <p>Enter the number for the transaction type of this transaction.</p> |

| IF...                                                                                                                                                                                                                                                                                                                            | THEN...                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <p>GARS provides one of the following authorization responses:</p> <ul style="list-style-type: none"> <li>• "The approval code is XXXXXX. To repeat the approval code, press 1. To end this call, press 2."</li> <li>• "This transaction has been declined. Thank you for calling Mastercard International. Goodbye."</li> </ul> | <p>Provide the merchant with Stand-In Authorization's authorization response.</p> |

## Acquirer best practices

Mastercard recommends that acquirers promote efficient and accurate processing of call referrals by using the following practices when talking to issuers' agents:

- Be prepared to provide the following authorization request information to the issuer's agent:
  - Card account number
  - Expiration date
  - Acceptor business code (MCC)
  - Transaction amount in U.S. dollars
 

If the acquirer chooses to receive DE 5 (Amount, Settlement) in the Authorization Request Response/0110 message, the acquirer may use that amount, which is stated in U.S. dollars; or the acquirer may determine the USD amount in DE 4 (Amount, Transaction) or DE 6 (Amount, Cardholder Billing). If the USD amount is not provided, the acquirer must perform currency conversion to U.S. dollars.
  - Acquirer ICA number
- Initiate the conversation with slow, distinct pronunciation. Be aware that the issuer's agent might not be comfortable with the acquirer's primary language.
- Request clarification when uncertain of the communication.

If GARS routes the call referral to the Mastercard Stand-In Authorization, acquirers should allow additional time to complete the call and be prepared to provide additional authorization information, as required by Stand-In Authorization.

Acquirers should monitor call referral response rates to identify merchants that do not comply with the call referral process.

Acquirers receive reimbursement automatically when they use GARS in response to a refer to card issuer authorization response.

**NOTE: Reimbursement to acquirers using GARS is automatic. If acquirers use non-GARS methods of communication, they must request reimbursement through the Fee Collection Record.**

## Methods acquirers can use to improve GARS performance

The acquirer can improve GARS performance and ensure that it qualifies for reimbursement by doing the following:

- Test your phone system to ensure that it generates true touch-tones on outbound calls. The GARS connection to the issuer is dependent on touch-tones.
- Communicate the GARS procedures to your agents. Mastercard is available to assist you in your training efforts.
- Consider adding a conference call feature to your phone system so that you, your merchant, the cardholder, and the issuer can interact directly.
- Consider adding speed dialing to your phone system to save time when calling GARS.
- Communicate the GARS procedures and the importance of GARS to your merchants. Encourage them to call you (rather than request another form of payment) when a call referral message appears on their POI devices.

## Issuer use of GARS

GARS helps issuers because it provides them with a method to more quickly respond to acquirers requesting authorizations for call referral responses.

### Issuer procedures

When you issue a response of refer to card issuer and the acquirer contacts you via GARS, use the following procedure.

### Associated edits

| IF...                                                                                      | THEN ...                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GARS states "Incoming Mastercard referral call. Press the zero key and the pound key now." | press the required zero (0) and pound (#) keys within 30 seconds.<br><br>if you do not enter 0# within 30 seconds, GARS connects the acquirer to Stand-In Authorization for the authorization response. |
| GARS connects with the acquirer to discuss the transaction.                                | obtain the necessary authorization information from the acquirer and communicate your authorization response to the acquirer.                                                                           |

### Issuer best practices

Mastercard recommends that issuers promote efficient and accurate processing of call referrals by using the following practices when talking to acquirers' agents:

- Initiate the conversation with slow, distinct pronunciation. Be aware that the acquirer's agent might not be comfortable with the issuer's primary language.
- Request clarification when uncertain of the communication.

To improve their call referral processes, issuers should monitor:



- The number of call referrals they generate in response to Authorization Request/0100 messages. Issuers should use call referrals only for high-risk situations because call referrals result in an expensive and time-consuming process, which can be detrimental to cardholder relationships.
- The rate of call referrals going to Stand-In Authorization. Issuers should ensure that they have sufficient call center staff and a stable telecommunications infrastructure to be able to support the number of call referrals they generate. Mastercard strongly urges issuers to register for GARS. Acquirer GARS calls will be routed to the Stand-In Authorization for issuers not registered.
- The average duration of call referrals. Issuers should try to meet the goal of keeping call referrals to five minutes or less. Short call durations result in reduced transaction times, improve the call referral process for merchants and cardholders, and provide the best use of resources.

Issuers should establish Stand-In Authorization transaction limits that are specific to transactions approved through GARS call referrals. The Mastercard Stand-In Authorization facility ensures a timely authorization decision when the issuer is not available.

Mastercard suggests that GARS issuers establish two phone numbers for GARS call referral processing. Issuers can have all intracountry call referrals routed to one phone number and all intercountry call referrals routed to another phone number. Issuers can better anticipate potential language differences and offer enhanced services to cardholders traveling outside of their countries.

Charges to the issuer apply for issuing each call referral and for completed GARS calls initiated by the acquirer. Issuers also pay a noncompliance assessment for Stand-In Authorization if they fail to answer a GARS call within 30 seconds.

### **Methods issuers can use to improve GARS performance**

Issuers can use the following methods to improve GARS performance:

- Ensure that your phone system can generate true touch-tone signals. If you use an automated system such as a Voice Response Unit to answer incoming calls, be sure the system responds with a zero key followed by a pound key (0, #). This allows the acquirer's agent to hear the instruction of your automated system to expedite your call referral response.
- Ensure that your system is prepared to answer phone calls within 30 seconds. Calls not answered within these time frames will be routed to Stand-In Authorization.
- Determine whether to establish specific Stand-In transaction limits for GARS. Complete the *GARS Initiation and Change Form* (Form 334) to establish specific limits.
- Determine whether to establish one phone number for all incoming GARS calls or two phone numbers: one for GARS requests from acquirers located outside of the issuer country and one for domestic GARS requests. Indicate the phone numbers on the *GARS Initiation and Change Form* (Form 334).

Submit any changes to your customer representative at least 15 days before using a new phone number or new parameters:

- Determine your ability to handle the volume of completed call referrals using GARS. Because GARS call referrals are fast and easy, there may be an increase in the ratio of calls completed to referrals issued. You may want to consider adjusting staffing and equipment levels to ensure higher control and risk reduction and avoid excessive Stand-In Authorization fees.
- Communicate to your employees the importance of answering GARS calls quickly to avoid additional Stand-In Authorization fees. Mastercard is available to assist you in your training efforts.

## Monitoring call referral activity

The minimum response and talk-time standards apply to all call referrals, both domestic and international, whether processed through GARS or direct customer-to-customer communications.

### Issuer and acquirer performance reports

Issuers and acquirers can monitor call referrals completed through GARS using the following reports:

- Issuer Authorization Detail Report and Issuer Call Referral Detail Report: shows issuer charges for call referrals
- Acquirer Authorization Detail Report and Acquirer Call Referral Detail Report: shows acquirer reimbursement for completed call referrals
- Call Referral Summary Report: shows tiered pricing for excessive issuer call referral activity
- Guaranteed Issuer Availability Report

Refer to the **Pricing and Billing Resource Center** application on Mastercard Connect® for fees and billing information.

### Store and forward messages

Issuers also can monitor call referrals completed through GARS via store-and-forward (SAF) messages. Mastercard generates Authorization Advice/0120 messages for GARS transactions that Stand-In Authorization. Mastercard uses a promotion code of MCGARS to identify GARS Stand-In activity.

## Requesting GARS or changing GARS parameters

To begin to use GARS or to change GARS parameters, complete the *GARS Initiation and Change Form* (Form 334).

## Incremental Preauthorization

Incremental Preauthorization provides a means to associate incremental, or additional, preauthorizations to a single clearing presentment. It is a common practice for merchant categories to submit incremental preauthorizations and settle those through a single clearing presentment. These standards provide a method for issuers, with certainty, to associate

incremental preauthorizations to a single presentment providing them the means to more effectively manage cardholder open-to-buy balances.

## Incremental Preauthorization Transaction Processing

Each incremental preauthorization transaction associated with a single cardholder event must reference the original preauthorization. When incremental Authorization Request/0100 messages are sent for the same transaction, acquirers must include DE 63 (Network Data) and DE 15 (Date, Settlement) information from the original Authorization Request Response/0110 message in DE 48 (Additional Data: Private Use), subelement 63 (Trace ID) as follows:

- Positions 1–3 = value from DE 63, subfield 1 (Financial Network Code)
- Positions 4–9 = value from DE 63, subfield 2 (Banknet Reference Number)
- Positions 10–13 = value from DE 15 (Date, Settlement)
- Positions 14–15 = blank filled

The First Presentment/1240 message submitted by the acquirer must include information from the original Authorization Request Response/0110 message in Integrated Product Message DE 63 (Transaction Life Cycle ID), subfield 2 (Trace ID). Upon receipt of the clearing record, the issuer must release holds created by all authorizations properly related to the original preauthorization.

## Guiding Principles

This section provides insight into the intent of the Incremental Preauthorization process.

- The primary intent of an incremental preauthorization is to increase the amount of an original preauthorization. The authorized amount is equal to the cumulative amount of all authorizations (plus tolerance if applicable). The allowable tolerances are described in the *Chargeback Guide*.
- The original preauthorization is the reference. All attributes associated with the original are inherited by the incremental preauthorizations. Transactions that would have incremental preauthorizations will likely be submitted as preauthorizations.
- The usage of incremental preauthorizations is optional. Acquirers can, as an alternative, submit clearing records (presentments) for each individual authorization.
- Incremental preauthorizations are recognized for all MCCs.
- Transactions that may have incremental preauthorizations are recommended to be submitted as preauthorizations. Incremental preauthorizations must be coded as preauthorizations.

## Specific Scenarios

### Typical Usage

The following table depicts the process of a typical incremental preauthorization. The merchant successfully completes an original preauthorization for 100, followed by an incremental preauthorization of 50. This total amount is then processed as a first presentment.

| Message Type                                                                                                                                                   | DE 63 (Banknet Reference Number) | DE 15 (Settlement Date) | Trace ID<br>(Authorization: DE 48, subelement 63)<br>(Clearing: DE 63, subfield 2) |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|------------------------------------------------------------------------------------|-----|
|                                                                                                                                                                |                                  |                         | DE 4 (Amount, Transaction)                                                         |     |
| Original Preauthorization (0100/0110)<br><br>The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.    | MCC123ABC                        | 1105                    | N/A                                                                                | 100 |
| Incremental Preauthorization (0100/0110)<br><br>The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer. | MCC456ABC                        | 1106                    | MCC123ABC1105                                                                      | 50  |
| First Presentment (1240–200)                                                                                                                                   | N/A                              | N/A                     | MCC123ABC1105                                                                      | 150 |

### Incremental Preauthorization Reversals

Acquirers are encouraged to recognize the original preauthorization and all related incremental preauthorizations in aggregate in the event they need to reverse part or the entire transaction amount. The recommended method of submitting reversals is illustrated in the following table.

### Incremental Preauthorization Reversals

In the following scenario, the merchant successfully completes an original preauthorization for 100, followed by an incremental preauthorization of 50, but then wants to cancel the transaction.

|                                                                                                                                                                |                                  |                         | Trace ID                              |                            |                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|---------------------------------------|----------------------------|----------------------------|
| Message Type                                                                                                                                                   | DE 63 (Banknet Reference Number) | DE 15 (Settlement Date) | (Authorization: DE 48, subelement 63) | DE 4 (Amount, Transaction) | DE 95 (Replacement Amount) |
|                                                                                                                                                                |                                  |                         | (Clearing: DE 63, subfield 2)         |                            |                            |
| Original Preauthorization (0100/0110)<br><br>The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.    | MCC123ABC                        | 1105                    | N/A                                   | 100                        | N/A                        |
| Incremental Preauthorization (0100/0110)<br><br>The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer. | MCC456ABC                        | 1106                    | MCC123ABC1105                         | 50                         | N/A                        |

| Message Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DE 63 (Banknet Reference Number) | DE 15 (Settlement Date) | Trace ID                              |                               |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|---------------------------------------|-------------------------------|----------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  |                         | (Authorization: DE 48, subelement 63) | (Clearing: DE 63, subfield 2) | DE 4 (Amount, Transaction) |
| Reversal (0400/0410)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MCC789ABC                        | 1106                    | MCC123ABC11                           | 150                           | 0                          |
| In the Reversal transaction, DE 90 (Original Data Elements) will contain DE 11 (System Trace Audit Number [STAN]), DE 7 (Transmission Date and Time), DE 32 (Acquiring Institution ID Code), and DE 33 (Forwarding Institution ID Code) from the Authorization Request Response/0110 message that corresponds to the original preauthorization request. The Transaction Amount (DE 4) is recommended to contain 150, while the Replacement Amount (DE 95) is recommended to contain 0, indicating a full reversal. |                                  |                         |                                       |                               |                            |

Incremental Preauthorization Reversals

In the next scenario, the merchant successfully completes an original preauthorization for 100, followed by an incremental preauthorization of 50. The merchant then reduces the preauthorization by 10 using a partial reversal.

| Message Type                                                                                                                                                   | DE 63 (Banknet Reference Number) | DE 15 (Settlement Date) | Trace ID                              |                               |                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|---------------------------------------|-------------------------------|----------------------------|
|                                                                                                                                                                |                                  |                         | (Authorization: DE 48, subelement 63) | (Clearing: DE 63, subfield 2) | DE 4 (Amount, Transaction) |
| Original Preauthorization (0100/0110)<br><br>The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.    | MCC123ABC                        | 1105                    | N/A                                   | 100                           | N/A                        |
| Incremental Preauthorization (0100/0110)<br><br>The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer. | MCC456ABC                        | 1106                    | MCC123ABC1105                         | 50                            | N/A                        |

| Message Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DE 63 (Banknet<br>Reference<br>Number) | DE 15<br>(Settlement<br>Date) | Trace ID                                    |                                  |                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------|---------------------------------------------|----------------------------------|-------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                        |                               | (Authorization:<br>DE 48,<br>subelement 63) | (Clearing: DE<br>63, subfield 2) | DE 4 (Amount,<br>Transaction) |
| Reversal<br>(0400/0410)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MCC789ABC                              | 1106                          | MCC123ABC11                                 | 150                              | 140                           |
| In the Reversal transaction, DE 90 (Original Data Elements) will contain DE 11 (System Trace Audit Number [STAN]), DE 7 (Transmission Date and Time), DE 32 (Acquiring Institution ID Code), and DE 33 (Forwarding Institution ID Code) from the Authorization Request Response/0110 message that corresponds to the original preauthorization request. The Transaction Amount (DE 4) is recommended to contain 150, while the Replacement Amount (DE 95) is recommended to contain 0, indicating a full reversal. |                                        |                               |                                             |                                  |                               |



| Message Type        | DE 63 (Banknet Reference Number) | DE 15 (Settlement Date) | Trace ID                              |                               |                            |
|---------------------|----------------------------------|-------------------------|---------------------------------------|-------------------------------|----------------------------|
|                     |                                  |                         | (Authorization: DE 48, subelement 63) | (Clearing: DE 63, subfield 2) | DE 4 (Amount, Transaction) |
| Clearing (1240–200) | N/A                              | N/A                     | MCC123ABC1105                         | 140                           | N/A                        |

Acquirers are expected to send reversals once they become aware that an adjustment is necessary. An exception to this is if the acquirer will be processing the first presentment within 24 hours of knowing a reversal is necessary.

### Guaranteed Reservations

Guaranteed reservations are common in the hotel, motel, vehicle rental, and cruise line industries. Guaranteed reservations are managed uniquely for each of these segments. The recommended practice for each of these is as follows:

- **Vehicle Rental:** An account status inquiry is issued at the time of the reservation to confirm the account is in good standing. No advance authorization is issued nor is any amount held or liability assumed for no-shows.
- **Cruise Lines:** Typically, cruise lines charge the entire amount of the cruise at the time of the reservation. Any charges incurred during the cruise are submitted as a new authorization with incremental preauthorizations as needed.
- **Hotel Reservation:** A hotel, motel, resort, or other lodging merchant participating in the Guaranteed Reservations service has the option of utilizing the rules in regard to no-show policies. The cardholder is obligated to cancel a confirmed reservation before 18:00 at the hotel, motel, or resort (merchant's local time). If the cardholder fails to cancel the confirmed reservation, the merchant can charge the cardholder a no-show charge equal to one night's lodging. In this case, it is appropriate for the merchant to authorize and clear the amount that would be charged for one night's lodging if they so choose. Any amount above that would be a violation of Mastercard rules and be at risk for a chargeback.

### Authorization Chargeback Protection Period Extension Request

In support of the time limit for chargeback protection related to Dual Message System authorizations, Mastercard provides an Authorization Chargeback Protection Period Extension Request for use by Dual Message System acquirers, which is a non-financial (zero amount) incremental preauthorization Authorization Request/0100 message.

Authorizations originally coded as a preauthorization may require a longer chargeback protection period. To increase the effective duration of the chargeback protection period, the merchant may submit incremental preauthorization requests for either an additional amount or zero amount for the same transaction on later dates. Incremental preauthorizations for an additional amount may be used to increase the authorized amount held against the card.

account and to extend the chargeback protection period associated with the original preauthorization. Incremental preauthorizations for a zero amount may be used to extend only the chargeback protection period associated with the original preauthorization.

If the issuer declines the chargeback extension request and the original preauthorization subsequently expires, the acquirer must request a new authorization.

**NOTE: Each approved preauthorization has an extended payment guarantee period of 30 calendar days from the authorization approval date. The issuer payment guarantee period is limited to a maximum period counting from the authorization or preauthorization date. This maximum period is 30 calendar days for Mastercard authorizations properly coded as preauthorizations and is seven calendar days for all other Mastercard authorizations and for all Maestro and Cirrus authorizations and preauthorizations.**

Issuers must be prepared to receive and process a chargeback protection period extension Authorization Request/0100 message.

To increase only the effective duration of the chargeback protection period of an original preauthorization, the merchant may submit an incremental preauthorization request for a zero amount. The acquirer must submit the Authorization Request/0100 as follows.

- DE 3 (Processing Code), subfield 1 (Transaction Type) = 00 (Purchase)
- DE 4 (Transaction Amount) = Zero (0) (Additional Amount)
- DE 61 (POS Data), subelement 7 (POS Transaction Status) = 4 (Preauthorized request)
- DE 61, subfield 4 (POS Cardholder Presence) is not equal to 4 (Recurring transactions)
- DE 48 (Additional Data), subfield 63 (Trace ID) must be present and contain the Trace ID data (DE 15 and DE 63) from the original preauthorization request in positions 1–15

**NOTE: Authorization Chargeback Protection Period Extension Requests are applicable to all types of purchase preauthorization transactions, except Mastercard contactless transit aggregated or transit debt recovery transactions nor to merchant-financed or acquirer-financed installment payment transactions properly coded as preauthorizations. If submitted, the message will be declined.**

Authorization Advice/0120: acquirer-generated and Reversal Request/0400 messages are not applicable for Authorization Chargeback Protection Period Extension Requests. If submitted, the message will be declined.

The issuer, at its discretion, will send the acquirer an Authorization Request Response/0110 message where DE 39 may contain one of the following values.

- 00 (Approved)
- 05 (Do not honor)
- 85 (Not declined)
- Other valid business declines  
Examples of other acceptable business declines include values 41 (Lost Card), 43 (Stolen Card), and 54 (Expired Card)

If the issuer is unable to reply, the acquirer will receive a response code of 91 (Authorization System or issuer system inoperative).

When the chargeback protection period of a preauthorization is extended because of an incremental preauthorization, it is extended for 30 calendar days from the date of the latest approved incremental preauthorization. Such requests, if approved by the issuer and when properly coded as an incremental preauthorization, will give rise to an extended chargeback protection period and optionally additional approved amounts that may be cleared under the conditions that would apply to the security parameters when applied to the original authorization. In other words:

- To the extent that interchange levels are determined by the security parameters of the authorization, then the parameters of the original authorization will be considered.
- To the extent that chargebacks consider the security parameters of the authorization, then the parameters of the original authorization will be taken into account.

When the chargeback protection period expires, issuers must release any block they may have placed on the cardholder's account in relation to the authorization.

Several other options may be used when the authorization lifecycle expires before the transaction is finalized:

- Submit a new preauthorization within 30 calendar days of the planned date of the stay. This creates the risk of a declined transaction. If the original preauthorization has not expired, Mastercard recommends that a reversal is sent of the first transaction to instruct the issuer to release the funds in advance of receiving the second transaction.
- Periodically use the Account Status Inquiry Service to confirm that the card is in good standing prior to the stay. An authorization for the anticipated amount may be submitted in advance of the stay (but no earlier than 30 calendar days before).
- Use the Advance Resort Deposit process as described in the [Chargeback Guide](#) to authorize and clear the transaction at the time of the reservation.

## Long Running Transactions

Occasionally, transactions have an extended life. Examples of these scenarios are vehicle rentals for multiple weeks, extended resort stays, or long cruises. Mastercard expects merchants to authorize the estimated amount at the initiation of the service. Incremental preauthorizations would be requested if the permitted tolerance (if applicable) is exceeded. For more information about transactions with permissible tolerances, refer to [Transaction Processing Rules](#), Merchant Acceptance section.

With the specific intent of extending the authorization lifecycle, an incremental preauthorization is recommended. As noted earlier, the authorization lifecycle is based on the date of the latest incremental preauthorization.

## Extending Security Features

Mastercard® Identity Check™ protocol authentication relates only to the transaction (and any related incremental preauthorizations if applicable) for which they were originally provided. An acquirer may retain these features by submitting an incremental preauthorization transaction with an amount greater than zero.

## Maestro preauthorized transaction processing

The Dual Message Authorization System allows Europe region acquirers to request preauthorization on transactions for which the final transaction amount is not yet determined. For automated fuel dispenser (AFD) transactions, once the issuer approves the preauthorization request and after the transaction completes at the point-of-interaction and the final amount is determined, the acquirer sends an authorization advice message to the issuer.

**NOTE: Applies only to the Europe region.**

For card-not-present (CNP) and post-authorized aggregated Maestro contactless transit transactions, if the final amount is not equal to the preauthorized amount, a partial reversal is required and clearing must be submitted for the new amount identified in the partial reversal.

The Clearing message must contain the final amount, which is equal to the amount contained the authorization advice message or in the new amount of a partial reversal message.

Preauthorization on Maestro transactions is permitted only at unattended point-of-sale (POS) terminals at petrol stations (MCC 5542, Fuel Dispenser, Automated). Preauthorization on Maestro transactions is also permitted for CNP transactions and for post-authorized aggregated Maestro transactions. Preauthorizations must not be used in any other acceptance environment.

Acquirers must ensure that an authorization request for an amount greater than zero is properly identified as a preauthorization when a CNP transaction might not be completed for reasons other than technical failure or lack of full issuer approval. For example, the transaction might not be completed when the cardholder will be offered the choice at a later time to complete the transaction with another payment means, or when the goods ordered by the cardholder might be later found to be out of stock. The risk of technical failures such as telecommunications failure or POI terminal failure should not be taken into account to determine if an authorization must be coded as a preauthorization.

For a CNP transaction, the transaction amount must equal the preauthorized amount. Preauthorizations for an estimated amount are only permitted for unattended petrol terminal (MCC 5542) transactions.

### How it works

Maestro preauthorized transaction processing works as follows:

- The acquirer submits an Authorization Request/0100 (preauthorization format) message containing the maximum amount determined by the acquirer or the merchant where DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains value 4 (Preauthorized request). Each approved preauthorization has a payment guarantee period of seven calendar days from the authorization approval date.
- The issuer generates an Authorization Request Response/0110 message for the full amount of the preauthorization or for a lesser amount, as determined by the issuer, if the merchant's

terminal supports partial approval. The transaction is guaranteed up to the amount authorized by the issuer.

- The acquirer must send Authorization Advice/0120: Acquirer-generated messages containing DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code), value 191 (Acquirer Processing System [APS] Completed Authorization Transaction) at the completion of Maestro CAT Level 1 petrol transactions.
- The issuer acknowledges the transaction using an Authorization Advice Response/0130 message.
- The acquirer must send Reversal Request/0400 messages containing DE 4 (Amount, Transaction) from the Authorization Request/0100 message or DE 4 from the Authorization Request Response/0110 message if a partial approval response and DE 95 (Replacement Amounts) that contains the actual amount subfield necessary to perform a partial reversal if the final amount is less than the preauthorized amount for card-not-present (CNP) and post-authorized aggregated Maestro contactless transit transactions.
- If any transaction is terminated after the preauthorization, the acquirer must send the issuer a Reversal Request/0400 message containing DE 4 (Amount, Transaction) from the Authorization Request/0100 message or DE 4 from the Authorization Request Response/0110 message if a partial approval response.
- The issuer acknowledges full and partial reversal requests using a Reversal Request Response/0410 message.

## Authorization Request/0100: Maestro Preauthorization

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                                             | Org | Sys | Dst | Values/Comments                                                   |
|--------------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------|
| DE 4 (Amount, Transaction)                                               | M   | •   | M   | Maximum amount determined by the acquirer or merchant.            |
| DE 18 (Merchant Type)                                                    | M   | •   | M   | Must contain value 5542 = Fuel Dispenser, Automated               |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) | M   | •   | M   | Must contain value 4 = Preauthorized request.                     |
| DE 61, subfield 10 (Card-Activated Terminal Level)                       | M   | •   | M   | 1 = Authorized Level 1 CAT: Automated dispensing machine with PIN |

## Authorization Advice/0120: Maestro Preauthorization Completion

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

### Data elements

| Data Element                                                                                                | Org | Sys | Dst | Values/Comments                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 4 (Amount, Transaction)                                                                                  | M   | •   | M   | Must contain the final transaction amount                                                                                                                    |
| DE 48 (Additional Data—Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time) | •   | X   | M   | Dual Message Authorization System supplies date and time.                                                                                                    |
| DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)                                                 | M   | •   | M   | 191 = Acquirer Processing System (APS) completed<br><br>Note: Issuers must also be prepared to receive value 190 (Acquirer Processing System (APS) approved) |

### Matching Request and Advice Messages

To match the original Authorization Request/0100 with the Authorization Advice/0120, the following DEs should be used:

- DE 2 (Primary Account Number [PAN])
- DE 7 (Transmission Date and Time)
- DE 11 (System Trace Audit Number [STAN])
- DE 32 (Acquiring Institution ID Code)
- DE 33 (Forwarding Institution ID Code)

## Maestro Recurring Payments Program

The Maestro® Recurring Payments Program enables the Maestro brand to accept recurring payments for electronic commerce (e-commerce) transactions. This program enables Maestro-branded products to provide enhanced value to customers.

### Acquirer requirements

- Acquirers in the Europe region that process Maestro e-commerce transactions may participate in the Maestro Recurring Payments Program.
- Participating acquirers must submit the following values in Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400: acquirer-

generated messages to identify Maestro e-commerce recurring payment transactions received from enrolled merchants:

- DE 22 (Point of Service Data Code), subfield 1 (Terminal Data: Card Data Input Capability), value 81 (PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 [UCAF] or DSRP cryptogram in DE 104, subelement 001 [Digital Payment Cryptogram])
- DE 48 (Additional Data: Private Use), subelement 32 (Mastercard Assigned ID)
- DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator), value 3 (UCAF data collection is supported by the merchant, and UCAF (Mastercard Assigned Static AAV) data must be present)
- DE 48, subelement 43 (Static AAV)
- DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 4 (Recurring transactions)
- DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level), value 6 (Authorized Level 6 CAT: Electronic commerce)

### Issuer requirements

- Maestro issuers must support receipt and processing of recurring payment e-commerce transactions, as identified by the presence of the following data elements in Authorization Request/0100, Authorization Advice/0120: acquirer-generated, Authorization Advice/0120: system-generated, Reversal Request/0400: acquirer-generated, and Reversal Request/0400: system-generated messages:
  - DE 22, subfield 1, value 81
  - DE 48, subelement 32
  - DE 48, subelement 42, subfield 1, position 3, value 3
  - DE 48, subelement 43
  - DE 61, subfield 4, value 4
  - DE 61, subfield 10, value 6
- Maestro issuers also must provide the following data elements in Authorization Request Response/0110, Authorization Advice Response/0130: system-generated, and Reversal Request Response/0410 messages:
  - DE 48, subelement 42
  - DE 48, subelement 43

## Authorization Request/0100: Maestro Recurring Payment

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element ID and Name                                                                                                                                                               | Org | Sys | Dst | Comments                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)                                                                                                    | M   | •   | M   | 81 = PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram).                                                          |
| DE 48 (Additional Data—Private Use), subelement 32 (Mastercard Assigned ID)                                                                                                            | C   | •   | C   | The value assigned by Mastercard                                                                                                                                                                                                                 |
| DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) | C   | •   | C   | 3 = UCAF data collection is supported by the merchant, and UCAF (Mastercard assigned static Accountholder Authentication Value) data must be present.<br><br><b>NOTE: DE 48, subelements 32 and 43 are required for Static AAV transactions.</b> |
| DE 48, subelement 43 (Static AAV)                                                                                                                                                      | C   | •   | C   | Contains the contents of positions 1–28. When DE 48, subelement 43 contains a static AAV, DE 48, subelement 32 is mandatory.                                                                                                                     |
| DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence)                                                                                                              | M   | •   | M   | 4 = Recurring transactions                                                                                                                                                                                                                       |
| DE 61, subfield 10 (Cardholder-Activated Terminal Level)                                                                                                                               | M   | •   | M   | 6 = Authorized Level 6 CAT: Electronic Commerce                                                                                                                                                                                                  |

## Dual Message Authorization System Edits

The following edits are performed on Authorization Request/0100 messages for Maestro Recurring Payment Program transactions.

Mastercard uses the following edit to verify participation in the Maestro Recurring Payments Program. Dual Message Authorization System uses this edit to validate the value in DE 48,



subelement 32 for Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages.

**NOTE: The same Static AAV value can be used for the Maestro Recurring Payments Program, the Mastercard Utility Payment Program, and the Maestro Low Risk Merchant Program.**

### Applicable edits

| WHEN...                                                                                                                                         | THEN the Dual Message Authorization System...                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| an Authorization Request/0100 or Authorization Advice/0120: system-generated message with DE 48, subelement 42, subfield 1, position 3, value 3 | validates that DE 48, subelement 32 and subelement 43 are present and contain valid values. |

### Existing Edits

The following existing edits will be applied to the Maestro Recurring Payments Program in the Dual Message Authorization System where the DE 48, subelement 42, subfield 1, position 3, value 3 (UCAF data collection is supported by the merchant, and UCAF [Mastercard assigned Static Accountholder Authentication Value] data must be present) data must be present for Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages submitted under the Maestro Recurring Payments Program for e-commerce initiated transactions.

### Applicable edits

| WHEN...                                                                                                                                                                                                                                                                                                                         | THEN...                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) has the following value: 3 (UCAF data collection supported by merchant, and UCAF [Mastercard Assigned Static AAV Value] data is present) | DE 48, subelement 43 (Static AAV) must contain the static AAV assigned by Mastercard for the Maestro Recurring Payments Program for e-commerce transactions.                                                                                                                                                                   |
| DE 48, subelement 43 (Static AAV) contains the static AAV assigned by Mastercard for the Maestro Recurring Payments Program for e-commerce transactions                                                                                                                                                                         | DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) must contain the value 3 (UCAF data collection supported by merchant, and UCAF [Mastercard-assigned Static AAV Value] data is present). |

| WHEN...                                                                                                                                                                                                | THEN...                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 42, subfield 1, position 3 (UCAF Collection Indicator) has the value 3 (UCAF data collection supported by merchant, and UCAF [Mastercard Assigned Static AAV Value] data is present) | for recurring payment e-commerce transactions: <ul style="list-style-type: none"> <li>DE 61 (Point-of-Service Data) subfield 4 (POS Cardholder Presence) must contain a value of 4 (Recurring transactions) and</li> <li>DE 61, subfield 10 (Cardholder-Activated Terminal Level) must contain a value of 6 (Authorized Level 6 CAT: Electronic commerce)</li> </ul> |

## Mastercard ATM Network

The Mastercard® ATM Network provides cash access around the world to credit and ATM debit cardholders by accepting cards bearing the Mastercard, Maestro, and Cirrus logos.

All Mastercard issuers must participate in the Mastercard ATM Network to provide cash access to their cardholders. For more information about ATM participation, refer to the [Security Rules and Procedures](#).

Where permitted, issuers may provide to their cardholders the option of conducting various transactions, such as balance inquiries and merchandise purchases, using the Mastercard ATM Network.

For information about ATM transaction processing capabilities for Europe region acquirers that process through the Mastercard Network, refer to [ATM processing for Europe region acquirers](#).

## Data security

Evaluation of a number of possible security schemes, such as Internet Protocol Security (IPSec), Secure Sockets Layer (SSL) or Transport Layer Security (TLS) shows that these introduce too great a timing overhead on transactions for the transit environment.

Accordingly, an alternative security scheme should be used to overcome these time issues. The scheme should take into consideration whether the terminal can support software or hardware encryption.

If software encryption is supported, the use of the Advanced Encryption Standard (AES) and symmetric session keys is recommended to encrypt and decrypt data.

If hardware encryption is supported, AES may be slower and 3-key Triple DES (3DES) and symmetric session keys may be an alternative viable option to encrypt and decrypt data. 3-key 3DES is preferable over 2-key 3DES.

Both methods have the advantage of introducing almost no time overhead.

Each session key is associated with a session key ID to uniquely identify the key which has been used in securing the data.

The session keys and corresponding key IDs shall be loaded securely into the terminal in a secure loading area and should be changed periodically and securely through a Maintenance Management System. The terminal should therefore allow for the remote distribution of key files.

## Global ATM Locator

The Mastercard Global ATM Locator is a set of websites and toll-free automated voice recognition systems that cardholders use to search for ATMs.

Cardholders access the Mastercard Global ATM Locator using:

- The Internet
- Mobile phones
- Personal digital assistants
- Voice response telephone numbers
- Onboard vehicle navigation systems
- Bank websites

Maps and directions to ATMs are available in many countries.

## Location Administration Tool

The Location Administration Tool enables customers and processors to efficiently manage and market their ATM location information and service-related terminal attributes to help drive transaction volume and increase revenue at the point-of-interaction. Services include ATMs, Mastercard® contactless terminals, Pre-Paid Travel, and Mastercard® *rePower*™.

## Supported transactions

The Mastercard ATM Network supports a variety of transactions.

- Cash withdrawal from debit accounts
- Cash disbursement from credit accounts
- Account balance inquiry from debit accounts
- Available balance inquiry from credit accounts
- Tokenized transactions at contactless ATMs initiated by Near Field Communications (NFC)-enabled devices and wallets
- Chip PIN management transactions (PIN Change and PIN Unblock for Chip cards)
- Magnetic Stripe PIN management (PIN Change only) transactions
- Merchandise purchase from debit accounts for items, such as postage stamps and other items approved by Mastercard.

Acquirers must successfully complete testing for ATM merchandise transaction before submitting merchandise purchase transactions to Mastercard.

- Mastercard® transactions.

- ATM Bill Payment is allowed for domestic transactions only when the service is offered domestically. When ATM Bill Payment is not allowed, acquirers will receive a response of Transaction not permitted to acquirer/terminal.
- Credit Card Cash Advance in Installments transactions at the ATM are allowed when the service is offered domestically.

## Restrict cash access and ATM balance inquiry transactions

Issuers can restrict cash access at the ATM to align the processing systems with Mastercard product rules. When the issuer has restricted cash access, acquirers receive DE 39 (Response Code), value 57 (Transaction not permitted to issuer/cardholder) in the Authorization Request Response/0110 message.

In addition to cash access restrictions, Mastercard allows issuers to restrict ATM balance inquiry transactions. When the issuer has restricted ATM balance inquiries, acquirers receive DE 39, value 57 in the Authorization Request Response/0110 message.

## Interfaces to the Mastercard ATM Network

Mastercard issuers can choose either of the following options to receive authorization requests for ATM transactions.

- A direct interface to the Mastercard ATM Network
- An interface via the Mastercard Network to the Mastercard ATM Network

### Direct interface

Issuers that choose to interface directly with the Mastercard ATM Network receive ATM transactions via a Financial Transaction Request/0200 message.

For a description of the 0200 messages, refer to [Mastercard Rules](#), and [Transaction Processing Rules](#).

### Interface via the Mastercard Network

If the issuer chooses to receive ATM transactions via the Mastercard Network, the issuer receives ATM transactions in an Authorization Request/0100 message.

Choosing to receive ATM transactions via the Mastercard Network in Authorization Request/0100 messages gives issuers the following additional options:

- Mastercard validation of the PIN
- Application of Mastercard Stand-In Authorization when the issuer is unavailable (only if Mastercard also validates the PIN)

### PIN validation

Issuers can choose to have Mastercard validate the PINs or to validate the PINs themselves.

### Applicable edits

| WHEN...                      | THEN...                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------|
| Mastercard validates the PIN | Mastercard performs this service before forwarding the Authorization Request/0100 message to the issuer. |
| the issuer validates the PIN | Mastercard forwards the PIN to the issuer with the Authorization Request/0100 message.                   |

Issuers that validate their own PINs must send Mastercard the encryption key to translate encrypted PINs in the online request message.

### Stand-In Authorization

Issuers can choose whether to have Stand-In Authorization process online request ATM transactions when they are not available. Issuers may choose whether to allow transactions with unverified PINs to be processed in the Stand-In Authorization or to have Mastercard validate a PIN if a PIN is present in an online request message when the issuer is unavailable.

### Applicable edits

| WHEN the issuer requests that...                                                                                    | THEN Mastercard...                                            |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Mastercard allow transactions with an unverified PIN in Stand-In Authorization                                      | processes the online request message.                         |
| Mastercard perform Stand-In Authorization and PIN validation                                                        | processes the online request message and validates the PIN.   |
| Mastercard not allow transactions with unverified PINs and not perform PIN validation during Stand-In Authorization | returns a response of decline when the issuer is unavailable. |

## ATM processing for Europe region acquirers

Mastercard provides ATM transaction processing capabilities for Europe region acquirers that process through the Mastercard Network.

**NOTE: Applies only to the Europe region.**

This processing only allows Europe region acquirers to submit ATM transactions via their dual message connection to the Mastercard Network.

ATM transactions are identified by:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) containing one of the following values:
  - 00 (Purchase)
  - 01 (Withdrawal)
  - 30 (Balance Inquiry)
  - 91 (PIN Unblock)
  - 92 (PIN Change)
- DE 48 (Additional Data: Private Use), position 1 (Transaction Category Code [TCC]) containing value Z (ATM Cash Disbursement)

Only specified acquirers have this functionality. To ensure the transactions are submitted by eligible acquirers, the Dual Message Authorization System verifies the acquirer parameter when an Authorization Request/0100 transaction is determined to originate from an ATM. ATM transactions provided by eligible acquirers are forwarded to the issuer. The Dual Message Authorization System will respond to all other requests with an Authorization Request Response/0110 message containing DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal).

### **ATM access fee**

Acquirers in approved Europe region countries may optionally apply an ATM access fee to domestic cash withdrawal transactions. Acquirers may charge ATM access fees on Mastercard, Maestro, or Cirrus cross-border transactions, as long as the access fee charged on those transactions is not higher than the access fee charged in connection with a transaction of any network accepted at that ATM. If another network accepted at the ATM prohibits the acquirer from charging an ATM access fee on cross-border ATM transactions, the acquirer must not charge an ATM access fee on Mastercard, Maestro or Cirrus cross-border transactions at the ATM. If all other networks accepted at the ATM permit the acquirer to charge access fees on cross-border ATM transactions, the acquirer will be permitted to charge an ATM access fee on cross-border Mastercard, Maestro, and Cirrus ATM transactions.

Acquirers include the transaction amount and access fee in DE 4 (Amount, Transaction) and the access fee amount (if applicable) in DE 28 (Amount, Transaction Fee) of Authorization Request/0100 and Reversal Request/0400 messages.

## **Mastercard One Credential**

Mastercard One Credential is a single, digitally connected credential that offers cardholders a choice of multiple payment methods. It allows customers to choose the payment option (debit, credit, or prepaid, including installments) that best suits their needs through a seamless digital experience in their banking app or website. Mastercard One Credential connects multiple existing funding accounts and lets consumers set their payment preferences.

Overall benefits of the Mastercard One Credential are:

- Consumer benefits: Mastercard One Credential addresses the consumer's need for flexibility and control, enabling consumers to personalize their spending preferences and access various payment methods. This results in greater choice and convenience.
- Issuer benefits: Issuers can provide a differentiated offering, enhancing customer loyalty, stickiness and engagement, solidifying their status as the consumer's primary financial provider. Mastercard One Credential allows issuers to offer new ways to access credit (subject to consumer being credit qualified) to existing customers and reach new customers.
- Acquirer benefits: Acquirers may benefit from increased transaction volumes and incremental revenue with minimal development required as Mastercard One Credential adoption grows.
- Merchant benefits: Availability of multiple funding methods may allow merchants to experience higher average transaction amounts and volumes, better conversion rates, and improved consumer retention through personalized mechanisms.

For more information on Mastercard One Credential, refer to the *Mastercard One Credential Technical Implementation Guide*.

## Product Status Inquiry Authorization Request/0100 message

This is the message layout for the Product Status Inquiry Authorization Request/0100 message.

### Message data element layout

| DE ID                             | DE Name                                             | Org | Sys | Dst | Comments                                           |
|-----------------------------------|-----------------------------------------------------|-----|-----|-----|----------------------------------------------------|
| DE 3, subfield 1                  | Cardholder Transaction Type Code                    | M   | .   | .   | Must contain 00                                    |
| DE 4                              | Amount, Transaction                                 | M   | .   | .   | Must contain all zeroes                            |
| DE 48, subelement 02, subfield 01 | Mastercard One Credential Exclusion Indicator       | C   | X   | .   | If present, must contain C, D or M                 |
| DE 48, subelement 22, subfield 05 | Cardholder/Merchant Initiated Transaction Indicator | C   | .   | .   |                                                    |
| DE 54, subfield 1                 | Account Type                                        | M   | .   | .   | Must match DE 3, subfield 2                        |
| DE 54, subfield 2                 | Amount Type                                         | M   | .   | .   | Must contain 06                                    |
| DE 54, subfield 3                 | Currency Code                                       | M   | .   | .   | Must contain the numeric transaction currency code |

| DE ID             | DE Name                           | Org | Sys | Dst | Comments                            |
|-------------------|-----------------------------------|-----|-----|-----|-------------------------------------|
| DE 54, subfield 4 | Debit or Credit Indicator         | M   | •   | •   | Must contain D                      |
| DE 54, subfield 5 | Amount                            | M   | •   | •   | Must contain the transaction amount |
| DE 61, subfield 7 | POS Transaction Status            | M   | •   | •   | Must contain 0                      |
| DE 61, subfield 9 | POS Transaction Status - Extended | M   | •   | •   | Must contain 2                      |

## Account Status Inquiry with Product Status Inquiry Authorization Request/0100 message

This is the message layout for the Account Status Inquiry with Product Status Inquiry Authorization Request/0100 message.

### Account Status Inquiry with Product Status Inquiry Authorization Request/0100 message

| DE ID                             | DE Name                                                  | Org | Sys | Dst | Comments                                                          |
|-----------------------------------|----------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------|
| DE 3, subfield 1                  | Cardholder Transaction Type Code                         | M   | P   | M   | Must contain 00                                                   |
| DE 4                              | Amount, Transaction                                      | M   | P   | M   | Must contain all zeros                                            |
| DE 48, subelement 02, subfield 01 | Mastercard One Credential Exclusion Indicator            | C   | X   | C   | If present, must contain C, D or M                                |
| DE 48, subelement 09, subfield 11 | Mastercard One Credential Account Number Indicator       | •   | X   | C   | Contains Mastercard One Credential account number indicator       |
| DE 48, subelement 09, subfield 12 | Mastercard One Credential Account Number                 | •   | X   | C   | Contains Mastercard One Credential account number (Funding PAN)   |
| DE 48, subelement 09, subfield 13 | Mastercard One Credential Account Number Expiration Date | •   | X   | C   | Contains Mastercard One Credential account number expiration date |



System functionality and services

Account Status Inquiry with Product Status Inquiry Authorization Request/0100 message

| DE ID                                      | DE Name                                                      | Org | Sys | Dst | Comments                                                                                                                                                                        |
|--------------------------------------------|--------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48,<br>subelement<br>22, subfield<br>05 | Cardholder/Merchant<br>Initiated Transaction<br>Indicator    | C   | P   | C   |                                                                                                                                                                                 |
| DE 48,<br>subelement<br>24, subfield<br>1  | ALM Service Code                                             | •   | X   | C   | Indicates the ALM<br>program and<br>Mastercard One<br>Credential service                                                                                                        |
| DE 48,<br>subelement<br>24, subfield<br>2  | ALM Product<br>Graduation Plus or<br>Registered Product Code | •   | X   | C   | Contains the licensed<br>graduated product code<br>or the registered product<br>code or the Mastercard<br>One Credential product<br>code                                        |
| DE 48,<br>subelement<br>24, subfield<br>3  | ALM Product Class                                            | •   | X   | C   | Indicates the Product<br>Class Override for the<br>applicable ALM service<br>or Mastercard One<br>Credential service                                                            |
| DE 48,<br>subelement<br>24, subfield<br>4  | ALM Rate Type                                                | •   | X   | C   | Rate Type number for<br>the applicable ALM<br>service and if not<br>supported for the ALM<br>service of the registered<br>PAN, the field will be<br>populated with value<br>XXX |
| DE 48,<br>subelement<br>24, subfield<br>5  | Mastercard One<br>Credential Acceptance<br>Brand             | •   | X   | C   | Acceptance brand of the<br>Mastercard One<br>Credential Funding PAN                                                                                                             |

System functionality and services

Account Status Inquiry with Product Status Inquiry Authorization Request/0100 message

| DE ID                                     | DE Name                                                                  | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48,<br>subelement<br>24, subfield<br>6 | Mastercard One<br>Credential Static/<br>Dynamic Interchange<br>Indicator | •   | X   | C   | Possible values: <ul style="list-style-type: none"> <li>D = Dynamic interchange: the transaction's interchange will be appropriate for the Mastercard One Credential Funding PAN instead of the DE 2 primary PAN</li> <li>S = Static interchange: the transaction's interchange will be appropriate for the DE 2 primary PAN</li> </ul> |
| DE 48,<br>subelement<br>24, subfield<br>7 | Mastercard One<br>Credential Issuer<br>Account Range                     | •   | X   | C   | Issuer account range of the Mastercard One Credential Funding PAN                                                                                                                                                                                                                                                                       |
| DE 54,<br>subfield 1                      | Account Type                                                             | M   | •   | •   | Must match DE 3, subfield 2                                                                                                                                                                                                                                                                                                             |
| DE 54,<br>subfield 2                      | Amount Type                                                              | M   | •   | •   | Must contain 06 if submitting for the Mastercard One Credential Product Status Inquiry service                                                                                                                                                                                                                                          |
| DE 54,<br>subfield 3                      | Currency Code                                                            | M   | •   | •   | Must contain the numeric transaction currency code                                                                                                                                                                                                                                                                                      |
| DE 54,<br>subfield 4                      | Debit or Credit Indicator                                                | M   | •   | •   | Must contain D                                                                                                                                                                                                                                                                                                                          |
| DE 54,<br>subfield 5                      | Amount                                                                   | M   | •   | •   | Must contain the transaction amount                                                                                                                                                                                                                                                                                                     |
| DE 61,<br>subfield 7                      | POS Transaction Status                                                   | M   | P   | M   | Must contain 8                                                                                                                                                                                                                                                                                                                          |
| DE 61,<br>subfield 9                      | POS Transaction Status<br>- Extended                                     | M   | P   | M   | Must contain 0 or 1                                                                                                                                                                                                                                                                                                                     |

| DE ID                | DE Name                | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 63,<br>subfield 1 | Financial Network Code | •   | X   | M   | <p>Will contain the graduated product when the issuer's cardholder account participates in the Product Graduation and Mastercard One Credential Service.</p> <p>For Mastercard One Credential, it will be based on DE 2 PAN for Static Interchange and DE 48, subelement 24, subfield 7 for Dynamic Interchange, which could be the graduated product code for a registered PAN.</p> |

## Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Request/0100 message

This is the message layout for the Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Request/0100 message.

### Message data element layout

| DE ID                                | DE Name                                            | Org | Sys | Dst | Comments                                                    |
|--------------------------------------|----------------------------------------------------|-----|-----|-----|-------------------------------------------------------------|
| DE 3,<br>subfield 1                  | Cardholder Transaction Type Code                   | M   | P   | M   | Must contain 00                                             |
| DE 4                                 | Amount, Transaction                                | M   | P   | M   | Must contain all zeros                                      |
| DE 48,<br>subelement 02, subfield 01 | Mastercard One Credential Exclusion Indicator      | C   | X   | C   | If present, must contain C, D or M                          |
| DE 48,<br>subelement 09, subfield 11 | Mastercard One Credential Account Number Indicator | •   | X   | C   | Contains Mastercard One Credential account number indicator |

System functionality and services  
Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry  
Authorization Request/0100 message

| DE ID                             | DE Name                                                  | Org | Sys | Dst | Comments                                                                                                                                                   |
|-----------------------------------|----------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 09, subfield 12 | Mastercard One Credential Account Number                 | •   | X   | C   | Contains Mastercard One Credential account number (Funding PAN)                                                                                            |
| DE 48, subelement 09, subfield 13 | Mastercard One Credential Account Number Expiration Date | •   | X   | C   | Contains Mastercard One Credential account number expiration date                                                                                          |
| DE 48, subelement 22, subfield 05 | Cardholder/Merchant Initiated Transaction Indicator      | C   | P   | C   |                                                                                                                                                            |
| DE 48, subelement 24, subfield 1  | ALM Service Code                                         | •   | X   | C   | Indicates the ALM program and Mastercard One Credential service                                                                                            |
| DE 48, subelement 24, subfield 2  | ALM Product Graduation Plus or Registered Product Code   | •   | X   | C   | Contains the licensed graduated product code or the registered product code or the Mastercard One Credential product code                                  |
| DE 48, subelement 24, subfield 3  | ALM Product Class                                        | •   | X   | C   | Indicates the Product Class Override for the applicable ALM service or Mastercard One Credential service                                                   |
| DE 48, subelement 24, subfield 4  | ALM Rate Type                                            | •   | X   | C   | Rate Type number for the applicable ALM service and if not supported for the ALM service of the registered PAN, the field will be populated with value XXX |
| DE 48, subelement 24, subfield 5  | Mastercard One Credential Acceptance Brand               | •   | X   | C   | Acceptance brand of the Mastercard One Credential Funding PAN                                                                                              |

System functionality and services  
Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry  
Authorization Request/0100 message

| DE ID                                     | DE Name                                                                  | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48,<br>subelement<br>24, subfield<br>6 | Mastercard One<br>Credential Static/<br>Dynamic Interchange<br>Indicator | •   | X   | C   | Possible values: <ul style="list-style-type: none"> <li>D = Dynamic interchange: the transaction's interchange will be appropriate for the Mastercard One Credential Funding PAN instead of the DE 2 primary PAN</li> <li>S = Static interchange: the transaction's interchange will be appropriate for the DE 2 primary PAN</li> </ul> |
| DE 48,<br>subelement<br>24, subfield<br>7 | Mastercard One<br>Credential Issuer<br>Account Range                     | •   | X   | C   | Issuer account range of the Mastercard One Credential Funding PAN                                                                                                                                                                                                                                                                       |
| DE 54,<br>subfield 1,<br>occurrence<br>1  | Account Type                                                             | M   | •   | •   | Must match DE 3, subfield 2                                                                                                                                                                                                                                                                                                             |
| DE 54,<br>subfield 2,<br>occurrence<br>1  | Amount Type                                                              | M   | •   | •   | Must contain 06 if submitting for the Mastercard One Credential Product Status Inquiry service                                                                                                                                                                                                                                          |
| DE 54,<br>subfield 3,<br>occurrence<br>1  | Currency Code                                                            | M   | •   | •   | Must contain the numeric transaction currency code                                                                                                                                                                                                                                                                                      |
| DE 54,<br>subfield 4,<br>occurrence<br>1  | Debit or Credit Indicator                                                | M   | •   | •   | Must contain D                                                                                                                                                                                                                                                                                                                          |
| DE 54,<br>subfield 5,<br>occurrence<br>1  | Amount                                                                   | M   | •   | •   | Must contain the transaction amount                                                                                                                                                                                                                                                                                                     |

System functionality and services  
Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry  
Authorization Request/0100 message

| DE ID                                    | DE Name                              | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------|--------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 54,<br>subfield 1,<br>occurrence<br>2 | Account Type                         | M   | •   | •   | Must match DE 3,<br>subfield 2                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 54,<br>subfield 2,<br>occurrence<br>2 | Amount Type                          | M   | •   | •   | Must contain 05                                                                                                                                                                                                                                                                                                                                                                                                            |
| DE 54,<br>subfield 3,<br>occurrence<br>2 | Currency Code                        | M   | •   | •   | Must contain the<br>numeric transaction<br>currency code                                                                                                                                                                                                                                                                                                                                                                   |
| DE 54,<br>subfield 4,<br>occurrence<br>2 | Debit or Credit Indicator            | M   | •   | •   | Must contain D                                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 54,<br>subfield 5,<br>occurrence<br>2 | Amount                               | M   | •   | •   | Must contain the<br>transaction amount                                                                                                                                                                                                                                                                                                                                                                                     |
| DE 61,<br>subfield 7                     | POS Transaction Status               | M   | P   | M   | Must contain 8                                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 61,<br>subfield 9                     | POS Transaction Status<br>- Extended | M   | P   | M   | Must contain 0 or 1                                                                                                                                                                                                                                                                                                                                                                                                        |
| DE 63,<br>subfield 1                     | Financial Network Code               | •   | X   | M   | Will contain the<br>graduated product when<br>the issuer's cardholder<br>account participates in<br>the Product Graduation<br>and Mastercard One<br>Credential Service.<br><br>For Mastercard One<br>Credential, it will be<br>based on DE 2 PAN for<br>Static Interchange and<br>DE 48, subelement 24,<br>subfield 7 for Dynamic<br>Interchange, which could<br>be the graduated<br>product code for a<br>registered PAN. |

## Product Status Inquiry Authorization Response/0110 (Mastercard-Generated) message

This is the message layout for the Product Status Inquiry Authorization Response/0110 (Mastercard-Generated) message.

### Message data element layout

| Data Element ID and Name         |                                                        | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|--------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39                            | Response Code                                          | •   | X   | M   | <p>Mastercard responds to the transaction with a DE 39 Response Code value of 85 = Not Declined</p> <p>If the issuer account range is not participating in this service, Mastercard will reject with DE 39 Response Code value of 12 = Invalid Transaction</p> <p>If the ALM/Mastercard One Credential Service is not available, Mastercard will decline with DE 39 Response Code value of 91 = Authorization System or Issuer System Inoperative</p> |
| DE 48, subelement 24, subfield 1 | ALM Service Code                                       | •   | X   | M   | Indicates the ALM program and/or Mastercard One Credential Service                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 48, subelement 24, subfield 2 | ALM Product Graduation Plus or Registered Product Code | •   | X   | M   | Contains the licensed graduated product code or the registered product code or the Mastercard One Credential Product Code                                                                                                                                                                                                                                                                                                                             |

| Data Element ID and Name         |                                                                | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                |
|----------------------------------|----------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 24, subfield 3 | ALM Product Class                                              | •   | X   | M   | Indicates the Product Class Override for the applicable ALM Service or Mastercard One Credential Service                                                                                                                                                                                                                                |
| DE 48, subelement 24, subfield 4 | ALM Rate Type                                                  | •   | X   | M   | Rate Type number for the applicable ALM Service and if not supported for the ALM service of the registered PAN, the field will be populated with value XXX                                                                                                                                                                              |
| DE 48, subelement 24, subfield 5 | Mastercard One Credential Acceptance Brand                     | •   | X   | C   | Acceptance brand of the Mastercard One Credential Funding PAN                                                                                                                                                                                                                                                                           |
| DE 48, subelement 24, subfield 6 | Mastercard One Credential Static/Dynamic Interchange Indicator | •   | X   | C   | Possible values: <ul style="list-style-type: none"> <li>D = Dynamic interchange: the transaction's interchange will be appropriate for the Mastercard One Credential Funding PAN instead of the DE 2 primary PAN</li> <li>S = Static interchange: the transaction's interchange will be appropriate for the DE 2 primary PAN</li> </ul> |
| DE 48, subelement 24, subfield 7 | Mastercard One Credential Issuer Account Range                 | •   | X   | C   | Issuer account range of the Mastercard One Credential Funding PAN                                                                                                                                                                                                                                                                       |



| Data Element ID and Name |                        | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 63,<br>subfield 1     | Financial Network Code | •   | X   | M   | <p>Will contain the graduated product when the issuer's cardholder account participates in the Product Graduation and Mastercard One Credential Service.</p> <p>For Mastercard One Credential, it will be based on DE 2 PAN for Static Interchange and DE 48, subelement 24, subfield 7 for Dynamic Interchange, which could be the graduated product code for a registered PAN.</p> |

## Account Status Inquiry with Product Status Inquiry Authorization Response/0110 message

This is the message layout for the Account Status Inquiry with Product Status Inquiry Authorization Response/0110 message.

### Message data element layout

| DE ID | DE Name       | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                             |
|-------|---------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 | Response Code | M   | X   | M   | <p>DE 39 (Response Code) may contain value 00 (Approved or completed successfully) 85 (Not declined), or a valid business decline response.</p> <p>If the issuer is unable to reply, the acquirer receives a DE 39 Response Code value of 91 (Authorization System or issuer system inoperative)</p> |

| DE ID                            | DE Name                                                        | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|----------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 24, subfield 1 | ALM Service Code                                               | •   | X   | M   | Indicates the ALM program and/or Mastercard One Credential Service                                                                                                                                                                                                                                                                    |
| DE 48, subelement 24, subfield 2 | ALM Product Graduation Plus or Registered Product Code         | •   | X   | M   | Contains the licensed graduated product code or the registered product code or the Mastercard One Credential Product Code                                                                                                                                                                                                             |
| DE 48, subelement 24, subfield 3 | ALM Product Class                                              | •   | X   | M   | Indicates the Product Class Override for the applicable ALM Service or Mastercard One Credential Service                                                                                                                                                                                                                              |
| DE 48, subelement 24, subfield 4 | ALM Rate Type                                                  | •   | X   | M   | Rate Type number for the applicable ALM Service and if not supported for the ALM service of the registered PAN, the field will be populated with value XXX                                                                                                                                                                            |
| DE 48, subelement 24, subfield 5 | Mastercard One Credential Acceptance Brand                     | •   | X   | C   | Acceptance brand of the Mastercard One Credential Funding PAN                                                                                                                                                                                                                                                                         |
| DE 48, subelement 24, subfield 6 | Mastercard One Credential Static/Dynamic Interchange Indicator | •   | X   | C   | Possible values: <ul style="list-style-type: none"> <li>D: Dynamic interchange: the transaction's interchange will be appropriate for the Mastercard One Credential Funding PAN instead of the DE 2 primary PAN</li> <li>S: Static interchange: the transaction's interchange will be appropriate for the DE 2 primary PAN</li> </ul> |

System functionality and services  
Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry  
Authorization Response/0110 message

| DE ID                                     | DE Name                                              | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------|------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48,<br>subelement<br>24, subfield<br>7 | Mastercard One<br>Credential Issuer<br>Account Range | .   | X   | C   | Issuer account range of<br>the Mastercard One<br>Credential Funding PAN                                                                                                                                                                                                                                                                                                                                                    |
| DE 63,<br>subfield 1                      | Financial Network Code                               | ME  | X   | M   | Will contain the<br>graduated product when<br>the issuer's cardholder<br>account participates in<br>the Product Graduation<br>and Mastercard One<br>Credential Service.<br><br>For Mastercard One<br>Credential, it will be<br>based on DE 2 PAN for<br>Static Interchange and<br>DE 48, subelement 24,<br>subfield 7 for Dynamic<br>Interchange, which could<br>be the graduated<br>product code for a<br>registered PAN. |

## Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Response/0110 message

This is the message layout for the Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry Authorization Response/0110 message.

### Message data element layout

| DE ID | DE Name          | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                               |
|-------|------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 | Response<br>Code | M   | X   | M   | DE 39 (Response Code) may<br>contain value 00 (Approved<br>or completed successfully),<br>85 (Not declined), or a valid<br>business decline response.<br><br>If the issuer is unable to<br>reply, the acquirer receives a<br>DE 39 Response Code value<br>of 91 (Authorization System<br>or issuer system inoperative) |

System functionality and services  
Account Status Inquiry with Approval Probability Indicator and Product Status Inquiry  
Authorization Response/0110 message

| DE ID                            | DE Name                                                        | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------|----------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 24, subfield 1 | ALM Service Code                                               | •   | X   | M   | Indicates the ALM program and/or Mastercard One Credential Service                                                                                                                                                                                                                                                                        |
| DE 48, subelement 24, subfield 2 | ALM Product Graduation Plus or Registered Product Code         | •   | X   | M   | Contains the licensed graduated product code or the registered product code or the Mastercard One Credential Product Code                                                                                                                                                                                                                 |
| DE 48, subelement 24, subfield 3 | ALM Product Class                                              | •   | X   | M   | Indicates the Product Class Override for the applicable ALM Service or Mastercard One Credential Service                                                                                                                                                                                                                                  |
| DE 48, subelement 24, subfield 4 | ALM Rate Type                                                  | •   | X   | M   | Rate Type number for the applicable ALM Service and if not supported for the ALM service of the registered PAN, the field will be populated with value XXX.                                                                                                                                                                               |
| DE 48, subelement 24, subfield 5 | Mastercard One Credential Acceptance Brand                     | •   | X   | C   | Acceptance brand of the Mastercard One Credential Funding PAN                                                                                                                                                                                                                                                                             |
| DE 48, subelement 24, subfield 6 | Mastercard One Credential Static/Dynamic Interchange Indicator | •   | X   | C   | Possible values: <ul style="list-style-type: none"> <li>• D: Dynamic interchange: the transaction's interchange will be appropriate for the Mastercard One Credential Funding PAN instead of the DE 2 primary PAN</li> <li>• S: Static interchange: the transaction's interchange will be appropriate for the DE 2 primary PAN</li> </ul> |
| DE 48, subelement 24, subfield 7 | Mastercard One Credential Issuer Account Range                 | •   | X   | C   | Issuer account range of the Mastercard One Credential Funding PAN                                                                                                                                                                                                                                                                         |

| DE ID                | DE Name                      | Org | Sys | Dst | Comments                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 63,<br>subfield 1 | Financial<br>Network<br>Code | ME  | X   | M   | <p>Will contain the graduated product when the issuer's cardholder account participates in the Product Graduation and Mastercard One Credential Service.</p> <p>For Mastercard One Credential, it will be based on DE 2 PAN for Static Interchange and DE 48, subelement 24, subfield 7 for Dynamic Interchange, which could be the graduated product code for a registered PAN.</p> |

## Member-defined data

Mastercard allows customers to exchange as many as 299 bytes of member-defined data to and from another customer in authorization messages.

Customers may use member-defined data to transmit information such as:

- Loyalty program information
- Promotional information
- Details about a purchase or money transfer
- Any other business-related data

### Structure of member-defined data

Acquirers may provide as many as 299 bytes of member-defined data in DE 124 of the Authorization Request/0100 message.

Issuers may provide as many as 299 bytes of member-defined data in DE 124 of the Authorization Request Response/0110 message. Issuers will receive DE 124 in Authorization Advice/0120 messages if DE 124 was present in the Authorization Request/0100 sent by the acquirer and the transaction was processed by Stand-In Authorization or X-Code Processing.

The Dual Message Authorization System will not edit the content of DE 124 as long as it is alphanumeric/special character data and is only as many as 299 bytes in length. The Dual Message Authorization System will limit the length of the member-defined data in DE 124 to 299 bytes in Authorization Request/0100 messages and 299 bytes in Authorization Request Response/0110 messages.

For additional details on providing member-defined data in DE 124, refer to [Data element definitions and details](#).

## Merchant Advice codes

Mastercard supports the use of the Merchant Advice Code field for issuers and Mastercard to communicate with merchants.

Issuers can use Merchant Advice Codes to provide specific direction to acquirers:

- Identify product type to inform credential on file management
- The actions merchants can take to continue to serve their customers

## Mastercard Authorization Optimizer

Customers should be aware that Mastercard also leverages the Merchant Advice Code field. Mastercard Authorization Optimizer is a global network-based service that provides near real-time intelligence to acquirers and merchants on digital commerce payments, also known as card-not-present (CNP) transactions.

This authorization insight is sent through Authorization Request Response/0110 messages in:

- DE 39 (Response Code)
- DE 48 (Additional Data: Private Use), subelement 84 (Merchant Advice Code)

DE 39 and DE 48, subelement 84 must be used in combination to take full benefit of the service. For more information on Mastercard Authorization Optimizer, refer to the *Mastercard Authorization Optimizer Product Guide*.

## Partial approvals

The partial approval enables an issuer to approve a portion of the transaction amount in the authorization request when the transaction amount exceeds the amount of funds available on the debit card or prepaid card and the merchant terminal supports partial approvals.

An issuer can choose to provide a partial approval response amount that is less than or equal to the requested amount. An issuer can provide a partial approval response equal to the amount requested when the cardholder's account does not have a sufficient balance to cover any allowed amount over the requested amount for tips or service charges. The following scenario demonstrates the successful completion of a partial approval equal to the requested amount.

A cardholder pays USD 50 at a restaurant with a debit card that has an available balance of USD 50. The gratuity allowance for the restaurant transaction is 20 percent. The issuer may use a partial approval response in the amount of USD 50 to indicate there are no additional funds and to obtain chargeback protection for any clearing amount submitted above USD 50 that was partially approved. The merchant must request another form of payment for any amount the cardholder may want to provide.

**NOTE: For automated fuel dispenser (AFD) transactions (MCC 5542), an issuer may respond with a partial approval amount that is less than, equal to, or greater than the requested amount.**

When the acquirer transmits an indicator in the authorization request that the merchant terminal supports partial approvals, the issuer has the option to respond with the partial approval amount and partial approval response code. The cardholder can then choose to use a supplemental payment method (split tender) to pay the balance and complete the purchase when the final transaction amount exceeds the partial approval amount.

All card issuers (including prepaid cards) in the U.S. and Canada regions must support partial approvals and updates to the cardholder's open-to-buy balance upon receipt of a reversal (full or partial).

All acquirers in the U.S. and Canada regions that support merchants within select acceptor business codes (MCCs) must support partial approvals for all account ranges, including all prepaid Debit Mastercard account ranges. This requirement applies only to card-present transactions occurring at attended terminals and at transactions identified with MCC 5542 (Fuel Dispenser, Automated).

For more information, refer to the [Transaction Processing Rules](#).

**NOTE: In the Europe region, a customer must support partial approvals at merchants identified with MCC 5542 (Fuel Dispenser, Automated) for all Mastercard account ranges if the customer supports partial approvals for Maestro or any other debit brand.**

A customer must support partial approvals on Mastercard and Maestro if it supports them on other brands, for the same product types and merchant types as on the other brands. To the extent that support for partial approvals is not required on other brands, then it is not required on Mastercard or Maestro, with the exception of support at merchants identified with MCC 5542 as set out in the preceding paragraph.

## How it works

When a merchant terminal indicates support of partial approval transactions, the acquirer sends an Authorization Request/0100 message containing DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 1 (Partial Approval Terminal Support Indicator), value 1 (Merchant terminal supports receipt of partial approvals).

When a merchant terminal indicates support of partial approval transactions and the cardholder's open-to-buy balance does not cover the full transaction amount, the issuer may respond to the authorization request with an Authorization Request Response/0110 message containing data elements necessary for a partial approval response, with DE 6 (Amount, Cardholder Billing) representing the cardholder's available purchase amount:

- DE 6 (Amount, Cardholder Billing) with the partial approval amount specified in the issuer's cardholder billing currency
- DE 38 (Authorization ID Response)
- DE 39 (Response Code), value 10 (Partial Approval)

When the issuer responds with DE 39, value 10 and the transaction is an AFD transaction, the amount in DE 6 may be less than, greater than, or equal to the requested amount in DE 6 of the Authorization Request/0100 message.

- DE 51 (Currency Code, Cardholder Billing)

Upon receipt of a partial approval, acquirers often send a balance due message to the integrated terminal. This message indicates the difference between the original purchase amount and the partial amount approved by the issuer. By displaying the remaining balance due amount, the cashier can quickly and accurately prompt the consumer to complete the split-tender purchase.

Acquirers should also advise the merchant of a partial approval that is equal to the request amount, so that the merchant does not add any gratuity.

### Alternate processing

The Stand-In Authorization and X-Code Processing do not provide partial approval responses to authorization requests because these systems do not maintain card balances.

## Authorization Request/0100: Partial Approval

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                                                                                                               | Org | Sys | Dst | Values/Comments                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------|
| DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 1 (Partial Approval Terminal Indicator) | C   | .   | C   | 1 = Merchant terminal supports receipt of partial approvals |

## Authorization Request Response/0110: Partial Approval

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Data elements

| Data Element               | Org | Sys | Dst | Values/Comments                                           |
|----------------------------|-----|-----|-----|-----------------------------------------------------------|
| DE 4 (Amount, Transaction) | .   | X   | M   | Partial approval amount in acquirer transaction currency. |



| Data Element                                           | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 5 (Amount, Settlement)                              | .   | X   | C   | Partial approval amount in the settlement currency.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DE 6 (Amount, Cardholder Billing)                      | M   | .   | M   | Partial approval amount in issuer cardholder billing currency. This amount can be less than or equal to the original amount present in the Authorization Request/0100 message in the amount data element that corresponds to the issuer cardholder billing currency.<br><br>This amount may be less than, greater than, or equal to the amount present in the Authorization Request/0100 message when the DE 18 (Merchant Type) is value 5542 (Fuel Dispenser, Automated) |
| DE 38 (Authorization ID Response)                      | M   | .   | M   | Authorization Code.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DE 39 (Response Code)                                  | M   | .   | M   | 10 = Partial Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DE 51 (Currency Code, Cardholder Billing)              | M   | .   | M   | Issuer cardholder billing currency code.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| DE 54 (Additional Amounts), subfield 1 (Account Type)  | .   | X   | M   | 00 = Default Account (not specified or not applicable)                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 54 (Additional Amounts), subfield 2 (Amount Type)   | .   | X   | M   | 57 = Original Amount                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DE 54 (Additional Amounts), subfield 3 (Currency Code) | .   | X   | M   | Refer to the <a href="#">Quick Reference Booklet</a>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DE 54, subfield 4 (Debit or Credit Indicator)          | .   | X   | M   | C plus 12 digits                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**NOTE: Issuers responding with DE 39, value 10 will not be required to echo DE 4 (Amount, Transaction) in the Authorization Request Response/0110 message. Likewise, if DE 5 (Amount, Settlement) was present in the Authorization Request/0100 message to the issuer, the issuer will not be required to echo DE 5 in the Authorization Request Response/0110 message when responding with DE 39, value 10. The issuer will provide the partial approval amount in DE 6 (Amount, Cardholder Billing).**

The Dual Message Authorization System additionally provides the partial approval amount to the acquirer in the following data elements of the Authorization Request Response/0110 message:

- DE 4 in acquirer's transaction currency
- DE 5 in U.S. dollars if the acquirer receives settlement amount-related data elements
- DE 6 in the issuer's cardholder billing currency

The Dual Message Authorization System will provide two additional occurrences of the original amount in DE 54 of the Authorization Request Response/0110 message to the acquirer; one in the acquirer's currency and one in the issuer cardholder billing currency.

## Reversal Request/0400: Partial Approval

In some cases, the cardholder or merchant may elect not to complete the transaction after receiving the partial approval response from the issuer. Mastercard supports full acquirer-generated reversal messages to allow the merchant to cancel the partial approval.

In addition to all other applicable data elements for the Reversal Request/0400 message, acquirers should submit Reversal Request/0400 messages with the following data elements for a partial approval:

- The partial approval amount in DE 4 that was present in the Authorization Request Response/0110 message to the acquirer, not the original amount present in DE 4 of the Authorization Request/0100 message from the acquirer
- DE 39, value 10

When processing a Reversal Request/0400 message for a partial approval (DE 39, value 10), the issuer should increase the cardholder open-to-buy using the partial approval amount present in the amount data element of the Reversal Request/0400 message that corresponds to the issuer cardholder billing currency.

The Reversal Request/0400 message will not contain DE 48, subelement 61 or DE 54, subelement 2, value 57.

## Reversal Advice/0420: Partial Approval

Following are data elements applicable to this message in addition to the required data elements.

If the Dual Message Authorization System generates a Reversal Advice/0420 message after the issuer has responded to the Authorization Request Response/0110 message with DE 39, value 10, the Dual Message Authorization System will provide the following data elements:

- Partial approval amount in DE 4 in the acquirer transaction currency
- Partial approval amount in DE 5 if the issuer has opted to receive amounts in the settlement currency (U.S. dollars)
- Partial approval amount in DE 6 in the issuer cardholder billing currency
- DE 39, value 10
- Original amount in DE 54 in the issuer cardholder billing currency (and acquirer transaction currency if different from issuer cardholder billing currency)

When processing an Reversal Advice/0420 message for a partial approval (DE 39, value 10), the issuer should increase the cardholder open-to-buy using the partial approval amount present in the amount data element of the Reversal Advice/0420 message that corresponds to the issuer's cardholder billing currency.

## Authorization Advice/0120: Acquirer-Generated

The following data elements apply to this message in addition to the required data elements.

When an acquirer creates an Authorization Advice/0120: acquirer-generated message to advise the issuer of an approved authorization performed by the acquirer, it may include DE 48, subelement 61 in the Authorization Advice/0120: acquirer-generated message if it was present in the original Authorization Request/0100 message.

DE 39, value 10 is not a valid value for Authorization Advice/0120: acquirer-generated messages. If an Authorization Advice/0120: acquirer-generated message contains DE 39, value 10 and the advice is not for an Automated Fuel Dispenser Completion (DE 18 = 5542), the Dual Message Authorization System will generate an Authorization Advice Response/0130 message containing DE 39, value 30 and DE 44 (Additional Data), value 039.

If an Authorization Advice/0120: acquirer-generated message contains DE 39, value 10 and the advice is for an Automated Fuel Dispenser Completion, the Dual Message Authorization System will change the DE 39 value to 00 (Approved or completed successfully) before forwarding the advice to the issuer.

## Alternate Processing

The Stand-In Authorization and the X-Code Processing will not provide a partial approval response to an authorization request because these systems do not maintain card balances.

However, if an Authorization Request/0100 message containing DE 48, subelement 61 is processed by the Stand-In Authorization or the X-Code Processing, the corresponding Authorization Advice/0120: system-generated message will contain DE 48, subelement 61.

## Payment Transactions

A Payment Transaction facilitates the movement of funds between two parties: a payer (sender) and a payee (recipient). This transaction can be used to support several business opportunities, such as person-to-person payments, merchant rebates and rewards, loading value to a debit or prepaid account, or issuer rebates and rewards.

The Payment Transaction is identified by DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction), DE 48 (Additional Data: Private Use), Transaction Category Code (TCC), value P (Payment Transaction), and the presence of DE 48, subelement 77 (Transaction Type Identifier).

Payment Transactions are identified in the following messages:

- Authorization Request/0100
- Authorization Advice/0120: system-generated
- Reversal Request/0400
- Reversal Advice/0420

Refer to [Subelement 77 \(Transaction Type Identifier\)](#) for more information that acquirers can use to identify specific types of Payment Transactions.

## Payment Transaction mandate

Mastercard has mandated that all acquirers and issuers support Payment Transactions.

For more information, refer to the Payment Transactions according to the specifications documented in the [Transaction Processing Rules](#), Section 6.1.

For information about Payment Account Status Inquiry (Payment ASI) transactions, refer to [Payment Account Status Inquiry](#).

## Payment Transaction blocking

Issuers that have legal restrictions that would prevent their participation in Payment Transactions can request to block Payment Transactions by completing and submitting the *Payment Transaction Blocking Form* (Form 530).

Mastercard must approve all issuer requests to block Payment Transaction authorizations.

For account ranges that issuers have arranged to have blocked, the Dual Message Authorization System rejects Payment Transactions containing any account number within the account range listed in the Payment Transaction Blocked BIN file. The Dual Message Authorization System returns a response code of Transaction not permitted to issuer/cardholder (DE 39 = 57) to the acquirer.

Issuers can block Payment Transaction authorizations at the account number range level (as many as 11 digits).

### Optional acquirer use of the Payment Transaction Blocked BIN File

Acquirers also have the option to receive the Payment Transaction Blocked BIN file (T114 [Payment Transaction Blocked BIN File] and T115 [Test: Payment Transaction Blocked BIN File]) and may deny Payment Transactions to cardholders with accounts in the blocked account ranges.

## Alternate processing

Customers can choose to use Stand-In Authorization for Payment Transactions. Customers cannot use X-Code Processing for Payment Transactions.

**NOTE: Mastercard does not encourage customers to use Stand-In Authorization for Payment Transaction authorizations.**

### Stand-In Authorization

If customers choose to use Stand-In Authorization for Payment Transaction authorizations, Mastercard recommends that customers use discretion for these transactions.

For more information, refer to the [Stand-In Authorization Reference Guide](#).

### X-Code Processing

Acquirer MIP X-Code Processing will not approve Payment Transactions.

When X-Code Processing processes the authorization, it will generate an authorization response of refer to card issuer (DE 39 = 01) for the Payment Transaction. During acquirer host X-Code Processing, the acquirer must issue a refer to card issuer response to the merchant for Payment Transaction authorizations.

## Authorization Request/0100: Payment Transaction Message

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                                                                       | Org | Sys | Dst | Values/Comments                                                                            |
|----------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) (Cardholder account credits) | M   | •   | M   | 28 = Payment Transaction                                                                   |
| DE 3, subfield 2 (Cardholder "From Account" Type Code)                                             | M   | •   | M   | 00 = Default account (account not specified or not applicable)                             |
| DE 3, subfield 3 (Cardholder "To Account" Type Code)                                               | M   | •   | M   | Any valid value for subfield 3.                                                            |
| DE 18 (Merchant Type)                                                                              | M   | •   | M   | Valid merchant type value for Payment Transaction.                                         |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                               | M   | •   | M   | P = Payment Transaction                                                                    |
| DE 48, subelement 77 (Transaction Type Identifier)                                                 | M   | •   | M   | Refer to <a href="#">Subelement 77 (Transaction Type Identifier)</a> for more information. |

## Authorization Request Response/0110: Payment Transaction

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Data elements

| Data Element                                                         | Org | Sys | Dst | Values/Comments                                                           |
|----------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------|
| DE 3 (Processing Code)                                               | ME  | .   | ME  | Processing Code value in the original Authorization Request/0100 message. |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code) | CE  | .   | CE  | P = Payment Transaction                                                   |

## Dual Message Authorization System Edits

Issuers must process transaction amounts in a Payment Transaction as a credit to the cardholder account. Following are Authorization Request/0100 and Reversal Request/0400 edits.

### Applicable edits

| WHEN...                                                                                                                                                         | THEN the Dual Message Authorization System...                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1, value 28 and TCC value P are not both present                                                                                                 | sends the acquirer a response where:<br>DE 39 = 30 (Format error)<br>DE 44 = 048000                                                                                                                       |
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) does not contain value 28 (Payment Transaction)<br>and<br>DE 48, subelement 77 is present | removes DE 48, subelement 77 from the following messages: <ul style="list-style-type: none"> <li>• Authorization Request/0100</li> <li>• Reversal Request/0400</li> <li>• Reversal Advice/0420</li> </ul> |
| DE 3, subelement 1 is value 28<br>and<br>DE 48, subelement 77 is not included                                                                                   | sends the acquirer a response where:<br>DE 39 = 30 (Format error)<br>DE 44 = 048077                                                                                                                       |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | THEN the Dual Message Authorization System...                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 28 (Payment Transaction), and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), contains value C09 (Card Activation) and DE 61 (Point of Sale [POS] Data) subfields does not contain the following values: <ul style="list-style-type: none"><li>• Subfield 1 (POS Terminal Attendance) = 0 (Attended Terminal)</li><li>• Subfield 3 (POS Terminal Location) = 0 (On premises of acceptor facility)</li><li>• Subfield 4 (POS Cardholder Presence) = 0 (Cardholder present)</li><li>• Subfield 5 (POS Card Presence) = 0 (Card present)</li><li>• Subfield 10 (Cardholder-Activated Terminal [CAT] Level) = 0 (Not a CAT transaction)</li></ul> | the Dual Message Authorization System declines the request with a format error response where:<br><br>DE 39 (Response Code) = 30<br><br>DE 44 (Response Data) = 061 |

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 28 (Payment Transaction), and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier), contains value C09 (Card Activation) and</p> <p>DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) does not contain one of the following values:</p> <ul style="list-style-type: none"> <li>• 02 = Merchant terminal PAN auto-entry via magnetic stripe</li> <li>• 05 = Merchant terminal PAN auto-entry via integrated circuit card</li> <li>• 07 = Merchant terminal PAN auto-entry via contactless M/Chip</li> <li>• 80 = Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in DE 45 (Track 1 Data) or DE 35 (Track 2 Data) without alteration or truncation. To use this value, the acquirer must be qualified to use value 90.</li> <li>• 90 = PAN auto-entry via magnetic stripe, the full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.</li> <li>• 91 = PAN auto-entry via contactless magnetic stripe, the full track data has been read from the data on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.</li> </ul> | <p>the Dual Message Authorization System declines the request with a format error response where:</p> <p>DE 39 (Response Code) = 30</p> <p>DE 44 (Response Data) = 022</p> |

**NOTE: The Dual Message Authorization System will reject the transaction with a format error on DE 3 if the acquirer sends an Authorization Advice/0120: acquirer-generated message where DE 3, subfield 1 contains value 28.**



## PIN management services

Mastercard supports two PIN management services.

- Chip PIN Management Service: Supports both PIN change and PIN unblock functionality
- Magnetic Stripe PIN Management Service: Supports PIN change functionality only

Issuers that participate in the Chip PIN Management Service also must support the Magnetic Stripe PIN Management service.

Acquirers in the Europe region that process dual message transactions and that participate in the PIN Management service may optionally choose to support PIN change transactions for magnetic stripe cards.

## Chip PIN management service

The Chip PIN Management service allows cardholders to perform the following transactions at ATMs that support Mastercard, Maestro, or Cirrus chip cards.

- PIN change: Allows cardholders to change the PIN code on their chip card.
- PIN unblock: Allows cardholders to unblock the PIN code on their chip card by resetting the PIN try counter.

The Chip PIN Management service is available only to full grade acquirers and chip grade issuers. For example, the Chip PIN Management service is not available for:

- Magnetic stripe-only cards.
- Transactions initiated with a chip card using magnetic stripe technology.
- Issuers that use the Chip to Magnetic Stripe Conversion service.

**NOTE: Only Europe region acquirers are allowed to submit PIN change and PIN unblock transactions to the Dual Message Authorization System.**

The Chip PIN Management service is an optional service. For more information on Chip PIN Management flows, refer to [Introduction to messages](#).

## Chip PIN management transactions

A cardholder can request a PIN change or PIN unblock only if submitting a valid online PIN. The online PIN is always transmitted to the issuer for validation.

**NOTE: Participating issuers must define their own policies for handling forgotten PINs.**

### PIN change process

1. The cardholder enters a valid, online PIN at an ATM that supports Mastercard, Maestro, or Cirrus chip cards.

2. The cardholder must enter the new PIN twice. The ATM/acquirer compares the two new PIN blocks and sends a single encrypted new PIN block in DE 125 (New PIN Data) along with the old PIN encrypted in DE 52 (Personal ID Number [PIN] Data) to the issuer as part of the authorization request.
3. The ATM initiates an EMV chip transaction with the card requesting an Application Cryptogram (AC) for a transaction value of zero, unless an access fee has been applied by the acquirer for an ATM transaction in countries where application of an ATM access fee is allowed.
4. The acquirer sends the issuer an Authorization Request/0100 message containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 92 (PIN Change) and the data from steps 2 and 3.
5. The issuer validates the online PIN and cryptogram, and then checks any other required information (for example, whether the unblock PIN status is set for the card in their system).
6. The issuer sends an Authorization Request Response/0110 message containing DE 39 (Response Code), which indicates whether to approve or decline the request. If the issuer approves the request, the 0110 message contains DE 39, value 85 (Not declined). In addition, the 0110 message may contain DE 55 with an issuer script. The issuer script contains instructions to the chip card:
  - If the issuer approves the PIN change request, the script message instructs the chip card to change the PIN on the card.
  - If the issuer declines the PIN change request, the issuer may optionally provide additional instructions to the chip card (for example, to block the card or the card application) using the related script.
7. The ATM confirms the status of the cardholder's request and the transaction is complete (that is, the script is applied successfully to the card). The cardholder will be advised if the transaction did not complete properly so that the cardholder understands the status of his or her card.

**NOTE: For PIN change requests, it is important that incomplete transactions are properly reversed. A Reversal Request/0400 message must be initiated when the issuer times out or when the script is not successfully processed by the card. If not, there is a risk that the offline PIN on the card, the online PIN in the issuer host system, and the PIN thought to be correct by the cardholder become unsynchronized.**

### **PIN unblock process**

1. The cardholder enters a valid, online PIN at an ATM that supports Mastercard, Maestro, or Cirrus chip cards.
2. The ATM initiates an EMV chip transaction with the card, requesting an Application Cryptogram (AC) for a transaction value of zero, unless an access fee has been applied by the acquirer for an ATM transaction in countries where application of an ATM access fee is allowed. The Authorization Request/0100 message must include DE 52 to transmit the online PIN.
3. The issuer validates the online PIN and cryptogram, and then checks any other required information (for example, whether the unblock PIN status is set for the card in their system).
4. The issuer generates an Authorization Request Response/0110 message containing DE 39 (Response Code) that indicates whether to approve or decline the request. If the issuer

approves the request, the 0110 message contains DE 39, value 85 (Not declined). In addition, the 0110 message will contain DE 55 with the system-generated Authorization Response Cryptogram (ARPC). Depending on the card application, DE 55 may contain an issuer script. The script message contains instructions to the chip card:

- If the issuer approves the PIN unblock request, the script message instructs the chip card to unblock the PIN on the card.
  - If the issuer declines the PIN unblock request, the issuer may optionally provide additional instructions to block the chip card (for example, block the card or the card application) using the related issuer script.
5. The ATM confirms the status of the cardholder's request and the transaction is complete. The cardholder will be advised if the transaction did not complete properly so that the cardholder understands the status of his or her card.

### **Alternate processing**

The Dual Message Authorization System does not support alternate processing via an alternate issuer host, Stand-In Authorization, or X-Code Processing for Chip PIN Management transactions.

### **Authorization reports**

The following reports provide information that supports the Chip PIN Management service:

- The Authorization Parameter Summary Report (SI737010-AA) includes a Chip PIN Management participant parameter in the Global Parameters section.
- The Authorization Summary Report (AB505010-AA) includes the Chip PIN Management Service values in the Response Code Decline category.

For a sample of the Authorization Summary Report (AB505010-AA) and Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#).

### **Acquirer requirements**

Acquirers using the service are responsible for:

- Offering the PIN change option on their ATMs for chip cards
- Offering the PIN unblock option on their ATMs for chip cards
- Implementing the appropriate dialog between ATMs, chip cards, and cardholders to support the Chip PIN management options

Acquirers must ensure that the new PIN entered by the cardholder is correct. The new PIN is entered twice at the ATM, and the two encrypted PIN blocks must be compared either at the ATM or in the acquirer system. Acquirers only send one new encrypted PIN block to the issuer. If the two PIN blocks do not agree, the acquirer should invite the cardholder to re-enter their new PIN twice.

Acquirers must follow the normal EMV transaction flow at the ATM. The ATM is not expected to process the offline PIN in any way.

Acquirers must support script messages of any length, up to the maximum of 128 bytes.

Acquirers must ensure that the cardholder is accurately advised of the outcome of the PIN change or unblock transaction whether successful or not. This advice must take into account the response from the issuer, the network, and/or the response from the chip card once the script has been delivered to the terminal.

Acquirers send a Reversal Request/0400 message when the issuer times out or when the script is not successfully processed by the chip card.

### Issuer requirements

Issuers using the Chip PIN Management service must validate the ARQC (Authorization Request Cryptogram) transmitted in the Authorization Request/0100. Issuers should also check that the:

- Transaction amount is zero, unless an access fee has been applied by the acquirer for an ATM transaction in countries where application of an ATM access fee is allowed
- ATC (Application Transaction Counter) is greater than the last one received

An Authorization Request/0100 for a PIN unblock transaction may contain an AAC (Application Authentication Cryptogram) instead of an ARQC (depending on the issuer chip card application and card setup). The issuer must carry out the same validation checks on the AAC.

### To participate

Participating acquirers and issuers must notify Mastercard to participate in the Chip PIN Management service. To request participation, contact the Global Customer Service team.

Acquirers and issuers must complete the *PIN Management Service Request* (Form 888), and provide the form to the Global Customer Service team.

When the acquirer does not participate in the PIN Management service, the Dual Message Authorization System sends the acquirer an Authorization Request Response/0110 message containing DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal). When the issuer does not participate in the service, the Dual Message Authorization System sends the acquirer an Authorization Request Response/0110 message containing, DE 39, value 57 (Transaction not permitted to issuer/cardholder).

## Authorization Request/0100: PIN Change or PIN Unblock (Chip Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                        | Org | Sys | Dst | Values/Comments |
|-------------------------------------|-----|-----|-----|-----------------|
| DE 2 (Primary Account Number [PAN]) | M   | .   | M   |                 |

| Data Element                                                                        | Org | Sys | Dst | Values/Comments                                                                                                                                                        |
|-------------------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1                                                  | M   | •   | M   | 91 = PIN Unblock<br>or<br>92 = PIN Change                                                                                                                              |
| DE 3, subfield 2 (Cardholder Transaction Type Code)                                 | M   | •   | M   | 00 = Default Account                                                                                                                                                   |
| DE 3, subfield 3 (Cardholder "To Account" Type Code)                                | M   | •   | M   | 00 = Default Account                                                                                                                                                   |
| DE 4 (Amount, Transaction)                                                          | M   | •   | M   | Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed. |
| DE 18 (Merchant Type)                                                               | M   | •   | M   | 6011 = Member Financial Institution—Automated Cash Disbursements                                                                                                       |
| DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | M   | •   | M   | 05 = Merchant terminal PAN auto-entry via integrated circuit card                                                                                                      |
| DE 22, subfield 2 (POS Terminal PIN Entry Mode)                                     | M   | •   | M   | 1 = Terminal has PIN entry capability                                                                                                                                  |
| DE 28 (Amount, Transaction Fee)                                                     | C   | •   | C   | May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.                                |
| DE 32 (Acquiring Institution ID Code)                                               | M   | •   | M   |                                                                                                                                                                        |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                | M   | •   | M   | Space<br>or<br>Z = ATM Cash Disbursement                                                                                                                               |
| DE 52 (Personal ID Number [PIN] Data)                                               | C   | X   | C   | Required for ATM transactions                                                                                                                                          |
| DE 55 (Integrated Circuit Card (ICC) System-Related Data)                           | C   | •   | C   | Conditional data for chip-based transactions                                                                                                                           |

| Data Element          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 125 (New PIN Data) | C   | X   | C   | Must be present for all online chip card PIN change transactions; otherwise not present<br><br>DE 125 contains the new PIN, which is formatted into one of the supported PIN block formats and is then encrypted. The PIN block format and encryption method used must be the same as the one used for the existing PIN that is stored in DE 52. |

## Authorization Request Response/0110: PIN Change or PIN Unblock (Chip Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Data elements

| Data Element          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                           |
|-----------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code) | M   | .   | M   | 70 = Contact Card Issuer<br><br>71 = PIN Not Changed<br><br>85 = Not Declined (Successful PIN Change or PIN unblock) (recommended for chip cards)<br><br>89 = Unacceptable PIN—Transaction Declined—Retry |

| Data Element                                              | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 55 (Integrated Circuit Card [ICC] System-Related Data) | C   | •   | C   | <p>Contains system-generated Authorization Response Cryptogram (ARPC).</p> <p>Depending on the card application, DE 55 may contain an issuer script. The issuer script message contains instructions to the chip card:</p> <ul style="list-style-type: none"> <li>• If the issuer approves the PIN unblock request, the script message instructs the chip card to unblock the PIN on the card.</li> <li>• If the issuer declines the PIN unblock request, the issuer may optionally provide additional instructions to block the chip card (for example, block the card or the card application) using the related issuer script.</li> </ul> |

## Reversal Request/0400: PIN Change (Chip Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

### Data elements

| Data Element                                         | Org | Sys | Dst | Values/Comments                                                                                                                                                        |
|------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number [PAN])                  | M   | •   | M   |                                                                                                                                                                        |
| DE 3 (Processing Code), subfield 1                   | M   | •   | M   | 92 = PIN Change                                                                                                                                                        |
| DE 3, subfield 2 (Cardholder Transaction Type Code)  | M   | •   | M   | 00 = Default Account                                                                                                                                                   |
| DE 3, subfield 3 (Cardholder "To Account" Type Code) | M   | •   | M   | 00 = Default Account                                                                                                                                                   |
| DE 4 (Amount, Transaction)                           | M   | •   | M   | Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed. |
| DE 18 (Merchant Type)                                | M   | •   | M   | 6011 = Member Financial Institution: Automated Cash Disbursements                                                                                                      |

| Data Element                                                                        | Org | Sys | Dst | Values/Comments                                                                                                                         |
|-------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | M   | .   | M   | 05 = Merchant terminal PAN auto-entry via integrated circuit card                                                                       |
| DE 22, subfield 2 (POS Terminal PIN Entry Mode)                                     | M   | .   | M   | 1 = Terminal has PIN entry capability                                                                                                   |
| DE 28 (Amount, Transaction Fee)                                                     | C   | .   | C   | May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed. |
| DE 32 (Acquiring Institution ID Code)                                               | M   | .   | M   |                                                                                                                                         |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                | M   | X   | M   | Space<br>or<br>Z = ATM Cash Disbursement                                                                                                |
| DE 48, subelement 20 (Cardholder Verification Method)                               | M   | X   | M   | P = Online PIN Verification                                                                                                             |

## PIN processing for non-Europe region customers

The Dual Message Authorization System supports processing of Mastercard purchase transactions that contain a personal identification number (PIN).

Cardholders may enter a PIN at the point-of-interaction as an alternative to signing a sales slip. The information in this section describes processing capabilities for non-Europe region acquirers and issuers that process through the Mastercard Network.

For information about PIN processing capabilities, refer to [Transaction Processing Rules](#) and [Security Rules and Procedures](#).

### Stand-In Authorization and acquirer MIP X-Code Processing

Issuers may choose to have the Dual Message Authorization System verify PIN data on their behalf. If the Dual Message Authorization System performs PIN verification, it can perform Stand-In Authorization and X-Code Processing using the existing authorization-qualifying criteria (issuer or Mastercard), when applicable.

If the Dual Message Authorization System does not verify the PIN data and the issuer is unavailable or unable to process the Authorization Request/0100 message, the Dual Message Authorization System responds to the acquirer with an Authorization Request Response/0110 message indicating the issuer could not process the Authorization Request/0100 message, except in situations where an issuer chooses to allow transactions with unverified PINs in Stand-In Authorization. For more information about the option to allow transactions with unverified PINs in Stand-In Authorization, contact the Global Customer Service team.



## Acquirer requirements

Acquirers that choose to support PIN authorization requests must use their existing Dual Message Authorization System with no changes in authorization message routing. Acquirers must comply with the following steps before processing Mastercard purchase transactions that contain a PIN in Authorization/01xx messages.

1. Determine whether to support static or dynamic PIN Encryption Key (PEK) exchanges. The Dual Message Authorization System and customers' systems use PEKs to encrypt or decrypt PINs. PEKs provide a secure means of passing PIN information in Authorization/01xx messages. Acquirers must determine whether they will support either static or dynamic PEK exchanges. For increased security, Mastercard strongly recommends using dynamic PEK exchanges.
2. Only acquirers that provide complete and unaltered track data in their Authorization Request/0100 messages can support PIN processing.
3. Ensure that applicable data elements are present in Authorization Request/0100 messages. For PIN authorization requests, Authorization Request/0100 messages must contain the following data elements:
  - DE 22 (Point-of-Service [POS] Entry Mode)
  - DE 26 (POS PIN Capture Code) (conditional)
  - DE 45 (Track 1 Data) or DE 35 (Track 2 Data)
  - DE 52 (PIN Data)
4. Ensure the ability to process applicable data elements in Authorization Request Response/0110 messages.

## Issuer requirements

Issuers must comply with the following steps before verifying the PINs in Mastercard purchase transactions that contain a PIN in Authorization Request/0100 messages.

1. Determine the method for receiving purchase transactions that contain a PIN from the following options:
  - Through a Dual Message System Authorization Request/0100 message interface: This method is the default method for issuers that currently receive their Mastercard ATM cash disbursement and ATM balance inquiry activity using this interface. Static PEKs that are already active and operational at the Dual Message Authorization System will be used for each of the issuer's bank identification number (BIN)/card ranges. Issuers may choose to support dynamic PEK exchanges instead.
  - Through a Single Message System Financial Transaction Request/0200 message interface: This method is the default method for issuers that currently receive their card activity, ATM activity, or both using this interface. These transactions are identified as preauthorization requests with DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 4 (Preauthorized request) because they must be cleared through the Mastercard clearing facility (the Global Clearing Management System) with all other purchase activity. PEKs that are already active and operational at the Dual Message Authorization System will be used for each of the issuer's BIN/card ranges.

2. Determine whether to support dynamic PEK exchanges.  
For dynamic PEK exchanges, the Dual Message Authorization System and the issuer must establish a new single Key Encryption Key (KEK) for all BINs that the issuer will process. This new KEK must be associated with the issuer's Member Group ID that the issuer uses for sign-on, sign-off, and store-and-forward (SAF) sessions for the associated BINs.  
  
For increased security, Mastercard strongly recommends using dynamic PEKs.
3. Ensure the ability to process applicable data elements in Authorization Request/0100 messages.
4. Ensure the ability to process applicable data elements in Authorization Advice/0120 messages.
5. Ensure the ability to process applicable data elements in Reversal Advice/0420 messages.

**NOTE: If an issuer does not comply with the requirements for verifying PIN data, the Dual Message Authorization System will forward all authorization requests for purchase transactions that contain a PIN to the issuer with PIN data containing zeros. The issuer can then make the authorization decision based on other data within the authorization message.**

## Support for both acquiring and issuing processing

Customers that support both acquiring and issuing authorization processing may use the same PEK for all purchase transactions that contain a PIN.

To use the same PEK, customers must use the same Customer ID as follows:

- As the Member Group ID for establishing the static PEK or the KEK.
- In DE 32 or DE 33 in all acquired transactions.
- In Member Group ID (DE 2) and DE 33 of the Network Management Request (PEK exchange: On Demand)/0800 message.
- In DE 2 of the Network Management Request/0800 group sign-in message.

## Security requirements

Within the Dual Message Authorization System environment, security includes ensuring message security and integrity as well as protection against cardholder PIN disclosure. Mastercard uses secure PIN encryption in the Dual Message Authorization System to protect all PINs.

### Encryption

The approach that the Dual Message Authorization System uses for network security management is PIN encryption using data encryption standard (DES). Customers must use Triple DES. The acquirer must send the entered PIN to the Single Message System, encrypted in an ANSI block format.

For more information about PIN encryption, refer to the *Mastercard Network Processing Single Message System Guide* and [Security Rules and Procedures](#).

For information on supporting or enabling PIN as a Cardholder Verification Method (CVM) or Multi-Factor Authentication (MFA) Methods for Remote Commerce Token Transactions, refer to the [Security Rules and Procedures](#). Refer to the applicable technical specifications and the [Transaction Processing Rules](#) manual for additional Terminal and Transaction processing requirements relating to the use of a PIN.

PIN verification

Mastercard provides a PIN verification service for all purchase transactions that contain a PIN. There are two PIN verification methods to choose from: DES/IBM 3624 or ABA.

The PIN verification service is an optional service.

**NOTE: Track data must be present for Mastercard to perform PIN verification because the Dual Message Authorization System must have a means to obtain offset data from the issuer.**

If the issuer chooses to have the Dual Message Authorization System perform PIN verification, it must provide PIN processing parameters to Mastercard. These parameters determine placement of PIN verification track data.

If the issuer currently uses the PIN Verification Service that the Single Message System provides for ATM transactions, the issuer may choose to have the Dual Message Authorization System perform PIN verification on all PIN-based purchase activity using the same parameters.

PIN verification in Stand-In Authorization

Mastercard provides a PIN verification service for transactions processed by Stand-In Authorization that contain unverified PIN data in Authorization Request/0100 messages. There are two PIN verification methods to choose from: DES/IBM 3624 or ABA.

Stand-In Authorization will not perform the PIN Verification Test, unless issuers participate in a PIN validation service. If no PIN validation is performed on a transaction, Stand-In Authorization will bypass the PIN Verification Test and proceed to the next sequential Stand-In Test.

Issuers that want Mastercard to perform the optional PIN Verification on transactions processed by Stand-In Authorization must participate in a PIN Pre-validation or PIN Validation in Stand-In service.

Issuers that want Mastercard to verify the PIN provided in DE 52 (Personal ID Number [PIN] Data) on transactions processed by Stand-In Authorization must provide PIN processing parameters to Mastercard.

Associated edits

| WHEN the PIN data is... | THEN Stand-In Authorization...                                |
|-------------------------|---------------------------------------------------------------|
| valid                   | proceeds to the next appropriate Stand-In Authorization test. |

| WHEN the PIN data is... | THEN Stand-In Authorization...                                                                  |
|-------------------------|-------------------------------------------------------------------------------------------------|
| invalid                 | sends a response of decline in the Authorization Request Response/0110 message to the acquirer. |

Mastercard uses DE 48, subelement 80 (PIN Service Code) and DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) in the Authorization Advice/0120 message to communicate the results of the PIN verification test to the issuer.

### Associated edits

| WHEN...                                                                     | THEN DE 48, subelement 80 contains...                                   | AND DE 60, subfield 2 contains...        |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------|
| the PIN is valid                                                            | PV (the Dual Message Authorization System verified the PIN)             | not applicable                           |
| the PIN is invalid                                                          | PI (the Dual Message Authorization System was unable to verify the PIN) | 0004 (Reject: Invalid PIN)               |
| the Dual Message Authorization System is unable to perform PIN verification | PI (the Dual Message Authorization System was unable to verify the PIN) | 0030 (Reject: Unable to verify PIN data) |

Mastercard will perform the PIN Verification in Stand-In service using track data provided in the message, unless the issuer has chosen to participate in the PIN Verification Value (PVV) on File Service.

### Authorization report

The Authorization Parameter Summary Report (SI737010-AA) includes a Stand-In PIN Verification Bypass indicator that identifies whether the Stand-In Authorization will bypass the PIN Verification Test or perform the PIN Verification Test.

This indicator will be present on the report for all customers, not just those customers that elect to participate in a PIN validation service.

## Magnetic stripe PIN management service

Mastercard® credit, , Maestro®, or Cirrus® cardholders can change their PINs at any participating ATM for non-chip (magnetic stripe) cards that originate from the Mastercard Network.

Issuers that currently support PIN change for chip cards must support receiving PIN change transactions for magnetic stripe cards. Acquirers in the Europe region that currently support PIN

change for chip cards and process dual message transactions may optionally choose to support PIN change transactions for magnetic stripe cards.

### PIN change process

Participating acquirers in the Europe region that process transactions through the Dual Message Authorization System and issuers that choose to support ATM PIN change for magnetic stripe cards must process Authorization Request/0100 messages containing:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 92 (PIN Change)
- DE 3, subfields 2 (Cardholder From Account Type Code) and 3 (Cardholder To Account Type Code), value 00 (Default Account [not specified or not applicable])
- DE 4 (Amount, Transaction) must contain a value of zero for PIN change transactions. When ATM transaction fees are allowed, DE 4 may have a value other than zero. In this case, the amount in DE 28 (Amount, Transaction Fee) is included in DE 4.
- DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) may contain a value of either:
  - 02 (Merchant terminal PAN auto-entry via magnetic stripe: track data is not required). Value 02 is acceptable, as long as the track data is present.
  - 90 (PAN auto-entry via magnetic stripe: the full track data has been read)
- DE 52 (Personal ID Number [PIN] Data)
- DE 125 (New PIN Data) will consist of a binary block containing a derived encrypted value that corresponds with the cardholder's new PIN.

### PIN change transactions

For PIN change transactions, the issuer may respond using one of the following methods:

- If the card is a non-chip (magnetic stripe) card or if the card is a chip card not supporting offline PIN and the issuer approves, the issuer sends an Authorization Request Response/0110 message containing DE 39 (Response Code), value 00 (Approved or completed successfully) or DE 39, value 85 (Not declined).

**NOTE: When approving a PIN change transaction, Mastercard recommends that issuers send DE 39, value 85 in the Authorization Request Response/0110 message.**

- If the card is a non-chip (magnetic stripe) card or if the card is a chip card not supporting offline PIN and the issuer declines, the issuer sends an Authorization Request Response/0110 message containing DE 39, value 70 (Contact Card Issuer), DE 39, value 71 (PIN Not Changed), or DE 39, value 89 (Unacceptable PIN: Transaction Declined: Retry).
- If the card is a chip card supporting offline PIN, the issuer must decline and send an Authorization Request Response/0110 message containing DE 39, value 57 (Transaction not permitted to issuer/cardholder).

Issuers that have chip cards personalized with both online and offline PIN must not approve PIN change transactions when DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) contains value 80 (chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-

read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in DE 45 [Track 1 Data] or DE 35 [Track 2 Data without alteration or truncation. To use this value, the acquirer must be qualified to use value 90) or value 90. Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in DE 45 (Track 1 Data) or DE 35 (Track 2 Data) without alteration or truncation. To use this value, the acquirer must be qualified to use value 90. If these transactions are approved, the offline PIN will be out-of-sync with the online PIN, and subsequent offline transactions may be declined due to invalid PIN.

If an issuer's account range participates in Chip to Magnetic Stripe Conversion or M/Chip Cryptogram Pre-validation on-behalf services, that account range cannot participate in the Chip PIN Management service, even if the account range also contains magnetic stripe cards.

### Alternate processing

The Dual Message Authorization System does not support alternate processing via an alternate issuer host, the Stand-In Authorization, or X-Code Processing for PIN change transactions performed on magnetic stripe cards.

### To participate

Registration to participate in the Magnetic Stripe PIN Management service is not required for acquirers and issuers currently participating in the Chip PIN Management service. However, changes to add or delete account ranges must be submitted on the *PIN Management Service Request* (Form 888).

Acquirers in the Europe region and issuers that do not currently support PIN change transactions, but choose to support PIN change transactions performed at the ATM for magnetic stripe cards, must complete the *PIN Management Service Request* (Form 888). For more information about registration and testing, contact the Global Customer Service team.

## Authorization Request/0100: PIN Change (Magnetic Stripe Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                        | Org | Sys | Dst | Values/Comments      |
|-----------------------------------------------------|-----|-----|-----|----------------------|
| DE 2 (Primary Account Number [PAN])                 | M   | •   | M   |                      |
| DE 3 (Processing Code), subfield 1                  | M   | •   | M   | 92 = PIN Change      |
| DE 3, subfield 2 (Cardholder Transaction Type Code) | M   | •   | M   | 00 = Default Account |

| Data Element                                                                        | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3, subfield 3 (Cardholder "To Account" Type Code)                                | M   | •   | M   | 00 = Default Account                                                                                                                                                                                                                                                                                                                              |
| DE 4 (Amount, Transaction)                                                          | M   | •   | M   | Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed.                                                                                                                                                                            |
| DE 18 (Merchant Type)                                                               | M   | •   | M   | 6011 = Member Financial Institution—Automated Cash Disbursements                                                                                                                                                                                                                                                                                  |
| DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | M   | •   | M   | 02 = Merchant terminal PAN auto-entry via magnetic stripe—track data is not required<br><br>or<br>90 = PAN auto-entry via magnetic stripe—the full track data has been read                                                                                                                                                                       |
| DE 22, subfield 2 (POS Terminal PIN Entry Mode)                                     | M   | •   | M   | 1 = Terminal has PIN entry capability                                                                                                                                                                                                                                                                                                             |
| DE 28 (Amount, Transaction Fee)                                                     | C   | •   | C   | May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.                                                                                                                                                                                                           |
| DE 32 (Acquiring Institution ID Code)                                               | M   | •   | M   |                                                                                                                                                                                                                                                                                                                                                   |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                | M   | •   | M   | Space<br><br>or<br>Z = ATM Cash Disbursement                                                                                                                                                                                                                                                                                                      |
| DE 52 (Personal ID Number [PIN] Data)                                               | C   | X   | C   | Required for ATM transactions.                                                                                                                                                                                                                                                                                                                    |
| DE 125 (New PIN Data)                                                               | C   | X   | C   | Must be present for all online chip card PIN change transactions; otherwise not present.<br><br>DE 125 contains the new PIN, which is formatted into one of the supported PIN block formats and is then encrypted. The PIN block format and encryption method used must be the same as the one used for the existing PIN that is stored in DE 52. |

**NOTE:** In the event of a transaction failure or time out, the acquirer must reverse the PIN change transaction so that the issuer is aware that the PIN change was not completed at the ATM. Consequently, the new PIN should not be considered active.

## Authorization Request Response/0110: PIN Change (Magnetic Stripe Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

### Data elements

| Data Element          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                   |
|-----------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code) | M   | .   | M   | 00 = Approved or completed successfully (Successful PIN Change)<br>70 = Contact Card Issuer<br>71 = PIN Not Changed<br>85 = Not Declined (Successful PIN Change) (recommended for chip cards)<br>89 = Unacceptable PIN—Transaction Declined—Retry |

## Reversal Request/0400: PIN Change (Magnetic Stripe Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

### Data elements

| Data Element                                         | Org | Sys | Dst | Values/Comments      |
|------------------------------------------------------|-----|-----|-----|----------------------|
| DE 3 (Processing Code), subfield 1                   | M   | .   | M   | 92 = PIN Change      |
| DE 3, subfield 2 (Cardholder Transaction Type Code)  | M   | .   | M   | 00 = Default Account |
| DE 3, subfield 3 (Cardholder "To Account" Type Code) | M   | .   | M   | 00 = Default Account |



| Data Element                                                                        | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 4 (Amount, Transaction)                                                          | M   | •   | M   | Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed.                                                                                                                                                                                 |
| DE 18 (Merchant Type)                                                               | M   | •   | M   | 6011 = Member Financial Institution: Automated Cash Disbursements                                                                                                                                                                                                                                                                                      |
| DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | M   | •   | M   | 02 = Merchant terminal PAN auto-entry via magnetic stripe: track data is not required<br><br>or<br><br>90 = PAN auto-entry via magnetic stripe: the full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation. |
| DE 22, subfield 2 (POS Terminal PIN Entry Mode)                                     | M   | •   | M   | 1 = Terminal has PIN entry capability                                                                                                                                                                                                                                                                                                                  |
| DE 28 (Amount, Transaction Fee)                                                     | C   | •   | C   | May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.                                                                                                                                                                                                                |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)                | M   | X   | M   | Space<br><br>or<br><br>Z = ATM Cash Disbursement                                                                                                                                                                                                                                                                                                       |
| DE 48, subelement 20 (Cardholder Verification Method)                               | M   | X   | M   | P = Online PIN Verification                                                                                                                                                                                                                                                                                                                            |

## Authorization Request/0100: Edits (Magnetic Stripe Card)

The following edits apply for PIN Management magnetic stripe transactions.

### Applicable edits

| WHEN...                                                                                                                                                                                                                                                                                                                | THEN the Dual Message Authorization System...                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1 contains value 91 (PIN Unblock)<br>and<br>DE 22, subfield 1 does not contain value 05 (PAN auto-entry via chip)                                                                                                                                                                                       | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 (Response Code) = 57 (Transaction not permitted to issuer/cardholder)                                    |
| DE 3, subfield 1 contains value 92 (PIN Change)<br>and<br>DE 125 is not present                                                                                                                                                                                                                                        | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 = 30 (Format error)<br><br>DE 44 (Additional Response Data) = 125 (indicating the data element in error) |
| DE 3, subfield 1 contains value 92<br>and<br>DE 22, subfield 1, does not contain one of the following values: <ul style="list-style-type: none"><li>• 02 (PAN auto-entry via magnetic stripe—track data is not required)</li><li>• 90 (PAN auto-entry via magnetic stripe—the full track data has been read)</li></ul> | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 = 30<br>DE 44 = 022                                                                                      |
| DE 3, subfield 1 contains value 92<br>and<br>DE 22, subfield 2 does not contain value 1 (Terminal has PIN entry capability)                                                                                                                                                                                            | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 = 30<br>DE 44 = 022                                                                                      |
| DE 3, subfield 1 contains value 92<br>and<br>DE 52 is not present                                                                                                                                                                                                                                                      | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 = 30<br>DE 44 = 052                                                                                      |

| WHEN...                                                                                                                                                         | THEN the Dual Message Authorization System...                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1 contains value 92<br>and<br>DE 4 does not contain a value of all zeros,<br>indicating a zero-amount transaction<br>and<br>DE 28 is not present | sends the acquirer an Authorization Request<br>Response/0110 message where:<br><br>DE 39 = 30<br>DE 44 = 004                                         |
| DE 3, subfield 1 contains value 92<br>and<br>DE 48, TCC contains a value other than Z (ATM<br>Cash Disbursement) or space                                       | sends the acquirer an Authorization Request<br>Response/0110 message where:<br><br>DE 39 = 30<br>DE 44 = 048000                                      |
| the acquirer does not participate in the PIN<br>Management service                                                                                              | sends the acquirer an Authorization Request<br>Response/0110 message where:<br><br>DE 39 = 58 (Transaction not permitted to acquirer/<br>terminal)   |
| the issuer does not participate in the PIN<br>Management service                                                                                                | sends the acquirer an Authorization Request<br>Response/0110 message where:<br><br>DE 39 = 57 (Transaction not permitted to issuer/<br>cardholder)   |
| the issuer is not able to respond to the PIN<br>Management transaction on time                                                                                  | sends the acquirer an Authorization Request<br>Response/0110 message where:<br><br>DE 39 = 91 (Authorization System or issuer system<br>inoperative) |

## Authorization Advice/0120: Acquirer-Generated Edits (Magnetic Stripe Card)

The following edits apply to PIN Management magnetic stripe card transactions.

### Applicable edits

| WHEN...                            | THEN the Dual Message Authorization System...                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1 contains value 92 | sends the acquirer an Authorization Advice Response/0130 message where:<br><br>DE 39 = 30<br>DE 44 = 003 |

## Reversal Request/0400: Edits (Magnetic Stripe Card)

The following edits apply to PIN Management, magnetic stripe card transactions.

### Applicable edits

| WHEN...                                                                                                                                                                                                          | THEN the Dual Message Authorization System...                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| DE 3, subfield 1 contains value 92<br>and<br>DE 22, subfield 1, does not contain one of the following values: <ul style="list-style-type: none"><li>• 02</li><li>• 90</li></ul>                                  | sends the acquirer a Reversal Request Response/0410 message where:<br><br>DE 39 = 30<br>DE 44 = 022 |
| DE 3, subfield 1 contains value 92<br>and<br>DE 22, subfield 2 does not contain value 1                                                                                                                          | sends the acquirer a Reversal Request Response/0410 message where:<br><br>DE 39 = 30<br>DE 44 = 022 |
| DE 3 (Processing Code) subfield 1 (Cardholder Transaction Type Code) contains value 92<br>and<br>DE 4 does not contain a value of all zeros, indicating a zero-amount transaction<br>and<br>DE 28 is not present | sends the acquirer a Reversal Request Response/0410 message where:<br><br>DE 39 = 30<br>DE 44 = 004 |

| WHEN...                                                                                        | THEN the Dual Message Authorization System...                                                                                        |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1 contains value 92<br>and<br>DE 48, TCC contains a value other than Z or space | sends the acquirer a Reversal Request Response/<br>0410 message where:<br>DE 39 = 30<br>DE 44 = 048000                               |
| the acquirer does not participate in the PIN<br>Management service                             | sends the acquirer a Reversal Request Response/<br>0410 message where DE 39 = 58 (Transaction not<br>permitted to acquirer/terminal) |
| the issuer does not participate in the PIN<br>Management service                               | sends the acquirer a Reversal Request Response/<br>0410 message where DE 39 = 57 (Transaction not<br>permitted to issuer/cardholder) |

## Issuer Response Options to a Magnetic Stripe PIN Change Request

For PIN change transactions, the issuer may respond using one of the following methods.

- If the card is a non-chip (magnetic stripe) card or if the card is a chip card not supporting offline PIN and the issuer approves, the issuer sends an Authorization Request Response/0110 message containing DE 39 (Response Code), value 00 (Approved or completed successfully) or DE 39, value 85 (Not declined).

**NOTE: When approving a PIN change transaction, Mastercard recommends that issuers of chip cards send DE 39, value 85 in the Authorization Request Response/0110 message.**

- If the card is a non-chip (magnetic stripe) card or if the card is a chip card not supporting offline PIN and the issuer declines, the issuer sends an Authorization Request Response/0110 message containing DE 39, value 70 (Contact Card Issuer), DE 39, value 71 (PIN Not Changed), or DE 39, value 89 (Unacceptable PIN—Transaction Declined—Retry).
- If the card is a chip card supporting offline PIN, the issuer must decline and send an Authorization Request Response/0110 message containing DE 39, value 57 (Transaction not permitted to issuer/cardholder).

Issuers that have chip cards personalized with both online and offline PIN must not approve PIN change transactions when DE 22 (Point of Service [POS] Entry Mode, subfield 1 (POS Terminal PAN Entry Mode) contains value 02 or value 90. If these transactions are approved, the offline PIN will be out-of-sync with the online PIN, and subsequent offline transactions may be declined due to invalid PIN.

If an issuer's account range participates in Chip to Magnetic Stripe Conversion or M/Chip Cryptogram Pre-validation on-behalf services, that account range cannot participate in the Chip PIN Management service, even if the account range also contains magnetic stripe cards.

## PIN Processing for Europe Region Customers

Mastercard supports PIN translation for acquirers and issuers and issuer PIN validation for Track 1 and Track 2 on the Mastercard Network for customers in the Europe Region.

For information about PIN processing capabilities, refer to [Transaction Processing Rules](#).

### PIN Translation

The Dual Message Authorization System performs PIN translation on DE 52 (Personal ID Number [PIN] Data) and DE 125 (New PIN Data) for acquirers and issuers that use the Mastercard Network.

When issuers perform PIN processing in-house or when they participate in the Online PIN Validation in Stand-In Authorization, issuers must specify DE 53, subfield 4 (PIN key index number) in the Network Management Request/0800: sign-on/sign-off message, all remaining DE 53, subfields may be zero filled.

Issuers participating in the Online PIN Pre-validation Service are not required to send DE 53 in the Network Management Request/0800: sign-on/sign-off message.

Acquirers specify the key used to encrypt the PIN using the Authorization Request/0100 message.

The Key Management Services (KMS) group provides customers a Member Key ID (MKID) for each service. The MKID is used to notify the KMS group of new security keys. Each customer will associate the PIN key index number to identify a specific security key. Customers may define a maximum of 99 security keys for use during PIN translation.

### DE 53 (Security-related Control Information)

Acquirers will specify DE 53, subfield 3 (PIN Block Format Code) when sending the Authorization Request/0100 message.

Issuers that use the PIN translation service to translate DE 52 and DE 125 (New PIN Data) will receive DE 53, subfield 4 (PIN Key Index Number) that identifies the PIN key used for translation.

PIN translation also will be performed on DE 125, if present, when the PIN translation was successfully completed for DE 52.

## PIN Translation Edits

The following edits apply to Authorization Request/0100 messages (for acquirers only).

### Associated edits

| WHEN...                                                                                                                                                                                               | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer uses the Dual Message Authorization System for PIN translation and DE 52 (Personal ID Number [PIN] Data) is present and DE 53 (Security-related Control Information) is not present      | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 = 30 (Format error)</li> <li>DE 44 = 053 (DE 53 is the data element in error)</li> </ul>                                                                                                             |
| the acquirer uses the Dual Message Authorization System for PIN translation and DE 52 is present and DE 53, subfield 3 (PIN Block Format Code) is not valid. (Valid values are 01 and 10.)            | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 = 30</li> <li>DE 44 = 053</li> </ul>                                                                                                                                                                 |
| the acquirer uses the Dual Message Authorization System for PIN translation and DE 52 is present and DE 53, subfield 4 (PIN Key Index Number) contains a PIN Key Index Number not known to Mastercard | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 = 88 (Cryptographic failure)</li> <li>DE 48 (Additional Data: Private Use), subelement 80 (PIN Service Code) = TI (The Dual Message Authorization System was unable to translate the PIN)</li> </ul> |
| the acquirer does not use the Dual Message Authorization System for PIN translation and DE 53 is present                                                                                              | removes DE 53 and forwards the Authorization Request/0100 message to the Single Message System                                                                                                                                                                                                                             |
| the Dual Message Authorization System Security Translation Platform fails                                                                                                                             | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 = 88</li> <li>DE 48, subelement 80 = TI</li> </ul>                                                                                                                                                   |

The following edits apply to Network Management Request/0800 sign-on messages.

Associated edits

| WHEN...                                                                                                                                                                                                               | THEN the Dual Message Authorization System...                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The issuer uses the Dual Message Authorization System for PIN translation and DE 53 (Security-Related Control Information), subfield 4 (PIN Key Index Number) contains a PIN Key Index Number not known to Mastercard | <div>Sends the issuer a Network Management Request Response/0810 message where:</div> <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 053</li></ul> |

PIN Validation

For information about PIN processing capabilities, refer to [Transaction Processing Rules](#).

PIN validation results will be provided in DE 48, subelement 71 (On-behalf Services) in Authorization Request/0100 and Authorization Advice/0120: system-generated messages when PIN validation has been performed on behalf of the issuer. DE 48, subelement 71 will contain the following values related to PIN validation:

- Subfield 1 (On-behalf [OB] Service) will contain the following values:
  - 08 (Online PIN Pre-validation)
  - 09 (Online PIN Validation in Stand-In)
- Subfield 2 (On-behalf [OB] Result 1) when subfield 1 is value 08 or 09:
  - I (Invalid)
  - P (Mandatory PVV not on file)
  - R (PIN retry exceeded [invalid PIN])
  - U (Unable to process)
  - V (Valid)
- Subfield 3 (On-behalf [OB] Result 2) is blank when it is sent to the issuer

The Dual Message Authorization System will manage tracking the number of PIN failed attempts at the card level (for example, PAN, card sequence number from the track and expiration date).

DE 52 (Personal ID Number [PIN] Data) is not included in the Authorization Request/0100 message when PIN validation is performed.



### Associated edits

| WHEN the issuer subscribes to the...            | THEN the issuer...                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Online PIN Validation in Stand-In Authorization | <p>chooses the PIN Failed Attempts limit, which must be less than or equal to five.</p> <p>Stand-In Authorization will reset the PIN Failed Attempts counter to zero when a valid PIN is entered and the count has not exceeded the issuer defined maximum.</p> <p>Stand-In Authorization will reset the PIN Failed Attempts counter to zero after the Stand-In Authorization maintenance.</p> |
| Online PIN Pre-validation                       | <p>does not choose the PIN Failed Attempts.</p> <p>Mastercard will allow five PIN Failed Attempts. The Dual Message Authorization System will reset the PIN Failed Attempts counter to zero when a valid PIN is entered and the count has not exceeded five.</p> <p>The Dual Message Authorization System will reset the PIN Failed Attempts counter to zero after 24 hours has passed.</p>    |

## PIN Validation edits

The following edits apply to Authorization Request/0100 messages (for issuers only).

### Associated edits

| WHEN...                                                                                                                                                       | THEN the Dual Message Authorization System...                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer uses the Dual Message Authorization System for PIN pre-validation and the PIN is valid and the number of PIN failed attempts were not yet exceeded | <p>forwards the Authorization Request/0100 message where:</p> <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1 = 08 (Online PIN Pre-validation)</li> <li>DE 48, subelement 71, subfield 2 = V (Valid)</li> </ul> |
| the issuer uses the Dual Message Authorization System for PIN pre-validation and the PIN validation cannot be performed                                       | <p>forwards the Authorization Request/0100 message where:</p> <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1 = 08</li> <li>DE 48, subelement 71, subfield 2 = U (Unable to process)</li> </ul>                 |

| WHEN...                                                                                                                                                                       | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer uses the Dual Message Authorization System for PIN pre-validation and the number of PIN retries were exceeded                                                      | forwards the Authorization Request/0100 message where: <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1 = 08</li> <li>DE 48, subelement 71, subfield 2 = R (PIN retry exceeded)</li> </ul>                                                                                                                                                                                                                                                                                                                                          |
| the issuer uses the Dual Message Authorization System for PIN pre-validation and the PIN validation fails                                                                     | forwards the Authorization Request/0100 message where: <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1 = 08</li> <li>DE 48, subelement 71, subfield 2 = I (Invalid)</li> </ul>                                                                                                                                                                                                                                                                                                                                                     |
| the issuer uses the Dual Message Authorization System for PIN pre-validation and has specified Mandatory PVV On File and the PVV was not found                                | forwards the Authorization Request/0100 message where: <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1 = 08</li> <li>DE 48, subelement 71, subfield 2 = P (Mandatory PVV not on file)</li> </ul>                                                                                                                                                                                                                                                                                                                                   |
| the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In Authorization, PIN validation fails, and the number of PIN retries were not yet exceeded | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 = issuer defined</li> <li>DE 48, subelement 80 = PI (Dual Message Authorization System unable to process PIN)</li> </ul> sends the issuer an Authorization Advice/0120: system-generated message where: <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1 = 09 (PIN Validation in Stand-In)</li> <li>DE 48, subelement 71, subfield 2 = I (Invalid)</li> <li>DE 60, subfield 2 = 0051 (invalid PIN)</li> </ul> |
| the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the PIN validation fails                                                             | sends the acquirer an Authorization Request Response/0110 message where: <ul style="list-style-type: none"> <li>DE 39 = issuer defined</li> <li>DE 48, subelement 80 = PI</li> </ul> sends the issuer an Authorization Advice/0120 (System-generated) message where: <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1 = 09</li> <li>DE 48, subelement 71, subfield 2 = U (Unable to process)</li> <li>DE 60, subfield 2 = 0050 (Unable to process)</li> </ul>                                                                       |

| WHEN...                                                                                                                                                       | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the PIN is valid and the number of PIN retries were not yet exceeded | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 = determined by remaining Stand-In Authorization tests</li> <li>• DE 48, subelement 80 = PV (Valid PIN)</li> </ul> <p>sends the issuer an Authorization Advice/0120 (System-generated) message where:</p> <ul style="list-style-type: none"> <li>• DE 48, subelement 71, subfield 1 = 09</li> <li>• DE 48, subelement 71, subfield 2 = V (Valid)</li> </ul>                                              |
| the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the number of PIN retries were exceeded                              | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 = issuer defined response code</li> <li>• DE 48, subelement 80 = PI</li> </ul> <p>sends the issuer an Authorization Advice/0120: system-generated message where:</p> <ul style="list-style-type: none"> <li>• DE 48, subelement 71, subfield 1 = 09</li> <li>• DE 48, subelement 71, subfield 2 = R (PIN retry exceeded)</li> <li>• DE 60, subfield 2 = 0052 (PIN Retry Exceeded-invalid PIN)</li> </ul> |
| the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and has specified Mandatory PVV On File and the PVV was not found        | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 = issuer defined value</li> <li>• DE 48, subelement 80 = PI</li> </ul> <p>sends the issuer an Authorization Advice/0120: system-generated message where:</p> <ul style="list-style-type: none"> <li>• DE 48, subelement 71, subfield 1 = 09</li> <li>• DE 48, subelement 71, subfield 2 = P (Mandatory PVV not on file)</li> <li>• DE 60, subfield 2 = 0052 (Mandatory PVV not on file)</li> </ul>       |

| WHEN...                                                                                                            | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the PIN translation fails | <p>sends the acquirer an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 88</li> <li>• DE 48, subelement 80 = TI</li> </ul> <p>sends the issuer an Authorization Advice/0120: system-generated message where:</p> <ul style="list-style-type: none"> <li>• DE 48, subelement 71, subfield 1 = 09</li> <li>• DE 48, subelement 71, subfield 2 = U (Unable to process)</li> <li>• DE 60, subfield 2 = 0050 (Unable to process)</li> </ul> |

## PIN Key Management

The Key Management Services (KMS) group will manage the security keys for customers using the PIN translation and PIN validation services supported by the Dual Message Authorization System on the Mastercard Network.

## PIN Verification Value/PIN Offset on File Service

The PIN Verification Value (PVV)/PIN Offset used in PIN validation is usually located on Track 1 or Track 2 of a card's magnetic stripe. Mastercard offers a service by which the issuer may optionally send the PVV/PIN offsets, in a file to Mastercard in order to override the information encoded in the track of a card's magnetic stripe.

Mastercard activates the PVV/PIN Offset data from an issuer upon receipt of the PVV/PIN Offset File. Issuers may send a full file replacements weekly.

## Processing Transactions Using PVV/PIN Offset

Issuers that use the optional PVV/PIN Offset on File processing must request participation in the service for each card range and (optionally) expiry date range.

To process a transaction, the Dual Message Authorization System checks that PVV/PIN Offset details are held on file for the relevant card range and expiry date range. If details are on file for the card range and expiry date range, the Dual Message Authorization System checks for the individual card's details within the issuer's stored file.

The Dual Message Authorization System checks the entry in the PVV/PIN Offset file to ensure that the card's PAN, card sequence number, and expiry date match. If they do match, the Dual Message Authorization System can use the PVV value on file. If the PAN, card sequence number, and expiry date do not match, the Dual Message Authorization System processes the transaction according to the processing parameters that the issuer specifies for the relevant card range and expiry date.

## Processing Parameters

Mastercard stores the issuer PVV/PIN Offset files for use in the Online PIN Pre-validation and the Online PIN Validation in Stand-In on-behalf services. When the Dual Message Authorization System cannot find a matching entry (with the correct PAN, card sequence number, and expiry date values) in the issuer PVV/PIN Offset file, it needs further instructions to process the transaction properly.

The issuer must select one of the following processing options for each card range and expiry date:

- Optional on file: With this option, if no matching PVV/PIN Offset entry is found in the PVV/PIN Offset file, the Dual Message Authorization System retrieves the PVV value from Track 1 or Track 2 of the relevant card.
- Mandatory on file: With this option, the Dual Message Authorization System, having checked the Track 1 or Track 2 information for the issuer card against the entries held in the issuer PVV/PIN Offset file, performs one of the following actions:
  - For Stand-In services, the Dual Message Authorization System returns an Authorization Request Response/0110 message to the acquirer and an Authorization Advice/0120: system-generated message to the issuer with the appropriate response code, detailing the result of the processing.
  - For Pre-validation services, the Dual Message Authorization System forwards the results of the PIN validation to the issuer in an Authorization Request/0100 message and to the acquirer in an Authorization Request Response/0110 message.

The [Global Parameters](#) section provides an indicator to identify issuer participation.

## PVV/PIN Offset File Format

Following is the PVV/PIN Offset File format that has a fixed length of 64 characters. Issuers can use Bulk ID RA85 or CONNECT:Direct to send the PVV/PIN Offset file to Mastercard.

### PVV/PIN Offset File Header Record

| Field Name          | Attribute    | Length | Comments/Values                                                                                                           |
|---------------------|--------------|--------|---------------------------------------------------------------------------------------------------------------------------|
| Record type ID      | Alphanumeric | 3      | HDR—header record                                                                                                         |
| File version number | Numeric      | 3      | 100 (initial version)                                                                                                     |
| Customer ID         | Alphanumeric | 11     | First six digits must contain the issuers ICA with leading zeros<br><br>Seventh digit and beyond contains trailing spaces |

| Field Name            | Attribute    | Length | Comments/Values                                                     |
|-----------------------|--------------|--------|---------------------------------------------------------------------|
| Transmission code     | Alphanumeric | 10     | Member assigned transmission code. May contain all spaces or zeros. |
| Transmission sequence | Numeric      | 4      | 0000–9999<br>Wraps at 9999                                          |
| Transmission date     | Numeric      | 6      | YYMMDD in UTC                                                       |
| Transmission time     | Numeric      | 6      | hhmmss in UTC                                                       |
| Input file type       | Alphanumeric | 1      | F: Full file replace                                                |
| Filler                | Alphanumeric | 20     | Spaces                                                              |

#### PVV/PIN Offset File Detail Record

| Field Name           | Attribute    | Length | Comments/Values                                                                                                                                                                   |
|----------------------|--------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record type ID       | Alphanumeric | 3      | DTL—detail record                                                                                                                                                                 |
| Update code          | Alphanumeric | 1      | A = Add/Update                                                                                                                                                                    |
| PAN                  | Numeric      | 19     | With trailing spaces                                                                                                                                                              |
| Card expiry date     | Numeric      | 4      | YYMM (must contain a valid year and month)                                                                                                                                        |
| Card sequence number | Numeric      | 1      | Must contain the Sequence Number associated with the PAN or 0. A value of 0 indicates that the card sequence number should not be used as criteria when matching to the PVV file. |
| PVV/PIN Offset       | Numeric      | 6      | Trailing spaces                                                                                                                                                                   |
| Filler               | Alphanumeric | 30     | Spaces                                                                                                                                                                            |

#### PVV/PIN Offset File Trailer Record

| Field Name               | Attribute    | Length | Comments/Values     |
|--------------------------|--------------|--------|---------------------|
| Record type ID           | Alphanumeric | 3      | TRL—trailer record  |
| Number of detail records | Numeric      | 11     | Detail record count |

| Field Name | Attribute    | Length | Comments/Values |
|------------|--------------|--------|-----------------|
| Filler     | Alphanumeric | 50     | Spaces          |

## Alternate Processing

Mastercard will support issuers subscribing to the Online PIN Pre-validation service (OB service code value 08) that are not available to respond to the Authorization Request/0100 message and issuers subscribing to the PIN Validation in Stand-In service (OB service code value 09) by responding to the Authorization Request/0100 message on behalf of the issuer. Mastercard will consider the results of the PIN validation in DE 48, subelement 71 when responding to the Authorization Request/0100 message.

The values contained in DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) are dependent on the values contained in DE 48, subelement 71, subfield 2 (OB Result 1) as described in the following table.

### Values

| WHEN DE 48, subelement 71, subfield 2 contains the value... | THEN DE 60, subfield 2 will contain... |
|-------------------------------------------------------------|----------------------------------------|
| U (Unable to Process)                                       | 0050                                   |
| I (Invalid)                                                 | 0051                                   |
| P (Mandatory PVV not on file)                               | 0052                                   |
| R (PIN Retry Exceeded [invalid PIN])                        | 0052                                   |

## PIN Processing for Non-Europe Customers

Personal identification number (PIN) is a proven technology that is not easily circumvented. It is secure within actual market conditions. Mastercard permits acquirers to use PIN technology at the point of sale that prompts Mastercard cardholders for a PIN.

This section contains PIN processing procedures for non-Europe customers. For information about PIN processing procedures for Europe region acquirers and issuers that process through the Mastercard Network, refer to [PIN Processing for Europe Region Customers](#).

## Acquirer Requirements

Acquirers must comply with the following before processing Mastercard purchase transactions that contain a PIN (such as CAT Level 1 transactions) in Authorization/01xx messages.

- Support either static or dynamic PIN Encryption Key (PEK) exchanges
- Comply with the Mastercard Magnetic Stripe Compliance Program
- Correctly format Authorization Request/0100—PIN messages
- Correctly format Authorization Request Response/0110—PIN messages

### Support either Static or Dynamic PIN Encryption Key (PEK) exchanges

The Dual Message Authorization System and customers' system use PEK to encrypt or decrypt PINs. PEKs provide a secure means of passing PIN information in Authorization/01xx messages. Acquirers must support either static or dynamic PEK exchanges.

**NOTE: For increased security, Mastercard strongly recommends using dynamic PEK exchanges.**

Following are the differences between the two options.

#### PEK options

| PEK     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Static  | <ul style="list-style-type: none"><li>• Customers and the Dual Message Authorization System establish these keys offline.</li><li>• Customers and the Dual Message Authorization System must establish a single PEK. This PEK must be associated with the acquirer's customer ID.</li></ul>                                                                                                                                                                                                                                                                                                                                       |
| Dynamic | <ul style="list-style-type: none"><li>• Customers and the Dual Message Authorization System exchange these keys automatically online every 24 hours or every 2,500 transactions, whichever occurs first.</li><li>• Customers and the Dual Message Authorization System use Key Encryption Keys (KEKs) to encrypt and decrypt the PEKs during PEK exchanges between the acquirer and the Dual Message Authorization System. Each acquirer must establish a KEK with Mastercard to exchange the PEKs dynamically with the Dual Message Authorization System. This KEK must be associated with the acquirer's customer ID.</li></ul> |



## Mastercard Magnetic Stripe Compliance Program Compliance

PIN verification requires valid track data, only acquirers that comply with the Mastercard magnetic stripe compliance program and provide complete and unaltered track data in their Authorization Request/0100 messages are able to support PIN processing.

### Authorization Request/0100: PIN Transactions

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

#### Data elements

| Data Element                                                                        | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | M   | .   | M   | Must contain one of the following values: <ul style="list-style-type: none"> <li>05 = Merchant terminal PAN auto-entry via integrated circuit card</li> <li>07 = Merchant terminal PAN auto-entry via contactless M/Chip</li> <li>80 = Chip card at chip-capable terminal was unable to process transaction using data on them chip; therefore, the terminal defaulted to the magnetic stripe-read PAN</li> <li>90 = PAN auto-entry via magnetic stripe: the full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation</li> </ul> |
| DE 26 (POS PIN Capture Code)                                                        | C   | .   | C   | Must be present when DE 52 (PIN Data) is present and the maximum PIN characters that the terminal accepts is something other (more or less) than 12 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DE 32 (Acquiring Institution ID Code)                                               | M   | .   | M   | If present, must be the same ID as the Member Group ID for the Dual Message Authorization System to translate the PIN data that the acquirer provides.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 33 (Forwarding Institution ID Code)                                              | C   | .   | C   | If present, must be the same ID as the Member Group ID for the Dual Message Authorization System to translate the PIN data that the acquirer provides.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Data Element         | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                               |
|----------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 35 (Track-2 Data) | C   | .   | C   | Must be present and represent the information as encoded on the magnetic stripe of the card (or the equivalent data if a chip card and chip-capable terminal are used at the point of interaction)<br>or DE 45 (Track-1 Data) |
| DE 45 (Track-1 Data) | C   | .   | C   | Must be present and represent the information as encoded on the magnetic stripe of the card (or the equivalent data if a chip card and chip-capable terminal are used at the point of interaction)                            |
| DE 52 (PIN Data)     | C   | X   | C   | The 16-digit number Data Encryption Standard (DES) hexadecimal number in the ANSI PIN block format (also referred to as "ISO Format-0")                                                                                       |

## Authorization Request Response/0110: PIN Transactions

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                             | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                         |
|------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code)                    | M   | .   | M   | Contains response code indicating the Dual Message Authorization System or the issuer could or could not verify the PIN entered at the point of interaction.<br><br>Refer to the following table that identifies possible responses based on PIN processing conditions. |
| DE 48 (Additional Data—Private Use), TCC | C   | .   | C   | Must contain the appropriate TCC code.                                                                                                                                                                                                                                  |

| Data Element                            | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 80 (PIN Service Code) | C   | .   | C   | <p>Identifies whether the Dual Message Authorization System attempted to verify or translate the PIN. Valid values:</p> <ul style="list-style-type: none"> <li>PV = The Dual Message Authorization System verified the PIN</li> <li>TV = The Dual Message Authorization System translated the PIN for issuer verification</li> <li>PI = The Dual Message Authorization System was unable to verify the PIN</li> <li>TI = The Dual Message Authorization System was unable to translate the PIN</li> </ul> |

Acquirer-related PIN processing errors, indicated by DE 39 (Response Code), value 30 (Format Error) and DE 44 (Additional Response Data) value 052 (DE 52 PIN Data), occur if the Dual Message Authorization System is unable to decrypt the PIN data. Possible causes are:

- The PEK that the Dual Message Authorization System and customer share is not synchronized. It should be the same key.
- The Dual Message Authorization System is unable to determine the proper PEK because DE 32 or DE 33 is incorrect.
- The acquirer is not qualified to be in the Mastercard Magnetic Stripe Compliance Program.
- The acquirer did not correctly establish the static PEK or KEK with Mastercard.
- The DE 22, subfield 1 value is not valid for transactions that contain a PIN.

Following are possible Authorization Request Response/0110 message responses:

#### Associated edits

| IF...                                                                                                  | THEN...                                                                  |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| The Dual Message Authorization System on behalf of the issuer was unable to verify the PIN.            | DE 39 = 55 (Invalid PIN)<br>DE 48, subelement 80 (PIN Service Code) = PI |
| The Dual Message Authorization System translated the PIN, but the issuer was unable to verify the PIN. | DE 39 = 55<br>DE 48, subelement 80 = TV                                  |
| The Dual Message Authorization System was unable to decrypt the PIN for the acquirer.                  | DE 39 = 30 (Format Error)<br>DE 44 = 52<br>DE 48, subelement 80 = TI     |

| IF...                                                                               | THEN...                                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| PIN processing was successful (verification or translation).                        | DE 39 = any valid code<br>DE 48, subelement 80 = TV or PV                                   |
| The Dual Message Authorization System was unable to decrypt the PIN for the issuer. | DE 39 = 91 (Authorization System or issuer system inoperative)<br>DE 48, subelement 80 = TI |

## Issuer Requirements

Issuers must comply with the following to process Mastercard transactions that contain a PIN in Authorization Request/0100 messages.

- Receive purchase or ATM transactions that contain a PIN
- Support Static or Dynamic PEK exchanges
- Process applicable data elements in Authorization Request/0100 messages
- Process applicable data elements in Authorization Request Response/0110 messages
- Process applicable data elements in Authorization Advice/0120: system-generated messages
- Process applicable data elements in Reversal Advice/0420: system-generated messages

For information, refer to *Transaction Processing Rules*.

## Receive Purchase Transactions that Contain a PIN

Issuers must be able to receive purchase transactions that contain a PIN using one of the following network options.

- Through a Mastercard Network connection using Authorization Request/0100 messages
- The default method for issuers that currently receive their Mastercard ATM cash disbursement and ATM balance inquiry activity using this interface. Static PEKs that are already active and operational at the Single Message System will be used for each of the issuer's bank identification number (BIN)/card ranges.

Issuers may choose to support dynamic PEK exchanges instead.

- Through a Single Message System connection using Financial Transaction Request/0200 messages

The default method for issuers that currently receive their Debit Mastercard activity, ATM activity, or both using this interface. These transactions will be identified as preauthorization requests with a DE 61, subfield 7, value 4 because they must be cleared through the Mastercard Global Clearing Management System, with all other purchase activity. PEKs that already are active and operational at the Single Message System will be used for each of the issuer's BIN/card ranges.

## Support Static or Dynamic PEK exchanges

For dynamic PEK exchanges, the Dual Message Authorization System and the issuer must establish a new single KEK for all BINs that the issuer will process. This KEK must be associated with the issuer's Group ID that the issuer uses for sign-on, sign-off, and store-and-forward (SAF) sessions for the associated BINs.

**NOTE: For increased security, Mastercard strongly recommends using dynamic PEKs.**

## Authorization Request/0100: PIN Messages

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                             | Org | Sys | Dst | Values/Comments                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|-----|-----|-----|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), TCC | M   | .   | M   | Must contain the appropriate TCC code.                             |                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 48, subelement 80 (PIN Service Code)  | .   | X   | C   | This subelement contains the results of PIN processing as follows: |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                          |     |     |     | IF the issuer chooses to...                                        | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                      |
|                                          |     |     |     | Use the Mastercard PIN Verification Service                        | <ul style="list-style-type: none"><li>• Omits DE 52 from the Authorization Request/0100 message sent to the issuer.</li><li>• Sends DE 48, subelement 80, value PV (The Dual Message Authorization System verified the PIN).</li><li>• If the PIN is invalid, the Dual Message Authorization System sends the acquirer an Authorization Request Response/0110 message where DE 39 contains the value 55.</li></ul> |

| Data Element                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 52 (Personal ID Number (PIN) Data) | C   | X   | C   | <p>Will be present if the Dual Message Authorization System does not perform PIN verification. The issuer verifies the PIN data in DE 52.</p> <p>Either DE 45 or DE 35 will be present in the message to provide the PIN verification value (PVV) or offset, depending on the issuer's verification method.</p> |

### Values

| IF...                | THEN...                          |
|----------------------|----------------------------------|
| DE 52 is present     | DE 48, subelement 80 is value TV |
| DE 52 is not present | DE 48, subelement 80 is value PV |

## Authorization Advice/0120: PIN Messages

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

### Data elements

| Data Element                                              | Org                                                                                                                                                                                                                                              | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |                                               |                                                           |                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), TCC                  | .                                                                                                                                                                                                                                                | M   | M   | Must contain the appropriate TCC code.                                                                                                                                                                                                                                                                                                                                                                                                                 |                             |                                               |                                                           |                                                                                                                                                                                                                                                  |
| DE 48, subelement 80 (PIN Service Code)                   |                                                                                                                                                                                                                                                  |     |     | This subelement contains the results of PIN processing as follows:                                                                                                                                                                                                                                                                                                                                                                                     |                             |                                               |                                                           |                                                                                                                                                                                                                                                  |
|                                                           |                                                                                                                                                                                                                                                  |     |     | <table><tr><td>IF the issuer chooses to...</td><td>THEN the Dual Message Authorization System...</td></tr><tr><td>Have the Dual Message Authorization System verify the PIN</td><td><ul style="list-style-type: none"><li>Omits DE 52 from the Authorization Advice/0120 message sent to the issuer.</li><li>Sends DE 48, subelement 80, value TI (The Dual Message Authorization System was unable to translate the PIN).</li></ul></td></tr></table> | IF the issuer chooses to... | THEN the Dual Message Authorization System... | Have the Dual Message Authorization System verify the PIN | <ul style="list-style-type: none"><li>Omits DE 52 from the Authorization Advice/0120 message sent to the issuer.</li><li>Sends DE 48, subelement 80, value TI (The Dual Message Authorization System was unable to translate the PIN).</li></ul> |
| IF the issuer chooses to...                               | THEN the Dual Message Authorization System...                                                                                                                                                                                                    |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |                                               |                                                           |                                                                                                                                                                                                                                                  |
| Have the Dual Message Authorization System verify the PIN | <ul style="list-style-type: none"><li>Omits DE 52 from the Authorization Advice/0120 message sent to the issuer.</li><li>Sends DE 48, subelement 80, value TI (The Dual Message Authorization System was unable to translate the PIN).</li></ul> |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |                                               |                                                           |                                                                                                                                                                                                                                                  |

| Data Element                                                | Org | Sys | Dst | Values/Comments                                                                                                                                  |
|-------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) | •   | M   | M   | 107 (PIN Processing Error)                                                                                                                       |
| DE 60, subfield 2 (Advice Detail Code)                      | •   | M   | M   | Provides additional details as follows:<br><br>0030 = Reject: Unable to verify PIN data<br><br>0031 = Reject: Unable to decrypt/encrypt PIN data |

## Reversal Advice/0420: PIN Messages

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

The issuer must be able to receive Reversal Advice/0420 messages with DE 60 values in addition to all other applicable data elements.

The Reversal Advice/0420 message identifies the corresponding Authorization Request Response/0110 message being reversed that contained PIN data.

### Data elements

| Data Element                                                | Org | Sys | Dst | Values/Comments                                                       |
|-------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------|
| DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) | •   | M   | M   | 400 = Banknet advice: APS error; unable to deliver response           |
| DE 60, subfield 2 (Advice Detail Code)                      | •   | M   | M   | 2000 = Reject: PIN data present in Authorization Request/0100 message |

## Alternate Processing

The issuer may choose to have the Dual Message Authorization System verify PIN data on their behalf. If the Dual Message Authorization System performs PIN verification, it will perform Stand-In Authorization and X-Code Processing using the issuer or Mastercard existing authorization-qualifying criteria, when applicable.

If the Dual Message Authorization System does not verify the PIN data and the issuer is unavailable or unable to process the Authorization Request/0100 message, the Dual Message Authorization System will respond to the acquirer with an Authorization Request Response/0110 message indicating the issuer could not process the Authorization Request/0100 message, except in situations where an issuer chooses to allow transactions with unverified PINs in Stand-In Authorization.

## Support for Both Acquiring and Issuing Processing

Customers that support both acquiring and issuing authorization processing may use the same PEK for all purchase transactions that contain a PIN. To accomplish this, the customer must use the same customer ID as follows:

- As the Group ID for establishing the static PEK or the KEK.
- In DE 32 (Acquiring Institution ID Code) or DE 33 (Forwarding Institution ID Code) in all acquired transactions.
- In DE 2 and DE 33 of the Network Management Request/0800: PEK exchange—on demand message.
- In DE 2 of the Network Management Request/0800: group sign-on message.

## Cleartext Use Prohibited

If there is a major problem with security equipment (for example, a faulty tamper-resistant security module [TRSM] or DES circuit board), the Dual Message Authorization System suspends all transaction processing with that customer. The customer must not send unencrypted (cleartext) PIN data.

For more information on PIN encipherment, refer to the [Security Rules and Procedures Guide](#).

## Emergency Static PEK or Emergency KEK Process

Following are the stages of the emergency static PEK or KEK process.

1. Mastercard considers the faulty customer as down.
2. Authorized Mastercard personnel randomly generate both parts of an emergency static PEK or KEK.
3. Mastercard personnel call the security or operations staff of the customer. Mastercard gives the emergency static PEK or KEK verbally to the customer.
4. Both the Mastercard personnel and the customer insert the new emergency static PEK or KEK in their TRSMs.
5. The Dual Message Authorization System initiates dynamic PEK exchange with a Network Management Request/0800: PEK exchange, using the new emergency static PEK or KEK.

Mastercard and the customer are to use the emergency static PEK or KEK process only as an interim measure to get the customer up as quickly as possible following a key exchange failure. Mastercard limits the use of the emergency static PEK or KEK in any one occurrence to six business days. The customer must notify their security officers responsible for key management immediately of the security failure situation and must conduct a secure key exchange at the earliest possible time.

## Previous PEKs

After exchanging new PEKs statically or dynamically, the Dual Message Authorization System and the customer are responsible for preserving the previous PEK for five minutes. They do this



in the event that the current PEK becomes inoperative. If the current PEK has a Sanity Check error during this five-minute interval, the Dual Message Authorization System or the customer should attempt to use the previous PEK. If the previous PEK is also inoperative, then refer to the steps previously discussed to determine action.

## PIN Verification Value on File Service

For issuers that use the Single Message System for PIN-processing services and elect to use the PIN Validation in Stand-In service, the Single Message System will perform the PIN validation service using the PVV for a card based on the PVV value provided in a file by the issuer.

Issuers that choose to register for this service must send a PVV/PIN Offset file to the Single Message System to update cardholder PINs and must use Single Message System-managed security keys. A Customer Implementation Services specialist will assist issuers with setting up a file or adding new files for testing and production.

Participating issuers must send a PVV/PIN Offset file to the Single Message System to provide and update cardholder PIN information. This file must be sent using the secure Global File Transfer (GFT) methods approved by Mastercard. The following bulk file IDs have been established for use when submitting PVV/PIN Offset files: RM29 (Production), RM31 (MTF).

Issuers also must ensure that they have exchanged the PIN verification keys (PVKs) with the Single Message System before using this service.

## PVV/PIN Offset File Format

The PVV/PIN Offset file has a fixed length of 64 characters. Issuers may use Bulk ID RM29 (Production), RM31 (MTF) or CONNECT:Direct to send the PVV/PIN Offset file to Mastercard. Refer to the following tables for information about the file header, detail, and trailer records.

### PVV/PIN Offset File Header Record

| Field Name          | Attribute    | Length | Comments and Values                                                                                                          |
|---------------------|--------------|--------|------------------------------------------------------------------------------------------------------------------------------|
| Record Type ID      | Alphanumeric | 3      | HDR—header record                                                                                                            |
| File Version Number | Numeric      | 3      | 100 (initial version)                                                                                                        |
| Customer ID         | Alphanumeric | 11     | The first six digits must contain the issuers ICA with leading zeros. The seventh digit and beyond contains trailing spaces. |
| Transmission code   | Alphanumeric | 10     | Member-assigned transmission code. May contain all spaces or zeros.                                                          |

| Field Name            | Attribute    | Length | Comments and Values        |
|-----------------------|--------------|--------|----------------------------|
| Transmission sequence | Numeric      | 4      | 0000–9999<br>Wraps at 9999 |
| Transmission date     | Numeric      | 6      | YYMMDD in UTC              |
| Transmission time     | Numeric      | 6      | hhmmss in UTC              |
| Input file type       | Alphanumeric | 1      | F = Full file replacement  |
| Filler                | Alphanumeric | 20     | Spaces                     |

### PVV/PIN Offset File Detail Record

| Field Name           | Attribute    | Length | Comments and Values                                                                                                                                                                                                        |
|----------------------|--------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record Type ID       | Alphanumeric | 3      | DTL = detail record                                                                                                                                                                                                        |
| Update code          | Alphanumeric | 1      | A = add/update                                                                                                                                                                                                             |
| PAN                  | Alphanumeric | 19     | With trailing spaces                                                                                                                                                                                                       |
| Card expiry date     | Numeric      | 4      | YYMM (must contain a valid year and month)                                                                                                                                                                                 |
| Card sequence number | Numeric      | 1      | Values can be in the range of 0–9<br><br>Must contain the Sequence Number associated with the PAN or 0. A value of 0 indicates that the card sequence number should not be used as criteria when matching to the PVV file. |
| PVV/PIN Offset       | Numeric      | 6      | Trailing spaces                                                                                                                                                                                                            |
| Filler               | Alphanumeric | 30     | Spaces                                                                                                                                                                                                                     |

### PVV/PIN Offset File Trailer Record

| Field Name               | Attribute    | Length | Comments and Values  |
|--------------------------|--------------|--------|----------------------|
| Record type ID           | Alphanumeric | 3      | TRL = trailer record |
| Number of detail records | Numeric      | 11     | Detail record count  |
| Filler                   | Alphanumeric | 50     | Spaces               |

## Alternate Processing

In instances where Stand-In Authorization is not accessible, X-Code Processing is initiated. X-Code Processing does not perform PIN verification. Therefore, authorization requests received with an unverified PIN will be declined.

## PIN Translation and Verification Process

The following describes the key exchanges during the PIN translation and verification process.

1. The cardholder enters the PIN at the point of interaction. The PIN is encrypted by the terminal's hardware under a PIN encryption key and is then sent to the acquirer.
2. The acquirer receives the encrypted PIN, which the acquirer then decrypts using the terminal PEK stored in a TRSM. The acquirer then creates the ANSI PIN block and encrypts it using the DES algorithm as follows:

First Block: PIN Data: The first digit of this block contains the control character 0, followed by a number representing the PIN length (maximum 12), and then the PIN itself. The acquirer then fills the remaining digits of the block on the right with hexadecimal F's to complete the 16-digit account number.

Second Block: PAN Data: The first four digits of the second block contain 0000, followed by the 12 right-most digits of the PAN, excluding the check digit. The acquirer then pads zeros to the left to complete the 16-digit data element.

In formatting an ANSI block, the acquirer performs an XOR function on the two 16-digit blocks.

After creating the PIN block, the acquirer sends it through the DES algorithm along with the 16-digit PEK that the acquirer and the Dual Message Authorization System share, producing the translated PIN block as follows. The acquirer may encrypt the PIN using:

- a. Single DES algorithm

Encrypt PIN information using a single DES key algorithm with single-length PEKs.

- b. Triple DES algorithm with double length PEKs

When a customer chooses to support triple DES, the customer must support both sending and receiving double-length (16-byte) PEKs.

- c. Triple DES algorithm with triple length PEKs

3. The acquirer sends the translated PIN block to the Dual Message Authorization System. The Dual Message Authorization System translates it from the acquirer PEK to the issuer PEK using the procedures described in step 2. The Dual Message Authorization System then sends the translated PIN block to the issuer for verification.  
DE 52 (PIN Data) is not in the Authorization Request/0100 message going to the issuer if the Dual Message Authorization System is performing PIN verification on the issuer's behalf.
4. The issuer then translates the PIN block and verifies the PIN using:
  - a. Single DES algorithm
  - b. Triple DES algorithm with double length PEKs

- c. Triple DES algorithm with triple length PEKs

## Detection of PEK Corruption Using Sanity Checks

Issuers and the Dual Message Authorization System perform Sanity Checks on PIN blocks as part of the PIN encryption/decryption process. This subsection describes Sanity Check errors that indicate PEK out-of-synchronization conditions that may occur during PIN decryption. It also explains ensuing steps that the Dual Message Authorization System and customer take.

All Tamper Resistant Security Modules (TRSMs) must be able to detect possible corruption of PEKs by performing Sanity Checks on the PIN block as part of decryption. When Sanity Check errors occur, customers should use the following procedures to resolve the problem.

## Dual Message Authorization System Sanity Check Error

The following describes the process when the Dual Message Authorization System TRSM discovers a PEK or KEK error.

1. The Dual Message Authorization System's Sanity Check reveals an out-of-synchronization condition with the PEK that it shares with the acquirer.
2. If five consecutive transactions fail the Sanity Check, and the Dual Message Authorization System has performed steps 1–3 with each of the previous five transactions from the acquirer, the Dual Message Authorization System takes the following action:
  - For a static PEK, Mastercard starts emergency procedures to create a new static PEK.
  - For a dynamic PEK, the Dual Message Authorization System sends the acquirer a Network Management Request/0800: PEK exchange message with a new PEK. If there are problems with the new PEK, the Dual Message Authorization System will initiate procedures to establish a new KEK.
3. For a dynamic PEK, the acquirer sends a Network Management Request Response/0810: PEK exchange to the Dual Message Authorization System acknowledging receipt of the Network Management Request/0800: PEK exchange message exchanging the PEK. If the message does not contain a DE 39 value of 00, completed successfully, the Dual Message Authorization System sends another Network Management Request/0800: PEK exchange message.
4. For a dynamic PEK, the Dual Message Authorization System sends a Network Management Advice/0820: PEK exchange message to the acquirer that notifies the customer that the new PEK is active and operational. Subsequent Authorization Request/0100 messages must use the new PEK to encrypt the PIN in DE 52.

## Issuer Sanity Check Error

The issuer TRSMs must be able to detect possible corruption of PEKs. The following describes the process when the issuer TRSM discovers a PEK error using a Sanity Check.

1. The issuer's Sanity Check reveals an out-of-synchronization condition with the PEK that it shares with the Dual Message Authorization System.
2. The issuer attempts to decrypt the PIN block using the prior PEK, if available.

3. If the Sanity Check fails using the prior PEK, the issuer sends an Authorization Request Response/0110 message.
4. If five consecutive transactions fail the Sanity Check, and the issuer has performed steps 1–3 with each of the five transactions, the issuer takes the following appropriate action:
  - For a static PEK, the issuer contacts Mastercard to start emergency procedures to create a new static PEK.
  - For a dynamic PEK, the issuer sends the Dual Message Authorization System a Network Management Request/0800: PEK exchange–On Demand message to create a new PEK. If there are problems with the new PEK, the issuer contacts Mastercard to start emergency procedures to establish a new KEK.
5. For a dynamic PEK, the Dual Message Authorization System acknowledges receipt of the request by sending a Network Management Request Response/0810:PEK exchange message.
6. For a dynamic PEK, the Dual Message Authorization System sends to the issuer a Network Management Request/0800: PEK exchange message with a new PEK.
7. For a dynamic PEK, the issuer sends a Network Management Request Response/0810: PEK exchange message to the Dual Message Authorization System acknowledging receipt of the Network Management Request/0800: PEK exchange message exchanging the PEK. If the message does not contain a DE 39 value of 00, completed successfully, the Dual Message Authorization System sends another Network Management Request/0800: PEK exchange message.
8. For a dynamic PEK, the issuer receives a Network Management Advice/0820: PEK exchange message from the Dual Message Authorization System that notifies the customer that the new PEK is active and operational. Subsequent Authorization Request/0100 messages must use the new PEK to encrypt the PIN in DE 52.

## Product Inquiry Service

The Product Inquiry Service allows an acquirer to send a product inquiry authorization request message to Mastercard.

**NOTE: Applies only to the U.S. region.**

As part of the authorization response to an acquirer, Mastercard will provide an acquirer with the product code associated with the particular Mastercard card number. Additionally, because product codes for various card programs potentially fall under different interchange rate structures, Mastercard will also populate the authorization response message with the applicable account category code as defined by the Account Level Management Service.

The information received by the acquirer through a Product Inquiry Service request, along with the published Mastercard interchange rate schedule and rate criteria, can be used by an acquirer and merchant to determine the product and associated interchange rate that may be applied to a purchase transaction for that card.

## Acquirers

Acquirers that choose to support Product Inquiry Service transactions must send Authorization Request/0100 messages containing DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and DE 4 (Amount, Transaction) with a transaction amount of zero when submitting Product Inquiry Service transaction requests.

**NOTE: Product Inquiry Service messages that include address verification and/or CVC 2 validation requests will be deemed to be an attempt to mitigate fraud. As such, they will be designated as an Account Status Inquiry Service transaction and billed accordingly.**

## Issuers

Issuers will receive Authorization Request/0100 messages containing DE 61, subfield 7, value 8, and DE 4 with a transaction amount of zero. Issuers must respond to these transactions with value 00 (Approved or completed successfully), 85 (Not Declined), 05 (Do Not Honor) or other valid business decline responses in DE 39 (Response Code). Invalid business declines include values 03 (Invalid merchant), 12 (Invalid transaction), 13 (Invalid amount), 30 (Format error), 51 (Insufficient funds/over credit limit), 57 (Transaction not permitted to issuer/cardholder), and 58 (Transaction not permitted to acquirer/terminal).

## For More Information

For more information about the Product Inquiry Service, contact the Global Customer Service team.

# Authorization Request/0100: Product Inquiry Service

Following is the list of the data elements applicable to this message. All mandatory Authorization Request/0100 data elements apply.

## Data elements

| DE ID | Data Element Name                                                                                                | Org | Sys | Dst | Comments                                                                             |
|-------|------------------------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------|
| 4     | Amount, Transaction                                                                                              | M   | .   | M   | Transaction amount of zero, in the acquirer's currency, at the point of interaction. |
| 61    | Point-of-Service (POS) Data, subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) | M   | .   | M   |                                                                                      |

## Promotion code

The promotion code is an alphanumeric code that customers can include in the Authorization Request/0100 message.

The promotion code service is an optional service.

### About promotion codes

Issuers may decide to offer a promotion to merchants. This promotion may require issuers to use different authorization criteria than they do with most transactions. The promotion code allows the issuer to recognize that this transaction uses the different criteria to determine authorization responses for transactions that occur at the participating merchants.

Mastercard may also specify other uses of the promotion code.

### How it works

1. The merchant includes the promotion code when it sends transaction data to the acquirer.
2. The acquirer sends the promotion code in DE 48 (Additional Data: Private Use) of the Authorization Request/0100 message to the issuer for the authorization decision.
3. The issuer can use the promotion code to identify transactions that occur at a merchant that participates in that issuer's promotion program.

For more information on data elements, refer to [Data element definitions and details](#).

### To participate

To participate in the Promotion Code service, issuers must notify Mastercard to participate in this service. To participate, contact the Global Customer Service team.

## Proximity Payments

The Mastercard Proximity Payments solution, which includes Mastercard contactless magnetic stripe and Mastercard contactless M/Chip, is part of the global Proximity Payments Program and is designed to enrich the traditional card with a contactless interface.

The contactless interface provides cardholder and merchant benefits that are particularly relevant in environments such as:

- Unattended point-of-service (POS) devices (for example, gas pumps and vending machines)
- High-traffic environments (for example, quick service and drive-through restaurants)

Proximity payments do not require cardholders holding a contactless Mastercard chip card to swipe or insert the card into a card reader or terminal. Instead, cardholders place the contactless card in proximity of a specially equipped merchant terminal to make a payment.

## Authorization Request/0100: Proximity Payments

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                                                        | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | M   | •   | M   | Contains one of the following values: <ul style="list-style-type: none"> <li>07 = Merchant terminal PAN auto-entry via contactless M/Chip</li> <li>91 = PAN auto-entry via contactless magnetic stripe—the full track data has been read from the data on the card and transmitted within the Authorization Request/0100 message in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.</li> </ul>                                                                                                                                                                      |
| DE 61 (Point-of-Service [POS] Data), subfield 11 (POS Card Data Terminal Input)     | M   | •   | M   | Contains one of the following values: <ul style="list-style-type: none"> <li>3 = Terminal supports contactless EMV input. The terminal may also support one or more other card input types, including contactless magstripe input, EMV contact chip input, magnetic stripe input and key entry input.</li> <li>4 = Contactless Mag Stripe (Proximity Chip) Terminal supports contactless magstripe input but not contactless EMV input. The terminal also may support one or more other card input types, including EMV contact chip input, magnetic stripe input, and key entry input.</li> </ul> |



## Purchase of Goods or Services with Cash Back

Mastercard allows the use of Purchase of Goods or Services with Cash Back in Authorization/01xx and Reversal/04xx messages for Debit Mastercard and Maestro cards.

### Participation Mandate

All issuers of Debit Mastercard and Maestro cards are required to support the receipt of authorization and reversal requests for Purchase of Goods or Services with Cash Back transactions. The Purchase of Goods or Services with Cash Back service is automatically associated with all Debit Mastercard and Maestro account ranges.

Issuers will continue to approve or decline Purchase of Goods or Services with Cash Back transactions at their discretion.

### Terminal Support Indicator

The Dual Message Authorization System allows acquirers to indicate whether the merchant terminal supports receipt of the purchase amount only approval response code in an authorization message.

The Authorization Request/0100 message must contain DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 09 (Purchase with Cash Back) and DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 2 (Purchase Amount Only Terminal Support Indicator), value 1 (Merchant terminal supports receipt of purchase-only approval).

For more information, refer to the [Transaction Processing Rules](#).

## Authorization Request/0100: Purchase of Goods or Services with Cash Back

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

### Data elements

| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                         |
|-----------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) | M   | .   | M   | Must contain value 09 = Purchase of Goods or Services with Cash Back    |
| DE 4 (Amount, Transaction)                                            | M   | .   | M   | Must contain the transaction amount, inclusive of the amount cash back. |
| DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)  | M   | .   | M   |                                                                         |

| Data Element                                                                                                            | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 61 (POS Data, Extended Condition Codes), subfield 2 (Purchase Amount Only Terminal Support Indicator) | C   | .   | C   | 1 = Merchant terminal supports receipt of purchase only approvals                                                                                                                                                                                                                                                                    |
| DE 54 (Additional Amounts), subfield 1 (Account Type)                                                                   | M   | X   | M   | <p>The Dual Message Authorization System provides a second occurrence of DE 54 to the issuer if billing currency is different than transaction currency.</p> <p>Must contain one of the following values:</p> <p>00 = Default account (not specified or not applicable)</p> <p>10 = Savings Account</p> <p>20 = Checking Account</p> |
| DE 54, subfield 2 (Amount Type)                                                                                         | M   | X   | M   | 40 = Amount Cash Back                                                                                                                                                                                                                                                                                                                |
| DE 54, subfield 3 (Currency Code)                                                                                       | M   | X   | M   | Must contain the valid three-digit numeric value present in DE 49 (Currency Code, Transaction)                                                                                                                                                                                                                                       |
| DE 54, subfield 4 (Debit or Credit Indicator)                                                                           | M   | X   | M   | D = (Debit amount) plus 12 digits right-justified with leading zeros                                                                                                                                                                                                                                                                 |

## Issuer Response Options

Issuers have several options in how they respond to a Purchase of Goods or Services with Cash Back request.

When the issuer receives an Authorization Request 0100/message containing DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 2 (Purchase Amount Only Terminal Support Indicator), value 1 (Merchant terminal supports receipt of purchase only approvals), the issuer has the option to:

- Approve the entire transaction amount
- Decline the entire transaction amount
- Respond with a purchase amount only approval

The issuer must approve the entire purchase amount. Partial approval (DE 39, value 10) of the purchase amount is not allowed.

The following table describes the information that the issuer must provide in the Authorization Request Response/0110 message when responding with a purchase-only approval and the information that the acquirer will receive for a purchase-only approval.

## Issuer Response Options

| The Issuer must provide...                                                                                                                                                                                                                                                                                                                                                                                                                                           | The Acquirer will receive...                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the approved amount (purchase amount) in DE 6 (Amount, Cardholder Billing) in the issuer's cardholder billing currency. This amount must be the purchase amount as calculated by subtracting the DE 54 (Additional Amounts) cash back amount and, if applicable, the DE 28 (Amount, Transaction Fee) from the DE 6 amount present in the Authorization Request/0100 message in the amount data element that corresponds to the issuer's cardholder billing currency. | the purchase-only approval amount in the acquirer's transaction currency in DE 4                                                                                                                                                                                                                                   |
| DE 38 (Authorization ID Response)                                                                                                                                                                                                                                                                                                                                                                                                                                    | DE 38                                                                                                                                                                                                                                                                                                              |
| DE 39 value 87 (Purchase only, no cash back allowed)                                                                                                                                                                                                                                                                                                                                                                                                                 | DE 39, value 87                                                                                                                                                                                                                                                                                                    |
| DE 51 (Currency Code, Cardholder Billing) with the issuer's cardholder billing currency code                                                                                                                                                                                                                                                                                                                                                                         | an occurrence of the original amount of the transaction in DE 54 (Additional Amounts) in the acquirer's transaction currency. The original amount is identified by DE 54, subfield 2 (Amount Type), value 57 (Original Amount), and subfield 4 (Debit or Credit Indicator), value C plus 12-digit original amount. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | an occurrence of the original amount of the transaction in DE 54 in the issuer's cardholder billing currency. The original amount is identified by DE 54, subfield 2, value 57, and subfield 4, value C plus 12-digit original amount.                                                                             |

**NOTE: Issuers responding with DE 39, value 87 will not be required to echo DE 4 (Amount, Transaction) in the Authorization Request Response/0110. Likewise, if DE 5 (Amount, Settlement) was present in the Authorization Request/0100 message to the issuer, the issuer will not be required to echo DE 5 in the Authorization Request Response/0110 when responding with DE 39, value 87. The issuer will provide the purchase-only approval amount in DE 6 and the issuer currency code in DE 51.**

## Reversal Request/0400

In some cases, the cardholder or merchant may elect not to complete the transaction after receiving the purchase-only approval response from the issuer. Mastercard supports full reversal messages to allow the merchant to cancel the transaction.

In addition to all other applicable data elements for the Reversal Request/0400 message, acquirers should submit Reversal Request/0400 messages with the following data elements for a reversal of a purchase-only approval:

- Purchase-only approval amount in DE 4 that was present in the Authorization Request Response/0110 message to the acquirer, not the original amount present in DE 4 of the Authorization Request/0100 message from the acquirer
- DE 39, value 87

The Dual Message Authorization System will perform currency conversion if appropriate and will provide the following data elements in the Reversal Request/0400 message to the issuer:

- Purchase-only approval amount in DE 4 in the acquirer's transaction currency
- Purchase-only approval amount in DE 5 in U.S. dollars if the issuer has opted to receive this data element in the message
- Purchase-only approval amount in DE 6 in the issuer's cardholder billing currency
- DE 39, value 87

When processing a Reversal Request/0400 for a purchase-only approval (DE 39, value 87), the issuer should increase the cardholder's open-to-buy.

## Reversal Advice/0420

If the Dual Message Authorization System generates a Reversal Advice/0420 message after the issuer has responded to the Authorization Request Response/0110 message with DE 39, value 87, the Dual Message Authorization System will provide the following data elements.

- Purchase-only approval amount in the acquirer's transaction currency in DE 4
- Purchase-only approval amount in DE 5 in U.S. dollars, if the issuer has opted to receive this data element in the message
- Purchase-only approval amount in DE 6 in the issuer's cardholder billing currency
- DE 39, value 87
- Original amount in DE 54 in the issuer's cardholder billing currency and acquirer's transaction currency

When processing a Reversal Advice/0420 for a purchase-only approval (DE 39, value 87), the issuer should increase the cardholder's open-to-buy.

## Authorization Advice/0120

In addition to the standard data elements that are part of the system-generated Authorization Advice/0120 message, these messages should include the following data elements for purchase-only approvals.

- Purchase-only approval amount in DE 4 in the acquirer's transaction currency
- Purchase-only approval amount in DE 5 in the settlement currency (US dollars)
- Purchase only-approval amount in DE 6 in the issuer's cardholder billing currency
- DE 39, value 87 as provided by the issuer or Stand-In in the Authorization Request Response/0110
- Original amount in DE 54 in the issuer's cardholder billing currency and acquirer's transaction currency

## Authorization Advice/0120: acquirer-generated

DE 39, value 87 is not a valid value for Authorization Advice/0120: acquirer-generated messages. If an Authorization Advice/0120: acquirer-generated message contains DE 39, value

87, the Dual Message Authorization System will generate an Authorization Advice Response/0130 message where DE 39 contains value 30 and DE 44 contains value 039.

## Alternate Processing

The Mastercard Network will invoke Stand-In Authorization or X-code Processing in accordance with issuer parameters and Mastercard rules and may subsequently approve the transaction for the following situations:

- If a format error is detected on incoming Authorization Request Response/0110 messages for all brands, regardless of the response code provided by the issuer in DE 39 or response code is invalid.
- If the issuer responds to an Authorization Request/0100 message with an Authorization Request Response/0110 message with one of the following values in DE 39 (Response Code) that indicates a system error, the transaction was unable to route, or the issuer is unavailable:
  - DE 39, value of 91 (Destination processor [customer or *INF*] not available)
  - DE 39, value of 92 (Authorization system or issuer system inoperative)
  - DE 39, value of 96 (System error)
- The Authorization Request/0100 message to the issuer times out or the issuer is marked down.

Issuers have the option to choose whether Purchase with Cash Back (PWCB) transactions are eligible for processing by Stand-In Authorization. An issuer may choose to allow PWCB for only PIN-based transactions, only Signature-based transactions, or allow both PIN and signature-based transactions be processed by Stand-In Authorization.

Mastercard provides issuers with parameters to define whether or not a Purchase With Cash Back transaction should be forwarded to Stand-In Authorization for processing. For a transaction to qualify for the PIN-based category, PIN Data will have to be present in the Authorization Request/0100 message. If PIN Data was not present in the incoming message, the transactions will be categorized as a signature-based transaction.

If the issuer chooses to have PIN, Signature or both PIN and Signature Purchase With Cash Back transactions excluded from Stand-In Authorization, the Dual Message Authorization System will provide the acquirer with an Authorization Request Response/0110 message with a DE 39 Response of 91 if the issuer is not able to respond to the Authorization Request/0100 message.

If the issuer chooses to process PIN, Signature or both PIN and Signature Purchase with Cash Back transactions in Stand-In, Stand-In Authorization may provide the purchase-only amount response DE 39, value 87 when the Authorization Request/0100 message contains DE 3, subfield 1, value 09, and DE 48, subelement 61, subfield 2 contains value 1 and the cash back transaction limit or the cash accumulation limits have been exceeded but the purchase limits have not.

Purchase with Cash Back transactions will not be processed with X-Code Processing.

For more information, refer to the [Stand-In Authorization Reference Guide](#).

## Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following edits on Purchase of Goods or Services with Cash Back transactions.

### Authorization Request/0100

#### Associated edits

| WHEN...                                                                                                                                      | THEN the Dual Message Authorization System...                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1 with the value 09 and DE 54 is not present                                                                                  | DE 39 (Response Code) = 30<br>DE 44 (Additional Response Data) = 003                                 |
| DE 3, subfield 1 with the value 09 and DE 54, subfield 1 is not a valid two-digit numeric value                                              | DE 39 = 30<br>DE 44 = 054                                                                            |
| DE 3, subfield 1 with the value 09 and DE 54, subfield 2 is not 40                                                                           | DE 39 = 30<br>DE 44 = 003                                                                            |
| DE 3, subfield 1 with the value 09 and DE 54, subfield 3 is not the same value as in DE 49                                                   | DE 39 = 30<br>DE 44 = 054                                                                            |
| DE 3, subfield 1 with the value 09 and DE 54, subfield 4 is not D followed by 12 numeric digits                                              | DE 39 = 30<br>DE 44 = 054                                                                            |
| DE 3, subfield 1 with the value 09 and the cash back amount in DE 54 is greater than the amount in DE 4 (Amount, Transaction)                | DE 39 = 30<br>DE 44 = 054                                                                            |
| DE 3, subfield 1 with the value 09 and the Primary Account Number (PAN) is not within an account range that supports Purchase With Cash Back | DE 39 = 57                                                                                           |
| DE 48, subelement 61, subfield 2 contains a value other than 0 or 1                                                                          | Generates an Authorization Request Response/0110 message containing:<br>DE 39 = 30<br>DE 44 = 048061 |
| DE 54 is present in the Authorization Request/0100 message where DE 54, subfield 2 contains value 57 (Original Amount)                       | Generates an Authorization Request Response/0110 message containing:<br>DE 39 = 30<br>DE 44 = 054    |

| WHEN...                                                            | THEN the Dual Message Authorization System... |
|--------------------------------------------------------------------|-----------------------------------------------|
| DE 54, subfield 2 with the value 40 and DE 3, subfield 1 is not 09 | DE 39 = 30<br>DE 44 = 003                     |

If the Authorization Request/0100 message passes all Dual Message Authorization System edits, the Dual Message Authorization System will forward the Authorization Request/0100 message to the issuer.

**NOTE: Acquirers should send only one occurrence of DE 54, subfields 1–4 in Authorization Request/0100 and Reversal Request/0400 messages.**

### Authorization Request Response/0110

The Dual Message Authorization System will perform the following edits on the Authorization Request Response/0110 message when the issuer has provided DE 39 with value 87.

The Dual Message Authorization System will provide two additional occurrences of DE 54, subfields 1–4 in the Authorization Request/0100 message to the issuer; one in the acquirer's transaction currency and one in the issuer's cardholder billing currency. This additional occurrence will be appended to the end of DE 54 before sending to the issuer.

If the issuer provides DE 54, subfield 2 with the value 40 in the Authorization Request Response/0110 message, Mastercard will not forward the cash back amount to the acquirer in the Authorization Request Response/0110 message.

### Associated edits

| WHEN...                                                                                                                                                                                     | THEN the Dual Message Authorization System...                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 61 is not present or DE 48, subelement 61, subfield 2 does not contain value 1 or DE 3, subfield 1 is not 09 in the Authorization Request/0100 message sent to the issuer | Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 039 |
| DE 38 is not present                                                                                                                                                                        | Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 038 |
| DE 6 is not present                                                                                                                                                                         | Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 006 |

| WHEN...                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| The amount in DE 6 of the Authorization Request Response/0110 message does not equal the purchase amount based on the original amounts present in the Authorization Request/0100 message. | Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 039 |
| The amount in DE 6 of the Authorization Request Response/0110 message is greater than the original amount in the Authorization Request/0100 message.                                      | Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 006 |
| DE 51 is not present or is not the issuer's correct cardholder billing currency code                                                                                                      | Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where:<br>DE 39 = 30<br>DE 44 = 051 |

## Authorization Advice/0120: acquirer-generated

### Associated edits

| WHEN...                 | THEN the Dual Message Authorization System...                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| DE 39 contains value 87 | Creates an Authorization Advice Response/0130—System-generated message containing:<br>DE 39 = 30<br>DE 44 = 039 |

## Reversal Request/0400

### Associated edits

| WHEN...                                                                                       | THEN the Dual Message Authorization System...                                                     |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| DE 54 is not formatted correctly in relation to alphanumeric specifications for the subfields | Returns the acquirer a Reversal Request Response/0410 message where:<br>DE 39 = 30<br>DE 44 = 054 |



| WHEN...                                                                                                                | THEN the Dual Message Authorization System...                                                             |
|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1, contains value 09 and DE 54, subfield 2, value 40 is not present and DE 39 does not contain value 87 | Returns the acquirer a Reversal Request Response/0410 message where:<br><br>DE 39 = 30<br><br>DE 44 = 003 |
| DE 3, subfield 1 does not contain value 09 and DE 54, subfield 2, value 40 is present                                  | Returns the acquirer a Reversal Request Response/0410 message where:<br><br>DE 39 = 30<br><br>DE 44 = 003 |

## Intracountry cash back transactions

For country-specific rule information, refer to the [Transaction Processing Rules](#) guide.

## Real-time Substantiation

The Real-time Substantiation (formerly referred to as Auto Substantiation) service supports substantiation at the point-of-sale (POS) for qualified expenses incurred on Flexible Spending Account (FSA), Healthcare Reimbursement Arrangement (HRA), or Health Savings Accounts (HSA) cards when used at a merchant with a qualifying Inventory Information Approval System (IIAS).

**NOTE: Applies only to the U.S. region.**

## Background

Internal Revenue Service Notice 2006-69 requires flexible spending account plan administrators to block card transactions occurring at non-medical MCCs (for example, grocery stores, discount stores, wholesale clubs), unless the merchant has implemented an Inventory Information Approval System (IIAS).

If FSA/HRA/HSA plan administrators allowed a transaction to be approved that contained non-verified and non-eligible merchandise, they run the risk of having their entire plan disqualified by the IRS. All grocery stores, discount stores, and online pharmacies that accept FSA debit cards must reference an IIAS at the point of sale.

Internal Revenue Service Notice 2006-69 also requires strict recordkeeping of IIAS-processed transactions. Health plan sponsors and administrators must negotiate agreements with merchants for the merchant to maintain the transaction amount, date the expense was incurred, and nature of the expense, and make this information accessible upon request, or alternatively send the information to the employer, administrator, or other applicable entity.

Affected acquirers are responsible for determining which merchants are IIAS-compliant, as defined by SIGIS.

## Merchant validation for real-time substantiated transactions

The Dual Message Authorization System performs a real-time authorization check to ensure that an Authorization Request/0100 message is from an IIAS-compliant merchant.

The Dual Message Authorization System identifies whether a transaction submitted as real-time substantiated is coming from a SIGIS (Special Interest Group for IIAS Standards)-compliant merchant. The Internal Revenue Service (IRS) placed additional requirements on merchants that do not use a healthcare acceptor business code (MCC) to accept healthcare cards requiring transaction substantiation.

### IIAS-compliant merchants

Mastercard supports the issuance of Mastercard Assigned IDs for merchants.

To indicate an IIAS-compliant merchant, acquirers must provide DE 48 (Additional Data: Private Use), subelement 32 (Mastercard Assigned ID) in Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages when the transaction is indicated as real-time substantiated (DE 48, subelement 61 [POS Data Extended Condition Codes], position 3 contains value 1).

To obtain Mastercard Assigned IDs for IIAS merchant validation, acquirers should send an email message to [sigis\\_merchant\\_setup@Mastercard.com](mailto:sigis_merchant_setup@Mastercard.com). The request should specify if it is an addition or update of the Mastercard Assigned ID, and at a minimum include the acquirer name, telephone number, email address, acquirer ID, processor ID, merchant parent/owner name (if applicable), Mastercard Assigned ID (if the request is for an update to the existing Mastercard Assigned ID), and merchant contact information.

Acquirers with existing Mastercard Assigned IDs for their merchants should continue to use those values, but must notify Mastercard that those merchants are SIGIS compliant.

### Non-IIAS compliant merchants

To indicate to issuers participating in real-time substantiation that the transaction was submitted as IIAS, but from a non-IIAS compliant merchant, Mastercard will define DE 48, subelement 61 (POS Data, Extended Condition Codes), subfield 3 (Real-time Substantiation Indicator) as value 4 (Transaction was submitted as real-time substantiated but from a non-IIAS certified merchant).

If the issuer is not participating in real-time substantiation, the original value of 1 (Merchant terminal verified the purchased items against the IIAS) is changed to value 0 (Merchant terminal did not verify the purchased items against the IIAS).

### 90% rule Merchant exemption

If the merchant generated 90 percent of its total sales from FSA/HSA/HRA-certified healthcare products in the previous fiscal year, that merchant is exempt from IIAS rules.

To indicate that the merchant is exempt from the IIAS rule based on the 90% rule, the acquirer must provide DE 48, subelement 61 (POS Data Extended Condition Codes), subfield 3 (Real-time Substantiation Indicator), value 2 (Merchant claims exemption from using an IIAS based on the IRS 90 percent rule) in Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages.

## Merchant terminal verification

Mastercard defines DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 3 (Real-time Substantiation Indicator) for use in the Authorization Request/0100 message to allow merchants to indicate whether the merchant terminal verified the purchased items against an IIAS.

For accurate processing of IIAS real-time substantiated transactions, Mastercard recommends that issuers validate the acceptor business code (MCC) (DE 18) and Real-time Substantiation Indicator (DE 48, subelement 61, subfield 3).

When the merchant terminal has verified the purchased items against an IIAS, the acquirer should populate the Authorization Request/0100 message with DE 48, subelement 61 and the following subfield values:

- Subfield 1 (Partial Approval Terminal Support Indicator) and subfield 2 (Purchase Amount Only Terminal Support Indicator) must contain values of zero (0) or 1
- Subfield 3 must contain a value of 1
- Subfields 4 and 5 are reserved for future use and must contain values of zero (0)

Acquirers do not receive DE 48, subelement 61 in the Authorization Request Response/0110 message.

When an acquirer creates an Authorization Advice/0120 message to advise the issuer of an approved authorization performed by the acquirer, DE 48, subelement 61 should be present if it was present in the original Authorization Request/0100 message.

## Real-time substantiation amounts

Mastercard supports the processing of real-time substantiation transactions with an indicator to specify healthcare transactions with the healthcare amount type values 10 (Healthcare Eligibility Amount), 11 (Prescription Eligibility Amount), and 12 (Vision Rx Eligibility Amount) in DE 54 (Additional Amounts), subfield 2 (Amount Type).

### Healthcare Eligibility Amount

When the acquirer provides the real-time substantiation indicator in DE 48 (Additional Data: Private Use), subelement 61 (POS Data Extended Condition Codes), subfield 3 (Real-time Substantiation Indicator), amount type 10 (Healthcare Eligibility Amount) allows the acquirer to indicate the portion of DE 4 (Amount, Transaction) that is eligible for real-time substantiation.

## Prescription Eligibility Amount

In addition to the Healthcare Eligibility Amount, Mastercard supports DE 54, subfield 2, value 11 (Prescription Eligibility Amount), which allows the acquirer to indicate the portion of the healthcare eligibility amount that includes the amount spent for prescriptions.

Subfield 2, value 11 must only be present when the acquirer provides subfield 2, value 10 in the Authorization Request/0100 or the Authorization Advice/0120 message. In addition, the amount in subfield 2, value 11 must be less than or equal to the amount of subfield 2, value 10.

DE 54 occurrences with subfield 2, values 10 and 11 are not returned to the acquirer in the Authorization Request Response/0110 message or the Authorization Advice Response/0130 message.

## Vision Rx Eligibility Amount (for U.S. region healthcare customers only)

U.S. region healthcare customers may use value 12 (Vision Rx Eligibility Amount) to indicate the portion of the transaction amount that is spent for vision prescriptions (Rx).

DE 54, subfield 2, value 12 can be provided by itself in the Authorization Request/0100 or the Authorization Advice/0120 messages or in combination with value 10 or values 10 and 11.

## Examples

Examples of Real-time Substantiation transaction processing are listed in table format.

Because Real-time Substantiation transactions can be used in combination with partial approvals, when the issuer receives an Authorization Request/0100 message containing DE 48, subelement 61, subfield 1, value 1, the issuer has the option to:

- Approve the entire transaction amount.
- Decline the entire transaction amount.
- Respond with a partial approval.
- For the issuer to respond with a partial approval, DE 48, subelement 61, subfield 1 must contain a value of 1; otherwise, the issuer must approve or decline the entire transaction amount.
- Partial approvals are not valid for Authorization Advice/0120 messages and will be rejected with a format error.

### Example 1: partial approval: entire Healthcare Eligibility Amount

This example illustrates that the issuer approved the entire Healthcare Eligibility Amount, which in this case is a partial approval of the total transaction amount. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 60 or remove some items from the purchase.

| Authorization Request/0100                                                                                                                                                                                                           |  | Authorization Request Response/<br>0110                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                      |  |                                                                                                                                                      |
|                                                                                                                                                                                                                                      |  | From Issuer                                                                                                                                          |
|                                                                                                                                                                                                                                      |  | To Acquirer                                                                                                                                          |
| <ul style="list-style-type: none"> <li>DE 48, subelement 61 = 10100 (indicates terminal can handle partial approvals and has verified against IIAS)</li> <li>DE 4 = USD 100</li> <li>DE 54, subfield 2, value 10 = USD 40</li> </ul> |  | <ul style="list-style-type: none"> <li>DE 6 = USD 40</li> <li>DE 39 = 10</li> </ul>                                                                  |
|                                                                                                                                                                                                                                      |  | <ul style="list-style-type: none"> <li>DE 4 = USD 40</li> <li>DE 54, subfield 2, value 57 (Original Amount) = USD 100</li> <li>DE 39 = 10</li> </ul> |

### Example 2: partial approval: partial Healthcare Eligibility amount

This example illustrates that the issuer approved only a portion of the Healthcare Eligibility Amount, which in this case is a partial approval of the total transaction amount. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 80 or remove some items from the purchase.

| Authorization Request/0100                                                                                                                           |  | Authorization Request Response/<br>0110                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                      |  |                                                                                                                                    |
|                                                                                                                                                      |  | From Issuer                                                                                                                        |
|                                                                                                                                                      |  | To Acquirer                                                                                                                        |
| <ul style="list-style-type: none"> <li>DE 48, subelement 61 = 10100</li> <li>DE 4 = USD 100</li> <li>DE 54, subfield 2, value 10 = USD 40</li> </ul> |  | <ul style="list-style-type: none"> <li>DE 6 = USD 20</li> <li>DE 39 = 10</li> </ul>                                                |
|                                                                                                                                                      |  | <ul style="list-style-type: none"> <li>DE 4 = USD 20</li> <li>DE 54, subfield 2, value 57 = USD 100</li> <li>DE 39 = 10</li> </ul> |

### Example 3: full approval: entire Healthcare Eligibility Amount

In this example, the merchant terminal is indicating that they do not support partial approvals; therefore, a full approval or decline is required. The issuer approves the full transaction; thereby approving the entire Healthcare Eligibility Amount.

| Authorization Request/0100 |  | Authorization Request Response/<br>0110 |
|----------------------------|--|-----------------------------------------|
|                            |  |                                         |
|                            |  | From Issuer                             |
|                            |  | To Acquirer                             |

| Authorization Request/0100                                                                                                                            | Authorization Request Response/<br>0110                                              |                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>DE 48, subelement 61 = 00100</li> <li>DE 4 = USD 100</li> <li>DE 54, subfield 2, value 10 = USD 100</li> </ul> | <ul style="list-style-type: none"> <li>DE 4 = USD 100</li> <li>DE 39 = 00</li> </ul> | <ul style="list-style-type: none"> <li>DE 4 = USD 100</li> <li>DE 39 = 00</li> </ul> <p>No DE 54, subfield 2, value 57 (Original Amount) is provided in this scenario because the issuer responded with DE 39 = 00</p> |

#### Example 4: partial approval: partial Healthcare Eligibility Amount, including prescriptions

This example illustrates that the issuer approved only a portion of the Healthcare and Prescription Eligibility amounts. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 80 or remove some items from the purchase.

| Authorization Request/0100                                                                                                                                                                         | Authorization Request Response/<br>0110                                             |                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                    | From Issuer                                                                         | To Acquirer                                                                                                                        |
| <ul style="list-style-type: none"> <li>DE 48, subelement 61 = 10100</li> <li>DE 4 = USD 100</li> <li>DE 54, subfield 2, value 10 = USD 40</li> <li>DE 54, subfield 2, value 11 = USD 30</li> </ul> | <ul style="list-style-type: none"> <li>DE 6 = USD 20</li> <li>DE 39 = 10</li> </ul> | <ul style="list-style-type: none"> <li>DE 4 = USD 20</li> <li>DE 54, subfield 2, value 57 = USD 100</li> <li>DE 39 = 10</li> </ul> |

#### Example 5: full approval: entire Healthcare Eligibility Amount, including prescriptions

In this example, the merchant terminal is indicating that they do not support partial approvals; therefore, a full approval or decline is required. The issuer approves the full transaction; thereby approving the entire Healthcare Eligibility Amount, including the Prescription Eligibility Amount.

| Authorization Request/0100 | Authorization Request Response/<br>0110 |             |
|----------------------------|-----------------------------------------|-------------|
|                            | From Issuer                             | To Acquirer |

| Authorization Request/0100                                                                                                                                                                          | Authorization Request Response/0110                                                  |                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>DE 48, subelement 61 = 00100</li> <li>DE 4 = USD 100</li> <li>DE 54, subfield 2, value 10 = USD 100</li> <li>DE 54, subfield 2, value 11 = USD 60</li> </ul> | <ul style="list-style-type: none"> <li>DE 4 = USD 100</li> <li>DE 39 = 00</li> </ul> | <ul style="list-style-type: none"> <li>DE 4 = USD 100</li> <li>DE 39 = 00</li> </ul> <p>No DE 54, subfield 2, value 57 (Original Amount) is provided in this scenario since the issuer responded with DE 39=00</p> |

## Authorization reports

The Authorization Parameter Summary Report (SI737010-AA) includes a Real-time Substantiation participant parameter in the [Global Parameters](#) section.

For a sample of this report and field descriptions, refer to [Reports and files](#).

## Recurring payments

Recurring payments are applicable to the following systems and products.

- Dual Message Authorization System: acquirers
- Dual Message Authorization System: issuers
- Product: Credit, Prepaid and Debit Mastercard

Mastercard supports recurring payment functionality and has defined additional message elements to allow more recurring payment information to be communicated between the acquirer and the issuer.

### Definition

Recurring payments are payments by an issuer to an acquirer on behalf of a cardholder who authorizes a merchant to bill the cardholder's account on a continued, periodic basis (such as monthly, quarterly, or annually) without a specified end date. The amount of each payment may be the same or may fluctuate.

Typical recurring payment merchants include, but are not limited to:

- Newspaper and magazine subscriptions
- Insurance Companies
- Utilities
- Internet Service Providers
- Club memberships
- Charities

For more information, refer to [Transaction Processing Rules](#).

## Indicating a recurring payment

Acquirers can indicate to issuers that a transaction is a recurring payment by including in DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence).

1. For the first transaction of a recurring payment series, which occurs when the cardholder authorizes the merchant to store a payment credential on file and use the payment credential for future recurring payments, a value of one of the following:
  - 0 (Cardholder present)
  - 4 (Recurring transactions) if the first transaction occurs in a card-not-present environment, such as on a merchant website
2. For any subsequent recurring payment transaction, a value of 4 (Recurring transactions)

For Intra-EEA Transactions and Intracountry transactions in the EEA, the initial authorization in a recurring payment arrangement must be identified as a recurring payment using the appropriate values in the fields specified by the registered switch of the customer's choice.

**NOTE: All recurring payment transactions which follow the first transaction where the cardholder authorizes the merchant to store their credential must be identified as credential-on-file. Such transactions must follow all credential-on-file requirements.**

For more information about determining the status of the cardholder account, refer to the [Purchase Account Status Inquiry/Recurring Payment Account Status Inquiry](#) section.

## Maestro Recurring Payments Program

The Maestro® Recurring Payments Program allows enrolled merchants to begin accepting Maestro cards for recurring payment electronic commerce (e-commerce) transactions. The initial transaction of a recurring payment arrangement may be completed face-to-face, with the subsequent transactions being submitted as standard non-face-to-face recurring payment transactions.

Maestro recurring payment e-commerce transactions are identified in Authorization Request/0100, Authorization Advice/0120: Acquirer-generated, and Reversal Request/0400: acquirer-generated messages by DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator), value 3 (UCAF data collection is supported by the merchant, and UCAF (Mastercard Assigned Static AAV). In addition, DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 4 (Recurring transactions) identifies Maestro e-commerce recurring payment transactions.

If the first transaction of a recurring payment arrangement is completed face-to-face, it must be identified as a face-to-face transaction and must not contain the recurring payment indicator in DE 61, subfield 4.

Issuers must not validate DE 48, subelement 43 (Universal Cardholder Authentication Field™), as that validation is performed during authorization processing.



DE 48, subelement 32 (Mastercard Assigned ID) must be present, and the Dual Message Authorization System edits DE 48, subelement 32 and subelement 43 (Static AAV) for valid values.

### Alternate processing

The Mastercard® Identity Check™ AAV Verification Service and Mastercard® Identity Check™ AAV Verification in Stand-In Authorization service will not be performed on Maestro e-commerce transactions that are processed under the Maestro Recurring Payments Program for e-commerce transactions.

## Refund transactions

A refund transaction allows a cardholder to return goods or services previously purchased from a merchant. The Dual Message Authorization System accepts refund transactions for Mastercard® and Maestro® transactions, in addition to Private Label transactions.

An acquirer sends an Authorization Request/0100 message with DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) containing value 20xx00 for the refund (where xx indicates the type of account to be credited). The issuer approves or declines the refund transaction using an Authorization Request Response/0110 message.

The refund Authorization Request/0100 message can be reversed using a Reversal Request/0400 message, if the merchant or cardholder decides not to complete the refund.

All refund transactions are routed to the same issuer host as for point-of-service (POS) transactions.

Refund transactions are not processed by the Stand-In Authorization or X-Code alternate processing.

- If the issuer is not available to respond to a refund Authorization Request/0100 message, the Dual Message Authorization System sends an Authorization Request Response/0110 message to the acquirer with DE 39 (Response Code), value 91 (Authorization System or Issuer System Inoperative).
- If the issuer is not available to respond to a refund Reversal Request/0400 message, the Dual Message Authorization System sends a Reversal Request Response/0410 message to the acquirer with DE 39, value 00 (Approved or completed successfully). The Dual Message Authorization System also sends a Reversal Advice/0420 message to the issuer as notification of the reversal that was responded to on their behalf.

## Reversal Processing

Mastercard supports the reversal of a full transaction amount or a partial transaction amount using reversal processing.

### Guiding principles for Authorization Reversal Processing

Mastercard provides the following update to its best practices for the management of authorization reversals. These best practices are intended to guide dual message acquirers, issuers, and processors in the usage of these transactions.

#### Background

Merchant-initiated authorization reversal messages are intended to assist issuers in managing "open-to-buy." This processing function is especially critical to debit and prepaid card issuers. Merchants and acquirers submit authorization reversal messages when:

- The final transaction amount was less than the authorized amount
- The cardholder cancels or chooses not to complete all or part of the transaction
- All or a portion of the goods or services could not be provided (for example, out-of-stock items)
- The authorization was submitted in error (for example, duplicate request or technical failures)

#### Guiding principles

The following is provided as guidance for reversal processing. Note that unless otherwise specified, these are best practices and are not mandatory:

- Authorization reversal messages are intended to fully or partially release a hold of funds prior to clearing the transaction.
- Purchase returns (or refunds) may be submitted by the merchant for authorization. If the merchant or cardholder wishes to cancel the transaction, the merchant should initiate a full reversal.
- Mastercard recommends that merchants submit reversals as soon as an adjustment to the original authorization amount is known. Only approved authorization requests or transactions that have timed out are required to be reversed.
  - Merchants and acquirers must submit a full or partial reversal (as applicable) within seven calendar days of an original undefined authorization or final authorization request and within 30 calendar days of an original preauthorization request.
  - Merchants and acquirers must submit a full or partial reversal within 24 hours of transaction cancellation or of the transaction completing for an amount different from the authorized amount.

**NOTE:** Refer to the [Incremental Preauthorization](#) section in this chapter of this manual for more information about revised Standards for authorizations and preauthorizations.

- Reversals must contain the mandatory matching data elements from the original authorization for an issuer to release the appropriate funds, subject to the issuer's risk or fraud control policies.

## Specific Scenarios

Specific Scenarios are described as follows:

- If an incremental authorization has been submitted, a reversal should reference original transaction data, including the original Trace ID, and DE 4 (Amount, Transaction) should be the sum of the original authorized amount plus any incremental amounts.
- Issuers will release any hold of funds once a clearing presentment has been matched (using the Trace ID in addition to other data elements) to the original authorization. Accordingly, a merchant is not required to submit a partial reversal if the lower amount is processed by Mastercard clearing within 24 hours of finalization of the transaction—assuming multi-clearing processing is not utilized.
- Partial reversals must be used when multi-clearing processing is utilized, since issuers will maintain a hold of funds for subsequent presentments.

## Reversal Message Data Element Details

As follows are details about reversal messaging data elements:

- Mandatory Reversal Request/0400 data elements used for issuer matching, which must contain the same value from the original Authorization 0100/0110 message:
  - DE 2 (Primary Account Number [PAN])
  - DE 3 (Processing Code)
  - DE 4 (Amount, Transaction)
  - DE 38 (Authorization ID Response)
  - DE 48, subelement 63 (Trace ID) (from original DE 15 [Date, Settlement] and DE 63 [Network Data])
  - DE 49 (Currency Code, Transaction)
  - DE 90 (Original Data Elements) (from original DE 7 [Transmission Date and Time], DE 11 [System Trace Audit Number (STAN)], and DE 32 [Acquiring Institution ID Code] and DE 33 [Forwarding Institution ID Code])

Reversals submitted for technical failures (DE 39 reversal reason code 06—Error) will not contain DE 38 (Authorization ID) or DE 48, subelement 63 (Trace ID), because no Authorization Request Response/0110 message was received. If Trace ID data from the original authorization is not available due to technical failures, DE 48, subelement 63 in the reversal must contain zeros.

DE 90, subfield 1 must contain the Message Type Identifier (MTI) of the original authorization message. The remaining subfields may contain valid matching values from the original authorization or may be zero-filled if they are not available.

- Mandatory Reversal Request/0400 message data elements that may use default values consistent with the merchant initiating the reversal, unless supplied directly from the merchant point-of-interaction (POI) (for example, merchants with multiple acceptor business codes [MCCs]):
  - DE 18 (Merchant Type)
  - DE 22 (Point-of-Service [POS] Entry Mode)
  - DE 32 (Acquiring Institution ID Code) and DE 33 (Forwarding Institution ID Code) by using default values consistent with the merchant, DE 32 and DE 33 will be the same values as in the original 0100 message
  - DE 39 (Response Code)
  - DE 41 (Acceptor Terminal ID)
  - DE 42 (Acceptor ID)
  - DE 43 (Acceptor Name and Location)
  - DE 61 (Point-of-Service [POS] Data)

Mastercard recommends that DE 39 (Response Code) be populated with one of the following Reversal Request/0400 reason codes:

- 06 is Error
- 17 is Customer Cancellation
- 32 is Partial Reversal
- 34 is Suspect Fraud
- 68 is Response Received Late
- Minimum list of original Authorization 0100/0110 data elements to be saved for submitting a Reversal Request/0400 message:
  - DE 2 (Primary Account Number [PAN])
  - DE 3 (Processing Code)
  - DE 4 (Amount, Transaction)
  - DE 7 (Transmission Date and Time)
  - DE 11 (System Trace Audit Number [STAN])
  - DE 15 (Date, Settlement)
  - DE 38 (Authorization ID Response)
  - DE 49 (Currency Code, Transaction)
  - DE 63 (Network Data)

**NOTE:**

**DE 15, DE 38, and DE 63 will only be available if an approved authorization response was received.**

**If DE 7 and DE 11 from the original authorization cannot be saved, DE 90, subelements 2 and 3 may be zero-filled within the Reversal Request/0400 message.**

- Optional/Conditional data elements that must contain the same value from the original Authorization 0100 if included in the Reversal Request/0400 message:
  - DE 12 (Time, Local Transaction)
  - DE 13 (Date, Local Transaction)

- DE 14 (Date, Expiration)
- DE 23 (Card Sequence Number)
- DE 28 (Amount, Transaction Fee)
- DE 37 (Retrieval Reference Number)
- DE 54 (Additional Amounts)

## Partial Reversals

The Dual Message Authorization System supports partial reversal functionality using the Reversal/04xx messages.

Partial reversal functionality is useful in adjusting a portion of the original authorization amount in the following scenarios:

- A merchant ships only a portion of merchandise.
- A cardholder returns a rental vehicle earlier than originally reserved.
- A cardholder checks out of a hotel earlier than originally reserved.
- A cardholder cancels a portion of the transaction.

Mastercard supports Reversal Request/0400 and Reversal Advice/0420 messages where DE 95 (Replacement Amounts), subfield 1 (Actual Amount, Transaction) contains a value other than all zeros. DE 95, subfield 1, in the originating reversal, must be a lesser amount than the amount in DE 4 (Amount, Transaction).

Refer to the [Chargeback Guide](#) for rules related to partial reversal functionality.

## Reversals of Balance Inquiry Transactions

Mastercard supports Reversal Request/0400 and Reversal Advice/0420 messages containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 30 (Balance Inquiry). In inquiry transactions, DE 4 must contain a value of all zeros unless an ATM transaction fee DE 28 (Amount, Transaction Fee) has been applied by an acquirer for an ATM transaction in a country where of an ATM transaction fee is allowed.

## Reversals of Purchase of Goods or Services with Cash Back Transactions

Mastercard provides DE 54 (Additional Amounts), subfield 2 (Amount Type), value 40 (Amount Cash Back) to issuers in the Reversal Request/0400 message if DE 54, subfield 2, value 40 was contained in the Reversal Request/0400 message from the acquirer. Acquirers provide DE 4 (Amount, Transaction) in the Reversal Request/0400 message for the full, original amount of the Authorization Request/0100 message. With Purchase of Goods and Services with Cash Back transactions, DE 4 contains the cash back amount, as defined in DE 54.

As with Authorization Request/0100 message processing, the Dual Message Authorization System edits Reversal Request/0400 messages when DE 54, subfield 2, value 40 is present. In this case, the Dual Message Authorization System checks to ensure that DE 3 (Processing

Code), subfield 1 (Cardholder Transaction Type Code) contains value 09 (Purchase with Cash Back).

If DE 54, subfield 2, value 40 is not present, the Dual Message Authorization System performs an additional check to ensure that the Reversal Request/0400 message contains DE 39 (Response Code), value 87 (Purchase only, no cash back allowed).

When an acquirer submits a full or partial reversal for a Purchase Amount Only approval where DE 39 (Response Code) of the Authorization Request Response/0110 message contains value 87, Mastercard recommends that the acquirer not send DE 54 in the reversal message. If the acquirer intends the Reversal Request/0400 message as a partial reversal, DE 95 (Replacement Amounts) will also be present, containing the adjusted amount of the original authorization.

DE 54, subfield 2, value 40 is the only instance of DE 54 that Mastercard will forward to issuers in a Reversal Request/0400 message. All other instances of DE 54 will be removed from the Reversal Request/0400 message before forwarding the message to the issuer.

DE 54, subfield 2, value 40 also may be included in the Reversal Advice/0420 message.

Alternate Processing

While no Stand-In Authorization processing tests will be applied to the Reversal Request/0400 message, the Dual Message Authorization System supports alternate processing of Reversal Request/0400 and Reversal Request Response/0410 messages when the issuer is unable or unavailable to respond to the Reversal Request/0400 message.

The Dual Message Authorization System will perform the following processing when an issuer is unable or is unavailable to respond to the Reversal Request/0400 message:

Associated edits

| WHEN...                                                                                                                                                                                                                                                                                   | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Dual Message Authorization System times out<br>or<br>The issuer is signed-out<br>or<br>the issuer responded with a Reversal Request Response/0410 message with an error<br>or<br>the Dual Message Authorization System cannot deliver the Reversal Request/0400 message to the issuer | sends the acquirer a Reversal Request Response/0410 message where:<br><br>DE 39 = 00 (Approved or Completed successfully)<br><br>sends a Reversal Advice/0420 message to SAF for immediate availability to the issuer where:<br><ul style="list-style-type: none"><li>DE 39 is The value from the acquirer's Reversal Request/0400 message</li><li>DE 60 contains one of the following values as applicable:<ul style="list-style-type: none"><li>402 = Issuer Time-out</li><li>403 = Issuer Sign-out</li><li>409 = Issuer Response Error</li><li>413 = Issuer Undelivered</li></ul></li></ul> |

| WHEN...                                                                                                         | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer responds with an error in the Reversal Request Response/0410 message                                 | <p>responds to the issuer with a Authorization Response Negative Acknowledgment/0190 message where:</p> <ul style="list-style-type: none"> <li>DE 39 = 30</li> <li>DE 44 = Data element in error</li> </ul> <p>provides a Reversal Advice/0420 message to the issuer from SAF for immediate availability to the issuer (as defined above)</p> |
| the Dual Message Authorization System cannot deliver the Reversal Request Response/0410 message to the acquirer | <p>will not send a Reversal Advice/0420 message to the issuer.</p> <p>in this case, acquirers have the responsibility to resend the reversal to the issuer.</p>                                                                                                                                                                               |

**NOTE: Customers in the Europe region that route to an alternate issuer host for alternate processing instead of Stand-In, will still receive an Authorization Advice/0120: acquirer-generated as described here. Alternate issuer host processing does not send Reversal Request/0400 or Authorization Advice/0120 messages to the alternate host.**

## Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on reversal transaction processing.

### Reversal Request/0400

The Dual Message Authorization System will perform the following edits on the Reversal Request/0400 message as it relates to DE 95 (Replacement Amounts) and partial reversals.

#### Associated edits

| WHEN...                                                                            | THEN the Dual Message Authorization System...                                                                          |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| DE 95 (Replacement Amounts) is equal to or greater than DE 4 (Amount, Transaction) | <p>sends the acquirer a Reversal Request Response/0410 message where:</p> <p>DE 39 = 30</p> <p>DE 44 = 095</p>         |
| DE 95 (Replacement Amounts) is equal to zero                                       | removes DE 95 from the Reversal Request/0400 message before providing the Reversal Request/0400 message to the issuer. |

The Dual Message Authorization System will perform the following edits on the Reversal Request/0400 message as it relates to DE 54 (Additional Amounts) and reversals of Purchase of Goods or Services with Cash Back transactions.

#### Associated edits

| WHEN...                                                                                                                | THEN the Dual Message Authorization System...                                                     |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| DE 54 is not formatted correctly in relation to alphanumeric specifications for the subfields                          | returns the acquirer a Reversal Request Response/0410 message where:<br>DE 39 = 30<br>DE 44 = 054 |
| DE 3, subfield 1, contains value 09 and DE 54, subfield 2, value 40 is not present and DE 39 does not contain value 87 | returns the acquirer a Reversal Request Response/0410 message where:<br>DE 39 = 30<br>DE 44 = 003 |
| DE 3, subfield 1 does not contain value 09 and DE 54, subfield 2, value 40 is present                                  | returns the acquirer a Reversal Request Response/0410 message where:<br>DE 39 = 30<br>DE 44 = 003 |

The Dual Message Authorization System will perform the following edits on the Reversal Request/0400 message as it relates to reversals of Balance Inquiry transactions.

#### Associated edits

| WHEN the Reversal Request/0400 message...                                                                                                                                                                 | THEN the Dual Message Authorization System...                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| contains a value of all zeros in DE 4 (Amount, Transaction) and DE 3, subfield 1, does not contain a value of 30 (Balance Inquiry)                                                                        | returns the acquirer a Reversal Request Response/0410 message where:<br>DE 39 = 30<br>DE 44 = 004 |
| contains a value of 30 (Balance Inquiry) in DE 3, subfield 1 then DE 4 (Amount, Transaction) must contain a value of all zeros unless the transaction contains DE 28 (Amount, Transaction Fee), otherwise | returns the acquirer a Reversal Request Response/0410 message where:<br>DE 39 = 30<br>DE 44 = 004 |



## Full Reversals

The Dual Message Authorization System supports full reversal functionality using the Reversal 04xx messages.

Full reversal functionality may be used for, but is not limited to, the following scenarios:

- Acquirer cannot deliver a response message 0110 or 0410 to the merchant
- Acquirer cannot match the response message 0110 or 0410 to the original request
- The authorization response message received contains errors
- The authorization response message was received too late

## Visa transaction processing

The Mastercard Dual Message Authorization System allows acquirers to process Visa transactions via the Mastercard Network. It supports both transactions that qualify for the Visa Custom Payment Service Request and transactions that do not.

## Issuer options

Issuers may choose to accept Visa activity using their connection to the Mastercard Network. This connection is called peer-to-peer. If the issuer is not peer-to-peer, Mastercard routes authorization requests to the Visa network.

## Custom Payment Service Request transactions

For Visa transactions that indicate Custom Payment Service Request, Mastercard routes the authorization request to the Visa network.

Mastercard-acquired Custom Payment Service Request transactions have specific requirements for the Authorization Request/0100 and the Authorization Request Response/0110 messages.

For the data requirements as they apply to the authorization message formats, refer to [Data element definitions and details](#).

## Indicators

Mastercard provides support of various Visa programs using specific indicators in Authorization Request/0100 messages.

### Authorization Characteristics Indicator

Mastercard customers that acquire Visa transactions can receive valid Visa response values for the Authorization Characteristics Indicator (ACI).

Mastercard forwards the ACI value provided by Visa in DE 48 (Additional Data: Private Use), subelement 90 (Visa Custom Payment Service Request) of Authorization Request Response/0110 messages.

**Mail/telephone or electronic commerce indicator**

Visa allows acquirers to participate in its interregional interchange reimbursement fee (IRF) program in cases where the merchant and acquirer met and complied with certificate serial number requirements even though the cardholder did not.

Mastercard acquirers forwarding Visa e-commerce transactions must have the following values in Authorization Request/0100 messages.

**Data elements**

| Data Element                                  | Value                                | Description                                                                                                                                                |
|-----------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22, subfield 1                             | 10                                   | Credential on File<br><br><b>NOTE: Mastercard mandates support of DE 22, subfield 1, value 10 for all issuers globally.</b>                                |
| DE 22, subfield 1                             | 81                                   | PAN/Token entry via electronic commerce with optional Identity Check AAV in the UCAF field or DSRP cryptogram in the Digital Payment Data field            |
| DE 22, subfield 1                             | 82                                   | PAN Auto Entry via Server (issuer, acquirer, or third party vendor system)                                                                                 |
| DE 61, subfield 10                            | 6                                    | Electronic Commerce                                                                                                                                        |
| DE 48, subelement 40, subfield 01             | Variable binary value 1-16 positions | To indicate that the merchant is certificate-compliant                                                                                                     |
| DE 48, subelement 40, subfield 02             | Not present                          | To indicate that the cardholder is not certificate-compliant                                                                                               |
| DE 48, subelement 42, subfield 01, position 2 | 1                                    | To indicate cardholder certificate not used. For details about Visa usage for subelement 42, refer to <a href="#">Data element definitions and details</a> |
| DE 48, subelement 43, 44, 45                  |                                      | For details about Visa usage for subelements 43, 44, and 45, refer to <a href="#">Data element definitions and details</a> .                               |

### **Market-specific Data Identifier (Visa-only)**

The Visa Market-Specific Data Identifier is a value used in Visa transactions to show the market-specific data identifier. Acquirers provide the appropriate Visa market-specific data identifier in DE 48 (Additional Data: Private Use), subelement 96 (Visa Market-Specific Data Identifier).

### **Prestigious Properties Indicator**

The Prestigious Properties Indicator is a value used in Visa transactions to specify participants in the Visa Prestigious Lodging program. Acquirers provide the appropriate Visa Prestigious properties indicator in DE 48 (Additional Data: Private Use), subelement 97 (Prestigious Properties Indicator).

### **Relationship Participant Indicator**

The Relationship Participant Indicator is a value used in Visa transactions to show that merchants have had long-term relationships with a cardholder from whom they regularly collect recurring payments.

Mastercard acquirers forwarding Visa transactions with a Relationship Participant Indicator must have value R (Merchant/Cardholder Relationship) or value I (Installment Payment) in DE 48 (Additional Data: Private Use), subelement 86 (Relationship Participant Indicator [Visa Only]) of Authorization Request/0100 messages.

### **Transponder Indicator**

Cardholders may use transponders, which use radio frequency signals, to exchange identification information with cardholder-activated terminals or other point-of-interaction (POI) devices to initiate a transaction. These non-face-to-face transactions occur when the card is not present (when the transponder is the device activating the transaction).

Mastercard customers acquiring Visa transactions with a transponder must have value 7 in DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level) in Authorization Request/0100 messages. This value indicates that a transponder was used to exchange authorization information.

### **U.S. Existing Debt Indicator**

When Mastercard acquirers in the U.S. region forward to Visa, through the Mastercard Network, transactions that pay an existing debt, they must indicate such in Authorization Request/0100 messages.

Acquirers must include value 9 in DE 48, subelement 85 (Visa U.S. Existing Debt Indicator) to indicate that the transaction is a payment for an existing debt.

### **Visa Payment Service Indicators**

Mastercard has modified and expanded the existing structure of DE 48 (Additional Data: Private Use), subelement 78 (Payment Service Indicators [Visa Only]) (formerly subelement 78 [Visa U.S. Deferred Billing Indicator]) to update the Data Representation from a-1 to ans-6.

In addition, Mastercard has introduced six new subfields, three of which are defined below, and an additional three for future Visa data use.

- DE 48, subelement 78, subfield 1 (Spend Qualified Indicator) in the Authorization Request/0110 message contains the Spend Qualified Indicator for Visa-branded, dual-message authorization transactions.
- DE 48, subelement 78, subfield 2 (Dynamic Currency Conversion Indicator) in the Authorization Request/0100 message contains the Dynamic Currency Conversion indicator for Visa-branded, dual-message authorization transactions.
- DE 48, subelement 78, subfield 3 (U.S. Deferred Billing Indicator) in the Authorization Request/0100 message contains the U.S. Deferred Billing indicator for Visa-branded, dual-message authorization transactions.
- DE 48, subelement 78, subfields 4-6 are reserved for future use.

The U.S. Deferred Billing Indicator is a value used by U.S. acquirers in Visa transactions to notify U.S. issuers that the transaction is being submitted to bill the cardholder for merchandise that was received within the past 90 days.

Mastercard acquirers forwarding Visa transactions with the U.S. Deferred Billing Indicator must have value D (U.S. Deferred Billing Indicator) in DE 48, subelement 78, subfield 3 in Authorization Request/0100 messages.

## Retail Key Entry program

Mastercard acquirers forwarding Visa Retail Key Entry transactions must have the following values in Authorization Request/0100 messages.

### Data elements

| Data Element                                                                        | Value         | Description                                                       |
|-------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------|
| DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) | 01            | Indicates that the card number was keyed into the POS device      |
| DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence)           | 0             | Indicates that the cardholder is present                          |
| DE 61, subfield 5 (POS Card Presence)                                               | 0             | Indicates that the card is present                                |
| DE 61, subfield 11 (POS Card Data Terminal Input Capability)                        | 2, 5, 7, or 8 | Indicates that the POS terminal can transmit magnetic stripe data |

**NOTE: For a transaction to be forwarded automatically to the Visa gateway, acquirers must identify transactions as Visa Custom Payment Services transactions. Meeting Retail Key Entry criteria will not force a transaction to the Visa gateway if the issuer is a Mastercard peer-to-peer participant.**

## Secure electronic commerce verification service

The Mastercard Dual Message Authorization System supports the Visa secure e-commerce verification service.

Mastercard will forward to the Visa gateway Authorization Request/0100 messages including secure e-commerce data that also contain a Visa account number in DE 2 (Primary Account Number [PAN]).

To participate, acquirers must provide all e-commerce data elements, including the data elements for the secure e-commerce verification, in Authorization Request/0100 messages.

Acquirers must use DE 48 (Additional Data: Private Use), subelement 43 (UCAF), and subelement 44 (Electronic Commerce Transaction Identifier) to send the secure e-commerce data in Authorization Request/0100 messages. If Visa responds with a Cardholder Authentication Verification Value (CAVV) Results Code, Mastercard forwards the code values to the acquirer in DE 48, subelement 45 (Electronic Commerce Transaction Response Code) of the Authorization Request Response/0110 message.

## Visa product ID

Acquirers that process Visa transactions through the Dual Message Authorization System receive DE 48 (Additional Data: Private Use), subelement 46 (Product ID) in Authorization Request Response/0110 and Reversal Request Response/0410 messages. The associated Visa field for DE 48, subelement 46 is field 62.23 (Product ID).

## Visa Commercial Card inquiry

Acquirers can inquire on Visa Commercial Cards using specific values in DE 48 (Additional Data: Private Use), subelement 94 (Visa Commercial Card inquiry request and response indicator).

## Visa Fleet Card

Mastercard supports processing of Visa Fleet Card transactions through the Dual Message Authorization System in DE 48, subelement 93 (Visa Fleet Card ID Request Data) indicator.

## Visa-assigned Merchant Verification Value

Mastercard supports the Visa mandate requiring that the Merchant Verification Value (MVV) be populated in all processing messages when an MVV has been assigned to the merchant by Visa.

Mastercard only includes the Visa-assigned MVV value on those transactions that are routed to the Visa network using the Mastercard Visa Gateway.

Acquirers of Visa-branded transactions will provide, when applicable, an alphanumeric Merchant Verification Value (MVV) in DE 48 (Additional Data: Private Use), subelement 36 (Visa Defined Data), subfield 1 in Authorization Request/0100 and Reversal Request/0400 messages. If it is provided in any other message types, the Dual Message Authorization System will remove it before sending the transaction to the Visa network.

Mastercard is not aware of which merchants are assigned an MVV by Visa, and therefore will not know which acquirers provide or not provide the MVV in a message. If present, the Mastercard Network forwards the value in the transaction to the Visa network.

The Dual Message Authorization System removes DE 48, subelement 36, subfield 1 from any transaction provided by an acquirer that is not a Visa-branded transaction.

The MVV is not provided back to the acquirer in any messages. If DE 48, subelement 36, subfield 1 is provided in any response messages, the Dual Message Authorization System will remove it before sending the message to the acquirer.

## Visa token processing

Mastercard has expanded the functionality of the Visa Gateway to support tokenization processing for partial shipments, recurring payments, and e-commerce mobile transactions for Visa-branded, dual-message authorization transactions submitted to the Mastercard Network.

These enhancements affect acquirers that use the Mastercard Network to submit Visa-branded dual-message authorization transactions.

### Tokenized Recurring Payment Messages

Acquirers must be prepared to identify tokenized recurring payment transactions in Authorization Request/0100 and Reversal Request/0400 messages.

- Existing value 4 (Standing order/recurring transactions) in DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence)
- Presence of DE 48 (Additional Data: Private Use), subelement 43 (Secure Electronic Commerce Verification Service) containing the partial shipment verbiage as follows: PARTIALbSHIPMENTbbbbbbbbbbbb or PARTIALSHIPMENT00000000000000 where b represents a space. This field contains 28 positions.
- Existing values 2 (Channel) in position 1 (Security Protocol), 1 (Cardholder certificate not used) in position 2 (Cardholder Authentication), and 0 (UCAF data collection is not supported by the merchant or an Identity Check merchant has chosen not to undertake Identity Check on this transaction) in position 3 (UCAF Collection Indicator) in DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator)
- Existing value 81 (PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 [UCAF] or DSRP cryptogram in DE 104, subelement 001 [Digital Payment Cryptogram]) in DE 22 (Point-of-Service Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)

### Tokenized partial shipment messages

Acquirers must be prepared to identify tokenized partial shipment transactions in Authorization Request/0100 and Reversal Request/0400 messages whose original transaction was token-based with a valid cryptogram.

- DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 5 (Electronic order [home PC, Internet, mobile phone, PDA])
- DE 48 (Additional Data: Private Use), subelement 43 containing the cryptographic data from the original transaction
- DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator):
  - Position 1 (Security Protocol), value 2 (Channel)
  - Position 2 (Cardholder Authentication), value 1 (Cardholder certificate not used)
  - Position 3 (UCAF Collection Indicator), value 2 (UCAF data collection is supported by the merchant, and UCAF data must be present [DE 48, subelement 43 must contain a fully authenticated AAV])
- DE 22 (Point-of-Service Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), values 10 (Credential on File) or value 81 (PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 [UCAF] or DSRP cryptogram in DE 104, subelement 001 [Digital Payment Cryptogram])

**NOTE: DE 48, subelement 43 is not required in Reversal Request/0400 messages for partial shipments.**

### Tokenized e-Commerce with mobile device messages

Acquirers must be prepared to receive the tokenized e-commerce with mobile device transaction identifier in Authorization Request Response/0110 and Reversal Request Response/0410 messages.

- Value B (Tokenized e-commerce with mobile device) in DE 48, subelement 90 (Custom Payment Service Request Response [Visa Only])
- Value Y (Transaction is processed through Visa Checkout) in DE 48, subelement 78 (Payment Service Indicators [Visa Only]), subfield 4 (Visa Checkout Indicator)

## Chapter 8 Reports and files

*The Reports and files section contains information about the reports and logs related to the Mastercard Dual Message Authorization System.*

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## Presentation of reports

Each authorization report is introduced with an overview that contains the following information.

- Title: name or title of the report
- Generated by: system generating the report
- Purpose: reason for providing the report
- Description: type of information included in the report
- Frequency: how often the report is generated
- Distribution methods: distribution options and associated bulk file or Mastercard File Express application identification numbers

## Report header information

Report header information is contained in all reports generated by the Dual Message Authorization System. This information is not described in the individual report descriptions.

Common report header details are:

- Report number appears in the upper left-hand corner of each report and identifies the report by number.
- Report header is in the upper center of each report and identifies the report by name.
- Page number appears in the upper right-hand corner of each report and identifies the page number within the report.
- Date appears in the upper right-hand corner of each report and identifies the date the report was generated. All times are Central Time (CT) dates.
- Time, when it appears, is in the upper right-hand corner of each report and identifies the time the report was generated. All are CT times.

## Authorization Summary Report (AB505010-AA)

An overview of the Authorization Summary Report (AB505010-AA) is listed in table format.

### Report overview

|              |                                                                           |
|--------------|---------------------------------------------------------------------------|
| Title        | Authorization Summary Report                                              |
| Generated by | Dual Message Authorization System                                         |
| Purpose      | To provide customers with a tool for analyzing authorization performance. |

|                      |                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description          | This report shows weekly authorization transaction processing volumes and percentages by BIN. It reflects transactions processed by the Dual Message Authorization System and billable items. |
| Frequency            | Weekly                                                                                                                                                                                        |
| Distribution Methods | Paper (Sent to the authorization contact established on the Principal Member Questionnaire); Mastercard Operational Reports, Mastercard File Express, complex-to-complex, bulk ID (T829)      |



## Authorization Summary Report (AB505010-AA), page 2

| REPORT AB505010-AA                         |                  | MASTERCARD INTERNATIONAL, INC.                               |                  |                      | PAGE 2                      |
|--------------------------------------------|------------------|--------------------------------------------------------------|------------------|----------------------|-----------------------------|
|                                            |                  | AUTHORIZATION SUMMARY                                        |                  |                      | DATE 10/20/24 TIME 12:58:08 |
|                                            |                  | FOR MCBS BILLING WEEK BEGINNING 10/12/24 AND ENDING 10/18/24 |                  |                      |                             |
|                                            |                  | ICA 019827 PREFIX 51121390100 THRU 51121390100               |                  |                      |                             |
| STAND-IN PROCESSED<br>ACTIVITY             | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME                                         | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME | WEEKLY<br>RANK              |
| ACCEPT                                     | 0                | 0                                                            | 2,391,187        | 34,606,671           | N/A                         |
| PERCENT OF TOTAL                           | 0.0              | 0.0                                                          | 52.3             | 34.4                 |                             |
| DECLINE                                    | 0                | 0                                                            | 2,164,285        | 65,583,792           | N/A                         |
| PERCENT OF TOTAL                           | 0.0              | 0.0                                                          | 47.3             | 65.2                 |                             |
| CALL-ME                                    | 0                | 0                                                            | 1,747            | 19,025               | N/A                         |
| PERCENT OF TOTAL                           | 0.0              | 0.0                                                          | 0.0              | 0.0                  |                             |
| PICK-UP                                    | 0                | 0                                                            | 16,545           | 348,489              | N/A                         |
| PERCENT OF TOTAL                           | 0.0              | 0.0                                                          | 0.4              | 0.3                  |                             |
| TOTAL STAND-IN PROCESSED                   | 0                | 0                                                            | 4,573,764        | 100,557,977          |                             |
| TIMED OUT                                  | 0                | 0                                                            | 1,045,513        | 39,609,904           | N/A                         |
| PERCENT OF TOTAL                           | 0.0              | 0.0                                                          | 22.9             | 39.4                 |                             |
| MEMBER DOWN                                | 0                | 0                                                            | 3,099,840        | 50,692,039           | N/A                         |
| PERCENT OF TOTAL                           | 0.0              | 0.0                                                          | 67.8             | 50.4                 |                             |
| SIGNED OUT                                 | 0                | 0                                                            | 430,733          | 10,316,814           | N/A                         |
| PERCENT OF TOTAL                           | 0.0              | 0.0                                                          | 9.4              | 10.3                 |                             |
| NON-ACCEPTED AUTHORIZATIONS BY FAIL REASON |                  |                                                              |                  |                      |                             |
| FAIL REASON                                | MEMBER<br>VOLUME | DOLLAR<br>VOLUME                                             |                  |                      |                             |
| AUTH FILE                                  | 0                | .00                                                          |                  |                      |                             |
| TRANS LIMIT                                | 0                | .00                                                          |                  |                      |                             |
| DAY 1 TRANS                                | 0                | .00                                                          |                  |                      |                             |
| DAY 1 DOLLARS                              | 0                | .00                                                          |                  |                      |                             |
| DAY 2 TRANS                                | 0                | .00                                                          |                  |                      |                             |
| DAY 2 DOLLARS                              | 0                | .00                                                          |                  |                      |                             |
| DAY 3 TRANS                                | 0                | .00                                                          |                  |                      |                             |
| DAY 3 DOLLARS                              | 0                | .00                                                          |                  |                      |                             |
| DAY 4 TRANS                                | 0                | .00                                                          |                  |                      |                             |
| DAY 4 DOLLARS                              | 0                | .00                                                          |                  |                      |                             |
| PRM ACCUM LIMIT                            | 0                | .00                                                          |                  |                      |                             |
| CASH ADV ACCUMS                            | 0                | .00                                                          |                  |                      |                             |
| MERCH SUSPIC                               | 0                | .00                                                          |                  |                      |                             |
| INVALID CHIP/CVC                           | 0                | .00                                                          |                  |                      |                             |
| ACCEPTED DOLLARS                           | \$ .00           | NON-ACCEPTED DOLLARS                                         | \$ .00           |                      |                             |

### Authorization Summary Report (AB505010-AA), page 3

REPORT AB505010-AA

MASTERCARD INTERNATIONAL, INC.

PAGE 3

AUTHORIZATION SUMMARY

DATE 10/20/24 TIME 12:58:08

FOR MCBS BILLING WEEK BEGINNING 10/12/24 AND ENDING 10/18/24  
ICA 019827 PREFIX 51121390100 THRU 51121390100

| LIMIT-1 MIP PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME | WEEKLY<br>RANK |
|-----------------------------------|------------------|----------------------|------------------|----------------------|----------------|
| ACCEPT                            | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| DECLINE                           | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| CALL-ME                           | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| PICK-UP                           | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| TOTAL LIMIT-1 PROCESSED           | 0                | 0                    | 0                | 0                    |                |
| <hr/>                             |                  |                      |                  |                      |                |
| MAIL/TELEPHONE                    | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| CASH ADVANCE                      | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| RETAIL                            | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| TRAVEL/ENTERTAINMENT              | 0                | 0                    | 0                | 0                    | N/A            |
| PERCENT OF TOTAL                  | 0.0              | 0.0                  | 0.0              | 0.0                  |                |

## Authorization Summary Report (AB505010-AA), page 4

| REPORT AB505010-AA               | MASTERCARD INTERNATIONAL, INC.                               |                      |                  |                      | PAGE 4                      |
|----------------------------------|--------------------------------------------------------------|----------------------|------------------|----------------------|-----------------------------|
|                                  | AUTHORIZATION SUMMARY                                        |                      |                  |                      | DATE 10/20/24 TIME 12:58:08 |
|                                  | FOR MCBS BILLING WEEK BEGINNING 10/12/24 AND ENDING 10/18/24 |                      |                  |                      |                             |
|                                  | ICA 019827 PREFIX 51121390100 THRU 51121390100               |                      |                  |                      |                             |
| X-CODE MIP PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME                                             | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME | WEEKLY<br>RANK              |
| ACCEPT                           | 0                                                            | 0                    | 5,196            | 341,228              | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 5.7              | 4.7                  |                             |
| DECLINE                          | 0                                                            | 0                    | 85,800           | 6,972,778            | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 94.2             | 95.3                 |                             |
| CALL-ME                          | 0                                                            | 0                    | 100              | 346                  | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 0.1              | 0.0                  |                             |
| PICK-UP                          | 0                                                            | 0                    | 1                | 287                  | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 0.0              | 0.0                  |                             |
| TOTAL X-CODE PROCESSED           | 0                                                            | 0                    | 91,097           | 7,314,639            |                             |
| -----                            |                                                              |                      |                  |                      |                             |
| MIP PROCESSED<br>ACTIVITY        | MEMBER<br>VOLUME                                             | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME | WEEKLY<br>RANK              |
| ACCEPT                           | 0                                                            | 0                    | 214              | 7,218                | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 45.4             | 51.6                 |                             |
| DECLINE                          | 0                                                            | 0                    | 257              | 6,767                | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 54.6             | 48.4                 |                             |
| CALL-ME                          | 0                                                            | 0                    | 0                | 5                    | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 0.0              | 0.0                  |                             |
| PICK-UP                          | 0                                                            | 0                    | 0                | 0                    | N/A                         |
| PERCENT OF TOTAL                 | 0.0                                                          | 0.0                  | 0.0              | 0.0                  |                             |
| TOTAL MIP PROCESSED              | 0                                                            | 0                    | 471              | 13,990               |                             |

## Authorization Summary Report (AB505010-AA), page 5

REPORT AB505010-AA      MASTERCARD INTERNATIONAL, INC.      PAGE 5  
AUTHORIZATION SUMMARY      DATE 10/20/24 TIME 12:58:08  
FOR MCBS BILLING WEEK BEGINNING 10/12/24 AND ENDING 10/18/24  
ICA 019827      PREFIX 51121390100 THRU 51121390100

### W E E K L Y   T O T A L S

| WEEKLY TOTAL<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME | WEEKLY<br>RANK |
|--------------------------|------------------|----------------------|------------------|----------------------|----------------|
| ACCEPT                   | 0                | 0                    | 2,800,192,737    | 16,879,927,849       | N/A            |
| PERCENT OF TOTAL         | 0.0              | 0.0                  | 87.9             | 51.3                 |                |
| DECLINE                  | 0                | 0                    | 362,120,924      | 14,940,045,804       | N/A            |
| PERCENT OF TOTAL         | 0.0              | 0.0                  | 11.4             | 45.4                 |                |
| CALL-ME                  | 0                | 0                    | 157,813          | 6,588,517            | N/A            |
| PERCENT OF TOTAL         | 0.0              | 0.0                  | 0.0              | 0.0                  |                |
| PICK-UP                  | 0                | 0                    | 21,590,529       | 1,061,597,114        | N/A            |
| PERCENT OF TOTAL         | 0.0              | 0.0                  | 0.7              | 3.2                  |                |
| TOTAL PROCESSED          | 0                | 0                    | 3,184,062,003    | 32,888,159,284       |                |

## Field descriptions

The fields on the Authorization Summary Report (AB505010-AA) are listed in table format.

### Field descriptions

| Field                     | Description                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member Processed Activity | Summary of all authorization activity received and processed by the issuer, listed by the following response categories: <ul style="list-style-type: none"> <li>Accept (approved)</li> <li>Decline</li> <li>Call-Me (refer to card issuer)</li> <li>Pick-Up (capture card)</li> </ul> |
|                           | Member Volume                                                                                                                                                                                                                                                                         |
|                           | Total transaction volume processed by the customer that generated the specific response and the percentage it represents of total member transaction volume.                                                                                                                          |

| Field                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Member YTD Volume</p> <p>Summary of the customer's authorization volume for the calendar year-to-date (by response category) and the percentage it represents of total member volume for year-to-date.</p>                                                                                                                                                                                                                                                                   |
|                                      | <p>System Volume</p> <p>Volume of authorization transactions processed by the entire membership that week and the percentage it represents of total volume for the membership for the week.</p>                                                                                                                                                                                                                                                                                 |
|                                      | <p>System YTD Volume</p> <p>Volume of authorization transactions processed by the entire membership for the calendar year-to-date and the percentage it represents of total volume for the membership year-to-date.</p>                                                                                                                                                                                                                                                         |
|                                      | <p>Weekly Rank</p> <p>Member volume, by bank identification number (BIN), ranked within the membership (compared to total system volume). This field shows where each BIN stands in relation to other Mastercard BINs. In the sample, the Weekly Rank column at the Accept Category line shows 00017/07153. This indicates that this BIN ranks 17th out of a total of 7,153 BINs in issuing approvals; 16 BINs issued more approvals and 7,136 BINs issued fewer approvals.</p> |
| Member Approvals by Transaction Type | <p>Summary of all approved responses returned, broken down by the following transaction categories:</p> <ul style="list-style-type: none"> <li>• Mail/Telephone</li> <li>• Retail</li> <li>• Cash Advance</li> <li>• College/Hospital</li> <li>• Hotel/Motel</li> <li>• Vehicle Rental</li> <li>• Transportation</li> <li>• Restaurant</li> <li>• Cirrus/ATM</li> <li>• Unique</li> <li>• Payment Transaction</li> </ul>                                                        |



| Field | Description                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <p>Average Dollars</p> <p>Average dollar amount approved for the week, listed by transaction type.</p>                                                                                                                                                                                 |
|       | <p>Total Dollars</p> <p>Total dollar amount approved for the week, listed by transaction type.</p>                                                                                                                                                                                     |
|       | <p>Total Volume</p> <p>Total transaction volume processed for the customer for the week, listed by transaction type. In the sample, the total volume for MO/TO transactions is 72,276. This indicates that out of a total of 564,630 approved transactions, 72,276 were for MO/TO.</p> |
|       | <p>YTD Average Dollars</p> <p>Average U.S. dollar amount approved, by transaction type, for the calendar year-to-date. This amount is the result of dividing the YTD TOTAL DOLLARS by the YTD TOTAL VOLUME.</p>                                                                        |
|       | <p>YTD Total Dollars</p> <p>Total U.S. dollar amount approved by the customer, listed by transaction type, for the calendar year-to-date.</p>                                                                                                                                          |
|       | <p>YTD Total Volume</p> <p>Total transaction volume approved by the customer, listed by transaction type, for the calendar year-to-date.</p>                                                                                                                                           |
|       | <p>Total Member Approvals</p> <p>Average U.S. dollar amount, total U.S. dollar amount, total transaction volume, and year-to-date totals for all transactions approved by the customer.</p>                                                                                            |

| Field                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stand-In Processed Activity                | <p>Summary of all authorization activity received and processed by Stand-In Authorization, listed by the following response categories:</p> <ul style="list-style-type: none"> <li>• Accept (approved)</li> <li>• Decline</li> <li>• Call-Me (refer to card issuer)</li> <li>• Pick-Up (capture card)</li> </ul> <p>Refer to field descriptions for an explanation of Member Volume, Member YTD Volume, System Volume, System YTD Volume, and Weekly Rank. All values are for authorization activity processed by Stand-In Authorization on behalf of the customer.</p> |
|                                            | <p>Timed Out (Percent of Total)</p> <p>Number of times the Mastercard Network authorization application sent the original authorization request to the issuer for processing, but the acquiring MIP did not receive a response in the allocated time. The second line indicates the percentage of total authorization requests that timed out in this manner.</p>                                                                                                                                                                                                       |
|                                            | <p>Member Down (Percent of Total)</p> <p>Number of authorization requests sent to Stand-In Authorization because the Mastercard Network authorization application recognized that the issuer's host was down. Transactions that left the acquiring node but did not reach the issuing MIP or did reach the issuing MIP but could not communicate with the issuer's host, are included in this count. The second line indicates the percentage of total authorization requests the application handled in this manner.</p>                                               |
|                                            | <p>Signed Out (Percent of Total)</p> <p>Number of transactions sent to Stand-In Authorization because the issuer was signed out. The second line indicates the percentage of total authorization requests that were sent to Stand-In Authorization for this reason.</p>                                                                                                                                                                                                                                                                                                 |
| Non-accepted Authorizations by Fail Reason | <p>Summary of all authorization activity processed by Stand-In that Stand-In did not approve because it failed the indicated tests performed during Stand-In Authorization.</p>                                                                                                                                                                                                                                                                                                                                                                                         |

| Field                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Fail Reason</p> <p>List of the Stand-In tests that authorization requests must pass to be approved.</p> <p><b>NOTE: The Fail Reason INVALID CHIP/CVC includes both transactions that failed for invalid CHIP/CVC in Stand-In and transactions that failed for magnetic stripe/CVC error (codes A–J) at the MIP.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                               | <p>Member Volume</p> <p>Total volume of transactions declined authorization because of failure to pass the Stand-In test listed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                               | <p>Dollar Volume</p> <p>Total U.S. dollar amount declined authorization because of failure to pass the Stand-In test listed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                               | <p>Accepted Dollars</p> <p>Total U.S. dollar amount that Stand-In authorized on behalf of the issuer for the week reported.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                               | <p>Non-Accepted Dollars</p> <p>Total U.S. dollar amount that Stand-In Authorization did not authorize when processing on behalf of the issuer for the week reported.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| X-Code MIP Processed Activity | <p>Summary of all authorization activity processed as X-Code transactions at the acquirer MIP. These transactions are transactions that should have gone to the issuer or to Stand-In Authorization or processing, but the authorization application could not obtain a response from either one.</p> <p>Transactions are listed by the following response categories:</p> <ul style="list-style-type: none"> <li>• Accept (approved)</li> <li>• Decline</li> <li>• Call-Me (refer to card issuer)</li> <li>• Pick-Up (capture card)</li> </ul> <p>Refer to field descriptions for an explanation of Member Volume, Member YTD Volume, System Volume, System YTD Volume, and Weekly Rank. All values are for authorization activity processed at the acquirer MIP using X-Code Processing parameters.</p> |

| Field                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MIP Processed Activity               | <p>Summary of all authorization activity involving certain cardholder-activated terminals processed at the MIP that would not be processed by an issuer or Stand-In Authorization.</p> <p>Transactions are listed by the following response categories:</p> <ul style="list-style-type: none"> <li>• Accept (approved)</li> <li>• Decline</li> <li>• Call-Me (refer to card issuer)</li> <li>• Pick-Up (capture card)</li> </ul> <p>Refer to field descriptions for an explanation of Member Volume, Member YTD Volume, System Volume, System YTD Volume, and Weekly Rank. All values are for authorization activity processed at the MIP using X-Code parameters or those transactions declined by the MIP Transaction Blocking service.</p> |
| Weekly Total Activity                | <p>Summary of total transaction volume processed by the customer, Stand-In, X-Code, MIP, and BAT on behalf of the customer for the week reported.</p> <p>Refer to field descriptions for an explanation of Member Volume, Member YTD Volume, System Volume, System YTD Volume, and Weekly Rank.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Acquirer Reversal Processed Activity | <p>Summary of Reversal Request/0400 message activity received and processed by the issuer, listed by the following response categories:</p> <ul style="list-style-type: none"> <li>• Accept (approved)</li> <li>• Decline</li> <li>• Call-Me (refer to card issuer)</li> <li>• Pick-Up (capture card)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>Reversal Volume</p> <p>Total reversal volume processed by the customer that generated the specific response and the percentage it represents of total member transaction volume.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      | <p>Reversal YTD Volume</p> <p>Summary of the customer's reversal volume for the calendar year-to-date (by response category) and the percentage it represents of total member volume for the year-to-date.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Field                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>System Volume</p> <p>Volume of reversal transactions processed by the entire membership that week and the percentage it represents of total volume for the membership for the week.</p>                                                                                                                                                                                                                               |
|                                        | <p>System YTD Volume</p> <p>Volume of reversal transactions processed by the entire membership for the calendar year-to-date and the percentage it represents of total volume for the membership for the year-to-date.</p>                                                                                                                                                                                               |
|                                        | <p>Weekly Rank</p> <p>Member volume, by bank identification number (BIN), ranked within the membership (compared to total system volume). This field shows where each BIN stands in relation to other BINs.</p>                                                                                                                                                                                                          |
|                                        | <p>Total Reversal Processed</p> <p>Average U.S. dollar amount, total U.S. dollar amount, total transaction volume, and year-to-date totals for all transactions approved by the customer.</p>                                                                                                                                                                                                                            |
| Reversal Approvals by Transaction Type | <p>Summary of all approved responses returned, broken down by the following transaction categories:</p> <ul style="list-style-type: none"> <li>• Mail/Telephone</li> <li>• Retail</li> <li>• Cash Advance</li> <li>• College/Hospital</li> <li>• Hotel/Motel</li> <li>• Vehicle Rental</li> <li>• Transportation</li> <li>• Restaurant</li> <li>• Cirrus/ATM</li> <li>• Unique</li> <li>• Payment Transaction</li> </ul> |
|                                        | <p>Average Dollars</p> <p>Average dollar amount approved for the week, listed by transaction type.</p>                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Total Dollars</p> <p>Total dollar amount approved for the week, listed by transaction type.</p>                                                                                                                                                                                                                                                                                                                       |

| Field                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Total Volume</p> <p>Total transaction volume processed by the customer for the week, listed by transaction type.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                 | <p>YTD Average Dollars</p> <p>Average U.S. dollar amount approved, by transaction type, for the calendar year-to-date. This amount is the result of dividing the YTD TOTAL DOLLARS by the YTD TOTAL VOLUME.</p>                                                                                                                                                                                                                                                                                                                                          |
|                                 | <p>YTD Total Dollars</p> <p>Total U.S. dollar amount approved by the customer, listed by transaction type, for the calendar year-to-date.</p>                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                 | <p>YTD Total Volume</p> <p>Total transaction volume approved by the customer, listed by transaction type, for the calendar year-to-date.</p>                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                 | <p>Total Reversals Approvals</p> <p>Average U.S. dollar amount, total U.S. dollar amount, total transaction volume, and year-to-date totals for all transactions approved by the customer.</p>                                                                                                                                                                                                                                                                                                                                                           |
| MIP Reversal Processed Activity | <p>Summary of all reversal activity received and processed by Stand-In, listed by the following response categories:</p> <ul style="list-style-type: none"> <li>• Accept (approved)</li> <li>• Decline</li> <li>• Call-Me (refer to card issuer)</li> <li>• Pick-Up (capture card)</li> </ul> <p>Refer to field descriptions for an explanation of Reversal Volume, Reversal YTD Volume, System Volume, System YTD Volume, and Weekly Rank. All values are for authorization activity processed by Stand-In Authorization on behalf of the customer.</p> |
|                                 | <p>Total MIP Reversal Processed</p> <p>Total reversal transaction volume, and year-to-date totals for all reversals approved by the customer.</p>                                                                                                                                                                                                                                                                                                                                                                                                        |

| Field                 | Description                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Weekly Total Activity | <p>Summary of total reversal transaction volume processed by the customer, Stand-In, X-Code, MIP, and BAT on behalf of the customer for the week reported.</p> <p>Refer to field descriptions for an explanation of Reversal Volume, Reversal YTD Volume, System Volume, System YTD Volume, and Weekly Rank.</p> |

## Authorization Processing Integrity Acquirer Detail Report (AB605010-AA and AB605010-FF)

The Authorization Processing Integrity Acquirer Detail Report for acquirers enables enhanced merchant reporting of the processing integrity fees.

The report provides acquirers with detailed transaction activity for each customer ID/ICA number identifying authorizations that were assessed a processing integrity fee. Acquirers may optionally choose to receive this report. Mastercard generates the report weekly. The report is delivered each Monday at 18:30 (Central Time). Acquirers can request to receive the report by contacting Global Customer Service.

### AB605010-AA image report

The report is available via bulk file in data or image file format. The image file is also available via Mastercard Connect® Operational Reports Online Reporting.

**NOTE: There is a size restriction applicable to any operational report; therefore, if the report is likely to be large (more than 1,000,000 lines), then delivery via bulk file is required.**

Report ID: AB605010-AA: Authorization Processing Integrity Acquirer detail Report (image file format):

- Bulk ID: T852
- Mastercard Connect Operational Reports: Authorization Processing Integrity Acquirer Detail Report

**NOTE: If a subscribing acquirer has no non-compliant authorizations to report within a billing period, the image format report (AB605010-AA) is not generated.**

## Field descriptions

| Field name displayed in report | Field Name (expanded)            | Data Element (DE)    | Field Description                                                                                                                                          |
|--------------------------------|----------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACQUIRER ICA/CUST ID           | Acquirer Institution ID          | DE 32                | Left-justified with trailing spaces (for example, 9744).                                                                                                   |
| CARDHOLDER NUMBER              | Cardholder Number                | DE 2                 | Masked (obfuscated) PAN. First six and last four digits of the PAN will be shown (for example, 999999XXXXXX9999 ).<br>Left-justified with trailing spaces. |
| AUTH AMT                       | Authorization Transaction Amount | DE 4                 | Right-justified with leading zeros (for example, 00000001211).                                                                                             |
| CLRNG AMT                      | Clearing Transaction Amount      | DE 4                 | Right-justified with leading zeros (for example, 00000001211).<br>Blank if not applicable for billing event.                                               |
| POS CNTRY                      | POS Country Code                 | DE 61, subfield 13   | Alpha country code.                                                                                                                                        |
| FWD-INST ID                    | Forwarding Institution ID        | DE 33                | Right-justified with leading zeros (for example, 00000008434).                                                                                             |
| AUTH DATE                      | Authorization Date               | DE 15                | YYYYMMDD Derived YYYY, DE 15 MMDD                                                                                                                          |
| CLRNG DATE                     | Clearing Date                    | PDS 0158, subfield 5 | YYYYMMDD Derived YYYY, PDS 0158 subfield 5 YYMMDD.<br>Blank if not applicable for billing event.                                                           |
| AUTH ID                        | Authorization ID                 | DE 38                |                                                                                                                                                            |
| FIN-NET CODE                   | Financial Network Code           | DE 63, subfield 1    |                                                                                                                                                            |
| AUTH BRN                       | Banknet Reference Number         | DE 63, subfield 2    | Left-justified.                                                                                                                                            |
| BRAND                          | Acceptance Brand                 |                      |                                                                                                                                                            |
| MCC                            | Merchant Category Code           | DE 18                |                                                                                                                                                            |



| Field name displayed in report | Field Name (expanded)               | Data Element (DE)                | Field Description                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRAN STATUS                    | Point of Service Transaction Status | DE 61, subfield 7                |                                                                                                                                                                                                                                                                                                                                                                   |
| MERCHANT ID                    | Merchant ID                         | DE 42                            | Left-justified with trailing spaces.                                                                                                                                                                                                                                                                                                                              |
| MERCHANT NAME                  | Merchant Name Location              | DE 43                            | Left-justified with trailing spaces.                                                                                                                                                                                                                                                                                                                              |
| FINAL AUTH IND                 | Final Authorization Indicator       | DE 48, subelement 61, subfield 5 |                                                                                                                                                                                                                                                                                                                                                                   |
| RESPONSE CODE                  | Response Code                       | DE 39                            |                                                                                                                                                                                                                                                                                                                                                                   |
| AUTH-CURR CODE                 | Authorization Currency Code         | DE 49                            |                                                                                                                                                                                                                                                                                                                                                                   |
| CLRNG-CURR CODE                | Clearing Currency Code              | DE 49                            |                                                                                                                                                                                                                                                                                                                                                                   |
| PROC CODE                      | Processing Code                     | DE 3, subfield 1                 |                                                                                                                                                                                                                                                                                                                                                                   |
| CLRNG BRN                      | Clearing Banknet Reference Number   | DE 63, subfield 2                | Left-justified with trailing spaces.<br>Blank if not applicable for billing event.                                                                                                                                                                                                                                                                                |
| CLRNG MATCH                    | Clearing Match                      |                                  | Identifies if a matching clearing transaction was found for an authorization.<br>Values are: <ul style="list-style-type: none"> <li>• Y = Matching clearing found</li> <li>• N = Matching clearing not found</li> <li>• M = Multiple clearing records processed for a single authorization request</li> <li>• Blank = Not applicable for billing event</li> </ul> |

| Field name displayed in report | Field Name (expanded)         | Data Element (DE) | Field Description                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|-------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FINAL AUTH RVSL IND            | Final Auth Reversal Indicator |                   | <p>Identifies if a final authorization was reversed, which resulted in a matching clearing transaction not being found for an authorization.</p> <ul style="list-style-type: none"> <li>Y = Authorization was reversed</li> <li>N = Authorization was not reversed</li> <li>Blank = Not applicable for billing event</li> </ul>                |
| CURR MISMATCH IND              | Currency Mismatch Indicator   |                   | <p>Identifies if the authorization and clearing Transaction Currency Code (DE 49) are not the same.</p> <p>Values are:</p> <ul style="list-style-type: none"> <li>Y = Authorization and clearing currencies do not match</li> <li>N = Authorization and clearing currencies match</li> <li>Blank = Not applicable for billing event</li> </ul> |
| AMT MISMATCH IND               | Amount Mismatch Indicator     |                   | <p>Identifies if the authorization and clearing Transaction Amount (DE 4) are not the same.</p> <p>Values are:</p> <ul style="list-style-type: none"> <li>Y = Authorization and clearing amounts do not match</li> <li>Blank = Not applicable for billing event</li> </ul>                                                                     |
| BILL-CURR CODE                 | Billing Currency Code         |                   | Identifies the currency code of the assessed fee.                                                                                                                                                                                                                                                                                              |

| Field name displayed in report | Field Name (expanded)                  | Data Element (DE) | Field Description                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BILL CONV AUTH AMT             | Billing Converted Authorization Amount |                   | Authorization Transaction Amount converted to the currency of the assessed fee.<br><br>Left-justified with trailing spaces, and may contain an embedded decimal point based on the currency exponent (for example, 1234.56). |
| MIP OWNER ICA                  | MIP Owner ICA                          |                   | ICA Code obtained from Auth BNA table. Space filled, left-justified (for example 9744 ).                                                                                                                                     |
| ADDTL RESP 1                   | FRMT Error Data element no             |                   | Data element where a format error was found (for example, 048 indicates DE 48).                                                                                                                                              |
| ADDTL RESP 2                   | FRMT Error Sub element no              |                   | Sub element where format error was found (for example 042 indicates sub element 42 from the data element identified in the previous field).                                                                                  |
| BILLING-EVENT                  | Billing Event                          |                   | Identifies which fee was assessed.<br><br>Left-justified.                                                                                                                                                                    |
| EXCESSIVE AUTH ATTEMPTS        | Excessive Auth Attempts                |                   | Example 0012                                                                                                                                                                                                                 |
| SUB PROGRAM INDICATOR          | Sub Program Indicator                  |                   | Sub information on specified Transaction Processing Excellence (TPE) programs (for example, D).                                                                                                                              |
| TERMINAL ID                    | Terminal Id                            | DE 41             | Acceptor terminal ID                                                                                                                                                                                                         |
| DEVICE SERIAL NUMBER           | Device Serial Number                   | DE 55, Tag 9F1E   | The serial number of the terminal used for a transaction indicated by DE 55 Tag 9F1E.                                                                                                                                        |

AB605010-FF data file

Report ID: AB605010-FF: Authorization Processing Integrity Acquirer Detail Data File (data file format):

- Bulk ID: TKR8

**NOTE: If a subscribing acquirer has no non-complaint authorizations to report within a billing period, the data file format report (AB605010-FF) will be generated and include a trailer record with a record count of zero.**

- Seven TKR8 bulk files are delivered each Monday following Sunday’s weekly billing cycle. A separate file of non-compliant transactions is generated for each day (Sunday-Saturday) of the prior week containing non-compliant transactions.

Field descriptions

The format and layout of the data file follows. It is a fixed format file. All fields are delimited by ','.

Field descriptions

| Field Name                       | Start | End | Length | Data Element (DE) | Field Description                                                                                                                                         | Field Data Type |
|----------------------------------|-------|-----|--------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Cardholder Number                | 1     | 19  | 19     | DE 2              | Masked (obfuscated) PAN. First six and last four digits of the PAN will be shown (for example, 99999999XX XXXXXX99). Left-justified with trailing spaces. | Alphanumeric    |
| Delimiter                        | 20    | 20  | 1      |                   | All fields delimited by ','.                                                                                                                              | Alphanumeric    |
| Authorization Transaction Amount | 21    | 32  | 12     | DE 4              | Right-justified with leading zeros (for example, 00000001211 ).                                                                                           | Numeric         |

| Field Name                  | Start | End | Length | Data Element (DE)  | Field Description                                                                                                | Field Data Type |
|-----------------------------|-------|-----|--------|--------------------|------------------------------------------------------------------------------------------------------------------|-----------------|
| Delimiter                   | 33    | 33  | 1      |                    | All fields delimited by "<br>,".                                                                                 | Alphanumeric    |
| Clearing Transaction Amount | 34    | 45  | 12     | DE 4               | Right-justified with leading zeros (for example, 00000001211).<br><br>Blank if not applicable for billing event. | Numeric         |
| Delimiter                   | 46    | 46  | 1      |                    | All fields delimited by "<br>,".                                                                                 | Alphanumeric    |
| Acquirer Institution ID     | 47    | 57  | 11     | DE 32              | Left-justified with trailing zeros. (for example, 9744).                                                         | Alphanumeric    |
| Delimiter                   | 58    | 58  | 1      |                    | All fields delimited by "<br>,".                                                                                 | Alphanumeric    |
| POS Country Code            | 59    | 61  | 3      | DE 61, subfield 13 | Alpha country code.                                                                                              | Alphanumeric    |
| Delimiter                   | 62    | 62  | 1      |                    | All fields delimited by "<br>,".                                                                                 | Alphanumeric    |
| Forwarding Institution ID   | 63    | 73  | 11     | DE 33              | Right-justified with leading zeros (for example, 00000008434).                                                   | Numeric         |
| Delimiter                   | 74    | 74  | 1      |                    | All fields delimited by "<br>,".                                                                                 | Alphanumeric    |
| Authorization Date          | 75    | 82  | 8      | DE 15              | YYYYMMDD<br>Derived YYYY,<br>DE 15 MMDD                                                                          | Alphanumeric    |

| Field Name               | Start | End | Length | Data Element (DE)    | Field Description                                                                                       | Field Data Type |
|--------------------------|-------|-----|--------|----------------------|---------------------------------------------------------------------------------------------------------|-----------------|
| Delimiter                | 83    | 83  | 1      |                      | All fields delimited by "<br>,".                                                                        | Alphanumeric    |
| Clearing Date            | 84    | 91  | 8      | PDS 0158, subfield 5 | YYYYMMDD<br>Derived YYYY, PDS 0158 subfield 5 YYMMDD.<br><br>Blank if not applicable for billing event. | Alphanumeric    |
| Delimiter                | 92    | 92  | 1      |                      | All fields delimited by "<br>,".                                                                        | Alphanumeric    |
| Merchant ID              | 93    | 107 | 15     | DE 42                | Left-justified with trailing zeros.                                                                     | Alphanumeric    |
| Delimiter                | 108   | 108 | 1      |                      | All fields delimited by "<br>,".                                                                        | Alphanumeric    |
| Merchant Name Location   | 109   | 148 | 40     | DE 43                | Left-justified with trailing zeros.                                                                     | Alphanumeric    |
| Delimiter                | 149   | 149 | 1      |                      | All fields delimited by "<br>,".                                                                        | Alphanumeric    |
| Authorization ID         | 150   | 155 | 6      | DE 38                |                                                                                                         | Alphanumeric    |
| Delimiter                | 156   | 156 | 1      |                      | All fields delimited by "<br>,".                                                                        | Alphanumeric    |
| Financial Network Code   | 157   | 159 | 3      | DE 63, subfield 1    |                                                                                                         | Alphanumeric    |
| Delimiter                | 160   | 160 | 1      |                      | All fields delimited by "<br>,".                                                                        | Alphanumeric    |
| Banknet Reference Number | 161   | 169 | 9      | DE 63, subfield 2    | Left-justified.                                                                                         | Alphanumeric    |

| Field Name                          | Start | End | Length | Data Element (DE)                | Field Description                    | Field Data Type |
|-------------------------------------|-------|-----|--------|----------------------------------|--------------------------------------|-----------------|
| Delimiter                           | 170   | 170 | 1      |                                  | All fields delimited by<br>",",<br>/ | Alphanumeric    |
| Acceptance Brand                    | 171   | 173 | 3      |                                  |                                      | Alphanumeric    |
| Delimiter                           | 174   | 174 | 1      |                                  | All fields delimited by<br>",",<br>/ | Alphanumeric    |
| Merchant Category Code              | 175   | 178 | 4      | DE 18                            |                                      | Alphanumeric    |
| Delimiter                           | 179   | 179 | 1      |                                  | All fields delimited by<br>",",<br>/ | Alphanumeric    |
| Point of Service Transaction Status | 180   | 180 | 1      | DE 61, subfield 7                |                                      | Alphanumeric    |
| Delimiter                           | 181   | 181 | 1      |                                  | All fields delimited by<br>",",<br>/ | Alphanumeric    |
| Final Authorization Indicator       | 182   | 182 | 1      | DE 48, subelement 61, subfield 5 |                                      | Alphanumeric    |
| Delimiter                           | 183   | 183 | 1      |                                  | All fields delimited by<br>",",<br>/ | Alphanumeric    |
| Response Code                       | 184   | 185 | 2      | DE 39                            |                                      | Alphanumeric    |
| Delimiter                           | 186   | 186 | 1      |                                  | All fields delimited by<br>",",<br>/ | Alphanumeric    |
| Authorization Currency Code         | 187   | 189 | 3      | DE 49                            |                                      | Alphanumeric    |
| Delimiter                           | 190   | 190 | 1      |                                  | All fields delimited by<br>",",<br>/ | Alphanumeric    |

| Field Name                        | Start | End | Length | Data Element (DE) | Field Description                                                        | Field Data Type |
|-----------------------------------|-------|-----|--------|-------------------|--------------------------------------------------------------------------|-----------------|
| Clearing Currency Code            | 191   | 193 | 3      | DE 49             |                                                                          | Alphanumeric    |
| Delimiter                         | 194   | 194 | 1      |                   | All fields delimited by "<br>,"                                          | Alphanumeric    |
| Processing Code                   | 195   | 196 | 2      | DE 3, subfield 1  |                                                                          | Alphanumeric    |
| Delimiter                         | 197   | 197 | 1      |                   | All fields delimited by "<br>,"                                          | Alphanumeric    |
| Clearing Banknet Reference Number | 198   | 206 | 9      | DE 63, subfield 2 | Space filled, left-justified. Blank if not applicable for billing event. | Alphanumeric    |
| Delimiter                         | 207   | 207 | 1      |                   | All fields delimited by "<br>,"                                          | Alphanumeric    |



| Field Name     | Start | End | Length | Data Element (DE) | Field Description                                                                                                                                                                                                                                                                                                                                          | Field Data Type |
|----------------|-------|-----|--------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Clearing Match | 208   | 208 | 1      |                   | Identifies if a matching clearing transaction was found for an authorization.<br>Values are:<br><ul style="list-style-type: none"> <li>Y = Matching clearing found</li> <li>N=Matching clearing not found</li> <li>M = Multiple clearing records processed for a single authorization request</li> <li>Blank = Not applicable for billing event</li> </ul> | Alphanumeric    |
| Delimiter      | 209   | 209 | 1      |                   | All fields delimited by<br>";" .                                                                                                                                                                                                                                                                                                                           | Alphanumeric    |

| Field Name                    | Start | End | Length | Data Element (DE) | Field Description                                                                                                                                                                                                                                                                                                                     | Field Data Type |
|-------------------------------|-------|-----|--------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Final Auth Reversal Indicator | 210   | 210 | 1      |                   | <p>Identifies if a final authorization was reversed, which resulted in a matching clearing transaction not being found for an authorization.</p> <ul style="list-style-type: none"> <li>• Y = Authorization was reversed</li> <li>• N = Authorization was not reversed</li> <li>• Blank = Not applicable for billing event</li> </ul> | Alphanumeric    |
| Delimiter                     | 211   | 211 | 1      |                   | All fields delimited by<br>",",<br>/ ,                                                                                                                                                                                                                                                                                                | Alphanumeric    |

| Field Name                  | Start | End | Length | Data Element (DE) | Field Description                                                                                                                                                                                                                                                                                                                      | Field Data Type |
|-----------------------------|-------|-----|--------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Currency Mismatch Indicator | 212   | 212 | 1      |                   | Identifies if the authorization and clearing Transaction Currency Code (DE 49) are not the same.<br>Values are:<br><ul style="list-style-type: none"> <li>Y = Authorization and clearing currencies do not match</li> <li>N = Authorization and clearing currencies match</li> <li>Blank = Not applicable for billing event</li> </ul> | Alphanumeric    |
| Delimiter                   | 213   | 213 | 1      |                   | All fields delimited by<br>","                                                                                                                                                                                                                                                                                                         | Alphanumeric    |

| Field Name                | Start | End | Length | Data Element (DE) | Field Description                                                                                                                                                                                                                                                     | Field Data Type |
|---------------------------|-------|-----|--------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Amount Mismatch Indicator | 214   | 214 | 1      |                   | Identifies if the authorization and clearing Transaction Amount (DE 4) are not the same.<br>Values are:<br><ul style="list-style-type: none"> <li>Y = Authorization and clearing currencies do not match</li> <li>Blank = Not applicable for billing event</li> </ul> | Alphanumeric    |
| Delimiter                 | 215   | 215 | 1      |                   | All fields delimited by<br>","                                                                                                                                                                                                                                        | Alphanumeric    |
| Billing Currency Code     | 216   | 218 | 3      |                   | Identifies the currency code of the assessed fee.                                                                                                                                                                                                                     | Alphanumeric    |
| Delimiter                 | 219   | 219 | 1      |                   | All fields delimited by<br>","                                                                                                                                                                                                                                        | Alphanumeric    |

| Field Name                                      | Start | End | Length | Data Element (DE) | Field Description                                                                                                                                                                                                    | Field Data Type |
|-------------------------------------------------|-------|-----|--------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Billing<br>Converted<br>Authorization<br>Amount | 220   | 231 | 12     |                   | Authorization Transaction Amount converted to the currency of the assessed fee. Left-justified with trailing zeros, and may contain an embedded decimal point based on the currency exponent (for example, 1234.56). | Alphanumeric    |
| Delimiter                                       | 232   | 232 | 1      |                   | All fields delimited by ",".                                                                                                                                                                                         | Alphanumeric    |
| MIP Owner<br>ICA                                | 233   | 243 | 11     |                   | ICA Code. Space filled, left-justified (for example 9744)                                                                                                                                                            | Alphanumeric    |
| Delimiter                                       | 244   | 244 | 1      |                   | All fields delimited by ",".                                                                                                                                                                                         | Alphanumeric    |
| FRMT Error<br>Data element<br>no                | 245   | 247 | 3      |                   | Data element where a format error was found (for example 048 indicates DE 48).                                                                                                                                       | Alphanumeric    |
| Delimiter                                       | 248   | 248 | 1      |                   | All fields delimited by ",".                                                                                                                                                                                         | Alphanumeric    |

| Field Name                      | Start | End | Length | Data Element (DE) | Field Description                                                                                                                           | Field Data Type |
|---------------------------------|-------|-----|--------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| FRMT Error<br>Sub element<br>no | 249   | 251 | 3      |                   | Sub element where format error was found (for example 042 indicates sub element 42 from the data element identified in the previous field). | Alphanumeric    |
| Delimiter                       | 252   | 252 | 1      |                   | All fields delimited by<br>",".                                                                                                             | Alphanumeric    |
| Billing Event                   | 253   | 263 | 11     |                   | Identifies which fee was assessed.<br>Left-justified.                                                                                       | Alphanumeric    |
| Delimiter                       | 264   | 264 | 1      |                   | All fields delimited by<br>",".                                                                                                             | Alphanumeric    |
| Excess Auth Attempts            | 265   | 268 | 4      |                   | Example 0012                                                                                                                                | Alphanumeric    |
| Delimiter                       | 269   | 269 | 1      |                   | All fields delimited by<br>",".                                                                                                             | Alphanumeric    |
| Sub Program Indicator           | 270   | 270 | 1      |                   | Sub information on specified TPE programs (for example D).                                                                                  | Alphanumeric    |
| Delimiter                       | 271   | 271 | 1      |                   | All fields delimited by<br>",".                                                                                                             | Alphanumeric    |
| Terminal ID                     | 272   | 279 | 8      | DE 41             | Acceptor terminal ID indicated by DE 41.                                                                                                    | Alphanumeric    |

| Field Name           | Start | End | Length | Data Element (DE) | Field Description                                                                     | Field Data Type |
|----------------------|-------|-----|--------|-------------------|---------------------------------------------------------------------------------------|-----------------|
| Delimiter            | 280   | 280 | 1      |                   | All fields delimited by<br>","                                                        | Alphanumeric    |
| Device Serial Number | 281   | 288 | 8      | DE 55 Tag 9F1E    | The serial number of the terminal used for a transaction indicated by DE 55 Tag 9F1E. | Alphanumeric    |
| Delimiter            | 289   | 289 | 1      |                   | All fields delimited by<br>","                                                        | Alphanumeric    |
| Filler               | 290   | 300 | 11     |                   | Blanks                                                                                |                 |

### AB605010-FF file trailer

#### Field descriptions

| Field Name   | Start | End | Length | Data Element (DE) | Field Description                                                                                     | Field Data Type |
|--------------|-------|-----|--------|-------------------|-------------------------------------------------------------------------------------------------------|-----------------|
| Record Type  | 1     | 7   | 7      |                   | Identifies the trailer record.<br>Contains the value TRAILER.                                         | Alphanumeric    |
| Endpoint     | 8     | 14  | 7      |                   | ID of the endpoint to which the bulk file was delivered.                                              | Alphanumeric    |
| Record Count | 15    | 26  | 12     |                   | Count of transaction detail records included in the bulk file.<br>Left-justified with trailing zeros. | Alphanumeric    |
| Filler       | 27    | 300 | 274    |                   | Blanks                                                                                                |                 |

## Transaction Processing Excellence Issuer Detail Report (AB705010-AA and AB705010-FF)

The following section describes the AB705010-AA and AB705010-FF detail reports.

### AB705010-AA image report

The report is available via bulk file in data or image file format. The image file is also available via Mastercard Connect® Operational Reports Online Reporting. Report ID AB705010-AA: Transaction Processing Excellence Issuer Detail Report (image file format):

- Bulk ID: T852
- Mastercard Connect Operational Reports: Transaction Processing Excellence Issuer Detail Report

**NOTE: There is a size restriction applicable to any operational report; therefore, if the report is likely to be large (more than 1,000,000 lines), then delivery via bulk file is required. If a subscribing issuer has no non-compliant authorizations to report within a billing period, then the image format report (AB705010-AA) is not generated.**

### Transaction Processing Excellence Issuer Detail Report AB705010-AA

The Transaction Processing Excellence Issuer Detail Report for issuers enables enhanced reporting of transaction processing excellence (TPE) fees. The report provides issuers with detailed transaction activity for each customer ID/ICA number identifying authorizations that were assessed a fee. The report is optional and Mastercard generates the report weekly. The report is delivered each Monday at 18:30 (Central Time). Issuers can request to receive the report by contacting Global Customer Service.

### Field descriptions

| Field Name Displayed in Report | Field Name (expanded)  | Data Element (DE) | Field Description                                                                                                                              |
|--------------------------------|------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| ISSUER ICA/CUST ID             | Issuer ICA Number      |                   | Left-justified with trailing spaces(for example, 9744)                                                                                         |
| CARDHOLDER NUMBER              | Cardholder Number      | DE 2              | Masked (obfuscated) PAN, first six and last four digits of the PAN will be shown (for example, 999999XXXXXX9999), space filled, left-justified |
| FIN_NET CODE                   | Financial Network Code | DE 63, subfield 1 |                                                                                                                                                |



| Field Name Displayed in Report | Field Name (expanded)                  | Data Element (DE)                | Field Description                                                                        |
|--------------------------------|----------------------------------------|----------------------------------|------------------------------------------------------------------------------------------|
| AUTH BRN                       | Authorization Banknet Reference Number | DE 63, subfield 2                |                                                                                          |
| AUTH ID                        | Authorization ID                       | DE 38                            |                                                                                          |
| AUTH DATE                      | Authorization Date                     |                                  | YYYYMMDD                                                                                 |
| AUTH TS                        | Authorization Timestamp                |                                  | HHMMSS                                                                                   |
| SETL DATE                      | Settlement Date                        | DE 15                            | YYYYMMDD                                                                                 |
| RESP CODE                      | Response Code                          | DE 39                            |                                                                                          |
| TPE-MATCH AUTH-DT              | TPE Matching Auth Date and Time        |                                  | YYYYMMDDHHMMSS                                                                           |
| MIP-OWNER ICA                  | MIP Owner ICA                          |                                  | ICA Code obtained from Auth BNA table, space filled, left-justified (for example, 9744 ) |
| BRAND                          | Acceptance Brand                       |                                  |                                                                                          |
| ACQ-INST ID                    | Acquirer Institution ID                | DE 32                            |                                                                                          |
| FWD-INST ID                    | Forwarding Institution ID              | DE 33                            | Right-justified with leading zeros (for example, 00000008434)                            |
| MCC                            | Merchant Category Code                 | DE 18                            |                                                                                          |
| POS CNTRY                      | POS Country                            | DE 61, subfield 13               | Alpha country code.                                                                      |
| MERCHANT ID                    | Merchant ID                            | DE 42                            |                                                                                          |
| MERCHANT NAME                  | Merchant Name Location                 | DE 43                            |                                                                                          |
| PROC CODE                      | Processing Code                        | DE 3, subfield 1                 |                                                                                          |
| TRAN STATUS                    | POS Transaction Status                 | DE 61, subfield 7                |                                                                                          |
| FINAL AUTH-IND                 | Final Authorization Indicator          | DE 48, subelement 61, subfield 5 |                                                                                          |
| AUTH-CURR CODE                 | Authorization Currency Code            | DE 49                            |                                                                                          |

| Field Name Displayed in Report | Field Name (expanded)                    | Data Element (DE)    | Field Description                                                                                                                                                                                                                                                                                                           |
|--------------------------------|------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUTH AMT                       | Authorization Transaction Amount         | DE 4                 | Right-justified with leading zeros (for example, 00000001211), blank if not applicable for billing event                                                                                                                                                                                                                    |
| AUTH-AMT BILL-CURR             | Authorization amount in billing currency |                      | Authorization Transaction Amount converted to the currency of the assessed fee, space filled, left-justified, and may contain an embedded decimal point based on the currency exponent (for example, 1234.56)                                                                                                               |
| CLRNG AMT                      | Clearing Transaction Amount              | DE 4                 | Right-justified with leading zeros (for example, 00000001211), blank if not applicable for billing event                                                                                                                                                                                                                    |
| CLRNG DATE                     | Clearing Date                            | PDS 0158, subfield 5 | YYYYMMDD derived YYYY, PDS 0158 subfield 5 YYMMDD, blank if not applicable for billing event                                                                                                                                                                                                                                |
| CLRNG-CURR CODE                | Clearing Currency Code                   | DE 49                | Contains XXX if multiple clearing records are processed for a single authorization and one clearing record currency code differs from the authorization currency code, otherwise contains the clearing currency code of the last clearing record processed for the authorization, blank if not applicable for billing event |

| Field Name Displayed in Report | Field Name (expanded)             | Data Element (DE) | Field Description                                                              |
|--------------------------------|-----------------------------------|-------------------|--------------------------------------------------------------------------------|
| CLRNG BRN                      | Clearing Banknet Reference Number | DE 63, subfield 2 | Left-justified with trailing spaces, blank if not applicable for billing event |
| BILL-CURR CODE                 | Billing Currency Code             |                   | Identifies the currency code of the assessed fee                               |
| BILLING-EVENT                  | Billing Event ID                  |                   | Identifies which fee was assessed, left justified                              |

### AB705010-FF data file

Report ID AB705010-FF: Transaction Processing Excellence Issuer Detail Report (data file format):

- Bulk ID: T34P
- Seven T34P bulk files are delivered each Monday following Sunday's weekly billing cycle. A separate file of non-compliant transactions is generated for each day, (Sunday - Saturday) of the prior week containing non-compliant transactions.

**NOTE: If a subscribing issuer has no non-compliant authorizations to report within a billing period, the data file format report (AB705010-FF) will be generated and will include a trailer record with a record count of zero.**

### Field descriptions

The format and layout of the AB705010-FF data file is a fixed format file, all fields are delimited by ";".

### Field descriptions

| Field name        | Start | End | Length | Data element (DE) | Field description           |
|-------------------|-------|-----|--------|-------------------|-----------------------------|
| Issuer ICA number | 1     | 11  | 11     |                   |                             |
| Delimiter         | 12    | 12  | 1      |                   | All fields delimited by ";" |

| Field name               | Start | End | Length | Data element (DE) | Field description                                                                                                                             |
|--------------------------|-------|-----|--------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Cardholder number        | 13    | 31  | 19     | DE 2              | Masked (Obfuscated) PAN. First 8 and last 2 digits of the PAN will be shown (for example, 99999999XXX XXXXX99 ), space filled, left-justified |
| Delimiter                | 32    | 32  | 1      |                   | All fields delimited by ";"                                                                                                                   |
| Financial network code   | 33    | 35  | 3      | DE 63 subfield 1  |                                                                                                                                               |
| Delimiter                | 36    | 36  | 1      |                   | All fields delimited by ";"                                                                                                                   |
| Banknet reference number | 37    | 45  | 9      | DE 63 subfield 2  | Left-justified                                                                                                                                |
| Delimiter                | 46    | 46  | 1      |                   | All fields delimited by ";"                                                                                                                   |
| Authorization ID         | 47    | 52  | 6      | DE 38             |                                                                                                                                               |
| Delimiter                | 53    | 53  | 1      |                   | All fields delimited by ";"                                                                                                                   |
| Authorization date       | 54    | 61  | 8      |                   |                                                                                                                                               |
| Delimiter                | 62    | 62  | 1      |                   | All fields delimited by ";"                                                                                                                   |
| Authorization time stamp | 63    | 68  | 6      |                   |                                                                                                                                               |
| Delimiter                | 69    | 69  | 1      |                   | All fields delimited by ";"                                                                                                                   |
| Settlement date          | 70    | 77  | 8      | DE 15             | YYYYMMDD derived YYYY, DE 15 MMDD                                                                                                             |
| Delimiter                | 78    | 78  | 1      |                   | All fields delimited by ";"                                                                                                                   |

| Field name                         | Start | End | Length | Data element (DE) | Field description                                                                       |
|------------------------------------|-------|-----|--------|-------------------|-----------------------------------------------------------------------------------------|
| Response code                      | 79    | 80  | 2      | DE 39             |                                                                                         |
| Delimiter                          | 81    | 81  | 1      |                   | All fields delimited by ";"                                                             |
| ASI TPE Fin Auth - Auth Date       | 82    | 95  | 14     |                   | YYYYMMDDHHMMSS                                                                          |
| Delimiter                          | 96    | 96  | 1      |                   | All fields delimited by ";"                                                             |
| MIP owner ICA                      | 97    | 107 | 11     |                   | ICA Code obtained from Auth BNA table, space filled, left-justified (for example, 9744) |
| Delimiter                          | 108   | 108 | 1      |                   | All fields delimited by ";"                                                             |
| Acceptance brand                   | 109   | 111 | 3      |                   |                                                                                         |
| Delimiter                          | 112   | 112 | 1      |                   | All fields delimited by ";"                                                             |
| Acquirer institution ID            | 113   | 123 | 11     | DE 32             | Left-justified with trailing spaces (for example, 9744)                                 |
| Delimiter                          | 124   | 124 | 1      |                   | All fields delimited by ";"                                                             |
| Acquirer forwarding institution ID | 125   | 135 | 11     | DE 33             | Right-justified with leading zeros (for example, 00000008434)                           |
| Delimiter                          | 136   | 136 | 1      |                   | All fields delimited by ";"                                                             |
| Merchant category code             | 137   | 140 | 4      | DE 18             |                                                                                         |
| Delimiter                          | 141   | 141 | 1      |                   | All fields delimited by ";"                                                             |

| Field name                    | Start | End | Length | Data element (DE)              | Field description                                              |
|-------------------------------|-------|-----|--------|--------------------------------|----------------------------------------------------------------|
| POS country code              | 142   | 144 | 3      | DE 61 subfield 13              | Alpha country code                                             |
| Delimiter                     | 145   | 145 | 1      |                                | All fields delimited by ";"                                    |
| Merchant ID                   | 146   | 160 | 15     | DE 42                          | Left-justified with trailing spaces                            |
| Delimiter                     | 161   | 161 | 1      |                                | All fields delimited by ";"                                    |
| Merchant name location        | 162   | 201 | 40     | DE 43                          | Left-justified with trailing spaces                            |
| Delimiter                     | 202   | 202 | 1      |                                | All fields delimited by ";"                                    |
| Processing code               | 203   | 204 | 2      | DE 3 subfield 1                |                                                                |
| Delimiter                     | 205   | 205 | 1      |                                | All fields delimited by ";"                                    |
| POS transaction status        | 206   | 206 | 1      | DE 61 subfield 7               |                                                                |
| Delimiter                     | 207   | 207 | 1      |                                | All fields delimited by ";"                                    |
| Final authorization indicator | 208   | 208 | 1      | DE 48 subelement 61 subfield 5 |                                                                |
| Delimiter                     | 209   | 209 | 1      |                                | All fields delimited by ";"                                    |
| Authorization currency code   | 210   | 212 | 3      | DE 49                          |                                                                |
| Delimiter                     | 213   | 213 | 1      |                                | All fields delimited by ";"                                    |
| Authorization amount          | 214   | 225 | 12     | DE 4                           | Right-justified with leading zeros, (for example, 00000001211) |
| Delimiter                     | 226   | 226 | 1      |                                | All fields delimited by ";"                                    |

| Field name                               | Start | End | Length | Data element (DE)   | Field description                                                                                                                                                                                                     |
|------------------------------------------|-------|-----|--------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authorization amount in billing currency | 227   | 238 | 12     |                     | Authorization Transaction Amount converted to the currency of the assessed fee. Left-justified with trailing spaces, and may contain an embedded decimal point based on the currency exponent (for example, 1234.56). |
| Delimiter                                | 239   | 239 | 1      |                     | All fields delimited by ";"                                                                                                                                                                                           |
| Clearing amount                          | 240   | 251 | 12     | DE 4                | Right-justified with leading zeros (for example, 00000001211), blank if not applicable for billing event                                                                                                              |
| Delimiter                                | 252   | 252 | 1      |                     | All fields delimited by ";"                                                                                                                                                                                           |
| Clearing date                            | 253   | 260 | 8      | PDS 0158 subfield 5 | YYYYMMDD Derived YYYY, PDS 0158 subfield 5 YYMMDD, blank if not applicable for billing event                                                                                                                          |
| Delimiter                                | 261   | 261 | 1      |                     | All fields delimited by ";"                                                                                                                                                                                           |
| Clearing currency code                   | 262   | 264 | 3      | DE 49               |                                                                                                                                                                                                                       |

| Field name                        | Start | End | Length | Data element (DE) | Field description                                 |
|-----------------------------------|-------|-----|--------|-------------------|---------------------------------------------------|
| Delimiter                         | 265   | 265 | 1      |                   | All fields delimited by ";"                       |
| Clearing Banknet reference number | 266   | 274 | 9      | DE 63 subfield 2  |                                                   |
| Delimiter                         | 275   | 275 | 1      |                   | All fields delimited by ";"                       |
| Billing currency code             | 276   | 278 | 3      |                   | Identifies the currency code of the assessed fee. |
| Delimiter                         | 279   | 279 | 1      |                   | All fields delimited by ";"                       |
| Billing event                     | 280   | 290 | 11     |                   | Identifies which fee was assessed, left-justified |
| Delimiter                         | 291   | 291 | 1      |                   | All fields delimited by ";"                       |

## Authorization Parameter Summary Report (SI737010-AA)

An overview of the Authorization Parameter Summary Report (SI737010-AA) is listed in table format.

### Report overview

|              |                                                               |
|--------------|---------------------------------------------------------------|
| Title        | Authorization Parameter Summary Report                        |
| Generated by | Mastercard Dual Message Authorization System                  |
| Purpose      | To provide customers with a list of authorization parameters. |



|                      |                                                                                                                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description          | <p>This report includes a comprehensive list of the parameters that the customer uses. It includes global parameters and Stand-In parameters.</p> <p>Mastercard provides updates to this report when customers' parameters change.</p> |
| Frequency            | Weekly                                                                                                                                                                                                                                 |
| Distribution Methods | <p>Paper (Sent to the authorization contact established on the Principal Member Questionnaire); Mastercard Operational Reports, Mastercard File Express, complex-to-complex, bulk ID (T300)</p>                                        |

## Report sample

Report sample of the Authorization Parameter Summary Report (SI737010-AA).

| SI737010-AA                            |                                                     |                                         |                       | MASTERCARD INTERNATIONAL               |                         | RUN DATE 10/20/24 |  |
|----------------------------------------|-----------------------------------------------------|-----------------------------------------|-----------------------|----------------------------------------|-------------------------|-------------------|--|
| ICA: 1001                              |                                                     |                                         |                       | AUTHORIZATION PARAMETER SUMMARY REPORT |                         | PAGE: 1           |  |
| ACQUIRER PARAMETERS N/A                |                                                     |                                         |                       |                                        |                         |                   |  |
| ISSUER PARAMETERS                      |                                                     |                                         |                       |                                        |                         |                   |  |
| ACCOUNT RANGE: FROM: 51477900000       |                                                     |                                         |                       |                                        |                         |                   |  |
| THRU: 51477999999                      |                                                     |                                         |                       |                                        |                         |                   |  |
| GLOBAL PARAMETERS                      |                                                     |                                         |                       |                                        |                         |                   |  |
| PRODUCT CODE                           | CARDHOLDER BILLING CURRENCY CODE                    | CVC2 PARTICIPANT                        | AVS LEVEL PARTICIPANT | RPCS PARTICIPANT                       | PAN MAPPING PARTICIPANT |                   |  |
| MPC                                    | 840                                                 | Y                                       | 4                     | Y                                      | N                       |                   |  |
| EXPIRATION DATE PARTICIPANT            | POS BALANCE INQUIRY PARTICIPANT                     | ACQUIRER REVERSAL MESSAGE PARTICIPATION |                       |                                        |                         |                   |  |
| Y                                      | N                                                   | Y ISSUER RECEIVES THE 0400 MESSAGE      |                       |                                        |                         |                   |  |
| REAL-TIME SUBSTANTIATION PARTICIPANT   | INCONTROL REAL CARD SPEND CONTROL                   |                                         |                       |                                        |                         |                   |  |
| N                                      | N                                                   |                                         |                       |                                        |                         |                   |  |
| RECEIVE SETTLEMENT AMOUNTS             | MPG BLOCKING PARTICIPANT                            | OPTIONAL NON-VALID CVC3 PROCESSING      |                       |                                        | CVC3 TRUNCATION IND     |                   |  |
| N                                      | N                                                   | N                                       |                       |                                        | N                       |                   |  |
| ISSUER PIN BLOCK FORMAT CODE           | BANKNET PIN PROCESSING                              | PVV ON-FILE-PARTICIPATION               |                       | MASTERCARD INCONTROL MAPPING           |                         |                   |  |
| 01                                     | M                                                   | MANDATORY                               |                       | N                                      |                         |                   |  |
| MERCHANT VERIFICATION SERVICE          | MASTERCARD ONLINE CHECKOUT SERVICE                  | ADC EVENT PARTICIPATION                 |                       | TPP IDENTIFICATION SERVICE             |                         |                   |  |
| N                                      | N                                                   | N                                       |                       | N                                      |                         |                   |  |
| INCONTROL FLEX CARD                    | INCONTROL ENHANCED ROUTING                          | MONEYSEND BLOCKING                      |                       | PAY WITH REWARDS                       |                         |                   |  |
| N                                      | N                                                   | Y                                       |                       | N                                      |                         |                   |  |
| STAND-IN PIN VERIFICATION BYPASS       | MASTERCARD DIGITAL ENABLEMENT SERVICE PARTICIPATION |                                         |                       | AUTOMATED NEGATIVE LISTING SERVICE     |                         |                   |  |
| Y                                      | N                                                   |                                         |                       | N                                      |                         |                   |  |
| POS ENABLED INSTALLMENTS PARTICIPATION | NAME VALIDATION SERVICE PARTICIPATION               |                                         |                       |                                        |                         |                   |  |
| N                                      | Y                                                   |                                         |                       |                                        |                         |                   |  |
| IN-FLIGHT COMMERCE BLOCKED RANGE       |                                                     |                                         |                       |                                        |                         |                   |  |
| PAYMENT TRANSACTION BLOCKED RANGE      |                                                     |                                         |                       |                                        |                         |                   |  |
| TRANSACTION BLOCKING SERVICE           |                                                     |                                         |                       |                                        |                         |                   |  |
| RANGE FROM:                            |                                                     | 51477900000                             |                       |                                        |                         |                   |  |
| RANGE TO:                              |                                                     | 51477999999                             |                       |                                        |                         |                   |  |
| ISSUER ICA NUMBER:                     |                                                     | 001001                                  |                       |                                        |                         |                   |  |
| COUNTRY CODE:                          |                                                     | 356                                     |                       |                                        |                         |                   |  |
| PROCESSING CODE:                       |                                                     | 09                                      |                       |                                        |                         |                   |  |
| RESPONSE CODE:                         |                                                     | 58                                      |                       |                                        |                         |                   |  |
| BILLING ICA:                           |                                                     | 001996                                  |                       |                                        |                         |                   |  |
| GROUP ID:                              |                                                     | 00227882217                             |                       |                                        |                         |                   |  |
| SCOPE:                                 |                                                     | ACQUIRER                                |                       |                                        |                         |                   |  |
| RANGE FROM:                            |                                                     | 51477900000                             |                       |                                        |                         |                   |  |
| RANGE TO:                              |                                                     | 51477999999                             |                       |                                        |                         |                   |  |
| ISSUER ICA NUMBER:                     |                                                     | 001001                                  |                       |                                        |                         |                   |  |
| COUNTRY CODE:                          |                                                     | 192                                     |                       |                                        |                         |                   |  |
| RESPONSE CODE:                         |                                                     | 05                                      |                       |                                        |                         |                   |  |
| BILLING ICA:                           |                                                     | 001001                                  |                       |                                        |                         |                   |  |
| GROUP ID:                              |                                                     | 00165458848                             |                       |                                        |                         |                   |  |
| SCOPE:                                 |                                                     | ACQUIRER                                |                       |                                        |                         |                   |  |
| GARS PARTICIPANT                       | AUTHORIZATION REFERRAL                              | INTERNATIONAL GARS                      |                       |                                        | SECURITY NUMBER         |                   |  |
| Y                                      | 8009557070                                          | 8047477200                              |                       |                                        |                         |                   |  |
| LOST/STOLEN                            |                                                     | CUSTOMER SERVICE                        |                       |                                        |                         |                   |  |

## Authorization Parameter Summary Report (SI737010-AA), page 2

SI737010-AA  
ICA: 1001  
ACQUIRER PARAMETERS N/A  
ISSUER PARAMETERS

MASTERCARD INTERNATIONAL  
AUTHORIZATION PARAMETER SUMMARY REPORT

RUN DATE 10/20/24  
PAGE: 2

ACCOUNT RANGE: FROM: 51477900000  
THRU: 51477999999

### STAND-IN PARAMETERS

UTC. OFFSET: 0000 + OR - + D.S.T. START DATE: 000000 TIME: 0000  
END DATE: 000000 TIME: 0000

| LOCAL TIME | OPEN | CLSE | OPEN | CLSE | OPEN | CLSE |
|------------|------|------|------|------|------|------|
| SUN        | 0000 | 2400 | 0000 | 0000 | 0000 | 0000 |
| MON        | 0000 | 2400 | 0000 | 0000 | 0000 | 0000 |
| TUE        | 0000 | 2400 | 0000 | 0000 | 0000 | 0000 |
| WED        | 0000 | 2400 | 0000 | 0000 | 0000 | 0000 |
| THU        | 0000 | 2400 | 0000 | 0000 | 0000 | 0000 |
| FRI        | 0000 | 2400 | 0000 | 0000 | 0000 | 0000 |
| SAT        | 0000 | 2400 | 0000 | 0000 | 0000 | 0000 |

### HOLIDAYS FOR CALL REFERRAL CENTER:

| DATE    | OPEN | CLSE | OPEN | CLSE | DATE    | OPEN | CLSE | OPEN | CLSE |
|---------|------|------|------|------|---------|------|------|------|------|
| 01. N/A | N/A  | N/A  | N/A  | N/A  | 14. N/A | N/A  | N/A  | N/A  | N/A  |
| 02. N/A | N/A  | N/A  | N/A  | N/A  | 15. N/A | N/A  | N/A  | N/A  | N/A  |
| 03. N/A | N/A  | N/A  | N/A  | N/A  | 16. N/A | N/A  | N/A  | N/A  | N/A  |
| 04. N/A | N/A  | N/A  | N/A  | N/A  | 17. N/A | N/A  | N/A  | N/A  | N/A  |
| 05. N/A | N/A  | N/A  | N/A  | N/A  | 18. N/A | N/A  | N/A  | N/A  | N/A  |
| 06. N/A | N/A  | N/A  | N/A  | N/A  | 19. N/A | N/A  | N/A  | N/A  | N/A  |
| 07. N/A | N/A  | N/A  | N/A  | N/A  | 20. N/A | N/A  | N/A  | N/A  | N/A  |
| 08. N/A | N/A  | N/A  | N/A  | N/A  | 21. N/A | N/A  | N/A  | N/A  | N/A  |
| 09. N/A | N/A  | N/A  | N/A  | N/A  | 22. N/A | N/A  | N/A  | N/A  | N/A  |
| 10. N/A | N/A  | N/A  | N/A  | N/A  | 23. N/A | N/A  | N/A  | N/A  | N/A  |
| 11. N/A | N/A  | N/A  | N/A  | N/A  | 24. N/A | N/A  | N/A  | N/A  | N/A  |
| 12. N/A | N/A  | N/A  | N/A  | N/A  | 25. N/A | N/A  | N/A  | N/A  | N/A  |
| 13. N/A | N/A  | N/A  | N/A  | N/A  | 26. N/A | N/A  | N/A  | N/A  | N/A  |

### Authorization Parameter Summary Report (SI737010-AA), page 3

|                                           |                                        |                  |                      |          |
|-------------------------------------------|----------------------------------------|------------------|----------------------|----------|
| SI737010-AA                               | MASTERCARD INTERNATIONAL               |                  | RUN DATE             | 10/20/24 |
| ICA: 1001                                 | AUTHORIZATION PARAMETER SUMMARY REPORT |                  | PAGE:                | 3        |
| ACQUIRER PARAMETERS N/A                   |                                        |                  |                      |          |
| ISSUER PARAMETERS                         |                                        |                  |                      |          |
| ACCOUNT RANGE: FROM: 51477900000          |                                        |                  |                      |          |
| THRU: 51477999999                         |                                        |                  |                      |          |
| STAND-IN PARAMETERS                       |                                        |                  |                      |          |
| OPEN SIGNED-IN:                           | ACCOUNT FILE                           | TRANSACTION      | DECISION MATRIX:     | PREMIUM  |
|                                           | C                                      | C                | ACCUMULATIVE         | C        |
| OPEN SIGNED-OUT:                          | ACCOUNT FILE                           | TRANSACTION      | ACCUMULATIVE         | PREMIUM  |
|                                           | C                                      | C                | C                    | C        |
| CLOSED SIGNED-IN:                         | ACCOUNT FILE                           | TRANSACTION      | ACCUMULATIVE         | PREMIUM  |
|                                           | N                                      | N                | N                    | N        |
| CLOSED SIGNED-OUT:                        | ACCOUNT FILE                           | TRANSACTION      | ACCUMULATIVE         | PREMIUM  |
|                                           | N                                      | N                | N                    | N        |
| -----                                     |                                        |                  |                      |          |
| CARD LEVEL SUPPORT PARTICIPANT            | MULTI-CURRENCY LIMITS PARTICIPANT      |                  | STAND-IN PIN RETRIES |          |
| N                                         | N                                      |                  | 00000                |          |
| -----                                     |                                        |                  |                      |          |
| STORE AND FORWARD CONTROLS                |                                        |                  |                      |          |
| DELIVERY TYPE:                            |                                        |                  |                      |          |
| NUMBER OF SIMULTANEOUS SESSIONS: 10       |                                        |                  |                      |          |
| DELAY TIME BETWEEN SESSIONS: 01.0 SECONDS |                                        |                  |                      |          |
| ADVICE RESPONSE WAIT TIME: 030 SECONDS    |                                        |                  |                      |          |
| AUTOMATIC DELETION OPTION: N              |                                        |                  |                      |          |
| -----                                     |                                        |                  |                      |          |
| TRANSACTION PROCESSING PARAMETERS:        |                                        |                  |                      |          |
| TARGET COUNTRY                            | PROMO CODE                             | CAT LEVEL        | PRODUCT ID           | MCC/TCC  |
| GLOBAL                                    | *****                                  | 0                | MPC                  | MCC      |
| MCC LIMITS:                               |                                        |                  |                      |          |
| MCC                                       | CARD PRESENT                           | CARD NOT PRESENT | CURRENCY CODE        |          |
| 6538                                      | 0                                      | 0                | 840                  |          |
| -----                                     |                                        |                  |                      |          |
| TRANSACTION PROCESSING PARAMETERS:        |                                        |                  |                      |          |
| TARGET COUNTRY                            | PROMO CODE                             | CAT LEVEL        | PRODUCT ID           | MCC/TCC  |
| GLOBAL                                    | *****                                  | 4                | MPC                  | MCC      |
| MCC LIMITS:                               |                                        |                  |                      |          |
| MCC                                       | CARD PRESENT                           | CARD NOT PRESENT | CURRENCY CODE        |          |
| 7995                                      | 350                                    | 0                | 840                  |          |

## Authorization Parameter Summary Report (SI737010-AA), page 4

SI737010-AA  
ICA: 1001  
ACQUIRER PARAMETERS N/A  
ISSUER PARAMETERS

MASTERCARD INTERNATIONAL  
AUTHORIZATION PARAMETER SUMMARY REPORT

RUN DATE 10/20/24  
PAGE: 4

ACCOUNT RANGE: FROM: 51477900000  
THRU: 51477999999

### TRANSACTION PROCESSING PARAMETERS:

| TARGET<br>COUNTRY<br>GLOBAL | PROMO<br>CODE<br>***** | CAT<br>LEVEL<br>2 | PRODUCT<br>ID<br>MPC | MCC/TCC<br>TCC | CURRENCY<br>CODE<br>840 |
|-----------------------------|------------------------|-------------------|----------------------|----------------|-------------------------|
| TCC VALUES                  |                        |                   |                      |                |                         |
| T - MAIL/PHONE              |                        |                   | CARD PRESENT         |                | CARD NOT PRESENT        |
| R - RETAIL                  |                        | 100               | 0                    | 0              | 0                       |
| F - RESTAURANT              |                        | 0                 | 0                    | 0              | 0                       |
| O - COLLEGE/HOSPITAL        |                        | 0                 | 0                    | 0              | 0                       |
| H - HOTEL/MOTEL             |                        | 100               | 0                    | 0              | 0                       |
| A - AUTOMOBILE              |                        | 100               | 0                    | 0              | 0                       |
| X - OTHER TRANSPORT         |                        | 100               | 0                    | 0              | 0                       |
| Z - ATM                     |                        | 0                 | 0                    | 0              | 0                       |
| U - UNIQUE                  |                        | 0                 | 0                    | 0              | 0                       |
| C - CASH ADVANCE            |                        | 0                 | 0                    | 0              | 0                       |
| P - PAYMENT                 |                        | 0                 | 0                    | 0              | 0                       |

## Authorization Parameter Summary Report (SI737010-AA), page 5

SI737010-AA  
ICA: 1001  
ACQUIRER PARAMETERS N/A  
ISSUER PARAMETERS

MASTERCARD INTERNATIONAL  
AUTHORIZATION PARAMETER SUMMARY REPORT

RUN DATE 10/20/24  
PAGE: 5

ACCOUNT RANGE: FROM: 51477900000  
THRU: 51477999999

### TRANSACTION PROCESSING PARAMETERS:

| TARGET<br>COUNTRY<br>GLOBAL | PROMO<br>CODE<br>***** | CAT<br>LEVEL<br>* | PRODUCT<br>ID<br>MPC | MCC/TCC<br>TCC | CURRENCY<br>CODE<br>840 |
|-----------------------------|------------------------|-------------------|----------------------|----------------|-------------------------|
| TCC VALUES                  |                        |                   |                      |                |                         |
| T - MAIL/PHONE              |                        |                   | CARD PRESENT         |                | CARD NOT PRESENT        |
| R - RETAIL                  |                        | 500               | 500                  | 500            | 500                     |
| F - RESTAURANT              |                        | 750               | 750                  | 750            | 750                     |
| O - COLLEGE/HOSPITAL        |                        | 450               | 450                  | 450            | 450                     |
| H - HOTEL/MOTEL             |                        | 1200              | 1200                 | 1200           | 1200                    |
| A - AUTOMOBILE              |                        | 1000              | 1000                 | 1000           | 1000                    |
| X - OTHER TRANSPORT         |                        | 1600              | 1600                 | 1600           | 1600                    |
| Z - ATM                     |                        | 0                 | 0                    | 0              | 0                       |
| U - UNIQUE                  |                        | 150               | 150                  | 150            | 150                     |
| C - CASH ADVANCE            |                        | 500               | 500                  | 500            | 500                     |
| P - PAYMENT                 |                        | 0                 | 0                    | 0              | 0                       |

### Authorization Parameter Summary Report (SI737010-AA), page 6

SI737010-AA  
ICA: 1001  
ACQUIRER PARAMETERS N/A  
ISSUER PARAMETERS

MASTERCARD INTERNATIONAL  
AUTHORIZATION PARAMETER SUMMARY REPORT

RUN DATE 10/20/24  
PAGE: 6

ACCOUNT RANGE: FROM: 51477900000  
THRU: 51477999999

-----

CURRENCY CODE: 840

ACCUMULATIVE LIMITS:

| DAYS | COUNT | AMOUNT |
|------|-------|--------|
| 01   | 6     | 2000   |
| 02   | 8     | 2000   |
| 03   | 10    | 2000   |
| 04   | 12    | 2000   |

CASH ADVANCE: DAYS AMOUNT  
0 0

-----

PREMIUM CONTROLS:

| DAYS | COUNT |
|------|-------|
| 4    | 999   |

-----

STAND-IN BLOCKED RANGE:

FROM: 51477900000000000000 THRU: 51477999999999999999

### Authorization Parameter Summary Report (SI737010-AA), page 7

SI737010-AA  
ICA: 1001  
ACQUIRER PARAMETERS N/A  
ISSUER PARAMETERS

MASTERCARD INTERNATIONAL  
AUTHORIZATION PARAMETER SUMMARY REPORT

RUN DATE 10/20/24  
PAGE: 7

ACCOUNT RANGE: FROM: 51477900000  
THRU: 51477999999

### Authorization Parameter Summary Report (SI737010-AA), page 8

SI737010-AA  
ICA: 1001  
ACQUIRER PARAMETERS N/A  
ISSUER PARAMETERS

MASTERCARD INTERNATIONAL  
AUTHORIZATION PARAMETER SUMMARY REPORT

RUN DATE 10/20/24  
PAGE: 8

ACCOUNT RANGE: FROM: 51477900000  
THRU: 51477999999

-----

ON-BEHALF SERVICES

| DESCRIPTION                         |
|-------------------------------------|
| EMS REAL-TIME AUTHORIZATION REQUEST |
| UNKNOWN SERVICE                     |

## Field descriptions

The fields on the Authorization Parameter Summary Report (SI737010-AA) are listed in table format.

The Authorization Parameter Summary Report (SI737010-AA) includes:

- Header Information
- Global Parameters
- Stand-In Parameters

**NOTE: Mastercard provides an updated version of this report for customers when their parameters change.**

## Global Parameters

The Global Parameters fields on the Authorization Parameter Summary Report (SI737010-AA) are listed in table format.

The fields in the Global Parameters section may include the following data.

### Report fields

| Field                            | Description                                                                                                                                                                                                               |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product Code                     | Three-position field that identifies the financial network code.                                                                                                                                                          |
| Cardholder Billing Currency Code | Three-digit number that identifies the participant's currency code.                                                                                                                                                       |
| CVC 2 Participant                | A one-character flag that indicates whether a customer participates in CVC 2. <ul style="list-style-type: none"><li>• Y = Customer participates in CVC 2.</li><li>• N = Customer does not participate in CVC 2.</li></ul> |

| Field                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AVS Level Participant            | <p>A one-position field that indicates a customer's AVS service level.</p> <ul style="list-style-type: none"> <li>• 0 = AVS not supported.</li> <li>• 1 = Issuer provides AVS, receives full address data.</li> <li>• 2 = Issuer provides AVS, receives condensed address data. This level supports the algorithm that uses the first five numeric digits in an address (when scanning the address from left to right).</li> <li>• 3 = Issuer provides AVS, receives condensed address data. This level supports the algorithm that uses the first five numeric digits in an address. This algorithm stops searching for numbers after it encounters an alphabetic character or space (when scanning the address from left to right).</li> <li>• 4 = Issuer provides AVS, receives condensed postal/ZIP code and address data. This level supports the algorithm that uses only numeric digits (with a maximum of nine digits) in the cardholder postal/ZIP code, and the first five numeric digits in an address (when scanning the address from left to right).</li> </ul> |
| Payment Cancellation Participant | <p>A one-character flag that indicates whether a customer participates in Payment Cancellation.</p> <ul style="list-style-type: none"> <li>• Y = Customer participates in Payment Cancellation.</li> <li>• N = Customer does not participate in Payment Cancellation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| PAN Mapping Participant          | <p>A one-character flag that indicates whether a customer participates in the Contactless Mapping Service.</p> <ul style="list-style-type: none"> <li>• Y = Customer participates in the Contactless Mapping Service.</li> <li>• N = Customer does not participate in the Contactless Mapping Service.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expiration Date Participant      | <p>A one-character flag that indicates whether a customer participates in the Expired Card Override service.</p> <ul style="list-style-type: none"> <li>• Y = Customer participates in the Expired Card Override service.</li> <li>• N = Customer does not participate in the Expired Card Override service.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



| Field                                   | Description                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POS Balance Inquiry Participant         | <p>A one-character flag that indicates whether the account range participates in POS balance inquiries.</p> <ul style="list-style-type: none"> <li>Y = Account range participates in POS balance inquiries.</li> <li>N = Account range does not participate in POS balance inquiries.</li> </ul>                    |
| Acquirer Reversal Message Participation | <p>A one-character flag that indicates whether the issuer participates in the reversal request message functionality.</p> <ul style="list-style-type: none"> <li>Y = Issuer receives the 0400 message.</li> <li>N = Issuer does not receive the 0400 message.</li> </ul>                                            |
| Real-time Substantiation                | <p>A one-character flag that indicates whether the issuer supports real-time substantiation at the POS.</p> <ul style="list-style-type: none"> <li>Y = Issuer supports real-time substantiation.</li> <li>N = Issuer does not support real-time substantiation.</li> </ul>                                          |
| In Control Real Card Spend Control      | <p>A one-character flag that indicates whether the issuer participates in the Consumer Controls services.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in Consumer Controls services.</li> <li>N = Issuer does not participate in Consumer Controls services.</li> </ul>                      |
| Mastercard Spend Alerts Service         | <p>A one-character flag that indicates whether the issuer participates in the Mastercard Spend Alerts Service.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in Mastercard Spend Alerts Service.</li> <li>N = Issuer does not participate in Mastercard Spend Alerts Service.</li> </ul>       |
| Mastercard Spend Controls Service       | <p>A one-character flag that indicates whether the issuer participates in the Mastercard Spend Controls Service.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in Mastercard Spend Controls Service.</li> <li>N = Issuer does not participate in Mastercard Spend Controls Service.</li> </ul> |

| Field                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Receive Settlement Amounts          | <p>A one-character flag that indicates whether the issuer receives settlement amount-related data elements.</p> <ul style="list-style-type: none"> <li>Y = Issuer receives settlement amount-related data elements.</li> <li>N = Issuer does not receive settlement amount-related data elements.</li> </ul>                                                                                                                                                                                       |
| MPG Blocking Participant            | <p>A one-character flag that indicates whether the issuer participates in the Mastercard Payment Gateway (MPG) Authorization Blocking service.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in the MPG Blocking service.</li> <li>N = Issuer does not participate in the MPG Blocking service.</li> </ul>                                                                                                                                                                    |
| Optional Non-Valid CVC 3 Processing | <p>A one-character flag that indicates whether the issuer participates in Optional Non-Valid CVC 3 Processing.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in the Optional Non-Valid CVC 3 Processing.</li> <li>N = Issuer does not participate in the Optional Non-Valid CVC 3 Processing.</li> </ul>                                                                                                                                                                      |
| CVC 3 Truncation Ind                | <p>A one-character flag that indicates whether the CVC 3 transaction is valid, that is, the value in DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 2 (On-behalf [OB] Result 1). This indicator is available to issuers participating in the Non-Valid CVC Processing option and participating in one of the CVC 3 services.</p> <ul style="list-style-type: none"> <li>Y = VC 3 transaction is valid.</li> <li>N = CVC 3 transaction is not valid.</li> </ul> |
| Issuer PIN Block Format Code        | <p>Applies only to the Europe region. A two-position field that indicates the issuer's PIN block format.</p> <ul style="list-style-type: none"> <li>01 = ANSI 1</li> <li>10 = ISO Format 0</li> </ul>                                                                                                                                                                                                                                                                                              |
| Banknet PIN Processing              | <p>A one-position field that indicates the service platform on which an issuer's PIN processing will be performed.</p> <ul style="list-style-type: none"> <li>B = Banknet PIN processing</li> <li>M = MDS PIN processing</li> </ul>                                                                                                                                                                                                                                                                |

| Field                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PVV On-File Participation                         | <p>Applies only to the Europe region. Supports PIN Validation using PIN Verification Value (PVV) on-file that issuers submit as basis to override PVV data provided in the card's magnetic stripe. Valid values are as follows:</p> <ul style="list-style-type: none"> <li>No = Does not participate in PVV/PIN Offset on File service.</li> <li>Optional = On File, the Authorization System retrieves the PVV value. If the PVV value is found, validates using that value. If the PVV value is not found, the data in the track is used.</li> <li>Mandatory = On File, the Authorization System uses the data on file if found to validate the PIN.</li> </ul> <p>If the PVV value is not found, PIN validation indicates mandatory PVV not on file as result.</p> |
| Mastercard In Control Mapping                     | <p>A one-character flag that indicates whether the issuer participates in the Mastercard In Control™ service.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in the Mastercard In Control service.</li> <li>N = Issuer does not participate in the Mastercard In Control service.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Mastercard Online Checkout Service                | <p>A one-character flag that indicates whether the issuer participates in Mastercard online checkout service.</p> <p>N = Issuer does not participate in Mastercard online checkout service.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| ADC Event Participation (Account Data Compromise) | <p>A one-character flag that indicates whether compromised event data has been found.</p> <ul style="list-style-type: none"> <li>Y = Compromised Event Data Found.</li> <li>N = Compromised Event Data Not Found.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| TPP Identification Service                        | <p>A one-character flag that indicates whether the issuer participates in the Third Party Processor (TPP) service.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in the TPP service.</li> <li>N = Issuer does not participate in the TPP service.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Field                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MoneySend Blocking            | <p>A one-character flag that identifies whether a receiving institution (RI) account range is blocked from processing transactions via the MoneySend platform.</p> <ul style="list-style-type: none"> <li>Y = MoneySend Payment Transactions are not allowed because the country, RI, or RI's account range associated with the ICA/account range is blocked from processing transactions via the MoneySend platform.</li> <li>N = MoneySend Payment Transactions are allowed to the ICA/account range.</li> </ul> |
| IPC Fraud Control             | <p>A one-character flag that indicates whether the issuer participates in IPC Fraud Control service.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in the IPC Fraud Control service.</li> <li>N = Issuer does not participate in the IPC Fraud Control service.</li> </ul>                                                                                                                                                                                                                    |
| Receive ATM Additional Data   | <p>Applies only to the Europe region. A one-character flag that indicates support of ATM additional data in DE 48, subelement 58 (ATM Additional Data) by customers participating in the Sweden Domestic Authorization Switching Service (SASS).</p> <ul style="list-style-type: none"> <li>Y = Issuer supports ATM additional data in DE 48, subelement 58.</li> <li>N = Issuer does not support ATM additional data in DE 48, subelement 58.</li> </ul>                                                          |
| PIN Management Participant    | <p>A one-character flag that indicates whether the issuer participates in the Chip PIN Management service.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in the PIN Management service.</li> <li>N = Issuer does not participate in the PIN Management service.</li> </ul>                                                                                                                                                                                                                    |
| Domestic Debit ATM Processing | <p>A one-character flag that indicates whether the issuer participates in domestic Debit ATM processing.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in domestic Debit ATM processing.</li> <li>N = Issuer does not participate in the domestic Debit ATM processing.</li> </ul>                                                                                                                                                                                                            |

| Field                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In-Flight Commerce Blocked Range  | <p>Two 19-digit account range numbers that indicate the from and the to account ranges to block for in-flight commerce service.</p> <p>In-flight commerce (IFC) is a service that identifies transactions conducted via interactive video terminals by passengers on a long-distance airplane flight.</p>                                                                                                                  |
| Stand-In PIN Verification Bypass  | <p>A one-character flag indicating whether the Stand-In Authorization will bypass or perform a PIN Verification Test. Issuers that want Stand-In to perform this test are required to participate in a PIN validation service.</p> <ul style="list-style-type: none"> <li>Y = Stand-In Authorization will bypass PIN Verification Test.</li> <li>N = Stand-In Authorization will perform PIN Verification Test.</li> </ul> |
| MDES Participation                | <p>A one-character flag indicating whether the issuer participates in MDES.</p> <ul style="list-style-type: none"> <li>Y = Issuer participates in MDES.</li> <li>N = Issuer does not participate in MDES.</li> </ul>                                                                                                                                                                                                       |
| Payment Transaction Blocked Range | <p>Two 19-digit numeric account range numbers that indicate the from and the to account ranges to block for this transaction type.</p> <p>A Payment Transaction is a transaction in which a Payment Service Provider facilitates the transfer of funds to an issuer for posting to a cardholder's Mastercard account.</p>                                                                                                  |
| MIP Transaction Blocking Service  | <p>MIP Transaction Blocking is a service that enables an issuer to decline specified types of transaction requests by issuer ICA number or account range, in combination with additional transaction parameters.</p>                                                                                                                                                                                                       |
|                                   | <p>Range From Range To</p> <p>Two 11-digit account range numbers that indicate the from and the to account ranges to block for the MIP Transaction Blocking service.</p>                                                                                                                                                                                                                                                   |

| Field             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Response Code</p> <p>Indicates the response code applicable to the transaction block, based on the response codes available for the Authorization Request Response/0110 message. The following list identifies the valid response code values:</p> <ul style="list-style-type: none"> <li>• 03 (Invalid merchant)</li> <li>• 04 (Capture card)</li> <li>• 05 (Do not honor)</li> <li>• 12 (Invalid transaction)</li> <li>• 57 (Transaction not permitted to issuer/ cardholder)</li> <li>• 58 (Transaction not permitted to acquirer/ terminal)</li> </ul>                                                                                                                                                                                                                                                                                                            |
|                   | <p>Scope</p> <p>Indicates whether transactions are blocked for all routes or transactions are only blocked from going to the Stand-In Authorization when the issuer is unavailable.</p> <ul style="list-style-type: none"> <li>• All</li> <li>• Stand-In</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Blocking Criteria | <p>Transaction blocking criteria. Indicates that transactions will be blocked based on the issuing ICA number or account range, and one or more of the following transaction parameters:</p> <ul style="list-style-type: none"> <li>• DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)</li> <li>• DE 18 (Merchant Type)</li> <li>• DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)</li> <li>• DE 61 (Point-of-Service [POS] Data), subfield 10 (Card Activated Terminal Level [CAT])</li> <li>• DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code)</li> </ul> <p>In the first example, the report sample shows that the Merchant Type, value 5555 is the transaction blocking criteria; in the second example, POS Terminal PAN Entry Mode, value 02 is the transaction blocking criteria.</p> |

| Field                  | Description                                                                                                                                                                                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GARS Participant       | <p>A one-character flag that indicates whether the customer is a Global Automated Referral System (GARS) participant.</p> <ul style="list-style-type: none"> <li>Y = Customer participates in the GARS service.</li> <li>N = Customer does not participate in the GARS service.</li> </ul> |
| Authorization Referral | Indicates the phone number GARS uses to contact the issuer on domestic-call referral responses.                                                                                                                                                                                            |
| International GARS     | Indicates the phone number GARS uses to contact the issuer on international call referral responses.                                                                                                                                                                                       |
| Security Number        | This phone number is not displayed.                                                                                                                                                                                                                                                        |
| Lost/Stolen            | This phone number is not displayed.                                                                                                                                                                                                                                                        |
| Customer Service       | This phone number is not displayed.                                                                                                                                                                                                                                                        |

## Stand-In Parameters

Descriptions of the Stand-In Parameter fields in the Authorization Parameter Summary Report (SI737010-AA).

The Authorization Parameter Summary Report (SI737010-AA) Stand-In parameters include the following data.

### Report fields

| Field              | Description                                                                                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hours of Operation | Indicates the hours of operation for the issuer's authorization system.                                                                                                                                                                     |
|                    | <p>Time difference between the customer's local time and Coordinated Universal Time (UTC).</p> <p>UTC Offset represents whether the local time is ahead (+) or behind (-) UTC by the number of hours indicated in the UTC Offset field.</p> |
| D.S.T.             | Daylight Saving Time (D.S.T.) represents the daylight-saving time start and end dates and times for the customer.                                                                                                                           |

| Field                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local Time                         | Indicates the open and close local time for the customer.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Holidays for Call Referral Center  | Definition of 26 holidays for the issuer's Call Referral Center, with up to two openings and closings per holiday.                                                                                                                                                                                                                                                                                                                                                                                                 |
| Decision Matrix                    | <p>A matrix that contains values that represent a customer's choice of Stand-In Authorization responses to transactions if a transaction goes to Stand-In Authorization in a given state.</p> <ul style="list-style-type: none"> <li>• C = Call referral</li> <li>• N = Decline</li> <li>• A = Continue with Stand-In Authorization</li> <li>• P = Capture card</li> </ul> <p>Open Signed In, Closed Signed In</p> <p>Indicates times when the issuer's authorization center is open versus when it is closed.</p> |
| Open Signed Out, Closed Signed Out | Indicates times when the issuer's authorization center is signed in versus when it is signed out (for online issuers).                                                                                                                                                                                                                                                                                                                                                                                             |
| Card Level Support Participant     | <p>A one-character flag that indicates if the customer is a Card Level Support participant.</p> <ul style="list-style-type: none"> <li>• Y = Customer participates in Card Level Support.</li> <li>• N = Customer does not participate in Card Level Support.</li> </ul>                                                                                                                                                                                                                                           |
| Multi-Currency Limits Participant  | <p>A one-character flag that indicates if the customer participates in multi-currency limits in Stand-In Authorization.</p> <ul style="list-style-type: none"> <li>• Y = Customer participates in multi-currency limits.</li> <li>• N = Customer does not participate in multi-currency limits.</li> </ul>                                                                                                                                                                                                         |
| Stand-In PIN Retries               | Applies only to the Europe region. Indicates the number of invalid PIN attempts allowed using a value of 1-5. This parameter applies only to the Online PIN Validation in Stand-In Authorization.                                                                                                                                                                                                                                                                                                                  |



| Field                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Store-and-Forward (SAF) Controls | <p>Additional parameters by which issuers can receive their SAF messages.</p> <p>Delivery Type</p> <p>Unsolicited: Issuer receives SAF messages after signing on to the Mastercard Network using a Network Management Request/0800 message containing DE 70 (Network Management Information), value 001 (Sign-on [by BIN]) or value 061 (Group sign on [by Mastercard group sign on]).</p> <p>Number of Simultaneous Sessions</p> <p>Represents the Number of Simultaneous Advice Messages that can be processed simultaneously. There is a maximum of 10 advice messages that can be processed simultaneously. Default is 10.</p> |
| Advice Response Wait Time        | <p>Advice response wait time in seconds (20 seconds) before SAF session termination (Default is a maximum of 20 seconds).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Automatic Deletion Option        | <p>Indicates whether the Stand-In System should delete the SAF message if a response is not received from the issuer after a given number of delivery attempts. The Stand-In Authorization will attempt to resend the SAF messages until:</p> <ul style="list-style-type: none"> <li>• A response is received</li> <li>• The resend limit has been reached</li> <li>• The advice is on file for four days (Default is No)</li> </ul>                                                                                                                                                                                               |
| No Response Resend Limit         | <p>Number of times (1–20) that the Stand-In Authorization attempts to resend a SAF message without response from the issuer before auto-deleting SAF messages. If Automatic Deletion Option is selected, default is 3.</p>                                                                                                                                                                                                                                                                                                                                                                                                         |

| Field                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transaction Processing Parameters | <p>This section may vary depending on the number of rules that apply to a customer. The report reflects all rules that apply to that particular customer.</p> <p>Target Country</p> <p>Indicates the acquirer country for which the limits are applied.</p> <p>Mastercard has established minimum and maximum values for certain parameters. Mastercard will set the minimum limits as default values to establish customer's limits.</p> <ul style="list-style-type: none"> <li>Foreign Default: Established limits outside of the country of issuance.</li> <li>Global: Established limits for all customers by product code, acceptor business code/merchant category code (MCC), and/or cardholder-activated terminal (CAT) level.</li> </ul> |
| Promo Code (Promotion Code)       | <p>Indicates the promotion code for which the limits are applied. The six-character alphanumeric codes an issuer establishes to identify transactions meeting the requirements for an issuer's promotional program on a card range.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| CAT Level                         | <p>Lists a one-character numeric code that identifies transactions that occur at a CAT. A CAT is a cardholder-activated terminal that is usually unattended and accepts payment cards, debit, charge, and proprietary cards for a card range. CATs are defined by levels.</p> <ul style="list-style-type: none"> <li>0 = Not a CAT transaction</li> <li>1 = Automated dispensing machines with PIN</li> <li>2 = Self-service terminals</li> <li>3 = Limited-amount terminals</li> <li>4 = In-flight commerce terminals</li> <li>6 = Electronic commerce</li> <li>7 = Transponder</li> </ul>                                                                                                                                                       |
| Product ID                        | <p>Lists the three-character code that identifies the type of card program for an account, such as Gold Mastercard® card.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| MCC/TCC                           | <p>Acceptor business code/merchant category code (MCC) and the transaction category code (TCC) for which limits are applied.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Field                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Currency Code                                               | Three-digit number that identifies the customer's chosen currency code for amount limits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Card Present Amount                                         | Transaction amount limit when a card is present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Card Not Present Amount                                     | Transaction amount limit when a card is not present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Product Graduation Transaction Limits Processing Parameters | This section of the report depends on whether the account range has product graduated limits. If the account range does not have product graduated limits, this section does not display.                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                             | <p>Target Country</p> <p>Indicates the acquirer country for which the limits are applied.</p> <p>Mastercard has established minimum and maximum values for certain parameters. Mastercard will set the minimum limits as default values to establish customer's limits.</p> <ul style="list-style-type: none"> <li>• Foreign Default: Established limits outside of the country of issuance.</li> <li>• Global: Established limits for all customers by product code, MCC, and/or CAT level.</li> </ul>                                                                                       |
|                                                             | <p>Promo Code (Promotion Code)</p> <p>Indicates the promotion code for which the limits are applied. The six-character alphanumeric codes an issuer establishes to identify transactions meeting the requirements for an issuer's promotional program on a card range.</p>                                                                                                                                                                                                                                                                                                                    |
|                                                             | <p>One-character numeric code that identifies transactions that occur at a CAT. A CAT is a cardholder-activated terminal that is usually unattended and accepts bankcards, debit, charge, and proprietary cards for a card range. CATs are defined by levels.</p> <ul style="list-style-type: none"> <li>• 0 = Not a CAT transaction</li> <li>• 1 = Automated dispensing machines with PIN</li> <li>• 2 = Self-service terminals</li> <li>• 3 = Limited-amount terminals</li> <li>• 4 = In-flight commerce terminals</li> <li>• 6 = Electronic commerce</li> <li>• 7 = Transponder</li> </ul> |

| Field                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>Product ID</p> <p>Lists the three-character code that identifies the type of card program for an account, such as Gold Mastercard® card.</p>                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>MCC/TCC</p> <p>Lists the acceptor business code (MCC) and the transaction category code (TCC) for which limits are applied.</p>                                                                                                                                                                                                                                                                                                                                                 |
|                                                           | <p>Currency Code</p> <p>Three-digit number that identifies the customer's chosen currency code for amount limits.</p>                                                                                                                                                                                                                                                                                                                                                              |
|                                                           | <p>Card Present Amount</p> <p>Transaction amount limit when a card is present.</p>                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                           | <p>Card Not Present Amount</p> <p>Transaction amount limit when a card is not present.</p>                                                                                                                                                                                                                                                                                                                                                                                         |
| Product Graduation Limits Authorization Parameter Summary | This section of the report may vary depending on whether the account range has product graduated limits. If the account range does not have product graduated limits, this section does not display.                                                                                                                                                                                                                                                                               |
| Accumulative Limits                                       | <p>Issuer established limits for accumulated transaction counts and amounts over a specified number of days. The accumulation limits are for the number of transactions and amounts. Stand-In Authorization will not approve any amount that raises the accumulative amount already approved above the specified limit for the specified period of days.</p> <p>Currency Code</p> <p>Three-digit number that identifies the customer's chosen currency code for amount limits.</p> |
| Premium Controls                                          | <p>Indicates specific limits that Stand-In Authorization applies to accounts listed in the Account File as Premium Listings. The Premium Listings designation is a method of identifying cardholder accounts that should have higher authorization limits than standard limits for the card type.</p> <p>During the Accumulative Limits test for Premium accounts, Stand-In Authorization applies the Days and Count Limit parameters.</p>                                         |

| Field                  | Description                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group and Series       | Identifies a range of card account numbers listed in the Electronic Warning Bulletin file submitted by Mastercard on behalf of an issuer to the Account Management System for the purpose of notifying a merchant to capture any card bearing an account number that falls within that range.                                                                   |
| Stand-In Blocked Range | Identifies a range of card numbers within a BIN that have not been issued to cardholders or card ranges previously blocked and should be unblocked. Range Blocking only applies for transactions processed by Stand-In. Card numbers within blocked ranges do not automatically appear in the Account Management System (AMS) Electronic Warning Bulletin file. |
| On-behalf Services     | Lists the on-behalf service description for the on-behalf services that are enabled for an account range. There is a maximum of 10 on-behalf services.                                                                                                                                                                                                          |

## MIP Transaction Blocking ICA Level Block Summary (SI738010-AA)

An overview of the MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) is listed in table format.

### Report overview

|              |                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title        | MIP Transaction Blocking ICA Level Block Summary                                                                                                                                                                                            |
| Generated by | Mastercard Dual Message Authorization System                                                                                                                                                                                                |
| Purpose      | To provide customers with a tool to identify all BINs that are associated with ICAs that are subscribed to the MIP Transaction Blocking Service.                                                                                            |
| Description  | <p>This report includes a comprehensive list of the parameters for each BIN when the MIP Transaction Blocking Service is invoked at the ICA level.</p> <p>Mastercard provides updates to this report when customers' parameters change.</p> |
| Frequency    | Weekly                                                                                                                                                                                                                                      |

|                      |                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distribution Methods | Paper (Sent to the authorization contact established on the Principal Member Questionnaire); Mastercard Operational Reports, Mastercard File Express, complex-to-complex, bulk ID (T300) |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Report sample

Report sample of the MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) report.

MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) report

|                              |                              |                                              |             |              |             |                    |  |
|------------------------------|------------------------------|----------------------------------------------|-------------|--------------|-------------|--------------------|--|
| SI738010-AA                  |                              | MASTERCARD WORLDWIDE                         |             |              |             | RUN DATE: 01/03/24 |  |
| BILLING ICA: 999705          |                              | TRANSACTION BLOCKING ICA LEVEL BLOCK SUMMARY |             |              |             | PAGE NUMBER: 1     |  |
| ISSUER ICA NUMBER:           | 999705                       | 999705                                       | 999705      | 999705       | 999705      |                    |  |
| COUNTRY CODE:                |                              |                                              |             |              |             |                    |  |
| MERCHANT TYPE:               |                              |                                              |             |              |             |                    |  |
| CAT LEVEL CODE:              |                              |                                              |             |              |             |                    |  |
| POS TERMINAL PAN ENTRY MODE: | 01                           | 09                                           | 81          | 81           | 00          |                    |  |
| RESPONSE CODE:               | 57                           | 57                                           | 57          | 57           | 57          |                    |  |
| GROUP ID:                    | 00028284908                  | 00046095436                                  | 00047058959 | 00052674749  | 00055876285 |                    |  |
| SCOPE:                       | ACQUIRER                     | ACQUIRER                                     | ACQUIRER    | ACQUIRER     | ACQUIRER    |                    |  |
| ISSUER ICA NUMBER:           | 999705                       | 999705                                       | 999705      | 999705       | 999705      |                    |  |
| COUNTRY CODE:                |                              |                                              |             |              |             |                    |  |
| MERCHANT TYPE:               |                              |                                              |             |              |             |                    |  |
| CAT LEVEL CODE:              |                              |                                              |             |              |             |                    |  |
| POS TERMINAL PAN ENTRY MODE: | 81                           | 09                                           | 01          | 82           | 00          |                    |  |
| RESPONSE CODE:               | 57                           | 57                                           | 57          | 57           | 57          |                    |  |
| GROUP ID:                    | 00067167567                  | 00082566681                                  | 00084291304 | 00110269742  | 00181250795 |                    |  |
| SCOPE:                       | ACQUIRER                     | ACQUIRER                                     | ACQUIRER    | ACQUIRER     | ACQUIRER    |                    |  |
| ISSUER ICA NUMBER:           | 999705                       | 999705                                       | 999705      | 999705       | 999705      |                    |  |
| COUNTRY CODE:                |                              |                                              |             |              |             |                    |  |
| MERCHANT TYPE:               |                              |                                              |             |              |             |                    |  |
| CAT LEVEL CODE:              |                              |                                              |             |              |             |                    |  |
| POS TERMINAL PAN ENTRY MODE: | 01                           | 82                                           | 82          | 81           | 81          |                    |  |
| RESPONSE CODE:               | 57                           | 57                                           | 57          | 57           | 57          |                    |  |
| GROUP ID:                    | 00198301017                  | 00219857322                                  | 00223715828 | 00228861180  | 00234949396 |                    |  |
| SCOPE:                       | ACQUIRER                     | ACQUIRER                                     | ACQUIRER    | ACQUIRER     | ACQUIRER    |                    |  |
| ISSUER ICA NUMBER:           | 999705                       | 999705                                       | 999705      | 999705       | 999705      |                    |  |
| COUNTRY CODE:                |                              |                                              |             |              |             |                    |  |
| MERCHANT TYPE:               |                              |                                              |             |              |             |                    |  |
| CAT LEVEL CODE:              |                              |                                              |             |              |             |                    |  |
| POS TERMINAL PAN ENTRY MODE: | 09                           | 09                                           | 82          | 01           | 00          |                    |  |
| RESPONSE CODE:               | 57                           | 57                                           | 57          | 57           | 57          |                    |  |
| GROUP ID:                    | 00253644112                  | 00304199625                                  | 00330388807 | 003666147197 | 00370166599 |                    |  |
| SCOPE:                       | ACQUIRER                     | ACQUIRER                                     | ACQUIRER    | ACQUIRER     | ACQUIRER    |                    |  |
| ISSUER ICA NUMBER:           | 999705                       | 999705                                       | 999705      | 999705       | 999705      |                    |  |
| COUNTRY CODE:                |                              |                                              |             |              |             |                    |  |
| MERCHANT TYPE:               |                              |                                              |             |              |             |                    |  |
| CAT LEVEL CODE:              |                              |                                              |             |              |             |                    |  |
| POS TERMINAL PAN ENTRY MODE: | 00                           | 01                                           | 09          | 82           | 00          |                    |  |
| RESPONSE CODE:               | 57                           | 57                                           | 57          | 57           | 57          |                    |  |
| GROUP ID:                    | 00418862001                  | 00458249365                                  | 00470029468 | 00477192627  | 00735387291 |                    |  |
| SCOPE:                       | ACQUIRER                     | ACQUIRER                                     | ACQUIRER    | ACQUIRER     | ACQUIRER    |                    |  |
| -----                        |                              |                                              |             |              |             |                    |  |
| ASSOCIATED BIN RANGES:       |                              |                                              |             |              |             |                    |  |
| 56562110300 THRU 56562110399 | 56562110400 THRU 56562110499 | 56562110500 THRU 56562110599                 |             |              |             |                    |  |
| 56562111300 THRU 56562111399 | 56562111400 THRU 56562111499 | 56562111500 THRU 56562111599                 |             |              |             |                    |  |
| 56562113300 THRU 56562113399 | 56562113400 THRU 56562113499 | 56562113500 THRU 56562113599                 |             |              |             |                    |  |
| 56562114300 THRU 56562114399 | 56562114400 THRU 56562114499 | 56562114500 THRU 56562114599                 |             |              |             |                    |  |
| 56562116300 THRU 56562116399 | 56562116400 THRU 56562116499 | 56562116500 THRU 56562116599                 |             |              |             |                    |  |
| 56562117300 THRU 56562117399 | 56562117400 THRU 56562117499 | 56562117500 THRU 56562117599                 |             |              |             |                    |  |
| -----                        |                              |                                              |             |              |             |                    |  |
| * END OF DETAIL RECORDS *    |                              |                                              |             |              |             |                    |  |

## Field descriptions

The fields on the MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) report are listed in table format.

### Field descriptions

| Field                       | Description                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Billing ICA                 | ICA that is billed for the Issuer ICA Number Transaction Blocking Service activity. This ICA may be the same or different than the Issuer ICA number.                                                                                                                                                                                                                     |
| Issuer ICA                  | Unique six-digit ID number assigned by Mastercard to identify a customer or processing endpoint. Sequenced by ICA and listed numerically. This ICA may be the same or different than the Billing ICA number.                                                                                                                                                              |
| Country Code                | A three-digit, numeric code that identifies the country. Issuers may use these codes to establish higher or lower transaction limits for a country or countries where the transaction is acquired.                                                                                                                                                                        |
| Merchant Type               | DE 18 (Merchant Type) is the classification (acceptor business code [MCC]) of the merchant's type of business or service.                                                                                                                                                                                                                                                 |
| CAT Level Code              | DE 61 (Point-of Service [POS] Data), subfield 10 (Card Activated Terminal Level [CAT]) indicates whether the cardholder activated the terminal with the use of the card and the CAT security level.                                                                                                                                                                       |
| POS Terminal PAN Entry Mode | DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) indicates how the PAN was entered at the terminal.                                                                                                                                                                                                                                    |
| Response Code               | DE 39 (Response Code) defines the disposition of a previous message, or an action taken as a result of receipt of a previous message.<br><br>Response codes also are used to indicate approval or decline of a transaction. If an authorization is declined, the response code indicates the reason for rejection and may indicate an action to be taken at the acceptor. |
| Group ID                    | Identifies a grouping of issuer account ranges that use the same transaction criteria for routing to the same destination route.                                                                                                                                                                                                                                          |



| Field                 | Description                                                                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope                 | <p>Indicates where the block occurs.</p> <ul style="list-style-type: none"> <li>Acquirer: Indicates that the block occurred at the acquirer MIP.</li> <li>Stand-In: Indicates that the block occurred during Stand-In Authorization.</li> </ul> |
| Associated BIN Ranges | <p>Identifies a range of card numbers within a BIN that are associated to that blocking ICA. These blocks occur at the MIP and not in Stand-In Authorization. These ranges may or may not have been issued to cardholders.</p>                  |

## Authorization Summary by CAT Level Report (SI458010-AA)

An overview of the Authorization Summary by CAT Level Report (SI458010-AA) is listed in table format.

### Report overview

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title        | Authorization Summary by CAT Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Generated by | Mastercard Dual Message Authorization System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Purpose      | To provide customers with a tool for analyzing authorization performance for cardholder-activated transactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description  | <p>Mastercard produces this report for each CAT level. The report shows authorization transaction processing volumes and percentages by BIN for the week for the CAT level of the report. It reflects network usage; that is, transactions that travel across the Mastercard Network, not only billable items, which are listed apart from other transactions in the authorization detail reports located in <i>Mastercard Consolidated Billing System</i>.</p> <p>The report displays activity by authorization response: Accept (approved), Decline (decline), Call-Me (refer to card issuer), and Pick-Up (capture card). It also displays transactions by transaction category, such as mail/telephone and retail.</p> |
| Frequency    | Weekly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Distribution Methods

Paper (Sent to the authorization contact established on the Principal Member Questionnaire); Mastercard Operational Reports

## Report sample

Report sample of the Authorization Summary by CAT Level Report (SI458010-AA).

Authorization Summary by CAT Level Report (SI458010-AA), page 1

|                                  |                                                                                                                                                                |                      |                  |                        |                                       |                     |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------|------------------------|---------------------------------------|---------------------|
| REPORT SI458010-AA               | MASTERCARD INTERNATIONAL, INC.<br>AUTHORIZATION SUMMARY BY CAT LEVEL<br>FOR CAT 1 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24<br>ICA 001001 PREFIX 51567600000 |                      |                  |                        | PAGE 1<br>DATE 10/20/24 TIME 15:10:47 |                     |
| CAT 1 PROCESSED<br>ACTIVITY      | MEMBER<br>VOLUME                                                                                                                                               | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME   |                                       |                     |
| ACCEPT                           | 2,441,325                                                                                                                                                      | 91,015,173           | 84,414,707       | 3,579,930,134          |                                       |                     |
| PERCENT OF TOTAL                 | 97.9                                                                                                                                                           | 97.8                 | 95.3             | 94.9                   |                                       |                     |
| DECLINE                          | 51,351                                                                                                                                                         | 2,006,954            | 4,153,016        | 190,455,204            |                                       |                     |
| PERCENT OF TOTAL                 | 2.1                                                                                                                                                            | 2.2                  | 4.7              | 5.0                    |                                       |                     |
| CALL-ME                          | 0                                                                                                                                                              | 0                    | 6,549            | 340,056                |                                       |                     |
| PERCENT OF TOTAL                 | 0.0                                                                                                                                                            | 0.0                  | 0.0              | 0.0                    |                                       |                     |
| PICK-UP                          | 0                                                                                                                                                              | 0                    | 20,813           | 990,160                |                                       |                     |
| PERCENT OF TOTAL                 | 0.0                                                                                                                                                            | 0.0                  | 0.0              | 0.0                    |                                       |                     |
| TOTAL MEMBER PROCESSED           | 2,492,676                                                                                                                                                      | 93,022,127           | 88,595,085       | 3,771,715,554          |                                       |                     |
| -----                            |                                                                                                                                                                |                      |                  |                        |                                       |                     |
| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS                                                                                                                                             | TOTAL<br>DOLLARS     | TOTAL<br>VOLUME  | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS                  | YTD TOTAL<br>VOLUME |
| MAIL/TELEPHONE                   | 10.95                                                                                                                                                          | 131.50               | 12               | 24.46                  | 7,974.84                              | 326                 |
| RETAIL                           | 18.78                                                                                                                                                          | 45,325,186.88        | 2,412,815        | 8.62                   | 774,982,997.07                        | 89,894,072          |
| CASH ADVANCE                     | 0.00                                                                                                                                                           | 0.00                 | 0                | 0.00                   | 0.00                                  | 0                   |
| COLLEGE/HOSPITAL                 | 32.86                                                                                                                                                          | 65.72                | 2                | 91.51                  | 12,903.85                             | 141                 |
| HOTEL/MOTEL                      | 62.23                                                                                                                                                          | 2,676.10             | 43               | 73.59                  | 131,358.45                            | 1,785               |
| VEHICLE RENTAL                   | 185.56                                                                                                                                                         | 1,484.50             | 8                | 144.70                 | 54,262.86                             | 375                 |
| TRANSPORTATION                   | 24.07                                                                                                                                                          | 68,091.06            | 2,828            | 23.72                  | 2,425,723.83                          | 102,255             |
| RESTAURANT                       | 17.65                                                                                                                                                          | 8,775.74             | 497              | 21.00                  | 375,211.31                            | 17,860              |
| CIRRUS/ATM                       | 138.75                                                                                                                                                         | 3,485,584.85         | 25,120           | 138.74                 | 138,514,368.12                        | 998,334             |
| UNIQUE                           | 0.00                                                                                                                                                           | 0.00                 | 0                | 237.21                 | 5,930.34                              | 25                  |
| PAYMENT                          | 0.00                                                                                                                                                           | 0.00                 | 0                | 0.00                   | 0.00                                  | 0                   |
| TOTAL MEMBER APPROVALS           | 20.03                                                                                                                                                          | 48,891,997.15        | 2,441,325        | 10.07                  | 916,510,730.67                        | 91,015,173          |

## Authorization Summary by CAT Level Report (SI458010-AA), page 2

REPORT SI458010-AA

MASTERCARD INTERNATIONAL, INC.  
AUTHORIZATION SUMMARY BY CAT LEVEL  
FOR CAT 2 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24  
ICA 001001 PREFIX 51567600000

PAGE 1  
DATE 10/20/24 TIME 15:10:47

| CAT 2 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME  | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME    | SYSTEM YTD<br>VOLUME  |
|-----------------------------|-------------------|----------------------|---------------------|-----------------------|
| ACCEPT<br>PERCENT OF TOTAL  | 1,628,519<br>98.8 | 62,954,910<br>98.9   | 104,161,720<br>97.0 | 4,202,493,252<br>97.0 |
| DECLINE<br>PERCENT OF TOTAL | 19,186<br>1.2     | 700,623<br>1.1       | 3,148,207<br>2.9    | 127,592,842<br>2.9    |
| CALL-ME<br>PERCENT OF TOTAL | 0<br>0.0          | 0<br>0.0             | 536<br>0.0          | 22,320<br>0.0         |
| PICK-UP<br>PERCENT OF TOTAL | 0<br>0.0          | 0<br>0.0             | 37,977<br>0.0       | 1,688,880<br>0.0      |
| TOTAL MEMBER PROCESSED      | 1,647,705         | 63,655,533           | 107,348,440         | 4,331,797,294         |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 19.87              | 756,183.67       | 38,045          | 33.10                  | 20,544,202.29        | 620,598             |
| RETAIL                           | 20.70              | 17,026,015.58    | 822,116         | 21.32                  | 742,535,116.52       | 34,819,349          |
| CASH ADVANCE                     | 500.00             | 500.00           | 1               | 1,108.47               | 24,386.39            | 22                  |
| COLLEGE/HOSPITAL                 | 7.21               | 19,205.43        | 2,661           | 7.76                   | 666,340.77           | 85,850              |
| HOTEL/MOTEL                      | 97.74              | 42,715.41        | 437             | 72.34                  | 1,239,610.62         | 17,135              |
| VEHICLE RENTAL                   | 19.43              | 1,185.53         | 61              | 59.74                  | 129,356.21           | 2,165               |
| TRANSPORTATION                   | 5.31               | 1,158,905.75     | 218,135         | 5.48                   | 41,287,408.85        | 7,528,367           |
| RESTAURANT                       | 3.71               | 2,031,672.69     | 546,469         | 3.82                   | 75,942,074.70        | 19,866,558          |
| CIRRUS/ATM                       | 94.75              | 852.75           | 9               | 72.24                  | 18,061.82            | 250                 |
| UNIQUE                           | 316.56             | 185,191.00       | 585             | 493.18                 | 7,208,400.79         | 14,616              |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 13.03              | 21,222,427.81    | 1,628,519       | 14.13                  | 889,594,958.96       | 62,954,910          |

### Authorization Summary by CAT Level Report (SI458010-AA), page 3

REPORT SI458010-AA      **MASTERCARD INTERNATIONAL, INC.**      **PAGE 1**  
**AUTHORIZATION SUMMARY BY CAT LEVEL**  
 FOR CAT 3 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24      **DATE 10/20/24 TIME 15:10:47**  
 ICA 001001      PREFIX 51567600000

| CAT 3 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |
|-----------------------------|------------------|----------------------|------------------|----------------------|
| ACCEPT                      | 526              | 20,845               | 826,275          | 22,077,797           |
| PERCENT OF TOTAL            | 99.2             | 99.2                 | 97.0             | 96.6                 |
| DECLINE                     | 4                | 176                  | 25,320           | 776,309              |
| PERCENT OF TOTAL            | 0.8              | 0.8                  | 3.0              | 3.4                  |
| CALL-ME                     | 0                | 0                    | 4                | 53                   |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| PICK-UP                     | 0                | 0                    | 62               | 2,101                |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| TOTAL MEMBER PROCESSED      | 530              | 21,021               | 851,661          | 22,856,260           |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 0.00               | 0.00             | 0               | 4.78                   | 43.05                | 9                   |
| RETAIL                           | 9.07               | 4,519.18         | 498             | 10.70                  | 208,225.71           | 19,445              |
| CASH ADVANCE                     | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| COLLEGE/HOSPITAL                 | 0.00               | 0.00             | 0               | 26.00                  | 52.00                | 2                   |
| HOTEL/MOTEL                      | 0.00               | 0.00             | 0               | 16.86                  | 269.76               | 16                  |
| VEHICLE RENTAL                   | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TRANSPORTATION                   | 3.07               | 86.13            | 28              | 6.10                   | 8,121.86             | 1,330               |
| RESTAURANT                       | 0.00               | 0.00             | 0               | 348.78                 | 7,324.42             | 21                  |
| CIRRUS/ATM                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| UNIQUE                           | 0.00               | 0.00             | 0               | 67.19                  | 1,478.33             | 22                  |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 8.76               | 4,605.31         | 526             | 10.82                  | 225,515.13           | 20,845              |

## Authorization Summary by CAT Level Report (SI458010-AA), page 4

|                                  |                    |                                                       |                  |                        |                             |                     |
|----------------------------------|--------------------|-------------------------------------------------------|------------------|------------------------|-----------------------------|---------------------|
| REPORT SI458010-AA               |                    | MASTERCARD INTERNATIONAL, INC.                        |                  |                        | PAGE 1                      |                     |
|                                  |                    | AUTHORIZATION SUMMARY BY CAT LEVEL                    |                  |                        | DATE 10/20/24 TIME 15:10:47 |                     |
|                                  |                    | FOR CAT 4 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24 |                  |                        |                             |                     |
|                                  |                    | ICA 001001 PREFIX 51567600000                         |                  |                        |                             |                     |
| CAT 4 PROCESSED<br>ACTIVITY      | MEMBER<br>VOLUME   | MEMBER YTD<br>VOLUME                                  | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME   |                             |                     |
| ACCEPT                           | 24                 | 2,499                                                 | 436,888          | 19,253,772             |                             |                     |
| PERCENT OF TOTAL                 | 100.0              | 56.4                                                  | 97.7             | 97.6                   |                             |                     |
| DECLINE                          | 0                  | 1,928                                                 | 10,227           | 481,330                |                             |                     |
| PERCENT OF TOTAL                 | 0.0                | 43.6                                                  | 2.3              | 2.4                    |                             |                     |
| CALL-ME                          | 0                  | 0                                                     | 0                | 150                    |                             |                     |
| PERCENT OF TOTAL                 | 0.0                | 0.0                                                   | 0.0              | 0.0                    |                             |                     |
| PICK-UP                          | 0                  | 0                                                     | 11               | 913                    |                             |                     |
| PERCENT OF TOTAL                 | 0.0                | 0.0                                                   | 0.0              | 0.0                    |                             |                     |
| TOTAL MEMBER PROCESSED           | 24                 | 4,427                                                 | 447,126          | 19,736,165             |                             |                     |
| -----                            |                    |                                                       |                  |                        |                             |                     |
| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS                                      | TOTAL<br>VOLUME  | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS        | YTD TOTAL<br>VOLUME |
| MAIL/TELEPHONE                   | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| RETAIL                           | 15.62              | 375.04                                                | 24               | 11.28                  | 26,720.71                   | 2,368               |
| CASH ADVANCE                     | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| COLLEGE/HOSPITAL                 | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| HOTEL/MOTEL                      | 0.00               | 0.00                                                  | 0                | 103.61                 | 103.61                      | 1                   |
| VEHICLE RENTAL                   | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| TRANSPORTATION                   | 0.00               | 0.00                                                  | 0                | 0.19                   | 24.36                       | 123                 |
| RESTAURANT                       | 0.00               | 0.00                                                  | 0                | 35.43                  | 248.01                      | 7                   |
| CIRRUS/ATM                       | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| UNIQUE                           | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| PAYMENT                          | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| TOTAL MEMBER APPROVALS           | 15.63              | 375.04                                                | 24               | 10.84                  | 27,096.69                   | 2,499               |

## Authorization Summary by CAT Level Report (SI458010-AA), page 5

|                             |                                                                                                                                                                |                      |                  |                      |                                       |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------|----------------------|---------------------------------------|
| REPORT SI458010-AA          | MASTERCARD INTERNATIONAL, INC.<br>AUTHORIZATION SUMMARY BY CAT LEVEL<br>FOR CAT 6 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24<br>ICA 001001 PREFIX 51567600000 |                      |                  |                      | PAGE 1<br>DATE 10/20/24 TIME 15:10:47 |
| CAT 6 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME                                                                                                                                               | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |                                       |
| ACCEPT                      | 9,938,499                                                                                                                                                      | 375,570,815          | 980,342,032      | 40,266,946,306       |                                       |
| PERCENT OF TOTAL            | 88.9                                                                                                                                                           | 89.3                 | 78.0             | 78.3                 |                                       |
| DECLINE                     | 1,243,192                                                                                                                                                      | 44,898,366           | 262,455,006      | 10,502,291,833       |                                       |
| PERCENT OF TOTAL            | 11.1                                                                                                                                                           | 10.7                 | 20.9             | 20.4                 |                                       |
| CALL-ME                     | 0                                                                                                                                                              | 0                    | 139,249          | 4,251,454            |                                       |
| PERCENT OF TOTAL            | 0.0                                                                                                                                                            | 0.0                  | 0.0              | 0.0                  |                                       |
| PICK-UP                     | 0                                                                                                                                                              | 0                    | 14,392,487       | 676,394,265          |                                       |
| PERCENT OF TOTAL            | 0.0                                                                                                                                                            | 0.0                  | 1.1              | 1.3                  |                                       |
| TOTAL MEMBER PROCESSED      | 11,181,691                                                                                                                                                     | 420,469,181          | 1,257,328,774    | 51,449,883,858       |                                       |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 63.52              | 608,082,774.10   | 9,572,168       | 2.33                   | 846,239,844.29       | 362,401,540         |
| RETAIL                           | 56.37              | 5,816,995.60     | 103,183         | 54.60                  | 206,429,754.51       | 3,780,080           |
| CASH ADVANCE                     | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| COLLEGE/HOSPITAL                 | 133.17             | 132,771.18       | 997             | 166.37                 | 7,343,327.46         | 44,137              |
| HOTEL/MOTEL                      | 146.18             | 2,049,284.37     | 14,018          | 176.31                 | 65,585,845.60        | 371,971             |
| VEHICLE RENTAL                   | 217.26             | 109,934.29       | 506             | 260.92                 | 5,659,961.25         | 21,692              |
| TRANSPORTATION                   | 148.81             | 6,196,902.90     | 41,642          | 147.93                 | 267,787,326.61       | 1,810,177           |
| RESTAURANT                       | 15.11              | 3,013,833.66     | 199,332         | 14.90                  | 103,106,014.86       | 6,917,903           |
| CIRRUS/ATM                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| UNIQUE                           | 134.73             | 896,376.53       | 6,653           | 135.12                 | 30,174,419.46        | 223,315             |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 63.02              | 626,298,872.63   | 9,938,499       | 1.42                   | 532,326,494.04       | 375,570,815         |



## Authorization Summary by CAT Level Report (SI458010-AA), page 6

|                                  |                    |                                                       |                  |                        |                             |                     |
|----------------------------------|--------------------|-------------------------------------------------------|------------------|------------------------|-----------------------------|---------------------|
| REPORT SI458010-AA               |                    | MASTERCARD INTERNATIONAL, INC.                        |                  |                        | PAGE 1                      |                     |
|                                  |                    | AUTHORIZATION SUMMARY BY CAT LEVEL                    |                  |                        | DATE 10/20/24 TIME 15:10:47 |                     |
|                                  |                    | FOR CAT 7 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24 |                  |                        |                             |                     |
|                                  |                    | ICA 001001 PREFIX 51567600000                         |                  |                        |                             |                     |
| CAT 7 PROCESSED<br>ACTIVITY      | MEMBER<br>VOLUME   | MEMBER YTD<br>VOLUME                                  | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME   |                             |                     |
| ACCEPT                           | 2,011              | 72,127                                                | 738,764          | 28,464,971             |                             |                     |
| PERCENT OF TOTAL                 | 99.4               | 99.5                                                  | 98.6             | 98.2                   |                             |                     |
| DECLINE                          | 13                 | 329                                                   | 10,112           | 489,491                |                             |                     |
| PERCENT OF TOTAL                 | 0.6                | 0.5                                                   | 1.4              | 1.7                    |                             |                     |
| CALL-ME                          | 0                  | 0                                                     | 91               | 3,995                  |                             |                     |
| PERCENT OF TOTAL                 | 0.0                | 0.0                                                   | 0.0              | 0.0                    |                             |                     |
| PICK-UP                          | 0                  | 0                                                     | 53               | 17,125                 |                             |                     |
| PERCENT OF TOTAL                 | 0.0                | 0.0                                                   | 0.0              | 0.1                    |                             |                     |
| TOTAL MEMBER PROCESSED           | 2,024              | 72,456                                                | 749,020          | 28,975,582             |                             |                     |
| -----                            |                    |                                                       |                  |                        |                             |                     |
| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS                                      | TOTAL<br>VOLUME  | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS        | YTD TOTAL<br>VOLUME |
| MAIL/TELEPHONE                   | 23.50              | 23.50                                                 | 1                | 52.21                  | 6,892.93                    | 132                 |
| RETAIL                           | 34.59              | 47,946.07                                             | 1,386            | 34.75                  | 1,761,803.89                | 50,691              |
| CASH ADVANCE                     | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| COLLEGE/HOSPITAL                 | 94.74              | 473.71                                                | 5                | 173.28                 | 29,631.38                   | 171                 |
| HOTEL/MOTEL                      | 159.94             | 6,557.84                                              | 41               | 146.32                 | 197,388.53                  | 1,349               |
| VEHICLE RENTAL                   | 27.06              | 162.39                                                | 6                | 101.40                 | 22,308.67                   | 220                 |
| TRANSPORTATION                   | 21.34              | 1,088.65                                              | 51               | 41.47                  | 60,963.65                   | 1,470               |
| RESTAURANT                       | 28.19              | 14,692.18                                             | 521              | 26.82                  | 485,438.79                  | 18,094              |
| CIRRUS/ATM                       | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| UNIQUE                           | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| PAYMENT                          | 0.00               | 0.00                                                  | 0                | 0.00                   | 0.00                        | 0                   |
| TOTAL MEMBER APPROVALS           | 35.28              | 70,944.34                                             | 2,011            | 35.55                  | 2,564,427.84                | 72,127              |

## Authorization Summary by CAT Level Report (SI458010-AA), page 7

REPORT SI458010-AA

MASTERCARD INTERNATIONAL, INC.  
AUTHORIZATION SUMMARY BY CAT LEVEL  
FOR CAT 2 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24  
ICA 001001 - TOTAL FOR ALL PREFIXES

PAGE 1  
DATE 10/20/24 TIME 15:10:47

| CAT 2 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |
|-----------------------------|------------------|----------------------|------------------|----------------------|
| ACCEPT                      | 4,031,997        | 168,354,345          | 104,161,720      | 4,202,493,252        |
| PERCENT OF TOTAL            | 98.0             | 98.1                 | 97.0             | 97.0                 |
| DECLINE                     | 82,346           | 3,256,874            | 3,148,207        | 127,592,842          |
| PERCENT OF TOTAL            | 2.0              | 1.9                  | 2.9              | 2.9                  |
| CALL-ME                     | 0                | 0                    | 536              | 22,320               |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| PICK-UP                     | 0                | 0                    | 37,977           | 1,688,880            |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| TOTAL MEMBER PROCESSED      | 4,114,343        | 171,611,219          | 107,348,440      | 4,331,797,294        |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 20.00              | 1,804,805.17     | 90,240          | 34.61                  | 52,085,614.20        | 1,504,705           |
| RETAIL                           | 20.28              | 40,323,484.67    | 1,987,877       | 9.99                   | 911,252,801.79       | 91,165,932          |
| CASH ADVANCE                     | 500.00             | 500.00           | 1               | 631.27                 | 49,239.58            | 78                  |
| COLLEGE/HOSPITAL                 | 7.83               | 41,425.67        | 5,285           | 8.08                   | 1,510,590.30         | 186,797             |
| HOTEL/MOTEL                      | 92.09              | 81,504.48        | 885             | 74.82                  | 2,810,848.56         | 37,565              |
| VEHICLE RENTAL                   | 21.28              | 2,660.43         | 125             | 69.10                  | 276,974.06           | 4,008               |
| TRANSPORTATION                   | 5.39               | 2,327,484.06     | 431,504         | 5.59                   | 89,037,589.34        | 15,924,153          |
| RESTAURANT                       | 3.61               | 5,481,443.23     | 1,514,375       | 3.69                   | 219,614,521.79       | 59,478,373          |
| CIRRUS/ATM                       | 170.76             | 2,049.12         | 12              | 118.72                 | 61,737.17            | 520                 |
| UNIQUE                           | 355.70             | 602,201.35       | 1,693           | 462.16                 | 24,131,675.38        | 52,214              |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 12.57              | 50,667,558.18    | 4,031,997       | 1.79                   | 300,831,592.17       | 168,354,345         |



## Authorization Summary by CAT Level Report (SI458010-AA), page 8

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MASTERCARD INTERNATIONAL, INC.  
AUTHORIZATION SUMMARY BY CAT LEVEL  
FOR CAT 3 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24  
ICA 001001 - TOTAL FOR ALL PREFIXES

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| CAT 3 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |
|-----------------------------|------------------|----------------------|------------------|----------------------|
| ACCEPT                      | 1,139            | 53,026               | 826,275          | 22,077,797           |
| PERCENT OF TOTAL            | 97.7             | 98.4                 | 97.0             | 96.6                 |
| DECLINE                     | 27               | 861                  | 25,320           | 776,309              |
| PERCENT OF TOTAL            | 2.3              | 1.6                  | 3.0              | 3.4                  |
| CALL-ME                     | 0                | 0                    | 4                | 53                   |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| PICK-UP                     | 0                | 0                    | 62               | 2,101                |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| TOTAL MEMBER PROCESSED      | 1,166            | 53,887               | 851,661          | 22,856,260           |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 0.00               | 0.00             | 0               | 4.39                   | 153.83               | 35                  |
| RETAIL                           | 10.05              | 10,800.62        | 1,082           | 11.08                  | 558,381.25           | 50,382              |
| CASH ADVANCE                     | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| COLLEGE/HOSPITAL                 | 0.00               | 0.00             | 0               | 108.00                 | 544.00               | 5                   |
| HOTEL/MOTEL                      | 0.00               | 0.00             | 0               | 46.14                  | 1,384.37             | 30                  |
| VEHICLE RENTAL                   | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TRANSPORTATION                   | 3.30               | 188.41           | 57              | 5.70                   | 14,188.26            | 2,485               |
| RESTAURANT                       | 0.00               | 0.00             | 0               | 163.76                 | 7,860.54             | 48                  |
| CIRRUS/ATM                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| UNIQUE                           | 0.00               | 0.00             | 0               | 61.76                  | 2,532.46             | 41                  |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 9.72               | 11,069.03        | 1,139           | 11.03                  | 585,044.71           | 53,026              |

## Authorization Summary by CAT Level Report (SI458010-AA), page 9

REPORT SI458010-AA      MASTERCARD INTERNATIONAL, INC.      PAGE 1  
 AUTHORIZATION SUMMARY BY CAT LEVEL  
 FOR CAT 4 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24      DATE 10/20/24 TIME 15:10:47  
 ICA 001001 - TOTAL FOR ALL PREFIXES

| CAT 4 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |
|-----------------------------|------------------|----------------------|------------------|----------------------|
| ACCEPT                      | 55               | 5,062                | 436,888          | 19,253,772           |
| PERCENT OF TOTAL            | 100.0            | 57.2                 | 97.7             | 97.6                 |
| DECLINE                     | 0                | 3,795                | 10,227           | 481,330              |
| PERCENT OF TOTAL            | 0.0              | 42.8                 | 2.3              | 2.4                  |
| CALL-ME                     | 0                | 0                    | 0                | 150                  |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| PICK-UP                     | 0                | 0                    | 11               | 913                  |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| TOTAL MEMBER PROCESSED      | 55               | 8,857                | 447,126          | 19,736,165           |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| RETAIL                           | 12.89              | 709.18           | 55              | 12.03                  | 57,079.29            | 4,742               |
| CASH ADVANCE                     | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| COLLEGE/HOSPITAL                 | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| HOTEL/MOTEL                      | 0.00               | 0.00             | 0               | 415.44                 | 830.89               | 2                   |
| VEHICLE RENTAL                   | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TRANSPORTATION                   | 0.00               | 0.00             | 0               | 0.19                   | 58.24                | 292                 |
| RESTAURANT                       | 0.00               | 0.00             | 0               | 25.74                  | 669.44               | 26                  |
| CIRRUS/ATM                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| UNIQUE                           | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 12.89              | 709.18           | 55              | 11.58                  | 58,637.86            | 5,062               |

## Authorization Summary by CAT Level Report (SI458010-AA), page 10

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AUTHORIZATION SUMMARY BY CAT LEVEL  
FOR CAT 6 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24  
ICA 001001 - TOTAL FOR ALL PREFIXES      DATE 10/20/24 TIME 15:10:47

| CAT 6 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |
|-----------------------------|------------------|----------------------|------------------|----------------------|
| ACCEPT                      | 27,214,962       | 1,111,893,070        | 980,342,032      | 40,266,946,306       |
| PERCENT OF TOTAL            | 80.2             | 81.0                 | 78.0             | 78.3                 |
| DECLINE                     | 6,728,098        | 260,486,531          | 262,455,006      | 10,502,291,833       |
| PERCENT OF TOTAL            | 19.8             | 19.0                 | 20.9             | 20.4                 |
| CALL-ME                     | 0                | 0                    | 139,249          | 4,251,454            |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| PICK-UP                     | 0                | 0                    | 14,392,487       | 676,394,265          |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 1.1              | 1.3                  |
| TOTAL MEMBER PROCESSED      | 33,943,060       | 1,372,379,601        | 1,257,328,774    | 51,449,883,858       |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 19.13              | 503,708,534.41   | 26,321,083      | 0.35                   | 378,507,815.60       | 1,077,281,244       |
| RETAIL                           | 48.81              | 14,511,036.11    | 297,260         | 47.54                  | 560,398,769.74       | 11,787,055          |
| CASH ADVANCE                     | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| COLLEGE/HOSPITAL                 | 122.08             | 263,581.79       | 2,159           | 146.82                 | 16,364,141.15        | 111,455             |
| HOTEL/MOTEL                      | 145.62             | 4,987,857.78     | 34,251          | 164.68                 | 161,333,999.41       | 979,641             |
| VEHICLE RENTAL                   | 254.38             | 297,120.11       | 1,168           | 279.22                 | 15,764,010.53        | 56,457              |
| TRANSPORTATION                   | 128.17             | 11,723,547.44    | 91,466          | 127.84                 | 559,013,450.29       | 4,372,727           |
| RESTAURANT                       | 15.05              | 6,668,072.73     | 442,816         | 14.80                  | 243,708,273.88       | 16,465,389          |
| CIRRUS/ATM                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| UNIQUE                           | 108.59             | 2,688,715.27     | 24,759          | 116.05                 | 97,379,606.94        | 839,102             |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 20.02              | 544,848,465.64   | 27,214,962      | 0.03                   | 32,470,067.54        | 1,111,893,070       |

## Authorization Summary by CAT Level Report (SI458010-AA), page 11

REPORT SI458010-AA      MASTERCARD INTERNATIONAL, INC.      PAGE 1  
AUTHORIZATION SUMMARY BY CAT LEVEL  
FOR CAT 7 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24      DATE 10/20/24 TIME 15:10:47  
ICA 001001 - TOTAL FOR ALL PREFIXES

| CAT 7 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |
|-----------------------------|------------------|----------------------|------------------|----------------------|
| ACCEPT<br>PERCENT OF TOTAL  | 2,831<br>99.2    | 106,029<br>99.3      | 738,764<br>98.6  | 28,464,971<br>98.2   |
| DECLINE<br>PERCENT OF TOTAL | 22<br>0.8        | 712<br>0.7           | 10,112<br>1.4    | 489,491<br>1.7       |
| CALL-ME<br>PERCENT OF TOTAL | 0<br>0.0         | 0<br>0.0             | 91<br>0.0        | 3,995<br>0.0         |
| PICK-UP<br>PERCENT OF TOTAL | 0<br>0.0         | 0<br>0.0             | 53<br>0.0        | 17,125<br>0.1        |
| TOTAL MEMBER PROCESSED      | 2,853            | 106,741              | 749,020          | 28,975,582           |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 23.50              | 23.50            | 1               | 47.26                  | 21,459.36            | 454                 |
| RETAIL                           | 39.33              | 76,906.21        | 1,955           | 38.68                  | 2,879,990.64         | 74,447              |
| CASH ADVANCE                     | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| COLLEGE/HOSPITAL                 | 211.05             | 2,321.63         | 11              | 171.93                 | 51,924.89            | 302                 |
| HOTEL/MOTEL                      | 130.77             | 8,893.02         | 68              | 143.33                 | 304,159.19           | 2,122               |
| VEHICLE RENTAL                   | 59.01              | 531.09           | 9               | 112.06                 | 41,799.55            | 373                 |
| TRANSPORTATION                   | 26.45              | 1,852.08         | 70              | 52.93                  | 115,982.04           | 2,191               |
| RESTAURANT                       | 27.75              | 19,898.56        | 717             | 27.19                  | 710,886.92           | 26,140              |
| CIRRUS/ATM                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| UNIQUE                           | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 39.01              | 110,426.09       | 2,831           | 38.92                  | 4,126,202.59         | 106,029             |

## Authorization Summary by CAT Level Report (SI458010-AA), page 12

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AUTHORIZATION SUMMARY BY CAT LEVEL  
FOR CAT 2 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24      DATE 10/20/24 TIME 15:10:47  
GRAND TOTAL FOR ALL ICA'S

| CAT 2 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME    | MEMBER YTD<br>VOLUME  | SYSTEM<br>VOLUME    | SYSTEM YTD<br>VOLUME  |
|-----------------------------|---------------------|-----------------------|---------------------|-----------------------|
| ACCEPT<br>PERCENT OF TOTAL  | 104,161,720<br>97.0 | 4,202,493,252<br>97.0 | 104,161,720<br>97.0 | 4,202,493,252<br>97.0 |
| DECLINE<br>PERCENT OF TOTAL | 3,148,207<br>2.9    | 127,592,842<br>2.9    | 3,148,207<br>2.9    | 127,592,842<br>2.9    |
| CALL-ME<br>PERCENT OF TOTAL | 536<br>0.0          | 22,320<br>0.0         | 536<br>0.0          | 22,320<br>0.0         |
| PICK-UP<br>PERCENT OF TOTAL | 37,977<br>0.0       | 1,688,880<br>0.0      | 37,977<br>0.0       | 1,688,880<br>0.0      |
| TOTAL MEMBER PROCESSED      | 107,348,440         | 4,331,797,294         | 107,348,440         | 4,331,797,294         |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 28.83              | 36,048,973.00    | 1,250,166       | 10.44                  | 316,430,546.30       | 30,294,439          |
| RETAIL                           | 3.48               | 215,625,718.61   | 61,875,774      | 0.07                   | 202,633,648.30       | 2,538,133,735       |
| CASH ADVANCE                     | 413.86             | 28,970.41        | 70              | 566.66                 | 2,961,412.56         | 5,226               |
| COLLEGE/HOSPITAL                 | 9.67               | 2,077,737.03     | 214,776         | 9.89                   | 73,570,932.72        | 7,434,871           |
| HOTEL/MOTEL                      | 87.57              | 6,357,113.76     | 72,587          | 84.93                  | 251,564,565.28       | 2,961,696           |
| VEHICLE RENTAL                   | 12.38              | 778,196.93       | 62,826          | 13.91                  | 36,644,598.78        | 2,632,729           |
| TRANSPORTATION                   | 5.51               | 77,330,433.40    | 14,013,752      | 0.89                   | 503,532,563.37       | 562,197,185         |
| RESTAURANT                       | 3.48               | 92,181,453.35    | 26,459,927      | 0.76                   | 798,237,778.19       | 1,049,617,070       |
| CIRRUS/ATM                       | 224.06             | 8,593,167.42     | 38,352          | 219.49                 | 353,958,154.85       | 1,612,597           |
| UNIQUE                           | 44.78              | 6,658,634.07     | 148,680         | 46.18                  | 289,869,103.21       | 6,276,221           |
| PAYMENT                          | 81.69              | 2,026,918.59     | 24,810          | 71.14                  | 94,438,299.58        | 1,327,483           |
| TOTAL MEMBER APPROVALS           | 4.30               | 447,707,317.37   | 104,161,720     | 0.22                   | 923,841,603.14       | 4,202,493,252       |

## Authorization Summary by CAT Level Report (SI458010-AA), page 13

|                                  |                                                       |                      |                  |                        |                             |
|----------------------------------|-------------------------------------------------------|----------------------|------------------|------------------------|-----------------------------|
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|                                  | AUTHORIZATION SUMMARY BY CAT LEVEL                    |                      |                  |                        | DATE 10/20/24 TIME 15:10:47 |
|                                  | FOR CAT 3 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24 |                      |                  |                        |                             |
|                                  | GRAND TOTAL FOR ALL ICA'S                             |                      |                  |                        |                             |
| CAT 3 PROCESSED<br>ACTIVITY      | MEMBER<br>VOLUME                                      | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME   |                             |
| ACCEPT                           | 826,275                                               | 22,077,797           | 826,275          | 22,077,797             |                             |
| PERCENT OF TOTAL                 | 97.0                                                  | 96.6                 | 97.0             | 96.6                   |                             |
| DECLINE                          | 25,320                                                | 776,309              | 25,320           | 776,309                |                             |
| PERCENT OF TOTAL                 | 3.0                                                   | 3.4                  | 3.0              | 3.4                    |                             |
| CALL-ME                          | 4                                                     | 53                   | 4                | 53                     |                             |
| PERCENT OF TOTAL                 | 0.0                                                   | 0.0                  | 0.0              | 0.0                    |                             |
| PICK-UP                          | 62                                                    | 2,101                | 62               | 2,101                  |                             |
| PERCENT OF TOTAL                 | 0.0                                                   | 0.0                  | 0.0              | 0.0                    |                             |
| TOTAL MEMBER PROCESSED           | 851,661                                               | 22,856,260           | 851,661          | 22,856,260             |                             |
| -----                            |                                                       |                      |                  |                        |                             |
| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS                                    | TOTAL<br>DOLLARS     | TOTAL<br>VOLUME  | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS        |
| MAIL/TELEPHONE                   | 7.50                                                  | 15.00                | 2                | 3.76                   | 10,003.67                   |
| RETAIL                           | 2.20                                                  | 1,381,479.23         | 625,691          | 5.23                   | 64,927,192.39               |
| CASH ADVANCE                     | 106.13                                                | 212.27               | 2                | 119.06                 | 9,287.34                    |
| COLLEGE/HOSPITAL                 | 55.67                                                 | 167.02               | 3                | 215.52                 | 68,537.81                   |
| HOTEL/MOTEL                      | 41.34                                                 | 5,043.84             | 122              | 84.95                  | 399,960.52                  |
| VEHICLE RENTAL                   | 0.00                                                  | 0.00                 | 0                | 510.37                 | 8,676.36                    |
| TRANSPORTATION                   | 2.17                                                  | 435,290.81           | 199,754          | 1.84                   | 17,768,266.27               |
| RESTAURANT                       | 20.92                                                 | 14,019.92            | 670              | 21.46                  | 738,709.41                  |
| CIRRUS/ATM                       | 0.00                                                  | 0.00                 | 0                | 0.00                   | 0.00                        |
| UNIQUE                           | 9.17                                                  | 45.86                | 5                | 171.09                 | 33,706.56                   |
| PAYMENT                          | 132.76                                                | 3,451.96             | 26               | 149.24                 | 122,981.92                  |
| TOTAL MEMBER APPROVALS           | 2.23                                                  | 1,839,725.91         | 826,275          | 3.81                   | 84,087,322.25               |



Authorization Summary by CAT Level Report (SI458010-AA), page 14

|                                  |  |                                                       |                      |                  |                        |                             |                     |
|----------------------------------|--|-------------------------------------------------------|----------------------|------------------|------------------------|-----------------------------|---------------------|
| REPORT SI458010-AA               |  | MASTERCARD INTERNATIONAL, INC.                        |                      |                  |                        | PAGE 1                      |                     |
|                                  |  | AUTHORIZATION SUMMARY BY CAT LEVEL                    |                      |                  |                        | DATE 10/20/24 TIME 15:10:47 |                     |
|                                  |  | FOR CAT 4 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24 |                      |                  |                        |                             |                     |
|                                  |  | GRAND TOTAL FOR ALL ICA'S                             |                      |                  |                        |                             |                     |
| CAT 4 PROCESSED<br>ACTIVITY      |  | MEMBER<br>VOLUME                                      | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME   |                             |                     |
| ACCEPT                           |  | 436,888                                               | 19,253,772           | 436,888          | 19,253,772             |                             |                     |
| PERCENT OF TOTAL                 |  | 97.7                                                  | 97.6                 | 97.7             | 97.6                   |                             |                     |
| DECLINE                          |  | 10,227                                                | 481,330              | 10,227           | 481,330                |                             |                     |
| PERCENT OF TOTAL                 |  | 2.3                                                   | 2.4                  | 2.3              | 2.4                    |                             |                     |
| CALL-ME                          |  | 0                                                     | 150                  | 0                | 150                    |                             |                     |
| PERCENT OF TOTAL                 |  | 0.0                                                   | 0.0                  | 0.0              | 0.0                    |                             |                     |
| PICK-UP                          |  | 11                                                    | 913                  | 11               | 913                    |                             |                     |
| PERCENT OF TOTAL                 |  | 0.0                                                   | 0.0                  | 0.0              | 0.0                    |                             |                     |
| TOTAL MEMBER PROCESSED           |  | 447,126                                               | 19,736,165           | 447,126          | 19,736,165             |                             |                     |
| -----                            |  |                                                       |                      |                  |                        |                             |                     |
| APPROVALS BY<br>TRANSACTION TYPE |  | AVERAGE<br>DOLLARS                                    | TOTAL<br>DOLLARS     | TOTAL<br>VOLUME  | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS        | YTD TOTAL<br>VOLUME |
| MAIL/TELEPHONE                   |  | 0.00                                                  | 0.00                 | 0                | 0.00                   | 0.00                        | 0                   |
| RETAIL                           |  | 0.81                                                  | 355,780.17           | 436,777          | 4.70                   | 14,570,994.91               | 3,098,251           |
| CASH ADVANCE                     |  | 0.00                                                  | 0.00                 | 0                | 0.00                   | 0.00                        | 0                   |
| COLLEGE/HOSPITAL                 |  | 0.00                                                  | 0.00                 | 0                | 0.00                   | 0.00                        | 0                   |
| HOTEL/MOTEL                      |  | 94.84                                                 | 758.78               | 8                | 82.48                  | 29,612.75                   | 359                 |
| VEHICLE RENTAL                   |  | 0.00                                                  | 0.00                 | 0                | 636.99                 | 14,650.94                   | 23                  |
| TRANSPORTATION                   |  | 17.59                                                 | 35.18                | 2                | 0.20                   | 3,279,921.72                | 16,148,968          |
| RESTAURANT                       |  | 27.73                                                 | 2,801.64             | 101              | 27.25                  | 168,189.55                  | 6,171               |
| CIRRUS/ATM                       |  | 0.00                                                  | 0.00                 | 0                | 0.00                   | 0.00                        | 0                   |
| UNIQUE                           |  | 0.00                                                  | 0.00                 | 0                | 0.00                   | 0.00                        | 0                   |
| PAYMENT                          |  | 0.00                                                  | 0.00                 | 0                | 0.00                   | 0.00                        | 0                   |
| TOTAL MEMBER APPROVALS           |  | 0.82                                                  | 359,375.77           | 436,888          | 0.94                   | 18,063,369.87               | 19,253,772          |

## Authorization Summary by CAT Level Report (SI458010-AA), page 15

REPORT SI458010-AA

MASTERCARD INTERNATIONAL, INC.  
AUTHORIZATION SUMMARY BY CAT LEVEL  
FOR CAT 5 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24  
GRAND TOTAL FOR ALL ICA'S

PAGE 1  
DATE 10/20/24 TIME 15:10:47

| CAT 5 PROCESSED<br>ACTIVITY | MEMBER<br>VOLUME | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME |
|-----------------------------|------------------|----------------------|------------------|----------------------|
| ACCEPT                      | 0                | 0                    | 0                | 0                    |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| DECLINE                     | 0                | 0                    | 0                | 0                    |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| CALL-ME                     | 0                | 0                    | 0                | 0                    |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| PICK-UP                     | 0                | 0                    | 0                | 0                    |
| PERCENT OF TOTAL            | 0.0              | 0.0                  | 0.0              | 0.0                  |
| TOTAL MEMBER PROCESSED      | 0                | 0                    | 0                | 0                    |

| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS | TOTAL<br>VOLUME | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME |
|----------------------------------|--------------------|------------------|-----------------|------------------------|----------------------|---------------------|
| MAIL/TELEPHONE                   | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| RETAIL                           | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| CASH ADVANCE                     | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| COLLEGE/HOSPITAL                 | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| HOTEL/MOTEL                      | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| VEHICLE RENTAL                   | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TRANSPORTATION                   | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| RESTAURANT                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| CIRRUS/ATM                       | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| UNIQUE                           | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| PAYMENT                          | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |
| TOTAL MEMBER APPROVALS           | 0.00               | 0.00             | 0               | 0.00                   | 0.00                 | 0                   |



## Authorization Summary by CAT Level Report (SI458010-AA), page 16

|                                  |                                                                                                                                                            |                      |                  |                        |                                                 |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------|------------------------|-------------------------------------------------|
| REPORT SI458010-AA               | MASTERCARD INTERNATIONAL, INC.<br>AUTHORIZATION SUMMARY BY CAT LEVEL<br>FOR CAT 6 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24<br>GRAND TOTAL FOR ALL ICA'S |                      |                  |                        | PAGE 1                                          |
|                                  |                                                                                                                                                            |                      |                  |                        | DATE 10/20/24 TIME 15:10:47                     |
| CAT 6 PROCESSED<br>ACTIVITY      | MEMBER<br>VOLUME                                                                                                                                           | MEMBER YTD<br>VOLUME | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME   |                                                 |
| ACCEPT                           | 980,342,032                                                                                                                                                | 40,266,946,306       | 980,342,032      | 40,266,946,306         |                                                 |
| PERCENT OF TOTAL                 | 78.0                                                                                                                                                       | 78.3                 | 78.0             | 78.3                   |                                                 |
| DECLINE                          | 262,455,006                                                                                                                                                | 10,502,291,833       | 262,455,006      | 10,502,291,833         |                                                 |
| PERCENT OF TOTAL                 | 20.9                                                                                                                                                       | 20.4                 | 20.9             | 20.4                   |                                                 |
| CALL-ME                          | 139,249                                                                                                                                                    | 4,251,454            | 139,249          | 4,251,454              |                                                 |
| PERCENT OF TOTAL                 | 0.0                                                                                                                                                        | 0.0                  | 0.0              | 0.0                    |                                                 |
| PICK-UP                          | 14,392,487                                                                                                                                                 | 676,394,265          | 14,392,487       | 676,394,265            |                                                 |
| PERCENT OF TOTAL                 | 1.1                                                                                                                                                        | 1.3                  | 1.1              | 1.3                    |                                                 |
| TOTAL MEMBER PROCESSED           | 1,257,328,774                                                                                                                                              | 51,449,883,858       | 1,257,328,774    | 51,449,883,858         |                                                 |
| -----                            |                                                                                                                                                            |                      |                  |                        |                                                 |
| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS                                                                                                                                         | TOTAL<br>DOLLARS     | TOTAL<br>VOLUME  | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS<br><br>YTD TOTAL<br>VOLUME |
| MAIL/TELEPHONE                   | 0.17                                                                                                                                                       | 142,592,594.15       | 823,581,218      | 0.02                   | 919,186,737.32 33,743,046,753                   |
| RETAIL                           | 3.01                                                                                                                                                       | 134,121,699.49       | 44,461,967       | 0.40                   | 774,276,529.59 1,911,245,611                    |
| CASH ADVANCE                     | 0.00                                                                                                                                                       | 0.00                 | 0                | 51.22                  | 5,224.63 102                                    |
| COLLEGE/HOSPITAL                 | 261.00                                                                                                                                                     | 71,262,506.55        | 273,027          | 28.10                  | 334,934,690.82 11,918,373                       |
| HOTEL/MOTEL                      | 234.20                                                                                                                                                     | 172,168,311.85       | 735,117          | 30.34                  | 730,914,267.93 24,083,590                       |
| VEHICLE RENTAL                   | 83.56                                                                                                                                                      | 18,491,953.19        | 221,285          | 93.50                  | 786,619,837.23 8,412,386                        |
| TRANSPORTATION                   | 58.03                                                                                                                                                      | 528,233,195.77       | 9,102,201        | 0.45                   | 180,554,165.79 395,187,074                      |
| RESTAURANT                       | 21.26                                                                                                                                                      | 298,944,238.82       | 14,056,906       | 1.31                   | 796,448,510.91 604,648,810                      |
| CIRRUS/ATM                       | 134.59                                                                                                                                                     | 3,572,530.45         | 26,542           | 141.70                 | 160,430,089.41 1,132,175                        |
| UNIQUE                           | 7.95                                                                                                                                                       | 424,287,731.97       | 53,359,779       | 0.41                   | 950,734,705.91 2,291,898,258                    |
| PAYMENT                          | 15.67                                                                                                                                                      | 541,212,225.61       | 34,523,990       | 0.71                   | 912,972,432.93 1,275,373,174                    |
| TOTAL MEMBER APPROVALS           | 0.34                                                                                                                                                       | 334,886,987.85       | 980,342,032      | 0.01                   | 547,077,192.47 40,266,946,306                   |

# Authorization Summary by CAT Level Report (SI458010-AA), page 17

|                                  |                    |                                                                                                                                                            |                  |                        |                      |                                       |                |
|----------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------|----------------------|---------------------------------------|----------------|
| TOTAL MEMBER APPROVALS           |                    | 61.24                                                                                                                                                      | 324,880,787.62   | 388,342,626            | 61.64                | 347,577,426.47                        | 40,400,340,360 |
| REPORT SI458010-AA               |                    | MASTERCARD INTERNATIONAL, INC.<br>AUTHORIZATION SUMMARY BY CAT LEVEL<br>FOR CAT 7 WEEK BEGINNING 10/12/24 AND ENDING 10/18/24<br>GRAND TOTAL FOR ALL ICA'S |                  |                        |                      | PAGE 1<br>DATE 10/20/24 TIME 15:10:47 |                |
| CAT 7 PROCESSED<br>ACTIVITY      | MEMBER<br>VOLUME   | MEMBER YTD<br>VOLUME                                                                                                                                       | SYSTEM<br>VOLUME | SYSTEM YTD<br>VOLUME   |                      |                                       |                |
| ACCEPT                           | 738,764            | 28,464,971                                                                                                                                                 | 738,764          | 28,464,971             |                      |                                       |                |
| PERCENT OF TOTAL                 | 98.6               | 98.2                                                                                                                                                       | 98.6             | 98.2                   |                      |                                       |                |
| DECLINE                          | 10,112             | 489,491                                                                                                                                                    | 10,112           | 489,491                |                      |                                       |                |
| PERCENT OF TOTAL                 | 1.4                | 1.7                                                                                                                                                        | 1.4              | 1.7                    |                      |                                       |                |
| CALL-ME                          | 91                 | 3,995                                                                                                                                                      | 91               | 3,995                  |                      |                                       |                |
| PERCENT OF TOTAL                 | 0.0                | 0.0                                                                                                                                                        | 0.0              | 0.0                    |                      |                                       |                |
| PICK-UP                          | 53                 | 17,125                                                                                                                                                     | 53               | 17,125                 |                      |                                       |                |
| PERCENT OF TOTAL                 | 0.0                | 0.1                                                                                                                                                        | 0.0              | 0.1                    |                      |                                       |                |
| TOTAL MEMBER PROCESSED           | 749,020            | 28,975,582                                                                                                                                                 | 749,020          | 28,975,582             |                      |                                       |                |
| -----                            |                    |                                                                                                                                                            |                  |                        |                      |                                       |                |
| APPROVALS BY<br>TRANSACTION TYPE | AVERAGE<br>DOLLARS | TOTAL<br>DOLLARS                                                                                                                                           | TOTAL<br>VOLUME  | YTD AVERAGE<br>DOLLARS | YTD TOTAL<br>DOLLARS | YTD TOTAL<br>VOLUME                   |                |
| MAIL/TELEPHONE                   | 70.01              | 196,589.61                                                                                                                                                 | 2,808            | 18.33                  | 5,534,624.93         | 301,850                               |                |
| RETAIL                           | 47.04              | 26,069,239.56                                                                                                                                              | 554,110          | 47.10                  | 999,266,361.37       | 21,214,991                            |                |
| CASH ADVANCE                     | 0.00               | 0.00                                                                                                                                                       | 0                | 0.00                   | 0.00                 | 0                                     |                |
| COLLEGE/HOSPITAL                 | 160.23             | 178,817.83                                                                                                                                                 | 1,116            | 150.55                 | 5,737,380.73         | 38,109                                |                |
| HOTEL/MOTEL                      | 149.33             | 1,152,736.41                                                                                                                                               | 7,719            | 160.94                 | 44,871,433.51        | 278,794                               |                |
| VEHICLE RENTAL                   | 167.96             | 173,673.19                                                                                                                                                 | 1,034            | 170.24                 | 6,783,855.14         | 39,848                                |                |
| TRANSPORTATION                   | 25.62              | 395,020.23                                                                                                                                                 | 15,418           | 31.16                  | 17,435,211.94        | 559,424                               |                |
| RESTAURANT                       | 23.63              | 3,663,059.00                                                                                                                                               | 155,001          | 25.61                  | 154,444,930.74       | 6,030,217                             |                |
| CIRRUS/ATM                       | 0.00               | 0.00                                                                                                                                                       | 0                | 131.32                 | 1,575.88             | 12                                    |                |
| UNIQUE                           | 16.55              | 25,791.29                                                                                                                                                  | 1,558            | 16.36                  | 28,239.87            | 1,726                                 |                |
| PAYMENT                          | 0.00               | 0.00                                                                                                                                                       | 0                | 0.00                   | 0.00                 | 0                                     |                |
| TOTAL MEMBER APPROVALS           | 43.12              | 31,854,927.12                                                                                                                                              | 738,764          | 8.22                   | 234,103,614.11       | 28,464,971                            |                |

## Field descriptions

The fields on the Authorization Summary by CAT Level Report (SI458010-AA) are listed in table format.

### Field descriptions

| Field                  | Description                                                                                                                                                                                                                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAT Processed Activity | Summary of all authorization activity received and processed by the issuer for the CAT level on the report, listed by the following response categories: <ul style="list-style-type: none"> <li>• Accept (approved)</li> <li>• Decline</li> <li>• Call-Me (refer to card issuer)</li> <li>• Pick-Up (capture card)</li> </ul> |
| Member Volume          | Total CAT transaction volume processed by the customer that generated the specific response and the percentage it represents of total customer CAT transaction volume.                                                                                                                                                        |
| Member YTD Volume      | Summary of the customer's CAT authorization volume for the calendar year-to-date (by response category) and the percentage it represents of total member CAT volume for the year-to-date.                                                                                                                                     |
| System Volume          | Volume of CAT authorization transactions processed by the entire membership that week and the percentage it represents of total CAT volume for the membership for the week.                                                                                                                                                   |
| System YTD Volume      | Volume of CAT authorization transactions processed by the entire membership for the calendar year-to-date and the percentage it represents of total CAT volume for the membership for the year-to-date.                                                                                                                       |
| Total Member Processed | Total member CAT volume, year-to-date volume, system volume, and system year-to-date volume for all authorization responses.                                                                                                                                                                                                  |

| Field                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approvals by Transaction Type | <p>Summary of all approved responses returned for CAT transactions, divided by the following transaction categories:</p> <ul style="list-style-type: none"> <li>• Mail/Telephone</li> <li>• Retail</li> <li>• Cash Advance</li> <li>• College/Hospital</li> <li>• Hotel/Motel</li> <li>• Vehicle Rental</li> <li>• Transportation</li> <li>• Restaurant</li> <li>• ATM</li> <li>• Unique</li> <li>• Payment Transaction</li> </ul> |
| Average Dollars               | Average dollar amount approved for CAT transactions for the week, listed by transaction type.                                                                                                                                                                                                                                                                                                                                      |
| Total Dollars                 | Total dollar amount approved for CAT transactions for the week, listed by transaction type.                                                                                                                                                                                                                                                                                                                                        |
| Total Volume                  | Total CAT transaction volume processed by the customer for the week, listed by transaction type. In the sample, the total volume for Restaurant transactions is 3. This indicates that out of a total of 86 approved CAT 2 transactions, 3 were for Restaurant.                                                                                                                                                                    |
| YTD Average Dollars           | Average U.S. dollar amount approved, by transaction type, for CAT transactions for the calendar year-to-date. This amount is the result of dividing the YTD TOTAL DOLLARS by the YTD TOTAL VOLUME.                                                                                                                                                                                                                                 |
| YTD Total Dollars             | Total U.S. dollar amount of CAT transactions approved by the customer for the calendar year-to-date.                                                                                                                                                                                                                                                                                                                               |
| YTD Total Volume              | Total CAT transaction volume approved by the customer for the calendar year-to-date.                                                                                                                                                                                                                                                                                                                                               |
| Total Member Approvals        | Average U.S. dollar amount, total U.S. dollar amount, total transaction volume, and year-to-date totals for all CAT transactions approved by the customer.                                                                                                                                                                                                                                                                         |

## Related reports in other guides

Information concerning authorization detail reports, billing reports, and layouts of all reports related to listing accounts or merchants listed in the Merchant File are provided in other manuals.

Authorization detail reports and billing reports, including the following reports, are found in the **Pricing and Billing Resource Center** application on Mastercard Connect®.

### Related reports

| Report Name                          | Report Number |
|--------------------------------------|---------------|
| Issuer Authorization Detail Report   | AB201010-A2   |
| Acquirer Authorization Detail Report | AB202010-A1   |
| Billable Customer Summary            | GB071010-AA   |
| Customer Service Summary             | GB073010-AA   |
| Customer Service Roll-Up             | GB074010-AA   |

Refer to the [Account Management System User Manual](#) for layouts of all reports related to listing accounts in the Account Management System (AMS) or merchants listed in the Merchant File.

## BIN Table Resource File

The BIN Table Resource File is a fixed-block format with a record size of 150 bytes and a block size of 27900 bytes. The file is identified as MCI.AR.TR52.

Each file consists of the following:

- Header Record
- Detail Record for each account range
- Trailer Record

The BIN Table Resource File is sent to acquirers either as a TR52 production bulk type or as a TR54 test bulk type. Standard Mastercard Global File Transfer methods are used to distribute the BIN Table Resource File.

The BIN Table Resource File layout contains the account ranges (low range and high range) along with the associated issuing member ID (that is, ICA number) and all token account ranges in use.

### BIN Table Resource File layout

| Field Name                  | Position | Attribute | Comments/Values                                                                                                                                                                                                                                           |
|-----------------------------|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-reloadable Indicator    | 332-333  | an-2      | Valid values:<br><br>01 = non-reloadable prepaid program<br><br>spaces = reloadable prepaid program or non-prepaid program<br><br>99 = the account range or BIN represents a non-Mastercard product                                                       |
| Anonymous Prepaid Indicator | 334      | ans-1     | Valid values:<br><br>A = Anonymous prepaid program and not anti-money laundering directives (AMLDs) compliant<br><br>E = Anonymous prepaid program and AMLDs compliant<br><br>N = Not prepaid or non-anonymous prepaid program/default<br><br>U = Unknown |

| Field Name                    | Position | Attribute | Comments/Values                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cardholder Currency Indicator | 335      | an-1      | <p>Valid values:</p> <p>C = Do not provide Point of Interaction (POI) currency conversion (or Dynamic Currency Conversion [DCC]) – corporate client opt-out</p> <p>D = POI currency conversion (or DCC) is permissible for account range per Mastercard Rules</p> <p>T = Do not provide POI currency conversion (or DCC) per Mastercard Rules – travel cards or multi-currency accounts</p> <p>U = Not a Mastercard-branded account range</p> |

#### BIN Table Resource File header record

| Field Name             | Attribute | Comments                                                                                            |
|------------------------|-----------|-----------------------------------------------------------------------------------------------------|
| Record Type Identifier | an-1      | <p>Code indicating the type of record.</p> <p>Valid value: H (Header)</p>                           |
| Transmission Date      | n-6       | <p>Date the file was created. Format: YYMMDD</p> <p>YY = Year</p> <p>MM = Month</p> <p>DD = Day</p> |
| Transmission Time      | n-4       | <p>Time the file was created. Format: HHMM</p> <p>HH = Hour</p> <p>MM = Minute</p>                  |

| Field Name | Attribute | Comments             |
|------------|-----------|----------------------|
| Filler     | ans-339   | Valid values: spaces |

#### BIN Table Resource File detail record

| Field Name                 | Attribute | Comments                                                                                                                                                                                                                                                 |
|----------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record Type Identifier     | an-1      | Code indicating the type of record.<br>Valid value: D (Detail)                                                                                                                                                                                           |
| Low Primary Account Range  | an-19     | The lowest possible account number that would be identified within this authorization account range.                                                                                                                                                     |
| High Primary Account Range | an-19     | The highest possible account number that would be identified within this authorization account range.                                                                                                                                                    |
| Acceptance Brand           | an-3      | Mastercard or other proprietary service mark.<br>Valid values: <ul style="list-style-type: none"> <li>• MCC (Mastercard®)</li> <li>• DMC Debit Mastercard®}</li> <li>• MSI (Maestro®)</li> <li>• CIR (Cirrus®)</li> <li>• PVL (Private Label)</li> </ul> |
| Customer ID                | n-11      | Mastercard ICA number associated with this authorization account range. Format: right-justified, leading zeros.                                                                                                                                          |
| Customer Name              | an-70     | The customer name.                                                                                                                                                                                                                                       |



| Field Name                | Attribute | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Issuing Country Code      | n-3       | <p>Licensed Country Code of the account range.</p> <p>Valuable in e-commerce fraud management to help detect inconsistencies between the IP address of the originating purchase and the cardholder billing address that may warrant additional analysis. When used in conjunction with the Local Use flag, a merchant can identify when a card is being used outside of its intended allowed area of use.</p>                                                                                                  |
| Local Use                 | n-1       | <p>Indicates whether cards within the authorization account range may be used outside of the country of issuance.</p> <p>Y = The range is intended for domestic use only.</p> <p>N = The range is intended for use outside the country of issuance.</p>                                                                                                                                                                                                                                                        |
| Authorization only        | n-1       | <p>Indicates account range is only valid for authorization processing.</p> <p>Y = Authorization only.</p> <p>N = Authorization and Clearing.</p> <p><b>NOTE: Accounts identified as Authorization Only that are submitted for clearing will reject with Clearing error code 0011 DE 2 PRIMARY ACCOUNT NUMBER (PAN) ACCOUNT RANGE INVALID. This information is provided in Message Exception/1644-691 messages to the acquirer and reported on report IP857010-AA Error Detail – Input Sequence Report.</b></p> |
| Brand Product Code        | ans-3     | The Brand Product Code associated with the account range.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Brand Product Description | ans-200   | The description of the Brand Product Code named above.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Field Name                  | Attribute | Comments                                                                                                                                                                                                                  |
|-----------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-reloadable indicator    | an-2      | Indicates whether the account range is registered as a non-reloadable prepaid card program.<br>Valid values:<br><br>01 = non-reloadable prepaid program<br><br>spaces = reloadable prepaid program or non-prepaid program |
| Anonymous Prepaid Indicator | ans-1     | Valid values:<br><br>A = Anonymous prepaid program and not AMLDS compliant<br><br>E = Anonymous prepaid program and AMLDS compliant<br><br>N = Not prepaid or non-anonymous prepaid program/default<br><br>U = Unknown    |
| Filler                      | ans-18    | Valid value: spaces                                                                                                                                                                                                       |

#### BIN Table Resource File trailer record

| Field Name                | Attribute | Comments                                                                                                              |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------|
| Record Type Identifier    | an-1      | Code indicating the type of record.<br>Valid value: T (Trailer)                                                       |
| Number of Records in File | n-12      | Total number of records in the file, including header and trailer records.<br>Format: right-justified, leading zeros. |
| Filler                    | ans-337   | Valid values: spaces                                                                                                  |

## PVV/PIN Offset File

The PVV/PIN Offset File has a fixed length of 64 characters.

Each file consists of the following:

- Header record
- Detail records
- Trailer record

Issuers sending their files to the Dual Message Authorization System can use Bulk ID RA85 or CONNECT:Direct to send the PVV/PIN Offset file to Mastercard.

Issuers sending their files to the Single Message System can use Bulk ID RM29 (Production), RM31 (MTF), or CONNECT:Direct to send the PVV/PIN Offset file to Mastercard.

### PVV/PIN Offset File header record

| Field Name            | Attribute    | Length | Comments/Values                                                                                               |
|-----------------------|--------------|--------|---------------------------------------------------------------------------------------------------------------|
| Record type ID        | Alphanumeric | 3      | HDR: header record                                                                                            |
| File version number   | Numeric      | 3      | 100 (initial version)                                                                                         |
| Customer ID           | Alphanumeric | 11     | First six digits must contain ICA with leading zeros<br><br>Seventh digit and beyond contains trailing spaces |
| Transmission code     | Alphanumeric | 10     | Member assigned transmission code. May contain all spaces or zeros.                                           |
| Transmission sequence | Numeric      | 4      | 0000-9999<br><br>Wraps at 9999                                                                                |
| Transmission date     | Numeric      | 6      | YYMMDD in UTC                                                                                                 |
| Transmission time     | Numeric      | 6      | hhmmss in UTC                                                                                                 |
| Input file type       | Alphanumeric | 1      | F: Full file replace<br><br>U: Update (not supported)                                                         |
| Filler                | Alphanumeric | 20     | Spaces                                                                                                        |

### PVV/PIN Offset File detail record

| Field Name           | Attribute    | Length | Comments/Values                                                                                                                                                                   |
|----------------------|--------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record type ID       | Alphanumeric | 3      | DTL: detail record                                                                                                                                                                |
| Update code          | Alphanumeric | 1      | A = add/update<br>D = delete (not supported)                                                                                                                                      |
| PAN                  | Alphanumeric | 19     | With trailing spaces                                                                                                                                                              |
| Card expiry date     | Numeric      | 4      | YYMM (must contain a valid year and month)                                                                                                                                        |
| Card sequence number | Numeric      | 1      | Must contain the sequence number associated with the PAN or 0. A value of 0 indicates that the card sequence number should not be used as criteria when matching to the PVV file. |
| PVV/PIN Offset       | Numeric      | 6      | Trailing spaces                                                                                                                                                                   |
| Filler               | Alphanumeric | 30     | Spaces                                                                                                                                                                            |

### PVV/PIN Offset File trailer record

| Field Name               | Attribute    | Length | Comments/Values     |
|--------------------------|--------------|--------|---------------------|
| Record type ID           | Alphanumeric | 3      | TRL: trailer record |
| Number of detail records | Numeric      | 11     | Detail record count |
| Filler                   | Alphanumeric | 50     | Spaces              |

## In-flight Commerce Blocked Gaming File

The In-flight Commerce (IFC) Blocked Gaming File has these specifications.

### Report overview

|             |                                                                  |
|-------------|------------------------------------------------------------------|
| Frequency:  | Generated twice per month                                        |
| Media:      | Bulk File (bulk type T104) or Mastercard File Express (ID AM005) |
| LRECL:      | 100                                                              |
| Format:     | Fixed block                                                      |
| Block Size: | 1,000                                                            |

Refer to the following file layouts for both bulk file and Mastercard File Express.

### IFC Blocked Gaming File header record

| Field Name      | Position | Length   | Comments/Valid Values  |
|-----------------|----------|----------|------------------------|
| Record Type     | 1-3      | 3        | Constant RHD           |
| Processing Date | 4-11     | 8        | Date of file creation. |
| Format:         | =        | YYYYMMDD | Must be numeric        |
| YYYY            | =        | year     |                        |
| MM              | =        | month    |                        |
| DD              | =        | day      |                        |
| Filler          | 12-100   | 89       | Valid values: spaces   |

### IFC Blocked Gaming File Financial detail record

| Field Name  | Position | Length | Comments/Valid Values |
|-------------|----------|--------|-----------------------|
| Record Type | 1-3      | 3      | Constant DTL          |

| Field Name     | Position | Length | Comments/Valid Values                                                                                            |
|----------------|----------|--------|------------------------------------------------------------------------------------------------------------------|
| From BIN       | 4-22     | 19     | First account number in the BIN range to be blocked.<br><br>Format: numeric, left-justified with trailing zeros. |
| To BIN         | 23-41    | 19     | Last account number in the BIN range to be blocked.<br><br>Format: numeric, left-justified with trailing nines.  |
| ICA            | 42-46    | 6      | ICA number associated with the blocked BIN.<br><br>Format: numeric, left-justified with trailing zeros.          |
| Effective Date | 47-54    | 8      | Date the BIN should begin being blocked.                                                                         |
| YYYY           | =        | year   | Must be numeric                                                                                                  |
| MM             | =        | month  |                                                                                                                  |
| DD             | =        | day    |                                                                                                                  |
| Filler         | 55-100   | 46     | Valid values: spaces                                                                                             |

#### IFC Blocked Gaming File trailer record

| Field Name   | Position | Length | Comments/Valid Values |
|--------------|----------|--------|-----------------------|
| Record Type  | 1-3      | 3      | Constant TRL          |
| Record Count | 4-12     | 9      | Numeric               |
| Filler       | 13-100   | 88     | Valid values: spaces  |

## Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB)

The Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB) contains all tokenized device account listings for a customer. This report shows device tokens currently on file with the digitization service and their associated status (active, suspended, or deactivated).

The Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB) separates all tokenized device account listings from the Detail Account Report (AM730010-AA). This allows customers to view device tokens alone in one report every month, if they opt to receive it. This report eliminates the need to scan through the list of all types of accounts (including non-tokens) to check a specific device token.

The Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB) uses the same structure as the Detail Account Report (AM730010-AA).

**NOTE: Tokenized account listings are no longer located on the Detail Account Report (AM730010-AA).**

The Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB) contains the following sections:

- Secure Element Tokens
- Mastercard Cloud-Based Payments Tokens

These two sections identify the type of device token provisioned via the MDES.

The Secure Element Tokens section is present when Mastercard has one or more secure element tokens on file for the issuer in the Account Management System.

The Mastercard Cloud-Based Payments Tokens section is present when Mastercard has one or more Mastercard Cloud-Based Payments tokens on file for the issuer in the Account Management System.

The Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB) is provided monthly via the T318 bulk file or eService OnLine Reports.

For a sample of the Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB), refer to the [Account Management System User Manual](#).

# Appendix A Message data elements

*These tables list all data elements (DEs) implemented within the ISO 8583:1987 message standard, both used and not used by the Dual Message Authorization System, in numeric order.*

---

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## Data elements currently used by the Dual Message Authorization System

This table lists all the data elements used by the Dual Message Authorization System. For more details on a specific data element, refer to [Data element definitions and details](#).

### Data elements currently used

| Data element | Data element title                                              |
|--------------|-----------------------------------------------------------------|
| DE 1         | Bit Map, Secondary                                              |
| DE 2         | Primary Account Number [PAN]                                    |
| DE 3         | Processing Code                                                 |
| DE 4         | Amount, Transaction                                             |
| DE 5         | Amount, Settlement                                              |
| DE 6         | Amount, Cardholder Billing                                      |
| DE 7         | Transmission Date and Time                                      |
| DE 9         | Conversion Rate, Settlement                                     |
| DE 10        | Conversion Rate, Cardholder Billing                             |
| DE 11        | System Trace Audit Number [STAN]                                |
| DE 12        | Time, Local Transaction                                         |
| DE 13        | Date, Local Transaction                                         |
| DE 14        | Date, Expiration                                                |
| DE 15        | Date, Settlement                                                |
| DE 16        | Date, Conversion                                                |
| DE 18        | Merchant Type                                                   |
| DE 19        | Acquiring Institution Country Code                              |
| DE 20        | Primary Account Number [PAN] Country Code                       |
| DE 22        | Point-of-Service [POS] Entry Mode                               |
| DE 23        | Card Sequence Number                                            |
| DE 26        | Point-of-Service (POS) Personal ID Number [PIN]<br>Capture Code |
| DE 28        | Amount, Transaction Fee                                         |
| DE 32        | Acquiring Institution ID Code                                   |
| DE 33        | Forwarding Institution ID Code                                  |
| DE 35        | Track 2 Data                                                    |

| <b>Data element</b> | <b>Data element title</b>                         |
|---------------------|---------------------------------------------------|
| DE 37               | Retrieval Reference Number                        |
| DE 38               | Authorization ID Response                         |
| DE 39               | Response Code                                     |
| DE 41               | Acceptor Terminal ID                              |
| DE 42               | Acceptor ID                                       |
| DE 43               | Acceptor Name and Location                        |
| DE 44               | Additional Response Data                          |
| DE 45               | Track 1 Data                                      |
| DE 48               | Additional Data: Private Use                      |
| DE 49               | Currency Code, Transaction                        |
| DE 50               | Currency Code, Settlement                         |
| DE 51               | Currency Code, Cardholder Billing                 |
| DE 52               | Personal ID Number [PIN] Data                     |
| DE 53               | Security-Related Control Information              |
| DE 54               | Additional Amounts                                |
| DE 55               | Integrated Circuit Card [ICC] System-Related Data |
| DE 56               | Payment Account Data                              |
| DE 60               | Advice Reason Code                                |
| DE 61               | Point-of-Service [POS] Data                       |
| DE 62               | Intermediate Network Facility [INF] Data          |
| DE 63               | Network Data                                      |
| DE 64               | Message Authentication Code                       |
| DE 65               | Bit Map, Extended                                 |
| DE 67               | Extended Payment Code                             |
| DE 70               | Network Management Information Code               |
| DE 90               | Original Data Elements                            |
| DE 91               | Issuer File Update Code                           |
| DE 94               | Service Indicator                                 |
| DE 95               | Replacement Amounts                               |
| DE 96               | Message Security Code                             |

| Data element | Data element title                        |
|--------------|-------------------------------------------|
| DE 100       | Receiving Institution ID Code             |
| DE 101       | File Name                                 |
| DE 102       | Account ID 1                              |
| DE 103       | Account ID 2                              |
| DE 104       | Digital Payment Data                      |
| DE 105       | Multi-Use Transaction Identification Data |
| DE 106       | Fleet Service Data                        |
| DE 108       | Additional Transaction Reference Data     |
| DE 110       | Additional Data–2                         |
| DE 110       | Encryption Data                           |
| DE 112       | Additional Data [National Use]            |
|              |                                           |
|              |                                           |
|              |                                           |
|              |                                           |
|              |                                           |
| DE 119       | Additional Data: Private Use 2            |
| DE 120       | Record Data                               |
| DE 121       | Authorizing Agent ID Code                 |
| DE 122       | Additional Record Data                    |
| DE 123       | Receipt Free Text                         |
| DE 124       | Member-Defined Data                       |
| DE 125       | New PIN Data                              |
| DE 126       | Private Data                              |
| DE 127       | Private Data                              |
| DE 128       | Message Authentication Code               |

## Data elements currently not used by Dual Message Authorization System

This table lists all the data elements not used by the Dual Message Authorization System. For more details on a specific data element, refer to [Data element definitions and details](#).

### Data elements currently not used

| Data element | Data element title                     |
|--------------|----------------------------------------|
| DE 8         | Amount, Cardholder Billing Fee         |
| DE 17        | Date, Capture                          |
| DE 21        | Forwarding Institution Country Code    |
| DE 24        | Network International ID               |
| DE 25        | Point-of-Service [POS] Condition Code  |
| DE 27        | Authorization ID Response Length       |
| DE 29        | Amount, Settlement Fee                 |
| DE 30        | Amount, Transaction Processing Fee     |
| DE 31        | Amount, Settlement Processing Fee      |
| DE 34        | Primary Account Number [PAN], Extended |
| DE 36        | Track 3 Data                           |
| DE 40        | Service Restriction Code               |
| DE 46        | Expanded Additional Amounts            |
| DE 47        | Additional Data: National Use          |
| DE 57- DE 59 | Reserved for National Use              |
| DE 66        | Settlement Code                        |
| DE 68        | Receiving Institution Country Code     |
| DE 69        | Settlement Institution Country Code    |
| DE 71        | Message Number                         |
| DE 72        | Message Number Last                    |
| DE 73        | Date, Action                           |
| DE 74        | Credits, Number                        |
| DE 75        | Credits, Reversal Number               |
| DE 76        | Debits, Number                         |

Message data elements

Data elements currently not used by Dual Message Authorization System

| Data element | Data element title              |
|--------------|---------------------------------|
| DE 77        | Debits, Reversal Number         |
| DE 78        | Transfers, Number               |
| DE 79        | Transfers, Reversal Number      |
| DE 80        | Inquiries, Number               |
| DE 81        | Authorizations, Number          |
| DE 82        | Credits, Processing Fee Amount  |
| DE 83        | Credits, Transaction Fee Amount |
| DE 84        | Debits, Processing Fee Amount   |
| DE 85        | Debits, Transaction Fee Amount  |
| DE 86        | Credits, Amount                 |
| DE 87        | Credits, Reversal Amount        |
| DE 88        | Debits, Amount                  |
| DE 89        | Debits, Reversal Amount         |
| DE 92        | File Security Code              |
| DE 93        | Response Indicator              |
| DE 97        | Amount, Net Settlement          |
| DE 98        | Payee                           |
| DE 99        | Settlement Institution ID Code  |
| DE 107       | Reserved for Mastercard Use     |
| DE 109       | Reserved for ISO Use            |
| DE 111       | Reserved for ISO Use            |
| DE 113       | Reserved for National Use       |
| DE 114       | Reserved for National Use       |
| DE 115       | Reserved for National Use       |
| DE 116       | Reserved for National Use       |
| DE 117       | Reserved for National Use       |
| DE 118       | Reserved for National Use       |

## Appendix B Services or products leveraging the system

*Mastercard offers services and products that leverage the Dual Message Authorization System.*

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Some of the products in this chapter are optional. Mastercard does not require acquirers or issuers to enroll in these optional products and services unless Participation Requirements state otherwise. Acquirers and issuers must code their systems to support these fields to leverage such product and services in a timely manner in the event of a security issue.

## Account Level Management

Mastercard Account Level Management is a platform that enables specialized processing so that Mastercard can manage capabilities at the individual card account level.

Mastercard Account Level Management provides issuers the flexibility of qualifying cardholder accounts for competitive interchange as well as upgrading cardholder accounts to different products.

The following are the key capabilities of Account Level Management functionality:

- Enhanced Value
- Product Graduation Plus
- Product Graduation Select
- High Value
- Spend Shortfall (World Mastercard and World Elite Mastercard)
- Small Business Spend Processing

For more information, refer to the [Account Level Management Manual](#).

## Alternate Processing

Both Stand-In Authorization and X-Code verify that DE 38, position 6 in the Authorization Request Response/0110 message contains the value provided by the Dual Message Authorization System in DE 48, subelement 38 of the Authorization Request/0100 message.

### Stand-In Authorization for Product Graduation

Stand-In Authorization supports Product Graduation. For product graduated accounts, Stand-In Authorization uses the product code in DE 63, subfield 1 instead of the product code associated with the authorization account range when applying Stand-In Authorization limits at the product code level.

### Graduated Product Code Limits

Mastercard offers issuers graduated product code limits as an option for defining Stand-In Authorization parameters for graduated accounts. Issuers can establish limits at the authorization account range level for each product code supported for Product Graduation. These limits will apply specifically to accounts within that account range that are graduated to that product and can be different from the limits that the issuer establishes for an account range that is associated to the same product code. If the issuer takes no action to establish limits for product codes supported for Product Graduation, the Mastercard default limits for the graduated product will apply. Mastercard provides the *Stand-In Processing Transaction Category*

Code *Global Parameters* (Form 041g) available on Mastercard Connect™ for issuer defined limits.

### X-Code Processing

If the product code in DE 63, subfield 1 is different than the product code assigned to the authorization account range, X-Code applies product code limits using the product code in DE 63, subfield 1.

## Account Management System

Account Management System (AMS) services include account-level services that issuers use to designate special handling during authorization. Issuers can list cardholder accounts as exception accounts or accounts with a preferred status.

For more information, refer to the [Account Management System User Manual](#).

## Cross-Border Fee Manager Service

The Cross-Border Fee Manager service is an optional service that gives issuers the flexibility to set markup rates for both currency-conversion and cross-border transactions, taking into account a variety of parameters.

Through this service, Mastercard offers two different methods, Currency Conversion Method and Cross Border Flex Method. Issuers may use these methods to add a markup.

Participating issuers may implement any combination of the following methods and add the associated fees to the total cardholder billing amount.

- Currency Conversion Method: Allows issuers to charge cardholders a percentage rate, fixed amount, or both by applying a markup to currency-converted transactions that may be further filtered by setup parameters.
- Cross-Border Flex Method: Allows issuers to charge cardholders an additional percentage rate, fixed amount, or both by applying a markup to cross-border transactions that may be further filtered by setup parameters.

**NOTE: Issuers may only participate in the Currency Conversion Method and the Cross-Border Flex Method where permitted by local regulations.**

For more information, refer to the [Cross-Border Fee Manager Service Program Guide](#).

## Debit Online Low Risk Merchant (DOLM) program

The DOLM Program is a program for Belgium that is available to e-commerce merchants that have their transactions processed by acquirers located in the European Economic Area (EEA) countries, United Kingdom, and Gibraltar and to issuers of Maestro cards supporting e-

commerce in Belgium. Eligible acquirers must have a licensed area of use in the EEA countries, United Kingdom, and Gibraltar.

**NOTE: Applies only to cards issued in Belgium.**

The request to participate in the program must be submitted by the merchant and its acquirer.

For participating merchants, the program facilitates acceptance of Maestro cards issued in Belgium.

### **DOLM eligibility requirements**

For each merchant, the acquirer must contact its local Mastercard representative to obtain the DOLM participation request form and submit the completed form to its Mastercard representative. Mastercard staff will review the participation request form and notify the acquirer if the merchant has been enrolled in the program. The DOLM Program does not apply to Mastercard branded cards issued in Belgium.

Merchant participation in the program is subject to an eligibility check process to verify that the e-commerce merchant fulfills the following program requirements:

- Maintains fraud rates of less than or equal to 10 basis points for the two consecutive quarters preceding the application for Mastercard and Maestro transactions
- Supports 3D-Secure (3DS) to enable Strong Customer Authentication (SCA) performed by the issuer
- Uses 3DS per the DOLM Operations Requirements section of this bulletin announcement
- Maintains an acquired combined Mastercard and Maestro volume exceeding EUR 100,000 per month for the last 12 months
- For new merchants, completes a Franchise Management Program (FMP) questionnaire to describe its internal risk management processes and tools

### **DOLM operational requirements**

Approved merchants must obtain SCA from the issuer through 3DS at the time of registration of the card payment data on the merchant account. Issuers of Maestro cards in Belgium will require SCA on every recurring payment and Merchant-Initiated Transaction (MIT) coming from every merchant that is not registered into DOLM.

On subsequent transactions, approved merchants must populate the following:

- Data element (DE) 48 (Additional Data: Private Use), subelement 32 (MasterCard Assigned ID) in the Authorization Request/0100 message with a Mastercard-assigned ID
- DE 48, subelement 43 (Universal Cardholder Authentication Field [UCAF]) in the Authorization Request/0100 message with the Mastercard-assigned static AAV of 5MUPPPROGRAM9999999999999999
- DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator) in the Authorization Request/0100 message with the value 213

- Private data subelement (PDS) 0052 (Electronic Commerce Security Level Indicator), subfield 3 (UCAF Collection Indicator) in the clearing record with a value of 3.
- PDS 0176 (Mastercard Assigned ID) in the clearing record submitted to GCMS for processing with a Mastercard-assigned ID

In addition, approved merchants must populate the following requirements to comply with PSD2 to process MITs or recurring payments in a controlled way:

- MIT or recurring payment exemption flag in DE 48, subelement 22 (Multi-Purpose Merchant Indicator), subfield 01 (Low-Risk Merchant Indicator)
- Trace ID in DE 48, subelement 63 (Trace ID), including either the:
  - Trace ID of the initial authorization
  - Value of MCC9999991231\_\_ if the initial authorization was sent before PSD2 became effective (known as grandfathering under PSD2)

### DOLM reviews by Fraud Management Program (FMP)

Merchant fraud levels are measured by acquirer ICA number on a six-month basis. Should the merchant's fraud threshold of 10 basis points be exceeded on Mastercard and Maestro transactions together, the acquirer and the merchant will receive an advisory letter from Mastercard and must undertake a FMP review.

The Mastercard FMP is dedicated to supporting healthy acquirer and merchant growth. The program works with customers to help ensure that they understand and operate within Mastercard Standards to minimize operational, financial, reputational, and compliance risk.

Mastercard is committed to working closely with its customers to fully understand their business dynamics and rationale, while helping to ensure that customers conduct their business in a risk-averse manner. In addition, the FMP provides industry best practices to support business growth by enhancing the overall operational efficiency and profitability while maintaining losses at an acceptable level.

The FMP review format for under-performing merchants is as follows:

- Mastercard will forward the DOLM FMP notification and questionnaire to the acquirer to complete and return within an agreed time frame.
- Mastercard will validate the fraud basis points provided by the merchant and acquirer to help ensure compliance with the DOLM fraud basis point criteria.
- Mastercard will review the completed questionnaire and provide a report to the merchant identifying potential gaps within the merchant's existing fraud control measures and, where appropriate, provide industry best practice guidelines.
- Mastercard will bill the registering acquirer via the Mastercard Consolidated Billing System (MCBS) upon completion of the review. Refer to the online digital *Pricing Guide*, available in the **Pricing and Billing Resource Center** on Mastercard Connect, for details.
- If a merchant's fraud basis points are found to exceed DOLM criteria, Mastercard will agree to an action plan with the merchant or acquirer to reduce the fraud basis points to a level less than the DOLM criteria within an agreed time frame.

## DOLM transactions and fraud liability

Liability and chargeback rights as defined in the [Chargeback Guide](#) will apply. Refer to the [Chargeback Guide](#) for more information. Generally speaking:

- When an approved merchant does not authenticate the cardholder using 3DS and submits a transaction under the DOLM program with a SLI of 213, the issuer has the right to charge back the transaction for fraud if the cardholder states he or she did not authorize the transaction.
- In the initial recurring payments and MITs set up, SCA is a regulatory requirement. The issuer bears liability in case of fraud, as 3DS was used and the issuer applied SCA. The SLI for this scenario can be either 212 or 217.
- For subsequent recurring payments and MITs, the merchant bears liability in case of fraud if the acquirer applied an exemption or exclusion and no 3DS was used. The SLIs for this scenario can be either 210 or 213 or 214 or 216. In the context of DOLM, the SLI will be 213. The acquirer may be able to defend fraud-related chargebacks by providing compelling evidence documenting authentication of the cardholder using 3DS at the start of a recurring transaction arrangement. However, the issuer then still may consider addressing the dispute using a non-fraud chargeback reason code if the issuer can fulfill the conditions (for example, the cardholder notified the merchant to cancel the recurring transaction and the merchant continued to bill the cardholder, or the cardholder was not aware that the cardholder was agreeing to a recurring transaction).

Pre-arbitration case filing is not available when the acquirer provided compelling evidence documenting authentication of the cardholder using Mastercard® SecureCode™ at the start of a recurring transaction arrangement.

## Digital Secure Remote Payment

A Digital Secure Remote Payment (DSRP) transaction is an electronic commerce transaction that contains cryptographic information, in the form of either full EMV chip data passed in DE 55 or a cryptographic value derived from an M/Chip cryptogram passed in the Universal Cardholder Authentication Field (UCAF) or in the Digital Payment Data field.

After the initial DSRP transaction, a related transaction for a partial shipment may occur, in which case, cryptographic information is not passed. When a DSRP transaction contains tokenized account information, Mastercard Digital Enablement Service (MDES) performs token mapping and cryptographic validation services.

DSRP transactions have the capability to include dynamic data to provide evidence that cardholder verification has been performed.

For more information, refer to the [MDES Digital Secure Remote Payment Acquirer Implementation Guide](#) and [MDES Technical Specifications for Dual and Single Message Systems](#).

## Fleet Card transactions

The Mastercard Corporate Fleet Card® is designed to monitor and control fuel and maintenance expenses. A Mastercard Corporate Fleet Card program can be assigned to an individual employee or to a vehicle.

Mastercard Corporate Fleet Card transactions are identified with a specific dedicated Primary Account Number BIN range and a Financial Network Code of MCF (Mastercard Corporate Fleet Card) and MGF (Mastercard Government Commercial Card).

For more information, refer to the [Fleet Implementation Guide Global](#) and [Fleet Implementation Guide For the U.S. Region](#).

### Incentive interchange rates

Acquirers receive incentive interchange rates for transactions that support the data collection requirements of Mastercard Corporate Fleet Card® transactions.

To receive the incentive interchange rates for Fleet Card transactions, acquirers must capture specific data to support submission requirements for clearing records.

For clearing record requirements, refer to the appropriate regional interchange manual.

### Transmitting Fleet Card data in Authorization Request/0100 and Authorization Advice/0120 messages

To support Fleet Card transactions and qualify for specific incentive interchange programs, acquirers must include specific data in Authorization Request/0100 and Authorization Advice/0120 - Acquirer-generated messages.

The Product Type Code determines which data is required from merchant terminals that recognize the Product Type Code on the magnetic stripe or magnetic stripe equivalent data on the EMV chip.

The Product Type Code determines the:

- Terminal prompt at the POI location
- Data that the acquirer must include in the DE 48 (Additional Data: Private Use) field of the Authorization Request/0100 message and Authorization Advice/0120 - Acquirer-generated message

### Terminal prompt at the POI location

When the merchant terminal reads the Product Type Code, the terminal responds by prompting for certain information.

## Product type code values

| Product type code value | Terminal prompt: acquirer may receive from merchant |
|-------------------------|-----------------------------------------------------|
| 1                       | ID Number and Odometer                              |
| 2                       | Vehicle Number and Odometer                         |
| 3                       | Driver Number and Odometer                          |
| 4                       | Odometer Only                                       |
| 5                       | No Prompt                                           |

### Data in the Authorization Request/0100 and Authorization Advice/0120 messages

When the acquirer receives the Fleet Card data from the merchant, the acquirer must include some of the data in the DE 48 (Additional Data: Private Use) field of the Authorization Request/0100 message and Authorization Advice/0120: acquirer-generated message.

### Product type code value details

| Product type code value | Acquirer must include in the 0100 message and 0120 message | Position in the Authorization Request/0100 and Authorization Advice/0120 messages |
|-------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1                       | ID Number                                                  | DE 48, subelement 98                                                              |
| 2                       | Vehicle Number                                             | DE 48, subelement 99                                                              |
| 3                       | Driver Number                                              | DE 48, subelement 98                                                              |
| 4                       | None                                                       | None                                                                              |
| 5                       | None                                                       | None                                                                              |

### Fleet Card Authorization Response

Acquirers that support the Fleet Card program must also be able to interpret the additional data provided by the issuer in DE 44 (Additional Response Data) of the Authorization Request Response/0110 message.



If the issuer declines a transaction because of an invalid driver, ID, or vehicle number, the issuer returns an Authorization Request Response/0110 message with value 12 in DE 39 (Response Code). The issuer returns additional data in DE 44 that explains the reason for the decline. The terminal at the point-of-interaction must display this additional data.

**Transmitting Fleet Card data in Authorization Request/0100 and Authorization Advice/0120 messages for transactions supporting Connexus/IFSF Standards**

The Fleet Prompted Data determines which data is required from merchant terminals that recognize the Fleet Prompted Data information on Connexus/International Forecourt Standards Forum (IFSF) industry standard EMV chip fleet cards.

The Fleet Prompted Data determines the:

- Terminal prompt at the POI location
- Data that the acquirer must include in DE 106 (Fleet Service Data), subelement 1 (Fleet Prompted Data) field of the Authorization Request/0100 message and Authorization Advice/0120: acquirer-generated message

**Terminal prompt at the POI location**

When the merchant terminal reads the Fleet Prompted Data, the terminal responds prompting the cardholder to input specific information at the point of interaction. For a complete list of current prompt codes, refer to DE 106 (Fleet Service Data), subelement 1 (Fleet Prompted Data).

**Data in the Authorization Request/0100 and Authorization Advice/0120 messages**

When the acquirer receives the Fleet Card prompted data from the merchant, the acquirer must include the data in the DE 106 (Fleet Service Data) field of the Authorization Request/0100 message and Authorization Advice/0120: acquirer-generated message.

**Fleet prompt number details**

| <b>Fleet Prompt Number</b> | <b>Acquirer must include in the Authorization Request/0100 message and optionally in the Authorization Advice/0120 message</b> | <b>Position in the Authorization Request/0100 or Authorization Advice/0120 message</b> |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Fleet Prompt 1             | Prompt Code 1                                                                                                                  | DE 106, subelement 1, subfield 1, position 1-3                                         |
| Fleet Prompt 1             | Driver-Related Data 1                                                                                                          | DE 106, subelement 1, subfield 1, position 4-38                                        |
| Fleet Prompt 2             | Prompt Code 2                                                                                                                  | DE 106, subelement 1, subfield 2, position 1-3                                         |

| <b>Fleet Prompt Number</b> | <b>Acquirer must include in the Authorization Request/0100 message and optionally in the Authorization Advice/0120 message</b> | <b>Position in the Authorization Request/0100 or Authorization Advice/0120 message</b> |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Fleet Prompt 2             | Driver-Related Data 2                                                                                                          | DE 106, subelement 1, subfield 2, position 4-38                                        |
| Fleet Prompt 3             | Prompt Code 3                                                                                                                  | DE 106, subelement 1, subfield 3, position 1-3                                         |
| Fleet Prompt 3             | Driver-Related Data 3                                                                                                          | DE 106, subelement 1, subfield 3, position 4-38                                        |
| Fleet Prompt 4             | Prompt Code 4                                                                                                                  | DE 106, subelement 1, subfield 4, position 1-3                                         |
| Fleet Prompt 4             | Driver-Related Data 4                                                                                                          | DE 106, subelement 1, subfield 4, position 4-38                                        |
| Fleet Prompt 5             | Prompt Code 5                                                                                                                  | DE 106, subelement 1, subfield 5, position 1-3                                         |
| Fleet Prompt 5             | Driver-Related Data 5                                                                                                          | DE 106, subelement 1, subfield 5, position 4-38                                        |

When the acquirer receives the details of fuel items purchased with a Fleet Card, they should provide that information in DE 106 (Fleet Service Data), subelement 4 (Fleet Fuel Information).

When the acquirer receives the details of non-fuel items purchased with a Fleet Card, they should provide that information in DE 106 (Fleet Service Data), subelement 5 (Fleet Non-Fuel Information).

### Initiating Fleet Card Support

To learn more about supporting the Fleet Card program, contact Global Customer Service.

## Issuer Currency Conversion Rate (ICCR) service

The ICCR is a parameter-based service that charges cardholders for currency conversion by applying a mark-up in clearing.

This supplemental fee is added to the transaction amount when the transaction currency is different from the cardholder billing currency. This fee is included in DE 6 (Amount, Cardholder Billing).

Issuers set the additional fee for the ICCR by account range, which applies to First Presentment messages.

Issuers can receive daily audit report files of the clearing transactions where the ICCR was applied. Bulk File TN70 (On Request Report Data File Transfer Raw Format) supports the daily audit report files.

ICCR applies to POS and ATM transactions acquired worldwide.

In the Dual Message Authorization System, all payment transactions (processing code = 28) will be bypassed for ICCR.

## Issuer Security Solutions

Mastercard offers best-in-class fraud management solutions to help issuers tackle fraud from every angle.

These solutions were developed to control the cost of fraud, protect cardholders, and minimize potential reputational damage to issuers. Mastercard addresses multi-channel fraud by layering security strategies, policies, and technologies at each stage of the payment life cycle.

For more information, refer to the [Mastercard Security Solutions Guide for Issuer Services](#).

## Decision Intelligence

Decision Intelligence is a real-time authorization decisioning solution that applies thousands of data points and sophisticated modeling techniques to each transaction, simplifying these insights into a single transaction decision score. This score helps issuers fine-tune their authorization decisions with the goal of approving genuine transactions and declining fraudulent ones.

Decision Intelligence enables smarter decisioning, giving issuers a multi-dimensional authorization tool to assess both the risks and rewards of every transaction, so they can enhance the consumer experience by approving more genuine transactions without increasing risk. Decision Intelligence evaluates information about consumers, merchants, and issuers during the shopping experience to help issuers decide whether a transaction makes sense and if it should be approved or declined.

## Decision Intelligence: Authorization IQ feature

In an overall effort to reduce and prevent fraud, Mastercard provides Authorization IQ, an actively managed segmentation feature of the Decision Intelligence service that can help issuers increase approvals and ensure a more satisfying consumer experience without incurring greater risk.

Authorization IQ gives issuers targeted, real-time segmentation insights to help them identify best-spending or most frequent-spending cardholders who show the greatest revenue opportunity at the time of card usage. The service adds a critical dimension to decision management strategies, supplementing and expanding fraud detection, by providing tangible data points to enable data-driven authorization strategies.

## Digital Transaction Insights

Through the Decision Intelligence service, Mastercard provides Digital Transaction Insights, an information exchange framework that enables merchants to share electronic commerce (ecommerce) data with issuers in card-not-present (CNP) environments.

Digital Transaction Insights (DTI) helps improve authorization risk decisioning for card not present (CNP) transactions by connecting billions of data points from the Mastercard payment network and Identity Network merchant risk assessments, and our proprietary models trained on global and region-specific data.

DTI is shared in the authorization message when a merchant submits a transaction through the Mastercard Identity Check network, either as a payment authentication or an Identity Check Insights request. Issuers can work with their authorization processor to integrate DTI into approval or decline decisioning.

## Mastercard ATM Cash Pick-Up service

The Mastercard ATM Cash Pick-Up service allows money to be sent to individual recipients who obtain their funds from an ATM which has been upgraded with the ATM Cash Pick-Up software.

The service recipients receive a code (via, for example, a Short Message Service [SMS] text message) that they can enter into the ATM; therefore, they do not need a debit card or relationship with a bank to receive funds.

The Mastercard ATM Cash Pick-Up service provides the following use cases for customers engaging in the service:

- Relief Fund Disbursement
- Emergency Cash Assistance
- Cash Rewards
- Person to Person Payments
- Small Insurance Claims
- Government Aid Payments/Social Benefits
- Compensation

For more information, refer to the [Mastercard Cash Pick-up Program Guide](#).

## Mastercard Commercial Payments Account

This section describes Dual Message Authorization System edits supporting the introduction of the MAP (Mastercard Commercial Payments Account) product code in Brazil and the Mastercard European Economic Area subregion.

### Dual Message Authorization System Edit to Support MAP in Brazil

Mastercard offers a domestic card-not-present (CNP) commercial payment solution in Brazil using the product code MAP.

The Dual Message Authorization System performs the following system edit to ensure product code MAP is used only in CNP transactions in Brazil.

#### Edits

| When...                                                                                                                                                                                  | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PAN in DE 2 contains the issuer account range associated with brand product MAP and the merchant country code is NOT EQUAL to Brazil (BRA) and DE 61, subfield 5 is NOT EQUAL to value 1 | rejects Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages with: <ul style="list-style-type: none"><li>DE 39 (Response Code) = 12 (Invalid Transactions) is returned in Authorization Request Response/0110 and Authorization Advice Response/0130 messages.</li></ul> |

## Dual Message Authorization System Edits to Support MAP in Mastercard European Economic Area Subregion

The Dual Message Authorization System will perform the following system edits.

### Edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the product code is MAP and merchant country code is not in the below country list: <ul style="list-style-type: none"><li>• Austria (040)</li><li>• Belgium (056)</li><li>• Bulgaria (100)</li><li>• Croatia (191)</li><li>• Czech Republic (203)</li><li>• France (250)</li><li>• Germany (280)</li><li>• Greece (300)</li><li>• Iceland (352)</li><li>• Ireland (372)</li><li>• Italy (380)</li><li>• Liechtenstein (438)</li><li>• Luxembourg (442)</li><li>• Malta (470)</li><li>• Netherlands (528)</li><li>• Norway (578)</li><li>• Portugal (620)</li><li>• Slovakia (703)</li><li>• Slovenia (705)</li><li>• Sweden (752)</li><li>• United Kingdom (826)</li></ul> | will reject Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages<br><br>with: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 12 (Invalid Transaction) returned in Authorization Request Response/0110 and Authorization Advice Response/0130 messages.</li></ul> |

Edits

| WHEN...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>PAN in DE 2 contains issuer account range associated with brand product MAP and Card Present Indicator DE 61 (Point-of-Service [POS] Data), subfield 5 (POS Card Presence) is not equal to value 1 (Card not present), and merchant country code is not in the below country list:</p> <ul style="list-style-type: none"><li>• Austria (040)</li><li>• Belgium (056)</li><li>• Bulgaria (100)</li><li>• Croatia (191)</li><li>• Czech Republic (203)</li><li>• France (250)</li><li>• Germany (280)</li><li>• Greece (300)</li><li>• Iceland (352)</li><li>• Ireland (372)</li><li>• Italy (380)</li><li>• Liechtenstein (438)</li><li>• Luxembourg (442)</li><li>• Malta (470)</li><li>• Netherlands (528)</li><li>• Norway (578)</li><li>• Portugal (620)</li><li>• Slovakia (703)</li><li>• Slovenia (705)</li><li>• Sweden (752)</li><li>• United Kingdom (826)</li></ul> | <p>will reject Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages</p> <p>with:</p> <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 12 (Invalid Transaction) returned in Authorization Request Response/0110 and Authorization Advice Response/0130 messages.</li></ul> <p><b>NOTE: This edit is also applicable with Region C to Region C messages.</b></p> |

Mastercard Contactless Mapping service

Mastercard® Contactless Mapping service uses proximity chip payment technology so consumers can pay for their purchases at the point of sale by simply tapping their contactless card or device on a contactless terminal.

The contactless card or device transmits the card information to the terminal. The contactless terminal interacts with existing authorization mechanisms to initiate and complete the authorization. The increased speed and convenience make contactless an attractive way to pay,

especially in high- traffic environments such as mass transit stations, movie theaters, and fast food restaurants.

The Contactless Mapping Service is not available to issuers that support the use of online PIN with a card or device product. If PAN mapping is performed in a transaction using online PIN, the PIN validation will fail.

For more information, refer to the [Contactless On-behalf Services Guide](#).

## Contactless Mapping Service description

Mastercard recommends that a unique account number be used for contactless transactions. This added layer of security protects issuers and cardholders against the risk of fraudulent acquisitions and use of a contactless account number.

One cardholder account is assigned two numbers:

- A primary account number (PAN), which is encoded on the magnetic stripe and printed on the card.
- A different contactless account number, which is personalized on the contactless chip of the same card or on a companion contactless device.

Implementation of a separate contactless account number requires the issuer to:

- Allocate a contactless account number and provide Contactless Mapping functionality to the existing PAN.
- Establish and maintain a cross-reference or mapping database between the PAN and the contactless account number.
- Upgrade the issuer host system to recognize that both account numbers must debit the same cardholder account and that:
  - The contactless account number must be accepted only when the transaction is originated at a contactless terminal.
  - The PAN must be accepted only when the card is swiped or when the PAN is provided by manual entry on a POS, on the Internet, or over the phone.

A contactless account number is a unique number encoded on the proximity chip of a contactless card or device that the Contactless Mapping Service associates to a PAN. A primary account number (PAN) is the number that is printed, encoded, or both, on a Mastercard card that identifies the issuer and the particular cardholder account.

Mastercard recognizes that the use of a separate contactless account number for some issuers may not be viable. For this reason, Mastercard offers the Contactless Mapping Service, which is an optional service that helps issuers process contactless transactions by translating contactless account numbers into primary account numbers that issuers can process with minimal impact.

Mastercard offers two options for participating in for the Contactless Mapping Service:



- Processing-only: At setup, the participating issuer provides Mastercard with a mapping file of PANs and contactless account numbers created and personalized on contactless cards or devices.
- Processing and Contactless Mobile Over-the-Air Provisioning: Mastercard offers secure web services that allow issuers to request both the contactless personalization of a near-field communication (NFC) mobile phone and its activation. Through the activation process, the registered PAN and contactless account number are mapped together. This option is available depending on the issuer market.

Processing that is common to both participation options consists of the Dual Message Authorization System translating contactless authorization requests into messages that the issuers' non-contactless-enabled authorization systems can understand and process.

## Contactless Mapping Service components

The Contactless Mapping Service consists of certain Dual Message Authorization System components.

- PAN Mapping File (MCC106)
- Processing of contactless account number
- Transaction history

**NOTE: Dynamic CVC 3 pre-validation is used when the Contactless Mapping Service is performed on contactless magnetic stripe transactions.**

## Contactless account number

After performing edits, the Dual Message Authorization System maps the contactless account number to the cardholder's PAN if indicated for the service-eligible contactless account range.

The following information provides the Dual Message Authorization System processing details when the contactless account number is in a Contactless Mapping Service-eligible account range and listed or not listed in the PAN Mapping File (MCC106).

### When the PAN Mapping File (MCC106) contains a contactless account number

If the contactless account number is in the PAN Mapping File (MCC106), the Dual Message Authorization System performs the following actions on Authorization Request/0100, Authorization Advice/0120, Reversal Request/0400, and Reversal Advice/0420 messages forwarded to the issuer, Stand-In Authorization, or X-Code.

- Inserts the cardholder's PAN in DE 2 (Primary Account Number [PAN])
- Inserts the PAN card expiration date in DE 14 (Date, Expiration) if available; otherwise, DE 14 is not present
- Performs dynamic CVC 3 pre-validation using CVC 3 validation key and algorithm only on the Authorization Request/0100 message where DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) contains value 91 (PAN auto-entry through contactless magnetic stripe)

- Performs Chip Cryptogram pre-validation (if the issuer is enrolled) on the Authorization Request/0100 message where DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) contains value 07 (PAN auto-entry through contactless M/Chip)
- Inserts the results of the CVC 3 validation in DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services) if DE 22, subfield 1 contains value 91 or, if applicable, the result of the Chip Cryptogram if DE 22, subfield 1 contains value 07
- If DE 22, subfield 1 contains value 91, inserts DE 48, subelement 34 (Dynamic CVC 3 ATC Information) when the CVC 3 validation result was value V (Valid), A (ATC outside allowed range), or E (CVC 3 ATC Replay)
- Changes DE 22, subfield 1, value 91 (PAN auto-entry through contactless magnetic stripe) to value 92 (Contactless input, Contactless Mapping Service applied when acquirer DE 22, subfield 1 = 91); or value 07 (PAN auto-entry through contactless M/Chip) to value 08 (PAN auto-entry through contactless M/Chip Contactless Mapping Service applied)
- Removes DE 35 (Track 2 Data) or DE 45 (Track 1 Data) or both
- Inserts DE 48, subelement 71 (On-behalf Services) results for Contactless Mapping Service

**NOTE: In the contactless magnetic stripe Authorization Request/0100 message to the issuer, DE 48, subelement 71 contains two service results: one for the Dynamic CVC 3 Pre-validation service and one for the Contactless Mapping Service.**

For Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 messages forwarded to acquirers, the Dual Message Authorization System:

- Inserts DE 48, subelement 33 (PAN Mapping File Information)
- Inserts DE 48, subelement 87 (Card Validation Code Result) only in the contactless magnetic stripe Authorization Request Response/0110 response if there were issues with the validation of the CVC 3

Acquirer response messages contain the same DE 2 value that was submitted by the acquirer.

### **When the PAN Mapping File (MCC106) does not contain a contactless account number**

If the contactless account number is not in the PAN Mapping File (MCC106), the Dual Message Authorization System:

- Rejects the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message and sends the acquirer the corresponding Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message with DE 39 (Response Code), value 14 (Invalid card number)
- Creates and sends an Authorization Advice/0120 message with DE 60 (Advice Reason Code), value 140 (Unable to Convert Contactless Account Number) to the issuer

## Alternate processing

When applied to contactless magnetic stripe or contactless M/Chip transactions, the Contactless Mapping Service is performed before any alternate routing to Stand-In Authorization or X-Code.

Mastercard supports issuers that are not available to respond to the Authorization Request/0100 message by considering the results of the CVC 3 or Chip Cryptogram validation in DE 48, subelement 71 when Stand-In Authorization responds to the Authorization Request/0100 message. The options available for Stand-In Authorization processing of CVC 3 results are detailed in [On-behalf Key Management Procedures](#) and [On-behalf Key Management Interface Specifications](#).

If Stand-In Authorization rejects a contactless magnetic stripe Authorization Request/0100 message because of issuer instructions based on CVC 3 validation results, Stand-In Authorization includes in the contactless magnetic stripe Authorization Advice/0120 message sent to the issuer the following values in DE 60, subfield 2:

- 0042 (CVC 3 Unable to Process)
- 0043 (CVC 3 ATC outside allowed range)
- 0044 (CVC 3 Invalid)
- 0045 (CVC 3 Unpredictable number mismatch)
- 0046 (CVC 3 ATC Replay)

In addition, the contactless magnetic stripe Authorization Advice/0120 message includes:

- DE 48, subelement 71
- DE 48, subelement 34, if DE 48, subelement 71, value V, A, or E
- DE 48, subelement 33
- DE 48, subelement 87, if there was a problem with the CVC validation

For more information about additional values sent if Stand-In Authorization rejects a contactless M/Chip Authorization Request/0100 message because of issuer instructions based on the Chip Cryptogram validation results, refer to the [M/Chip Processing Services Service Description Guide](#).

Mastercard does not consider the CVC 3 or Chip Cryptogram validation results in X-Code Processing and may approve an authorization request with an invalid CVC 3 or Chip Cryptogram.

## Authorization reports

The Authorization Parameter Summary Report (SI737010-AA) and Daily Account File Activity Report (AM700010-DD) reports provide information that supports the Contactless Mapping Service.

- The Authorization Parameter Summary Report (SI737010-AA) provides an indicator in the Global Parameters section to identify the account ranges that are participating in the Contactless Mapping Service. Additionally, the On-behalf Services section lists the Dynamic

CVC 3 Pre-validation Service for the participating Contactless Mapping Service account ranges and one of the Chip Cryptogram Validation Services if the issuer is enrolled in these services.

- The Daily Account File Activity Report (AM700010-DD) contains the PAN Mapping File activity.

For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#). For a sample of the Daily Account File Activity Report (AM700010-DD), refer to the [Account Management System User Manual](#).

## Mastercard Digital Enablement Service

The Mastercard Digital Enablement Service (MDES) is a secure, globally-scalable digitization platform for the management, generation, and provisioning of digital payment credentials onto mobile devices, PCs, servers, and other form factors. This provides simpler, more secure digital payment experiences.

MDES was developed to facilitate the financial industry transition from consumer account credentials stored on payment cards to digital credentials provisioned into mobile devices. These digitized credentials enable consumers to perform:

- Payments via existing contactless POS, other POS systems, and newer remote payment methods, such as in-app payments.
- Cash withdrawals or other financial transactions occurring at contactless ATMs with the use of Near Field Communications (NFC)-enabled devices and wallets.

MDES technology has been enhanced to allow the digitization of cardholder credentials stored on servers, facilitating more secure browser-based electronic commerce (e-commerce) and in-app transactions.

For more information about issuer participation in MDES, refer to the [MDES Issuer Implementation Guide](#).

## MDES authorization reports

The Authorization Parameter Summary Report, Daily Account File Activity Report, and Detail Account Report contain information about MDES.

- The Authorization Parameter Summary Report (SI737010-AA) includes a MDES participant parameter in the Global Parameters section.
- The Daily Account File Activity Report (AM700010-AA) includes activity for MDES within AM700010-CC.
- The Detail Account Report (AM730010-AA) includes a section for MDES accounts.

**NOTE: Tokenized device account listings are now listed on the Detail Account Report: Mastercard Digital Enablement Service Device Tokens (AM730010-BB). Refer to the [MDES Register for Issuer Pre-digitization Network Message Values](#) section.**

For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#). For samples of the Daily Account File Activity Report (AM700010-CC) and Detail Account Report (AM730010-AA), refer to the [Account Management System User Manual](#).

## MDES Register for Issuer Pre-digitization Network Message Values

The following Dual Message Authorization System message layouts support real-time messages for tokenization.

MDES Pre-Digitization API documentation that describes the API messages and parameters is available on the Mastercard Developers site.

### Tokenization Eligibility Request (TER)

Following is a list of the data elements, values, and subfields applicable for Tokenization Eligibility Request (TER).

- [Authorization Request/0100 for TER](#)
- [DE 124 Subfields in Authorization Request/0100 for TER](#)
- [Authorization Request Response/0110 for TER](#)
- [DE 124 Subfields for Authorization Request Response/0110 for TER](#)

### Authorization Request/0100 for TER

All mandatory Authorization Request/0100 data elements apply.

#### Authorization Request/0100 for TER

| Data Element ID and Name      | Org | Sys | Dst | Values/Comments                                                                                                                           |
|-------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number) | •   | X   | M   | Cardholder's primary account number                                                                                                       |
| DE 3 (Processing Code)        | •   | X   | M   | 00 = Purchase                                                                                                                             |
| DE 4 (Amount, Transaction)    | •   | X   | M   | Will be zero                                                                                                                              |
| DE 14 (Date, Expiration)      | •   | X   | C   | Cardholder's primary account expiration date                                                                                              |
| DE 22 (POS Entry Mode)        | •   | X   | M   | Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry)<br>Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown) |

| Data Element ID and Name                                                                                                 | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 35 (Track 2 Data)                                                                                                     | •   | X   | C   | The Authorization System creates and provides Track 2 data for Maestro tokenization messages.                                                                                                                                                                                           |
| DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)              | •   | X   | M   | <p>The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA".</p> <p>For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer.</p> |
| DE 48 Transaction Category Code                                                                                          | •   | X   | C   | T (Phone, Mail, or Electronic Commerce Order)                                                                                                                                                                                                                                           |
| DE 48 (Additional Data—Private Use), subelement 23 (Payment Initiation Channel)                                          | •   | X   | C   | Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.                                                                                                                                                                      |
| DE 48 (Additional Data—Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)                 | •   | X   | C   | Contains the identifier associated with the Wallet Provider                                                                                                                                                                                                                             |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator) | •   | X   | C   | <p>Contains the type of PAN mapping account.</p> <ul style="list-style-type: none"> <li>• C = When MDES Secure Element Device Token</li> <li>• H = When MDES Cloud-based Payments Device Token</li> <li>• F = When MDES Static Token</li> </ul>                                         |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 2 (Account Number)           | •   | X   | C   | Indicates the PAN mapping account                                                                                                                                                                                                                                                       |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 6 (Token Requestor ID)       | •   | X   | C   | Contains the ID assigned by the Token Service Provider to the Token Requestor.                                                                                                                                                                                                          |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 8 (Storage Technology)       | •   | X   | C   | Contains a value indicating the Storage Technology of the requested token.                                                                                                                                                                                                              |

| Data Element ID and Name                                                 | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 49 (Currency Code, Transaction)                                       | •   | X   | M   | Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA.<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country. |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) | •   | X   | M   | 9 = Tokenization Request/Notification                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 61 (Additional POS Data), subfield 13 (POS Country Code)              | •   | X   | M   | Indicates the country of the POS location (not the acquirer location) using ISO-specified codes.<br><br>The POS Country Code is set to "840", corresponding to USA.<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.                                                                                                                                  |
| DE 124 (Member-defined Data)                                             | •   | X   | M   | See layout for DE 124.                                                                                                                                                                                                                                                                                                                                                                                                            |

## DE 124 Subfields in Authorization Request/0100 for TER

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

### DE 124 Subfields in Authorization Request/0100 for TER

| Subfield Name  | Attributes | Values/Comments                                                                                    |
|----------------|------------|----------------------------------------------------------------------------------------------------|
| Message Type   | ans-2      | TE = Tokenization Eligibility Request (TER)                                                        |
| Correlation ID | an-14      | Identifier assigned by Mastercard to associate related tokenization request/notification messages. |

| Subfield Name                                          | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Account Number Source                          | an-1                                       | Identifies the method by which the cardholder is attempting to tokenize a primary account number.<br><br>1 = Card on file<br>2 = Card added manually<br>3 = Card added via application<br>4 = Existing token credential<br>5 = Account added via browser                                                             |
| Payment Application Instance ID                        | ans-48, left-justified, padded with spaces | Identifier associated with the payment application instance installed onto a device.                                                                                                                                                                                                                                 |
| Number of Active Tokens for the Primary Account Number | ans-2, leading zeros                       | Number of active or suspended tokens for the primary account number provisioned to a consumer. Space-filled when requested token digitized to a server.<br><br><b>NOTE: Space-filled means if data is not available then length of that field is filled with space; this is specific to ISO message not for API.</b> |



| Subfield Name                   | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wallet Provider Account ID Hash | ans-64     | <p>When provided by the Wallet Provider, the issuer may use this hash value to match against known identifiers for the cardholder; for example, their email addresses on file. If the hash values match, this may aid an issuer's digitization decision by providing additional factors to help verify that the Wallet Provider account holder is indeed their cardholder, or to differentiate between primary and secondary cardholders. The Wallet Provider computes the hash over an email address; MDES receives this and includes it in the data sent to the issuer. The issuer computes the hash of the email address on file for the cardholder; if it matches the hash of the one received from the Wallet Provider, the comparison with the hash value received from the Wallet Provider may be used by the issuer to assist determining the digitization decision. Hashing is used to help the issuer verify the cardholder email (or other relevant identifiers that the issuer may have for the cardholder) without the Wallet Provider providing the full email for privacy and security reasons. When the Wallet Provider is Apple Pay, the hash is generated using the PBKDF2 algorithm (PKCS #5). PBKDF2 is performed using 10 iterations, a salt, and the lowercase account ID as the password. The salt is calculated by taking the lower case UTF-8 bytes from the account ID and performing a SHA-256. Hash calculation example:</p> <p>Example: Input (Password): csharp@walletprovider.com</p> <p>Salt:<br/>41404d1bca85ddb59ab21466e277ac1ac5f61470be120c82a21b1e45b5248123</p> <p>Count: 10</p> <p>Output:<br/>7098014b646d44c6f3b454c5d54f7a32b3b46e2b0c8e2367f3e5307e3036dfe6</p> <p>For all other Wallet Providers, the field contains the hash resulting from the following "accountIdHash" algorithm, rightpadded with spaces. "accountId" is the lower case UTF-8 bytes of the account ID:</p> <pre>public String accountIdHash (String accountId) {String random8Bytes = 123CCB2F30BA420B return random8Bytes + lessSignificant24bytes(strongerHash(accountId + random8Bytes))} public String strongerHash(String dataToHash) {String currentHash = dataToHash; for (int i = 0; i &lt; 5000; i++) {currentHash = sha256(currentHash);} return sha256(sha256(dataToHash) + currentHash);}</pre> |

| Subfield Name                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>NOTE: String random8Bytes = 123CCB2F30BA420B" is a fixed value for all Wallet Providers.</b></p> <p>Hash calculation example:</p> <ul style="list-style-type: none"> <li>email address: cardholdername@walletprovider.com</li> <li>accountId: 63617264686F6C6465726E616D654077616C6C657470726F76696465722E636F6D</li> <li>random8Bytes: 123CCB2F30BA420B</li> <li>output: 123CCB2F30BA420B17F837DF60E2FC9D6965A74476849FC D43A640F 792A2B358</li> </ul> |                                            |                                                                                                                                                                                                                                                                                                                                                              |
| Cardholder Name                                                                                                                                                                                                                                                                                                                                                                                                                                               | ans-27, left-justified, padded with spaces | <p>This field may be present and contain the name of the cardholder. The format is either LASTNAME/FIRSTNAME with the names delimited by a slash "/" (Example: SMITH/JOE) or the format is FIRSTNAME LASTNAME (Example: JOE SMITH).</p> <p>If the cardholder's name is longer than 27 positions, the data will be truncated to the maximum length of 27.</p> |
| Token Type                                                                                                                                                                                                                                                                                                                                                                                                                                                    | an-1                                       | <p>Contains a value indicating the type of requested token.</p> <p>C = Mastercard Cloud-based Payments</p> <p>F = Static Token</p> <p>S = Embedded Secure Element</p>                                                                                                                                                                                        |

## Authorization Request Response/0110 for TER

All mandatory Authorization Request Response/0110 data elements apply.

### Authorization Request Response/0110 for TER

| Data Element ID and Name   | Org | Sys | Dst | Values/Comments                                                   |
|----------------------------|-----|-----|-----|-------------------------------------------------------------------|
| DE 3 (Processing Code)     | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |
| DE 4 (Amount, Transaction) | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |

| Data Element ID and Name     | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code)        | M   | .   | .   | <p>One of the following values indicating how the issuer wants Mastercard to proceed with the tokenization process:</p> <ul style="list-style-type: none"> <li>• 00 = Continue</li> <li>• 85 = Continue, but require additional authentication</li> <li>• 05 or any other value = Decline</li> </ul> <p><b>NOTE: Issuers must send the appropriate decline response codes when declining a provisioning attempt. This should comply with <i>Mastercard Rules</i> which state the limited use of the 05-Do Not Honor response code on all Mastercard and Maestro cards not present decline responses.</b></p> |
| DE 124 (Member-defined Data) | C   | .   | .   | <p>See layout for DE 124. If the issuer responds with DE 39 = 00 or 85, the issuer may optionally provide DE 124. If the issuer responds with DE 39 = 05, DE 124 must not be present.</p>                                                                                                                                                                                                                                                                                                                                                                                                                    |

## DE 124 Subfields for Authorization Request Response/0110 for TER

Absolute positioning of data in DE 124 subfields is required. Each value must be padded with spaces if the length does not fit exactly the length of the subfield. Subfields not containing values must be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

## DE 124 Subfields for Authorization Request Response/0110 for TER

| Subfield Name                               | Attributes           | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |            |                 |                        |     |                                                                                                                                                                                                                     |
|---------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|-----------------|------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Issuer Product Configuration ID             | ans-10               | <p>The unique product configuration identifier provided by the issuer that identifies a particular set of card art, texts and other product related data, provided during the issuer enablement or maintenance process. This ID is case-sensitive.</p> <p>By specifying the Issuer Product Configuration ID on the Token Eligibility Request Response, an issuer may override the usual Product Configuration assigned as the default at Account Range level. This allows greater control over exactly which card art and related card product data gets applied as part of a specific tokenization process. However, this does not provide a mechanism to override the Issuer Terms and Conditions file which always comes from the default Product Configuration(s) assigned at Account Range level.</p> |      |            |                 |                        |     |                                                                                                                                                                                                                     |
| Primary Account Number Card Sequence Number | ans-3, leading zeros | The card sequence number associated with the primary account number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |            |                 |                        |     |                                                                                                                                                                                                                     |
| Token Data and Activation Method(s)         | ans...186            | <p>When the issuer responds with DE 39 = 85 (Continue, but require additional authentication), the issuer may provide one or more methods by which the consumer can retrieve the activation code. If the issuer responds with DE 39 = 00 (Continue) or 05 (Decline), or if the issuer responds with DE 39 = 85 but uses an issuer-specified default for the activation method, this field will not be present. Each method will consist of:</p>                                                                                                                                                                                                                                                                                                                                                            |      |            |                 |                        |     |                                                                                                                                                                                                                     |
|                                             |                      | <table> <tr> <th>Name</th><th>Attributes</th><th>Values/Comments</th></tr> <tr> <td>Activation Method Type</td><td>n-1</td><td> 1 = Masked mobile phone number<br/> 2 = Masked email address<br/> 3 = Automated call center phone number<br/> 4 = Call center phone number<br/> 5 = Website<br/> 6 = Mobile application<br/> 7 = Masked voice call phone number </td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                        | Name | Attributes | Values/Comments | Activation Method Type | n-1 | 1 = Masked mobile phone number<br>2 = Masked email address<br>3 = Automated call center phone number<br>4 = Call center phone number<br>5 = Website<br>6 = Mobile application<br>7 = Masked voice call phone number |
| Name                                        | Attributes           | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |            |                 |                        |     |                                                                                                                                                                                                                     |
| Activation Method Type                      | n-1                  | 1 = Masked mobile phone number<br>2 = Masked email address<br>3 = Automated call center phone number<br>4 = Call center phone number<br>5 = Website<br>6 = Mobile application<br>7 = Masked voice call phone number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |            |                 |                        |     |                                                                                                                                                                                                                     |

| Subfield Name | Attributes              | Values/Comments                                                                                                                 |
|---------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|               |                         | 9 = Identity check EMV 3DS authentication                                                                                       |
|               | Activation Method Value | ans...183 Refer to the following examples.                                                                                      |
|               | Delimiter               | ans...2 If multiple activation methods are provided by the issuer, the issuer must separate each method with the " " delimiter. |
|               |                         | Following the last character of the final method in the field, the issuer should include two " " delimiters.                    |
|               |                         | Values:                                                                                                                         |
|               |                         | " " indicates another Activation Method follows.                                                                                |
|               |                         | "  " indicates the last Activation Method.                                                                                      |

## Examples

1(###) ### 4567 |

1 = Masked mobile phone number

The 1 will be followed by the masked mobile phone number, then the delimiter.

2a\*\*\*d@anymail.com |

2 = Masked email address

The 2 will be followed by the consumer's masked email address (the issuer will mask according to their own format), then the delimiter.

3(555) 123 4567 |

3 = Automated call center phone number

The 3 will be followed by the phone number, then the delimiter. This phone number is not masked.

|                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4(555) 123 8901                                                                                                                                          |
| 4 = Call center phone number                                                                                                                             |
| The 4 will be followed by the phone number, then the delimiter. This phone number is not masked.                                                         |
| 5http://www.anybank.com                                                                                                                                  |
| 5 = Website                                                                                                                                              |
| The 5 will be followed by the issuer’s website URL, then the delimiter.                                                                                  |
| 6com.anybank.mobileapp                                                                                                                                   |
| 6 = Mobile app                                                                                                                                           |
| The 6 will be followed by the issuer’s mobile app information, the content of which depends upon the mobile device operating system, then the delimiter. |
| 7(###) ### 2345                                                                                                                                          |
| 7 = Masked voice call phone number                                                                                                                       |
| The 7 will be followed by the masked voice call phone number. If this were the last Activation Method provided, the value is followed by two delimiters. |
| 9 = Identity check - EMV 3DS                                                                                                                             |

Tokenization Authorization Request (TAR)

Following is a list of the data elements, values, and subfields applicable for Tokenization Authorization Request (TAR).

- [Authorization Request/0100 for TAR](#)
- [DE 124 Subfields in Authorization Request/0100 for TAR](#)
- [Authorization Request Response/0110 for TAR](#)
- [DE 124 Subfields in Authorization Request Response/0110 for TAR](#)

Authorization Request/0100 for TAR

All mandatory Authorization Request/0100 data elements apply.

Authorization Request/0100 for TAR

| Data Element ID and Name      | Org | Sys | Dst | Values/Comments                     |
|-------------------------------|-----|-----|-----|-------------------------------------|
| DE 2 (Primary Account Number) | •   | X   | M   | Cardholder’s primary account number |
| DE 3 (Processing Code)        | •   | X   | M   | 00 = Purchase                       |

| Data Element ID and Name               | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 4 (Amount, Transaction)             | •   | X   | M   | Will be zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 14 (Date, Expiration)               | •   | X   | C   | Cardholder's primary account expiration date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 22 (POS Entry Mode)                 | •   | X   | M   | <p>Subfield 1 (POS Terminal PAN Entry Mode)</p> <p>Values:</p> <ul style="list-style-type: none"> <li>• 01 = PAN manual entry</li> <li>• 05 = Merchant terminal PAN auto-entry through integrated circuit card (reserved for Card DIP)</li> <li>• 07 = Merchant terminal PAN auto-entry through Contactless M/Chip</li> </ul> <p>Subfield 2 (POS Terminal PIN Entry Mode)</p> <p>Values:</p> <ul style="list-style-type: none"> <li>• 0 = (Unspecified or unknown)</li> <li>• 3 = mPOS Software-based PIN Entry Capability, if PIN is present</li> </ul> |
| DE 23 (Card Sequence Number)           | •   | X   | C   | <p>Distinguishes among separate cards having the same DE 2 (Primary Account Number [PAN])</p> <p>This is present only when issuer has opted to validate chip data for tap to add card provisioning.</p>                                                                                                                                                                                                                                                                                                                                                  |
| DE 33 (Forwarding Institution ID Code) | C   | •   | C   | Identifies the token requestor or wallet for PIN block translation. Example - 201115                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DE 35 (Track 2 Data)                   | •   | X   | C   | <p>The Authorization System creates and provides Track 2 data for Maestro tokenization messages.</p> <p>This is present only when issuer has opted to validate chip data for tap to add card provisioning.</p>                                                                                                                                                                                                                                                                                                                                           |

| Data Element ID and Name                                                                                                   | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)                | •   | X   | M   | The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA".<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer. |
| DE 48 Transaction Category Code                                                                                            | •   | X   | C   | T (Phone, Mail, or Electronic Commerce Order)                                                                                                                                                                                                                                    |
| DE 48 (Additional Data - Private Use), subelement 23 (Payment Initiation Channel)                                          | •   | X   | C   | Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.                                                                                                                                                               |
| DE 48 (Additional Data - Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)                 | •   | X   | C   | Contains the identifier associated with the Wallet Provider.                                                                                                                                                                                                                     |
| DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator) | •   | X   | C   | Contains the type of PAN mapping account. Possible values: <ul style="list-style-type: none"> <li>• C = when MDES Secure Element Device Token</li> <li>• H = when MDES Cloud-based Payments Device Token</li> <li>• F = when MDES Static Token</li> </ul>                        |
| DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 2 (Account Number)           | •   | X   | C   | Indicates the PAN mapping account.                                                                                                                                                                                                                                               |
| DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 6 (Token Requestor ID)       | •   | X   | C   | Contains the ID assigned by the Token Service Provider to the Token Requestor.                                                                                                                                                                                                   |
| DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 8 (Storage Technology)       | •   | X   | C   | Contains a value indicating the storage technology of the requested token.                                                                                                                                                                                                       |



| Data Element ID and Name                                                                                             | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data - Private Use), subelement 71 (On-behalf Services [OBS]) subfield 1 (On-behalf [OB] Service)  | •   | X   | C   | <p>This value notifies the issuer of the OBS performed by Mastercard and the result of chip data validation.</p> <p>This is present only when issuer has opted to use OBS for tap to add card provisioning.</p>                                                                                                                                                                                                                                                                                                                                   |
| DE 48 (Additional Data - Private Use), subelement 71 (On-behalf Services [OBS]) subfield 2 (On-behalf [OB] Result 1) | •   | X   | C   | <p>Indicates the results of the service processing.</p> <p>This is present only when issuer has opted to use OBS for tap to add card provisioning.</p>                                                                                                                                                                                                                                                                                                                                                                                            |
| DE 48 (Additional Data: Private Use), Subelement 80 (PIN Service Code)                                               | •   | X   | C   | <p>Indicates the results of PIN processing by the Authorization Platform.</p> <ul style="list-style-type: none"> <li>• PD = The Authorization Platform dropped the PIN (Mastercard Use Only for Credit Transactions with PIN)</li> <li>• PV = The Authorization Platform verified the PIN.</li> <li>• TV = The Authorization Platform translated the PIN for issuer verification.</li> <li>• PI = The Authorization Platform was unable to verify the PIN.</li> <li>• TI = The Authorization Platform was unable to translate the PIN.</li> </ul> |
| DE 48 (Additional Data - Private Use), subelement 82 (Address Verification Service)                                  | •   | X   | C   | 52 = AVS and Authorization Request/0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DE 48 (Additional Data - Private Use), subelement 92 (CVC 2)                                                         | •   | X   | C   | CVC 2 value from the signature panel of the card when applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Data Element ID and Name                                                           | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 49 (Currency Code, Transaction)                                                 | •   | X   | M   | <p>Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA.</p> <p>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.</p> |
| DE 52 (Personal ID Number [PIN] Data) - PIN Block Format 0                         | C   | X   | C   | Online PIN = It contains a number assigned to a cardholder intended to uniquely identify that cardholder at the point of interaction.                                                                                                                                                                                                                                                                                                    |
| DE 55 (Integrated Circuit Card [ICC] System-Related Data)                          | •   | X   | C   | <p>Integrated Circuit Card [ICC] System-Related Data, binary data captured from chip data.</p> <p>This is present only when issuer has opted to validate chip data for tap to add card provisioning.</p>                                                                                                                                                                                                                                 |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)           | •   | X   | M   | 9 = Tokenization Request/Notification                                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 61 (Additional POS Data), subfield 13 (POS Country Code)                        | •   | X   | M   | <p>Indicates the country of the POS location (not the acquirer location) using ISO-specified codes.</p> <p>The POS Country Code is set to "840", corresponding to USA.</p> <p>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.</p>                                                                                                                                  |
| DE 110 (Encryption Data), Dataset 01 (PIN Encryption), TAG 87 (PIN Block Format 4) | C   | X   | C   | <p>Value = 14</p> <p>Online PIN = It contains a number assigned to a cardholder intended to uniquely identify that cardholder at the point of interaction.</p>                                                                                                                                                                                                                                                                           |
| DE 120 (Record Data)                                                               | •   | X   | C   | AVS postal code and address data in the format specified by the issuer                                                                                                                                                                                                                                                                                                                                                                   |

| Data Element ID and Name     | Org | Sys | Dst | Values/Comments        |
|------------------------------|-----|-----|-----|------------------------|
| DE 124 (Member-defined Data) | .   | X   | M   | See layout for DE 124. |

## DE 124 Subfields in Authorization Request/0100 for TAR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

### DE 124 Subfields in Authorization Request/0100 for TAR

| Subfield Name                   | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                          |
|---------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Message Type                    | an-2                                       | TA = Tokenization Authorization Request (TAR)                                                                                                                                                                                                                                            |
| Correlation ID                  | an-14                                      | Identifier assigned by Mastercard that can be used by the issuer to associate related tokenization request/notification messages.                                                                                                                                                        |
| Primary Account Number Source   | an-1                                       | Identifies the method which the cardholder is attempting to tokenize a primary account number.<br><br>1 = Card on file<br>2 = Card added manually<br>3 = Card added via application<br>4 = Existing token credential<br>5 = Account added via browser<br>6 = Account added via chip data |
| Payment Application Instance ID | ans-48, left-justified, padded with spaces | Identifier associated with the payment application instance installed onto a device.                                                                                                                                                                                                     |
| Device Source IP Address        | ans-12, left-justified, padded with spaces | Variable length IP address. Each octet of the IP address is converted to hex, and joined into one string, with the order maintained.                                                                                                                                                     |

| Subfield Name                                                         | Attributes                                  | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wallet Service Provider<br>Account ID Hash                            | ans-64                                      | <p>This field may be present and contain PBKDF2 hash of the consumer's account ID with the Wallet Provider. The Account ID is typically expected to be an email address. PBKDF2 is performed using 10 iterations, a salt, and the lowercase account ID as the password. The salt is calculated by taking the lower case UTF-8 bytes from the account ID and performing a SHA-256.</p> <p>Example: Input (Password):<br/>csharp@walletprovider.com</p> <p>Salt:<br/>41404d1bca85ddb59ab21466e277ac1ac5f61470be<br/>120c82a21b1e45b5248123</p> <p>Count: 10</p> <p>Output:<br/>7098014b646d44c6f3b454c5d54f7a32b3b46e2b0c<br/>8e2367f3e5307e3036dfe6</p> |
| Cardholder Name                                                       | ans-27, left-justified,<br>trailing spaces  | <p>This field may be present and contain the name of the cardholder. The format is either LASTNAME/FIRSTNAME with the names delimited by a slash "/" (Example: SMITH/JOE) or the format is FIRSTNAME LASTNAME (Example: JOE SMITH).</p> <p>If the cardholder's name is longer than 27 positions, the data will be truncated to the maximum length of 27.</p>                                                                                                                                                                                                                                                                                           |
| Wallet Provider<br>Tokenization<br>Recommendation                     | an-1                                        | <p>Tokenization decision suggested by the Wallet Provider. One of the following values:</p> <p>0 = Decline</p> <p>1 = Approve</p> <p>2 = Require additional authentication</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Wallet Provider<br>Tokenization<br>Recommendation<br>Standard Version | an-2, left-justified,<br>padded with spaces | <p>The version of the standards the Wallet Provider is using to determine the suggested tokenization recommendation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Wallet Provider Device<br>Score                                       | n-1                                         | <p>Score assigned by Wallet Provider for the device. Value between 1 and 5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Wallet Provider Account<br>Score                                      | n-1                                         | <p>Score assigned by Wallet Provider for the primary account number. Value between 1 and 5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Subfield Name                                                    | Attributes           | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of Active Tokens for the Primary Account Number           | ans-2, leading zeros | <p>Number of active or suspended tokens for the primary account number provisioned to a consumer. Space-filled when requested token digitized to a server.</p> <p><b>NOTE: Space-filled means if data is not available then length of that field is filled with space; this is specific to ISO message not for API.</b></p>                                                                                                                                                                                                                                                                                                                       |
| Wallet Service Provider Tokenization Recommendation Reason Codes | ans-6                | <p>Code indicating the specific reason the Wallet Provider is suggesting the tokenization recommendation.</p> <p>The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code.</p> <p>The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01, 05, and 16 were encoded, the bitmap would be 0000000010000000000010001 and the hex value of this field would be 008011.</p> <p>If the Wallet Provider has no reason, this field will contain spaces.</p> |

| Subfield Name   | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Location | ans-9      | <p>Latitude and longitude where the device the consumer is attempting to tokenize a card onto is located.</p> <p>Device Location Latitude - an-4; hexadecimal encoded degrees with 2 decimal places. The first two characters represent the latitude value before the decimal place, and the last two characters represent the latitude after the decimal place.</p> <p>Device Location Longitude - an-4; hexadecimal encoded degrees with 2 decimal places. The first two characters represent the longitude value before the decimal place, and the last two characters represent the longitude after the decimal place.</p> <p>Device Location Lat/Long Sector - n-1; one of the following values:</p> <p>1 = Latitude = N, Longitude = W (positive latitude, negative longitude)</p> <p>2 = Latitude = N, Longitude = E (positive latitude, positive longitude)</p> <p>3 = Latitude = S, Longitude = W (negative latitude, negative longitude)</p> <p>4 = Latitude = S, Longitude = E (negative latitude, positive longitude)</p> <p>For example, a value of 264B5A4B1 would be interpreted as:</p> <ul style="list-style-type: none"> <li>Latitude (first four characters) 264B <ul style="list-style-type: none"> <li>First hex digit (for value before decimal place) = 0x26 = 38 (decimal)</li> <li>Second hex digit (for value after decimal place) = 0x4B = 75 (decimal)</li> </ul> </li> <li>Longitude (next four characters) 5A4B <ul style="list-style-type: none"> <li>First hex digit (for value before decimal place) = 0x5A = 90 (decimal)</li> <li>Second hex digit (for value after decimal place) = 0x4B = 75 (decimal)</li> </ul> </li> <li>Device Location Sector (final digit) 1 = NW</li> <li>Final location = 38.75N, 90.75W</li> </ul> <p>This field will contain spaces if the Wallet Provider has not provided this information.</p> |

| Subfield Name               | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobile Number Last 4 Digits | ans-4      | <p>Last four digits of the consumer's mobile phone number</p> <p>This field will contain spaces if the Wallet Provider has not provided this information.</p>                                                                                                                                                                                                                                                            |
| Token Type                  | an-1       | <p>Contains a value indicating the type of requested token.</p> <p>C = Mastercard Cloud-based Payments</p> <p>F = Static</p> <p>S = Embedded Secure Element</p>                                                                                                                                                                                                                                                          |
| Consumer Identifier         | an-88      | <p>Unique identifier for each consumer, which financial institutions will then need to verify before a consumer can add their credentials to the payment-enabled devices. Position 212–299 in the DE 124 of the Authorization Request/0100 - Tokenization Authorization message only.</p> <p>Issuers and processors will not receive the new Consumer Identifier in DE 124 unless they explicitly opt to receive it.</p> |

## Authorization Request Response/0110 for TAR

All mandatory Authorization Request Response/0110 data elements apply.

### Authorization Request Response/0110 for TAR

| Data Element ID and Name   | Org | Sys | Dst | Values/Comments                                                   |
|----------------------------|-----|-----|-----|-------------------------------------------------------------------|
| DE 3 (Processing Code)     | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |
| DE 4 (Amount, Transaction) | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |

| Data Element ID and Name                                                                     | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code)                                                                        | M   | .   | .   | <p>One of the following values indicating how the issuer wants Mastercard to proceed with the tokenization process:</p> <ul style="list-style-type: none"> <li>• 00 = Approve</li> <li>• 85 = Approve, but require additional authentication</li> <li>• 05 or any other value = Decline</li> </ul> <p><b>NOTE: Issuers must send the appropriate decline response codes when declining a provisioning attempt. This should comply with Mastercard Rules which state the limited use of the 05-Do Not Honor response code on all Mastercard and Maestro cards not present decline responses.</b></p> |
| DE 48 Transaction Category Code                                                              | CE  | CE  | .   | Must be the same value as the original Authorization Request/0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DE 48 (Additional Data - Private Use), subelement 82 (Address Verification Service)          | CE  | CE  | .   | 52 = AVS and Authorization Request/0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 48 (Additional Data - Private Use), subelement 83 (Address Verification Service Response) | C   | .   | .   | The AVS response code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| DE 48 (Additional Data - Private Use), subelement 87 (Card Validation Code Result)           | C   | .   | .   | The CVC 2 result code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| DE 48 (Additional Data - Private Use), subelement 92 (CVC 2)                                 | CE  | CE  | .   | Must be the same value as the original Authorization Request/0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DE 48, (Additional Data: Private Use), subelement 09 (Additional PAN Data)                   | C   | C   | .   | These actions must be taken when an issuer-generated consumer VCN is used in DE 2 (Primary Account Number) of the tokenization request message                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DE 56 (Payment Account Data)                                                                 | C   | X   | C   | PAR value provided by the Issuer when Issuer or its designated service provider is BIN Controller (Mastercard is not BIN Controller).                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



| Data Element ID and Name     | Org | Sys | Dst | Values/Comments                                                                                                                                                                    |
|------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 124 (Member-defined Data) | C   | .   | .   | See layout for DE 124. If the issuer responds with DE 39 = 00 or 85, the issuer may optionally provide DE 124. If the issuer responds with DE 39 = 05, DE 124 must not be present. |

## DE 124 Subfields in Authorization Request Response/0110 for TAR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values must be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

### DE 124 Subfields in Authorization Request Response/0110 for TAR

| Subfield Name                               | Attributes           | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Issuer Product Configuration ID             | ans-10               | <p>The unique product configuration identifier provided by the issuer that identifies a particular set of card art, texts and other product related data, provided during the issuer enablement or maintenance process. This ID is case-sensitive.</p> <p>By specifying the Issuer Product Configuration ID on the Tokenization Authorization Request Response, an issuer may override the usual Product Configuration assigned as the default at Account Range level. This allows greater control over exactly which card art and related card product data gets applied as part of a specific tokenization process. However, this does not provide a mechanism to override the Issuer Terms and Conditions file which always comes from the default Product Configuration(s) assigned at Account Range level.</p> |
| Primary Account Number Card Sequence Number | ans-3, leading zeros | The card sequence number associated with the primary account number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Subfield Name                         | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Token Data and Activation Method(s)   | ans...286  | <p>The subfield can have zero to multiple groupings of three Token Data and Activation Method Type, Token Data and Activation Method Value, and Delimiter.</p> <p>When the issuer responds with DE 39 = 85 (Approve, but require additional authentication), the issuer may provide one or more activation methods by which the consumer can retrieve the activation code. If the issuer responds with DE 39 = 00 (Approve) or 05 (Decline) activation methods will not be present. If the issuer responds with DE 39 = 85 but uses an issuer-specified default for the activation method, activation methods may not be present.</p> <p>When the issuer responds with DE 39 = 00 (Approve) or DE 39 = 85 (Approved, but require additional authentication), the issuer may provide zero to multiple token personalization data.</p> |
| Token Data and Activation Method Type | an-1       | <p>1 = Masked mobile phone number (activation method)</p> <p>2 = Masked email address (activation method)</p> <p>3 = Automated call center phone number (activation method)</p> <p>4 = Call center phone number (activation method)</p> <p>5 = Website (activation method)</p> <p>6 = Mobile application (activation method)</p> <p>7 = Masked voice call phone number (activation method)</p> <p>9 = Identity check EMV 3DS (activation method)</p> <p>a = Alternate account identifier</p> <p>t = One or multiple token personalization data items</p>                                                                                                                                                                                                                                                                             |

| Subfield Name                          | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Token Data and Activation Method Value | ans...283  | <p>When the Token Data and Activation Method Type is 1, 2, 3, 4, 5, 6, 7 or 9 the issuer must provide values used to specify an activation method. See Examples 1, 2, 3, and 4 for examples of Token Data and Activation Method(s) containing activation method values.</p> <p>Issuers have the option of providing an alternate account identifier in Tokenization Authorization Response messages (Token Data and Activation Method Type = a).</p> <p>An Alternate Account Identifier is a cardholder-friendly reference to a bank account, for example an IBAN (International Bank Account Number). It is typically useful for a Wallet Provider to display a suffix of this identifier to the cardholder to help them identify their tokenized card when they are not aware of their Account PAN.</p> <p>This will enable wallets to display alternate account identifier suffixes depending on the wallet functionality. Issuers must refer to each wallet's documentation to determine if/how the alternate account identifier suffixes are displayed to cardholders.</p> <p>An alternate account identifier is in the format of a string with minimum 9 and maximum 64 characters, with no spaces allowed. An IBAN is an example of an alternate account identifier, as for example, GB82WEST12345698765432.</p> <p>When the Token Data and Activation Method Type is t, the issuer must provide one or multiple proprietary token personalization data items that MDES will include in the token profile. See Example 8 for an example of Token Data and Activation Method(s) containing multiple token personalization data items.</p> <p>Each token personalization data item can have one to multiple instances of couples Token Personalization Data and Separator.</p> <p>Token Personalization Data: Base64 encoded data. Structure and content of data is defined separately per <i>Mastercard Digital Enablement Service (MDES) Issuer Implementation Guide</i>. See Examples 6 and 7.</p> <p>Separator</p> <p>ans...1</p> <p>Each separate token personalization data (including after the last) will be delimited by a "~" character.</p> |

| Subfield Name | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |            | See Example 5 for an example of Token Data and Activation Method(s) containing multiple activation method values and multiple token personalization data items.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Delimiter     | ans-2      | <p>If multiple Token Data and Activation Methods are provided by the issuer, the issuer must separate each method with the " " delimiter.</p> <p><b>NOTE: When at least another grouping of three Token Data and Activation Method Type, Token Data and Activation Method Value, and Delimiter follows, then the Delimiter has attributes of ans-1.</b></p> <p>Delimiter Values:</p> <p>" " indicates that another Token Data or Activation Method follows.</p> <p>Following the last character of the final Token Data and Activation Method in the field, the issuer should include two " " delimiters.</p> <p>"  " indicates the last Token Data and Activation Method.</p> |

## Examples

| Example | Description                                                                                                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | An example of Token Data and Activation Method(s) subfield containing only a masked mobile phone number activation method: 1(###)###4567                                                                                                                                   |
| 2       | An example of Token Data and Activation Method(s) subfield containing the masked email address activation method and the automated call center phone number activation method: 2a***d@anymail.com 3(555)1234567                                                            |
| 3       | An example of Token Data and Activation Method(s) subfield containing the call center phone number activation method, the website activation method and the masked voice call phone number activation method: 4(555)1238901 5http://www.anybank.com 6com.anybank.mobileapp |
| 4       | An example of Token Data and Activation Method(s) subfield containing only a masked voice call phone number activation method: 7(###)###4567                                                                                                                               |

| Example | Description                                                                                                                                                                                                                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5       | An example of Token Data and Activation Method(s) subfield containing mobile application activation method, masked email address activation method and some personalization data for the token profile:<br><br>6com.anybank.mobileapp 2a***d@anymail.com <br>tBJ9uCwJQAAEwOGECAwQFX1UCR1 =~CwECAwQFBgc=~                     |
| 6       | An example of Token Personalization Data: "BJ9uCwJQAAEwOGECAwQFX1UCR1 =" that is "049F6E 0B 0250 0001 3039 0102030405 5F55 02 4652" base64 encoded. The purpose of Token Personalization Data is to replace default Third Party Data (9F6E) and add Issuer Country Code (5F55) in Contactless FCI Issuer Discretionary Data. |
| 7       | An example of Token Personalization Data: "CwECAwQFBgc=" that is "0B01020304050607" base64 encoded. The purpose of Token Personalization Data is to create a new tag "9F70" (Protected Data Envelope) readable through GET DATA on both Management and Contactless Payment interfaces.                                       |
| 8       | An example of Token Data and Activation Method(s) subfield containing multiple token personalization data items:<br>tBJ9uCwJQAAEwOGECAwQFX1UCR1 =~CwECAwQFBgc=~                                                                                                                                                              |

## Authentication Methods Request (AMR)

Following is a list of the data elements, values, and subfields applicable for Authentication Methods Request (AMR).

- [Authorization Request/0100 for AMR](#)
- [DE 124 Subfields in Authorization Request/0100 for AMR](#)
- [Authorization Request Response/0110 for AMR](#)
- [DE 124 Subfields in Authorization Request Response/0110 for AMR](#)

### Authorization Request/0100 for AMR

All mandatory Authorization Request/0100 data elements apply.

#### Authorization Request/0100 for AMR

| Data Element ID and Name      | Org | Sys | Dst | Values/Comments                              |
|-------------------------------|-----|-----|-----|----------------------------------------------|
| DE 2 (Primary Account Number) | •   | X   | M   | Cardholder's primary account number          |
| DE 3 (Processing Code)        | •   | X   | M   | 00 = Purchase                                |
| DE 4 (Amount, Transaction)    | •   | X   | M   | Will be zero                                 |
| DE 14 (Date, Expiration)      | •   | X   | C   | Cardholder's primary account expiration date |

| Data Element ID and Name                                                                                    | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (POS Entry Mode)                                                                                      | •   | X   | M   | Subfield 1 (POS Terminal PAN Entry Mode) = 01 PAN manual entry<br><br>Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)                                                                                                                                      |
| DE 35 (Track 2 Data)                                                                                        | •   | X   | C   | The Authorization System creates and provides Track 2 data for Maestro tokenization messages.                                                                                                                                                                                    |
| DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code) | •   | X   | M   | The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA".<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer. |
| DE 48 Transaction Category Code                                                                             | •   | X   | C   | T (Phone, Mail, or Electronic Commerce Order)                                                                                                                                                                                                                                    |

| Data Element ID and Name                                                                                                                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data - Private Use) subelement 06 (Advanced Digital Transaction Data 1)<br>Subfield 1 (Binding ID)<br>Subfield 2 (Phone Number)<br>Subfield 3 (Device Location) | •   | X   | C   | <p>Subfield 1 contains the Bind ID - Helps to uniquely identify the component of data that is bound to the credential.</p> <p>Subfield 2 contains the Phone Number - The full phone number or partial phone number of the consumer.</p> <p>Subfield 3 contains the Device Location - Latitude and longitude where the device the cardholder is using is located for authentication.</p> <p>Device Location Latitude - an-4; hexadecimal encoded degrees with 2 decimal places.</p> <p>Device Location Longitude - an-4; hexadecimal encoded degrees with 2 decimal places.</p> <p>Device Location Lat/Long Sector - n-1; one of the following values:</p> <p>1 = Latitude = N, Longitude = W</p> <p>2 = Latitude = N, Longitude = E</p> <p>3 = Latitude = S, Longitude = W</p> <p>4 = Latitude = S, Longitude = E</p> <p>This field will contain spaces if the Wallet Provider has not provided this information.</p> |
| DE 48 (Additional Data - Private Use), subelement 07 (Advanced Digital Transaction Data 2)<br>Subfield 1 (Email Address Format)<br>Subfield 2 (Email Address)                     | •   | X   | C   | <p>Subfield 1 Email Address Format - Value 0 indicates - Contains a hashed email address. Value 1 indicates - Contains a clear email address.</p> <p>Subfield 1 Email Address - The clear email address or hashed email address.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| DE 48 (Additional Data - Private Use), subelement 08 (Advanced Digital Transaction Data 2), subfield 1 (IP Address)                                                               | •   | X   | C   | Subfield 1 IP Address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DE 48 (Additional Data - Private Use), subelement 23 (Payment Initiation Channel)                                                                                                 | •   | X   | C   | Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Data Element ID and Name                                                                                                                                                                                                                                                                                 | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data - Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)                                                                                                                                                                                               | •   | X   | C   | Contains the identifier associated with the Wallet Provider.                                                                                                                                                                                                                                                                                                                                                                      |
| DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information)<br>Subfield 1 (Account Number Indicator)<br>Subfield 2 (Account Number)<br>Subfield 3 (Expiration Date)<br>Subfield 5 (Token Assurance Method)<br>Subfield 6 (Token Requestor ID)<br>Subfield 8 (Storage Technology) | •   | X   | C   | Subfield 1 Indicates the type of PAN mapping account.<br>Subfield 2 Indicates the PAN mapping account.<br>Subfield 3 Indicates the expiration date of the PAN mapping account.<br>Subfield 5 Indicates Token Assurance Method<br>Subfield 6 Contains the ID assigned by the Token Service Provider to the Token Requestor.<br>Subfield 8 Indicates Storage Technology of the token.                                               |
| DE 49 (Currency Code, Transaction)                                                                                                                                                                                                                                                                       | •   | X   | M   | Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA.<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country. |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)                                                                                                                                                                                                                                 | •   | X   | M   | 9 = Tokenization Request/Notification                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 61 (Additional POS Data), subfield 13 (POS Country Code)                                                                                                                                                                                                                                              | •   | X   | M   | Indicates the country of the POS location (not the acquirer location) using ISO-specified codes.<br><br>The POS Country Code is set to "840", corresponding to USA.<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.                                                                                                                                  |



| Data Element ID and Name                                                      | Org | Sys | Dst | Values/Comments                                                                              |
|-------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------|
| DE 104 (Digital Payment Data)<br>Subelement 004 (Digital Service Provider)    | •   | X   | C   | Subfield 001 (Digital Service Provider ID 1)<br>Subfield 002 (Digital Service Provider ID 2) |
| DE 104 (Digital Payment Data)<br>Subelement 005 (Digital Authentication Data) | •   | X   | C   | Subfield 001 (Cardholder Verification)<br>Subfield 002 (Authentication Factors)              |
| DE 124 (Member-defined Data)                                                  | •   | X   | M   | See layout for DE 124.                                                                       |

## DE 124 Subfields in Authorization Request/0100 for AMR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

### DE 124 Subfields in Authorization Request/0100 for AMR

| Subfield Name          | Attributes                                 | Values/Comments                                                                                                                                        |
|------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Message Type           | an-2                                       | AM = Authentication Methods Request (AMR)                                                                                                              |
| Correlation ID         | an-14                                      | Identifier assigned by Mastercard that can be used by the issuer to associate related tokenization request/notification messages.                      |
| Token Unique Reference | ans-48, left-justified, padded with spaces | Service-allocated unique reference to the token.                                                                                                       |
| Token Type             | an-1                                       | Contains a value indicating the type of requested token.<br><br>C = Mastercard Cloud-based Payments<br>F = Static Token<br>S = Embedded Secure Element |

| Subfield Name                | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reason Codes                 | ans-6      | <p>The reason for the cardholder authentication. The reason could impact the generation and validation of the authentication code in a number of ways including but not limited to the time period of validity of the code and the number of attempts allowed to validate the code. New reason codes can be added at any time and should not result in a failure.</p> <ul style="list-style-type: none"> <li>• 01 = ADD_CARD – The account holder is being authenticated to verify account ownership at the time of adding a card in merchant system.</li> <li>• 02 = VERIFY_ACCOUNT - The account holder is being authenticated to verify account ownership to use a service</li> <li>• 03 = OTHER</li> <li>• 04 = PAYMENT_TRANSACTION - The account holder is being authenticated to initiate payment transaction</li> <li>• 05 = TOKEN_BINDING - The account holder is being authenticated to bind a token to bind id.</li> <li>• 06 = ACCOUNT_BINDING - The account holder is being authenticated to bind associated account with bind id (device).</li> </ul> <p>The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code.</p> <p>The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01 and 02 were encoded, the bitmap would be 00000000000000000000000011 and the hex value of this field would be 000003.</p> |
| Token Request Account Score  | n-1        | Score assigned by Token Requestor for the primary account number/Token. Value between 1 and 5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Token Requestor Device Score | n-1        | <p>Score assigned by Token Requestor for the device. Value between 1 and 5.</p> <p>Device used at the time of authentication.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Account Lifetime             | n-2        | The lifetime (in months) of the account with the Token Requestor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| authRequestCorrelationId     | ans-64     | Value linking different authentication requests/messages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Subfield Name                                                                 | Attributes                              | Values/Comments                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device OS Name                                                                | an-1                                    | Operating System of the device that is used for cardholder authentication.<br><br><ul style="list-style-type: none"> <li>0 = Android</li> <li>1 = iOS</li> <li>2 = Windows</li> <li>3 = Tizen</li> <li>4 = PAGER_EMBEDDED</li> <li>5 = ANDROID_WEAR</li> <li>6 = EMBEDDED_OS</li> <li>z = None of the above</li> </ul> |
| Device Name                                                                   | ans-20 left-justified padded with space | Device Name set by the consumer or manufacturer.                                                                                                                                                                                                                                                                       |
| Recently authenticated by trusted authenticating entity ID                    | an-5                                    | Identification of Mastercard trusted authenticator name who recently authenticated cardholder.                                                                                                                                                                                                                         |
| Recent authentication proof authenticated by trusted authenticating entity ID | ans-35                                  | Authentication Proof shared by the trusted authenticator.                                                                                                                                                                                                                                                              |
| Recent authentication time, authenticated by trusted authenticating entity ID | n-10                                    | Date and time when this authentication happened.<br><br>Time is in UTC in the following format: YYMMDDhhmm.                                                                                                                                                                                                            |

## Authorization Request Response/0110 for AMR

All mandatory Authorization Request Response/0110 data elements apply.

### Authorization Request Response/0110 for AMR

| Data Element ID and Name   | Org | Sys | Dst | Values/Comments                                                   |
|----------------------------|-----|-----|-----|-------------------------------------------------------------------|
| DE 3 (Processing Code)     | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |
| DE 4 (Amount, Transaction) | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |

| Data Element ID and Name        | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 39 (Response Code)           | M   | •   | •   | <p>00 - Requested use case is approved and Cardholder Authentication is not required.</p> <p>85 - Cardholder Authentication is required.</p> <p>All other: Requested use case is not approved.</p> <p><b>NOTE: Issuers must send the appropriate decline response codes when declining a provisioning attempt. This should comply with <i>Mastercard Rules</i> which state the limited use of the 05 - Do Not Honor response code on all Mastercard and Maestro cards not present decline responses.</b></p> |
| DE 48 Transaction Category Code | CE  | CE  | •   | Must be the same value as the original Authorization Request/0100                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DE 124 (Member-defined Data)    | M   | •   | •   | See layout for DE 124.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## DE 124 Subfields in Authorization Request Response/0110 for AMR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values must be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

### DE 124 Subfields in Authorization Request Response/0110 for AMR

| Subfield Name                           | Attributes | Values/Comments                                                                                                                                                 |
|-----------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Token Data and Authentication Method(s) | ans...199  | The subfield can have zero to multiple groupings of three Token Data and Authentication Method Type, Token Data and Authentication Method Value, and Delimiter. |

| Subfield Name                              | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Token Data and Authentication Method Type  | an-1       | <ul style="list-style-type: none"> <li>1 = Masked mobile phone number</li> <li>2 = Masked email address</li> <li>3 = Automated call center phone number</li> <li>4 = Call center phone number</li> <li>5 = Website</li> <li>6 = Mobile application</li> <li>7 = Masked voice call phone number</li> <li>8 = Push Notification (means issuer supports push notification to deliver Authentication Code in issuer app)</li> <li>9 = EMV 3DS (Identity Check)</li> </ul>                                                                                                                                                                      |
| Token Data and Authentication Method Value | ans...196  | <p>Authentication Value is required for authentication type - 1, 2, 5.</p> <p>Authentication Value is not required for authentication type - 6, 8, 9.</p> <p>Authentication Value guidance for authentication type 6: value will be PUSH_NOTIFICATION if an issuer wants to their application to open through push notification using Android intent and iOS deep link URL.</p> <p>Issuer will make sure that they configure iOS deep link URL and Android intent information through MDES Manager.</p> <p>PUSH_NOTIFICATION is not applicable to activate a device token, it is only applicable for post-tokenization authentication.</p> |

| Subfield Name | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Delimiter     | ans-2      | <p>If multiple Token Data and Activation Methods are provided by the issuer, the issuer must separate each method with the " " delimiter.</p> <p><b>NOTE: When at least another grouping of three Token Data and Activation Method Type, Token Data and Activation Method Value, and Delimiter follows, then the Delimiter has attributes of ans-1.</b></p> <p>Delimiter Values:</p> <p>" " indicates that another Token Data or Activation Method follows.</p> <p>Following the last character of the final Token Data and Activation Method in the field, the issuer should include two " " delimiters.</p> <p>"  " indicates the last Token Data and Activation Method.</p> |

## Activation Code Notification (ACN)

Following is a list of the data elements, values, and subfields applicable for Activation Code Notification (ACN).

- [Authorization Request/0100 for ACN](#)
- [DE 124 Subfields in Authorization Request/0100 for ACN](#)
- [Authorization Request Response/0110 for ACN](#)

### Authorization Request/0100 for ACN

All mandatory Authorization Request/0100 data elements apply.

#### Authorization Request/0100 for ACN

| Data Element ID and Name      | Org | Sys | Dst | Values/Comments                              |
|-------------------------------|-----|-----|-----|----------------------------------------------|
| DE 2 (Primary Account Number) | •   | X   | M   | Cardholder's primary account number          |
| DE 3 (Processing Code)        | •   | X   | M   | 00 = Purchase                                |
| DE 4 (Amount, Transaction)    | •   | X   | M   | Will be zero                                 |
| DE 14 (Date, Expiration)      | •   | X   | C   | Cardholder's primary account expiration date |

| Data Element ID and Name                                                                                                 | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 22 (POS Entry Mode)                                                                                                   | •   | X   | M   | Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry)<br><br>Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)                                                                                                                                    |
| DE 35 (Track 2 Data)                                                                                                     | •   | X   | C   | The Authorization System creates and provides Track 2 data for Maestro tokenization messages.                                                                                                                                                                                    |
| DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)              | •   | X   | M   | The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA".<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer. |
| DE 48 Transaction Category Code                                                                                          | •   | X   | C   | T (Phone, Mail, or Electronic Commerce Order)                                                                                                                                                                                                                                    |
| DE 48 (Additional Data—Private Use), subelement 23 (Payment Initiation Channel)                                          | •   | X   | C   | Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.                                                                                                                                                               |
| DE 48, subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)                                               | •   | X   | C   | Contains the identifier associated with the Wallet Provider                                                                                                                                                                                                                      |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator) | •   | X   | C   | Contains the type of PAN mapping account. Possible values:<br><ul style="list-style-type: none"> <li>• C = When MDES Secure Element Device Token</li> <li>• H = When MDES Cloud-based Payments Device Token</li> <li>• F = When MDES Static Token</li> </ul>                     |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 2 (Account Number)           | •   | X   | C   | Indicates the PAN mapping account                                                                                                                                                                                                                                                |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 6 (Token Requestor ID)       | •   | X   | C   | Contains the ID assigned by the Token Service Provider to the Token Requestor.                                                                                                                                                                                                   |

| Data Element ID and Name                                                 | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 49 (Currency Code, Transaction)                                       | •   | X   | M   | Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA.<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country. |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) | •   | X   | M   | 9 = Tokenization Request/Notification                                                                                                                                                                                                                                                                                                                                                                                             |
| DE 61 (Additional POS Data), subfield 13 (POS Country Code)              | •   | X   | M   | Indicates the country of the POS location (not the acquirer location) using ISO-specified codes.<br><br>The POS Country Code is set to "840", corresponding to USA.<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.                                                                                                                                  |
| DE 124 (Member-defined Data)                                             | •   | X   | M   | See layout for DE 124.                                                                                                                                                                                                                                                                                                                                                                                                            |

## DE 124 Subfields in Authorization Request/0100 for ACN

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfields will not be present and the total length of DE 124 will be reduced by the subfield lengths.

### DE 124 Subfields in Authorization Request/0100 for ACN

| Subfield Name  | Attributes | Values/Comments                                                                                        |
|----------------|------------|--------------------------------------------------------------------------------------------------------|
| Message Type   | an-2       | AC = Activation Code Notification (ACN)                                                                |
| Correlation ID | an-14      | The Identifier assigned by Mastercard to associate related tokenization request/notification messages. |



| Subfield Name                            | Attributes                                                                                                                                                  | Values/Comments                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activation Code                          | ans-8, left-justified, padded with spaces                                                                                                                   | Activation code assigned by Mastercard that will be provided to the issuer for delivery to the consumer to complete the tokenization process.                                                                                                                                                                                                                       |
| Activation Code Expiration Date and Time | n-10                                                                                                                                                        | The date and time that the Activation Code expires specified in UTC units.<br>Format: YYMMDDhhmm.                                                                                                                                                                                                                                                                   |
| Consumer's Activation Method Preference  | ans-165, left-justified, padded with spaces<br><br><b>NOTE: For post-tokenization authentication flow type: ans - 65, left justified, padded with space</b> | This field contains the activation method selected by the consumer, if only one was offered by the issuer, then that activation method. There will be only one method contained within this field. This field will only be present if the cardholder provides a choice. If the reason code is not present, then the additional space will not be padded with space. |

| Name                   | Attributes | Values/Comments                                                                                                                                                                                                                                                                         |
|------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activation Method Type | n-1        | 1 = Masked mobile phone number                                                                                                                                                                                                                                                          |
|                        |            | 2 = Masked email address                                                                                                                                                                                                                                                                |
|                        |            | 3 = Automated call center phone number                                                                                                                                                                                                                                                  |
|                        |            | 4 = Call center phone number                                                                                                                                                                                                                                                            |
|                        |            | 5 = Website                                                                                                                                                                                                                                                                             |
|                        |            | 6 = Mobile application. If the value is PUSH_NOTIFICATION, then the issuer should send a push notification to the cardholder to open the application and complete authentication. In this case, the issuer must use the <b>Customer Service API</b> to share the authentication result. |

| Subfield Name | Attributes | Values/Comments         |                                                                                                           |                                                                                                               |
|---------------|------------|-------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|               |            |                         |                                                                                                           | 7 = Masked voice call phone number                                                                            |
|               |            |                         |                                                                                                           | 8 = Push Notification. The issuer must share the activation code to the cardholder using a push notification. |
|               |            |                         |                                                                                                           | 9 = EMV 3DS (Identity Check)                                                                                  |
|               |            | Activation Method Value | ans...164                                                                                                 | Refer to the following examples                                                                               |
|               |            |                         | <b>NOTE: For post-tokenization authentication flow type: ans - 64, left justified, padded with space.</b> |                                                                                                               |

| Subfield Name          | Attributes                                    | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reason Code            | ans-6                                         | <p>The reason the account holder is being authenticated. The reason could impact the generation and validation of the code in various ways including but not limited to the time period of validity of the code and the number of attempts allowed to validate the code. New reason codes can be added at any time and should not result in a failure.</p> <p>Bit Position Meaning</p> <p>01 "ADD_CARD" - The account holder is being authenticated to verify account ownership at the time of adding a card in the consent requestor system</p> <p>Note: In the context of MDES - token might be in an active state.</p> <p>02 "VERIFY_ACCOUNT" - The account holder is being authenticated to verify account ownership to use a service</p> <p>03 OTHER - Requires Token Binding</p> <p>04 - PAYMENT_TRANSACTION – The account holder is being authenticated to initiate a payment transaction</p> <p>05 - TOKEN_BINDING – The account holder is being authenticated to bind a token to bind ID.</p> <p>06 - ACCOUNT_BINDING – The account holder is being authenticated to bind associated account with bind id (device)</p> <p>The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code. The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01 and 02 were encoded, the bitmap would be 0000000000000000000000000011 and the hex value of this field would be 000003.</p> <p><b>NOTE: This parameter is present only if an issuer has opted for Authentication Service for Click to Pay and Secure Card on File Tokens.</b></p> |
| Token Unique Reference | ans-48, left-justified,<br>padded with spaces | <p>Service-allocated unique reference to the token.</p> <p><b>NOTE: This parameter is present only if the issuer has opted for Authentication Service for Click to Pay and Secure Card on File Tokens.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Subfield Name            | Attributes                                               | Values/Comments                                                                                                                                |
|--------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| authRequestCorrelationId | ans-64 - Value linking different authentication messages | <b>NOTE: This parameter is present only if an issuer has opted for Authentication Service for Click to Pay and Secure Card on File Tokens.</b> |

## Examples

1(###) ### 4567

1 = Masked mobile phone number

The 1 will be followed by the masked mobile phone number.

2a\*\*\*d@anymail.com

2 = Masked email address

The 2 will be followed by the consumer's masked email address (the issuer will mask according to their own format).

3(555) 123 4567

3 = Automated call center phone number

The 3 will be followed by the phone number. This phone number is not masked.

4(555) 123 8901

4 = Call center phone number

The 4 will be followed by the phone number. This phone number is not masked.

5http://www.anybank.com

5 = Website

The 5 will be followed by the issuer's website URL.

6PUSH\_NOTIFICATION

6 = Mobile app

The 6 will be followed by the PUSH\_NOTIFICATION, that informs issuer that send a push notification to cardholder mobile application instance.

---

7(###) ### 2345

7 = Masked voice call phone number

The 7 will be followed by the masked voice call phone number.

---

8

8 = Push notification

No value is required for 8. Issuer must send a push notification to cardholder mobile banking application with authentication code.

---

## Authorization Request Response/0110 for ACN

All mandatory Authorization Request Response/0110 data elements apply.

### Authorization Request Response/0110 for ACN

| Data Element ID and Name   | Org | Sys | Dst | Values/Comments                                                   |
|----------------------------|-----|-----|-----|-------------------------------------------------------------------|
| DE 3 (Processing Code)     | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |
| DE 4 (Amount, Transaction) | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |
| DE 39 (Response Code)      | M   | •   | •   | 00 = Approved                                                     |

**NOTE: DE 124 is not present.**

## Tokenization Complete Notification (TCN)

Following is a list of the data elements, values, and subfields applicable for Tokenization Complete Notification (TCN).

- [Authorization Request/0100 for TCN](#)
- [DE 124 Subfields in Authorization Request/0100 for TCN](#)
- [Authorization Request Response/0110 for TCN](#)

## Authorization Request/0100 for TCN

All mandatory Authorization Request/0100 data elements apply.

### Authorization Request/0100 for TCN

| Data Element ID and Name                                                                                    | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number)                                                                               | •   | X   | M   | Cardholder's primary account number                                                                                                                                                                                                                                              |
| DE 3 (Processing Code)                                                                                      | •   | X   | M   | 00 = Purchase                                                                                                                                                                                                                                                                    |
| DE 4 (Amount, Transaction)                                                                                  | •   | X   | M   | Will be zero                                                                                                                                                                                                                                                                     |
| DE 14 (Date, Expiration)                                                                                    | •   | X   | C   | Cardholder's primary account expiration date                                                                                                                                                                                                                                     |
| DE 22 (POS Entry Mode)                                                                                      | •   | X   | M   | Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry)<br><br>Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)                                                                                                                                    |
| DE 35 (Track 2 Data)                                                                                        | •   | X   | C   | The Authorization System creates and provides Track 2 data for Maestro tokenization messages.                                                                                                                                                                                    |
| DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code) | •   | X   | M   | The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA".<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer. |
| DE 48 Transaction Category Code                                                                             | •   | X   | C   | T (Phone, Mail, or Electronic Commerce Order)                                                                                                                                                                                                                                    |
| DE 48 (Additional Data—Private Use), subelement 23 (Payment Initiation Channel)                             | •   | X   | C   | Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.                                                                                                                                                               |
| DE 48, subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)                                  | •   | X   | C   | Contains the identifier associated with the Wallet Provider.                                                                                                                                                                                                                     |

| Data Element ID and Name                                                                                                                                                                                                                                                                       | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator), subfield 2 (Account Number), subfield 3 (Expiration Date), subfield 5 (Token Assurance Method), subfield 6 (Token Requestor ID), and subfield 8 (Storage Technology) | •   | X   | C   | <p>Subfield 1 indicates the type of PAN mapping account.</p> <p>Subfield 2 indicates the PAN mapping account number.</p> <p>Subfield 3 indicates the expiration date of the PAN mapping account.</p> <p>Subfield 5 indicates Token Assurance Method.</p> <p>Subfield 6 contains the ID assigned by the Token Service Provider to the Token Requestor.</p> <p>Subfield 8 indicates Storage Technology of the token.</p>                   |
| DE 49 (Currency Code, Transaction)                                                                                                                                                                                                                                                             | •   | X   | M   | <p>Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA.</p> <p>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.</p> |
| DE 56 (Payment Account Data)                                                                                                                                                                                                                                                                   | •   | X   | C   | This is only for MDES Tokenization Complete Notifications.                                                                                                                                                                                                                                                                                                                                                                               |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)                                                                                                                                                                                                                       | •   | X   | M   | 9 = Tokenization Request/Notification                                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 61 (Additional POS Data), subfield 13 (POS Country Code)                                                                                                                                                                                                                                    | •   | X   | M   | <p>Indicates the country of the POS location (not the acquirer location) using ISO-specified codes.</p> <p>The POS Country Code is set to "840", corresponding to USA.</p> <p>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.</p>                                                                                                                                  |
| DE 124 (Member-defined Data)                                                                                                                                                                                                                                                                   | •   | X   | M   | See layout for DE 124.                                                                                                                                                                                                                                                                                                                                                                                                                   |

## DE 124 Subfields in Authorization Request/0100 for TCN

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

### DE 124 Subfields in Authorization Request/0100 for TCN

| Subfield Name                                          | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Message Type                                           | an-2                                       | TC = Tokenization Complete Notification (TCN)                                                                                                                                                                                                                                                                        |
| Correlation ID                                         | an-14                                      | Identifier assigned by Mastercard to associate related tokenization request/notification messages.                                                                                                                                                                                                                   |
| Number of Active Tokens for the Primary Account Number | ans-2, leading zeros                       | Number of active or suspended tokens for the primary account number provisioned to a consumer. Space-filled when requested token digitized to a server.<br><br><b>NOTE: Space-filled means if data is not available then length of that field is filled with space; this is specific to ISO message not for API.</b> |
| Issuer Product Configuration ID                        | ans-10                                     | The unique product configuration identifier provided by the issuer that identifies a particular set of card art, texts and other product related data, provided during the issuer enablement or maintenance process. This ID is case-sensitive.                                                                      |
| Consumer Language                                      | a-2                                        | Language preference selected by the consumer.                                                                                                                                                                                                                                                                        |
| Device Name                                            | ans-20, left-justified, padded with spaces | Name that the consumer has associated to the device with the Wallet Provider.                                                                                                                                                                                                                                        |
| Final Tokenization Decision                            | ans-1                                      | The final tokenization decision that was used in the tokenization of the card:<br><br>1 = Approve<br>2 = Approve, but require additional authentication                                                                                                                                                              |
| Final Tokenization Decision Indicator                  | ans-1                                      | The element of the Service that was responsible for determining the final tokenization decision:<br><br>1 = Tokenization Eligibility Response<br>2 = Tokenization Authorization Response<br>3 = Issuer predefined tokenization rules<br>4 = Mobile Application                                                       |



| Subfield Name                           | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                 |
|-----------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T&C Identifier                          | ans-32, left-justified, padded with spaces | Identifier associated with the version of terms and conditions accepted by the consumer.                                                                                                                                                                        |
| T&C Date and Time                       | ans-10                                     | Date and time that the consumer accepted the terms and conditions of the Service specified in UTC units.<br><br>Format: YYMMDDhhmm                                                                                                                              |
| Number of Activation Attempts           | ans-1                                      | Number of activation code entry attempts by the cardholder. Space-filled when DE 124, subfield 14 (Token Type) value is F.                                                                                                                                      |
| Token Unique Reference                  | ans-48, left-justified, padded with spaces | Service-allocated unique reference to the token.                                                                                                                                                                                                                |
| Primary Account Number Unique Reference | ans-48, left-justified, padded with spaces | Service-allocated unique reference to the tokenized Primary Account Number at the wallet level.                                                                                                                                                                 |
| Token Type                              | an-1                                       | Contains a value indicating the type of token present in DE 48, subelement 33, subfield 2 (Account Number) in an 0100 Tokenization Complete Notification message.<br><br>C = Mastercard Cloud-based Payments<br>F = Static Token<br>S = Embedded Secure Element |

## Authorization Request Response/0110 for TCN

All mandatory Authorization Request Response/0110 data elements apply.

### Authorization Request Response/0110 for TCN

| Data Element ID and Name   | Org | Sys | Dst | Values/Comments                                                    |
|----------------------------|-----|-----|-----|--------------------------------------------------------------------|
| DE 3 (Processing Code)     | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100. |
| DE 4 (Amount, Transaction) | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100. |
| DE 39 (Response Code)      | M   | •   | •   | 00 = Approved                                                      |

**NOTE: DE 124 is not present.**

## Tokenization Event Notification (TVN)

Following is a list of the data elements, values, and subfields applicable for Tokenization Event Notification (TVN).

- [Authorization Request/0100 for TVN](#)
- [DE 124 Subfields in Authorization Request/0100 for TVN](#)
- [Authorization Request Response/0110 for TVN](#)

### Authorization Request/0100 for TVN

All mandatory Authorization Request/0100 data elements apply.

#### Authorization Request/0100 for TVN

| Data Element ID and Name                                                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number)                                                                                     | •   | X   | M   | Cardholder's primary account number                                                                                                                                                                                                                                              |
| DE 3 (Processing Code)                                                                                            | •   | X   | M   | 00 = Purchase                                                                                                                                                                                                                                                                    |
| DE 4 (Amount, Transaction)                                                                                        | •   | X   | M   | Will be zero                                                                                                                                                                                                                                                                     |
| DE 14 (Date, Expiration)                                                                                          | •   | X   | C   | Cardholder's primary account expiration date                                                                                                                                                                                                                                     |
| DE 22 (POS Entry Mode)                                                                                            | •   | X   | M   | Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry)<br><br>Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)                                                                                                                                    |
| DE 35 (Track 2 Data)                                                                                              | •   | X   | C   | The Authorization System creates and provides Track 2 data for Maestro tokenization messages.                                                                                                                                                                                    |
| DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)       | •   | X   | M   | The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA".<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer. |
| DE 48 Transaction Category Code                                                                                   | •   | X   | C   | T (Phone, Mail, or Electronic Commerce Order)                                                                                                                                                                                                                                    |
| DE 48 (Additional Data - Private Use) subelement 06 (Advanced Digital Transaction Data 1) subfield 1 (Binding ID) | •   | X   | C   | Subfield 1 contains the Binding Id - The unique and fixed identifier to a component of data bound to the credential.                                                                                                                                                             |

| Data Element ID and Name                                                                                                                                                                                                                                                                         | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data - Private Use), subelement 23 (Payment Initiation Channel)                                                                                                                                                                                                                | •   | X   | C   | Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.                                                                                                                                                                                                                                                                                                                       |
| DE 48, subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)                                                                                                                                                                                                                       | •   | X   | C   | Contains the identifier associated with the Wallet Provider.                                                                                                                                                                                                                                                                                                                                                                             |
| DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator), subfield 2 (Account Number), subfield 3 (Expiration Date), subfield 5 (Token Assurance Method), subfield 6 (Token Requestor ID), and subfield 8 (Storage Technology) | •   | X   | C   | <p>Subfield 1 indicates the type of PAN mapping account.</p> <p>Subfield 2 indicates the PAN mapping account number.</p> <p>Subfield 3 indicates the expiration date of the PAN mapping account.</p> <p>Subfield 5 indicates Token Assurance Method.</p> <p>Subfield 6 contains the ID assigned by the Token Service Provider to the Token Requestor.</p> <p>Subfield 8 indicates Storage Technology of the token.</p>                   |
| DE 49 (Currency Code, Transaction)                                                                                                                                                                                                                                                               | •   | X   | M   | <p>Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA.</p> <p>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.</p> |
| DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)                                                                                                                                                                                                                         | •   | X   | M   | 9 = Tokenization Request/Notification                                                                                                                                                                                                                                                                                                                                                                                                    |

| Data Element ID and Name                                                         | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 61 (Additional POS Data),<br>subfield 13 (POS Country Code)                   | •   | X   | M   | Indicates the country of the POS location (not the acquirer location) using ISO-specified codes.<br><br>The POS Country Code is set to "840", corresponding to USA.<br><br>For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code. |
| DE 104 (Digital Payment Data),<br>subelement 004 (Digital Service Provider)      | •   | X   | C   | Subfield 001 (Digital Service Provider ID 1)<br>Subfield 002 (Digital Service Provider ID 2)                                                                                                                                                                                                     |
| DE 104 (Digital Payment Data),<br>subelement 005 ((Digital Authentication Data)) | •   | X   | C   | Subfield 001 (Cardholder Verification)<br>Subfield 002 (Authentication Factors)                                                                                                                                                                                                                  |
| DE 124 (Member-defined Data)                                                     | •   | X   | M   | See layout for DE 124.                                                                                                                                                                                                                                                                           |

## DE 124 Subfields in Authorization Request/0100 for TVN

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

### DE 124 Subfields in Authorization Request/0100 for TVN

| Subfield Name  | Attributes | Values/Comments                                                                                    |
|----------------|------------|----------------------------------------------------------------------------------------------------|
| Message Type   | an-2       | TV = Tokenization Event Notification (TVN)                                                         |
| Correlation ID | an-14      | Identifier assigned by Mastercard to associate related tokenization request/notification messages. |

| Subfield Name                  | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tokenization Event Indicator   | an-1       | <p>Value indicating the event that has occurred on the Mastercard Digital Enablement Service for the token.</p> <p>1 = Funding Account Update</p> <p>2 = Payment Channel Preference Updated</p> <p>3 = Deactivate</p> <p>4 = Deleted from consumer device</p> <p>5 = Authentication Performed</p> <p><b>NOTE: Only applicable for wallets supporting the enrollment using a physical card in a physical store.</b></p> <p>6 = Suspend</p> <p>7 = Resume</p> <p>8 = Tokenization Exception Event</p> <p>9 = Replacement (token re-digitization)</p> <p>a = Token Binding</p> <p>b = Account Binding</p> <p>c = Payment Account Reference</p> |
| Tokenization Event Reason Code | an-2       | <p>If Tokenization Event Indicator contains value 8 (Tokenization Exception Event), this field contains a value indicating the event reason. If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will not be present.</p> <p>00 = Activation code retries exceeded</p> <p>01 = Activation code expired or invalidated</p> <p>02 = Activation code entered incorrectly by cardholder</p>                                                                                                                                                                                          |

| Subfield Name                | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event Requestor              | ans-1                                      | <p>If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will contain a value indicating the party that requested the event. If the Tokenization Event Indicator contains a value of 8 (Tokenization Exception Event) this field will be space filled.</p> <p>0 = Indicates the Tokenization Event was requested by the Wallet Provider or Token Requestor</p> <p>1 = Indicates the Tokenization Event was requested by the Funding Account issuer</p> <p>2 = Indicates the Tokenization Event was requested by the Cardholder</p> <p>3 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by Mobile PIN Validation security (applicable to Tokenization Event Indicator value of 6 (Suspend), or 7 (Resume) only)</p> <p>4 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by the deletion of tokens, which did not have a transaction in the set (default or issuer-defined) number of days and were deleted by consumer app.</p> <p>5 = Reserved for future use</p> <p>6 = Reserved for future use</p> <p>7 = Reserved for future use</p> <p>8 = Reserved for future use</p> <p>9 = Reserved for future use</p> |
| Token Unique Reference (TUR) | ans-48, left justified, padded with spaces | Service-allocated unique reference to the token.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Subfield Name                                       | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cardholder Disabled<br>Payment Channel (India only) | an-3       | <p>This field is present only if Tokenization Event Indicator contains value 2 (Payment Channel Preference Updated).</p> <p>The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific payment channel disabled by the cardholder.</p> <p>The bitmap is big-endian with the least significant bit corresponding to Payment Channel 01, with the next least significant bit corresponding to Payment Channel 02 and so on. For example, if Payment Channels 01, 03, and 05 were encoded, the bitmap would be 0000 0001 0101 and the hex value of this field would be 15.</p> <p>MDES will pass 0 if all payment channels are enabled.</p> <p>Bit Position and Payment Channel</p> <p>1 = Contactless</p> <p>2 = Merchant Presented QR (MPQR)</p> <p>3 = Customer Presented QR (CPQR)</p> <p>4 = Digital Secure Remote Payment (DSRP) In-App</p> <p>5 = Person to Person (P2P)</p> <p>6 = Dynamic Magnetic Stripe Data (DMSD)</p> <p>7 to 12 = Reserved for future use</p> |
| Binding Status                                      | n-1        | <p>If the Tokenization Event Indicator contains a value of a (Token Binding), b (Account Binding) this field will contain a value indicating the status of Binding</p> <ul style="list-style-type: none"> <li>1 = Active</li> <li>2 = Deactivated (deleted)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Authorization Request Response/0110 for TVN

All mandatory Authorization Request Response/0110 data elements apply.

### Authorization Request Response/0110 for TVN

| Data Element ID and Name   | Org | Sys | Dst | Values/Comments                                                   |
|----------------------------|-----|-----|-----|-------------------------------------------------------------------|
| DE 3 (Processing Code)     | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |
| DE 4 (Amount, Transaction) | ME  | ME  | •   | Must be the same value as the original Authorization Request/0100 |

| Data Element ID and Name | Org | Sys | Dst | Values/Comments |
|--------------------------|-----|-----|-----|-----------------|
| DE 39 (Response Code)    | M   | .   | .   | 00 = Approved   |

**NOTE: DE 124 will not be present.**

## DE 120 Layout for MCC106 Mastercard Digital Enablement Service

The following tables provides the technical details of DE 120 (Record Data) for Token Update and PAN Update - Deactivate/Suspend/Resume Token.

- [Issuer File Update Request/0302 - Maintenance \(Token/PAN Update\)](#)
- [Issuer File Update Request Response/0312 - Issuer Token Maintenance Response \(Token/PAN Update\)](#)
- [Token Update](#)
- [PAN Update - Deactivate/Suspend/Resume Token](#)

**NOTE: For In Control customers, check the status of the In Control VCN before updating a token status. Refer to the *Mastercard In Control for Commercial Payments User and Company Guide*.**

### Issuer File Update Request/0302 - Maintenance (Token/PAN Update)

Following is a list of the data elements and values applicable to this message type. All mandatory Issuer File Update Request/0302 data elements apply.

#### Issuer File Update Request/0302 - Maintenance (Token/PAN Update)

| Data Element and Name           | Org | Sys | Dst | Values/Comments                                                                |
|---------------------------------|-----|-----|-----|--------------------------------------------------------------------------------|
| DE 2 (Primary Account Number)   | M   | M   | .   | Contains the cardholder's primary account number                               |
| DE 91 (Issuer File Update Code) | M   | M   | .   | Contains the Issuer File Update function code. Valid value will be 2 (Update). |
| DE 101 (File Name)              | M   | M   | .   | MCC106 (PAN Mapping File)                                                      |
| DE 120 (Record Data)            | M   | M   | .   | Contains token and PAN information. Refer to the layout of DE 120.             |



## Token Update

The following table provides the technical details of DE 120 (Record Data) when DE 101 (File Name) contains the value MCC106 (Mastercard Digital Enablement Service Mapping File PAN Update) in the Issuer File Update Request/0302 message.

### Token Update

| Field ID | Field Name                      | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|---------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Mapping File Indicator          | an-1       | M = Mastercard Digital Enablement Service Tokens excluding Static Tokens<br><br>A = All Mastercard Digital Enablement Service Tokens including Static Tokens<br><br>Best practice for the issuer to use value A.                                                                                                                                                                                                                                                                                                                              |
| 2        | Replacement PAN                 | ans-19     | Number replacing the number that is embossed, encoded, or both on the card (the primary account number). Customers may input only account numbers for BINs assigned to the associated customer ID assigned by Mastercard.<br><br>The issuer has the option to provide a replacement primary account number for association to the token.<br><br>Format: Left-justified, with trailing spaces                                                                                                                                                  |
| 3        | Replacement PAN Expiration Date | n-4        | Expiration date that is associated with the number replacing the number that is embossed, encoded, or both on the card that represents the cardholder primary account number.<br><br>If the issuer has provided a replacement primary account number, this field contains the expiration date that is associated with the replacement primary account number.<br><br>If the issuer has not provided a replacement primary account number, this field contains the expiration date of the existing primary account number.<br><br>Format: YYMM |

| Field ID | Field Name                               | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4        | Primary Account Card Sequence Number     | ans-3      | <p>If the issuer has provided a replacement primary account number, this field will contain the card sequence number associated with the replacement primary account number.</p> <p>If the issuer has not provided a replacement primary account number, this field will contain the card sequence number associated with the original primary account number.</p> <p>Format: Left-justified, padded with leading zeros.</p>                                                                                                                                                                                                                                                  |
| 5        | Notify Wallet Service Provider Indicator | n-1        | <p>If the issuer has provided a replacement primary account number and replacement primary account number expiration date, this field indicates whether the Wallet Provider should be notified of the change.</p> <p>Values:</p> <ul style="list-style-type: none"> <li>• 0 = Update token mapping information and notify the Wallet Provider with the primary account number information</li> <li>• 1 = Update token mapping information, but do not notify the Wallet Provider with the primary account number information</li> <li>• 2 = Do not update token mapping information, but do update the Wallet Provider with the primary account number information</li> </ul> |
| 6        | Token—If replacing a specific token      | ans-19     | <p>Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check.</p> <p>Format: Left-justified, with trailing spaces</p> <p>If field not present, update shall apply to all token to PAN mappings associated with the PAN.</p>                                                                                                                                                                                                                                                                                                          |

## PAN Update - Deactivate/Suspend/Resume Token

The following table provides the technical details of DE 120 (Record Data) when DE 101 (File Name) contains the value MCC106 (Mastercard Digital Enablement Service Mapping File PAN Update) in the Issuer File Update Request/0302 message.

### PAN Update - Deactivate/Suspend/Resume Token

| Field ID | Field Name                               | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Mapping File Indicator                   | an-1       | <p>M = Mastercard Digital Enablement Service tokens excluding Static Tokens.</p> <p>A = All Mastercard Digital Enablement Service tokens including Static Tokens.</p> <p>Best practice for the issuer to use value A.</p>                                                                                                                                                                                                         |
| 2        | Action Required                          | an-1       | <p>S = Suspend token</p> <p>D = Deactivate token</p> <p>C = Resume token</p>                                                                                                                                                                                                                                                                                                                                                      |
| 3        | Notify Wallet Service Provider indicator | n-1        | <p>This field indicates whether the Wallet Provider should be notified of the change.</p> <p>Value:</p> <p>0 = Update token mapping information and notify the Wallet Provider.</p>                                                                                                                                                                                                                                               |
| 4        | Token - If updating a specific token     | ans-19     | <p>Surrogate value for a PAN that is consistent with ISO message requirements and is a 13-19-digit numeric value that passes basic validation rules of an account number, including the LUHN check.</p> <p>Format: Left-justified, with trailing spaces.</p> <p>If not present, do across the board switch - based on Field 1. If field not present, update shall apply to all token to PAN mappings associated with the PAN.</p> |
| 5        | Date-Last-Update-Activity                | n-6        | <p>Format: MMDDYY (Month, Day, Year)</p> <p>Format: Spaces must be provided if there are fields present after subfield 6.</p>                                                                                                                                                                                                                                                                                                     |
| 6        | Time-Last-Update-Activity                | n-4        | <p>Format: hhmm (hour, minute)</p> <p>Format: Spaces must be provided if there are fields present after subfield 7.</p>                                                                                                                                                                                                                                                                                                           |

## Issuer File Update Request Response/0312 - Issuer Token Maintenance Response (Token/PAN Update)

Following is a list of the data elements and values applicable to this message type. All mandatory Issuer File Update Request Response/0312 data elements apply.

### Issuer File Update Request Response/0312 - Issuer Token Maintenance Response (Token/PAN Update)

| Data Element and Name            | Org | Sys | Dst | Values/Comments                                                                                                          |
|----------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number)    | •   | ME  | ME  | Must be the same value as in the original Issuer File Update Request/0302.                                               |
| DE 39 (Response Code)            | •   | M   | M   | Indicates disposition of Issuer File Update Request/0302.                                                                |
| DE 44 (Additional Response Data) | •   | C   | C   | May contain additional response information, based on DE 39. Error messages should spell out field in error (ex: 120XXX) |
| DE 91 (Issuer File Update Code)  | •   | ME  | ME  | Must be the same value as in the original Issuer File Update Request/0302.                                               |
| DE 101 (File Name)               | •   | ME  | ME  | Must be the same value as in the original Issuer File Update Request/0302.                                               |
| DE 120 (Record Data)             | •   | ME  | ME  | Must be the same value as in the original Issuer File Update Request/0302.                                               |

## DE 120 Layout for Administrative Advice/0620 Issuer Token Notification Advice

The following tables provides the technical details of DE 120 (Record Data) for Issuer Token Notification Advice for Activation Code Notification, Tokenization Complete Notification and Tokenization Event Notification.

- [Activation Code Notification](#)
- [Tokenization Complete Notification](#)
- [Tokenization Event Notification](#)

### Activation Code Notification

The following table provides the technical details of DE 120 (Record Data) when DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) contains the value 141 (Mastercard Digital Enablement Service Advice to Issuer) and DE 60, subfield 2 (Advice Detail Code) contains the value 0250 (Activation Code Notification) in the Administrative Advice/0620 message.

Absolute positioning of data in DE 120 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value the subfields will not be present and the total length of DE 120 will be reduced by the subfield lengths.

### Activation Code Notification

| Field ID | Field Name                               | Attributes                                  | Values/Comments                                                                                                                                                                                                                                                                        |
|----------|------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Primary Account Number                   | an-19                                       | Cardholder's primary account number                                                                                                                                                                                                                                                    |
| 2        | Primary Account Number Expiry Date       | n-4                                         | Cardholder's primary account number expiration date                                                                                                                                                                                                                                    |
| 3        | Correlation ID                           | an-14                                       | The identifier assigned by Mastercard to associate related tokenization request/notification messages.                                                                                                                                                                                 |
| 4        | Activation Code                          | ans-8, left-justified, padded with spaces   | Activation code assigned by Mastercard that will be provided to the issuer for delivery to the consumer to complete the tokenization process.                                                                                                                                          |
| 5        | Activation Code Expiration Date and Time | n-10                                        | The date and time that the Activation Code expires specified in UTC units.<br>Format: YYMMDDhhmm.                                                                                                                                                                                      |
| 6        | Customer's Activation Method Preference  | ans-165, left-justified, padded with spaces | This field contains the activation method selected by the consumer, if only one was offered by the issuer, then that activation method will be present. There will be only one method contained within this field. This field will only be present if the cardholder made a selection. |
|          |                                          | Name                                        | Attributes                                                                                                                                                                                                                                                                             |
|          |                                          | Activation Method Type                      | n-1                                                                                                                                                                                                                                                                                    |
|          |                                          |                                             | 1 = Masked mobile phone number                                                                                                                                                                                                                                                         |
|          |                                          |                                             | 2 = Masked email address                                                                                                                                                                                                                                                               |
|          |                                          |                                             | 3 = Automated call center phone number                                                                                                                                                                                                                                                 |
|          |                                          |                                             | 4 = Call center phone number                                                                                                                                                                                                                                                           |
|          |                                          |                                             | 5 = Website                                                                                                                                                                                                                                                                            |

| Field ID | Field Name         | Attributes              | Values/Comments                                                                                                                                                                                                                                                                         |
|----------|--------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                    |                         | 6 = Mobile application. If the value is PUSH_NOTIFICATION, then the issuer should send a push notification to the cardholder to open the application and complete authentication. In this case, the issuer must use the <b>Customer Service</b> API to share the authentication result. |
|          |                    |                         | 7 = Masked voice call phone number                                                                                                                                                                                                                                                      |
|          |                    |                         | 8 = Push Notification. The issuer must share the activation code with the cardholder using a push notification.                                                                                                                                                                         |
|          |                    |                         | 9 = EMV 3DS (Identity Check)                                                                                                                                                                                                                                                            |
|          |                    | Activation Method Value | ans...164 See examples below.                                                                                                                                                                                                                                                           |
| 7        | Token Requestor ID | n-11                    | Contains the identifier assigned by the Token Service Provider to the Token Requestor.                                                                                                                                                                                                  |
| 8        | Wallet ID          | ans-3                   | Contains the identifier associated with the Wallet Provider. The presence of this field is conditional.                                                                                                                                                                                 |
| 9        | Device Type        | ans-2                   | Indicates the type of device used. The presence of this field is conditional.                                                                                                                                                                                                           |

| Field ID | Field Name             | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10       | Reason Code            | ans-6                                      | <p>The reason the account holder is being authenticated. The reason could impact the generation and validation of the code in various ways including but not limited to the time period of validity of the code and the number of attempts allowed to validate the code. New reason codes can be added at any time and should not result in a failure.</p> <p>Bit Position Meaning</p> <p>01 ADD_CARD - The account holder is being authenticated to verify account ownership at the time of adding a card in the consent requestor system</p> <p><b>NOTE: In the context of MDES - token might be in an active state.</b></p> <p>02 VERIFY_ACCOUNT - The account holder is being authenticated to verify account ownership to use a service</p> <p>03 OTHER -</p> <p>04 PAYMENT_TRANSACTION - The account holder is being authenticated to initiate a payment transaction</p> <p>05 TOKEN_BINDING - The account holder is being authenticated to bind a token to bind ID.</p> <p>06 ACCOUNT_BINDING - The account holder is being authenticated to bind associated account with bind ID (device).</p> <p>The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code. The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01, and 02 the bitmap would be 0000000000000000000011 and the hex value of this field would be 000003.</p> <p><b>NOTE: This parameter is present only if an issuer has opted for Authentication Service for Click to Pay and Secure Card on File Tokens.</b></p> |
| 11       | Token Unique Reference | ans-48, left-justified, padded with spaces | Service-allocated unique reference to the token.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Field ID | Field Name               | Attributes | Values/Comments                                  |
|----------|--------------------------|------------|--------------------------------------------------|
| 12       | authRequestCorrelationId | ans-64     | Value linking different authentication messages. |

Examples

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| "1(###) ### 4567"                                                                                                                      |
| 1 = Masked mobile phone number                                                                                                         |
| The "1" will be followed by the masked mobile phone number.                                                                            |
| "2a***d@anymail.com"                                                                                                                   |
| 2 = Masked email address                                                                                                               |
| The "2" will be followed by the consumer's masked email address (the issuer will mask according to their own format).                  |
| "3(555) 123 4567"                                                                                                                      |
| 3 = Automated call center phone number                                                                                                 |
| The "3" will be followed by the phone number. This phone number is not masked.                                                         |
| "4(555) 123 8901"                                                                                                                      |
| 4 = Call center phone number                                                                                                           |
| The "4" will be followed by the phone number. This phone number is not masked.                                                         |
| "5http://www.anybank.com"                                                                                                              |
| 5 = Website                                                                                                                            |
| The "5" will be followed by the issuer's website URL.                                                                                  |
| "6com.anybank.mobileapp"                                                                                                               |
| 6 = Mobile app                                                                                                                         |
| The "6" will be followed by the issuer's mobile app information, the content of which depends upon the mobile device operating system. |
| "7(###) ### 2345"                                                                                                                      |
| 7 = Masked voice call phone number                                                                                                     |
| The "7" will be followed by the masked voice call phone number.                                                                        |



## Tokenization Complete Notification

The following table provides the technical details of DE 120 (Record Data) when DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) contains the value 141 (Mastercard Digital Enablement Service Advice to Issuer) and DE 60, subfield 2 (Advice Detail Code) contains the value 0251 (Tokenization Complete Notification) in the Administrative Advice/0620 message.

Absolute positioning of data in DE 120 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value the subfield(s) will not be present and the total length of DE 120 will be reduced by the subfield length(s).

### Tokenization Complete Notification

| Field ID | Field Name                            | Attributes | Values/Comments                                                                                                                                                                                                                                                        |
|----------|---------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Token                                 | an-19      | Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check.<br><br>Format: Left-justified, with trailing spaces                |
| 2        | Token Expiration Date                 | n-4        | Expiration date associated with the token.                                                                                                                                                                                                                             |
| 3        | PAN                                   | an-19      | Number that is embossed, encoded, or both on the card (also known as the primary account number). Customers may input only account numbers for BINs assigned to the associated customer ID assigned by Mastercard.<br><br>Format: Left-justified, with trailing spaces |
| 4        | PAN Expiration Date                   | n-4        | Expiration date that is embossed, encoded, or both on the card that represents the cardholder primary account number (primary account number).<br><br>Format: YYMM                                                                                                     |
| 5        | Token Service Provider Identification | a-1        | M = Mastercard Digital Enablement Service                                                                                                                                                                                                                              |
| 6        | Token Assurance Method                | n-2        | Assurance method assigned to the token (value between 00 and 99).                                                                                                                                                                                                      |
| 7        | Token Requestor ID                    | n-11       | The ID assigned by Mastercard to the token requestor.                                                                                                                                                                                                                  |

| Field ID | Field Name                                             | Attributes           | Values/Comments                                                                                                                                                                                                                                                                                                                                                                               |
|----------|--------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8        | Contactless Usage                                      | n-1                  | <p>Contains value indicating if the token is permitted for use in contactless transactions.</p> <p>Values:</p> <ul style="list-style-type: none"> <li>0 = Token is not permitted for use in contactless transactions</li> <li>1 = Token is permitted for use in contactless transactions</li> </ul>                                                                                           |
| 9        | Card on File Electronic Commerce Usage                 | n-1                  | <p>Contains value indicating if the token is permitted for use in card on file electronic commerce transactions.</p> <p>Values:</p> <ul style="list-style-type: none"> <li>0 = Token is not permitted for use in Card on File electronic commerce transactions</li> <li>1 = Token is permitted for use in card on file electronic commerce transactions</li> </ul>                            |
| 10       | Mobile/Digital Wallet Electronic Commerce Usage        | n-1                  | <p>Contains value indicating if the token is permitted for use in mobile/digital wallet electronic commerce transactions.</p> <p>Values:</p> <ul style="list-style-type: none"> <li>0 = Token is not permitted for use in mobile/digital wallet electronic commerce transactions</li> <li>1 = Token is permitted for use in mobile/digital wallet electronic commerce transactions</li> </ul> |
| 11       | Correlation ID                                         | an-14                | Identifier assigned by Mastercard to associate related tokenization request/notification messages.                                                                                                                                                                                                                                                                                            |
| 12       | Number of Active Tokens for the Primary Account Number | ans-2, leading zeros | <p>Number of active or suspended tokens for the primary account number digitized to consumer devices. Space-filled when token present in DE 48, subelement 33, subfield 2 (Account Number) in an 0100 Tokenization Complete Notification message is provisioned to a server. Presence of this field is conditional.</p>                                                                       |

| Field ID | Field Name                            | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13       | Issuer Product Configuration ID       | ans-10                                     | The unique product configuration identifier applied to the token, as provided by the issuer, identifying a particular set of card art, texts, and other product related data, that were provided during the issuer enablement or maintenance process. Presence of this field is conditional. This ID is case-sensitive.                                                                   |
| 14       | Consumer Language                     | a-2                                        | Language preference selected by the consumer. Presence of this field is conditional.                                                                                                                                                                                                                                                                                                      |
| 15       | Device Name                           | ans-20, left-justified, padded with spaces | Name that the consumer has associated to the device with the Wallet Provider. Presence of this field is conditional.                                                                                                                                                                                                                                                                      |
| 16       | Final Tokenization Decision           | ans-1                                      | <p>The final tokenization decision that was used in the tokenization of the card.</p> <ul style="list-style-type: none"> <li>• 1 = Approve</li> <li>• 2 = Approve but requires additional authentication</li> </ul> <p>Presence of this field is conditional.</p>                                                                                                                         |
| 17       | Final Tokenization Decision Indicator | ans-1                                      | <p>The element of the Service that was responsible for determining the final tokenization decision:</p> <ul style="list-style-type: none"> <li>• 1 = Tokenization Eligibility Response</li> <li>• 2 = Tokenization Authorization Response</li> <li>• 3 = Issuer pre-defined tokenization rules</li> <li>• 4 = Mobile Application</li> </ul> <p>Presence of this field is conditional.</p> |
| 18       | T&C Identifier                        | ans-32, left justified, padded with spaces | Identifier associated with the version of terms and conditions accepted by the consumer. Presence of this field is conditional.                                                                                                                                                                                                                                                           |
| 19       | T&C Date and Time                     | ans-10                                     | <p>Date and time that the consumer accepted the terms and conditions of the Service, specified in UTC units.</p> <p>Format: YYMMDDhhmm</p>                                                                                                                                                                                                                                                |

| Field ID | Field Name                              | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                    |
|----------|-----------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20       | Number of Activation Attempts           | ans-1                                      | Number of activation code entry attempts by the cardholder. Space-filled when DE124, SF14 (Token Type) value is F. Presence of this field is conditional.                                                                                                                                          |
| 21       | Token Unique Reference                  | ans-48, left justified, padded with spaces | Service-allocated unique reference to the token.                                                                                                                                                                                                                                                   |
| 22       | Primary Account Number Unique Reference | ans-48, left justified, padded with spaces | Service-allocated unique reference to the tokenized Primary Account Number at the wallet level.                                                                                                                                                                                                    |
| 23       | Token Type                              | an-1                                       | Contains a value indicating the type of token present in DE 120, field 1 (Token) in an 0620 Tokenization Complete Notification message. <ul style="list-style-type: none"> <li>• C = Mastercard Cloud-Based Payments</li> <li>• F = Static Token</li> <li>• S = Embedded Secure Element</li> </ul> |
| 24       | Wallet ID                               | ans-3                                      | Contains the identifier associated with the Wallet Provider. Presence of this field is conditional.                                                                                                                                                                                                |
| 25       | Device Type                             | ans-2                                      | Indicates the type of device used. Presence of this field is conditional.                                                                                                                                                                                                                          |
| 26       | Storage Technology                      | an-2                                       | Contains a value indicating the storage technology of the token. Presence of this field is conditional.                                                                                                                                                                                            |

**NOTE: The Token Requestor ID is already being included in DE 120, field ID 7 of the Administrative Advice/0620—Issuer Token Notification Advice for Tokenization Complete Notification message.**

## Tokenization Event Notification

The following table provides the technical details of DE 120 (Record Data) when DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) contains the value 141 (Mastercard Digital Enablement Service Advice to Issuer) and DE 60, subfield 2 (Advice Detail Code) contains the value 0252 (Tokenization Event Notification) in the Administrative Advice/0620 message.

Absolute positioning of data in DE 120 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value the subfield(s) will not be present and the total length of DE 120 will be reduced by the subfield length(s).

## Tokenization Event Notification

| Field ID | Field Name                                           | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Primary Account Number                               | an-19      | Cardholder's primary account number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2        | Primary Account Number Expiry Date                   | n-4        | Cardholder's primary account expiration date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3        | Token (PAN mapping file information)                 | an-19      | Token                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4        | Token Expiration Date (PAN mapping file information) | an-4       | Token expiration date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5        | Token Service Provider Identification                | an-1       | M = Mastercard Digital Enablement Service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6        | Correlation ID                                       | an-14      | Identifier assigned by Mastercard to associate related tokenization request/notification messages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7        | Tokenization Event Indicator                         | n-1        | <p>Value indicating the event that has occurred on the Mastercard Digital Enablement Service for the token</p> <p>1 = Funding Account Update</p> <p>2 = Payment Channel Preference Updated (India only)</p> <p>3 = Deactivate</p> <p>4 = Deleted from consumer device</p> <p>5 = Authentication Performed</p> <p><b>NOTE: Only applicable for wallets supporting the enrollment using a physical card in a physical store.</b></p> <p>6 = Suspend</p> <p>7 = Resume</p> <p>8 = Tokenization Exception Event</p> <p>9 = Replacement</p> <ul style="list-style-type: none"> <li>a = Token Binding</li> <li>b = Account Binding</li> </ul> |

| Field ID | Field Name                                      | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8        | Tokenization Event Reason Code                  | an-2       | <p>If the Tokenization Event Indicator contains value 8 (Tokenization Exception Event), this field contains a value indicating the event reason. If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will not be present.</p> <ul style="list-style-type: none"> <li>00 = Activation code retries exceeded</li> <li>01 = Activation code expired or invalidated</li> <li>02 = Activation code entered incorrectly by cardholder</li> </ul> |
| 9        | Contactless Usage                               | n-1        | <p>Contains value indicating if the token is permitted for use in contactless transactions.</p> <p>Values:</p> <ul style="list-style-type: none"> <li>0 = Token is not permitted for use in contactless transactions</li> <li>1 = Token is permitted for use in contactless transactions</li> </ul>                                                                                                                                                                                                   |
| 10       | Card on File Electronic Commerce Usage          | n-1        | <p>Contains value indicating if the token is permitted for use in card on file electronic commerce transactions.</p> <p>Values:</p> <ul style="list-style-type: none"> <li>0 = Token is not permitted for use in Card on File electronic commerce transactions</li> <li>1 = Token is permitted for use in card on file electronic commerce transactions</li> </ul>                                                                                                                                    |
| 11       | Mobile/Digital Wallet Electronic Commerce Usage | n-1        | <p>Contains value indicating if the token is permitted for use in mobile/digital wallet electronic commerce transactions.</p> <p>Values:</p> <ul style="list-style-type: none"> <li>0 = Token is not permitted for use in mobile/digital wallet electronic commerce transactions</li> <li>1 = Token is permitted for use in mobile/digital wallet electronic commerce transactions</li> </ul>                                                                                                         |

| Field ID | Field Name                   | Attributes                                 | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12       | Event Requestor              | ans-1                                      | <p>If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will contain a value indicating the party that requested the event. If the Tokenization Event Indicator contains a value of 8 (Tokenization Exception Event) this field will be space filled.</p> <ul style="list-style-type: none"> <li>• 0 = Indicates the Tokenization Event was requested by the Wallet Provider or Token Requestor</li> <li>• 1 = Indicates the Tokenization Event was requested by the Funding Account issuer</li> <li>• 2 = Indicates the Tokenization Event was requested by the Cardholder</li> <li>• 3 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by Mobile PIN Validation security (applicable to Tokenization Event Indicator value of 6 (Suspend), or 7 (Resume) only)</li> <li>• 4 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by the deletion of tokens, which did not have a transaction in the set (default or issuer-defined) number of days and were deleted by consumer app.</li> <li>• 5 = Reserved for future use</li> <li>• 6 = Reserved for future use</li> <li>• 7 = Reserved for future use</li> <li>• 8 = Reserved for future use</li> <li>• 9 = Reserved for future use</li> </ul> |
| 13       | Token Requestor ID           | n-11                                       | Contains the identifier assigned by the Token Service Provider to the Token Requestor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 14       | Wallet ID                    | ans-3                                      | Contains the identifier associated with the Wallet Provider. Presence of this field is conditional.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 15       | Device Type                  | ans-2                                      | Indicates the type of device used. Presence of this field is conditional.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 16       | Token Unique Reference (TUR) | ans-48, left justified, padded with spaces | Service-allocated unique reference to the token.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Field ID | Field Name                                       | Attributes | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|--------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17       | Cardholder Disabled Payment Channel (India only) | an-3       | <p>This field is present only if Tokenization Event Indicator contains value 2 (Payment Channel Preference Updated).</p> <p>The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific payment channel disabled by the cardholder.</p> <p>The bitmap is big-endian with the least significant bit corresponding to Payment Channel 01, with the next least significant bit corresponding to Payment Channel 02 and so on. For example, if Payment Channels 01, 03, and 05 were encoded, the bitmap would be 0000 0001 0101 and the hex value of this field would be 15.</p> <p>MDES will pass 0 if all payment channels are enabled.</p> <p>Bit Position and Payment Channel</p> <p>1 = Contactless</p> <p>2 = Merchant Presented QR (MPQR)</p> <p>3 = Customer Presented QR (CPQR)</p> <p>4 = Digital Secure Remote Payment (DSRP) In-App</p> <p>5 = Person to Person (P2P)</p> <p>6 = Dynamic Magnetic Stripe Data (DMSD)</p> <p>7 to 12 = Reserved for future use</p> |
| 18       | Binding Status                                   | n-1        | <p>If the Tokenization Event Indicator contains a value of a (Token Binding), b (Account Binding) this field will contain a value indicating the status of Binding 1= Active, 2 = Deactivated (deleted)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 19       | Digital Service Provider ID 1                    | an-5       | <p>Contains the alphanumeric value that identifies the Digital Service Provider type and entity. This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated the cardholder in a remote commerce transaction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



| Field ID | Field Name                    | Attributes | Values/Comments                                                                                                                                                                                                                           |
|----------|-------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20       | Digital Service Provider ID 2 | an-5       | Contains the alphanumeric value that identifies the Digital Service Provider type and entity. This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated. |
| 21       | Cardholder Verification       | an-5       | Contains an alphanumeric value to indicate the cardholder will be authenticated and specifies the Cardholder Verification Method (CVM) will be used to authenticate the cardholder during a contactless or remote commerce transaction.   |
| 22       | Authentication Factors        | an-5       | Contains the alphanumeric value that identifies the authentication factors will be used to authenticate the cardholder during a contactless or remote commerce transaction.                                                               |
| 23       | Binding ID                    | ans-48     | The unique and fixed identifier to a component of data bound to the credential.                                                                                                                                                           |

## Administrative Advice/0620 - Issuer Token Notification Advice

Mastercard will support new layouts for Administrative Advice/0620 messages to communicate activation codes to issuers in support of token issuance and to notify issuers that Mastercard has assigned a token on their behalf. Following is a list of the data elements and values applicable to this message type. All mandatory Administrative Advice/0620 - System-generated data elements apply.

### Administrative Advice/0620 - Issuer Token Notification Advice

| Data Element and Name                     | Org | Sys | Dst | Values/Comments                                                                              |
|-------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------|
| Message Type Identifier (MTI)             | •   | X   | M   | Constant—0620 (Administrative Advice)                                                        |
| Bit Map, Primary                          | •   | X   | M   | Mandatory                                                                                    |
| Bit Map, Secondary                        | •   | X   | M   | Mandatory                                                                                    |
| DE 7 (Transmission Date and Time)         | •   | X   | M   | Transaction time stamp; UTC date and time that this transaction was entered into interchange |
| DE 11 (Systems Trace Audit Number [STAN]) | •   | X   | M   | Transaction trace number; must be unique value for transaction initiator within each UTC day |

| Data Element and Name                     | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 33 (Forwarding Institution ID Code)    | •   | X   | M   | Identifies the customer, institution, or Authorization System facility originating the Administrative Advice/0620 message                                                                                                                                                                                                                                                                                                                                        |
| DE 42 (Card Acceptor Identification Code) | •   | X   | M   | Identifies the card acceptor.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DE 56 (Payment Account Data)              | •   | X   | C   | This is only for MDES Tokenization Complete Notifications.                                                                                                                                                                                                                                                                                                                                                                                                       |
| DE 60 (Advice Reason Code)                | •   | X   | M   | <p>Indicates the specific reason for the transmission of the Advice message</p> <p>Subfield 1 (Advice Reason Code) = 141 (Mastercard Digital Enablement Service Advice to Issuer)</p> <p>Subfield 2 (Advice Detail Code) = one of the following values:</p> <ul style="list-style-type: none"> <li>• 0250 = Activation Code Notification (ACN)</li> <li>• 0251 = Tokenization Complete Notification</li> <li>• 0252 = Tokenization Event Notification</li> </ul> |
| DE 63 (Network Data)                      | •   | X   | M   | Contains the Banknet Reference Number that the Authorization System provides as a unique transaction ID.                                                                                                                                                                                                                                                                                                                                                         |
| DE 100 (Receiving Institution ID Code)    | •   | X   | M   | Identifies the customer, institution, or Authorization System facility that will receive this Administrative Advice/0620.                                                                                                                                                                                                                                                                                                                                        |
| DE 120 (Record Data)                      | •   | X   | M   | Contains activation code information. Refer to the associated DE 120 layout.                                                                                                                                                                                                                                                                                                                                                                                     |

## Administrative Advice Response/0630 - Issuer Token Notification Advice Response

Following is a list of the data elements and values applicable to this message type. All mandatory Administrative Advice Response/0630 data elements apply.

### Administrative Advice Response/0630 - Issuer Token Notification Advice Response

| Data Element and Name                     | Org | Sys | Dst | Values/Comments                                                                   |
|-------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------|
| Message Type Identifier (MTI)             | M   | M   | •   | Constant—0630 (Administrative Advice Response)                                    |
| Bit Map, Primary                          | M   | M   | •   | Mandatory                                                                         |
| Bit Map, Secondary                        | M   | M   | •   | Mandatory                                                                         |
| DE 7 (Transmission Date and Time)         | ME  | ME  | •   | Must be the same value as in the original Administrative Advice/0620              |
| DE 11 (Systems Trace Audit Number [STAN]) | ME  | ME  | •   | Must be the same value as in the original Administrative Advice/0620              |
| DE 33 (Forwarding Institution ID Code)    | ME  | ME  | •   | Must be the same value as in the original Administrative Advice/0620              |
| DE 39 (Response Code)                     | M   | M   | •   | Indicates the disposition of the original Administrative Advice/0620              |
| DE 44 (Additional Response Data)          | C   | C   | •   | May contain the additional error code information depending on the value in DE 39 |
| DE 63 (Network Data)                      | ME  | ME  | •   | Must be the same value as in the original Administrative Advice/0620              |
| DE 100 (Receiving Institution ID Code)    | ME  | ME  | •   | Must be the same value as in the original Administrative Advice/0620              |

## MDES for Merchants

This section describes enhancements to the Mastercard Digital Enablement Service (MDES) for merchants program that provides a network tokenization service for merchants. The section provides the technical requirements for issuers choosing to participate in the program and for acquirers supporting merchants that are enrolled in the program.

Mastercard offers specific values to existing data elements in the Dual Message Authorization System to help acquirers and issuers identify the MDES for merchants program and participants throughout the authorization flow.

The following process explains an authorization transaction request when a token transaction is generated by an MDES for merchant participant.

1. When a merchant initiates the token transaction, the tokenized transaction data is transmitted to the acquirer.
2. The acquirer initiates an Authorization Request/0100 message containing the following data and sends it to the Mastercard Network for initial purchase:
  - DE 2 (Primary Account Number [PAN]) with the token
  - DE 22 (POS Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 10 (Credential on File)
  - DE 48 (Additional Data: Private Use), subelement 42 containing value 246 indicating Merchant Risk Management Decisioning
  - DE 48 (Additional Data: Private Use), subelement 43 (Universal Cardholder Authentication Field [UCAF]) with cryptographic data
  - DE 61 (Point of Service [POS] Data), subfield 3 (POS Terminal Location)
    - Value 2 (Off premises), or
    - Value 4 (On premises of acceptor facility [cardholder terminal including home PC, mobile phone, PDA])

Acquirers submit a merchant-initiated Authorization Request/0100 message containing the following data:

- DE 2 (Primary Account Number [PAN]) with the token
  - DE 22 (POS Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 10 (Credential on File)
  - DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 containing values:
    - Position 1, value 2 (Channel)
    - Position 2, value 4 (Digital Secure Remote Payment [DSRP] with UCAF data)
    - Position 3, value 7 (Merchant-initiated transaction)
  - DE 48 (Additional Data: Private Use), subelement 43 not present
  - DE 61 (Point-of-Service [POS] Data), subfield 3 = 2 (Off premises of acceptor facility [cardholder terminal including home PC, mobile phone, PDA])
  - DE 61 (Point-of-Service [POS] Data), subfield 4 = 4 (Standing order/recurring transactions) for recurring payment transactions or = 5 (Electronic order [home PC, Internet, mobile phone, PDA]) for partial shipments.
3. The Dual Message Authorization System attempts to identify a PAN mapping relationship between the token sent by the acquirer, and a PAN in the PAN mapping database.
    - If the mapping or validation fails, MDES rejects the transaction back to the acquirer and merchant with the appropriate response, and then generates an Authorization Advice/0120 response message to inform the issuer of the action.
    - If the MDES services are successful, the Dual Message Authorization System sends the issuer the authorization transaction request containing:
      - DE 2 (Primary Account Number [PAN]) with the cardholder's primary account number
      - DE 14 (Date, Expiration) with the primary account number's expiration date

- DE 22 (Point-of-Service [POS] Entry Mode), subfield 1, value 10 (Credential on File)
- DE 48 (Additional Data: Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier) value = 327
- DE 48 (Additional Data: Private Use), subelement 30 (Token Transaction Identifier)
- DE 48 (Additional Data: Private Use), subelement 33 containing:
  - Subfield 1 (Account Number Indicator), value H (Cloud-Based Token)
  - Subfield 2 (Account Number) containing the token
  - Subfield 3 (Expiration Date) containing the token expiration date
  - Subfield 5 (Token Assurance Method) static value 00 to 99
  - Subfield 6 (Token Requestor ID)
  - Subfield 8 (Storage Technology)
- DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 containing value 246 indicating Merchant Risk Management Decisioning
- DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 containing value 247 for merchant-initiated transactions

**NOTE: The Dual Message Authorization System will validate or modify to the correct security level indicator (SLI) if needed. See the following for more details.**

- DE 48 (Additional Data: Private Use), subelement 43 (Universal Cardholder Authentication Field [UCAF]) may be present if the issuer has opted-in to receive cryptographic data
  - DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services) including:
    - 50C = MDES PAN Mapping service
    - 61V = MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)
4. The issuer responds with Authorization Request Response/0110 message. The Dual Message Authorization System provides the response to the acquirer containing the following:
- DE 2 (Primary Account Number [PAN]) contains the token
  - DE 48 (Additional Data: Private Use), subelement 33, containing:
    - Subfield 1 (Account Number Indicator), value M (Primary Account Number)
    - Subfield 2 (Account Number)
    - Subfield 3 (Expiration Date) containing the expiration date
    - Subfield 5 (Token Assurance Method) static value 00 to 99
    - Subfield 6 (Token Requestor ID)
5. The acquirer forwards the Authorization Request Response/0110 including the token to the merchant.

## Mastercard Fraud Scoring Services

This section describes the Mastercard Fraud Scoring Services, which includes Fraud Rule Manager.

### Fraud Rule Manager

Fraud Rule Manager is an optional service that allows participating issuers to publish and enact business rules specific to their portfolio of accounts. Fraud Rule Manager enables issuers to automate precise, highly variable fraud decisions during authorization to respond to evolving fraud trends quickly and easily.

For more information on the Mastercard Fraud Scoring Services, refer to [Mastercard Security Solutions Guide for Issuer Services](#).

## Mastercard Identity Check

The Mastercard Identity Check™ program encompasses varying solution-oriented protocols that build upon the infrastructure requirements for channel encryption with the additional benefit of cardholder authentication. When used with components of the Mastercard Dual Message Authorization System, this program provides a mechanism for online merchants to potentially receive an enhanced payment guarantee similar to what retailers receive with qualifying physical point-of-sale transactions.

When participating in the Mastercard Identity Check program, the SLI and UCAF data must be included in the authorization to the issuer when it is available in DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators) and 43 (Universal Cardholder Authentication Field [UCAF]). It is a variable length (maximum 32 characters) field with a flexible data structure that can be tailored to support the needs of issuer security and authentication approaches. Mastercard AAV UCAF data is always a base64 value with a length of 28 characters.

For more information on the Mastercard Identity Check program, refer to the *Mastercard Identity Check Program Guide*.

## Mastercard In Control services

The Mastercard In Control™ platform provides a number of advanced authorization, transaction routing, and alert notification capabilities designed to assist issuers in creating new and enhanced payment products for their consumer and commercial cardholders.

Mastercard In Control fully integrates patented payment technologies with unparalleled global processing capabilities. With the integration of this capability into the Dual Message Authorization System, issuers can benefit from cost efficiencies and speed to market for new product development.

For more information, refer to the [Mastercard In Control Program Guide](#), [Mastercard In Control for Mobile Payments Issuer Implementation Guide](#), [Mastercard In Control for Business Travel](#), and [Mastercard In Control for Commercial Payments Quick Start Guide](#).

## Mastercard In Control Purchase Controls

The Mastercard In Control™ Purchase Controls service allows cardholders to establish spend control rules for their pre-existing payment cards that are enforced during the authorization process.

## Mastercard In Control Real Card Spend Control services

The Mastercard In Control™ Real Card Spend Control service targets the consumer cardholder segment and offers a broad suite of authorization control categories that provide increased security and budgetary monitoring capabilities coupled with an optional Short Message Service (SMS) alerting service.

### Authorization processing

**NOTE: Spend control rules can only be applied to transactions that flow through the Mastercard Network. Spend control rules are not applied to Authorization Advice/0120: Acquirer-generated and Reversal Request/0400 messages.**

1. The acquirer sends the Authorization Request/0100 message containing the Mastercard In Control real card number in DE 2 (Primary Account Number [PAN]) to the Dual Message Authorization System.
2. The Dual Message Authorization System sends the transaction to Mastercard In Control for verification against the cardholder's registered spend control rules.

### Associated edits

| When the transaction...                               | Then Mastercard In Control...                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| meets the control rules established by the cardholder | sends the issuer the Authorization Request/0100 message where DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) contains value 20 (In Control RCN Spend Control) and subfield 2 (On-behalf [OB] Result 1) contains value V (Valid). |

| When the transaction...                                                                                                                                                                                           | Then Mastercard In Control...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>does not meet the control rules established by the cardholder</p> <p>and</p> <p>the cardholder has registered an action response of decline and alert notification upon rule failure</p>                       | <p>declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response.</p> <p>and</p> <p>sends the issuer an Authorization Advice/0120: System-generated message where:</p> <ul style="list-style-type: none"> <li>– DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services): <ul style="list-style-type: none"> <li>– Subfield 1 (On-behalf [OB] Service) contains value 20</li> <li>– Subfield 2 (On-behalf [OB] Result 1) contains the spend control rule that failed</li> </ul> </li> <li>– DE 60 (Advice Reason Code): <ul style="list-style-type: none"> <li>– Subfield 1 (Advice Reason Code) contains value 200 (In Control Processing Advice to issuer)</li> <li>– Subfield 2 (Advice Detail Code) contains a valid advice reason code</li> </ul> </li> </ul> <p>and</p> <p>Sends the cardholder an alert notification indicating rule failure.</p> |
| <p>does not meet the spend control rules established by the cardholder</p> <p>and</p> <p>the cardholder has registered an action response of alert notification only upon rule failure, but no decline action</p> | <p>sends the issuer an Authorization Request/0100 message where DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) contains value 20 (In Control RCN Spend Control) and subfield 2 (On-behalf [OB] Result 1) indicates the spend control rule that failed.</p> <p>and</p> <p>sends the cardholder an alert notification indicating rule failure.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**NOTE: If the transaction is declined (by the issuer, alternate issuer, Stand-In Authorization, or X-Code Processing), or the issuer/alternate issuer performs a partial approval or purchase amount only approval, the Dual Message Authorization System sends an Authorization Advice/0120: system-generated message to Mastercard In Control to update the disposition of the transaction.**



### Associated edits

| When...                                                                                                           | Then the Dual Message Authorization System...                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mastercard In Control is unavailable or responds late<br>or<br>the spend control rules have recently been removed | populates DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) value 20 (In Control RCN Spend Control) and subfield 2 (On-behalf [OB] Result 1) value U (Unable to process), and forwards the transaction to the issuer. |

## Mastercard In Control Virtual Card Mapping and Spend Control Service

The Mastercard In Control™ Virtual Card Mapping and Spend Control Service is for cardholders to request a virtual card number, optionally establish spend controls on that virtual card, and receive alerts.

Mastercard In Control Virtual Card Mapping and Spend Control Service Authorization Request/0100 messages are identified by:

- DE 2 (Primary Account Number [PAN]) contains the real card number
- DE 48 (Additional Data: Private Use), subelement 33 (PAN Mapping File Information) contains the virtual account data
- DE 48 (Additional Data: Private Use), subelement 71 (On-Behalf Services) contains the on-behalf service performed on the transaction

The following information describes exception processing for a Mastercard In Control Virtual Card Mapping and Spend Control Service Authorization Request/0100 message that was declined by In Control processing. Other possible scenarios for a virtual card number and virtual card number with spend controls are also described.

### Virtual card number scenarios

| When the...                                                                                | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| virtual card number successfully mapped and a valid expiration date and CVC 2 are included | forwards the Authorization Request/0100 message to the issuer.                                                                                                                                                                                                                                     |
| virtual card number was not successfully mapped                                            | <ul style="list-style-type: none"><li>• declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response and</li><li>• sends an Authorization Advice/0120: system-generated message to the issuer.</li></ul> |

| When the...                                                                                       | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| virtual card number successfully mapped, but contains an invalid expiration date or invalid CVC 2 | <ul style="list-style-type: none"> <li>declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response and</li> <li>sends an Authorization Advice/0120: system-generated message to the issuer.</li> </ul> |

### Virtual card number with Spend Controls scenarios

| When the...                                                                                                                                                                                                                | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| virtual card number successfully mapped, valid expiration date and CVC 2 are included, and the transaction passes spend controls                                                                                           | forwards the Authorization Request/0100 message to the issuer.                                                                                                                                                                                                                                    |
| virtual card number was not successfully mapped                                                                                                                                                                            | <ul style="list-style-type: none"> <li>declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response and</li> <li>sends an Authorization Advice/0120: system-generated message to the issuer.</li> </ul> |
| virtual card number successfully mapped, but contains an invalid expiration date or CVC 2<br>and<br>the cardholder wants optional alerts                                                                                   | <ul style="list-style-type: none"> <li>declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response and</li> <li>sends an Authorization Advice/0120: system-generated message to the issuer.</li> </ul> |
| virtual card number successfully mapped, valid expiration date and CVC 2 are included, but the transaction fails spend controls<br>and<br>the cardholder wants optional alerts, but does not want the transaction rejected | forwards the Authorization Request/0100 message to the issuer, and then sends an optional alert to the cardholder.                                                                                                                                                                                |
| virtual card number successfully mapped, valid expiration date and CVC 2 are included, but the transaction fails spend controls<br>and<br>the cardholder wants optional alerts, but wants the transaction declined         | <ul style="list-style-type: none"> <li>declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response and</li> <li>sends an Authorization Advice/0120: system-generated message to the issuer.</li> </ul> |

## Mastercard Consumer Controls

Consumer Controls allows issuers to offer their cardholders popular transaction alerts and controls.

Issuers can implement alerts and controls quickly for their cardholders based on:

- Transaction amount
- Channel (for example, ATM or online)
- Country in which a transaction originates (as in, cardholders may opt-in to alerts and controls for cross-border transactions)
- Static geographic location (include/exclude specific countries)
- Acceptor business code (MCC) (single MCC or multiple MCCs)
- Budget limits
- Travel (included and excluded countries between valid to and valid from dates)

Also, cardholders can:

- Opt-in to receive alerts for all transactions.
- Disable their card to stop (decline) all payments or transactions temporarily, excluding recurring payments.
- Set decline controls based on any combination of transaction amount, channel, country in which a transaction originates, static geolocation, or MCCs.

For more information on Consumer Controls, refer to the [Consumer Controls Program Guide](#) and the [Consumer Controls Product Application Specifications Guide](#).

## Mastercard In Control Virtual Card Service

The Mastercard In Control Virtual Card Service allows issuers to process partial approvals.

Issuers must opt-in to the Mastercard In Control Virtual Card Service when choosing to process partial approvals by registering with Mastercard.

## Authorization Request/0100: In Control Purchase Control

The following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply. This layout applies to participating issuers only.

### Data elements

| Data Element                  | Org | Sys | Dst | Values/Comments                               |
|-------------------------------|-----|-----|-----|-----------------------------------------------|
| DE 2 (Primary Account Number) | M   | •   | M   | Contains the cardholder's real account number |

| Data Element                                                                                                             | Org | Sys | Dst | Values/Comments                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 14 (Date, Expiration)                                                                                                 | C   | .   | C   | The cardholder's real account number expiration date                                                                                                      |
| DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator) | .   | X   | C   | Indicates Mastercard In Control virtual card number<br>V = Virtual Card Number                                                                            |
| DE 48, subelement 33, subfield 2 (Account Number)                                                                        | .   | X   | C   | Virtual card number                                                                                                                                       |
| DE 48, subelement 33, subfield 3 (Expiration Date)                                                                       | .   | X   | C   | VCN Expiration date                                                                                                                                       |
| DE 48, subelement 71 (On-Behalf Services), subfield 1 (On-Behalf [OB] Service)                                           | .   | X   | C   | 17 = In Control Virtual Card Service                                                                                                                      |
| DE 48, subelement 71 (On-Behalf Services), subfield 2 (On-Behalf [OB] Result 1)                                          | .   | X   | C   | V = Valid<br><br><b>NOTE: If the Purchase Control service is unsuccessful, refer to Subfield 2 (On-behalf Result 1) for a list of other valid values.</b> |

## Mastercard In Control Real Card Spend Control

The Mastercard In Control Real Card Spend Control service allows cardholders to establish spend control rules for their preexisting payment cards to be enforced during the authorization process.

These rules can be defined for both card-present and card-not-present transaction environments, and depending on the cardholder's predefined response rules, may result in an alert notification being generated to the cardholder or the transaction being declined on their behalf.

**NOTE: Spend control rules can only be applied to transactions that flow through the Mastercard Network. Spend control rules are not applied to Authorization Advice/0120: acquirer-generated and Reversal Request/0400 messages.**

## Processing a Mastercard In Control service-eligible transaction

This process describes the stages of the Dual Message Authorization System processing for a Mastercard In Control Service-eligible transaction.

The acquirer sends the Authorization Request/0100 message containing the Mastercard In Control real card number in DE 2 (Primary Account Number [PAN]) to the Mastercard Network.

### Associated edits

| When...                                                                                                                                                                                             | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the transaction complies with the control rules established by the cardholder                                                                                                                       | sends the issuer the Authorization Request/0100 message where DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) contains value 20 (In Control—RCN Spend Control) and subfield 2 (On-behalf [OB] Result 1) contains value V (Valid).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| the transaction does not comply with the control rules established by the cardholder<br>and<br>the cardholder has registered an action response of decline and alert notification upon rule failure | declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response.<br>and<br>sends the issuer an Authorization Advice/0120: system-generated message where: <ul style="list-style-type: none"><li>• DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services):<ul style="list-style-type: none"><li>– Subfield 1 (On-behalf [OB] Service) contains value 20</li><li>– Subfield 2 (On-behalf Result 1) contains the spend control rule that failed</li></ul></li><li>• DE 60 (Advice Reason Code):<ul style="list-style-type: none"><li>– Subfield 1 (Advice Reason Code) contains value 200 (In Control Processing Advice to Issuer)</li><li>– Subfield 2 (Advice Detail Code) contains a valid advice reason code</li></ul></li></ul> and<br>sends the cardholder an alert notification indicating rule failure. |

| When...                                                                                                                                                                                                                     | Then the Dual Message Authorization System...                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the transaction does not comply with the spend control rules established by the cardholder<br>and<br>the cardholder has registered an action response of "alert notification only" upon rule failure, but no decline action | sends the issuer an Authorization Request/0100 message where DE 48, subelement 71, subfield 1 is value 20 and subfield 2 indicates the spend control rule that failed.<br>and<br>sends the cardholder an alert notification indicating rule failure. |

**NOTE: If the transaction is declined (by the issuer, alternate issuer, Stand-In Authorization, or X-Code Processing), or the issuer/alternate issuer performs a partial approval or purchase amount only approval, the Dual Message Authorization System sends an Authorization Advice/0120: system-generated message to Mastercard In Control to update the disposition of the transaction.**

#### Associated edits

| When...                                                                                                           | Then the Dual Message Authorization System...                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mastercard In Control is unavailable or responds late<br>or<br>the spend control rules have recently been removed | populates DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) contains value 20 (In Control—RCN Spend Control) and Subfield 2 (On-behalf Result 1), value U (Unable to process) and forwards the transaction to issuer. |

## Authorization Request/0100: In Control Real Card Spend Control

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

#### Data elements

| Data Element                                                                                                 | Org | Sys | Dst | Values/Comments                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number)                                                                                | M   | .   | M   | Account range must participate in Mastercard In Control Real Card Spend Control Service; DE 2 contains the Mastercard In Control real card number |
| DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) | .   | X   | C   | 20 = In Control RCN Spend Control Service                                                                                                         |

| Data Element                                                                                                  | Org | Sys | Dst | Values/Comments                                                                                                 |
|---------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------|
| DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 2 (On-behalf [OB] Result 1) | •   | X   | C   | Indicates the results of the service processing. Valid subfield 2 values: D, E, F, G, H, J, K, L, M, P, U, or V |

## Authorization Advice/0120: In Control Real Card Spend Control

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

### Data elements

| Data Element                                                                                                  | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number)                                                                                 | •   | M   | M   | Account range must participate in Mastercard In Control Real Card Spend Control Service                                                                                                                                                                            |
| DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service)  | •   | M   | M   | 20 = In Control RCN Spend Control Service                                                                                                                                                                                                                          |
| DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 2 (On-behalf [OB] Result 1) | •   | M   | M   | Indicates the results of the service processing. Valid subfield 2 values: D, E, F, G, H, J, K, L, M, P, U, or V                                                                                                                                                    |
| DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)                                                   | •   | M   | M   | 200 = In Control Processing Advice to Issuer                                                                                                                                                                                                                       |
| DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code)                                                   | •   | M   | M   | Reject reason                                                                                                                                                                                                                                                      |
| DE 121 (Authorizing Agent ID Code)                                                                            | •   | C   | C   | 000003 = Applicable only to issuers participating in the In Control services and will only be present in Authorization Advice/0120: system-generated messages notifying issuers of a declined request due to a failed spend control rule defined by the cardholder |

## Mastercard In Control Virtual Card Mapping and Spend Control Service

The Mastercard In Control platform leverages the same authorization and clearing data elements for mapping virtual card numbers to their real card account numbers as in the Mastercard In Control Purchase Control service and the Mastercard Contactless Mapping service. Issuers currently supporting the Mastercard Contactless Mapping service need only to recognize new data values in existing data elements to support the Mastercard In Control Virtual Card Mapping and Spend Control service.

### Authorization Request/0100: In Control Virtual Card Mapping and Spend Control Service

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

#### Data elements

| Data Element                                                                      | Org | Sys | Dst | Values/Comments                                    |
|-----------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------|
| DE 2 (Primary Account Number [PAN])                                               | M   | .   | M   | The real card number                               |
| DE 48 (Additional Data—Private Use), Subelement 33 (PAN Mapping File Information) | .   | X   | C   | The virtual account data                           |
| DE 48, Subelement 71                                                              | .   | X   | C   | The on-behalf service performed on the transaction |

### Exception Processing

The following message flow describes the Dual Message Authorization System processing for a Mastercard In Control Virtual Card Mapping and Spend Control service authorization transaction that was declined by In Control processing.

1. The acquirer sends an Authorization Request/0100 message to the Dual Message Authorization System containing the In Control virtual card number in DE 2 (Primary Account Number [PAN]).
2. The Dual Message Authorization System applies unique controls for the In Control virtual card number and performs mapping to the cardholder's primary account number.
3. If the transaction fails mapping:
  - The Dual Message Authorization System declines the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/400 message and sends the acquirer an



- Authorization Request Response/0110, Authorization Advice Response/0130: system-generated, or a Reversal Request Response/0410 message.
- If the transaction fails spend controls, the Dual Message Authorization System declines the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/400 message and sends the acquirer an Authorization Request Response/0110 message.
4. If the transaction fails spend controls but PAN mapping is successful, the Dual Message Authorization System sends an Authorization Advice/0120: system-generated message to the issuer containing:
- DE 2 (Primary Account Number [PAN])
  - DE 48 (Additional Data: Private Use), subelement 33 (PAN Mapping File Information)
  - DE 48 (Additional Data: Private Use), subelement 71 (On-Behalf Services), subfield 1 (On-Behalf Services), value 17 (In Control Virtual Card Service) and subfield 2 (On-Behalf [OB] Result 1) containing the appropriate result value
  - DE 60 (Advice Reason code), subfield 1 (Advice Reason Code) = 200 (In Control Processing Advice to Issuer), subfield 2 (Advice Detail Code) = appropriate advice reason code

## Mastercard In Control Authorization reports

The following reports provide information about the Mastercard In Control service.

- The Authorization Parameter Summary Report (SI737010-AA) includes a Mastercard In Control Mapping participant parameter and a Mastercard In Control Real Card Spend Control participant parameter indicate issuer account range participation in the Global Parameters section.
- The Daily Account File Activity Report (AM700010-EE) displays activity associated with Mastercard In Control.
- The Detail Account Report (AM730010-AA) displays Mastercard In Control accounts.

For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#). For a sample of the Daily Account File Activity Report (AM700010-EE) and Detail Account Report (AM730010-AA), refer to the [Account Management System User Manual](#).

## Mastercard Installment Payment Service

The Mastercard Installment Payment Service is a service offered by Mastercard to Dual Message Authorization System-connected issuers to support installment payments for authorization and clearing processing. This service enables cardholders to choose to pay for a purchase in installment payments and select the installment payment terms from choices provided at the time of purchase.

For additional guidance on the Mastercard Installment Payment Service, refer to the [Mastercard Installments Payment Services Program Guide](#).

## Mastercard Merchant Presented QR

This service offers a mobile payment option for person-to-merchant transactions in markets where conventional transactions rely heavily on cash for retail purchases.

Mastercard Merchant Presented QR is a consumer-initiated, push payment transaction initiated with a mobile device. Using a smartphone, the consumer makes a cashless payment by scanning a Mastercard Merchant Presented QR code at any merchant accepting Mastercard Merchant Presented QR transactions. This product fulfills the need to make payments instantly to merchants in a direct, convenient, and secure method for both the merchant and cardholder.

This service operates in the four-party business model, which enables funds to be transferred from a licensed Mastercard Originating Institution (OI), the consumer's bank, to a Receiving Institution (RI), the merchant's bank. The transaction originates at the OI and routes to the merchant after successfully authenticating and authorizing the debit from the consumer's account. The RI will process the payment transaction and notify the merchant that the payment has been received successfully. The RI will credit the merchant account after processing the payment transaction.

For more information about Mastercard Merchant Presented QR, refer to the [Mastercard QR: Send Program Guide](#).

## Dual Message Authorization System Edits

The Dual Message Authorization System performs the following system edits.

### Mastercard Merchant Presented QR Funding Transaction Processing

The Dual Message Authorization System validates Mastercard Merchant Presented QR transactions that are identified as:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase)
- DE 48 (Additional Data), subelement 77 (Transaction Type Identifier), value C67 (Mastercard Merchant Presented QR)

### Mastercard Merchant Presented QR Payment Transaction

The Dual Message Authorization System validates Merchant Presented QR Payment transactions that are identified as:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction)
- DE 48 (Additional Data), subelement 77 (Funding/Payment Transaction Type Identifier), value C67 (Masterpass QR)

The Dual Message Authorization System validates Merchant Presented QR Refund Payment transactions that are identified as:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction)
- DE 48 (Additional Data), subelement 77 (Funding/Payment Transaction Type Identifier), new value C68 (Masterpass QR Refund)

### Validate Merchant Code edits

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Then the Dual Message Authorization System...                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>DE 18 (Merchant Code) contains any one of the following MCCs in an Authorization Request/0100 message for a Mastercard Merchant Presented QR Payment Transaction, Mastercard Merchant Presented QR Refund Payment Transaction, or for a Mastercard Merchant Presented QR Funding Transaction:</p> <ul style="list-style-type: none"> <li>• 4829 = Money Transfer</li> <li>• 6010 = Manual Cash</li> <li>• 6011 = ATM</li> <li>• 6532 = Payment Transaction—Customer Financial Institution</li> <li>• 6533 = Payment Transaction: Merchant</li> <li>• 6538 = Mastercard Send Funding</li> <li>• 6540 = Funding (exclusive of Mastercard Send)</li> <li>• 6555 = MA Initiated Rewards/Rebate</li> <li>• 6050 = Quasi Cash</li> <li>• 6051 = Quasi Cash—merchant</li> </ul> | <p>sends the Originating Institution an Authorization Request Response/0110 with:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 018 (indicating the data element in error)</li> </ul> |

### Validate the presence of mandatory data edits

| When...                                                                                                                                                                                           | Then the Dual Message Authorization System...                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 11 (Sender Account Number) is not present, not numeric, contains all zeros, or exceeds 20 characters</p> | <p>sends the Originating Institution an Authorization Request Response/0110 with</p> <ul style="list-style-type: none"> <li>• DE 39 = 30 (format error)</li> <li>• DE 44 (Additional Response Data) = 108 (indicating the data element in error)</li> </ul> |

### Validate the presence of mandatory data edits

| When...                                                                                                                                                                                                                                                                                                     | Then the Dual Message Authorization System...                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 108 (Additional Transaction Reference Data), subelement 02 (Transaction Reference Data), subfield 3 (Funding Source) is not present or does not contain one of the following values: 01 (Credit), 02 (Debit), 03 (Prepaid), 04 (Deposit Account), 05 (Mobile Money Account), 06 (Cash), or 07 (Reserved) | sends the Originating Institution an Authorization Request Response/0110 with <ul style="list-style-type: none"> <li>• DE 39 = 30 (format error)</li> <li>• DE 44 (Additional Response Data) = 108 (indicating the data element in error)</li> </ul> |

### Validate transaction amount limit edits

| When...                                  | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the transaction amount limit is exceeded | sends the Originating Institution an Authorization Request Response/0110 with DE 39 = 05 (Do not honor) <p>an Authorization Advice/0120 message is sent to the Receiving Institution with:</p> <ul style="list-style-type: none"> <li>• DE 60 (Advice Reason Code), subfield 1 (Values, in Authorization Advice/0120), value 151 (Blocking Processing Advice to Issuer [Mastercard Merchant Presented QR]).</li> <li>• DE 60, subfield 2 (Values, in Mastercard In Control Service), code 0064 (Reject: Transaction Limit Check).</li> <li>• DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations), value 37D (Mastercard Merchant Presented QR Blocking: Transaction Amount Limit Exceeded).</li> </ul> |

### Validate cumulative transaction amount limit edits

| When...                                             | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the cumulative transaction amount limit is exceeded | <p>sends the Originating Institution an Authorization Request Response/0110 with DE 39 = 05 (Do not honor)</p> <p>an Authorization Advice/0120 message is sent to the Receiving Institution with:</p> <ul style="list-style-type: none"><li>• DE 60 (Advice Reason Code), subfield 1 (Values, in Authorization Advice/0120), value 151 (Blocking Processing Advice to Issuer [Mastercard Merchant Presented QR]).</li><li>• DE 60, subfield 2 (Values, in Mastercard In Control Service), code 0065 (Reject: Cumulative Limit Check).</li><li>• DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations), value 37E (Mastercard Merchant Presented QR Blocking: Cumulative Transaction Amount Limit Exceeded).</li></ul> |

### Validate the Originating Institution edits

| When...                                                                                                                                                                                                                         | Then the Dual Message Authorization System...                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| the Originating Institution, not registered for participation in the Mastercard Merchant Presented QR service, submits a transaction containing DE 48 (Additional Data), subelement 77 (Transaction Type Identifier), value C67 | sends the Originating Institution an Authorization Request Response/0110 with DE 39 = 58 (Transaction not permitted to acquirer/terminal) |

### Domestic activity edits

| When...                                                                                                                                                                                                                                                          | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| if the Mastercard Merchant Presented QR Receiving Institution has chosen to limit Mastercard Merchant Presented QR to domestic activity, then if the PAN country code does not equal the country code value in DE 32 (Acquiring Institution Identification Code) | <p>sends the originating institution an Authorization Request Response/0110 with DE 39 = 05 (Do not honor)</p> <p>an Authorization Advice/0120 message is sent to the Receiving Institution with</p> <ul style="list-style-type: none"> <li>DE 60 (Advice Reason Code), subfield 1 (Values, in Authorization Advice/0120), code 151 (In Control Processing Advice to Issuer [Mastercard Merchant Presented QR]).</li> <li>DE 60, subfield 2 (Values, in Mastercard In Control Service) code 0072 (Reject: Geographic Restriction).</li> <li>DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations), value 37F (Mastercard Merchant Presented QR Blocking—Domestic Activity Only).</li> </ul> |

### Valid authorization messages edits

| When...                                                                                                                                                                           | Then the Dual Message Authorization System...                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Originating Institution submits an Authorization Request/0100 message for a valid Mastercard Merchant Presented QR transaction                                                | sends the Authorization Request/0100 message to the Receiving Institution with DE 48, subelement 71, subfield 1, subfield 2, containing value 37V (Valid).             |
| the Originating Institution submits an Authorization Request/0100 message for any other system issue that prevents successful Mastercard Merchant Presented QR service processing | sends the Authorization Request/0100 message to the Receiving Institution with DE 48, subelement 71, subfield 1, subfield 2, containing value 37U (Unable to process). |

### Reject reversal messages

| When...                                                                                                                | Then the Dual Message Authorization System...                                                                                       |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| the Originating Institution submits a Reversal Request/0400 message for a Mastercard Merchant Presented QR transaction | sends the Originating Institution a Reversal Request Response/0410 with DE 39 = 58 (Transaction not permitted to acquirer/terminal) |

### Edits on payment transactions

| When...                                                                                                                                                                                                           | Then the Dual Message Authorization System...                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in an Authorization Request/0100 message does not contain a valid PAN | sends the Originating Institution an Authorization Request Response/0110 message that contains DE 39 (Response Code) = 12 (Invalid) |

### Edits on payment transactions

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Then the Dual Message Authorization System...                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in an Authorization Request/0100 message is not properly formatted <ul style="list-style-type: none"><li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction)</li><li>DE 48 (Additional Data), subelement 77 (Transaction Type Identifier), value C67 (Mastercard Merchant Presented QR)</li></ul> | sends the Originating Institution an Authorization Request Response/0110 message that contains DE 39 (Response Code) = 12 (Invalid Transaction) |

### Edits on payment transactions

| When...                                                                                                                                                                                                                                                                                               | Then the Dual Message Authorization System...                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in an Authorization Request/0100 message contains a Digital Account Reference Number that is assigned already or is mapped to a different funding account | sends the Originating Institution with an Authorization Request Response/0110 message that contains DE 39 (Response Code) = 12 (Invalid) |

### Edits on payment transactions

| When...                                                                                                                                                                                                                                                                                                                                                            | Then the Dual Message Authorization System...                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) is not contained in an Authorization Request/0100 message, and the value in DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 11 (Sender Account Number) is a valid Mastercard PAN | populates DE 108, subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in the Authorization Request/0100 message and forwards the message to the Receiving Institution/merchant bank. |

### Edits on payment transactions

| When...                                                                                                                                                                                                                                                                              | Then the Dual Message Authorization System...                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) is not contained in an Authorization Request/0100 message, the value in DE 108, subelement 02, subfield 11 is not a valid Mastercard PAN | sends the message to the Receiving Institution without the PAN field populated.<br><br>The merchant handles the refund as a manual cash distribution. |



### Edits on payment transactions

| When...                                                                                                                       | Then the Dual Message Authorization System...                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Originating Institution submits a Reversal Request/0400 message for a Mastercard Merchant Presented QR Refund transaction | sends the Originating Institution with a Reversal Request Response/0410 message that contains DE 39 = 58 (Transaction not permitted to acquirer/terminal) |

### Edit on funding transactions

| When...                                                                                                                                                                                                                                                                                                                                                                                                                             | Then the Dual Message Authorization System...                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Originating Institution is not registered for participation in the Mastercard Merchant Presented QR service and submits a funding transaction in an Authorization Request/0100 message containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), value C67 (Mastercard Merchant Presented QR) | sends the Originating Institution an Authorization Request Response/0110 message that contains DE 39 = 58 (Transaction not permitted to acquirer/terminal) |

### Edit on refund payment transactions

| When...                                                                                                                                                                                                                                                                                                                                   | Then the Dual Message Authorization System...                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Receiving Institution is not registered for participation in the Mastercard Merchant Presented QR service and submits a refund transaction in an Authorization Request/0100 message containing DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier), value C68 (Mastercard Merchant Presented QR Refund) | sends the Originating Institution an Authorization Request Response/0110 message that contains DE 39 = 58 (Transaction not permitted to acquirer/terminal) |

### Mastercard Merchant Presented QR Payment Transaction edits

| When...                                                                                                                                                                                              | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the individual transaction limit or the cumulative transaction amount or count limit is exceeded in an Authorization Request/0100 message for a Mastercard Merchant Presented QR Payment Transaction | <p>forwards the Originating Institution an Authorization Request Response/0110 message with: DE 39 (Response Code) = 05 (Do Not Honor)</p> <p>sends to the Receiving Institution an Authorization Advice/0120: system-generated message with:</p> <ul style="list-style-type: none"> <li>• DE 48 (Additional Data), subelement 71 (On-behalf Service [OBS]), subfield 1 (On-behalf Service Indicator), value 37 (Mastercard Merchant Presented QR Blocking Service)</li> <li>• DE 48, subelement 71, subfield 2 (On-behalf Result 1), value E (Mastercard Merchant Presented QR Blocking: Cumulative Transaction Amount or Count Limit Exceeded)</li> <li>• DE 60 (Advice Reason Code), in the table Subfield 1 Values, in Authorization Advice/0120, value 151 (In Control Processing Advice to Issuer [Mastercard Merchant Presented QR])</li> <li>• DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0064 (Reject: Transaction Limit Check) when the individual transaction limit is exceeded<br/>or</li> <li>• DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0065 (Reject: Cumulative Limit Check) when the cumulative counts or amounts limit is exceeded</li> </ul> |

### Mastercard Merchant Presented QR Funding Transaction edits

| When...                                                                                                                                                                                                                                                  | Then the Dual Message Authorization System...                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an Authorization Request/0100 message for a Mastercard Merchant Presented QR Funding Transaction does not contain DE 108 (Additional Transaction Reference Data), subelement 03 (Transaction Reference Data), subfield 01 (Unique Transaction Reference) | <p>forwards the Originating Institution an Authorization Request Response/0110 with:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108 (indicating the data element)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Then the Dual Message Authorization System...                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>an Authorization Request/0100 message for a Mastercard Merchant Presented QR Funding Transaction contains DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 02 (Mastercard Merchant Presented QR Receiving Account Number):</p> <ul style="list-style-type: none"><li>• That is not a valid length</li><li>• That contains leading spaces</li><li>• That contains all zeros</li><li>• That does not contain all spaces or all zeros</li></ul> | <p>forwards the Originating Institution an Authorization Request Response/0110 with:</p> <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 108 (indicating the data element in error)</li></ul> |
| <p>an Authorization Request/0100 message for a Mastercard Merchant Presented QR Funding Transaction contains DE 108 (Additional Transaction Reference Data), subelement 06 (Dynamic QR Data):</p> <ul style="list-style-type: none"><li>• That is not a valid length</li></ul>                                                                                                                                                                                                                       | <p>forwards the Originating Institution an Authorization Request Response/0110 with:</p> <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 108 (indicating the data element in error)</li></ul> |

Mastercard Merchant Presented QR Refund Transaction edits

| When...                                                                                                                                                                                             | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the individual transaction limit or the cumulative transaction amount or count limit is exceeded in an Authorization Request/0100 message for a Mastercard Merchant Presented QR Refund Transaction | <p>forwards to the Receiving Institution an Authorization Request Response/0110 message with: DE 39 (Response Code) = 05 (Do Not Honor)</p> <p>forwards to the Originating Institution an Authorization Advice/0120: system-generated message with:</p> <ul style="list-style-type: none"><li>• DE 48 (Additional Data), subelement 71 (On-behalf Service [OBS]), subfield 1 (On-behalf Service Indicator), value 37 (Mastercard Merchant Presented QR Blocking Service)</li><li>• DE 48, subelement 71, subfield 2 (On-behalf Result 1), value E (Mastercard Merchant Presented QR Blocking—Cumulative Transaction Amount or Count Limit Exceeded)</li><li>• DE 60 (Advice Reason Code), in the table Subfield 1 Values, in Authorization Advice/0120, value 151 (In Control Processing Advice to Issuer [Mastercard Merchant Presented QR])</li><li>• DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0064 (Reject: Transaction Limit Check) when the individual transaction limit is exceeded<br/>or</li><li>• DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0065 (Reject: Cumulative Limit Check) when the cumulative counts or amounts limit is exceeded</li></ul> |

### Mastercard Merchant Presented QR Refund Payment Processing edits

| When...                                                                                                                                                                                                                       | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Receiving Institution submits an Authorization Request/0100 message for a Mastercard Merchant Presented QR refund transaction it will contain the Digital Account Reference Number in DE 2 (Primary Account Number [PAN]) | <p>includes this value in DE 2 of the Authorization Request/0100 message sent to the Originating Institution</p> <p>and the following data is provided in DE 108 (Additional Transaction Reference Data):</p> <ul style="list-style-type: none"><li>• Subelement 02 (Sender Data), subfield 11 (Sender Account Number)</li><li>• Subelement 03 (Transaction Reference Data), subfield 03 (Funding Source)</li></ul> |

## Alternate Processing

If the receiving institution is unable to respond, Mastercard Merchant Presented QR transactions are declined.

If the Mastercard Merchant Presented QR Blocking Service is not available:

- The platform that called the service receives notice that the service is not available. The following information is sent to the Receiving Institution in the Authorization Request/0100 message for an authorization decision.
  - DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Service), subfield 1 (On-behalf Service Indicator), value 37 (Mastercard Merchant Presented QR Blocking Service)
  - DE 48, subelement 71, subfield 2 (On-behalf Result 1), value U (Unable to process)

### Alternate Processing for Refund Payment Transactions

If the consumer's bank is unable to respond to a payment or refund payment request, the Mastercard Merchant Presented QR refund transaction is declined.

Stand-In limits are set to zero.

### X-Code Processing

Mastercard Merchant Presented QR transactions is declined by the Dual Message Authorization System in X-Code Processing with DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

For more information about Mastercard Merchant Presented QR, refer to the [Mastercard QR: Send Program Guide](#).

## Mastercard Send programs

Mastercard Send includes the Mastercard Send Payments Transaction program, the Mastercard Funding Transaction program, and the Mastercard Gaming and Gambling Payments program.

The Mastercard Send programs allow processing of funding and payments transactions on the Dual Message Authorization System to facilitate various money transfer and remittance flows globally for person to person, person to merchant, business to consumer and business to business processing. Consumers can leverage their Mastercard®, Debit® Mastercard®, Cirrus®, or Maestro® card to send and access funds for both domestic and cross-border transactions. Funding and payment transactions can be enabled through multiple channels such as an ATM, a bank branch, a stand-alone kiosk, mobile, or remote.

### Mastercard Funding Transaction program

The funding transaction program facilitates both intracountry and cross-border funding transactions from Mastercard, Maestro, and Cirrus-branded cards.

A funding transaction is a point-of-sale (POS) transaction for the purchase of funds transfer services that involves the transfer of funds from an eligible account (by an acquirer or an originating institution on behalf of the cardholder or the sender) for the purpose of either: (a) funding a subsequent transactions and linking funds transfer from the Sender to another person or entity; or (b) transferring funds into another eligible financial account held by that sender.

### Mastercard Send Payment program

The Mastercard Send Payment program enables person-to-person transfers, account-to-account transfers, agent cash out, credit card bill payment, business to consumer disbursement, government/non-government organization to consumer disbursement, and business money transfers by allowing consumers to use their Mastercard, Debit Mastercard, Cirrus, or Maestro card to send and access funds. The Mastercard Send Payment program also allows the use of multiple channels to initiate transactions such as an ATM, a bank branch, a stand-alone kiosk, mobile phone, or over the Internet.

### Mastercard Gaming and Gambling Payments program

A gaming payment transaction is a type of payment transaction that transfers winnings or value usable for gambling or gaming to a Mastercard or Maestro account. It is affected pursuant to, and subject to, the Gaming and Gambling Payments Standards. The funds are transferred by an Originating Institution, or an acquirer, to the recipient's receiving account at the Receiving Institution, or an issuer. Gaming payment transactions serve the growing demand by originating institutions and receiving institutions for fast, secure, and convenient ways for customers to send and receive gaming payments, respectively.

For a list of countries that process and receive Gaming and Gambling payment transactions, refer to [Mastercard Gaming and Gambling Payments Program Standards](#).

For a list of MCCs supported under the Gaming and Gambling program, refer to [Mastercard Gaming and Gambling Payments Program Standards](#).

### Gaming and gambling payment transaction identification

| DE 3, Subfield 1 (Cardholder Transaction Type Code) | DE 48, Subelement 77 (Transaction Type Identifier) | DE 18 (Merchant Type)                               |
|-----------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| 28                                                  | C04                                                | MCC Supported under the Gaming and Gambling program |

For more information about implementing the Mastercard Send Payment Transaction Program and Gaming and Gambling Payments Program, refer to the [Mastercard Send and Funding Transactions Program Standards](#) or contact the Global Customer Service team.

## Authorization Request/0100: Mastercard Funding Transactions

Originating Institutions (OIs) that are registered for the Mastercard Funding Transaction program will be required to send an Authorization Request/0100 and Reversal Request/0400 message with the required data elements in addition to the data elements that are specific to this service.

### Applicable data elements

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                         |
|-----------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) | M   | .   | M   | 00 = Funding Transaction                                                                                                                                                                                |
| DE 18 (Merchant Type)                                                 | M   | .   | M   | MCC 4829 (Money Transfer)<br>MCC 6538 (Mastercard Send Funding)<br>MCC 6540 (Funding Transaction)                                                                                                       |
| DE 48 (Additional Data—Private Use), TCC                              | M   | .   | M   | R = Face-to-face retail (applicable to MCCs 6538 and 6540)<br>T = Non-face-to-face retail (applicable to MCCs 6538 and 6540)<br>U = Unique Transaction—Quasi Cash Disbursement (applicable to MCC 4829) |

| Data Element                                       | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 48, subelement 77 (Transaction Type Identifier) | M   | .   | M   | C07 = General Person-to-Person Transfer<br>C52 = General Transfer to Own Account<br>C53 = Agent Cash Out<br>C54 = Payment of Own Credit Card Bill<br>C55 = Business Disbursement<br>C56 = Government/Non-profit Disbursement<br>C57 = Rapid Merchant Settlement<br>C65 = General Business to Business Transfer<br>F07 = General Person-to-Person Transfer<br>F08 = Person-to-Person Transfer to Card Account<br>F52 = General Transfer to Own Account<br>F53 = Agent Cash Out<br>F54 = Payment of Own Credit Card Bill<br>F55 = Business Disbursement<br>F61 = Transfer to Own Staged Digital Wallet Account<br>F64 = Transfer to Own Debit or Prepaid Account<br>F65 = General Business-to-Business Transfer |
| DE 108, Additional Transaction Reference Data      | O   | .   | C   | Required for all Mastercard Funding Transactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



## Reversal Request/0400: Mastercard Funding Transactions

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

### Applicable data elements

| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                                                                                                                                                         |
|-----------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) | M   | •   | M   | 00 = Funding Transaction                                                                                                                                                                                |
| DE 18 (Merchant Type)                                                 | M   | •   | M   | MCC 4829 (Money Transfer)<br>MCC 6538 (Mastercard Send Funding)<br>MCC 6540 (Funding Transaction)                                                                                                       |
| DE 48 (Additional Data—Private Use), TCC                              | M   | •   | M   | R = Face-to-face retail (applicable to MCCs 6538 and 6540)<br>T = Non-face-to-face retail (applicable to MCCs 6538 and 6540)<br>U = Unique Transaction—Quasi Cash Disbursement (applicable to MCC 4829) |
| DE 108 (Additional Transaction Reference Data)                        | O   | •   | C   | Required for all Mastercard Funding Transactions.                                                                                                                                                       |

## Dual Message Authorization System edits

The Dual Message Authorization System performs the following edits on the Authorization Request/0100 and Reversal Request/0400 messages for funding transactions.

**NOTE:** Refer to the [Mastercard Gaming and Gambling Payments Program Standards](#) for the full list of mandatory fields and subelements.

Associated edits

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund</p> <p>and</p> <p>The originating institution identified by the ICA provided in DE 32 (Acquiring Institution ID Code) has not registered to process transactions via the Mastercard Send platform</p>                              | <p>sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 58 (Transaction not permitted to issuer/cardholder).</p>                                                                          |
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund</p> <p>and</p> <p>the brand product code associated with the transaction does not participate in the Mastercard Send transaction</p>                                                                                                | <p>sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 12 (Invalid Transaction).</p>                                                                                                     |
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund</p> <p>and</p> <p>the Unique Transaction reference number (DE 108 [Additional Transaction Reference Data], subelement 03 [Transaction Reference Data], subfield 1 [Unique Transaction Reference]) contains all zeros or spaces</p> | <p>sends the originating institution an Authorization Request Response/0110 message to include the following:</p> <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 108003 (Additional Data—Private Use)</li></ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund</p> <p>and</p> <p>The issuer account range does not participate in the Mastercard Send Payment Transaction Program, Mastercard Funding Transaction Program, or both.</p>                                                                                                                                                                                             | <p>sends the originating institution an Authorization Request Response/ 0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 57 (Transaction not permitted to issuer or cardholder)</p>                                                                    |
| <p>the originating institution is registered for the Mastercard Send service and is submitting Authorization Request/0100 messages for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li> <li>DE 18 (Merchant Type) does not contain MCC 6538</li> <li>DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65</li> </ul>                                   | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 018 (indicating the data element in error)</li> </ul> |
| <p>the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li> <li>DE 18 (Merchant Type) does not contain MCC 4829 or MCC 6540</li> <li>DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65</li> </ul> | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 018 (indicating the data element in error)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Then the Dual Message Authorization System...                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send service and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"><li>• DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li><li>• DE 18 (Merchant Type) contains MCC 6538</li><li>• DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65</li><li>• DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction—Quasi Cash Disbursement).</li></ul>                    | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)</li></ul> |
| <p>the originating institution is registered for the Mastercard Send service and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where:</p> <ul style="list-style-type: none"><li>• DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20</li><li>• DE 18 (Merchant Type) contains MCC 4829 or MCC 6540</li><li>• DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65</li><li>• DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction: Quasi Cash Disbursement).</li></ul> | <p>sends the originating institution an Authorization Request Response/0110 to include the following:</p> <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 30 (Format error)</li><li>• DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)</li></ul> |

## Authorization Request/0100: Mastercard Send payment transactions

Originating Institutions (OIs) that are registered for the Mastercard Send Payments Transaction program are required to send Authorization Request/0100 and Reversal Request/0400

messages with the required data elements in addition to the data elements that are specific to this program.

### Applicable edits

The following table is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

| Data Element                                                                           | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)                  | M   | •   | M   | 28 = Payment Transaction                                                                                                                                                                                                                                                                                                                                                                                       |
| DE 18 (Merchant Type)                                                                  | M   | •   | M   | Must be one of the following values:<br>MCC 6536 (Mastercard Send Intracountry)<br>MCC 6537 (Mastercard Send Intercountry)                                                                                                                                                                                                                                                                                     |
| DE 48 (Additional Data: Private Use), TCC                                              | M   | •   | M   | P = Payment Transaction<br><br><b>NOTE: If TCC is not supplied by the acquirer, Mastercard will insert a "P".</b>                                                                                                                                                                                                                                                                                              |
| DE 48, subelement 77 (Transaction Type Identifier)                                     | C   | •   | C   | C07 = General Person-to-Person Transfer<br>C52 = General Transfer to Own Account<br>C53 = Agent Cash Out<br>C54 = Payment of Own Credit Card Bill<br>C55 = Business Disbursement<br>C56 = Government/Non-profit Disbursement<br>C57 = Rapid Merchant Settlement<br>C58 = Cash in at ATM<br>C59 = Cash in at point of sale<br>C60 = Fast Refund to Original Card<br>C65 = General Business to Business Transfer |
| DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level) | M   | •   | M   | The Dual Message Authorization System validates the POI to ensure DE 61, subfield 10 is one of the following values:                                                                                                                                                                                                                                                                                           |
|                                                                                        |     |     |     | IF... THEN...                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                        |     |     |     | at a bank branch 0 = Not a CAT transaction                                                                                                                                                                                                                                                                                                                                                                     |

| Data Element                                   | Org | Sys | Dst | Values/Comments                                                                                                         |
|------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------|
|                                                |     |     |     | at an ATM<br>1 = Authorized Level 1 CAT: Automated dispensing machine with PIN                                          |
|                                                |     |     |     | at an unmanned kiosk<br>2 = Authorized Level 2 CAT: Self-service terminal                                               |
|                                                |     |     |     | on the Internet<br>6 = Authorized Level 6 CAT: Electronic commerce<br>8 = Reserved for future use                       |
| DE 108 (Additional Transaction Reference Data) | M   | X   | C   | Edits are applied to specific subelements and subfields within DE 108. These edits are described later in this section. |

## Reversal Request/0400: Mastercard Send Payment Transaction

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

### Applicable edits

| Data Element                                                          | Org | Sys | Dst | Values/Comments                                                                                                            |
|-----------------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) | M   | •   | M   | 28 = Payment Transaction                                                                                                   |
| DE 18 (Merchant Type)                                                 | M   | •   | M   | Must be one of the following values:<br>MCC 6536 (Mastercard Send Intracountry)<br>MCC 6537 (Mastercard Send Intercountry) |
| DE 48 (Additional Data: Private Use), TCC                             | M   | •   | M   | P = Payment Transaction                                                                                                    |

| Data Element                                                                           | Org | Sys | Dst | Values/Comments                                                                                                                                                                                                                                                                                                                                                          |                                                                      |
|----------------------------------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| DE 48, subelement 77<br>(Transaction Type Identifier)                                  | C   | .   | C   | C07 = General Person-to-Person Transfer<br>C52 = General Transfer to Own Account<br>C53 = Agent Cash Out<br>C54 = Payment of Own Credit Card Bill<br>C55 = Business Disbursement<br>C56 = Government/Non-profit Disbursement<br>C57 = Rapid Merchant Settlement<br>C58 = Cash in at ATM<br>C59 = Cash in at point of sale<br>C65 = General Business to Business Transfer |                                                                      |
| DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level) | M   | .   | M   | The Dual Message Authorization System will validate the POI to ensure DE 61, subfield 10 is one of the following values: C07, C52–C59, C65, C67                                                                                                                                                                                                                          |                                                                      |
|                                                                                        |     |     |     | IF...                                                                                                                                                                                                                                                                                                                                                                    | THEN...                                                              |
|                                                                                        |     |     |     | at a bank branch                                                                                                                                                                                                                                                                                                                                                         | 0 = Not a CAT transaction                                            |
|                                                                                        |     |     |     | at an ATM                                                                                                                                                                                                                                                                                                                                                                | 1 = Authorized Level 1<br>CAT: Automated dispensing machine with PIN |
|                                                                                        |     |     |     | at an unmanned kiosk                                                                                                                                                                                                                                                                                                                                                     | 2 = Authorized Level 2<br>CAT: Self-service terminal                 |
|                                                                                        |     |     |     | on the Internet                                                                                                                                                                                                                                                                                                                                                          | 6 = Authorized Level 6<br>CAT: Electronic commerce                   |
| DE 108 (Additional Transaction Reference Data)                                         | C   | X   | C   | Mandatory for acquirers to send for all Mastercard Send Payment and Gaming and Gambling Payment Transactions and optional for all Funding Transactions.<br><br>Issuers are required to send DE 108 in the response message; issuers must echo, unedited, the information sent by the originator/acquirer.                                                                |                                                                      |

**NOTE: A Mastercard Send Payment transaction may only be reversed by the acquirer for reason of a documented clerical error and upon agreement with the issuer. In such an event, the error must be reversed within 60 seconds of the time the Mastercard Send Payment transaction was authorized.**

## Dual Message Authorization System edits

The Dual Message Authorization System performs the following edits on the Authorization Request/0100 and Reversal Request/0400 messages for Mastercard Send Payment and Gaming and Gambling Payment transactions.

### Associated edits

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Then the Dual Message Authorization System...                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an originating institution identified by the ICA number provided in DE 32 (Acquiring Institution ID Code) attempts to submit a Mastercard Send Payment Transaction or Gaming and Gambling Payment transactions but has not registered for the Mastercard Send Payment Transaction Program or Gaming and Gambling payment program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal). |
| the originating institution is registered for the Mastercard Send service and is submitting a Mastercard Send Payment Transaction where: <ul style="list-style-type: none"><li>• DE 3, subfield 1 contains value 28</li><li>• DE 18 does not contain MCC 6536 or MCC 6537</li><li>• DE 48, subelement 77 contains value C07, C52, C53, C54, C55, C56, C57, C58, C59, C60, or C65</li><li>• DE 61, subfield 10 criteria are met for Moneysend Payment transaction<ul style="list-style-type: none"><li>– 0 = Not a CAT transaction</li><li>– 1 = Authorized Level 1 CAT: Automated dispensing machine with PIN</li><li>– 2 = Authorized Level 2 CAT: Self-service terminal</li><li>– 6 = Authorized Level 6 CAT: Electronic commerce</li><li>– 8 = Remote terminal</li></ul></li><li>• DE 108 is present</li></ul> | sends the originating institution an Authorization Request Response/0110 message where: <ul style="list-style-type: none"><li>• DE 39 = 30</li><li>• DE 44 = 018 (Merchant Type)</li></ul>                      |



| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Then the Dual Message Authorization System...                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program and is submitting a Mastercard Send Payment Transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains value MCC associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment</li> <li>• DE 61, subfield 10 criteria is not met</li> <li>• DE 108 is present</li> </ul>                                                                                                              | <p>sends the originating institution an Authorization Request Response/0110 where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 061010 (Point-of-Service [POS] Data)</li> </ul> |
| <p>the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting a Mastercard Send Payment Transaction or Gaming and Gambling Payment transactions where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 48, subelement 77 contains value C07, C52, C53, C54, C55, C56, C57, C58, C59, C60, or C65</li> <li>• DE 61, subfield 10 criteria are met on Moneysend Payment transaction</li> <li>• DE 108 is not present</li> </ul> | <p>sends the originating institution an Authorization Request Response/0110 where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 108</li> </ul>                                  |
| <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program</li> <li>• DE 61, subfield 13 (POS Country Code), and the licensed country of DE 2 (Primary Account Number) do not match (indicating a cross-border transaction)</li> <li>• DE 108, subelement 01, subfield 01 (Receiver/Recipient First Name) is missing, all spaces, all numbers, or any combination of spaces and numbers</li> </ul>                                                                                                                | <p>sends the originating institution an Authorization Request Response/0110 where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 108 (Receiver/Recipient Data)</li> </ul>        |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Then the Dual Message Authorization System...                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 3, subfield 1 contains value 28 and DE 61, subfield 10 contains value 1 and DE 52 and DE 55 are not present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | forwards the Authorization Request/0100 message to the issuer.                                                                                                                                                   |
| the Mastercard Send platform does not support an acquiring country                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | declines the request and sends the originating institution an Authorization Request Response/0110 message where DE 39 = 58 (Transaction not permitted to acquirer/terminal)                                      |
| <ul style="list-style-type: none"> <li>DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment)</li> <li>DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program and The account range is not registered for the Mastercard Send Payment Transaction Program Mastercard Gaming and Gambling Payments Program</li> </ul>                                                                                                                                                                                                                                                                                                          | declines the request and sends the originating institution an Authorization Request Response/0110 message where DE 39 = 57 (Transaction not permitted to issuer/cardholder)                                      |
| <p>the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>DE 3, subfield 1 contains value 28</li> <li>DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling Payments Program</li> <li>DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments program</li> <li>DE108, subelement 02, subfield 04 (Sender Street Address) is not present or contains all zeros, all spaces, or any combination of zeros and spaces</li> </ul> | <p>sends the Originating Institution an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>DE 39 = 30 (Format Error)</li> <li>DE 44 = 108002 (Sender Data)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment or Gaming and Gambling Payment transaction where:</p> <ul style="list-style-type: none"> <li>DE 3, subfield 1 contains value 28</li> <li>DE 18 contains MCC values associated with Mastercard Send Payment or Gaming and Gambling Payments program</li> <li>DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments Program</li> <li>DE 108, subelement 02, subfield 07 Sender Country code is present but not a valid ISO recognized alpha Country code (Refer to the <a href="#">Quick Reference Booklet</a> for valid Alpha [3 Characters] Country code)</li> </ul> | <p>sends the Originating Institution an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>DE 39 = 30</li> <li>DE 44 = 108002 (Sender Data)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send payment transaction or Gaming and Gambling payment transaction where:</p> <ul style="list-style-type: none"> <li>DE 3, subfield 1 contains value 28</li> <li>DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling Payments Program</li> <li>DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments Program</li> <li>DE 108, subelement 01, subfield 07 (Receiver/Recipient Country) is valid but blocked (part of the Mastercard Send Blocking Country List)</li> </ul>                                                                          | <p>sends the Originating Institution an Authorization Request Response/0110 to include the following:<br/>DE 39 (Response Code) = 57 (Transaction not permitted to issuer/cardholder)</p> <p>and</p> <p>sends the RFI an Authorization Advice Message/0120 where:</p> <ul style="list-style-type: none"> <li>DE 48, subelement 71, subfield 1, value 33 (Send Blocking Service)</li> <li>DE 48, subelement 71, subfield 2, value W (Send Mastercard Blocking—Country not allowed for the Send Transaction)</li> <li>DE 60, subfield 1, value 200 (In Control Processing Advice to Issuer)</li> <li>DE 60, subfield 2, value 0072 (Reject: Geographic Restriction)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Then the Dual Message Authorization System...                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send payment transaction or Gaming and Gambling payment transaction where:</p> <ul style="list-style-type: none"> <li>• DE 3, subfield 1 contains value 28</li> <li>• DE 18 contains MCC value associated with Mastercard Send Payment or Gaming and Gambling Payments Program</li> <li>• DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments Program</li> <li>• DE 108, subelement 02, subfield 07 (Sender Country) is USA or CAN</li> <li>• DE 108, subelement 02, subfield 06 (Sender State/Province Code) is not present or contains all zeros or spaces or invalid code for USA and CAN</li> </ul> | <p>sends the Originating Institution an Authorization Request Response/0110 message where:</p> <ul style="list-style-type: none"> <li>• DE 39 = 30</li> <li>• DE 44 = 108002 (Sender Data)</li> </ul>                                                                                        |
| <p>an Originating Institution that participates in Payment Transfer Activity (PTA) submits a non-Mastercard Send or non-Mastercard Merchant Presented QR transaction that does not contain DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type), value 28 (Payment Transaction) in the Authorization Request/0100 message</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: DE 39 (Response Code) = 12 (Decline: Invalid Transaction)</p>                                                                                                                    |
| <p>Mastercard Send or Gaming and Gambling payment transactions containing DE 108 (Additional Transaction Reference Data), subelement 01 (Receiver/Recipient Data), subfield 07 (Receiver/Recipient Country) is USA or CAN and DE 108, subelement 01 (Receiver/Recipient Data), subfield 06 (Receiver/Recipient State/Province Code) contains all spaces or all zeros in the Authorization Request/0100 message</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)</li> </ul> |
| <p>DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 04 (Sender Street Address) for Mastercard Send payment transactions is not present or contains all zeros, all spaces, or any combination of these two values in the Authorization Request/0100 message</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where:</p> <ul style="list-style-type: none"> <li>• DE 39 (Response Code) = 30 (Format error)</li> <li>• DE 44 (Additional Response Data) = 108002</li> </ul>                           |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Then the Dual Message Authorization System...                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 07 (Sender Country) for Mastercard Send payment transactions or Gaming and Gambling Payment transactions is not present or contains all zeros, all spaces, or any combination of these two values in the Authorization Request/0100 message                                                                                                                                                | rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format error)</li> <li>DE 44 (Additional Response Data) = 108002</li> </ul> |
| Gaming and Gambling Payment transactions where: DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) equals C04 (Gaming Re-pay) and DE 61, subfield 13 (POS Country Code) and the licensed country of DE 2 (Primary Account Number) does not match and cross-border is not permitted as per Mastercard Gaming and Gambling Payments Program Standards | rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: DE 39 (Response Code) = 12 (Invalid Transaction)                                                                                               |

## Mastercard MoneySend Issuer Transaction Controls

The Mastercard® MoneySend™ Issuer Transaction Controls is an On-behalf Service that Receiving Institutions can use to manage their risk. Monitoring and blocking options include velocity by number of transactions and cumulative dollar amount, and sanctions score.

## Network Blocking

Mastercard sets single transaction amounts and 30-day accumulative amount limits to receiving accounts for Mastercard Send Payment, Gaming and Gambling payment transactions. The maximum amount limit for single transactions and 30-day accumulative amount limits vary by program type. Mastercard declines transactions over the set limits with DE 39, value 05 (Do not honor).

## Sanctions Scoring

Mastercard provides a Sanctions Scoring service to assist with meeting Anti-Money Laundering (AML) obligations.

**NOTE: Mastercard has suspended the Mastercard Send Sanctions Scoring Service effective 30 April 2021. Customers will be advised accordingly should Mastercard reinstate the service at a later date.**

The sanctions score is provided to acquirers and issuers and is available for all cross-border Mastercard Send Payment transactions and intracountry Mastercard Send payment transactions in Egypt, Canada, across Europe, and the U.S. This service provides a score based

on screening the individual sender name (consumer, business, government, and non-government) in the authorization message to support real-time decision making.

## Stand-In Authorization and X-Code Processing

Mastercard Send payment and Gaming and Gambling payment transactions are declined by Stand-In Authorization and X-Code Processing. Mastercard Send funding transactions are evaluated against limits and processing in accordance with the issuer Stand-In limits or X-Code controls pertinent to the applicable information in the message (such as Brand, MCC, Purchase (DE 3 = 00), and so forth).

## Mastercard Send payment transaction blocking criteria

The Dual Message Authorization System supports transaction blocking for Mastercard® payment transactions at the country and Receiving Institution (RI).

Transaction blocking may occur at the following levels:

- Country level: When Mastercard has determined that processing through Mastercard Send is not supported in a particular country, the Dual Message Authorization System automatically declines all Mastercard Send payment transaction authorization requests sent to or received from that country.
- Receiving financial institution level: When Mastercard has determined that an RI or associated RI account range is not eligible to process transactions through Mastercard Send, the Dual Message Authorization System automatically declines all Mastercard Send Payment Transaction authorization requests with that RI.

### Block all transactions

Mastercard has enhanced the On-Behalf opt-in service Mastercard Send Issuer Transaction controls service by adding a control to block all Mastercard Send payment and funding transactions. The Block All control is limited to issuers on a case-by-case basis and only available upon request. The block all control will deny all the Mastercard Send Payment and Funding authorization transactions that are submitted in both the Dual Message Authorization System for specific account ranges.

### Authorization report

The Authorization Parameter Summary Report (SI737010-AA) includes a Mastercard Send Blocking flag in the Global Parameters section that identifies whether an RI's account range is blocked from processing transactions through the Mastercard Send platform.

For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#).

## Mastercard Send Issuer Transaction Controls

The Mastercard® Issuer Transaction Controls is an on-behalf service that Receiving Institutions can use to manage their risk. Monitoring and blocking options include velocity by number of transactions and cumulative dollar amount, along with sanctions score.

## Mastercard Pay By Account Program

The Mastercard® Pay by Account for retail payments (referred to as Pay by Account Program) helps issuers, referred to as Account Holding Institutions (AHIs) such as banks, credit unions, and digital players, enrolled in the Pay by Account Program, accelerate the digital transformation of everyday payments by offering consumers additional payment choices, using a secure retail payment solution.

The Pay by Account Program is designed to function as a traditional Debit Mastercard to leverage the Mastercard global acceptance network across in-store (contactless and QR) and e-commerce channels.

The Pay by Account Program enables:

- Account holders to use a digital account-based payment solution through an AHI's online and mobile platforms to process retail transactions. The solution enables an account holder to select an eligible consumer account to make a payment to a merchant across a range of use cases including online, in-app, and in-store purchases; and
- A merchant, through its acquirer, which is referred to as an Acquiring Institution, enrolled in the Pay by Account Program to (1) opt-in and receive real-time settlement of the merchant's proceeds from the Acquiring Institution or (2) process the transaction as a Debit Mastercard transaction and receive settlement of the merchant's proceeds from the Acquiring Institution through its usual processes for the Dual Message Authorization System.

### To Participate

Acquiring Institutions and AHIs that choose to participate in the Mastercard Pay by Account Program must register by contacting their Mastercard account representative or Global Customer Service.

## MCC 106: MDES PAN Mapping File

Account Holding Institution (AHI) can request two distinct types of maintenance for Mastercard Pay by Account tokens stored in the MDES PAN Mapping File: Token status changes and financial account information replacement.

Requests for Mastercard Pay by Account token record maintenance are made using the Issuer File Update Request/0302 message. This message meets all the general message requirements and the following Mastercard Pay by Account requirements:

- DE 2 (Primary Account Number) must not be present
- DE 91 (Issuer File Update Code) must contain value 2 (Update)

- DE 101 (File Name) must contain the value MCC106 (MDES PAN Mapping File)
- DE 120 must be present and formatted as described in the following table

### Field values

| Field ID | Field Name             | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Mapping File Indicator | an-1       | Required<br>F = Financial Account Maintenance                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2        | ICA                    | n-6        | Required<br>ICA of the Account Holding Institution (AHI) at which the funding account resides. Cannot be all zeros; must contain a valid ICA.<br>Must be the same as the ICA currently associated with the account.<br>Format: Right-justified with leading zeros for ICAs less than six-digits (for example, 001234).                                                                                                                                    |
| 3        | Country Code           | an-3       | Required<br>Numeric three-digit country code of the bank at which the funding account resides.<br>Cannot be all spaces nor all zeros; must be a valid three-digit Mastercard-supported country code. Alpha country codes are not supported in this field.<br>Must be the same as the country code currently associated with the account.<br>Customers can reference the <i>Mastercard Quick Reference Booklet</i> for the listing of valid country codes. |
| 4        | Action Required        | an-1       | Required<br>C = Resume Token<br>D = Deactivate Token<br>S = Suspend Token<br>U = Update (Replace) Financial Account Information                                                                                                                                                                                                                                                                                                                           |



| Field ID | Field Name                            | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|---------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5        | Current Financial Account Information | ans-64     | <p>Required</p> <p>The financial account information currently associated with one or more Mastercard Pay by Account tokens.</p> <p>The value provided must be a minimum of 9 characters, and a maximum of 64 characters.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6        | Token                                 | an-19      | <p>Conditional</p> <p>A token associated with the financial account information.</p> <p>Contains all spaces when requesting an update action to replace the current financial account information.</p> <p>Subfield 6 must contain:</p> <ul style="list-style-type: none"> <li>A valid token or all spaces when subfield 4 (Action Required) contains C (Resume Token), D (Deactivate Token), or S (Suspend Token). The presence of a token indicates a request to change the status of only the token provided. The presence of all spaces indicates a request to change the status of all tokens associated with the financial account information in subfield 5.</li> <li>All spaces when subfield 4 contains U (Update Financial Account Information).</li> </ul> |

| Field ID | Field Name                                | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7        | Replacement Financial Account Information | ans-64     | <p>Conditional</p> <p>The new financial account information that is associated with all Mastercard Pay by Account tokens with the same ICA, country code, and related current financial account information.</p> <p>When requesting an update to the financial account information (subfield 4, value U), all tokens associated with the current financial account information are updated with the new financial account information; financial account information for a specific token (in a multi-token scenario) cannot be replaced. A specific token cannot be provided when subfield 4 contains U.</p> <p>The value provided must be a minimum of 9 characters, and a maximum of 64 characters.</p> <p><b>NOTE: The update to Financial Account Information will proceed regardless of the Replacement Financial Account Information already being associated to existing tokens. Those tokens that are already associated with new financial accounts are not impacted by this operation.</b></p> |

### Field values

The following DE 120 layout applies to Issuer File Update Request Response/0312 messages created by Mastercard in response to MCC106 maintenance for Mastercard Pay by Account Token-PAN Financial Account information mapping records.

| Field ID | Field Name             | Attributes | Comments/Valid Values                                                                                                                                                       |
|----------|------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Mapping File Indicator | an-1       | <p>F = Financial Account Maintenance</p> <p>The value is echoed from the Issuer File Maintenance Request/0302 Message.</p>                                                  |
| 2        | ICA                    | n-6        | <p>ICA of the Account Holding Institution (AHI) at which the funding account resides.</p> <p>The value is echoed from the Issuer File Maintenance Request/0302 Message.</p> |

| Field ID | Field Name                                | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                       |
|----------|-------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3        | Country Code                              | an-3       | Numeric three-digit country code of the bank at which the funding account resides.<br><br>The value is echoed from the Issuer File Maintenance Request/0302 Message.                                                                                                                                        |
| 4        | Action Required                           | an-1       | C = Resume Token<br><br>D = Deactivate Token<br><br>S = Suspend Token<br><br>U = Update (Replace) Financial Account Information<br><br>The value is echoed from the Issuer File Maintenance Request/0302 Message.                                                                                           |
| 5        | Current Financial Account Information     | ans-64     | The financial account information provided in the original request message.<br><br>The value is echoed from the Issuer File Maintenance Request/0302 Message.                                                                                                                                               |
| 6        | Token                                     | an-19      | The token or space value provided in the original request message.<br><br>The value is echoed from the Issuer File Maintenance Request/0302 Message.                                                                                                                                                        |
| 7        | Replacement Financial Account Information | ans-64     | The financial account information provided in the original request message.<br><br>The value is echoed from the Issuer File Maintenance Request/0302 Message.                                                                                                                                               |
| 8        | Date of Last Update Activity              | n-6        | Contains the date on which the record was most recently maintained.<br><br>Format: MMDDYY (Month, Day, Year)<br><br>Present only in the Issuer File Update Request Response/0312 Message sent by Mastercard. It must not be present in the Issuer File Maintenance Request/0302 Message sent to Mastercard. |

| Field ID | Field Name                   | Attributes | Comments/Valid Values                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9        | Time of Last Update Activity | n-4        | Contains the time at which the record was updated on the most recent maintenance date.<br><br>Format: HHMM (hour, minute)<br><br>Present only in the Issuer File Update Request Response/0312 Message sent by Mastercard. It must not be present in the Issuer File Maintenance Request/0302 Message sent to Mastercard. |

## Dual Message Authorization System Edits

The Dual Message Authorization System performs the following system edits related to Mastercard Pay By Account transactions.

### Associated edits

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Then the Dual Message Authorization System                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 2 (Primary Account Number [PAN]) is present in the Issuer File Update Request/0302 Message and contains a Mastercard Pay by Account PAN<br><br>And<br><br>DE 120 (Record Data), field 1 contains value M (Mastercard Digital Enablement Service Tokens excluding Static Tokens) or A (All Mastercard Digital Enablement Service Tokens including Static Tokens)<br><br>And<br><br>DE 120 (Record Data), subfield 2 (Action Required) does not contain value C (Resume Token), D (Deactivate Token), or S (Suspend Token) | sends an Issuer File Update Request Response/ 0312 with: <ul style="list-style-type: none"><li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li><li>DE 44 (Additional Response Data) = 002 (indicating the data element in error)</li></ul> |

| When...                                                                                                                                                                                                                                                                                                       | Then the Dual Message Authorization System                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>DE 91 (Issuer File Update Code) contains value 1 (Add), 3 (Delete), or 5 (Inquiry)</p> <p>And</p> <p>DE 101 (File Name) = MCC106 (MDES PAN Mapping File)</p> <p>and</p> <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p>                  | <p>send an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 40 (Requested function not supported)</li> <li>DE 44 (Additional Response Data) = 101 (indicating the data element in error)</li> </ul>        |
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains a value other than F, M, or A</p>                                                                                                                                                                                                       | <p>sends an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120001 (indicating the data element in error)</li> </ul> |
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator), contains value F (Financial Account Maintenance)</p> <p>And</p> <p>DE 2 (Primary Account Number [PAN]) is present in the Issuer File Update Request/0302 Message containing a Mastercard Pay by Account PAN</p>                                 | <p>rejects the message with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 30 (Format Error)</li> <li>DE 44 (Additional Response Data) = 002 (indicating the data element in error)</li> </ul>                                                         |
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p> <p>and</p> <p>DE 120, subfield 2 (ICA) is zero-filled or does not contain a valid ICA</p> <p>or</p> <p>DE 120, subfield 2 does not match the ICA present in the existing mapping record</p> | <p>sends an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120002 (indicating the data element in error)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Then the Dual Message Authorization System                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p> <p>and</p> <p>DE 120, subfield 3 (Country Code) contains any non-numeric characters, or is zero-filled, or does not contain a valid Mastercard-supported three-digit country code value</p> <p>or</p> <p>DE 120, subfield 3 does not match the country code present in the existing mapping record</p>                                                                                                              | <p>sends an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120003 (indicating the data element in error)</li> </ul> |
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p> <p>and</p> <p>DE 120 (Record Data), subfield 4 (Action Required) contains a space or blank, or is other than value C (Resume Token), D (Deactivate Token), S (Suspend Token), or U (Update Financial Account Information)</p> <p>or</p> <p>DE 120 (Record Data), subfield 4 (Action Required) contains value C, D, or S and subfield 7 (Replacement Financial Account Information) is not space or blank filled</p> | <p>sends an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120004 (indicating the data element in error)</li> </ul> |
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p> <p>and</p> <p>DE 120 (Record Data), subfield 4 (Action Required) contains value U (Update Financial Account Information) and subfield 7 (Replacement Financial Account Information) is space or blank filled</p>                                                                                                                                                                                                    | <p>sends an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120004 (indicating the data element in error)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Then the Dual Message Authorization System                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p> <p>and</p> <p>DE 120 (Record Data), subfield 4 (Action Required) contains value C (Resume Token), D (Deactivate Token), or S (Suspend Token) and subfield 6 (Token) contains a value other than spaces or blanks or a valid numeric Mastercard Pay by Account token</p> <p>or</p> <p>DE 120 (Record Data), subfield 4 (Action Required) contains value U (Update Financial Account Information) and subfield 6 (Token) is not space or blank filled</p> <p>or</p> <p>valid numeric Mastercard Pay by Account token</p> | <p>sends an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 25 (Unable to locate record on file [no action taken])</li> <li>DE 44 (Additional Response Data) = 120006 (indicating the data element in error)</li> </ul> |
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p> <p>and</p> <p>DE 120 (Record Data), subfield 5 (Current Financial Account Information) contains a space or blank</p> <p>or</p> <p>DE 120 (Record Data), subfield 5 (Current Financial Account Information) cannot be located in the Token-PAN mapping file</p>                                                                                                                                                                                                                                                         | <p>valid numeric Mastercard Pay by Account token</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120005 (indicating the data element in error)</li> </ul>                         |
| <p>DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)</p> <p>and</p> <p>DE 120 (Record Data), subfield 4 (Action Required) contains value U (Update Financial Account information)</p> <p>and</p> <p>DE 120 (Record Data), subfield 6 (Token) cannot be located in the Token-PAN mapping file</p>                                                                                                                                                                                                                                                                                | <p>sends an Issuer File Update Request Response/0312 with:</p> <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 25 (Unable to locate record on file [no action taken])</li> <li>DE 44 (Additional Response Data) = 120006 (indicating the data element in error)</li> </ul> |

| When...                                                                                                                                                                                                                                                                                                                                                                     | Then the Dual Message Authorization System                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)<br>and<br>DE 120 (Record Data), subfield 5 (Current Financial Account Information) contains the same value as subfield 7 (Replacement Financial Account Information)                                                                                             | sends an Issuer File Update Request Response/0312 with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> </ul>                                                                                           |
| DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)<br>and<br>DE 120 (Record Data), subfield 4 (Action Required) contains value of U (Update Financial Account information) and subfield 5 (Current Financial Account Information) contains the same value as subfield 7 (Replacement Financial Account Information) | sends an Issuer File Update Request Response/0312 with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120007 (indicating the data element in error)</li> </ul> |
| DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)<br>and<br>DE 120 (Record Data), subfield 4 (Action Required) contains value U (Update Financial Account Information) and subfield 7 (Replacement Financial Account Information) value provided is fewer than 9 characters                                        | sends an Issuer File Update Request Response/0312 with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120007 (indicating the data element in error)</li> </ul> |
| DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)<br>and<br>DE 120 (Record Data), subfield 7 (Replacement Financial Account Information) contains the same financial account information as the Token-PAN Mapping File                                                                                             | sends an Issuer File Update Request Response/0312 with: <ul style="list-style-type: none"> <li>DE 39 (Response Code) = 27 (Issuer File Update field edit error)</li> <li>DE 44 (Additional Response Data) = 120004 (indicating the data element in error)</li> </ul> |



| When...                                                                                                                                                                                                                                                                                                                                                                  | Then the Dual Message Authorization System                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DE 120 (Record Data), subfield 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)<br>and<br>DE 120 (Record Data), subfield 4 (Action Required) contains value U (Update Financial Account information) and subfield 7 (Replacement Financial Account Information) contains the same financial account information as the Token-PAN Mapping File | sends an Issuer File Update Request Response/ 0312 with: <ul style="list-style-type: none"><li>• DE 39 (Response Code) = 80 (Duplicate Add, action not performed)</li><li>• DE 44 (Additional Response Data) = 120007 (indicating the data element in error)</li></ul> |

## Mastercard Payment Gateway

The Mastercard Payment Gateway (Payment Gateway) is a Mastercard hosted, centralized, modular gateway for routing commercial payments between buyers and suppliers.

**NOTE: Applies only to the U.S. region.**

The Payment Gateway collects, transforms, and routes commercial payment instruction orders. The Payment Gateway accepts files containing payment instructions and remittance advice information from commercial buyers. Payment instructions are generated directly from the buyer's Enterprise Resource Planning (ERP) or Accounts Payable system.

### Benefits

Key features of the Payment Gateway include:

- Data translation and routing
- Bank connectivity that is available anywhere and anytime
- Universal supplier directory
- Remittance advices to buyers and suppliers
- Payment views and reports
- Straight Through Payment Processing
- Web-based portal

### How it works

The Payment Gateway supports two payment types:

- Payables Account (virtual Purchasing Card) payments
- Electronic Funds Transfer (EFT) payments

The authorization blocking feature supports Payables Account (virtual Purchasing Card) payments only.

The Payment Gateway does not replace the financial institution's existing commercial payment offerings. Rather, it provides the entryway for receiving a buyer's payment files, thus providing a single destination or interface for business-to-business payment processing. In this way, Mastercard does all the work of file translation and data mapping, but the actual payment processing remains the domain of the financial institutions.

The Payment Gateway uses a modular approach. Financial institutions can market and package distinct modules to augment their existing payment services.

Issuers that offer commercial cards may choose to restrict the authorization of certain account ranges to occur through the Payment Gateway by participating in the Mastercard Payment Gateway Authorization Blocking service.

The Dual Message Authorization System determines whether to continue processing or reject Payment Gateway authorization requests based on the presence of DE 48 (Additional Data: Private Use), subelement 47 (Mastercard payment Gateway Transaction Indicator), value MC-MPG/W and the issuer's participation in the Mastercard Payment Gateway Authorization Blocking service.

### **Authorization reports**

The Authorization Parameter Summary Report (SI737010-AA) includes a Mastercard Payment Gateway participant parameter in the Global Parameters section, which indicates when an issuer participates in the Mastercard Payment Gateway Authorization Blocking service. For a sample of this report and field descriptions, refer to [Reports and files](#).

### **Mastercard Payment Gateway Authorization Blocking service**

The Mastercard Payment Gateway Authorization Blocking service allows issuers to restrict the authorization of transactions to only those originating from the Mastercard Payment Gateway platform.

## **Mastercard Safety Net**

This section provides a brief overview of the Mastercard Safety Net service, part of the Mastercard Global Safety and Security Standards suite of services.

### **Summary and Background**

Mastercard Safety Net provides issuers with a second line of defense to limit the impact of a large-scale fraud attack on one or more of their payment channels (for example, in ATM or e-commerce) when their payment systems are breached. The service also identifies when large-scale fraud attacks are occurring, using insights from the Mastercard Network, so that appropriate action can be taken by the issuer. It does not replace an issuer's primary fraud prevention system.

As applicable, Mastercard requires issuers to participate in Mastercard Safety Net across their entire consumer and commercial credit and debit portfolios, without the availability of an opt-out.

Issuers may enroll in the Mastercard Safety Net Alert Only option if desired through **Manage My Accounts** in Mastercard Connect™.

### Data Requirements

If a transaction is declined by the Safety Net service, issuers will receive an Authorization Advice/0120 message with the following values:

- DE 48, subelement 71, subfield 1 = 18 (Fraud Scoring Service)
- DE 48, subelement 71, subfield 2 = C (Fraud Scoring Service was performed successfully)
- DE 48, subelement 75, subfield 1 = 998 or 000
- DE 48, subelement 75, subfield 2 = NM (Network Monitor)
- DE 60, subfield 1 = 120 (Transaction Blocking)
- DE 121 = 000003 (Decline occurred due to an on-behalf service)

For more information about these data elements and values, refer to [Data element definitions and details](#).

### For More Information

For more information, refer to the [Safety Net User Guide](#).

## Mastercard transit transactions

The demands of the transit environment make it different from the traditional retail payment environment.

Specifically, the transit environment requires Fast and secure access to the transit system. Control of access to the transit system (generally referred to as "permission to travel"). Ticketing, beyond simple system access, and ticket inspection establishes that the individual is authorized to take a specific trip. A flexible relationship between access and payment, as the final transaction amount is often not known until the journey is complete. Mastercard delivers on all of these requirements through a range of transit models and related business rules that support acceptance of a broad range of contactless cards.

For more information, refer to the [Mastercard Global Transit Implementation Guide](#).

## Pre-funded transit transactions

A pre-funded transit transaction occurs when the cardholder purchases value redeemable for future travel with a transit merchant. The purchased value is held by the transit merchant in an account linked to the cardholder's card or device. Any usage of the card or device following the purchase of value will reduce the value held with the transit merchant.

The transit merchant follows its usual authorization procedures by generating an Authorization Request/0100 message for the full amount of the transaction.

## Real-time authorized transit transactions

A real-time authorized transit transaction occurs when the transit merchant generates an Authorization Request/0100 message each time a cardholder uses his or her card or device at the transit merchant's terminal.

The transit merchant follows its usual authorization procedures by generating an Authorization Request/0100 message for the full amount of the transaction.

## Post-authorized aggregated contactless transit transactions

A post-authorized aggregated contactless transit transaction occurs when the transit merchant generates a First Presentment/1240 message combining one or more contactless taps performed with one contactless account number and occurring with one transit merchant.

For the contactless transit merchant to receive chargeback protection all of the following must occur:

- The transit merchant must send a properly identified Authorization Request/0100 message, which can be for any amount not exceeding the cardholder verification method (CVM) limit amount, as published in the *Chargeback Guide* on the day of the transaction.
- The issuer must have approved the transaction.
- The combined amount of the contactless taps must be equal to, or less than, the cardholder verification method (CVM) limit amount, as published in the *Chargeback Guide*.
- The maximum time period from the first contactless tap until the First Presentment/1240 message is generated must be 14 calendar days or less.

**NOTE: These transit transactions are limited to MCCs 4111, 4131, and 4784.**

Upon the cardholder's request, the transit merchant must provide a list of the contactless taps that were combined into a First Presentment/1240 message.

**NOTE: The phrases "tap the contactless card or device" and "contactless tap" refer to the same series of actions: a cardholder touching the contactless card or device to a contactless terminal, the contactless terminal reading the card data, and then the contactless terminal flashing a light and making a sound to indicate the outcome of the tap to the cardholder.**

## Authorized-aggregated split clearing transactions

An authorized-aggregated split clearing transaction is subject to the following restrictions.

**NOTE: Applies only to domestic United Kingdom (U.K.) transactions.**

- The contactless-enabled card or device must indicate that an M/Chip profile is supported.
- Tag 5F28 (Issuer Country Code) must have a value of 826.

- The denominated primary currency is GBP.
- The transaction is properly identified in the same manner as described in Post-authorized Aggregated Contactless Transit Transactions, with the exception that the value in the transit indicator field is value 4, instead of value 3.

## Post-authorized aggregated Maestro contactless transit transactions

A post-authorized aggregated Maestro contactless transit transaction occurs when the acquirer generates an Authorization Request/0100 message for an estimated or maximum amount in connection with the use of one Maestro contactless account number at one transit merchant.

The transit merchant may combine multiple Maestro contactless taps into a single transaction amount and reverse any unused funds by the cardholder at the end of the aggregation time period.

The transactions are processed as follows:

- The transit merchant sends an Authorization Request/0100 (Europe-acquired only) message containing value 06 in DE 48, subelement 64, subfield 1 (Transit Transaction Type Indicator) for an estimated or maximum amount not to exceed the applicable contactless transit aggregated CVM amount.
- The issuer must have approved the transaction.
- The cardholder may make subsequent taps for additional rides. These taps will not be sent to the issuer for authorization. The combined amount of the taps must be equal to or less than the applicable contactless aggregated CVM limit amount.
- When the authorized amount is reached or within 3 calendar days (3 days in Mexico), the transit merchant totals the value of all taps and generates a partial Reversal Request/0400 or an Authorization Advice/0120 message to reverse any unused funds. The Authorization Advice/0120 message option is only available for Europe region acquirers.

The transaction is properly identified in the same manner as described in Post-authorized Aggregated Contactless Transit Transactions, with the exception that the transit transaction type identifier (DE 48, subelement 64) value is 06.

## Post-authorized aggregated Maestro contactless transit transactions criteria

All post-authorized aggregated Maestro contactless transit transactions must be identified properly.

Post-authorized aggregated Maestro contactless transit Authorization Request/0100 messages submitted by a transit merchant for Maestro contactless cards or devices must contain the following data element values.

# Differences between post-authorized aggregated and authorized-aggregated split clearing transit transaction rules

Differences between the current Mastercard post-authorized aggregated transit rule and the authorized-aggregated split clearing transit rule for U.K. transit transactions are listed in table format.

## Transaction rules

| Post-authorized Aggregated Transit Transaction Rule                                                                                                                                                                                                                                                                       | Authorized-aggregated Split Clearing Transaction Rule in the U.K.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The transit agency receives no chargeback protection in the event that an issuer declines the authorization.                                                                                                                                                                                                              | Extends the transit agency's chargeback protection rights to include transactions for which the authorization on a U.K. issued credit, debit or prepaid card or device is declined, provided the authorization and clearing amounts are GBP 10 or less.<br><br><b>NOTE: Cards that do not meet the criteria to be eligible for this liability protection follow the Post-Authorized Aggregated Transit Transaction Rule, that is, an overseas issuer, issuing cards not denominated in GBP would not have any additional liability.</b> |
| Each aggregation period (up to 14 days) is initiated by a single authorization and completed by a single clearing message.<br><br>The aggregation period when processing Maestro transactions will be different. Time periods range from 1-3 days, depending on the region and country the transaction is being acquired. | Accommodates the UKCA transaction model by allowing the transit agency to submit a clearing message at the end of every day on which the card incurred transit spend (subject to either an approved authorization message or a declined authorization message and the spend being up to and including GBP 10.00).<br><br>Therefore, there can be multiple clearing messages submitted between successive authorizations for the same card.                                                                                              |

## ATC update request

Authorization Request/0100 messages containing DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 6 (ATC Update) are used to notify the issuer of the most recent ATC (Application Transaction Counter) value for the contactless card or device.

All ATC update authorization requests will have a zero value for the transaction amount. ATC update transactions may be sent regularly, and issuers must ensure that these transactions do not affect the issuer's velocity checks. In addition, issuers should authorize these transactions using DE 39 (Response Code), value 00 (Approved or completed successfully) or 85 (Not

declined), or decline these transactions using DE 39, value 05 (Do not honor) based on the fact that the ATC provided is within or outside of the issuer's ATC tolerance.

For approved transactions where the application cryptogram has been successfully validated, issuers should keep a record of the last seen ATC and set a feasible range. Subsequent transactions received that contain an ATC outside of this range should be treated as suspicious but should not be routinely declined.

Acquirers supporting merchants that implement the Mastercard contactless aggregated models may support zero amount Application Transaction Counter (ATC) update Authorization Request/0100 messages to notify issuers of the most recent ATC values for the offline cardholder aggregation activity.

## Transit debt recovery transactions

Transit merchants typically work solely in a prepaid environment in which the passenger purchases a ticket or a prepaid balance before travel.

With transit merchants migrating to open loop payments where the card presented for payment is also used as a token at the point of entry, there is some risk in allowing a cardholder to travel before receiving payment or even knowing if the account linked to the card being presented for travel is in good standing.

Mastercard Rules allow transit merchants to recover an amount owed to them (a transit debt recovery transaction) following a declined authorization request. A transit debt recovery transaction occurs when transit fare spend has been incurred, in granting a Mastercard contactless card access to travel, and as a result of the issuer declining the authorization request, the transit merchant is initially unable to collect payment for the fare spend. When this situation occurs, the transit merchant may add the card to a Deny List, thereby blocking that card from access to travel until such time as the money owed by the cardholder has been recovered.

A mechanism by which a cardholder's debt is cleared for the cardholder to access the transit system to resume travel is through debt recovery transactions. The Mastercard Contactless Transit Rules create the opportunity for this situation to arise:

- Post-authorized aggregated (Global)
- Authorized aggregated split clearing (U.K. and Republic of Ireland only)
- Post-authorized aggregated Maestro
- Any other transit implementation where the cardholder is allowed to travel before the authorization taking place (deferred authorization)

There are several methods of debt recovery that a transit merchant can implement and each method allows debt recovery transactions to recover a debt from a card or device. Mastercard and Maestro contactless products allow debt recovery to be performed through the following mechanisms:

- Automatically: Presented as a card-not-present (CNP)/PAN key-entered transaction
- Call Centers: Presented as a CNP/MOTO (Mail Order/Telephone Order) transaction

- Websites: Presented as a CNP/e-commerce transaction
- Ticket-vending machines and other ticket office processing: Presented as a card present transaction

## Support for U.K. transit transactions

Mastercard® and Maestro® contactless customers that choose to implement contactless cards and devices in the U.K. must process transit transactions according to the U.K. Cards Association Transit Transaction Model, published in [U.K. Domestic Rules](#).

Customers must ensure that their contactless cardholders are able to use their cards successfully for travel on public transport operated by Transport for London and other U.K. transit operators.

## PAN-association requirements for transit

Issuers that issue alternate account numbers for contactless products and respond to Authorization Request/0100 messages from MCCs 4111, 4131, 4784, and 7523 must provide the following values in DE 48 (Additional Data: Private Use), subelement 33 (PAN Mapping File Information) when sending the Authorization Response/0110 message.

- Subfield 1 (Account Number Indicator), value E (Printed Account Number Provided by Issuer)
- Subfield 2 (Account Number), embossed PAN
- Subfield 3 (Expiration Date), expiration date of the embossed PAN

For issuers participating in the Mastercard® Contactless Mapping Service, the Dual Message Authorization System will provide these values on their behalf in the authorization response messages. This helps ensure that the pre-purchased fares by cardholders using the PAN printed on the card are properly associated with their turnstile requests. This will also help ensure proper customer service.

Acquirers processing for merchants belonging to MCC 4111, 4131, 4784, and 7523 must provide the DE 48, subelement 33 details when present in the authorization response message back to the merchants belonging to these MCCs upon their request.

Refer to the [Mastercard Global Transit Implementation Guide](#) or [Transaction Processing Rules](#) for more information.

## Merchant Risk Predict

Merchant Risk Predict provides predictive, AI generated merchant fraud detection scores to acquirers on dual message authorization transactions. The seven scores sit within the authorization message to acquirers and provide immediate feedback on merchant health, compliance, and chargeback risks.

Merchant Risk Predict provides an alphanumeric code for each score within the Authorization Request Response/0110 message in authorization in DE 48, subelement 57, and is used by



acquirers. The alphanumeric code indicates the likelihood of fraud across seven different scores that provide immediate feedback on merchant health, compliance, and chargeback risks.

For more information, refer to the [Mastercard Merchant Risk Predict Program Guide](#).

## Scores and values

### Scores and values

| Score                              | Predictive insight                                                                                                                                                                          | Customer value                                                                                                                                            |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Merchant Fraud Risk                | Likelihood that merchant encounters fraudulent activities as a portion of overall business                                                                                                  | Understand merchant risk profiles for liability purposes, enforcement of higher security standards, and to evaluate business relationships with merchants |
| Ghost Merchant Risk                | Likelihood that merchant is set up temporarily to collect fraudulent transactions (new merchants only) and subsequently reports high fraud and chargeback bps within 90 days before closure |                                                                                                                                                           |
| Merchant Fraud Attrition Risk      | Likelihood of merchant attrition due to increasing fraud or chargeback in the next 3 months (for tenured merchants)                                                                         | Understand fraud liability due to future merchant inactivity risk                                                                                         |
| Merchant Alternative Identity Risk | Likelihood that a merchant is operating under an alternative category not representative of the business                                                                                    | Ensure merchant credentials are established with legitimate business purposes                                                                             |
| Merchant Account Takeover Risk     | Indicates likelihood of account takeover with fraud testing due to weak authentication and fraud prevention systems                                                                         | Understand risk for specific merchants to be breached by fraudsters                                                                                       |
| Excessive Fraud Chargeback Risk    | Likelihood that the merchant experiences excessive chargebacks as a share of total gross dollar volume (GDV) within next 3 months                                                           | Manage expected chargeback volumes related to fraud over time and take mitigating actions with merchants at highest risk and highest anticipated volumes  |
| Fraud Chargeback GDV               | Estimated revenue segment merchant is likely to owe in fraud related chargebacks in the next 3 months                                                                                       |                                                                                                                                                           |

## Interpretation guidance

Sample Output: GHS123ATT456ATO001AIS789MFR555CBR777CBV112

### Interpretation guidance

| Output | Identifier                                              | Description                                                                                                          |
|--------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 123    | Model output                                            | 000 - 999 from lowest to highest likelihood of a merchant presenting ghost merchant risk                             |
| ATT    | Identifier for Merchant Attrition Risk Score            | Static field                                                                                                         |
| 456    | Model output                                            | 000 - 999 from lowest to highest likelihood for a specific merchant to attrite due to fraud                          |
| ATO    | Identifier for Merchant Account Takeover Risk Score     | Static field                                                                                                         |
| 001    | Model output                                            | 999 if the merchant is likely to have a takeover risk, otherwise 000                                                 |
| AIS    | Identifier for Merchant Alternative Identity Risk Score | Static field                                                                                                         |
| 789    | Model output                                            | 000 - 999 from lowest to highest likelihood of a merchant presenting alternate identity risk                         |
| MFR    | Identifier for Merchant Fraud Risk Score                | Static field                                                                                                         |
| 555    | Model output                                            | 000 - 999 from lowest to highest likelihood of a merchant encountering fraudulent activity                           |
| CBR    | Identifier for Excessive Chargeback Risk Score          | Static field                                                                                                         |
| 777    | Model output                                            | 000 - 999 from lowest to highest likelihood of a merchant to experience excessive chargebacks in relation to its GDV |
| CBV    | Identifier for Merchant Fraud Chargeback GDV Score      | Static field                                                                                                         |

|     |              |                                                                                                                                                                                                                                                                           |
|-----|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 112 | Model output | 000 - 999 is assigned, with the first digit indicating the predicted fraud chargeback GDV risk segment (0 - lowest, 9 - highest) and last 2 digits predicting the mean fraud chargeback GDV expected on the merchants in the given segment expressed as an exponent of 10 |
|-----|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Threshold for review**

Acquirers can determine at which threshold they wish to evaluate the result scores. For example, a customer may choose to review only the top 10 percent highest scored merchants for each category.

**M/Chip processing services**

Mastercard provides M/Chip processing services to help issuers that want to migrate to Mastercard M/Chip technology without having to make an initial investment to completely upgrade their host systems for M/Chip processing.

The M/Chip processing services support issuers using M/Chip technology by performing all or part of the M/Chip-related authorization processing on behalf of the issuer for a designated account range.

The M/Chip processing services are available to all brands: Mastercard, Maestro, Cirrus, and Debit Mastercard. There are three M/Chip processing services:

- Chip CVC to CVC1 Conversion Service
- Chip to Magnetic Stripe Conversion
- M/Chip Cryptogram Pre-validation
- M/Chip Cryptogram Validation in Stand-In Authorization

**NOTE: All issuers globally that participate in Stand-In Authorization must have M/Chip Cryptogram Validation in Stand-In Authorization performed during Stand-In Authorization.**

The Chip to Magnetic Stripe Conversion and the M/Chip Cryptogram Pre-validation services are provided on a permanent basis. The M/Chip Cryptogram Validation in Stand-In Authorization service is provided on a dynamic basis if an issuer is not able to respond to an authorization request.

In addition to these services, issuers can combine the M/Chip Cryptogram Pre-validation Service and the Chip to Magnetic Stripe Conversion Service (referred to as the Combined Service Option).

Issuers that participate in the M/Chip Cryptogram Pre-validation service, the M/Chip Cryptogram Validation in Stand-In Authorization service, or the Combined Service Option may optionally support M/Chip Advance.

For more detailed information about these M/Chip processing services, including information about the Chip Processing Services Member Requirements and Parameters Form, refer to the [M/Chip Processing Services Service Description Guide](#).

## Chip to Magnetic Stripe Conversion

Mastercard provides Chip to Magnetic Stripe Conversion service to issuers by performing all or part of the M/Chip-related authorization processing on behalf of the issuer for a designated account range.

The Chip to Magnetic Stripe Conversion service is optional.

**NOTE: The Global Clearing Management System (GCMS) also offers the Chip to Magnetic Stripe Conversion service.**

## Chip CVC to CVC 1 Conversion service

The Chip CVC to CVC 1 Conversion service offers issuers the option of having Mastercard pre-validate the Chip CVC value and, when valid, replace it with the appropriate CVC 1 value before delivering the authorization request message to the issuer during normal authorization processing.

Pre-validation is performed to ensure that a CVC 1 value is generated only for transactions containing a valid Chip CVC. The Chip CVC to CVC 1 conversion service validates the Chip CVC value and, if valid, replaces it with the appropriate CVC 1 value before delivering the authorization request message to the issuer during normal authorization processing.

The Chip CVC to CVC 1 Conversion service is optional.

### Service options

Two service options are available to issuers:

- Single key and decision matrix for both CVC 1 and Chip CVC
- Different keys and decision matrices for CVC 1 versus Chip CVC

The service first attempts to validate the Chip CVC value using issuer-provided keys (validation is performed to help ensure that only transactions containing a valid Chip CVC are serviced). If the validation is successful, the service uses issuer-provided keys to calculate the CVC 1 value and replace the Chip CVC with the CVC 1 value.

Regardless of the Chip CVC validation result (successful or unsuccessful), the service always attempts to send the authorization request message to the issuer for an authorization decision (if the issuer is unavailable and Stand-In Authorization is enabled for the account range, the Stand-In Authorization will make the authorization decision). The service does not offer an option to decline on behalf of the issuer.

The service supports issuers with authorization host systems that do not support Chip CVC validation. Issuers may choose to use the service as part of their magnetic stripe to chip technology migration path, or as a component of their long-term chip issuance strategy.

**NOTE: Issuers that use the service must ensure CVC 1 and Chip CVC are located in the same positions of the Track 2 data, in the physical magnetic stripe Track 2 and in the Track 2 equivalent data encoded in the chip.**

**Validation of Chip CVC using issuer-provided keys**

Before generating CVC 1, the service will first attempt to validate the Chip CVC using issuer-provided keys.

**Edits**

| WHEN...                    | THEN...                                                                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Validation is successful   | The service will generate CVC 1, which will replace the Chip CVC in the Track 1 DE 45 or Track 2 DE 35 data as applicable.                                                                                                                                                                                                         |
| Validation is unsuccessful | <p>The service will not generate CVC 1, and the Chip CVC will remain in the Track 1 DE 45 or Track 2 DE 35 data as applicable.</p> <p>Mastercard recommends that issuers include DE 48, subelement 87 (CVC Result) value "Y" (Invalid CVC 1) in the authorization response message whenever CVC 1 is determined to be invalid.</p> |

The appropriate On-behalf service IDs and service result code will be added to the message in DE 48, subelement 71.

**U.S. Chip-Enabled Travel Card program**

The U.S. Chip-Enabled Travel Card program leverages the functionality of the Chip to Magnetic Stripe Conversion service and allows issuers that participate in the Chip to Magnetic Stripe Conversion service the option not to receive notification that the conversion has taken place (service results) in authorization and financial request messages.

**Benefits**

The U.S. Chip-Enabled Travel Card program benefits include the following:

- Creates new revenue streams and greater operational efficiency
- Provides the opportunity to capture more payments in new markets, displacing many cash purchases made around the world
- Promotes the globalization of EMV chip technology

How it works

Issuers have the option not to receive notification that the Chip to Magnetic Stripe Conversion has taken place, removing the impact to their authorization and clearing systems when implementing chip. Existing functionality of the Chip to Magnetic Stripe Conversion service, as well as the Combined Service Option (Chip to Magnetic Stripe Conversion and M/Chip Cryptogram Pre-validation services) are not affected. Service results are retained by Mastercard for billing and reporting purposes.

Transactions are tracked at the account range. The source of the track data provided is from Track 2 Equivalent Data (tag 57). Issuers should be aware that the CVC value contained therein is the Chip CVC and not CVC 1; therefore, it will impact any CVC validation that an issuer performs.

For a chip transaction received for an account within the account range, the chip transaction is converted to the magnetic stripe equivalent. Although participation is set at the account range, instead of at the specific account, other accounts that the issuer does not want to participate will not be chip enabled.

Chip-enabled U.S. travel cards include a hybrid chip and magnetic stripe card with signature, as the card verification method and online PIN capabilities. Chip-enabled U.S. Travel cards allow chip acceptance and continue to function similar to existing issuer magnetic stripe cards.

Authorization processing

The following information describes the Dual Message Authorization System of U.S. chip-enabled travel cards transaction processing.

Associated edits

| WHEN...                                                                       | THEN the Dual Message Authorization System...                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the Chip to Magnetic Stripe Conversion Service is requested for a transaction | verifies whether the issuer wants to receive Chip to Magnetic Stripe Conversion Service results in Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages for an account range that is eligible to participate in the service. |

| WHEN...                                                                                                                                                    | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the issuer for the account range does not want to receive the Chip to Magnetic Stripe Conversion Service results in the Authorization Request/0100 message | removes the Chip to Magnetic Stripe Conversion Service results contained in DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services [OBS]), subfield 1 (On-behalf [OB] Service) and subfield 2 (On-behalf Result 1) from the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message. The Chip to Magnetic Stripe Conversion service removes chip-related data in DE 55 (Integrated Circuit Card [ICC] System-Related Data) and DE 23 (Card Sequence Number) if present when DE 14 (Date Expiration) is greater than or equal to the floor expiration date. |
| the issuer for the account range wants to receive the Chip to Magnetic Stripe Conversion Service results in the Authorization Request/0100 message         | follows business-as-usual processing by providing the Chip to Magnetic Stripe Conversion Service results in the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message.                                                                                                                                                                                                                                                                                                                                                                                                    |

**To participate**

Issuers that want to participate in the U.S. chip-enabled travel card program must register for the Chip to Magnetic Stripe Conversion Service. In addition, issuers must contact their regional Mastercard representative to request this service and identify the account range for which the Chip to Magnetic Stripe Conversion service will support.

**M/Chip Cryptogram Pre-validation service**

The M/Chip Cryptogram Pre-validation service is for issuers that use the M/Chip Select 2.0, M/Chip Lite 2.1, M/Chip 4.0 (Mastercard, EMV CSK, and EMV2000 session key derivation methods), *M/Chip™ Advance*, and EMV CCD-compliant chip cards.

This service:

- Validates the Authorization Request Cryptogram (ARQC) and Transaction Certificate Cryptogram (TC), and generates the Authorization Response Cryptogram (ARPC) on behalf of an issuer.
- Validates critical chip information included in the Card Verification Results (CVR) and Terminal Verification Results (TVR).

After performing this service, Mastercard notifies issuers of the results of the cryptogram validation. Issuers may use these results in the authorization approval process.

The M/Chip Cryptogram Pre-validation service is mandatory.

## Decline option for issuers

Mastercard has enhanced the M/Chip Cryptogram Pre-validation service with a decline option indicator to decline transactions on-behalf of participating issuers when the Application Request Cryptogram or the chip data validation for the transaction is not successful.

For the decline option indicator, Mastercard will forward the transaction to the issuer or decline the transaction on-behalf of the issuer on the basis of the information included in the decision matrix. For each value of the On-behalf Service (OBS) results, the decision matrix includes which action to take (forward or decline the transaction).

Mastercard supports the following four options for issuers registering to the OBS 02 service.

- Option 1: Do not use the decline option for neither contact nor contactless transactions.
- Option 2: Use the decline option for contact and contactless transactions.
- Option 3: Use the decline option for contact transactions and do not use the decline option for contactless transactions.
- Option 4: Use the decline option for contactless transactions and do not use the decline option for contact transactions.

## Combined service option

Issuers can choose an option that combines the Chip to Magnetic Stripe Conversion and M/Chip Cryptogram Pre-validation services.

This option allows two M/Chip processing services to be performed on a single transaction, providing issuers with a bridge to maximize the benefits of chip card processing capabilities while minimizing impacts on their authorization systems. These two services also are available individually.

The Combined Service Option is optional.

## M/Chip Cryptogram Validation in Stand-In Authorization

M/Chip Cryptogram Validation in Stand-In Authorization is available for issuers that use the M/Chip Select 2.0, M/Chip Lite 2.1, and M/Chip 4.0 (Mastercard, EMV CSK, and EMV2000 session key derivation methods), M/Chip Advance, and EMV CCD-compliant chip cards.

**NOTE: All issuers globally that participate in Stand-In Authorization must have M/Chip Cryptogram Validation in Stand-In Authorization performed during Stand-In Authorization.**

This service authorizes M/Chip transactions within Stand-In Authorization when an issuer host is not available. When Stand-In Authorization authorizes M/Chip transactions, Mastercard validates the ARQC, TC, and critical (CVR and TVR) chip information, processes the authorization request based on the issuer's Stand-In parameters, and generates the ARPC.

M/Chip Cryptogram Validation in Stand-In Authorization is mandatory for all chip card issuers.

Issuers should consider participating in PIN Validation services for successful Stand-In Authorization of chip transactions with online PIN.



## M/Chip Advance

Mastercard® M/Chip™ Advance is a dual interface chip card specification for credit, debit, prepaid, and Mastercard® contactless programs.

M/Chip Advance:

- Integrates contactless products: Fully integrates contact and contactless cards and simplifies the adoption and implementation of contactless products.
- Provides enhanced on-card risk management: Increases the volume of approved transactions with more informed card risk management decision making.
- Supports new third-party transit, loyalty, voucher programs, and ticketing (for example, music concerts, cinema, theatre, or sports events): Provides the functionality to extend card payment services into new markets.
- Provides enhanced payment features. M/Chip 2 and M/Chip 4 have a single set of counters to manage the risk of all offline transactions. This means that the risk management functionality of the application is not able to differentiate between CVM and no-CVM transactions or contact and contactless transactions.
- Provides backwards compatibility with existing M/Chip platforms: Facilitates the migration from M/Chip 2 and M/Chip 4 to M/Chip Advance.

The M/Chip Advance card application specification is available to card developers under a specific Mastercard development agreement.

With M/Chip Advance, DE 55 (Integrated Circuit Card [ICC] System-Related Data), subelement 9F10 (Issuer Application Data [IAD]) contains expanded counters. These expanded counters are used in the ARQC when the value of the right-most bit of the CVN is 1. The IAD can be configured to include accumulators, counters, or both, in plain text or encrypted. The value of an accumulator or counter included in the IAD may be either its absolute value (for example, Accumulator Amount or Counter Number) or the value of their balance. If so, the IAD is extended with one or two, 8-byte fields.

M/Chip issuers may optionally choose to support M/Chip Advance but are not required to migrate to M/Chip Advance specifications to continue using these services.

Issuers that participate in the M/Chip Cryptogram Pre-validation service, the M/Chip Cryptogram Validation in Stand-In Authorization service, or the Combined Service Option may optionally support M/Chip Advance.

## MIP Transaction Blocking

MIP Transaction Blocking provides issuers with easy-to-manage, flexible controls that supplement their authorization strategy. MIP Transaction Blocking extends an issuer's authorization capabilities, augmenting their defenses to protect portfolios from fraud attacks, as well as manage specialized payment programs with ease. An issuer's participation in the MIP Transaction Blocking Service is optional.

Issuers may want to apply MIP Transaction Blocking to:

- Prevent the authorization of fraudulent transactions for compromised account ranges
- Prevent the authorization of transactions due to operational emergencies (for example, the issuer encounters problems supporting certain processing codes or POS PAN entry modes)
- Decline authorizations for an account range in combination with the point-of-interaction (POI) country in the event of local issues that necessitate preventing transactions from being authorized until those issues are resolved
- Protect assigned BIN ranges before they go into production or account ranges in production that are not actively used

MIP Transaction Blocking provides issuers the ability to decline authorizations in the Dual Message Authorization System based upon the issuing ICA number or account range, and one or more of the following transaction parameters:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)
- DE 18 (Merchant Type)
- DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)
- DE 32 (Acquiring Institution ID Code)
- DE 61 (Point-of-Service [POS] Data), subfield 10 (Card Activated Terminal Level [CAT])
- DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code)

MIP Transaction Blocking can also provide Full BIN Blocking on authorization requests for an entire BIN range or a segment of a BIN range, defined up to 11 digits, and is available globally on both the Dual Message Authorization and Single Message Systems.

### MIP transaction block setup

The following information is required for each transaction block setup:

- ICA or Routing & Transit Number and account range
- The data elements identifying the type of transaction to be blocked
- Effective date
- The response code that will be returned in the Authorization Request Response/0110 message to the acquirer when the transaction matches the blocking criteria. Issuers may choose a response code from the following list:
  - 03 (Invalid merchant)
  - 04 (Capture card)
  - 05 (Do not honor)
  - 12 (Invalid transaction)
  - 57 (Transaction not permitted to issuer/cardholder)
  - 58 (Transaction not permitted to acquirer/terminal)

**NOTE: If issuers do not specify a decline response code, the default response code is 05 (Do not honor).**

- Registration for the optional MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) and delivery endpoint.

MIP Transaction Blocking ICA Level Block Summary (SI738010-AA)

The MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) is an optional report showing each ICA level block for the issuer and the account ranges associated to that block. The ICA Blocking Summary report will assist issuers in determining the account ranges that are blocked via ICA level MIP Transaction Blocking records.

For more information, refer to [Reports and files](#).

To participate

Issuers may choose to participate in and customize their MIP Transaction Blocking settings by completing the *MIP Transaction Blocking Service Request Form* (Form 810) and submitting the form to Global Customer Service.

For questions about participating in MIP Transaction Blocking, contact Global Customer Service.

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on Authorization Request/0100: MIP Transaction Blocking transactions.

Associated edits

| WHEN...                                                                                                             | THEN the Dual Message Authorization System...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer sends an Authorization Request/0100 message and the transaction matches a MIP transaction block record | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 = The response code identified in the MIP transaction block setup<br><br>and<br><br>may send the issuer an Authorization Advice/0120 message where <ul style="list-style-type: none"><li>DE 39 = The response code sent in the Authorization Request Response/0110</li><li>DE 60 (Advice Reason Code), subfield 1 (Advice Detail Code) will contain 120</li><li>DE 121 (Authorizing Agent ID Code) will contain 000003 (Decline occurred due to an on-behalf service)</li></ul> |
| the transaction does not match any MIP transaction block                                                            | forwards the Authorization Request/0100 message to the issuer or to the Stand-In Authorization if the issuer is unavailable.                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Full BIN Block

The Full BIN Block service blocks authorization requests for an entire BIN range or a segment of a BIN range, defined up to 11 digits.

Through this service, issuers can protect assigned BIN ranges before they go into production. Issuers also can protect account ranges in production that are not actively used. The Full BIN Block service will decline any authorization request in the identified account range with response code 14 (Invalid card number).

The Full BIN Block service is available globally on both the Dual Message Authorization and Single Message Systems. This service is recommended when a BIN has been assigned by Mastercard for use by the issuer, but that is not yet supported by the issuer's authorization system.

Participation in the Full BIN Block service is optional.

### Benefits

Issuers may want to apply authorization blocking for inactive BINs to:

- Block an entire BIN range or a segment of a BIN range.
- Combat fraud attacks on inactive BIN ranges, defined as "live" on Mastercard systems but not in production on the issuer host system.

### To Participate

To sign up for the Full BIN Block service, issuers must complete the *MIP Transaction Blocking Service Request Form* (Form 810).

## Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on Authorization Request/0100—Full BIN Block transactions.

### Associated edits

| WHEN...                                                                                                    | THEN the Dual Message Authorization System...                                                                                                  |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| the acquirer sends an Authorization Request/0100 message and the transaction falls within a Full BIN Block | sends the acquirer an Authorization Request Response/0110 message where:<br><br>DE 39 = 14 (Invalid card number. The transaction is declined.) |

## Name Validation Service

The Name Validation Service validates that the name provided by a registered user matches the name associated to the card at the issuer. Customers can validate cardholder name information by including a name validation request in an Account Status Inquiry (ASI) message.

## Participation requirements

Acquirers should be aware that name validation can be included in an ASI message.

Issuers will receive ASI service messages with a request for cardholder name validation. Issuers must validate the name information provided in the request message and respond with a result indicating the likelihood of a match.

The availability and effectiveness of name validation may vary, based on the issuer's capability to perform the validation.

## Name Validation Process

1. The acquirer creates an ASI Authorization Request/0100 message, when submitting name validation requests, the message will contain:
  - a. Data Element (DE) 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 01 (Sender First name), subfield 02 (Sender Middle Name), and subfield 03 (Sender Last Name), DE 108 subelement 07 (Data Verification Type), subfield 01 (Data Verification Type) = 11 (Sender Data Verification Request)
  - b. DE 108, subelement 01 (Receiver/Recipient Data), subfield 01 (Receiver/Recipient First name), subfield 02 (Receiver/Recipient Middle Name), and subfield 03 (Receiver/Recipient Last Name), DE 108, subelement 07, subfield 01 = 10 (Receiver/Recipient Data Verification Request)
2. The issuer (or issuer processor) compares the name information provided in DE 108 to the cardholder name(s) on file
3. The issuer generates an ASI Authorization Response/0110 message with the name validation response included:
  - a. When sender name validation was requested, DE 108, subelement 08, subfield 02 (Sender Service Response) is populated with one of the following validation result values: A (match), B (partial match), C (no match), or U (unverified)
  - b. When receiver name validation was requested, DE 108, subelement 08 (Data Verification Response), subfield 01 (Receiver/Recipient Service Response) is populated with one of the following validation result values: A (match), B (partial match), C (no match), or U (unverified)

### For more information

For more information on the Name Validation Service, refer to *Name Validation Service Issuer Name Matching Guide*.

## Payment Cancellation Service

Payment Cancellation Service enables issuers to block unwanted card-not-present payment transactions identified in pre-authorization messages and completion records.

The cancellation of a payment will result in Mastercard declining future payment transactions for a specific account, if specified, from a particular merchant when requested by the cardholder.

Mastercard uses the issuer's payment blocking criteria to decline online authorization requests for card-not-present and recurring payments when appropriate. Also, the Global Clearing Management System (GCMS) will use the same issuer payment blocking criteria to reject the clearing records.

For more information, refer to the *Payment Cancellation Program Guide*.

Acquirers receive Authorization Request Response/0110 messages containing DE 48 (Additional Data: Private Use), subelement 84 (Merchant Advice Code) with value 21 (Payment Cancellation [Mastercard use only]) indicating Payment Cancellation.

**NOTE: The Dual Message Authorization System, not the issuer, sends the Authorization Request Response/0110 message containing DE 48, subelement 84, value 21.**

Issuers receive a store-and-forward Authorization Advice/0120 message indicating that the authorization request was blocked by the Dual Message Authorization System.

### Authorization reports

The following reports provide information that support Payment Cancellation:

- The Authorization Parameter Summary Report (SI737010-AA) includes a Payment Cancellation participant parameter in the Global Parameters section.
- The Daily Account File Activity Report (AM700010-AA) includes a section to report the Payment Cancellation account maintenance activity. This section groups Payment Cancellation activity by rejected, accepted, and purged.
- The Delete Preview Report (AM710010-AA) includes a section to report the Payment Cancellation accounts that are due to be deleted in the next 3 weeks. A total field by ICA of Payment Cancellation accounts will also display.

For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#). For a sample of the Daily Account File Activity Report (AM700010-AA) and Delete Preview Report (AM710010-AA), refer to the *Account Management System User Manual*.

### To participate

To request participation in the Payment Cancellation Service, contact Global Customer Service.

## Portfolio sales support

Mastercard offers support for customers when they buy or sell a portfolio.

### Portfolio sales support

| Service                                                                                                                                                                               | Service Available for |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|
|                                                                                                                                                                                       | Full BIN Transfer     | Partial BIN Transfer |
| Transfer of BINs from one customer to another. After Mastercard transfers the BIN, the Dual Message Authorization System processes all transactions within the BIN under the new ICA. | √                     | √                    |
| Transfer of AMS account listings from one customer to another. After Mastercard transfers the listings, the new ICA maintains the listings.                                           | √                     |                      |

## Full BIN transfer

To transfer an entire BIN from one ICA number to another, contact the Global Customer Service team and request support for a portfolio transfer.

You will need to provide the following information:

- Old ICA number
- New ICA number
- Time and date on which the transfer will be effective
- BIN that you are buying or selling

### Automatic AMS transfer

If you are requesting an automatic AMS transfer as part of the BIN transfer, complete section "e" of the Member Conversion Questionnaire. The Global Customer Service team will provide the questionnaire.

For examples of the automatic transfer of AMS accounts and for billing information for that service, refer to the *Account Management System User Manual*.

The following table shows the transfer of BINs and AMS accounts.

### Automatic AMS transfer

| When it Occurs                                                              | Description                                                                                           |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Before the planned transfer date                                            | The customer (new ICA number or old ICA number) requests support for a portfolio transfer.            |
| At the time and date when the transfer is effective                         | The Dual Message Authorization System moves authorization processing from the old ICA to the new ICA. |
| On the transfer effective date, after the authorization processing transfer | Mastercard automatically transfers the ownership of the AMS accounts from the old ICA to the new ICA. |
| On the evening of the transfer effective date                               | The Daily Account File Activity Report for the new ICA shows the account listings under the new ICA.  |

### Partial BIN transfer

To transfer a part of a BIN from one customer to another, notify the Global Customer Service team in writing.

You will need to provide the following information:

- Old ICA number
- New ICA number
- Time and date on which the transfer will be effective
- Which portion of the BIN requires the new routing

#### Transfer process for a partial BIN

### Transfer process for a partial BIN

The following table shows the transfer process for a partial transfer of a BIN.

| When it Occurs                   | Description                                                          |
|----------------------------------|----------------------------------------------------------------------|
| Before the planned transfer date | The customer (new ICA or old ICA) requests support for the transfer. |



| When it Occurs                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At the time and date when the transfer is effective | <ul style="list-style-type: none"> <li>The Dual Message Authorization System routes authorization requests for the transferred part of the BIN to the new endpoint.</li> <li>The Stand-In parameters of the new ICA number and BIN govern processing for the transferred part of the BIN.</li> <li>Mastercard bills all issuing authorization charges to the new ICA number.</li> <li>The new ICA number can add accounts within the BIN to AMS and to the Account File.</li> </ul> |

## Private Label Transaction processing

The Mastercard Private Label Program uses the existing Mastercard Network to provide efficient authorization, clearing, and settlement services for private label card transactions, in addition to value-added products and services that issuers may use with their private label programs.

Mastercard has extensive network reach, acceptance, and standardized formats that provide issuers with a network processing solution for private label card transactions, thereby addressing the infrastructure limitations of a closed-loop processing model.

In addition to providing a superior private label transaction processing solution, the Mastercard Private Label Program provides issuers with the ability to use many of the world-class, value-added services offered by Mastercard. Issuers may optionally benefit from the array of Mastercard risk management solutions, enhanced transaction processing services, and portfolio performance management expertise to facilitate financial improvement of their portfolios.

With the Private Label Program, issuers can use the Activation and Initial Load of Private Label Prepaid Card service, which is provided exclusively for transactions originating from the approved private label card portfolios of Mastercard issuers.

For more information, refer to the *Private Label Program Guide for Acquirers* and *Private Label Program Guide for Issuers*.

### Authorization Processing

Based on the issuer's private label prepaid card program, the private label prepaid cards can be magnetic stripe or chip and may or may not require the entry of a PIN.

Stages for prepaid card activation at the POS terminal:

1. The cardholder purchases a private label prepaid card and requests to have it activated at the POS terminal.
2. The merchant enters a request to activate a private label prepaid card from the POS terminal.

3. The POS terminal sends an Authorization Request/0100 message to the acquirer.
4. The acquirer forwards the Authorization Request/0100 message through the Mastercard Network to the issuer for a private label prepaid card activation request.
5. The issuer validates the request and sends an approval in an Authorization Request Response/0110 message through the Mastercard Network to the acquirer. Mastercard suggests that private label issuers approve card activation transactions using response code 00 (Approved or completed successfully).
6. The acquirer delivers the response to the POS terminal.
7. The issuer activates the private label prepaid card on its system.

Acquirers supporting the private label prepaid card activation request message must be able to send a Reversal Request/0400 message to deactivate the card if they are not able to deliver the Authorization Request Response/0110 message to the merchant's terminal or time-out waiting for a response from the Dual Message Authorization System.

Private label issuers may want to define policies and rules with their merchants for governing exceptions that may occur during the activation process, such as what actions to take if there is a decline response.

### Alternate Processing

Private label prepaid card activation plus initial load transactions are not eligible for alternate (Stand-In or alternate issuer host routing) or X-Code Processing. If the primary issuer is not available to respond to a card activation request, an Authorization Request Response/0110 message is returned to the acquirer with DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

## Activation and Initial Load of Private Label Prepaid Cards

The Dual Message Authorization System supports card activation or deactivation requests between acquirers and private label issuers in the Europe region.

**NOTE: Applies only to the Europe region for private label card programs.**

### Benefits

The benefits of providing private label issuers with the ability to allow their cardholders to activate and to set the amount to load on their private label prepaid cards at the moment of purchase and activation at the POS terminal are as follows:

- Allows merchants to:
  - Activate prepaid cards over the Dual Message Authorization System as cards are being sold
  - Provide refunds to consumers with a private label prepaid card (typically a gift card) loaded with the full amount of the refund, rather than giving cash that could be spent at another merchant location.

- Allows issuers to keep prepaid accounts inactive until an activation request is received from the merchant and to help reduce fraud by enabling the activation of prepaid cards with or without a predefined amount.
- Provides a security measure to ensure that prepaid cards cannot be used before activation.

### **Transaction Type Identifier C09**

The Dual Message Authorization System supports Transaction Type Identifier value C09 (Card Activation) to specify private label prepaid card activation and initial load requests.

### **Authorization Processing**

For private label prepaid cards, the Dual Message Authorization System processes one transaction for both activating and initially loading a card. The activation request uses an Authorization Request/0100 message with a Payment Type processing code and a valid amount in DE 4 (Amount, Transaction) for private label prepaid card activation authorization messages. The activation also can be canceled by a Reversal Request/0400 message.

Based on the issuer's private label prepaid card program, the private label prepaid cards can be magnetic stripe or chip and may or may not require the entry of a PIN.

Stages for prepaid card activation at the POS terminal:

1. The cardholder purchases a private label prepaid card and requests to have it activated at the POS terminal.
2. The merchant enters a request to activate a private label prepaid card from the POS terminal.
3. The POS terminal sends an Authorization Request/0100 message to the acquirer.
4. The acquirer forwards the Authorization Request/0100 message via the Mastercard Network to the issuer for a private label prepaid card activation request.
5. The issuer validates the request and sends an approval in an Authorization Request Response/0110 message via the Mastercard Network to the acquirer. Mastercard suggests private label issuers approve card activation transactions using response code 00 (Approved or completed successfully).
6. The acquirer delivers the response to the POS terminal.
7. The issuer activates the private label prepaid card on its system.

Acquirers supporting the private label prepaid card activation request message must be able to send a Reversal Request/0400 message to deactivate the card if they are not able to deliver the Authorization Request Response/0110 message to the merchant's terminal or time-out waiting for a response from the Dual Message Authorization System.

Private label issuers may want to define policies and rules with their merchants for governing exceptions that may occur during the activation process, such as what actions to take if there is a decline response.

### **Alternate Processing**

Private label prepaid card activation plus initial load transactions are not eligible for alternate (Stand-In or alternate issuer host routing) or X-Code Processing. If the primary issuer is not

available to respond to a card activation request, an Authorization Request Response/0110 message is returned to the acquirer with DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

## Private Label Non-Financial Service

The Private Label Non-Financial Service uses the capabilities with the Mastercard Private Label Program and Dual Message Authorization System to offer a non-financial service.

The Private Label Non-Financial Service provides purchase point-of-sale (POS) information to manufacturers of luxury goods for use in various applications, such as warranty registration or inventory forecasting.

For more information, refer to the *Private Label Program Guide for Acquirers* and *Private Label Program Guide for Issuers*.

### Authorization processing

The Authorization Request/0100 message that is triggered by this event includes all mandatory data elements needed to carry the product information. The Authorization Request/0100 message, capturing the purchase POS information, is initiated for purchased goods of participating merchants regardless of what form of payment was used to purchase the goods.

Product code PVA (Private Label A) is used for identification of POS information from luxury goods transactions. Product code PVA is set up in account ranges for authorization processing only. Any clearing messages received with product code PVA is rejected.

### Alternate processing

The Authorization Request/0100 message containing value PVA in DE 63 (Network Data), subfield 1 (Financial Network Code) is routed to and declined by Stand-In Authorization with a response code of 05 (Do Not Honor) in DE 39 (Response Code). X-Code also responds with value 05 if authorization cannot be obtained from Stand-In Authorization.

### For more information

For more information about the Mastercard Private Label Non-Financial Service, contact the Global Customer Service team.

## QR code payments

The quick response (QR) code payment is a fast, convenient payment option for consumers. These transactions may originate when either a consumer mobile wallet generates a QR code that can be scanned by a merchant to receive payments or a merchant generates QR code that can be scanned by a consumer's mobile device to complete payments.

For more information, refer to the [Mastercard QR: Send Program Guide](#).

## Consumer Presented QR transactions

Consumer Presented QR transactions originate when a consumer mobile wallet generates a quick response (QR) code that can be scanned by a merchant to receive payments.

Mastercard identifies:

- Mastercard Consumer Presented QR transactions when value 03 (PAN auto-entry via barcode reader) in DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)
- Mastercard Consumer Presented QR terminal capabilities through subfield 02 (Additional Terminal Capability Indicator) in DE 48 (Additional Data: Private Use), subelement 21 (Acceptance Data)

These enhancements will help ensure that tokenized Mastercard Consumer Presented QR transactions are properly identified and securely processed on the Mastercard Network.

### To participate

To submit or process tokenized Mastercard Consumer Presented QR transactions, acquirers and issuers must contact their Customer Implementation Specialist or account representatives.

## Merchant Presented QR transactions

Mastercard® Merchant Presented QR™ transactions are consumer-initiated, push payment transactions with a mobile payment device through a low-cost, point-of-interaction device.

Using a mobile payment device, the consumer makes a cashless payment by scanning a QR code at any location that accepts Mastercard Merchant Presented QR transactions.

This program extends payment services to merchants currently excluded from electronic payments and is similar to a cash experience with funds immediately available to the merchant. Merchant Presented QR provides a viable, cost-effective alternative to merchants who are cash focused, but would like an electronic option.

Merchant Presented QR operates in an alternate four-party business model that enables funds to be transferred from a licensed Mastercard customer acting as an Originating Institution (OI), to a licensed Mastercard customer acting as a Receiving Institution (RI). The transaction originates at the OI, consumer bank or third-party wallet providers and routes to the merchant after successfully authenticating and authorizing the debit from the consumer's account. The RI will credit the merchant account upon authorization and notify the merchant that the payment has been successfully received.

The QR code-based program is a payment transaction, the transfer of funds from a Mastercard Account or Maestro Account to a Mastercard Receiving Account using the Mastercard Network payment transaction, defined on the Dual Message Authorization System by using:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) value 28 (Payment Transaction)
- A unique payment transaction type identifier DE 48 (Additional Data: Private Use), Transaction Category Code (TCC)=P (Payment Transaction), subelement 77 (Funding Transaction or Payment Transaction Type Identifier), value of C67 (Merchant Presented QR) to support the payment to the merchant.

**NOTE: The QR code will include the actual merchant MCC. The MCC is a mandatory field on the QR code. The OI must obtain the MCC from the merchant's QR code and include in the ISO message or the API.**

If Merchant Presented QR Transactions are enabled through a third-party wallet provider, the financial institution supporting the wallet provider will secure funds from a consumer account by initiating a funding transaction before processing the payment transaction.

For more information, refer to the [Mastercard QR: Send Program Guide](#).

### To participate

To submit or process Mastercard Merchant Presented QR transactions, OIs and RIs must complete and submit the *Registration for Mastercard Merchant Presented QR: Global* (Form 1274) to their regional account representative.

## Sweden Domestic Authorization Switching Service

The Dual Message Authorization System supports the authorization processing for customers that participate in Sweden Domestic Authorization Switching Service (SASS).

**NOTE: Applies only to the Europe region.**

The Dual Message Authorization System supports the following authorization processing for customers that participate in SASS:

- ATM Additional Data
- Forgotten Card at ATM
- Receipt Free Text
- Refund Transactions

### Forgotten Card at ATM

Forgotten card at ATM service invokes when a cardholder leaves the terminal without taking his or her card or when the cardholder cannot retrieve the card from the terminal for a technical

reason. This service is available only to customers that participate in Swedish Domestic Authorization Switching Service (SASS).

Reversal Request/0400: Forgotten Card

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

Applicable data elements

| Data Element                                                             | Org | Sys | Dst | Values/Comments                                                                          |
|--------------------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------------------|
| DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)    | M   | •   | M   | Contains one of the following values:<br><br>01 = Withdrawal<br><br>30 = Balance Inquiry |
| DE 4 (Amount, Transaction)                                               | M   | •   | M   | Contains value zero                                                                      |
| DE 48 (Additional Data—Private Use), subelement 58 (ATM Additional Data) | O   | X   | C   | Subfield data may be present for Swedish ATMs                                            |

**NOTE: If the issuer does not support DE 48, subelement 58, the Dual Message Authorization System will send the acquirer a Reversal Request Response/0410 message containing DE 39 (Response Code), value 57 (Transaction not permitted to issuer/cardholder).**

Third-Party Processor identification

The Third-Party Processor (TPP) Identification service assists issuers in identifying an acquirer’s third-party processor. This optional service provides issuers additional information in their fraud analysis during an account data compromise event.

Issuers that choose to register for this service will receive additional information to identify the entity involved in acquiring the transaction and presenting the transaction to the acquiring MIP.

How it works

The Dual Message Authorization System requires acquirers to submit DE 32 (Acquiring Institution ID Code) in all Authorization Request/0100 transactions. DE 32 contains the ICA number for the institution acting as the acquiring bank or merchant bank for the transaction. If an institution is acting as a processor for the acquiring bank, DE 32 should represent the acquiring bank and DE 33 (Forwarding Institution ID Code) should contain the ICA number for the forwarding or processing institution.

In addition to the values in DE 32 and DE 33 provided by the acquirer or processor in Authorization Request/0100 and Authorization Advice/0120 messages, the Dual Message Authorization System includes the additional entity identifier submitting the message in DE 48 (Additional Data: Private Use), subelement 16 (Processor Pseudo ICA). DE 48, subelement 16 contains a unique identifier from the Mastercard Registration Program (MRP) database for the TPP processing the Authorization Request/0100 or Authorization Advice/0120 message.

This data is sent in the Authorization Request/0100 and Authorization Advice/0120 messages only for participating account ranges.

### **Authorization reports**

The Authorization Parameter Summary Report (SI737010-AA) includes a TPP participant parameter in the Global Parameters section to indicate issuers that have registered to receive unique identifiers for TPPs. For a sample of the Authorization Parameter Summary Report (SI737010-AA), refer to [Reports and files](#).

### **To participate**

Issuers must register for this optional service. To request participation, contact Global Customer Service.



## Appendix C Data reference tables

*This section provides details on character and extended character sets.*

---

|                                                             |      |
|-------------------------------------------------------------|------|
| Extended character sets.....                                | 1815 |
| Swedish Domestic Authorization Switching character set..... | 1822 |

## Character sets

Character sets define how letters, numbers, symbols, and special characters are displayed in computer text.

### Which character sets does Mastercard support?

Mastercard supports ASCII (ISO 8859-1) and EBCDIC (Code Page 1047) code pages for both standard and extended character sets.

Mastercard supports standard EBCDIC formatted messages and bulk files. In addition to standard EBCDIC, Mastercard supports the extended EBCDIC format as well as the standard and extended ASCII format for both authorization messages and selected bulk files. Customers using the MIP to transmit bulk files still need to adhere to the existing Mastercard File Transfer Protocol format, that is, EBCDIC (for Header and Trailer information).

**NOTE: Extended characters: Within the full character set, the alphabet is represented as used in the English language (with no character enhancements) and characters that may be found, for example, in the Spanish alphabet where the tilde is used over the Ñ.**

Mastercard uses extended EBCDIC as the default character set type for sending messages unless otherwise specified by the customer.

Customers also have the option to send and receive either standard or extended character sets for both ASCII and EBCDIC formats in specific data elements that contain text. Following are examples of some data elements that may contain standard or extended text characters:

- DE 42 (Acceptor ID)
- DE 43 (Acceptor Name and Location)
- DE 120 (Record Data)
- DE 123 (Receipt Free Text)
- DE 124 (Member-defined Data)

The following bulk files will be supported in ASCII:

- R311: AMS File Updates
- R361: AMS File Updates (TEST)

Customers that want to receive messages in standard EBCDIC or extended or standard ASCII must inform Mastercard.

## Character sets

The following tables contain valid ASCII to EBCDIC character sets.

**NOTE: For ISO 8583, the <space> character is treated as a special character (attribute "s"). It may, however, also be used in fields designated as alphanumeric (attribute "an") but only in the form of trailing spaces used to pad out significant data to fill a fixed-length field.**

**ASCII to EBCDIC character set: ASCII hex order**

| Decimal value | ASCII hex | EBCDIC hex | ASCII symbol | Meaning                        | EBCDIC symbol | Meaning                     |
|---------------|-----------|------------|--------------|--------------------------------|---------------|-----------------------------|
| 032           | 20        | 40         | < space >    | Space                          | DS            | digit select                |
| 033           | 21        | 5A         | !            | exclamation mark               | SOS           | start of significance       |
| 034           | 22        | 7F         | "            | straight double quotation mark | FS            | field separator             |
| 035           | 23        | 7B         | #            | number sign                    | WUS           | word underscore             |
| 036           | 24        | 5B         | \$           | dollar sign                    | BYP/INP       | bypass/inhibit presentation |
| 037           | 25        | 6C         | %            | percent sign                   | LF            | line feed                   |
| 038           | 26        | 50         | &            | ampersand                      | ETB           | end of transmission block   |
| 039           | 27        | 7D         | '            | apostrophe                     | ESC           | escape                      |
| 040           | 28        | 4D         | (            | left parenthesis               | SA            | set attribute               |
| 041           | 29        | 5D         | )            | right parenthesis              | SFE           |                             |
| 042           | 2A        | 5C         | *            | asterisk                       | SM/SW         | set mode switch             |
| 043           | 2B        | 4E         | +            | addition sign                  | CSP           | control sequence prefix     |
| 044           | 2C        | 6B         | ,            | comma                          | MFA           | modify field attribute      |
| 045           | 2D        | 60         | -            | subtraction sign               | ENQ           | enquiry                     |
| 046           | 2E        | 4B         | .            | period                         | ACK           | acknowledge                 |
| 047           | 2F        | 61         | /            | right slash                    | BEL           | bell                        |
| 048           | 30        | F0         | 0            |                                |               |                             |
| 049           | 31        | F1         | 1            |                                |               |                             |

| Decimal value | ASCII hex | EBCDIC hex | ASCII symbol | Meaning       | EBCDIC symbol | Meaning               |
|---------------|-----------|------------|--------------|---------------|---------------|-----------------------|
| 050           | 32        | F2         | 2            |               | SYN           | synchronous idle      |
| 051           | 33        | F3         | 3            |               | IR            | index return          |
| 052           | 34        | F4         | 4            |               | PP            | presentation position |
| 053           | 35        | F5         | 5            |               | TRN           |                       |
| 054           | 36        | F6         | 6            |               | NBS           | numeric backspace     |
| 055           | 37        | F7         | 7            |               | EOT           | end of transmission   |
| 056           | 38        | F8         | 8            |               | SBS           | subscript             |
| 057           | 39        | F9         | 9            |               | IT            | indent tab            |
| 058           | 3A        | 7A         | :            | colon         | RFF           | required form feed    |
| 059           | 3B        | 5E         | ;            | semicolon     | CU3           | customer use 3        |
| 060           | 3C        | 4C         | <            | less than     | DC4           | device control 4      |
| 061           | 3D        | 7E         | =            | equal         | NAK           | negative acknowledge  |
| 062           | 3E        | 6E         | >            | greater than  |               |                       |
| 063           | 3F        | 6F         | ?            | question mark | SUB           | substitute            |
| 064           | 40        | 7C         | @            | at symbol     | SP            | space                 |
| 065           | 41        | C1         | A            |               | RSP           |                       |
| 066           | 42        | C2         | B            |               | â             |                       |
| 067           | 43        | C3         | C            |               | ä             |                       |
| 068           | 44        | C4         | D            |               | à             |                       |
| 069           | 45        | C5         | E            |               | á             |                       |
| 070           | 46        | C6         | F            |               | ã             |                       |
| 071           | 47        | C7         | G            |               | å             |                       |
| 072           | 48        | C8         | H            |               | ç             |                       |
| 073           | 49        | C9         | I            |               | ñ             |                       |
| 074           | 4A        | D1         | J            |               | □             | cent                  |

| Decimal value | ASCII hex | EBCDIC hex | ASCII symbol | Meaning         | EBCDIC symbol | Meaning           |
|---------------|-----------|------------|--------------|-----------------|---------------|-------------------|
| 075           | 4B        | D2         | K            |                 | .             | period            |
| 076           | 4C        | D3         | L            |                 | <             | less than         |
| 077           | 4D        | D4         | M            |                 | (             | left parenthesis  |
| 078           | 4E        | D5         | N            |                 | +             | addition sign     |
| 079           | 4F        | D6         | O            |                 |               | logical or        |
| 080           | 50        | D7         | P            |                 | &             | ampersand         |
| 081           | 51        | D8         | Q            |                 | é             |                   |
| 082           | 52        | D9         | R            |                 | ê             |                   |
| 083           | 53        | E2         | S            |                 | ë             |                   |
| 084           | 54        | E3         | T            |                 | è             |                   |
| 085           | 55        | E4         | U            |                 | í             |                   |
| 086           | 56        | E5         | V            |                 | î             |                   |
| 087           | 57        | E6         | W            |                 | ï             |                   |
| 088           | 58        | E7         | X            |                 | ì             |                   |
| 089           | 59        | E8         | Y            |                 | þ             |                   |
| 090           | 5A        | E9         | Z            |                 | !             | exclamation point |
| 091           | 5B        | AD         | [            | left bracket    | \$            | dollar sign       |
| 092           | 5C        | E0         | \            | left slash      | *             | asterisk          |
| 093           | 5D        | BD         | ]            | right bracket   | )             | right parenthesis |
| 094           | 5E        | 5F         | ^            | hat, circumflex | ;             | semicolon         |
| 095           | 5F        | 6D         | _            | underscore      | ^             | hat, circumflex   |
| 096           | 60        | 79         | `            | grave           | -             | subtraction sign  |
| 097           | 61        | 81         | a            |                 | /             | right slash       |
| 098           | 62        | 82         | b            |                 | Â             |                   |
| 099           | 63        | 83         | c            |                 | Ä             |                   |
| 100           | 64        | 84         | d            |                 | À             |                   |

| Decimal value | ASCII hex | EBCDIC hex | ASCII symbol | Meaning        | EBCDIC symbol | Meaning            |
|---------------|-----------|------------|--------------|----------------|---------------|--------------------|
| 101           | 65        | 85         | e            |                | Á             |                    |
| 102           | 66        | 86         | f            |                | Ã             |                    |
| 103           | 67        | 87         | g            |                | Å             |                    |
| 104           | 68        | 88         | h            |                | Ç             |                    |
| 105           | 69        | 89         | i            |                | Ñ             |                    |
| 106           | 6A        | 91         | j            |                | ¡             | split vertical bar |
| 107           | 6B        | 92         | k            |                | ,             | comma              |
| 108           | 6C        | 93         | l            |                | %             | percent sign       |
| 109           | 6D        | 94         | m            |                | –             | underscore         |
| 110           | 6E        | 95         | n            |                | >             | greater than       |
| 111           | 6F        | 96         | o            |                | ?             | question mark      |
| 112           | 70        | 97         | p            |                | Ø             |                    |
| 113           | 71        | 98         | q            |                | É             |                    |
| 114           | 72        | 99         | r            |                | Ê             |                    |
| 115           | 73        | A2         | s            |                | Ë             |                    |
| 116           | 74        | A3         | t            |                | È             |                    |
| 117           | 75        | A4         | u            |                | Í             |                    |
| 118           | 76        | A5         | v            |                | Î             |                    |
| 119           | 77        | A6         | w            |                | Ï             |                    |
| 120           | 78        | A7         | x            |                | Ì             |                    |
| 121           | 79        | A8         | y            |                | `             | grave              |
| 122           | 7A        | A9         | z            |                | :             | colon              |
| 123           | 7B        | C0         | {            | left brace     | #             | number sign        |
| 124           | 7C        | 6A         |              | logical or     | @             | at symbol          |
| 125           | 7D        | D0         | }            | right brace    | '             | apostrophe         |
| 126           | 7E        | A1         | ~            | similar, tilde | =             | equal              |

**NOTE:** For ISO 8583, the < space > character is treated as a special character (attribute "s"). It may, however, also be used in fields designated as alphanumeric (attribute "an") but only in the form of trailing spaces used to pad out significant data to fill a fixed-length field.

## ASCII to EBCDIC character set: EBCDIC hex order

| Decimal value | EBCDIC hex | ASCII hex | EBCDIC symbol | Meaning                         | ASCII symbol | Meaning              |
|---------------|------------|-----------|---------------|---------------------------------|--------------|----------------------|
| 032           | 40         | 20        | DS            | digit select                    | < space >    | Space                |
| 046           | 4B         | 2E        | ACK           | acknowledge                     | .            | period               |
| 060           | 4C         | 3C        | DC4           | device control<br>4             | <            | less than            |
| 040           | 4D         | 28        | SA            | set attribute                   | (            | left<br>parenthesis  |
| 043           | 4E         | 2B        | CSP           | control<br>sequence<br>prefix   | +            | addition sign        |
| 038           | 50         | 26        | ETB           | end of<br>transmission<br>block | &            | ampersand            |
| 033           | 5A         | 21        | SOS           | start of<br>significance        | !            | exclamation<br>mark  |
| 036           | 5B         | 24        | BYP/INP       | bypass/inhibit<br>presentation  | \$           | dollar sign          |
| 042           | 5C         | 2A        | SM/SW         | set mode<br>switch              | *            | asterisk             |
| 041           | 5D         | 29        | SFE           |                                 | )            | right<br>parenthesis |
| 059           | 5E         | 3B        | CU3           | customer use<br>3               | ;            | semicolon            |
| 094           | 5F         | 5E        | ;             | semicolon                       | ^            | hat,<br>circumflex   |
| 045           | 60         | 2D        | ENQ           | enquiry                         | -            | subtraction<br>sign  |
| 047           | 61         | 2F        | BEL           | bell                            | /            | right slash          |
| 124           | 6A         | 7C        | @             | at symbol                       |              | logical or           |
| 044           | 6B         | 2C        | MFA           | modify field<br>attribute       | ,            | comma                |
| 037           | 6C         | 25        | LF            | line feed                       | %            | percent sign         |
| 095           | 6D         | 5F        | ^             | hat,<br>circumflex              | _            | underscore           |

| Decimal value | EBCDIC hex | ASCII hex | EBCDIC symbol | Meaning              | ASCII symbol | Meaning                        |
|---------------|------------|-----------|---------------|----------------------|--------------|--------------------------------|
| 062           | 6E         | 3E        |               |                      | >            | greater than                   |
| 063           | 6F         | 3F        | SUB           | substitute           | ?            | question mark                  |
| 096           | 79         | 60        | -             | subtraction sign     | `            | grave                          |
| 058           | 7A         | 3A        | RFF           | required form feed   | :            | colon                          |
| 035           | 7B         | 23        | WUS           | word underscore      | #            | number sign                    |
| 064           | 7C         | 40        | SP            | space                | @            | at symbol                      |
| 039           | 7D         | 27        | ESC           | escape               | '            | apostrophe                     |
| 061           | 7E         | 3D        | NAK           | negative acknowledge | =            | equal                          |
| 034           | 7F         | 22        | FS            | field separator      | "            | straight double quotation mark |
| 097           | 81         | 61        | /             | right slash          | a            |                                |
| 098           | 82         | 62        | Â             |                      | b            |                                |
| 099           | 83         | 63        | Ä             |                      | c            |                                |
| 100           | 84         | 64        | À             |                      | d            |                                |
| 101           | 85         | 65        | Á             |                      | e            |                                |
| 102           | 86         | 66        | Ã             |                      | f            |                                |
| 103           | 87         | 67        | Å             |                      | g            |                                |
| 104           | 88         | 68        | Ç             |                      | h            |                                |
| 105           | 89         | 69        | Ñ             |                      | i            |                                |
| 106           | 91         | 6A        | ¡             | split vertical bar   | j            |                                |
| 107           | 92         | 6B        | ,             | comma                | k            |                                |
| 108           | 93         | 6C        | %             | percent sign         | l            |                                |
| 109           | 94         | 6D        | _             | underscore           | m            |                                |
| 110           | 95         | 6E        | >             | greater than         | n            |                                |
| 111           | 96         | 6F        | ?             | question mark        | o            |                                |
| 112           | 97         | 70        | Ø             |                      | p            |                                |



| Decimal value | EBCDIC hex | ASCII hex | EBCDIC symbol | Meaning           | ASCII symbol | Meaning        |
|---------------|------------|-----------|---------------|-------------------|--------------|----------------|
| 113           | 98         | 71        | É             |                   | q            |                |
| 114           | 99         | 72        | Ê             |                   | r            |                |
| 126           | A1         | 7E        | =             | equal             | ~            | similar, tilde |
| 115           | A2         | 73        | Ë             |                   | s            |                |
| 116           | A3         | 74        | È             |                   | t            |                |
| 117           | A4         | 75        | Í             |                   | u            |                |
| 118           | A5         | 76        | Î             |                   | v            |                |
| 119           | A6         | 77        | Ï             |                   | w            |                |
| 120           | A7         | 78        | Ì             |                   | x            |                |
| 121           | A8         | 79        | `             | grave             | y            |                |
| 122           | A9         | 7A        | :             | colon             | z            |                |
| 091           | AD         | 5B        | \$            | dollar sign       | [            | left bracket   |
| 093           | BD         | 5D        | )             | right parenthesis | ]            | right bracket  |
| 123           | C0         | 7B        | #             | number sign       | {            | left brace     |
| 065           | C1         | 41        | RSP           |                   | A            |                |
| 066           | C2         | 42        | â             |                   | B            |                |
| 067           | C3         | 43        | ä             |                   | C            |                |
| 068           | C4         | 44        | à             |                   | D            |                |
| 069           | C5         | 45        | á             |                   | E            |                |
| 070           | C6         | 46        | ã             |                   | F            |                |
| 071           | C7         | 47        | å             |                   | G            |                |
| 072           | C8         | 48        | ç             |                   | H            |                |
| 073           | C9         | 49        | ñ             |                   | I            |                |
| 125           | D0         | 7D        | '             | apostrophe        | }            | right brace    |
| 074           | D1         | 4A        | ¢             | cent              | J            |                |
| 075           | D2         | 4B        | .             | period            | K            |                |
| 076           | D3         | 4C        | <             | less than         | L            |                |
| 077           | D4         | 4D        | (             | left parenthesis  | M            |                |
| 078           | D5         | 4E        | +             | addition sign     | N            |                |

| Decimal value | EBCDIC hex | ASCII hex | EBCDIC symbol | Meaning               | ASCII symbol | Meaning    |
|---------------|------------|-----------|---------------|-----------------------|--------------|------------|
| 079           | D6         | 4F        |               | logical or            | O            |            |
| 080           | D7         | 50        | &             | ampersand             | P            |            |
| 081           | D8         | 51        | é             |                       | Q            |            |
| 082           | D9         | 52        | ê             |                       | R            |            |
| 092           | E0         | 5C        | *             | asterisk              | \            | left slash |
| 083           | E2         | 53        | ë             |                       | S            |            |
| 084           | E3         | 54        | è             |                       | T            |            |
| 085           | E4         | 55        | í             |                       | U            |            |
| 086           | E5         | 56        | î             |                       | V            |            |
| 087           | E6         | 57        | ï             |                       | W            |            |
| 088           | E7         | 58        | ì             |                       | X            |            |
| 089           | E8         | 59        | β             |                       | Y            |            |
| 090           | E9         | 5A        | !             | exclamation point     | Z            |            |
| 048           | F0         | 30        |               |                       | 0            |            |
| 049           | F1         | 31        |               |                       | 1            |            |
| 050           | F2         | 32        | SYN           | synchronous idle      | 2            |            |
| 051           | F3         | 33        | IR            | index return          | 3            |            |
| 052           | F4         | 34        | PP            | presentation position | 4            |            |
| 053           | F5         | 35        | TRN           |                       | 5            |            |
| 054           | F6         | 36        | NBS           | numeric backspace     | 6            |            |
| 055           | F7         | 37        | EOT           | end of transmission   | 7            |            |
| 056           | F8         | 38        | SBS           | subscript             | 8            |            |
| 057           | F9         | 39        | IT            | indent tab            | 9            |            |

## Extended character sets

These tables contain valid ASCII to EBCDIC extended character sets.

### Extended Character Set: ASCII Hex order

| Decimal value | ASCII hex | EBCDIC hex | ASCII character representation | Meaning               | Mapped character | Mapping hex |
|---------------|-----------|------------|--------------------------------|-----------------------|------------------|-------------|
| 192           | C0        | 64         | À                              | capital A, grave      | A                | 41          |
| 193           | C1        | 65         | Á                              | capital A, acute      | A                | 41          |
| 194           | C2        | 62         | Â                              | capital A, circumflex | A                | 41          |
| 195           | C3        | 66         | Ã                              | capital A, tilde      | A                | 41          |
| 196           | C4        | 63         | Ä                              | capital A, umlaut     | A                | 41          |
| 197           | C5        | 67         | Å                              | capital A, ring       | A                | 41          |
| 198           | C6        | 9E         | Æ                              | capital AE, diphthong | E                | 45          |
| 199           | C7        | 68         | Ç                              | capital C, cedilla    | C                | 43          |
| 200           | C8        | 74         | È                              | capital E, grave      | E                | 45          |
| 201           | C9        | 71         | É                              | capital E, acute      | E                | 45          |
| 202           | CA        | 72         | Ê                              | capital E, circumflex | E                | 45          |
| 203           | CB        | 73         | Ë                              | capital E, umlaut     | E                | 45          |
| 204           | CC        | 78         | Ì                              | capital I, grave      | I                | 49          |
| 205           | CD        | 75         | Í                              | capital I, acute      | I                | 49          |
| 206           | CE        | 76         | Î                              | capital I, circumflex | I                | 49          |

| Decimal value | ASCII hex | EBCDIC hex | ASCII character representation | Meaning                  | Mapped character | Mapping hex |
|---------------|-----------|------------|--------------------------------|--------------------------|------------------|-------------|
| 207           | CF        | 77         | İ                              | capital I, umlaut        | I                | 49          |
| 208           | D0        | AC         | Ð                              | capital Eth, Icelandic   | D                | 44          |
| 209           | D1        | 69         | Ñ                              | capital N, tilde         | N                | 4E          |
| 210           | D2        | ED         | Ò                              | capital O, tilde         | O                | 4F          |
| 211           | D3        | EE         | Ó                              | capital O, acute         | O                | 4F          |
| 212           | D4        | EB         | Ô                              | capital O, circumflex    | O                | 4F          |
| 213           | D5        | EF         | Õ                              | capital O, tilde         | O                | 4F          |
| 214           | D6        | EC         | Ö                              | capital O, umlaut        | O                | 4F          |
| 215           | D7        | BF         | ×                              | multiply sign            | ×                | 78          |
| 216           | D8        | 80         | Ø                              | capital O, slash         | O                | 4F          |
| 217           | D9        | FD         | Ù                              | capital U, grave         | U                | 55          |
| 218           | DA        | FE         | Ú                              | capital U, acute         | U                | 55          |
| 219           | DB        | FB         | Û                              | capital U, circumflex    | U                | 55          |
| 220           | DC        | FC         | Ü                              | capital U, umlaut        | U                | 55          |
| 221           | DD        | BA         | Ý                              | capital Y, acute         | Y                | 59          |
| 222           | DE        | AE         | Þ                              | capital THORN, Icelandic | P                | 50          |
| 223           | DF        | 59         | ß                              | small sap s, German      | S                | 53          |
| 224           | E0        | 44         | à                              | small a, grave           | a                | 61          |
| 225           | E1        | 45         | á                              | small a, acute           | a                | 61          |

| Decimal value | ASCII hex | EBCDIC hex | ASCII character representation | Meaning              | Mapped character | Mapping hex |
|---------------|-----------|------------|--------------------------------|----------------------|------------------|-------------|
| 226           | E2        | 42         | â                              | small a, circumflex  | a                | 61          |
| 227           | E3        | 46         | ã                              | small a, tilde       | a                | 61          |
| 228           | E4        | 43         | ä                              | small a, umlaut      | a                | 61          |
| 229           | E5        | 47         | å                              | small a, ring        | a                | 61          |
| 230           | E6        | 9C         | æ                              | small ae diphthong   | e                | 65          |
| 231           | E7        | 48         | ç                              | small c, cedilla     | c                | 63          |
| 232           | E8        | 54         | è                              | small e, grave       | e                | 65          |
| 233           | E9        | 51         | é                              | small e, acute       | e                | 65          |
| 234           | EA        | 52         | ê                              | small e, circumflex  | e                | 65          |
| 235           | EB        | 53         | ë                              | small e, umlaut      | e                | 65          |
| 236           | EC        | 58         | ì                              | small i, grave       | i                | 69          |
| 237           | ED        | 55         | í                              | small i, acute       | i                | 69          |
| 238           | EE        | 56         | î                              | small i, circumflex  | i                | 69          |
| 239           | EF        | 57         | ï                              | small i, umlaut      | i                | 69          |
| 240           | F0        | 8C         | ð                              | small eth, Icelandic | d                | 64          |
| 241           | F1        | 49         | ñ                              | small n, tilde       | n                | 6E          |
| 242           | F2        | CD         | ò                              | small o, grave       | o                | 6F          |
| 243           | F3        | CE         | ó                              | small o, acute       | o                | 6F          |
| 244           | F4        | CB         | ô                              | small o, circumflex  | o                | 6F          |
| 245           | F5        | CF         | õ                              | small o, tilde       | o                | 6F          |
| 246           | F6        | CC         | ö                              | small o, umlaut      | o                | 6F          |
| 247           | F7        | E1         | ÷                              | division sign        | /                | 2F          |
| 248           | F8        | 70         | ø                              | small o, slash       | o                | 6F          |

| Decimal value | ASCII hex | EBCDIC hex | ASCII character representation | Meaning                | Mapped character | Mapping hex |
|---------------|-----------|------------|--------------------------------|------------------------|------------------|-------------|
| 249           | F9        | DD         | ù                              | small u, grave         | u                | 75          |
| 250           | FA        | DE         | ú                              | small u, acute         | u                | 75          |
| 251           | FB        | DB         | û                              | small u, circumflex    | u                | 75          |
| 252           | FC        | DC         | ü                              | small u, umlaut        | u                | 75          |
| 253           | FD        | 8D         | ý                              | small y, acute         | y                | 79          |
| 254           | FE        | 8E         | þ                              | small thorn, Icelandic | p                | 70          |
| 255           | FF        | DF         | ÿ                              | small y, umlaut        | y                | 79          |

#### Extended Character Set: EBCDIC Hex order

| Decimal value | EBCDIC hex | ASCII hex | ASCII character representation | Meaning             | Mapped character | Mapping hex |
|---------------|------------|-----------|--------------------------------|---------------------|------------------|-------------|
| 226           | 42         | E2        | â                              | small a, circumflex | a                | 61          |
| 228           | 43         | E4        | ä                              | small a, umlaut     | a                | 61          |
| 224           | 44         | E0        | à                              | small a, grave      | a                | 61          |
| 225           | 45         | E1        | á                              | small a, acute      | a                | 61          |
| 227           | 46         | E3        | ã                              | small a, tilde      | a                | 61          |
| 229           | 47         | E5        | å                              | small a, ring       | a                | 61          |
| 231           | 48         | E7        | ç                              | small c, cedilla    | c                | 63          |
| 241           | 49         | F1        | ñ                              | small n, tilde      | n                | 6E          |
| 233           | 51         | E9        | é                              | small e, acute      | e                | 65          |
| 234           | 52         | EA        | ê                              | small e, circumflex | e                | 65          |
| 235           | 53         | EB        | ë                              | small e, umlaut     | e                | 65          |

| Decimal value | EBCDIC hex | ASCII hex | ASCII character representation | Meaning               | Mapped character | Mapping hex |
|---------------|------------|-----------|--------------------------------|-----------------------|------------------|-------------|
| 232           | 54         | E8        | è                              | small e, grave        | e                | 65          |
| 237           | 55         | ED        | í                              | small i, acute        | i                | 69          |
| 238           | 56         | EE        | î                              | small i, circumflex   | i                | 69          |
| 239           | 57         | EF        | ï                              | small i, umlaut       | i                | 69          |
| 236           | 58         | EC        | ì                              | small i, grave        | i                | 69          |
| 223           | 59         | DF        | ß                              | small sap s, German   | S                | 53          |
| 194           | 62         | C2        | Â                              | capital A, circumflex | A                | 41          |
| 196           | 63         | C4        | Ä                              | capital A, umlaut     | A                | 41          |
| 192           | 64         | C0        | À                              | capital A, grave      | A                | 41          |
| 193           | 65         | C1        | Á                              | capital A, acute      | A                | 41          |
| 195           | 66         | C3        | Ã                              | capital A, tilde      | A                | 41          |
| 197           | 67         | C5        | Å                              | capital A, ring       | A                | 41          |
| 199           | 68         | C7        | Ç                              | capital C, cedilla    | C                | 43          |
| 209           | 69         | D1        | Ñ                              | capital N, tilde      | N                | 4E          |
| 248           | 70         | F8        | ø                              | small o, slash        | o                | 6F          |
| 201           | 71         | C9        | É                              | capital E, acute      | E                | 45          |
| 202           | 72         | CA        | Ê                              | capital E, circumflex | E                | 45          |
| 203           | 73         | CB        | Ë                              | capital E, umlaut     | E                | 45          |
| 200           | 74         | C8        | È                              | capital E, grave      | E                | 45          |
| 205           | 75         | CE        | Í                              | capital I, acute      | I                | 49          |

| Decimal value | EBCDIC hex | ASCII hex | ASCII character representation | Meaning                  | Mapped character | Mapping hex |
|---------------|------------|-----------|--------------------------------|--------------------------|------------------|-------------|
| 206           | 76         | CD        | î                              | capital I, circumflex    | I                | 49          |
| 207           | 77         | CF        | ï                              | capital I, umlaut        | I                | 49          |
| 204           | 78         | CC        | ì                              | capital I, grave         | I                | 49          |
| 216           | 80         | D8        | Ø                              | capital O, slash         | O                | 4F          |
| 240           | 8C         | F0        | ð                              | small eth, Icelandic     | d                | 64          |
| 253           | 8D         | FD        | ý                              | small y, acute           | y                | 79          |
| 254           | 8E         | FE        | þ                              | small thorn, Icelandic   | p                | 70          |
| 230           | 9C         | E6        | æ                              | small ae diphthong       | e                | 65          |
| 198           | 9E         | C6        | Æ                              | capital AE, diphthong    | E                | 45          |
| 208           | AC         | D0        | Ð                              | capital Eth, Icelandic   | D                | 44          |
| 222           | AE         | DE        | Þ                              | capital THORN, Icelandic | P                | 50          |
| 221           | BA         | DD        | Ý                              | capital Y, acute         | Y                | 59          |
| 215           | BF         | D7        | ×                              | multiply sign            | ×                | 78          |
| 244           | CB         | F4        | ô                              | small o, circumflex      | o                | 6F          |
| 246           | CC         | F6        | ö                              | small o, umlaut          | o                | 6F          |
| 242           | CD         | F2        | ò                              | small o, grave           | o                | 6F          |
| 243           | CE         | F3        | ó                              | small o, acute           | o                | 6F          |
| 245           | CF         | F5        | õ                              | small o, tilde           | o                | 6F          |
| 251           | DB         | FB        | û                              | small u, circumflex      | u                | 75          |



| Decimal value | EBCDIC hex | ASCII hex | ASCII character representation | Meaning               | Mapped character | Mapping hex |
|---------------|------------|-----------|--------------------------------|-----------------------|------------------|-------------|
| 252           | DC         | FC        | ü                              | small u, umlaut       | u                | 75          |
| 249           | DD         | F9        | ù                              | small u, grave        | u                | 75          |
| 250           | DE         | FA        | ú                              | small u, acute        | u                | 75          |
| 255           | DF         | FF        | ÿ                              | small y, umlaut       | y                | 79          |
| 247           | E1         | F7        | ÷                              | division sign         | /                | 2F          |
| 212           | EB         | D4        | Ô                              | capital O, circumflex | O                | 4F          |
| 214           | EC         | D6        | Ö                              | capital O, umlaut     | O                | 4F          |
| 210           | ED         | D2        | Ò                              | capital O, tilde      | O                | 4F          |
| 211           | EE         | D3        | Ó                              | capital O, acute      | O                | 4F          |
| 213           | EF         | D5        | Õ                              | capital O, tilde      | O                | 4F          |
| 219           | FB         | DB        | Û                              | capital U, circumflex | U                | 55          |
| 220           | FC         | DC        | Ü                              | capital U, umlaut     | U                | 55          |
| 217           | FD         | D9        | Ù                              | capital U, grave      | U                | 55          |
| 218           | FE         | DA        | Ú                              | capital U, acute      | U                | 55          |

## Swedish Domestic Authorization Switching character set

Customers using the Swedish Domestic Authorization Switching Service (SASS) also may support Swedish national characters in EBCDIC as described in the following table.

Swedish Domestic Authorization Switching character set

| EBCDIC IBM-278 | EBCDIC<br>(Hexadecimal) | ASCII<br>(Hexadecimal) | Graphic | Attribute |
|----------------|-------------------------|------------------------|---------|-----------|
| Å              | 5B                      | 24                     | \$      | s         |
| Ä              | 7B                      | 23                     | #       | s         |
| Ö              | 7C                      | 40                     | @       | s         |
| å              | D0                      | 7d                     | }       | s         |
| ä              | C0                      | 7b                     | {       | s         |

# Appendix D System access information

*This section provides details on accessing the Mastercard Dual Message Authorization System. For more information, refer to the [Secured Data Communications](#).*

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|                       |      |
|-----------------------|------|
| Acquirer methods..... | 1824 |
| Issuer methods.....   | 1825 |

## Accessing the Dual Message Authorization System

To process authorization activity, acquirers and issuers must access the Mastercard Network. This network is the path by which they send and receive authorization messages and related data.

Selecting a particular authorization processing access method depends on several factors, including:

- Physical installations present at the point-of-interaction and at the customer site
- Volume of incoming and outgoing authorizations
- Customer location

For help in selecting an appropriate access method, contact a Customer Implementation Services specialist.

## Acquirer methods

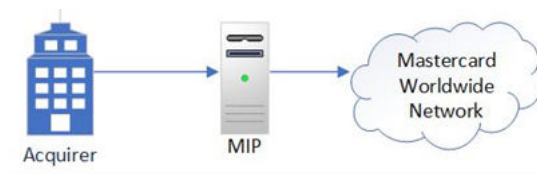
Online host-to-MIP access is available to acquirers. There are several configurations for connectivity to a MIP or MIPs for online access.

### Online host-to-MIP access

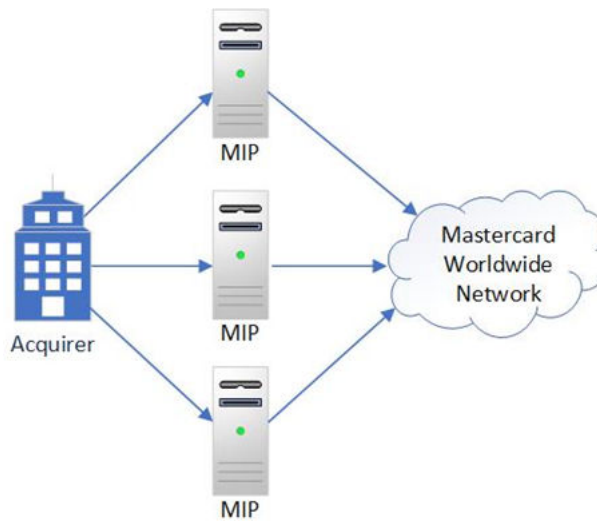
These diagrams show configurations for connectivity to a MIP or MIPS for online access.

To have direct connectivity to a MIP, the acquirer must have a host computer that can support this type of connectivity.

#### Acquirer online: direct host to MIP connectivity



### Acquirer online: direct host to multiple MIP connectivity



Mastercard offers other interfaces to acquirers that do not have online connectivity. Mastercard converts messages sent through these other interfaces into online messages.

## Issuer methods

Issuers may choose the online host-to-MIP access to the Mastercard Network.

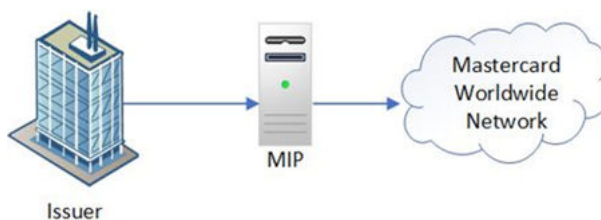
### Online host-to-MIP access

Online host-to-MIP issuers receive authorization requests from the Dual Message Authorization System via a MIP connected to the issuer's host.

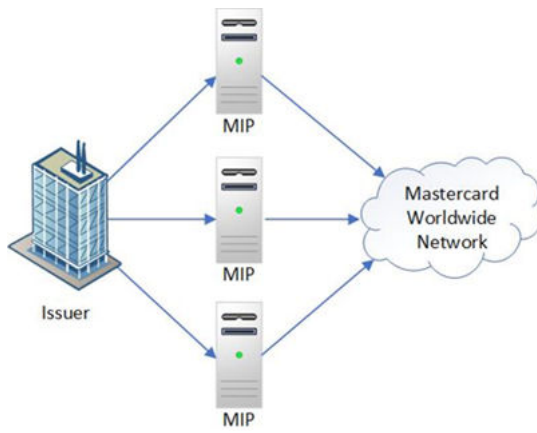
These diagrams show configurations for connectivity to a MIP or MIPs for online access.

To have direct connectivity to a MIP, the issuer must have a host computer that can support this type of connectivity.

### Issuer online: direct host to MIP connectivity



**Issuer online: direct host to multiple MIP connectivity**



## Appendix E Message timer table

This section provides information on the message routing time values.

### Dual Message Authorization System message routing timers

Following are the specific Mastercard Network message routing timer values applicable to acquirer-generated 0100, 0120, and 0400 messages. All system-generated Advice/0120, Issuer File Update/03xx, system-generated Reversal/0420, Administrative/06xx, and Network Management/08xx messages remain unchanged and retain their current timer values

**NOTE: Mastercard reserves the right to adjust local market timer values based on specific market conditions and as additional exception countries are added, they will be announced in a global announcement.**

### Dual Message Authorization System message routing timers

| Product and Transaction Type   | Issuer Response Time (in seconds) | Alternate Authorization Service Provider (if applicable) | Acquirer Minimum Wait Time (in seconds) |
|--------------------------------|-----------------------------------|----------------------------------------------------------|-----------------------------------------|
| Mastercard Credit: POS         | 9                                 | 3                                                        | 12                                      |
| Mastercard: POS PIN for Credit | 18                                | 6                                                        | 24                                      |
| Mastercard: ATM                | 12                                | 6                                                        | 18                                      |
| Debit Mastercard: POS          | Same as Mastercard Credit: POS    | Same as Mastercard Credit: POS                           | Same as Mastercard Credit: POS          |
| Maestro: All                   | 18                                | 6                                                        | 24                                      |
| Cirrus: ATM                    | 18                                | 6                                                        | 24                                      |
| Private Label: POS             | 12                                | 6                                                        | 18                                      |
| AMEX: POS                      | 12                                | 6                                                        | 18                                      |
| Visa: POS                      | 12                                | 6                                                        | 18                                      |

For transactions acquired in the countries of Brazil, Canada, the United Kingdom, and the United States, Mastercard has reduced the timers by 2 additional seconds. In these countries, the maximum time Mastercard will wait before invoking alternate authorization provider processing is reduced to 7 seconds and the minimum time that an acquirer must wait is reduced to 10 seconds.

For Maestro POS transactions acquired in the Netherlands, Mastercard has reduced the timer to 7 seconds, which is the maximum time Mastercard will wait before invoking alternate authorization provider processing. If no response is received within 10 seconds, Mastercard will send a time-out response to the acquirer.

Because the Mastercard Network does not track session status information for an acquirer, sign-on/sign-off for acquirers is not required. However, acquirers optionally may send Mastercard Network sign-on/sign-off messages (for example, if required by vendor software).

A transaction is considered to be PIN for Credit if it is Dual Message Authorization System acquired, but the issuer keys are managed by the Single Message System. If the issuer is Dual Message Authorization System managed, the transaction is not flagged as being PIN for Credit and defaults to the standard POS timer of 9 seconds.

### Routing timer values for POS transactions in Switzerland

Acquirers that support transactions submitted by merchants in Switzerland and issuers globally must take into account POS routing timer response times to provide authorization response messages in a timely manner and to avoid timeouts at the network level.

Routing response times in Switzerland that apply to acquirer-generated Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages are listed in table format. All Network Management/08xx, Issuer File Update/03xx, Administrative/06xx, and system-generated Authorization Advice messages remain unchanged and retain their current timer attributes.

#### Routing timers for Switzerland

| Product and transaction type | Issuer response time (in seconds) | Alternate authorization service provider, if applicable (in seconds) | Acquirer minimum wait time (in seconds) |
|------------------------------|-----------------------------------|----------------------------------------------------------------------|-----------------------------------------|
| Mastercard Credit: POS       | 6                                 | 3                                                                    | 9                                       |
| Debit Mastercard: POS        | 6                                 | 3                                                                    | 9                                       |
| Maestro: POS                 | 6                                 | 3                                                                    | 9                                       |



# Appendix F Transaction research information

*A transaction research request is a request by a customer for Mastercard to provide information about a transaction.*

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|                                                   |      |
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Customers can use either of the following two methods to research information about previous transactions:

- Transaction Investigator tool to conduct research about specific transactions or a range of transactions for an account number. The report generated provides customers with transactional data such as card number, card validation code 2 (CVC 2) response, point-of-sale entry mode, cardholder present information, merchant advice code, and message type identifier (MTI).
- If the transaction is over 180 days, customers can submit a request using the *Authorization Research Request* (Form 074). The Global Customer Service team will perform the research for you. The team provides information about authorization transactions, duplicate reports, microfiche re-creation, currency conversion rates, BIN and ICA numbers, and conversions.

## About the Transaction Research Request Tool

Customers can use the *Authorization Research Request* (Form 0074) to request the Global Customer Service team to perform research for you.

If you submit a transaction research request by completing and sending the *Authorization Research Request* (Form 0704) to Mastercard, Mastercard sends you a letter, with the authorization information, that you can use for chargeback purposes. If you request the services informally via a customer representative, you will not receive a letter.

## Request the Transaction Research Tool

Complete the following steps to request the Transaction Research Tool.

### Procedure

1. Log on to Mastercard Connect®.
2. Enter your **User ID** and **Password**.
3. Click **Store** on the main menu (located in the upper right corner) on the Mastercard Connect® home page.
4. Scroll to or search for **Transaction Research**.
5. Click **Add to Cart**.  
A confirmation message appears.
6. Click **Cart** to display the cart, and then click **Check Out** to display the Order Details window.

7. Under **Order Details**, click **Review Order** > **Place Order**.

Once the Security Administrator approves the request, Transaction Research will appear under the Applications menu on the Mastercard Connect® home page.

## How to Access the Transaction Research Tool

Complete the following steps to access the Transaction Research Tool.

1. Log on to Mastercard Connect®.
2. On the **Applications** menu, click **Transaction Research**. The Mastercard Authorization Transaction Research page appears.

## Submit an Authorization request

Complete the following steps to submit a request through the Transaction Research Tool.

### Procedure

1. In the **Card Number** box, enter the cardholder account number that you want to research.

**NOTE: Mastercard uses the optional ICA number for your tracking purposes only. If you enter an ICA number, it appears on your billing report to help you identify transactions.**

2. In the **Date Range** area, enter a starting transaction date and an ending transaction date using a MM/DD/YYYY format or click the **Calendar** link to select a date.
3. To identify a transaction more specifically by its approval code, enter the **Authorization Code**.
4. To identify a transaction more specifically by its amount, select one of the following options:
  - Click **All Transactions** to retrieve all transactions.
  - Click **Single Transaction**, and then enter the amount in U.S. dollars to identify a single transaction.
  - Click **Transaction Range**, and then enter a starting transaction amount in U.S. dollars and the ending transaction amount in U.S. dollars to identify a range of transactions.
5. To view recurring payments only, select the **View Only Recurring Payments** check box.

6. Click **Search**. The search results page displays.
7. To display the detailed results for a specific transaction, select the transaction you want, and then click **Detail**.

## Results

The Authorization Transaction Research tool displays a presence indicator for the following data elements in which they are present: DE 35 (Track 2 Data), DE 45 (Track 1 Data), and CVC 2 values in DE 48 (Additional Data: Private Use), subelement 92 (CVC 2).

## About fees for transaction research requests

A general research fee applies to each customer inquiry requesting information that is readily available to the customer in a Mastercard publication, report, service, or other source.

Refer to the **Pricing and Billing Resource Center** application on Mastercard Connect® for fees and billing information.

## Product Value Constraints

Product value constraints list types of transactions permitted by card program, and the interrelated value constraints for certain data elements carried in the Dual Message Authorization System messages.

The business rules related to the card programs Mastercard supports differentiate between the nature of the transactions permitted for:

- Mastercard (MCC)
- Debit Mastercard (DMC)
- Maestro (MSI)
- Cirrus (CIR)
- Private Label (PVL)

## Permitted Transactions by card program

Following are the types of transactions permitted by card program and the interrelated value constraints for certain data elements in Dual Message Authorization System messages.

The interrelated data elements are:

- DE 3 (Processing Code)
- DE 22 (Point-of-Service [POS] Entry Mode)
- DE 52 (Personal ID Number [PIN] Data)
- DE 61 (Point-of-Service [POS] Data)

### Permitted Transactions by Card Program

Use the reference number in the Ref column in the following tables with the specific value constraints in the corresponding [Value Constraints by Transaction Type](#) table.

#### ATM Withdrawal

| Ref | Transaction Type | PAN Entry Mode  | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                                                                  |
|-----|------------------|-----------------|------------|-----|-----|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1 | ATM              | Magnetic Stripe | Online PIN | X   | X   | X   | X   |     | <p>For this message, DE 18 must contain value 6011 (Automated Cash Disbursements).</p> <p>For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.</p> |

| Ref | Transaction Type | PAN Entry Mode | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                                                                |
|-----|------------------|----------------|------------|-----|-----|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.2 | ATM              | Chip           | Online PIN | X   | X   | X   | X   |     | For this message, DE 18 must contain value 6011 (Automated Cash Disbursements).<br><br>For MCC, DMC, MSI, contactless are transactions permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |

### Cash Advance

| Ref | Transaction Type | PAN Entry Mode | CVM       | MCC | DMC | MSI | CIR | PVL | Comments                                                                    |
|-----|------------------|----------------|-----------|-----|-----|-----|-----|-----|-----------------------------------------------------------------------------|
| 2.1 | Cash Advance     | Manual         | Signature | X   | X   |     |     |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) |

| Ref | Transaction Type | PAN Entry Mode  | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                                                       |
|-----|------------------|-----------------|-------------|-----|-----|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.2 | Cash Advance     | Magnetic Stripe | Signature   | X   | X   |     |     |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)<br><br>For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 2.3 | Cash Advance     | Magnetic Stripe | Online PIN  | X   | X   |     |     |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)<br><br>For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 2.4 | Cash Advance     | Chip            | Offline PIN | X   | X   |     |     |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)                                                                                                                                                    |

| Ref | Transaction Type        | PAN Entry Mode  | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                                                       |
|-----|-------------------------|-----------------|------------|-----|-----|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.5 | Cash Advance            | Chip            | Online PIN | X   | X   |     |     |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)<br><br>For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 2.6 | Cash Advance            | Chip            | Signature  | X   | X   |     |     |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)<br><br>For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 2.7 | Electronic Cash Advance | Magnetic Stripe | Online PIN |     |     | X   | X   |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)                                                                                                                                                    |



| Ref | Transaction Type        | PAN Entry Mode | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                      |
|-----|-------------------------|----------------|-------------|-----|-----|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 2.8 | Electronic Cash Advance | Chip           | Offline PIN |     |     | X   | X   |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)<br><br>A waiver is required for this item. Issuers must register. |
| 2.9 | Electronic Cash Advance | Chip           | Online PIN  |     |     | X   | X   |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)                                                                   |

### Purchase of Goods and Services

| Ref | Transaction Type | PAN Entry Mode  | CVM       | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                |
|-----|------------------|-----------------|-----------|-----|-----|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1 | Purchase         | Manual          | Signature | X   | X   |     |     | X   |                                                                                                                                                                                         |
| 3.2 | Purchase         | Magnetic Stripe | Signature | X   | X   | X   |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.<br><br>For MSI, country restrictions apply. |

| Ref | Transaction Type | PAN Entry Mode  | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                    |
|-----|------------------|-----------------|-------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.3 | Purchase         | Magnetic Stripe | Online PIN  | X   | X   | X   |     |     | For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.<br><br>For MSI, country restrictions apply. |
| 3.4 | Purchase         | Chip            | Offline PIN | X   | X   | X   |     |     | For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.<br><br>For MSI, country restrictions apply. |
| 3.5 | Purchase         | Chip            | Signature   | X   | X   |     |     | X   |                                                                                                                                                                                             |
| 3.6 | Purchase         | Chip            | Online PIN  | X   | X   | X   |     |     | For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.<br><br>For MSI, country restrictions apply. |

| Ref  | Transaction Type  | PAN Entry Mode     | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                        |
|------|-------------------|--------------------|-------------|-----|-----|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.7  | Purchase          | Chip               | No CVM      | X   | X   | X   |     |     | For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 3.8  | Purchase          | Credential on File | No CVM      | X   | X   | X   |     |     |                                                                                                                                                 |
| 3.9  | MO/TO             | Manual             | No CVM      | X   | X   | X   |     | X   |                                                                                                                                                 |
| 3.10 | MO/TO             | Credential on File | No CVM      | X   | X   | X   |     |     |                                                                                                                                                 |
| 3.11 | Recurring Payment | Credential on File | N/A         | X   | X   | X   |     |     |                                                                                                                                                 |
| 3.12 | CAT Level 1       | Magnetic Stripe    | Online PIN  | X   | X   |     |     |     | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.     |
| 3.13 | CAT Level 1       | Chip               | Online PIN  | X   | X   |     |     |     | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.     |
| 3.14 | CAT Level 1       | Chip               | Offline PIN |     |     |     |     |     |                                                                                                                                                 |

| Ref  | Transaction Type | PAN Entry Mode  | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                    |
|------|------------------|-----------------|-------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| 3.15 | CAT Level 2      | Magnetic Stripe | No CVM      | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 3.16 | CAT Level 2      | Chip            | No CVM      | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 3.17 | CAT Level 3      | Magnetic Stripe | No CVM      | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
|      | CAT Level 3      | Chip            | Offline PIN |     |     |     |     |     |                                                                                                                                             |
| 3.18 | CAT Level 3      | Chip            | No CVM      | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |

| Ref  | Transaction Type | PAN Entry Mode      | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                    |
|------|------------------|---------------------|-------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| 3.19 | CAT Level 4      | Magnetic Stripe     | No CVM      | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 3.20 | CAT Level 4      | Chip                | No CVM      | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 3.21 | CAT Level 6      | Credential on File  | No CVM      | X   | X   | X   |     |     |                                                                                                                                             |
| 3.22 | CAT Level 6      | Electronic Commerce | No CVM      | X   | X   | X   |     | X   |                                                                                                                                             |
| 3.23 | CAT Level 6      | Electronic Commerce | Offline PIN | X   | X   | X   |     |     |                                                                                                                                             |
| 3.24 | CAT Level 6      | Electronic Commerce | Online PIN  | X   | X   | X   |     |     |                                                                                                                                             |
| 3.25 | CAT Level 7      | N/A                 | N/A         | X   | X   |     |     | X   | No specific constraints defined.                                                                                                            |

## Payments

| Ref | Transaction Type     | PAN Entry Mode | CVM    | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                            |
|-----|----------------------|----------------|--------|-----|-----|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.1 | Payment Transactions | N/A            | No CVM | X   | X   | X   |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.<br><br>No specific constraints defined. |

## PIN Change Management

| Ref | Transaction Type | PAN Entry Mode | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                    |
|-----|------------------|----------------|------------|-----|-----|-----|-----|-----|-----------------------------------------------------------------------------|
| 5.1 | ATM PIN unblock  | Chip           | Online PIN | X   | X   | X   | X   |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) |
| 5.2 | ATM PIN change   | Chip           | Online PIN | X   | X   | X   | X   |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) |

| Ref | Transaction Type | PAN Entry Mode  | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                    |
|-----|------------------|-----------------|------------|-----|-----|-----|-----|-----|-----------------------------------------------------------------------------|
| 5.3 | ATM PIN change   | Magnetic Stripe | Online PIN | X   | X   | X   | X   |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) |

### Balance Inquiry

| Ref | Transaction Type    | PAN Entry Mode  | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                                                           |
|-----|---------------------|-----------------|------------|-----|-----|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.1 | ATM Balance Inquiry | Magnetic Stripe | Online PIN | X   | X   | X   | X   |     | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)<br><br>For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |

| Ref | Transaction Type    | PAN Entry Mode  | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                                                                                                        |
|-----|---------------------|-----------------|-------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.2 | ATM Balance Inquiry | Chip            | Online PIN  | X   | X   | X   | X   |     | <p>For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)</p> <p>For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.</p> <p>Applies to prepaid cards only.</p> |
| 6.3 | POS Balance Inquiry | Magnetic Stripe | Signature   | X   | X   | X   |     | X   | <p>For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.</p>                                                                                                                          |
| 6.4 | POS Balance Inquiry | Magnetic Stripe | Online PIN  | X   | X   | X   | X   | X   | <p>For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.</p>                                                                                                                              |
| 6.5 | POS Balance Inquiry | Chip            | Offline PIN | X   | X   | X   |     | X   |                                                                                                                                                                                                                                                                                 |



| Ref | Transaction Type    | PAN Entry Mode | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                    |
|-----|---------------------|----------------|------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| 6.6 | POS Balance Inquiry | Chip           | Signature  | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 6.7 | POS Balance Inquiry | Chip           | Online PIN | X   | X   |     |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 6.8 | SMS Balance Inquiry | eCommerce      | No CVM     | X   | X   |     |     |     |                                                                                                                                             |

#### Purchase with Cash Back (PWCB)

| Ref | Transaction Type | PAN Entry Mode  | CVM       | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                                                                                                          |
|-----|------------------|-----------------|-----------|-----|-----|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.1 | PWCB             | Magnetic Stripe | Signature |     | X   | X   |     |     | Approved intracountry countries, as indicated in <a href="#">Mastercard Rules</a> .<br><br>For MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |

| Ref | Transaction Type | PAN Entry Mode  | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                    |
|-----|------------------|-----------------|-------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| 7.2 | PWCB             | Magnetic Stripe | Online PIN  |     | X   | X   |     |     | Approved intracountry countries, as indicated in <a href="#">Mastercard Rules</a> .                                                         |
| 7.3 | PWCB             | Chip            | Offline PIN |     | X   | X   |     |     | Approved intracountry countries, as indicated in <a href="#">Mastercard Rules</a> .                                                         |
| 7.4 | PWCB             | Chip            | Online PIN  |     | X   | X   |     |     | For DMC, MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 7.5 | PWCB             | Manual          | Signature   |     |     | X   |     |     |                                                                                                                                             |

## Refund

| Ref | Transaction Type | PAN Entry Mode  | CVM        | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                    |
|-----|------------------|-----------------|------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| 8.1 | Refund           | Manual          | Signature  | X   | X   |     |     | X   |                                                                                                                                             |
| 8.2 | Refund           | Magnetic Stripe | Signature  | X   | X   | X   |     | X   | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 8.3 | Refund           | Magnetic Stripe | Online PIN |     |     | X   |     |     |                                                                                                                                             |

| Ref | Transaction Type | PAN Entry Mode | CVM         | MCC | DMC | MSI | CIR | PVL | Comments                                                                                                                                    |
|-----|------------------|----------------|-------------|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| 8.4 | Refund           | Chip           | Signature   | X   | X   |     |     |     | For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. |
| 8.5 | Refund           | Chip           | Online PIN  |     |     | X   |     |     | For MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.      |
| 8.6 | Refund           | Chip           | Offline PIN | X   | X   |     |     |     |                                                                                                                                             |
| 8.7 | Refund           | Chip           | No CVM      |     |     | X   |     |     | For MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.      |

## Value Constraints by Transaction Type

Following are value constraints by transaction type.

### Value Constraints by Transaction Type

#### ATM Withdrawal

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values      | CVM Method | DE 61, SF 10 values | Comments                                                                        |
|-----|----------------------------------------------------------------------|-------------|-------------------|------------|---------------------|---------------------------------------------------------------------------------|
| 1.1 | ATM, Magnetic Stripe, Online PIN                                     | 010000      | 901, 91x (or 801) | P          | 0 or 1              | For this message, DE 18 must contain value 6011 (Automated Cash Disbursements). |
| 1.2 | ATM, Chip, Online PIN                                                | 010000      | 051, 071          | P          | 0 or 1              | For this message, DE 18 must contain value 6011 (Automated Cash Disbursements). |

#### Cash Advance

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values      | CVM Method | DE 61, SF 10 values | Comments                                                                     |
|-----|----------------------------------------------------------------------|-------------|-------------------|------------|---------------------|------------------------------------------------------------------------------|
| 2.1 | Cash Advance, Manual, Signature                                      | 170000      | 00x, 01x (or 79x) | S          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values      | CVM Method | DE 61, SF 10 values | Comments                                                                     |
|-----|----------------------------------------------------------------------|-------------|-------------------|------------|---------------------|------------------------------------------------------------------------------|
| 2.2 | Cash Advance, Magnetic Stripe, Signature                             | 170000      | 90x, 91x (or 80x) | S          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |
| 2.3 | Cash Advance, Magnetic Strip, Online PIN                             | 170000      | 90x, 91x (or 80x) | P          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |
| 2.4 | Cash Advance, Chip, Offline PIN                                      | 170000      | 051               | S          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |
| 2.5 | Cash Advance, Chip Online PIN                                        | 170000      | 051, 071          | P          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |
| 2.6 | Cash Advance, Chip, Signature                                        | 170000      | 05x, 07x          | S          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |
| 2.7 | Electronic Cash Advance, Magnetic Stripe, Online PIN                 | 010000      | 901               | P          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values | CVM Method | DE 61, SF 10 values | Comments                                                                     |
|-----|----------------------------------------------------------------------|-------------|--------------|------------|---------------------|------------------------------------------------------------------------------|
| 2.8 | Electronic Cash Advance, Chip, Offline PIN                           | 010000      | 051          | S          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |
| 2.9 | Electronic Cash Advance, Chip Online PIN                             | 010000      | 051          | P          | 0                   | For this message, DE 18 must contain value 6010 (Manual Cash Disbursements). |

### Purchase of Goods and Services

| Ref  | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values      | CVM Method | DE 61, SF 10 values |
|------|----------------------------------------------------------------------|-------------|-------------------|------------|---------------------|
| 3.1  | Purchase, Manual, Signature                                          | 00xx00      | 00x, 01x (or 79x) | S          | 0                   |
| 3.2  | Purchase, Magnetic Stripe, Signature                                 | 00xx00      | 90x, 91x (or 80x) | S          | 0                   |
| 3.3  | Purchase, Magnetic Stripe, Online PIN                                | 00xx00      | 901, 911 (or 80x) | P          | 0                   |
| 3.4  | Purchase, Chip, Offline PIN                                          | 00xx00      | 051               | S          | 0                   |
| 3.5  | Purchase, Chip, Signature                                            | 00xx00      | 05x, 07x          | S          | 0                   |
| 3.6  | Purchase, Chip, Online PIN                                           | 00xx00      | 051, 071          | P          | 0                   |
| 3.7  | Purchase, Chip, No CVM                                               | 00xx00      | 05x, 07x          | None       | 0                   |
| 3.8  | Purchase, Credential on File, No CVM                                 | 00xx00      | 10x               | None       | 0                   |
| 3.9  | MO/TO, Manual, No CVM                                                | 00xx00      | 01x, 10x          | None       | 0                   |
| 3.10 | MO/TO, Credential on File, No CVM                                    | 00xx00      | 10x               | None       | 0                   |

| Ref  | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values      | CVM Method      | DE 61, SF 10 values |
|------|----------------------------------------------------------------------|-------------|-------------------|-----------------|---------------------|
| 3.11 | Recurring Payment, Credential on File, No CVM                        | 00xx00      | 10x               | No CVM          | 0                   |
| 3.12 | CAT Level 1, Magnetic Stripe, Online PIN                             | 00xx00      | 901, 911 (or 801) | P               | 1                   |
| 3.13 | CAT Level 1, Chip, Online PIN                                        | 00xx00      | 051, 071          | P               | 1                   |
| 3.14 | CAT Level 1, Chip, Offline PIN                                       | 00xx00      | 051               | S               | 1                   |
| 3.15 | CAT Level 2, Magnetic Stripe, No CVM                                 | 00xx00      | 90x, 91x (or 80x) | None            | 2                   |
| 3.16 | CAT Level 2, Chip, No CVM                                            | 00xx00      | 05x, 07x          | None            | 2                   |
| 3.17 | CAT Level 3, Magnetic Stripe, No CVM                                 | 00xx00      | 90x, 91x (or 80x) | None            | 3                   |
| 3.18 | CAT Level 3, Chip, No CVM                                            | 00xx00      | 05x, 07x          | None            | 3                   |
| 3.19 | CAT Level 4, Magnetic Stripe, No CVM                                 | 00xx00      | 90x (or 80x)      | None            | 4                   |
| 3.20 | CAT Level 4, Chip, No CVM                                            | 00xx00      | 05x, 07x          | None            | 4                   |
| 3.21 | CAT Level 6, Credential on File, No CVM                              | 00xx00      | 10x               | None            | 6                   |
| 3.22 | CAT Level 6, Electronic Commerce, No CVM                             | 00xx00      | 10x, 81x, 82x     | None            | 6                   |
| 3.23 | CAT Level 6, Electronic Commerce, Offline PIN                        | 00xx00      | 10x, 81x          | S               | 6                   |
| 3.24 | CAT Level 6, Electronic Commerce, Online PIN                         | 00xx00      | 10x, 81x, 82x     | P               | 6                   |
| 3.25 | CAT Level 7                                                          | 00xx00      | No constraints    | Any constraints | 7                   |

## Payments

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values   | CVM Method | DE 61, SF 10 values |
|-----|----------------------------------------------------------------------|-------------|----------------|------------|---------------------|
| 4.1 | Payment Transactions, No CVM                                         | 280000      | No constraints | None       | No constraints      |

## PIN Change Management

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values | CVM Method | DE 61, SF 10 values | Comments                                                                        |
|-----|----------------------------------------------------------------------|-------------|--------------|------------|---------------------|---------------------------------------------------------------------------------|
| 5.1 | ATM PIN unblock, Chip, Online PIN                                    | 910000      | 051          | P          | 0 or 1              | For this message, DE 18 must contain value 6011 (Automated Cash Disbursements). |
| 5.2 | ATM PIN change, Chip, Online PIN                                     | 920000      | 051          | P          | 0 or 1              | For this message, DE 18 must contain value 6011 (Automated Cash Disbursements). |
| 5.3 | ATM PIN change Magnetic Stripe, Online PIN                           | 920000      | 021, 901     | P          | 0 or 1              |                                                                                 |



## Balance Inquiry

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values            | DE 22 values      | CVM Method | DE 61, SF 10 values | Comments                                                                       |
|-----|----------------------------------------------------------------------|------------------------|-------------------|------------|---------------------|--------------------------------------------------------------------------------|
| 6.1 | ATM Balance Inquiry, Magnetic Stripe, Online PIN                     | 30xx00                 | 901, 911          | P          | 0 or 1              | For this message, DE 18 must contain value 6011 (Automated Cash Disbursements) |
| 6.2 | ATM Balance Inquiry, Chip, Online PIN                                | 30xx00                 | 051, 071          | P          | 0 or 1              |                                                                                |
| 6.3 | POS Balance Inquiry, Magnetic Stripe, Signature                      | 300000<br>or<br>303000 | 90x, 91x (or 80x) | S          | 0                   |                                                                                |
| 6.4 | POS Balance Inquiry, Magnetic Stripe, Online PIN                     | 300000<br>or<br>303000 | 901, 911 (or 801) | P          | 0                   |                                                                                |
| 6.5 | POS Balance Inquiry, Chip, Offline PIN                               | 300000<br>or<br>303000 | 051               | S          | 0                   |                                                                                |

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values            | DE 22 values | CVM Method | DE 61, SF 10 values | Comments |
|-----|----------------------------------------------------------------------|------------------------|--------------|------------|---------------------|----------|
| 6.6 | POS Balance Inquiry, Chip, Signature                                 | 300000<br>or<br>303000 | 05x, 07x     | S          | 0                   |          |
| 6.7 | POS Balance Inquiry, Chip, Online PIN                                | 300000<br>or<br>303000 | 051, 071     | P          | 0                   |          |
| 6.8 | SMS Balance Inquiry                                                  |                        |              |            |                     |          |

#### Purchase with Cash Back (PWCB)

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values | CVM Method | DE 61, SF 10 values |
|-----|----------------------------------------------------------------------|-------------|--------------|------------|---------------------|
| 7.1 | PWCB, Magnetic Stripe, Signature                                     | 09xx00      | 90x (or 80x) | S          | 0                   |
| 7.2 | PWCB, Magnetic Stripe, Online PIN                                    | 09xx00      | 901 (or 801) | P          | 0                   |
| 7.3 | PWCB, Chip Offline PIN                                               | 09xx00      | 051          | S          | 0                   |
| 7.4 | PWCB, Chip Online PIN                                                | 09xx00      | 051          | P          | 0                   |
| 7.5 | PWCB, Manual, Signature                                              | 09xx00      | 01x          | S          | 0                   |

## Refund

| Ref | Transaction Type, PAN Entry Mode, and Cardholder Verification Method | DE 3 values | DE 22 values      | CVM Method | DE 61, SF 10 values |
|-----|----------------------------------------------------------------------|-------------|-------------------|------------|---------------------|
| 8.1 | Refund, Manual, Signature                                            | 20xx00      | 00x, 01x          | S          | 0                   |
| 8.2 | Refund, Magnetic Stripe, Signature                                   | 20xx00      | 90x, 91x (or 80x) | S          | 0                   |
| 8.3 | Refund, Magnetic Stripe, Online PIN                                  | 20xx00      | 901 (or 801)      | P          | 0                   |
| 8.4 | Refund, Chip, Signature                                              | 20xx00      | 05x, 07x          | S          | 0                   |
| 8.5 | Refund, Chip, Online PIN                                             | 20xx00      | 051               | P          | 0                   |
| 8.6 | Refund, Chip, Offline PIN                                            | 20xx00      | 051               | S          | 0                   |

## Appendix G Transaction data classifications table

This section provides details on the transaction data classifications table.

Category One: Transaction data that, when received by a customer or Service Provider, must only be used, or disclosed by the customer or Service Provider for the purpose of carrying out the customer's issuing program, acquiring program, or digital activities (for example, as relating to digital wallets and tokenization).

Category Two: Transaction data that, when received by a customer or Service Provider, must only be used, or disclosed in connection with the specific purpose for which it was provided (for example, as relating to participation in a program or service).

### Transaction data classifications

| Data element | Data element description                         | Transaction data category |
|--------------|--------------------------------------------------|---------------------------|
| DE 1         | Bit Map, Secondary                               | One                       |
| DE 2         | Primary Account Number [PAN]                     | One                       |
| DE 3         | Processing Code                                  | One                       |
| DE 3         | Subfield 1 (Cardholder Transaction Type Code)    | One                       |
| DE 3         | Subfield 2 (Cardholder "From Account" Type Code) | One                       |
| DE 3         | Subfield 3 (Cardholder "To Account" Type Code)   | One                       |
| DE 4         | Amount, Transaction                              | One                       |
| DE 5         | Amount, Settlement                               | One                       |
| DE 6         | Amount, Cardholder Billing                       | One                       |
| DE 7         | Transmission Date and Time                       | One                       |
| DE 7         | Subfield 1 (Date)                                | One                       |
| DE 7         | Subfield 2 (Time)                                | One                       |
| DE 8         | Amount, Cardholder Billing Fee                   | One                       |
| DE 9         | Conversion Rate, Settlement                      | One                       |
| DE 9         | Subfield 1 (Decimal Indicator)                   | One                       |
| DE 9         | Subfield 2 (Conversion Rate)                     | One                       |
| DE 10        | Conversion Rate, Cardholder Billing              | One                       |
| DE 10        | Subfield 1 (Decimal Indicator)                   | One                       |

| Data element | Data element description                                     | Transaction data category |
|--------------|--------------------------------------------------------------|---------------------------|
| DE 10        | Subfield 2 (Cardholder Billing Conversion Rate)              | One                       |
| DE 11        | System Trace Audit Number [STAN]                             | One                       |
| DE 12        | Time, Local Transaction                                      | One                       |
| DE 13        | Date, Local Transaction                                      | One                       |
| DE 14        | Date, Expiration                                             | One                       |
| DE 15        | Date, Settlement                                             | One                       |
| DE 16        | Date, Conversion                                             | One                       |
| DE 17        | Date, Capture                                                | One                       |
| DE 18        | Merchant Type                                                | One                       |
| DE 19        | Acquiring Institution Country Code                           | One                       |
| DE 20        | Primary Account Number [PAN]<br>Country Code                 | One                       |
| DE 21        | Forwarding Institution Country Code                          | One                       |
| DE 22        | Point-of-Service [POS] Entry Mode                            | One                       |
| DE 22        | Subfield 1 (POS Terminal PAN Entry Mode)                     | One                       |
| DE 22        | Subfield 2 (POS Terminal PIN Entry Mode)                     | One                       |
| DE 23        | Card Sequence Number                                         | One                       |
| DE 24        | Network International ID                                     | Two                       |
| DE 25        | Point-of-Service [POS] Condition Code                        | One                       |
| DE 26        | Point-of-Service (POS) Personal ID Number [PIN] Capture Code | One                       |
| DE 27        | Authorization ID Response Length                             | Two                       |
| DE 28        | Amount, Transaction Fee                                      | One                       |
| DE 28        | Subfield 1 (Debit/Credit Indicator)                          | One                       |
| DE 28        | Subfield 2 (Amount)                                          | One                       |
| DE 32        | Acquiring Institution ID Code                                | One                       |
| DE 33        | Forwarding Institution ID Code                               | One                       |
| DE 34        | Primary Account Number [PAN],<br>Extended                    | One                       |

| Data element | Data element description                                           | Transaction data category |
|--------------|--------------------------------------------------------------------|---------------------------|
| DE 35        | Track 2 Data                                                       | One                       |
| DE 37        | Retrieval Reference Number                                         | Two                       |
| DE 37        | Subfield 1 (Transaction Date and Initiator Discretionary Data)     | Two                       |
| DE 37        | Subfield 2 (Terminal Transaction Number)                           | Two                       |
| DE 38        | Authorization ID Response                                          | One                       |
| DE 39        | Response Code                                                      | One                       |
| DE 41        | Acceptor Terminal ID                                               | One                       |
| DE 42        | Acceptor ID                                                        | One                       |
| DE 43        | Acceptor Name and Location                                         | One                       |
| DE 43        | Subfield 1 (Acceptor Name)                                         | One                       |
| DE 43        | Subfield 2 (Space)                                                 | One                       |
| DE 43        | Subfield 3 (Acceptor City)                                         | One                       |
| DE 43        | Subfield 4 (Space)                                                 | One                       |
| DE 43        | Subfield 5 (Acceptor Country Code)                                 | One                       |
| DE 44        | Additional Response Data                                           | One                       |
| DE 45        | Track 1 Data                                                       | One                       |
| DE 48        | Additional Data: Private Use                                       | see subelements           |
| DE 48        | Subelement 09 (Virtual Card Number Data)                           | One                       |
| DE 48        | Subfield 01 (Virtual Card Number Indicator)                        | One                       |
| DE 48        | Subfield 02 (Virtual Card Number)                                  | One                       |
| DE 48        | Subfield 03 (Virtual Card Number Expiration Date)                  | One                       |
| DE 48        | Subelement 11 (Key Exchange Block Data [Double-Length Keys])       | Two                       |
| DE 48        | Subelement 11 (Key Exchange Block Data [Triple-Length Keys])       | Two                       |
| DE 48        | Subelement 12 (Routing Indicator)                                  | Two                       |
| DE 48        | Subelement 13 (Mastercard Hosted Mobile Phone Top-Up Request Data) | Two                       |

| Data element | Data element description                                          | Transaction data category |
|--------------|-------------------------------------------------------------------|---------------------------|
| DE 48        | Subfield 1 (Mobile Phone Number)                                  | Two                       |
| DE 48        | Subfield 2 (Mobile Phone Service Provider Name)                   | Two                       |
| DE 48        | Subelement 14 (Account Type Indicator)                            | Two                       |
| DE 48        | Subelement 15 (Authorization System Advice Date and Time)         | One                       |
| DE 48        | Subfield 1 (Date)                                                 | One                       |
| DE 48        | Subfield 2 (Time)                                                 | One                       |
| DE 48        | Subelement 16 (Processor Pseudo ICA)                              | One                       |
| DE 48        | Subelement 17 (Authentication Indicator)                          | Two                       |
| DE 48        | Subelement 18 (Service Parameters)                                | One                       |
| DE 48        | Subfield 01 (Canada Domestic Indicator)                           | One                       |
| DE 48        | Subelement 20 (Cardholder Verification Method)                    | One                       |
| DE 48        | Subelement 21 (Acceptance Data)                                   | One                       |
| DE 48        | Subfield 01 (mPOS Acceptance Device Type)                         | One                       |
| DE 48        | Subfield 2 (Additional Terminal Capability Indicator)             | One                       |
| DE 48        | Subelement 22 (Multi-Purpose Merchant Indicator)                  | Two                       |
| DE 48        | Subfield 01 (Low-Risk Merchant Indicator)                         | Two                       |
| DE 48        | Subfield 02 (Single Tap Indicator)                                | Two                       |
| DE 48        | Subfield 03 (Response to PIN Request)                             | Two                       |
| DE 48        | Subfield 04 (Issuer PIN Request in a Single Tap Mode)             | Two                       |
| DE 48        | Subfield 05 (Cardholder/Merchant Initiated Transaction Indicator) | Two                       |
| DE 48        | Subfield 06 (Message Format Version Code: Acquirer)               | Two                       |

| Data element | Data element description                                            | Transaction data category |
|--------------|---------------------------------------------------------------------|---------------------------|
| DE 48        | Subelement 23 (Payment Initiation Channel)                          | One                       |
| DE 48        | Subfield 1 (Device Type)                                            | One                       |
| DE 48        | Subelement 24 (Account Level Management [ALM] Service Data)         | Two                       |
| DE 48        | Subfield 1 (ALM Service Code)                                       | Two                       |
| DE 48        | Subfield 2 (ALM Product Graduation Plus or Registered Product Code) | Two                       |
| DE 48        | Subfield 3 (ALM Product Class)                                      | Two                       |
| DE 48        | Subfield 4 (ALM Rate Type)                                          | Two                       |
| DE 48        | Subelement 25 (Mastercard Cash Program Data)                        | Two                       |
| DE 48        | Subfield 01 (Message Identifier)                                    | Two                       |
| DE 48        | Subelement 26 (Wallet Program Data)                                 | Two                       |
| DE 48        | Subfield 1 (Wallet Identifier)                                      | Two                       |
| DE 48        | Subelement 27 (Transaction Analysis)                                | Two                       |
| DE 48        | Subfield 1 (Overview)                                               | Two                       |
| DE 48        | Subfield 2 (Test Results)                                           | Two                       |
| DE 48        | Subelement 28 (Cardless ATM Order ID)                               | Two                       |
| DE 48        | Subelement 29 (Additional POS Terminal Locations)                   | One                       |
| DE 48        | Subelement 30 (Token Transaction Identifier)                        | Two                       |
| DE 48        | Subelement 32 (Mastercard Assigned ID)                              | Two                       |
| DE 48        | Subelement 33 (PAN Mapping File Information)                        | Two                       |
| DE 48        | Subfield 1 (Account Number Indicator)                               | Two                       |
| DE 48        | Subfield 2 (Account Number)                                         | Two                       |
| DE 48        | Subfield 3 (Expiration Date)                                        | Two                       |
| DE 48        | Subfield 4 (Product Code)                                           | Two                       |



| Data element | Data element description                                                                      | Transaction data category |
|--------------|-----------------------------------------------------------------------------------------------|---------------------------|
| DE 48        | Subfield 5 (Token Assurance Method)                                                           | Two                       |
| DE 48        | Subfield 6 (Token Requestor ID)                                                               | Two                       |
| DE 48        | Subfield 7 (Primary Account Number, Account Range)                                            | Two                       |
| DE 48        | Subfield 8 (Storage Technology)                                                               | Two                       |
| DE 48        | Subelement 34 (ATC Information)                                                               | Two                       |
| DE 48        | Subfield 1 (ATC Value)                                                                        | Two                       |
| DE 48        | Subfield 2 (ATC Discrepancy Value)                                                            | Two                       |
| DE 48        | Subfield 3 (ATC Discrepancy Indicator)                                                        | Two                       |
| DE 48        | Subelement 35 (Contactless Non-Card Form Factor Request/Response)                             | Two                       |
| DE 48        | Subelement 36 (Additional Visa Request Data [Visa Only])                                      | Two                       |
| DE 48        | Subelement 37 (Additional Merchant Data)                                                      | Two                       |
| DE 48        | Subfield 1 (Payment Facilitator ID)                                                           | Two                       |
| DE 48        | Subfield 2 (Independent Sales Organization ID)                                                | Two                       |
| DE 48        | Subfield 3 (Sponsored Merchant ID)                                                            | Two                       |
| DE 48        | Subfield 4 (Merchant Country of Origin)                                                       | Two                       |
| DE 48        | Subfield 5 (Merchant Payment Gateway ID)                                                      | Two                       |
| DE 48        | Subelement 38 (Account Category)                                                              | Two                       |
| DE 48        | Subelement 39 (Account Data Compromise Information)                                           | Two                       |
| DE 48        | Subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only]) | Two                       |
| DE 48        | Subfield 1 (Merchant Certificate Serial Number)                                               | Two                       |

| Data element | Data element description                                               | Transaction data category |
|--------------|------------------------------------------------------------------------|---------------------------|
| DE 48        | Subfield 2 (Cardholder Certificate Serial Number)                      | Two                       |
| DE 48        | Subelement 41 (Electronic Commerce Certificate Qualifying Information) | Two                       |
| DE 48        | Subfield 1 (Additional Recurring Payments Information)                 | Two                       |
| DE 48        | Subfield 2 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 3 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 4 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 5 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 6 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 7 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 8 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 9 (Reserved for Future Use)                                   | Two                       |
| DE 48        | Subfield 10 (Reserved for Future Use)                                  | Two                       |
| DE 48        | Subfield 11 (Citizen ID)                                               | Two                       |
| DE 48        | Subfield 12 (Reserved for Future Use)                                  | Two                       |
| DE 48        | Subfield 13 (Reserved for Future Use)                                  | Two                       |
| DE 48        | Subfield 14 (Reserved for Future Use)                                  | Two                       |
| DE 48        | Subfield 15 (Reserved for Future Use)                                  | Two                       |
| DE 48        | Subfield 16 (Reserved for Future Use)                                  | Two                       |
| DE 48        | Subfield 17 (Reserved for Future Use)                                  | Two                       |

| Data element | Data element description                                                                                           | Transaction data category |
|--------------|--------------------------------------------------------------------------------------------------------------------|---------------------------|
| DE 48        | Subfield 18 (Reserved for Future Use)                                                                              | Two                       |
| DE 48        | Subelement 42 (Electronic Commerce Indicators)                                                                     | One                       |
| DE 48        | Subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator)                            | One                       |
| DE 48        | Subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator)                   | Two                       |
| DE 48        | Subfield 3 (Reason for UCAF Collection Indicator Downgrade)                                                        | Two                       |
| DE 48        | Subelement 43 (Universal Cardholder Authentication Field [UCAF])                                                   | Two                       |
| DE 48        | Subelement 43 (3-D Secure for Mastercard Identity Check)                                                           | Two                       |
| DE 48        | Subelement 43 (Digital Secure Remote Payment Universal Cardholder Authentication Field [UCAF])                     | Two                       |
| DE 48        | Subelement 43 (Static AAV)                                                                                         | Two                       |
| DE 48        | Subelement 43 (3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express]) | Two                       |
| DE 48        | Subelement 44 (3-D Secure Electronic Commerce Transaction Identifier [XID][Visa and American Express])             | Two                       |
| DE 48        | Subelement 45 (3-D Secure Electronic Commerce Transaction Response Code [Visa and American Express])               | Two                       |
| DE 48        | Subelement 46 (Product ID [Visa Only])                                                                             | Two                       |
| DE 48        | Subelement 47 (Mastercard Payment Gateway Transaction Indicator)                                                   | One                       |

| Data element | Data element description                                     | Transaction data category |
|--------------|--------------------------------------------------------------|---------------------------|
| DE 48        | Subelement 48 (Digital Commerce Solutions Indicators)        | Two                       |
| DE 48        | Subfield 01 (Digital Commerce Domain Identifier)             | Two                       |
| DE 48        | Subfield 02 (Digital Commerce Program Type)                  | Two                       |
| DE 48        | Subfield 03 (Digital Commerce Check-In Entity Type and Name) | Two                       |
| DE 48        | Subfield 04 (Digital Commerce Check-In Method)               | Two                       |
| DE 48        | Subelement 49 (Time Validation Information)                  | One                       |
| DE 48        | Subfield 1 (Time Value)                                      | One                       |
| DE 48        | Subfield 2 (Time Discrepancy Value)                          | One                       |
| DE 48        | Subfield 3 (Time Discrepancy Indicator)                      | One                       |
| DE 48        | Subelement 50 (Embedded Interchange Data)                    | Two                       |
| DE 48        | Subfield 01 (Interchange Indicator)                          | Two                       |
| DE 48        | Subfield 02 (Interchange Percentage)                         | Two                       |
| DE 48        | Subfield 03 (Unit Fee Amount)                                | Two                       |
| DE 48        | Subfield 04 (Flat Interchange Amount)                        | Two                       |
| DE 48        | Subfield 05 (Payment Party)                                  | Two                       |
| DE 48        | Subfield 06 (Supplemental Interchange Data)                  | Two                       |
| DE 48        | Subelement 51 (Merchant On-behalf Services)                  | Two                       |
| DE 48        | Subfield 1 (Merchant On-behalf [OB] Service)                 | Two                       |
| DE 48        | Subfield 2 (Merchant On-behalf [OB] Result 1)                | Two                       |
| DE 48        | Subfield 3 (Additional Information)                          | Two                       |
| DE 48        | Subelement 52 (Interchange Structure Indicator)              | Two                       |

| Data element | Data element description                                                   | Transaction data category |
|--------------|----------------------------------------------------------------------------|---------------------------|
| DE 48        | Subelement 53 (E-ID Request Code)                                          | Two                       |
| DE 48        | Subfield 1 (E-ID Request Value)                                            | Two                       |
| DE 48        | Subelement 55 (Merchant Fraud Scoring Data/Approval Probability Indicator) | Two                       |
| DE 48        | Subfield 1 (Merchant Fraud Score/ASI Probability Indicator)                | Two                       |
| DE 48        | Subfield 2 (Reason Code)                                                   | Two                       |
| DE 48        | Subfield 3 (Reserved for Future Use)                                       | Two                       |
| DE 48        | Subfield 4 (Reserved for Future Use)                                       | Two                       |
| DE 48        | Subfield 5 (Reserved for Future Use)                                       | Two                       |
| DE 48        | Subelement 56 (Security Services Additional Data for Issuers)              | Two                       |
| DE 48        | Subfield 1 (Security Services Indicator)                                   | Two                       |
| DE 48        | Subfield 2 (Security Services Data)                                        | Two                       |
| DE 48        | Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)         | Two                       |
| DE 48        | Subelement 57 (Security Services Additional Data for Acquirers)            | Two                       |
| DE 48        | Subfield 1 (Security Services Indicator)                                   | Two                       |
| DE 48        | Subfield 2 (Security Services Data)                                        | Two                       |
| DE 48        | Subelement 58 (ATM Additional Data)                                        | One                       |
| DE 48        | Subfield 1 (ATM Time)                                                      | One                       |
| DE 48        | Subfield 2 (ATM Date)                                                      | One                       |
| DE 48        | Subfield 3 (Watermark)                                                     | One                       |
| DE 48        | Subfield 4 (Mark 1)                                                        | One                       |
| DE 48        | Subfield 5 (Mark 2)                                                        | One                       |
| DE 48        | Subfield 6 (Mark 3)                                                        | One                       |

| Data element | Data element description                                            | Transaction data category |
|--------------|---------------------------------------------------------------------|---------------------------|
| DE 48        | Subfield 7 (Card Swallowed Status)                                  | One                       |
| DE 48        | Subfield 8 (Posting Date)                                           | One                       |
| DE 48        | Subelement 60 (Additional Service Data For Issuers)                 | One                       |
| DE 48        | Subfield 01 (Reference Conversion Rate)                             | One                       |
| DE 48        | Subfield 02 (Reference Conversion Rate Differential Sign Indicator) | One                       |
| DE 48        | Subfield 03 (Reference Conversion Rate Percent Differential)        | One                       |
| DE 48        | Subfield 04 (Currency Rate Unique Identifier)                       | One                       |
| DE 48        | Subelement 61 (POS Data Extended Condition Codes)                   | One                       |
| DE 48        | Subfield 1 (Partial Approval Terminal Support Indicator)            | One                       |
| DE 48        | Subfield 2 (Purchase Amount Only Terminal Support Indicator)        | One                       |
| DE 48        | Subfield 3 (Real-time Substantiation Indicator)                     | One                       |
| DE 48        | Subfield 4 (Merchant Transaction Fraud Scoring Indicator)           | One                       |
| DE 48        | Subfield 5 (Final Authorization Indicator)                          | One                       |
| DE 48        | Subelement 62 (Real-Time Payment Information)                       | Two                       |
| DE 48        | Subfield 1 (Merchant Direct Participation)                          | Two                       |
| DE 48        | Subfield 2 (Type of Transaction)                                    | Two                       |
| DE 48        | Subfield 3 (Real-Time Payment Rules)                                | Two                       |
| DE 48        | Subfield 4 (Real-Time Payment Settlement Network ID and Name)       | Two                       |
| DE 48        | Subfield 5 (Real-Time Payment Funds Transfer Position)              | Two                       |

| Data element | Data element description                          | Transaction data category |
|--------------|---------------------------------------------------|---------------------------|
| DE 48        | Subfield 6 (DSS Reference Number)                 | Two                       |
| DE 48        | Subelement 63 (Trace ID)                          | One                       |
| DE 48        | Subelement 64 (Transit Program)                   | One                       |
| DE 48        | Subfield 1 (Transit Transaction Type Indicator)   | One                       |
| DE 48        | Subfield 2 (Transportation Mode Indicator)        | One                       |
| DE 48        | Subelement 65 (Terminal Compliant Indicator)      | One                       |
| DE 48        | Subfield 1 (TLE Compliant)                        | One                       |
| DE 48        | Subfield 2 (UKPT/DUKPT Compliant)                 | One                       |
| DE 48        | Subelement 66 (Authentication Data)               | Two                       |
| DE 48        | Subfield 1 (Program Protocol)                     | Two                       |
| DE 48        | Subfield 2 (Directory Server Transaction ID)      | Two                       |
| DE 48        | Subelement 67 (Mastercard Send Information)       | Two                       |
| DE 48        | Subfield 1 (Sanctions Score)                      | Two                       |
| DE 48        | Subelement 68 (Financial Account Information)     | Two                       |
| DE 48        | Subfield 1 (Financial Account Identifier)         | Two                       |
| DE 48        | Subelement 71 (On-behalf Services)                | Two                       |
| DE 48        | Subfield 1 (On-behalf (OB) Service)               | Two                       |
| DE 48        | Subfield 2 (On-behalf Result 1)                   | Two                       |
| DE 48        | Subfield 3 (On-behalf Result 2)                   | Two                       |
| DE 48        | Subelement 72 (Issuer Chip Authentication)        | Two                       |
| DE 48        | Subelement 74 (Additional Processing Information) | One                       |
| DE 48        | Subfield 1 (Processing Indicator)                 | One                       |
| DE 48        | Subfield 2 (Processing Information)               | One                       |

| Data element | Data element description                                       | Transaction data category |
|--------------|----------------------------------------------------------------|---------------------------|
| DE 48        | Subelement 75 (Fraud Scoring Data)                             | Two                       |
| DE 48        | Subfield 1 (Fraud Score)                                       | Two                       |
| DE 48        | Subfield 2 (Fraud Reason Code)                                 | Two                       |
| DE 48        | Subfield 3 (Model Score)                                       | Two                       |
| DE 48        | Subfield 4 (Model Reason Code)                                 | Two                       |
| DE 48        | Subfield 5 (Reserved for Future Use)                           | Two                       |
| DE 48        | Subelement 76 (Mastercard Electronic Acceptance Indicator)     | One                       |
| DE 48        | Subelement 77 (Transaction Type Identifier)                    | One                       |
| DE 48        | Subelement 78 (Payment Service Indicators [Visa Only])         | Two                       |
| DE 48        | Subelement 79 (Chip CVR/TVR Bit Error Results)                 | Two                       |
| DE 48        | Subfield 1 (CVR or TVR Identifier)                             | Two                       |
| DE 48        | Subfield 2 (Byte ID)                                           | Two                       |
| DE 48        | Subfield 3 (Bit Identifier)                                    | Two                       |
| DE 48        | Subfield 4 (Value of Bit in Error)                             | Two                       |
| DE 48        | Subelement 80 (PIN Service Code)                               | Two                       |
| DE 48        | Subelement 82 (Address Verification Service Request)           | One                       |
| DE 48        | Subelement 83 (Address Verification Service Response)          | One                       |
| DE 48        | Subelement 84 (Merchant Advice Code)                           | One                       |
| DE 48        | Subelement 84 (Visa Response Codes [Visa Only])                | One                       |
| DE 48        | Subelement 85 (Account Status [Visa Only])                     | One                       |
| DE 48        | Subelement 86 (Relationship Participant Indicator [Visa Only]) | One                       |
| DE 48        | Subelement 87 (Card Validation Code Result)                    | One                       |



| Data element | Data element description                                                   | Transaction data category |
|--------------|----------------------------------------------------------------------------|---------------------------|
| DE 48        | Subelement 87 (CVV2 Response [Visa Only])                                  | One                       |
| DE 48        | Subelement 88 (Magnetic Stripe Compliance Status Indicator)                | Two                       |
| DE 48        | Subelement 89 (Magnetic Stripe Compliance Error Indicator)                 | Two                       |
| DE 48        | Subelement 90 (Lodging and Auto Rental Indicator)                          | One                       |
| DE 48        | Subelement 90 (Custom Payment Service Request [Visa Only])                 | One                       |
| DE 48        | Subelement 90 (Custom Payment Service Request Response [Visa Only])        | One                       |
| DE 48        | Subelement 91 (Acquirer Reference Data [American Express Only])            | One                       |
| DE 48        | Subelement 91 (Custom Payment Service Request/Transaction ID [Visa Only])  | One                       |
| DE 48        | Subelement 91 (Custom Payment Service Response/Transaction ID [Visa Only]) | One                       |
| DE 48        | Subelement 92 (CVC 2)                                                      | One                       |
| DE 48        | Subelement 92 (CVV2 Data [Visa Only])                                      | One                       |
| DE 48        | Subelement 93 (Fleet Card ID Request Data [Visa Only])                     | One                       |
| DE 48        | Subelement 94 (Commercial Card Inquiry Request [Visa Only])                | One                       |
| DE 48        | Subelement 94 (Commercial Card Inquiry Response [Visa Only])               | One                       |
| DE 48        | Subelement 95 (Mastercard Promotion Code)                                  | Two                       |
| DE 48        | Subelement 96 (Visa Market-Specific Data Identifier [Visa Only])           | One                       |
| DE 48        | Subelement 97 (Prestigious Properties Indicator [Visa Only])               | One                       |

| Data element | Data element description                                         | Transaction data category |
|--------------|------------------------------------------------------------------|---------------------------|
| DE 48        | Subelement 98 (Mastercard Corporate Fleet Card ID/Driver Number) | Two                       |
| DE 48        | Subelement 99 (Mastercard Corporate Fleet Card Vehicle Number)   | Two                       |
| DE 49        | Currency Code, Transaction                                       | One                       |
| DE 50        | Currency Code, Settlement                                        | One                       |
| DE 51        | Currency Code, Cardholder Billing                                | One                       |
| DE 52        | Personal ID Number [PIN] Data                                    | One                       |
| DE 53        | Security-Related Control Information                             | One                       |
| DE 53        | Subfield 1 (PIN Security Type Code)                              | One                       |
| DE 53        | Subfield 2 (PIN Encryption Type Code)                            | One                       |
| DE 53        | Subfield 3 (PIN Block Format Code)                               | One                       |
| DE 53        | Subfield 4 (PIN Key Index Number)                                | One                       |
| DE 53        | Subfield 5 (Reserved for Future Use)                             | One                       |
| DE 53        | Subfield 6 (Reserved for Future Use)                             | One                       |
| DE 54        | Additional Amounts                                               | One                       |
| DE 54        | Subfield 1 (Account Type)                                        | One                       |
| DE 54        | Subfield 2 (Amount Type)                                         | One                       |
| DE 54        | Subfield 3 (Currency Code)                                       | One                       |
| DE 54        | Subfield 4 (Amount)                                              | One                       |
| DE 55        | Integrated Circuit Card [ICC] System-Related Data                | One                       |
| DE 56        | Payment Account Data                                             | One                       |
| DE 56        | Subelement 01 (Payment Account Data)                             | One                       |
| DE 56        | Subfield 01 (Payment Account Reference [PAR])                    | One                       |
| DE 60        | Advice Reason Code                                               | One                       |
| DE 60        | Subfield 1 (Advice Reason Code)                                  | One                       |

| Data element | Data element description                                                          | Transaction data category |
|--------------|-----------------------------------------------------------------------------------|---------------------------|
| DE 60        | Subfield 2 (Advice Detail Code)                                                   | One                       |
| DE 60        | Subfield 3 (Advice Detail Text)                                                   | One                       |
| DE 61        | Point-of-Service [POS] Data                                                       | One                       |
| DE 61        | Subfield 1 (POS Terminal Attendance)                                              | One                       |
| DE 61        | Subfield 2 (Reserved for Future Use)                                              | One                       |
| DE 61        | Subfield 3 (POS Terminal Location)                                                | One                       |
| DE 61        | Subfield 4 (POS Cardholder Presence)                                              | One                       |
| DE 61        | Subfield 5 (POS Card Presence)                                                    | One                       |
| DE 61        | Subfield 6 (POS Card Capture Capabilities)                                        | One                       |
| DE 61        | Subfield 7 (POS Transaction Status)                                               | One                       |
| DE 61        | Subfield 8 (POS Transaction Security)                                             | One                       |
| DE 61        | Subfield 9 (POS Transaction Status - Extended)                                    | One                       |
| DE 61        | Subfield 10 (Cardholder-Activated Terminal Level)                                 | One                       |
| DE 61        | Subfield 11 (POS Card Data Terminal Input Capability Indicator)                   | One                       |
| DE 61        | Subfield 12 (POS Authorization Life Cycle)                                        | One                       |
| DE 61        | Subfield 13 (POS Country Code [or Sponsored Merchant Information, if applicable]) | One                       |
| DE 61        | Subfield 14 (Acceptor Postal Code)                                                | One                       |
| DE 62        | Intermediate Network Facility [INF] Data                                          | Two                       |
| DE 63        | Network Data                                                                      | Two                       |
| DE 63        | Subfield 1 (Financial Network Code)                                               | One                       |
| DE 63        | Subfield 2 (Banknet Reference Number)                                             | One                       |

| Data element | Data element description                                     | Transaction data category |
|--------------|--------------------------------------------------------------|---------------------------|
| DE 64        | Message Authentication Code                                  | Two                       |
| DE 65        | Bit Map, Extended                                            | One                       |
| DE 67        | Extended Payment Code                                        | One                       |
| DE 70        | Network Management Information Code                          | Two                       |
| DE 90        | Original Data Elements                                       | One                       |
| DE 90        | Subfield 1 (Original Message Type Identifier)                | One                       |
| DE 90        | Subfield 2 (Original DE 11 [Systems Trace Audit Number])     | One                       |
| DE 90        | Subfield 3 (Original DE 7 [Transmission Date and Time])      | One                       |
| DE 90        | Subfield 4 (Original DE 32 [Acquiring Institution ID Code])  | One                       |
| DE 90        | Subfield 5 (Original DE 33 [Forwarding Institution ID Code]) | One                       |
| DE 91        | Issuer File Update Code                                      | Two                       |
| DE 94        | Service Indicator                                            | Two                       |
| DE 94        | Subfield 1 (Reserved for Future Use)                         | Two                       |
| DE 94        | Subfield 2 (Acquirer/Issuer Indicator)                       | Two                       |
| DE 94        | Subfield 3 (Address Data Indicator)                          | Two                       |
| DE 95        | Replacement Amounts                                          | One                       |
| DE 95        | Subfield 1 (Actual Amount, Transaction)                      | One                       |
| DE 95        | Subfield 2 (Actual Amount, Settlement)                       | One                       |
| DE 95        | Subfield 3 (Actual Amount, Cardholder Billing)               | One                       |
| DE 95        | Subfield 4 (Zero Fill)                                       | One                       |
| DE 96        | Message Security Code                                        | Two                       |
| DE 100       | Receiving Institution ID Code                                | One                       |
| DE 101       | File Name                                                    | Two                       |
| DE 102       | Account ID 1                                                 | One                       |

| Data element | Data element description                                                    | Transaction data category |
|--------------|-----------------------------------------------------------------------------|---------------------------|
| DE 103       | Account ID 2                                                                | One                       |
| DE 104       | Digital Payment Data                                                        | Two                       |
| DE 104       | Subelement 001 (Digital Payment Cryptogram)                                 | Two                       |
| DE 104       | Subelement 002 (Estimated Amount)                                           | Two                       |
| DE 104       | Subelement 003 (Remote Commerce Acceptor Identifier)                        | Two                       |
| DE 104       | Subelement 004 (Digital Service Provider)                                   | Two                       |
| DE 104       | Subfield 001 (Digital Service Provider ID 1)                                | Two                       |
| DE 104       | Subfield 002 (Digital Service Provider ID 2)                                | Two                       |
| DE 104       | Subelement 005 (Digital Authentication Data)                                | Two                       |
| DE 104       | Subfield 001 (Cardholder Verification)                                      | Two                       |
| DE 104       | Subfield 002 (Authentication Factors)                                       | Two                       |
| DE 105       | Multi-Use Transaction Identification Data                                   | One                       |
| DE 105       | Subelement 001 (Transaction Link ID [TLID])                                 | One                       |
| DE 106       | Fleet Service Data                                                          | Two                       |
| DE 106       | Subelement 001 (Fleet Prompted Data)                                        | Two                       |
| DE 106       | Subfield 001 (Fleet Prompt 1)                                               | Two                       |
| DE 106       | Subfield 002 (Fleet Prompt 2)                                               | Two                       |
| DE 106       | Subfield 003 (Fleet Prompt 3)                                               | Two                       |
| DE 106       | Subfield 004 (Fleet Prompt 4)                                               | Two                       |
| DE 106       | Subfield 005 (Fleet Prompt 5)                                               | Two                       |
| DE 106       | Subelement 002 (Merchant Fleet Spend Control Override Capability Indicator) | Two                       |

| Data element | Data element description                            | Transaction data category |
|--------------|-----------------------------------------------------|---------------------------|
| DE 106       | Subelement 003 (Fleet Spend Control Override Items) | Two                       |
| DE 106       | Subelement 004 (Fleet Fuel Information)             | Two                       |
| DE 106       | Subfield 001 (Fuel Item Purchased 1)                | Two                       |
| DE 106       | Subfield 002 (Fuel Item Purchased 2)                | Two                       |
| DE 106       | Subfield 003 (Fuel Item Purchased 3)                | Two                       |
| DE 106       | Subelement 005 (Fleet Non-Fuel Information)         | Two                       |
| DE 106       | Subfield 001 (Non-Fuel Item Purchased 1)            | Two                       |
| DE 106       | Subfield 002 (Non-Fuel Item Purchased 2)            | Two                       |
| DE 106       | Subfield 003 (Non-Fuel Item Purchased 3)            | Two                       |
| DE 106       | Subfield 004 (Non-Fuel Item Purchased 4)            | Two                       |
| DE 106       | Subfield 005 (Non-Fuel Item Purchased 5)            | Two                       |
| DE 106       | Subfield 006 (Non-Fuel Item Purchased 6)            | Two                       |
| DE 106       | Subfield 007 (Non-Fuel Item Purchased 7)            | Two                       |
| DE 106       | Subfield 008 (Non-Fuel Item Purchased 8)            | Two                       |
| DE 106       | Subfield 009 (Non-Fuel Item Purchased 9)            | Two                       |
| DE 106       | Subfield 010 (Non-Fuel Item Purchased 10)           | Two                       |
| DE 106       | Subfield 011 (Non-Fuel Item Purchased 11)           | Two                       |
| DE 106       | Subfield 012 (Non-Fuel Item Purchased 12)           | Two                       |
| DE 108       | Additional Transaction Reference Data               | Two                       |

| Data element | Data element description                                           | Transaction data category |
|--------------|--------------------------------------------------------------------|---------------------------|
| DE 108       | Subelement 01 (Receiver/Recipient Data)                            | Two                       |
| DE 108       | Subfield 01 (Receiver/Recipient First Name)                        | Two                       |
| DE 108       | Subfield 02 (Receiver/Recipient Middle Name)                       | Two                       |
| DE 108       | Subfield 03 (Receiver/Recipient Last Name)                         | Two                       |
| DE 108       | Subfield 04 (Receiver/Recipient Street Address)                    | Two                       |
| DE 108       | Subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only]) |                           |
| DE 108       | Subfield 05 (Receiver/Recipient City)                              | Two                       |
| DE 108       | Subfield 06 (Receiver/Recipient State/Province Code)               | Two                       |
| DE 108       | Subfield 07 (Receiver/Recipient Country)                           | Two                       |
| DE 108       | Subfield 08 (Receiver/Recipient Postal Code)                       | Two                       |
| DE 108       | Subfield 08 (Receiver/Recipient Postal Code [Visa Only])           | Two                       |
| DE 108       | Subfield 09 (Receiver/Recipient Phone Number)                      | Two                       |
| DE 108       | Subfield 10 (Receiver/Recipient Date of Birth)                     | Two                       |
| DE 108       | Subfield 11 (Receiver/Recipient Account Number)                    | Two                       |
| DE 108       | Subfield 12 (Receiver/Recipient Identification Type)               | Two                       |
| DE 108       | Subfield 13 (Receiver/Recipient Identification Number)             | Two                       |
| DE 108       | Subfield 13 (Receiver/Recipient Identification Number [Visa Only]) | Two                       |
| DE 108       | Subfield 14 (Receiver/Recipient Identification Country Code)       | Two                       |
| DE 108       | Subfield 15 (Receiver/Recipient Identification Expiration Date)    | Two                       |

| Data element | Data element description                                                     | Transaction data category |
|--------------|------------------------------------------------------------------------------|---------------------------|
| DE 108       | Subfield 16 (Receiver/Recipient Nationality)                                 | Two                       |
| DE 108       | Subfield 17 (Receiver/Recipient Country of Birth)                            | Two                       |
| DE 108       | Subfield 18 (Receiver/Recipient Account Number Type)                         | Two                       |
| DE 108       | Subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only])           | Two                       |
| DE 108       | Subfield 20 (Receiver/Recipient Street Name [Visa Only])                     | Two                       |
| DE 108       | Subfield 21 (Receiver/Recipient Building Number [Visa Only])                 | Two                       |
| DE 108       | Subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only]) | Two                       |
| DE 108       | Subelement 02 (Sender Data)                                                  | Two                       |
| DE 108       | Subfield 01 (Sender First Name)                                              | Two                       |
| DE 108       | Subfield 02 (Sender Middle Name)                                             | Two                       |
| DE 108       | Subfield 03 (Sender Last Name)                                               | Two                       |
| DE 108       | Subfield 04 (Sender Street Address)                                          | Two                       |
| DE 108       | Subfield 05 (Sender City)                                                    | Two                       |
| DE 108       | Subfield 06 (Sender State/Province Code)                                     | Two                       |
| DE 108       | Subfield 07 (Sender Country)                                                 | Two                       |
| DE 108       | Subfield 08 (Postal Code)                                                    | Two                       |
| DE 108       | Subfield 09 (Sender Phone Number)                                            | Two                       |
| DE 108       | Subfield 10 (Sender Date of Birth)                                           | Two                       |
| DE 108       | Subfield 11 (Sender Account Number)                                          | Two                       |
| DE 108       | Subfield 12 (Sender Identification Type)                                     | Two                       |
| DE 108       | Subfield 13 (Sender Identification Number)                                   | Two                       |



| Data element | Data element description                                                | Transaction data category |
|--------------|-------------------------------------------------------------------------|---------------------------|
| DE 108       | Subfield 13 (Sender Identification Number [Visa Only])                  | Two                       |
| DE 108       | Subfield 14 (Sender Identification Country Code)                        | Two                       |
| DE 108       | Subfield 15 (Sender Identification Expiration Date)                     | Two                       |
| DE 108       | Subfield 16 (Sender Nationality)                                        | Two                       |
| DE 108       | Subfield 17 (Sender Country of Birth)                                   | Two                       |
| DE 108       | Subfield 18 (Sender Account Number Type)                                | Two                       |
| DE 108       | Subelement 03 (Transaction Reference Data)                              | Two                       |
| DE 108       | Subfield 01 (Unique Transaction Reference)                              | Two                       |
| DE 108       | Subfield 01 (Sender Reference Number [Visa Only])                       | Two                       |
| DE 108       | Subfield 02 (Additional Message)                                        | Two                       |
| DE 108       | Subfield 03 (Funding Source)                                            | Two                       |
| DE 108       | Subfield 04 (Participation ID)                                          | Two                       |
| DE 108       | Subfield 05 (Transaction Purpose)                                       | Two                       |
| DE 108       | Subelement 04 (Language Description)                                    | Two                       |
| DE 108       | Subfield 01 (Language Identification)                                   | Two                       |
| DE 108       | Subfield 02 (Language Data)                                             | Two                       |
| DE 108       | Subelement 05 (Digital Account Information)                             | Two                       |
| DE 108       | Subfield 01 (Digital Account Reference Number)                          | Two                       |
| DE 108       | Subfield 02 (Mastercard Merchant Presented QR Receiving Account Number) | Two                       |
| DE 108       | Subelement 06 (QR Dynamic Code Data)                                    | Two                       |
| DE 108       | Subelement 07 (Data Validation Request)                                 | Two                       |

| Data element | Data element description                                                      | Transaction data category |
|--------------|-------------------------------------------------------------------------------|---------------------------|
| DE 108       | Subfield 01 (Name Validation Request)                                         | Two                       |
| DE 108       | Subelement 08 (Data Validation Response)                                      | Two                       |
| DE 108       | Subfield 01 (Receiver Name Validation Response)                               | Two                       |
| DE 108       | Subfield 02 (Sender Name Validation Response)                                 | Two                       |
| DE 108       | Subelement 09 (Account Owner Information 1)                                   | Two                       |
| DE 108       | Subfield 01 (Account Owner First Name)                                        | Two                       |
| DE 108       | Subfield 02 (Account Owner Middle Name)                                       | Two                       |
| DE 108       | Subfield 03 (Account Owner Last Name)                                         | Two                       |
| DE 108       | Subfield 04 (Account Owner Sequence Number)                                   | Two                       |
| DE 108       | Subfield 05 (Account Owner Classification)                                    | Two                       |
| DE 108       | Subelement 10 (Account Owner Information 2)                                   | Two                       |
| DE 108       | Subfield 01 (Account Owner First Name)                                        | Two                       |
| DE 108       | Subfield 02 (Account Owner Middle Name)                                       | Two                       |
| DE 108       | Subfield 03 (Account Owner Last Name)                                         | Two                       |
| DE 108       | Subfield 04 (Account Owner Sequence Number)                                   | Two                       |
| DE 108       | Subfield 05 (Account Owner Classification)                                    | Two                       |
| DE 110       | Additional Data-2                                                             | Two                       |
| DE 110       | Subelement 9 (ANSI X9 TR-31 Key Block Key [128-bit Key Block Protection Key]) | Two                       |

| Data element | Data element description                                                      | Transaction data category |
|--------------|-------------------------------------------------------------------------------|---------------------------|
| DE 110       | Subelement 9 (ANSI X9 TR-31 Key Block Key [192-bit Key Block Protection Key]) | Two                       |
| DE 110       | Subelement 10 (Key Check Value)                                               | Two                       |
| DE 110       | Encryption Data                                                               | Two                       |
| DE 110       | Dataset 01 (PIN Encryption)                                                   | Two                       |
| DE 110       | TAG 80 (Control)                                                              | Two                       |
| DE 110       | TAG 81 (Key-set Identifier)                                                   | Two                       |
| DE 110       | TAG 83 (Algorithm)                                                            | Two                       |
| DE 110       | TAG 86 (Key Index)                                                            | Two                       |
| DE 110       | TAG 87 (PIN Block Format)                                                     | Two                       |
| DE 110       | TAG 88 (Encrypted PIN Block)                                                  | Two                       |
| DE 110       | TAG 89 (Additional Encrypted PIN Block)                                       | Two                       |
| DE 110       | Dataset 04 (Key Exchange)                                                     | Two                       |
| DE 110       | TAG 80 (Control)                                                              | Two                       |
| DE 110       | TAG 81 (Key-set Identifier)                                                   | Two                       |
| DE 110       | TAG 83 (Algorithm)                                                            | Two                       |
| DE 110       | TAG 86 (Key Index)                                                            | Two                       |
| DE 112       | Additional Data [National Use]                                                | Two                       |
| DE 112       | Subelement 042 (Acceptor Tax ID)                                              | Two                       |
| DE 112       | Subelement 045 (Additional Commercial Issuer Data)                            | Two                       |
| DE 112       | Subfield 01 (Taxpayer Country Code)                                           | Two                       |
| DE 112       | Subfield 02 (Taxpayer Identification Number)                                  | Two                       |
| DE 112       | Subfield 03 (Tax Registration Reason Code)                                    | Two                       |
| DE 112       | Subelement 055 (Brazil Benefits Payment Data)                                 | Two                       |
| DE 112       | Subfield 01 (PAT Eligibility)                                                 | Two                       |
| DE 112       | Subfield 02 (Benefit Program Code)                                            | Two                       |

| Data element | Data element description                                                                                 | Transaction data category |
|--------------|----------------------------------------------------------------------------------------------------------|---------------------------|
| DE 112       | Subelement 60 (Mastercard Network Gateway Services Platform [MNGS] Transport Data) (Mastercard use only) | Two                       |
| DE 112       | All Regions: Installment Payment Transactions                                                            | One                       |
| DE 112       | Subelement 022 (Installment Payment Data)                                                                | One                       |
| DE 112       | Subfield 01 (Installment Plan ID)                                                                        | One                       |
| DE 112       | Brazil: Payment Transactions                                                                             | One                       |
| DE 112       | Subelement 012 (Brazil Commercial and Financing Data)                                                    | One                       |
| DE 112       | Subelement 013 (Crediário First Simulation)                                                              | One                       |
| DE 112       | Subelement 014 (Crediário Second Simulation)                                                             | One                       |
| DE 112       | Subelement 015 (Crediário Third Simulation)                                                              | One                       |
| DE 112       | Subelement 018 (Brazil Post-Dated Transaction Data)                                                      | One                       |
| DE 112       | Subelement 019 (Original Purchase Amount)                                                                | One                       |
| DE 112       | Brazil: Merchant Fraud Scoring Data                                                                      | Two                       |
| DE 112       | Subelement 028 (Merchant Fraud Score Data)                                                               | Two                       |
| DE 112       | Canada Domestic: Transactions                                                                            | Two                       |
| DE 112       | Subelement 047 (Payment Consent ID)                                                                      | Two                       |
| DE 112       | Chile: Payment Transactions                                                                              | One                       |
| DE 112       | Subelement 010 (Installment Payment Data)                                                                | One                       |
| DE 112       | Subelement 042 (Acceptor Tax ID)                                                                         | One                       |
| DE 112       | Colombia: Domestic Transactions                                                                          | One                       |
| DE 112       | Subelement 035 (Issuer Fee Inquiry Indicator)                                                            | One                       |

| Data element | Data element description                                   | Transaction data category |
|--------------|------------------------------------------------------------|---------------------------|
| DE 112       | Subelement 036 (Issuer Fee Amount)                         | One                       |
| DE 112       | Colombia: Payment Transactions                             | One                       |
| DE 112       | Subelement 010 (Installment Payment Data)                  | One                       |
| DE 112       | Subelement 011 (Customer ID)                               | One                       |
| DE 112       | Cuotas: Payment Transactions                               | One                       |
| DE 112       | Subelement 001 (Installment Payment Data)                  | One                       |
| DE 112       | Subelement 003 (Installment Payment Response Data)         | One                       |
| DE 112       | Subelement 027 (ATM Credit Card Cash Advance Installments) | One                       |
| DE 112       | Subfield 1 (Transaction Type)                              | One                       |
| DE 112       | Subfield 2 (Requested Number of Installments)              | One                       |
| DE 112       | Subfield 3 (Approved Number of Installments)               | One                       |
| DE 112       | Subfield 4 (Installment Amount)                            | One                       |
| DE 112       | Subfield 5 (Total Transaction Amount)                      | One                       |
| DE 112       | Subfield 6 (Yearly Interest Rate)                          | One                       |
| DE 112       | Subfield 7 (Currency Code)                                 | One                       |
| DE 112       | Subfield 8 (Member-Defined Data)                           | One                       |
| DE 112       | Subfield 9 (Member-Defined Data)                           | One                       |
| DE 112       | Europe Region and Philippines: Payment Transactions        | One                       |
| DE 112       | Subelement 009 (Installment Payment Data)                  | One                       |
| DE 112       | Subelement 020 (Domestic Acceptor Tax ID)                  | One                       |
| DE 112       | France: Payment Transactions                               | One                       |
| DE 112       | Subelement 020 (Domestic Acceptor Tax ID)                  | One                       |
| DE 112       | Subelement 096 (Transit Offer ID)                          | One                       |

| Data element | Data element description                                           | Transaction data category |
|--------------|--------------------------------------------------------------------|---------------------------|
| DE 112       | Subelement 097 (Application Selection ID)                          | One                       |
| DE 112       | Subelement 098 (Brand ID)                                          | One                       |
| DE 112       | Greece: Payment Transactions                                       | One                       |
| DE 112       | Subelement 006 (Installment Payment Data)                          | One                       |
| DE 112       | Subelement 008 (Installment Payment Response Data)                 | One                       |
| DE 112       | Hungary: Benefits Program                                          | One                       |
| DE 112       | Subelement 013 (Program ID)                                        | One                       |
| DE 112       | India: Payment Transactions                                        | One                       |
| DE 112       | Subelement 044 (Installment and Recurring Payment Data: Visa Only) | One                       |
| DE 112       | Subfield 01 (Transaction Type)                                     | One                       |
| DE 112       | Subfield 02 (Recurring Payment Validation)                         | One                       |
| DE 112       | Subfield 03 (Maximum Recurring Payment Amount)                     | One                       |
| DE 112       | Subfield 04 (Registered Transaction Identifier)                    | One                       |
| DE 112       | Subfield 05 (Number of Payments)                                   | One                       |
| DE 112       | Subfield 06 (Payment Frequency)                                    | One                       |
| DE 112       | Subfield 07 (Total Installment Amount)                             | One                       |
| DE 112       | Subfield 08 (Currency Code)                                        | One                       |
| DE 112       | Subfield 09 (Installment Interest Rate)                            | One                       |
| DE 112       | Subfield 10 (Installment Payment Entity)                           | One                       |
| DE 112       | Subfield 11 (Installment Interest Charged)                         | One                       |
| DE 112       | Subfield 12 (Member-Defined Data)                                  | One                       |
| DE 112       | Japan: Payment Transactions                                        | One                       |

| Data element | Data element description                                         | Transaction data category |
|--------------|------------------------------------------------------------------|---------------------------|
| DE 112       | Subelement 030 (Japan Domestic POS Data)                         | One                       |
| DE 112       | Subelement 031 (Japan Domestic Response Code)                    | One                       |
| DE 112       | Subelement 032 (Japan Payment Options)                           | One                       |
| DE 112       | Subfield 1 (Payment Option Code)                                 | One                       |
| DE 112       | Subfield 2 (Bonuses Per Year)                                    | One                       |
| DE 112       | Subfield 3 (First Bonus Month)                                   | One                       |
| DE 112       | Subfield 4 (First Bonus Amount)                                  | One                       |
| DE 112       | Subfield 5 (Second Bonus Month)                                  | One                       |
| DE 112       | Subfield 6 (Second Bonus Amount)                                 | One                       |
| DE 112       | Subfield 7 (Total Number of Installments)                        | One                       |
| DE 112       | Subfield 8 (First Installment Month)                             | One                       |
| DE 112       | Mexcta: Payment Transactions                                     | One                       |
| DE 112       | Subelement 004 (Credit Line Usage Fee (CLUF))                    | One                       |
| DE 112       | Subelement 005 (Issuing Bank Name (AKA Doing Business As [DBA])) | One                       |
| DE 112       | Subelement 006 (Financial Institution ID (FIID))                 | One                       |
| DE 112       | Subelement 007 (Installment Payment Data)                        | One                       |
| DE 112       | Subelement 008 (Installment Payment Response Data)               | One                       |
| DE 112       | Netherlands: IBAN (Account Inquiry)                              | One                       |
| DE 112       | Subelement 037 (Additional Cardholder Information)               | One                       |
| DE 112       | Subfield 01 (Primary Cardholder Identifier)                      | One                       |
| DE 112       | Subfield 02 (Secondary Cardholder Identifier)                    | One                       |

| Data element | Data element description                                      | Transaction data category |
|--------------|---------------------------------------------------------------|---------------------------|
| DE 112       | Paraguay: Payment Transactions                                | One                       |
| DE 112       | Subelement 042 (Acceptor Tax ID)                              | One                       |
| DE 112       | Parcelas: Payment Transactions                                | One                       |
| DE 112       | Subelement 001 (Installment Payment Data)                     | One                       |
| DE 112       | Subelement 002 (Installment Payment Response Data)            | One                       |
| DE 112       | Subelement 016 (Additional Installment Payment Response Data) | One                       |
| DE 112       | Percta: Payment Transactions                                  | One                       |
| DE 112       | Subelement 007 (Installment Payment Data)                     | One                       |
| DE 112       | Subelement 008 (Installment Payment Response Data)            | One                       |
| DE 112       | Poland: Value-Added Tax specifications                        | One                       |
| DE 112       | Subelement 041 (Poland VAT Data)                              | One                       |
| DE 112       | Spain: Domestic ATM Transactions                              | One                       |
| DE 112       | Subelement 017 (ATM Domestic Fee)                             | One                       |
| DE 112       | Subfield 1 (ATM Service Fee)                                  | One                       |
| DE 112       | Subfield 2 (ATM Disloyalty Fee)                               | One                       |
| DE 112       | Subfield 3 (Credit Card Fee)                                  | One                       |
| DE 112       | Taiwan: National Data                                         | One                       |
| DE 112       | Subelement 038 (Taiwan National Data)                         | One                       |
| DE 112       | United Kingdom: Debt Repayment Transactions                   | One                       |
| DE 112       | Subelement 033 (UK Recipient Details)                         | One                       |
| DE 112       | Subfield 1 (Recipient Last Name)                              | One                       |
| DE 112       | Subfield 2 (Recipient Postal Code)                            | One                       |
| DE 112       | Subfield 3 (Recipient Date of Birth)                          | One                       |



| Data element | Data element description                                           | Transaction data category |
|--------------|--------------------------------------------------------------------|---------------------------|
| DE 112       | Subfield 4 (Recipient Account Number)                              | One                       |
| DE 119       | Additional Data: Private Use 2                                     | Two                       |
| DE 119       | Subelement 003 (MDES Service Indicators for Issuers)               | Two                       |
| DE 119       | Subelement 004 (MDES Service Indicators for Acquirers)             | Two                       |
| DE 119       | Subelement 005 (MDES Service Indicators for Issuers and Acquirers) | Two                       |
| DE 120       | Record Data                                                        | Two                       |
| DE 120       | Subfield 01 (AVS Service Indicator 1)                              | Two                       |
| DE 120       | Subfield 02 (AVS Service Indicator 2)                              | Two                       |
| DE 120       | Subfield 03 (AVS Service Indicator 3)                              | Two                       |
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| DE 121       | Authorizing Agent ID Code                                          | Two                       |
| DE 122       | Additional Record Data                                             | One                       |
| DE 122       | Subelement 001 (Additional Acceptor Data)                          | One                       |
| DE 122       | Subfield 001 (Acceptor Street Address)                             | One                       |
| DE 122       | Subfield 002 (Acceptor Country Subdivision Code)                   | One                       |
| DE 122       | Subfield 003 (Acceptor URL address)                                | One                       |
| DE 122       | Subfield 004 (Acceptor Customer Service Phone Number)              | One                       |
| DE 122       | Subfield 005 (Acceptor Phone Number)                               | One                       |
| DE 122       | Subfield 006 (Acceptor Additional Contact Information)             | One                       |
| DE 122       | Subfield 007 (Acceptor Tax ID)                                     | Two                       |
| DE 122       | Subfield 008 (Partner ID Code)                                     | One                       |

| Data element | Data element description                                   | Transaction data category |
|--------------|------------------------------------------------------------|---------------------------|
| DE 122       | Subfield 009 (Service Location City Name)                  | One                       |
| DE 122       | Subfield 010 (Service Location Country Subdivision Code)   | One                       |
| DE 122       | Subfield 011 (Service Location Country Code)               | One                       |
| DE 122       | Subfield 012 (Service Location Postal Code)                | One                       |
| DE 122       | Subfield 013 (Acceptor Geographic Coordinates)             | One                       |
| DE 122       | Subfield 014 (Service Location Geographic Coordinates)     | One                       |
| DE 122       | Subfield 015 (Acceptor Tax ID Name)                        | One                       |
| DE 123       | Receipt Free Text                                          | Two                       |
| DE 124       | Member-Defined Data                                        | Two                       |
| DE 124       | Member-Defined Data: General Use                           | Two                       |
| DE 124       | Member-Defined Data: Prepaid Voucher                       | Two                       |
| DE 124       | Subfield 1 (Merchant Direct Settlement Indicator)          | Two                       |
| DE 124       | Member-Defined Data: Mastercard Send                       | Two                       |
| DE 124       | Member-Defined Data: Brazil Maestro Only                   | Two                       |
| DE 124       | Subfield 1 (Unique Reference Number)                       | Two                       |
| DE 124       | Subfield 2 (Sender/Payer/User ID)                          | Two                       |
| DE 124       | Subfield 3 (Sender/Payer Address)                          | Two                       |
| DE 124       | Subfield 4 (Additional Sender Information)                 | Two                       |
| DE 124       | Subfield 6 (Discretionary Message on Sales Slip Supported) | Two                       |
| DE 124       | Subfield 7 (Discretionary Message on Sales Slip Code)      | Two                       |

| Data element | Data element description                                 | Transaction data category |
|--------------|----------------------------------------------------------|---------------------------|
| DE 124       | Subfield 8 (Discretionary Message on Sales Slip Content) | Two                       |
| DE 124       | Subfield 9 (Phoneshop [Phone Company ID])                | Two                       |
| DE 124       | Subfield 10 (Phoneshop [Cell Phone Number])              | Two                       |
| DE 124       | Subfield 11 (Phoneshop [Message Security Code])          | Two                       |
| DE 124       | Subfield 12 (Merchant CNPJ Number)                       | Two                       |
| DE 124       | Subfield 13 (Total Annual Effective Cost)                | Two                       |
| DE 124       | Member-Defined Data [Colombia Domestic Use Only])        | Two                       |
| DE 124       | Subfield 1 (Card Issuer Data)                            | Two                       |
| DE 124       | Subfield 2 (Tax [IVA])                                   | Two                       |
| DE 124       | Subfield 3 (Tax Amount Base)                             | Two                       |
| DE 124       | Subfield 4 (Retailer Data)                               | Two                       |
| DE 124       | Subfield 5 (Terminal Acquirer Data)                      | Two                       |
| DE 124       | Subfield 6 (Acquirer Original Processing Code)           | Two                       |
| DE 124       | Subfield 7 (Bill Payment/Top up Data)                    | Two                       |
| DE 124       | Subfield 8 (Local POS Data)                              | Two                       |
| DE 124       | Subfield 9 (Local Response Codes)                        | Two                       |
| DE 124       | Subfield 10 (Original Transaction Data)                  | Two                       |
| DE 124       | Subfield 11 (IAC Tax Amount)                             | Two                       |
| DE 124       | Member-Defined Data: Ukraine Domestic Transactions       | Two                       |
| DE 125       | New PIN Data                                             | Two                       |
| DE 126       | Private Data                                             | Two                       |
| DE 127       | Private Data                                             | Two                       |
| DE 128       | Message Authentication Code                              | Two                       |

## Appendix H Glossary

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|                                                               |      |
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X-Code, X-Code Processing..... 1898

## acceptor

A person or entity that agrees to accept payment cards or devices bearing a particular brand or in respect to an agreement with the issuer. Also referred to as a merchant, ATM, bank, merchant ATM, or other entity authorized to accept brand transactions.

**Plural Form:** acceptors

## acceptance brand

Acceptance brands are Mastercard and other private label brands, such as cards specific to a particular department store or airline.

**Plural Form:** acceptance brands

## acquirer

A licensed Mastercard customer that acquires the data relating to a transaction from the acceptor and submits that data for authorization. The acquirer also supports clearing and settlement functions to exchange funds between the issuer and the acceptor. Also referred to as an acquiring bank, acquirer bank, merchant bank, or acquiring financial institution.

**Plural Form:** acquirers

## acquirer host

A term referring to customer host processing at the acquirer's location.

## advice message

A message that notifies a party of an action that has been taken, requiring no response.

**Plural Form:** advice messages

## advisement

A message that identifies the net settlement position of one or more ICA numbers (or customers) and is transmitted to the destination specified by the member Net Settlement Agreement (NSA) form.

**Plural Form:** advisements

## alternate issuer host

Alternate issuer authorization route used instead of the Stand-In Authorization; available in Europe region.

## Authorization

Real-time communication between acquirers and issuers to approve or decline the use of a Mastercard-branded card, access device, or account to conduct a transaction. Approval from issuers will create a financial obligation from the issuer to the acquirer (for a transaction that debits funds from a cardholder's account) or from the acquirer to the issuer (for a transaction that credits funds to a cardholder's account).

## authorization request

A request by an acquirer for issuer approval to complete a transaction involving a payment card.

**Plural Form:** authorization requests

## back-office process

Non-client-facing tasks that keep the business running and provide support for front-end operations.

## batch processing

A type of data processing operation and data communications transmission where related transactions are grouped together and transmitted for processing, usually by the same computer and under the same application; generally regarded as non-real-time data traffic comprising large files; type of data traffic where response time is not critical.

## cardholder

A cardholder is an authorized person to whom a payment card or a device payment has been issued. Also referred to as a consumer or account holder.

**Plural Form:** cardholders

## Clearing

Multiple clearing batch processes per day (that is, clearing cycles) determine and calculate processing costs, resulting in one net amount per acquirer and issuer per clearing cycle.

## Dual Message Authorization System

The Mastercard Dual Message Authorization System is a global real-time message processing system and is a component of the Mastercard Network. The Mastercard Network supports real-time authorization processing followed by batch reconciliation leading to settlement of funds between acquirers and issuers.

The Authorization System processing is integrated with the acquirer and issuer processing system communicates in real time to process authorization data to get an issuer decision.

Mastercard acquirers and issuers leverage the Authorization System to exchange financial transaction data captured at the point-of-interaction device to obtain an authorization decision.

The Authorization System consists of both hardware and software components. The physical computing infrastructure includes servers, routers, switches, and cabling. This hardware is configured with Mastercard's proprietary technology, which makes it a Mastercard interface processor (MIP) and is physically set up and connected with an acquirer or issuer processing system in their data centers.

**Plural Form:** Dual Message Authorization System

## Dual Message Clearing System

The Dual Message Clearing System or Mastercard Global Clearing Management System (GCMS) is a component of the Mastercard Network that provides a central mechanism for the exchange of offline financial clearing messages. All participants are both senders and receivers of clearing data.

## echo

Data repeated back to the sender by the recipient to verify that it was received correctly



## encryption

The process by which data is translated into secret code. Encryption is the most effective way to secure data. To read an encrypted file, the user must have access to a key or password that enables them to decrypt it.

## exception processing

A procedure for managing transactions that cannot be processed or which are interrupted.

## fraud

A deception deliberately practiced to secure unlawful gain.

## fraudulent transaction

A transaction effected using a cardholder's account number which is not engaged in or authorized by the cardholder. Such transactions are categorized as lost, stolen, not received, issued on a fraudulent application, counterfeit, fraudulent processing of transactions, account takeover (ATO), or other fraudulent conditions.

**Plural Form:** fraudulent transactions

## host

An intelligent processor or device that is connected to a network and satisfies the needs of remote users.

## International Organization for Standardization (ISO)

### International Organization for Standardization 8583:1987

The international standard for bankcard-originated messages and interchange message specifications.

**Plural Form:** International Organization for Standardization (ISO) 8583:1987

**Abbreviation:** ISO 8583:1987

## issuer

An issuer is a licensed Mastercard customer that issues cards or accounts and is responsible for approving or declining authorization requests. The issuer also bills cardholders and supports clearing and settlement functions to exchange funds between the cardholders account and the acquirer. Also referred to as an issuing bank, issuer bank, cardholder bank, or issuing financial institution.

**Plural Form:** issuers

## issuer host

A term referring to customer host processing at the issuer's location.

## Mastercard Network

The computer hardware and software operated by and on behalf of the Corporation for the routing, processing, and settlement of Transactions and PTA Transactions including, without limitation, the Mastercard Network, the Mastercard ATM Network, the Dual Message System, the Single Message System, the Global Clearing Management System (GCMS), the Settlement Account Management (SAM) system, and the China Switch system.

**Plural Form:** Mastercard Network

## Mastercard interface processor (MIP)

Front-end communications processor placed on-site at a Mastercard customer's facility or at a processor or hub site. The MIP provides access to the Mastercard Network.

**Plural Form:** Mastercard interface processors

**Abbreviation:** MIP

## Network

See Switch.

## Network Enablement Partner

A service provider that has a direct agreement with Mastercard and performs certain functions on behalf of one or more customers.

**Plural Form:** Network Enablement Partners

## primary issuer

Banking primary issuer, also known as an issuing bank or card issuer, is a financial institution that provides a cardholder with a payment card

## processor

A processor provides authorization, clearing, or settlement services on behalf of a customer. Also referred to as a third-party processor or payment processor.

**Plural Form:** processors

## reason code

Also known as message reason code. For each chargeback, an issuer selects and submits a numeric reason code. This code assist the merchant and acquirer diagnose errors and improve customer experience.

**Plural Form:** reason codes

## receipt

A hard copy document recording a transaction that took place at the point of interaction, with a description that usually includes the date, merchant name/location, primary account number, amount, and reference number.

## reconciliation

Process by which settlement payments are balanced against amounts and totals as provided by the association's processing systems.

## request

A message originated an interactive series of messages.

**Plural Form:** requests

## response code

Response codes indicate the disposition of a previous message or an action taken as a result of receipt of a previous message.

**Plural Form:** response codes

## response time

The elapsed time between the generation of the last character of a message at a terminal and the receipt of the first character of the reply. It includes terminal delay and network delay.

**Plural Form:** response times

## reversal message

A message relating to a reversal.

**Plural Form:** reversal messages

## service provider

A person or entity that is registered with Mastercard by a customer to perform services related to the customers issuing or an acquiring activity.

**Plural Form:** service providers

## Settlement

Aggregates financial positions from all clearing settlement information into one overall daily net settlement amount for each acquirer and issuer. Creating and distributing advisements (notifications) to acquirers and issuers along with Transfer Funds Orders to Settlement Banks for amounts to be transferred to or collected from acquirers and issuers.

## Single Message System

A global real-time message processing system and is a component of the Mastercard Network. The Mastercard Network supports real-time authorization processing followed by batch reconciliation leading to settlement of funds between acquirers and issuers.

Mastercard acquirers and issuers leverage the Single Message System to exchange financial transaction data captured at the point-of-interaction device. The Single Message System, through a single financial message interaction, fulfills authorization processing, reconciliation, and settlement functions.

The Single Message System consists of both hardware and software components. The physical computing infrastructure includes servers, routers, switches, and cabling. This hardware is configured with Mastercard's proprietary technology, which makes it a Mastercard interface processor (MIP) and is physically set up and connected with an acquirer or issuer processing system in their data centers.

**Plural Form:** Single Message System

**Abbreviation:** Single Message System

## Stand-In Authorization

Authorization processing provided by Mastercard for online issuers that are unavailable or that cannot be reached.

## Switch

The computer hardware and software operated by and on behalf of a payment scheme for the routing, processing, and settlement of transactions bearing the payment scheme brand in accordance with such payment scheme's rules and regulations, and which may also be used to route transactions to another payment scheme's switch or to route, process, and settle non-branded (also called private label) transactions as agreed by the participating parties. Also called a Network.

**Plural Form:** Switches

## Dual Message Switch

A Switch used to process transactions for which authorization and clearing occurs separately in a dual message format. The Mastercard Dual Message System (both authorization and clearing) is a Dual Message Switch.

## Single Message Switch

A Switch used to process transactions for which authorization and clearing occurs in a single message format. The Mastercard Single Message System is a Single Message Switch.

## X-Code, X-Code Processing

A system that performs issuer functions by receiving and responding to authorization requests at the acquirer Mastercard interface processor (MIP).

## Appendix I Abbreviations and acronyms

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|                                                    |      |
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| account data compromise (ADC).....                 | 1903 |
| Account Holding Institution (AHI).....             | 1903 |
| Account Management System (AMS).....               | 1903 |
| account status inquiry (ASI).....                  | 1903 |
| Accountholder Authentication Value (AAV).....      | 1903 |
| acquirer processing system (APS).....              | 1903 |
| Address Verification Service (AVS).....            | 1903 |
| advanced encryption standard (AES).....            | 1904 |
| Advice Reason Code (ARC).....                      | 1904 |
| annual equivalent charge rate (CAE).....           | 1904 |
| annual percentage rate (APR).....                  | 1904 |
| Application Authentication Cryptogram (AAC).....   | 1904 |
| Application Cryptogram (AC).....                   | 1904 |
| Application Transaction Counter (ATC).....         | 1904 |
| Authorization Characteristics Indicator (ACI)..... | 1904 |
| Automated Referral Service (ARS).....              | 1905 |
| Authorization Response Cryptogram (ARPC).....      | 1905 |
| automated fuel dispenser (AFD).....                | 1905 |
| automated response unit (ARU).....                 | 1905 |
| Automated Teller Machine (ATM).....                | 1905 |
| Bank Account Number (BAN).....                     | 1905 |
| Bank Identification Code (BIC).....                | 1905 |
| Bank Identification Number (BIN).....              | 1905 |
| Cadastro de Pessoa Física (CPF).....               | 1906 |
| Cadastro Nacional da Pessoa Jurídica (CNPJ).....   | 1906 |
| Card Identifier (CID).....                         | 1906 |
| Card Validation Code (CVC).....                    | 1906 |
| Card Verification Results (CVR).....               | 1906 |
| card-not-present (CNP).....                        | 1906 |
| cardholder activated terminal (CAT).....           | 1906 |
| cardholder verification method (CVM).....          | 1906 |
| conditional echo (CE).....                         | 1907 |
| Cross-border fee manager (CBFM).....               | 1907 |
| Customer Implementation Services (CIS).....        | 1907 |

|                                                    |      |
|----------------------------------------------------|------|
| customer processing system (CPS).....              | 1907 |
| data element (DE).....                             | 1907 |
| data encryption algorithm (DEA).....               | 1907 |
| Data Encryption System (DES).....                  | 1907 |
| Debit Online Low Risk Merchant (DOLM).....         | 1907 |
| Digital Secure Remote Payments (DSRP).....         | 1908 |
| dynamic currency conversion (DCC).....             | 1908 |
| dynamic token verification code (DTVC).....        | 1908 |
| electronic funds transfer (EFT).....               | 1908 |
| electronic warning bulletin (EWB).....             | 1908 |
| embedded flexible interchange (EFI).....           | 1908 |
| end-of-file (EOF).....                             | 1908 |
| Enhanced Issuer Identification (EII).....          | 1908 |
| European Economic Area (EEA).....                  | 1909 |
| Financial Institution ID (FIID).....               | 1909 |
| flexible spending account (FSA).....               | 1909 |
| Global Automated Referral Service (GARS).....      | 1909 |
| Global Clearing Management System (GCMS).....      | 1909 |
| global file transfer (GFT).....                    | 1909 |
| health savings account (HSA).....                  | 1909 |
| healthcare reimbursement arrangement (HRA).....    | 1909 |
| Hypertext Transfer Protocol Secure (HTTPS).....    | 1910 |
| Integrated Circuit Card (ICC).....                 | 1910 |
| interface device (IFD).....                        | 1910 |
| Intermediate Network Facilities (INF).....         | 1910 |
| international bank account number (IBAN).....      | 1910 |
| Inventory Information Approval System (IIAS).....  | 1910 |
| Issuer Currency Conversion Rate (ICCR).....        | 1910 |
| Issuer identification number (IIN).....            | 1910 |
| Integrated Product Message (IPM).....              | 1911 |
| Internet Protocol Security (IPSec).....            | 1911 |
| Internal Revenue Service (IRS).....                | 1911 |
| Key Encryption Key (KEK).....                      | 1911 |
| Key Management Services (KMS).....                 | 1911 |
| Longitudinal Redundancy Check (LRC).....           | 1911 |
| Message Authentication Code (MAC).....             | 1911 |
| Mastercard Brazilian Development Bank (BNDES)..... | 1911 |
| Mastercard Consolidated Billing System (MCBS)..... | 1912 |
| Mastercard Digital Enablement Service (MDES).....  | 1912 |



|                                                              |      |
|--------------------------------------------------------------|------|
| Mastercard Debit Switch (MDS).....                           | 1912 |
| multi-factor authentication (MFA).....                       | 1912 |
| master file key (MFK).....                                   | 1912 |
| Maestro Low Risk Merchant Program (MLRMP).....               | 1912 |
| Mastercard Network Gateway Services Platform (MNGS).....     | 1912 |
| member key ID (MKID).....                                    | 1912 |
| Mastercard Parameter Extract (MPE).....                      | 1913 |
| Message Type Indicator (MTI).....                            | 1913 |
| Maestro Utility Payment Program (MUPP).....                  | 1913 |
| On-Behalf Key Management (OBKM).....                         | 1913 |
| optical character reader (OCR).....                          | 1913 |
| Primary Account Number (PAN).....                            | 1913 |
| Programa de Alimentação do Trabalhador (PAT).....            | 1913 |
| PIN Encryption Key (PEK).....                                | 1913 |
| point-of-interaction (POI).....                              | 1914 |
| purchase with cash back (PWCB).....                          | 1914 |
| PIN verification keys (PVKs).....                            | 1914 |
| PIN verification value (PVV).....                            | 1914 |
| Quick Payment Service (QPS).....                             | 1914 |
| Settlement Account Management (S.A.M.).....                  | 1914 |
| secure payment application (SPA).....                        | 1914 |
| Secure Sockets Layer (SSL).....                              | 1914 |
| Short Message Service (SMS).....                             | 1915 |
| Single Message Transaction Manager (SMTM).....               | 1915 |
| Special Interest Group of IIAS Standards (SIGIS).....        | 1915 |
| store-and-forward (SAF).....                                 | 1915 |
| strong customer authentication (SCA).....                    | 1915 |
| switch serial number.....                                    | 1915 |
| Systems Trace Audit Number (STAN).....                       | 1915 |
| Swedish Domestic Authorization Switching Service (SASS)..... | 1915 |
| Transaction Category Code (TCC).....                         | 1916 |
| triple data encryption algorithm (TDEA).....                 | 1916 |
| Triple Data Encryption Standard (3DES).....                  | 1916 |
| Trusted Execution Environment (TEE).....                     | 1916 |
| Transaction Identifier (TID).....                            | 1916 |
| Transaction Link ID (TLID).....                              | 1916 |
| Transport Layer Security (TLS).....                          | 1916 |
| Tamper-Resistant Security Module (TRSM).....                 | 1916 |
| Terminal Verification Results (TVR).....                     | 1917 |

|                                                                           |      |
|---------------------------------------------------------------------------|------|
| Identification Number (UID).....                                          | 1917 |
| United States (U.S.).....                                                 | 1917 |
| Universal Cardholder Authentication Field (UCAF).....                     | 1917 |
| unique key per terminal/derived unique key per terminal (UKPT/DUKPT)..... | 1917 |
| unpredictable number (UN).....                                            | 1917 |
| uniform resource locator (URL).....                                       | 1917 |
| Universal Time (UTC).....                                                 | 1917 |
| value-added tax (VAT).....                                                | 1918 |
| SLI.....                                                                  | 1918 |
| EFI.....                                                                  | 1918 |
| CPS.....                                                                  | 1918 |
| EMS.....                                                                  | 1918 |
| CVV.....                                                                  | 1918 |

## acceptor business code (MCC)

**Abbreviation:** MCC

## account data compromise (ADC)

**Abbreviation:** ADC

## Account Holding Institution (AHI)

**Abbreviation:** AHI

## Account Management System (AMS)

**Abbreviation:** AMS

## account status inquiry (ASI)

**Abbreviation:** ASI

## Accountholder Authentication Value (AAV)

**Abbreviation:** AAV

## acquirer processing system (APS)

**Abbreviation:** APS

## Address Verification Service (AVS)

**Abbreviation:** AVS

## advanced encryption standard (AES)

**Abbreviation:** AES

## Advice Reason Code (ARC)

**Abbreviation:** ARC

## annual equivalent charge rate (CAE)

**Abbreviation:** CAE

## annual percentage rate (APR)

**Abbreviation:** APR

## Application Authentication Cryptogram (AAC)

**Abbreviation:** AAC

## Application Cryptogram (AC)

**Abbreviation:** AC

## Application Transaction Counter (ATC)

**Abbreviation:** ATC

## Authorization Characteristics Indicator (ACI)

**Abbreviation:** ACI

## Automated Referral Service (ARS)

**Abbreviation:** ARS

## Authorization Response Cryptogram (ARPC)

**Abbreviation:** ARPC

## automated fuel dispenser (AFD)

**Abbreviation:** AFD

## automated response unit (ARU)

**Abbreviation:** ARU

## Automated Teller Machine (ATM)

**Plural Form:** Automated Teller Machine (ATM)

**Abbreviation:** ATM

## Bank Account Number (BAN)

**Abbreviation:** BAN

## Bank Identification Code (BIC)

**Abbreviation:** BIC

## Bank Identification Number (BIN)

**Abbreviation:** BIN

## Cadastro de Pessoa Física (CPF)

**Abbreviation:** CPF

## Cadastro Nacional da Pessoa Jurídica (CNPJ)

**Abbreviation:** CNPJ

## Card Identifier (CID)

**Abbreviation:** CID

## Card Validation Code (CVC)

**Abbreviation:** CVC

## Card Verification Results (CVR)

**Abbreviation:** CVR

## card-not-present (CNP)

**Abbreviation:** CNP

## cardholder activated terminal (CAT)

**Abbreviation:** CAT

## cardholder verification method (CVM)

**Abbreviation:** CVM

## conditional echo (CE)

**Abbreviation:** CE

## Cross-border fee manager (CBFM)

**Abbreviation:** CBFM

## Customer Implementation Services (CIS)

**Abbreviation:** CIS

## customer processing system (CPS)

**Abbreviation:** CPS

## data element (DE)

**Abbreviation:** DE

## data encryption algorithm (DEA)

**Abbreviation:** DEA

## Data Encryption System (DES)

**Abbreviation:** DES

## Debit Online Low Risk Merchant (DOLM)

**Abbreviation:** DOLM

## Digital Secure Remote Payments (DSRP)

**Abbreviation:** DSRP

## dynamic currency conversion (DCC)

**Abbreviation:** DCC

## dynamic token verification code (DTVVC)

**Abbreviation:** DTVVC

## electronic funds transfer (EFT)

**Abbreviation:** EFT

## electronic warning bulletin (EWB)

**Abbreviation:** EWB

## embedded flexible interchange (EFI)

**Abbreviation:** EFI

## end-of-file (EOF)

**Abbreviation:** EOF

## Enhanced Issuer Identification (EII)

**Abbreviation:** EII



## European Economic Area (EEA)

**Abbreviation:** EEA

## Financial Institution ID (FIID)

**Abbreviation:** FIID

## flexible spending account (FSA)

**Abbreviation:** FSA

## Global Automated Referral Service (GARS)

**Abbreviation:** GARS

## Global Clearing Management System (GCMS)

**Abbreviation:** GCMS

## global file transfer (GFT)

**Abbreviation:** GFT

## health savings account (HSA)

**Abbreviation:** HSA

## healthcare reimbursement arrangement (HRA)

**Abbreviation:** HRA

## Hypertext Transfer Protocol Secure (HTTPS)

**Abbreviation:** HTTPS

## Integrated Circuit Card (ICC)

**Abbreviation:** ICC

## interface device (IFD)

**Abbreviation:** IFD

## Intermediate Network Facilities (INF)

**Abbreviation:** INF

## international bank account number (IBAN)

**Abbreviation:** IBAN

## Inventory Information Approval System (IIAS)

**Abbreviation:** IIAS

## Issuer Currency Conversion Rate (ICCR)

**Abbreviation:** ICCR

## Issuer identification number (IIN)

**Abbreviation:** IIN

## Integrated Product Message (IPM)

**Abbreviation:** IPM

## Internet Protocol Security (IPSec)

**Abbreviation:** IPSec

## Internal Revenue Service (IRS)

**Abbreviation:** IRS

## Key Encryption Key (KEK)

**Abbreviation:** KEK

## Key Management Services (KMS)

**Abbreviation:** KMS

## Longitudinal Redundancy Check (LRC)

**Abbreviation:** LRC

## Message Authentication Code (MAC)

**Abbreviation:** MAC

## Mastercard Brazilian Development Bank (BNDES)

**Abbreviation:** BNDES

## Mastercard Consolidated Billing System (MCBS)

**Abbreviation:** MCBS

## Mastercard Digital Enablement Service (MDES)

**Abbreviation:** MDES

## Mastercard Debit Switch (MDS)

**Abbreviation:** MDS

## multi-factor authentication (MFA)

**Abbreviation:** MFA

## master file key (MFK)

**Abbreviation:** MFK

## Maestro Low Risk Merchant Program (MLRMP)

**Abbreviation:** MLRMP

## Mastercard Network Gateway Services Platform (MNGS)

**Abbreviation:** MNGS

## member key ID (MKID)

**Abbreviation:** MKID

## Mastercard Parameter Extract (MPE)

**Abbreviation:** MPE

## Message Type Indicator (MTI)

**Plural Form:** Message Type Indicators

**Abbreviation:** MTI

## Maestro Utility Payment Program (MUPP)

**Abbreviation:** MUPP

## On-Behalf Key Management (OBKM)

**Abbreviation:** OBKM

## optical character reader (OCR)

**Abbreviation:** OCR

## Primary Account Number (PAN)

**Abbreviation:** PAN

## Programa de Alimentação do Trabalhador (PAT)

**Abbreviation:** PAT

## PIN Encryption Key (PEK)

**Abbreviation:** PEK

## point-of-interaction (POI)

**Abbreviation:** POI

## purchase with cash back (PWCB)

**Abbreviation:** PWCB

## PIN verification keys (PVKs)

**Abbreviation:** PVKs

## PIN verification value (PVV)

**Abbreviation:** PVV

## Quick Payment Service (QPS)

**Abbreviation:** QPS

## Settlement Account Management (S.A.M.)

**Abbreviation:** S.A.M.

## secure payment application (SPA)

**Abbreviation:** SPA

## Secure Sockets Layer (SSL)

**Abbreviation:** SSL

## Short Message Service (SMS)

**Abbreviation:** SMS

## Single Message Transaction Manager (SMTM)

**Abbreviation:** SMTM

## Special Interest Group of IAS Standards (SIGIS)

**Abbreviation:** SIGIS

## store-and-forward (SAF)

**Abbreviation:** SAF

## strong customer authentication (SCA)

**Abbreviation:** SCA

## switch serial number

**Abbreviation:** SSN

## Systems Trace Audit Number (STAN)

**Abbreviation:** STAN

## Swedish Domestic Authorization Switching Service (SASS)

**Abbreviation:** SASS

## Transaction Category Code (TCC)

**Abbreviation:** TCC

## triple data encryption algorithm (TDEA)

**Abbreviation:** TDEA

## Triple Data Encryption Standard (3DES)

**Abbreviation:** 3DES

## Trusted Execution Environment (TEE)

**Abbreviation:** TEE

## Transaction Identifier (TID)

**Abbreviation:** TID

## Transaction Link ID (TLID)

**Abbreviation:** TLID

## Transport Layer Security (TLS)

**Abbreviation:** TLS

## Tamper-Resistant Security Module (TRSM)

**Abbreviation:** TRSM



## Terminal Verification Results (TVR)

**Abbreviation:** TVR

## Identification Number (UID)

**Abbreviation:** UID

## United States (U.S.)

**Abbreviation:** U.S.

## Universal Cardholder Authentication Field (UCAF)

**Abbreviation:** UCAF

## unique key per terminal/derived unique key per terminal (UKPT/ DUKPT)

**Abbreviation:** UKPT/DUKPT

## unpredictable number (UN)

**Abbreviation:** UN

## uniform resource locator (URL)

**Abbreviation:** URL

## Universal Time (UTC)

**Abbreviation:** UTC

## value-added tax (VAT)

**Abbreviation:** VAT

SLI

EFI

CPS

EMS

CVV

## Appendix J Dual Message System data element abbreviations

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|                                                                           |      |
|---------------------------------------------------------------------------|------|
| DE 1 (Bit Map, Secondary).....                                            | 1923 |
| DE 2 (Primary Account Number [PAN]).....                                  | 1923 |
| DE 3 (Processing Code).....                                               | 1923 |
| DE 4 (Amount, Transaction).....                                           | 1923 |
| DE 5 (Amount, Settlement).....                                            | 1923 |
| DE 6 (Amount, Cardholder Billing).....                                    | 1924 |
| DE 7 (Transmission Date and Time).....                                    | 1924 |
| DE 8 (Amount, Cardholder Billing Fee).....                                | 1924 |
| DE 9 (Conversion Rate, Settlement).....                                   | 1924 |
| DE 10 (Conversion Rate, Cardholder Billing).....                          | 1925 |
| DE 11 (System Trace Audit Number [STAN]).....                             | 1925 |
| DE 12 (Time, Local Transaction).....                                      | 1925 |
| DE 13 (Date, Local Transaction).....                                      | 1925 |
| DE 14 (Date, Expiration).....                                             | 1925 |
| DE 15 (Date, Settlement).....                                             | 1926 |
| DE 16 (Date, Conversion).....                                             | 1926 |
| DE 17 (Date, Capture).....                                                | 1926 |
| DE 18 (Merchant Type).....                                                | 1926 |
| DE 19 (Acquiring Institution Country Code).....                           | 1926 |
| DE 20 (Primary Account Number [PAN] Country Code).....                    | 1927 |
| DE 21 (Forwarding Institution Country Code).....                          | 1927 |
| DE 22 (Point-of-Service [POS] Entry Mode).....                            | 1927 |
| DE 23 (Card Sequence Number).....                                         | 1927 |
| DE 24 (Network International ID).....                                     | 1927 |
| DE 25 (Point-of-Service [POS] Condition Code).....                        | 1928 |
| DE 26 (Point-of-Service (POS) Personal ID Number [PIN] Capture Code)..... | 1928 |
| DE 27 (Authorization ID Response Length).....                             | 1928 |
| DE 28 (Amount, Transaction Fee).....                                      | 1928 |
| DE 29 (Amount, Settlement Fee).....                                       | 1929 |
| DE 30 (Amount, Transaction Processing Fee).....                           | 1929 |
| DE 31 (Amount, Settlement Processing Fee).....                            | 1929 |
| DE 32 (Acquiring Institution ID Code).....                                | 1929 |
| DE 33 (Forwarding Institution ID Code).....                               | 1929 |
| DE 34 (Primary Account Number [PAN], Extended).....                       | 1930 |

|                                                                |      |
|----------------------------------------------------------------|------|
| DE 35 (Track 2 Data).....                                      | 1930 |
| DE 36 (Track 3 Data).....                                      | 1930 |
| DE 37 (Retrieval Reference Number).....                        | 1930 |
| DE 38 (Authorization ID Response).....                         | 1931 |
| DE 39 (Response Code).....                                     | 1931 |
| DE 40 (Service Restriction Code).....                          | 1931 |
| DE 41 (Acceptor Terminal ID).....                              | 1931 |
| DE 42 (Acceptor ID).....                                       | 1932 |
| DE 43 (Acceptor Name and Location).....                        | 1932 |
| DE 44 (Additional Response Data).....                          | 1932 |
| DE 45 (Track 1 Data).....                                      | 1932 |
| DE 46 (Expanded Additional Amounts).....                       | 1933 |
| DE 47 (Additional Data: National Use).....                     | 1933 |
| DE 48 (Additional Data: Private Use).....                      | 1933 |
| DE 49 (Currency Code, Transaction).....                        | 1933 |
| DE 50 (Currency Code, Settlement).....                         | 1933 |
| DE 51 (Currency Code, Cardholder Billing).....                 | 1934 |
| DE 52 (Personal ID Number [PIN] Data).....                     | 1934 |
| DE 53 (Security-Related Control Information).....              | 1934 |
| DE 54 (Additional Amounts).....                                | 1934 |
| DE 55 (Integrated Circuit Card [ICC] System-Related Data)..... | 1935 |
| DE 56 (Payment Account Data).....                              | 1935 |
| DE 57–DE 59 (Reserved for National Use).....                   | 1935 |
| DE 60 (Advice Reason Code).....                                | 1935 |
| DE 61 (Point-of-Service [POS] Data).....                       | 1935 |
| DE 62 (Intermediate Network Facility [INF] Data).....          | 1936 |
| DE 63 (Network Data).....                                      | 1936 |
| DE 64 (Message Authentication Code).....                       | 1936 |
| DE 65 (Bit Map, Extended).....                                 | 1936 |
| DE 66 (Settlement Code).....                                   | 1937 |
| DE 67 (Extended Payment Code).....                             | 1937 |
| DE 68 (Receiving Institution Country Code).....                | 1937 |
| DE 69 (Settlement Institution Country Code).....               | 1937 |
| DE 70 (Network Management Information Code).....               | 1937 |
| DE 71 (Message Number).....                                    | 1938 |
| DE 72 (Message Number Last).....                               | 1938 |
| DE 73 (Date, Action).....                                      | 1938 |
| DE 74 (Credits, Number).....                                   | 1938 |
| DE 75 (Credits, Reversal Number).....                          | 1938 |

|                                                         |      |
|---------------------------------------------------------|------|
| DE 76 (Debits, Number).....                             | 1939 |
| DE 77 (Debits, Reversal Number).....                    | 1939 |
| DE 78 (Transfers, Number).....                          | 1939 |
| DE 79 (Transfers, Reversal Number).....                 | 1939 |
| DE 80 (Inquiries, Number).....                          | 1939 |
| DE 81 (Authorizations, Number).....                     | 1940 |
| DE 82 (Credits, Processing Fee Amount).....             | 1940 |
| DE 83 (Credits, Transaction Fee Amount).....            | 1940 |
| DE 84 (Debits, Processing Fee Amount).....              | 1940 |
| DE 85 (Debits, Transaction Fee Amount).....             | 1940 |
| DE 86 (Credits, Amount).....                            | 1941 |
| DE 87 (Credits, Reversal Amount).....                   | 1941 |
| DE 88 (Debits, Amount).....                             | 1941 |
| DE 89 (Debits, Reversal Amount).....                    | 1941 |
| DE 90 (Original Data Elements).....                     | 1941 |
| DE 91 (Issuer File Update Code).....                    | 1942 |
| DE 92 (File Security Code).....                         | 1942 |
| DE 93 (Response Indicator).....                         | 1942 |
| DE 94 (Service Indicator).....                          | 1942 |
| DE 95 (Replacement Amounts).....                        | 1942 |
| DE 96 (Message Security Code).....                      | 1943 |
| DE 97 (Amount, Net Settlement).....                     | 1943 |
| DE 98 (Payee).....                                      | 1943 |
| DE 99 (Settlement Institution ID Code).....             | 1943 |
| DE 100 (Receiving Institution ID Code).....             | 1943 |
| DE 101 (File Name).....                                 | 1944 |
| DE 102 (Account ID 1).....                              | 1944 |
| DE 103 (Account ID 2).....                              | 1944 |
| DE 104 (Digital Payment Data).....                      | 1944 |
| DE 105 (Multi-Use Transaction Identification Data)..... | 1944 |
| DE 106 (Fleet Service Data).....                        | 1945 |
| DE 107 (Reserved for Mastercard Use).....               | 1945 |
| DE 108 (Additional Transaction Reference Data).....     | 1945 |
| DE 109 (Reserved for ISO Use).....                      | 1945 |
| DE 110 (Additional Data–2).....                         | 1945 |
| DE 110 (Encryption Data).....                           | 1946 |
| DE 111 (Reserved for ISO Use).....                      | 1946 |
| DE 112 (Additional Data [National Use]).....            | 1946 |
| DE 113 (Reserved for National Use).....                 | 1946 |

|                                                       |      |
|-------------------------------------------------------|------|
| DE 114 (Reserved for National Use).....               | 1947 |
| DE 115 (Reserved for National Use).....               | 1947 |
| DE 116 (Reserved For National Use).....               | 1947 |
| DE 117 (Reserved for National Use).....               | 1947 |
| DE 118 (Reserved for National Use).....               | 1947 |
| DE 119 (Additional Data: Private Use 2).....          | 1948 |
| Using DE 113–119 in Administrative 06xx Messages..... | 1948 |
| DE 120 (Record Data).....                             | 1948 |
| DE 121 (Authorizing Agent ID Code).....               | 1948 |
| DE 122 (Additional Record Data).....                  | 1949 |
| DE 123 (Receipt Free Text).....                       | 1949 |
| DE 124 (Member-Defined Data).....                     | 1949 |
| DE 125 (New PIN Data).....                            | 1949 |
| DE 126 (Private Data).....                            | 1949 |
| DE 127 (Private Data).....                            | 1950 |
| DE 128 (Message Authentication Code).....             | 1950 |

## DE 1 (Bit Map, Secondary)

DE 1 (Bit Map, Secondary) is a series of 64 bits that identifies the presence (1) or absence (0) of each data element in the second segment of a message. This would include DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]).

**Plural Form:** DE 1 (Bit Map, Secondary)

**Abbreviation:** DE 1

## DE 2 (Primary Account Number [PAN])

DE 2 (Primary Account Number [PAN]) is a series of digits used to identify a customer account or relationship.

**Plural Form:** DE 2 (Primary Account Number [PAN])

**Abbreviation:** DE 2

## DE 3 (Processing Code)

DE 3 (Processing Code) describes the effect of a transaction on the customer account and the type of accounts affected.

**Plural Form:** DE 3 (Processing Code)

**Abbreviation:** DE 3

## DE 4 (Amount, Transaction)

DE 4 (Amount, Transaction) is the amount of funds the cardholder requested in the local currency of the acquirer or source location of the transaction.

**Plural Form:** DE 4 (Amount, Transaction)

**Abbreviation:** DE 4

## DE 5 (Amount, Settlement)

DE 5 (Amount, Settlement) is the amount of funds to be transferred between the acquirer and the issuer equal to DE 4 (Amount, Transaction) in the settlement currency. Mastercard programs and services use U.S. dollars as the currency of settlement.

**Plural Form:** DE 5 (Amount, Settlement)

**Abbreviation:** DE 5

## DE 6 (Amount, Cardholder Billing)

DE 6 (Amount, Cardholder Billing) indicates the transaction amount in the issuer's currency. It is the amount billed to the cardholder in the cardholder account currency, excluding cardholder billing fees.

**Plural Form:** DE 6 (Amount, Cardholder Billing)

**Abbreviation:** DE 6

## DE 7 (Transmission Date and Time)

DE 7 (Transmission Date and Time) is the date and time that a message is entered into the Mastercard Network. Date and time must be expressed in Coordinated Universal Time (UTC).

**Plural Form:** DE 7 (Transmission Date and Time)

**Abbreviation:** DE 7

## DE 8 (Amount, Cardholder Billing Fee)

DE 8 (Amount, Cardholder Billing Fee) is the fee the issuer bills to the cardholder in the same currency as DE 6 (Amount, Cardholder Billing).

**Plural Form:** DE 8 (Amount, Cardholder Billing Fee)

**Abbreviation:** DE 8

## DE 9 (Conversion Rate, Settlement)

DE 9 (Conversion Rate, Settlement) is the factor used in the conversion from transaction to settlement amount. DE 4 (Amount, Transaction) is multiplied by DE 9 to determine DE 5 (Amount, Settlement).

**Plural Form:** DE 9 (Conversion Rate, Settlement)

**Abbreviation:** DE 9



## DE 10 (Conversion Rate, Cardholder Billing)

DE 10 (Conversion Rate, Cardholder Billing) is the factor used in the conversion from transaction to cardholder billing amount. DE 4 (Amount, Transaction) is multiplied by DE 10 to determine DE 6 (Amount, Cardholder Billing).

**Plural Form:** DE 10 (Conversion Rate, Cardholder Billing)

**Abbreviation:** DE 10

## DE 11 (System Trace Audit Number [STAN])

DE 11 (Systems Trace Audit Number [STAN]) is a number a message initiator assigns to uniquely identify a transaction.

**Plural Form:** DE 11 (System Trace Audit Number [STAN])

**Abbreviation:** DE 11

## DE 12 (Time, Local Transaction)

DE 12 (Time, Local Transaction) is the local time at which the transaction takes place at the point of acceptor location.

**Plural Form:** DE 12 (Time, Local Transaction)

**Abbreviation:** DE 12

## DE 13 (Date, Local Transaction)

DE 13 (Date, Local Transaction) is the local month and day on which the transaction takes place at the point of acceptor location.

**Plural Form:** DE 13 (Date, Local Transaction)

**Abbreviation:** DE 13

## DE 14 (Date, Expiration)

DE 14 (Date, Expiration) specifies the year and month after which an issuer designates a cardholder's card to be expired.

**Plural Form:** DE 14 (Date, Expiration)

**Abbreviation:** DE 14

## DE 15 (Date, Settlement)

DE 15 (Date, Settlement) is the date (month and day) that funds will be transferred between an acquirer and an issuer or an appropriate intermediate network facility (INF).

**Plural Form:** DE 15 (Date, Settlement)

**Abbreviation:** DE 15

## DE 16 (Date, Conversion)

DE 16 (Date, Conversion) indicates the effective date of DE 9 (Conversion Rate, Settlement) and also DE 10 (Conversion Rate, Cardholder Billing) whenever these data elements are present within a message.

**Plural Form:** DE 16 (Date, Conversion)

**Abbreviation:** DE 16

## DE 17 (Date, Capture)

DE 17 (Date, Capture) is the month and day the acquirer processed the transaction data.

**Plural Form:** DE 17 (Date, Capture)

**Abbreviation:** DE 17

## DE 18 (Merchant Type)

DE 18 (Merchant Type) is the classification (acceptor business code/merchant category code [MCC]) of the merchant's type of business or service.

**Plural Form:** DE 18 (Merchant Type)

**Abbreviation:** DE 18

## DE 19 (Acquiring Institution Country Code)

DE 19 (Acquiring Institution Country Code) is based on Mastercard franchise data information.

**Plural Form:** DE 19 (Acquiring Institution Country Code)

**Abbreviation:** DE 19

## DE 20 (Primary Account Number [PAN] Country Code)

DE 20 (Primary Account Number [PAN] Country Code) is a code identifying the country where the card issuer is located.

**Plural Form:** DE 20 (Primary Account Number [PAN] Country Code)

**Abbreviation:** DE 20

## DE 21 (Forwarding Institution Country Code)

DE 21 (Forwarding Institution Country Code) is the code of the country where the forwarding institution is located.

**Plural Form:** DE 21 (Forwarding Institution Country Code)

**Abbreviation:** DE 21

## DE 22 (Point-of-Service [POS] Entry Mode)

DE 22 (Point-of-Service [POS] Entry Mode) indicates the method used for PAN entry to initiate a transaction and the PIN entry capabilities.

**Plural Form:** DE 22 (Point-of-Service [POS] Entry Mode)

**Abbreviation:** DE 22

## DE 23 (Card Sequence Number)

DE 23 (Card Sequence Number) distinguishes among separate cards having the same DE 2 (Primary Account Number [PAN]) or DE 34 (Primary Account Number [PAN] Extended). Issuers may encode chip cards with Card Sequence Numbers. Acquirers with chip-reading capability may pass this information encoded on the chip in DE 23 of Authorization Request/0100 messages.

**Plural Form:** DE 23 (Card Sequence Number)

**Abbreviation:** DE 23

## DE 24 (Network International ID)

DE 24 (Network International ID) identifies a single international network of card issuers.

**Plural Form:** DE 24 (Network International ID)

**Abbreviation:** DE 24

## DE 25 (Point-of-Service [POS] Condition Code)

DE 25 (Point-of-Service [POS] Condition Code) is an ID of the condition under which the transaction takes place at the point of interaction.

**Plural Form:** DE 25 (Point-of-Service [POS] Condition Code)

**Abbreviation:** DE 25

## DE 26 (Point-of-Service (POS) Personal ID Number [PIN] Capture Code)

DE 26 (Point-of-Service [POS] Personal ID Number [PIN] Capture Code) indicates the technique, maximum number, or both of PIN characters that can be accepted by the POS terminal used to construct the PIN data.

**Plural Form:** DE 26 (Point-of-Service (POS) Personal ID Number [PIN] Capture Code)

**Abbreviation:** DE 26

## DE 27 (Authorization ID Response Length)

DE 27 (Authorization ID Response Length) is the maximum length of the authorization response that the acquirer can accommodate. The issuer or its agent is expected to limit response to this length.

**Plural Form:** DE 27 (Authorization ID Response Length)

**Abbreviation:** DE 27

## DE 28 (Amount, Transaction Fee)

DE 28 (Amount, Transaction Fee) is the fee charged (for example, by the acquirer) for transaction activity in the currency of DE 4 (Amount, Transaction).

**Plural Form:** DE 28 (Amount, Transaction Fee)

**Abbreviation:** DE 28

## DE 29 (Amount, Settlement Fee)

DE 29 (Amount, Settlement Fee) is the fee to be transferred between the acquirer and the issuer equal to DE 28 (Amount, Transaction Fee) in the currency of DE 5 (Amount, Settlement).

**Plural Form:** DE 29 (Amount, Settlement Fee)

**Abbreviation:** DE 29

## DE 30 (Amount, Transaction Processing Fee)

DE 30 (Amount, Transaction Processing Fee) is the fee charged (for example, by the acquirer, issuer, or INF) for the handling and routing of messages in the currency of DE 4 (Amount, Transaction).

**Plural Form:** DE 30 (Amount, Transaction Processing Fee)

**Abbreviation:** DE 30

## DE 31 (Amount, Settlement Processing Fee)

DE 31 (Amount, Settlement Processing Fee) is the fee charged (for example, by the acquirer, issuer, or INF) for the handling and routing of messages in the currency of DE 5 (Amount, Settlement).

**Plural Form:** DE 31 (Amount, Settlement Processing Fee)

**Abbreviation:** DE 31

## DE 32 (Acquiring Institution ID Code)

DE 32 (Acquiring Institution ID Code) identifies the acquiring institution (for example, merchant bank) or its agent.

**Plural Form:** DE 32 (Acquiring Institution ID Code)

**Abbreviation:** DE 32

## DE 33 (Forwarding Institution ID Code)

DE 33 (Forwarding Institution ID Code) identifies the institution forwarding a Request or Advice message in an interchange system if it is not the same institution as specified in DE 32 (Acquiring Institution ID Code). DE 33 is used within a message to contain the Mastercard six-

digit customer ID number of the customer or INF responsible for directly routing that message to the Dual Message Authorization System.

**Plural Form:** DE 33 (Forwarding Institution ID Code)

**Abbreviation:** DE 33

## DE 34 (Primary Account Number [PAN], Extended)

DE 34 (Primary Account Number [PAN], Extended) identifies a customer account or relationship, and is used only when PAN begins with a 59 BIN.

**Plural Form:** DE 34 (Primary Account Number [PAN], Extended)

**Abbreviation:** DE 34

## DE 35 (Track 2 Data)

DE 35 (Track 2 Data) is the information encoded on track 2 of the card magnetic stripe as defined in the ISO 7813 specification, including data element separators but excluding beginning and ending sentinels and longitudinal redundancy check (LRC) characters as defined therein.

**Plural Form:** DE 35 (Track 2 Data)

**Abbreviation:** DE 35

## DE 36 (Track 3 Data)

DE 36 (Track 3 Data) is the information encoded on track 3 of the card magnetic stripe as defined in the ISO 4909:1986 specification, including data element separators but excluding beginning and ending sentinels and LRC characters as defined therein.

**Plural Form:** DE 36 (Track 3 Data)

**Abbreviation:** DE 36

## DE 37 (Retrieval Reference Number)

DE 37 (Retrieval Reference Number) is a document reference number supplied by the system retaining the original source document of the transaction and assists in locating that source document or a copy thereof. DE 37 is made available for use by automated merchant POS systems that may be interconnected into the interchange system. Merchant POS systems may assign a unique receipt or sales document ID to be used to satisfy regulatory or legal requirements when the merchant performs source document capture and truncation. DE 37

may be used to relay source document reference numbers to the issuer at the time each transaction is processed.

**Plural Form:** DE 37 (Retrieval Reference Number)

**Abbreviation:** DE 37

## DE 38 (Authorization ID Response)

DE 38 (Authorization ID Response) is a transaction response ID code that the authorizing institution assigns. DE 38 is used to transmit a card issuer's "authorization code" for Authorization transactions.

**Plural Form:** DE 38 (Authorization ID Response)

**Abbreviation:** DE 38

## DE 39 (Response Code)

DE 39 (Response Code) defines the disposition of a previous message or an action taken as a result of receipt of a previous message. Response codes also are used to indicate approval or decline of a transaction. In the event an authorization is declined, the response code indicates the reason for rejection and may indicate an action to be taken at the acceptor (for example, capture card).

**Plural Form:** DE 39 (Response Code)

**Abbreviation:** DE 39

## DE 40 (Service Restriction Code)

DE 40 (Service Restriction Code) identifies geographic or service availability.

**Plural Form:** DE 40 (Service Restriction Code)

**Abbreviation:**

## DE 41 (Acceptor Terminal ID)

DE 41 (Acceptor Terminal ID) is the identifier assigned by the acceptor or the acquirer to the terminal, payment gateway, or other acceptance device used to capture account data to affect a transaction at the acceptor location.

**Plural Form:** DE 41 (Acceptor Terminal ID)

**Abbreviation:** DE 41

## DE 42 (Acceptor ID)

DE 42 (Acceptor ID) contains the identifier assigned by the acquirer to the acceptor and its location, or to the payment facilitator.

**Plural Form:** DE 42 (Acceptor ID)

**Abbreviation:** DE 42

## DE 43 (Acceptor Name and Location)

DE 43 (Acceptor Name and Location) contains the name and location of the acceptor point of interaction.

**Plural Form:** DE 43 (Acceptor Name and Location)

**Abbreviation:** DE 43

## DE 44 (Additional Response Data)

DE 44 (Additional Response Data) provides other supplemental data that may be required in response to an authorization or other type of transaction request. This data element may also be present in any response message when DE 39 (Response Code) contains the value 30, indicating that a Format Error condition was detected in the preceding message. In this case, DE 44 will contain either a three- or six-digit value that indicates the data element number (three digits) only or the data element plus subelement numbers (six digits) in which the format error occurred. If the format error occurred in a data element that does not have subelements, DE 44 will contain a three-digit numeric value.

**Plural Form:** DE 44 (Additional Response Data)

**Abbreviation:** DE 44

## DE 45 (Track 1 Data)

DE 45 (Track 1 Data) is the information encoded on track 1 of the card's magnetic stripe as defined in the ISO 7813 specification, including data element separators but excluding beginning and ending sentinels and LRC characters as defined in this data element definition.

**Plural Form:** DE 45 (Track 1 Data)

**Abbreviation:** DE 45



## DE 46 (Expanded Additional Amounts)

DE 46 (Additional Data: ISO Use) provides data supplemental to that already conveyed in the specific data elements in the message.

**Plural Form:** DE 46 (Expanded Additional Amounts)

**Abbreviation:** DE 46

## DE 47 (Additional Data: National Use)

DE 47 (Additional Data—National Use) is reserved for national organizations to define data unique to country applications.

**Plural Form:** DE 47 (Additional Data: National Use)

**Abbreviation:** DE 47

## DE 48 (Additional Data: Private Use)

DE 48 (Additional Data: Private Use) contains data associated with various Mastercard programs, products, and services. DE 48 is used to convey data that is not related to any other ISO-defined data elements. It is a variable-length data element used for multiple purposes.

**Plural Form:** DE 48 (Additional Data: Private Use)

**Abbreviation:** DE 48

## DE 49 (Currency Code, Transaction)

DE 49 (Currency Code, Transaction) is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction) and DE 95 (Replacement Amounts), subfield 1 (Actual Amount, Transaction).

**Plural Form:** DE 49 (Currency Code, Transaction)

**Abbreviation:** DE 49

## DE 50 (Currency Code, Settlement)

DE 50 (Currency Code, Settlement) defines the currency of DE 5 (Amount, Settlement), DE 29 (Amount, Settlement Fee), and DE 95 (Replacement Amounts), subfield 2 (Actual Amount, Settlement).

**Plural Form:** DE 50 (Currency Code, Settlement)

**Abbreviation:** DE 50

## DE 51 (Currency Code, Cardholder Billing)

DE 51 (Currency Code, Cardholder Billing) defines the currency of DE 6 (Amount, Cardholder Billing), DE 8 (Amount, Cardholder Billing Fee), and DE 95 (Replacement Amounts), subfield 3 (Actual Amount, Cardholder Billing).

**Plural Form:** DE 51 (Currency Code, Cardholder Billing)

**Abbreviation:** DE 51

## DE 52 (Personal ID Number [PIN] Data)

DE 52 (Personal ID Number [PIN] Data) contains a number assigned to a cardholder intended to uniquely identify that cardholder at the point of interaction. The use of the PIN is subject to bilateral agreement. The data element may contain the PIN itself or a derivative. This data element transmits PIN information from acquirers to issuers (or to the network) for PIN verification or validation.

**Plural Form:** DE 52 (Personal ID Number [PIN] Data)

**Abbreviation:** DE 52

## DE 53 (Security-Related Control Information)

DE 53 (Security-Related Control Information) is used with PIN data to provide specific information about PIN block encoding and PIN data encryption to assist the issuer (or its agent) in processing PINs entered at the point of interaction.

**Plural Form:** DE 53 (Security-Related Control Information)

**Abbreviation:** DE 53

## DE 54 (Additional Amounts)

DE 54 (Additional Amounts) provides information on up to three amount types and related account data.

**Plural Form:** DE 54 (Additional Amounts)

**Abbreviation:** DE 54

## DE 55 (Integrated Circuit Card [ICC] System-Related Data)

DE 55 (Integrated Circuit Card [ICC] System-Related Data) contains binary data that only the issuer, the issuer agent, or MDES processes; it is used locally by the payment application on the chip at a chip-capable terminal. This data element is present in chip full-grade transactions and can be present in DSRP transactions.

**Plural Form:** DE 55 (Integrated Circuit Card [ICC] System-Related Data)

**Abbreviation:** DE 55

## DE 56 (Payment Account Data)

DE 56 (Payment Account Data) contains unique, non-financial reference information associated with the PAN or token used to initiate the transaction.

**Plural Form:** DE 56 (Payment Account Data)

**Abbreviation:** DE 56

## DE 57–DE 59 (Reserved for National Use)

DE 57–DE 59 are reserved for future use.

**Plural Form:** DE 57–DE 59 (Reserved for National Use)

**Abbreviation:** DE 57–DE 59

## DE 60 (Advice Reason Code)

DE 60 (Advice Reason Code) indicates to the receiver of an Advice message the specific reason for the transmission of the Advice message.

**Plural Form:** DE 60 (Advice Reason Code)

**Abbreviation:** DE 60

## DE 61 (Point-of-Service [POS] Data)

DE 61 (Point-of-Service [POS] Data) indicates the conditions that exist at the point of service at the time of the transaction. Note that DE 61 supersedes and replaces the ISO-specified DE 25 (Point-of-Service [POS] Condition Code) which is not used in the *Mastercard Network Processing Dual Message Authorization System Guide*.

**Plural Form:** DE 61 (Point-of-Service [POS] Data)

**Abbreviation:** DE 61

## DE 62 (Intermediate Network Facility [INF] Data)

DE 62 (Intermediate Network Facility [INF] Data), contains "acquiring network trace information" that INFs may require to quickly and accurately route Administrative Advice/0620 messages back to the original acquiring institution. DE 62 assists acquiring INF facilities that connect directly to the Dual Message Authorization System. It allows these INFs to maintain sufficient information within a message to permit immediate online routing of chargebacks and retrieval requests without the requirement of maintaining large online reference databases containing the original transactions.

**Plural Form:** DE 62 (Intermediate Network Facility [INF] Data)

**Abbreviation:** DE 62

## DE 63 (Network Data)

DE 63 (Network Data) is generated by the Dual Message Authorization System for each originating message routed through the network. The receiver must retain the data element and use it in any response or acknowledgment message associated with the originating message.

**Plural Form:** DE 63 (Network Data)

**Abbreviation:** DE 63

## DE 64 (Message Authentication Code)

DE 64 (Message Authentication Code [MAC]) validates the source and the text of the message between the sender and the receiver.

**Plural Form:** DE 64 (Message Authentication Code)

**Abbreviation:** DE 64

## DE 65 (Bit Map, Extended)

DE 65 (Bit Map, Extended) is a series of eight bytes (64 bits) used to identify the presence (denoted by 1) or the absence (denoted by 0) of each data element in an extended (third) message segment.

**Plural Form:** DE 65 (Bit Map, Extended)

**Abbreviation:** DE 65

## DE 66 (Settlement Code)

DE 66 (Settlement Code) indicates the result of a reconciliation request.

**Plural Form:** DE 66 (Settlement Code)

**Abbreviation:** DE 66

## DE 67 (Extended Payment Code)

DE 67 (Extended Payment Code) indicates the number of months that the cardholder prefers to pay for an item (the item purchased during the course of this transaction) if permitted by the card issuer.

**Plural Form:** DE 67 (Extended Payment Code)

**Abbreviation:** DE 67

## DE 68 (Receiving Institution Country Code)

DE 68 (Receiving Institution Country Code) is the code of the country where the receiving institution is located.

**Plural Form:** DE 68 (Receiving Institution Country Code)

**Abbreviation:** DE 68

## DE 69 (Settlement Institution Country Code)

DE 69 (Settlement Institution Country Code) is the code of the country where the settlement institution is located.

**Plural Form:** DE 69 (Settlement Institution Country Code)

**Abbreviation:** DE 69

## DE 70 (Network Management Information Code)

DE 70 (Network Management Information Code) identifies network status.

**Plural Form:** DE 70 (Network Management Information Code)

**Abbreviation:** DE 70

## DE 71 (Message Number)

DE 71 (Message Number) is a sequential, cyclic number the message initiator assigns to a message. Message Number is used to monitor the integrity of interchange.

**Plural Form:** DE 71 (Message Number)

**Abbreviation:** DE 71

## DE 72 (Message Number Last)

DE 72 (Message Number Last) is a sequential, cyclic number the message initiator assigns to a message, used to monitor the integrity of interchange.

**Plural Form:** DE 72 (Message Number Last)

**Abbreviation:** DE 72

## DE 73 (Date, Action)

DE 73 (Date, Action) specifies the date (year, month, and day) of a future action. In addition, a message originator may use it as a static time such as a birthdate.

**Plural Form:** DE 73 (Date, Action)

**Abbreviation:** DE 73

## DE 74 (Credits, Number)

DE 74 (Credits, Number) is the numeric sum of credit transactions processed.

**Plural Form:** DE 74 (Credits, Number)

**Abbreviation:** DE 74

## DE 75 (Credits, Reversal Number)

DE 75 (Credits, Reversal Number) is the sum number of reversal credit transactions.

**Plural Form:** DE 75 (Credits, Reversal Number)

**Abbreviation:** DE 75

## DE 76 (Debits, Number)

DE 76 (Debits, Number) is the sum number of debit transactions processed.

**Plural Form:** DE 76 (Debits, Number)

**Abbreviation:** DE 76

## DE 77 (Debits, Reversal Number)

DE 77 (Debits, Reversal Number) is the sum number of reversal debit transactions.

**Plural Form:** DE 77 (Debits, Reversal Number)

**Abbreviation:** DE 77

## DE 78 (Transfers, Number)

DE 78 (Transfers, Number) is the sum number of all transfer transactions processed.

**Plural Form:** DE 78 (Transfers, Number)

**Abbreviation:** DE 78

## DE 79 (Transfers, Reversal Number)

DE 79 (Transfers, Reversal Number) is the sum number of all transfer reversal transactions processed.

**Plural Form:** DE 79 (Transfers, Reversal Number)

**Abbreviation:** DE 79

## DE 80 (Inquiries, Number)

DE 80 (Inquiries, Number) is the sum number of inquiry transaction requests processed.

**Plural Form:** DE 80 (Inquiries, Number)

**Abbreviation:** DE 80

## DE 81 (Authorizations, Number)

DE 81 (Authorizations, Number) is the sum number of Authorization Request/0100 and Authorization Advice/0120 messages processed.

**Plural Form:** DE 81 (Authorizations, Number)

**Abbreviation:** DE 81

## DE 82 (Credits, Processing Fee Amount)

DE 82 (Credits, Processing Fee Amount) is the sum of all processing fees due to an institution or customer for services associated with handling and routing transactions. This Mastercard definition replaces the ISO standard definition.

**Plural Form:** DE 82 (Credits, Processing Fee Amount)

**Abbreviation:** DE 82

## DE 83 (Credits, Transaction Fee Amount)

DE 83 (Credits, Transaction Fee Amount) is the sum of all transaction fees due to an institution or customer for processing interchange transactions. This Mastercard definition replaces the ISO standard definition.

**Plural Form:** DE 83 (Credits, Transaction Fee Amount)

**Abbreviation:** DE 83

## DE 84 (Debits, Processing Fee Amount)

DE 84 (Debits, Processing Fee Amount) is the sum of all processing fees due from an institution or customer for services associated with handling and routing transactions. This Mastercard definition replaces the ISO standard definition.

**Plural Form:** DE 84 (Debits, Processing Fee Amount)

**Abbreviation:** DE 84

## DE 85 (Debits, Transaction Fee Amount)

DE 85 (Debits, Transaction Fee Amount) is the sum of all transaction fees due from an institution or customer for processing interchange transactions. This Mastercard definition replaces the ISO standard definition.



**Abbreviation:** DE 85

## DE 86 (Credits, Amount)

DE 86 (Credits, Amount) is the sum amount of all credit transactions processed exclusive of any fees.

**Plural Form:** DE 86 (Credits, Amount)

**Abbreviation:** DE 86

## DE 87 (Credits, Reversal Amount)

DE 87 (Credits, Reversal Amount) is the sum amount of reversal credits processed exclusive of any fees.

**Plural Form:** DE 87 (Credits, Reversal Amount)

**Abbreviation:** DE 87

## DE 88 (Debits, Amount)

DE 88 (Debits, Amount) is the sum amount of all debit transactions processed exclusive of any fees.

**Plural Form:** DE 88 (Debits, Amount)

**Abbreviation:** DE 88

## DE 89 (Debits, Reversal Amount)

DE 89 (Debits, Reversal Amount) is the sum amount of reversal debits processed exclusive of any fees.

**Plural Form:** DE 89 (Debits, Reversal Amount)

**Abbreviation:** DE 89

## DE 90 (Original Data Elements)

DE 90 (Original Data Elements) is the data elements in the original message, intended to identify a transaction for correction or reversal.

**Plural Form:** DE 90 (Original Data Elements)

**Abbreviation:** DE 90

## DE 91 (Issuer File Update Code)

DE 91 (Issuer File Update Code) indicates to the system maintaining a file which procedure to follow.

**Plural Form:** DE 91 (Issuer File Update Code)

**Abbreviation:** DE 91

## DE 92 (File Security Code)

DE 92 (File Security Code) is an Issuer File Update security code used to indicate that a message originator is authorized to update a file.

**Plural Form:** DE 92 (File Security Code)

**Abbreviation:** DE 92

## DE 93 (Response Indicator)

DE 93 (Response Indicator) indicates the update action a POS system takes.

**Plural Form:** DE 93 (Response Indicator)

**Abbreviation:** DE 93

## DE 94 (Service Indicator)

DE 94 (Service Indicator) indicates the service a message recipient requires.

**Plural Form:** DE 94 (Service Indicator)

**Abbreviation:** DE 94

## DE 95 (Replacement Amounts)

DE 95 (Replacement Amounts) contains the actual amount subfields necessary to perform a partial or full reversal of a financial transaction.

**Plural Form:** DE 95 (Replacement Amounts)

**Abbreviation:** DE 95

## DE 96 (Message Security Code)

DE 96 (Message Security Code) is a verification between an acceptor and a card issuer that a message is authorized to update or modify a special file.

**Plural Form:** DE 96 (Message Security Code)

**Abbreviation:** DE 96

## DE 97 (Amount, Net Settlement)

DE 97 (Amount, Net Settlement) is the net value of all gross amounts.

**Plural Form:** DE 97 (Amount, Net Settlement)

**Abbreviation:** DE 97

## DE 98 (Payee)

DE 98 (Payee), is the third-party beneficiary in a Payment Transaction.

**Plural Form:** DE 98 (Payee)

**Abbreviation:** DE 98

## DE 99 (Settlement Institution ID Code)

DE 99 (Settlement Institution ID Code) identifies the settlement institution or its agent.

**Plural Form:** DE 99 (Settlement Institution ID Code)

**Abbreviation:** DE 99

## DE 100 (Receiving Institution ID Code)

DE 100 (Receiving Institution ID Code) is the identity of the institution receiving a Request or Advice message in an interchange system if not the same as identified in DE 2 (Primary Account Number [PAN]) or DE 34 (Primary Account Number [PAN], Extended). The Dual Message Authorization System uses DE 100 to determine the destination routing of Administrative/06xx messages. For these messages, DE 33 (Forwarding Institution ID Code) identifies the sender of the message; DE 100 identifies the receiver of the message.

**Plural Form:** DE 100 (Receiving Institution ID Code)

**Abbreviation:** DE 100

## DE 101 (File Name)

DE 101 (File Name) is the actual or abbreviated name of the file that the issuer accesses. DE 101 is used in Issuer File Update/03xx messages to identify the specific name of a Dual Message Authorization System data file or program parameter table that is being updated by a customer's Issuer File Update Request/0302.

**Plural Form:** DE 101 (File Name)

**Abbreviation:** DE 101

## DE 102 (Account ID 1)

DE 102 (Account ID-1) is a series of digits that identify a customer account or relationship. Customers primarily use it for the from account in a transfer transaction. DE 102 may be used in Authorization Request Response/0110 messages to identify the specific from account that the transaction affected. DE 102 may be used for printing on cardholder transaction receipts.

**Plural Form:** DE 102 (Account ID 1)

**Abbreviation:** DE 102

## DE 103 (Account ID 2)

DE 103 (Account ID-2) is a series of digits that identify a customer account or relationship. Customers primarily use it for the "to" account in a transfer transaction.

**Plural Form:** DE 103 (Account ID 2)

**Abbreviation:** DE 103

## DE 104 (Digital Payment Data)

DE 104 (Digital Payment Data) is used in Authorization Request/0100 messages to send digital payment data for e-commerce transactions.

**Plural Form:** DE 104 (Digital Payment Data)

**Abbreviation:** DE 104

## DE 105 (Multi-Use Transaction Identification Data)

DE 105 is alphanumeric, variable length using a tag-length-data encoding scheme to allow for easy addition of new subelements.

**Plural Form:** DE 105 (Multi-Use Transaction Identification Data)

**Abbreviation:** DE 105

## DE 106 (Fleet Service Data)

DE 106 (Fleet Service Data) contains information pertaining to Fleet card transactions and the items being purchased.

**Plural Form:** DE 106 (Fleet Service Data)

**Abbreviation:** DE 106

## DE 107 (Reserved for Mastercard Use)

DE 107 (Reserved for Mastercard Use) is reserved for Mastercard use.

**Plural Form:** DE 107 (Reserved for Mastercard Use)

**Abbreviation:** DE 107

## DE 108 (Additional Transaction Reference Data)

DE 108 (Additional Transaction Reference Data) provides the capability for the acquirers to send sender, receiver, and transaction level data to the issuer in Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions), Mastercard Merchant Presented QR Transactions, and Visa Account Funding Transactions.

**Plural Form:** DE 108 (Additional Transaction Reference Data)

**Abbreviation:** DE 108

## DE 109 (Reserved for ISO Use)

DE 109 (Reserved for ISO Use) is reserved for ISO use.

**Plural Form:** DE 109 (Reserved for ISO Use)

**Abbreviation:** DE 109

## DE 110 (Additional Data-2)

DE 110 (Additional Data-2) is reserved for use based on product type.

**Plural Form:** DE 110 (Additional Data–2)

**Abbreviation:** DE 110

## DE 110 (Encryption Data)

DE 110 (Encryption Data) is used in PIN encryption data and key management encryption data.

**Plural Form:** DE 110 (Encryption Data)

**Abbreviation:** DE 110

## DE 111 (Reserved for ISO Use)

DE 111 (Reserved for ISO Use) is reserved for ISO use.

**Plural Form:** DE 111 (Reserved for ISO Use)

**Abbreviation:** DE 111

## DE 112 (Additional Data [National Use])

DE 112 (Additional Data [National Use]) is reserved for national organizations to define data unique to specific networks or specific programs and services. DE 112 provides other supplemental data in a message when a specific ISO-designated data element is not available. It is a free-format, variable-length, alphanumeric data element used for information on transactions between customers.

**Plural Form:** DE 112 (Additional Data [National Use])

**Abbreviation:** DE 112

## DE 113 (Reserved for National Use)

DE 113 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**Plural Form:** DE 113 (Reserved for National Use)

**Abbreviation:** DE 113

## DE 114 (Reserved for National Use)

DE 114 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**Plural Form:** DE 114 (Reserved for National Use)

**Abbreviation:** DE 114

## DE 115 (Reserved for National Use)

DE 115 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**Plural Form:** DE 115 (Reserved for National Use)

**Abbreviation:** DE 115

## DE 116 (Reserved For National Use)

DE 116 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**Plural Form:** DE 116 (Reserved For National Use)

**Abbreviation:** DE 116

## DE 117 (Reserved for National Use)

DE 117 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**Plural Form:** DE 117 (Reserved for National Use)

**Abbreviation:** DE 117

## DE 118 (Reserved for National Use)

DE 118 (Reserved for National Use) is reserved for national organizations to define data unique to specific networks or specific programs and services.

**Plural Form:** DE 118 (Reserved for National Use)

**Abbreviation:** DE 118

## DE 119 (Additional Data: Private Use 2)

DE 119 (Additional Data: Private Use 2) contains data associated with various Mastercard programs, products, and services. DE 119 is used to convey data that is not related to any other ISO-defined data elements. It is a variable-length data element used for multiple purposes.

**Plural Form:** DE 119 (Additional Data: Private Use 2)

**Abbreviation:** DE 119

## Using DE 113–119 in Administrative 06xx Messages

The following information explains how to use DE 113–119 in Administrative 0600/0610 messages. Mastercard recommends that the contents of DE 113–119 from the 0600 message not be returned in the 0610 message to avoid the maximum message length constraint of 8k bytes.

**Plural Form:** Using DE 113–119 in Administrative 06xx Messages

**Abbreviation:** Using DE 113–119

## DE 120 (Record Data)

DE 120 (Record Data) is a variable-length data element used for transmitting file record data or textual character string data in various message types.

**Plural Form:** DE 120 (Record Data)

**Abbreviation:** DE 120

## DE 121 (Authorizing Agent ID Code)

DE 121 (Authorizing Agent ID Code), when used, must contain the appropriate Mastercard-assigned customer ID number that uniquely identifies the Dual Message Authorization System Stand-In Authorization facility or alternate routing customer responsible for performing Stand-In Authorization on-behalf of the issuer.

**Plural Form:** DE 121 (Authorizing Agent ID Code)

**Abbreviation:** DE 121



## DE 122 (Additional Record Data)

DE 122 is a free-format, variable-length data element used for transmitting file record data in various message types.

**Plural Form:** DE 122 (Additional Record Data)

**Abbreviation:** DE 122

## DE 123 (Receipt Free Text)

DE 123 (Receipt Free Text) only applies to the Swedish Domestic Authorization Switching Service (SASS), Peru, and the Mastercard Installment Payment Service. For SASS, DE 123 contains information to be printed on a receipt (not displayed on the terminal screen) for balance inquiry and ATM transactions (where DE 3 [Processing Code] is value 01 [Withdrawal] or value 30 [Balance Inquiry]). For Peru and Mastercard Installment Payment Service, DE 123 contains a text message to be printed on point-of-sale (POS) sales receipts.

**Plural Form:** DE 123 (Receipt Free Text)

**Abbreviation:** DE 123

## DE 124 (Member-Defined Data)

DE 124 (Member-defined Data) is described for general use and Prepaid Voucher, Brazil Maestro only, Bill Payment at the ATM only, and Mastercard Digital Enablement Service use.

**Plural Form:** DE 124 (Member-Defined Data)

**Abbreviation:** DE 124

## DE 125 (New PIN Data)

DE 125 (New PIN Data) consists of a binary block containing a derived encrypted value calculated from the new PIN introduced by the cardholder at the ATM offering the PIN change service.

**Plural Form:** DE 125 (New PIN Data)

**Abbreviation:** DE 125

## DE 126 (Private Data)

DE 126 (Private Data) is reserved for future use.

**Abbreviation:** DE 126

## DE 127 (Private Data)

DE 127 (Private Data) may contain any private-use data that the customer may want to include in a message. Any Authorization System message originator may use DE 127.

**Plural Form:** DE 127 (Private Data)

**Abbreviation:** DE 127

## DE 128 (Message Authentication Code)

DE 128 (Message Authentication Code [MAC]) validates the source and the text of the message between the sender and the receiver.

**Plural Form:** DE 128 (Message Authentication Code)

**Abbreviation:** DE 128

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